



Statistics Reference Guide for UnityCloud

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Affirmed Networks Inc.

A Microsoft Company

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Summary

Metric Name	Type	Synopsis
amfcc_n1_init_registration_accept_rate	counter	Initial registration accept rate
amfcc_n1_init_registration_request_rate	counter	Initial registration request rate

amfcc_n1_init_registration_accept_rate

Description

The initial registration accept rate. This counter indicates the average initial registration accept rate on the AMF in the last 5 minutes

Type

counter

Prometheus Expression

$$\frac{((\text{sum}(\text{increase}(\text{amfcc_n1_init_registration_accept}[5\text{m}]))) >= 0 \text{ or on() vector}(0))}{((\text{sum}(\text{increase}(\text{amfcc_n1_init_registration_accept}[5\text{m}]))) >= 0 \text{ or on() vector}(0)) + ((\text{sum}(\text{increase}(\text{amfcc_n1_init_registration_reject}[5\text{m}]))) >= 0 \text{ or on() vector}(0)))} * 100$$

amfcc_n1_init_registration_request_rate

Description

The rate of initial registration requests per second. This counter indicates the average initial registration requests per second received by the AMF in the last 5 minutes

Type

counter

Prometheus Expression

$$\text{sum}(\text{rate}(\text{amfcc_n1_init_registration_request}[5\text{m}]))$$

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- [Summary](#)
- [amfn2_n2_uplink_message_rate](#)

Summary

Metric Name	Type	Synopsis
amfn2_n2_uplink_message_rate	counter	Rate of N2 uplink messages

amfn2_n2_uplink_message_rate

Description

The rate of N2 uplink messages per second. This counter indicates the average N2 uplink messages per second received by the AMF in the last 5 minutes

Type

counter

Prometheus Expression

sum(rate(amfn2_n2_uplink_message[5m]))

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Summary

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_put_requests	Counter	The total number of ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_requests_sent	Counter	The total number of ETCD Client PUT Requests sent.
ETCD Client Metrics	etcd_put_successes	Counter	The total number of successful ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_failures	Counter	The total number of failed ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_with_lease_requests	Counter	The total number of ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_requests_sent	Counter	The total number of ETCD Client PUT with Lease Requests sent.
ETCD Client Metrics	etcd_put_with_lease_successes	Counter	The total number of successful ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_failures	Counter	The total number of failed ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_range_requests	Counter	The total number of ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_requests_sent	Counter	The total number of ETCD Client Range Requests sent.
ETCD Client Metrics	etcd_range_successes	Counter	The total number of successful ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_failures	Counter	The total number of failed ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_delete_requests	Counter	The total number of ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_requests_sent	Counter	The total number of ETCD Client Range Delete Requests sent.
ETCD Client Metrics	etcd_range_delete_successes	Counter	The total number of successful ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_failures	Counter	The total number of failed ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_txn_requests	Counter	The total number of ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_requests_sent	Counter	The total number of ETCD Client Txn Requests sent.
ETCD Client Metrics	etcd_txn_successes	Counter	The total number of successful ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_failures	Counter	The total number of failed ETCD Client Txn Requests.
ETCD Client Metrics	etcd_watch_requests	Counter	The total number of ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_requests_sent	Counter	The total number of ETCD Client Watch Requests sent.
ETCD Client Metrics	etcd_watch_successes	Counter	The total number of successful ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_failures	Counter	The total number of failed ETCD Client Watch Requests.
ETCD Client Metrics	etcdwatcher_requests	Counter	The total number of ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_requests_sent	Counter	The total number of ETCD Watcher Requests sent.
ETCD Client Metrics	etcdwatcher_successes	Counter	The total number of successful ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_failures	Counter	The total number of failed ETCD Watcher Requests.

ETCD Client Metrics

- [etcd_put_requests](#)

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests.

Labels

keyPrefix

etcd_put_requests_sent
Type
64-bit Counter
Description
The total number of ETCD Client PUT Requests sent.
Labels
<i>keyPrefix</i>
etcd_put_successes
Type
64-bit Counter
Description
The total number of successful ETCD Client PUT Requests.
Labels
<i>keyPrefix</i>
etcd_put_failures
Type
64-bit Counter
Description
The total number of failed ETCD Client PUT Requests.
Labels
<i>keyPrefix, failureReason</i>
etcd_put_with_lease_requests
Type
64-bit Counter
Description
The total number of ETCD Client PUT with Lease Requests.
Labels
<i>keyPrefix</i>
etcd_put_with_lease_requests_sent
Type
64-bit Counter
Description
The total number of ETCD Client PUT with Lease Requests sent.
Labels
<i>keyPrefix</i>
etcd_put_with_lease_successes
Type
64-bit Counter
Description

The total number of successful ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT with Lease Requests.

Labels

keyPrefix, failureReason

- etcd_range_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests sent.

Labels

keyPrefix

- etcd_range_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Requests.

Labels

keyPrefix, failureReason

- etcd_range_delete_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests.

Labels

keyPrefix

- etcd_range_delete_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests sent.

Labels

keyPrefix

- etcd_range_delete_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Delete Requests.

Labels

keyPrefix

- etcd_range_delete_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Delete Requests.

Labels

keyPrefix, failureReason

- etcd_txn_requests

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests.

Labels

keyPrefix

- etcd_txn_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests sent.

Labels

keyPrefix

• etcd_txn_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Txn Requests.

Labels

keyPrefix

• etcd_txn_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Txn Requests.

Labels

keyPrefix, failureReason

• etcd_watch_requests

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests.

Labels

keyPrefix

• etcd_watch_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests sent.

Labels

keyPrefix

• etcd_watch_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Watch Requests.

Labels

keyPrefix

• etcd_watch_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Watch Requests.

Labels

keyPrefix, failureReason

• etcdwatcher_requests

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests.

Labels

keyPrefix

• etcdwatcher_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests sent.

Labels

keyPrefix

• etcdwatcher_successes

Type

64-bit Counter

Description

The total number of successful ETCD Watcher Requests.

Labels

keyPrefix

• etcdwatcher_failures

Type

64-bit Counter

Description

The total number of failed ETCD Watcher Requests.

Labels

keyPrefix, failureReason

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- [HTTP Punting Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
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Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_sent	Counter	The total number of HTTP requests successfully punted.
HTTP Punting Metrics	punting_http_requests_failed	Counter	The total number of HTTP requests punting failed.
HTTP Punting Metrics	punting_http_responses_received	Counter	The total number of HTTP responses received for the requests punted.
HTTP Punting Metrics	punting_http_responses_forwarded	Counter	The total number of HTTP responses forwarded after punting transaction completion.
HTTP Punting Metrics	punting_http_requests_aborted	Counter	The total number of HTTP requests aborted.
HTTP Punting Metrics	punting_http_request_duration_seconds	Histogram	Punting operation duration in seconds.

HTTP Punting Metrics

- [punting_http_requests_sent](#)

Type

64-bit Counter

Description

The total number of HTTP requests successfully punted.

Labels

message_type

- [punting_http_requests_failed](#)

Type

64-bit Counter

Description

The total number of HTTP requests for which the punting failed.

Labels

message_type

- [punting_http_responses_received](#)

Type

64-bit Counter

Description

The total number of HTTP responses received for the requests punted.

Labels

message_type, status_code

- [punting_http_responses_forwarded](#)

Type

64-bit Counter

Description

The total number of HTTP responses forwarded after punting transaction completion.

Labels

message_type, status_code

- [punting_http_requests_aborted](#)

Type

64-bit Counter

Description

The total number of HTTP requests aborted.

Labels

message_type
- [punting_http_request_duration_seconds](#)

Type

64-bit Histogram

Description

The punting transaction duration is measured from the time the HTTP request is sent out to the time the response is received for the punting request.

Labels

message_type

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

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- [Summary](#)
- [In memory bufferstatistics](#)

Summary

Group	Metric Name	Type	Synopsis
In memory bufferstatistics	inmemory_buffer_pdu_added	Counter	Count of the PDUs added to the inmemorybuffer
In memory bufferstatistics	inmemory_buffer_pdu_removed	Counter	Count of the PDUs removed from the inmemorybuffer
In memory bufferstatistics	inmemory_buffer_pdu_dropped	Counter	Count of the PDUs dropped from the inmemorybuffer
In memory bufferstatistics	inmemory_buffer_pdu_remove_all	Counter	Count of times removeAllPdu is called for the inmemorybuffer

In memory bufferstatistics

- [inmemory_buffer_pdu_added](#)

Type

64-bit Counter

Description

Count of the PDUs added to the inmemorybuffer

Labels

did, position
- [inmemory_buffer_pdu_removed](#)

Type

64-bit Counter

Description

Count of the PDUs removed from the inmemorybuffer

Labels

did, position

- inmemory_buffer_pdu_dropped

Type

64-bit Counter

Description

Count of the PDUs dropped from the inmemorybuffer

Labels

did, position

- inmemory_buffer_pdu_remove_all

Type

64-bit Counter

Description

Count of times removeAllPdu is called for the inmemorybuffer

Labels

did, reason

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- [LI PPE Statistics](#)
- [Reports API Statistics](#)
- [ETCD Statistics](#)

Summary

Group	Metric Name	Type	Synopsis
LI PPE Statistics	li_ppe_tcp_connection_attempts	Counter	Number of tcp connection open attempts from liapp to MDF
LI PPE Statistics	li_ppe_tcp_connection_success	Counter	Number of tcp connection open requests successful from liapp to MDF
LI PPE Statistics	li_ppe_tcp_connection_failures	Counter	Number of tcp connection open requests that failed from liapp to MDF
LI PPE Statistics	li_ppe_tcp_connection_status	Counter	Number of times a connection has flipped between up and down
LI PPE Statistics	li_ppe_tcp_socket_errors	Counter	TCP socket related errors
LI PPE Statistics	li_ppe_keep_alive_attempts	Counter	Keepalive PDUs sent towards MDF
LI PPE Statistics	li_ppe_keep_alive_success	Counter	Successful Keepalive PDUs sent towards MDF
LI PPE Statistics	li_ppe_keep_alive_failures	Counter	Failed Keepalive PDUs sent towards MDF
LI PPE Statistics	li_ppe_keep_alive_waitingack	Counter	LiApp Keepalive waiting acks received from MDF
LI PPE Statistics	li_ppe_switchback_timer_started	Counter	Number of times switchback timer is started for a DID towards MDF
LI PPE Statistics	li_ppe_switchback_timer_cancelled	Counter	Number of times switchback timer is cancelled for a DID towards MDF
LI PPE Statistics	li_ppe_switchback_timer_expired	Counter	Number of times switchback timer is expired for a DID towards MDF
LI PPE Statistics	li_ppe_primary_to_secondary_connection_switch	Counter	Number of times connection has switched from primary to secondary for a DID towards MDF
LI PPE Statistics	li_ppe_secondary_to_primary_connection_switch	Counter	Number of times connection has switched from secondary to primary for a DID towards MDF
LI PPE Statistics	li_ppe_buffered_bytes	Counter	Bytes buffered for a DID
LI PPE Statistics	li_ppe_buffered_bytes_sent	Counter	Bytes that were buffered for a DID and then sent to MDF
LI PPE Statistics	li_ppe_buffered_bytes_dropped	Counter	Bytes that were buffered for a DID and then dropped
LI PPE Statistics	li_ppe_buffered_pdus	Counter	PDUs buffered for a DID

Group	Metric Name	Type	Synopsis
LI PPE Statistics	li_ppe_buffered_pdus_sent	Counter	PDUs that were buffered for a DID and then sent to MDF
LI PPE Statistics	li_ppe_buffered_pdus_dropped	Counter	PDUs that were buffered for a DID and then dropped
LI PPE Statistics	li_ppe_loopback_not_found	Gauge	Set to one if loopback interface not present to establish connection to MDF
Reports API Statistics	li_ppe_report_task_issue_request_attempts	Counter	Report task issue requests received on report api interface
Reports API Statistics	li_ppe_report_task_issue_request_success	Counter	Report task issue requests successful on report api interface
Reports API Statistics	li_ppe_report_task_issue_request_failures	Counter	Report task issue requests failed on report api interface
Reports API Statistics	li_ppe_report_task_issue_internal_failures	Counter	Report task issue requests failed due to internal problems such as memory pressure or unknown XID
Reports API Statistics	li_ppe_report_task_stat_request_attempts	Counter	Report task stat requests received on report api interface
Reports API Statistics	li_ppe_report_task_stat_request_success	Counter	Report task stat requests successful on report api interface
Reports API Statistics	li_ppe_report_task_stat_request_failures	Counter	Report task stat requests failed on report api interface
Reports API Statistics	li_ppe_report_destination_issue_request_attempts	Counter	Report report destination issue requests received on report api interface
Reports API Statistics	li_ppe_report_destination_issue_request_success	Counter	Report report destination issue requests successful on report api interface
Reports API Statistics	li_ppe_report_destination_issue_request_failures	Counter	Report report destination issue requests failed on report api interface
ETCD Statistics	li_ppe_etcd_range_request_attempts	Counter	number of range requests sent out to ETCD
ETCD Statistics	li_ppe_etcd_range_request_success	Counter	number of successful range requests from LiApp
ETCD Statistics	li_ppe_etcd_range_request_failures	Counter	number of failed range requests from LiApp
ETCD Statistics	li_ppe_etcd_watch_request_attempts	Counter	total number of watch requests from LiApp
ETCD Statistics	li_ppe_etcd_watch_request_success	Counter	total number of successful etcd requests from LiApp
ETCD Statistics	li_ppe_etcd_watch_request_failures	Counter	total number of failed watch requests from LiApp

LI PPE Statistics

- li_ppe_tcp_connection_attempts

Type

64-bit Counter

Description

Count of number of tcp connections open request attempted from liapp to MDF

Labels

DID, deliveryType, peerType

- li_ppe_tcp_connection_success

Type

64-bit Counter

Description

Count of number of tcp connection open requests successful from liapp to MDF

Labels

DID, deliveryType, status, peerType

- li_ppe_tcp_connection_failures

Type

64-bit Counter

Description

Count of number of tcp connection open requests that failed from liapp to MDF

Labels

DID, deliveryType, status, peerType

• li_ppe_tcp_connection_status

Type

64-bit Counter

Description

Number of times a connection has flipped between up and down

Labels

DID, deliveryType, status, peerType

• li_ppe_tcp_socket_errors

Type

64-bit Counter

Description

Number of times any tcp socket related errors have occurred

Labels

DID, deliveryType, errortype, peerType

• li_ppe_keep_alive_attempts

Type

64-bit Counter

Description

Number of attempts made to send Keepalive PDUs towards MDF

Labels

DID, deliveryType, peerType

• li_ppe_keep_alive_success

Type

64-bit Counter

Description

Number of successful Keepalive PDUs sent towards MDF

Labels

DID, deliveryType, keepalive_status

• li_ppe_keep_alive_failures

Type

64-bit Counter

Description

Number of LiApp Keepalive PDUs sent towards MDF failed

Labels

DID, deliveryType, keepalive_status

- li_ppe_keep_alive_waitingack

Type

64-bit Counter

Description

Number of Keepalive waiting acks received from MDF

Labels

DID, deliveryType, keepalive_status

- li_ppe_switchback_timer_started

Type

64-bit Counter

Description

Number of times switchback timer is started for a DID towards MDF

Labels

- li_ppe_switchback_timer_cancelled

Type

64-bit Counter

Description

Number of times switchback timer is cancelled for a DID towards MDF

Labels

- li_ppe_switchback_timer_expired

Type

64-bit Counter

Description

Number of times switchback timer is expired for a DID towards MDF

Labels

- li_ppe_primary_to_secondary_connection_switch

Type

64-bit Counter

Description

Number of times connection has switched from primary to secondary for a DID towards MDF

Labels

- li_ppe_secondary_to_primary_connection_switch

Type

64-bit Counter

Description

Number of times connection has switched from secondary to primary for a DID towards MDF

Labels

- li_ppe_buffered_bytes

Type

64-bit Counter

Description

Number of bytes buffered for a DID due to MDF being unavailable or congested

Labels

DID, deliveryType

- li_ppe_buffered_bytes_sent

Type

64-bit Counter

Description

Number of bytes that were buffered for a DID and then successfully sent to MDF

Labels

DID, deliveryType

- li_ppe_buffered_bytes_dropped

Type

64-bit Counter

Description

Number of bytes that were buffered for a DID and then dropped due to buffer space being exceeded

Labels

DID, deliveryType

- li_ppe_buffered_pdus

Type

64-bit Counter

Description

Number of PDUs buffered for a DID due to MDF being unavailable or congested

Labels

DID, deliveryType

- li_ppe_buffered_pdus_sent

Type

64-bit Counter

Description

Number of PDUs that were buffered for a DID and then successfully sent to MDF

Labels

DID, deliveryType

- li_ppe_buffered_pdus_dropped

Type

64-bit Counter

Description

Number of PDUs that were buffered for a DID and then dropped due to buffer space being exceeded

Labels

DID, deliveryType

- li_ppe_loopback_not_found

Type

64-bit Gauge

Description

Indicates if loopback interface is not present to establish connection to MDF

Labels

DID, deliveryType, peerType

Reports API Statistics

- li_ppe_report_task_issue_request_attempts

Type

64-bit Counter

Description

Total number of report task issue requests received on report api interface

Labels

- li_ppe_report_task_issue_request_success

Type

64-bit Counter

Description

Total number of report task issue requests successful on report api interface

Labels

- li_ppe_report_task_issue_request_failures

Type

64-bit Counter

Description

Total number of report task issue requests failed on report api interface

Labels

- li_ppe_report_task_issue_internal_failures

Type

64-bit Counter

Description

Total number of report task issue failed due to internal issues such as memory pressure

Labels

• li_ppe_report_task_stat_request_attempts Type 64-bit Counter Description Total number of report task stat requests received on report api interface Labels
• li_ppe_report_task_stat_request_success Type 64-bit Counter Description Total number of report task stat requests successful on report api interface Labels
• li_ppe_report_task_stat_request_failures Type 64-bit Counter Description Total number of report activate task requests failed on report api interface Labels
• li_ppe_report_destination_issue_request_attempts Type 64-bit Counter Description Total number of report destination issue requests received on report api interface Labels
• li_ppe_report_destination_issue_request_success Type 64-bit Counter Description Total number of report report destination issue requests successful on report api interface Labels
• li_ppe_report_destination_issue_request_failures Type 64-bit Counter Description Total number of report report destination issue requests failed on report api interface Labels

ETCD Statistics

- li_ppe_etcd_range_request_attempts

Type
64-bit Counter
Description
Total number of range requests sent out to ETCD
Labels

- li_ppe_etcd_range_request_success

Type
64-bit Counter
Description
Total number of successful range requests
Labels
<i>responsemessage</i>

- li_ppe_etcd_range_request_failures

Type
64-bit Counter
Description
Total number of failed range requests
Labels
<i>responsemessage</i>

- li_ppe_etcd_watch_request_attempts

Type
64-bit Counter
Description
Total number of watch requests sent out to ETCD
Labels

- li_ppe_etcd_watch_request_success

Type
64-bit Counter
Description
Total number of successful watch requests
Labels
<i>responsemessage</i>

- li_ppe_etcd_watch_request_failures

Type
64-bit Counter
Description

Total number of failed watch requests

Labels

responsemessage

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- [LiCmn statistics](#)
- [Key Ciphering Statistics](#)

Summary

Group	Metric Name	Type	Synopsis
LiCmn statistics	svcmgr_service_available_ind_notification	Counter	How many times, Service Available Indication message received
LiCmn statistics	svcmgr_service_available_ind_parse_errs	Counter	How many times, Service Available Indication parse error happened
LiCmn statistics	svcmgr_service_available_ind_notification_ack	Counter	How many times, Service Available Indication message is acknowledged
LiCmn statistics	svcmgr_service_unavailable_ind_notification	Counter	How many times, Service Unavailable Indication message
LiCmn statistics	svcmgr_service_unavailable_ind_parse_errs	Counter	How many times, Service Unavailable Indication parse error happened
LiCmn statistics	svcmgr_service_unavailable_ind_notification_ack	Counter	How many times, Service Unavailable Indication message is acknowledged
LiCmn statistics	svcmgr_publish_msg_sent	Counter	How many times, Publication message sent
LiCmn statistics	svcmgr_publish_msg_success	Counter	How many times, Publication message success response received
LiCmn statistics	svcmgr_publish_msg_failure	Counter	How many times, failure received for publication message
LiCmn statistics	svcmgr_subscribe_msg_sent	Counter	How many times, Subscribe message sent
LiCmn statistics	svcmgr_subscribe_msg_success	Counter	How many times, subscribe message success
LiCmn statistics	svcmgr_subscribe_msg_failure	Counter	How many times, failure received for subscribe message
LiCmn statistics	svcmgr_unsubscribe_msg_sent	Counter	How many times, unsubscribe message sent
LiCmn statistics	svcmgr_unsubscribe_msg_failure	Counter	How many times, failure received for unsubscribe message
LiCmn statistics	svcmgr_config_req_sent	Counter	How many times, config request sent
LiCmn statistics	svcmgr_config_req_failure	Counter	How many times, failure received for config request
LiCmn statistics	svcmgr_config_req_success	Counter	How many times, success received for config request
LiCmn statistics	svcmgr_config_register_response	Counter	How many times, config register response received
LiCmn statistics	svcmgr_config_register_response_ack	Counter	How many times, config register response is acknowledged
LiCmn statistics	svcmgr_config_register_response_parse_err	Counter	How many times, parse error happened for config register response
LiCmn statistics	svcmgr_config_update_notification	Counter	How many times, config update notification received
LiCmn statistics	svcmgr_config_update_notification_response	Counter	How many times, config update response
LiCmn statistics	svcmgr_intf_schema_update_ind_notification	Counter	How many times, interface schema update indication message received
LiCmn statistics	svcmgr_intf_schema_update_ind_notification_ack	Counter	How many times, interface schema update indication message is acknowledged
LiCmn statistics	svcmgr_intf_schema_update_ind_parse_errs	Counter	How many times, interface schema update indication parse error happened
Key Ciphering Statistics	li_cmn_cipher_object_registered	Counter	Report number of cipher objects registered
Key Ciphering Statistics	li_cmn_secret_key_rotated	Counter	Report number of times secret key has been changed
Key Ciphering Statistics	li_cmn_enciphering_failed	Counter	Report number of times enciphering has failed
Key Ciphering Statistics	li_cmn_deciphering_failed	Counter	Report number of times deciphering has failed
Key Ciphering Statistics	li_cmn_key_version_not_available	Counter	Report number of times when the key corresponding to the key version is not available.

LiCmn statistics

- `svcmgr_service_available_ind_notification`

Type
64-bit Counter
Description
Number of Times Service Available Indication notification received
Labels
<i>servicename</i>

- `svcmgr_service_available_ind_parse_errs`

Type
64-bit Counter
Description
Number of Times Service Available Indication parse error happened
Labels

- `svcmgr_service_available_ind_notification_ack`

Type
64-bit Counter
Description
Number of Times Service Available Indication message is acknowledged
Labels
<i>servicename</i>

- `svcmgr_service_unavailable_ind_notification`

Type
64-bit Counter
Description
Number of Times Service Unavailable Indication notification received
Labels
<i>servicename</i>

- `svcmgr_service_unavailable_ind_parse_errs`

Type
64-bit Counter
Description
Number of Times Service Unavailable Indication parse error happened
Labels

- `svcmgr_service_unavailable_ind_notification_ack`

Type
64-bit Counter
Description
Number of Times Service Unavailable Indication message is acknowledged

Labels

servicename

- **svcmgr_publish_msg_sent**

Type

64-bit Counter

Description

Number of Times Publication message sent

Labels

servicename, retry

- **svcmgr_publish_msg_success**

Type

64-bit Counter

Description

Number of Times Publication message success response received

Labels

servicename, causecode, retry

- **svcmgr_publish_msg_failure**

Type

64-bit Counter

Description

Number of Times failure received for publication message

Labels

servicename, causecode, retry

- **svcmgr_subscribe_msg_sent**

Type

64-bit Counter

Description

Number of Times Subscribe message sent

Labels

servicename, retry

- **svcmgr_subscribe_msg_success**

Type

64-bit Counter

Description

Number of Times Send subscribe message success response received

Labels

servicename, causecode, retry

- **svcmgr_subscribe_msg_failure**

Type

64-bit Counter

Description

Number of time failure received for subscribe message

Labels

servicename, causecode, retry

• **svcmgr_unsubscribe_msg_sent**

Type

64-bit Counter

Description

Number of Times unsubscribe message sent

Labels

servicename

• **svcmgr_unsubscribe_msg_failure**

Type

64-bit Counter

Description

Number of Times failure received for unsubscribe message

Labels

servicename

• **svcmgr_config_req_sent**

Type

64-bit Counter

Description

Number of Times config request sent

Labels

retry

• **svcmgr_config_req_failure**

Type

64-bit Counter

Description

Number of Times failure received for config request

Labels

causecode, retry

• **svcmgr_config_req_success**

Type

64-bit Counter

Description

Number of Times success received for config request

Labels

causecode, retry

- `svcmgr_config_register_response`

Type

64-bit Counter

Description

Number of Times config register response received

Labels

statuscode

- `svcmgr_config_register_response_ack`

Type

64-bit Counter

Description

Number of Times config register response is acknowledged

Labels

- `svcmgr_config_register_response_parse_err`

Type

64-bit Counter

Description

Number of Times parse error happened for config register response

Labels

- `svcmgr_config_update_notification`

Type

64-bit Counter

Description

Number of Times config update notification received

Labels

- `svcmgr_config_update_notification_response`

Type

64-bit Counter

Description

Number of Times config update response sent

Labels

- `svcmgr_intf_schema_update_ind_notification`

Type

64-bit Counter

Description

Number of Times interface schema update indication message received

Labels

servicename

- svcmgr_intf_schema_update_ind_notification_ack

Type

64-bit Counter

Description

Number of Times interface schema update indication message is acknowledged

Labels

servicename

- svcmgr_intf_schema_update_ind_parse_errs

Type

64-bit Counter

Description

Number of Times Service interface schema update indication parse error happened

Labels

Key Cipherring Statistics

- li_cmnm_cipher_object_registered

Type

64-bit Counter

Description

Total number of cipher objects registered

Labels

key_version

- li_cmnm_secret_key_rotated

Type

64-bit Counter

Description

Total number of times secret keys has been rotated

Labels

- li_cmnm_enciphering_failed

Type

64-bit Counter

Description

Total number of times enciphering has failed

Labels

key_version

- li_cmnm_deciphering_failed

Type

64-bit Counter
Description
Total number of times deciphering has failed
Labels
key_version
li_cmn_key_version_not_available
Type
64-bit Counter
Description
Total number of times key corresponding to the key version is not available.
Labels
key_version

Table of Contents

- Summary - NfMonitor statistics - DebugNfMonitorCmn - NfManagement statistics - Debug statistics - ApiServer statistics - Feature Functionality statistics - Consul statistics

Summary

Group	Metric Name	Type	Synopsis
NfMonitor statistics	nfmonitor_unstable_nf_counter	Counter	Howmany times, the Nf is found to be Unstable
NfMonitor statistics	nfmonitor_force_restart_due_to_nf_unstability	Counter	Numbe of times the VNFRestart triggered due to Unstability of the NF
NfMonitor statistics	nfmonitor_cumulative_readiness_failure_count	Counter	Numbe of times Service sound to have insuffucient count due to Readiness probe
NfMonitor statistics	nfmonitor_cumulative_readiness_failure_alerts	Counter	Numbe of times Service sound to have insuffucient count due to Readiness probe
NfMonitor statistics	nfmonitor_cumulative_k8api_get_query_failures	Counter	Numbe of times the k8 get Api query failures
DebugNfMonitorCmn	nfmonitor_force_restart_counter	Counter	Number of times force-restart triggered
NfManagement statistics	nfmonitor_nfmgmt_service_request_attempts	Counter	Howmany times, the Nf is attempting to send NRF Requests
NfManagement statistics	nfmonitor_nfmgmt_service_request_success	Counter	Numbe of times Success response is received for the requests to NRF
NfManagement statistics	nfmonitor_nfmgmt_service_request_failures	Counter	Numbe of times Failure response is received for the requests to NRF
NfManagement statistics	nfmonitor_nfmgmt_aborted_request_msgs	Counter	Numbe of times the requests to NRF is Aborted/Cancelled
Debug statistics	nfmonitor_vnf_restart_trigger	Counter	Howmany times, the Force Restart is received
Debug statistics	nfmonitor_fsm_state_changes	Counter	The statistics for the the FSM state changes
Debug statistics	nfmonitor_deactivation_triggers	Counter	Numbe of times the NF deactivation trigger is received
Debug statistics	nfmonitor_critical_service_up_notification	Counter	Numbe of times the Critical Service Up Notification received
Debug statistics	nfmonitor_critical_service_down_notification	Counter	Numbe of times the Critical Service Down Notification received
Debug statistics	nfmonitor_critical_services_overall_up_notification	Counter	Numbe of times the Up notification received when all monitored Critical Services are available

Group	Metric Name	Type	Synopsis
Debug statistics	nfmonitor_critical_services_overall_down_notification	Counter	Numbe of times the Down notification received when at least one Critical Service was unavailable
Debug statistics	nfmonitor_hysterisis_timer_started	Counter	Numbe of times the Hysterisis Timer is started due to missing Critical Service
Debug statistics	nfmonitor_hysterisis_timer_stopped	Counter	Numbe of times the Hysterisis Timer is stopped due to Critical Service become available
Debug statistics	nfmonitor_hysterisis_timer_expired	Counter	Numbe of times the Hysterisis Timer is expired due to missing Critical Service
Debug statistics	nfmonitor_personality_update_triggers	Counter	Numbe of times the Nf Personality update is received
Debug statistics	nfmonitor_maintenance_trigger_received	Counter	Numbe of times the Maintenance Mode is activated
Debug statistics	nfmonitor_profile_update_triggers	Counter	Numbe of times the Nf Profile Update happened
Debug statistics	nfmonitor_critical_service_down_on_register_entry	Counter	Numbe of times the Critical Services were down while starting with RegisterFsmState
ApiServer statistics	nfmonitor_deployment_create_watch_request	Counter	Howmany times, the deployment watch requests are created
ApiServer statistics	nfmonitor_deployment_delete_watch_request	Counter	Howmany times, the deployment watch requests are deleted
ApiServer statistics	nfmonitor_deployment_watch_notification_success	Counter	Howmany times, the watch notifications are successful
ApiServer statistics	nfmonitor_deployment_watch_notification_error	Counter	Howmany times, the watch notifications are failed
ApiServer statistics	nfmonitor_limit_range_query_attempts	Counter	Howmany times, the limit-range query attempts are made
ApiServer statistics	nfmonitor_limit_range_query_failures	Counter	Howmany times, the limit-range query attempts are failed
ApiServer statistics	nfmonitor_limit_range_query_success	Counter	Howmany times, the limit-range query attempts are succeeded
ApiServer statistics	nfmonitor_hpa_query_attempts	Counter	Howmany times, the Hpa query attempts are made
ApiServer statistics	nfmonitor_hpa_query_failures	Counter	Howmany times, the Hpa query attempts are failed
ApiServer statistics	nfmonitor_hpa_query_success	Counter	Howmany times, the Hpa query attempts are succeeded
ApiServer statistics	nfmonitor_readiness_probe_query_attempts	Counter	Howmany times, the Readiness Probe query attempts are made
ApiServer statistics	nfmonitor_readiness_probe_query_failures	Counter	Howmany times, the Readiness Probe query attempts are failed
ApiServer statistics	nfmonitor_readiness_probe_query_success	Counter	Howmany times, the Readiness Probe query attempts are succeeded
Feature Functionality statistics	nfmonitor_critical_service_frequent_restart	Counter	Howmany times, minimum count is not met for the stable Critical Services
Feature Functionality statistics	nfmonitor_alarm_due_to_frequent_restart	Counter	Howmany times alarm raised, due to minimum count is not met for the stable Critical Services
Feature Functionality statistics	nfmonitor_force_restart_due_to_frequent_restart	Counter	Howmany times, force restart triggerd because of the unstable Critical Services
Feature Functionality statistics	nfmonitor_critical_service_readiness_failure	Counter	Howmany times, minimum count is not met for the readiness succeeded Critical Services
Feature Functionality statistics	nfmonitor_alarm_due_to_readiness_failure	Counter	Alarm raised, due to minimum count is not met for the readiness succeeded Critical Services
Feature Functionality statistics	nfmonitor_force_restart_due_to_readiness_failure	Counter	Force restart triggerd because of Critical Service Readiness failure
Consul statistics	nfmonitor_publish_status_into_consul	Counter	Number of times Publish message is sent to Consul
Consul statistics	nfmonitor_consul_read_attempts	Counter	Number of times Consul read attempts
Consul statistics	nfmonitor_consul_read_failures	Counter	Number of times Consul read attempts failed
Consul statistics	nfmonitor_consul_read_success	Counter	Number of times Consul read attempts succeeded

NfMonitor statistics

- [nfmonitor_unstable_nf_counter](#)

Type
64-bit Counter
Description

Number of Times Nf is found Unstable

Labels

- nfmonitor_force_restart_due_to_nf_unstability

Type

64-bit Counter

Description

The number of times the Force Restart triggered due to NF-Unstability

Labels

- nfmonitor_cumulative_readiness_failure_count

Type

64-bit Counter

Description

The number of times the service is found to be failing the Readiness probe

Labels

critical_service_name, pod_name

- nfmonitor_cumulative_readiness_failure_alerts

Type

64-bit Counter

Description

Total number of EventAlerts generated due to Readiness probe failure

Labels

critical_service_name, pod_name

- nfmonitor_cumulative_k8api_get_query_failures

Type

64-bit Counter

Description

Total number of k8 api get pod query failures

Labels

critical_service_name, pod_name

DebugNfMonitorCmn

- nfmonitor_force_restart_counter

Type

64-bit Counter

Description

Number of times force-restart triggered

Labels

agent, name

NfManagement statistics

- nfmonitor_nfmgmt_service_request_attempts

Type
64-bit Counter
Description
Number of Requests sent to NRF
Labels

instance_id, msg_type, method, retry

- nfmonitor_nfmgmt_service_request_success

Type
64-bit Counter
Description
The number of times the Request to NRF is Successful
Labels

instance_id, msg_type, method, retry, status_code

- nfmonitor_nfmgmt_service_request_failures

Type
64-bit Counter
Description
The number of times the Request to NRF is Failure
Labels

instance_id, msg_type, method, retry, status_code

- nfmonitor_nfmgmt_aborted_request_msgs

Type
64-bit Counter
Description
The number of times the Request to NRF is Aborted
Labels

instance_id, msg_type

Debug statistics

- nfmonitor_vnf_restart_trigger

Type
64-bit Counter
Description
Number of VNF Restart triggers received
Labels

curr_nf_status, nfmgmt_state, nfmonitor_fsm_state, up_count, timestamp, trigger, type

- nfmonitor_fsm_state_changes

Type

64-bit Counter

Description

The FSM State Change events stats

Labels

curr_nf_status, nfmgmt_state, up_count, from_state, to_state

• nfmonitor_deactivation_triggers

Type

64-bit Counter

Description

The number of times the deactivation trigger received

Labels

nf_status, nfmgmt_state, up_count, nfmonitor_fsm_state

• nfmonitor_critical_service_up_notification

Type

64-bit Counter

Description

The number of times the CriticalService Up notification received

Labels

service, criticality, nf_status, nfmgmt_state, up_count, fsm_state

• nfmonitor_critical_service_down_notification

Type

64-bit Counter

Description

The number of times the CriticalService Down notification received

Labels

service, criticality, nf_status, nfmgmt_state, up_count, fsm_state

• nfmonitor_critical_services_overall_up_notification

Type

64-bit Counter

Description

The number of times the Overall CriticalServices Up notification received

Labels

nf_status, nfmgmt_state, up_count, fsm_state

• nfmonitor_critical_services_overall_down_notification

Type

64-bit Counter

Description

The number of times the Overall CriticalServices Down notification received

Labels

nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_hysteresis_timer_started

Type

64-bit Counter

Description

The number of times the Hysteresis Timer is started

Labels

nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_hysteresis_timer_stopped

Type

64-bit Counter

Description

The number of times the Hysteresis Timer is stopped

Labels

nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_hysteresis_timer_expired

Type

64-bit Counter

Description

The number of times the Hysteresis Timer is expired

Labels

nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_personality_update_triggers

Type

64-bit Counter

Description

The number of times the Nf Personality Update is received

Labels

nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_maintenance_trigger_received

Type

64-bit Counter

Description

The number of times the Maintenance Mode is activated

Labels

nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_profile_update_triggers

Type

64-bit Counter

Description

The number of times the Nf Profile Update happened

Labels

nf_status, nfmgmt_state, up_count, fsm_state, update_operation

- nfmonitor_critical_service_down_on_register_entry

Type

64-bit Counter

Description

The number of times the Critical Services were down while starting with RegisterFsmState

Labels

nf_status, up_count

ApiServer statistics

- nfmonitor_deployment_create_watch_request

Type

64-bit Counter

Description

The number of Watch Reqs created for deployment type

Labels

resource

- nfmonitor_deployment_delete_watch_request

Type

64-bit Counter

Description

The number of Watch Reqs deleted for deployment type

Labels

resource

- nfmonitor_deployment_watch_notification_success

Type

64-bit Counter

Description

The number of times Watch notifications are successful

Labels

resource

- nfmonitor_deployment_watch_notification_error

Type

64-bit Counter

Description

The number of times Watch notifications are failed

Labels

resource, reason

- nfmonitor_limit_range_query_attempts

Type

64-bit Counter

Description

The number of Limit Range query attempts

Labels

- nfmonitor_limit_range_query_failures

Type

64-bit Counter

Description

The number of Limit Range query failures

Labels

status_code

- nfmonitor_limit_range_query_success

Type

64-bit Counter

Description

The number of Limit Range query succeeded

Labels

status_code

- nfmonitor_hpa_query_attempts

Type

64-bit Counter

Description

The number of Hpa query attempts

Labels

resource, continue_param

- nfmonitor_hpa_query_failures

Type

64-bit Counter

Description

The number of Hpa query failures

Labels

resource, continue_param, status_code

- nfmonitor_hpa_query_success

Type

64-bit Counter

Description

The number of Hpa query succeeded

Labels

resource, continue_param, status_code

- nfmonitor_readiness_probe_query_attempts

Type

64-bit Counter

Description

The number of Readiness Probe query attempts

Labels

resource, continue_param

- nfmonitor_readiness_probe_query_failures

Type

64-bit Counter

Description

The number of Readiness Probe query failures

Labels

resource, continue_param, status_code

- nfmonitor_readiness_probe_query_success

Type

64-bit Counter

Description

The number of Readiness Probe query succeeded

Labels

resource, continue_param, status_code

Feature Functionality statistics

- nfmonitor_critical_service_frequent_restart

Type

64-bit Counter

Description

Howmany times the minimum required stable CriticalService count is not met

Labels

service

- nfmonitor_alarm_due_to_frequent_restart

Type

64-bit Counter

Description

Howmany times Alarm is raised due to minimum required stable CriticalService count is not met

Labels

service, alarm_status

- nfmonitor_force_restart_due_to_frequent_restart

Type

64-bit Counter

Description

Howmany times Force Restart initiated due to missing stable CriticalService count

Labels

service

- nfmonitor_critical_service_readiness_failure

Type

64-bit Counter

Description

Howmany times the minimum required readiness succeeded CriticalService count is not met

Labels

service

- nfmonitor_alarm_due_to_readiness_failure

Type

64-bit Counter

Description

Alarm raised due to minimum count is not met for the readiness succeeded CriticalService

Labels

service, alarm_status

- nfmonitor_force_restart_due_to_readiness_failure

Type

64-bit Counter

Description

Howmany times Force Restart initiated due to readiness failure of CriticalService

Labels

service

Consul statistics

- nfmonitor_publish_status_into_consul

Type

64-bit Counter

Description

Howmany times Publish is sent to Consul

Labels

nf_status, nf_state, up_count, fsm_state

- nfmonitor_consul_read_attempts

Type

64-bit Counter

Description

Howmany times Consul read attempt is made

Labels

fsm_state, retry

- nfmonitor_consul_read_failures

Type

64-bit Counter

Description

Howmany times Consul read attempt are failed

Labels

fsm_state, retry, status_code

- nfmonitor_consul_read_success

Type

64-bit Counter

Description

Howmany times Consul read attempt are succeeded

Labels

fsm_state, retry, status_code

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Summary

Group	Metric Name	Type	Synopsis
Packet Streaming Metrics	num_packet_streaming_attempted	Counter	The total number of attempted streaming of packets
Packet Streaming Metrics	num_packet_streaming_success	Counter	The total number of packets streamed successfully
Packet Streaming Metrics	num_packet_streaming_failed	Counter	The total number of failed streaming of packets
Packet Streaming Metrics	num_packet_streaming_bytes	Counter	The total number of bytes successfully streamed

Packet Streaming Metrics

- num_packet_streaming_attempted

Type

64-bit Counter

Description

The total number of attempted streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_success

Type

64-bit Counter

Description

The total number of packets streamed successfully

Labels

interface_name, server_fqdn, kafka_topic

• num_packet_streaming_failed

Type

64-bit Counter

Description

The total number of failed streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

• num_packet_streaming_bytes

Type

64-bit Counter

Description

The total number of bytes successfully streamed

Labels

interface_name, server_fqdn, kafka_topic

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- [Deep Packet Inspection Perf statistics](#)
- [Deep Packet Inspection Tethering Statistics](#)
- [Deep Packet Inspection Application statistics](#)
- [Deep Packet Inspection Protocol Statistics](#)

Summary

Group	Metric Name	Type	Synopsis
Deep Packet Inspection General Statistics	total_connections	Counter	Total connections processed by DPI detection.
Deep Packet Inspection General Statistics	detection_total_packets	Counter	Total packets on flows processed by DPI detection.
Deep Packet Inspection General Statistics	detection_total_bytes	Counter	Total bytes on flows processed by DPI detection.
Deep Packet Inspection General Statistics	detection_uplink_packets	Counter	Uplink packets processed by DPI detection.
Deep Packet Inspection General Statistics	detection_downlink_packets	Counter	Downlink packets processed by DPI detection.
Deep Packet Inspection General Statistics	detection_uplink_bytes	Counter	Uplink bytes processed by DPI detection.
Deep Packet Inspection General Statistics	detection_downlink_bytes	Counter	Downlink bytes processed by DPI detection.
Deep Packet Inspection General Statistics	total_packets_on_dpi_flows	Counter	Total packets on the flows sent to DPI-Engine.
Deep Packet Inspection General Statistics	total_bytes_on_dpi_flows	Counter	Total bytes on the flows sent to DPI-Engine.
Deep Packet Inspection General Statistics	uplink_packets_on_dpi_flows	Counter	Total uplink packets on the flows sent to DPI-Engine.

Group	Metric Name	Type	Synopsis
Deep Packet Inspection General Statistics	uplink_bytes_on_dpi_flows	Counter	Total uplink bytes on the flows sent to DPI-Engine.
Deep Packet Inspection General Statistics	downlink_packets_on_dpi_flows	Counter	Total downlink packets on the flows sent to DPI-Engine.
Deep Packet Inspection General Statistics	downlink_bytes_on_dpi_flows	Counter	Total downlink bytes on the flows sent to DPI-Engine.
Deep Packet Inspection General Statistics	no_dpi_flows	Counter	Total flows excluded from DPI-Engine.
Deep Packet Inspection General Statistics	no_dpi_total_packets	Counter	Total packets excluded from DPI-Engine.
Deep Packet Inspection General Statistics	no_dpi_total_bytes	Counter	Total bytes excluded from DPI-Engine.
Deep Packet Inspection Dynamic Software Upgrade statistics	total_dpi_upgrade_trigger	Counter	Total number of DPI DSU triggered
Deep Packet Inspection Dynamic Software Upgrade statistics	total_dpi_upgrade_success	Counter	Total number of DPI DSU succeeded
Deep Packet Inspection Dynamic Software Upgrade statistics	total_dpi_upgrade_failure	Counter	Total number of DPI DSU failed
Deep Packet Inspection OS statistics	os_total_packets	Counter	Total packets seen for a particular OS as detected by DPI.
Deep Packet Inspection OS statistics	os_total_bytes	Counter	Total bytes seen for a particular OS as detected by DPI.
Deep Packet Inspection OS statistics	os_uplink_packets	Counter	Total uplink packets seen for a particular OS as detected by DPI.
Deep Packet Inspection OS statistics	os_uplink_bytes	Counter	Total uplink bytes seen for a particular OS as detected by DPI.
Deep Packet Inspection OS statistics	os_downlink_packets	Counter	Total downlink packets seen for a particular OS as detected by DPI.
Deep Packet Inspection OS statistics	os_downlink_bytes	Counter	Total downlink bytes seen for a particular OS as detected by DPI.
Deep Packet Inspection OS statistics	os_num_flows	Counter	Total flows seen for a particular OS as detected by DPI.
Deep Packet Inspection Rulematch statistics	total_reclassifications	Counter	Total number of Reclassifications
Deep Packet Inspection Rulematch statistics	parg_evaluated	Counter	Total number of PARG(protocol-application-rule-group) rules evaluated
Deep Packet Inspection Rulematch statistics	parg_bypassed	Counter	Total number of PARG(protocol-application-rule-group) rules bypassed
Deep Packet Inspection Rulematch statistics	parg_matched	Counter	Total number of PARG(protocol-application-rule-group) rules matched
Deep Packet Inspection Rulematch statistics	hrg_evaluated	Counter	Total number of HRG(http-rule-group) rules evaluated
Deep Packet Inspection Rulematch statistics	hrg_bypassed	Counter	Total number of HRG(http-rule-group) rules bypassed
Deep Packet Inspection Rulematch statistics	hrg_matched	Counter	Total number of HRG(http-rule-group) rules matched
Deep Packet Inspection Rulematch statistics	trg_evaluated	Counter	Total number of TRG(tls-rule-group) rules evaluated
Deep Packet Inspection Rulematch statistics	trg_bypassed	Counter	Total number of TRG(tls-rule-group) rules bypassed
Deep Packet Inspection Rulematch statistics	trg_matched	Counter	Total number of TRG(tls-rule-group) rules matched
Deep Packet Inspection Rulematch statistics	drg_evaluated	Counter	Total number of DRG(domain-rule-group) rules evaluated
Deep Packet Inspection Rulematch statistics	drg_bypassed	Counter	Total number of DRG(domain-rule-group) rules bypassed

Group	Metric Name	Type	Synopsis
Deep Packet Inspection Rulematch statistics	drg_matched	Counter	Total number of DRG(domain-rule-group) rules matched
Deep Packet Inspection Rulematch statistics	drg_snoop_matched	Counter	Total number of DRG Snoop(domain-rule-group) rules matched
Deep Packet Inspection Rulematch statistics	pfd_evaluated	Counter	Total number of PFD(packet-flow-description) rules evaluated
Deep Packet Inspection Rulematch statistics	pfd_matched	Counter	Total number of PFD(packet-flow-description) rules matched
Deep Packet Inspection Rulematch statistics	packet_filters_evaluated	Counter	Total number of packet-filters evaluated
Deep Packet Inspection Rulematch statistics	packet_filters_matched	Counter	Total number of packet-filters matched
Deep Packet Inspection Perf statistics	total_tethering_detection_ul_pkts	Counter	Total tethering detection uplink packets
Deep Packet Inspection Perf statistics	total_tethering_detection_dl_pkts	Counter	Total tethering detection downlink packets
Deep Packet Inspection Perf statistics	total_flows_unk_app_within_6pkts	Counter	Total flows with unk app[marked using 1-6 packets]
Deep Packet Inspection Perf statistics	total_flows_unk_app_within_7to10pkts	Counter	Total flows with unk app[marked using 7-10 packets]
Deep Packet Inspection Perf statistics	total_flows_unk_app_within_11to20pkts	Counter	Total flows with unk app[marked using 11-20 packets]
Deep Packet Inspection Perf statistics	total_flows_unk_app_above_20pkts	Counter	Total flows with unk app[marked using > 20 packets]
Deep Packet Inspection Perf statistics	total_flows_known_app_within_firstpkt	Counter	Total flows with known app[marked using 1st packet]
Deep Packet Inspection Perf statistics	total_flows_known_app_within_2to4pkts	Counter	Total flows with known app[marked using 2-4 packets]
Deep Packet Inspection Perf statistics	total_flows_known_app_within_5to10pkts	Counter	Total flows with known app[marked using 5-10 packets]
Deep Packet Inspection Perf statistics	total_flows_known_app_within_11to50pkts	Counter	Total flows with known app[marked using 11-50 packets]
Deep Packet Inspection Perf statistics	total_flows_known_app_above_50pkts	Counter	Total flows with known app[marked using > 50 packets]
Deep Packet Inspection Perf statistics	total_flows_unk_proto_within_6pkts	Counter	Total flows with unk proto[marked using 1-6 packets]
Deep Packet Inspection Perf statistics	total_flows_unk_proto_within_7to10pkts	Counter	Total flows with unk proto[marked using 7-10 packets]
Deep Packet Inspection Perf statistics	total_flows_unk_proto_within_11to20pkts	Counter	Total flows with unk proto[marked using 11-20 packets]
Deep Packet Inspection Perf statistics	total_flows_unk_proto_above_20pkts	Counter	Total flows with unk proto[marked using > 20 packets]
Deep Packet Inspection Perf statistics	total_flows_known_proto_within_firstpkt	Counter	Total flows with known proto[marked using 1st packet]
Deep Packet Inspection Perf statistics	total_flows_known_proto_within_2to4pkts	Counter	Total flows with known proto[marked using 2-4 packets]
Deep Packet Inspection Perf statistics	total_flows_known_proto_within_5to10pkts	Counter	Total flows with known proto[marked using 5-10 packets]
Deep Packet Inspection Perf statistics	total_flows_known_proto_within_11to50pkts	Counter	Total flows with known proto[marked using 11-50 packets]
Deep Packet Inspection Perf statistics	total_flows_known_proto_above_50pkts	Counter	Total flows with known proto[marked using > 50 packets]
Deep Packet Inspection Perf statistics	total_http_video_bytes	Counter	Total HTTP Video Bytes
Deep Packet Inspection Perf statistics	total_ssl_video_bytes	Counter	Total SSL Video Bytes
Deep Packet Inspection Perf statistics	total_quic_video_bytes	Counter	Total QUIC Video Bytes
Deep Packet Inspection Perf statistics	total_rtp_video_bytes	Counter	Total RTP Video Bytes

Group	Metric Name	Type	Synopsis
Deep Packet Inspection Perf statistics	total_other_proto_video_bytes	Counter	Total Other Protocols Video Bytes
Deep Packet Inspection Perf statistics	total_uplink_classifications	Counter	Total uplink classifications
Deep Packet Inspection Perf statistics	total_downlink_classifications	Counter	Total downlink classifications
Deep Packet Inspection Perf statistics	reclassification_count	Counter	Total Reclassifications
Deep Packet Inspection Perf statistics	avg_tethering_tcp_ts_ul_pkts	Gauge	Average uplink packets for TCP-TS tethering detection
Deep Packet Inspection Perf statistics	avg_tethering_gquic_ua_ul_pkts	Gauge	Average uplink packets for GQUIC-UA tethering detection
Deep Packet Inspection Perf statistics	avg_video_classification_pkts	Gauge	Average video classification packets
Deep Packet Inspection Perf statistics	avg_audio_classification_pkts	Gauge	Average audio classification packets
Deep Packet Inspection Perf statistics	avg_stream_classification_pkts	Gauge	Average stream classification packets
Deep Packet Inspection Perf statistics	avg_chat_classification_pkts	Gauge	Average chat classification packets
Deep Packet Inspection Perf statistics	avg_filetransfer_classification_pkts	Gauge	Average file-transfer classification packets
Deep Packet Inspection Tethering Statistics	connected_device_flows	Counter	Total flows seen from connected devices.
Deep Packet Inspection Tethering Statistics	connected_device_uplink_packets	Counter	Total uplink packets seen on flows from connected devices.
Deep Packet Inspection Tethering Statistics	connected_device_uplink_bytes	Counter	Total uplink bytes seen on flows from connected devices.
Deep Packet Inspection Tethering Statistics	connected_device_downlink_packets	Counter	Total downlink packets seen on flows from connected devices.
Deep Packet Inspection Tethering Statistics	connected_device_downlink_bytes	Counter	Total downlink bytes seen on flows from connected devices.
Deep Packet Inspection Tethering Statistics	tethered_device_flows	Counter	Total flows seen from tethered devices.
Deep Packet Inspection Tethering Statistics	tethered_device_uplink_packets	Counter	Total uplink packets seen on flows from tethered devices.
Deep Packet Inspection Tethering Statistics	tethered_device_uplink_bytes	Counter	Total uplink bytes seen on flows from tethered devices.
Deep Packet Inspection Tethering Statistics	tethered_device_downlink_packets	Counter	Total downlink packets seen on flows from tethered devices.
Deep Packet Inspection Tethering Statistics	tethered_device_downlink_bytes	Counter	Total downlink bytes seen on flows from tethered devices.
Deep Packet Inspection Tethering Statistics	tethered_flows_detected_user_agent	Counter	Total flows detected as tethered using user agent algorithm.
Deep Packet Inspection Tethering Statistics	tethered_flows_detected_gquic_user_agent	Counter	Total flows detected as tethered using gquic user agent algorithm.
Deep Packet Inspection Tethering Statistics	tethered_flows_detected_tcpfingerprinting	Counter	Total flows detected as tethered using tcp fingerprinting algorithm.
Deep Packet Inspection Tethering Statistics	tethered_flows_detected_ttl	Counter	Total flows detected as tethered using ttl algorithm.
Deep Packet Inspection Tethering Statistics	tethered_flows_detected_user_agent_report_only	Counter	Total flows detected as tethered using user agent algorithm and report only mode.
Deep Packet Inspection Tethering Statistics	tethered_flows_detected_gquic_user_agent_report_only	Counter	Total flows detected as tethered using gquic user agent algorithm and report only mode.
Deep Packet Inspection Tethering Statistics	tethered_flows_detected_tcpfingerprinting_report_only	Counter	Total flows detected as tethered using tcp fingerprinting algorithm and report only mode.
Deep Packet Inspection Tethering Statistics	tethered_flows_detected_ttl_report_only	Counter	Total flows detected as tethered using ttl algorithm and report only mode.
Deep Packet Inspection Application statistics	application_total_packets	Counter	Total packets seen from the application classified by attribute detected by DPI.
Deep Packet Inspection Application statistics	application_total_bytes	Counter	Total bytes seen from the application classified by attribute detected by DPI.

Group	Metric Name	Type	Synopsis
Deep Packet Inspection Application statistics	application_uplink_packets	Counter	Uplink packets seen from the application classified by attribute detected by DPI.
Deep Packet Inspection Application statistics	application_uplink_bytes	Counter	Uplink bytes seen from the application classified by attribute detected by DPI.
Deep Packet Inspection Application statistics	application_downlink_packets	Counter	Downlink packets seen from the application classified by attribute detected by DPI.
Deep Packet Inspection Application statistics	application_downlink_bytes	Counter	Downlink bytes seen from the application classified by attribute detected by DPI.
Deep Packet Inspection Application statistics	application_flows	Counter	Number of flows seen from the application classified by attribute detected by DPI.
Deep Packet Inspection Protocol Statistics	protocol_total_packets	Counter	Total packets seen from the protocol classified by attribute detected by DPI.
Deep Packet Inspection Protocol Statistics	protocol_total_bytes	Counter	Total bytes seen from the protocol classified by attribute detected by DPI.
Deep Packet Inspection Protocol Statistics	protocol_uplink_packets	Counter	Uplink packets seen from the protocol classified by attribute detected by DPI.
Deep Packet Inspection Protocol Statistics	protocol_uplink_bytes	Counter	Uplink bytes seen from the protocol classified by attribute detected by DPI.
Deep Packet Inspection Protocol Statistics	protocol_downlink_packets	Counter	Downlink packets seen from the protocol classified by attribute detected by DPI.
Deep Packet Inspection Protocol Statistics	protocol_downlink_bytes	Counter	Downlink bytes seen from the protocol classified by attribute detected by DPI.
Deep Packet Inspection Protocol Statistics	protocol_flows	Counter	Number of flows seen from the protocol classified by attribute detected by DPI.
Deep Packet Inspection Protocol Statistics	tcp_total_packets	Counter	Total tcp packets seen for flows sent to DPI.
Deep Packet Inspection Protocol Statistics	tcp_total_bytes	Counter	Total tcp bytes seen for flows sent to DPI.
Deep Packet Inspection Protocol Statistics	tcp_uplink_packets	Counter	Total tcp uplink packets seen for flows sent to DPI.
Deep Packet Inspection Protocol Statistics	tcp_uplink_bytes	Counter	Total tcp uplink bytes seen for flows sent to DPI.
Deep Packet Inspection Protocol Statistics	tcp_downlink_packets	Counter	Total tcp downlink packets seen for flows sent to DPI.
Deep Packet Inspection Protocol Statistics	tcp_downlink_bytes	Counter	Total tcp downlink bytes seen for flows sent to DPI.
Deep Packet Inspection Protocol Statistics	tcp_flows	Counter	Total tcp flows sent to DPI.
Deep Packet Inspection Protocol Statistics	udp_total_packets	Counter	Total udp packets seen for flows sent to DPI.
Deep Packet Inspection Protocol Statistics	udp_total_bytes	Counter	Total udp bytes seen for flows sent to DPI.
Deep Packet Inspection Protocol Statistics	udp_uplink_packets	Counter	Total udp uplink packets seen for flows sent to DPI.
Deep Packet Inspection Protocol Statistics	udp_uplink_bytes	Counter	Total udp uplink bytes seen for flows sent to DPI.
Deep Packet Inspection Protocol Statistics	udp_downlink_packets	Counter	Total udp downlink packets seen for flows sent to DPI.
Deep Packet Inspection Protocol Statistics	udp_downlink_bytes	Counter	Total udp downlink bytes seen for flows sent to DPI.
Deep Packet Inspection Protocol Statistics	udp_flows	Counter	Total udp flows sent to DPI.
Deep Packet Inspection Protocol Statistics	icmp_total_packets	Counter	Total icmp packets seen for flows sent to DPI.
Deep Packet Inspection Protocol Statistics	icmp_total_bytes	Counter	Total icmp bytes seen for flows sent to DPI.
Deep Packet Inspection Protocol Statistics	icmp_uplink_packets	Counter	Total icmp uplink packets seen for flows sent to DPI.
Deep Packet Inspection Protocol Statistics	icmp_uplink_bytes	Counter	Total icmp uplink bytes seen for flows sent to DPI.

Group	Metric Name	Type	Synopsis
Deep Packet Inspection Protocol Statistics	icmp_downlink_packets	Counter	Total icmp downlink packets seen for flows sent to DPI.
Deep Packet Inspection Protocol Statistics	icmp_downlink_bytes	Counter	Total icmp downlink bytes seen for flows sent to DPI.
Deep Packet Inspection Protocol Statistics	icmp_flows	Counter	Total icmp flows sent to DPI.

Deep Packet Inspection General Statistics

- total_connections

Type

64-bit Counter

Description

Total connections processed by DPI detection.

Metrics

Instances	Labels	Description
1	stats_group: dpi_general_stats	Total connections processed by DPI detection.

- detection_total_packets

Type

64-bit Counter

Description

Total packets on flows processed by DPI detection.

Metrics

Instances	Labels	Description
1	stats_group: dpi_general_stats	Total packets on flows processed by DPI detection.

- detection_total_bytes

Type

64-bit Counter

Description

Total bytes on flows processed by DPI detection.

Metrics

Instances	Labels	Description
1	stats_group: dpi_general_stats	Total bytes on flows processed by DPI detection.

- detection_uplink_packets

Type

64-bit Counter

Description

Uplink packets processed by DPI detection.

Metrics

Instances	Labels	Description
1	stats_group: dpi_general_stats	Uplink packets processed by DPI detection.

- detection_downlink_packets

Type

64-bit Counter

Description

Downlink packets processed by DPI detection.

Metrics

Instances	Labels	Description
1	stats_group: dpi_general_stats	Downlink packets processed by DPI detection.

- detection_uplink_bytes

Type

64-bit Counter

Description

Uplink bytes processed by DPI detection.

Metrics

Instances	Labels	Description
1	stats_group: dpi_general_stats	Uplink bytes processed by DPI detection.

- detection_downlink_bytes

Type

64-bit Counter

Description

Downlink bytes processed by DPI detection.

Metrics

Instances	Labels	Description
1	stats_group: dpi_general_stats	Downlink bytes processed by DPI detection.

- total_packets_on_dpi_flows

Type

64-bit Counter

Description

Total packets on the flows sent to DPI-Engine.

Metrics

Instances	Labels	Description
1	stats_group: dpi_general_stats	Total packets on the flows sent to DPI-Engine.

- total_bytes_on_dpi_flows

Type

64-bit Counter

Description

Total bytes on the flows sent to DPI-Engine.

Metrics

Instances	Labels	Description
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Instances	Labels	Description
1	stats_group: dpi_general_stats	Total bytes on the flows sent to DPI-Engine.
• uplink_packets_on_dpi_flows		
Type		
64-bit Counter		
Description		
Total uplink packets on the flows sent to DPI-Engine.		
Metrics		

Instances	Labels	Description
1	stats_group: dpi_general_stats	Total uplink packets on the flows sent to DPI-Engine.
• uplink_bytes_on_dpi_flows		
Type		
64-bit Counter		
Description		
Total uplink bytes on the flows sent to DPI-Engine.		
Metrics		

Instances	Labels	Description
1	stats_group: dpi_general_stats	Total uplink bytes on the flows sent to DPI-Engine.
• downlink_packets_on_dpi_flows		
Type		
64-bit Counter		
Description		
Total downlink packets on the flows sent to DPI-Engine.		
Metrics		

Instances	Labels	Description
1	stats_group: dpi_general_stats	Total downlink packets on the flows sent to DPI-Engine.
• downlink_bytes_on_dpi_flows		
Type		
64-bit Counter		
Description		
Total downlink bytes on the flows sent to DPI-Engine.		
Metrics		

Instances	Labels	Description
1	stats_group: dpi_general_stats	Total downlink bytes on the flows sent to DPI-Engine.
• no_dpi_flows		
Type		
64-bit Counter		
Description		
Total flows excluded from DPI-Engine.		

Metrics

Instances	Labels	Description
1	stats_group: dpi_general_stats	Total flows excluded from DPI-Engine.
• no_dpi_total_packets		
Type		
64-bit Counter		
Description		
Total packets excluded from DPI-Engine.		
Metrics		

Instances	Labels	Description
1	stats_group: dpi_general_stats	Total packets excluded from DPI-Engine.
• no_dpi_total_bytes		
Type		
64-bit Counter		
Description		
Total bytes excluded from DPI-Engine.		
Metrics		

Instances	Labels	Description
1	stats_group: dpi_general_stats	Total bytes excluded from DPI-Engine.

Deep Packet Inspection Dynamic Software Upgrade statistics

• total_dpi_upgrade_trigger

Type

64-bit Counter

Description

Total number of DPI DSU triggered

Labels

• total_dpi_upgrade_success

Type

64-bit Counter

Description

Total number of DPI DSU succeeded

Labels

• total_dpi_upgrade_failure

Type

64-bit Counter

Description

Total number of DPI DSU failed

Labels

Deep Packet Inspection OS statistics

- os_total_packets

Type

64-bit Counter

Description

Total packets seen for a particular OS as detected by DPI.

Labels

os_name

- os_total_bytes

Type

64-bit Counter

Description

Total bytes seen for a particular OS as detected by DPI.

Labels

os_name

- os_uplink_packets

Type

64-bit Counter

Description

Total uplink packets seen for a particular OS as detected by DPI.

Labels

os_name

- os_uplink_bytes

Type

64-bit Counter

Description

Total uplink bytes seen for a particular OS as detected by DPI.

Labels

os_name

- os_downlink_packets

Type

64-bit Counter

Description

Total downlink packets seen for a particular OS as detected by DPI.

Labels

os_name

- os_downlink_bytes

Type

64-bit Counter

Description

Total downlink bytes seen for a particular OS as detected by DPI.

Labels

os_name

- os_num_flows

Type

64-bit Counter

Description

Total flows seen for a particular OS as detected by DPI.

Labels

os_name

Deep Packet Inspection Rulematch statistics

- total_reclassifications

Type

64-bit Counter

Description

Total number of Reclassifications

Labels

- parg_evaluated

Type

64-bit Counter

Description

Total number of PARG(protocol-application-rule-group) rules evaluated

Labels

- parg_bypassed

Type

64-bit Counter

Description

Total number of PARG(protocol-application-rule-group) rules bypassed

Labels

- parg_matched

Type

64-bit Counter

Description

Total number of PARG(protocol-application-rule-group) rules matched

Labels

- **hrg_evaluated**

Type

64-bit Counter

Description

Total number of HRG(http-rule-group) rules evaluated

Labels

- **hrg_bypassed**

Type

64-bit Counter

Description

Total number of HRG(http-rule-group) rules bypassed

Labels

- **hrg_matched**

Type

64-bit Counter

Description

Total number of HRG(http-rule-group) rules matched

Labels

- **trg_evaluated**

Type

64-bit Counter

Description

Total number of TRG(tls-rule-group) rules evaluated

Labels

- **trg_bypassed**

Type

64-bit Counter

Description

Total number of TRG(tls-rule-group) rules bypassed

Labels

- **trg_matched**

Type

64-bit Counter

Description

Total number of TRG(tls-rule-group) rules matched

Labels

- drg_evaluated

Type

64-bit Counter

Description

Total number of DRG(domain-rule-group) rules evaluated

Labels

- drg_bypassed

Type

64-bit Counter

Description

Total number of DRG(domain-rule-group) rules bypassed

Labels

- drg_matched

Type

64-bit Counter

Description

Total number of DRG(domain-rule-group) rules matched

Labels

- drg_snoop_matched

Type

64-bit Counter

Description

Total number of DRG Snoop(domain-rule-group) rules matched

Labels

- pfd_evaluated

Type

64-bit Counter

Description

Total number of PFD(packet-flow-description) rules evaluated

Labels

- pfd_matched

Type

64-bit Counter

Description

Total number of PFD(packet-flow-description) rules matched

Labels

- packet_filters_evaluated

Type

64-bit Counter

Description

Total number of packet-filters evaluated

Labels

- packet_filters_matched

Type

64-bit Counter

Description

Total number of packet-filters matched

Labels

Deep Packet Inspection Perf statistics

- total_tethering_detection_ul_pkts

Type

64-bit Counter

Description

Total tethering detection uplink packets

Labels

- total_tethering_detection_dl_pkts

Type

64-bit Counter

Description

Total tethering detection downlink packets

Labels

- total_flows_unk_app_within_6pkts

Type

64-bit Counter

Description

Total flows with unk app[marked using 1-6 packets]

Labels

- total_flows_unk_app_within_7to10pkts

Type

64-bit Counter

Description

Total flows with unk app[marked using 7-10 packets]

Labels

- total_flows_unk_app_within_11to20pkts

Type

64-bit Counter

Description

Total flows with unk app[marked using 11-20 packets]

Labels

- total_flows_unk_app_above_20pkts

Type

64-bit Counter

Description

Total flows with unk app[marked using > 20 packets]

Labels

- total_flows_known_app_within_firstpkt

Type

64-bit Counter

Description

Total flows with known app[marked using 1st packet]

Labels

- total_flows_known_app_within_2to4pkts

Type

64-bit Counter

Description

Total flows with known app[marked using 2-4 packets]

Labels

- total_flows_known_app_within_5to10pkts

Type

64-bit Counter

Description

Total flows with known app[marked using 5-10 packets]

Labels

- total_flows_known_app_within_11to50pkts

Type

64-bit Counter

Description

Total flows with known app[marked using 11-50 packets]

Labels

- total_flows_known_app_above_50pkts

Type

64-bit Counter

Description

Total flows with known app[marked using > 50 packets]

Labels

- total_flows_unk_proto_within_6pkts

Type

64-bit Counter

Description

Total flows with unk proto[marked using 1-6 packets]

Labels

- total_flows_unk_proto_within_7to10pkts

Type

64-bit Counter

Description

Total flows with unk proto[marked using 7-10 packets]

Labels

- total_flows_unk_proto_within_11to20pkts

Type

64-bit Counter

Description

Total flows with unk proto[marked using 11-20 packets]

Labels

- total_flows_unk_proto_above_20pkts

Type

64-bit Counter

Description

Total flows with unk proto[marked using > 20 packets]

Labels

- total_flows_known_proto_within_firstpkt

Type

64-bit Counter

Description

Total flows with known proto[marked using 1st packet]

Labels

- total_flows_known_proto_within_2to4pkts

Type

64-bit Counter

Description

Total flows with known proto[marked using 2-4 packets]

Labels

- total_flows_known_proto_within_5to10pkts

Type

64-bit Counter

Description

Total flows with known proto[marked using 5-10 packets]

Labels

- total_flows_known_proto_within_11to50pkts

Type

64-bit Counter

Description

Total flows with known proto[marked using 11-50 packets]

Labels

- total_flows_known_proto_above_50pkts

Type

64-bit Counter

Description

Total flows with known proto[marked using > 50 packets]

Labels

- total_http_video_bytes

Type

64-bit Counter

Description

Total HTTP Video Bytes

Labels

- total_ssl_video_bytes

Type

64-bit Counter

Description

Total SSL Video Bytes

Labels

- total_quic_video_bytes

Type

64-bit Counter

Description

Total QUIC Video Bytes

Labels

- total_rtp_video_bytes

Type

64-bit Counter

Description

Total RTP Video Bytes

Labels

- total_other_proto_video_bytes

Type

64-bit Counter

Description

Total Other Protocols Video Bytes

Labels

- total_uplink_classifications

Type

64-bit Counter

Description

Total uplink classifications

Labels

- total_downlink_classifications

Type

64-bit Counter

Description

Total downlink classifications

Labels

- reclassification_count

Type

64-bit Counter

Description

Total Reclassifications

Labels

trigger_name

• avg_tethering_tcp_ts_ul_pkts

Type

64-bit Gauge

Description

Average uplink packets for TCP-TS tethering detection

Labels

• avg_tethering_gquic_ua_ul_pkts

Type

64-bit Gauge

Description

Average uplink packets for GQUIC-UA tethering detection

Labels

• avg_video_classification_pkts

Type

64-bit Gauge

Description

Average video classification packets

Labels

• avg_audio_classification_pkts

Type

64-bit Gauge

Description

Average audio classification packets

Labels

• avg_stream_classification_pkts

Type

64-bit Gauge

Description

Average stream classification packets

Labels

• avg_chat_classification_pkts

Type

64-bit Gauge

Description

Average chat classification packets

Labels

- avg_filetransfer_classification_pkts

Type

64-bit Gauge

Description

Average file-transfer classification packets

Labels

Deep Packet Inspection Tethering Statistics

- connected_device_flows

Type

64-bit Counter

Description

Total flows seen from connected devices.

Metrics

Instances	Labels	Description
1	stats_group: dpi_tethering_stats	Total flows seen from connected devices.

- connected_device_uplink_packets

Type

64-bit Counter

Description

Total uplink packets seen on flows from connected devices.

Metrics

Instances	Labels	Description
1	stats_group: dpi_tethering_stats	Total uplink packets seen on flows from connected devices.

- connected_device_uplink_bytes

Type

64-bit Counter

Description

Total uplink bytes seen on flows from connected devices.

Metrics

Instances	Labels	Description
1	stats_group: dpi_tethering_stats	Total uplink bytes seen on flows from connected devices.

- connected_device_downlink_packets

Type

64-bit Counter

Description		
Total downlink packets seen on flows from connected devices.		
Metrics		
Instances	Labels	Description
1	stats_group: dpi_tethering_stats	Total downlink packets seen on flows from connected devices.
connected_device_downlink_bytes		
Type		
64-bit Counter		
Description		
Total downlink bytes seen on flows from connected devices.		
Metrics		
Instances	Labels	Description
1	stats_group: dpi_tethering_stats	Total downlink bytes seen on flows from connected devices.
tethered_device_flows		
Type		
64-bit Counter		
Description		
Total flows seen from tethered devices.		
Metrics		
Instances	Labels	Description
1	stats_group: dpi_tethering_stats	Total flows seen from tethered devices.
tethered_device_uplink_packets		
Type		
64-bit Counter		
Description		
Total uplink packets seen on flows from tethered devices.		
Metrics		
Instances	Labels	Description
1	stats_group: dpi_tethering_stats	Total uplink packets seen on flows from tethered devices.
tethered_device_uplink_bytes		
Type		
64-bit Counter		
Description		
Total uplink bytes seen on flows from tethered devices.		
Metrics		
Instances	Labels	Description
1	stats_group: dpi_tethering_stats	Total uplink bytes seen on flows from tethered devices.
tethered_device_downlink_packets		
Type		

64-bit Counter
Description
Total downlink packets seen on flows from tethered devices.
Metrics

Instances	Labels	Description
1	stats_group: dpi_tethering_stats	Total downlink packets seen on flows from tethered devices.

- tethered_device_downlink_bytes

Type
64-bit Counter
Description
Total downlink bytes seen on flows from tethered devices.
Metrics

Instances	Labels	Description
1	stats_group: dpi_tethering_stats	Total downlink bytes seen on flows from tethered devices.

- tethered_flows_detected_user_agent

Type
64-bit Counter
Description
Total flows detected as tethered using user agent algorithm.
Metrics

Instances	Labels	Description
1	stats_group: dpi_tethering_stats	Total flows detected as tethered using user agent algorithm.

- tethered_flows_detected_gquic_user_agent

Type
64-bit Counter
Description
Total flows detected as tethered using gquic user agent algorithm.
Metrics

Instances	Labels	Description
1	stats_group: dpi_tethering_stats	Total flows detected as tethered using gquic user agent algorithm.

- tethered_flows_detected_tcpfingerprinting

Type
64-bit Counter
Description
Total flows detected as tethered using tcp fingerprinting algorithm.
Metrics

Instances	Labels	Description
1	stats_group: dpi_tethering_stats	Total flows detected as tethered using tcp fingerprinting algorithm.

- tethered_flows_detected_ttl

Type
64-bit Counter
Description
Total flows detected as tethered using ttl algorithm.
Metrics

Instances	Labels	Description
1	stats_group: dpi_tethering_stats	Total flows detected as tethered using ttl algorithm.

- tethered_flows_detected_user_agent_report_only

Type
64-bit Counter
Description
Total flows detected as tethered using user agent algorithm and report only mode.
Metrics

Instances	Labels	Description
1	stats_group: dpi_tethering_stats	Total flows detected as tethered using user agent algorithm and report only mode.

- tethered_flows_detected_gquic_user_agent_report_only

Type
64-bit Counter
Description
Total flows detected as tethered using gquic user agent algorithm and report only mode.
Metrics

Instances	Labels	Description
1	stats_group: dpi_tethering_stats	Total flows detected as tethered using gquic user agent algorithm and report only mode.

- tethered_flows_detected_tcpfingerprinting_report_only

Type
64-bit Counter
Description
Total flows detected as tethered using tcp fingerprinting algorithm and report only mode.
Metrics

Instances	Labels	Description
1	stats_group: dpi_tethering_stats	Total flows detected as tethered using tcp fingerprinting algorithm and report only mode.

- tethered_flows_detected_ttl_report_only

Type
64-bit Counter
Description
Total flows detected as tethered using ttl algorithm and report only mode.
Metrics

Instances	Labels	Description
1	stats_group: dpi_tethering_stats	Total flows detected as tethered using ttl algorithm and report only mode.

Deep Packet Inspection Application statistics

- application_total_packets
 - Type
 - 64-bit Counter
 - Description
 - Total packets seen from the application classified by attribute detected by DPI.
 - Labels
 - group_name, application_name, attribute_name**
- application_total_bytes
 - Type
 - 64-bit Counter
 - Description
 - Total bytes seen from the application classified by attribute detected by DPI.
 - Labels
 - group_name, application_name, attribute_name**
- application_uplink_packets
 - Type
 - 64-bit Counter
 - Description
 - Uplink packets seen from the application classified by attribute detected by DPI.
 - Labels
 - group_name, application_name, attribute_name**
- application_uplink_bytes
 - Type
 - 64-bit Counter
 - Description
 - Uplink bytes seen from the application classified by attribute detected by DPI.
 - Labels
 - group_name, application_name, attribute_name**
- application_downlink_packets
 - Type
 - 64-bit Counter
 - Description
 - Downlink packets seen from the application classified by attribute detected by DPI.
 - Labels
 - group_name, application_name, attribute_name**
- application_downlink_bytes
 - Type
 - 64-bit Counter

Description
Downlink bytes seen from the application classified by attribute detected by DPI.
Labels
<i>group_name, application_name, attribute_name</i>
• application_flows
Type
64-bit Counter
Description
Number of flows seen from the application classified by attribute detected by DPI.
Labels
<i>group_name, application_name, attribute_name</i>

Deep Packet Inspection Protocol Statistics

• protocol_total_packets
Type
64-bit Counter
Description
Total packets seen from the protocol classified by attribute detected by DPI.
Labels
<i>group_name, protocol_name, attribute_name</i>
• protocol_total_bytes
Type
64-bit Counter
Description
Total bytes seen from the protocol classified by attribute detected by DPI.
Labels
<i>group_name, protocol_name, attribute_name</i>
• protocol_uplink_packets
Type
64-bit Counter
Description
Uplink packets seen from the protocol classified by attribute detected by DPI.
Labels
<i>group_name, protocol_name, attribute_name</i>
• protocol_uplink_bytes
Type
64-bit Counter
Description
Uplink bytes seen from the protocol classified by attribute detected by DPI.
Labels

group_name, protocol_name, attribute_name

- protocol_downlink_packets

Type

64-bit Counter

Description

Downlink packets seen from the protocol classified by attribute detected by DPI.

Labels

group_name, protocol_name, attribute_name

- protocol_downlink_bytes

Type

64-bit Counter

Description

Downlink bytes seen from the protocol classified by attribute detected by DPI.

Labels

group_name, protocol_name, attribute_name

- protocol_flows

Type

64-bit Counter

Description

Number of flows seen from the protocol classified by attribute detected by DPI.

Labels

group_name, protocol_name, attribute_name

- tcp_total_packets

Type

64-bit Counter

Description

Total tcp packets seen for flows sent to DPI.

Labels

attribute_name

- tcp_total_bytes

Type

64-bit Counter

Description

Total tcp bytes seen for flows sent to DPI.

Labels

attribute_name

- tcp_uplink_packets

Type

64-bit Counter

Description

Total tcp uplink packets seen for flows sent to DPI.

Labels

attribute_name

- tcp_uplink_bytes

Type

64-bit Counter

Description

Total tcp uplink bytes seen for flows sent to DPI.

Labels

attribute_name

- tcp_downlink_packets

Type

64-bit Counter

Description

Total tcp downlink packets seen for flows sent to DPI.

Labels

attribute_name

- tcp_downlink_bytes

Type

64-bit Counter

Description

Total tcp downlink bytes seen for flows sent to DPI.

Labels

attribute_name

- tcp_flows

Type

64-bit Counter

Description

Total tcp flows sent to DPI.

Labels

attribute_name

- udp_total_packets

Type

64-bit Counter

Description

Total udp packets seen for flows sent to DPI.

Labels

- udp_total_bytes

Type

64-bit Counter

Description

Total udp bytes seen for flows sent to DPI.

Labels

- **udp_uplink_packets**

Type

64-bit Counter

Description

Total udp uplink packets seen for flows sent to DPI.

Labels

- **udp_uplink_bytes**

Type

64-bit Counter

Description

Total udp uplink bytes seen for flows sent to DPI.

Labels

- **udp_downlink_packets**

Type

64-bit Counter

Description

Total udp downlink packets seen for flows sent to DPI.

Labels

- **udp_downlink_bytes**

Type

64-bit Counter

Description

Total udp downlink bytes seen for flows sent to DPI.

Labels

- **udp_flows**

Type

64-bit Counter

Description

Total udp flows sent to DPI.

Labels

- **icmp_total_packets**

Type

64-bit Counter

Description

Total icmp packets seen for flows sent to DPI.

Labels

attribute_name

- icmp_total_bytes

Type

64-bit Counter

Description

Total icmp bytes seen for flows sent to DPI.

Labels

attribute_name

- icmp_uplink_packets

Type

64-bit Counter

Description

Total icmp uplink packets seen for flows sent to DPI.

Labels

attribute_name

- icmp_uplink_bytes

Type

64-bit Counter

Description

Total icmp uplink bytes seen for flows sent to DPI.

Labels

attribute_name

- icmp_downlink_packets

Type

64-bit Counter

Description

Total icmp downlink packets seen for flows sent to DPI.

Labels

attribute_name

- icmp_downlink_bytes

Type

64-bit Counter

Description

Total icmp downlink bytes seen for flows sent to DPI.

Labels

attribute_name

- icmp_flows

Type

64-bit Counter

Description

Total icmp flows sent to DPI.

Labels

attribute_name

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Summary

Group	Metric Name	Type	Synopsis
Network Address Translation Module Statistics	total_nat_sessions_created	Counter	Total sessions created by NAT.
Network Address Translation Module Statistics	total_nat_sessions_deleted	Counter	Total sessions deleted by NAT.
Network Address Translation Module Statistics	total_nat_connections_created	Counter	Total connections created by NAT.
Network Address Translation Module Statistics	total_nat_connections_deleted	Counter	Total connections deleted by NAT.
Network Address Translation Module Statistics	total_nat_tcp_connections_created	Counter	Total TCP connections created by NAT.
Network Address Translation Module Statistics	total_nat_udp_connections_created	Counter	Total UDP connections created by NAT.
Network Address Translation Module Statistics	total_nat_icmp_connections_created	Counter	Total ICMP connections created by NAT.
Network Address Translation Module Statistics	total_nat_tcp_connections_deleted	Counter	Total TCP connections deleted by NAT.
Network Address Translation Module Statistics	total_nat_udp_connections_deleted	Counter	Total UDP connections deleted by NAT.
Network Address Translation Module Statistics	total_nat_icmp_connections_deleted	Counter	Total ICMP connections deleted by NAT.
Network Address Translation Module Statistics	total_nat_connections_created_action_pass	Counter	Total connections created by NAT with action PASS.
Network Address Translation Module Statistics	total_nat_connections_created_action_drop	Counter	Total connections created by NAT with action DROP.
Network Address Translation Module Statistics	total_nat_connections_created_invalid_proto_and_alg_type	Counter	Total connections created by NAT with protocol and ALG type mismatch.

Group	Metric Name	Type	Synopsis
Network Address Translation Module Statistics	connection_create_failure_natpool_not_found	Counter	Total connection create failures due to natpool not found.
Network Address Translation Module Statistics	connection_create_failure_natpool_pending_delete	Counter	Total connection create failures due to natpool is pending deletion.
Network Address Translation Module Statistics	connection_create_failure_nat_ip_pool_not_configured	Counter	Total connection create failures due to nat ip pool not configured.
Network Address Translation Module Statistics	connection_create_failure_exceeded_per_session_nat_connections_quota	Counter	Total connection create failures due to session nat connections quota exhaustion.
Network Address Translation Module Statistics	connection_create_failure_nat_port_allocation_failures	Counter	Total connection create failures due to nat port allocation failures.
Network Address Translation Module Statistics	nat_port_allocation_success	Counter	Total nat port allocation success.
Network Address Translation Module Statistics	nat_port_deallocation_success	Counter	Total nat port deallocation success.
Network Address Translation Module Statistics	nat_port_allocation_failure_internal_error	Counter	Total nat port allocation failures due to internal error.
Network Address Translation Module Statistics	nat_port_allocation_failure_resource_exhaustion	Counter	Total nat port allocation failures due to port unavailability.
Network Address Translation Module Statistics	total_nat_connection_creation_failure_system_resource_exhaustion	Counter	Total nat connection allocation failures due to system resource unavailability.
Network Address Translation Module Statistics	nat_port_allocation_failure_unsupported_protocol	Counter	Total nat port allocation failures due to unsupported protocol.
Network Address Translation Module Statistics	total_nat_connection_creation_failure_due_to_unknown_icmp_type	Counter	Total nat connection creation failure due to unknown icmp type.
Network Address Translation Module Statistics	total_nat_connection_creation_ignored_due_to_unsupp_proto	Counter	Total nat connection creation ignored due to unsupported proto.
Network Address Translation Module Statistics	nat_port_deallocation_failure_internal_error	Counter	Total nat port deallocation failures due to internal error.
Network Address Translation Module Statistics	nat_port_deallocation_failure_unsupported_protocol	Counter	Total nat port deallocation failures due to unsupported protocol.
Network Address Translation Module Statistics	nat_total_uplink_packets	Counter	Total nat uplink packets
Network Address Translation Module Statistics	nat_total_uplink_bytes	Counter	Total nat uplink bytes
Network Address Translation Module Statistics	nat_total_downlink_packets	Counter	Total nat downlink packets
Network Address Translation Module Statistics	nat_total_downlink_bytes	Counter	Total nat downlink bytes
Network Address Translation Module Statistics	nat_11_port_alloc	Counter	NAT1:1 port allocation
Network Address Translation Module Statistics	nat_11_port_alloc_zero	Counter	NAT1:1 port allocation with zero
Network Address Translation Module Statistics	nat_11_address_reuse	Counter	NAT1:1 address reuse
Network Address Translation Module Statistics	nat_11_address_alloc	Counter	NAT1:1 address allocation
Network Address Translation Module Statistics	nat_11_stale_address_return	Counter	NAT1:1 stale address return
Network Address Translation Module Statistics	nat_11_address_release	Counter	NAT1:1 address release
Network Address Translation Module Statistics	nat_11_no_address	Counter	NAT1:1 no address available

Group	Metric Name	Type	Synopsis
Network Address Translation Module Statistics	nat_11_update_pinhole_failure	Counter	NAT1:1 Update pinhole failure
Network Address Translation Module Statistics	nat_11_update_pinhole_success	Counter	NAT1:1 Update pinhole success
Network Address Translation Module Statistics	nat_11_open_pinhole	Counter	NAT1:1 Open Pinhole
Network Address Translation Module Statistics	nat_11_known_ueport_open_pinhole	Counter	NAT1:1 Known UE Port Open Pinhole
Network Address Translation Module Statistics	gnat_port_alloc	Counter	Guaranteed NAT port allocation
Network Address Translation Module Statistics	gnat_port_dealloc	Counter	Guaranteed NAT port deallocation
Network Address Translation Module Statistics	gnat_handle_session_ip_failure	Counter	Guaranteed NAT handle session IP failure
Network Address Translation Module Statistics	gnat_handle_session_ip_no_natpool	Counter	Guaranteed NAT handle session IP no NAT pool
Network Address Translation Module Statistics	max_users_per_nat_ip_limit_reached	Counter	Max users per NAT IP limit exceeded
Network Address Translation Module Statistics	num_ip_released_from_subscriber_session	Counter	Number of IP's released from subscriber session
Network Address Translation Module Statistics	num_port_object_internal_access_failure	Counter	None
Network Address Translation Module Statistics	num_no_free_ports_on_ip	Counter	None
Network Address Translation Module Statistics	num_nat_port_allocation_collisions	Counter	None
Network Address Translation Module Statistics	total_nat_port_allocation_success_on_first_try	Counter	None
Network Address Translation Module Statistics	total_nat_port_allocation_not_success_on_first_try	Counter	None
Network Address Translation Module Statistics	total_nat_reverse_lookup_failure	Counter	None
Network Address Translation Module Statistics	total_nat_conn_fail_due_to_max_connection_limit_reached	Counter	None
Network Address Translation Module Statistics	nat_m1_update_pinhole_failure	Counter	NATM:1 Update pinhole failure
Network Address Translation Module Statistics	nat_m1_update_pinhole_success	Counter	NATM:1 Update pinhole success
Network Address Translation Module Statistics	conn_create_failure_sip_cpp_interface_failure	Counter	NATM:1 Update pinhole success
Network Address Translation Module Statistics	total_nat_active_connections	Gauge	Total present NAT connections.
Network Address Translation Module Statistics	total_nat_active_sessions	Gauge	Total present NAT Sessions.
Network Address Translation Module Statistics	total_nat_per_connection_average_port_collision_count	Gauge	None
Network Address Translation NatPool Statistics	total_natpool_sessions_created	Counter	Total sessions for which NAT resources were allocated from nat-pool.
Network Address Translation NatPool Statistics	total_natpool_sessions_deleted	Counter	Total sessions for which NAT resources were deallocated from nat-pool.
Network Address Translation NatPool Statistics	total_natpool_connections_created	Counter	Total connections created using NAT resources from nat-pool.

Group	Metric Name	Type	Synopsis
Network Address Translation NatPool Statistics	total_natpool_connections_deleted	Counter	Total connections deleted using NAT resources from nat-pool.
Network Address Translation NatPool Statistics	total_natpool_sessions_denied_resource	Counter	Total sessions denied resource using this nat-pool.
Network Address Translation NatPool Statistics	total_natpool_connections_create_failure	Counter	Total connections create failure using this nat-pool.
Network Address Translation NatPool Statistics	total_natpool_port_alloc_success_on_first_try	Counter	Total port alloc success on first try using this nat-pool.
Network Address Translation NatPool Statistics	total_natpool_port_alloc_not_success_on_first_try	Counter	Total port alloc success on first try using this nat-pool.
Network Address Translation NatPool Statistics	total_natpool_num_port_alloc_max_retry_reached	Counter	Port allocation max retry reached.
Network Address Translation NatPool Statistics	natpool_connection_create_failure_nat_port_allocation_failures	Counter	Total connection create failures due to nat port allocation failures.
Network Address Translation NatPool Statistics	natpool_connection_create_failure_nat_ip_pool_not_configured	Counter	Total connection create failures due to nat ip pool not configured.
Network Address Translation NatPool Statistics	natpool_connection_create_failure_exceeded_per_session_nat_connections_quota	Counter	Total connection create failures due to session nat connections quota exhaustion.
Network Address Translation NatPool Statistics	total_natpool_number_of_nat_port_allocation_collisions	Counter	Total Number of NAT port collision during NAT Connection creation.
Network Address Translation NatPool Statistics	total_natpool_active_sessions	Gauge	Total sessions present using NAT resources from nat-pool.
Network Address Translation NatPool Statistics	total_natpool_active_connections	Gauge	Total connections present using NAT resources from nat-pool.
Network Address Translation NatPool Statistics	total_natpool_average_port_collision_count	Gauge	Average NAT port collision during NAT Connection creation.
Network Address Translation NatPool Statistics	total_natpool_port_util_percentage	Gauge	Total free nat ports on this nat-pool.
Network Address Translation NatPool Statistics	total_natpool_used_nat_ports	Gauge	Total used nat ports on this nat-pool.
Network Address Translation NatPool Statistics	total_natpool_free_nat_ports	Gauge	Total free nat ports on this nat-pool.
Network Address Translation FTP ALG Statistics	total_port_command	Counter	Total port commands received by NAT.
Network Address Translation FTP ALG Statistics	total_eprt_command	Counter	Total EPRT command received by NAT.
Network Address Translation FTP ALG Statistics	total_pasv_command	Counter	Total PASV command received by NAT.
Network Address Translation FTP ALG Statistics	total_epasv_command	Counter	Total EPASV command received by NAT.
Network Address Translation FTP ALG Statistics	total_epasv_v6_command	Counter	Total EPASV v6 command received by NAT.
Network Address Translation FTP ALG Statistics	total_epasv_v6_cmd_resp_wrong_addr	Counter	Total EPASV command Response with wrong address received by NAT.
Network Address Translation FTP ALG Statistics	total_eprt_invalid_arg	Counter	Total EPRT command with invalid argument.
Network Address Translation FTP ALG Statistics	total_eprt_v6_invalid_arg	Counter	Total EPRT v6 command with invalid argument.
Network Address Translation FTP ALG Statistics	total_eprt_v6_command	Counter	Total EPRT v6 command received by NAT.
Network Address Translation FTP ALG Statistics	total_eprt_v6_cmd_wrong_addr	Counter	Total EPRT v6 command Response with wrong address received by NAT.

Group	Metric Name	Type	Synopsis
Network Address Translation FTP ALG Statistics	total_uplink_pinhole_connection_count	Counter	Total uplink pin-hole connections successfully created.
Network Address Translation FTP ALG Statistics	total_uplink_pinhole_connection_failure	Counter	Total uplink pin-hole connection create failures.
Network Address Translation FTP ALG Statistics	total_downlink_pinhole_connection_count	Counter	Total downlink pin-hole connections successfully created.
Network Address Translation FTP ALG Statistics	total_downlink_pinhole_connection_failure	Counter	Total downlink pin-hole connection create failures.
Network Address Translation FTP ALG Statistics	total_invalid_port_command	Counter	Total invalid port command received by NAT.
Network Address Translation FTP ALG Statistics	total_invalid_pasv_response	Counter	Total invalid PASV response received by NAT.
Network Address Translation FTP ALG Statistics	total_active_mode_wrong_address	Counter	Total active mode wrong address received by NAT.
Network Address Translation FTP ALG Statistics	total_pasv_wrong_address	Counter	Total PASV wrong address received by NAT.
Network Address Translation FTP ALG Statistics	total_multiline_response	Counter	Total multi-line response received by NAT.
Network Address Translation FTP ALG Statistics	total_invalid_argument_error	Counter	Total invalid argument error received by NAT.
Network Address Translation FTP ALG Statistics	total_operation_not_allowed_error	Counter	Total operation not allowed error received by NAT.
Network Address Translation FTP ALG Statistics	total_data_port_failed_error	Counter	Total data port failed error.
Network Address Translation FTP ALG Statistics	total_network_protocol_not_supported	Counter	Total network protocol not supported error.
Network Address Translation FTP ALG Statistics	total_command_not_supported	Counter	Total command not supported by NAT.
Network Address Translation FTP ALG Statistics	total_data_connection_open_error	Counter	Total data connection open error received by NAT.
Network Address Translation FTP ALG Statistics	total_active_mode_wrong_port	Counter	Total active mode wrong port received by NAT.
Network Address Translation TFTP ALG Statistics	buffer_parsing_failure	Counter	Total buffer parsing failure
Network Address Translation TFTP ALG Statistics	bad_relative_path_name	Counter	Access of relative path
Network Address Translation TFTP ALG Statistics	blocked_access_to_server_system_file	Counter	Accessing of system folders such as /etc on unix/linux and Windows folder on windows os is not allowed
Network Address Translation TFTP ALG Statistics	transfer_mode_not_present	Counter	Total transfer mode not present in messages
Network Address Translation TFTP ALG Statistics	good_tftp_packet	Counter	Total number of good tftp packets
Network Address Translation TFTP ALG Statistics	good_tftp_v6_packet	Counter	Total number of good tftp v6 packets
Network Address Translation TFTP ALG Statistics	bad_tftp_packet	Counter	Total number of bad tftp packets
Network Address Translation TFTP ALG Statistics	bad_tftp_v6_packet	Counter	Total number of bad tftp v6 packets
Network Address Translation TFTP ALG Statistics	invalid_message	Counter	Invalid message type(invalid opcode)
Network Address Translation TFTP ALG Statistics	pinhole_create_failure	Counter	Pinhole creation failure

Group	Metric Name	Type	Synopsis
Network Address Translation TFTP ALG Statistics	pinhole_create_success	Counter	Pinhole creation success
Network Address Translation TFTP ALG Statistics	file_name_not_present	Counter	File name not present
Network Address Translation TFTP ALG Statistics	ignore_tftp_request_sent_to_client	Counter	Invalid TFTP request sent to client
Network Address Translation TFTP ALG Statistics	malformed_tftp_packets	Counter	Total number of malformed tftp packets received
Network Address Translation SIP ALG Statistics	sip_mangled_up	Counter	Total sip mangled up
Network Address Translation SIP ALG Statistics	sip_mangled_dn	Counter	Total sip mangled down
Network Address Translation SIP ALG Statistics	sip_register_dn	Counter	Total sip register down
Network Address Translation SIP ALG Statistics	sip_invite_up	Counter	Total sip invite up
Network Address Translation SIP ALG Statistics	sip_invite_dn	Counter	Total sip invite down
Network Address Translation SIP ALG Statistics	sip_cancel_dn	Counter	Total sip cancel down
Network Address Translation SIP ALG Statistics	sip_cancel_up	Counter	Total sip cancel up
Network Address Translation SIP ALG Statistics	sip_bye_dn	Counter	Total sip bye down
Network Address Translation SIP ALG Statistics	sip_bye_up	Counter	Total sip bye up
Network Address Translation SIP ALG Statistics	create_ph_reg	Counter	Total create pinhole register
Network Address Translation SIP ALG Statistics	create_ph_aud_up	Counter	Total create pinhole audio up
Network Address Translation SIP ALG Statistics	create_ph_aud_dn	Counter	Total create pinhole audio down
Network Address Translation SIP ALG Statistics	create_ph_vid_up	Counter	Total create pinhole video up
Network Address Translation SIP ALG Statistics	create_ph_vid_dn	Counter	Total create pinhole video down
Network Address Translation SIP ALG Statistics	sip_parse	Counter	Total sip parse
Network Address Translation SIP ALG Statistics	epoll	Counter	Total epoll
Network Address Translation SIP ALG Statistics	epoll_add	Counter	Total epoll add
Network Address Translation SIP ALG Statistics	bind	Counter	Total bind
Network Address Translation SIP ALG Statistics	bind_out	Counter	Total bind out
Network Address Translation SIP ALG Statistics	create_ph_reg_failure	Counter	Total create pinhole register failure
Network Address Translation SIP ALG Statistics	replace_contact	Counter	Total replace contact
Network Address Translation SIP ALG Statistics	create_ph_aud_up_failure	Counter	Total create pinhole audio up failure
Network Address Translation SIP ALG Statistics	create_ph_aud_dn_failure	Counter	Total create pinhole audio down failure

Group	Metric Name	Type	Synopsis
Network Address Translation SIP ALG Statistics	create_ph_vid_up_failure	Counter	Total create pinhole video up failure
Network Address Translation SIP ALG Statistics	create_ph_vid_dn_failure	Counter	Total create pinhole video down failure
Network Address Translation SIP ALG Statistics	bye_no_binding	Counter	Total bye no binding
Network Address Translation SIP ALG Statistics	ack_no_binding	Counter	Total ack no binding
Network Address Translation SIP ALG Statistics	invite_no_binding	Counter	Total invite no binding
Network Address Translation SIP ALG Statistics	invalid_sip	Counter	Total invalid sip
Network Address Translation SIP ALG Statistics	pbx_rcv	Counter	Total pbx received
Network Address Translation SIP ALG Statistics	ue_rcv	Counter	Total ue received
Network Address Translation SIP ALG Statistics	inbound_socket_creation_failure	Counter	Total inbound socket creation failure
Network Address Translation SIP ALG Statistics	outbound_socket_creation_failure	Counter	Total outbound socket creation failure
Network Address Translation SIP ALG Statistics	listen	Counter	Total listen
Network Address Translation SIP ALG Statistics	accept	Counter	Total accept
Network Address Translation SIP ALG Statistics	contact_and_ue_addr_mismatch	Counter	Total contact and ue address mismatch
Network Address Translation SIP ALG Statistics	inbound_sock_creation_failed	Counter	Total inbound socket creation failed
Network Address Translation SIP ALG Statistics	uplink_unsupported_method	Counter	Total uplink unsupported method
Network Address Translation SIP ALG Statistics	downlink_unsupported_method	Counter	Total downlink unsupported method
Network Address Translation IPSEC ALG Statistics	ipsec_alg_new_ipsec_connections	Counter	Total number of,new IPSEC flows
Network Address Translation IPSEC ALG Statistics	ipsec_alg_conn_with_natt_support	Counter	Num flows with existing NAT-T support
Network Address Translation IPSEC ALG Statistics	ipsec_alg_conn_need_natt_assist	Counter	Number of new flows served by UPF where end points did not support NAT-T, and needed NAT-T assist from UPF.
Network Address Translation IPSEC ALG Statistics	ipsec_alg_existing_ike_during_esp_create	Counter	Number of times while creating a new ESP flow a corresponding IKE flow already existed.
Network Address Translation IPSEC ALG Statistics	ipsec_alg_existing_esp_during_ike_create	Counter	Number of times while creating an IKE flow corresponding ESP flow already existed
Network Address Translation IPSEC ALG Statistics	ipsec_alg_pinhole_create_failure	Counter	Pinhole creation failure
Network Address Translation IPSEC ALG Statistics	ipsec_alg_pinhole_create_success	Counter	Pinhole creation success
Network Address Translation IPSEC ALG Statistics	ipsec_alg_no_11_pool_config	Counter	Number of flows rejected because of no configuration of NAT 1:1 Pool
Network Address Translation IPSEC ALG Statistics	ipsec_alg_existing_conn_has_parent_conn	Counter	Number of flows rejected because existing flow had a parent connection

Group	Metric Name	Type	Synopsis
Network Address Translation IPSEC ALG Statistics	ipsec_alg_ul_ike_pkts	Counter	Number of IKE packets received in uplink direction
Network Address Translation IPSEC ALG Statistics	ipsec_alg_ul_esp_pkts	Counter	Number of ESP packets in uplink direction
Network Address Translation IPSEC ALG Statistics	ipsec_alg_ul_non_ike_esp_pkts	Counter	Number of IPSEC packets in Uplink direction that were neither IKE nor ESP
Network Address Translation IPSEC ALG Statistics	ipsec_alg_ike_packets_ul_success	Counter	Number of UL IKE packets successfully processed
Network Address Translation IPSEC ALG Statistics	ipsec_alg_ike_packets_ul_fail	Counter	Number of UL IKE packets which could not be successfully processed
Network Address Translation IPSEC ALG Statistics	ipsec_alg_esp_packets_ul_success	Counter	Number of UL ESP packets successfully processed
Network Address Translation IPSEC ALG Statistics	ipsec_alg_esp_packets_ul_fail	Counter	Number of UL ESP packets which could not be successfully processed
Network Address Translation IPSEC ALG Statistics	ipsec_alg_dl_ike_pkts	Counter	Number of IKE packets received in downlink direction
Network Address Translation IPSEC ALG Statistics	ipsec_alg_dl_esp_pkts	Counter	Number of ESP packets in downlink direction
Network Address Translation IPSEC ALG Statistics	ipsec_alg_ike_packets_dl_success	Counter	Number of dl IKE packets successfully processed
Network Address Translation IPSEC ALG Statistics	ipsec_alg_ike_packets_dl_fail	Counter	Number of dl IKE packets which could not be successfully processed
Network Address Translation IPSEC ALG Statistics	ipsec_alg_esp_packets_dl_success	Counter	Number of dl ESP packets successfully processed
Network Address Translation IPSEC ALG Statistics	ipsec_alg_esp_packets_dl_fail	Counter	Number of dl ESP packets which could not be successfully processed
Network Address Translation IPSEC ALG Statistics	ipsec_alg_buffer_parsing_failure	Counter	Number of times parsing IPSEC packet failed
Network Address Translation IPSEC ALG Statistics	ipsec_alg_ikev2_packets	Counter	Total number of IKEv2 packets received
Network Address Translation IPSEC ALG Statistics	ipsec_alg_ikev1_packets	Counter	Total number of IKEv1 packets received
Network Address Translation IPSEC ALG Statistics	ipsec_alg_informational_ikev2_msg	Counter	Total number of informational message IKEv2 received
Network Address Translation IPSEC ALG Statistics	ipsec_alg_informational_ikev1_msg	Counter	Total number of informational message IKEv1 received
Network Address Translation PPTP ALG Statistics	invalid_pptp_packet	Counter	Total number of invalid packets received
Network Address Translation PPTP ALG Statistics	buffer_dequeue_failed	Counter	Dequeue of buffer failed
Network Address Translation PPTP ALG Statistics	session_request	Counter	Number of session request message received
Network Address Translation PPTP ALG Statistics	invalid_session_request_packet	Counter	Number of invalid session request message received
Network Address Translation PPTP ALG Statistics	session_reply	Counter	Number of session reply message received
Network Address Translation PPTP ALG Statistics	invalid_session_reply_packet	Counter	Number of invalid session reply message received
Network Address Translation PPTP ALG Statistics	session_stop_request	Counter	Number of session stop request message received
Network Address Translation PPTP ALG Statistics	session_stop_reply	Counter	Number of session stop reply message received

Group	Metric Name	Type	Synopsis
Network Address Translation PPTP ALG Statistics	invalid_session_stop_reply_packet	Counter	Number of invalid session stop reply message received
Network Address Translation PPTP ALG Statistics	out_call_request	Counter	Number of out call request message received
Network Address Translation PPTP ALG Statistics	invalid_out_call_request_packet	Counter	Number of invalid out call request message received
Network Address Translation PPTP ALG Statistics	invalid_pac_call_id_received	Counter	Number of invalid pac callid received
Network Address Translation PPTP ALG Statistics	nat_connection_not_found	Counter	Nat connection not found
Network Address Translation PPTP ALG Statistics	pin_hole_create_failure	Counter	Number of pinhole create failure
Network Address Translation PPTP ALG Statistics	pin_hole_create_success	Counter	Number of pinhole create success
Network Address Translation PPTP ALG Statistics	out_call_reply	Counter	Number of out call reply message received
Network Address Translation PPTP ALG Statistics	invalid_out_call_reply_packet	Counter	Number of invalid out call reply message received
Network Address Translation PPTP ALG Statistics	pin_hole_patch_failure	Counter	Number of pinhole patch failure
Network Address Translation PPTP ALG Statistics	pin_hole_patch_success	Counter	Number of pinhole patch success
Network Address Translation PPTP ALG Statistics	in_call_request	Counter	Number of in call request message received
Network Address Translation PPTP ALG Statistics	invalid_in_call_request_packet	Counter	Number of invalid in call request message received
Network Address Translation PPTP ALG Statistics	in_call_reply	Counter	Number of in call reply message received
Network Address Translation PPTP ALG Statistics	invalid_in_call_reply_packet	Counter	Number of invalid in call reply message received
Network Address Translation PPTP ALG Statistics	in_call_connect	Counter	Number of in call connect message received
Network Address Translation PPTP ALG Statistics	invalid_in_call_connect_packet	Counter	Number of invalid in call connect message received
Network Address Translation PPTP ALG Statistics	nat_pac_call_id_not_present	Counter	Nat pac call id not present
Network Address Translation PPTP ALG Statistics	invalid_pns_call_id_received	Counter	Number of invalid pns callid received
Network Address Translation PPTP ALG Statistics	call_clear_request	Counter	Number of call clear request message received
Network Address Translation PPTP ALG Statistics	invalid_call_clear_request_packet	Counter	Number of invalid call clear request message received
Network Address Translation PPTP ALG Statistics	call_disconnect_notify	Counter	Number of call disconnect notify message received
Network Address Translation PPTP ALG Statistics	invalid_call_disconnect_notify_packet	Counter	Number of invalid call disconnect notify message received
Network Address Translation PPTP ALG Statistics	wan_error_notify	Counter	Number of wan error notify message received
Network Address Translation PPTP ALG Statistics	invalid_wan_error_notify_packet	Counter	Number of invalid wan error notify message received
Network Address Translation PPTP ALG Statistics	set_link_info	Counter	Number of set link info message received
Network Address Translation PPTP ALG Statistics	invalid_set_link_info_packet	Counter	Number of invalid set link info message received

Group	Metric Name	Type	Synopsis
Network Address Translation PPTP ALG Statistics	echo_request	Counter	Number of echo request message received
Network Address Translation PPTP ALG Statistics	echo_reply	Counter	Number of echo reply message received
Network Address Translation RTSP ALG Statistics	pinhole_patch_audio_count	Counter	Total server initiated PinHole for audio success
Network Address Translation RTSP ALG Statistics	pinhole_patch_video_count	Counter	Total server initiated PinHole for video success
Network Address Translation RTSP ALG Statistics	pinhole_patch_audio_error	Counter	Total server initiated PinHole creation failure for audio
Network Address Translation RTSP ALG Statistics	pinhole_patch_video_error	Counter	Total server initiated PinHole creation failure for video
Network Address Translation RTSP ALG Statistics	pinhole_audio_port_count	Counter	Total client initiated PinHole success for audio
Network Address Translation RTSP ALG Statistics	pinhole_video_port_count	Counter	Total client initiated PinHole success for video
Network Address Translation RTSP ALG Statistics	pinhole_audio_port_error_count	Counter	Total client initiated PinHole creation failure for audio
Network Address Translation RTSP ALG Statistics	pinhole_video_port_error_count	Counter	Total client initiated PinHole creation failure for video
Network Address Translation RTSP ALG Statistics	invalid_client_ports	Counter	Total invalid client Ports. Either client ports are not valid
Network Address Translation RTSP ALG Statistics	invalid_server_ports	Counter	Total invalid server Ports. Either server ports are not valid
Network Address Translation RTSP ALG Statistics	invalid_transaction	Counter	Total invalid transaction
Network Address Translation RTSP ALG Statistics	uplink_version_not_supported	Counter	RTSP version not supported for uplink interleave mode packet. Only RTSP/1.0 is supported
Network Address Translation RTSP ALG Statistics	downlink_version_not_supported	Counter	RTSP version not supported for Downlink interleave mode packet. Only RTSP/1.0 is supported
Network Address Translation RTSP ALG Statistics	uplink_interleave_mode_request	Counter	Uplink interleave mode request
Network Address Translation RTSP ALG Statistics	downlink_interleave_mode_request	Counter	Downlink interleave mode response
Network Address Translation RTSP ALG Statistics	uplink_options_request	Counter	OPTIONS request from UE
Network Address Translation RTSP ALG Statistics	uplink_describe_request	Counter	DESCRIBE request from UE
Network Address Translation RTSP ALG Statistics	uplink_setup_request	Counter	SETUP request from UE
Network Address Translation RTSP ALG Statistics	uplink_play_request	Counter	PLAY request from UE
Network Address Translation RTSP ALG Statistics	uplink_tearardown_request	Counter	TEARDOWN request from UE
Network Address Translation RTSP ALG Statistics	downlink_options_response	Counter	OPTIONS response from server
Network Address Translation RTSP ALG Statistics	downlink_describe_response	Counter	DESCRIBE response from server
Network Address Translation RTSP ALG Statistics	downlink_setup_response	Counter	SETUP response from server
Network Address Translation RTSP ALG Statistics	downlink_tearardown_response	Counter	TEARDOWN response from server

Group	Metric Name	Type	Synopsis
Network Address Translation RTSP ALG Statistics	rtsp_transport_update_failure	Counter	Updating rtsp transport header failed
Network Address Translation RTSP ALG Statistics	rtsp_transport_header_parsing_failed	Counter	Parsing rtsp transport header failed
Network Address Translation RTSP ALG Statistics	total_error_messages_send	Counter	Total error messages send due to unsupported rtsp

Network Address Translation Module Statistics

- total_nat_sessions_created

Type

64-bit Counter

Description

Total sessions created by NAT.

Metrics

Instances	Labels	Description
1	stats_group: nat_module_stats	Total sessions created by NAT.

- total_nat_sessions_deleted

Type

64-bit Counter

Description

Total sessions deleted by NAT.

Metrics

Instances	Labels	Description
1	stats_group: nat_module_stats	Total sessions deleted by NAT.

- total_nat_connections_created

Type

64-bit Counter

Description

Total connections created by NAT.

Metrics

Instances	Labels	Description
1	stats_group: nat_module_stats	Total connections created by NAT.

- total_nat_connections_deleted

Type

64-bit Counter

Description

Total connections deleted by NAT.

Metrics

Instances	Labels	Description
1	stats_group: nat_module_stats	Total connections deleted by NAT.

- total_nat_tcp_connections_created

Type

64-bit Counter

Description

Total TCP connections created by NAT.

Metrics

Instances	Labels	Description
1	stats_group: nat_module_stats	Total TCP connections created by NAT.

- total_nat_udp_connections_created

Type

64-bit Counter

Description

Total UDP connections created by NAT.

Metrics

Instances	Labels	Description
1	stats_group: nat_module_stats	Total UDP connections created by NAT.

- total_nat_icmp_connections_created

Type

64-bit Counter

Description

Total ICMP connections created by NAT.

Metrics

Instances	Labels	Description
1	stats_group: nat_module_stats	Total ICMP connections created by NAT.

- total_nat_tcp_connections_deleted

Type

64-bit Counter

Description

Total TCP connections deleted by NAT.

Metrics

Instances	Labels	Description
1	stats_group: nat_module_stats	Total TCP connections deleted by NAT.

- total_nat_udp_connections_deleted

Type

64-bit Counter

Description

Total UDP connections deleted by NAT.

Metrics

Instances	Labels	Description
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Instances	Labels	Description
1	stats_group: nat_module_stats	Total UDP connections deleted by NAT.
• total_nat_icmp_connections_deleted		
Type		
64-bit Counter		
Description		
Total ICMP connections deleted by NAT.		
Metrics		

Instances	Labels	Description
1	stats_group: nat_module_stats	Total ICMP connections deleted by NAT.
• total_nat_connections_created_action_pass		
Type		
64-bit Counter		
Description		
Total connections created by NAT with action PASS.		
Metrics		

Instances	Labels	Description
1	stats_group: nat_module_stats	Total connections created by NAT with action PASS.
• total_nat_connections_created_action_drop		
Type		
64-bit Counter		
Description		
Total connections created by NAT with action DROP.		
Metrics		

Instances	Labels	Description
1	stats_group: nat_module_stats	Total connections created by NAT with action DROP.
• total_nat_connections_created_invalid_proto_and_alg_type		
Type		
64-bit Counter		
Description		
Total connections created by NAT with protocol and ALG type mismatch.		
Metrics		

Instances	Labels	Description
1	stats_group: nat_module_stats	Total connections created by NAT with protocol and ALG type mismatch.
• connection_create_failure_natpool_not_found		
Type		
64-bit Counter		
Description		
Total connection create failures due to natpool not found.		

Metrics

Instances	Labels	Description
1	stats_group: nat_module_stats	Total connection create failures due to natpool not found.

- connection_create_failure_natpool_pending_delete

Type

64-bit Counter

Description

Total connection create failures due to natpool is pending deletion.

Metrics

Instances	Labels	Description
1	stats_group: nat_module_stats	Total connection create failures due to natpool is pending deletion.

- connection_create_failure_nat_ip_pool_not_configured

Type

64-bit Counter

Description

Total connection create failures due to nat ip pool not configured.

Metrics

Instances	Labels	Description
1	stats_group: nat_module_stats	Total connection create failures due to nat ip pool not configured.

- connection_create_failure_exceeded_per_session_nat_connections_quota

Type

64-bit Counter

Description

Total connection create failures due to session nat connections quota exhaustion.

Metrics

Instances	Labels	Description
1	stats_group: nat_module_stats	Total connection create failures due to session nat connections quota exhaustion.

- connection_create_failure_nat_port_allocation_failures

Type

64-bit Counter

Description

Total connection create failures due to nat port allocation failures.

Metrics

Instances	Labels	Description
1	stats_group: nat_module_stats	Total connection create failures due to nat port allocation failures.

- nat_port_allocation_success

Type

64-bit Counter

Description

Total nat port allocation success.

Metrics

Instances	Labels	Description
1	stats_group: nat_module_stats	Total nat port allocation success.

- nat_port_deallocation_success

Type

64-bit Counter

Description

Total nat port deallocation success.

Metrics

Instances	Labels	Description
1	stats_group: nat_module_stats	Total nat port deallocation success.

- nat_port_allocation_failure_internal_error

Type

64-bit Counter

Description

Total nat port allocation failures due to internal error.

Metrics

Instances	Labels	Description
1	stats_group: nat_module_stats	Total nat port allocation failures due to internal error.

- nat_port_allocation_failure_resource_exhaustion

Type

64-bit Counter

Description

Total nat port allocation failure due to port unavailability.

Metrics

Instances	Labels	Description
1	stats_group: nat_module_stats	Total nat port allocation failures due to port unavailability.

- total_nat_connection_creation_failure_system_resource_exhaustion

Type

64-bit Counter

Description

Total nat connection allocation failure due to system resource unavailability.

Metrics

Instances	Labels	Description
1	stats_group: nat_module_stats	Total nat connection allocation failures due to system resource unavailability.

- nat_port_allocation_failure_unsupported_protocol

Type

64-bit Counter

Description		
Total nat port allocation failure due to unsupported protocol.		
Metrics		
Instances	Labels	Description
1	stats_group: nat_module_stats	Total nat port allocation failures due to unsupported protocol.
total_nat_connection_creation_failure_due_to_unknown_icmp_type		
Type		
64-bit Counter		
Description		
Total nat connection creation failure due to unknown icmp type.		
Metrics		
Instances	Labels	Description
1	stats_group: nat_module_stats	Total nat connection creation failure due to unknown icmp type.
total_nat_connection_creation_ignored_due_to_unsupp_proto		
Type		
64-bit Counter		
Description		
Total nat connection creation ignored due to unsupported proto.		
Metrics		
Instances	Labels	Description
1	stats_group: nat_module_stats	Total nat connection creation ignored due to unsupported proto.
nat_port_deallocation_failure_internal_error		
Type		
64-bit Counter		
Description		
Total nat port deallocation failure due to internal error.		
Metrics		
Instances	Labels	Description
1	stats_group: nat_module_stats	Total nat port deallocation failures due to internal error.
nat_port_deallocation_failure_unsupported_protocol		
Type		
64-bit Counter		
Description		
Total nat port deallocation failure due to unsupported protocol.		
Metrics		
Instances	Labels	Description
1	stats_group: nat_module_stats	Total nat port deallocation failures due to unsupported protocol.
nat_total_uplink_packets		
Type		

64-bit Counter
Description
Total nat uplink packets
Metrics

Instances	Labels	Description
1	stats_group: nat_module_stats	Total nat uplink packets

- nat_total_uplink_bytes

Type
64-bit Counter
Description
Total nat uplink bytes
Metrics

Instances	Labels	Description
1	stats_group: nat_module_stats	Total nat uplink bytes

- nat_total_downlink_packets

Type
64-bit Counter
Description
Total nat downlink packets
Metrics

Instances	Labels	Description
1	stats_group: nat_module_stats	Total nat downlink packets

- nat_total_downlink_bytes

Type
64-bit Counter
Description
Total nat downlink bytes
Metrics

Instances	Labels	Description
1	stats_group: nat_module_stats	Total nat downlink bytes

- nat_11_port_alloc

Type
64-bit Counter
Description
NAT1:1 port allocation
Metrics

Instances	Labels	Description
1	stats_group: nat_module_stats	NAT1:1 port allocation

- nat_11_port_alloc_zero

Type
64-bit Counter
Description
NAT1:1 port allocation with zero
Metrics

Instances	Labels	Description
1	stats_group: nat_module_stats	NAT1:1 port allocation with zero

- nat_11_address_reuse

Type
64-bit Counter
Description
NAT1:1 address reuse
Metrics

Instances	Labels	Description
1	stats_group: nat_module_stats	NAT1:1 address reuse

- nat_11_address_alloc

Type
64-bit Counter
Description
NAT1:1 address allocation
Metrics

Instances	Labels	Description
1	stats_group: nat_module_stats	NAT1:1 address allocation

- nat_11_stale_address_return

Type
64-bit Counter
Description
NAT1:1 stale address return
Metrics

Instances	Labels	Description
1	stats_group: nat_module_stats	NAT1:1 stale address return

- nat_11_address_release

Type
64-bit Counter
Description
NAT1:1 address release
Metrics

Instances	Labels	Description
1	stats_group: nat_module_stats	NAT1:1 address release

• nat_11_no_address

Type
64-bit Counter
Description
NAT1:1 no address available
Metrics

Instances	Labels	Description
1	stats_group: nat_module_stats	NAT1:1 no address available

• nat_11_update_pinhole_failure

Type
64-bit Counter
Description
NAT1:1 Update pinhole failure
Metrics

Instances	Labels	Description
1	stats_group: nat_module_stats	NAT1:1 Update pinhole failure

• nat_11_update_pinhole_success

Type
64-bit Counter
Description
NAT1:1 Update pinhole success
Metrics

Instances	Labels	Description
1	stats_group: nat_module_stats	NAT1:1 Update pinhole success

• nat_11_open_pinhole

Type
64-bit Counter
Description
NAT1:1 Open Pinhole
Metrics

Instances	Labels	Description
1	stats_group: nat_module_stats	NAT1:1 Open Pinhole

• nat_11_known_ueport_open_pinhole

Type
64-bit Counter
Description
NAT1:1 Known UE Port Open Pinhole
Metrics

Instances	Labels	Description
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Instances	Labels	Description
1	stats_group: nat_module_stats	NAT1:1 Known UE Port Open Pinhole
• gnat_port_alloc		
Type		
64-bit Counter		
Description		
Guaranteed NAT port allocation		
Metrics		

Instances	Labels	Description
1	stats_group: nat_module_stats	Guaranteed NAT port allocation
• gnat_port_dealloc		
Type		
64-bit Counter		
Description		
Guaranteed NAT port deallocation		
Metrics		

Instances	Labels	Description
1	stats_group: nat_module_stats	Guaranteed NAT port deallocation
• gnat_handle_session_ip_failure		
Type		
64-bit Counter		
Description		
Guaranteed NAT handle session IP failure		
Metrics		

Instances	Labels	Description
1	stats_group: nat_module_stats	Guaranteed NAT handle session IP failure
• gnat_handle_session_ip_no_natpool		
Type		
64-bit Counter		
Description		
Guaranteed NAT handle session IP no NAT pool		
Metrics		

Instances	Labels	Description
1	stats_group: nat_module_stats	Guaranteed NAT handle session IP no NAT pool
• max_users_per_nat_ip_limit_reached		
Type		
64-bit Counter		
Description		
Max users per NAT IP limit exceeded		

Metrics		
Instances	Labels	Description
1	stats_group: nat_module_stats	Maximum subscribers per NAT IP limit reached
num_ip_released_from_subscriber_session		
Type		
64-bit Counter		
Description		
Number of IP's released from subscriber session		
Metrics		
Instances	Labels	Description
1	stats_group: nat_module_stats	Number of IP's released from subscriber session
num_port_object_internal_access_failure		
Type		
64-bit Counter		
Description		
Number of port object internal access failure		
Metrics		
Instances	Labels	Description
1	stats_group: nat_module_stats	Number of port object internal access failure
num_no_free_ports_on_ip		
Type		
64-bit Counter		
Description		
No Available free Ports available for IP		
Metrics		
Instances	Labels	Description
1	stats_group: nat_module_stats	No free Ports available for IP
num_nat_port_allocation_collisions		
Type		
64-bit Counter		
Description		
Number of NAT port allocation collisions		
Metrics		
Instances	Labels	Description
1	stats_group: nat_module_stats	Number of NAT port allocation collisions
total_nat_port_allocation_success_on_first_try		
Type		
64-bit Counter		
Description		

Number of NAT port allocation Success on first try

Metrics

Instances	Labels	Description
1	stats_group: nat_module_stats	Number of NAT port allocation Success on first try

- total_nat_port_allocation_not_success_on_first_try

Type

64-bit Counter

Description

Number of NAT port allocation not Success on first try

Metrics

Instances	Labels	Description
1	stats_group: nat_module_stats	Number of NAT port allocation not Success on first try

- total_nat_reverse_lookup_failure

Type

64-bit Counter

Description

NAT Reverse Lookup Failure

Metrics

Instances	Labels	Description
1	stats_group: nat_module_stats	NAT Reverse Lookup Failure

- total_nat_conn_fail_due_to_max_connection_limit_reached

Type

64-bit Counter

Description

Connection Failure due to max connection limit reached

Metrics

Instances	Labels	Description
1	stats_group: nat_module_stats	Connection Fail due to max connection limit reached

- nat_m1_update_pinhole_failure

Type

64-bit Counter

Description

NATM:1 Update pinhole failure

Metrics

Instances	Labels	Description
1	stats_group: nat_module_stats	NATM:1 Update pinhole failure

- nat_m1_update_pinhole_success

Type

64-bit Counter

Description

NATM:1 Update pinhole success

Metrics

Instances	Labels	Description
1	stats_group: nat_module_stats	NATM:1 Update pinhole success

• conn_create_failure_sip_cpp_interface_failure

Type

64-bit Counter

Description

NATM:1 Update pinhole success

Metrics

Instances

Labels

Description

1	stats_group: nat_module_stats	NATM:1 Update pinhole success
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• total_nat_active_connections

Type

64-bit Gauge

Description

Total present NAT connections.

Metrics

Instances

Labels

Description

1	stats_group: nat_module_stats	Total present NAT connections.
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• total_nat_active_sessions

Type

64-bit Gauge

Description

Total present NAT sessions

Metrics

Instances

Labels

Description

1	stats_group: nat_module_stats	Total present NAT stats.
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• total_nat_per_connection_average_port_collision_count

Type

64-bit Gauge

Description

Average Port Collision count per connection

Metrics

Instances

Labels

Description

1	stats_group: nat_module_stats	Average Port Collision count per connection
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Network Address Translation NatPool Statistics

- total_natpool_sessions_created
 - Type
 - 64-bit Counter
 - Description
 - Total sessions for which NAT resources were allocated from nat-pool.
 - Labels
 - networkcontext_name, natpool_name**
- total_natpool_sessions_deleted
 - Type
 - 64-bit Counter
 - Description
 - Total sessions for which NAT resources were deallocated from nat-pool.
 - Labels
 - networkcontext_name, natpool_name**
- total_natpool_connections_created
 - Type
 - 64-bit Counter
 - Description
 - Total connections created using NAT resources from nat-pool.
 - Labels
 - networkcontext_name, natpool_name**
- total_natpool_connections_deleted
 - Type
 - 64-bit Counter
 - Description
 - Total connections deleted using NAT resources from nat-pool.
 - Labels
 - networkcontext_name, natpool_name**
- total_natpool_sessions_denied_resource
 - Type
 - 64-bit Counter
 - Description
 - Total sessions denied resource using this nat-pool.
 - Labels
 - networkcontext_name, natpool_name**
- total_natpool_connections_create_failure
 - Type
 - 64-bit Counter
 - Description

Total connections create failure using this nat-pool.

Labels

networkcontext_name, natpool_name

- total_natpool_port_alloc_success_on_first_try

Type

64-bit Counter

Description

Total port alloc success on first try using this nat-pool.

Labels

networkcontext_name, natpool_name

- total_natpool_port_alloc_not_success_on_first_try

Type

64-bit Counter

Description

Total port alloc success on first try using this nat-pool.

Labels

networkcontext_name, natpool_name

- total_natpool_num_port_alloc_max_retry_reached

Type

64-bit Counter

Description

Port allocation max retry reached.

Labels

networkcontext_name, natpool_name

- natpool_connection_create_failure_nat_port_allocation_failures

Type

64-bit Counter

Description

Total connection create failures due to nat port allocation failures.

Labels

networkcontext_name, natpool_name

- natpool_connection_create_failure_nat_ip_pool_not_configured

Type

64-bit Counter

Description

Total connection create failures due to nat ip pool not configured.

Labels

networkcontext_name, natpool_name

- natpool_connection_create_failure_exceeded_per_session_nat_connections_quota

Type

64-bit Counter

Description

Total connection create failures due to session nat connections quota exhaustion.

Labels

networkcontext_name, natpool_name

- total_natpool_number_of_nat_port_allocation_collisions

Type

64-bit Counter

Description

Total Number of NAT port collision during NAT Connection creation.

Labels

networkcontext_name, natpool_name

- total_natpool_active_sessions

Type

64-bit Gauge

Description

Total sessions present using NAT resources from nat-pool.

Labels

networkcontext_name, natpool_name

- total_natpool_active_connections

Type

64-bit Gauge

Description

Total connections present using NAT resources from nat-pool.

Labels

networkcontext_name, natpool_name

- total_natpool_average_port_collision_count

Type

64-bit Gauge

Description

Average NAT port collision during NAT Connection creation.

Labels

networkcontext_name, natpool_name

- total_natpool_port_util_percentage

Type

64-bit Gauge

Description

Total free nat ports on this nat-pool.

Labels

networkcontext_name, natpool_name

- total_natpool_used_nat_ports

Type

64-bit Gauge

Description

Total used nat ports on this nat-pool.

Labels

networkcontext_name, natpool_name

- total_natpool_free_nat_ports

Type

64-bit Gauge

Description

Total free nat ports on this nat-pool.

Labels

networkcontext_name, natpool_name

Network Address Translation FTP ALG Statistics

- total_port_command

Type

64-bit Counter

Description

Total port commands received by NAT.

Metrics

Instances	Labels	Description
1	<i>stats_group: nat_ftp_alg_stats</i>	Total port commands received by NAT.

- total_eprt_command

Type

64-bit Counter

Description

Total EPRT command received by NAT.

Metrics

Instances	Labels	Description
1	<i>stats_group: nat_ftp_alg_stats</i>	Total EPRT command received by NAT.

- total_pasv_command

Type

64-bit Counter

Description

Total PASV command received by NAT.

Metrics		
Instances	Labels	Description
1	stats_group: nat_ftp_alg_stats	Total PASV command received by NAT.
total_epasv_command		
Type		
64-bit Counter		
Description		
Total EPASV command received by NAT.		
Metrics		
Instances	Labels	Description
1	stats_group: nat_ftp_alg_stats	Total EPASV command received by NAT.
total_epasv_v6_command		
Type		
64-bit Counter		
Description		
Total EPASV v6 command received by NAT.		
Metrics		
Instances	Labels	Description
1	stats_group: nat_ftp_alg_stats	Total EPASV v6 command received by NAT.
total_epasv_v6_cmd_resp_wrong_addr		
Type		
64-bit Counter		
Description		
Total EPASV command Response with wrong address received by NAT.		
Metrics		
Instances	Labels	Description
1	stats_group: nat_ftp_alg_stats	Total EPASV command Response with wrong address received by NAT.
total_eprt_invalid_arg		
Type		
64-bit Counter		
Description		
Total EPRT command with invalid argument.		
Metrics		
Instances	Labels	Description
1	stats_group: nat_ftp_alg_stats	Total EPRT command with invalid argument.
total_eprt_v6_invalid_arg		
Type		
64-bit Counter		
Description		

Total EPRT v6 command with invalid argument.

Metrics

Instances	Labels	Description
1	stats_group: nat_ftp_alg_stats	Total EPRT v6 command with invalid argument.

- total_eprt_v6_command

Type

64-bit Counter

Description

Total EPRT v6 command received by NAT.

Metrics

Instances	Labels	Description
1	stats_group: nat_ftp_alg_stats	Total EPRT v6 command received by NAT.

- total_eprt_v6_cmd_wrong_addr

Type

64-bit Counter

Description

Total EPRT v6 command Response with wrong address received by NAT.

Metrics

Instances	Labels	Description
1	stats_group: nat_ftp_alg_stats	Total EPRT v6 command Response with wrong address received by NAT.

- total_uplink_pinhole_connection_count

Type

64-bit Counter

Description

Total uplink pin-hole connections successfully created.

Metrics

Instances	Labels	Description
1	stats_group: nat_ftp_alg_stats	Total uplink pin-hole connections successfully created.

- total_uplink_pinhole_connection_failure

Type

64-bit Counter

Description

Total uplink pin-hole connection create failures.

Metrics

Instances	Labels	Description
1	stats_group: nat_ftp_alg_stats	Total uplink pin-hole connection create failures.

- total_downlink_pinhole_connection_count

Type

64-bit Counter

Description

Total downlink pin-hole connections successfully created.

Metrics

Instances	Labels	Description
1	stats_group: nat_ftp_alg_stats	Total downlink pin-hole connections successfully created.

- total_downlink_pinhole_connection_failure

Type

64-bit Counter

Description

Total downlink pin-hole connection create failures.

Metrics

Instances	Labels	Description
1	stats_group: nat_ftp_alg_stats	Total downlink pin-hole connection create failures.

- total_invalid_port_command

Type

64-bit Counter

Description

Total invalid port command received by NAT.

Metrics

Instances	Labels	Description
1	stats_group: nat_ftp_alg_stats	Total invalid port command received by NAT.

- total_invalid_pasv_response

Type

64-bit Counter

Description

Total invalid PASV response received by NAT.

Metrics

Instances	Labels	Description
1	stats_group: nat_ftp_alg_stats	Total invalid port command received by NAT.

- total_active_mode_wrong_address

Type

64-bit Counter

Description

Total active mode wrong address received by NAT.

Metrics

Instances	Labels	Description
1	stats_group: nat_ftp_alg_stats	Total invalid PASV response received by NAT.

- total_pasv_wrong_address

Type

Instances	Labels	Description
1	stats_group: nat_ftp_alg_stats	Total active mode wrong address received by NAT.

64-bit Counter
Description
Total PASV wrong address received by NAT.
Metrics

Instances	Labels	Description
1	stats_group: nat_ftp_alg_stats	Total PASV wrong address received by NAT.

- total_multiline_response
- | |
|--|
| Type |
| 64-bit Counter |
| Description |
| Total multi-line response received by NAT. |
| Metrics |

Instances	Labels	Description
1	stats_group: nat_ftp_alg_stats	Total multi-line response received by NAT.

- total_invalid_argument_error
- | |
|---|
| Type |
| 64-bit Counter |
| Description |
| Total invalid argument error received by NAT. |
| Metrics |

Instances	Labels	Description
1	stats_group: nat_ftp_alg_stats	Total invalid argument error received by NAT.

- total_operation_not_allowed_error
- | |
|--|
| Type |
| 64-bit Counter |
| Description |
| Total operation not allowed error received by NAT. |
| Metrics |

Instances	Labels	Description
1	stats_group: nat_ftp_alg_stats	Total operation not allowed error received by NAT.

- total_data_port_failed_error
- | |
|-------------------------------|
| Type |
| 64-bit Counter |
| Description |
| Total data port failed error. |
| Metrics |

Instances	Labels	Description
1	stats_group: nat_ftp_alg_stats	Total data port failed error.

- total_network_protocol_not_supported

Type
64-bit Counter
Description
Total network protocol not supported error.
Metrics

Instances	Labels	Description
1	stats_group: nat_ftp_alg_stats	Total network protocol not supported error.

- total_command_not_supported

Type
64-bit Counter
Description
Total command not supported by NAT.
Metrics

Instances	Labels	Description
1	stats_group: nat_ftp_alg_stats	Total command not supported by NAT.

- total_data_connection_open_error

Type
64-bit Counter
Description
Total data connection open error received by NAT.
Metrics

Instances	Labels	Description
1	stats_group: nat_ftp_alg_stats	Total data connection open error received by NAT.

- total_active_mode_wrong_port

Type
64-bit Counter
Description
Total active mode wrong port received by NAT.
Metrics

Instances	Labels	Description
1	stats_group: nat_ftp_alg_stats	Total active mode wrong port received by NAT.

Network Address Translation TFTP ALG Statistics

- buffer_parsing_failure

Type
64-bit Counter
Description
Total buffer parsing failure
Metrics

Instances	Labels	Description
1	stats_group: nat_tftp_alg_stats	Total buffer parsing failure
• bad_relative_path_name		
Type		
64-bit Counter		
Description		
Access of relative path		
Metrics		

Instances	Labels	Description
1	stats_group: nat_tftp_alg_stats	Access of relative path
• blocked_access_to_server_system_file		
Type		
64-bit Counter		
Description		
Accessing of system folders such as /etc on unix/linux and Windows folder on windows os is not allowed		
Metrics		

Instances	Labels	Description
1	stats_group: nat_tftp_alg_stats	Accessing of system folders such as /etc on unix/linux and Windows folder on windows os is not allowed
• transfer_mode_not_present		
Type		
64-bit Counter		
Description		
Total transfer mode not present in messages		
Metrics		

Instances	Labels	Description
1	stats_group: nat_tftp_alg_stats	Total transfer mode not present in messages
• good_tftp_packet		
Type		
64-bit Counter		
Description		
Total number of good tftp packets		
Metrics		

Instances	Labels	Description
1	stats_group: nat_tftp_alg_stats	Total number of good tftp packets
• good_tftp_v6_packet		
Type		
64-bit Counter		
Description		
Total number of good tftp v6 packets		

Metrics		
Instances	Labels	Description
1	stats_group: nat_tftp_alg_stats	Total number of good tftp v6 packets
bad_tftp_packet		
Type		
64-bit Counter		
Description		
Total number of bad tftp packets		
Metrics		
Instances	Labels	Description
1	stats_group: nat_tftp_alg_stats	Total number of bad tftp packets
bad_tftp_v6_packet		
Type		
64-bit Counter		
Description		
Total number of bad tftp v6 packets		
Metrics		
Instances	Labels	Description
1	stats_group: nat_tftp_alg_stats	Total number of bad tftp v6 packets
invalid_message		
Type		
64-bit Counter		
Description		
Invalid message type(invalid opcode)		
Metrics		
Instances	Labels	Description
1	stats_group: nat_tftp_alg_stats	Invalid message type(invalid opcode)
pinhole_create_failure		
Type		
64-bit Counter		
Description		
Pinhole creation failure		
Metrics		
Instances	Labels	Description
1	stats_group: nat_tftp_alg_stats	Pinhole creation failure
pinhole_create_success		
Type		
64-bit Counter		
Description		

Pinhole creation success

Metrics

Instances	Labels	Description
1	stats_group: nat_tftp_alg_stats	Pinhole creation success

- file_name_not_present

Type

64-bit Counter

Description

File name not present

Metrics

Instances	Labels	Description
1	stats_group: nat_tftp_alg_stats	File name not present

- ignore_tftp_request_sent_to_client

Type

64-bit Counter

Description

Invalid TFTP request sent to client

Metrics

Instances	Labels	Description
1	stats_group: nat_tftp_alg_stats	Invalid TFTP request sent to client

- malformed_tftp_packets

Type

64-bit Counter

Description

Total number of malformed tftp packets received

Metrics

Instances	Labels	Description
1	stats_group: nat_tftp_alg_stats	Total number of malformed tftp packets received

Network Address Translation SIP ALG Statistics

- sip_mangled_up

Type

64-bit Counter

Description

Total sip mangled up

Metrics

Instances	Labels	Description
1	stats_group: nat_sip_alg_stats	Total sip mangled up

- sip_mangled_dn

Type
64-bit Counter
Description
Total sip mangled down
Metrics

Instances	Labels	Description
1	stats_group: nat_sip_alg_stats	Total sip mangled down

• sip_register_dn

Type
64-bit Counter
Description
Total sip register down
Metrics

Instances	Labels	Description
1	stats_group: nat_sip_alg_stats	Total sip register down

• sip_invite_up

Type
64-bit Counter
Description
Total sip invite up
Metrics

Instances	Labels	Description
1	stats_group: nat_sip_alg_stats	Total sip invite up

• sip_invite_dn

Type
64-bit Counter
Description
Total sip invite down
Metrics

Instances	Labels	Description
1	stats_group: nat_sip_alg_stats	Total sip invite down

• sip_cancel_dn

Type
64-bit Counter
Description
Total sip cancel down
Metrics

Instances	Labels	Description
1	stats_group: nat_sip_alg_stats	Total sip cancel down

• sip_cancel_up

Type

64-bit Counter

Description

Total sip cancel up

Metrics

Instances	Labels	Description
1	<i>stats_group: nat_sip_alg_stats</i>	Total sip cancel up

• sip_bye_dn

Type

64-bit Counter

Description

Total sip bye down

Metrics

Instances	Labels	Description
1	<i>stats_group: nat_sip_alg_stats</i>	Total sip bye down

• sip_bye_up

Type

64-bit Counter

Description

Total sip bye up

Metrics

Instances	Labels	Description
1	<i>stats_group: nat_sip_alg_stats</i>	Total sip bye up

• create_ph_reg

Type		
64-bit Counter		
Description		
Total create pinhole register		
Metrics		
Instances	Labels	Description
1	<i>stats_group: nat_sip_alg_stats</i>	Total create pinhole register

• create_ph_aud_up

Type

64-bit Counter

Description

Total create pinhole audio up

Metrics

Instances	Labels	Description
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Instances	Labels	Description
1	stats_group: nat_sip_alg_stats	Total create pinhole audio up
• create_ph_aud_dn		
Type		
64-bit Counter		
Description		
Total create pinhole audio down		
Metrics		

Instances	Labels	Description
1	stats_group: nat_sip_alg_stats	Total create pinhole audio down
• create_ph_vid_up		
Type		
64-bit Counter		
Description		
Total create pinhole video up		
Metrics		

Instances	Labels	Description
1	stats_group: nat_sip_alg_stats	Total create pinhole video up
• create_ph_vid_dn		
Type		
64-bit Counter		
Description		
Total create pinhole video down		
Metrics		

Instances	Labels	Description
1	stats_group: nat_sip_alg_stats	Total create pinhole video down
• sip_parse		
Type		
64-bit Counter		
Description		
Total sip parse		
Metrics		

Instances	Labels	Description
1	stats_group: nat_sip_alg_stats	Total sip parse
• epoll		
Type		
64-bit Counter		
Description		
Total epoll		

Metrics		
Instances	Labels	Description
1	stats_group: nat_sip_alg_stats	Total epoll
• epoll_add		
Type		
64-bit Counter		
Description		
Total epoll add		
Metrics		
Instances	Labels	Description
1	stats_group: nat_sip_alg_stats	Total epoll add
• bind		
Type		
64-bit Counter		
Description		
Total bind		
Metrics		
Instances	Labels	Description
1	stats_group: nat_sip_alg_stats	Total bind
• bind_out		
Type		
64-bit Counter		
Description		
Total bind out		
Metrics		
Instances	Labels	Description
1	stats_group: nat_sip_alg_stats	Total bind out
• create_ph_reg_failure		
Type		
64-bit Counter		
Description		
Total create pinhole register failure		
Metrics		
Instances	Labels	Description
1	stats_group: nat_sip_alg_stats	Total create pinhole register failure
• replace_contact		
Type		
64-bit Counter		
Description		

Total replace contact		
Metrics		
Instances	Labels	Description
1	stats_group: nat_sip_alg_stats	Total replace contact
create_ph_aud_up_failure		
Type		
64-bit Counter		
Description		
Total create pinhole audio up failure		
Metrics		
Instances	Labels	Description
1	stats_group: nat_sip_alg_stats	Total create pinhole audio up failure
create_ph_aud_dn_failure		
Type		
64-bit Counter		
Description		
Total create pinhole audio down failure		
Metrics		
Instances	Labels	Description
1	stats_group: nat_sip_alg_stats	Total create pinhole audio down failure
create_ph_vid_up_failure		
Type		
64-bit Counter		
Description		
Total create pinhole video up failure		
Metrics		
Instances	Labels	Description
1	stats_group: nat_sip_alg_stats	Total create pinhole video up failure
create_ph_vid_dn_failure		
Type		
64-bit Counter		
Description		
Total create pinhole video down failure		
Metrics		
Instances	Labels	Description
1	stats_group: nat_sip_alg_stats	Total create pinhole video down failure
bye_no_binding		
Type		
64-bit Counter		

Description

Total bye no binding

Metrics

Instances	Labels	Description
1	stats_group: nat_sip_alg_stats	Total bye no binding

- ack_no_binding

Type

64-bit Counter

Description

Total ack no binding

Metrics

Instances	Labels	Description
1	stats_group: nat_sip_alg_stats	Total ack no binding

- invite_no_binding

Type

64-bit Counter

Description

Total invite no binding

Metrics

Instances	Labels	Description
1	stats_group: nat_sip_alg_stats	Total invite no binding

- invalid_sip

Type

64-bit Counter

Description

Total invalid sip

Metrics

Instances	Labels	Description
1	stats_group: nat_sip_alg_stats	Total invalid sip

- pbx_rcv

Type

64-bit Counter

Description

Total pbx received

Metrics

Instances	Labels	Description
1	stats_group: nat_sip_alg_stats	Total pbx received

- ue_rcv

Type

64-bit Counter

Description

Total ue received

Metrics

Instances	Labels	Description
1	stats_group: nat_sip_alg_stats	Total ue received

- inbound_socket_creation_failure

Type

64-bit Counter

Description

Total inbound socket creation failure

Metrics

Instances	Labels	Description
1	stats_group: nat_sip_alg_stats	Total inbound socket creation failure

- outbound_socket_creation_failure

Type

64-bit Counter

Description

Total outbound socket creation failure

Metrics

Instances	Labels	Description
1	stats_group: nat_sip_alg_stats	Total outbound socket creation failure

- listen

Type

64-bit Counter

Description

Total listen

Metrics

Instances	Labels	Description
1	stats_group: nat_sip_alg_stats	Total listen

- accept

Type

64-bit Counter

Description

Total accept

Metrics

Instances	Labels	Description
1	stats_group: nat_sip_alg_stats	Total accept

- contact_and_ue_addr_mismatch

Type
64-bit Counter
Description
Total contact and ue address mismatch
Metrics

Instances	Labels	Description
1	stats_group: nat_sip_alg_stats	Total contact and ue address mismatch

- inbound_sock_creation_failed

Type
64-bit Counter
Description
Total inbound socket creation failed
Metrics

Instances	Labels	Description
1	stats_group: nat_sip_alg_stats	Total inbound socket creation failed

- uplink_unsupported_method

Type
64-bit Counter
Description
Total uplink unsupported method
Metrics

Instances	Labels	Description
1	stats_group: nat_sip_alg_stats	Total uplink unsupported method

- downlink_unsupported_method

Type
64-bit Counter
Description
Total downlink unsupported method
Metrics

Instances	Labels	Description
1	stats_group: nat_sip_alg_stats	Total downlink unsupported method

Network Address Translation IPSEC ALG Statistics

- ipsec_alg_new_ipsec_connections

Type
64-bit Counter
Description
Total number of new IPSEC flows
Metrics

Instances	Labels	Description
1	stats_group: nat_ipsec_alg_stats	Total number of new IPSEC flows
• ipsec_alg_conn_with_natt_support		
Type		
64-bit Counter		
Description		
Num flows with existing NAT-T support		
Metrics		

Instances	Labels	Description
1	stats_group: nat_ipsec_alg_stats	Num Flows with existing NAT-T support
• ipsec_alg_conn_need_natt_assist		
Type		
64-bit Counter		
Description		
Number of new flows served by UPF where end points did not support NAT-T, and needed NAT-T assist from UPF.		
Metrics		

Instances	Labels	Description
1	stats_group: nat_ipsec_alg_stats	Number of new flows served by UPF where end points did not support NAT-T, and needed NAT-T assist from UPF.
• ipsec_alg_existing_ike_during_esp_create		
Type		
64-bit Counter		
Description		
Number of times while creating a new ESP flow a corresponding IKE flow already existed.		
Metrics		

Instances	Labels	Description
1	stats_group: nat_ipsec_alg_stats	Number of times while creating a new ESP flow a corresponding IKE flow already existed.
• ipsec_alg_existing_esp_during_ike_create		
Type		
64-bit Counter		
Description		
Number of times while creating an IKE flow corresponding ESP flow already existed		
Metrics		

Instances	Labels	Description
1	stats_group: nat_ipsec_alg_stats	Total number of good ipsec packets
• ipsec_alg_pinhole_create_failure		
Type		
64-bit Counter		
Description		
Pinhole creation failure		

Metrics

Instances	Labels	Description
1	stats_group: nat_ipsec_alg_stats	Pinhole creation failure
• ipsec_alg_pinhole_create_success		
Type		
64-bit Counter		
Description		
Pinhole creation success		
Metrics		

Instances	Labels	Description
1	stats_group: nat_ipsec_alg_stats	Pinhole creation success
• ipsec_alg_no_11_pool_config		
Type		
64-bit Counter		
Description		
Number of flows rejected because of no configuration of NAT 1:1 Pool		
Metrics		

Instances	Labels	Description
1	stats_group: nat_ipsec_alg_stats	Number of flows rejected because of no configuration of NAT 1:1 Pool
• ipsec_alg_existing_conn_has_parent_conn		
Type		
64-bit Counter		
Description		
Number of flows rejected because existing flow had a parent connection		
Metrics		

Instances	Labels	Description
1	stats_group: nat_ipsec_alg_stats	Number of flows rejected because existing flow had a parent connection
• ipsec_alg_ul_ike_pkts		
Type		
64-bit Counter		
Description		
Number of IKE packets received in uplink direction		
Metrics		

Instances	Labels	Description
1	stats_group: nat_ipsec_alg_stats	Number of IKE packets received in uplink direction
• ipsec_alg_ul_esp_pkts		
Type		
64-bit Counter		
Description		

Number of ESP packets in uplink direction

Metrics

Instances	Labels	Description
1	stats_group: nat_ipsec_alg_stats	Number of ESP packets in uplink direction

- ipsec_alg_ul_non_ike_esp_pkts

Type

64-bit Counter

Description

Number of IPSEC packets in Uplink direction that were neither IKE nor ESP

Metrics

Instances	Labels	Description
1	stats_group: nat_ipsec_alg_stats	Number of IPSEC packets in Uplink direction that were neither IKE nor ESP

- ipsec_alg_ike_packets_ul_success

Type

64-bit Counter

Description

Number of UL IKE packets successfully proccesed

Metrics

Instances	Labels	Description
1	stats_group: nat_ipsec_alg_stats	Number of UL IKE packets successfully proccesed

- ipsec_alg_ike_packets_ul_fail

Type

64-bit Counter

Description

Number of UL IKE packets which could not be successfully processed

Metrics

Instances	Labels	Description
1	stats_group: nat_ipsec_alg_stats	Number of UL IKE packets which could not be successfully processed

- ipsec_alg_esp_packets_ul_success

Type

64-bit Counter

Description

Number of UL ESP packets successfully proccesed

Metrics

Instances	Labels	Description
1	stats_group: nat_ipsec_alg_stats	Number of UL ESP packets successfully proccesed

- ipsec_alg_esp_packets_ul_fail

Type

64-bit Counter

Description

Number of UL ESP packets which could not be successfully processed

Metrics

Instances	Labels	Description
1	stats_group: nat_ipsec_alg_stats	Number of UL ESP packets which could not be successfully processed

- ipsec_alg_dl_ike_pkts

Type

64-bit Counter

Description

Number of IKE packets received in downlink direction

Metrics

Instances	Labels	Description
1	stats_group: nat_ipsec_alg_stats	Number of IKE packets received in downlink direction

- ipsec_alg_dl_esp_pkts

Type

64-bit Counter

Description

Number of ESP packets in downlink direction

Metrics

Instances	Labels	Description
1	stats_group: nat_ipsec_alg_stats	Number of IKE packets received in downlink direction

- ipsec_alg_ike_packets_dl_success

Type

64-bit Counter

Description

Number of DL IKE packets successfully processed

Metrics

Instances	Labels	Description
1	stats_group: nat_ipsec_alg_stats	Number of ESP packets in downlink direction

- ipsec_alg_ike_packets_dl_fail

Type

64-bit Counter

Description

Number of dl IKE packets which could not be successfully processed

Metrics

Instances	Labels	Description
1	stats_group: nat_ipsec_alg_stats	Number of dl IKE packets successfully processed

- ipsec_alg_esp_packets_dl_success

Type

Instances	Labels	Description
1	stats_group: nat_ipsec_alg_stats	Number of dl IKE packets which could not be successfully processed

64-bit Counter
Description
Number of dl ESP packets successfdlly proccesed
Metrics

Instances	Labels	Description
1	stats_group: nat_ipsec_alg_stats	Number of dl ESP packets successfdlly proccesed

• ipsec_alg_esp_packets_dl_fail
Type
64-bit Counter
Description
Number of dl ESP packets which could not be successfdlly processed
Metrics

Instances	Labels	Description
1	stats_group: nat_ipsec_alg_stats	Number of dl ESP packets which could not be successfdlly processed

• ipsec_alg_buffer_parsing_failure
Type
64-bit Counter
Description
Number of times parsing IPSEC packet failed
Metrics

Instances	Labels	Description
1	stats_group: nat_ipsec_alg_stats	Number of times parsing IPSEC packet failed

• ipsec_alg_ikev2_packets
Type
64-bit Counter
Description
Total number of IKEv2 packets received
Metrics

Instances	Labels	Description
1	stats_group: nat_ipsec_alg_stats	Total number of IKEv2 packets received

• ipsec_alg_ikev1_packets
Type
64-bit Counter
Description
Total number of IKEv1 packets received
Metrics

Instances	Labels	Description
1	stats_group: nat_ipsec_alg_stats	Total number of IKEv1 packets received

• ipsec_alg_informational_ikev2_msg

Type		
64-bit Counter		
Description		
Total number of informational message IKEv2 received		
Metrics		
Instances	Labels	Description
1	stats_group: nat_ipsec_alg_stats	Total number of informational message IKEv2 received
• ipsec_alg_informational_ikev1_msg		
Type		
64-bit Counter		
Description		
Total number of informational message IKEv1 received		
Metrics		

Instances	Labels	Description
1	stats_group: nat_ipsec_alg_stats	Total number of informational message IKEv1 received

Network Address Translation PPTP ALG Statistics

• invalid_pptp_packet		
Type		
64-bit Counter		
Description		
Total number of invalid packets received		
Metrics		
Instances	Labels	Description
1	stats_group: nat_pptp_alg_stats	Total number of invalid packets received
• buffer_dequeue_failed		
Type		
64-bit Counter		
Description		
Dequeue of buffer failed		
Metrics		
Instances	Labels	Description
1	stats_group: nat_pptp_alg_stats	Dequeue of buffer failed
• session_request		
Type		
64-bit Counter		
Description		
Number of session request message received		
Metrics		

Instances	Labels	Description
1	stats_group: nat_pptp_alg_stats	Number of session request message received
• invalid_session_request_packet		
Type		
64-bit Counter		
Description		
Number of invalid session request message received		
Metrics		

Instances	Labels	Description
1	stats_group: nat_pptp_alg_stats	Number of invalid session request message received
• session_reply		
Type		
64-bit Counter		
Description		
Number of session reply message received		
Metrics		

Instances	Labels	Description
1	stats_group: nat_pptp_alg_stats	Number of session reply message received
• invalid_session_reply_packet		
Type		
64-bit Counter		
Description		
Number of invalid session reply message received		
Metrics		

Instances	Labels	Description
1	stats_group: nat_pptp_alg_stats	Number of invalid session reply message received
• session_stop_request		
Type		
64-bit Counter		
Description		
Number of session stop request message received		
Metrics		

Instances	Labels	Description
1	stats_group: nat_pptp_alg_stats	Number of session stop request message received
• session_stop_reply		
Type		
64-bit Counter		
Description		
Number of session stop reply message received		

Metrics

Instances	Labels	Description
1	stats_group: nat_pttp_alg_stats	Number of session stop reply message received
invalid_session_stop_reply_packet		
Type		
64-bit Counter		
Description		
Number of invalid session stop reply message received		
Metrics		

Instances	Labels	Description
1	stats_group: nat_pttp_alg_stats	Pinhole creation success
out_call_request		
Type		
64-bit Counter		
Description		
Number of out call request message received		
Metrics		

Instances	Labels	Description
1	stats_group: nat_pttp_alg_stats	Number of out call request message received
invalid_out_call_request_packet		
Type		
64-bit Counter		
Description		
Number of invalid out call request message received		
Metrics		

Instances	Labels	Description
1	stats_group: nat_pttp_alg_stats	Number of invalid out call request message received
invalid_pac_call_id_received		
Type		
64-bit Counter		
Description		
Number of invalid pac callid received		
Metrics		

Instances	Labels	Description
1	stats_group: nat_pttp_alg_stats	Number of invalid pac callid received
nat_connection_not_found		
Type		
64-bit Counter		
Description		

Nat connection not found

Metrics

Instances	Labels	Description
1	stats_group: nat_pptp_alg_stats	Nat connection not found

- pin_hole_create_failure

Type

64-bit Counter

Description

Number of pinhole create failure

Metrics

Instances	Labels	Description
1	stats_group: nat_pptp_alg_stats	Number of pinhole create failure

- pin_hole_create_success

Type

64-bit Counter

Description

Number of pinhole create success

Metrics

Instances	Labels	Description
1	stats_group: nat_pptp_alg_stats	Number of pinhole create success

- out_call_reply

Type

64-bit Counter

Description

Number of out call reply message received

Metrics

Instances	Labels	Description
1	stats_group: nat_pptp_alg_stats	Number of out call reply message received

- invalid_out_call_reply_packet

Type

64-bit Counter

Description

Number of invalid out call reply message received

Metrics

Instances	Labels	Description
1	stats_group: nat_pptp_alg_stats	Number of invalid out call reply message received

- pin_hole_patch_failure

Type

64-bit Counter

Description

Number of pinhole patch failure

Metrics

Instances	Labels	Description
1	stats_group: nat_pptp_alg_stats	Number of pinhole patch failure

- pin_hole_patch_success

Type

64-bit Counter

Description

Number of pinhole patch success

Metrics

Instances	Labels	Description
1	stats_group: nat_pptp_alg_stats	Number of pinhole patch success

• in_call_request

Type

64-bit Counter

Description

Number of in call request message received

Metrics

Instances	Labels	Description
1	stats_group: nat_pptp_alg_stats	Number of in call request message received

• invalid_in_call_request_packet

Type

64-bit Counter

Description

Number of invalid in call request message received

Metrics

Instances	Labels	Description
1	stats_group: nat_pptp_alg_stats	Number of in call request message received

• in_call_reply

Type

64-bit Counter

Description

Number of in call reply message received

Metrics

Instances	Labels	Description
1	stats_group: nat_pptp_alg_stats	Number of invalid in call request message received

• invalid_in_call_reply_packet

Type

Instances	Labels	Description
1	stats_group: nat_pptp_alg_stats	Number of in call reply message received

64-bit Counter
Description
Number of invalid in call reply message received
Metrics

Instances	Labels	Description
1	stats_group: nat_pttp_alg_stats	Number of invalid in call reply message received

- in_call_connect

Type
64-bit Counter
Description
Number of in call connect message received
Metrics

Instances	Labels	Description
1	stats_group: nat_pttp_alg_stats	Number of in call connect message received

- invalid_in_call_connect_packet

Type
64-bit Counter
Description
Number of invalid in call connect message received
Metrics

Instances	Labels	Description
1	stats_group: nat_pttp_alg_stats	Number of invalid in call connect message received

- nat_pac_call_id_not_present

Type
64-bit Counter
Description
Nat pac call id not present
Metrics

Instances	Labels	Description
1	stats_group: nat_pttp_alg_stats	Nat pac call id not present

- invalid_pns_call_id_received

Type
64-bit Counter
Description
Number of invalid pns callid received
Metrics

Instances	Labels	Description
1	stats_group: nat_pttp_alg_stats	Number of invalid pns callid received

- call_clear_request

Type
64-bit Counter
Description
Number of call clear request message received
Metrics

Instances	Labels	Description
1	stats_group: nat_pptp_alg_stats	Number of call clear request message received

- invalid_call_clear_request_packet

Type
64-bit Counter
Description
Number of invalid call clear request message received
Metrics

Instances	Labels	Description
1	stats_group: nat_pptp_alg_stats	Number of invalid call clear request message received

- call_disconnect_notify

Type
64-bit Counter
Description
Number of call disconnect notify message received
Metrics

Instances	Labels	Description
1	stats_group: nat_pptp_alg_stats	Number of call disconnect notify message received

- invalid_call_disconnect_notify_packet

Type
64-bit Counter
Description
Number of invalid call disconnect notify message received
Metrics

Instances	Labels	Description
1	stats_group: nat_pptp_alg_stats	Number of invalid call disconnect notify message received

- wan_error_notify

Type
64-bit Counter
Description
Number of wan error notify message received
Metrics

Instances	Labels	Description
1	stats_group: nat_pptp_alg_stats	Number of wan error notify message received

- invalid_wan_error_notify_packet

Type

64-bit Counter

Description

Number of invalid wan error notify message received

Metrics

Instances	Labels	Description
1	stats_group: nat_pptp_alg_stats	Number of invalid wan error notify message received

- set_link_info

Type

64-bit Counter

Description

Number of set link info message received

Metrics

Instances	Labels	Description
1	stats_group: nat_pptp_alg_stats	Number of set link info message received

- invalid_set_link_info_packet

Type

64-bit Counter

Description

Number of invalid set link info message received

Metrics

Instances	Labels	Description
1	stats_group: nat_pptp_alg_stats	Number of invalid set link info message received

- echo_request

Type

64-bit Counter

Description

Number of echo request message received

Metrics

Instances	Labels	Description
1	stats_group: nat_pptp_alg_stats	Number of echo request message received

- echo_reply

Type

64-bit Counter

Description

Number of echo reply message received

Metrics

Instances	Labels	Description
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Instances	Labels	Description
1	stats_group: nat_pptp_alg_stats	Number of echo reply message received

Network Address Translation RTSP ALG Statistics

- pinhole_patch_audio_count

Type

64-bit Counter

Description

Total server initiated PinHole for audio success

Metrics

Instances	Labels	Description
1	stats_group: nat_rtsp_alg_stats	Total server initiated PinHole for audio success

- pinhole_patch_video_count

Type

64-bit Counter

Description

Total server initiated PinHole for video success

Metrics

Instances	Labels	Description
1	stats_group: nat_rtsp_alg_stats	Total server initiated PinHole for video success

- pinhole_patch_audio_error

Type

64-bit Counter

Description

Total server initiated PinHole creation failure for audio

Metrics

Instances	Labels	Description
1	stats_group: nat_rtsp_alg_stats	Total server initiated PinHole creation failure for audio

- pinhole_patch_video_error

Type

64-bit Counter

Description

Total server initiated PinHole creation failure for video

Metrics

Instances	Labels	Description
1	stats_group: nat_rtsp_alg_stats	Total server initiated PinHole creation failure for video

- pinhole_audio_port_count

Type

64-bit Counter

Description		
Total client initiated PinHole success for audio		
Metrics		
Instances	Labels	Description
1	stats_group: nat_rtsp_alg_stats	Total client initiated PinHole success for audio
pinhole_video_port_count		
Type		
64-bit Counter		
Description		
Total client initiated PinHole success for video		
Metrics		
Instances	Labels	Description
1	stats_group: nat_rtsp_alg_stats	Total client initiated PinHole success for video
pinhole_audio_port_error_count		
Type		
64-bit Counter		
Description		
Total client initiated PinHole creation failure for audio		
Metrics		
Instances	Labels	Description
1	stats_group: nat_rtsp_alg_stats	Total client initiated PinHole creation failure for audio
pinhole_video_port_error_count		
Type		
64-bit Counter		
Description		
Total client initiated PinHole creation failure for video		
Metrics		
Instances	Labels	Description
1	stats_group: nat_rtsp_alg_stats	Total client initiated PinHole creation failure for video
invalid_client_ports		
Type		
64-bit Counter		
Description		
Total invalid client Ports. Either client ports are not valid		
Metrics		
Instances	Labels	Description
1	stats_group: nat_rtsp_alg_stats	Total invalid client Ports. Either client ports are not valid
invalid_server_ports		
Type		

64-bit Counter
Description
Total invalid server Ports. Either server ports are not valid
Metrics

Instances	Labels	Description
1	stats_group: nat_rtsp_alg_stats	Total invalid server Ports. Either server ports are not valid

- invalid_transaction

Type
64-bit Counter
Description
Total invalid transaction
Metrics

Instances	Labels	Description
1	stats_group: nat_rtsp_alg_stats	Total invalid transaction

- uplink_version_not_supported

Type
64-bit Counter
Description
RTSP version not supported for uplink interleave mode packet. Only RTSP/1.0 is supported
Metrics

Instances	Labels	Description
1	stats_group: nat_rtsp_alg_stats	RTSP version not supported for uplink interleave mode packet. Only RTSP/1.0 is supported

- downlink_version_not_supported

Type
64-bit Counter
Description
RTSP version not supported for Downlink interleave mode packet. Only RTSP/1.0 is supported
Metrics

Instances	Labels	Description
1	stats_group: nat_rtsp_alg_stats	RTSP version not supported for Downlink interleave mode packet. Only RTSP/1.0 is supported

- uplink_interleave_mode_request

Type
64-bit Counter
Description
Uplink interleave mode request
Metrics

Instances	Labels	Description
1	stats_group: nat_rtsp_alg_stats	Uplink interleave mode request

- downlink_interleave_mode_request

Type
64-bit Counter
Description
Downlink interleave mode response
Metrics

Instances	Labels	Description
1	stats_group: nat_rtsp_alg_stats	Downlink interleave mode response

- uplink_options_request

Type
64-bit Counter
Description
OPTIONS request from UE
Metrics

Instances	Labels	Description
1	stats_group: nat_rtsp_alg_stats	OPTIONS request from UE

- uplink_describe_request

Type
64-bit Counter
Description
DESCRIBE request from UE
Metrics

Instances	Labels	Description
1	stats_group: nat_rtsp_alg_stats	DESCRIBE request from UE

- uplink_setup_request

Type
64-bit Counter
Description
SETUP request from UE
Metrics

Instances	Labels	Description
1	stats_group: nat_rtsp_alg_stats	SETUP request from UE

- uplink_play_request

Type
64-bit Counter
Description
PLAY request from UE
Metrics

Instances	Labels	Description
1	stats_group: nat_rtsp_alg_stats	PLAY request from UE

• uplink_tearardown_request

Type

64-bit Counter

Description

TEARDOWN request from UE

Metrics

Instances	Labels	Description
1	stats_group: nat_rtsp_alg_stats	TEARDOWN request from UE

• downlink_options_response

Type

64-bit Counter

Description

OPTIONS response from server

Metrics

Instances	Labels	Description
1	stats_group: nat_rtsp_alg_stats	OPTIONS response from server

• downlink_describe_response

Type

64-bit Counter

Description

DESCRIBE response from server

Metrics

Instances	Labels	Description
1	stats_group: nat_rtsp_alg_stats	DESCRIBE response from server

• downlink_setup_response

Type

64-bit Counter

Description

SETUP response from server

Metrics

Instances	Labels	Description
1	stats_group: nat_rtsp_alg_stats	SETUP response from server

• downlink_tearardown_response

Type

64-bit Counter

Description

TEARDOWN response from server

Metrics

Instances	Labels	Description
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Instances	Labels	Description
1	stats_group: nat_rtsp_alg_stats	TEARDOWN response from server
• rtsp_transport_update_failure		
Type		
64-bit Counter		
Description		
Updating rtsp transport header failed		
Metrics		

Instances	Labels	Description
1	stats_group: nat_rtsp_alg_stats	Updating rtsp transport header failed
• rtsp_transport_header_parsing_failed		
Type		
64-bit Counter		
Description		
Parsing rtsp transport header failed		
Metrics		

Instances	Labels	Description
1	stats_group: nat_rtsp_alg_stats	Parsing rtsp transport header failed
• total_error_messages_send		
Type		
64-bit Counter		
Description		
Total error messages send due to unsupported rtsp		
Metrics		

Instances	Labels	Description
1	stats_group: nat_rtsp_alg_stats	Total error messages send to client due to unsupported rtsp

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Summary

Group	Metric Name	Type	Synopsis
Config PDR Statistics	total_pdr_evaluations	Counter	Total number of evaluations performed for a PDR.
Config PDR Statistics	total_pdr_matches	Counter	Total number of times a PDR matched for the incoming traffic.
Ppe Pce Statistics	total_connections_created	Counter	VPP thread connections
Ppe Pce Statistics	total_cgnat_flows_created	Counter	VPP thread CGNAT flows
Ppe Pce Statistics	total_dns_flows_created	Counter	VPP thread DNS flows
Ppe Pce Statistics	total_tcp_flows_created	Counter	VPP thread TCP flows
Ppe Pce Statistics	total_udp_flows_created	Counter	VPP thread UDP flows

Group	Metric Name	Type	Synopsis
Ppe Pce Statistics	total_proxy_tcp_flows_created	Counter	VPP thread Proxy TCP flows
Ppe Pce Statistics	total_proxy_udp_flows_created	Counter	VPP thread Proxy UDP flows
Ppe Pce Statistics	total_tcp_flows_fast_aging_timed_out	Counter	VPP thread TCP flows fast aging timed out
Ppe Pce Statistics	total_tcp_flows_idle_timed_out	Counter	VPP thread TCP flows idle timed out
Ppe Pce Statistics	total_udp_flows_idle_timed_out	Counter	VPP thread UDP flows idle timed out
Ppe Pce Statistics	total_icmp_flows_idle_timed_out	Counter	VPP thread ICMP flows idle timed out
Ppe Pce Statistics	total_dns_tcp_flows_idle_timed_out	Counter	VPP thread DNS TCP flows idle timed out
Ppe Pce Statistics	total_dns_udp_flows_idle_timed_out	Counter	VPP thread DNS UDP flows idle timed out
Ppe Pce Statistics	total_other_flows_idle_timed_out	Counter	VPP thread other flows idle timed out
Ppe Pce Statistics	total_proxy_tcp_443_flows_created	Counter	VPP thread Proxy TCP 443 flows
Ppe Pce Statistics	total_proxy_udp_443_flows_created	Counter	VPP thread CGNAT flows
Ppe Pce Statistics	total_udp_quic_flows_created	Counter	VPP thread UDP QUIC flows
Ppe Pce Statistics	total_tcp_https_flows_created	Counter	VPP thread TCP HTTPS flows
Ppe Pce Statistics	total_tcp_443_uplink_packets	Counter	VPP thread TCP 443 Uplink Packets
Ppe Pce Statistics	total_tcp_443_uplink_bytes	Counter	VPP thread TCP 443 Uplink Bytes
Ppe Pce Statistics	total_tcp_443_downlink_packets	Counter	VPP thread TCP 443 Downlink Packets
Ppe Pce Statistics	total_tcp_443_downlink_bytes	Counter	VPP thread TCP 443 Downlink Bytes
Ppe Pce Statistics	total_udp_443_uplink_packets	Counter	VPP thread UDP 443 Uplink Packets
Ppe Pce Statistics	total_udp_443_uplink_bytes	Counter	VPP thread UDP 443 Uplink Bytes
Ppe Pce Statistics	total_udp_443_downlink_packets	Counter	VPP thread UDP 443 downlink Packets
Ppe Pce Statistics	total_udp_443_downlink_bytes	Counter	VPP thread UDP 443 downlink Bytes
Ppe Pce Statistics	total_tcp_80_uplink_packets	Counter	VPP thread TCP 80 uplink packets
Ppe Pce Statistics	total_tcp_80_uplink_bytes	Counter	VPP thread TCP 80 uplink bytes
Ppe Pce Statistics	total_tcp_80_downlink_packets	Counter	VPP thread TCP 80 downlink packets
Ppe Pce Statistics	total_tcp_80_downlink_bytes	Counter	VPP thread TCP 80 downlink bytes
Ppe Pce Statistics	Total_active_nat_connections	Gauge	VPP thread NAT active connections
Ppe Pce Statistics	Total_active_plain_connections	Gauge	VPP thread Plain active connections
Ppe Pce Statistics	Total_active_tdef_connections	Gauge	VPP thread active TDEF connections
Ppe Pce Statistics	Total_active_tvs_connections	Gauge	VPP thread active TVS connections

Config PDR Statistics

- total_pdr_evaluations

Type

64-bit Counter

Description

Total number of evaluations performed for a PDR.

Labels

pdr_name

- total_pdr_matches

Type

64-bit Counter

Description

Total number of times a PDR matched for the incoming traffic.

Labels

pdr_name

Ppe Pce Statistics

- total_connections_created
 - Type
 - 64-bit Counter
 - Description
 - Total connections created
 - Labels
 - thread_id***
- total_cgnat_flows_created
 - Type
 - 64-bit Counter
 - Description
 - Total NAT flows created
 - Labels
 - thread_id***
- total_dns_flows_created
 - Type
 - 64-bit Counter
 - Description
 - Total DNS flows created
 - Labels
 - thread_id***
- total_tcp_flows_created
 - Type
 - 64-bit Counter
 - Description
 - Total TCP flows created
 - Labels
 - thread_id***
- total_udp_flows_created
 - Type
 - 64-bit Counter
 - Description
 - Total UDP flows created
 - Labels
 - thread_id***
- total_proxy_tcp_flows_created

Type

64-bit Counter

Description

Total Proxy TCP flows created

Labels

thread_id

- total_proxy_udp_flows_created

Type

64-bit Counter

Description

Total Proxy UDP flows created

Labels

thread_id

- total_tcp_flows_fast_aging_timed_out

Type

64-bit Counter

Description

Total TCP flows fast aging timed out

Labels

thread_id

- total_tcp_flows_idle_timed_out

Type

64-bit Counter

Description

Total TCP flows idle timed out

Labels

thread_id

- total_udp_flows_idle_timed_out

Type

64-bit Counter

Description

Total UDP flows idle timed out

Labels

thread_id

- total_icmp_flows_idle_timed_out

Type

64-bit Counter

Description

Total ICMP flows idle timed out

Labels

thread_id

- total_dns_tcp_flows_idle_timed_out

Type

64-bit Counter

Description

Total DNS TCP flows idle timed out

Labels

thread_id

- total_dns_udp_flows_idle_timed_out

Type

64-bit Counter

Description

Total DNS UDP flows idle timed out

Labels

thread_id

- total_other_flows_idle_timed_out

Type

64-bit Counter

Description

Total other flows idle timed out

Labels

thread_id

- total_proxy_tcp_443_flows_created

Type

64-bit Counter

Description

Total Proxy TCP 443 flows created

Labels

thread_id

- total_proxy_udp_443_flows_created

Type

64-bit Counter

Description

Total NAT flows created

Labels

thread_id

- total_udp_quic_flows_created

Type

64-bit Counter

Description

Total UDP QUIC flows created

Labels

thread_id

- total_tcp_https_flows_created

Type

64-bit Counter

Description

Total TCP HTTPS flows created

Labels

thread_id

- total_tcp_443_uplink_packets

Type

64-bit Counter

Description

Total TCP 443 Uplink Packets

Labels

thread_id

- total_tcp_443_uplink_bytes

Type

64-bit Counter

Description

Total TCP 443 Uplink Bytes

Labels

thread_id

- total_tcp_443_downlink_packets

Type

64-bit Counter

Description

Total TCP 443 Downlink Packets

Labels

thread_id

- total_tcp_443_downlink_bytes

Type

64-bit Counter

Description

Total TCP 443 Downlink Bytes

Labels

thread_id

- total_udp_443_uplink_packets

Type

64-bit Counter

Description

Total UDP 443 Uplink Packets

Labels

thread_id

- total_udp_443_uplink_bytes

Type

64-bit Counter

Description

Total UDP 443 Uplink Bytes

Labels

thread_id

- total_udp_443_downlink_packets

Type

64-bit Counter

Description

Total UDP 443 downlink Packets

Labels

thread_id

- total_udp_443_downlink_bytes

Type

64-bit Counter

Description

Total UDP 443 downlink Bytes

Labels

thread_id

- total_tcp_80_uplink_packets

Type

64-bit Counter

Description

Total TCP 80 uplink packets

Labels

thread_id

- total_tcp_80_uplink_bytes

Type

64-bit Counter

Description

Total TCP 80 uplink bytes

Labels

thread_id

- total_tcp_80_downlink_packets

Type

64-bit Counter

Description

Total TCP 80 downlink packets

Labels

thread_id

- total_tcp_80_downlink_bytes

Type

64-bit Counter

Description

Total TCP 80 downlink bytes

Labels

thread_id

- Total_active_nat_connections

Type

64-bit Gauge

Description

Total Active NAT flows

Labels

thread_id

- Total_active_plain_connections

Type

64-bit Gauge

Description

Total Active Plain flows

Labels

thread_id

- Total_active_tdef_connections

Type

64-bit Gauge

Description

Total TDEF Plain flows

Labels

thread_id

• Total_active_tvs_connections

Type
64-bit Gauge
Description
Total TVS Plain flows
Labels
<i>thread_id</i>

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Summary

Group	Metric Name	Type	Synopsis
Session Manager control plane Statistics	upf_session_creation_request	Counter	Number of Session Create Attempts.
Session Manager control plane Statistics	upf_session_creation_request_success	Counter	Number of Session creates succeeded.
Session Manager control plane Statistics	upf_session_creation_request_failure	Counter	Number of Session creates failed.
Session Manager control plane Statistics	upf_stale_session_del_to_vpp_sess_id_in_use	Counter	Number of deletes initiated by SM towards VPP for stale sess id in use.
Session Manager control plane Statistics	upf_stale_session_del_to_vpp_ue_v4_in_use	Counter	Number of deletes initiated by SM towards VPP for stale ue v4 in use.
Session Manager control plane Statistics	upf_stale_session_del_to_vpp_ue_v6_in_use	Counter	Number of deletes initiated by SM towards VPP for stale ue v6 in use.
Session Manager control plane Statistics	upf_stale_session_del_to_vpp_remote_teid_in_use	Counter	Number of deletes initiated by SM towards VPP for stale remote teid in use.
Session Manager control plane Statistics	upf_session_creation_request_failure_resource_block_not_present	Counter	Number of Session creates failed due to no resource block present.
Session Manager control plane Statistics	upf_session_creation_request_failure_ue_ip_out_of_range	Counter	Number of Session creates failed due to ue ip out of range.
Session Manager control plane Statistics	upf_session_creation_request_allocate_ue_ip_success	Counter	Number of ue ip allocation succeeded.
Session Manager control plane Statistics	upf_session_creation_request_allocate_ue_ip_failure	Counter	Number of ue ip allocation failed.
Session Manager control plane Statistics	upf_session_creation_request_punt_success	Counter	Number of Session Create punted to the right ppe.
Session Manager control plane Statistics	upf_session_creation_request_punt_failure	Counter	Number of Session Create failed to punt to the right ppe.
Session Manager control plane Statistics	upf_session_creation_request_licensing_failure	Counter	Number of Session Create failed due to UPF license down.
Session Manager control plane Statistics	upf_session_creation_request_invalid_length_failure	Counter	Number of Session Create failed due to invalid PFCP message length.

Group	Metric Name	Type	Synopsis
Session Manager control plane Statistics	upf_session_creation_request_invalid_message_failure	Counter	Number of Session Create failed due to invalid PFCP message format.
Session Manager control plane Statistics	upf_session_creation_duplicate_request	Counter	Number of Session Creates that are duplicate requests.
Session Manager control plane Statistics	upf_session_creation_request_failure_no_association	Counter	Number of Session Create failed due to no association.
Session Manager control plane Statistics	upf_session_creation_request_local_resource_allocation_failure	Counter	Number of Session creates rejected due to local resource allocation failures.
Session Manager control plane Statistics	upf_session_creation_request_failures_with_crankback	Counter	Number of Session creates failed for which crankback was requested.
Session Manager control plane Statistics	upf_session_create_request_failure_remote_pool_in_disabled_mode	Counter	Number of Static UE IP Session Create failed due to remote pool in disabled mode.
Session Manager control plane Statistics	upf_session_create_request_failure_remote_pool_in_maintenance_mode	Counter	Number of Static UE IP Session Create failed due to remote pool in maintenance mode.
Session Manager control plane Statistics	upf_session_modify_request	Counter	Number of Session modify requests.
Session Manager control plane Statistics	upf_session_modify_request_success	Counter	Number of Session modify requests succeeded.
Session Manager control plane Statistics	upf_session_modify_request_node_id_update	Counter	Number of Session modify requests that updates cpfseid and nodeid.
Session Manager control plane Statistics	upf_session_modify_request_failure	Counter	Number of Session modify requests failed.
Session Manager control plane Statistics	upf_session_modify_request_failure_resource_block_not_present	Counter	Number of Session modify failed due to no resource block present.
Session Manager control plane Statistics	upf_session_modify_request_failure_upf_session_id_out_of_range	Counter	Number of Session modify failed due to upf session id out of range.
Session Manager control plane Statistics	upf_session_modify_request_failure_session_in_clear_pacer	Counter	Number of Session modify failed due to session present in clear pacer.
Session Manager control plane Statistics	upf_session_modify_request_punt_success	Counter	Number of Session Modify punted to the right ppe.
Session Manager control plane Statistics	upf_session_modify_request_punt_failure	Counter	Number of Session Modify failed to punt to the right ppe.
Session Manager control plane Statistics	upf_session_modify_request_invalid_length_failure	Counter	Number of Session Modify failed due to invalid PFCP message length.
Session Manager control plane Statistics	upf_session_modify_request_invalid_message_failure	Counter	Number of Session Modify failed due to invalid PFCP message format.
Session Manager control plane Statistics	upf_session_modify_request_no_session_failure	Counter	Number of Session Modify failed due to no matching session.
Session Manager control plane Statistics	upf_session_modify_request_fail_deleting_session	Counter	Number of Session Modify failed due to session being deleted.
Session Manager control plane Statistics	upf_session_modify_request_fail_session_not_in_vpp	Counter	Number of Session Modify failed due to session not in VPP deleted.
Session Manager control plane Statistics	upf_session_modify_request_dropped_due_to_rb_state_not_in_sync	Counter	Number of Session modify requests dropped.
Session Manager control plane Statistics	upf_session_delete_request	Counter	Number of Session delete requests.
Session Manager control plane Statistics	upf_session_delete_request_success	Counter	Number of Session delete requests succeeded.
Session Manager control plane Statistics	upf_session_delete_request_failure	Counter	Number of Session delete requests failed.
Session Manager control plane Statistics	upf_session_delete_request_failure_resource_block_not_present	Counter	Number of Session delete failed due to no resource block present.

Group	Metric Name	Type	Synopsis
Session Manager control plane Statistics	upf_session_delete_request_failure_upf_session_id_out_of_range	Counter	Number of Session delete failed due to upf session id out of range.
Session Manager control plane Statistics	upf_session_delete_request_punt_success	Counter	Number of Session Delete punted to the right ppe.
Session Manager control plane Statistics	upf_session_delete_request_punt_failure	Counter	Number of Session Delete failed to punt to the right ppe.
Session Manager control plane Statistics	upf_session_delete_request_invalid_session_failure	Counter	Number of Session Delete failed due to no matching session.
Session Manager control plane Statistics	upf_session_delete_request_invalid_message_failure	Counter	Number of Session Delete failed due to invalid PFCP message format.
Session Manager control plane Statistics	upf_session_delete_request_dropped_due_to_rb_state_not_in_sync	Counter	Number of Session delete requests dropped.
Session Manager control plane Statistics	upf_session_restore_request	Counter	Number of Session Restore Attempts.
Session Manager control plane Statistics	upf_session_restore_request_success	Counter	Number of Session Restore attempts succeeded.
Session Manager control plane Statistics	upf_session_restore_request_failure	Counter	Number of Session restores failed.
Session Manager control plane Statistics	upf_session_report_request	Counter	Number of Session report requests.
Session Manager control plane Statistics	upf_session_report_request_retry	Counter	Number of Session report requests retries.
Session Manager control plane Statistics	upf_session_report_request_success	Counter	Number of Session reports succeeded.
Session Manager control plane Statistics	upf_session_report_request_retry_success	Counter	Number of Session report retries succeeded.
Session Manager control plane Statistics	upf_session_report_request_failure	Counter	Number of Session reports failed.
Session Manager control plane Statistics	upf_session_report_request_retry_failure	Counter	Number of Session report retries failed.
Session Manager control plane Statistics	upf_session_report_response	Counter	Number of Session report responses recived.
Session Manager control plane Statistics	upf_session_report_response_success	Counter	Number of Session report responses succeeded.
Session Manager control plane Statistics	upf_session_report_response_node_id_update	Counter	Number of Session report responses that updates cpfseid and nodeid.
Session Manager control plane Statistics	upf_session_report_response_failure	Counter	Number of Session report failed.
Session Manager control plane Statistics	upf_session_report_response_context_not_found	Counter	Number of Session report response with context not found.
Session Manager control plane Statistics	upf_session_create_proxy_send_failure	Counter	Number of session create proxy send failures.
Session Manager control plane Statistics	upf_session_modify_proxy_send_failure	Counter	Number of session modify proxy send failures.
Session Manager control plane Statistics	upf_session_delete_proxy_send_failure	Counter	Number of session delete proxy send failures.
Session Manager control plane Statistics	upf_session_restore_proxy_send_failure	Counter	Number of session restore proxy send failures.
Session Manager control plane Statistics	upf_session_create_proxy_response_failure	Counter	Number of session create proxy response failures.
Session Manager control plane Statistics	upf_session_modify_proxy_response_failure	Counter	Number of session modify proxy response failures.
Session Manager control plane Statistics	upf_session_delete_proxy_response_failure	Counter	Number of session delete proxy response failures.

Group	Metric Name	Type	Synopsis
Session Manager control plane Statistics	upf_session_restore_proxy_response_failure	Counter	Number of session restore proxy response failures.
Session Manager control plane Statistics	upf_session_create_proxy_response_timeout	Counter	Number of session create proxy timeouts.
Session Manager control plane Statistics	upf_session_modify_proxy_response_timeout	Counter	Number of session modify proxy timeouts.
Session Manager control plane Statistics	upf_session_delete_proxy_response_timeout	Counter	Number of session delete proxy response timeouts.
Session Manager control plane Statistics	upf_session_restore_proxy_response_timeout	Counter	Number of session restore proxy timeouts.
Session Manager control plane Statistics	upf_session_create_vpp_send_failure	Counter	Number of session create vpp send failures.
Session Manager control plane Statistics	upf_session_modify_vpp_send_failure	Counter	Number of session modify vpp send failures.
Session Manager control plane Statistics	upf_session_delete_vpp_send_failure	Counter	Number of session delete vpp send failures.
Session Manager control plane Statistics	upf_session_restore_vpp_send_failure	Counter	Number of session restore vpp send failures.
Session Manager control plane Statistics	upf_session_create_vpp_response_failure	Counter	Number of session create vpp response failures.
Session Manager control plane Statistics	upf_session_modify_vpp_response_failure	Counter	Number of session modify vpp response failures.
Session Manager control plane Statistics	upf_session_delete_vpp_response_failure	Counter	Number of session delete vpp response failures.
Session Manager control plane Statistics	upf_session_restore_vpp_response_failure	Counter	Number of session restore vpp response failures.
Session Manager control plane Statistics	upf_session_create_vpp_response_timeout	Counter	Number of session create vpp response timeouts.
Session Manager control plane Statistics	upf_session_modify_vpp_response_timeout	Counter	Number of session modify vpp response timeouts.
Session Manager control plane Statistics	upf_session_delete_vpp_response_timeout	Counter	Number of session delete vpp response timeouts.
Session Manager control plane Statistics	upf_session_restore_vpp_response_timeout	Counter	Number of session restore vpp response timeouts.
Session Manager control plane Statistics	upf_session_create_vpp_procctx_notfound	Counter	Number of create vpp responses not processed due to proc context not found.
Session Manager control plane Statistics	upf_session_modify_vpp_procctx_notfound	Counter	Number of modify vpp responses not processed due to proc context not found.
Session Manager control plane Statistics	upf_session_delete_vpp_procctx_notfound	Counter	Number of session delete vpp responses not processed due to proc context not found.
Session Manager control plane Statistics	upf_session_restore_vpp_procctx_notfound	Counter	Number of session restore vpp responses not processed due to proc context not found .
Session Manager control plane Statistics	upf_session_create_crud_procctx_notfound	Counter	Number of create crud responses not processed due to proc context not found.
Session Manager control plane Statistics	upf_session_modify_crud_procctx_notfound	Counter	Number of modify crud responses not processed due to proc context not found.
Session Manager control plane Statistics	upf_session_delete_crud_procctx_notfound	Counter	Number of session delete crud responses not processed due to proc context not found.
Session Manager control plane Statistics	upf_session_report_req_procctx_notfound	Counter	Number of session report requests not processed due to proc context not found .

Group	Metric Name	Type	Synopsis
Session Manager control plane Statistics	upf_session_notfound_present_in_ucd	Counter	Number of Session not present locally but present in UCD found ucd audit process.
Session Manager control plane Statistics	upf_session_dup_est_req_ignored	Counter	Number of session create requests ignored as duplicate create requests .
Session Manager control plane Statistics	upf_session_dup_modify_req_ignored	Counter	Number of session create requests ignored as duplicate modify requests .
Session Manager control plane Statistics	upf_session_dup_delete_req_ignored	Counter	Number of session create requests ignored as duplicate delete requests .
Session Manager control plane Statistics	upf_session_pending_procctx_created	Counter	Number of pending proc contexts created since an existing session FSM is in progress.
Session Manager control plane Statistics	upf_session_local_deletes	Counter	Number of sessions torn down by session manager without trigger from SMF.
Session Manager control plane Statistics	upf_session_local_deletes_report_resp_error	Counter	Number of sessions torn down by session manager due to report request error from SMF.
Session Manager control plane Statistics	upf_session_local_deletes_session_clear	Counter	Number of sessions torn down by session manager due to session clear by the pacer.
Session Manager control plane Statistics	upf_session_local_association_down_request	Counter	Number of session removal requests due to association_down.
Session Manager control plane Statistics	upf_session_local_association_down_request_success	Counter	Number of sessions removed successfully due to association_down.
Session Manager control plane Statistics	upf_session_local_association_down_request_failure	Counter	Number of session removal requests due to association down failed.
Session Manager control plane Statistics	upf_session_create_fsm_expired	Counter	Number of session create failures due to fsm timer expiry.
Session Manager control plane Statistics	upf_session_modify_fsm_expired	Counter	Number of session modify failures due to fsm timer expiry.
Session Manager control plane Statistics	upf_session_delete_fsm_expired	Counter	Number of session delete failures due to fsm timer expiry.
Session Manager control plane Statistics	upf_session_restore_fsm_expired	Counter	Number of session restore failures due to fsm timer expiry.
Session Manager control plane Statistics	upf_session_bad_report_req_msgs	Counter	Number of session report responses dropped due to no cause code.
Session Manager control plane Statistics	upf_session_report_req_timeout	Counter	Number of session report requests timed out waiting for PFCP response.
Session Manager control plane Statistics	upf_session_report_req_dldr	Counter	Number of session report requests with Downlink Data Report.
Session Manager control plane Statistics	upf_session_report_req_usage_report	Counter	Number of session report requests with usage report.
Session Manager control plane Statistics	upf_session_report_req_error_ind	Counter	Number of session report requests with error indication report.
Session Manager control plane Statistics	upf_session_report_req_inactivity_report	Counter	Number of session report requests with inactivity report.
Session Manager control plane Statistics	upf_session_report_req_session_active_report	Counter	Number of session report requests with active report.
Session Manager control plane Statistics	upf_session_report_resp_wrong_ppe	Counter	Number of session report responses received in the wrong PPE.
Session Manager control plane Statistics	upf_session_report_resp_session_notfound	Counter	Number of session report responses dropped due to session not found.
Session Manager control plane Statistics	upf_session_clear_pacer_initiated	Counter	Number of session clear pacer initiated for a resource block.
Session Manager control plane Statistics	upf_session_clear_pacer_completed	Counter	Number of session clear pacer completed for a resource block.

Group	Metric Name	Type	Synopsis
Session Manager control plane Statistics	upf_session_partial_db_update	Counter	Number of partial DB update done for a modify session request.
Session Manager control plane Statistics	upf_session_full_db_update	Counter	Number of full DB update done for a modify session request.
Session Manager control plane Statistics	upf_session_report_idle_session_clear_req	Counter	Number of session report requests of type UP initiated session request.
Session Manager control plane Statistics	upf_session_report_triggered_from_session_clear_pacer	Counter	Number of session report requests triggered from session clear pacer.
Session Manager control plane Statistics	upf_standby_session_creation_request	Counter	Number of Standby Session Create Attempts.
Session Manager control plane Statistics	upf_standby_session_creation_request_success	Counter	Number of Standby Session creates succeeded.
Session Manager control plane Statistics	upf_standby_session_creation_request_failure	Counter	Number of Standby Session creates failed.
Session Manager control plane Statistics	upf_standby_session_creation_request_failure_resource_block_not_present	Counter	Number of Standby Session creates failed due to no resource block present.
Session Manager control plane Statistics	upf_standby_session_creation_request_failure_ue_ip_out_of_range	Counter	Number of Standby Session creates failed due to ue ip out of range.
Session Manager control plane Statistics	upf_standby_session_creation_request_invalid_length_failure	Counter	Number of Standby Session Create failed due to invalid PFCP message length.
Session Manager control plane Statistics	upf_standby_session_creation_request_invalid_message_failure	Counter	Number of Standby Session Create failed due to invalid PFCP message format.
Session Manager control plane Statistics	upf_standby_session_creation_duplicate_request	Counter	Number of Standby Session Creates that are duplicate requests.
Session Manager control plane Statistics	upf_standby_session_creation_request_failure_no_association	Counter	Number of Standby Session Create failed due to no association.
Session Manager control plane Statistics	upf_standby_session_creation_request_failure_cpseid_mismatch	Counter	Number of Standby Session Create failed due to cpseid mismatch.
Session Manager control plane Statistics	upf_standby_session_modify_request	Counter	Number of Standby Session modify requests.
Session Manager control plane Statistics	upf_standby_session_modify_request_success	Counter	Number of Standby Session modify requests succeeded.
Session Manager control plane Statistics	upf_standby_session_modify_request_node_id_update	Counter	Number of Standby Session modify requests that updates cpseid and nodeid.
Session Manager control plane Statistics	upf_standby_session_modify_request_failure	Counter	Number of Standby Session modify requests failed.
Session Manager control plane Statistics	upf_standby_session_modify_request_failure_resource_block_not_present	Counter	Number of Standby Session modify failed due to no resource block present.
Session Manager control plane Statistics	upf_standby_session_modify_request_failure_upf_session_id_out_of_range	Counter	Number of Standby Session modify failed due to upf session id out of range.
Session Manager control plane Statistics	upf_standby_session_modify_request_invalid_length_failure	Counter	Number of Standby Session Modify failed due to invalid PFCP message length.
Session Manager control plane Statistics	upf_standby_session_modify_request_invalid_message_failure	Counter	Number of Standby Session Modify failed due to invalid PFCP message format.
Session Manager control plane Statistics	upf_standby_session_modify_request_no_session_failure	Counter	Number of Standby Session Modify failed due to no matching session.
Session Manager control plane Statistics	upf_standby_session_modify_request_fail_deleting_session	Counter	Number of Standby Session Modify failed due to session being deleted.
Session Manager control plane Statistics	upf_standby_session_modify_request_fail_session_not_in_vpp	Counter	Number of Standby Session Modify failed due to session not in VPP deleted.
Session Manager control plane Statistics	upf_standby_session_report_response	Counter	Number of Standby Session report responses.

Group	Metric Name	Type	Synopsis
Session Manager control plane Statistics	upf_standby_session_report_response_node_id_update	Counter	Number of Standby Session report responses that updated cpfseid and nodeid.
Session Manager control plane Statistics	upf_standby_session_report_response_success	Counter	Number of Standby Session report responses succeeded.
Session Manager control plane Statistics	upf_standby_session_report_response_failure	Counter	Number of Standby Session modify requests failed.
Session Manager control plane Statistics	upf_standby_session_report_response_no_session_failure	Counter	Number of Standby Session Report responses failed due to no matching session.
Session Manager control plane Statistics	upf_standby_session_report_response_fail_deleting_session	Counter	Number of Standby Session Report responses failed due to session being deleted.
Session Manager control plane Statistics	upf_standby_session_report_response_fail_session_not_in_vpp	Counter	Number of Standby Session Report responses failed due to session not in VPP deleted.
Session Manager control plane Statistics	upf_standby_session_delete_request	Counter	Number of Standby Session delete requests.
Session Manager control plane Statistics	upf_standby_session_delete_request_success	Counter	Number of Standby Session delete requests succeeded.
Session Manager control plane Statistics	upf_standby_session_delete_request_failure	Counter	Number of Standby Session delete requests failed.
Session Manager control plane Statistics	upf_standby_session_delete_request_failure_resource_block_not_present	Counter	Number of Standby Session delete failed due to no resource block present.
Session Manager control plane Statistics	upf_standby_session_delete_request_failure_upf_session_id_out_of_range	Counter	Number of Standby Session delete failed due to upf session id out of range.
Session Manager control plane Statistics	upf_standby_session_delete_request_invalid_session_failure	Counter	Number of Standby Session Delete failed due to no matching session.
Session Manager control plane Statistics	upf_standby_session_delete_request_invalid_message_failure	Counter	Number of Standby Session Delete failed due to invalid PCFP message format.
Session Manager control plane Statistics	upf_standby_session_delete_response_success	Counter	Number of Standby Session delete response success.
Session Manager control plane Statistics	upf_standby_session_create_proxy_send_success	Counter	Number of Standby Session create proxy send success.
Session Manager control plane Statistics	upf_standby_session_modify_proxy_send_success	Counter	Number of Standby Session modify proxy send success.
Session Manager control plane Statistics	upf_standby_session_delete_proxy_send_success	Counter	Number of Standby Session delete proxy send success.
Session Manager control plane Statistics	upf_standby_session_create_proxy_response_success	Counter	Number of Standby Session create proxy response success.
Session Manager control plane Statistics	upf_standby_session_modify_proxy_response_success	Counter	Number of Standby Session modify proxy response success.
Session Manager control plane Statistics	upf_standby_session_delete_proxy_response_success	Counter	Number of Standby Session delete proxy response success.
Session Manager control plane Statistics	upf_standby_session_create_proxy_send_failure	Counter	Number of Standby Session create proxy send failures.
Session Manager control plane Statistics	upf_standby_session_modify_proxy_send_failure	Counter	Number of Standby Session modify proxy send failures.
Session Manager control plane Statistics	upf_standby_session_delete_proxy_send_failure	Counter	Number of Standby Session delete proxy send failures.
Session Manager control plane Statistics	upf_standby_session_create_proxy_response_failure	Counter	Number of Standby Session create proxy response failures.
Session Manager control plane Statistics	upf_standby_session_modify_proxy_response_failure	Counter	Number of Standby Session modify proxy response failures.

Group	Metric Name	Type	Synopsis
Session Manager control plane Statistics	upf_standby_session_delete_proxy_response_failure	Counter	Number of Standby Session delete proxy response failures.
Session Manager control plane Statistics	upf_standby_session_create_proxy_session_skipped	Counter	Number of Standby Proxy Session create skipped.
Session Manager control plane Statistics	upf_standby_session_modify_proxy_session_skipped	Counter	Number of Standby Session modify proxy skipped.
Session Manager control plane Statistics	upf_standby_session_delete_proxy_session_skipped	Counter	Number of Standby Proxy Session delete skipped
Session Manager control plane Statistics	upf_standby_session_create_proxy_response_timeout	Counter	Number of Standby Session create proxy timeouts.
Session Manager control plane Statistics	upf_standby_session_modify_proxy_response_timeout	Counter	Number of Standby Session modify proxy timeouts.
Session Manager control plane Statistics	upf_standby_session_delete_proxy_response_timeout	Counter	Number of Standby Session delete proxy response timeouts.
Session Manager control plane Statistics	upf_standby_session_create_vpp_send_success	Counter	Number of Standby Session create vpp send success.
Session Manager control plane Statistics	upf_standby_session_modify_vpp_send_success	Counter	Number of Standby Session modify vpp send success.
Session Manager control plane Statistics	upf_standby_session_delete_vpp_send_success	Counter	Number of Standby Session delete vpp send success.
Session Manager control plane Statistics	upf_standby_session_create_vpp_send_failure	Counter	Number of Standby Session create vpp send failures.
Session Manager control plane Statistics	upf_standby_session_modify_vpp_send_failure	Counter	Number of Standby Session modify vpp send failures.
Session Manager control plane Statistics	upf_standby_session_delete_vpp_send_failure	Counter	Number of Standby Session delete vpp send failures.
Session Manager control plane Statistics	upf_standby_session_create_vpp_response_success	Counter	Number of Standby Session create vpp response success.
Session Manager control plane Statistics	upf_standby_session_modify_vpp_response_success	Counter	Number of Standby Session modify vpp response success.
Session Manager control plane Statistics	upf_standby_session_delete_vpp_response_success	Counter	Number of Standby Session delete vpp response success.
Session Manager control plane Statistics	upf_standby_session_create_vpp_response_failure	Counter	Number of standby session create vpp response failures.
Session Manager control plane Statistics	upf_standby_session_modify_vpp_response_failure	Counter	Number of standby session modify vpp response failures.
Session Manager control plane Statistics	upf_standby_session_delete_vpp_response_failure	Counter	Number of standby session delete vpp response failures.
Session Manager control plane Statistics	upf_standby_session_create_vpp_response_timeout	Counter	Number of Standby Session create vpp response timeouts.
Session Manager control plane Statistics	upf_standby_session_modify_vpp_response_timeout	Counter	Number of Standby Session modify vpp response timeouts.
Session Manager control plane Statistics	upf_standby_session_delete_vpp_response_timeout	Counter	Number of Standby Session delete vpp response timeouts.
Session Manager control plane Statistics	upf_standby_session_create_vpp_retry_count	Counter	Number of Standby Session create vpp retries.
Session Manager control plane Statistics	upf_standby_session_create_vpp_retry_due_to_send_failure_count	Counter	Number of Standby Session create vpp retries due to send failure to vpp.
Session Manager control plane Statistics	upf_standby_session_create_vpp_send_failures_waiting_to_retry	Counter	Number of Standby Session create request sending failed and waiting to retry
Session Manager control plane Statistics	upf_standby_session_modify_vpp_retry_count	Counter	Number of Standby Session modify vpp retries.
Session Manager control plane Statistics	upf_session_modify_vpp_retry_count	Counter	Number of Session modify vpp retries.

Group	Metric Name	Type	Synopsis
Session Manager control plane Statistics	upf_standby_session_delete_vpp_retry_count	Counter	Number of Standby Session delete vpp retries.
Session Manager control plane Statistics	upf_standby_session_create_vpp_procctxt_notfound	Counter	Number of create vpp responses not processed due to proc context not found.
Session Manager control plane Statistics	upf_standby_session_modify_vpp_procctxt_notfound	Counter	Number of modify vpp responses not processed due to proc context not found.
Session Manager control plane Statistics	upf_standby_session_delete_vpp_procctxt_notfound	Counter	Number of Standby Session delete vpp responses not processed due to proc context not found.
Session Manager control plane Statistics	upf_standby_session_dup_est_req_ignored	Counter	Number of Standby Session create requests ignored as duplicate create requests .
Session Manager control plane Statistics	upf_standby_session_dup_modify_req_ignored	Counter	Number of Standby Session create requests ignored as duplicate modify requests .
Session Manager control plane Statistics	upf_standby_session_dup_del_req_ignored	Counter	Number of Standby Session create requests ignored as duplicate delete requests .
Session Manager control plane Statistics	upf_standby_session_fsm_expired	Counter	Number of Standby Session failures due to fsm timer expiry.
Session Manager control plane Statistics	upf_standby_session_create_fsm_expired	Counter	Number of Standby Session create failures due to fsm timer expiry.
Session Manager control plane Statistics	upf_standby_session_modify_fsm_expired	Counter	Number of Standby Session modify failures due to fsm timer expiry.
Session Manager control plane Statistics	upf_standby_session_delete_fsm_expired	Counter	Number of Standby Session delete failures due to fsm timer expiry.
Session Manager control plane Statistics	upf_standby_session_create_proxy_procctxt_notfound	Counter	Number of Standby Session Create Proxy procedure context not found
Session Manager control plane Statistics	upf_standby_session_proc_ctxt_alloc	Counter	Number of Standby Session procedure context allocated
Session Manager control plane Statistics	upf_standby_session_proc_ctxt_dealloc	Counter	Number of Standby Session Procedure Context Deallocated
Session Manager control plane Statistics	upf_standby_session_record_alloc	Counter	Number of Standby Session Record Allocated.
Session Manager control plane Statistics	upf_standby_session_record_dealloc	Counter	Number of Standby Session Record Deallocated.
Session Manager control plane Statistics	upf_standby_session_report_request_ignored	Counter	Number of Report Requests Ignored for Standby Session.
Session Manager control plane Statistics	upf_standby_fsm_stale_session_del_to_vpp_sess_id_in_use	Counter	Number of deletes initiated by SM towards VPP for stale sess id in use during standby FSM.
Session Manager control plane Statistics	upf_standby_fsm_stale_session_del_to_vpp_ue_v4_in_use	Counter	Number of deletes initiated by SM towards VPP for stale ue v4 in use during standby FSM.
Session Manager control plane Statistics	upf_standby_fsm_stale_session_del_to_vpp_ue_v6_in_use	Counter	Number of deletes initiated by SM towards VPP for stale ue v6 in use during standby FSM.
Session Manager control plane Statistics	upf_standby_fsm_stale_session_del_to_vpp_remote_teid_in_use	Counter	Number of deletes initiated by SM towards VPP for stale Tunnel Id in use during standby FSM.
Session Manager control plane Statistics	upf_session_start_record_generated	Counter	Number of Session Start EDR Record Generated
Session Manager control plane Statistics	upf_session_stop_record_generated	Counter	Number of Session Stop EDR Record Generated

Group	Metric Name	Type	Synopsis
Session Manager control plane Statistics	upf_session_record_send_success	Counter	Number of Session EDR Record Sent to Kafka Broker Successfully
Session Manager control plane Statistics	upf_session_record_send_failure	Counter	Number of Session EDR Record Sent to Kafka Broker Failure
Session Manager control plane Statistics	upf_session_record_delivery_success	Counter	Number of Session EDR Record Delivered to Kafka Broker Successfully
Session Manager control plane Statistics	upf_session_record_delivery_failure	Counter	Number of Session EDR Record Delivered to Kafka Broker Failure
Session Manager control plane Statistics	upf_num_total_sessions	Gauge	Number of total sessions in the PPE.
Session Manager control plane Statistics	upf_num_active_sessions	Gauge	Number of active sessions in the PPE.
Session Manager control plane Statistics	upf_session_pending_procctxt_count	Gauge	Number of pending Proc Contexts waiting for an FSM to complete.
Session Manager control plane Statistics	upf_num_standby_sessions	Gauge	Number of Standby Session count.
Session Manager control plane Statistics	upf_dnn_num_active_sessions	Gauge	Number of active sessions in the PPE for a DNN.
Session Manager control plane Statistics	upf_dnn_num_standby_sessions	Gauge	Number of standby sessions in the PPE for a DNN.
Session Manager control plane Statistics	upf_ue_ip_pool_total_ip_count	Gauge	Number of total UE IP count in the SM for a UE Ip Pool.
Session Manager control plane Statistics	upf_ue_ip_pool_free_ip_count	Gauge	Number of free UE IP count in the SM for a UE Ip Pool.
Session Manager control plane Statistics	upf_dnn_v4_total_ue_ip_count	Gauge	Number of total v4 UE IP count in the SM for a DNN.
Session Manager control plane Statistics	upf_dnn_v6_total_ue_ip_count	Gauge	Number of total v6 UE IP count in the SM for a DNN.
Session Manager control plane Statistics	upf_dnn_v4_free_ue_ip_count	Gauge	Number of total v4 UE IP count in the SM for a DNN.
Session Manager control plane Statistics	upf_dnn_v6_free_ue_ip_count	Gauge	Number of total v6 UE IP count in the SM for a DNN.
Session Manager control plane Statistics	upf_session_create_total_duration_milliseconds	Histogram	Histogram of the duration, in milliseconds, of create FSM for a PFCP sessEstReq message.
Session Manager control plane Statistics	upf_session_modify_total_duration_milliseconds	Histogram	Histogram of the duration, in milliseconds, of modify FSM for a PFCP sessModReq message.
Session Manager control plane Statistics	upf_session_delete_total_duration_milliseconds	Histogram	Histogram of the duration, in milliseconds, of delete FSM for a PFCP sessDelReq message.
Session Manager control plane Statistics	upf_session_restore_total_duration_milliseconds	Histogram	Histogram of the duration, in milliseconds, of restore FSM for a PFCP sessEstReq message.
Session Manager control plane Statistics	upf_session_create_proxy_duration_milliseconds	Histogram	Histogram of the duration, in milliseconds, of all create proxy requests sent and response received for a PFCP sessEstReq message.
Session Manager control plane Statistics	upf_session_create_vpp_duration_milliseconds	Histogram	Histogram of the duration, in milliseconds, of all create vpp requests sent and response received for a PFCP sessEstReq message.
Session Manager control plane Statistics	upf_session_modify_proxy_duration_milliseconds	Histogram	Histogram of the duration, in milliseconds, of all modify proxy requests sent and response received for a PFCP sessModReq message.

Group	Metric Name	Type	Synopsis
Session Manager control plane Statistics	upf_session_modify_vpp_duration_milliseconds	Histogram	Histogram of the duration, in milliseconds, of all modify vpp requests sent and response received for a PFCP sessModReq message.
Session Manager control plane Statistics	upf_session_delete_proxy_duration_milliseconds	Histogram	Histogram of the duration, in milliseconds, of all delete proxy requests sent and response received for a PFCP sessDelReq message.
Session Manager control plane Statistics	upf_session_delete_vpp_duration_milliseconds	Histogram	Histogram of the duration, in milliseconds, of all delete vpp requests sent and response received for a PFCP sessDelReq message.
Session Manager control plane Statistics	upf_session_restore_proxy_duration_milliseconds	Histogram	Histogram of the duration, in milliseconds, of all restore proxy requests sent and response received for a PFCP sessRestoreReq message.
Session Manager control plane Statistics	upf_session_restore_vpp_duration_milliseconds	Histogram	Histogram of the duration, in milliseconds, of all restore vpp requests sent and response received for a PFCP sessRestoreReq message.
Session Manager control plane Statistics	upf_session_report_pfcg_duration_milliseconds	Histogram	Histogram of the duration, in milliseconds, of all session report requests sent to PFCP.
Session Manager control plane Statistics	upf_stby_session_create_total_duration_milliseconds	Histogram	Histogram of the duration, in milliseconds, of standby create FSM for a PFCP sessEstReq message.
Session Manager control plane Statistics	upf_stby_session_modify_total_duration_milliseconds	Histogram	Histogram of the duration, in milliseconds, of standby modify FSM for a PFCP sessModReq message.
Session Manager control plane Statistics	upf_stby_session_delete_total_duration_milliseconds	Histogram	Histogram of the duration, in milliseconds, of standby delete FSM for a PFCP sessDelReq message.
Session Manager control plane Statistics	upf_stby_session_create_proxy_duration_milliseconds	Histogram	Histogram of the duration, in milliseconds, of all standby create proxy requests sent and response received for a PFCP sessEstReq message.
Session Manager control plane Statistics	upf_stby_session_create_vpp_duration_milliseconds	Histogram	Histogram of the duration, in milliseconds, of all standby create vpp requests sent and response received for a PFCP sessEstReq message.
Session Manager control plane Statistics	upf_stby_session_modify_proxy_duration_milliseconds	Histogram	Histogram of the duration, in milliseconds, of all standby modify proxy requests sent and response received for a PFCP sessModReq message.
Session Manager control plane Statistics	upf_stby_session_modify_vpp_duration_milliseconds	Histogram	Histogram of the duration, in milliseconds, of all standby modify vpp requests sent and response received for a PFCP sessModReq message.
Session Manager data plane Statistics	upf_session_uplink_packets_received	Counter	Uplink Packets Received.
Session Manager data plane Statistics	upf_session_uplink_packets_transmitted	Counter	Uplink Packets Transmitted.
Session Manager data plane Statistics	upf_session_uplink_packets_discarded	Counter	Uplink Packets Discarded.
Session Manager data plane Statistics	upf_session_uplink_bytes_received	Counter	Uplink Packets Received.
Session Manager data plane Statistics	upf_session_uplink_bytes_transmitted	Counter	Uplink Packets Transmitted.

Group	Metric Name	Type	Synopsis
Session Manager data plane Statistics	upf_session_uplink_bytes_discarded	Counter	Uplink Packets Discarded.
Session Manager data plane Statistics	upf_session_downlink_packets_received	Counter	Downlink Packets Received.
Session Manager data plane Statistics	upf_session_downlink_packets_transmitted	Counter	Downlink Packets Transmitted.
Session Manager data plane Statistics	upf_session_downlink_packets_discarded	Counter	Downlink Packets Discarded.
Session Manager data plane Statistics	upf_session_downlink_bytes_received	Counter	Downlink Packets Received.
Session Manager data plane Statistics	upf_session_downlink_bytes_transmitted	Counter	Downlink Packets Transmitted.
Session Manager data plane Statistics	upf_session_downlink_bytes_discarded	Counter	Downlink Packets Discarded.
Session Manager Interface Statistics	upf_intf_packets_received	Counter	Packets Received on an interface.
Session Manager Interface Statistics	upf_intf_packets_transmitted	Counter	Packets Transmitted on an interface.
Session Manager Interface Statistics	upf_intf_bytes_received	Counter	Bytes Received on an interface.
Session Manager Interface Statistics	upf_intf_bytes_transmitted	Counter	Bytes Transmitted on an interface.
Session Manager Interface Statistics	upf_intf_rx_ipv4	Counter	IPv4 packets counts Rx on an interface.
Session Manager Interface Statistics	upf_intf_rx_ipv6	Counter	IPv6 packets counts Rx on an interface.
Session Manager Interface Statistics	upf_intf_tx_drops	Counter	Packets drop counts on an interface.
Session Manager Interface Statistics	upf_intf_rx_nobuff	Counter	No MBUF count on RX side of an interface.
Session Manager Interface Statistics	upf_intf_rx_miss	Counter	Interface dropped packets due to rx ring buffer full.
Session Manager Interface Statistics	upf_intf_rx_error	Counter	Any misc Rx errors on an interface.
Session Manager Interface Statistics	upf_intf_tx_error	Counter	Any other Tx errors on an interface.
Session Manager Session Database Statistics	upf_session_db_creation_request	Counter	Number of PFCP initiated attempts to add sessions to the CRUD database.
Session Manager Session Database Statistics	upf_session_db_creation_request_cancel	Counter	Number of session add requests cancelled before completion.
Session Manager Session Database Statistics	upf_session_db_creation_request_success	Counter	Number of successful attempts to add sessions to the CRUD database.
Session Manager Session Database Statistics	upf_session_db_creation_request_failure	Counter	Number of unsuccessful attempts to add sessions to the CRUD database.
Session Manager Session Database Statistics	upf_session_db_modify_request	Counter	Number of attempts to update sessions in the CRUD database.
Session Manager Session Database Statistics	upf_session_db_modify_request_cancel	Counter	Number of session update requests cancelled before completion.
Session Manager Session Database Statistics	upf_session_db_modify_request_success	Counter	Number of successful attempts to update sessions in the CRUD database.
Session Manager Session Database Statistics	upf_session_db_modify_request_failure	Counter	Number of unsuccessful attempts to update sessions in the CRUD database.
Session Manager Session Database Statistics	upf_session_db_delete_best_effort	Counter	Number of best effort attempts to delete sessions from the CRUD database. Result from CRUD database is unknown

Group	Metric Name	Type	Synopsis
Session Manager Session Database Statistics	upf_session_db_delete_request	Counter	Number of attempts to delete sessions from the CRUD database.
Session Manager Session Database Statistics	upf_session_db_delete_request_cancel	Counter	Number of session delete requests cancelled before completion.
Session Manager Session Database Statistics	upf_session_db_delete_request_success	Counter	Number of successful attempts to delete sessions from the CRUD database.
Session Manager Session Database Statistics	upf_session_db_delete_request_failure	Counter	Number of unsuccessful attempts to delete sessions from the CRUD database.
Session Manager Session Database Statistics	upf_session_db_update_request	Counter	Number of data plane initiated attempts to update sessions in the CRUD database.
Session Manager Session Database Statistics	upf_session_db_update_request_success	Counter	Number of successful data plane initiated attempts to update sessions in the CRUD database.
Session Manager Session Database Statistics	upf_session_db_update_request_failure	Counter	Number of unsuccessful data plane initiated attempts to update sessions in the CRUD database.
Session Manager Session Database Statistics	upf_session_db_group_read_request	Counter	Number of attempts to read a group of sessions from the CRUD database.
Session Manager Session Database Statistics	upf_session_db_group_read_request_cancel	Counter	Number of session group reads cancelled before completion.
Session Manager Session Database Statistics	upf_session_db_group_read_request_success	Counter	Number of successful attempts to read groups of sessions from the CRUD database.
Session Manager Session Database Statistics	upf_session_db_group_read_request_failure	Counter	Number of unsuccessful attempts to read groups of sessions from the CRUD database, either due to DB retrieval failure, control-plane installation failure, or data plane installation failure.
Session Manager Session Database Statistics	upf_session_db_single_read_request	Counter	Number of attempts to read individual sessions from the CRUD database.
Session Manager Session Database Statistics	upf_session_db_single_read_request_success	Counter	Number of successful attempts to individual sessions from the CRUD database.
Session Manager Session Database Statistics	upf_session_db_single_read_request_failure	Counter	Number of unsuccessful attempts to read individual sessions from the CRUD database, either due to DB retrieval failure, control-plane installation failure, or data plane installation failure.
Session Manager Session Database Statistics	upf_session_db_group_delete_best_effort	Counter	Number of group best effort attempts to delete sessions from the CRUD database. Result from CRUD database is unknown
Session Manager Session Database Statistics	upf_session_db_group_delete_best_effort_request_failure	Counter	Number of failed group best effort attempts to delete records from CRUD database.
Session Manager Session Database Statistics	upf_session_db_creation_duration_milliseconds	Histogram	Histogram of duration, in milliseconds, of all database operations performed to successfully add a session to the CRUD database when processing a PFCP sessEstReq message.
Session Manager Session Database Statistics	upf_session_db_modify_duration_milliseconds	Histogram	Histogram of duration, in milliseconds, of all database operations performed to successfully update a session in the CRUD database when processing a PFCP sessModReq message.
Session Manager Session Database Statistics	upf_session_db_delete_duration_milliseconds	Histogram	Histogram of duration, in milliseconds, of all database operations performed to successfully delete a session from the CRUD database when processing a PFCP sessDelReq message.

Group	Metric Name	Type	Synopsis
Session Manager Session Database Statistics	upf_session_db_update_duration_milliseconds	Histogram	Histogram of duration, in milliseconds, of all database operations performed to successfully update a session in the CRUD database when processing a PFCP sessDelReq message.
Session Manager Session Database Statistics	upf_session_db_group_read_duration_milliseconds	Histogram	Histogram of duration, in milliseconds, of all database operations performed to successfully read a group of sessions from the CRUD database.
Session Manager Session Database Statistics	upf_session_db_single_read_duration_milliseconds	Histogram	Histogram of duration, in milliseconds, of all database operations performed to successfully read an individual session from the CRUD database.
Session Manager VPP Threads Cpu Utilization	upf_vpp_thread_cpu_utilization	Gauge	VPP thread CPU utilization reported as percentage.
Session Manager VPP Threads Instantaneous Cpu Utilization	upf_vpp_thread_instant_cpu_utilization	Gauge	VPP thread CPU utilization from intant sample reported as percentage.
Session Manager host route statistics	upf_session_create_host_route_advertise_request	Counter	Number of created session host route advertisement requests.
Session Manager host route statistics	upf_session_create_host_route_advertise_request_failure	Counter	Number of created session host route advertisement requests faileded.
Session Manager host route statistics	upf_session_restore_host_route_advertise_request	Counter	Number of restored session host route advertisement requests
Session Manager host route statistics	upf_session_restore_host_route_advertise_request_failure	Counter	Number of restored session host route advertisement requests faileded.
Session Manager host route statistics	upf_session_delete_host_route_withdraw_request	Counter	Number of deleted session host route withdrawal requests
Session Manager host route statistics	upf_session_delete_host_route_withdraw_request_failure	Counter	Number of deleted session host route withdrawal requests faileded.
DNN level statistics	upf_dnn_total_packets	Counter	Total packets per DNN.
DNN level statistics	upf_dnn_total_bytes	Counter	Total bytes per DNN.
DNN level statistics	upf_dnn_total_rx_packets	Counter	Total RX packets per DNN.
DNN level statistics	upf_dnn_total_rx_bytes	Counter	Total RX bytes per DNN.
DNN level statistics	upf_dnn_total_drop_packets	Counter	Total packets per DNN.
DNN level statistics	upf_dnn_total_drop_bytes	Counter	Total drop bytes per DNN.
DNN level statistics	upf_dnn_total_rx_drop_packets	Counter	Total RX drop packets per DNN.
DNN level statistics	upf_dnn_total_rx_drop_bytes	Counter	Total RX drop bytes per DNN.

Session Manager control plane Statistics

- upf_session_creation_request

Type

64-bit Counter

Description

Number of Session Create Attempts.

Labels

remote_node_id, apn_dnn, sm_instance_id

- upf_session_creation_request_success

Type

64-bit Counter

Description

Number of Session creates succeeded.

Labels

remote_node_id, apn_dnn, sm_instance_id

- upf_session_creation_request_failure

Type

64-bit Counter

Description

Number of Session creates failed.

Labels

remote_node_id, apn_dnn, sm_instance_id, cause

- upf_stale_session_del_to_vpp_sess_id_in_use

Type

64-bit Counter

Description

Number of deletes initiated by SM towards VPP for stale sess id in use.

Labels

remote_node_id, sm_instance_id

- upf_stale_session_del_to_vpp_ue_v4_in_use

Type

64-bit Counter

Description

Number of deletes initiated by SM towards VPP for stale ue v4 in use.

Labels

remote_node_id, sm_instance_id

- upf_stale_session_del_to_vpp_ue_v6_in_use

Type

64-bit Counter

Description

Number of deletes initiated by SM towards VPP for stale ue v6 in use.

Labels

remote_node_id, sm_instance_id

- upf_stale_session_del_to_vpp_remote_teid_in_use

Type

64-bit Counter

Description

Number of deletes initiated by SM towards VPP for stale remote teid in use.

Labels

remote_node_id, sm_instance_id

- upf_session_creation_request_failure_resource_block_not_present

Type

64-bit Counter

Description

Number of Session creates failed due to no resource block present.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_creation_request_failure_ue_ip_out_of_range

Type

64-bit Counter

Description

Number of Session creates failed due to ue ip out of range.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_creation_request_allocate_ue_ip_success

Type

64-bit Counter

Description

Number of ue ip allocation requests succeeded.

Labels

apn_dnn, sm_instance_id

- upf_session_creation_request_allocate_ue_ip_failure

Type

64-bit Counter

Description

Number of ue ip allocation requests failed.

Labels

apn_dnn, sm_instance_id

- upf_session_creation_request_punt_success

Type

64-bit Counter

Description

Number of Session Create request that are punted to the right ppe.

Labels

remote_node_id, sm_instance_id

- upf_session_creation_request_punt_failure

Type

64-bit Counter

Description

Number of Session Create request that are failed to punt to the right ppe.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_creation_request_licensing_failure

Type

64-bit Counter

Description

Number of Session Create request that are failed due to UPF license down.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_creation_request_invalid_length_failure

Type

64-bit Counter

Description

Number of Session Create request that are failed due to invalid PFCP message length.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_creation_request_invalid_message_failure

Type

64-bit Counter

Description

Number of Session Create request that are failed due to invalid PFCP message format.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_creation_duplicate_request

Type

64-bit Counter

Description

Number of Session Create requests that are duplicate requests.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_creation_request_failure_no_association

Type

64-bit Counter

Description

Number of Session Create request that are failed due to no association.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_creation_request_local_resource_allocation_failure

Type

64-bit Counter

Description

Number of Session creates rejected due to local resource allocation failures. Crankback might be requested for some failures

Labels

remote_node_id, apn_dnn, sm_instance_id, cause

- upf_session_creation_request_failures_with_crankback

Type

64-bit Counter

Description

Number of Session creates failed for which crankback was requested.

Labels

remote_node_id, sm_instance_id

- upf_session_create_request_failure_remote_pool_in_disabled_mode

Type

64-bit Counter

Description

Number of Static UE IP Session Create request that are failed, due to remote pool in disabled mode.

Labels

remote_node_id, remote_pool, sm_instance_id

- upf_session_create_request_failure_remote_pool_in_maintenance_mode

Type

64-bit Counter

Description

Number of Static UE IP Session Create request that are failed, due to remote pool in maintenance mode.

Labels

remote_node_id, remote_pool, sm_instance_id

- upf_session_modify_request

Type

64-bit Counter

Description

Number of Session modify requests.

Labels

remote_node_id, apn_dnn, sm_instance_id

- upf_session_modify_request_success

Type

64-bit Counter

Description

Number of Session modify requests succeeded.

Labels

remote_node_id, apn_dnn, sm_instance_id

- upf_session_modify_request_node_id_update

Type

64-bit Counter

Description

Number of Session modify requests that updates cpfseid and nodeid.

Labels

remote_node_id, apn_dnn, sm_instance_id

- upf_session_modify_request_failure

Type

64-bit Counter

Description

Number of Session modify requests failed.

Labels

remote_node_id, apn_dnn, sm_instance_id, cause

- upf_session_modify_request_failure_resource_block_not_present

Type

64-bit Counter

Description

Number of Session modify failed due to no resource block present.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_modify_request_failure_upf_session_id_out_of_range

Type

64-bit Counter

Description

Number of Session modify failed due to upf session id out of range.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_modify_request_failure_session_in_clear_pacer

Type

64-bit Counter

Description

Number of Session modify failed due to session present in clear pacer.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_modify_request_punt_success

Type

64-bit Counter

Description

Number of Session Modify request that are punted to the right ppe.

Labels

remote_node_id, sm_instance_id

- upf_session_modify_request_punt_failure

Type

64-bit Counter

Description

Number of Session Modify request that are failed to punt to the right ppe.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_modify_request_invalid_length_failure

Type

64-bit Counter

Description

Number of Session Modify request that are failed due to invalid PCFP message length.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_modify_request_invalid_message_failure

Type

64-bit Counter

Description

Number of Session Modify request that are failed due to invalid PCFP message format.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_modify_request_no_session_failure

Type

64-bit Counter

Description

Number of Session Modify request that are failed due to no matching session.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_modify_request_fail_deleting_session

Type

64-bit Counter

Description

Number of Session Modify request that are failed due to session being deleted.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_modify_request_fail_session_not_in_vpp

Type

64-bit Counter

Description

Number of Session Modify request that are failed due to session not in VPP.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_modify_request_dropped_due_to_rb_state_not_in_sync

Type

64-bit Counter

Description

Number of Session modify requests dropped.

Labels

remote_node_id, apn_dnn, sm_instance_id

- upf_session_delete_request

Type

64-bit Counter

Description

Number of Session delete requests.

Labels

remote_node_id, apn_dnn, sm_instance_id

- upf_session_delete_request_success

Type

64-bit Counter

Description

Number of Session delete requests succeeded.

Labels

remote_node_id, apn_dnn, sm_instance_id

- upf_session_delete_request_failure

Type

64-bit Counter

Description

Number of Session delete requests failed.

Labels

remote_node_id, apn_dnn, sm_instance_id, cause

- upf_session_delete_request_failure_resource_block_not_present

Type

64-bit Counter

Description

Number of Session delete failed due to no resource block present.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_delete_request_failure_upf_session_id_out_of_range

Type

64-bit Counter

Description

Number of Session delete failed due to upf session id out of range.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_delete_request_punt_success

Type

64-bit Counter

Description

Number of Session Delete request that are punted to the right ppe.

Labels

remote_node_id, sm_instance_id

- upf_session_delete_request_punt_failure

Type

64-bit Counter

Description

Number of Session Delete request that are failed to punt to the right ppe.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_delete_request_invalid_session_failure

Type

64-bit Counter

Description

Number of Session Delete request that are failed due to no matching session.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_delete_request_invalid_message_failure

Type

64-bit Counter

Description

Number of Session Delete request that are failed due to invalid PFCP message format.

Labels

remote_node_id, sm_instance_id, cause

• upf_session_delete_request_dropped_due_to_rb_state_not_in_sync
Type
64-bit Counter
Description
Number of Session delete dropped.
Labels
<i>remote_node_id, apn_dnn, sm_instance_id</i>
• upf_session_restore_request
Type
64-bit Counter
Description
Number of Session Restore Attempts.
Labels
<i>remote_node_id, apn_dnn, sm_instance_id</i>
• upf_session_restore_request_success
Type
64-bit Counter
Description
Number of Session restores succeeded.
Labels
<i>remote_node_id, apn_dnn, sm_instance_id</i>
• upf_session_restore_request_failure
Type
64-bit Counter
Description
Number of Session restore failed.
Labels
<i>remote_node_id, apn_dnn, sm_instance_id, cause</i>
• upf_session_report_request
Type
64-bit Counter
Description
Number of Session report requests.
Labels
<i>remote_node_id, apn_dnn, sm_instance_id</i>
• upf_session_report_request_retry
Type
64-bit Counter
Description

Number of Session report requests retries.

Labels

remote_node_id, sm_instance_id, retry_type

- upf_session_report_request_success

Type

64-bit Counter

Description

Number of Session reports succeeded.

Labels

remote_node_id, apn_dnn, sm_instance_id

- upf_session_report_request_retry_success

Type

64-bit Counter

Description

Number of Session report retries succeeded.

Labels

remote_node_id, sm_instance_id, retry_type

- upf_session_report_request_failure

Type

64-bit Counter

Description

Number of Session reports failed.

Labels

remote_node_id, apn_dnn, sm_instance_id, cause

- upf_session_report_request_retry_failure

Type

64-bit Counter

Description

Number of Session report retries failed.

Labels

remote_node_id, sm_instance_id, cause, retry_type

- upf_session_report_response

Type

64-bit Counter

Description

Number of Session report responses received.

Labels

remote_node_id, apn_dnn, sm_instance_id

- upf_session_report_response_success

Type

64-bit Counter

Description

Number of Session report responses succeeded.

Labels

remote_node_id, apn_dnn, sm_instance_id

- upf_session_report_response_node_id_update

Type

64-bit Counter

Description

Number of Session report responses that updates cpfseid and nodeid.

Labels

remote_node_id, apn_dnn, sm_instance_id

- upf_session_report_response_failure

Type

64-bit Counter

Description

Number of Session report failed.

Labels

remote_node_id, apn_dnn, sm_instance_id

- upf_session_report_response_context_not_found

Type

64-bit Counter

Description

Number of Session report response with context not found.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_create_proxy_send_failure

Type

64-bit Counter

Description

Number of Session create failures due to proxy send failure.

Labels

remote_node_id, apn_dnn, sm_instance_id, cause

- upf_session_modify_proxy_send_failure

Type

64-bit Counter

Description

Number of Session modify failures due to proxy send failure.

Labels

remote_node_id, apn_dnn, sm_instance_id, cause

- upf_session_delete_proxy_send_failure

Type

64-bit Counter

Description

Number of Session delete failures due to proxy send failure.

Labels

remote_node_id, apn_dnn, sm_instance_id, cause

- upf_session_restore_proxy_send_failure

Type

64-bit Counter

Description

Number of Session restore failures due to proxy send failure.

Labels

remote_node_id, apn_dnn, sm_instance_id, cause

- upf_session_create_proxy_response_failure

Type

64-bit Counter

Description

Number of Session create failures due to proxy response failure.

Labels

remote_node_id, apn_dnn, sm_instance_id, cause

- upf_session_modify_proxy_response_failure

Type

64-bit Counter

Description

Number of Session modify failures due to proxy response failure.

Labels

remote_node_id, apn_dnn, sm_instance_id, cause

- upf_session_delete_proxy_response_failure

Type

64-bit Counter

Description

Number of Session delete failures due to proxy response failure.

Labels

remote_node_id, apn_dnn, sm_instance_id, cause

- upf_session_restore_proxy_response_failure

Type

64-bit Counter

Description

Number of Session restore failures due to proxy response failure.

Labels

remote_node_id, apn_dnn, sm_instance_id, cause

• upf_session_create_proxy_response_timeout

Type

64-bit Counter

Description

Number of Session create failures due to proxy timeouts.

Labels

remote_node_id, apn_dnn, sm_instance_id, cause

• upf_session_modify_proxy_response_timeout

Type

64-bit Counter

Description

Number of Session modify failures due to proxy timeouts.

Labels

remote_node_id, apn_dnn, sm_instance_id, cause

• upf_session_delete_proxy_response_timeout

Type

64-bit Counter

Description

Number of Session delete failures due to proxy timeout.

Labels

remote_node_id, apn_dnn, sm_instance_id, cause

• upf_session_restore_proxy_response_timeout

Type

64-bit Counter

Description

Number of Session restore failures due to proxy timeouts failure.

Labels

remote_node_id, apn_dnn, sm_instance_id, cause

• upf_session_create_vpp_send_failure

Type

64-bit Counter

Description

Number of Session create failures due to vpp send failures.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_modify_vpp_send_failure

Type

64-bit Counter

Description

Number of Session modify failures due to vpp send failures.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_delete_vpp_send_failure

Type

64-bit Counter

Description

Number of Session delete failures due to vpp send failures.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_restore_vpp_send_failure

Type

64-bit Counter

Description

Number of Session restore failures due to vpp send failures.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_create_vpp_response_failure

Type

64-bit Counter

Description

Number of Session create failures due to vpp response failures.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_modify_vpp_response_failure

Type

64-bit Counter

Description

Number of Session modify failures due to vpp response failures.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_delete_vpp_response_failure

Type

64-bit Counter

Description

Number of Session delete failures due to vpp response failures.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_restore_vpp_response_failure

Type

64-bit Counter

Description

Number of Session restore failures due to vpp response failures.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_create_vpp_response_timeout

Type

64-bit Counter

Description

Number of Session create failures due to vpp response timeouts.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_modify_vpp_response_timeout

Type

64-bit Counter

Description

Number of Session modify failures due to vpp response timeouts.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_delete_vpp_response_timeout

Type

64-bit Counter

Description

Number of Session delete failures due to vpp response timeouts.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_restore_vpp_response_timeout

Type

64-bit Counter

Description

Number of Session restore failures due to vpp response timeouts.

Labels

remote_node_id, sm_instance_id, cause

upf_session_create_vpp_procctx_notfound
Type
64-bit Counter
Description
Number of create vpp responses not processed due to proc context not found.
Labels
<i>remote_node_id, sm_instance_id, cause</i>
upf_session_modify_vpp_procctx_notfound
Type
64-bit Counter
Description
Number of modify vpp responses not processed due to proc context not found.
Labels
<i>remote_node_id, sm_instance_id, cause</i>
upf_session_delete_vpp_procctx_notfound
Type
64-bit Counter
Description
Number of Session delete vpp responses not processed due to proc context not found.
Labels
<i>remote_node_id, sm_instance_id, cause</i>
upf_session_restore_vpp_procctx_notfound
Type
64-bit Counter
Description
Number of Session restore vpp responses not processed due to proc context not found.
Labels
<i>remote_node_id, sm_instance_id, cause</i>
upf_session_create_crud_procctx_notfound
Type
64-bit Counter
Description
Number of create crud responses not processed due to proc context not found.
Labels
<i>remote_node_id, sm_instance_id, cause</i>
upf_session_modify_crud_procctx_notfound
Type
64-bit Counter
Description

Number of modify crud responses not processed due to proc context not found.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_delete_crud_procctxt_notfound

Type

64-bit Counter

Description

Number of Session delete crud responses not processed due to proc context not found.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_report_req_procctxt_notfound

Type

64-bit Counter

Description

Number of Session report requests not processed due to proc context not found.

Labels

remote_node_id, apn_dnn, sm_instance_id, cause

- upf_session_notfound_present_in_ucd

Type

64-bit Counter

Description

Number of Session not present locally but present in UCD found ucd audit process.

Labels

sm_instance_id, rb_id

- upf_session_dup_est_req_ignored

Type

64-bit Counter

Description

Number of session create requests ignored as duplicate create requests.

Labels

remote_node_id, apn_dnn, sm_instance_id, cause

- upf_session_dup_modify_req_ignored

Type

64-bit Counter

Description

Number of session modify requests ignored as duplicate modify requests.

Labels

remote_node_id, apn_dnn, sm_instance_id, cause

- upf_session_dup_delete_req_ignored

Type

64-bit Counter

Description

Number of session delete requests ignored as duplicate delete requests.

Labels

remote_node_id, apn_dnn, sm_instance_id, cause

- upf_session_pending_procctx_created

Type

64-bit Counter

Description

Number of pending proc contexts created since an existing session FSM is in progress.

Labels

remote_node_id, apn_dnn, sm_instance_id, cause

- upf_session_local_deletes

Type

64-bit Counter

Description

Number of sessions removed by session manager without trigger from SMF.

Labels

remote_node_id, apn_dnn, sm_instance_id, cause

- upf_session_local_deletes_report_resp_error

Type

64-bit Counter

Description

Number of sessions removed by session manager due to report request error from SMF.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_local_deletes_session_clear

Type

64-bit Counter

Description

Number of sessions removed by session manager due to session clear by the Pacer.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_local_association_down_request

Type

64-bit Counter

Description

Number of session removal requests due to association down.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_local_association_down_request_success

Type

64-bit Counter

Description

Number of sessions removed successfully due to association down.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_local_association_down_request_failure

Type

64-bit Counter

Description

Number of session removal requests due to association down failed.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_create_fsm_expired

Type

64-bit Counter

Description

Number of session create failures due to fsm timer expiry.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_modify_fsm_expired

Type

64-bit Counter

Description

Number of session modify failures due to fsm timer expiry.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_delete_fsm_expired

Type

64-bit Counter

Description

Number of session delete failures due to fsm timer expiry.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_restore_fsm_expired

Type

64-bit Counter

Description

Number of session restore failures due to fsm timer expiry.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_bad_report_req_msgs

Type

64-bit Counter

Description

Number of session report responses dropped due to no cause code.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_report_req_timeout

Type

64-bit Counter

Description

Number of session report requests timed out waiting for PFCP response.

Labels

remote_node_id, sm_instance_id, cause

- upf_session_report_req_dldr

Type

64-bit Counter

Description

Number of session report requests with Downlink Data Report.

Labels

remote_node_id, sm_instance_id

- upf_session_report_req_usage_report

Type

64-bit Counter

Description

Number of session report requests with usage report.

Labels

remote_node_id, sm_instance_id

- upf_session_report_req_error_ind

Type

64-bit Counter

Description

Number of session report requests with error indication report.

Labels

remote_node_id, sm_instance_id

- upf_session_report_req_inactivity_report

Type

64-bit Counter

Description

Number of session report requests with inactivity report.

Labels

remote_node_id, sm_instance_id

- upf_session_report_req_session_active_report

Type

64-bit Counter

Description

Number of session report requests with session active report.

Labels

remote_node_id, sm_instance_id

- upf_session_report_resp_wrong_ppe

Type

64-bit Counter

Description

Number of session report responses received in the wrong PPE.

Labels

remote_node_id, sm_instance_id

- upf_session_report_resp_session_notfound

Type

64-bit Counter

Description

Number of session report responses dropped due to session not found.

Labels

remote_node_id, sm_instance_id

- upf_session_clear_pacer_initiated

Type

64-bit Counter

Description

Number of session clear pacer initiated for a resource block.

Labels

resource_block, sm_instance_id

- upf_session_clear_pacer_completed

Type

64-bit Counter

Description

Number of session clear pacer completed for a resource block.

Labels

resource_block, sm_instance_id

- upf_session_partial_db_update

Type

64-bit Counter

Description

Number of partial DB update done for a modify session.

Labels

remote_node_id, sm_instance_id

- upf_session_full_db_update

Type

64-bit Counter

Description

Number of full DB update done for a modify session request.

Labels

remote_node_id, sm_instance_id

- upf_session_report_idle_session_clear_req

Type

64-bit Counter

Description

Number of session report requests of type UP initiated session request.

Labels

remote_node_id, sm_instance_id

- upf_session_report_triggered_from_session_clear_pacer

Type

64-bit Counter

Description

Number of session report requests triggered from session clear pacer.

Labels

remote_node_id, apn_dnn, sm_instance_id

- upf_standby_session_creation_request

Type

64-bit Counter

Description

Number of Standby Session Create Attempts.

Labels

remote_node_id, sm_instance_id, rb_id

upf_standby_session_creation_request_success
Type
64-bit Counter
Description
Number of Standby Session creates succeeded.
Labels
remote_node_id, sm_instance_id, rb_id
upf_standby_session_creation_request_failure
Type
64-bit Counter
Description
Number of Standby Session creates failed.
Labels
remote_node_id, sm_instance_id, cause, rb_id
upf_standby_session_creation_request_failure_resource_block_not_present
Type
64-bit Counter
Description
Number of Standby Session creates failed due to no resource block present.
Labels
remote_node_id, sm_instance_id, cause, rb_id
upf_standby_session_creation_request_failure_ue_ip_out_of_range
Type
64-bit Counter
Description
Number of Standby Session creates failed due to ue ip out of range.
Labels
remote_node_id, sm_instance_id, cause, rb_id
upf_standby_session_creation_request_invalid_length_failure
Type
64-bit Counter
Description
Number of Standby Session Create request that are failed due to invalid PFCP message length.
Labels
remote_node_id, sm_instance_id, cause, rb_id
upf_standby_session_creation_request_invalid_message_failure
Type
64-bit Counter
Description

Number of Standby Session Create request that are failed due to invalid PFCP message format.

Labels

remote_node_id, sm_instance_id, cause, rb_id

- upf_standby_session_creation_duplicate_request

Type

64-bit Counter

Description

Number of Standby Session Create requests that are duplicate requests.

Labels

remote_node_id, sm_instance_id, cause, rb_id

- upf_standby_session_creation_request_failure_no_association

Type

64-bit Counter

Description

Number of Standby Session Create request that are failed due to no association.

Labels

remote_node_id, sm_instance_id, cause, rb_id

- upf_standby_session_creation_request_failure_cpseid_mismatch

Type

64-bit Counter

Description

Number of Standby Session Create request that are failed due to cpSeld in sessEstReq not matching cpSeld in sessionRecord.

Labels

remote_node_id, sm_instance_id, cause, rb_id

- upf_standby_session_modify_request

Type

64-bit Counter

Description

Number of Standby Session modify requests.

Labels

remote_node_id, sm_instance_id, rb_id

- upf_standby_session_modify_request_success

Type

64-bit Counter

Description

Number of Standby Session modify requests succeeded.

Labels

remote_node_id, sm_instance_id, rb_id

- upf_standby_session_modify_request_node_id_update

Type

64-bit Counter

Description

Number of Standby Session modify requests that updates cpfseid and nodeid.

Labels

remote_node_id, sm_instance_id, rb_id

- upf_standby_session_modify_request_failure

Type

64-bit Counter

Description

Number of Standby Session modify requests failed.

Labels

remote_node_id, sm_instance_id, cause, rb_id

- upf_standby_session_modify_request_failure_resource_block_not_present

Type

64-bit Counter

Description

Number of Standby Session modify failed due to no resource block present.

Labels

remote_node_id, sm_instance_id, cause, rb_id

- upf_standby_session_modify_request_failure_upf_session_id_out_of_range

Type

64-bit Counter

Description

Number of Standby Session modify failed due to upf session id out of range.

Labels

remote_node_id, sm_instance_id, cause, rb_id

- upf_standby_session_modify_request_invalid_length_failure

Type

64-bit Counter

Description

Number of Standby Session Modify request that are failed due to invalid PFCP message length.

Labels

remote_node_id, sm_instance_id, cause, rb_id

- upf_standby_session_modify_request_invalid_message_failure

Type

64-bit Counter

Description

Number of Standby Session Modify request that are failed due to invalid PFCP message format.

Labels

remote_node_id, sm_instance_id, cause, rb_id

- upf_standby_session_modify_request_no_session_failure

Type

64-bit Counter

Description

Number of Standby Session Modify request that are failed due to no matching session.

Labels

remote_node_id, sm_instance_id, cause, rb_id

- upf_standby_session_modify_request_fail_deleting_session

Type

64-bit Counter

Description

Number of Standby Session Modify request that are failed due to session being deleted.

Labels

remote_node_id, sm_instance_id, cause, rb_id

- upf_standby_session_modify_request_fail_session_not_in_vpp

Type

64-bit Counter

Description

Number of Standby Session Modify request that are failed due to session not in VPP.

Labels

remote_node_id, sm_instance_id, cause, rb_id

- upf_standby_session_report_response

Type

64-bit Counter

Description

Number of Standby Session Report Responses from primary.

Labels

remote_node_id, apn_dnn, sm_instance_id, rb_id

- upf_standby_session_report_response_node_id_update

Type

64-bit Counter

Description

Number of Standby Session Report Responses updated cpfseid and nodeid.

Labels

remote_node_id, apn_dnn, sm_instance_id, rb_id

- upf_standby_session_report_response_success

Type

64-bit Counter

Description

Number of Standby Session Report Responses succeeded.

Labels

remote_node_id, apn_dnn, sm_instance_id, rb_id

• upf_standby_session_report_response_failure

Type

64-bit Counter

Description

Number of Standby Session Report Responses failed.

Labels

remote_node_id, apn_dnn, sm_instance_id, cause, rb_id

• upf_standby_session_report_response_no_session_failure

Type

64-bit Counter

Description

Number of Standby Session Report Responses that are failed due to no matching session.

Labels

remote_node_id, sm_instance_id, cause, rb_id

• upf_standby_session_report_response_fail_deleting_session

Type

64-bit Counter

Description

Number of Standby Session Report Responses that are failed due to session being deleted.

Labels

remote_node_id, sm_instance_id, cause, rb_id

• upf_standby_session_report_response_fail_session_not_in_vpp

Type

64-bit Counter

Description

Number of Standby Session Report Responses that are failed due to session not in VPP.

Labels

remote_node_id, sm_instance_id, cause, rb_id

• upf_standby_session_delete_request

Type

64-bit Counter

Description

Number of Standby Session delete requests.

Labels

remote_node_id, apn_dnn, sm_instance_id, rb_id

- upf_standby_session_delete_request_success

Type

64-bit Counter

Description

Number of Standby Session delete requests succeeded.

Labels

remote_node_id, apn_dnn, sm_instance_id, rb_id

- upf_standby_session_delete_request_failure

Type

64-bit Counter

Description

Number of Standby Session delete requests failed.

Labels

remote_node_id, apn_dnn, sm_instance_id, cause, rb_id

- upf_standby_session_delete_request_failure_resource_block_not_present

Type

64-bit Counter

Description

Number of Standby Session delete failed due to no resource block present.

Labels

remote_node_id, sm_instance_id, cause, rb_id

- upf_standby_session_delete_request_failure_upf_session_id_out_of_range

Type

64-bit Counter

Description

Number of Standby Session delete failed due to upf session id out of range.

Labels

remote_node_id, sm_instance_id, cause, rb_id

- upf_standby_session_delete_request_invalid_session_failure

Type

64-bit Counter

Description

Number of Standby Session Delete request that are failed due to no matching session.

Labels

remote_node_id, sm_instance_id, cause, rb_id

- upf_standby_session_delete_request_invalid_message_failure

Type

64-bit Counter

Description

Number of Standby Session Delete request that are failed due to invalid PFCP message format.

Labels

remote_node_id, sm_instance_id, cause, rb_id

- upf_standby_session_delete_response_success

Type

64-bit Counter

Description

Number of Standby Session Delete Response success.

Labels

remote_node_id, sm_instance_id, rb_id

- upf_standby_session_create_proxy_send_success

Type

64-bit Counter

Description

Number of Standby Session create success due to proxy send success.

Labels

remote_node_id, sm_instance_id, rb_id

- upf_standby_session_modify_proxy_send_success

Type

64-bit Counter

Description

Number of Standby Session modify success due to proxy send success.

Labels

remote_node_id, sm_instance_id, rb_id

- upf_standby_session_delete_proxy_send_success

Type

64-bit Counter

Description

Number of Standby Session delete success due to proxy send success.

Labels

remote_node_id, sm_instance_id, rb_id

- upf_standby_session_create_proxy_response_success

Type

64-bit Counter

Description

Number of Standby Session create success due to proxy response success.

Labels

remote_node_id, sm_instance_id, rb_id

- upf_standby_session_modify_proxy_response_success

Type

64-bit Counter

Description

Number of Standby Session modify success due to proxy response success.

Labels

remote_node_id, sm_instance_id, rb_id
- upf_standby_session_delete_proxy_response_success

Type

64-bit Counter

Description

Number of Standby Session delete success due to proxy response success.

Labels

remote_node_id, sm_instance_id, rb_id
- upf_standby_session_create_proxy_send_failure

Type

64-bit Counter

Description

Number of Standby Session create failures due to proxy send failure.

Labels

remote_node_id, sm_instance_id, cause, rb_id
- upf_standby_session_modify_proxy_send_failure

Type

64-bit Counter

Description

Number of Standby Session modify failures due to proxy send failure.

Labels

remote_node_id, sm_instance_id, cause, rb_id
- upf_standby_session_delete_proxy_send_failure

Type

64-bit Counter

Description

Number of Standby Session delete failures due to proxy send failure.

Labels

remote_node_id, sm_instance_id, cause, rb_id
- upf_standby_session_create_proxy_response_failure

Type

64-bit Counter

Description

Number of Standby Session create failures due to proxy response failure.

Labels

remote_node_id, sm_instance_id, cause, rb_id

- upf_standby_session_modify_proxy_response_failure

Type

64-bit Counter

Description

Number of Standby Session modify failures due to proxy response failure.

Labels

remote_node_id, sm_instance_id, cause, rb_id

- upf_standby_session_delete_proxy_response_failure

Type

64-bit Counter

Description

Number of Standby Session delete failures due to proxy response failure.

Labels

remote_node_id, sm_instance_id, cause, rb_id

- upf_standby_session_create_proxy_session_skipped

Type

64-bit Counter

Description

Number of Standby Proxy Session create skipped.

Labels

remote_node_id, sm_instance_id, cause, rb_id

- upf_standby_session_modify_proxy_session_skipped

Type

64-bit Counter

Description

Number of Standby Proxy Session skipped.

Labels

remote_node_id, sm_instance_id, cause, rb_id

- upf_standby_session_delete_proxy_session_skipped

Type

64-bit Counter

Description

Number of Standby Proxy Session delete skipped.

Labels

remote_node_id, sm_instance_id, cause, rb_id

- upf_standby_session_create_proxy_response_timeout

Type

64-bit Counter

Description

Number of Standby Session create failures due to proxy timeouts.

Labels

remote_node_id, sm_instance_id, cause, rb_id

- upf_standby_session_modify_proxy_response_timeout

Type

64-bit Counter

Description

Number of Standby Session modify failures due to proxy timeouts.

Labels

remote_node_id, sm_instance_id, cause, rb_id

- upf_standby_session_delete_proxy_response_timeout

Type

64-bit Counter

Description

Number of Standby Session delete failures due to proxy timeout.

Labels

remote_node_id, sm_instance_id, cause, rb_id

- upf_standby_session_create_vpp_send_success

Type

64-bit Counter

Description

Number of Standby Session create success due to vpp send success.

Labels

remote_node_id, sm_instance_id, rb_id

- upf_standby_session_modify_vpp_send_success

Type

64-bit Counter

Description

Number of Standby Session modify success due to vpp send success.

Labels

remote_node_id, sm_instance_id, rb_id

- upf_standby_session_delete_vpp_send_success

Type

64-bit Counter

Description

Number of Standby Session delete success due to vpp send success.

Labels

remote_node_id, sm_instance_id, rb_id

- upf_standby_session_create_vpp_send_failure

Type

64-bit Counter

Description

Number of Standby Session create failures due to vpp send failures.

Labels

remote_node_id, sm_instance_id, cause, rb_id

- upf_standby_session_modify_vpp_send_failure

Type

64-bit Counter

Description

Number of Standby Session modify failures due to vpp send failures.

Labels

remote_node_id, sm_instance_id, cause, rb_id

- upf_standby_session_delete_vpp_send_failure

Type

64-bit Counter

Description

Number of Standby Session delete failures due to vpp send failures.

Labels

remote_node_id, sm_instance_id, cause, rb_id

- upf_standby_session_create_vpp_response_success

Type

64-bit Counter

Description

Number of Standby Session create success due to vpp response success.

Labels

remote_node_id, sm_instance_id, rb_id

- upf_standby_session_modify_vpp_response_success

Type

64-bit Counter

Description

Number of Standby Session modify success due to vpp response success.

Labels

remote_node_id, sm_instance_id, rb_id

- upf_standby_session_delete_vpp_response_success

Type

64-bit Counter

Description

Number of Standby Session delete success due to vpp response success.

Labels

remote_node_id, sm_instance_id, rb_id

• upf_standby_session_create_vpp_response_failure

Type

64-bit Counter

Description

Number of Standby Session create failures due to vpp response failures.

Labels

remote_node_id, sm_instance_id, cause, rb_id

• upf_standby_session_modify_vpp_response_failure

Type

64-bit Counter

Description

Number of Standby Session modify failures due to vpp response failures.

Labels

remote_node_id, sm_instance_id, cause, rb_id

• upf_standby_session_delete_vpp_response_failure

Type

64-bit Counter

Description

Number of Standby Session delete failures due to vpp response failures.

Labels

remote_node_id, sm_instance_id, cause, rb_id

• upf_standby_session_create_vpp_response_timeout

Type

64-bit Counter

Description

Number of Standby Session create failures due to vpp response timeouts.

Labels

remote_node_id, sm_instance_id, cause, rb_id

• upf_standby_session_modify_vpp_response_timeout

Type

64-bit Counter

Description

Number of Standby Session modify failures due to vpp response timeouts.

Labels

remote_node_id, sm_instance_id, cause, rb_id

- upf_standby_session_delete_vpp_response_timeout

Type

64-bit Counter

Description

Number of Standby Session delete failures due to vpp response timeouts.

Labels

remote_node_id, sm_instance_id, cause, rb_id

- upf_standby_session_create_vpp_retry_count

Type

64-bit Counter

Description

Number of Standby Session create request VPP retries.

Labels

remote_node_id, sm_instance_id, cause, rb_id

- upf_standby_session_create_vpp_retry_due_to_send_failure_count

Type

64-bit Counter

Description

Number of Standby Session create request vpp retries due to reason that SM couldn't send request to VPP

Labels

remote_node_id, cause, rb_id

- upf_standby_session_create_vpp_send_failures_waiting_to_retry

Type

64-bit Counter

Description

Number of Standby Session create request failed to send to vpp and is waiting to retry again

Labels

remote_node_id, cause, rb_id

- upf_standby_session_modify_vpp_retry_count

Type

64-bit Counter

Description

Number of Standby Session modify request vpp retries.

Labels

remote_node_id, sm_instance_id, cause, rb_id

- upf_session_modify_vpp_retry_count

Type

64-bit Counter

	Description Number of Session modify request vpp retries. Labels <i>remote_node_id, apn_dnn, sm_instance_id</i>
• upf_standby_session_delete_vpp_retry_count	Type 64-bit Counter Description Number of Standby Session delete request vpp retries. Labels <i>remote_node_id, sm_instance_id, cause, rb_id</i>
• upf_standby_session_create_vpp_procctx_notfound	Type 64-bit Counter Description Number of create vpp responses not processed due to proc context not found. Labels <i>remote_node_id, sm_instance_id, cause, rb_id</i>
• upf_standby_session_modify_vpp_procctx_notfound	Type 64-bit Counter Description Number of modify vpp responses not processed due to proc context not found. Labels <i>remote_node_id, sm_instance_id, cause, rb_id</i>
• upf_standby_session_delete_vpp_procctx_notfound	Type 64-bit Counter Description Number of Standby Session delete vpp responses not processed due to proc context not found. Labels <i>remote_node_id, sm_instance_id, cause, rb_id</i>
• upf_standby_session_dup_est_req_ignored	Type 64-bit Counter Description Number of Standby Session create requests ignored as duplicate create requests. Labels <i>remote_node_id, sm_instance_id, cause, rb_id</i>

- upf_standby_session_dup_modify_req_ignored

Type

64-bit Counter

Description

Number of Standby Session modify requests ignored as duplicate modify requests.

Labels

remote_node_id, sm_instance_id, cause, rb_id
- upf_standby_session_dup_del_req_ignored

Type

64-bit Counter

Description

Number of Standby Session delete requests ignored as duplicate delete requests.

Labels

remote_node_id, sm_instance_id, cause, rb_id
- upf_standby_session_fsm_expired

Type

64-bit Counter

Description

Number of Standby Session failures due to fsm timer expiry.

Labels

remote_node_id, sm_instance_id, cause, rb_id
- upf_standby_session_create_fsm_expired

Type

64-bit Counter

Description

Number of Standby Session create failures due to fsm timer expiry.

Labels

remote_node_id, sm_instance_id, cause, rb_id
- upf_standby_session_modify_fsm_expired

Type

64-bit Counter

Description

Number of Standby Session modify failures due to fsm timer expiry.

Labels

remote_node_id, sm_instance_id, cause, rb_id
- upf_standby_session_delete_fsm_expired

Type

64-bit Counter

Description

Number of Standby Session delete failures due to fsm timer expiry.

Labels

remote_node_id, sm_instance_id, cause, rb_id

- upf_standby_session_create_proxy_procctx_notfound

Type

64-bit Counter

Description

Number of Standby Session Create Proxy Procedure Context not found.

Labels

remote_node_id, sm_instance_id, cause, rb_id

- upf_standby_session_proc_ctxt_alloc

Type

64-bit Counter

Description

Number of Standby Session Procedure Context allocated.

Labels

remote_node_id, sm_instance_id

- upf_standby_session_proc_ctxt_dealloc

Type

64-bit Counter

Description

Number of Standby Session Procedure Context Deallocated.

Labels

remote_node_id, sm_instance_id

- upf_standby_session_record_alloc

Type

64-bit Counter

Description

Number of Standby Session Record Allocated.

Labels

remote_node_id, sm_instance_id

- upf_standby_session_record_dealloc

Type

64-bit Counter

Description

Number of Standby Session Record Deallocated.

Labels

remote_node_id, sm_instance_id

- upf_standby_session_report_request_ignored

Type

64-bit Counter

Description

Number of Report Requests Ignored for Standby Session.

Labels

remote_node_id, sm_instance_id, rb_id

- upf_standby_fsm_stale_session_del_to_vpp_sess_id_in_use

Type

64-bit Counter

Description

Number of deletes initiated by SM towards VPP for stale sess id in use during standby FSM.

Labels

remote_node_id, sm_instance_id

- upf_standby_fsm_stale_session_del_to_vpp_ue_v4_in_use

Type

64-bit Counter

Description

Number of deletes initiated by SM towards VPP for stale ue v4 in use during standby FSM.

Labels

remote_node_id, sm_instance_id

- upf_standby_fsm_stale_session_del_to_vpp_ue_v6_in_use

Type

64-bit Counter

Description

Number of deletes initiated by SM towards VPP for stale ue v6 in use during standby FSM.

Labels

remote_node_id, sm_instance_id

- upf_standby_fsm_stale_session_del_to_vpp_remote_teid_in_use

Type

64-bit Counter

Description

Number of deletes initiated by SM towards VPP for stale Tunnel Id in use during standby FSM.

Labels

remote_node_id, sm_instance_id

- upf_session_start_record_generated

Type

64-bit Counter

Description

Number of Session Start EDR Record Generated

Labels

sm_instance_id

- upf_session_stop_record_generated

Type

64-bit Counter

Description

Number of Session Stop EDR Record Generated

Labels

sm_instance_id

- upf_session_record_send_success

Type

64-bit Counter

Description

Number of Session EDR Record Sent to Kafka Broker Successfully

Labels

sm_instance_id

- upf_session_record_send_failure

Type

64-bit Counter

Description

Number of Session EDR Record Sent to Kafka Broker Failure

Labels

sm_instance_id

- upf_session_record_delivery_success

Type

64-bit Counter

Description

Number of Session EDR Record Delivered to Kafka Broker Successfully

Labels

sm_instance_id

- upf_session_record_delivery_failure

Type

64-bit Counter

Description

Number of Session EDR Record Delivered to Kafka Broker Failure

Labels

sm_instance_id

- upf_num_total_sessions

Type

64-bit Gauge

Description

Number of total sessions.

Labels

sm_instance_id

- upf_num_active_sessions

Type

64-bit Gauge

Description

Number of active sessions.

Labels

sm_instance_id

- upf_session_pending_procctxt_count

Type

64-bit Gauge

Description

Number of pending Proc Contexts.

Labels

net_ctxt

- upf_num_standby_sessions

Type

64-bit Gauge

Description

Number of Standby Sessions.

Labels

sm_instance_id

- upf_dnn_num_active_sessions

Type

64-bit Gauge

Description

Number of active sessions for a DNN.

Labels

apn_dnn, sm_instance_id

- upf_dnn_num_standby_sessions

Type

64-bit Gauge

Description

Number of standby sessions for a DNN.

Labels

apn_dnn, sm_instance_id

- upf_ue_ip_pool_total_ip_count
- Type
- 64-bit Gauge
- Description
- Number of total UE IP count for a UE Ip Pool.
- Labels

ue_pool, sm_instance_id

- upf_ue_ip_pool_free_ip_count
- Type
- 64-bit Gauge
- Description
- Number of free UE IP count for a UE Ip Pool.
- Labels

ue_pool, sm_instance_id

- upf_dnn_v4_total_ue_ip_count
- Type
- 64-bit Gauge
- Description
- Number of total v4 UE IP count for a DNN.
- Labels

apn_dnn, sm_instance_id

- upf_dnn_v6_total_ue_ip_count
- Type
- 64-bit Gauge
- Description
- Number of total v6 UE IP count for a DNN.
- Labels

apn_dnn, sm_instance_id

- upf_dnn_v4_free_ue_ip_count
- Type
- 64-bit Gauge
- Description
- Number of free v4 UE IP count for a DNN.
- Labels

apn_dnn, sm_instance_id

- upf_dnn_v6_free_ue_ip_count
- Type
- 64-bit Gauge

Description

Number of free v6 UE IP count for a DNN.

Labels

apn_dnn, sm_instance_id

- upf_session_create_total_duration_milliseconds

Type

64-bit Histogram

Description

Histogram of the duration, in milliseconds, of create FSM for a PFCP sessEstReq message.

Labels

sm_instance_id

Boundaries

[5, 10, 15, 20, 30, 50, 100, 500, 1000, 2500]

- upf_session_modify_total_duration_milliseconds

Type

64-bit Histogram

Description

Histogram of the duration, in milliseconds, of modify FSM for a PFCP sessModReq message.

Labels

sm_instance_id

Boundaries

[5, 50, 100, 200, 300, 400, 500, 600, 750, 1000, 2500]

- upf_session_delete_total_duration_milliseconds

Type

64-bit Histogram

Description

Histogram of the duration, in milliseconds, of delete FSM for a PFCP sessDelReq message.

Labels

sm_instance_id

Boundaries

[5, 10, 15, 20, 30, 50, 100, 500, 1000, 2500]

- upf_session_restore_total_duration_milliseconds

Type

64-bit Histogram

Description

Histogram of the duration, in milliseconds, of restore FSM for a PFCP sessEstReq message.

Labels

sm_instance_id

Boundaries

[5, 10, 15, 20, 30, 50, 100, 500, 1000, 2500]

- upf_session_create_proxy_duration_milliseconds

Type
64-bit Histogram
Description
Histogram of the duration, in milliseconds, of all create proxy requests sent and response received for a PFCP sessEstReq message.
Labels
sm_instance_id
Boundaries
[5, 10, 15, 20, 30, 50, 100, 500, 1000, 2500]

- upf_session_create_vpp_duration_milliseconds

Type
64-bit Histogram
Description
Histogram of the duration, in milliseconds, of all create vpp requests sent and response received for a PFCP sessEstReq message.
Labels
sm_instance_id
Boundaries
[5, 10, 15, 20, 30, 50, 100, 500, 1000, 2500]

- upf_session_modify_proxy_duration_milliseconds

Type
64-bit Histogram
Description
Histogram of the duration, in milliseconds, of all modify proxy requests sent and response received for a PFCP sessModReq message.
Labels
sm_instance_id
Boundaries
[5, 50, 100, 200, 300, 400, 500, 600, 750, 1000, 2500]

- upf_session_modify_vpp_duration_milliseconds

Type
64-bit Histogram
Description
Histogram of the duration, in milliseconds, of all modify vpp requests sent and response received for a PFCP sessModReq message.
Labels
sm_instance_id
Boundaries
[5, 50, 100, 200, 300, 400, 500, 600, 750, 1000, 2500]

- upf_session_delete_proxy_duration_milliseconds

Type
64-bit Histogram

Description

Histogram of the duration, in milliseconds, of all delete proxy requests sent and response received for a PFCP sessDelReq message.

Labels

sm_instance_id

Boundaries

[5, 10, 15, 20, 30, 50, 100, 500, 1000, 2500]

- upf_session_delete_vpp_duration_milliseconds

Type

64-bit Histogram

Description

Histogram of the duration, in milliseconds, of all delete vpp requests sent and response received for a PFCP sessDelReq message.

Labels

sm_instance_id

Boundaries

[5, 10, 15, 20, 30, 50, 100, 500, 1000, 2500]

- upf_session_restore_proxy_duration_milliseconds

Type

64-bit Histogram

Description

Histogram of the duration, in milliseconds, of all restore proxy requests sent and response received for a PFCP sessRestoreReq message.

Labels

sm_instance_id

Boundaries

[5, 10, 15, 20, 30, 50, 100, 500, 1000, 2500]

- upf_session_restore_vpp_duration_milliseconds

Type

64-bit Histogram

Description

Histogram of the duration, in milliseconds, of all restore vpp requests sent and response received for a PFCP sessRestoreReq message.

Labels

sm_instance_id

Boundaries

[5, 10, 15, 20, 30, 50, 100, 500, 1000, 2500]

- upf_session_report_pfcps_duration_milliseconds

Type

64-bit Histogram

Description

Histogram of the duration, in milliseconds, of all session report requests sent to PFCP.

Labels

sm_instance_id

Boundaries

[5, 10, 15, 25, 50, 100, 500, 1000, 2500, 5000]

• upf_stby_session_create_total_duration_milliseconds

Type

64-bit Histogram

Description

Histogram of the duration, in milliseconds, of standby create FSM for a PFCP sessEstReq message.

Labels

sm_instance_id

Boundaries

[5, 10, 15, 20, 30, 50, 100, 500, 1000, 2500]

• upf_stby_session_modify_total_duration_milliseconds

Type

64-bit Histogram

Description

Histogram of the duration, in milliseconds, of standby modify FSM for a PFCP sessModReq message.

Labels

sm_instance_id

Boundaries

[5, 10, 15, 20, 30, 50, 100, 500, 1000, 2500]

• upf_stby_session_delete_total_duration_milliseconds

Type

64-bit Histogram

Description

Histogram of the duration, in milliseconds, of standby delete FSM for a PFCP sessDelReq message.

Labels

sm_instance_id

Boundaries

[5, 10, 15, 20, 30, 50, 100, 500, 1000, 2500]

• upf_stby_session_create_proxy_duration_milliseconds

Type

64-bit Histogram

Description

Histogram of the duration, in milliseconds, of all standby create proxy requests sent and response received for a PFCP sessEstReq message.

Labels

sm_instance_id

Boundaries

[5, 10, 15, 20, 30, 50, 100, 500, 1000, 2500]

• upf_stby_session_create_vpp_duration_milliseconds

Type

64-bit Histogram

Description

Histogram of the duration, in milliseconds, of all standby create vpp requests sent and response received for a PFCP sessEstReq message.

Labels

sm_instance_id

Boundaries

[5, 10, 15, 20, 30, 50, 100, 500, 1000, 2500]

- upf_stby_session_modify_proxy_duration_milliseconds

Type

64-bit Histogram

Description

Histogram of the duration, in milliseconds, of all standby modify proxy requests sent and response received for a PFCP sessModReq message.

Labels

sm_instance_id

Boundaries

[5, 10, 15, 20, 30, 50, 100, 500, 1000, 2500]

- upf_stby_session_modify_vpp_duration_milliseconds

Type

64-bit Histogram

Description

Histogram of the duration, in milliseconds, of all standby modify vpp requests sent and response received for a PFCP sessModReq message.

Labels

sm_instance_id

Boundaries

[5, 10, 15, 20, 30, 50, 100, 500, 1000, 2500]

Session Manager data plane Statistics

- upf_session_uplink_packets_received

Type

64-bit Counter

Description

Uplink Packets Received

Labels

net_ctxt, remote_node_id

- upf_session_uplink_packets_transmitted

Type

64-bit Counter

Description

Uplink Packets Transmitted

Labels

net_ctxt, remote_node_id

- upf_session_uplink_packets_discarded

Type

64-bit Counter

Description

Uplink Packets Discarded

Labels

net_ctxt, remote_node_id, cause

- upf_session_uplink_bytes_received

Type

64-bit Counter

Description

Uplink Packets Received

Labels

net_ctxt, remote_node_id

- upf_session_uplink_bytes_transmitted

Type

64-bit Counter

Description

Uplink Packets Transmitted

Labels

net_ctxt, remote_node_id

- upf_session_uplink_bytes_discarded

Type

64-bit Counter

Description

Uplink Packets Discarded

Labels

net_ctxt, remote_node_id, cause

- upf_session_downlink_packets_received

Type

64-bit Counter

Description

Downlink Packets Received

Labels

net_ctxt, remote_node_id

- upf_session_downlink_packets_transmitted

Type

64-bit Counter

Description

Downlink Packets Transmitted

Labels

net_ctxt, remote_node_id

- upf_session_downlink_packets_discarded

Type

64-bit Counter

Description

Downlink Packets Discarded

Labels

net_ctxt, remote_node_id, cause

- upf_session_downlink_bytes_received

Type

64-bit Counter

Description

Downlink Packets Received

Labels

net_ctxt, remote_node_id

- upf_session_downlink_bytes_transmitted

Type

64-bit Counter

Description

Downlink Packets Transmitted

Labels

net_ctxt, remote_node_id

- upf_session_downlink_bytes_discarded

Type

64-bit Counter

Description

Downlink Packets Discarded

Labels

net_ctxt, remote_node_id, cause

Session Manager Interface Statistics

- upf_intf_packets_received

Type

64-bit Counter

Description

Packets Received on an interface

Labels

interfaceName, coreId, networkContextName

• upf_intf_packets_transmitted

Type

64-bit Counter

Description

Packets Transmitted on an interface

Labels

interfaceName, coreId, networkContextName

• upf_intf_bytes_received

Type

64-bit Counter

Description

Bytes Received on an interface

Labels

interfaceName, coreId, networkContextName

• upf_intf_bytes_transmitted

Type

64-bit Counter

Description

Bytes Transmitted on an interface

Labels

interfaceName, coreId, networkContextName

• upf_intf_rx_ipv4

Type

64-bit Counter

Description

IPv4 packets counts Rx on an interface

Labels

interfaceName, coreId, networkContextName

• upf_intf_rx_ipv6

Type

64-bit Counter

Description

IPv6 packets counts Rx on an interface

Labels

interfaceName, coreId, networkContextName

• upf_intf_tx_drops

Type

64-bit Counter

Description

Packets drop counts on an interface

Labels

interfaceName, coreId, networkContextName

• upf_intf_rx_nobuff

Type

64-bit Counter

Description

No MBUF count on RX side of an interface

Labels

interfaceName, coreId, networkContextName

• upf_intf_rx_miss

Type

64-bit Counter

Description

Interface dropped packets due to rx ring buffer full

Labels

interfaceName, coreId, networkContextName

• upf_intf_rx_error

Type

64-bit Counter

Description

Any misc Rx errors on an interface

Labels

interfaceName, coreId, networkContextName

• upf_intf_tx_error

Type

64-bit Counter

Description

Any other Tx errors on an interface

Labels

interfaceName, coreId, networkContextName

Session Manager Session Database Statistics

• upf_session_db_creation_request

Type

64-bit Counter

Description

Number of attempts to add sessions to the CRUD database.

Labels

sm_instance_id

- upf_session_db_creation_request_cancel

Type

64-bit Counter

Description

Number of active session add requests converted to best effort.

Labels

sm_instance_id

- upf_session_db_creation_request_success

Type

64-bit Counter

Description

Number of sessions successfully added to the CRUD database.

Labels

sm_instance_id

- upf_session_db_creation_request_failure

Type

64-bit Counter

Description

Number of sessions that failed to be added to the CRUD database.

Labels

sm_instance_id, cause

- upf_session_db_modify_request

Type

64-bit Counter

Description

Number of attempts to update sessions in the CRUD database.

Labels

sm_instance_id

- upf_session_db_modify_request_cancel

Type

64-bit Counter

Description

Number of active session update requests converted to best effort.

Labels

sm_instance_id

- upf_session_db_modify_request_success

Type

64-bit Counter

Description

Number of sessions successfully updated in the CRUD database.

Labels

sm_instance_id

- upf_session_db_modify_request_failure

Type

64-bit Counter

Description

Number of sessions that failed to be updated in the CRUD database.

Labels

sm_instance_id, cause

- upf_session_db_delete_best_effort

Type

64-bit Counter

Description

Number of best effort attempts to delete sessions from the CRUD database. Result from CRUD database is unknown.

Labels

sm_instance_id

- upf_session_db_delete_request

Type

64-bit Counter

Description

Number of attempts to delete sessions from the CRUD database.

Labels

sm_instance_id

- upf_session_db_delete_request_cancel

Type

64-bit Counter

Description

Number of active session delete requests converted to best effort.

Labels

sm_instance_id

- upf_session_db_delete_request_success

Type

64-bit Counter

Description

Number of sessions successfully deleted from the CRUD database.

Labels

sm_instance_id

- upf_session_db_delete_request_failure

Type

64-bit Counter

Description

Number of sessions that failed to be deleted from the CRUD database.

Labels

sm_instance_id, cause

- upf_session_db_update_request

Type

64-bit Counter

Description

Number of attempts to update sessions in the CRUD database in response to data plane activity.

Labels

sm_instance_id

- upf_session_db_update_request_success

Type

64-bit Counter

Description

Number of sessions successfully updated in the CRUD database in response to data plane activity.

Labels

sm_instance_id

- upf_session_db_update_request_failure

Type

64-bit Counter

Description

Number of sessions that failed to be updated in the CRUD database in response to data plane activity.

Labels

sm_instance_id, cause

- upf_session_db_group_read_request

Type

64-bit Counter

Description

Number of attempts to read a groups of sessions from the CRUD database.

Labels

sm_instance_id

- upf_session_db_group_read_request_cancel

Type

64-bit Counter

Description

Number of session group reads cancelled before completion.

Labels

sm_instance_id

- upf_session_db_group_read_request_success

Type

64-bit Counter

Description

Number of groups of sessions successfully read from the CRUD database.

Labels

sm_instance_id

- upf_session_db_group_read_request_failure

Type

64-bit Counter

Description

Number of groups of sessions that failed to be read from the CRUD database, either due to DB retrieval failure, control-plane installation failure, or data plane installation failure.

Labels

sm_instance_id, cause

- upf_session_db_single_read_request

Type

64-bit Counter

Description

Number of attempts to read individual sessions from the CRUD database.

Labels

sm_instance_id

- upf_session_db_single_read_request_success

Type

64-bit Counter

Description

Number of individual sessions successfully read from the CRUD database.

Labels

sm_instance_id

- upf_session_db_single_read_request_failure

Type

64-bit Counter

Description

Number of individual sessions that failed to be read from the CRUD database, either due to DB retrieval failure, control-plane installation failure, or data plane installation failure.

Labels

sm_instance_id, cause

- upf_session_db_group_delete_best_effort

Type

64-bit Counter

Description

Number of group best effort attempts to delete sessions from the CRUD database. Result from CRUD database is unknown.

Labels

sm_instance_id

- upf_session_db_group_delete_best_effort_request_failure

Type

64-bit Counter

Description

Number of failed group best effort attempts to delete records from CRUD database.

Labels

sm_instance_id, cause

- upf_session_db_creation_duration_milliseconds

Type

64-bit Histogram

Description

Histogram of the duration, in milliseconds, of all database operations performed when a session is successfully added to the CRUD database in response to a PFCP sessEstReq message.

Labels

sm_instance_id

Boundaries

[10, 25, 50, 100, 250, 500, 1000, 2500, 5000, 10000]

- upf_session_db_modify_duration_milliseconds

Type

64-bit Histogram

Description

Histogram of the duration, in milliseconds, of all database operations performed when a session is successfully updated in the CRUD database in response to a PFCP sessModReq message.

Labels

sm_instance_id

Boundaries

[10, 25, 50, 100, 250, 500, 1000, 2500, 5000, 10000]

- upf_session_db_delete_duration_milliseconds

Type

64-bit Histogram

Description

Histogram of the duration, in milliseconds, of all database operations performed when a session is successfully deleted from the CRUD database in response to a PFCP sessDelReq message.

Labels
<i>sm_instance_id</i>
Boundaries
[10, 25, 50, 100, 250, 500, 1000, 2500, 5000, 10000]
• upf_session_db_update_duration_milliseconds
Type
64-bit Histogram
Description
Histogram of the duration, in milliseconds, of all database operations performed when a session is successfully updated in the CRUD database in response to data plane activity.
Labels
<i>sm_instance_id</i>
Boundaries
[10, 25, 50, 100, 250, 500, 1000, 2500, 5000, 10000]
• upf_session_db_group_read_duration_milliseconds
Type
64-bit Histogram
Description
Histogram of the duration, in milliseconds, of all database operations performed when a group of sessions is successfully read from the CRUD database.
Labels
<i>sm_instance_id</i>
Boundaries
[10, 25, 50, 100, 250, 500, 1000, 2500, 5000, 10000]
• upf_session_db_single_read_duration_milliseconds
Type
64-bit Histogram
Description
Histogram of the duration, in milliseconds, of all database operations performed when an individual session is successfully read from the CRUD database.
Labels
<i>sm_instance_id</i>
Boundaries
[10, 25, 50, 100, 250, 500, 1000, 2500, 5000, 10000]

Session Manager VPP Threads Cpu Utilization

• upf_vpp_thread_cpu_utilization
Type
64-bit Gauge
Description
VPP thread CPU utilization reported as percentage.
Labels

thread_id, thread_type

Session Manager VPP Threads Instantaneous Cpu Utilization

- upf_vpp_thread_instant_cpu_utilization

Type

64-bit Gauge

Description

VPP thread CPU utilization from instant sample reported as percentage.

Labels

thread_id, thread_type

Session Manager host route statistics

- upf_session_create_host_route_advertise_request

Type

64-bit Counter

Description

Number of created session host route advertisement requests.

Labels

net_ctxt, sm_instance_id

- upf_session_create_host_route_advertise_request_failure

Type

64-bit Counter

Description

Number of created session host route advertisement requests failed.

Labels

net_ctxt, sm_instance_id

- upf_session_restore_host_route_advertise_request

Type

64-bit Counter

Description

Number of restored session host route advertisement requests

Labels

net_ctxt, sm_instance_id

- upf_session_restore_host_route_advertise_request_failure

Type

64-bit Counter

Description

Number of restored session host route advertisement requests failed.

Labels

net_ctxt, sm_instance_id

- upf_session_delete_host_route_withdraw_request
 - Type
 - 64-bit Counter
 - Description
 - Number of deleted session host route withdrawal requests
 - Labels
 - net_ctxt, sm_instance_id**
- upf_session_delete_host_route_withdraw_request_failure
 - Type
 - 64-bit Counter
 - Description
 - Number of deleted session host route withdrawal requests failed.
 - Labels
 - net_ctxt, sm_instance_id**

DNN level statistics

- upf_dnn_total_packets
 - Type
 - 64-bit Counter
 - Description
 - Total packets per DNN.
 - Labels
 - apn_dnn**
- upf_dnn_total_bytes
 - Type
 - 64-bit Counter
 - Description
 - Total bytes per DNN.
 - Labels
 - apn_dnn**
- upf_dnn_total_rx_packets
 - Type
 - 64-bit Counter
 - Description
 - Total RX packets per DNN.
 - Labels
 - apn_dnn**
- upf_dnn_total_rx_bytes
 - Type
 - 64-bit Counter

Description

Total RX bytes per DNN.

Labels

apn_dnn

- upf_dnn_total_drop_packets

Type

64-bit Counter

Description

Total drop packets per DNN.

Labels

apn_dnn

- upf_dnn_total_drop_bytes

Type

64-bit Counter

Description

Total drop bytes per DNN.

Labels

apn_dnn

- upf_dnn_total_rx_drop_packets

Type

64-bit Counter

Description

Total RX drop packets per DNN.

Labels

apn_dnn

- upf_dnn_total_rx_drop_bytes

Type

64-bit Counter

Description

Total RX drop bytes per DNN.

Labels

apn_dnn

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- [Summary](#)
- [UE Hairpin Statistics](#)

Summary

Group	Metric Name	Type	Synopsis
UE Hairpin Statistics	total_number_UE_Hairpin_packets	Counter	Total number of UE Hairpin packets.
UE Hairpin Statistics	total_number_UE_hairpin_next_node_arp	Counter	Total number of UE Hairpin next node arp.

Group	Metric Name	Type	Synopsis
UE Hairpin Statistics	total_number_UE_hairpin_next_node_rewrite	Counter	Total number of UE Hairpin next node rewrite.
UE Hairpin Statistics	total_number_UE_hairpin_next_node_other	Counter	Total number of UE Hairpin next node other.
UE Hairpin Statistics	total_number_UE_hairpin_find_no_rb	Counter	Total number of UE Hairpin find no resource block.
UE Hairpin Statistics	total_number_UE_hairpin_error_fill_hdr	Counter	Total number of UE Hairpin error fill hdr.
UE Hairpin Statistics	total_number_UE_hairpin_find_no_rb_lpbk	Counter	Total number of UE Hairpin find no loopback for a resource block.

UE Hairpin Statistics

- [total_number_UE_Hairpin_packets](#)

Type

64-bit Counter

Description

Total number of UE Hairpin packets.

Labels

thread_id

- [total_number_UE_hairpin_next_node_arp](#)

Type

64-bit Counter

Description

Total number of UE Hairpin next node arp.

Labels

thread_id

- [total_number_UE_hairpin_next_node_rewrite](#)

Type

64-bit Counter

Description

Total number of UE Hairpin next node rewrite.

Labels

thread_id

- [total_number_UE_hairpin_next_node_other](#)

Type

64-bit Counter

Description

Total number of UE Hairpin next node other.

Labels

thread_id

- [total_number_UE_hairpin_find_no_rb](#)

Type

64-bit Counter

Description

Total number of UE Hairpin find no resource block.

Labels

thread_id

- total_number_UE_hairpin_error_fill_hdr

Type

64-bit Counter

Description

Total number of UE Hairpin error fill hdr.

Labels

thread_id

- total_number_UE_hairpin_find_no_rb_lpbk

Type

64-bit Counter

Description

Total number of UE Hairpin find no loopback for a resource block.

Labels

thread_id

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- [Summary](#)
- [Vpp Node counter statistics](#)

Summary

Group	Metric Name	Type	Synopsis
Vpp Node counter statistics	node_counters	Counter	Vpp Node Counters

Vpp Node counter statistics

- node_counters

Type

64-bit Counter

Description

Vpp Node counters

Labels

thread_name, node_name, reason

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- [Summary](#)
- [UDSF DBClient Metrics](#)
- [UDSF API latency statistics](#)

Summary

Group	Metric Name	Type	Synopsis
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Group	Metric Name	Type	Synopsis
UDSF DBClient Metrics	udsf_db_operation_request	Counter	The total number of udsf dbclient requests received.
UDSF DBClient Metrics	udsf_db_operation_response	Counter	The total number of udsf dbclient responses sent.
UDSF DBClient Metrics	udsf_dbc_total_init_success	Counter	The total number of UDSF INIT successes.
UDSF DBClient Metrics	udsf_dbc_total_init_failure	Counter	The total number of UDSF INIT failures.
UDSF DBClient Metrics	udsf_dbc_total_deinit_success	Counter	The total number of UDSF DEINIT successes.
UDSF DBClient Metrics	udsf_dbc_total_deinit_failure	Counter	The total number of UDSF DEINIT failures.
UDSF DBClient Metrics	udsf_db_transaction_counter	Counter	The total number of CRUD operations.
UDSF API latency statistics	udsf_db_transaction_duration	Histogram	Roundtrip duration taken by transaction in seconds.
UDSF API latency statistics	udsf_db_execution_duration	Histogram	Execution time taken by transaction in microseconds.
UDSF API latency statistics	udsf_hiredis_bufferwrite_duration	Histogram	Duration of time taken in hiredis to write data to socket in seconds

UDSF DBClient Metrics

- [udsf_db_operation_request](#)

Type

64-bit Counter

Description

The total number of udsf dbclient requests received from worker. Labels- 'app_name' : Originating thread name, ex:smfcc, 'operation_type' : Type of CRUD operation/API type involved, ex:CREATE_OR_MODIFY_RECORD_ASYNC

Labels

app_name, operation_type

- [udsf_db_operation_response](#)

Type

64-bit Counter

Description

The total number of udsf dbclient responses sent. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE 'operation_type' : Type of CRUD operation/API type involved., ex:CREATE_OR_MODIFY_RECORD_ASYNC

Labels

app_name, status, operation_type

- [udsf_dbc_total_init_success](#)

Type

64-bit Counter

Description

The total number of times UDSF API init was success. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: SUCCESS

Labels

app_name, status

- [udsf_dbc_total_init_failure](#)

Type

64-bit Counter

Description

The total number of times UDSF API init failed. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: FAILURE

Labels
<i>app_name, status</i>
udsf_dbc_total_deinit_success
Type
64-bit Counter
Description
The total number of times UDSF API Deinit succeeded. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: SUCCESS
Labels
<i>app_name, status</i>
udsf_dbc_total_deinit_failure
Type
64-bit Counter
Description
The total number of times UDSF API Deinit failed. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: FAILURE
Labels
<i>app_name, status</i>
udsf_db_transaction_counter
Type
64-bit Counter
Description
The total number of times CRUD operations are incrementing. Labels- 'app_name' : Originating thread name, ex:smfcc, 'crud_op' : The categorical CRUD operation in which this sub-transaction falls into, ex: CREATE_OR_MODIFY
Labels
<i>app_name, crud_op</i>

UDSF API latency statistics

udsf_db_transaction_duration
Type
64-bit Histogram
Description
Duration observed on total journey/roundtrip taken by a transaction after receiving the request. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE 'operation_type' : Type of CRUD operation/API type involved., ex:CREATE_OR_MODIFY_RECORD_ASYNC, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE, 'crud_op' : The categorical CRUD operation in which this transaction falls into, ex: CREATE_OR_MODIFY
Labels
<i>app_name,operation_type,status,crud_op</i>
Boundaries
[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]
udsf_db_execution_duration
Type

64-bit Histogram

Description

Processing time taken by UDSF client. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE 'operation_type' : Type of CRUD operation/API type involved., ex:CREATE_OR_MODIFY_RECORD_ASYNC, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE, 'crud_op' : The categorical CRUD operation in which this transation falls into, ex: CREATE_OR_MODIFY

Labels

app_name,operation_type,status,crud_op

Boundaries

[1, 2, 4, 8, 16, 32, 64, 128, 256, 384, 512, 768, 1024, 1536, 2048, 4096, 8192, 16384]

- [udsf_hiredis_bufferwrite_duration](#)

Type

64-bit Histogram

Description

Duration observed between hiredis buffer append and write to socket. This is fair approximation for time spent before data is written to socket as the buffer threshold accounted for append and write is dynamic and not fixed size. Per socket, the buffer append start-time is recorded along with size intended to transmit and end-time recorded once data greater than recorded size is written on the socket

Labels

app_name

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

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- [Summary](#)
- [Vprobe Agent Module Statistics](#)

Summary

Group	Metric Name	Type	Synopsis
Vprobe Agent Module Statistics	total_start_flow_edrs_generated	Counter	Total START flow EDRs generated by PPE.
Vprobe Agent Module Statistics	total_start_flow_edrs_failed_to_parse	Counter	Total START flow EDRs that could not be parsed in PPE.
Vprobe Agent Module Statistics	total_start_flow_edrs_successfully_sent_to_external_consumer	Counter	Total START flow EDRs successfully sent to external consumer by PPE.
Vprobe Agent Module Statistics	total_start_flow_edrs_dropped_due_to_service_unavailable	Counter	Total START flow EDRs generated by PPE that were dropped as external service is unavailable.
Vprobe Agent Module Statistics	total_start_flow_edrs_failed_to_send_to_external_consumer	Counter	Total START flow EDRs generated by PPE that could not be set to external consumer.
Vprobe Agent Module Statistics	total_interim_flow_edrs_generated	Counter	Total INTERIM flow EDRs generated by PPE.
Vprobe Agent Module Statistics	total_interim_flow_edrs_failed_to_parse	Counter	Total INTERIM flow EDRs that could not be parsed in PPE.
Vprobe Agent Module Statistics	total_interim_flow_edrs_successfully_sent_to_external_consumer	Counter	Total INTERIM flow EDRs successfully sent to external consumer by PPE.
Vprobe Agent Module Statistics	total_interim_flow_edrs_dropped_due_to_service_unavailable	Counter	Total INTERIM flow EDRs generated by PPE that were dropped as external service is unavailable.
Vprobe Agent Module Statistics	total_interim_flow_edrs_failed_to_send_to_external_consumer	Counter	Total INTERIM flow EDRs generated by PPE that could not be set to external consumer.

Group	Metric Name	Type	Synopsis
Vprobe Agent Module Statistics	total_stop_flow_edrs_generated	Counter	Total STOP flow EDRs generated by PPE.
Vprobe Agent Module Statistics	total_stop_flow_edrs_failed_to_parse	Counter	Total STOP flow EDRs that could not be parsed in PPE.
Vprobe Agent Module Statistics	total_stop_flow_edrs_successfully_sent_to_external_consumer	Counter	Total STOP flow EDRs successfully sent to external consumer by PPE.
Vprobe Agent Module Statistics	total_stop_flow_edrs_dropped_due_to_service_unavailable	Counter	Total STOP flow EDRs generated by PPE that were dropped as external service is unavailable.
Vprobe Agent Module Statistics	total_stop_flow_edrs_failed_to_send_to_external_consumer	Counter	Total STOP flow EDRs generated by PPE that could not be set to external consumer.
Vprobe Agent Module Statistics	total_start_stop_flow_edrs_generated	Counter	Total START STOP flow EDRs generated by PPE.
Vprobe Agent Module Statistics	total_start_stop_flow_edrs_failed_to_parse	Counter	Total START STOP flow EDRs that could not be parsed in PPE.
Vprobe Agent Module Statistics	total_start_stop_flow_edrs_successfully_sent_to_external_consumer	Counter	Total START STOP flow EDRs successfully sent to external consumer by PPE.
Vprobe Agent Module Statistics	total_start_stop_flow_edrs_dropped_due_to_service_unavailable	Counter	Total START STOP flow EDRs generated by PPE that were dropped as external service is unavailable.
Vprobe Agent Module Statistics	total_start_stop_flow_edrs_failed_to_send_to_external_consumer	Counter	Total START STOP flow EDRs generated by PPE that could not be set to external consumer.
Vprobe Agent Module Statistics	total_flow_edrs_rejected_by_external_consumer	Counter	Total flow EDRs rejected by by external consumer.
Vprobe Agent Module Statistics	total_proto_buf_failed_to_parse	Counter	Total protobuf failed to parse

Vprobe Agent Module Statistics

- total_start_flow_edrs_generated

Type

64-bit Counter

Description

Total START type flow EDRs generated by PPE.

Metrics

Instances	Labels	Description
1	stats_group: vprobe_module_stats	Total START Flow EDRs generatd by PPE.

- total_start_flow_edrs_failed_to_parse [Deprecated]

Type

64-bit Counter

Description

Total START type flow EDRs that could not be parsed in PPE

Metrics

Instances	Labels	Description
1	stats_group: vprobe_module_stats	Total START Flow EDRs that could not be parsed in PPE.

- total_start_flow_edrs_successfully_sent_to_external_consumer

Type

64-bit Counter

Description

Total START type flow EDRs successfully sent to external consumer by PPE.

Metrics

Instances	Labels	Description
1	stats_group: vprobe_module_stats	Total START Flow EDRs successfully sent to external consumer by PPE.

- total_start_flow_edrs_dropped_due_to_service_unavailable

Type

64-bit Counter

Description

Total START type flow EDRs generated by PPE that were dropped as external service is unavailable.

Metrics

Instances	Labels	Description
1	stats_group: vprobe_module_stats	Total START Flow EDRs generatd by PPE that were dropped as external service is unavailable.

- total_start_flow_edrs_failed_to_send_to_external_consumer

Type

64-bit Counter

Description

Total START type flow EDRs generated by PPE that could not be set to external consumer.

Metrics

Instances	Labels	Description
1	stats_group: vprobe_module_stats	Total START Flow EDRs generatd by PPE that could not be set to external consumer.

- total_interim_flow_edrs_generated

Type

64-bit Counter

Description

Total INTERIM type flow EDRs generated by PPE.

Metrics

Instances	Labels	Description
1	stats_group: vprobe_module_stats	Total START Flow EDRs generatd by PPE that could not be set to external consumer.

- total_interim_flow_edrs_failed_to_parse [Deprecated]

Type

64-bit Counter

Description

Total INTERIM type flow EDRs that could not be parsed in PPE

Metrics

Instances	Labels	Description
1	stats_group: vprobe_module_stats	Total INTERIM Flow EDRs generatd by PPE.

- total_interim_flow_edrs_successfully_sent_to_external_consumer

Type

Instances	Labels	Description
1	stats_group: vprobe_module_stats	Total INTERIM Flow EDRs that could not be parsed in PPE.

64-bit Counter
Description
Total INTERIM type flow EDRs successfully sent to external consumer by PPE.
Metrics

Instances	Labels	Description
1	stats_group: vprobe_module_stats	Total INTERIM Flow EDRs successfully sent to external consumer by PPE.

• total_interim_flow_edrs_dropped_due_to_service_unavailable
Type
64-bit Counter
Description
Total INTERIM type flow EDRs generated by PPE that were dropped as external service is unavailable.
Metrics

Instances	Labels	Description
1	stats_group: vprobe_module_stats	Total INTERIM Flow EDRs generatd by PPE that were dropped as external service is unavailable.

• total_interim_flow_edrs_failed_to_send_to_external_consumer
Type
64-bit Counter
Description
Total INTERIM type flow EDRs generated by PPE that could not be set to external consumer.
Metrics

Instances	Labels	Description
1	stats_group: vprobe_module_stats	Total INTERIM Flow EDRs generatd by PPE that could not be set to external consumer.

• total_stop_flow_edrs_generated
Type
64-bit Counter
Description
Total STOP type flow EDRs generated by PPE.
Metrics

Instances	Labels	Description
1	stats_group: vprobe_module_stats	Total STOP Flow EDRs generatd by PPE.

• total_stop_flow_edrs_failed_to_parse [Deprecated]
Type
64-bit Counter
Description
Total STOP type flow EDRs that could not be parsed in PPE
Metrics

Instances	Labels	Description
1	stats_group: vprobe_module_stats	Total STOP Flow EDRs that could not be parsed in PPE.

• total_stop_flow_edrs_successfully_sent_to_external_consumer

Type
64-bit Counter
Description
Total STOP type flow EDRs successfully sent to external consumer by PPE.
Metrics

Instances	Labels	Description
1	stats_group: vprobe_module_stats	Total STOP Flow EDRs successfully sent to external consumer by PPE.
• total_stop_flow_edrs_dropped_due_to_service_unavailable		

Type
64-bit Counter
Description
Total STOP type flow EDRs generated by PPE that were dropped as external service is unavailable.
Metrics

Instances	Labels	Description
1	stats_group: vprobe_module_stats	Total STOP Flow EDRs generatd by PPE that were dropped as external service is unavailable.
• total_stop_flow_edrs_failed_to_send_to_external_consumer		

Type
64-bit Counter
Description
Total STOP type flow EDRs generated by PPE that could not be set to external consumer.
Metrics

Instances	Labels	Description
1	stats_group: vprobe_module_stats	Total STOP Flow EDRs generatd by PPE that could not be set to external consumer.
• total_start_stop_flow_edrs_generated		

Type
64-bit Counter
Description
Total START STOP type flow EDRs generated by PPE.
Metrics

Instances	Labels	Description
1	stats_group: vprobe_module_stats	Total START STOP Flow EDRs generatd by PPE.
• total_start_stop_flow_edrs_failed_to_parse [Deprecated]		

Type
64-bit Counter
Description
Total START STOP type flow EDRs that could not be parsed in PPE
Metrics

Instances	Labels	Description
1	stats_group: vprobe_module_stats	Total START STOP Flow EDRs that could not be parsed in PPE.

- total_start_stop_flow_edrs_successfully_sent_to_external_consumer

Type

64-bit Counter

Description

Total START STOP type flow EDRs successfully sent to external consumer by PPE.

Metrics

Instances	Labels	Description
1	stats_group: vprobe_module_stats	Total START STOP Flow EDRs successfully sent to external consumer by PPE.

- total_start_stop_flow_edrs_dropped_due_to_service_unavailable

Type

64-bit Counter

Description

Total START STOP type flow EDRs generated by PPE that were dropped as external service is unavailable.

Metrics

Instances	Labels	Description
1	stats_group: vprobe_module_stats	Total START STOP Flow EDRs generatd by PPE that were dropped as external service is unavailable.

- total_start_stop_flow_edrs_failed_to_send_to_external_consumer

Type

64-bit Counter

Description

Total START STOP type flow EDRs generated by PPE that could not be set to external consumer.

Metrics

Instances	Labels	Description
1	stats_group: vprobe_module_stats	Total START STOP Flow EDRs generatd by PPE that could not be set to external consumer.

- total_flow_edrs_rejected_by_external_consumer

Type

64-bit Counter

Description

Total flow EDRs rejected by external consumer

Metrics

Instances	Labels	Description
1	stats_group: vprobe_module_stats	Total Flow EDRs rejected by external consumer.

- total_proto_buf_failed_to_parse

Type

64-bit Counter

Description

Total protobuf failed to parse

Metrics

Instances	Labels	Description
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Instances	Labels	Description
1	stats_group: vprobe_module_stats	Total protobuf failed to parse

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- [charging_local_cdr_files_generation_rate](#)
- [charging_local_cdrs_generation_rate](#)

Summary

Metric Name	Type	Synopsis
charging_local_cdr_files_generation_rate	counter	Local CDR files generation rate
charging_local_cdrs_generation_rate	counter	Total Local CDRs generation rate

charging_local_cdr_files_generation_rate

Description

Rate of local CDR files generation for every interval of 15 minutes per record type, file format, gw type, cdr record type

Type

counter

Prometheus Expression

sum(rate(charging_local_cdrs_files_count [15m])) by (record_type, file_format, gw_type, cdr_record_Type)

charging_local_cdrs_generation_rate

Description

Rate of local CDRs generation for every interval of 15 minutes per record type, file format, gw type, cdr record type

Type

counter

Prometheus Expression

sum(rate(charging_local_cdrs_count_total [15m])) by (record_type, file_format, gw_type, cdr_record_Type)

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- [Summary](#)
- [charging_nchf_server_message_success_percentage](#)
- [charging_nchf_session_message_success_percentage](#)

Summary

Metric Name	Type	Synopsis
charging_nchf_server_message_success_percentage	counter	Nchf Session Message Success Percentage
charging_nchf_session_message_success_percentage	counter	Nchf+ Session Message Success Percentage

charging_nchf_server_message_success_percentage

Description

This counter indicates the success percentage of Nchf session messages in the last 15 minutes per Nchf message type namely SessionCreate, SessionUpdate, SessionRelease and SessionNotify.

Type

counter

Prometheus Expression

sum(sum_over_time(charging_nchf_server_convergedcharging_client_response_total{status=~"2.."} [15m]) / ignoring(status) group_left sum_over_time(charging_nchf_server_convergedcharging_client_request_total [15m]) * 100) by (message_type)

charging_nchf_session_message_success_percentage

Description

This counter indicates the success percentage of Nchf+ session messages in the last 15 minutes per Nchf message type namely SessionCreate, SessionUpdate, SessionRelease and SessionNotify.

Type

counter

Prometheus Expression

sum(sum_over_time(charging_nchf_convergedcharging_response_total{status=~"2.."} [15m]) / ignoring(status) group_left sum_over_time(charging_nchf_convergedcharging_request_total [15m]) * 100) by (message_type)

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- [Event Data Records for AMF Call Control Service](#)

Summary

Group	Metric Name	Type	Synopsis
Session Management layers statistics for AMF Call Control Service	amfcc_nsmf_pduession_create_sm_context_request	Counter	Create SM context requests sent by AMF to SMF
Session Management layers statistics for AMF Call Control Service	amfcc_nsmf_pduession_create_sm_context_response	Counter	Create SM context responses received by AMF from SMF
Session Management layers statistics for AMF Call Control Service	amfcc_nsmf_pduession_update_sm_context_request	Counter	Update SM context requests sent by AMF to SMF
Session Management layers statistics for AMF Call Control Service	amfcc_nsmf_pduession_update_sm_context_response	Counter	Update SM context responses received by AMF from SMF
Session Management layers statistics for AMF Call Control Service	amfcc_nsmf_pduession_notify_sm_context_status_request_cb	Counter	SM context status notify requests received by AMF from SMF
Session Management layers statistics for AMF Call Control Service	amfcc_nsmf_pduession_notify_sm_context_status_response	Counter	SM context status notify responses sent by AMF to SMF
Session Management layers statistics for AMF Call Control Service	amfcc_nsmf_pduession_release_sm_context_request	Counter	Release SM context requests sent by AMF to SMF

Group	Metric Name	Type	Synopsis
Session Management layers statistics for AMF Call Control Service	amfcc_nsmf_pdusection_release_sm_context_response	Counter	Release SM context responses received by AMF from SMF
Session Management layers statistics for AMF Call Control Service	amfcc_nsmf_pdusection_retrieve_sm_context_request	Counter	Retrieve SM context requests sent by AMF to SMF
Session Management layers statistics for AMF Call Control Service	amfcc_nsmf_pdusection_retrieve_sm_context_response	Counter	Retrieve SM context responses received by AMF from SMF
Session Management layers statistics for AMF Call Control Service	amfcc_sm_smf_selection_attempt	Counter	SMF selection attempts
Session Management layers statistics for AMF Call Control Service	amfcc_sm_smf_selection_success	Counter	SMF selection success
Session Management layers statistics for AMF Call Control Service	amfcc_sm_smf_selection_failure	Counter	SMF selection failures
Mobility Management layers statistics for AMF Call Control Service	amfcc_n1_init_registration_request	Counter	Initial registration requests received by AMF from UE
Mobility Management layers statistics for AMF Call Control Service	amfcc_n1_init_registration_accept	Counter	Initial registration accepts sent by AMF to UE
Mobility Management layers statistics for AMF Call Control Service	amfcc_n1_init_registration_complete	Counter	Initial registration complete received by AMF from UE
Mobility Management layers statistics for AMF Call Control Service	amfcc_n1_init_registration_reject	Counter	Initial registration rejects sent by AMF to UE
Mobility Management layers statistics for AMF Call Control Service	amfcc_n1_mobility_registration_request	Counter	Mobility registration requests received by AMF from UE
Mobility Management layers statistics for AMF Call Control Service	amfcc_n1_mobility_registration_accept	Counter	Mobility registration accepts sent by AMF to UE
Mobility Management layers statistics for AMF Call Control Service	amfcc_n1_mobility_registration_complete	Counter	Mobility registration complete received by AMF from UE
Mobility Management layers statistics for AMF Call Control Service	amfcc_n1_mobility_registration_reject	Counter	Mobility registration rejects sent by AMF to UE
Mobility Management layers statistics for AMF Call Control Service	amfcc_n1_periodic_registration_request	Counter	Periodic registration requests received by AMF from UE
Mobility Management layers statistics for AMF Call Control Service	amfcc_n1_periodic_registration_accept	Counter	Periodic registration accepts sent by AMF to UE
Mobility Management layers statistics for AMF Call Control Service	amfcc_n1_periodic_registration_complete	Counter	Periodic registration complete received by AMF from UE
Mobility Management layers statistics for AMF Call Control Service	amfcc_n1_periodic_registration_reject	Counter	Periodic registration rejects sent by AMF to UE
Mobility Management layers statistics for AMF Call Control Service	amfcc_n1_auth_request	Counter	Authentication requests sent by AMF to UE

Group	Metric Name	Type	Synopsis
Mobility Management layers statistics for AMF Call Control Service	amfcc_n1_auth_response	Counter	Authentication responses received by AMF from UE
Mobility Management layers statistics for AMF Call Control Service	amfcc_n1_auth_failure	Counter	Authentication failures received by AMF from UE
Mobility Management layers statistics for AMF Call Control Service	amfcc_n1_auth_reject	Counter	Authentication rejects sent by AMF to UE
Mobility Management layers statistics for AMF Call Control Service	amfcc_n1_security_mode_command	Counter	Security mode commands sent by AMF to UE
Mobility Management layers statistics for AMF Call Control Service	amfcc_n1_security_mode_complete	Counter	Security mode completes received by AMF from UE
Mobility Management layers statistics for AMF Call Control Service	amfcc_n1_security_mode_reject	Counter	Security mode reject sends by AMF to UE
Mobility Management layers statistics for AMF Call Control Service	amfcc_n1_identity_request	Counter	Identity requests sent by AMF to UE
Mobility Management layers statistics for AMF Call Control Service	amfcc_n1_identity_response	Counter	Identity responses received by AMF from UE
Mobility Management layers statistics for AMF Call Control Service	amfcc_n1_uplink_nas_transport	Counter	Uplink nas transport received by AMF from UE
Mobility Management layers statistics for AMF Call Control Service	amfcc_n1_downlink_nas_transport	Counter	Downlink nas transport sent by AMF to UE
Mobility Management layers statistics for AMF Call Control Service	amfcc_n1_ue_deregistration_request	Counter	Deregistration requests received by AMF from UE
Mobility Management layers statistics for AMF Call Control Service	amfcc_n1_network_deregistration_accept	Counter	Deregistration accepts sent by AMF to UE
Mobility Management layers statistics for AMF Call Control Service	amfcc_n1_network_deregistration_request	Counter	Deregistration requests sent by AMF to UE
Mobility Management layers statistics for AMF Call Control Service	amfcc_n1_ue_deregistration_accept	Counter	Deregistration accepts received by AMF from UE
Mobility Management layers statistics for AMF Call Control Service	amfcc_n1_service_request	Counter	Service requests received by AMF from UE
Mobility Management layers statistics for AMF Call Control Service	amfcc_n1_service_accept	Counter	Service accepts sent by AMF to UE
Mobility Management layers statistics for AMF Call Control Service	amfcc_n1_service_reject	Counter	Service rejects sent by AMF to UE
Mobility Management layers statistics for AMF Call Control Service	amfcc_n1_configuration_update_command	Counter	Configuration update commands sent by AMF to UE
Mobility Management layers statistics for AMF Call Control Service	amfcc_n1_configuration_update_complete	Counter	Configuration update completes received by AMF from UE

Group	Metric Name	Type	Synopsis
Mobility Management layers statistics for AMF Call Control Service	amfcc_n1_network_mm_status	Counter	5GMM Status sent by AMF to UE
Mobility Management layers statistics for AMF Call Control Service	amfcc_nausf_ueau_request	Counter	UE authentication requests sent by AMF to AUSF
Mobility Management layers statistics for AMF Call Control Service	amfcc_nausf_ueau_response	Counter	UE authentication responses received by AMF from AUSF
Mobility Management layers statistics for AMF Call Control Service	amfcc_nausf_ueau_confirm_request	Counter	UE authentication confirmation requests sent by AMF to AUSF
Mobility Management layers statistics for AMF Call Control Service	amfcc_nausf_ueau_confirm_response	Counter	UE authentication confirmation responses received by AMF from AUSF
Mobility Management layers statistics for AMF Call Control Service	amfcc_nausf_ueau_eap_session_request	Counter	EAP session requests sent by AMF to AUSF
Mobility Management layers statistics for AMF Call Control Service	amfcc_nausf_ueau_eap_session_response	Counter	EAP session responses received by AMF from AUSF
Mobility Management layers statistics for AMF Call Control Service	amfcc_nausf_rgau_request	Counter	RG authentication requests sent by AMF to AUSF
Mobility Management layers statistics for AMF Call Control Service	amfcc_nausf_rgau_response	Counter	RG authentication responses received by AMF from AUSF
Mobility Management layers statistics for AMF Call Control Service	amfcc_nudm_uecm_registration_3gpp_request	Counter	UE context management requests sent by AMF to UDM for 3GPP registration
Mobility Management layers statistics for AMF Call Control Service	amfcc_nudm_uecm_registration_3gpp_response	Counter	UE context management responses received by AMF from UDM for 3GPP registration
Mobility Management layers statistics for AMF Call Control Service	amfcc_nudm_uecm_registration_update_3gpp_request	Counter	UE context management requests sent by AMF to UDM for 3GPP registration update
Mobility Management layers statistics for AMF Call Control Service	amfcc_nudm_uecm_registration_update_3gpp_response	Counter	UE context management responses received by AMF from UDM for 3GPP registration update
Mobility Management layers statistics for AMF Call Control Service	amfcc_nudm_uecm_deregistration_notification	Counter	UE context management messages received by old AMF from UDM for 3GPP de-registration following registration of new AMF
Mobility Management layers statistics for AMF Call Control Service	amfcc_nudm_sdm_get_request	Counter	Subscriber data management requests sent by AMF to UDM
Mobility Management layers statistics for AMF Call Control Service	amfcc_nudm_sdm_get_response	Counter	Subscriber data management responses received by AMF from UDM
Mobility Management layers statistics for AMF Call Control Service	amfcc_nudm_sdm_get_nssai_request	Counter	Subscriber data management requests for NSSAI sent by AMF to UDM
Mobility Management layers statistics for AMF Call Control Service	amfcc_nudm_sdm_get_nssai_response	Counter	Subscriber data management responses for NSSAI received by AMF from UDM
Mobility Management layers statistics for AMF Call Control Service	amfcc_nudm_sdm_subscription_request	Counter	Subscriber data management subscription requests sent by AMF to UDM

Group	Metric Name	Type	Synopsis
Mobility Management layers statistics for AMF Call Control Service	amfcc_nudm_sdm_subscription_response	Counter	Subscriber data management subscription responses received by AMF from UDM
Mobility Management layers statistics for AMF Call Control Service	amfcc_nudm_sdm_change_notification_request	Counter	Change notification requests for Subscriber data management resources received by AMF from UDM.
Mobility Management layers statistics for AMF Call Control Service	amfcc_nudm_sdm_change_notification_response	Counter	Change notification responses for Subscriber data management resources sent by AMF to UDM.
Mobility Management layers statistics for AMF Call Control Service	amfcc_nudm_sdm_info_request	Counter	Subscriber data management info requests sent by AMF to UDM
Mobility Management layers statistics for AMF Call Control Service	amfcc_nudm_sdm_info_response	Counter	Subscriber data management info responses received by AMF from UDM
Mobility Management layers statistics for AMF Call Control Service	amfcc_nnsf_nsselection_request_for_registration	Counter	Network slice selection requests sent by AMF to NSSF for registration procedure
Mobility Management layers statistics for AMF Call Control Service	amfcc_nnsf_nsselection_response_for_registration	Counter	Network slice selection reponses received by AMF from NSSF for registration procedure
Mobility Management layers statistics for AMF Call Control Service	amfcc_nnsf_nsselection_request_for_pdu_session_establishment	Counter	Network slice selection requests sent by AMF to NSSF for PDU session establishment
Mobility Management layers statistics for AMF Call Control Service	amfcc_nnsf_nsselection_response_for_pdu_session_establishment	Counter	Network slice selection reponses received by AMF from NSSF for PDU session establishment
Mobility Management layers statistics for AMF Call Control Service	amfcc_nnsf_nsselection_request_for_ue_configuration_update	Counter	Network slice selection requests sent by AMF to NSSF for UE configuration update procedure
Mobility Management layers statistics for AMF Call Control Service	amfcc_nnsf_nsselection_response_for_ue_configuration_update	Counter	Network slice selection responses received by AMF from NSSF for UE configuration update procedure
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_initial_registration_attempt	Counter	Initial registration procedure attempts
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_initial_registration_success	Counter	Initial registration procedure success
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_inter_new_amf_initial_registration_attempt	Counter	Inter new AMF initial registration procedure attempts
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_inter_new_amf_initial_registration_success	Counter	Inter new AMF initial registration procedure success
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_inter_new_amf_initial_registration_failure	Counter	Inter new AMF initial registration procedure failures
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_initial_registration_failure	Counter	Initial registration procedure failures
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_intra_mobility_registration_attempt	Counter	Mobility registration procedure attempts
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_intra_mobility_registration_success	Counter	Mobility registration procedure success

Group	Metric Name	Type	Synopsis
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_intra_mobility_registration_failure	Counter	Mobility registration procedure failures
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_inter_new_amf_mobility_registration_attempt	Counter	Inter new AMF mobility registration procedure attempts
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_inter_new_amf_mobility_registration_success	Counter	Inter new AMF mobility registration procedure success
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_inter_new_amf_mobility_registration_failure	Counter	Inter new AMF mobility registration procedure failures
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_periodic_registration_attempt	Counter	Periodic registration procedure attempts
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_periodic_registration_success	Counter	Periodic registration procedure success
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_periodic_registration_failure	Counter	Periodic registration procedure failures
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_ue_deregistration_attempt	Counter	Deregistration procedure attempts
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_ue_deregistration_success	Counter	Deregistration procedure success
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_ue_deregistration_failure	Counter	Deregistration procedure failures
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_nwk_deregistration_attempt	Counter	Deregistration procedure attempts by network
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_auth_attempt	Counter	Authentication procedure attempts
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_auth_success	Counter	Authentication procedure success
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_auth_failure	Counter	Authentication procedure failures
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_service_request_attempt	Counter	Service request procedure attempts
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_service_request_success	Counter	Service request procedure success
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_service_request_failure	Counter	Service request procedure failures
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_inter_old_amf_initial_registration_attempt	Counter	Inter source AMF UE context transfer procedure attempts with reason initial registration
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_inter_old_amf_initial_registration_success	Counter	Inter source AMF UE context transfer procedure success with reason initial registration

Group	Metric Name	Type	Synopsis
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_inter_old_amf_initial_registration_failure	Counter	Inter source AMF UE context transfer procedure failures with reason initial registration
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_inter_old_amf_mobility_registration_attempt	Counter	Inter source AMF UE context transfer procedure attempts with reason mobility registration
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_inter_old_amf_mobility_registration_success	Counter	Inter source AMF UE context transfer procedure success with reason mobility registration
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_inter_old_amf_mobility_registration_failure	Counter	Inter source AMF UE context transfer procedure failures with reason mobility registration
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_n26_inter_eps_new_amf_mobility_registration_attempt	Counter	EPS to 5GS with N26 mobility registration procedure attempts on new AMF
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_n26_inter_eps_new_amf_mobility_registration_success	Counter	EPS to 5GS with N26 mobility registration procedure success on new AMF
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_n26_inter_eps_new_amf_mobility_registration_failure	Counter	EPS to 5GS with N26 mobility registration procedure failures on new AMF
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_inter_eps_new_amf_initial_registration_attempt	Counter	EPS to 5GS without N26 initial registration procedure attempts on new AMF
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_inter_eps_new_amf_initial_registration_success	Counter	EPS to 5GS without N26 initial registration procedure success on new AMF
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_inter_eps_new_amf_initial_registration_failure	Counter	EPS to 5GS without N26 initial registration procedure failures on new AMF
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_inter_eps_new_amf_mobility_registration_attempt	Counter	EPS to 5GS without N26 mobility registration procedure attempts on new AMF
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_inter_eps_new_amf_mobility_registration_success	Counter	EPS to 5GS without N26 mobility registration procedure success on new AMF
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_inter_eps_new_amf_mobility_registration_failure	Counter	EPS to 5GS without N26 mobility registration procedure failures on new AMF
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_inter_eps_old_amf_mobility_registration_attempt	Counter	5GS to EPS mobility registration procedure attempts on old AMF
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_inter_eps_old_amf_mobility_registration_success	Counter	5GS to EPS mobility registration procedure success on old AMF
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_inter_eps_old_amf_mobility_registration_failure	Counter	5GS to EPS mobility registration procedure failures on old AMF
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_paging_attempt	Counter	MM paging procedure to UE
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_paging_first_retry_attempt	Counter	MM paging attempt first retry to UE
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_paging_second_retry_attempt	Counter	MM paging attempt second retry to UE

Group	Metric Name	Type	Synopsis
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_paging_third_retry_attempt	Counter	MM paging attempt third retry to UE
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_paging_failure	Counter	MM paging procedure failure
Mobility Management layers statistics for AMF Call Control Service	amfcc_npcf_ue_policy_control_create_request	Counter	Send UE policy control create Requests from AMF to PCF
Mobility Management layers statistics for AMF Call Control Service	amfcc_npcf_ue_policy_control_create_response	Counter	Received UE policy control create Responses from PCF to AMF
Mobility Management layers statistics for AMF Call Control Service	amfcc_npcf_ue_policy_control_update_request	Counter	Send UE policy control update Requests from AMF to PCF
Mobility Management layers statistics for AMF Call Control Service	amfcc_npcf_ue_policy_control_update_response	Counter	Received UE policy control update Responses from PCF to AMF
Mobility Management layers statistics for AMF Call Control Service	amfcc_npcf_ue_policy_control_update_notify_request	Counter	Send UE policy control update notify Requests from PCF to AMF
Mobility Management layers statistics for AMF Call Control Service	amfcc_npcf_ue_policy_control_update_notify_response	Counter	Received UE policy control update notify Responses from AMF to PCF
Mobility Management layers statistics for AMF Call Control Service	amfcc_npcf_ue_policy_control_delete_request	Counter	Send UE policy control delete Requests from AMF to PCF
Mobility Management layers statistics for AMF Call Control Service	amfcc_npcf_ue_policy_control_delete_response	Counter	Received UE policy control delete Responses from PCF to AMF
Mobility Management layers statistics for AMF Call Control Service	amfcc_npcf_am_policy_control_create_request	Counter	Send AM policy control create requests from AMF to PCF
Mobility Management layers statistics for AMF Call Control Service	amfcc_npcf_am_policy_control_create_response	Counter	Received AM policy control create responses from PCF to AMF
Mobility Management layers statistics for AMF Call Control Service	amfcc_npcf_am_policy_control_update_request	Counter	Send AM policy control update requests from AMF to PCF
Mobility Management layers statistics for AMF Call Control Service	amfcc_npcf_am_policy_control_update_response	Counter	Received AM policy control update responses from PCF to AMF
Mobility Management layers statistics for AMF Call Control Service	amfcc_npcf_am_policy_control_delete_request	Counter	Send AM policy control delete requests from AMF to PCF
Mobility Management layers statistics for AMF Call Control Service	amfcc_npcf_am_policy_control_delete_response	Counter	Received AM policy control delete responses from PCF to AMF
Mobility Management layers statistics for AMF Call Control Service	amfcc_nsmsf_sms_activation_request	Counter	Send SMS Service Activation Requests from AMF to SMSF
Mobility Management layers statistics for AMF Call Control Service	amfcc_nsmsf_sms_activation_response	Counter	Send SMS Service Activation Responses from SMSF to AMF
Mobility Management layers statistics for AMF Call Control Service	amfcc_nsmsf_sms_deactivation_request	Counter	Send SMS Service Deactivation Requests from AMF to SMSF

Group	Metric Name	Type	Synopsis
Mobility Management layers statistics for AMF Call Control Service	amfcc_nsmsf_sms_deactivation_response	Counter	Send SMS Service Deactivation Responses from SMSF to AMF
Mobility Management layers statistics for AMF Call Control Service	amfcc_nsmsf_uplink_sms_request	Counter	Send Uplink SMS Requests from AMF to SMSF
Mobility Management layers statistics for AMF Call Control Service	amfcc_nsmsf_uplink_sms_response	Counter	Send Uplink SMS Responses from SMSF to AMF
Mobility Management layers statistics for AMF Call Control Service	amfcc_mm_procedure_duration	Histogram	AMFCC MM procedure execution time in seconds
Handover statistics for AMF Call Control Service	amfcc_xn_handover_attempt	Counter	Xn handover procedure attempts
Handover statistics for AMF Call Control Service	amfcc_xn_handover_success	Counter	Xn handover procedure success
Handover statistics for AMF Call Control Service	amfcc_xn_handover_failure	Counter	Xn handover procedure failures
Handover statistics for AMF Call Control Service	amfcc_n2_intra_handover_attempt	Counter	N2 intra-AMF handover procedure attempts
Handover statistics for AMF Call Control Service	amfcc_n2_intra_handover_success	Counter	N2 intra-AMF handover procedure success
Handover statistics for AMF Call Control Service	amfcc_n2_intra_handover_failure	Counter	N2 intra-AMF handover procedure failures
Handover statistics for AMF Call Control Service	amfcc_n2_intra_handover_cancel	Counter	N2 intra-AMF handover procedure cancellations
Handover statistics for AMF Call Control Service	amfcc_n2_inter_source_amf_handover_attempt	Counter	N2 inter-AMF handover procedure attempts on source AMF
Handover statistics for AMF Call Control Service	amfcc_n2_inter_source_amf_handover_success	Counter	N2 inter-AMF handover procedure success on source AMF
Handover statistics for AMF Call Control Service	amfcc_n2_inter_source_amf_handover_failure	Counter	N2 inter-AMF handover procedure failures on source AMF
Handover statistics for AMF Call Control Service	amfcc_n2_inter_target_amf_handover_attempt	Counter	N2 inter-AMF handover procedure attempts on target AMF
Handover statistics for AMF Call Control Service	amfcc_n2_inter_target_amf_handover_success	Counter	N2 inter-AMF handover procedure success on target AMF
Handover statistics for AMF Call Control Service	amfcc_n2_inter_target_amf_handover_failure	Counter	N2 inter-AMF handover procedure failures on target AMF
Handover statistics for AMF Call Control Service	amfcc_n2_inter_target_amf_handover_cancel	Counter	N2 inter-AMF handover procedure cancellations on target AMF
Handover statistics for AMF Call Control Service	amfcc_n26_inter_eps_source_amf_handover_attempt	Counter	N26 handover procedure attempts on source AMF
Handover statistics for AMF Call Control Service	amfcc_n26_inter_eps_source_amf_handover_success	Counter	N26 handover procedure successes on source AMF
Handover statistics for AMF Call Control Service	amfcc_n26_inter_eps_source_amf_handover_failure	Counter	N26 handover procedure failures on source AMF
Handover statistics for AMF Call Control Service	amfcc_n26_inter_eps_target_amf_handover_attempt	Counter	N26 handover procedure attempts on target AMF
Handover statistics for AMF Call Control Service	amfcc_n26_inter_eps_target_amf_handover_success	Counter	N26 handover procedure successes on target AMF
Handover statistics for AMF Call Control Service	amfcc_n26_inter_eps_target_amf_handover_failure	Counter	N26 handover procedure failures on target AMF
Handover statistics for AMF Call Control Service	amfcc_ho_procedure_duration	Histogram	AMFCC HO procedure execution time in seconds

Group	Metric Name	Type	Synopsis
gNB layer statistics for AMF Call Control Service	amfcc_gnb_init_modify_attempt	Counter	gNB initiated modify procedure attempts
gNB layer statistics for AMF Call Control Service	amfcc_gnb_init_modify_success	Counter	gNB initiated modify procedure success
gNB layer statistics for AMF Call Control Service	amfcc_gnb_init_modify_failure	Counter	gNB initiated modify procedure failures
gNB layer statistics for AMF Call Control Service	amfcc_n2_intra_ran_config_transfer_attempt	Counter	N2 intra-AMF RAN configuration transfer attempts
gNB layer statistics for AMF Call Control Service	amfcc_n2_intra_ran_config_transfer_success	Counter	N2 intra-AMF RAN configuration transfer success
gNB layer statistics for AMF Call Control Service	amfcc_n2_intra_ran_config_transfer_failure	Counter	N2 intra-AMF RAN configuration transfer failures
gNB layer statistics for AMF Call Control Service	amfcc_n2_inter_target_amf_ran_config_transfer_attempt	Counter	N2 inter-AMF RAN configuration transfer attempts on target AMF
gNB layer statistics for AMF Call Control Service	amfcc_n2_inter_target_amf_ran_config_transfer_success	Counter	N2 inter-AMF RAN configuration transfer success on target AMF
gNB layer statistics for AMF Call Control Service	amfcc_n2_inter_target_amf_ran_config_transfer_failure	Counter	N2 inter-AMF RAN configuration transfer failure on target AMF
gNB layer statistics for AMF Call Control Service	amfcc_n2_inter_source_amf_ran_config_transfer_attempt	Counter	N2 inter-AMF RAN configuration transfer attempts on source AMF
gNB layer statistics for AMF Call Control Service	amfcc_n2_inter_source_amf_ran_config_transfer_success	Counter	N2 inter-AMF RAN configuration transfer success on source AMF
gNB layer statistics for AMF Call Control Service	amfcc_n2_inter_source_amf_ran_config_transfer_failure	Counter	N2 inter-AMF RAN configuration transfer failures on source AMF
gNB layer statistics for AMF Call Control Service	amfcc_n2_inter_eps_target_amf_ran_config_transfer_attempt	Counter	N2 inter-EPS RAN configuration transfer attempts on target AMF
gNB layer statistics for AMF Call Control Service	amfcc_n2_inter_eps_target_amf_ran_config_transfer_success	Counter	N2 inter-EPS RAN configuration transfer success on target AMF
gNB layer statistics for AMF Call Control Service	amfcc_n2_inter_eps_target_amf_ran_config_transfer_failure	Counter	N2 inter-EPS RAN configuration transfer failure on target AMF
gNB layer statistics for AMF Call Control Service	amfcc_n2_inter_eps_source_amf_ran_config_transfer_attempt	Counter	N2 inter-EPS RAN configuration transfer attempt on source AMF
gNB layer statistics for AMF Call Control Service	amfcc_n2_inter_eps_source_amf_ran_config_transfer_success	Counter	N2 inter-EPS RAN configuration transfer success on source AMF
gNB layer statistics for AMF Call Control Service	amfcc_n2_inter_eps_source_amf_ran_config_transfer_failure	Counter	N2 inter-EPS RAN configuration transfer failure on source AMF
gNB layer statistics for AMF Call Control Service	amfcc_gnb_procedure_duration	Histogram	AMFCC GNB procedure execution time in seconds
Namf interface statistics for AMF Call Control Service	amfcc_client_namf_comm_n1n2_message_transfer_request	Counter	N1N2 message transfer requests sent by source AMF to target AMF during inter-AMF handover procedure as counted on the source side
Namf interface statistics for AMF Call Control Service	amfcc_client_namf_comm_n1n2_message_transfer_response	Counter	N1N2 message transfer responses received by source AMF from target AMF during inter-AMF handover procedure
Namf interface statistics for AMF Call Control Service	amfcc_namf_comm_n1n2_message_transfer_request	Counter	N1 N2 transfer requests received by AMF from NF service consumers
Namf interface statistics for AMF Call Control Service	amfcc_namf_comm_n1n2_message_transfer_response	Counter	N1 N2 transfer responses sent by AMF to NF service consumers
Namf interface statistics for AMF Call Control Service	amfcc_namf_comm_ebi_assignment_request	Counter	EBI assignment requests received by AMF from NF service consumers
Namf interface statistics for AMF Call Control Service	amfcc_namf_comm_ebi_assignment_response	Counter	EBI assignment responses sent by AMF to NF service consumers

Group	Metric Name	Type	Synopsis
Namf interface statistics for AMF Call Control Service	amfcc_client_namf_comm_ue_context_transfer_request	Counter	UE context transfer requests sent by target AMF to source AMF for idle mode inter-AMF mobility as counted on the target side
Namf interface statistics for AMF Call Control Service	amfcc_client_namf_comm_ue_context_transfer_response	Counter	UE context transfer responses sent by target AMF to source AMF for idle mode inter-AMF mobility
Namf interface statistics for AMF Call Control Service	amfcc_namf_comm_ue_context_transfer_request	Counter	UE context transfer requests received by source AMF for idle mode inter-AMF mobility as counted on the source side
Namf interface statistics for AMF Call Control Service	amfcc_namf_comm_ue_context_transfer_response	Counter	UE context transfer responses sent by source AMF to target AMF for idle mode inter-AMF mobility
Namf interface statistics for AMF Call Control Service	amfcc_client_namf_comm_ue_registration_status_update_request	Counter	UE registration status update requests sent by target AMF to source AMF for idle mode inter-AMF mobility as counted on the target side
Namf interface statistics for AMF Call Control Service	amfcc_client_namf_comm_ue_registration_status_update_response	Counter	UE registration status update responses sent by target AMF to source AMF for idle mode inter-AMF mobility
Namf interface statistics for AMF Call Control Service	amfcc_namf_comm_ue_registration_status_update_request	Counter	UE registration status update requests sent by source AMF to target AMF for idle mode inter-AMF mobility as counted on the source side
Namf interface statistics for AMF Call Control Service	amfcc_namf_comm_ue_registration_status_update_response	Counter	UE registration status update responses sent by source AMF to target AMF for idle mode inter-AMF mobility
Namf interface statistics for AMF Call Control Service	amfcc_client_namf_comm_create_ue_context_request	Counter	UE context creation requests sent by source AMF to target AMF during inter-AMF N2 handover procedure as counted on the source side
Namf interface statistics for AMF Call Control Service	amfcc_client_namf_comm_create_ue_context_response	Counter	UE context creation responses received by source AMF from target AMF during inter-AMF N2 handover procedure
Namf interface statistics for AMF Call Control Service	amfcc_namf_comm_create_ue_context_request	Counter	UE context creation requests received by target AMF from source AMF during inter-AMF N2 handover procedure as counted on the target side
Namf interface statistics for AMF Call Control Service	amfcc_namf_comm_create_ue_context_response	Counter	UE context creation responses sent by target AMF to source AMF during inter-AMF N2 handover procedure
Namf interface statistics for AMF Call Control Service	amfcc_client_namf_comm_release_ue_context_request	Counter	UE context release requests sent by source AMF to target AMF during inter-AMF N2 handover procedure as counted on the source side
Namf interface statistics for AMF Call Control Service	amfcc_client_namf_comm_release_ue_context_response	Counter	UE context release responses received by source AMF from target AMF during inter-AMF N2 handover procedure
Namf interface statistics for AMF Call Control Service	amfcc_namf_comm_release_ue_context_request	Counter	UE context release requests received by target AMF from source AMF during inter-AMF N2 handover procedure as counted on the target side
Namf interface statistics for AMF Call Control Service	amfcc_namf_comm_release_ue_context_response	Counter	UE context release responses sent by target AMF to source AMF during inter-AMF N2 handover procedure
Namf interface statistics for AMF Call Control Service	amfcc_client_namf_comm_n2_info_notification	Counter	N2 information notification requests sent by target AMF to source AMF during inter-AMF N2 handover procedure as counted on the target side

Group	Metric Name	Type	Synopsis
Namf interface statistics for AMF Call Control Service	amfcc_client_namf_comm_n2_info_notification_response	Counter	N2 information notification request acknowledgements received by target AMF from source AMF during inter-AMF N2 handover procedure
Namf interface statistics for AMF Call Control Service	amfcc_namf_comm_n2_info_notification	Counter	N2 information notification requests received by source AMF from target AMF during inter-AMF N2 handover procedure as counted on the source side
Namf interface statistics for AMF Call Control Service	amfcc_namf_comm_n2_info_notification_response	Counter	N2 information notification request acknowledgements sent by source AMF to target AMF during inter-AMF N2 handover procedure
Namf interface statistics for AMF Call Control Service	amfcc_client_namf_comm_relocate_ue_context_request	Counter	UE context relocate requests sent by source AMF to target AMF during N26 handover procedure as counted on the source side
Namf interface statistics for AMF Call Control Service	amfcc_client_namf_comm_relocate_ue_context_response	Counter	UE context relocate responses received by source AMF from target AMF during N26 handover procedure
Namf interface statistics for AMF Call Control Service	amfcc_namf_comm_relocate_ue_context_request	Counter	UE context relocate requests received by target AMF from source AMF during N26 handover procedure as counted on the target side
Namf interface statistics for AMF Call Control Service	amfcc_namf_comm_relocate_ue_context_response	Counter	UE context relocate responses sent by target AMF to source AMF during N26 handover procedure
Namf interface statistics for AMF Call Control Service	amfcc_client_namf_comm_cancel_relocate_ue_context_request	Counter	UE context cancel relocate requests sent by source AMF to target AMF during N26 handover procedure as counted on the source side
Namf interface statistics for AMF Call Control Service	amfcc_client_namf_comm_cancel_relocate_ue_context_response	Counter	UE context cancel relocate responses received by source AMF from target AMF during N26 handover procedure
Namf interface statistics for AMF Call Control Service	amfcc_namf_comm_cancel_relocate_ue_context_request	Counter	UE context cancel relocate requests received by target AMF from source AMF during N26 handover procedure as counted on the target side
Namf interface statistics for AMF Call Control Service	amfcc_namf_comm_cancel_relocate_ue_context_response	Counter	UE context cancel relocate responses sent by target AMF to source AMF during N26 handover procedure
Namf interface statistics for AMF Call Control Service	amfcc_namf_mt_enable_reachability_request	Counter	MT EnableReachability request received by AMF from NF service consumers
Namf interface statistics for AMF Call Control Service	amfcc_namf_mt_enable_reachability_response	Counter	MT EnableReachability responses sent by AMF to NF service consumers
Namf interface statistics for AMF Call Control Service	amfcc_namf_mt_provide_domain_selection_info_request	Counter	MT ProvideDomainSelectionInfo request received by AMF from NF service consumers
Namf interface statistics for AMF Call Control Service	amfcc_namf_mt_provide_domain_selection_info_response	Counter	MT ProvideDomainSelectionInfo responses sent by AMF to NF service consumers
Namf interface statistics for AMF Call Control Service	amfcc_namf_comm_n1_info_notification	Counter	AMF sends n1 information notification to LMF
Namf interface statistics for AMF Call Control Service	amfcc_namf_comm_n1_info_notification_response	Counter	LMF sends n1 information notification response to AMF
Namf interface statistics for AMF Call Control Service	amfcc_namf_location_event_notification	Counter	AMF notifies the NF about an UE location related event information
Namf interface statistics for AMF Call Control Service	amfcc_namf_location_event_notification_response	Counter	Event notification response sent by NF to AMF

Group	Metric Name	Type	Synopsis
Namf interface statistics for AMF Call Control Service	amfcc_namf_location_provide_postioning_info_request	Counter	Postioning info request invoked by an NF to AMF inorder obtain current location of UE
Namf interface statistics for AMF Call Control Service	amfcc_namf_location_provide_postioning_info_response	Counter	Postioning info response sent by an AMF to NF
Nlmf interface statistics for AMF Call Control Service	amfcc_nlmf_location_determine_location_request	Counter	Location determine request sent from AMF to LMF.
Nlmf interface statistics for AMF Call Control Service	amfcc_nlmf_location_determine_location_response	Counter	Location determine response sent from LMF to AMF
Location Services statistics for AMF Call Control Service	amfcc_lcs_network_induced_location_request_attempt	Counter	serving AMF for a UE initiates location of the UE for some regulatory service
Location Services statistics for AMF Call Control Service	amfcc_lcs_network_induced_location_request_success	Counter	network induced location request success
Location Services statistics for AMF Call Control Service	amfcc_lcs_network_induced_location_request_failure	Counter	network induced location request failed
Location Services statistics for AMF Call Control Service	amfcc_lcs_mobile_terminated_location_request_attempt	Counter	LCS client sends a location request to the PLMN for the location of a target UE
Location Services statistics for AMF Call Control Service	amfcc_lcs_mobile_terminated_location_request_success	Counter	mobile terminated location request success
Location Services statistics for AMF Call Control Service	amfcc_lcs_mobile_terminated_location_request_failure	Counter	mobile terminated location request failure
Event Data Records for AMF Call Control Service	amfcc_edr_generated	Counter	EDR generated and sent on AMF
Event Data Records for AMF Call Control Service	amfcc_edr_callback_rcvd	Counter	EDR callback received on AMF

Session Management layers statistics for AMF Call Control Service

- [amfcc_nsmf_pdusession_create_sm_context_request](#)

Type

64-bit Counter

Description

The number of create SM context requests sent by AMF. The request message body carries SMContextCreateData structure as defined in section 6.1.6.2.2 of 3GPP TS 29.502.

Labels

dnn, slice, peer

- [amfcc_nsmf_pdusession_create_sm_context_response](#)

Type

64-bit Counter

Description

The number of create SM context responses received by AMF. On success, response message body contains SMContextCreatedData structure as defined in section 6.1.6.2.3 of 3GPP TS 29.502. On failure, response message body contains SMContextCreateError structure as defined in 6.1.6.2.35 of 3GPP TS 29.502.

Labels

dnn, slice, peer, result

- [amfcc_nsmf_pdusession_update_sm_context_request](#)

Type

64-bit Counter

Description

The number of update SM context requests sent by AMF for pdu message transfer(payload container received from UE in uplink nas transport message), user plane activation and user plane deactivation. The request message body carries SMContextUpdateData structure as defined in section 6.1.6.2.4 of 3GPP TS 29.502.

Labels

dnn, slice, peer, msg_type

- amfcc_nsmf_pdusession_update_sm_context_response

Type

64-bit Counter

Description

The number of update SM context responses received by AMF for pdu message transfer(payload container received from UE in uplink nas transport message), user plane activation and user plane deactivation. On success, request message body contains SMContextUpdatedData structure as defined in section 6.1.6.2.4 of 3GPP TS 29.502. On failure, response message body contains SMContextUpdateError structure as defined in 6.1.6.2.36 of 3GPP TS 29.502.

Labels

dnn, slice, peer, msg_type, result

- amfcc_nsmf_pdusession_notify_sm_context_status_request_cb

Type

64-bit Counter

Description

The number of SM context status notification request received by AMF. The request message body carries StatusNotification structure as defined in section 6.1.6.2.17 of 3GPP TS 29.502.

Labels

dnn, slice, peer

- amfcc_nsmf_pdusession_notify_sm_context_status_response

Type

64-bit Counter

Description

The number of SM context status notification responses sent by AMF. On Success empty payload body is received in POST response. On failure, response message body contains ProblemDetails structure as defined in section 5.2.4.1 of 3GPP TS 29.571.

Labels

dnn, slice, peer, result

- amfcc_nsmf_pdusession_release_sm_context_request

Type

64-bit Counter

Description

The number of release SM context requests sent by AMF. The request message body carries SMContextReleaseData structure as defined in section 6.1.6.2.6 of 3GPP TS 29.502.

Labels

dnn, slice, peer

- amfcc_nsmf_pdusession_release_sm_context_response

Type

64-bit Counter

Description

The number of release SM context responses received by AMF. On Success empty payload body is received in POST response. On failure, response message body contains ProblemDetails structure as defined in section 5.2.4.1 of 3GPP TS 29.571.

Labels

dnn, slice, peer, result

• amfcc_nsmf_pdusession_retrieve_sm_context_request

Type

64-bit Counter

Description

The number of retrieve SM context requests sent by AMF. The request message body carries SMContextRetrieveData structure as defined in section 6.1.6.2.7 of 3GPP TS 29.502.

Labels

dnn, slice, peer

• amfcc_nsmf_pdusession_retrieve_sm_context_response

Type

64-bit Counter

Description

The number of retrieve SM context responses received by AMF. On success, request message body contains SMContextRetrievedData structure as defined in section 6.1.6.2.27 of 3GPP TS 29.502. On failure, response message body contains ProblemDetails structure as defined in section 5.2.4.1 of 3GPP TS 29.571.

Labels

dnn, slice, peer, result

• amfcc_sm_smf_selection_attempt

Type

64-bit Counter

Description

The number of PDU establishment requests received from UE for new PDU sessions.

Labels

dnn, slice

• amfcc_sm_smf_selection_success

Type

64-bit Counter

Description

The number of successful SMF selection procedures.

Labels

dnn, slice

• amfcc_sm_smf_selection_failure

Type

64-bit Counter

Description

The number of failed SMF selection procedures

Labels

dnn, slice, result

Mobility Management layers statistics for AMF Call Control Service

- amfcc_n1_init_registration_request

Type

64-bit Counter

Description

The number of UE initiated registration requests received by the AMF with a registration type of "initial registration". The REGISTRATION REQUEST message is defined in section 8.2.6 of 3GPP TS 24.501.

Labels

tai

- amfcc_n1_init_registration_accept

Type

64-bit Counter

Description

The number of registration accepts sent by AMF for UE initiated registration requests with a registration type of "initial registration". The REGISTRATION ACCEPT message is defined in section 8.2.7 of 3GPP TS 24.501.

Labels

tai

- amfcc_n1_init_registration_complete

Type

64-bit Counter

Description

The number of registration completes received by AMF for UE initiated registration requests with a registration type of "initial registration". The REGISTRATION COMPLETE message is defined in section 8.2.8 of 3GPP TS 24.501.

Labels

tai

- amfcc_n1_init_registration_reject

Type

64-bit Counter

Description

The number of registration rejects sent by AMF for UE initiated registration requests with a registration type of "initial registration". The REGISTRATION REJECT message is defined in section 8.2.9 of 3GPP TS 24.501.

Labels

tai, result

- amfcc_n1_mobility_registration_request

Type

64-bit Counter

Description

The number of UE initiated registration requests received by the AMF with a registration type of "mobility registration". The REGISTRATION REQUEST message is defined in section 8.2.6 of 3GPP TS 24.501.

Labels

tai

- amfcc_n1_mobility_registration_accept

Type

64-bit Counter

Description

The number of registration accepts sent by AMF for UE initiated registration requests with a registration type of "mobility registration". The REGISTRATION ACCEPT message is defined in section 8.2.7 of 3GPP TS 24.501.

Labels

tai

- amfcc_n1_mobility_registration_complete

Type

64-bit Counter

Description

The number of registration completes received by AMF for UE initiated registration requests with a registration type of "mobility registration". The REGISTRATION COMPLETE message is defined in section 8.2.8 of 3GPP TS 24.501.

Labels

tai

- amfcc_n1_mobility_registration_reject

Type

64-bit Counter

Description

The number of registration rejects sent by AMF for UE initiated registration requests with a registration type of "mobility registration". The REGISTRATION REJECT message is defined in section 8.2.9 of 3GPP TS 24.501.

Labels

tai, result

- amfcc_n1_periodic_registration_request

Type

64-bit Counter

Description

The number of UE initiated registration requests received by the AMF with a registration type of "periodic registration". The REGISTRATION REQUEST message is defined in section 8.2.6 of 3GPP TS 24.501.

Labels

tai

- amfcc_n1_periodic_registration_accept

Type

64-bit Counter

Description

The number of registration accepts sent by AMF for UE initiated registration requests with a registration type of "periodic registration". The REGISTRATION ACCEPT message is defined in section 8.2.7 of 3GPP TS 24.501.

Labels

tai

- amfcc_n1_periodic_registration_complete

Type

64-bit Counter

Description

The number of registration completes received by AMF for UE initiated registration requests with a registration type of "periodic registration". The REGISTRATION COMPLETE message is defined in section 8.2.8 of 3GPP TS 24.501.

Labels

tai

- amfcc_n1_periodic_registration_reject

Type

64-bit Counter

Description

The number of registration rejects sent by AMF for UE initiated registration requests with a registration type of "periodic registration". The REGISTRATION REJECT message is defined in section 8.2.9 of 3GPP TS 24.501.

Labels

tai, result

- amfcc_n1_auth_request

Type

64-bit Counter

Description

The number of authentication requests sent by AMF. The AUTHENTICATION REQUEST message is defined in section 8.2.1 of 3GPP TS 24.501.

Labels

tai

- amfcc_n1_auth_response

Type

64-bit Counter

Description

The number of authentication responses received by AMF. The AUTHENTICATION RESPONSE message is defined in section 8.2.2 of 3GPP TS 24.501.

Labels

tai

- amfcc_n1_auth_failure

Type

64-bit Counter

Description

The number of authentication failures received by AMF. The AUTHENTICATION FAILURE message is defined in section 8.2.4 of 3GPP TS 24.501.

Labels

tai, result

- amfcc_n1_auth_reject

Type

64-bit Counter

Description

The number of authentication rejects sent by AMF. The AUTHENTICATION REJECT message is defined in section 8.2.5 of 3GPP TS 24.501.

Labels

tai

- amfcc_n1_security_mode_command

Type

64-bit Counter

Description

The number of security mode commands sent by AMF. The SECURITY MODE COMMAND message is defined in section 8.2.25 of 3GPP TS 24.501.

Labels

tai

- amfcc_n1_security_mode_complete

Type

64-bit Counter

Description

The number of security mode completes received by AMF. The SECURITY MODE COMPLETE message is defined in section 8.2.26 of 3GPP TS 24.501.

Labels

tai

- amfcc_n1_security_mode_reject

Type

64-bit Counter

Description

The number of security mode rejects sent by AMF. The SECURITY MODE REJECT message is defined in section 8.2.27 of 3GPP TS 24.501.

Labels

tai, result

- amfcc_n1_identity_request

Type

64-bit Counter

Description

The number of identity requests sent by AMF. The IDENTITY REQUEST message is defined in section 8.2.21 of 3GPP TS 24.501.

Labels

tai

- amfcc_n1_identity_response

Type

64-bit Counter

Description

The number of identity responses received by AMF. The IDENTITY RESPONSE message is defined in section 8.2.22 of 3GPP TS 24.501.

Labels

tai

- amfcc_n1_uplink_nas_transport

Type

64-bit Counter

Description

The number of uplink NAS transports received by AMF for transferring the payload container from UE. The UL NAS TRANSPORT message is defined in section 8.2.10 of 3GPP TS 24.501. Payload types include: n1SmlInformation, sms, lppContainer, transparentContainer, uePolicyContainer, lcsContainer, mulPayloadContainer

Labels

tai, payload_type

- amfcc_n1_downlink_nas_transport

Type

64-bit Counter

Description

The number of downlink NAS transports sent by AMF for transferring the payload container to UE. The DL NAS TRANSPORT message is defined in section 8.2.11 of 3GPP TS 24.501.

Labels

tai, payload_type, result

- amfcc_n1_ue_deregistration_request

Type

64-bit Counter

Description

The number of UE initiated deregistration requests received by AMF. The DEREGISTRATION REQUEST message is defined in section 8.2.12 of 3GPP TS 24.501.

Labels

tai

- amfcc_n1_network_deregistration_accept

Type

64-bit Counter

Description

The number of deregistration accepts sent by AMF for UE initiated deregistration requests. The DEREGISTRATION ACCEPT message is defined in section 8.2.13 of 3GPP TS 24.501

Labels

tai

- amfcc_n1_network_deregistration_request

Type

64-bit Counter

Description

The number of network initiated deregistration requested sent by the AMF to the UE. The DEREGISTRATION REQUEST message is defined in section 8.2.14 of 3GPP TS 24.501.

Labels

tai

- amfcc_n1_ue_deregistration_accept

Type

64-bit Counter

Description

The number of deregistration accepts received by AMF from UE for network initiated deregistration requests. The DEREGISTRATION ACCEPT message is defined in section 8.2.15 of 3GPP TS 24.501

Labels

tai

- amfcc_n1_service_request

Type

64-bit Counter

Description

The number of service requests received by AMF. The SERVICE REQUEST message is defined in section 8.2.16 of 3GPP TS 24.501.

Labels

tai

- amfcc_n1_service_accept

Type

64-bit Counter

Description

The number of service accepts sent by AMF. The SERVICE ACCEPT message is defined in section 8.2.17 of 3GPP TS 24.501.

Labels

tai

- amfcc_n1_service_reject

Type

64-bit Counter

Description

The number of service rejects sent by AMF. The SERVICE REJECT message is defined in section 8.2.18 of 3GPP TS 24.501.

Labels

tai, result

- amfcc_n1_configuration_update_command

Type

64-bit Counter

Description

The number of configuration update commands sent by AMF. The CONFIGURATION UPDATE COMMAND message is defined in section 8.2.19 of 3GPP TS 24.501.

Labels

tai

- amfcc_n1_configuration_update_complete

Type

64-bit Counter

Description

The number of configuration update completes received by AMF. The CONFIGURATION UPDATE COMPLETE message is defined in section 8.2.20 of 3GPP TS 24.501.

Labels

tai

• amfcc_n1_network_mm_status

Type

64-bit Counter

Description

The number of 5GMM Status sent by AMF. The 5GMM status message is defined in section 8.2.29 of 3GPP TS 24.501.

Labels

tai, result

• amfcc_nausf_ueau_request

Type

64-bit Counter

Description

The number UE authentication requests sent by AMF. The request message body carries AuthenticationInfo structure as defined in section 6.1.6.2.2 of 3GPP TS 29.509.

Labels

peer

• amfcc_nausf_ueau_response

Type

64-bit Counter

Description

The number of UE authentication responses received by AMF. On success, response message body contains UEAuthenticationCtx structure as defined in section 6.1.6.2.3 of 3GPP TS 29.509. On failure, response message body contains ProblemDetails structure as defined in section 5.2.4.1 of 3GPP TS 29.571.

Labels

peer, result

• amfcc_nausf_ueau_confirm_request

Type

64-bit Counter

Description

The number of UE authentication confirmation requests sent by AMF. The message body carries ConfirmationData structure as defined in section 6.1.6.2.6 of 3GPP TS 29.509.

Labels

peer

• amfcc_nausf_ueau_confirm_response

Type

64-bit Counter

Description

The number of UE authentication confirmation responses received by AMF. On Success empty payload body is received in POST response. On failure, response message body may contain ProblemDetails structure as defined in section 5.2.4.1 of 3GPP TS 29.571.

Labels

peer, result

• amfcc_nausf_ueau_eap_session_request

Type

64-bit Counter

Description

The number of EAP session requests sent by AMF. The message body carries EapSessionData structure as defined in section 6.1.6.2.7 of 3GPP TS 29.509.

Labels

peer

- amfcc_nausf_ueau_eap_session_response

Type

64-bit Counter

Description

The number of EAP session responses received by AMF. On Success, KSeaf shall be present in POST response. On failure, authResult shall be set to AUTHENTICATION_FAILURE and the response body may contain ProblemDetails structure as defined in section 5.2.4.1 of 3GPP TS 29.571.

Labels

peer, result

- amfcc_nausf_rgau_request

Type

64-bit Counter

Description

The number of RG authentication requests sent by AMF. The request message body carries RgAuthenticationInfo structure as defined in section 6.1.6.2.9 of 3GPP TS 29.509.

Labels

peer

- amfcc_nausf_rgau_response

Type

64-bit Counter

Description

The number of RG authentication responses received by AMF. On success, response message body contains RgAuthCtx structure as defined in section 6.1.6.2.10 of 3GPP TS 29.509. On failure, response message body contains ProblemDetails structure as defined in section 5.2.4.1 of 3GPP TS 29.571.

Labels

peer, result

- amfcc_nudm_uecm_registration_3gpp_request

Type

64-bit Counter

Description

The number of UE context management requests sent by AMF for 3GPP registration. The request message body carries Amf3GppAccessRegistration structure as defined in section 6.1.6.2.2 of 3GPP TS 29.503.

Labels

peer

- amfcc_nudm_uecm_registration_3gpp_response

Type

64-bit Counter

Description

The number of UE context management responses received by AMF for 3GPP registration. On Success empty payload body is received in POST response. On failure, response message body may contain ProblemDetails structure as defined in section 5.2.4.1 of 3GPP TS 29.571.

Labels

peer, result

- amfcc_nudm_uecm_registration_update_3gpp_request

Type

64-bit Counter

Description

The number of UE context management requests sent by AMF for 3GPP registration update. The request message body carries Amf3GppAccessRegistrationModification structure as defined in section 6.2.6.2.7 of 3GPP TS 29.503.

Labels

peer, trigger

- amfcc_nudm_uecm_registration_update_3gpp_response

Type

64-bit Counter

Description

The number of UE context management responses received by AMF for 3GPP registration update. On Success empty payload body is received in PATCH response. On failure, response message body may contain ProblemDetails structure as defined in section 5.2.4.1 of 3GPP TS 29.571.

Labels

peer, trigger, result

- amfcc_nudm_uecm_deregistration_notification

Type

64-bit Counter

Description

The number of UE context management messages sent by UDM to old AMF for 3GPP de-registration after new AMF is registered for 3GPP access. The notification message body carries DeregistrationData, as defined in section 6.2.6.2.5 of 3GPP TS 29.503.

Labels

type

- amfcc_nudm_sdm_get_request

Type

64-bit Counter

Description

The number of subscriber data management requests sent by AMF. The request message can request for multiple data sets like am-data, nssai, smf-select-data etc as defined in section 6.1.3.1 of 3GPP TS 29.503.

Labels

peer

- amfcc_nudm_sdm_get_response

Type

64-bit Counter

Description

The number of subscriber data management responses received by AMF. The response message can contain multiple data structures based on the requested data sets in subscriber data management request sent to UDM as defined in section 6.1.6.2 of 3GPP TS 29.503.

Labels

peer, result

- amfcc_nudm_sdm_get_nssai_request

Type

64-bit Counter

Description

The number of subscriber data management requests for NSSAI sent by AMF. The request message contains a request to retrieve the UE's subscribed NSSAI as defined in section 6.1.3.2 of 3GPP TS 29.503.

Labels

peer

- amfcc_nudm_sdm_get_nssai_response

Type

64-bit Counter

Description

The number of subscriber data management responses for NSSAI received by AMF. On success, response message body contains NSSAI structure as defined in section 6.1.6.2.2 of 3GPP TS 29.503.

Labels

peer, result

- amfcc_nudm_sdm_subscription_request

Type

64-bit Counter

Description

The number of subscriber data management subscription requests sent by AMF. The request message can request for data change notification for multiple resources like am-data, nssai, smf-select-data etc as defined in SdmSubscription structure in section 6.1.3.3 of 3GPP TS 29.503.

Labels

peer

- amfcc_nudm_sdm_subscription_response

Type

64-bit Counter

Description

The number of subscriber data management subscription responses received by AMF. On success, response message body contains the resources subscribed on UDM for change notification as defined in SdmSubscription structure in section 6.1.6.2.3 of 3GPP TS 29.503.

Labels

peer, result

- amfcc_nudm_sdm_change_notification_request

Type

64-bit Counter

Description

The number of change notification requests for subscriber data management resources received by AMF. The request message contains the resourceId and multiple changeItems as defined in ModificationNotification structure in section 6.1.5.2 of 3GPP TS 29.503.

Labels

peer

- amfcc_nudm_sdm_change_notification_response

Type

64-bit Counter

Description

The number of change notification requests for subscriber data management resources sent by AMF. On success, response message is sent by AMF with no content as defined in section 6.1.5.2 of 3GPP TS 29.503.

Labels

peer, result

- amfcc_nudm_sdm_info_request

Type

64-bit Counter

Description

The number of subscriber data management info requests sent by AMF. The info message contains acknowledgement from UE as defined in AcknowledgeInfo structure in section 6.1.6.25 of 3GPP TS 29.503.

Labels

peer

- amfcc_nudm_sdm_info_response

Type

64-bit Counter

Description

The number of subscriber data management info responses received by AMF. On success, response message is sent by UDM with no content as defined in section 6.1.3.21 of 3GPP TS 29.503.

Labels

peer, result

- amfcc_nnsf_nsselection_request_for_registration

Type

64-bit Counter

Description

The number of network slice selection requests sent by AMF for registration procedure. The GET request URI is defined in section 6.1.3.1 of 3GPP TS 29.531.

Labels

slice, peer

- amfcc_nnsf_nsselection_response_for_registration

Type

64-bit Counter

Description

The number of network slice selection responses received by AMF for registration procedure. On success, response message body contains AuthorizedNetworkSliceInfo structure as defined in section 6.1.6.2.2 of 3GPP TS 29.531. On failure, response message body contains ProblemDetails structure as defined in section 5.2.4.1 of 3GPP TS 29.571.

Labels

slice, peer, result

- amfcc_nnssf_nsselection_request_for_pdu_session_establishment

Type

64-bit Counter

Description

The number of network slice selection requests sent by AMF for PDU session establishment. The GET request URI is defined in section 6.1.3.1 of 3GPP TS 29.531.

Labels

slice, peer

- amfcc_nnssf_nsselection_response_for_pdu_session_establishment

Type

64-bit Counter

Description

The number of network slice selection responses received by AMF for PDU session establishment. On success, response message body contains AuthorizedNetworkSliceInfo structure as defined in section 6.1.6.2.2 of 3GPP TS 29.531. On failure, response message body contains ProblemDetails structure as defined in section 5.2.4.1 of 3GPP TS 29.571.

Labels

slice, peer, result

- amfcc_nnssf_nsselection_request_for_ue_configuration_update

Type

64-bit Counter

Description

The number of network slice selection requests sent by AMF for UE configuration update procedure. The GET request URI is defined in section 6.1.3.1 of 3GPP TS 29.531.

Labels

slice, peer

- amfcc_nnssf_nsselection_response_for_ue_configuration_update

Type

64-bit Counter

Description

The number of network slice selection responses received by AMF for UE configuration update procedure. On success, response message body contains AuthorizedNetworkSliceInfo structure as defined in section 6.1.6.2.2 of 3GPP TS 29.531. On failure, response message body contains ProblemDetails structure as defined in section 5.2.4.1 of 3GPP TS 29.571.

Labels

slice, peer, result

- amfcc_mm_initial_registration_attempt

Type

64-bit Counter

Description

The number of attempted registration requests with registration type "initial registration". The counter is pegged on receiving registration request from UE with registration type "initial registration".

Labels

tai, ue_service

- amfcc_mm_initial_registration_success

Type

64-bit Counter

Description

The number of successful registration requests with registration type "initial registration". The counter is pegged on successfully completing the registration procedure and moving the UE to registered state.

Labels

tai, ue_service

- amfcc_mm_inter_new_amf_initial_registration_attempt

Type

64-bit Counter

Description

The number of attempted inter-AMF registration requests with registration type "initial registration". The counter is pegged on receiving registration request from UE with registration type "initial registration" for a GUTI unknown on this AMF.

Labels

old_amf_guami, tai

- amfcc_mm_inter_new_amf_initial_registration_success

Type

64-bit Counter

Description

The number of successful inter-AMF registration requests with registration type "initial registration". The counter is pegged on successfully completing the registration procedure and moving the UE to registered state.

Labels

old_amf_guami, tai

- amfcc_mm_inter_new_amf_initial_registration_failure

Type

64-bit Counter

Description

The number of failed inter-AMF registration procedures with registration type "initial registration". The counter is pegged if the registration procedure is aborted/rejected.

Labels

old_amf_guami, tai, result, source, reason

- amfcc_mm_initial_registration_failure

Type

64-bit Counter

Description

The number of failed registration procedures with registration type "initial registration". The counter is pegged if the registration procedure is aborted/rejected.

Labels

tai, ue_service, result, source, reason

- amfcc_mm_intra_mobility_registration_attempt

Type

64-bit Counter

Description

The number of attempted registration requests with registration type "mobility registration". The counter is pegged on receiving registration request from UE with registration type "initial registration".

Labels

tai, ue_service

- amfcc_mm_intra_mobility_registration_success

Type

64-bit Counter

Description

The number of successful registration requests with registration type "mobility registration". The counter is pegged on successfully completing the registration procedure and moving the UE to registered state.

Labels

tai, ue_service

- amfcc_mm_intra_mobility_registration_failure

Type

64-bit Counter

Description

The number of failed registration procedures with registration type "mobility registration". The counter is pegged if the registration procedure is aborted/rejected.

Labels

tai, ue_service, result, source, reason

- amfcc_mm_inter_new_amf_mobility_registration_attempt

Type

64-bit Counter

Description

The number of attempted inter-AMF registration requests with registration type "mobility registration". The counter is pegged on receiving registration request from UE with registration type "mobility registration" for a GUTI unknown on this AMF.

Labels

old_amf_guami, tai, ue_service

- amfcc_mm_inter_new_amf_mobility_registration_success

Type

64-bit Counter

Description

The number of successful inter-AMF registration requests with registration type "mobility registration". The counter is pegged on successfully completing the registration procedure and moving the UE to registered state.

Labels

old_amf_guami, tai, ue_service

- amfcc_mm_inter_new_amf_mobility_registration_failure

Type

64-bit Counter

Description

The number of failed inter-AMF registration procedures with registration type "mobility registration". The counter is pegged if the registration procedure is aborted/rejected.

Labels

old_amf_guami, tai, ue_service, result, source, reason

• amfcc_mm_periodic_registration_attempt

Type

64-bit Counter

Description

The number of attempted registration requests with registration type "periodic registration". The counter is pegged on receiving registration request from UE with registration type "initial registration".

Labels

tai

• amfcc_mm_periodic_registration_success

Type

64-bit Counter

Description

The number of successful registration requests with registration type "periodic registration". The counter is pegged on successfully completing the registration procedure and moving the UE to registered state.

Labels

tai

• amfcc_mm_periodic_registration_failure

Type

64-bit Counter

Description

The number of failed registration procedures with registration type "periodic registration". The counter is pegged if the registration procedure is aborted/rejected.

Labels

tai, result, source, reason

• amfcc_mm_ue_deregistration_attempt

Type

64-bit Counter

Description

The number of attempted deregistration procedures. The counter is pegged when a deregistration procedure is triggered to move the UE state to deregistered.

Labels

tai, type

• amfcc_mm_ue_deregistration_success

Type

64-bit Counter

Description

The number of successful deregistration procedures. The counter is pegged on successfully completing the deregistration procedure and moving the UE to deregistered state.

Labels

tai, type

• amfcc_mm_ue_deregistration_failure

Type

64-bit Counter

Description

The number of failed deregistration procedures. The counter is pegged if the deregistration procedure is aborted.

Labels

tai, type, source, reason

• amfcc_mm_nwk_deregistration_attempt

Type

64-bit Counter

Description

The number of attempted deregistration procedures initiated by the network. This counter is pegged when a deregistration procedure is triggered by the network to move the UE state to deregistered.

Labels

tai, type

• amfcc_mm_auth_attempt

Type

64-bit Counter

Description

The number of attempted authentication procedures. The counter is pegged on sending out an AUSF discovery request to NRF.

Labels

tai

• amfcc_mm_auth_success

Type

64-bit Counter

Description

The number of successful authentication procedures. The counter is pegged on receiving authentication confirmation response from AUSF.

Labels

tai

• amfcc_mm_auth_failure

Type

64-bit Counter

Description

The number of failed authentication procedures. The counter is pegged if there is any failure in discovery or sending/receiving authentication message to AUSF/UE.

Labels

tai, result

• amfcc_mm_service_request_attempt

Type

64-bit Counter

Description

The number of attempted service request procedures. The counter is pegged on receiving service request message from UE.

Labels

tai, ue_service

- amfcc_mm_service_request_success

Type

64-bit Counter

Description

The number of successful service request procedures. The counter is pegged on receiving gNB response if there are no session or after receiving update SM context response from SMF.

Labels

tai, ue_service

- amfcc_mm_service_request_failure

Type

64-bit Counter

Description

The number of failed service request procedures. The counter is pegged if there are any failures in processing service request procedure.

Labels

tai, ue_service, result, source, reason

- amfcc_mm_inter_old_amf_initial_registration_attempt

Type

64-bit Counter

Description

The number of attempted inter-AMF registration procedures on the source AMF with reason initial registration. The counter is pegged on receiving UE context transfer requests from the target AMF.

Labels

-
- amfcc_mm_inter_old_amf_initial_registration_success

Type

64-bit Counter

Description

The number of successful inter-AMF registration procedures on the source AMF with reason initial registration. The counter is pegged on successfully completing the UE context transfer procedure on the source AMF and releasing the UE context.

Labels

-
- amfcc_mm_inter_old_amf_initial_registration_failure

Type

64-bit Counter

Description

The number of failed inter-AMF registration procedures on the source AMF with reason initial registration. The counter is pegged if the UE context transfer procedure is aborted/rejected.

Labels

result, source, reason

• amfcc_mm_inter_old_amf_mobility_registration_attempt Type 64-bit Counter Description The number of attempted inter-AMF registration procedures on the source AMF with reason mobility registration. The counter is pegged on receiving UE context transfer requests from the target AMF. Labels
• amfcc_mm_inter_old_amf_mobility_registration_success Type 64-bit Counter Description The number of successful inter-AMF registration procedures on the source AMF with reason mobility registration. The counter is pegged on successfully completing the UE context transfer procedure on the source AMF and releasing the UE context. Labels
• amfcc_mm_inter_old_amf_mobility_registration_failure Type 64-bit Counter Description The number of failed inter-AMF registration procedures on the source AMF with reason mobility registration. The counter is pegged if the UE context transfer procedure aborted/rejected. Labels result, source, reason
• amfcc_mm_n26_inter_eps_new_amf_mobility_registration_attempt Type 64-bit Counter Description The number of attempted EPS to 5GS registration requests with N26 support, registration type "mobility registration" and the UE status indicates "EMM-registered". The counter is pegged on the new AMF on receiving registration request from UE with registration type "mobility registration" and the UE status indicates "EMM-registered", for a GUTI unknown on this AMF. Labels peer, tai, ue_service
• amfcc_mm_n26_inter_eps_new_amf_mobility_registration_success Type 64-bit Counter Description The number of successful EPS to 5GS registration requests with N26 support, registration type "mobility registration" and the UE status indicates "EMM-registered". The counter is pegged on the new AMF on successfully completing the registration procedure and moving the UE to registered state. Labels peer, tai, ue_service
• amfcc_mm_n26_inter_eps_new_amf_mobility_registration_failure Type

64-bit Counter

Description

The number of failed EPS to 5GS registration procedures with N26 support, registration type "mobility registration" and the UE status indicates "EMM-registered". The counter is pegged on the new AMF if the registration procedure is aborted/rejected.

Labels

peer, tai, ue_service, result

- amfcc_mm_inter_eps_new_amf_initial_registration_attempt

Type

64-bit Counter

Description

The number of attempted EPS to 5GS registration requests without N26 support, registration type "initial registration" and the UE status indicates "EMM-registered". The counter is pegged on the new AMF on receiving registration request from UE with registration type "initial registration" and the UE status indicates "EMM-registered".

Labels

peer, tai

- amfcc_mm_inter_eps_new_amf_initial_registration_success

Type

64-bit Counter

Description

The number of successful EPS to 5GS registration requests without N26 support, registration type "initial registration" and the UE status indicates "EMM-registered". The counter is pegged on the new AMF on successfully completing the registration procedure and moving the UE to registered state.

Labels

peer, tai

- amfcc_mm_inter_eps_new_amf_initial_registration_failure

Type

64-bit Counter

Description

The number of failed EPS to 5GS registration procedures without N26 support, registration type "initial registration" and the UE status indicates "EMM-registered". The counter is pegged on the new AMF if the registration procedure is aborted/rejected.

Labels

peer, tai, result

- amfcc_mm_inter_eps_new_amf_mobility_registration_attempt

Type

64-bit Counter

Description

The number of attempted EPS to 5GS registration requests without N26 support, registration type "mobility registration" and the UE status indicates "EMM-registered". The counter is pegged on the new AMF on receiving registration request from UE with registration type "mobility registration" and the UE status indicates "EMM-registered", for a GUTI unknown on this AMF.

Labels

peer, tai, ue_service

- amfcc_mm_inter_eps_new_amf_mobility_registration_success

Type

64-bit Counter

Description

The number of successful EPS to 5GS registration requests without N26 support, registration type "mobility registration" and the UE status indicates "EMM-registered". The counter is pegged on the new AMF on successfully completing the registration procedure and moving the UE to registered state.

Labels

peer, tai, ue_service

- amfcc_mm_inter_eps_new_amf_mobility_registration_failure

Type

64-bit Counter

Description

The number of failed EPS to 5GS registration procedures without N26 support, registration type "mobility registration" and the UE status indicates "EMM-registered". The counter is pegged on the new AMF if the registration procedure is aborted/rejected.

Labels

peer, tai, result, ue_service

- amfcc_mm_inter_eps_old_amf_mobility_registration_attempt

Type

64-bit Counter

Description

The number of attempted 5GS to EPS idle mode mobility registration on old AMF. The counter is pegged on the old AMF on receiving context request message from EPS.

Labels

peer

- amfcc_mm_inter_eps_old_amf_mobility_registration_success

Type

64-bit Counter

Description

The number of successful 5GS to EPS idle mode mobility registration on old AMF. The counter is pegged on successfully completing the context transfer to EPS and releasing the UE context

Labels

peer

- amfcc_mm_inter_eps_old_amf_mobility_registration_failure

Type

64-bit Counter

Description

The number of failed 5GS to EPS idle mode mobility registration on old AMF. The counter is pegged on the old AMF if the context transfer procedure is aborted/rejected.

Labels

peer, result

- amfcc_mm_paging_attempt

Type

64-bit Counter

Description

The number of attempted MM paging procedures. The counter is pegged when paging is sent successfully as result of UDM dereg or the request from SMF.

Labels

initiator, paging_profile_set, paging_profile

- amfcc_mm_paging_first_retry_attempt

Type

64-bit Counter

Description

The number of MM paging attempt first retry. The counter is pegged when paging attempt fails to get response and first retry starts.

Labels

initiator, paging_profile_set, paging_profile

- amfcc_mm_paging_second_retry_attempt

Type

64-bit Counter

Description

The number of MM paging attempt second retry. The counter is pegged when first paging retry fails to get response and second retry starts.

Labels

initiator, paging_profile_set, paging_profile

- amfcc_mm_paging_third_retry_attempt

Type

64-bit Counter

Description

The number of MM paging attempt third retry. The counter is pegged when second paging retry fails to get response and third retry starts.

Labels

initiator, paging_profile_set, paging_profile

- amfcc_mm_paging_failure

Type

64-bit Counter

Description

The number of failed MM paging procedure. The counter is pegged if all paging retries fail to get response.

Labels

initiator, paging_profile_set, paging_profile, source

- amfcc_npcf_ue_policy_control_create_request

Type

64-bit Counter

Description

The number of UE policy control create request sent by AMF to PCF. The request message contains the UE policy association data structure as specified in section 5.6.2.2 of 3GPP spec 29.525.

Labels

peer

- amfcc_npcf_ue_policy_control_create_response

Type

64-bit Counter

Description

The number of UE policy control create response received by AMF from PCF. On success response PCF sends associated UE Policy details as requested.

Labels

peer, result

- amfcc_npcf_ue_policy_control_update_request

Type

64-bit Counter

Description

The number of UE policy control update request sent by AMF to PCF. The request message contains the UE policy association update request data structure as specified in section 5.6.2.4 of 3GPP 29.525

Labels

peer

- amfcc_npcf_ue_policy_control_update_response

Type

64-bit Counter

Description

The number of UE policy control update response received by AMF from PCF. On successful response AMF receives the updated UE control policy details from PCF

Labels

peer, result

- amfcc_npcf_ue_policy_control_update_notify_request

Type

64-bit Counter

Description

The number of UE policy control update notify request sent by PCF to AMF. The request contains update or terminate of UE policy as per PolicyUpdate or terminationNotification specified in section 5.6 of 3GPP 29.525

Labels

peer, procedure

- amfcc_npcf_ue_policy_control_update_notify_response

Type

64-bit Counter

Description

The number of UE policy control update notify response received by PCF from AMF. On successful response, AMF is notified about the update or terminate for the requested UE policy

Labels

peer, procedure, result

- amfcc_npcf_ue_policy_control_delete_request

Type

64-bit Counter

Description

The number of UE policy control delete request sent by AMF to PCF. The request message contains the UE policy details to delete the policy association with PCF.

Labels

peer

- amfcc_npcf_ue_policy_control_delete_response

Type

64-bit Counter

Description

The number of UE policy control delete response received by AMF from PCF. On successful response PCF informs AMF about the removal of request UE policy association

Labels

peer, result

- amfcc_npcf_am_policy_control_create_request

Type

64-bit Counter

Description

The number of AM policy control create requests sent by AMF to PCF. The request message contains AM policy association data structure as specified in section 5.6.2.2 of 3GPP spec 29.507.

Labels

peer

- amfcc_npcf_am_policy_control_create_response

Type

64-bit Counter

Description

The number of AM policy control create responses received by AMF from PCF. On success response PCF sends associated AM Policy details as requested.

Labels

peer, result

- amfcc_npcf_am_policy_control_update_request

Type

64-bit Counter

Description

The number of AM policy control update requests sent by AMF to PCF. The request message contains the AM policy association update requests data structure as specified in section 5.6.2.5 of 3GPP spec 29.507.

Labels

peer

- amfcc_npcf_am_policy_control_update_response

Type

64-bit Counter

Description

The number of AM policy control update responses received by AMF from PCF. On successful response AMF receives the updated AM control policy details from PCF

Labels

peer, result

- amfcc_npcf_am_policy_control_delete_request

Type

64-bit Counter

Description

The number of AM policy control delete requests sent by AMF to PCF. The request message contains the AM policy details to delete the policy association with PCF.

Labels

peer

- amfcc_npcf_am_policy_control_delete_response

Type

64-bit Counter

Description

The number of AM policy control delete responses received by AMF from PCF. On successful response PCF informs AMF about the removal of request AM policy association

Labels

peer, result

- amfcc_nsmsf_sms_activation_request

Type

64-bit Counter

Description

The number of SMS activation requests sent by AMF to SMSF. The request contains information used for activating SMS service for a service user, or updating the parameters for SMS service specified in section 6.1.6.2.2 of 3GPP 29.540

Labels

peer

- amfcc_nsmsf_sms_activation_response

Type

64-bit Counter

Description

The number of SMS activation response received by AMF from SMSF. On successful response SMSF informs AMF about UE context as created or no content if already created

Labels

peer, result

- amfcc_nsmsf_sms_deactivation_request

Type

64-bit Counter

Description

The number of SMS deactivation request sent by AMF to SMSF. The request contains SUPI details as specified in section 6.1 of 3GPP 29.540

Labels

peer

amfcc_nsmsf_sms_deactivation_response
Type
64-bit Counter
Description
The number of SMS deactivation response received by AMF from SMSF. On successfull response SMSF sends deactivation responses to AMF
Labels
peer, result
amfcc_nsmsf_uplink_sms_request
Type
64-bit Counter
Description
The number of uplink SMS request sent by AMF to SMSF. The request contains information invoking UplinkSMS service operation, for delivering SMS payload specified in section 6.1.6.2.3 of 3GPP 29.540
Labels
peer
amfcc_nsmsf_uplink_sms_response
Type
64-bit Counter
Description
The number of uplink SMS response received by AMF from SMSF. Successfull response contains information for invoking UplinkSMS service operation specified in section 6.1.6.2.5 of 3GPP 29.540
Labels
peer, result
amfcc_mm_procedure_duration
Type
64-bit Histogram
Description
The time observed for AMFCC MM procedure execution in seconds
Labels
procedure, status
Boundaries
[0.5, 1, 1.5, 2, 3, 5, 10, 20]

Handover statistics for AMF Call Control Service

amfcc_xn_handover_attempt
Type
64-bit Counter
Description
The number of attempted Xn handover procedures. The counter is pegged on receiving Path Switch Request message from gNB.
Labels
tai, ue_service

- amfcc_xn_handover_success

Type

64-bit Counter

Description

The number of successful Xn handover procedures. The counter is pegged on sending Path Switch Acknowledgement to target gNB.

Labels

tai, ue_service

- amfcc_xn_handover_failure

Type

64-bit Counter

Description

The number of failed Xn handover procedures. The counter is pegged on sending Path Switch Request Failure message to gNB.

Labels

tai, ue_service, result, source, reason

- amfcc_n2_intra_handover_attempt

Type

64-bit Counter

Description

The number of attempted N2 intra-AMF handover procedures. The counter is pegged on receiving Handover Required from gNB.

Labels

tai, ue_service

- amfcc_n2_intra_handover_success

Type

64-bit Counter

Description

The number of successful N2 intra-AMF handover procedures. The counter is pegged on completing the handover procedure.

Labels

tai, ue_service

- amfcc_n2_intra_handover_failure

Type

64-bit Counter

Description

The number of failed N2 intra-AMF handover procedures. The counter is pegged if there are any failures processing N2 intra-AMF handover procedure.

Labels

tai, ue_service, result, source, reason

- amfcc_n2_intra_handover_cancel

Type

64-bit Counter

Description

The number of cancelled N2 intra-AMF handover procedures. The counter is pegged on cancelling the handover procedure.

Labels

tai

- amfcc_n2_inter_source_amf_handover_attempt

Type

64-bit Counter

Description

The number of attempted N2 inter-AMF handover procedures on the source AMF. The counter is pegged on receiving Handover Required from gNB.

Labels

peer

- amfcc_n2_inter_source_amf_handover_success

Type

64-bit Counter

Description

The number of successful N2 inter-AMF handover procedures on the source AMF. The counter is pegged on completing the handover procedure.

Labels

peer

- amfcc_n2_inter_source_amf_handover_failure

Type

64-bit Counter

Description

The number of failed N2 inter-AMF handover procedures on the source AMF. The counter is pegged if there are any failures processing N2 inter-AMF handover procedure.

Labels

peer, result, source, reason

- amfcc_n2_inter_target_amf_handover_attempt

Type

64-bit Counter

Description

The number of attempted N2 inter-AMF handover procedures on the target AMF. The counter is pegged on receiving Create UE Context Request from source AMF.

Labels

peer, ue_service

- amfcc_n2_inter_target_amf_handover_success

Type

64-bit Counter

Description

The number of successful N2 inter-AMF handover procedures on the target AMF. The counter is pegged on completing the handover procedure.

Labels

peer, ue_service

- amfcc_n2_inter_target_amf_handover_failure

Type

64-bit Counter

Description

The number of failed N2 inter-AMF handover procedures on the target AMF. The counter is pegged if there are any failures processing N2 inter-AMF handover procedure.

Labels

peer, ue_service, result, source, reason

- amfcc_n2_inter_target_amf_handover_cancel

Type

64-bit Counter

Description

The number of cancelled N2 inter-AMF handover procedures on the target AMF. The counter is pegged on cancelling the handover procedure.

Labels

peer

- amfcc_n26_inter_eps_source_amf_handover_attempt

Type

64-bit Counter

Description

The number of attempted N26 inter-EPS handover procedures on the source AMF. The counter is pegged on receiving the Handover Required with the Handover Type of 5GStoEPS, after the peer is identified.

Labels

peer

- amfcc_n26_inter_eps_source_amf_handover_success

Type

64-bit Counter

Description

The number of successful N26 handover procedures on the source AMF. The counter is pegged on completing the handover procedure.

Labels

peer

- amfcc_n26_inter_eps_source_amf_handover_failure

Type

64-bit Counter

Description

The number of failed N26 inter-EPS handover procedures on the source AMF. The counter is pegged if there are any failures processing N26 handover procedure.

Labels

peer, result

- amfcc_n26_inter_eps_target_amf_handover_attempt

Type

64-bit Counter

Description

The number of attempted N26 handover procedures on the target AMF. The counter is pegged on receiving Forward relocation request from EPS

Labels

peer, ue_service

- amfcc_n26_inter_eps_target_amf_handover_success

Type

64-bit Counter

Description

The number of successful N26 handover procedures on the target AMF. The counter is pegged on completing the handover procedure.

Labels

peer, ue_service

- amfcc_n26_inter_eps_target_amf_handover_failure

Type

64-bit Counter

Description

The number of failed N26 handover procedures on the target AMF. The counter is pegged if there are any failures processing N26 handover procedure.

Labels

peer, ue_service, result

- amfcc_ho_procedure_duration

Type

64-bit Histogram

Description

The time observed for AMFCC HO procedure execution in seconds

Labels

procedure,status

Boundaries

[0.5, 1, 1.5, 2, 3, 5, 10, 20]

gNB layer statistics for AMF Call Control Service

- amfcc_gnb_init_modify_attempt

Type

64-bit Counter

Description

The number of attempted gNB initiated modify procedures. The counter is pegged on receiving PDU session resource modify indication from gNB.

Labels

tai

- amfcc_gnb_init_modify_success

Type

64-bit Counter

Description

The number of successful gNB initiated modify procedures. The counter is pegged on sending PDU session resource modify confirm gNB.

Labels

tai

- amfcc_gnb_init_modify_failure

Type

64-bit Counter

Description

The number of failed gNB initiated modify procedures. The counter is pegged when PDU session resource modify indication processing fails for all PDU sessions.

Labels

tai, result

- amfcc_n2_intra_ran_config_transfer_attempt

Type

64-bit Counter

Description

The number of attempted intra-AMF RAN configuration transfer procedures. The counter is pegged on receiving uplink RAN configuration transfer message from gNB.

Labels

- amfcc_n2_intra_ran_config_transfer_success

Type

64-bit Counter

Description

The number of successful intra-AMF RAN configuration transfer procedures. The counter is pegged on completing the intra-AMF RAN configuration transfer procedure.

Labels

- amfcc_n2_intra_ran_config_transfer_failure

Type

64-bit Counter

Description

The number of failed intra-AMF RAN configuration transfer procedures. The counter is pegged if there are any failures processing the intra-AMF RAN configuration transfer procedure.

Labels

result

- amfcc_n2_inter_target_amf_ran_config_transfer_attempt

Type

64-bit Counter

Description

The number of attempted inter-AMF RAN configuration transfer procedures on target AMF. The counter is pegged on receiving inter-AMF RAN configuration transfer message from source AMF.

Labels

• amfcc_n2_inter_target_amf_ran_config_transfer_success Type 64-bit Counter Description The number of successful inter-AMF RAN configuration transfer procedures on target AMF. The counter is pegged on completing the inter-AMF RAN configuration transfer procedure from source AMF. Labels
• amfcc_n2_inter_target_amf_ran_config_transfer_failure Type 64-bit Counter Description The number of failed inter-AMF RAN configuration transfer procedures on target AMF. The counter is pegged if there are any failures processing the inter-AMF RAN configuration transfer procedure. Labels result
• amfcc_n2_inter_source_amf_ran_config_transfer_attempt Type 64-bit Counter Description The number of attempted inter-AMF RAN configuration transfer procedures on the source AMF. The counter is pegged on receiving inter-AMF RAN configuration transfer message from gNB. Labels
• amfcc_n2_inter_source_amf_ran_config_transfer_success Type 64-bit Counter Description The number of successful inter-AMF RAN configuration transfer procedures on the source AMF. The counter is pegged on completing the inter-AMF RAN configuration transfer procedure. Labels peer
• amfcc_n2_inter_source_amf_ran_config_transfer_failure Type 64-bit Counter Description The number of failed inter-AMF RAN configuration transfer procedures on the source AMF. The counter is pegged if there are any failures processing the inter-AMF RAN configuration transfer procedure. Labels peer, result
• amfcc_n2_inter_eps_target_amf_ran_config_transfer_attempt Type

64-bit Counter

Description

The number of attempted EPS to 5GS RAN configuration transfer on target AMF. This counter is pegged on receiving inter-AMF/MME RAN configuration transfer message from MME.

Labels

- amfcc_n2_inter_eps_target_amf_ran_config_transfer_success

Type

64-bit Counter

Description

The number of successful EPS to 5GS RAN configuration transfer on target AMF. The counter is pegged on completing the inter-AMF/MME RAN configuration transfer procedure.

Labels

peer

- amfcc_n2_inter_eps_target_amf_ran_config_transfer_failure

Type

64-bit Counter

Description

The number of failed EPS to 5GS RAN configuration transfer on target AMF. The counter is pegged if there are any failures processing the RAN configuration transfer procedure.

Labels

peer, result

- amfcc_n2_inter_eps_source_amf_ran_config_transfer_attempt

Type

64-bit Counter

Description

The number of attempted 5GS to EPS RAN configuration transfer on source AMF. This counter is pegged on receiving uplink RAN configuration transfer message from gNB.

Labels

- amfcc_n2_inter_eps_source_amf_ran_config_transfer_success

Type

64-bit Counter

Description

The number of successful 5GS to EPS RAN configuration transfer on source AMF. The counter is pegged on completing the RAN configuration transfer procedure.

Labels

peer

- amfcc_n2_inter_eps_source_amf_ran_config_transfer_failure

Type

64-bit Counter

Description

The number of failed 5GS to EPS RAN configuration transfer on source AMF. The counter is pegged if there are any failures processing the RAN configuration transfer procedure.

Labels

peer, result

- amfcc_gnb_procedure_duration

Type

64-bit Histogram

Description

The time observed for AMFCC GNB procedure execution in seconds

Labels

procedure, status

Boundaries

[0.5, 1, 1.5, 2, 3, 5, 10, 20]

Namf interface statistics for AMF Call Control Service

- amfcc_client_namf_comm_n1n2_message_transfer_request

Type

64-bit Counter

Description

The number of N1N2 message transfer requests sent by source AMF to target AMF during inter-AMF handover procedure as counted on the source side. The request message body carries N1N2MessageTransferReqData structure as defined in section 6.1.6.2.18 of 3GPP TS 29.518.

Labels

peer, peer_type, msg_type

- amfcc_client_namf_comm_n1n2_message_transfer_response

Type

64-bit Counter

Description

The number of N1N2 message transfer responses received by source AMF from target AMF during inter-AMF handover procedure. On success, the response message body contains N1N2MessageTransferRspData structure as defined in section 6.1.3.5.3.1-3 of 3GPP TS 29.518. On failure or redirection, the response message body contains N1N2MessageTransferError as defined in section 6.1.3.5.3.1-3 of 3GPP TS 29.518.

Labels

peer, peer_type, result

- amfcc_namf_comm_n1n2_message_transfer_request

Type

64-bit Counter

Description

The number of N1 N2 transfer requests received by AMF to transfer N1 message to UE or N2 information to 5G-AN or both. The request message body carries N1N2MessageTransferReqData structure as defined in section 6.1.6.2.18 of 3GPP TS 29.518.

Labels

dnn, slice, peer, peer_type, msg_type

- amfcc_namf_comm_n1n2_message_transfer_response

Type

64-bit Counter

Description

The number of N1 N2 transfer responses sent by AMF. On Success, response message body contains N1N2MessageTransferRspData structure as defined in section 6.1.6.2.19 of 3GPP TS 29.518. On failure, response message body contains N1N2MessageTransferError structure as defined in section 6.1.6.2.31 of 3GPP TS 29.518 or ProblemDetails structure as defined in section 5.2.4.1 of 3GPP TS 29.571.

Labels

dnn, slice, peer, peer_type, msg_type, result

- amfcc_namf_comm_ebi_assignment_request

Type

64-bit Counter

Description

The number of EBI Assignment requests received by AMF to allocate EPS bearer ID(s) towards EPS bearer(s) mapped from QoS flow(s) for an existing PDU Session for a given UE. The request message body carries AssignEbiData structure as defined in section 6.1.6.2.5 of 3GPP TS 29.518.

Labels

dnn, slice, peer, peer_type

- amfcc_namf_comm_ebi_assignment_response

Type

64-bit Counter

Description

The number of EBI Assignment responses sent by AMF. On Success, response message body contains AssignedEbiData structure as defined in 6.1.6.2.6 of 3GPP TS 29.518. On failure, response message body contains AssignEbiFailed structure as defined in section 6.1.6.2.7 of 3GPP TS 29.518 or ProblemDetails structure as defined in section 5.2.4.1 of 3GPP TS 29.571.

Labels

dnn, slice, peer, peer_type, result

- amfcc_client_namf_comm_ue_context_transfer_request

Type

64-bit Counter

Description

The number of UE context transfer requests sent by target AMF to source AMF during idle mode inter-AMF mobility as counted on the target side. The request message body carries UeContextTransferReqData structure as defined in section 6.1.6.2.23 of 3GPP TS 29.518.

Labels

peer

- amfcc_client_namf_comm_ue_context_transfer_response

Type

64-bit Counter

Description

The number of UE context transfer responses received by target AMF to source AMF during idle mode inter-AMF mobility. On Success, response message body contains UeContextTransferRspData structure as defined in section 6.1.6.2.24 of 3GPP TS 29.518. On failure, response message body may contain ProblemDetails structure as defined in section 6.1.3.2.4.4.2 of 3GPP TS 29.571.

Labels

peer, result

- amfcc_namf_comm_ue_context_transfer_request

Type

64-bit Counter

Description

The number of UE context transfer requests received by source AMF from target AMF during idle mode inter-AMF mobility as counted on the source side. The request message body carries UeContextTransferReqData structure as defined in section 6.1.6.2.23 of 3GPP TS 29.518.

Labels

- amfcc_namf_comm_ue_context_transfer_response

Type

64-bit Counter

Description

The number of UE context transfer responses sent by source AMF from target AMF during idle mode inter-AMF mobility. On Success, response message body contains UeContextTransferRspData structure as defined in section 6.1.6.2.24 of 3GPP TS 29.518. On failure, response message body may contain ProblemDetails structure as defined in section 6.1.3.2.4.4.2 of 3GPP TS 29.571.

Labels

result

- amfcc_client_namf_comm_ue_registration_status_update_request

Type

64-bit Counter

Description

The number of UE registration status update requests sent by target AMF to source AMF during idle mode inter-AMF mobility as counted on the target side. The request message body carries UeRegStatusUpdateReqData structure as defined in section 6.1.6.2.39 of 3GPP TS 29.518.

Labels

peer

- amfcc_client_namf_comm_ue_registration_status_update_response

Type

64-bit Counter

Description

The number of UE registration status update responses received by target AMF to source AMF during idle mode inter-AMF mobility. On Success, response message body contains UeRegStatusUpdateRspData structure as defined in section 6.1.6.2.52 of 3GPP TS 29.518. On failure, response message body may contain ProblemDetails structure as defined in section 6.1.3.2.4.5.2 of 3GPP TS 29.571.

Labels

peer, result

- amfcc_namf_comm_ue_registration_status_update_request

Type

64-bit Counter

Description

The number of UE registration status update requests received by source AMF from target AMF during idle mode inter-AMF mobility as counted on the source side. The request message body carries UeRegStatusUpdateReqData structure as defined in section 6.1.6.2.39 of 3GPP TS 29.518.

Labels

- amfcc_namf_comm_ue_registration_status_update_response

Type

64-bit Counter

Description

The number of UE registration status update responses sent by source AMF to target AMF during idle mode inter-AMF mobility. On Success, response message body contains UeRegStatusUpdateRspData structure as defined in section 6.1.6.2.52 of 3GPP TS 29.518. On failure, response message body may contain ProblemDetails structure as defined in section 6.1.3.2.4.5.2 of 3GPP TS 29.571.

Labels

result

- amfcc_client_namf_comm_create_ue_context_request

Type

64-bit Counter

Description

The number of UE context creation requests sent by source AMF to target AMF when source AMF cannot serve the UE and selects the target AMF during inter-AMF N2 handover procedure, as counted on the source side. The request message body carries UeContextCreateData structure as defined in section 6.1.6.2.41 of 3GPP TS 29.518.

Labels

peer

- amfcc_client_namf_comm_create_ue_context_response

Type

64-bit Counter

Description

The number of UE context creation responses received by source AMF from target AMF when source AMF cannot serve the UE and selects the target AMF during inter-AMF N2 handover procedure. On success, the response message body contains UeContextCreatedData structure as defined in section 6.1.6.2.42 of 3GPP TS 29.518. On failure or redirection, the response message body contains UeContextCreateError structure as defined in section 6.1.6.2.43 of 3GPP TS 29.518.

Labels

peer, result

- amfcc_namf_comm_create_ue_context_request

Type

64-bit Counter

Description

The number of UE context creation requests received by target AMF from source AMF during inter-AMF N2 handover procedure, as counted on the target side. The request message body carries UeContextCreateData structure as defined in section 6.1.6.2.41 of 3GPP TS 29.518.

Labels

peer

- amfcc_namf_comm_create_ue_context_response

Type

64-bit Counter

Description

The number of UE context creation responses sent by target AMF to source AMF during inter-AMF N2 handover procedure. On success, the response message body contains UeContextCreatedData structure as defined in section 6.1.6.2.42 of 3GPP TS 29.518. On failure or redirection, the response message body contains UeContextCreateError structure as defined in section 6.1.6.2.43 of 3GPP TS 29.518.

Labels

peer, result

- amfcc_client_namf_comm_release_ue_context_request

Type

64-bit Counter

Description

The number of UE context release requests sent by source AMF to target AMF during inter-AMF N2 handover procedure, as counted on the source side. The request message body carries UEContextRelease structure as defined in section 6.1.6.2.8 of 3GPP TS 29.518.

Labels

peer, reason

• amfcc_client_namf_comm_release_ue_context_response

Type

64-bit Counter

Description

The number of UE context release responses received by source AMF from target AMF during inter-AMF N2 handover procedure. On success, the response message body is empty. On failure or redirection, the response message body contains ProblemDetails structure as defined in section 6.1.3.2.4.2.2 of 3GPP TS 29.518.

Labels

result

• amfcc_namf_comm_release_ue_context_request

Type

64-bit Counter

Description

The number of UE context release requests received by target AMF from source AMF during inter-AMF N2 handover procedure, as counted on the target side. The request message body carries UEContextRelease structure as defined in section 6.1.6.2.8 of 3GPP TS 29.518.

Labels

peer

• amfcc_namf_comm_release_ue_context_response

Type

64-bit Counter

Description

The number of UE context release responses sent from target AMF to source AMF during inter-AMF N2 handover procedure. On success, the response message body is empty. On failure or redirection, the response message body contains ProblemDetails structure as defined in section 6.1.3.2.4.2.2 of 3GPP TS 29.518.

Labels

peer, result

• amfcc_client_namf_comm_n2_info_notification

Type

64-bit Counter

Description

The number of N2 information notification requests sent by target AMF to source AMF when inter-AMF N2 handover procedure is successful, as counted on the target side. The request message body carries N2InformationNotification structure as defined in section 6.1.6.2.14 of 3GPP TS 29.518.

Labels

peer

• amfcc_client_namf_comm_n2_info_notification_response

Type

64-bit Counter

Description

The number of N2 information notification request acknowledgements received by target AMF from source AMF during inter-AMF N2 handover procedure. On success, the acknowledgement message body is empty unless secondary RAT usage data is available at the source AMF, in which case the message body contains N2InfoNotificationRspData as defined in section 6.1.6.2.54 of 3GPP TS 29.518. On failure, the acknowledgement message body contains ProblemDetails structure as defined in section 5.2.2.3.6.1 of 3GPP TS 29.518.

Labels

peer, result

- amfcc_namf_comm_n2_info_notification

Type

64-bit Counter

Description

The number of N2 information notification requests received by source AMF from target AMF when inter-AMF N2 handover procedure is successful, as counted on the source side. The request message body carries N2InformationNotification structure as defined in section 6.1.6.2.14 of 3GPP TS 29.518.

Labels

-
- amfcc_namf_comm_n2_info_notification_response

Type

64-bit Counter

Description

The number of N2 information notification request acknowledgements sent by source AMF to target AMF during inter-AMF N2 handover procedure. On success, the acknowledgement message body is empty unless secondary RAT usage data is available at the source AMF, in which case the message body contains N2InfoNotificationRspData as defined in section 6.1.6.2.54 of 3GPP TS 29.518. On failure, the acknowledgement message body contains ProblemDetails structure as defined in section 5.2.2.3.6.1 of 3GPP TS 29.518.

Labels

result

- amfcc_client_namf_comm_relocate_ue_context_request

Type

64-bit Counter

Description

The number of UE context relocate requests sent by source AMF to target AMF when source AMF cannot serve the UE and selects the target AMF during N26 handover procedure, as counted on the source side. The request message body carries UeContextRelocateData structure as defined in section 6.1.6.2.61 of 3GPP TS 29.518.

Labels

peer

- amfcc_client_namf_comm_relocate_ue_context_response

Type

64-bit Counter

Description

The number of UE context relocate responses received by source AMF from target AMF when source AMF cannot serve the UE and selects the target AMF during N26 handover procedure. On success, the response message body contains UeContextRelocatedData structure as defined in section 6.1.6.2.62 of 3GPP TS 29.518. On failure or redirection, the response message body contains UeContextRelocateError structure as defined in section 6.1.6.2.63 of 3GPP TS 29.518.

Labels

peer, result

- amfcc_namf_comm_relocate_ue_context_request

Type

64-bit Counter

Description

The number of UE context relocate requests received by target AMF from source AMF during N26 handover procedure, as counted on the target side. The request message body carries UeContextRelocateData structure as defined in section 6.1.6.2.61 of 3GPP TS 29.518.

Labels

- amfcc_namf_comm_relocate_ue_context_response

Type

64-bit Counter

Description

The number of UE context relocate responses sent by target AMF to source AMF during N26 handover procedure. On success, the response message body contains UeContextRelocatedData structure as defined in section 6.1.6.2.62 of 3GPP TS 29.518. On failure or redirection, the response message body contains UeContextRelocateError structure as defined in section 6.1.6.2.63 of 3GPP TS 29.518.

Labels

result

- amfcc_client_namf_comm_cancel_relocate_ue_context_request

Type

64-bit Counter

Description

The number of UE context cancel relocate requests sent by source AMF to target AMF when source AMF receive handover cancel from source ran during N26 handover procedure, as counted on the source side. The request message body carries UeContextCancelRelocateData structure as defined in section 6.1.6.2.67 of 3GPP TS 29.518.

Labels

peer

- amfcc_client_namf_comm_cancel_relocate_ue_context_response

Type

64-bit Counter

Description

The number of UE context cancel relocate responses received by source AMF from target AMF when source AMF receives handover cancel from source ran during N26 handover procedure. On success, response message is sent by Target AMF with no content as defined in section as defined in section 5.2.2.2.6 of 3GPP TS 29.518. On failure, the acknowledgement message body contains ProblemDetails structure as defined in section 5.2.2.3.6.1 of 3GPP TS 29.518.

Labels

peer, result

- amfcc_namf_comm_cancel_relocate_ue_context_request

Type

64-bit Counter

Description

The number of UE context cancel relocate requests received by target AMF from source AMF during N26 handover procedure, as counted on the target side. The request message body carries UeContextCancelRelocateData structure as defined in section 6.1.6.2.67 of 3GPP TS 29.518.

Labels

- amfcc_namf_comm_cancel_relocate_ue_context_response

Type

64-bit Counter

Description

The number of UE context cancel relocate responses sent by target AMF to source AMF during N26 handover procedure. On success, response message is sent by Target AMF with no content as defined in section 5.2.2.2.6 of 3GPP TS 29.518. On failure, the acknowledgement message body contains ProblemDetails structure as defined in section 5.2.2.3.6.1 of 3GPP TS 29.518.

Labels

result

- amfcc_namf_mt_enable_reachability_request

Type

64-bit Counter

Description

The number of MT EnableReachability requests received by AMF to transfer N1 message to UE. The request message body carries EnableUeReachabilityReqData structure as defined in section 5.4.2.2 of 3GPP TS 29.518.

Labels

peer

- amfcc_namf_mt_enable_reachability_response

Type

64-bit Counter

Description

The number of MT EnableReachability responses sent by AMF. On Success, response message body contains EnableUeReachabilityRspData structure as defined in section 5.4.2.2 of 3GPP TS 29.518. On failure, response message body shall contain a ProblemDetails structure with the cause attribute set to one of the application error listed in Table 6.3.3.2.3.1-3.

Labels

peer, result

- amfcc_namf_mt_provide_domain_selection_info_request

Type

64-bit Counter

Description

The number of MT ProvideDomainSelectionInfo requests received by AMF. The request message body carries information as defined in section 5.4.2.3 of 3GPP TS 29.518.

Labels

peer

- amfcc_namf_mt_provide_domain_selection_info_response

Type

64-bit Counter

Description

The number of MT ProvideDomainSelectionInfo responses sent by AMF. On Success, response message body contains UeContextInfo structure as defined in section 5.4.2.3 of 3GPP TS 29.518. On failure, response message body shall contain a ProblemDetails structure with the cause attribute set to one of the application error listed in Table 6.3.3.3.3.1-3.

Labels

peer, result

- amfcc_namf_comm_n1_info_notification

Type

64-bit Counter

Description

The number of N1 information notification requests sent by AMF to LMF. The counter is pegged when AMF sends n1 notification to LMF

Labels

msg_class

- amfcc_namf_comm_n1_info_notification_response

Type

64-bit Counter

Description

The number of N1 information notification response sent by LMF to AMF. The counter is pegged when LMF sends n1 notification response to AMF

Labels

msg_class, result

- amfcc_namf_location_event_notification

Type

64-bit Counter

Description

The number of event notification sent by AMF to NF. This counter is pegged when an AMF sends event notification to NF

Labels

peer

- amfcc_namf_location_event_notification_response

Type

64-bit Counter

Description

The number of event notification response sent by an NF to AMF. This counter is pegged when NF sends event notification response sent by NF to AMF.

Labels

peer, result

- amfcc_namf_location_provide_positioning_info_request

Type

64-bit Counter

Description

The number of Positioning info request invoked by an NF to AMF in order to obtain current location of UE. This counter is pegged when an NF sends request to AMF

Labels

client_type

- amfcc_namf_location_provide_positioning_info_response

Type

64-bit Counter

Description

The number of positioning info response sent by an AMF to NF. This counter is pegged on receiving positioning info from amf

Labels

client_type, result

Nlmf interface statistics for AMF Call Control Service

- amfcc_nlmf_location_determine_location_request

Type

64-bit Counter

Description

The number of location determine request sent by an AMF to LMF.This counter is pegged on receiving location determine request from AMF.

Labels

peer

- amfcc_nlmf_location_determine_location_response

Type

64-bit Counter

Description

The number of location determine response sent by an LMF to AMF.This counter is pegged on receiving location determine response from LMF

Labels

peer, result

Location Services statistics for AMF Call Control Service

- amfcc_lcs_network_induced_location_request_attempt

Type

64-bit Counter

Description

The number of NI-LR sent by an AMF.This counter is pegged on receiving NI-LR request from AMF

Labels

location_event

- amfcc_lcs_network_induced_location_request_success

Type

64-bit Counter

Description

The number of times NI-LR request sent successfully.This counter is pegged on NI-LR request success

Labels

location_event

- amfcc_lcs_network_induced_location_request_failure

Type

64-bit Counter

Description

The number of times NI-LR request failed.This counter is pegged on NI-LR request failure

Labels

location_event, result

- amfcc_lcs_mobile_terminated_location_request_attempt

Type

64-bit Counter

Description

The number of MT-LR sent by LCS client to PLMN.This counter is pegged on receiving MT-LR request from LCS client

Labels

client_type

- amfcc_lcs_mobile_terminated_location_request_success

Type

64-bit Counter

Description

The number of times MT-LR request sent successfully.This counter is pegged on MT-LR request success

Labels

client_type

- amfcc_lcs_mobile_terminated_location_request_failure

Type

64-bit Counter

Description

The number of times MT-LR request failed.This counter is pegged on MT-LR request failure

Labels

client_type, result

Event Data Records for AMF Call Control Service

- amfcc_edr_generated

Type

64-bit Counter

Description

The number of EDR sent by an AMF. This counter is pegged on sending EDR from AMF

Labels

event_layer, event_type, outcome

- amfcc_edr_callback_rcvd

Type

64-bit Counter

Description

The number of times EDR callback was received successfully. This counter is pegged on receiving EDR callback

Labels

result

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- [Summary](#)
- [ETCD Client Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_put_requests	Counter	The total number of ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_requests_sent	Counter	The total number of ETCD Client PUT Requests sent.
ETCD Client Metrics	etcd_put_successes	Counter	The total number of successful ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_failures	Counter	The total number of failed ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_with_lease_requests	Counter	The total number of ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_requests_sent	Counter	The total number of ETCD Client PUT with Lease Requests sent.
ETCD Client Metrics	etcd_put_with_lease_successes	Counter	The total number of successful ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_failures	Counter	The total number of failed ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_range_requests	Counter	The total number of ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_requests_sent	Counter	The total number of ETCD Client Range Requests sent.
ETCD Client Metrics	etcd_range_successes	Counter	The total number of successful ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_failures	Counter	The total number of failed ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_delete_requests	Counter	The total number of ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_requests_sent	Counter	The total number of ETCD Client Range Delete Requests sent.
ETCD Client Metrics	etcd_range_delete_successes	Counter	The total number of successful ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_failures	Counter	The total number of failed ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_txn_requests	Counter	The total number of ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_requests_sent	Counter	The total number of ETCD Client Txn Requests sent.
ETCD Client Metrics	etcd_txn_successes	Counter	The total number of successful ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_failures	Counter	The total number of failed ETCD Client Txn Requests.
ETCD Client Metrics	etcd_watch_requests	Counter	The total number of ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_requests_sent	Counter	The total number of ETCD Client Watch Requests sent.
ETCD Client Metrics	etcd_watch_successes	Counter	The total number of successful ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_failures	Counter	The total number of failed ETCD Client Watch Requests.
ETCD Client Metrics	etcdwatcher_requests	Counter	The total number of ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_requests_sent	Counter	The total number of ETCD Watcher Requests sent.
ETCD Client Metrics	etcdwatcher_successes	Counter	The total number of successful ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_failures	Counter	The total number of failed ETCD Watcher Requests.

ETCD Client Metrics

- [etcd_put_requests](#)

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests.

Labels

keyPrefix

- [etcd_put_requests_sent](#)

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests sent.

Labels

keyPrefix

- etcd_put_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT Requests.

Labels

keyPrefix

- etcd_put_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT Requests.

Labels

keyPrefix, failureReason

- etcd_put_with_lease_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests sent.

Labels

keyPrefix

- etcd_put_with_lease_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT with Lease Requests.

Labels

keyPrefix, failureReason

• etcd_range_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests.

Labels

keyPrefix

• etcd_range_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests sent.

Labels

keyPrefix

• etcd_range_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Requests.

Labels

keyPrefix

• etcd_range_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Requests.

Labels

keyPrefix, failureReason

• etcd_range_delete_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests.

Labels

keyPrefix

• etcd_range_delete_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests sent.

Labels

keyPrefix

• etcd_range_delete_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Delete Requests.

Labels

keyPrefix

• etcd_range_delete_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Delete Requests.

Labels

keyPrefix, failureReason

• etcd_txn_requests

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests.

Labels

keyPrefix

• etcd_txn_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests sent.

Labels

keyPrefix

• etcd_txn_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Txn Requests.

Labels

keyPrefix

- etcd_txn_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Txn Requests.

Labels

keyPrefix, failureReason

- etcd_watch_requests

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests sent.

Labels

keyPrefix

- etcd_watch_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Watch Requests.

Labels

keyPrefix, failureReason

• etcdwatcher_requests

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests.

Labels

keyPrefix

• etcdwatcher_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests sent.

Labels

keyPrefix

• etcdwatcher_successes

Type

64-bit Counter

Description

The total number of successful ETCD Watcher Requests.

Labels

keyPrefix

• etcdwatcher_failures

Type

64-bit Counter

Description

The total number of failed ETCD Watcher Requests.

Labels

keyPrefix, failureReason

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- [Summary](#)
- [HTTP Punting Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_sent	Counter	The total number of HTTP requests successfully punted.
HTTP Punting Metrics	punting_http_requests_failed	Counter	The total number of HTTP requests punting failed.
HTTP Punting Metrics	punting_http_responses_received	Counter	The total number of HTTP responses received for the requests punted.

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_responses_forwarded	Counter	The total number of HTTP responses forwarded after punting transaction completion.
HTTP Punting Metrics	punting_http_requests_aborted	Counter	The total number of HTTP requests aborted.
HTTP Punting Metrics	punting_http_request_duration_seconds	Histogram	Punting operation duration in seconds.

HTTP Punting Metrics

- `punting_http_requests_sent`

Type

64-bit Counter

Description

The total number of HTTP requests successfully punted.

Labels

message_type

- `punting_http_requests_failed`

Type

64-bit Counter

Description

The total number of HTTP requests for which the punting failed.

Labels

message_type

- `punting_http_responses_received`

Type

64-bit Counter

Description

The total number of HTTP responses received for the requests punted.

Labels

message_type, status_code

- `punting_http_responses_forwarded`

Type

64-bit Counter

Description

The total number of HTTP responses forwarded after punting transaction completion.

Labels

message_type, status_code

- `punting_http_requests_aborted`

Type

64-bit Counter

Description

The total number of HTTP requests aborted.

Labels

message_type

- `punting_http_request_duration_seconds`

Type

64-bit Histogram

Description

The punting transaction duration is measured from the time the HTTP request is sent out to the time the response is received for the punting request.

Labels

message_type

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

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- [Summary](#)
- [In memory bufferstatistics](#)

Summary

Group	Metric Name	Type	Synopsis
In memory bufferstatistics	inmemory_buffer_pdu_added	Counter	Count of the PDUs added to the inmemorybuffer
In memory bufferstatistics	inmemory_buffer_pdu_removed	Counter	Count of the PDUs removed from the inmemorybuffer
In memory bufferstatistics	inmemory_buffer_pdu_dropped	Counter	Count of the PDUs dropped from the inmemorybuffer
In memory bufferstatistics	inmemory_buffer_pdu_remove_all	Counter	Count of times removeAllPdus is called for the inmemorybuffer

In memory bufferstatistics

- `inmemory_buffer_pdu_added`

Type

64-bit Counter

Description

Count of the PDUs added to the inmemorybuffer

Labels

did, position

- `inmemory_buffer_pdu_removed`

Type

64-bit Counter

Description

Count of the PDUs removed from the inmemorybuffer

Labels

did, position

- `inmemory_buffer_pdu_dropped`

Type

64-bit Counter

Description

Count of the PDUs dropped from the inmemorybuffer

Labels

did, position

- inmemory_buffer_pdu_remove_all

Type

64-bit Counter

Description

Count of times removeAllPdus is called for the inmemorybuffer

Labels

did, reason

Table of Contents

- [Summary](#)
- [LiCmn statistics](#)
- [Key Ciphering Statistics](#)

Summary

Group	Metric Name	Type	Synopsis
LiCmn statistics	svcmgr_service_available_ind_notification	Counter	How many times, Service Available Indication message received
LiCmn statistics	svcmgr_service_available_ind_parse_errs	Counter	How many times, Service Available Indication parse error happened
LiCmn statistics	svcmgr_service_available_ind_notification_ack	Counter	How many times, Service Available Indication message is acknowledged
LiCmn statistics	svcmgr_service_unavailable_ind_notification	Counter	How many times, Service Unavailable Indication message
LiCmn statistics	svcmgr_service_unavailable_ind_parse_errs	Counter	How many times, Service Unavailable Indication parse error happened
LiCmn statistics	svcmgr_service_unavailable_ind_notification_ack	Counter	How many times, Service Unavailable Indication message is acknowledged
LiCmn statistics	svcmgr_publish_msg_sent	Counter	How many times, Publication message sent
LiCmn statistics	svcmgr_publish_msg_success	Counter	How many times, Publication message success response received
LiCmn statistics	svcmgr_publish_msg_failure	Counter	How many times, failure received for publication message
LiCmn statistics	svcmgr_subscribe_msg_sent	Counter	How many times, Subscribe message sent
LiCmn statistics	svcmgr_subscribe_msg_success	Counter	How many times, subscribe message success
LiCmn statistics	svcmgr_subscribe_msg_failure	Counter	How many times, failure received for subscribe message
LiCmn statistics	svcmgr_unsubscribe_msg_sent	Counter	How many times, unsubscribe message sent
LiCmn statistics	svcmgr_unsubscribe_msg_failure	Counter	How many times, failure received for unsubscribe message
LiCmn statistics	svcmgr_config_req_sent	Counter	How many times, config request sent
LiCmn statistics	svcmgr_config_req_failure	Counter	How many times, failure received for config request
LiCmn statistics	svcmgr_config_req_success	Counter	How many times, success received for config request
LiCmn statistics	svcmgr_config_register_response	Counter	How many times, config register response received
LiCmn statistics	svcmgr_config_register_response_ack	Counter	How many times, config register response is acknowledged
LiCmn statistics	svcmgr_config_register_response_parse_err	Counter	How many times, parse error happened for config register response
LiCmn statistics	svcmgr_config_update_notification	Counter	How many times, config update notification received
LiCmn statistics	svcmgr_config_update_notification_response	Counter	How many times, config update response
LiCmn statistics	svcmgr_intf_schema_update_ind_notification	Counter	How many times, interface schema update indication message received
LiCmn statistics	svcmgr_intf_schema_update_ind_notification_ack	Counter	How many times, interface schema update indication message is acknowledged

Group	Metric Name	Type	Synopsis
LiCmn statistics	svcmgr_intf_schema_update_ind_parse_errs	Counter	How many times, interface schema update indication parse error happened
Key Ciphering Statistics	li_cmn_cipher_object_registered	Counter	Report number of cipher objects registered
Key Ciphering Statistics	li_cmn_secret_key_rotated	Counter	Report number of times secret key has been changed
Key Ciphering Statistics	li_cmn_enciphering_failed	Counter	Report number of times enciphering has failed
Key Ciphering Statistics	li_cmn_deciphering_failed	Counter	Report number of times deciphering has failed
Key Ciphering Statistics	li_cmn_key_version_not_available	Counter	Report number of times when the key corresponding to the key version is not available.

LiCmn statistics

- svcmgr_service_available_ind_notification

Type

64-bit Counter

Description

Number of Times Service Available Indication notification received

Labels

servicename

- svcmgr_service_available_ind_parse_errs

Type

64-bit Counter

Description

Number of Times Service Available Indication parse error happened

Labels

- svcmgr_service_available_ind_notification_ack

Type

64-bit Counter

Description

Number of Times Service Available Indication message is acknowledged

Labels

servicename

- svcmgr_service_unavailable_ind_notification

Type

64-bit Counter

Description

Number of Times Service Unavailable Indication notification received

Labels

servicename

- svcmgr_service_unavailable_ind_parse_errs

Type

64-bit Counter

Description

Number of Times Service Unavailable Indication parse error happened

Labels

- `svcmgr_service_unavailable_ind_notification_ack`

Type

64-bit Counter

Description

Number of Times Service Unavailable Indication message is acknowledged

Labels

servicename

- `svcmgr_publish_msg_sent`

Type

64-bit Counter

Description

Number of Times Publication message sent

Labels

servicename, retry

- `svcmgr_publish_msg_success`

Type

64-bit Counter

Description

Number of Times Publication message success response received

Labels

servicename, causecode, retry

- `svcmgr_publish_msg_failure`

Type

64-bit Counter

Description

Number of Times failure received for publication message

Labels

servicename, causecode, retry

- `svcmgr_subscribe_msg_sent`

Type

64-bit Counter

Description

Number of Times Subscribe message sent

Labels

servicename, retry

- `svcmgr_subscribe_msg_success`

Type

64-bit Counter

Description

Number of Times Send subscribe message success response received

Labels

servicename, causecode, retry

- `svcmgr_subscribe_msg_failure`

Type

64-bit Counter

Description

Number of time failure received for subscribe message

Labels

servicename, causecode, retry

- `svcmgr_unsubscribe_msg_sent`

Type

64-bit Counter

Description

Number of Times unsubscribe message sent

Labels

servicename

- `svcmgr_unsubscribe_msg_failure`

Type

64-bit Counter

Description

Number of Times failure received for unsubscribe message

Labels

servicename

- `svcmgr_config_req_sent`

Type

64-bit Counter

Description

Number of Times config request sent

Labels

retry

- `svcmgr_config_req_failure`

Type

64-bit Counter

Description

Number of Times failure received for config request

Labels

causecode, retry

- svcmgr_config_req_success

Type

64-bit Counter

Description

Number of Times success received for config request

Labels

causecode, retry

- svcmgr_config_register_response

Type

64-bit Counter

Description

Number of Times config register response received

Labels

statuscode

- svcmgr_config_register_response_ack

Type

64-bit Counter

Description

Number of Times config register response is acknowledged

Labels

- svcmgr_config_register_response_parse_err

Type

64-bit Counter

Description

Number of Times parse error happened for config register response

Labels

- svcmgr_config_update_notification

Type

64-bit Counter

Description

Number of Times config update notification received

Labels

- svcmgr_config_update_notification_response

Type

64-bit Counter

Description

Number of Times config update response sent

Labels

• svcmgr_intf_schema_update_ind_notification

Type

64-bit Counter

Description

Number of Times interface schema update indication message received

Labels

servicename

• svcmgr_intf_schema_update_ind_notification_ack

Type

64-bit Counter

Description

Number of Times interface schema update indication message is acknowledged

Labels

servicename

• svcmgr_intf_schema_update_ind_parse_errs

Type

64-bit Counter

Description

Number of Times Service interface schema update indication parse error happened

Labels

Key Cipherring Statistics

• li_cm_n_cipher_object_registered

Type

64-bit Counter

Description

Total number of cipher objects registered

Labels

key_version

• li_cm_n_secret_key_rotated

Type

64-bit Counter

Description

Total number of times secret keys has been rotated

Labels

• li_cmn_enciphering_failed			
Type			
64-bit Counter			
Description			
Total number of times enciphering has failed			
Labels			
key_version			
• li_cmn_deciphering_failed			
Type			
64-bit Counter			
Description			
Total number of times deciphering has failed			
Labels			
key_version			
• li_cmn_key_version_not_available			
Type			
64-bit Counter			
Description			
Total number of times key corresponding to the key version is not available.			
Labels			
key_version			

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- [Consul statistics](#)

Summary

Group	Metric Name	Type	Synopsis
NfMonitor statistics	nfmonitor_unstable_nf_counter	Counter	Howmany times, the Nf is found to be Unstable
NfMonitor statistics	nfmonitor_force_restart_due_to_nf_unstability	Counter	Numbe of times the VNFRestart triggered due to Unstability of the NF
NfMonitor statistics	nfmonitor_cumulative_readiness_failure_count	Counter	Numbe of times Service sound to have insuffucient count due to Readiness probe
NfMonitor statistics	nfmonitor_cumulative_readiness_failure_alerts	Counter	Numbe of times Service sound to have insuffucient count due to Readiness probe
NfMonitor statistics	nfmonitor_cumulative_k8api_get_query_failures	Counter	Numbe of times the k8 get Api query failures
DebugNfMonitorCmn	nfmonitor_force_restart_counter	Counter	Number of times force-restart triggered
NfManagement statistics	nfmonitor_nfmgmt_service_request_attempts	Counter	Howmany times, the Nf is attempting to send NRF Requests

Group	Metric Name	Type	Synopsis
NfManagement statistics	nfmonitor_nfmgmt_service_request_success	Counter	Numbe of times Success response is received for the requests to NRF
NfManagement statistics	nfmonitor_nfmgmt_service_request_failures	Counter	Numbe of times Failure response is received for the requests to NRF
NfManagement statistics	nfmonitor_nfmgmt_aborted_request_msgs	Counter	Numbe of times the requests to NRF is Aborted/Cancelled
Debug statistics	nfmonitor_vnf_restart_trigger	Counter	Howmany times, the Force Restart is received
Debug statistics	nfmonitor_fsm_state_changes	Counter	The statistics for the the FSM state changes
Debug statistics	nfmonitor_deactivation_triggers	Counter	Numbe of times the NF deactivation trigger is received
Debug statistics	nfmonitor_critical_service_up_notification	Counter	Numbe of times the Critical Service Up Notification received
Debug statistics	nfmonitor_critical_service_down_notification	Counter	Numbe of times the Critical Service Down Notification received
Debug statistics	nfmonitor_critical_services_overall_up_notification	Counter	Numbe of times the Up notification received when all monitored Critical Services are available
Debug statistics	nfmonitor_critical_services_overall_down_notification	Counter	Numbe of times the Down notification received when at least one Critical Service was unavailable
Debug statistics	nfmonitor_hysteresis_timer_started	Counter	Numbe of times the Hysteresis Timer is started due to missing Critical Service
Debug statistics	nfmonitor_hysteresis_timer_stopped	Counter	Numbe of times the Hysteresis Timer is stopped due to Critical Service become available
Debug statistics	nfmonitor_hysteresis_timer_expired	Counter	Numbe of times the Hysteresis Timer is expired due to missing Critical Service
Debug statistics	nfmonitor_personality_update_triggers	Counter	Numbe of times the Nf Personality update is received
Debug statistics	nfmonitor_maintenance_trigger_received	Counter	Numbe of times the Maintenance Mode is activated
Debug statistics	nfmonitor_profile_update_triggers	Counter	Numbe of times the Nf Profile Update happened
Debug statistics	nfmonitor_critical_service_down_on_register_entry	Counter	Numbe of times the Critical Services were down while starting with RegisterFsmState
ApiServer statistics	nfmonitor_deployment_create_watch_request	Counter	Howmany times, the deployment watch requests are created
ApiServer statistics	nfmonitor_deployment_delete_watch_request	Counter	Howmany times, the deployment watch requests are deleted
ApiServer statistics	nfmonitor_deployment_watch_notification_success	Counter	Howmany times, the watch notifications are successful
ApiServer statistics	nfmonitor_deployment_watch_notification_error	Counter	Howmany times, the watch notifications are failed
ApiServer statistics	nfmonitor_limit_range_query_attempts	Counter	Howmany times, the limit-range query attempts are made
ApiServer statistics	nfmonitor_limit_range_query_failures	Counter	Howmany times, the limit-range query attempts are failed
ApiServer statistics	nfmonitor_limit_range_query_success	Counter	Howmany times, the limit-range query attempts are succeeded
ApiServer statistics	nfmonitor_hpa_query_attempts	Counter	Howmany times, the Hpa query attempts are made
ApiServer statistics	nfmonitor_hpa_query_failures	Counter	Howmany times, the Hpa query attempts are failed
ApiServer statistics	nfmonitor_hpa_query_success	Counter	Howmany times, the Hpa query attempts are succeeded
ApiServer statistics	nfmonitor_readiness_probe_query_attempts	Counter	Howmany times, the Readiness Probe query attempts are made
ApiServer statistics	nfmonitor_readiness_probe_query_failures	Counter	Howmany times, the Readiness Probe query attempts are failed
ApiServer statistics	nfmonitor_readiness_probe_query_success	Counter	Howmany times, the Readiness Probe query attempts are succeeded
Feature Functionality statistics	nfmonitor_critical_service_frequent_restart	Counter	Howmany times, minimum count is not met for the stable Critical Services
Feature Functionality statistics	nfmonitor_alarm_due_to_frequent_restart	Counter	Howmany times alarm raised, due to minimum count is not met for the stable Critical Services
Feature Functionality statistics	nfmonitor_force_restart_due_to_frequent_restart	Counter	Howmany times, force restart triggerd because of the unstable Critical Services
Feature Functionality statistics	nfmonitor_critical_service_readiness_failure	Counter	Howmany times, minimum count is not met for the readiness succeeded Critical Services
Feature Functionality statistics	nfmonitor_alarm_due_to_readiness_failure	Counter	Alarm raised, due to minimum count is not met for the readiness succeeded Critical Services

Group	Metric Name	Type	Synopsis
Feature Functionality statistics	nfmonitor_force_restart_due_to_readiness_failure	Counter	Force restart triggerd because of Critical Service Readiness failure
Consul statistics	nfmonitor_publish_status_into_consul	Counter	Number of times Publish message is sent to Consul
Consul statistics	nfmonitor_consul_read_attempts	Counter	Number of times Consul read attempts
Consul statistics	nfmonitor_consul_read_failures	Counter	Number of times Consul read attempts failed
Consul statistics	nfmonitor_consul_read_success	Counter	Number of times Consul read attempts succeeded

NfMonitor statistics

- [nfmonitor_unstable_nf_counter](#)

Type

64-bit Counter

Description

Number of Times Nf is found Unstable

Labels
- [nfmonitor_force_restart_due_to_nf_unstability](#)

Type

64-bit Counter

Description

The number of times the Force Restart triggered due to NF-Unstability

Labels
- [nfmonitor_cumulative_readiness_failure_count](#)

Type

64-bit Counter

Description

The number of times the service is found to be failing the Readiness probe

Labels

critical_service_name, pod_name
- [nfmonitor_cumulative_readiness_failure_alerts](#)

Type

64-bit Counter

Description

Total number of EventAlerts generated due to Readiness probe failure

Labels

critical_service_name, pod_name
- [nfmonitor_cumulative_k8api_get_query_failures](#)

Type

64-bit Counter

Description

Total number of k8 api get pod query failures

Labels

critical_service_name, pod_name

DebugNfMonitorCmn

- nfmonitor_force_restart_counter
- Type
- 64-bit Counter
- Description
- Number of times force-restart triggered
- Labels
- agent, name*

NfManagement statistics

- nfmonitor_nfmgmt_service_request_attempts
- Type
- 64-bit Counter
- Description
- Number of Requests sent to NRF
- Labels
- instance_id, msg_type, method, retry*
- nfmonitor_nfmgmt_service_request_success
- Type
- 64-bit Counter
- Description
- The number of times the Request to NRF is Successful
- Labels
- instance_id, msg_type, method, retry, status_code*
- nfmonitor_nfmgmt_service_request_failures
- Type
- 64-bit Counter
- Description
- The number of times the Request to NRF is Failure
- Labels
- instance_id, msg_type, method, retry, status_code*
- nfmonitor_nfmgmt_aborted_request_msgs
- Type
- 64-bit Counter
- Description
- The number of times the Request to NRF is Aborted
- Labels

instance_id, msg_type

Debug statistics

- nfmonitor_vnf_restart_trigger

Type

64-bit Counter

Description

Number of VNF Restart triggers received

Labels

curr_nf_status, nfmgmt_state, nfmonitor_fsm_state, up_count, timestamp, trigger, type

- nfmonitor_fsm_state_changes

Type

64-bit Counter

Description

The FSM State Change events stats

Labels

curr_nf_status, nfmgmt_state, up_count, from_state, to_state

- nfmonitor_deactivation_triggers

Type

64-bit Counter

Description

The number of times the deactivation trigger received

Labels

nf_status, nfmgmt_state, up_count, nfmonitor_fsm_state

- nfmonitor_critical_service_up_notification

Type

64-bit Counter

Description

The number of times the CriticalService Up notification received

Labels

service, criticality, nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_critical_service_down_notification

Type

64-bit Counter

Description

The number of times the CriticalService Down notification received

Labels

service, criticality, nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_critical_services_overall_up_notification

Type

64-bit Counter

Description

The number of times the Overall CriticalServices Up notification received

Labels

nf_status, nfmgmt_state, up_count, fsm_state

• nfmonitor_critical_services_overall_down_notification

Type

64-bit Counter

Description

The number of times the Overall CriticalServices Down notification received

Labels

nf_status, nfmgmt_state, up_count, fsm_state

• nfmonitor_hysteresis_timer_started

Type

64-bit Counter

Description

The number of times the Hysteresis Timer is started

Labels

nf_status, nfmgmt_state, up_count, fsm_state

• nfmonitor_hysteresis_timer_stopped

Type

64-bit Counter

Description

The number of times the Hysteresis Timer is stopped

Labels

nf_status, nfmgmt_state, up_count, fsm_state

• nfmonitor_hysteresis_timer_expired

Type

64-bit Counter

Description

The number of times the Hysteresis Timer is expired

Labels

nf_status, nfmgmt_state, up_count, fsm_state

• nfmonitor_personality_update_triggers

Type

64-bit Counter

Description

The number of times the Nf Personality Update is received

Labels

nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_maintenance_trigger_received

Type

64-bit Counter

Description

The number of times the Maintenance Mode is activated

Labels

nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_profile_update_triggers

Type

64-bit Counter

Description

The number of times the Nf Profile Update happened

Labels

nf_status, nfmgmt_state, up_count, fsm_state, update_operation

- nfmonitor_critical_service_down_on_register_entry

Type

64-bit Counter

Description

The number of times the Critical Services were down while starting with RegisterFsmState

Labels

nf_status, up_count

ApiServer statistics

- nfmonitor_deployment_create_watch_request

Type

64-bit Counter

Description

The number of Watch Reqs created for deployment type

Labels

resource

- nfmonitor_deployment_delete_watch_request

Type

64-bit Counter

Description

The number of Watch Reqs deleted for deployment type

Labels

resource

- nfmonitor_deployment_watch_notification_success

Type

64-bit Counter

Description

The number of times Watch notifications are successful

Labels

resource

- nfmonitor_deployment_watch_notification_error

Type

64-bit Counter

Description

The number of times Watch notifications are failed

Labels

resource, reason

- nfmonitor_limit_range_query_attempts

Type

64-bit Counter

Description

The number of Limit Range query attempts

Labels

- nfmonitor_limit_range_query_failures

Type

64-bit Counter

Description

The number of Limit Range query failures

Labels

status_code

- nfmonitor_limit_range_query_success

Type

64-bit Counter

Description

The number of Limit Range query succeeded

Labels

status_code

- nfmonitor_hpa_query_attempts

Type

64-bit Counter

Description

The number of Hpa query attempts

Labels

resource, continue_param

- nfmonitor_hpa_query_failures
Type
64-bit Counter
Description
The number of Hpa query failures
Labels
resource, continue_param, status_code
- nfmonitor_hpa_query_success
Type
64-bit Counter
Description
The number of Hpa query succeeded
Labels
resource, continue_param, status_code
- nfmonitor_readiness_probe_query_attempts
Type
64-bit Counter
Description
The number of Readiness Probe query attempts
Labels
resource, continue_param
- nfmonitor_readiness_probe_query_failures
Type
64-bit Counter
Description
The number of Readiness Probe query failures
Labels
resource, continue_param, status_code
- nfmonitor_readiness_probe_query_success
Type
64-bit Counter
Description
The number of Readiness Probe query succeeded
Labels
resource, continue_param, status_code

Feature Functionality statistics

- nfmonitor_critical_service_frequent_restart
Type
64-bit Counter

Description

Howmany times the minimum required stable CriticalService count is not met

Labels

service

- nfmonitor_alarm_due_to_frequent_restart

Type

64-bit Counter

Description

Howmany times Alarm is raised due to minimum required stable CriticalService count is not met

Labels

service, alarm_status

- nfmonitor_force_restart_due_to_frequent_restart

Type

64-bit Counter

Description

Howmany times Force Restart initiated due to missing stable CriticalService count

Labels

service

- nfmonitor_critical_service_readiness_failure

Type

64-bit Counter

Description

Howmany times the minimum required readiness succeeded CriticalService count is not met

Labels

service

- nfmonitor_alarm_due_to_readiness_failure

Type

64-bit Counter

Description

Alarm raised due to minimum count is not met for the readiness succeeded CriticalService

Labels

service, alarm_status

- nfmonitor_force_restart_due_to_readiness_failure

Type

64-bit Counter

Description

Howmany times Force Restart initiated due to readiness failure of CriticalService

Labels

service

Consul statistics

- `nfmonitor_publish_status_into_consul`

Type
64-bit Counter
Description
Howmany times Publish is sent to Consul
Labels

nf_status, nf_state, up_count, fsm_state

- `nfmonitor_consul_read_attempts`

Type
64-bit Counter
Description
Howmany times Consul read attempt is made
Labels

fsm_state, retry

- `nfmonitor_consul_read_failures`

Type
64-bit Counter
Description
Howmany times Consul read attempt are failed
Labels

fsm_state, retry, status_code

- `nfmonitor_consul_read_success`

Type
64-bit Counter
Description
Howmany times Consul read attempt are succeeded
Labels

fsm_state, retry, status_code

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- [Packet Streaming Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Packet Streaming Metrics	num_packet_streaming_attempted	Counter	The total number of attempted streaming of packets
Packet Streaming Metrics	num_packet_streaming_success	Counter	The total number of packets streamed successfully
Packet Streaming Metrics	num_packet_streaming_failed	Counter	The total number of failed streaming of packets
Packet Streaming Metrics	num_packet_streaming_bytes	Counter	The total number of bytes successfully streamed

Packet Streaming Metrics

- num_packet_streaming_attempted

Type

64-bit Counter

Description

The total number of attempted streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_success

Type

64-bit Counter

Description

The total number of packets streamed successfully

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_failed

Type

64-bit Counter

Description

The total number of failed streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_bytes

Type

64-bit Counter

Description

The total number of bytes successfully streamed

Labels

interface_name, server_fqdn, kafka_topic

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- [Summary](#)
- [Circuit Breaker Metrics](#)
- [Service Endpoint Learner Metrics](#)
- [Service Endpoint Manager Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Circuit Breaker Metrics	sem_cb_endpoints	Gauge	Total number of endpoints for a service in respective states
Service Endpoint Learner Metrics	sem_num_dns_errors	Counter	Total number of DNS resolution errors for a given service

Group	Metric Name	Type	Synopsis
Service Endpoint Learner Metrics	sem_num_endpoints	Gauge	Total number of endpoints learnt for a service
Service Endpoint Manager Metrics	sem_num_endpoint_transactions_sent	Counter	Total number of times an endpoint is given to the application for communication with it
Service Endpoint Manager Metrics	sem_num_endpoint_transaction_status_received	Counter	Total number of transactions for a given endpoint received with the transaction status

Circuit Breaker Metrics

- [sem_cb_endpoints](#)

Type

64-bit Gauge

Description

Total number of endpoints for a service in respective states. There are 3 states viz. Active, Blocked, Restoring. In Active state, the endpoint is responding as expected, i.e. the response code doesn't indicate a service failure. The endpoint will be moved to Blocked state based on response failures which indicate service failure. After a timeout, the endpoint will be moved to Restoring state to check if the endpoint has recovered. In this state, if endpoint responds with response code other than service failure, it will be moved to Active state. Else, it will be moved back to Blocked state.

Labels

service_name, endpoint_state

Service Endpoint Learner Metrics

- [sem_num_dns_errors](#)

Type

64-bit Counter

Description

Total number of DNS resolution errors for a given service, labels are service name for DNS resolution and DNS error codes

Labels

service_name, error_code

- [sem_num_endpoints](#)

Type

64-bit Gauge

Description

Total number of endpoints learnt from the Kubernetes DNS server for a given service

Labels

service

Service Endpoint Manager Metrics

- [sem_num_endpoint_transactions_sent](#)

Type

64-bit Counter

Description

Total number of times an endpoint is given to the application for communication with it

Labels

service

- sem_num_endpoint_transaction_status_received

Type

64-bit Counter

Description

Total number of transactions for a given endpoint received with the transaction status

Labels

service, status

Table of Contents

- Summary
- UDSF DBClient Metrics
- UDSF API latency statistics

Summary

Group	Metric Name	Type	Synopsis
UDSF DBClient Metrics	udsf_db_operation_request	Counter	The total number of udsf dbclient requests received.
UDSF DBClient Metrics	udsf_db_operation_response	Counter	The total number of udsf dbclient responses sent.
UDSF DBClient Metrics	udsf_dbc_total_init_success	Counter	The total number of UDSF INIT successes.
UDSF DBClient Metrics	udsf_dbc_total_init_failure	Counter	The total number of UDSF INIT failures.
UDSF DBClient Metrics	udsf_dbc_total_deinit_success	Counter	The total number of UDSF DEINIT successes.
UDSF DBClient Metrics	udsf_dbc_total_deinit_failure	Counter	The total number of UDSF DEINIT failures.
UDSF DBClient Metrics	udsf_db_transaction_counter	Counter	The total number of CRUD operations.
UDSF API latency statistics	udsf_db_transaction_duration	Histogram	Roundtrip duration taken by transaction in seconds.
UDSF API latency statistics	udsf_db_execution_duration	Histogram	Execution time taken by transaction in microseconds.
UDSF API latency statistics	udsf_hiredis_bufferwrite_duration	Histogram	Duration of time taken in hiredis to write data to socket in seconds

UDSF DBClient Metrics

udsf_db_operation_request

Type

64-bit Counter

Description

The total number of udsf dbclient requests received from worker. Labels- 'app_name' : Originating thread name, ex:smfcc, 'operation_type' : Type of CRUD operation/API type involved, ex:CREATE_OR_MODIFY_RECORD_ASYNC

Labels

app_name, operation_type

udsf_db_operation_response

Type

64-bit Counter

Description

The total number of udsf dbclient responses sent. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE 'operation_type' : Type of CRUD operation/API type involved., ex:CREATE_OR_MODIFY_RECORD_ASYNC

Labels

app_name, status, operation_type

udsf_dbc_total_init_success
Type
64-bit Counter
Description
The total number of times UDSF API init was success. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: SUCCESS
Labels
<i>app_name, status</i>
udsf_dbc_total_init_failure
Type
64-bit Counter
Description
The total number of times UDSF API init failed. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: FAILURE
Labels
<i>app_name, status</i>
udsf_dbc_total_deinit_success
Type
64-bit Counter
Description
The total number of times UDSF API Deinit succeeded. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: SUCCESS
Labels
<i>app_name, status</i>
udsf_dbc_total_deinit_failure
Type
64-bit Counter
Description
The total number of times UDSF API Deinit failed. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: FAILURE
Labels
<i>app_name, status</i>
udsf_db_transaction_counter
Type
64-bit Counter
Description
The total number of times CRUD operations are incrementing. Labels- 'app_name' : Originating thread name, ex:smfcc, 'crud_op' : The categorical CRUD operation in which this sub-transation falls into, ex: CREATE_OR_MODIFY
Labels
<i>app_name, crud_op</i>

• udsf_db_transaction_duration

Type

64-bit Histogram

Description

Duration observed on total journey/roundtrip taken by a transaction after receiving the request. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE 'operation_type' : Type of CRUD operation/API type involved., ex:CREATE_OR_MODIFY_RECORD_ASYNC, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE, 'crud_op' : The categorical CRUD operation in which this transation falls into, ex: CREATE_OR_MODIFY

Labels

app_name,operation_type,status,crud_op

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

• udsf_db_execution_duration

Type

64-bit Histogram

Description

Processing time taken by UDSF client. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE 'operation_type' : Type of CRUD operation/API type involved., ex:CREATE_OR_MODIFY_RECORD_ASYNC, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE, 'crud_op' : The categorical CRUD operation in which this transation falls into, ex: CREATE_OR_MODIFY

Labels

app_name,operation_type,status,crud_op

Boundaries

[1, 2, 4, 8, 16, 32, 64, 128, 256, 384, 512, 768, 1024, 1536, 2048, 4096, 8192, 16384]

• udsf_hiredis_bufferwrite_duration

Type

64-bit Histogram

Description

Duration observed between hiredis buffer append and write to socket. This is fair approximation for time spent before data is written to socket as the buffer threshold accounted for append and write is dynamic and not fixed size. Per socket, the buffer append start-time is recorded along with size intended to transmit and end-time recorded once data greater than recorded size is written on the socket

Labels

app_name

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

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- [Summary](#)
- [AMF statistics](#)

Summary

Group	Metric Name	Type	Synopsis
AMF statistics	amf_registered_subscribers	Gauge	Current number of registered subscribers in AMF
AMF statistics	amf_registered_subscribers_idle	Gauge	Current number of registered subscribers in Idle State in AMF

Group	Metric Name	Type	Synopsis
AMF statistics	amf_registered_subscribers_connected	Gauge	Current number of registered subscribers in Connected State in AMF
AMF statistics	amf_registered_subscribers_with_session	Gauge	Current number of registered subscribers with at least one session in AMF
AMF statistics	amf_connected_gnb	Gauge	Current number of Connected GNBs in AMF
AMF statistics	amf_registered_tai	Gauge	Current number of registered TAIs in AMF
AMF statistics	amf_active_tnla	Gauge	Current number of Active TNLAs in AMF
AMF statistics	amf_connected_n3iwf	Gauge	Current number of Connected N3IWFs in AMF
AMF statistics	amf_connected_wagf	Gauge	Current number of Connected WAGFs in AMF
AMF statistics	amf_load_metric	Gauge	Current Calculated Load Metric for AMF

AMF statistics

- [amf_registered_subscribers](#)

Type64-bit GaugeDescriptionThe number of registered subscribers in AMFLabels
- [amf_registered_subscribers_idle](#)

Type64-bit GaugeDescriptionThe number of registered subscribers in Idle State in AMFLabels
- [amf_registered_subscribers_connected](#)

Type64-bit GaugeDescriptionThe number of registered subscribers in Connected State in AMFLabels
- [amf_registered_subscribers_with_session](#)

Type64-bit GaugeDescriptionThe number of registered subscribers with at least one session in AMFLabels
- [amf_connected_gnb](#)

Type64-bit Gauge

Description

The number of Connected GNBs in AMF

Labels

- amf_registered_tai

Type

64-bit Gauge

Description

The number of registered TAIs in AMF

Labels

- amf_active_tnla

Type

64-bit Gauge

Description

The number of Active TNLAs in AMF

Labels

- amf_connected_n3iwf

Type

64-bit Gauge

Description

The number of Connected N3IWFs in AMF

Labels

- amf_connected_wagf

Type

64-bit Gauge

Description

The number of Connected WAGFs in AMF

Labels

- amf_load_metric

Type

64-bit Gauge

Description

The calculated Load Metric for AMF

Labels

Table of Contents

- [Summary](#)
- [ChargingCmn statistics](#)

Summary

Group	Metric Name	Type	Synopsis
ChargingCmn statistics	svcmgr_service_available_ind_notification	Counter	How many times, Service Available Indication message received
ChargingCmn statistics	svcmgr_service_available_ind_parse_errs	Counter	How many times, Service Available Indication parse error happened
ChargingCmn statistics	svcmgr_service_available_ind_notification_ack	Counter	How many times, Service Available Indication message is acknowledged
ChargingCmn statistics	svcmgr_service_unavailable_ind_notification	Counter	How many times, Service Unavailable Indication message
ChargingCmn statistics	svcmgr_service_unavailable_ind_parse_errs	Counter	How many times, Service Unavailable Indication parse error happened
ChargingCmn statistics	svcmgr_service_unavailable_ind_notification_ack	Counter	How many times, Service Unavailable Indication message is acknowledged
ChargingCmn statistics	svcmgr_publish_msg_sent	Counter	How many times, Publication message sent
ChargingCmn statistics	svcmgr_publish_msg_success	Counter	How many times, Publication message success response received
ChargingCmn statistics	svcmgr_publish_msg_failure	Counter	How many times, failure received for publication message
ChargingCmn statistics	svcmgr_subscribe_msg_sent	Counter	How many times, Subscribe message sent
ChargingCmn statistics	svcmgr_subscribe_msg_success	Counter	How many times, subscribe message success
ChargingCmn statistics	svcmgr_subscribe_msg_failure	Counter	How many times, failure received for subscribe message
ChargingCmn statistics	svcmgr_unsubscribe_msg_sent	Counter	How many times, unsubscribe message sent
ChargingCmn statistics	svcmgr_unsubscribe_msg_failure	Counter	How many times, failure received for unsubscribe message
ChargingCmn statistics	svcmgr_config_req_sent	Counter	How many times, config request sent
ChargingCmn statistics	svcmgr_config_req_failure	Counter	How many times, failure received for config request
ChargingCmn statistics	svcmgr_config_req_success	Counter	How many times, success received for config request
ChargingCmn statistics	svcmgr_config_register_response	Counter	How many times, config register response received
ChargingCmn statistics	svcmgr_config_register_response_ack	Counter	How many times, config register response is acknowledged
ChargingCmn statistics	svcmgr_config_register_response_parse_err	Counter	How many times, parse error happened for config register response
ChargingCmn statistics	svcmgr_config_update_notification	Counter	How many times, config update notification received
ChargingCmn statistics	svcmgr_config_update_notification_response	Counter	How many times, config update response
ChargingCmn statistics	svcmgr_intf_schema_update_ind_notification	Counter	How many times, interface schema update indication message received
ChargingCmn statistics	svcmgr_intf_schema_update_ind_notification_ack	Counter	How many times, interface schema update indication message is acknowledged
ChargingCmn statistics	svcmgr_intf_schema_update_ind_parse_errs	Counter	How many times, interface schema update indication parse error happened

ChargingCmn statistics

- `svcmgr_service_available_ind_notification`

Type
64-bit Counter
Description
Number of Times Service Available Indication notification received
Labels
<i>servicename</i>

- `svcmgr_service_available_ind_parse_errs`

Type
64-bit Counter
Description
Number of Times Service Available Indication parse error happened
Labels

- `svcmgr_service_available_ind_notification_ack`

Type
64-bit Counter
Description
Number of Times Service Available Indication message is acknowledged
Labels
<i>servicename</i>

- `svcmgr_service_unavailable_ind_notification`

Type
64-bit Counter
Description
Number of Times Service Unavailable Indication notification received
Labels
<i>servicename</i>

- `svcmgr_service_unavailable_ind_parse_errs`

Type
64-bit Counter
Description
Number of Times Service Unavailable Indication parse error happened
Labels

- `svcmgr_service_unavailable_ind_notification_ack`

Type
64-bit Counter
Description
Number of Times Service Unavailable Indication message is acknowledged

Labels
<i>servicename</i>
• svcmgr_publish_msg_sent
Type
64-bit Counter
Description
Number of Times Publication message sent
Labels
<i>servicename, retry</i>
• svcmgr_publish_msg_success
Type
64-bit Counter
Description
Number of Times Publication message success response received
Labels
<i>servicename, causecode, retry</i>
• svcmgr_publish_msg_failure
Type
64-bit Counter
Description
Number of Times failure received for publication message
Labels
<i>servicename, causecode, retry</i>
• svcmgr_subscribe_msg_sent
Type
64-bit Counter
Description
Number of Times Subscribe message sent
Labels
<i>servicename, retry</i>
• svcmgr_subscribe_msg_success
Type
64-bit Counter
Description
Number of Times Send subscribe message success response received
Labels
<i>servicename, causecode, retry</i>
• svcmgr_subscribe_msg_failure
Type

64-bit Counter

Description

Number of time failure received for subscribe message

Labels

servicename, causecode, retry

- **svcmgr_unsubscribe_msg_sent**

Type

64-bit Counter

Description

Number of Times unsubscribe message sent

Labels

servicename

- **svcmgr_unsubscribe_msg_failure**

Type

64-bit Counter

Description

Number of Times failure received for unsubscribe message

Labels

servicename

- **svcmgr_config_req_sent**

Type

64-bit Counter

Description

Number of Times config request sent

Labels

retry

- **svcmgr_config_req_failure**

Type

64-bit Counter

Description

Number of Times failure received for config request

Labels

causecode, retry

- **svcmgr_config_req_success**

Type

64-bit Counter

Description

Number of Times success received for config request

Labels

causecode, retry

- `svcmgr_config_register_response`

Type

64-bit Counter

Description

Number of Times config register response received

Labels

statuscode

- `svcmgr_config_register_response_ack`

Type

64-bit Counter

Description

Number of Times config register response is acknowledged

Labels

- `svcmgr_config_register_response_parse_err`

Type

64-bit Counter

Description

Number of Times parse error happened for config register response

Labels

- `svcmgr_config_update_notification`

Type

64-bit Counter

Description

Number of Times config update notification received

Labels

- `svcmgr_config_update_notification_response`

Type

64-bit Counter

Description

Number of Times config update response sent

Labels

- `svcmgr_intf_schema_update_ind_notification`

Type

64-bit Counter

Description

Number of Times interface schema update indication message received

Labels

servicename

- `svcmgr_intf_schema_update_ind_notification_ack`

Type

64-bit Counter

Description

Number of Times interface schema update indication message is acknowledged

Labels

servicename

- `svcmgr_intf_schema_update_ind_parse_errs`

Type

64-bit Counter

Description

Number of Times Service interface schema update indication parse error happened

Labels

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- [Summary](#)
- [AMF/MME common statistics](#)

Summary

Group	Metric Name	Type	Synopsis
AMF/MME common statistics	ready_pods	Gauge	The number of ready pods in the federation.
AMF/MME common statistics	active_subscribers	Gauge	The number of attached and/or registered subscribers in the federation.

AMF/MME common statistics

- `ready_pods`

Type

64-bit Gauge

Description

The number of ready pods in the federation.

Labels

pod_type

- `active_subscribers`

Type

64-bit Gauge

Description

The number of attached and/or registered subscribers in the federation.

Labels

Table of Contents

- [Summary](#)
- [ETCD Client Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_put_requests	Counter	The total number of ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_requests_sent	Counter	The total number of ETCD Client PUT Requests sent.
ETCD Client Metrics	etcd_put_successes	Counter	The total number of successful ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_failures	Counter	The total number of failed ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_with_lease_requests	Counter	The total number of ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_requests_sent	Counter	The total number of ETCD Client PUT with Lease Requests sent.
ETCD Client Metrics	etcd_put_with_lease_successes	Counter	The total number of successful ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_failures	Counter	The total number of failed ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_range_requests	Counter	The total number of ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_requests_sent	Counter	The total number of ETCD Client Range Requests sent.
ETCD Client Metrics	etcd_range_successes	Counter	The total number of successful ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_failures	Counter	The total number of failed ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_delete_requests	Counter	The total number of ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_requests_sent	Counter	The total number of ETCD Client Range Delete Requests sent.
ETCD Client Metrics	etcd_range_delete_successes	Counter	The total number of successful ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_failures	Counter	The total number of failed ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_txn_requests	Counter	The total number of ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_requests_sent	Counter	The total number of ETCD Client Txn Requests sent.
ETCD Client Metrics	etcd_txn_successes	Counter	The total number of successful ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_failures	Counter	The total number of failed ETCD Client Txn Requests.
ETCD Client Metrics	etcd_watch_requests	Counter	The total number of ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_requests_sent	Counter	The total number of ETCD Client Watch Requests sent.
ETCD Client Metrics	etcd_watch_successes	Counter	The total number of successful ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_failures	Counter	The total number of failed ETCD Client Watch Requests.
ETCD Client Metrics	etcdwatcher_requests	Counter	The total number of ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_requests_sent	Counter	The total number of ETCD Watcher Requests sent.
ETCD Client Metrics	etcdwatcher_successes	Counter	The total number of successful ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_failures	Counter	The total number of failed ETCD Watcher Requests.

ETCD Client Metrics

- [etcd_put_requests](#)

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests.

Labels

keyPrefix

- [etcd_put_requests_sent](#)

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests sent.

Labels

keyPrefix

- etcd_put_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT Requests.

Labels

keyPrefix

- etcd_put_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT Requests.

Labels

keyPrefix, failureReason

- etcd_put_with_lease_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests sent.

Labels

keyPrefix

- etcd_put_with_lease_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT with Lease Requests.

Labels

keyPrefix, failureReason

- etcd_range_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests sent.

Labels

keyPrefix

- etcd_range_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Requests.

Labels

keyPrefix, failureReason

- etcd_range_delete_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests.

Labels

keyPrefix

- etcd_range_delete_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests sent.

Labels

keyPrefix

- etcd_range_delete_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Delete Requests.

Labels

keyPrefix

- etcd_range_delete_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Delete Requests.

Labels

keyPrefix, failureReason

- etcd_txn_requests

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests.

Labels

keyPrefix

- etcd_txn_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests sent.

Labels

keyPrefix

- etcd_txn_successes
 - Type
 - 64-bit Counter
 - Description
 - The total number of successful ETCD Client Txn Requests.
 - Labels
 - keyPrefix**
- etcd_txn_failures
 - Type
 - 64-bit Counter
 - Description
 - The total number of failed ETCD Client Txn Requests.
 - Labels
 - keyPrefix, failureReason**
- etcd_watch_requests
 - Type
 - 64-bit Counter
 - Description
 - The total number of ETCD Client Watch Requests.
 - Labels
 - keyPrefix**
- etcd_watch_requests_sent
 - Type
 - 64-bit Counter
 - Description
 - The total number of ETCD Client Watch Requests sent.
 - Labels
 - keyPrefix**
- etcd_watch_successes
 - Type
 - 64-bit Counter
 - Description
 - The total number of successful ETCD Client Watch Requests.
 - Labels
 - keyPrefix**
- etcd_watch_failures
 - Type
 - 64-bit Counter
 - Description

The total number of failed ETCD Client Watch Requests.

Labels

keyPrefix, failureReason

• etcdwatcher_requests

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests.

Labels

keyPrefix

• etcdwatcher_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests sent.

Labels

keyPrefix

• etcdwatcher_successes

Type

64-bit Counter

Description

The total number of successful ETCD Watcher Requests.

Labels

keyPrefix

• etcdwatcher_failures

Type

64-bit Counter

Description

The total number of failed ETCD Watcher Requests.

Labels

keyPrefix, failureReason

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- [Summary](#)
- [HTTP Punting Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_sent	Counter	The total number of HTTP requests successfully punted.
HTTP Punting Metrics	punting_http_requests_failed	Counter	The total number of HTTP requests punting failed.

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_responses_received	Counter	The total number of HTTP responses received for the requests punted.
HTTP Punting Metrics	punting_http_responses_forwarded	Counter	The total number of HTTP responses forwarded after punting transaction completion.
HTTP Punting Metrics	punting_http_requests_aborted	Counter	The total number of HTTP requests aborted.
HTTP Punting Metrics	punting_http_request_duration_seconds	Histogram	Punting operation duration in seconds.

HTTP Punting Metrics

- [punting_http_requests_sent](#)

Type

64-bit Counter

Description

The total number of HTTP requests successfully punted.

Labels

message_type

- [punting_http_requests_failed](#)

Type

64-bit Counter

Description

The total number of HTTP requests for which the punting failed.

Labels

message_type

- [punting_http_responses_received](#)

Type

64-bit Counter

Description

The total number of HTTP responses received for the requests punted.

Labels

message_type, status_code

- [punting_http_responses_forwarded](#)

Type

64-bit Counter

Description

The total number of HTTP responses forwarded after punting transaction completion.

Labels

message_type, status_code

- [punting_http_requests_aborted](#)

Type

64-bit Counter

Description

The total number of HTTP requests aborted.

Labels

message_type

- punting_http_request_duration_seconds

Type

64-bit Histogram

Description

The punting transaction duration is measured from the time the HTTP request is sent out to the time the response is received for the punting request.

Labels

message_type

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

Table of Contents

- [Summary](#)
- [MME statistics](#)

Summary

Group	Metric Name	Type	Synopsis
MME statistics	mme_attached_subscribers	Gauge	The number of attached subscribers.
MME statistics	mme_attached_subscribers_connected	Gauge	The number of attached subscribers in connected state.
MME statistics	mme_attached_subscribers_dcnr	Gauge	The number of attached DCNR capable subscribers.
MME statistics	mme_attached_subscribers_idle	Gauge	The number of attached subscribers in idle state.
MME statistics	mme_attached_subscribers_n1	Gauge	The number of attached N1 capable subscribers.
MME statistics	mme_attached_subscribers_visiting	Gauge	The number of attached visiting subscribers.
MME statistics	mme_attached_subscribers_combined	Gauge	The number of combined attached subscribers.
MME statistics	mme_connected_enb	Gauge	The number of eNBs connected to the MME.
MME statistics	mme_registered_tai	Gauge	The number of TAIs (Tracking Area Identifiers) learned by the MME.
MME statistics	mme_connected_enb_nbiot	Gauge	The number of NB-IoT capable eNBs connected to the MME.
MME statistics	mme_registered_tai_nbiot	Gauge	The number of NB-IoT TAIs (Tracking Area Identifiers) learned by the MME.
MME statistics	mme_load_metric	Gauge	The calculated load metric for MME.

MME statistics

- mme_attached_subscribers

Type

64-bit Gauge

Description

The number of attached subscribers.

Labels

- mme_attached_subscribers_connected

Type

64-bit Gauge

Description

The number of attached subscribers in connected state.

Labels

• mme_attached_subscribers_dcnr

Type

64-bit Gauge

Description

The number of attached DCNR capable subscribers. The UE network capability IE is indicated in the 4G NAS layer as defined by TS 24.301 section 9.9.3.34, where the DCNR bit indicates the capability for dual connectivity with NR (New Radio).

Labels

• mme_attached_subscribers_idle

Type

64-bit Gauge

Description

The number of attached subscribers in idle state.

Labels

• mme_attached_subscribers_n1

Type

64-bit Gauge

Description

The number of attached N1 capable subscribers. The UE network capability IE is indicated in the 4G NAS layer as defined by TS 24.301 section 9.9.3.34, where the UE supports N1 bit indicates the capability for N1 mode.

Labels

• mme_attached_subscribers_visiting

Type

64-bit Gauge

Description

The number of attached visiting subscribers.

Labels

ue_plmn

• mme_attached_subscribers_combined

Type

64-bit Gauge

Description

The number of combined attached subscribers.

Labels

mme_connected_enb
Type
64-bit Gauge
Description
The number of eNBs connected to the MME. A connected eNB is one that has one or more established SCTP connections to the MME and that has completed the S1AP setup procedure.
Labels
mme_registered_tai
Type
64-bit Gauge
Description
The number of TAIs (Tracking Area Identifiers) learned by the MME.
Labels
mme_connected_enb_nbiot
Type
64-bit Gauge
Description
The number of NB-IoT (Narrowband - Internet of Things) capable eNBs connected to the MME. A connected eNB is one that has one or more established SCTP connections to the MME and that has completed the S1AP setup procedure.
Labels
mme_registered_tai_nbiot
Type
64-bit Gauge
Description
The number of NB-IoT TAIs (Tracking Area Identifiers) learned by the MME.
Labels
mme_load_metric
Type
64-bit Gauge
Description
The calculated load metric for MME.
Labels

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Summary

Group	Metric Name	Type	Synopsis
NfMonitor statistics	nfmonitor_unstable_nf_counter	Counter	Howmany times, the Nf is found to be Unstable
NfMonitor statistics	nfmonitor_force_restart_due_to_nf_unstability	Counter	Numbe of times the VNFRestart triggered due to Unstability of the NF
NfMonitor statistics	nfmonitor_cumulative_readiness_failure_count	Counter	Numbe of times Service sound to have insuffucient count due to Readiness probe
NfMonitor statistics	nfmonitor_cumulative_readiness_failure_alerts	Counter	Numbe of times Service sound to have insuffucient count due to Readiness probe
NfMonitor statistics	nfmonitor_cumulative_k8api_get_query_failures	Counter	Numbe of times the k8 get Api query failures
DebugNfMonitorCmn	nfmonitor_force_restart_counter	Counter	Number of times force-restart triggered
NfManagement statistics	nfmonitor_nfmgmt_service_request_attempts	Counter	Howmany times, the Nf is attempting to send NRF Requests
NfManagement statistics	nfmonitor_nfmgmt_service_request_success	Counter	Numbe of times Success response is received for the requests to NRF
NfManagement statistics	nfmonitor_nfmgmt_service_request_failures	Counter	Numbe of times Failure response is received for the requests to NRF
NfManagement statistics	nfmonitor_nfmgmt_aborted_request_msgs	Counter	Numbe of times the requests to NRF is Aborted/Cancelled
Debug statistics	nfmonitor_vnf_restart_trigger	Counter	Howmany times, the Force Restart is received
Debug statistics	nfmonitor_fsm_state_changes	Counter	The statistics for the the FSM state changes
Debug statistics	nfmonitor_deactivation_triggers	Counter	Numbe of times the NF deactivation trigger is received
Debug statistics	nfmonitor_critical_service_up_notification	Counter	Numbe of times the Critical Service Up Notification received
Debug statistics	nfmonitor_critical_service_down_notification	Counter	Numbe of times the Critical Service Down Notification received
Debug statistics	nfmonitor_critical_services_overall_up_notification	Counter	Numbe of times the Up notification received when all monitored Critical Services are available
Debug statistics	nfmonitor_critical_services_overall_down_notification	Counter	Numbe of times the Down notification received when at least one Critical Service was unavailable
Debug statistics	nfmonitor_hysterisis_timer_started	Counter	Numbe of times the Hysterisis Timer is started due to missing Critical Service
Debug statistics	nfmonitor_hysterisis_timer_stopped	Counter	Numbe of times the Hysterisis Timer is stopped due to Critical Service become available
Debug statistics	nfmonitor_hysterisis_timer_expired	Counter	Numbe of times the Hysterisis Timer is expired due to missing Critical Service
Debug statistics	nfmonitor_personality_update_triggers	Counter	Numbe of times the Nf Personality update is received
Debug statistics	nfmonitor_maintenance_trigger_received	Counter	Numbe of times the Maintenance Mode is activated
Debug statistics	nfmonitor_profile_update_triggers	Counter	Numbe of times the Nf Profile Update happened
Debug statistics	nfmonitor_critical_service_down_on_register_entry	Counter	Numbe of times the Critical Services were down while starting with RegisterFsmState
ApiServer statistics	nfmonitor_deployment_create_watch_request	Counter	Howmany times, the deployment watch requests are created
ApiServer statistics	nfmonitor_deployment_delete_watch_request	Counter	Howmany times, the deployment watch requests are deleted
ApiServer statistics	nfmonitor_deployment_watch_notification_success	Counter	Howmany times, the watch notifications are successful
ApiServer statistics	nfmonitor_deployment_watch_notification_error	Counter	Howmany times, the watch notifications are failed
ApiServer statistics	nfmonitor_limit_range_query_attempts	Counter	Howmany times, the limit-range query attempts are made
ApiServer statistics	nfmonitor_limit_range_query_failures	Counter	Howmany times, the limit-range query attempts are failed
ApiServer statistics	nfmonitor_limit_range_query_success	Counter	Howmany times, the limit-range query attempts are succeeded
ApiServer statistics	nfmonitor_hpa_query_attempts	Counter	Howmany times, the Hpa query attempts are made
ApiServer statistics	nfmonitor_hpa_query_failures	Counter	Howmany times, the Hpa query attempts are failed
ApiServer statistics	nfmonitor_hpa_query_success	Counter	Howmany times, the Hpa query attempts are succeeded

Group	Metric Name	Type	Synopsis
ApiServer statistics	nfmonitor_readiness_probe_query_attempts	Counter	Howmany times, the Readiness Probe query attempts are made
ApiServer statistics	nfmonitor_readiness_probe_query_failures	Counter	Howmany times, the Readiness Probe query attempts are failed
ApiServer statistics	nfmonitor_readiness_probe_query_success	Counter	Howmany times, the Readiness Probe query attempts are succeeded
Feature Functionality statistics	nfmonitor_critical_service_frequent_restart	Counter	Howmany times, minimum count is not met for the stable Critical Services
Feature Functionality statistics	nfmonitor_alarm_due_to_frequent_restart	Counter	Howmany times alarm raised, due to minimum count is not met for the stable Critical Services
Feature Functionality statistics	nfmonitor_force_restart_due_to_frequent_restart	Counter	Howmany times, force restart triggerd because of the unstable Critical Services
Feature Functionality statistics	nfmonitor_critical_service_readiness_failure	Counter	Howmany times, minimum count is not met for the readiness succeeded Critical Services
Feature Functionality statistics	nfmonitor_alarm_due_to_readiness_failure	Counter	Alarm raised, due to minimum count is not met for the readiness succeeded Critical Services
Feature Functionality statistics	nfmonitor_force_restart_due_to_readiness_failure	Counter	Force restart triggerd because of Critical Service Readiness failure
Consul statistics	nfmonitor_publish_status_into_consul	Counter	Number of times Publish message is sent to Consul
Consul statistics	nfmonitor_consul_read_attempts	Counter	Number of times Consul read attempts
Consul statistics	nfmonitor_consul_read_failures	Counter	Number of times Consul read attempts failed
Consul statistics	nfmonitor_consul_read_success	Counter	Number of times Consul read attempts succeeded

NfMonitor statistics

- [nfmonitor_unstable_nf_counter](#)

Type

64-bit Counter

Description

Number of Times Nf is found Unstable

Labels

- [nfmonitor_force_restart_due_to_nf_unstability](#)

Type

64-bit Counter

Description

The number of times the Force Restart triggered due to NF-Unstability

Labels

- [nfmonitor_cumulative_readiness_failure_count](#)

Type

64-bit Counter

Description

The number of times the service is found to be failing the Readiness probe

Labels

critical_service_name, pod_name

- [nfmonitor_cumulative_readiness_failure_alerts](#)

Type

64-bit Counter

Description

Total number of EventAlerts generated due to Readiness probe failure

Labels

critical_service_name, pod_name

- nfmonitor_cumulative_k8api_get_query_failures

Type

64-bit Counter

Description

Total number of k8 api get pod query failures

Labels

critical_service_name, pod_name

DebugNfMonitorCmn

- nfmonitor_force_restart_counter

Type

64-bit Counter

Description

Number of times force-restart triggered

Labels

agent, name

NfManagement statistics

- nfmonitor_nfmgmt_service_request_attempts

Type

64-bit Counter

Description

Number of Requests sent to NRF

Labels

instance_id, msg_type, method, retry

- nfmonitor_nfmgmt_service_request_success

Type

64-bit Counter

Description

The number of times the Request to NRF is Successful

Labels

instance_id, msg_type, method, retry, status_code

- nfmonitor_nfmgmt_service_request_failures

Type

64-bit Counter

Description

The number of times the Request to NRF is Failure

Labels

instance_id, msg_type, method, retry, status_code

- nfmonitor_nfmgmt_aborted_request_msgs

Type

64-bit Counter

Description

The number of times the Request to NRF is Aborted

Labels

instance_id, msg_type

Debug statistics

- nfmonitor_vnf_restart_trigger

Type

64-bit Counter

Description

Number of VNF Restart triggers received

Labels

curr_nf_status, nfmgmt_state, nfmonitor_fsm_state, up_count, timestamp, trigger, type

- nfmonitor_fsm_state_changes

Type

64-bit Counter

Description

The FSM State Change events stats

Labels

curr_nf_status, nfmgmt_state, up_count, from_state, to_state

- nfmonitor_deactivation_triggers

Type

64-bit Counter

Description

The number of times the deactivation trigger received

Labels

nf_status, nfmgmt_state, up_count, nfmonitor_fsm_state

- nfmonitor_critical_service_up_notification

Type

64-bit Counter

Description

The number of times the CriticalService Up notification received

Labels

service, criticality, nf_status, nfmgmt_state, up_count, fsm_state

• nfmonitor_critical_service_down_notification

Type

64-bit Counter

Description

The number of times the CriticalService Down notification received

Labels

service, criticality, nf_status, nfmgmt_state, up_count, fsm_state

• nfmonitor_critical_services_overall_up_notification

Type

64-bit Counter

Description

The number of times the Overall CriticalServices Up notification received

Labels

nf_status, nfmgmt_state, up_count, fsm_state

• nfmonitor_critical_services_overall_down_notification

Type

64-bit Counter

Description

The number of times the Overall CriticalServices Down notification received

Labels

nf_status, nfmgmt_state, up_count, fsm_state

• nfmonitor_hysteresis_timer_started

Type

64-bit Counter

Description

The number of times the Hysteresis Timer is started

Labels

nf_status, nfmgmt_state, up_count, fsm_state

• nfmonitor_hysteresis_timer_stopped

Type

64-bit Counter

Description

The number of times the Hysteresis Timer is stopped

Labels

nf_status, nfmgmt_state, up_count, fsm_state

• nfmonitor_hysteresis_timer_expired

Type

64-bit Counter

Description

The number of times the Hysterisis Timer is expired

Labels

nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_personality_update_triggers

Type

64-bit Counter

Description

The number of times the Nf Personality Update is received

Labels

nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_maintenance_trigger_received

Type

64-bit Counter

Description

The number of times the Maintenance Mode is activated

Labels

nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_profile_update_triggers

Type

64-bit Counter

Description

The number of times the Nf Profile Update happened

Labels

nf_status, nfmgmt_state, up_count, fsm_state, update_operation

- nfmonitor_critical_service_down_on_register_entry

Type

64-bit Counter

Description

The number of times the Critical Services were down while starting with RegisterFsmState

Labels

nf_status, up_count

ApiServer statistics

- nfmonitor_deployment_create_watch_request

Type

64-bit Counter

Description

The number of Watch Reqs created for deployment type

Labels

resource

- nfmonitor_deployment_delete_watch_request

Type

64-bit Counter

Description

The number of Watch Reqs deleted for deployment type

Labels

resource

- nfmonitor_deployment_watch_notification_success

Type

64-bit Counter

Description

The number of times Watch notifications are successful

Labels

resource

- nfmonitor_deployment_watch_notification_error

Type

64-bit Counter

Description

The number of times Watch notifications are failed

Labels

resource, reason

- nfmonitor_limit_range_query_attempts

Type

64-bit Counter

Description

The number of Limit Range query attempts

Labels

-
- nfmonitor_limit_range_query_failures

Type

64-bit Counter

Description

The number of Limit Range query failures

Labels

status_code

- nfmonitor_limit_range_query_success

Type

64-bit Counter

Description

The number of Limit Range query succeeded

Labels

status_code

• nfmonitor_hpa_query_attempts

Type

64-bit Counter

Description

The number of Hpa query attempts

Labels

resource, continue_param

• nfmonitor_hpa_query_failures

Type

64-bit Counter

Description

The number of Hpa query failures

Labels

resource, continue_param, status_code

• nfmonitor_hpa_query_success

Type

64-bit Counter

Description

The number of Hpa query succeeded

Labels

resource, continue_param, status_code

• nfmonitor_readiness_probe_query_attempts

Type

64-bit Counter

Description

The number of Readiness Probe query attempts

Labels

resource, continue_param

• nfmonitor_readiness_probe_query_failures

Type

64-bit Counter

Description

The number of Readiness Probe query failures

Labels

resource, continue_param, status_code

• nfmonitor_readiness_probe_query_success

Type

64-bit Counter

Description

The number of Readiness Probe query succeeded

Labels

resource, continue_param, status_code

Feature Functionality statistics

- nfmonitor_critical_service_frequent_restart

Type

64-bit Counter

Description

Howmany times the minimum required stable CriticalService count is not met

Labels

service

- nfmonitor_alarm_due_to_frequent_restart

Type

64-bit Counter

Description

Howmany times Alarm is raised due to minimum required stable CriticalService count is not met

Labels

service, alarm_status

- nfmonitor_force_restart_due_to_frequent_restart

Type

64-bit Counter

Description

Howmany times Force Restart initiated due to missing stable CriticalService count

Labels

service

- nfmonitor_critical_service_readiness_failure

Type

64-bit Counter

Description

Howmany times the minimum required readiness succeeded CriticalService count is not met

Labels

service

- nfmonitor_alarm_due_to_readiness_failure

Type

64-bit Counter

Description

Alarm raised due to minimum count is not met for the readiness succeeded CriticalService

Labels

service, alarm_status

- nfmonitor_force_restart_due_to_readiness_failure

Type

64-bit Counter

Description

Howmany times Force Restart initiated due to readiness failure of CriticalService

Labels

service

Consul statistics

- nfmonitor_publish_status_into_consul

Type

64-bit Counter

Description

Howmany times Publish is sent to Consul

Labels

nf_status, nf_state, up_count, fsm_state

- nfmonitor_consul_read_attempts

Type

64-bit Counter

Description

Howmany times Consul read attempt is made

Labels

fsm_state, retry

- nfmonitor_consul_read_failures

Type

64-bit Counter

Description

Howmany times Consul read attempt are failed

Labels

fsm_state, retry, status_code

- nfmonitor_consul_read_success

Type

64-bit Counter

Description

Howmany times Consul read attempt are succeeded

Labels

fsm_state, retry, status_code

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- [Packet Streaming Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Packet Streaming Metrics	num_packet_streaming_attempted	Counter	The total number of attempted streaming of packets
Packet Streaming Metrics	num_packet_streaming_success	Counter	The total number of packets streamed successfully
Packet Streaming Metrics	num_packet_streaming_failed	Counter	The total number of failed streaming of packets
Packet Streaming Metrics	num_packet_streaming_bytes	Counter	The total number of bytes successfully streamed

Packet Streaming Metrics

- num_packet_streaming_attempted

Type

64-bit Counter

Description

The total number of attempted streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_success

Type

64-bit Counter

Description

The total number of packets streamed successfully

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_failed

Type

64-bit Counter

Description

The total number of failed streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_bytes

Type

64-bit Counter

Description

The total number of bytes successfully streamed

Labels

interface_name, server_fqdn, kafka_topic

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- [Summary](#)
- [Circuit Breaker Metrics](#)
- [Service Endpoint Learner Metrics](#)
- [Service Endpoint Manager Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Circuit Breaker Metrics	sem_cb_endpoints	Gauge	Total number of endpoints for a service in respective states
Service Endpoint Learner Metrics	sem_num_dns_errors	Counter	Total number of DNS resolution errors for a given service
Service Endpoint Learner Metrics	sem_num_endpoints	Gauge	Total number of endpoints learnt for a service
Service Endpoint Manager Metrics	sem_num_endpoint_transactions_sent	Counter	Total number of times an endpoint is given to the application for communication with it
Service Endpoint Manager Metrics	sem_num_endpoint_transaction_status_received	Counter	Total number of transactions for a given endpoint received with the transaction status

Circuit Breaker Metrics

• [sem_cb_endpoints](#)

Type

64-bit Gauge

Description

Total number of endpoints for a service in respective states. There are 3 states viz. Active, Blocked, Restoring. In Active state, the endpoint is responding as expected, i.e. the response code doesn't indicate a service failure. The endpoint will be moved to Blocked state based on response failures which indicate service failure. After a timeout, the endpoint will be moved to Restoring state to check if the endpoint has recovered. In this state, if endpoint responds with response code other than service failure, it will be moved to Active state. Else, it will be moved back to Blocked state.

Labels

service_name, endpoint_state

Service Endpoint Learner Metrics

• [sem_num_dns_errors](#)

Type

64-bit Counter

Description

Total number of DNS resolution errors for a given service, labels are service name for DNS resolution and DNS error codes

Labels

service_name, error_code

• [sem_num_endpoints](#)

Type

64-bit Gauge

Description

Total number of endpoints learnt from the Kubernetes DNS server for a given service

Labels

service

Service Endpoint Manager Metrics

- sem_num_endpoint_transactions_sent

Type

64-bit Counter

Description

Total number of times an endpoint is given to the application for communication with it

Labels

service

- sem_num_endpoint_transaction_status_received

Type

64-bit Counter

Description

Total number of transactions for a given endpoint received with the transaction status

Labels

service, status

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- [Summary](#)
- [N2 statistics](#)

Summary

Group	Metric Name	Type	Synopsis
N2 statistics	amfn2_n2_uplink_message	Counter	Uplink message for AMF N2 Interface
N2 statistics	amfn2_n2_downlink_message	Counter	Downlink message for AMF N2 Interface
N2 statistics	amfn2_n2_ng_setup_request	Counter	Uplink NG Setup Request for AMF N2 Interface
N2 statistics	amfn2_n2_ng_setup_response	Counter	Downlink NG Setup Response for AMF N2 Interface
N2 statistics	amfn2_n2_ng_setup_failure	Counter	Downlink NG Setup Failure for AMF N2 Interface
N2 statistics	amfn2_n2_ng_reset_rx	Counter	Uplink NG reset for AMF N2 Interface
N2 statistics	amfn2_n2_ng_reset_tx	Counter	Downlink NG reset for AMF N2 Interface
N2 statistics	amfn2_n2_ng_reset_ack_rx	Counter	Uplink NG reset acknowledge for AMF N2 Interface
N2 statistics	amfn2_n2_ng_reset_ack_tx	Counter	Downlink NG reset acknowledge for AMF N2 Interface
N2 statistics	amfn2_n2_initial_ue_message	Counter	Uplink Initial UE Message for AMF N2 Interface
N2 statistics	amfn2_n2_uplink_nas_transport	Counter	Uplink
N2 statistics	amfn2_n2_downlink_nas_transport	Counter	Downlink
N2 statistics	amfn2_n2_nas_non_delivery_indication	Counter	Uplink
N2 statistics	amfn2_n2_initial_context_setup_request	Counter	Downlink
N2 statistics	amfn2_n2_initial_context_setup_response	Counter	Uplink
N2 statistics	amfn2_n2_initial_context_setup_failure	Counter	Uplink
N2 statistics	amfn2_n2_ue_context_release_request	Counter	Uplink
N2 statistics	amfn2_n2_ue_context_release_command	Counter	Downlink
N2 statistics	amfn2_n2_ue_context_release_complete	Counter	Uplink
N2 statistics	amfn2_n2_ue_context_modification_request	Counter	Downlink
N2 statistics	amfn2_n2_ue_context_modification_response	Counter	Uplink

Group	Metric Name	Type	Synopsis
N2 statistics	amfn2_n2_ue_context_modification_failure	Counter	Uplink
N2 statistics	amfn2_n2_pdu_session_resource_setup_request	Counter	Downlink
N2 statistics	amfn2_n2_pdu_session_resource_setup_response	Counter	Uplink
N2 statistics	amfn2_n2_pdu_session_resource_modify_request	Counter	Downlink
N2 statistics	amfn2_n2_pdu_session_resource_modify_response	Counter	Uplink
N2 statistics	amfn2_n2_pdu_session_resource_release_command	Counter	Downlink
N2 statistics	amfn2_n2_pdu_session_resource_release_response	Counter	Uplink
N2 statistics	amfn2_n2_pdu_session_resource_notify	Counter	Uplink
N2 statistics	amfn2_n2_pdu_session_resource_modify_indication	Counter	Uplink
N2 statistics	amfn2_n2_pdu_session_resource_modify_confirm	Counter	Downlink
N2 statistics	amfn2_n2_uplink_ue_associated_nrppa_transport	Counter	Uplink
N2 statistics	amfn2_n2_uplink_non_ue_associated_nrppa_transport	Counter	Uplink
N2 statistics	amfn2_n2_downlink_ue_associated_nrppa_transport	Counter	Downlink
N2 statistics	amfn2_n2_downlink_non_ue_associated_nrppa_transport	Counter	Downlink
N2 statistics	amfn2_n2_write_replace_warning_request	Counter	Downlink
N2 statistics	amfn2_n2_write_replace_warning_response	Counter	Uplink
N2 statistics	amfn2_n2_pws_cancel_request	Counter	Downlink
N2 statistics	amfn2_n2_pws_cancel_response	Counter	Uplink
N2 statistics	amfn2_n2_pws_restart_indication	Counter	Uplink
N2 statistics	amfn2_n2_pws_failure_indiction	Counter	Uplink
N2 statistics	amfn2_n2_message_unsupported_uplink	Counter	Uplink
N2 statistics	amfn2_n2_message_unsupported_downlink	Counter	Downlink
N2 statistics	amfn2_n2_message_error_uplink	Counter	Uplink
N2 statistics	amfn2_n2_message_error_downlink	Counter	Downlink
N2 statistics	amfn2_n2_error_indication_rx	Counter	Uplink
N2 statistics	amfn2_n2_error_indication_tx	Counter	Downlink
N2 statistics	amfn2_n2_ue_radio_capability_info_indication	Counter	Uplink
N2 statistics	amfn2_n2_handover_required	Counter	Uplink
N2 statistics	amfn2_n2_handover_command	Counter	Downlink
N2 statistics	amfn2_n2_handover_prep_failure	Counter	Downlink
N2 statistics	amfn2_n2_handover_request	Counter	Downlink
N2 statistics	amfn2_n2_handover_request_ack	Counter	Uplink
N2 statistics	amfn2_n2_handover_failure	Counter	Uplink
N2 statistics	amfn2_n2_handover_notify	Counter	Uplink
N2 statistics	amfn2_n2_handover_cancel	Counter	Uplink
N2 statistics	amfn2_n2_handover_cancel_ack	Counter	Downlink
N2 statistics	amfn2_n2_pathswitch_request	Counter	Uplink
N2 statistics	amfn2_n2_pathswitch_request_ack	Counter	Downlink
N2 statistics	amfn2_n2_pathswitch_request_failure	Counter	Downlink
N2 statistics	amfn2_n2_uplink_ran_status_transfer	Counter	Uplink
N2 statistics	amfn2_n2_downlink_ran_status_transfer	Counter	Downlink
N2 statistics	amfn2_n2_uplink_ran_config_transfer	Counter	Uplink
N2 statistics	amfn2_n2_downlink_ran_config_transfer	Counter	Downlink
N2 statistics	amfn2_n2_paging	Counter	Downlink
N2 statistics	amfn2_ran_config_update	Counter	Uplink

Group	Metric Name	Type	Synopsis
N2 statistics	amfn2_ran_config_update_ack	Counter	Downlink
N2 statistics	amfn2_ran_config_update_failure	Counter	Downlink
N2 statistics	amfn2_amf_config_update	Counter	Downlink
N2 statistics	amfn2_amf_config_update_ack	Counter	Uplink
N2 statistics	amfn2_amf_config_update_failure	Counter	Uplink
N2 statistics	amfn2_reroute_nas_request	Counter	Downlink
N2 statistics	amfn2_trace_start	Counter	Downlink
N2 statistics	amfn2_trace_failure_indication	Counter	Uplink
N2 statistics	amfn2_deactivate_trace	Counter	Downlink
N2 statistics	amfn2_active_tnla_per_pod	Gauge	The number of active TNLAs per AMF_N2 pod

N2 statistics

- [amfn2_n2_uplink_message](#)

Type

64-bit Counter

Description

N2 uplink message counter

Labels

- [amfn2_n2_downlink_message](#)

Type

64-bit Counter

Description

N2 downlink message counter

Labels

- [amfn2_n2_ng_setup_request](#)

Type

64-bit Counter

Description

NG setup request counter

Labels

tnla, globalrannodeid

- [amfn2_n2_ng_setup_response](#)

Type

64-bit Counter

Description

NG setup response counter

Labels

tnla, globalrannodeid

- [amfn2_n2_ng_setup_failure](#)

Type

64-bit Counter

Description

NG setup failures counter

Labels

tnla, cause, globalrannodeid

• amfn2_n2_ng_reset_rx

Type

64-bit Counter

Description

NG reset counter

Labels

tnla, globalrannodeid

• amfn2_n2_ng_reset_tx

Type

64-bit Counter

Description

NG reset counter

Labels

tnla, globalrannodeid

• amfn2_n2_ng_reset_ack_rx

Type

64-bit Counter

Description

Uplink NG reset acknowledge counter

Labels

tnla, globalrannodeid

• amfn2_n2_ng_reset_ack_tx

Type

64-bit Counter

Description

Downlink NG reset acknowledge counter

Labels

tnla, globalrannodeid

• amfn2_n2_initial_ue_message

Type

64-bit Counter

Description

Initial UE message counter

Labels
<i>tnla, globalrannodeid</i>
amfn2_n2_uplink_nas_transport
Type
64-bit Counter
Description
Uplink Nas transport
Labels
<i>tnla, globalrannodeid</i>
amfn2_n2_downlink_nas_transport
Type
64-bit Counter
Description
Downlink Nas transport
Labels
<i>tnla, globalrannodeid</i>
amfn2_n2_nas_non_delivery_indication
Type
64-bit Counter
Description
Uplink Nas non delivery indication
Labels
<i>tnla, globalrannodeid</i>
amfn2_n2_initial_context_setup_request
Type
64-bit Counter
Description
Initial context setup request counter
Labels
<i>tnla, globalrannodeid</i>
amfn2_n2_initial_context_setup_response
Type
64-bit Counter
Description
Initial context setup response counter
Labels
<i>tnla, globalrannodeid</i>
amfn2_n2_initial_context_setup_failure
Type

64-bit Counter

Description

Initial context setup failure counter

Labels

tnla, cause, globalrannodeid

• amfn2_n2_ue_context_release_request

Type

64-bit Counter

Description

UE context release request counter

Labels

tnla, globalrannodeid

• amfn2_n2_ue_context_release_command

Type

64-bit Counter

Description

UE context release command counter

Labels

tnla, globalrannodeid

• amfn2_n2_ue_context_release_complete

Type

64-bit Counter

Description

UE context release complete counter

Labels

tnla, globalrannodeid

• amfn2_n2_ue_context_modification_request

Type

64-bit Counter

Description

UE context modification request counter

Labels

tnla, globalrannodeid

• amfn2_n2_ue_context_modification_response

Type

64-bit Counter

Description

UE context modification response counter

Labels

tnla, globalrannodeid

- amfn2_n2_ue_context_modification_failure

Type

64-bit Counter

Description

UE context modification failure counter

Labels

tnla, cause, globalrannodeid

- amfn2_n2_pdu_session_resource_setup_request

Type

64-bit Counter

Description

PDU session resource setup request counter

Labels

tnla, globalrannodeid

- amfn2_n2_pdu_session_resource_setup_response

Type

64-bit Counter

Description

PDU session resource setup response counter

Labels

tnla, globalrannodeid

- amfn2_n2_pdu_session_resource_modify_request

Type

64-bit Counter

Description

PDU session resource modify request counter

Labels

tnla, globalrannodeid

- amfn2_n2_pdu_session_resource_modify_response

Type

64-bit Counter

Description

PDU session resource modify response counter

Labels

tnla, globalrannodeid

- amfn2_n2_pdu_session_resource_release_command

Type

64-bit Counter

Description

PDU session resource release request counter

Labels

tnla, globalrannodeid

- amfn2_n2_pdu_session_resource_release_response

Type

64-bit Counter

Description

PDU session resource release response counter

Labels

tnla, globalrannodeid

- amfn2_n2_pdu_session_resource_notify

Type

64-bit Counter

Description

PDU session resource notify counter

Labels

tnla, globalrannodeid

- amfn2_n2_pdu_session_resource_modify_indication

Type

64-bit Counter

Description

PDU session resource modify indication counter

Labels

tnla, globalrannodeid

- amfn2_n2_pdu_session_resource_modify_confirm

Type

64-bit Counter

Description

PDU session resource modify confirm counter

Labels

tnla, globalrannodeid

- amfn2_n2_uplink_ue_associated_nrppa_transport

Type

64-bit Counter

Description

Uplink UE associated NRPPa transport

Labels

tnla, globalrannodeid

• amfn2_n2_uplink_non_ue_associated_nrppa_transport
Type
64-bit Counter
Description
Uplink non UE associated NRPPa transport
Labels
<i>tnla, globalrannodeid</i>
• amfn2_n2_downlink_ue_associated_nrppa_transport
Type
64-bit Counter
Description
Downlink UE associated NRPPa transport
Labels
<i>tnla, globalrannodeid</i>
• amfn2_n2_downlink_non_ue_associated_nrppa_transport
Type
64-bit Counter
Description
Downlink non UE associated NRPPa transport
Labels
<i>tnla, globalrannodeid</i>
• amfn2_n2_write_replace_warning_request
Type
64-bit Counter
Description
PWS write replace warning request counter
Labels
<i>tnla, globalrannodeid, msg_id</i>
• amfn2_n2_write_replace_warning_response
Type
64-bit Counter
Description
PWS write replace warning response counter
Labels
<i>tnla, globalrannodeid, msg_id</i>
• amfn2_n2_pws_cancel_request
Type
64-bit Counter
Description

PWS cancel request counter

Labels

tnla, globalrannodeid, msg_id

• amfn2_n2_pws_cancel_response

Type

64-bit Counter

Description

PWS cancel response counter

Labels

tnla, globalrannodeid, msg_id

• amfn2_n2_pws_restart_indication

Type

64-bit Counter

Description

PWS restart indication counter

Labels

tnla, globalrannodeid

• amfn2_n2_pws_failure_indiction

Type

64-bit Counter

Description

PWS failure indiction counter

Labels

tnla, globalrannodeid

• amfn2_n2_message_unsupported_uplink

Type

64-bit Counter

Description

Message unsupported uplink counter

Labels

tnla, type, procedure_code, globalrannodeid

• amfn2_n2_message_unsupported_downlink

Type

64-bit Counter

Description

Message unsupported downlink counter

Labels

tnla, type, procedure_code, globalrannodeid

• amfn2_n2_message_error_uplink

Type

64-bit Counter

Description

Message error uplink counter

Labels

tnla, type, procedure_code, globalrannodeid

- amfn2_n2_message_error_downlink

Type

64-bit Counter

Description

Message error downlink counter

Labels

tnla, type, procedure_code, globalrannodeid

- amfn2_n2_error_indication_rx

Type

64-bit Counter

Description

Uplink Error indication counter

Labels

tnla, type, procedure_code, globalrannodeid

- amfn2_n2_error_indication_tx

Type

64-bit Counter

Description

Downlink Error indication counter

Labels

tnla, type, procedure_code, globalrannodeid

- amfn2_n2_ue_radio_capability_info_indication

Type

64-bit Counter

Description

UE radio capability info indication counter

Labels

tnla, globalrannodeid

- amfn2_n2_handover_required

Type

64-bit Counter

Description

Handover required counter

Labels
<i>tnla, globalrannodeid</i>
• amfn2_n2_handover_command
Type
64-bit Counter
Description
Handover command counter
Labels
<i>tnla, globalrannodeid</i>
• amfn2_n2_handover_prep_failure
Type
64-bit Counter
Description
Handover prep failure counter
Labels
<i>tnla, globalrannodeid</i>
• amfn2_n2_handover_request
Type
64-bit Counter
Description
Handover request counter
Labels
<i>tnla, globalrannodeid</i>
• amfn2_n2_handover_request_ack
Type
64-bit Counter
Description
Handover request acknowledge counter
Labels
<i>tnla, globalrannodeid</i>
• amfn2_n2_handover_failure
Type
64-bit Counter
Description
Handover failure counter
Labels
<i>tnla, globalrannodeid</i>
• amfn2_n2_handover_notify
Type

64-bit Counter

Description

Handover notify counter

Labels

tnla, globalrannodeid

• amfn2_n2_handover_cancel

Type

64-bit Counter

Description

Handover cancel counter

Labels

tnla, globalrannodeid

• amfn2_n2_handover_cancel_ack

Type

64-bit Counter

Description

Handover cancel acknowledge counter

Labels

tnla, globalrannodeid

• amfn2_n2_pathswitch_request

Type

64-bit Counter

Description

Pathswitch request counter

Labels

tnla, globalrannodeid

• amfn2_n2_pathswitch_request_ack

Type

64-bit Counter

Description

Pathswitch request acknowledge counter

Labels

tnla, globalrannodeid

• amfn2_n2_pathswitch_request_failure

Type

64-bit Counter

Description

Pathswitch request failure counter

Labels

tnla, globalrannodeid

- amfn2_n2_uplink_ran_status_transfer

Type

64-bit Counter

Description

Uplink RAN status transfer counter

Labels

tnla, globalrannodeid

- amfn2_n2_downlink_ran_status_transfer

Type

64-bit Counter

Description

Downlink RAN status transfer counter

Labels

tnla, globalrannodeid

- amfn2_n2_uplink_ran_config_transfer

Type

64-bit Counter

Description

Uplink RAN configuration transfer counter

Labels

tnla, globalrannodeid

- amfn2_n2_downlink_ran_config_transfer

Type

64-bit Counter

Description

Downlink RAN configuration transfer counter

Labels

tnla, globalrannodeid

- amfn2_n2_paging

Type

64-bit Counter

Description

Paging counter

Labels

tnla, globalrannodeid

- amfn2_ran_config_update

Type

64-bit Counter

Description

RAN configuration update counter

Labels

tnla, globalrannodeid

- amfn2_ran_config_update_ack

Type

64-bit Counter

Description

RAN configuration update counter acknowledge

Labels

tnla, globalrannodeid

- amfn2_ran_config_update_failure

Type

64-bit Counter

Description

RAN configuration update failure counter

Labels

tnla, cause, globalrannodeid

- amfn2_amf_config_update

Type

64-bit Counter

Description

AMF configuration update counter

Labels

tnla, globalrannodeid

- amfn2_amf_config_update_ack

Type

64-bit Counter

Description

AMF configuration update acknowledge counter

Labels

tnla, globalrannodeid

- amfn2_amf_config_update_failure

Type

64-bit Counter

Description

AMF configuration update failure counter

Labels

tnla, cause, globalrannodeid

amfn2_reroute_nas_request
Type
64-bit Counter
Description
Reroute NAS request counter
Labels
tnla, globalrannodeid
amfn2_trace_start
Type
64-bit Counter
Description
Trace Start messages sent from AMF to NG-RAN node as defined by TS 38.413 Section 8.11.
Labels
tnla, globalrannodeid
amfn2_trace_failure_indication
Type
64-bit Counter
Description
Trace Failure Indication messages received by AMF from NG-RAN node as defined by TS 38.413 Section 8.11.
Labels
tnla, globalrannodeid
amfn2_deactivate_trace
Type
64-bit Counter
Description
Deactivate Trace messages sent from AMF to NG-RAN node as defined by TS 38.413 Section 8.11.
Labels
tnla, globalrannodeid
amfn2_active_tnla_per_pod
Type
64-bit Gauge
Description
The number of active TNLAs per AMF_N2 pod
Labels
pod_id

Table of Contents

- [Summary](#)
- [ETCD Client Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_put_requests	Counter	The total number of ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_requests_sent	Counter	The total number of ETCD Client PUT Requests sent.
ETCD Client Metrics	etcd_put_successes	Counter	The total number of successful ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_failures	Counter	The total number of failed ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_with_lease_requests	Counter	The total number of ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_requests_sent	Counter	The total number of ETCD Client PUT with Lease Requests sent.
ETCD Client Metrics	etcd_put_with_lease_successes	Counter	The total number of successful ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_failures	Counter	The total number of failed ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_range_requests	Counter	The total number of ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_requests_sent	Counter	The total number of ETCD Client Range Requests sent.
ETCD Client Metrics	etcd_range_successes	Counter	The total number of successful ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_failures	Counter	The total number of failed ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_delete_requests	Counter	The total number of ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_requests_sent	Counter	The total number of ETCD Client Range Delete Requests sent.
ETCD Client Metrics	etcd_range_delete_successes	Counter	The total number of successful ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_failures	Counter	The total number of failed ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_txn_requests	Counter	The total number of ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_requests_sent	Counter	The total number of ETCD Client Txn Requests sent.
ETCD Client Metrics	etcd_txn_successes	Counter	The total number of successful ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_failures	Counter	The total number of failed ETCD Client Txn Requests.
ETCD Client Metrics	etcd_watch_requests	Counter	The total number of ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_requests_sent	Counter	The total number of ETCD Client Watch Requests sent.
ETCD Client Metrics	etcd_watch_successes	Counter	The total number of successful ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_failures	Counter	The total number of failed ETCD Client Watch Requests.
ETCD Client Metrics	etcdwatcher_requests	Counter	The total number of ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_requests_sent	Counter	The total number of ETCD Watcher Requests sent.
ETCD Client Metrics	etcdwatcher_successes	Counter	The total number of successful ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_failures	Counter	The total number of failed ETCD Watcher Requests.

ETCD Client Metrics

- etcd_put_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests.

Labels

keyPrefix

- etcd_put_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests sent.

Labels

keyPrefix

- etcd_put_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT Requests.

Labels

keyPrefix

- etcd_put_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT Requests.

Labels

keyPrefix, failureReason

- etcd_put_with_lease_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests sent.

Labels

keyPrefix

- etcd_put_with_lease_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT with Lease Requests.

Labels

keyPrefix, failureReason

- etcd_range_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests sent.

Labels

keyPrefix

- etcd_range_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Requests.

Labels

keyPrefix, failureReason

- etcd_range_delete_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests.

Labels

keyPrefix

- etcd_range_delete_requests_sent
 - Type
 - 64-bit Counter
 - Description
 - The total number of ETCD Client Range Delete Requests sent.
 - Labels
 - keyPrefix**
- etcd_range_delete_successes
 - Type
 - 64-bit Counter
 - Description
 - The total number of successful ETCD Client Range Delete Requests.
 - Labels
 - keyPrefix**
- etcd_range_delete_failures
 - Type
 - 64-bit Counter
 - Description
 - The total number of failed ETCD Client Range Delete Requests.
 - Labels
 - keyPrefix, failureReason**
- etcd_txn_requests
 - Type
 - 64-bit Counter
 - Description
 - The total number of ETCD Client Txn Requests.
 - Labels
 - keyPrefix**
- etcd_txn_requests_sent
 - Type
 - 64-bit Counter
 - Description
 - The total number of ETCD Client Txn Requests sent.
 - Labels
 - keyPrefix**
- etcd_txn_successes
 - Type
 - 64-bit Counter
 - Description

The total number of successful ETCD Client Txn Requests.

Labels

keyPrefix

- etcd_txn_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Txn Requests.

Labels

keyPrefix, failureReason

- etcd_watch_requests

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests sent.

Labels

keyPrefix

- etcd_watch_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Watch Requests.

Labels

keyPrefix, failureReason

- etcdwatcher_requests

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests.

Labels

keyPrefix

• etcdwatcher_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests sent.

Labels

keyPrefix

• etcdwatcher_successes

Type

64-bit Counter

Description

The total number of successful ETCD Watcher Requests.

Labels

keyPrefix

• etcdwatcher_failures

Type

64-bit Counter

Description

The total number of failed ETCD Watcher Requests.

Labels

keyPrefix, failureReason

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- [Summary](#)
- [HTTP Punting Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_sent	Counter	The total number of HTTP requests successfully punted.
HTTP Punting Metrics	punting_http_requests_failed	Counter	The total number of HTTP requests punting failed.
HTTP Punting Metrics	punting_http_responses_received	Counter	The total number of HTTP responses received for the requests punted.
HTTP Punting Metrics	punting_http_responses_forwarded	Counter	The total number of HTTP responses forwarded after punting transaction completion.

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Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_aborted	Counter	The total number of HTTP requests aborted.
HTTP Punting Metrics	punting_http_request_duration_seconds	Histogram	Punting operation duration in seconds.

HTTP Punting Metrics

- [punting_http_requests_sent](#)

Type

64-bit Counter

Description

The total number of HTTP requests successfully punted.

Labels

message_type
- [punting_http_requests_failed](#)

Type

64-bit Counter

Description

The total number of HTTP requests for which the punting failed.

Labels

message_type
- [punting_http_responses_received](#)

Type

64-bit Counter

Description

The total number of HTTP responses received for the requests punted.

Labels

message_type, status_code
- [punting_http_responses_forwarded](#)

Type

64-bit Counter

Description

The total number of HTTP responses forwarded after punting transaction completion.

Labels

message_type, status_code
- [punting_http_requests_aborted](#)

Type

64-bit Counter

Description

The total number of HTTP requests aborted.

Labels

message_type

• punting_http_request_duration_seconds

Type

64-bit Histogram

Description

The punting transaction duration is measured from the time the HTTP request is sent out to the time the response is received for the punting request.

Labels

message_type

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

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- [NfMonitor statistics](#)
- [DebugNfMonitorCmn](#)
- [NfManagement statistics](#)
- [Debug statistics](#)
- [ApiServer statistics](#)
- [Feature Functionality statistics](#)
- [Consul statistics](#)

Summary

Group	Metric Name	Type	Synopsis
NfMonitor statistics	nfmonitor_unstable_nf_counter	Counter	Howmany times, the Nf is found to be Unstable
NfMonitor statistics	nfmonitor_force_restart_due_to_nf_unstability	Counter	Numbe of times the VNFRestart triggered due to Unstability of the NF
NfMonitor statistics	nfmonitor_cumulative_readiness_failure_count	Counter	Numbe of times Service sound to have insuffucient count due to Readiness probe
NfMonitor statistics	nfmonitor_cumulative_readiness_failure_alerts	Counter	Numbe of times Service sound to have insuffucient count due to Readiness probe
NfMonitor statistics	nfmonitor_cumulative_k8api_get_query_failures	Counter	Numbe of times the k8 get Api query failures
DebugNfMonitorCmn	nfmonitor_force_restart_counter	Counter	Number of times force-restart triggered
NfManagement statistics	nfmonitor_nfmgmt_service_request_attempts	Counter	Howmany times, the Nf is attempting to send NRF Requests
NfManagement statistics	nfmonitor_nfmgmt_service_request_success	Counter	Numbe of times Success response is received for the requests to NRF
NfManagement statistics	nfmonitor_nfmgmt_service_request_failures	Counter	Numbe of times Failure response is received for the requests to NRF
NfManagement statistics	nfmonitor_nfmgmt_aborted_request_msgs	Counter	Numbe of times the requests to NRF is Aborted/Cancelled
Debug statistics	nfmonitor_vnf_restart_trigger	Counter	Howmany times, the Force Restart is received
Debug statistics	nfmonitor_fsm_state_changes	Counter	The statistics for the the FSM state changes
Debug statistics	nfmonitor_deactivation_triggers	Counter	Numbe of times the NF deactivation trigger is received
Debug statistics	nfmonitor_critical_service_up_notification	Counter	Numbe of times the Critical Service Up Notification received
Debug statistics	nfmonitor_critical_service_down_notification	Counter	Numbe of times the Critical Service Down Notification received
Debug statistics	nfmonitor_critical_services_overall_up_notification	Counter	Numbe of times the Up notification received when all monitored Critical Services are available
Debug statistics	nfmonitor_critical_services_overall_down_notification	Counter	Numbe of times the Down notification received when at least one Critical Service was unavailable

Group	Metric Name	Type	Synopsis
Debug statistics	nfmonitor_hysteresis_timer_started	Counter	Numbe of times the Hysteresis Timer is started due to missing Critical Service
Debug statistics	nfmonitor_hysteresis_timer_stopped	Counter	Numbe of times the Hysteresis Timer is stopped due to Critical Service become available
Debug statistics	nfmonitor_hysteresis_timer_expired	Counter	Numbe of times the Hysteresis Timer is expired due to missing Critical Service
Debug statistics	nfmonitor_personality_update_triggers	Counter	Numbe of times the Nf Personality update is received
Debug statistics	nfmonitor_maintenance_trigger_received	Counter	Numbe of times the Maintenance Mode is activated
Debug statistics	nfmonitor_profile_update_triggers	Counter	Numbe of times the Nf Profile Update happened
Debug statistics	nfmonitor_critical_service_down_on_register_entry	Counter	Numbe of times the Critical Services were down while starting with RegisterFsmState
ApiServer statistics	nfmonitor_deployment_create_watch_request	Counter	Howmany times, the deployment watch requests are created
ApiServer statistics	nfmonitor_deployment_delete_watch_request	Counter	Howmany times, the deployment watch requests are deleted
ApiServer statistics	nfmonitor_deployment_watch_notification_success	Counter	Howmany times, the watch notifications are successful
ApiServer statistics	nfmonitor_deployment_watch_notification_error	Counter	Howmany times, the watch notifications are failed
ApiServer statistics	nfmonitor_limit_range_query_attempts	Counter	Howmany times, the limit-range query attempts are made
ApiServer statistics	nfmonitor_limit_range_query_failures	Counter	Howmany times, the limit-range query attempts are failed
ApiServer statistics	nfmonitor_limit_range_query_success	Counter	Howmany times, the limit-range query attempts are succeeded
ApiServer statistics	nfmonitor_hpa_query_attempts	Counter	Howmany times, the Hpa query attempts are made
ApiServer statistics	nfmonitor_hpa_query_failures	Counter	Howmany times, the Hpa query attempts are failed
ApiServer statistics	nfmonitor_hpa_query_success	Counter	Howmany times, the Hpa query attempts are succeeded
ApiServer statistics	nfmonitor_readiness_probe_query_attempts	Counter	Howmany times, the Readiness Probe query attempts are made
ApiServer statistics	nfmonitor_readiness_probe_query_failures	Counter	Howmany times, the Readiness Probe query attempts are failed
ApiServer statistics	nfmonitor_readiness_probe_query_success	Counter	Howmany times, the Readiness Probe query attempts are succeeded
Feature Functionality statistics	nfmonitor_critical_service_frequent_restart	Counter	Howmany times, minimum count is not met for the stable Critical Services
Feature Functionality statistics	nfmonitor_alarm_due_to_frequent_restart	Counter	Howmany times alarm raised, due to minimum count is not met for the stable Critical Services
Feature Functionality statistics	nfmonitor_force_restart_due_to_frequent_restart	Counter	Howmany times, force restart triggerd because of the unstable Critical Services
Feature Functionality statistics	nfmonitor_critical_service_readiness_failure	Counter	Howmany times, minimum count is not met for the readiness succeeded Critical Services
Feature Functionality statistics	nfmonitor_alarm_due_to_readiness_failure	Counter	Alarm raised, due to minimum count is not met for the readiness succeeded Critical Services
Feature Functionality statistics	nfmonitor_force_restart_due_to_readiness_failure	Counter	Force restart triggerd because of Critical Service Readiness failure
Consul statistics	nfmonitor_publish_status_into_consul	Counter	Number of times Publish message is sent to Consul
Consul statistics	nfmonitor_consul_read_attempts	Counter	Number of times Consul read attempts
Consul statistics	nfmonitor_consul_read_failures	Counter	Number of times Consul read attempts failed
Consul statistics	nfmonitor_consul_read_success	Counter	Number of times Consul read attempts succeeded

NfMonitor statistics

- [nfmonitor_unstable_nf_counter](#)

Type

64-bit Counter

Description

Number of Times Nf is found Unstable

Labels

nfmonitor_force_restart_due_to_nf_unstability Type 64-bit Counter Description The number of times the Force Restart triggered due to NF-Unstability Labels
nfmonitor_cumulative_readiness_failure_count Type 64-bit Counter Description The number of times the service is found to be failing the Readiness probe Labels critical_service_name, pod_name
nfmonitor_cumulative_readiness_failure_alerts Type 64-bit Counter Description Total number of EventAlerts generated due to Readiness probe failure Labels critical_service_name, pod_name
nfmonitor_cumulative_k8api_get_query_failures Type 64-bit Counter Description Total number of k8 api get pod query failures Labels critical_service_name, pod_name

DebugNfMonitorCmn

nfmonitor_force_restart_counter Type 64-bit Counter Description Number of times force-restart triggered Labels agent, name
--

NfManagement statistics

nfmonitor_nfmgmt_service_request_attempts
--

Type

64-bit Counter

Description

Number of Requests sent to NRF

Labels

instance_id, msg_type, method, retry

- nfmonitor_nfmgmt_service_request_success

Type

64-bit Counter

Description

The number of times the Request to NRF is Successful

Labels

instance_id, msg_type, method, retry, status_code

- nfmonitor_nfmgmt_service_request_failures

Type

64-bit Counter

Description

The number of times the Request to NRF is Failure

Labels

instance_id, msg_type, method, retry, status_code

- nfmonitor_nfmgmt_aborted_request_msgs

Type

64-bit Counter

Description

The number of times the Request to NRF is Aborted

Labels

instance_id, msg_type

Debug statistics

- nfmonitor_vnf_restart_trigger

Type

64-bit Counter

Description

Number of VNF Restart triggers received

Labels

curr_nf_status, nfmgmt_state, nfmonitor_fsm_state, up_count, timestamp, trigger, type

- nfmonitor_fsm_state_changes

Type

64-bit Counter

Description

The FSM State Change events stats

Labels

curr_nf_status, nfmgmt_state, up_count, from_state, to_state

- nfmonitor_deactivation_triggers

Type

64-bit Counter

Description

The number of times the deactivation trigger received

Labels

nf_status, nfmgmt_state, up_count, nfmonitor_fsm_state

- nfmonitor_critical_service_up_notification

Type

64-bit Counter

Description

The number of times the CriticalService Up notification received

Labels

service, criticality, nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_critical_service_down_notification

Type

64-bit Counter

Description

The number of times the CriticalService Down notification received

Labels

service, criticality, nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_critical_services_overall_up_notification

Type

64-bit Counter

Description

The number of times the Overall CriticalServices Up notification received

Labels

nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_critical_services_overall_down_notification

Type

64-bit Counter

Description

The number of times the Overall CriticalServices Down notification received

Labels

nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_hysterisis_timer_started

Type

64-bit Counter

Description

The number of times the Hysterisis Timer is started

Labels

nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_hysterisis_timer_stopped

Type

64-bit Counter

Description

The number of times the Hysterisis Timer is stopped

Labels

nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_hysterisis_timer_expired

Type

64-bit Counter

Description

The number of times the Hysterisis Timer is expired

Labels

nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_personality_update_triggers

Type

64-bit Counter

Description

The number of times the Nf Personality Update is received

Labels

nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_maintenance_trigger_received

Type

64-bit Counter

Description

The number of times the Maintenance Mode is activated

Labels

nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_profile_update_triggers

Type

64-bit Counter

Description

The number of times the Nf Profile Update happened

Labels

nf_status, nfmgmt_state, up_count, fsm_state, update_operation

- nfmonitor_critical_service_down_on_register_entry

Type

64-bit Counter

Description

The number of times the Critical Services were down while starting with RegisterFsmState

Labels

nf_status, up_count

ApiServer statistics

- nfmonitor_deployment_create_watch_request

Type

64-bit Counter

Description

The number of Watch Reqs created for deployment type

Labels

resource

- nfmonitor_deployment_delete_watch_request

Type

64-bit Counter

Description

The number of Watch Reqs deleted for deployment type

Labels

resource

- nfmonitor_deployment_watch_notification_success

Type

64-bit Counter

Description

The number of times Watch notifications are successful

Labels

resource

- nfmonitor_deployment_watch_notification_error

Type

64-bit Counter

Description

The number of times Watch notifications are failed

Labels

resource, reason

- nfmonitor_limit_range_query_attempts

Type

64-bit Counter

Description

The number of Limit Range query attempts

Labels

- `nfmonitor_limit_range_query_failures`

Type

64-bit Counter

Description

The number of Limit Range query failures

Labels

status_code

- `nfmonitor_limit_range_query_success`

Type

64-bit Counter

Description

The number of Limit Range query succeeded

Labels

status_code

- `nfmonitor_hpa_query_attempts`

Type

64-bit Counter

Description

The number of Hpa query attempts

Labels

resource, continue_param

- `nfmonitor_hpa_query_failures`

Type

64-bit Counter

Description

The number of Hpa query failures

Labels

resource, continue_param, status_code

- `nfmonitor_hpa_query_success`

Type

64-bit Counter

Description

The number of Hpa query succeeded

Labels

resource, continue_param, status_code

- nfmonitor_readiness_probe_query_attempts

Type

64-bit Counter

Description

The number of Readiness Probe query attempts

Labels

resource, continue_param

- nfmonitor_readiness_probe_query_failures

Type

64-bit Counter

Description

The number of Readiness Probe query failures

Labels

resource, continue_param, status_code

- nfmonitor_readiness_probe_query_success

Type

64-bit Counter

Description

The number of Readiness Probe query succeeded

Labels

resource, continue_param, status_code

Feature Functionality statistics

- nfmonitor_critical_service_frequent_restart

Type

64-bit Counter

Description

Howmany times the minimum required stable CriticalService count is not met

Labels

service

- nfmonitor_alarm_due_to_frequent_restart

Type

64-bit Counter

Description

Howmany times Alarm is raised due to minimum required stable CriticalService count is not met

Labels

service, alarm_status

- nfmonitor_force_restart_due_to_frequent_restart

Type

64-bit Counter

Description

Howmany times Force Restart initiated due to missing stable CriticalService count

Labels

service

- nfmonitor_critical_service_readiness_failure

Type

64-bit Counter

Description

Howmany times the minimum required readiness succeeded CriticalService count is not met

Labels

service

- nfmonitor_alarm_due_to_readiness_failure

Type

64-bit Counter

Description

Alarm raised due to minimum count is not met for the readiness succeeded CriticalService

Labels

service, alarm_status

- nfmonitor_force_restart_due_to_readiness_failure

Type

64-bit Counter

Description

Howmany times Force Restart initiated due to readiness failure of CriticalService

Labels

service

Consul statistics

- nfmonitor_publish_status_into_consul

Type

64-bit Counter

Description

Howmany times Publish is sent to Consul

Labels

nf_status, nf_state, up_count, fsm_state

- nfmonitor_consul_read_attempts

Type

64-bit Counter

Description

Howmany times Consul read attempt is made

Labels

fsm_state, retry

- nfmonitor_consul_read_failures

Type

64-bit Counter

Description

Howmany times Consul read attempt are failed

Labels

fsm_state, retry, status_code

- nfmonitor_consul_read_success

Type

64-bit Counter

Description

Howmany times Consul read attempt are succeeded

Labels

fsm_state, retry, status_code

Table of Contents

- [Summary](#)
- [Packet Streaming Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Packet Streaming Metrics	num_packet_streaming_attempted	Counter	The total number of attempted streaming of packets
Packet Streaming Metrics	num_packet_streaming_success	Counter	The total number of packets streamed successfully
Packet Streaming Metrics	num_packet_streaming_failed	Counter	The total number of failed streaming of packets
Packet Streaming Metrics	num_packet_streaming_bytes	Counter	The total number of bytes successfully streamed

Packet Streaming Metrics

- num_packet_streaming_attempted

Type

64-bit Counter

Description

The total number of attempted streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_success

Type

64-bit Counter

Description

The total number of packets streamed successfully

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_failed

Type

64-bit Counter

Description

The total number of failed streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_bytes

Type

64-bit Counter

Description

The total number of bytes successfully streamed

Labels

interface_name, server_fqdn, kafka_topic

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- [Summary](#)
- [Circuit Breaker Metrics](#)
- [Service Endpoint Learner Metrics](#)
- [Service Endpoint Manager Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Circuit Breaker Metrics	sem_cb_endpoints	Gauge	Total number of endpoints for a service in respective states
Service Endpoint Learner Metrics	sem_num_dns_errors	Counter	Total number of DNS resolution errors for a given service
Service Endpoint Learner Metrics	sem_num_endpoints	Gauge	Total number of endpoints learnt for a service
Service Endpoint Manager Metrics	sem_num_endpoint_transactions_sent	Counter	Total number of times an endpoint is given to the application for communication with it
Service Endpoint Manager Metrics	sem_num_endpoint_transaction_status_received	Counter	Total number of transactions for a given endpoint received with the transaction status

Circuit Breaker Metrics

- sem_cb_endpoints

Type

64-bit Gauge

Description

Total number of endpoints for a service in respective states. There are 3 states viz. Active, Blocked, Restoring. In Active state, the endpoint is responding as expected, i.e. the response code doesn't indicate a service failure. The endpoint will be moved to Blocked state based on response failures which indicate service failure. After a timeout, the endpoint will be moved to Restoring state to check if the endpoint has recovered. In this state, if endpoint responds with response code other than service failure, it will be moved to Active state. Else, it will be moved back to Blocked state.

Labels

service_name, endpoint_state

Service Endpoint Learner Metrics

- sem_num_dns_errors

Type

64-bit Counter

Description

Total number of DNS resolution errors for a given service, labels are service name for DNS resolution and DNS error codes

Labels

service_name, error_code

- sem_num_endpoints

Type

64-bit Gauge

Description

Total number of endpoints learnt from the Kubernetes DNS server for a given service

Labels

service

Service Endpoint Manager Metrics

- sem_num_endpoint_transactions_sent

Type

64-bit Counter

Description

Total number of times an endpoint is given to the application for communication with it

Labels

service

- sem_num_endpoint_transaction_status_received

Type

64-bit Counter

Description

Total number of transactions for a given endpoint received with the transaction status

Labels

service, status

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- [Summary](#)
- [Unified Load Balancer Endpoint Manager Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Unified Load Balancer Endpoint Manager Metrics	ulb_registration_request_sent	Counter	Total number of times an registration request is sent to an unified load balancer pod

Group	Metric Name	Type	Synopsis
Unified Load Balancer Endpoint Manager Metrics	ulb_registration_response_received	Counter	Total number of times registration response is received from unified load balancer pod
Unified Load Balancer Endpoint Manager Metrics	ulb_deregistration_request_sent	Counter	Total number of times an deregistration request is sent to an unified load balancer pod
Unified Load Balancer Endpoint Manager Metrics	ulb_deregistration_response_received	Counter	Total number of times deregistration response is received from unified load balancer pod
Unified Load Balancer Endpoint Manager Metrics	ulb_heartbeat_request_sent	Counter	Total number of times an heartbeat request is sent to an unified load balancer pod
Unified Load Balancer Endpoint Manager Metrics	ulb_heartbeat_response_received	Counter	Total number of times heartbeat response is received from unified load balancer pod

Unified Load Balancer Endpoint Manager Metrics

- ulb_registration_request_sent
 - Type
 - 64-bit Counter
 - Description
 - Total number of times registration request is sent to an unified load balancer pod
 - Labels
 - ulbName**
- ulb_registration_response_received
 - Type
 - 64-bit Counter
 - Description
 - Total number of times registration response is received from unified load balancer pod
 - Labels
 - UlbName, status**
- ulb_deregistration_request_sent
 - Type
 - 64-bit Counter
 - Description
 - Total number of times deregistration request is sent to an unified load balancer pod
 - Labels
 - ulbName**
- ulb_deregistration_response_received
 - Type
 - 64-bit Counter
 - Description
 - Total number of times deregistration response is received from unified load balancer pod
 - Labels
 - UlbName, status**
- ulb_heartbeat_request_sent

Type

64-bit Counter

Description

Total number of times heartbeat request is sent to an unified load balancer pod

Labels

ulbName

- ulb_heartbeat_response_received

Type

64-bit Counter

Description

Total number of times heartbeat response is received from unified load balancer pod

Labels

UlbName, status

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- [Summary](#)
- [SCTP statistics](#)

Summary

Group	Metric Name	Type	Synopsis
SCTP statistics	sctpActiveEstabs	Counter	The number of times associations have been established.
SCTP statistics	sctpPassiveEstabs	Counter	The number of times associations have been established.
SCTP statistics	sctpAbortedds	Counter	The number of times associations have been aborted.
SCTP statistics	sctpShutdowns	Counter	The number of times associations have been gracefully terminated.
SCTP statistics	sctpOutOfBlues	Counter	The number of out of the blue packets received.
SCTP statistics	sctpChecksumErrors	Counter	The number of SCTP packets received with an invalid checksum.
SCTP statistics	sctpOutCtrlChunks	Counter	The number of SCTP control chunks sent.
SCTP statistics	sctpOutOrderChunks	Counter	The number of SCTP ordered data chunks sent.
SCTP statistics	sctpOutUnorderChunks	Counter	The number of SCTP unordered data chunks sent.
SCTP statistics	sctpInCtrlChunks	Counter	The number of SCTP control chunks received.
SCTP statistics	sctpInOrderChunks	Counter	The number of SCTP ordered data chunks received.
SCTP statistics	sctpInUnorderChunks	Counter	The number of SCTP unordered data chunks received.
SCTP statistics	sctpOutSCTPPacks	Counter	The number of SCTP packets sent.
SCTP statistics	sctpInSCTPPacks	Counter	The number of SCTP packets received.
SCTP statistics	sctpCurrEstab	Gauge	The current number of established associations.

SCTP statistics

- sctpActiveEstabs

Type

64-bit Counter

Description

The number of times that associations have made a direct transition to the ESTABLISHED state from the COOKIE-ECHOED state: COOKIE-ECHOED -> ESTABLISHED. The upper layer initiated the association attempt.

Labels

- sctpPassiveEstabs

Type

64-bit Counter

Description

The number of times that associations have made a direct transition to the ESTABLISHED state from the CLOSED state: CLOSED -> ESTABLISHED. The remote endpoint initiated the association attempt.

Labels

- sctpAborted

Type

64-bit Counter

Description

The number of times that associations have made a direct transition to the CLOSED state from any state due to sending or receiving of an ABORT chunk: AnyState --Abort--> CLOSED. Ungraceful termination of the association.

Labels

- sctpShutdowns

Type

64-bit Counter

Description

The number of times that associations have made a direct transition to the CLOSED state from either the SHUTDOWN-SENT state or the SHUTDOWN-ACK-SENT state. Graceful termination of the association.

Labels

- sctpOutOfBlues

Type

64-bit Counter

Description

The number of out of the blue packets received by the host. An out of the blue packet is an SCTP packet correctly formed, including the proper checksum, but for which the receiver was unable to identify an appropriate association.

Labels

- sctpChecksumErrors

Type

64-bit Counter

Description

The number of SCTP packets received with an invalid checksum.

Labels

- sctpOutCtrlChunks

Type

64-bit Counter

Description

The number of SCTP control chunks sent (retransmissions are not included). Control chunks are those chunks different from DATA.

Labels

- **sctpOutOrderChunks**

Type

64-bit Counter

Description

The number of SCTP ordered data chunks sent (retransmissions are not included).

Labels

- **sctpOutUnorderChunks**

Type

64-bit Counter

Description

The number of SCTP unordered chunks (data chunks in which the U bit is set to 1) sent (retransmissions are not included).

Labels

- **sctpInCtrlChunks**

Type

64-bit Counter

Description

The number of SCTP control chunks received (no duplicate chunks included).

Labels

- **sctpInOrderChunks**

Type

64-bit Counter

Description

The number of SCTP ordered data chunks received (no duplicate chunks included).

Labels

- **sctpInUnorderChunks**

Type

64-bit Counter

Description

The number of SCTP unordered chunks (data chunks in which the U bit is set to 1) received (no duplicate chunks included).

Labels

- **sctpOutSCTPPacks**

Type

64-bit Counter

Description

The number of SCTP packets sent. Retransmitted DATA chunks are included.

Labels

• sctpInSCTPPacks

Type

64-bit Counter

Description

The number of SCTP packets received. Duplicates are included.

Labels

• sctpCurrEstab

Type

64-bit Gauge

Description

The number of associations for which the current state is either ESTABLISHED, SHUTDOWN-RECEIVED or SHUTDOWN-PENDING.

Labels

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- [Summary](#)
- [ETCD Client Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_put_requests	Counter	The total number of ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_requests_sent	Counter	The total number of ETCD Client PUT Requests sent.
ETCD Client Metrics	etcd_put_successes	Counter	The total number of successful ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_failures	Counter	The total number of failed ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_with_lease_requests	Counter	The total number of ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_requests_sent	Counter	The total number of ETCD Client PUT with Lease Requests sent.
ETCD Client Metrics	etcd_put_with_lease_successes	Counter	The total number of successful ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_failures	Counter	The total number of failed ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_range_requests	Counter	The total number of ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_requests_sent	Counter	The total number of ETCD Client Range Requests sent.
ETCD Client Metrics	etcd_range_successes	Counter	The total number of successful ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_failures	Counter	The total number of failed ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_delete_requests	Counter	The total number of ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_requests_sent	Counter	The total number of ETCD Client Range Delete Requests sent.
ETCD Client Metrics	etcd_range_delete_successes	Counter	The total number of successful ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_failures	Counter	The total number of failed ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_txn_requests	Counter	The total number of ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_requests_sent	Counter	The total number of ETCD Client Txn Requests sent.
ETCD Client Metrics	etcd_txn_successes	Counter	The total number of successful ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_failures	Counter	The total number of failed ETCD Client Txn Requests.
ETCD Client Metrics	etcd_watch_requests	Counter	The total number of ETCD Client Watch Requests.

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_watch_requests_sent	Counter	The total number of ETCD Client Watch Requests sent.
ETCD Client Metrics	etcd_watch_successes	Counter	The total number of successful ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_failures	Counter	The total number of failed ETCD Client Watch Requests.
ETCD Client Metrics	etcdwatcher_requests	Counter	The total number of ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_requests_sent	Counter	The total number of ETCD Watcher Requests sent.
ETCD Client Metrics	etcdwatcher_successes	Counter	The total number of successful ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_failures	Counter	The total number of failed ETCD Watcher Requests.

ETCD Client Metrics

- `etcd_put_requests`

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests.

Labels

keyPrefix

- `etcd_put_requests_sent`

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests sent.

Labels

keyPrefix

- `etcd_put_successes`

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT Requests.

Labels

keyPrefix

- `etcd_put_failures`

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT Requests.

Labels

keyPrefix, failureReason

- `etcd_put_with_lease_requests`

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests sent.

Labels

keyPrefix

- etcd_put_with_lease_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT with Lease Requests.

Labels

keyPrefix, failureReason

- etcd_range_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests sent.

Labels

keyPrefix

- etcd_range_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Requests.

Labels

keyPrefix, failureReason

- etcd_range_delete_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests.

Labels

keyPrefix

- etcd_range_delete_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests sent.

Labels

keyPrefix

- etcd_range_delete_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Delete Requests.

Labels

keyPrefix

- etcd_range_delete_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Delete Requests.

Labels

keyPrefix, failureReason

- etcd_txn_requests

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests.

Labels

keyPrefix

- etcd_txn_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests sent.

Labels

keyPrefix

- etcd_txn_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Txn Requests.

Labels

keyPrefix

- etcd_txn_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Txn Requests.

Labels

keyPrefix, failureReason

- etcd_watch_requests

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_requests_sent
 - Type
 - 64-bit Counter
 - Description
 - The total number of ETCD Client Watch Requests sent.
 - Labels
 - keyPrefix**
- etcd_watch_successes
 - Type
 - 64-bit Counter
 - Description
 - The total number of successful ETCD Client Watch Requests.
 - Labels
 - keyPrefix**
- etcd_watch_failures
 - Type
 - 64-bit Counter
 - Description
 - The total number of failed ETCD Client Watch Requests.
 - Labels
 - keyPrefix, failureReason**
- etcdwatcher_requests
 - Type
 - 64-bit Counter
 - Description
 - The total number of ETCD Watcher Requests.
 - Labels
 - keyPrefix**
- etcdwatcher_requests_sent
 - Type
 - 64-bit Counter
 - Description
 - The total number of ETCD Watcher Requests sent.
 - Labels
 - keyPrefix**
- etcdwatcher_successes
 - Type
 - 64-bit Counter
 - Description

The total number of successful ETCD Watcher Requests.

Labels

keyPrefix

- etcdwatcher_failures

Type

64-bit Counter

Description

The total number of failed ETCD Watcher Requests.

Labels

keyPrefix, failureReason

Table of Contents

- [Summary](#)
- [HTTP Punting Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_sent	Counter	The total number of HTTP requests successfully punted.
HTTP Punting Metrics	punting_http_requests_failed	Counter	The total number of HTTP requests punting failed.
HTTP Punting Metrics	punting_http_responses_received	Counter	The total number of HTTP responses received for the requests punted.
HTTP Punting Metrics	punting_http_responses_forwarded	Counter	The total number of HTTP responses forwarded after punting transaction completion.
HTTP Punting Metrics	punting_http_requests_aborted	Counter	The total number of HTTP requests aborted.
HTTP Punting Metrics	punting_http_request_duration_seconds	Histogram	Punting operation duration in seconds.

HTTP Punting Metrics

- punting_http_requests_sent

Type

64-bit Counter

Description

The total number of HTTP requests successfully punted.

Labels

message_type

- punting_http_requests_failed

Type

64-bit Counter

Description

The total number of HTTP requests for which the punting failed.

Labels

message_type

- `punting_http_responses_received`

Type

64-bit Counter

Description

The total number of HTTP responses received for the requests punted.

Labels

message_type, status_code

- `punting_http_responses_forwarded`

Type

64-bit Counter

Description

The total number of HTTP responses forwarded after punting transaction completion.

Labels

message_type, status_code

- `punting_http_requests_aborted`

Type

64-bit Counter

Description

The total number of HTTP requests aborted.

Labels

message_type

- `punting_http_request_duration_seconds`

Type

64-bit Histogram

Description

The punting transaction duration is measured from the time the HTTP request is sent out to the time the response is received for the punting request.

Labels

message_type

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

Table of Contents

- [Summary](#)
- [Packet Streaming Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Packet Streaming Metrics	num_packet_streaming_attempted	Counter	The total number of attempted streaming of packets
Packet Streaming Metrics	num_packet_streaming_success	Counter	The total number of packets streamed successfully
Packet Streaming Metrics	num_packet_streaming_failed	Counter	The total number of failed streaming of packets

Group	Metric Name	Type	Synopsis
Packet Streaming Metrics	num_packet_streaming_bytes	Counter	The total number of bytes successfully streamed

Packet Streaming Metrics

- [num_packet_streaming_attempted](#)

Type

64-bit Counter

Description

The total number of attempted streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- [num_packet_streaming_success](#)

Type

64-bit Counter

Description

The total number of packets streamed successfully

Labels

interface_name, server_fqdn, kafka_topic

- [num_packet_streaming_failed](#)

Type

64-bit Counter

Description

The total number of failed streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- [num_packet_streaming_bytes](#)

Type

64-bit Counter

Description

The total number of bytes successfully streamed

Labels

interface_name, server_fqdn, kafka_topic

Table of Contents

- [Summary](#)
- [Circuit Breaker Metrics](#)
- [Service Endpoint Learner Metrics](#)
- [Service Endpoint Manager Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
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Group	Metric Name	Type	Synopsis
Circuit Breaker Metrics	sem_cb_endpoints	Gauge	Total number of endpoints for a service in respective states
Service Endpoint Learner Metrics	sem_num_dns_errors	Counter	Total number of DNS resolution errors for a given service
Service Endpoint Learner Metrics	sem_num_endpoints	Gauge	Total number of endpoints learnt for a service
Service Endpoint Manager Metrics	sem_num_endpoint_transactions_sent	Counter	Total number of times an endpoint is given to the application for communication with it
Service Endpoint Manager Metrics	sem_num_endpoint_transaction_status_received	Counter	Total number of transactions for a given endpoint received with the transaction status

Circuit Breaker Metrics

- sem_cb_endpoints

Type

64-bit Gauge

Description

Total number of endpoints for a service in respective states. There are 3 states viz. Active, Blocked, Restoring. In Active state, the endpoint is responding as expected, i.e. the response code doesn't indicate a service failure. The endpoint will be moved to Blocked state based on response failures which indicate service failure. After a timeout, the endpoint will be moved to Restoring state to check if the endpoint has recovered. In this state, if endpoint responds with response code other than service failure, it will be moved to Active state. Else, it will be moved back to Blocked state.

Labels

service_name, endpoint_state

Service Endpoint Learner Metrics

- sem_num_dns_errors

Type

64-bit Counter

Description

Total number of DNS resolution errors for a given service, labels are service name for DNS resolution and DNS error codes

Labels

service_name, error_code

- sem_num_endpoints

Type

64-bit Gauge

Description

Total number of endpoints learnt from the Kubernetes DNS server for a given service

Labels

service

Service Endpoint Manager Metrics

- sem_num_endpoint_transactions_sent

Type

64-bit Counter

Description

Total number of times an endpoint is given to the application for communication with it

Labels

service

- sem_num_endpoint_transaction_status_received

Type

64-bit Counter

Description

Total number of transactions for a given endpoint received with the transaction status

Labels

service, status

Table of Contents

- [Summary](#)
- [CDR-Agent CDR Stats](#)

Summary

Group	Metric Name	Type	Synopsis
CDR-Agent CDR Stats	charging_local_cdrs_count	Counter	Total number of local CDRs generated
CDR-Agent CDR Stats	charging_local_cdr_dropped	Counter	Total number of local CDR generation service requests encoding failed
CDR-Agent CDR Stats	charging_local_cdrs_files_count	Counter	Total number of local CDR files generated

CDR-Agent CDR Stats

- charging_local_cdrs_count

Type

64-bit Counter

Description

Total number of local CDR generated by CdrAgent. The CDRs count is per RecordType/Profile/FileFormat/GwType/CdrRecordType. Profile, record_type, file_format gw_type and cdr_record_type are populated from the received ChargingRecord message from the service consumers.

Labels

profile, record_type, file_format, gw_type, cdr_record_type

- charging_local_cdr_dropped

Type

64-bit Counter

Description

Total number of local CDR generation service requests encoding failed in CdrAgent. The service requests count is per RecordType/Profile/FileFormat/GwType/CdrRecordType/Error. RecordType is the type of local CDR namely ChfRecord etc. Profile, PlmnId and Slice are populated from the received ChargingRecord message from the service consumers.

Labels

profile, record_type, file_format, gw_type, cdr_record_type, error

- charging_local_cdrs_files_count

Type

64-bit Counter

Description
Total number of local CDR files generated by CdrAgent. The CDR files count is per FileFormat/federation. FedName and FileFormat are populated from the received ChargingRecord message from the service consumers.
Labels
<i>fed_name, file_format</i>

Table of Contents

- Summary - ChargingCmn statistics
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Summary

Group	Metric Name	Type	Synopsis
ChargingCmn statistics	svcmgr_service_available_ind_notification	Counter	How many times, Service Available Indication message received
ChargingCmn statistics	svcmgr_service_available_ind_parse_errs	Counter	How many times, Service Available Indication parse error happened
ChargingCmn statistics	svcmgr_service_available_ind_notification_ack	Counter	How many times, Service Available Indication message is acknowledged
ChargingCmn statistics	svcmgr_service_unavailable_ind_notification	Counter	How many times, Service Unavailable Indication message
ChargingCmn statistics	svcmgr_service_unavailable_ind_parse_errs	Counter	How many times, Service Unavailable Indication parse error happened
ChargingCmn statistics	svcmgr_service_unavailable_ind_notification_ack	Counter	How many times, Service Unavailable Indication message is acknowledged
ChargingCmn statistics	svcmgr_publish_msg_sent	Counter	How many times, Publication message sent
ChargingCmn statistics	svcmgr_publish_msg_success	Counter	How many times, Publication message success response received
ChargingCmn statistics	svcmgr_publish_msg_failure	Counter	How many times, failure received for publication message
ChargingCmn statistics	svcmgr_subscribe_msg_sent	Counter	How many times, Subscribe message sent
ChargingCmn statistics	svcmgr_subscribe_msg_success	Counter	How many times, subscribe message success
ChargingCmn statistics	svcmgr_subscribe_msg_failure	Counter	How many times, failure received for subscribe message
ChargingCmn statistics	svcmgr_unsubscribe_msg_sent	Counter	How many times, unsubscribe message sent
ChargingCmn statistics	svcmgr_unsubscribe_msg_failure	Counter	How many times, failure received for unsubscribe message
ChargingCmn statistics	svcmgr_config_req_sent	Counter	How many times, config request sent
ChargingCmn statistics	svcmgr_config_req_failure	Counter	How many times, failure received for config request
ChargingCmn statistics	svcmgr_config_req_success	Counter	How many times, success received for config request
ChargingCmn statistics	svcmgr_config_register_response	Counter	How many times, config register response received
ChargingCmn statistics	svcmgr_config_register_response_ack	Counter	How many times, config register response is acknowledged
ChargingCmn statistics	svcmgr_config_register_response_parse_err	Counter	How many times, parse error happened for config register response

Group	Metric Name	Type	Synopsis
ChargingCmn statistics	svcmgr_config_update_notification	Counter	How many times, config update notification received
ChargingCmn statistics	svcmgr_config_update_notification_response	Counter	How many times, config update response
ChargingCmn statistics	svcmgr_intf_schema_update_ind_notification	Counter	How many times, interface schema update indication message received
ChargingCmn statistics	svcmgr_intf_schema_update_ind_notification_ack	Counter	How many times, interface schema update indication message is acknowledged
ChargingCmn statistics	svcmgr_intf_schema_update_ind_parse_errs	Counter	How many times, interface schema update indication parse error happened

ChargingCmn statistics

- svcmgr_service_available_ind_notification
 - Type
 - 64-bit Counter
 - Description
 - Number of Times Service Available Indication notification received
 - Labels
 - servicename***
- svcmgr_service_available_ind_parse_errs
 - Type
 - 64-bit Counter
 - Description
 - Number of Times Service Available Indication parse error happened
 - Labels
- svcmgr_service_available_ind_notification_ack
 - Type
 - 64-bit Counter
 - Description
 - Number of Times Service Available Indication message is acknowledged
 - Labels
 - servicename***
- svcmgr_service_unavailable_ind_notification
 - Type
 - 64-bit Counter
 - Description
 - Number of Times Service Unavailable Indication notification received
 - Labels
 - servicename***
- svcmgr_service_unavailable_ind_parse_errs
 - Type

64-bit Counter

Description

Number of Times Service Unavailable Indication parse error happened

Labels

• svcmgr_service_unavailable_ind_notification_ack

Type

64-bit Counter

Description

Number of Times Service Unavailable Indication message is acknowledged

Labels

servicename

• svcmgr_publish_msg_sent

Type

64-bit Counter

Description

Number of Times Publication message sent

Labels

servicename, retry

• svcmgr_publish_msg_success

Type

64-bit Counter

Description

Number of Times Publication message success response received

Labels

servicename, causecode, retry

• svcmgr_publish_msg_failure

Type

64-bit Counter

Description

Number of Times failure received for publication message

Labels

servicename, causecode, retry

• svcmgr_subscribe_msg_sent

Type

64-bit Counter

Description

Number of Times Subscribe message sent

Labels

servicename, retry

- **svcmgr_subscribe_msg_success**

Type

64-bit Counter

Description

Number of Times Send subscribe message success response received

Labels

servicename, causecode, retry

- **svcmgr_subscribe_msg_failure**

Type

64-bit Counter

Description

Number of time failure received for subscribe message

Labels

servicename, causecode, retry

- **svcmgr_unsubscribe_msg_sent**

Type

64-bit Counter

Description

Number of Times unsubscribe message sent

Labels

servicename

- **svcmgr_unsubscribe_msg_failure**

Type

64-bit Counter

Description

Number of Times failure received for unsubscribe message

Labels

servicename

- **svcmgr_config_req_sent**

Type

64-bit Counter

Description

Number of Times config request sent

Labels

retry

- **svcmgr_config_req_failure**

Type

64-bit Counter

Description

Number of Times failure received for config request

Labels

causecode, retry

- `svcmgr_config_req_success`

Type

64-bit Counter

Description

Number of Times success received for config request

Labels

causecode, retry

- `svcmgr_config_register_response`

Type

64-bit Counter

Description

Number of Times config register response received

Labels

statuscode

- `svcmgr_config_register_response_ack`

Type

64-bit Counter

Description

Number of Times config register response is acknowledged

Labels

- `svcmgr_config_register_response_parse_err`

Type

64-bit Counter

Description

Number of Times parse error happened for config register response

Labels

- `svcmgr_config_update_notification`

Type

64-bit Counter

Description

Number of Times config update notification received

Labels

- `svcmgr_config_update_notification_response`

Type

64-bit Counter

Description

Number of Times config update response sent

Labels

- [svcmgr_intf_schema_update_ind_notification](#)

Type

64-bit Counter

Description

Number of Times interface schema update indication message received

Labels

servicename

- [svcmgr_intf_schema_update_ind_notification_ack](#)

Type

64-bit Counter

Description

Number of Times interface schema update indication message is acknowledged

Labels

servicename

- [svcmgr_intf_schema_update_ind_parse_errs](#)

Type

64-bit Counter

Description

Number of Times Service interface schema update indication parse error happened

Labels

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- [Summary](#)
- [CDR-Agent CDR Stats for Disk Quota Computation](#)

Summary

Group	Metric Name	Type	Synopsis
CDR-Agent CDR Stats for Disk Quota Computation	cdr_agent_disk_usage_failure	Counter	Total number of failures in computing disk usage for CDR-Agent
CDR-Agent CDR Stats for Disk Quota Computation	cdr_agent_disk_quota_failure	Counter	Total number of failures in computing disk quota for CDR-Agent
CDR-Agent CDR Stats for Disk Quota Computation	cdr_agent_disk_usage_crossed_above_high_threshold	Counter	Total number of times disk usage crossed above high threshold for CDR-Agent
CDR-Agent CDR Stats for Disk Quota Computation	cdr_agent_disk_usage_crossed_below_low_threshold	Counter	Total number of times disk usage crossed below low threshold for CDR-Agent
CDR-Agent CDR Stats for Disk Quota Computation	cdr_agent_disk_management_profile_not_configured	Counter	Total number of times disk management profile not configured for CDR-Agent

Group	Metric Name	Type	Synopsis
CDR-Agent CDR Stats for Disk Quota Computation	cdr_agent_disk_quota_full_file_close_failure	Counter	Total number of times disk quota full file close failure for CDR-Agent
CDR-Agent CDR Stats for Disk Quota Computation	cdr_agent_disk_quota_failure_to_decode_timer_expiry_notification_message	Counter	Total number of times disk quota failure to decode timer expiry notification message for CDR-Agent
CDR-Agent CDR Stats for Disk Quota Computation	cdr_agent_disk_quota_notification_input_buffer_not_present	Counter	Total number of times disk quota notification input buffer not present for CDR-Agent

CDR-Agent CDR Stats for Disk Quota Computation

- cdr_agent_disk_usage_failure

Type64-bit CounterDescriptionTotal number of failures in computing disk usage for CDR-AgentLabels**configured_low_threshold, configured_high_threshold, alarm_raised**
- cdr_agent_disk_quota_failure

Type64-bit CounterDescriptionTotal number of failures in computing disk quota for CDR-AgentLabels**configured_low_threshold, configured_high_threshold, alarm_raised**
- cdr_agent_disk_usage_crossed_above_high_threshold

Type64-bit CounterDescriptionTotal number of times disk usage crossed above high threshold for CDR-AgentLabels**configured_high_threshold**
- cdr_agent_disk_usage_crossed_below_low_threshold

Type64-bit CounterDescriptionTotal number of times disk usage crossed below low threshold for CDR-AgentLabels**configured_low_threshold**
- cdr_agent_disk_management_profile_not_configured

Type64-bit Counter

Description

Total number of times disk management profile not configured for CDR-Agent

Labels

- `cdr_agent_disk_quota_full_file_close_failure`

Type

64-bit Counter

Description

Total number of times disk quota full file close failure for CDR-Agent

Labels

- `cdr_agent_disk_quota_failure_to_decode_timer_expiry_notification_message`

Type

64-bit Counter

Description

Total number of times disk quota failure to decode timer expiry notification message for CDR-Agent

Labels

- `cdr_agent_disk_quota_notification_input_buffer_not_present`

Type

64-bit Counter

Description

Total number of times disk quota notification input buffer not present for CDR-Agent

Labels

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- [Summary](#)
- [ETCD Client Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_put_requests	Counter	The total number of ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_requests_sent	Counter	The total number of ETCD Client PUT Requests sent.
ETCD Client Metrics	etcd_put_successes	Counter	The total number of successful ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_failures	Counter	The total number of failed ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_with_lease_requests	Counter	The total number of ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_requests_sent	Counter	The total number of ETCD Client PUT with Lease Requests sent.
ETCD Client Metrics	etcd_put_with_lease_successes	Counter	The total number of successful ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_failures	Counter	The total number of failed ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_range_requests	Counter	The total number of ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_requests_sent	Counter	The total number of ETCD Client Range Requests sent.
ETCD Client Metrics	etcd_range_successes	Counter	The total number of successful ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_failures	Counter	The total number of failed ETCD Client Range Requests.

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_range_delete_requests	Counter	The total number of ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_requests_sent	Counter	The total number of ETCD Client Range Delete Requests sent.
ETCD Client Metrics	etcd_range_delete_successes	Counter	The total number of successful ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_failures	Counter	The total number of failed ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_txn_requests	Counter	The total number of ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_requests_sent	Counter	The total number of ETCD Client Txn Requests sent.
ETCD Client Metrics	etcd_txn_successes	Counter	The total number of successful ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_failures	Counter	The total number of failed ETCD Client Txn Requests.
ETCD Client Metrics	etcd_watch_requests	Counter	The total number of ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_requests_sent	Counter	The total number of ETCD Client Watch Requests sent.
ETCD Client Metrics	etcd_watch_successes	Counter	The total number of successful ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_failures	Counter	The total number of failed ETCD Client Watch Requests.
ETCD Client Metrics	etcdwatcher_requests	Counter	The total number of ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_requests_sent	Counter	The total number of ETCD Watcher Requests sent.
ETCD Client Metrics	etcdwatcher_successes	Counter	The total number of successful ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_failures	Counter	The total number of failed ETCD Watcher Requests.

ETCD Client Metrics

- [etcd_put_requests](#)

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests.

Labels

keyPrefix

- [etcd_put_requests_sent](#)

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests sent.

Labels

keyPrefix

- [etcd_put_successes](#)

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT Requests.

Labels

keyPrefix

- [etcd_put_failures](#)

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT Requests.

Labels

keyPrefix, failureReason

- etcd_put_with_lease_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests sent.

Labels

keyPrefix

- etcd_put_with_lease_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT with Lease Requests.

Labels

keyPrefix, failureReason

- etcd_range_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests sent.

Labels

keyPrefix

- etcd_range_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Requests.

Labels

keyPrefix, failureReason

- etcd_range_delete_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests.

Labels

keyPrefix

- etcd_range_delete_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests sent.

Labels

keyPrefix

- etcd_range_delete_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Delete Requests.

Labels

keyPrefix

- etcd_range_delete_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Delete Requests.

Labels

keyPrefix, failureReason

- etcd_txn_requests

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests.

Labels

keyPrefix

- etcd_txn_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests sent.

Labels

keyPrefix

- etcd_txn_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Txn Requests.

Labels

keyPrefix

- etcd_txn_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Txn Requests.

Labels

keyPrefix, failureReason

etcd_watch_requests
Type
64-bit Counter
Description
The total number of ETCD Client Watch Requests.
Labels
keyPrefix
etcd_watch_requests_sent
Type
64-bit Counter
Description
The total number of ETCD Client Watch Requests sent.
Labels
keyPrefix
etcd_watch_successes
Type
64-bit Counter
Description
The total number of successful ETCD Client Watch Requests.
Labels
keyPrefix
etcd_watch_failures
Type
64-bit Counter
Description
The total number of failed ETCD Client Watch Requests.
Labels
keyPrefix, failureReason
etcdwatcher_requests
Type
64-bit Counter
Description
The total number of ETCD Watcher Requests.
Labels
keyPrefix
etcdwatcher_requests_sent
Type
64-bit Counter
Description

The total number of ETCD Watcher Requests sent.

Labels

keyPrefix

- etcdwatcher_successes

Type

64-bit Counter

Description

The total number of successful ETCD Watcher Requests.

Labels

keyPrefix

- etcdwatcher_failures

Type

64-bit Counter

Description

The total number of failed ETCD Watcher Requests.

Labels

keyPrefix, failureReason

Table of Contents

- [Summary](#)
- [HTTP Punting Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_sent	Counter	The total number of HTTP requests successfully punted.
HTTP Punting Metrics	punting_http_requests_failed	Counter	The total number of HTTP requests punting failed.
HTTP Punting Metrics	punting_http_responses_received	Counter	The total number of HTTP responses received for the requests punted.
HTTP Punting Metrics	punting_http_responses_forwarded	Counter	The total number of HTTP responses forwarded after punting transaction completion.
HTTP Punting Metrics	punting_http_requests_aborted	Counter	The total number of HTTP requests aborted.
HTTP Punting Metrics	punting_http_request_duration_seconds	Histogram	Punting operation duration in seconds.

HTTP Punting Metrics

- punting_http_requests_sent

Type

64-bit Counter

Description

The total number of HTTP requests successfully punted.

Labels

message_type

- `punting_http_requests_failed`

Type

64-bit Counter

Description

The total number of HTTP requests for which the punting failed.

Labels

message_type

- `punting_http_responses_received`

Type

64-bit Counter

Description

The total number of HTTP responses received for the requests punted.

Labels

message_type, status_code

- `punting_http_responses_forwarded`

Type

64-bit Counter

Description

The total number of HTTP responses forwarded after punting transaction completion.

Labels

message_type, status_code

- `punting_http_requests_aborted`

Type

64-bit Counter

Description

The total number of HTTP requests aborted.

Labels

message_type

- `punting_http_request_duration_seconds`

Type

64-bit Histogram

Description

The punting transaction duration is measured from the time the HTTP request is sent out to the time the response is received for the punting request.

Labels

message_type

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

- [Summary](#)
- [CDR-Agent Interface Stats](#)

Summary

Group	Metric Name	Type	Synopsis
CDR-Agent Interface Stats	charging_local_cdr_request	Counter	Total number of local CDR generation service requests received
CDR-Agent Interface Stats	charging_local_cdr_response	Counter	Total number of local CDR generation service responses sent

CDR-Agent Interface Stats

- charging_local_cdr_request
 - Type
 - 64-bit Counter
 - Description

Total number of local CDR generation service requests received by CdrAgent. The service requests count is per RecordType/Profile/FileFormat/GwType/CdrRecordType. RecordType is the type of local CDR namely ChfRecord etc. Profile, RecordType, FileFormat, GwType and CdrRecordType are populated from the received ChargingRecord message from the service consumers.
 - Labels

record_type, profile, file_format, gw_type, cdr_record_type
- charging_local_cdr_response
 - Type
 - 64-bit Counter
 - Description

Total number of local CDR generation service responses sent by CdrAgent. The service requests count is per RecordType/Profile/FileFormat/GwType/CdrRecordType. RecordType is the type of local CDR namely ChfRecord etc. Profile, RecordType, FileFormat, GwType and CdrRecordType are populated from the received ChargingRecord message from the service consumers.
 - Labels

record_type, profile, file_format, gw_type, cdr_record_type, status

Table of Contents

- [Summary](#)
- [Packet Streaming Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Packet Streaming Metrics	num_packet_streaming_attempted	Counter	The total number of attempted streaming of packets
Packet Streaming Metrics	num_packet_streaming_success	Counter	The total number of packets streamed successfully
Packet Streaming Metrics	num_packet_streaming_failed	Counter	The total number of failed streaming of packets
Packet Streaming Metrics	num_packet_streaming_bytes	Counter	The total number of bytes successfully streamed

Packet Streaming Metrics

- num_packet_streaming_attempted
 - Type
 - 64-bit Counter
 - Description

The total number of attempted streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_success

Type

64-bit Counter

Description

The total number of packets streamed successfully

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_failed

Type

64-bit Counter

Description

The total number of failed streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_bytes

Type

64-bit Counter

Description

The total number of bytes successfully streamed

Labels

interface_name, server_fqdn, kafka_topic

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- [Summary](#)
- [Circuit Breaker Metrics](#)
- [Service Endpoint Learner Metrics](#)
- [Service Endpoint Manager Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Circuit Breaker Metrics	sem_cb_endpoints	Gauge	Total number of endpoints for a service in respective states
Service Endpoint Learner Metrics	sem_num_dns_errors	Counter	Total number of DNS resolution errors for a given service
Service Endpoint Learner Metrics	sem_num_endpoints	Gauge	Total number of endpoints learnt for a service
Service Endpoint Manager Metrics	sem_num_endpoint_transactions_sent	Counter	Total number of times an endpoint is given to the application for communication with it
Service Endpoint Manager Metrics	sem_num_endpoint_transaction_status_received	Counter	Total number of transactions for a given endpoint received with the transaction status

Circuit Breaker Metrics

- **sem_cb_endpoints**

Type
64-bit Gauge
Description
Total number of endpoints for a service in respective states. There are 3 states viz. Active, Blocked, Restoring. In Active state, the endpoint is responding as expected, i.e. the response code doesn't indicate a service failure. The endpoint will be moved to Blocked state based on response failures which indicate service failure. After a timeout, the endpoint will be moved to Restoring state to check if the endpoint has recovered. In this state, if endpoint responds with response code other than service failure, it will be moved to Active state. Else, it will be moved back to Blocked state.
Labels
<i>service_name, endpoint_state</i>

Service Endpoint Learner Metrics

- **sem_num_dns_errors**

Type
64-bit Counter
Description
Total number of DNS resolution errors for a given service, labels are service name for DNS resolution and DNS error codes
Labels
<i>service_name, error_code</i>

- **sem_num_endpoints**

Type
64-bit Gauge
Description
Total number of endpoints learnt from the Kubernetes DNS server for a given service
Labels
<i>service</i>

Service Endpoint Manager Metrics

- **sem_num_endpoint_transactions_sent**

Type
64-bit Counter
Description
Total number of times an endpoint is given to the application for communication with it
Labels
<i>service</i>

- **sem_num_endpoint_transaction_status_received**

Type
64-bit Counter
Description
Total number of transactions for a given endpoint received with the transaction status
Labels

service, status

Table of Contents

- [Summary](#)
- [ConfigDataMgr Metrics](#)
- [ConfigSchemaMgr Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
ConfigDataMgr Metrics	cfgmgr_transactions_total	Counter	Cfgmgr schema transactions
ConfigDataMgr Metrics	cfgmgr_client_schema	Gauge	Cfgmgr client schema
ConfigDataMgr Metrics	cfgmgr_file_system_used_mbytes	Gauge	Cfgmgr file system disk space used.
ConfigDataMgr Metrics	cfgmgr_file_system_avail_mbytes	Gauge	Cfgmgr file system disk space available.
ConfigDataMgr Metrics	cfgmgr_file_system_usage_percent	Gauge	Cfgmgr file system disk space percent used.
ConfigSchemaMgr Metrics	cfgmgr_client_pods	Gauge	Running client pods.

ConfigDataMgr Metrics

- [cfgmgr_transactions_total](#)

Type
64-bit Counter
Description
None
Metrics

Instances	Labels	Description
1	type: schema_upgrade, status: success	Number of successful cfgmgr schema upgrades.
2	type: schema_upgrade, status: failure	Number of cfgmgr schema upgrade failures.
3	type: schema_remove, status: success	Number of successful cfgmgr schema removes.
4	type: schema_remove, status: failure	Number of cfgmgr schema remove failures.
5	type: commit, status: success	Number of cfgmgr commit transactions.

- [cfgmgr_file_system_used_mbytes](#)

Type
64-bit Gauge
Description
Disk total mbytes used.
Labels
path

- [cfgmgr_file_system_avail_mbytes](#)

Type
64-bit Gauge
Description
Disk space mbytes available.
Labels
path

- [cfgmgr_file_system_usage_percent](#)

Type

64-bit Gauge

Description

Disk space percent used.

Labels

path

- [cfgmgr_client_schema](#)

Type

64-bit Gauge

Description

None

Metrics

Instances	Labels	Description
1	<i>operId: loaded</i>	Number of loaded config schemas.

ConfigSchemaMgr Metrics

- [cfgmgr_client_pods](#)

Type

64-bit Gauge

Description

Number of running client pods.

Labels

podType

Table of Contents

- [Summary](#)
- [ChargingCmn statistics](#)

Summary

Group	Metric Name	Type	Synopsis
ChargingCmn statistics	svcmgr_service_available_ind_notification	Counter	How many times, Service Available Indication message received
ChargingCmn statistics	svcmgr_service_available_ind_parse_errs	Counter	How many times, Service Available Indication parse error happened
ChargingCmn statistics	svcmgr_service_available_ind_notification_ack	Counter	How many times, Service Available Indication message is acknowledged
ChargingCmn statistics	svcmgr_service_unavailable_ind_notification	Counter	How many times, Service Unavailable Indication message
ChargingCmn statistics	svcmgr_service_unavailable_ind_parse_errs	Counter	How many times, Service Unavailable Indication parse error happened
ChargingCmn statistics	svcmgr_service_unavailable_ind_notification_ack	Counter	How many times, Service Unavailable Indication message is acknowledged
ChargingCmn statistics	svcmgr_publish_msg_sent	Counter	How many times, Publication message sent

Group	Metric Name	Type	Synopsis
ChargingCmn statistics	svcmgr_publish_msg_success	Counter	How many times, Publication message success response received
ChargingCmn statistics	svcmgr_publish_msg_failure	Counter	How many times, failure received for publication message
ChargingCmn statistics	svcmgr_subscribe_msg_sent	Counter	How many times, Subscribe message sent
ChargingCmn statistics	svcmgr_subscribe_msg_success	Counter	How many times, subscribe message success
ChargingCmn statistics	svcmgr_subscribe_msg_failure	Counter	How many times, failure received for subscribe message
ChargingCmn statistics	svcmgr_unsubscribe_msg_sent	Counter	How many times, unsubscribe message sent
ChargingCmn statistics	svcmgr_unsubscribe_msg_failure	Counter	How many times, failure received for unsubscribe message
ChargingCmn statistics	svcmgr_config_req_sent	Counter	How many times, config request sent
ChargingCmn statistics	svcmgr_config_req_failure	Counter	How many times, failure received for config request
ChargingCmn statistics	svcmgr_config_req_success	Counter	How many times, success received for config request
ChargingCmn statistics	svcmgr_config_register_response	Counter	How many times, config register response received
ChargingCmn statistics	svcmgr_config_register_response_ack	Counter	How many times, config register response is acknowledged
ChargingCmn statistics	svcmgr_config_register_response_parse_err	Counter	How many times, parse error happened for config register response
ChargingCmn statistics	svcmgr_config_update_notification	Counter	How many times, config update notification received
ChargingCmn statistics	svcmgr_config_update_notification_response	Counter	How many times, config update response
ChargingCmn statistics	svcmgr_intf_schema_update_ind_notification	Counter	How many times, interface schema update indication message received
ChargingCmn statistics	svcmgr_intf_schema_update_ind_notification_ack	Counter	How many times, interface schema update indication message is acknowledged
ChargingCmn statistics	svcmgr_intf_schema_update_ind_parse_errs	Counter	How many times, interface schema update indication parse error happened

ChargingCmn statistics

- svcmgr_service_available_ind_notification

Type

64-bit Counter

Description

Number of Times Service Available Indication notification received

Labels

servicename

- svcmgr_service_available_ind_parse_errs

Type

64-bit Counter

Description

Number of Times Service Available Indication parse error happened

Labels

- `svcmgr_service_available_ind_notification_ack`

Type

64-bit Counter

Description

Number of Times Service Available Indication message is acknowledged

Labels

servicename

- `svcmgr_service_unavailable_ind_notification`

Type

64-bit Counter

Description

Number of Times Service Unavailable Indication notification received

Labels

servicename

- `svcmgr_service_unavailable_ind_parse_errs`

Type

64-bit Counter

Description

Number of Times Service Unavailable Indication parse error happened

Labels

- `svcmgr_service_unavailable_ind_notification_ack`

Type

64-bit Counter

Description

Number of Times Service Unavailable Indication message is acknowledged

Labels

servicename

- `svcmgr_publish_msg_sent`

Type

64-bit Counter

Description

Number of Times Publication message sent

Labels

servicename, retry

- `svcmgr_publish_msg_success`

Type

64-bit Counter

Description

Number of Times Publication message success response received

Labels

servicename, causecode, retry

- **svcmgr_publish_msg_failure**

Type

64-bit Counter

Description

Number of Times failure received for publication message

Labels

servicename, causecode, retry

- **svcmgr_subscribe_msg_sent**

Type

64-bit Counter

Description

Number of Times Subscribe message sent

Labels

servicename, retry

- **svcmgr_subscribe_msg_success**

Type

64-bit Counter

Description

Number of Times Send subscribe message success response received

Labels

servicename, causecode, retry

- **svcmgr_subscribe_msg_failure**

Type

64-bit Counter

Description

Number of time failure received for subscribe message

Labels

servicename, causecode, retry

- **svcmgr_unsubscribe_msg_sent**

Type

64-bit Counter

Description

Number of Times unsubscribe message sent

Labels

servicename

- `svcmgr_unsubscribe_msg_failure`

Type

64-bit Counter

Description

Number of Times failure received for unsubscribe message

Labels

servicename

- `svcmgr_config_req_sent`

Type

64-bit Counter

Description

Number of Times config request sent

Labels

retry

- `svcmgr_config_req_failure`

Type

64-bit Counter

Description

Number of Times failure received for config request

Labels

causecode, retry

- `svcmgr_config_req_success`

Type

64-bit Counter

Description

Number of Times success received for config request

Labels

causecode, retry

- `svcmgr_config_register_response`

Type

64-bit Counter

Description

Number of Times config register response received

Labels

statuscode

- `svcmgr_config_register_response_ack`

Type

64-bit Counter

Description

Number of Times config register response is acknowledged

Labels

- `svcmgr_config_register_response_parse_err`

Type

64-bit Counter

Description

Number of Times parse error happened for config register response

Labels

- `svcmgr_config_update_notification`

Type

64-bit Counter

Description

Number of Times config update notification received

Labels

- `svcmgr_config_update_notification_response`

Type

64-bit Counter

Description

Number of Times config update response sent

Labels

- `svcmgr_intf_schema_update_ind_notification`

Type

64-bit Counter

Description

Number of Times interface schema update indication message received

Labels

servicename

- `svcmgr_intf_schema_update_ind_notification_ack`

Type

64-bit Counter

Description

Number of Times interface schema update indication message is acknowledged

Labels

servicename

- `svcmgr_intf_schema_update_ind_parse_errs`

Type

64-bit Counter

Description

Number of Times Service interface schema update indication parse error happened

Labels

Table of Contents

- [Summary](#)
- [HTTP Punting Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_sent	Counter	The total number of HTTP requests successfully punted.
HTTP Punting Metrics	punting_http_requests_failed	Counter	The total number of HTTP requests punting failed.
HTTP Punting Metrics	punting_http_responses_received	Counter	The total number of HTTP responses received for the requests punted.
HTTP Punting Metrics	punting_http_responses_forwarded	Counter	The total number of HTTP responses forwarded after punting transaction completion.
HTTP Punting Metrics	punting_http_requests_aborted	Counter	The total number of HTTP requests aborted.
HTTP Punting Metrics	punting_http_request_duration_seconds	Histogram	Punting operation duration in seconds.

HTTP Punting Metrics

- [punting_http_requests_sent](#)

Type

64-bit Counter

Description

The total number of HTTP requests successfully punted.

Labels

message_type

- [punting_http_requests_failed](#)

Type

64-bit Counter

Description

The total number of HTTP requests for which the punting failed.

Labels

message_type

- [punting_http_responses_received](#)

Type

64-bit Counter

Description

The total number of HTTP responses received for the requests punted.

Labels

message_type, status_code

- `punting_http_responses_forwarded`

Type

64-bit Counter

Description

The total number of HTTP responses forwarded after punting transaction completion.

Labels

message_type, status_code

- `punting_http_requests_aborted`

Type

64-bit Counter

Description

The total number of HTTP requests aborted.

Labels

message_type

- `punting_http_request_duration_seconds`

Type

64-bit Histogram

Description

The punting transaction duration is measured from the time the HTTP request is sent out to the time the response is received for the punting request.

Labels

message_type

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

Summary

- 1. [cfgmgr.alert.rules Alerts](#)

cfgmgr.alert.rules

Alert Name	Description	Alert Info	Diagnostics
cfgmgr_file_system_usage_alert_{{.Release.Namespace}}	Cfgmgr file system disk space usage exceeded threshold of 85%	1. Summary: Cfgmgr file system disk space usage exceeded threshold Severity: major Realert Time: 1m Expression: cfgmgr_file_system_usage_percent{namespace="{{.Release.Namespace}}"} > 85	1. Potential Impact: Cfgmgr may become unusable. New pod schema may not be added and new config transactions may not be possible Repair Action: Check the file system for any large or unwanted files, especially core files. Consider mounting this path to a larger persistent volume

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- [Summary](#)
- [Packet Streaming Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Packet Streaming Metrics	num_packet_streaming_attempted	Counter	The total number of attempted streaming of packets
Packet Streaming Metrics	num_packet_streaming_success	Counter	The total number of packets streamed successfully
Packet Streaming Metrics	num_packet_streaming_failed	Counter	The total number of failed streaming of packets
Packet Streaming Metrics	num_packet_streaming_bytes	Counter	The total number of bytes successfully streamed

Packet Streaming Metrics

- num_packet_streaming_attempted

Type

64-bit Counter

Description

The total number of attempted streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_success

Type

64-bit Counter

Description

The total number of packets streamed successfully

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_failed

Type

64-bit Counter

Description

The total number of failed streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_bytes

Type

64-bit Counter

Description

The total number of bytes successfully streamed

Labels

interface_name, server_fqdn, kafka_topic

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- [Summary](#)
- [ChargingCmn statistics](#)

Summary

Group	Metric Name	Type	Synopsis
ChargingCmn statistics	svcmgr_service_available_ind_notification	Counter	How many times, Service Available Indication message received
ChargingCmn statistics	svcmgr_service_available_ind_parse_errs	Counter	How many times, Service Available Indication parse error happened
ChargingCmn statistics	svcmgr_service_available_ind_notification_ack	Counter	How many times, Service Available Indication message is acknowledged
ChargingCmn statistics	svcmgr_service_unavailable_ind_notification	Counter	How many times, Service Unavailable Indication message
ChargingCmn statistics	svcmgr_service_unavailable_ind_parse_errs	Counter	How many times, Service Unavailable Indication parse error happened
ChargingCmn statistics	svcmgr_service_unavailable_ind_notification_ack	Counter	How many times, Service Unavailable Indication message is acknowledged
ChargingCmn statistics	svcmgr_publish_msg_sent	Counter	How many times, Publication message sent
ChargingCmn statistics	svcmgr_publish_msg_success	Counter	How many times, Publication message success response received
ChargingCmn statistics	svcmgr_publish_msg_failure	Counter	How many times, failure received for publication message
ChargingCmn statistics	svcmgr_subscribe_msg_sent	Counter	How many times, Subscribe message sent
ChargingCmn statistics	svcmgr_subscribe_msg_success	Counter	How many times, subscribe message success
ChargingCmn statistics	svcmgr_subscribe_msg_failure	Counter	How many times, failure received for subscribe message
ChargingCmn statistics	svcmgr_unsubscribe_msg_sent	Counter	How many times, unsubscribe message sent
ChargingCmn statistics	svcmgr_unsubscribe_msg_failure	Counter	How many times, failure received for unsubscribe message
ChargingCmn statistics	svcmgr_config_req_sent	Counter	How many times, config request sent
ChargingCmn statistics	svcmgr_config_req_failure	Counter	How many times, failure received for config request
ChargingCmn statistics	svcmgr_config_req_success	Counter	How many times, success received for config request
ChargingCmn statistics	svcmgr_config_register_response	Counter	How many times, config register response received
ChargingCmn statistics	svcmgr_config_register_response_ack	Counter	How many times, config register response is acknowledged
ChargingCmn statistics	svcmgr_config_register_response_parse_err	Counter	How many times, parse error happened for config register response
ChargingCmn statistics	svcmgr_config_update_notification	Counter	How many times, config update notification received
ChargingCmn statistics	svcmgr_config_update_notification_response	Counter	How many times, config update response
ChargingCmn statistics	svcmgr_intf_schema_update_ind_notification	Counter	How many times, interface schema update indication message received
ChargingCmn statistics	svcmgr_intf_schema_update_ind_notification_ack	Counter	How many times, interface schema update indication message is acknowledged

Group	Metric Name	Type	Synopsis
ChargingCmn statistics	svcmgr_intf_schema_update_ind_parse_errs	Counter	How many times, interface schema update indication parse error happened

ChargingCmn statistics

- svcmgr_service_available_ind_notification

Type

64-bit Counter

Description

Number of Times Service Available Indication notification received

Labels

servicename

- svcmgr_service_available_ind_parse_errs

Type

64-bit Counter

Description

Number of Times Service Available Indication parse error happened

Labels

- svcmgr_service_available_ind_notification_ack

Type

64-bit Counter

Description

Number of Times Service Available Indication message is acknowledged

Labels

servicename

- svcmgr_service_unavailable_ind_notification

Type

64-bit Counter

Description

Number of Times Service Unavailable Indication notification received

Labels

servicename

- svcmgr_service_unavailable_ind_parse_errs

Type

64-bit Counter

Description

Number of Times Service Unavailable Indication parse error happened

Labels

- svcmgr_service_unavailable_ind_notification_ack

Type

64-bit Counter

Description

Number of Times Service Unavailable Indication message is acknowledged

Labels

servicename

- **svcmgr_publish_msg_sent**

Type

64-bit Counter

Description

Number of Times Publication message sent

Labels

servicename, retry

- **svcmgr_publish_msg_success**

Type

64-bit Counter

Description

Number of Times Publication message success response received

Labels

servicename, causecode, retry

- **svcmgr_publish_msg_failure**

Type

64-bit Counter

Description

Number of Times failure received for publication message

Labels

servicename, causecode, retry

- **svcmgr_subscribe_msg_sent**

Type

64-bit Counter

Description

Number of Times Subscribe message sent

Labels

servicename, retry

- **svcmgr_subscribe_msg_success**

Type

64-bit Counter

Description

Number of Times Send subscribe message success response received

Labels

servicename, causecode, retry

- **svcmgr_subscribe_msg_failure**

Type

64-bit Counter

Description

Number of time failure received for subscribe message

Labels

servicename, causecode, retry

- **svcmgr_unsubscribe_msg_sent**

Type

64-bit Counter

Description

Number of Times unsubscribe message sent

Labels

servicename

- **svcmgr_unsubscribe_msg_failure**

Type

64-bit Counter

Description

Number of Times failure received for unsubscribe message

Labels

servicename

- **svcmgr_config_req_sent**

Type

64-bit Counter

Description

Number of Times config request sent

Labels

retry

- **svcmgr_config_req_failure**

Type

64-bit Counter

Description

Number of Times failure received for config request

Labels

causecode, retry

- **svcmgr_config_req_success**

Type

64-bit Counter

Description

Number of Times success received for config request

Labels

causecode, retry

• svcmgr_config_register_response

Type

64-bit Counter

Description

Number of Times config register response received

Labels

statuscode

• svcmgr_config_register_response_ack

Type

64-bit Counter

Description

Number of Times config register response is acknowledged

Labels

• svcmgr_config_register_response_parse_err

Type

64-bit Counter

Description

Number of Times parse error happened for config register response

Labels

• svcmgr_config_update_notification

Type

64-bit Counter

Description

Number of Times config update notification received

Labels

• svcmgr_config_update_notification_response

Type

64-bit Counter

Description

Number of Times config update response sent

Labels

• svcmgr_intf_schema_update_ind_notification

Type

64-bit Counter

Description

Number of Times interface schema update indication message received

Labels

servicename

- [svcmgr_intf_schema_update_ind_notification_ack](#)

Type

64-bit Counter

Description

Number of Times interface schema update indication message is acknowledged

Labels

servicename

- [svcmgr_intf_schema_update_ind_parse_errs](#)

Type

64-bit Counter

Description

Number of Times Service interface schema update indication parse error happened

Labels

Table of Contents

- [Summary](#)
- [ETCD Client Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_put_requests	Counter	The total number of ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_requests_sent	Counter	The total number of ETCD Client PUT Requests sent.
ETCD Client Metrics	etcd_put_successes	Counter	The total number of successful ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_failures	Counter	The total number of failed ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_with_lease_requests	Counter	The total number of ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_requests_sent	Counter	The total number of ETCD Client PUT with Lease Requests sent.
ETCD Client Metrics	etcd_put_with_lease_successes	Counter	The total number of successful ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_failures	Counter	The total number of failed ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_range_requests	Counter	The total number of ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_requests_sent	Counter	The total number of ETCD Client Range Requests sent.
ETCD Client Metrics	etcd_range_successes	Counter	The total number of successful ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_failures	Counter	The total number of failed ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_delete_requests	Counter	The total number of ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_requests_sent	Counter	The total number of ETCD Client Range Delete Requests sent.
ETCD Client Metrics	etcd_range_delete_successes	Counter	The total number of successful ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_failures	Counter	The total number of failed ETCD Client Range Delete Requests.

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_txn_requests	Counter	The total number of ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_requests_sent	Counter	The total number of ETCD Client Txn Requests sent.
ETCD Client Metrics	etcd_txn_successes	Counter	The total number of successful ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_failures	Counter	The total number of failed ETCD Client Txn Requests.
ETCD Client Metrics	etcd_watch_requests	Counter	The total number of ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_requests_sent	Counter	The total number of ETCD Client Watch Requests sent.
ETCD Client Metrics	etcd_watch_successes	Counter	The total number of successful ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_failures	Counter	The total number of failed ETCD Client Watch Requests.
ETCD Client Metrics	etcdwatcher_requests	Counter	The total number of ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_requests_sent	Counter	The total number of ETCD Watcher Requests sent.
ETCD Client Metrics	etcdwatcher_successes	Counter	The total number of successful ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_failures	Counter	The total number of failed ETCD Watcher Requests.

ETCD Client Metrics

- `etcd_put_requests`

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests.

Labels

keyPrefix

- `etcd_put_requests_sent`

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests sent.

Labels

keyPrefix

- `etcd_put_successes`

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT Requests.

Labels

keyPrefix

- `etcd_put_failures`

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT Requests.

Labels

keyPrefix, failureReason

• etcd_put_with_lease_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests.

Labels

keyPrefix

• etcd_put_with_lease_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests sent.

Labels

keyPrefix

• etcd_put_with_lease_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT with Lease Requests.

Labels

keyPrefix

• etcd_put_with_lease_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT with Lease Requests.

Labels

keyPrefix, failureReason

• etcd_range_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests.

Labels

keyPrefix

• etcd_range_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests sent.

Labels

keyPrefix

- etcd_range_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Requests.

Labels

keyPrefix, failureReason

- etcd_range_delete_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests.

Labels

keyPrefix

- etcd_range_delete_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests sent.

Labels

keyPrefix

- etcd_range_delete_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Delete Requests.

Labels

keyPrefix

- etcd_range_delete_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Delete Requests.

Labels

keyPrefix, failureReason

- etcd_txn_requests

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests.

Labels

keyPrefix

- etcd_txn_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests sent.

Labels

keyPrefix

- etcd_txn_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Txn Requests.

Labels

keyPrefix

- etcd_txn_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Txn Requests.

Labels

keyPrefix, failureReason

- etcd_watch_requests

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests sent.

Labels

keyPrefix

- etcd_watch_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Watch Requests.

Labels

keyPrefix, failureReason

- etcdwatcher_requests

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests.

Labels

keyPrefix

- etcdwatcher_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests sent.

Labels

keyPrefix

- etcdwatcher_successes

Type

64-bit Counter

Description

The total number of successful ETCD Watcher Requests.

Labels

keyPrefix

- etcdwatcher_failures

Type

64-bit Counter

Description

The total number of failed ETCD Watcher Requests.

Labels

keyPrefix, failureReason

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- [Summary](#)
- [CHF Failover Stats](#)

Summary

Group	Metric Name	Type	Synopsis
CHF Failover Stats	failover_with_primary_server_not_attempted	Counter	The number of times failover attempted due to primary server unavailability.
CHF Failover Stats	failover_with_primary_server_attempted	Counter	The number of times failover attempted due to negative response from the primary server.
CHF Failover Stats	total_failover_attempted	Counter	The number of times failover attempted.

CHF Failover Stats

- failover_with_primary_server_not_attempted

Type

64-bit Counter

Description

The number of times failover attempted with the alternate server because the primary server was unreachable due to admin state or server is blocklisted etc.

Labels

alternate_server, message_type

- failover_with_primary_server_attempted

Type

64-bit Counter

Description

The number of times failover attempted with the alternate server after receiving configured failover status code from the primary server.

Labels

alternate_server, message_type

- total_failover_attempted

Type

64-bit Counter

Description

The total number of times failover attempted with alternate server.

Labels

alternate_server

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- [Summary](#)
- [HTTP Punting Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_sent	Counter	The total number of HTTP requests successfully punted.
HTTP Punting Metrics	punting_http_requests_failed	Counter	The total number of HTTP requests punting failed.
HTTP Punting Metrics	punting_http_responses_received	Counter	The total number of HTTP responses received for the requests punted.
HTTP Punting Metrics	punting_http_responses_forwarded	Counter	The total number of HTTP responses forwarded after punting transaction completion.
HTTP Punting Metrics	punting_http_requests_aborted	Counter	The total number of HTTP requests aborted.
HTTP Punting Metrics	punting_http_request_duration_seconds	Histogram	Punting operation duration in seconds.

HTTP Punting Metrics

- punting_http_requests_sent

Type

64-bit Counter

Description

The total number of HTTP requests successfully punted.

Labels

message_type

- punting_http_requests_failed

Type

64-bit Counter

Description

The total number of HTTP requests for which the punting failed.

Labels

message_type

- punting_http_responses_received

Type

64-bit Counter

Description

The total number of HTTP responses received for the requests punted.

Labels

message_type, status_code

- **punting_http_responses_forwarded**

Type

64-bit Counter

Description

The total number of HTTP responses forwarded after punting transaction completion.

Labels

message_type, status_code

- **punting_http_requests_aborted**

Type

64-bit Counter

Description

The total number of HTTP requests aborted.

Labels

message_type

- **punting_http_request_duration_seconds**

Type

64-bit Histogram

Description

The punting transaction duration is measured from the time the HTTP request is sent out to the time the response is received for the punting request.

Labels

message_type

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

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- [Summary](#)
- [CHF Agent Interface Stats](#)

Summary

Group	Metric Name	Type	Synopsis
CHF Agent Interface Stats	charging_nchf_convergedcharging_request	Counter	Total number of convergedcharging service requests received
CHF Agent Interface Stats	charging_nchf_convergedcharging_response	Counter	Total number of convergedcharging service responses sent
CHF Agent Interface Stats	charging_lcdr_client_request	Counter	Total number of Local CDR Charging requests sent

Group	Metric Name	Type	Synopsis
CHF Agent Interface Stats	charging_lcdr_client_response	Counter	Total number of Local CDR Charging responses received
CHF Agent Interface Stats	charging_nchf_server_convergedcharging_client_request	Counter	Total number of nchf-convergedcharging service requests sent to CHF
CHF Agent Interface Stats	charging_nchf_server_convergedcharging_client_response	Counter	Total number of nchf-convergedcharging service responses received from CHF
CHF Agent Interface Stats	charging_nchf_server_convergedcharging_client_internal_response	Counter	Total number of nchf-convergedcharging service responses internally generated from ChfAgent(460-499)
CHF Agent Interface Stats	charging_kafka_client_request	Counter	Total number of one time usage requests sent to Kafka Message Broker
CHF Agent Interface Stats	charging_kafka_client_response	Counter	Total number of one time usage responses received from Kafka Message Broker.
CHF Agent Interface Stats	charging_nchf_server_message_response_duration	Histogram	Charging message response time duration in seconds from CHF Server

CHF Agent Interface Stats

- charging_nchf_convergedcharging_request

Type

64-bit Counter

Description

Total number of convergedcharging service requests received by ChfAgent. The service requests count is per DNN/NodeFunctionality/RatType/MessageType. Dnn, NodeFunctionality and RatType are populated from the convergedcharging service message received from the NF consumers whereas MessageType is one of the service operation types namely create/update/release.

Labels

dnn, node_functionality, rat_type, slice, message_type

- charging_nchf_convergedcharging_response

Type

64-bit Counter

Description

Total number of convergedcharging service responses sent by ChfAgent. The service responses count is per DNN/NodeFunctionality/RatType/MessageType/HTTP Status Codes. Dnn, NodeFunctionality and RatType are populated from the convergedcharging service message received from the NF consumers whereas MessageType is one of the service operation types namely create/update/release. Status is any of the HTTP response codes sent in the response message.

Labels

dnn, node_functionality, rat_type, slice, message_type, status

- charging_lcdr_client_request

Type

64-bit Counter

Description

Total number of Local CDR Charging service requests sent by ChfAgent. The service requests count is per MessageType and RecordType. MessageType is one of the service operation types namely create/update/release. RecordType is any of the supported CDR Record types namely ChfRecord and PgwRecord.

Labels

message_type, record_type

- charging_lcdr_client_response

Type

64-bit Counter

Description

Total number of Local CDR Charging service responses received by ChfAgent. The service responses count is per RecordType and Status. RecordType is any of the supported CDR Record types namely ChfRecord and PgwRecord. Status is any of the HTTP response codes received in the response message.

Labels

record_type, status

- **charging_nchf_server_convergedcharging_client_request**

Type

64-bit Counter

Description

Total number of nchf-convergedcharging service requests sent to a charging function (CHF). The service requests count is per DNN/NodeFunctionality/RatType/MessageType. Dnn, NodeFunctionality and RatType are populated from the nchf-convergedcharging service message received from the NF consumers whereas MessageType is one of the service operation types namely create/update/release.

Labels

chf_server, dnn, node_functionality, rat_type, slice, message_type

- **charging_nchf_server_convergedcharging_client_response**

Type

64-bit Counter

Description

Total number of nchf-convergedcharging service responses received by ChfAgent from a CHF server. The service responses count is per DNN/NodeFunctionality/RatType/MessageType/HTTP Status Codes. Dnn, NodeFunctionality and RatType are populated from the nchf-convergedcharging service message received from the NF consumers whereas MessageType is one of the service operation types namely create/update/release. Status is any of the HTTP response codes sent in the response message.

Labels

chf_server, dnn, node_functionality, rat_type, slice, message_type, status

- **charging_nchf_server_convergedcharging_client_internal_response**

Type

64-bit Counter

Description

Total number of nchf-convergedcharging service responses internally generated from ChfAgent(Httpplib 460-499). The service responses count is per DNN/NodeFunctionality/RatType/MessageType/HTTP Status Codes. Dnn, NodeFunctionality and RatType are populated from the nchf-convergedcharging service message received from the NF consumers whereas MessageType is one of the service operation types namely create/update/release. Status is 460-499 HTTP response codes.

Labels

chf_server, dnn, node_functionality, rat_type, slice, message_type, status

- **charging_kafka_client_request**

Type

64-bit Counter

Description

Total number of one time usage requests sent to Kafka Message Broker.

Labels

topic_name

- **charging_kafka_client_response**

Type

64-bit Counter

Description

Total number of one time usage responses received from Kafka Message Broker.

Labels

topic_name, status

- charging_nchf_server_message_response_duration

Type

64-bit Histogram

Description

The duration is measured from charging message send to receipt of the response from CHF Server

Labels

message

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

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- [Summary](#)
- [Registration of a CHF FQDN in Istio Config Agent stats](#)

Summary

Group	Metric Name	Type	Synopsis
Registration of a CHF FQDN in Istio Config Agent stats	charging_nchf_istio_cfgAgt_operation	Counter	Number of times a particular CHF FQDN attempts to register and deregister with Istio Config Agent

Registration of a CHF FQDN in Istio Config Agent stats

- charging_nchf_istio_cfgAgt_operation

Type

64-bit Counter

Description

The total number of times a particular CHF FQDN attempts to register and deregister itself with Istio Config Agent

Labels

chf_server_fqdn, chf_server_host_name, chf_server_ip_address, istio_operation, status, service_id

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- [Summary](#)
- [CHF Server Reachability Stats](#)

Summary

Group	Metric Name	Type	Synopsis
CHF Server Reachability Stats	charging_num_nchf_server_blacklisted	Counter	Number of times a particular CHF server is blacklisted.
CHF Server Reachability Stats	charging_num_nchf_server_blocklisted	Counter	Number of times a particular CHF server is blocklisted.
CHF Server Reachability Stats	charging_num_nchf_server_whitelisted	Counter	Number of times a particular CHF server is whitelisted.
CHF Server Reachability Stats	charging_num_nchf_server_allowedlisted	Counter	Number of times a particular CHF server is allowedlisted.

CHF Server Reachability Stats

- charging_num_nchf_server_blacklisted [Deprecated]

Type64-bit CounterDescriptionThe total number of times a particular CHF server is considered as blacklisted for its reachability due to configured consecutive number of failures in reaching it.Labels**chf_server**
- charging_num_nchf_server_blocklisted

Type64-bit CounterDescriptionThe total number of times a particular CHF server is considered as blocklisted for its reachability due to configured consecutive number of failures in reaching it.Labels**chf_server**
- charging_num_nchf_server_whitelisted [Deprecated]

Type64-bit CounterDescriptionThe total number of times a particular CHF server is considered as whitelisted after which the server is considered as available for interfacing with it.Labels**chf_server**
- charging_num_nchf_server_allowedlisted

Type64-bit CounterDescriptionThe total number of times a particular CHF server is considered as allowedlisted after which the server is considered as available for interfacing with it.Labels**chf_server**

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- [Summary](#)
- [Packet Streaming Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Packet Streaming Metrics	num_packet_streaming_attempted	Counter	The total number of attempted streaming of packets
Packet Streaming Metrics	num_packet_streaming_success	Counter	The total number of packets streamed successfully
Packet Streaming Metrics	num_packet_streaming_failed	Counter	The total number of failed streaming of packets
Packet Streaming Metrics	num_packet_streaming_bytes	Counter	The total number of bytes successfully streamed

Packet Streaming Metrics

- num_packet_streaming_attempted

Type

64-bit Counter

Description

The total number of attempted streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_success

Type

64-bit Counter

Description

The total number of packets streamed successfully

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_failed

Type

64-bit Counter

Description

The total number of failed streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_bytes

Type

64-bit Counter

Description

The total number of bytes successfully streamed

Labels

interface_name, server_fqdn, kafka_topic

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- [Summary](#)
- [CHF Agent Procedure Stats](#)

Summary

Group	Metric Name	Type	Synopsis
CHF Agent Procedure Stats	charging_nchf_session_create_db_request_attempts	Counter	Total number of DB requests sent for charging session create event
CHF Agent Procedure Stats	charging_nchf_session_create_db_requests_success	Counter	Total number of DB responses received for charging session create event

Group	Metric Name	Type	Synopsis
CHF Agent Procedure Stats	charging_nchf_session_create_db_request_failures	Counter	Total number of DB requests sent failure for charging session create event
CHF Agent Procedure Stats	charging_nchf_session_update_db_request_attempts	Counter	Total number of DB requests sent for charging session Update event
CHF Agent Procedure Stats	charging_nchf_session_update_db_requests_success	Counter	Total number of DB responses received for charging session Update event
CHF Agent Procedure Stats	charging_nchf_session_update_db_request_failures	Counter	Total number of DB requests sent failure for charging session Update event
CHF Agent Procedure Stats	charging_nchf_session_release_db_request_attempts	Counter	Total number of DB requests sent for charging session Release event
CHF Agent Procedure Stats	charging_nchf_session_release_db_requests_success	Counter	Total number of DB responses received for charging session Release event
CHF Agent Procedure Stats	charging_nchf_session_release_db_request_failures	Counter	Total number of DB requests sent failure for charging session Release event
CHF Agent Procedure Stats	charging_nchf_session_unknown_db_request_transaction_attempts	Counter	Number of unknown db transactions received by CHF Agent in request flow
CHF Agent Procedure Stats	charging_nchf_session_unknown_db_request_transactions_success	Counter	Number of unknown db transactions received by CHF Agent in response flow
CHF Agent Procedure Stats	charging_nchf_session_unknown_db_request_transaction_failures	Counter	Number of unknown db failure transactions received by CHF Agent in request flow
CHF Agent Procedure Stats	charging_nchf_session_abort_db_requests	Counter	Total number of times DB requests are aborted
CHF Agent Procedure Stats	charging_nchf_session_udsf_connection_status	Counter	Number of times the connection to UDSF has gone up or down
CHF Agent Procedure Stats	charging_nchf_session_cnf_state_status	Counter	Number of times the state of the CNF has changed
CHF Agent Procedure Stats	charging_nchf_session_db_request_rejected_for_nf_state_down	Counter	Number of requests rejected due to the NF state being down
CHF Agent Procedure Stats	charging_nchf_session_db_request_rejected_for_udsf_clientmgr_not_found	Counter	Number of requests rejected due to the udsf clientmgr object not found
CHF Agent Procedure Stats	charging_nchf_session_db_request_rejected_for_swagger_record_not_found	Counter	Number of requests rejected due to protobuf to swagger record conversion is failed
CHF Agent Procedure Stats	charging_nchf_session_db_response_null_transaction_id_ptr	Counter	Number of responses for which the transaction id pointer is null
CHF Agent Procedure Stats	charging_nchf_session_db_response_null_egress_event_ptr	Counter	Number of responses for which the egress event pointer is null
CHF Agent Procedure Stats	charging_nchf_session_db_response_null_proc_ptr	Counter	Number of responses for which the proc pointer is null
CHF Agent Procedure Stats	charging_nchf_create_procedure_duration	Histogram	Charging Session create Procedure Execution Time
CHF Agent Procedure Stats	charging_nchf_update_procedure_duration	Histogram	Charging Session Update Procedure Execution Time
CHF Agent Procedure Stats	charging_nchf_release_procedure_duration	Histogram	Charging Session Release Procedure Execution Time

CHF Agent Procedure Stats

- [charging_nchf_session_create_db_request_attempts](#)

Type

64-bit Counter

Description

Total number of DB requests sent for charging session create event. The requests count is per DNN/NodeFunctionality/RatType/MessageType/TriggerEvent/OperationType/RetryStatus. Dnn, NodeFunctionality and RatType are populated from the

Nchf_ConvergedCharging service message received from the NF consumers whereas MessageType is one of the DB operation types. RetryStatus indicates whether this request is part of retry attempt or normal request.

Labels

dnn, node_functionality, rat_type, slice, message_type, trigevent, optype, retry

• charging_nchf_session_create_db_requests_success

Type

64-bit Counter

Description

Total number of DB responses received for charging session create event. The requests count is per DNN/NodeFunctionality/RatType/MessageType/TriggerEvent/OperationType/ResponseCodeMsg. Dnn, NodeFunctionality and RatType are populated from the Nchf_ConvergedCharging service message received from the NF consumers whereas MessageType is one of the DB operation types. ResponseCodeMsg is UDSF response message received in the lib API response.

Labels

dnn, node_functionality, rat_type, slice, message_type, trigevent, optype, responsecodemsg

• charging_nchf_session_create_db_request_failures

Type

64-bit Counter

Description

Total number of DB requests sent failure for charging session create event. The requests count is per DNN/NodeFunctionality/RatType/MessageType/TriggerEvent/OperationType/ResponseCodeMsg. Dnn, NodeFunctionality and RatType are populated from the Nchf_ConvergedCharging service message received from the NF consumers whereas MessageType is one of the DB operation types. ResponseCodeMsg is UDSF response message received in the lib API response.

Labels

dnn, node_functionality, rat_type, slice, message_type, trigevent, optype, responsecodemsg

• charging_nchf_session_update_db_request_attempts

Type

64-bit Counter

Description

Total number of DB requests sent for charging session Update event. The requests count is per DNN/NodeFunctionality/RatType/MessageType/TriggerEvent/OperationType/RetryStatus. Dnn, NodeFunctionality and RatType are populated from the Nchf_ConvergedCharging service message received from the NF consumers whereas MessageType is one of the DB operation types. RetryStatus indicates whether this request is part of retry attempt or normal request.

Labels

dnn, node_functionality, rat_type, slice, message_type, trigevent, optype, retry

• charging_nchf_session_update_db_requests_success

Type

64-bit Counter

Description

Total number of DB responses received for charging session Update event. The requests count is per DNN/NodeFunctionality/RatType/MessageType/TriggerEvent/OperationType/ResponseCodeMsg. Dnn, NodeFunctionality and RatType are populated from the Nchf_ConvergedCharging service message received from the NF consumers whereas MessageType is one of the DB operation types. ResponseCodeMsg is UDSF response message received in the lib API response.

Labels

dnn, node_functionality, rat_type, slice, message_type, trigevent, optype, responsecodemsg

• charging_nchf_session_update_db_request_failures

Type

64-bit Counter

Description

Total number of DB requests sent failure for charging session Update event. The requests count is per DNN/NodeFunctionality/RatType/MessageType/TriggerEvent/OperationType/ResponseCodeMsg. Dnn, NodeFunctionality and RatType are populated from the Nchf_ConvergedCharging service message received from the NF consumers whereas MessageType is one of the DB operation types. ResponseCodeMsg is UDSF response message received in the lib API response.

Labels

dnn, node_functionality, rat_type, slice, message_type, trigevent, optype, responsecodemsg

- charging_nchf_session_release_db_request_attempts

Type

64-bit Counter

Description

Total number of DB requests sent for charging session Release event. The requests count is per DNN/NodeFunctionality/RatType/MessageType/TriggerEvent/OperationType/RetryStatus. Dnn, NodeFunctionality and RatType are populated from the Nchf_ConvergedCharging service message received from the NF consumers whereas MessageType is one of the DB operation types. RetryStatus indicates whether this request is part of retry attempt or normal request.

Labels

dnn, node_functionality, rat_type, slice, message_type, trigevent, optype, retry

- charging_nchf_session_release_db_requests_success

Type

64-bit Counter

Description

Total number of DB responses received for charging session Release event. The requests count is per DNN/NodeFunctionality/RatType/MessageType/TriggerEvent/OperationType/ResponseCodeMsg. Dnn, NodeFunctionality and RatType are populated from the Nchf_ConvergedCharging service message received from the NF consumers whereas MessageType is one of the DB operation types. ResponseCodeMsg is UDSF response message received in the lib API response.

Labels

dnn, node_functionality, rat_type, slice, message_type, trigevent, optype, responsecodemsg

- charging_nchf_session_release_db_request_failures

Type

64-bit Counter

Description

Total number of DB requests sent failure for charging session Release event. The requests count is per DNN/NodeFunctionality/RatType/MessageType/TriggerEvent/OperationType/ResponseCodeMsg. Dnn, NodeFunctionality and RatType are populated from the Nchf_ConvergedCharging service message received from the NF consumers whereas MessageType is one of the DB operation types. ResponseCodeMsg is UDSF response message received in the lib API response.

Labels

dnn, node_functionality, rat_type, slice, message_type, trigevent, optype, responsecodemsg

- charging_nchf_session_unknown_db_request_transaction_attempts

Type

64-bit Counter

Description

Number of unknown db transactions request received by CHF Agent in request flow. The requests count is per DNN/NodeFunctionality/RatType/MessageType/TriggerEvent/OperationType/RetryStatus. Dnn, NodeFunctionality and RatType are populated from the

Nchf_ConvergedCharging service message received from the NF consumers whereas MessageType is one of the DB operation types. RetryStatus indicates whether this request is part of retry attempt or normal request.

Labels

dnn, node_functionality, rat_type, slice, message_type, trigevent, optype, retry

- charging_nchf_session_unknown_db_request_transactions_success

Type

64-bit Counter

Description

Number of unknown db transactions received by CHF Agent in response flow. The requests count is per DNN/NodeFunctionality/RatType/MessageType/TriggerEvent/OperationType/ResponseCodeMsg. Dnn, NodeFunctionality and RatType are populated from the Nchf_ConvergedCharging service message received from the NF consumers whereas MessageType is one of the DB operation types. ResponseCodeMsg is UDSF response message received in the lib API response.

Labels

dnn, node_functionality, rat_type, slice, message_type, trigevent, optype, responsecodemsg

- charging_nchf_session_unknown_db_request_transaction_failures

Type

64-bit Counter

Description

Number of unknown db failure transactions request received by CHF Agent in request flow. The requests count is per DNN/NodeFunctionality/RatType/MessageType/TriggerEvent/OperationType/ResponseCodeMsg. Dnn, NodeFunctionality and RatType are populated from the Nchf_ConvergedCharging service message received from the NF consumers whereas MessageType is one of the DB operation types. ResponseCodeMsg is UDSF response message received in the lib API response.

Labels

dnn, node_functionality, rat_type, slice, message_type, trigevent, optype, responsecodemsg

- charging_nchf_session_abort_db_requests

Type

64-bit Counter

Description

Total number of times DB requests are aborted. The requests count is per DNN/NodeFunctionality/RatType/MessageType/TriggerEvent/OperationType. Dnn, NodeFunctionality and RatType are populated from the Nchf_ConvergedCharging service message received from the NF consumers whereas MessageType is one of the DB operation types.

Labels

dnn, node_functionality, rat_type, slice, message_type, trigevent, optype

- charging_nchf_session_udsf_connection_status

Type

64-bit Counter

Description

Number of times the connection to UDSF has gone up or down. Status indicates link between CHF Agent and UDSF lib.

Labels

status

- charging_nchf_session_cnf_state_status

Type

64-bit Counter

Description

Number of times the state of the CNF has changed

Labels

nfstate

- charging_nchf_session_db_request_rejected_for_nf_state_down

Type

64-bit Counter

Description

Number of requests rejected due to the NF state being down. The requests count is per DNN/NodeFunctionality/RatType/MessageType/TriggerEvent/OperationType/RetryStatus. Dnn, NodeFunctionality and RatType are populated from the Nchf_ConvergedCharging service message received from the NF consumers whereas MessageType is one of the DB operation types. RetryStatus indicates whether this request is part of retry attempt or normal request.

Labels

dnn, node_functionality, rat_type, slice, message_type, trigevent, optype, retry

- charging_nchf_session_db_request_rejected_for_udsf_clientmgr_not_found

Type

64-bit Counter

Description

Number of requests rejected due to the udsf clientmgr object not found. The requests count is per DNN/NodeFunctionality/RatType/MessageType/TriggerEvent/OperationType/RetryStatus. Dnn, NodeFunctionality and RatType are populated from the Nchf_ConvergedCharging service message received from the NF consumers whereas MessageType is one of the DB operation types. RetryStatus indicates whether this request is part of retry attempt or normal request.

Labels

dnn, node_functionality, rat_type, slice, message_type, trigevent, optype, retry

- charging_nchf_session_db_request_rejected_for_swagger_record_not_found

Type

64-bit Counter

Description

Number of requests rejected due to protobuf to swagger record conversion is failed. The requests count is per DNN/NodeFunctionality/RatType/MessageType/TriggerEvent/OperationType/RetryStatus. Dnn, NodeFunctionality and RatType are populated from the Nchf_ConvergedCharging service message received from the NF consumers whereas MessageType is one of the DB operation types. RetryStatus indicates whether this request is part of retry attempt or normal request.

Labels

dnn, node_functionality, rat_type, slice, message_type, trigevent, optype, retry

- charging_nchf_session_db_response_null_transaction_id_ptr

Type

64-bit Counter

Description

Number of responses for which the transaction id pointer is null. ResponseCodeMsg is UDSF response message received in the lib API response.

Labels

responsecodemsg

- charging_nchf_session_db_response_null_egress_event_ptr

Type

64-bit Counter

Description

Number of responses for which the egress event pointer is null. ResponseCodeMsg is UDSF response message received in the lib API response.

Labels

responsecodemsg

- charging_nchf_session_db_response_null_proc_ptr

Type

64-bit Counter

Description

Number of responses for which the proc pointer is null. ResponseCodeMsg is UDSF response message received in the lib API response.

Labels

responsecodemsg

- charging_nchf_create_procedure_duration

Type

64-bit Histogram

Description

Execution Time observed for charging session create procedures

Metrics

Instances	Labels	Description	Boundaries
1	profile_type: default	Execution Time observed for charging session create procedure of default profile type	[0, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100]
2	profile_type: lcdr	Execution Time observed for charging session create procedure of Local CDR profile type	[0, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100]
3	profile_type: nchf	Execution Time observed for charging session create procedure of Nchf interface profile type	[0, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100]
4	profile_type: nchf-lcdr	Execution Time observed for charging session create procedure of Nchf and Local CDR interfaces profile type	[0, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100]

- charging_nchf_update_procedure_duration

Type

64-bit Histogram

Description

Execution Time observed for charging session Update procedures

Metrics

Instances	Labels	Description	Boundaries
1	profile_type: default	Execution Time observed for charging session update procedure of default profile type	[0, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100]
2	profile_type: lcdr	Execution Time observed for charging session update procedure of Local CDR profile type	[0, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100]
3	profile_type: nchf	Execution Time observed for charging session update procedure of Nchf interface profile type	[0, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100]
4	profile_type: nchf-lcdr	Execution Time observed for charging session update procedure of Nchf and Local CDR interfaces profile type	[0, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100]

- charging_nchf_release_procedure_duration

Type

64-bit Histogram

Description

Execution Time observed for charging session Release procedures

Metrics

Instances	Labels	Description	Boundaries
1	profile_type: default	Execution Time observed for charging session release procedure of default profile type	[0, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100]
2	profile_type: lcdr	Execution Time observed for charging session release procedure of Local CDR profile type	[0, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100]
3	profile_type: nchf	Execution Time observed for charging session release procedure of Nchf interface profile type	[0, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100]
4	profile_type: nchf-lcdr	Execution Time observed for charging session release procedure of Nchf and Local CDR interfaces profile type	[0, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100]

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- [Summary](#)
- [Circuit Breaker Metrics](#)
- [Service Endpoint Learner Metrics](#)
- [Service Endpoint Manager Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Circuit Breaker Metrics	sem_cb_endpoints	Gauge	Total number of endpoints for a service in respective states
Service Endpoint Learner Metrics	sem_num_dns_errors	Counter	Total number of DNS resolution errors for a given service
Service Endpoint Learner Metrics	sem_num_endpoints	Gauge	Total number of endpoints learnt for a service
Service Endpoint Manager Metrics	sem_num_endpoint_transactions_sent	Counter	Total number of times an endpoint is given to the application for communication with it
Service Endpoint Manager Metrics	sem_num_endpoint_transaction_status_received	Counter	Total number of transactions for a given endpoint received with the transaction status

Circuit Breaker Metrics

- sem_cb_endpoints

Type

64-bit Gauge

Description

Total number of endpoints for a service in respective states. There are 3 states viz. Active, Blocked, Restoring. In Active state, the endpoint is responding as expected, i.e. the response code doesn't indicate a service failure. The endpoint will be moved to Blocked state based on response failures which indicate service failure. After a timeout, the endpoint will be moved to Restoring state to check if the endpoint has recovered. In this state, if endpoint responds with response code other than service failure, it will be moved to Active state. Else, it will be moved back to Blocked state.

Labels

service_name, endpoint_state

Service Endpoint Learner Metrics

- sem_num_dns_errors

Type

64-bit Counter

Description

Total number of DNS resolution errors for a given service, labels are service name for DNS resolution and DNS error codes

Labels

service_name, error_code

• sem_num_endpoints

Type

64-bit Gauge

Description

Total number of endpoints learnt from the Kubernetes DNS server for a given service

Labels

service

Service Endpoint Manager Metrics

• sem_num_endpoint_transactions_sent

Type

64-bit Counter

Description

Total number of times an endpoint is given to the application for communication with it

Labels

service

• sem_num_endpoint_transaction_status_received

Type

64-bit Counter

Description

Total number of transactions for a given endpoint received with the transaction status

Labels

service, status

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- [Summary](#)
- [UDSF DBClient Metrics](#)
- [UDSF API latency statistics](#)

Summary

Group	Metric Name	Type	Synopsis
UDSF DBClient Metrics	udsf_db_operation_request	Counter	The total number of udsf dbclient requests received.
UDSF DBClient Metrics	udsf_db_operation_response	Counter	The total number of udsf dbclient responses sent.
UDSF DBClient Metrics	udsf_dbc_total_init_success	Counter	The total number of UDSF INIT successes.
UDSF DBClient Metrics	udsf_dbc_total_init_failure	Counter	The total number of UDSF INIT failures.
UDSF DBClient Metrics	udsf_dbc_total_deinit_success	Counter	The total number of UDSF DEINIT successes.
UDSF DBClient Metrics	udsf_dbc_total_deinit_failure	Counter	The total number of UDSF DEINIT failures.
UDSF DBClient Metrics	udsf_db_transaction_counter	Counter	The total number of CRUD operations.

Group	Metric Name	Type	Synopsis
UDSF API latency statistics	udsf_db_transaction_duration	Histogram	Roundtrip duration taken by transaction in seconds.
UDSF API latency statistics	udsf_db_execution_duration	Histogram	Execution time taken by transaction in microseconds.
UDSF API latency statistics	udsf_hiredis_bufferwrite_duration	Histogram	Duration of time taken in hiredis to write data to socket in seconds

UDSF DBClient Metrics

- udsf_db_operation_request

Type

64-bit Counter

Description

The total number of udsf dbclient requests received from worker. Labels- 'app_name' : Originating thread name, ex:smfcc, 'operation_type' : Type of CRUD operation/API type involved, ex:CREATE_OR_MODIFY_RECORD_ASYNC

Labels

app_name, operation_type

- udsf_db_operation_response

Type

64-bit Counter

Description

The total number of udsf dbclient responses sent. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE 'operation_type' : Type of CRUD operation/API type involved., ex:CREATE_OR_MODIFY_RECORD_ASYNC

Labels

app_name, status, operation_type

- udsf_dbc_total_init_success

Type

64-bit Counter

Description

The total number of times UDSF API init was success. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: SUCCESS

Labels

app_name, status

- udsf_dbc_total_init_failure

Type

64-bit Counter

Description

The total number of times UDSF API init failed. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: FAILURE

Labels

app_name, status

- udsf_dbc_total_deinit_success

Type

64-bit Counter

Description

The total number of times UDSF API Deinit succeeded. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: SUCCESS

Labels

app_name, status

• udsf_dbc_total_deinit_failure

Type

64-bit Counter

Description

The total number of times UDSF API Deinit failed. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: FAILURE

Labels

app_name, status

• udsf_db_transaction_counter

Type

64-bit Counter

Description

The total number of times CRUD operations are incrementing. Labels- 'app_name' : Originating thread name, ex:smfcc, 'crud_op' : The categorical CRUD operation in which this sub-transaction falls into, ex: CREATE_OR_MODIFY

Labels

app_name, crud_op

UDSF API latency statistics

• udsf_db_transaction_duration

Type

64-bit Histogram

Description

Duration observed on total journey/roundtrip taken by a transaction after receiving the request. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE 'operation_type' : Type of CRUD operation/API type involved., ex:CREATE_OR_MODIFY_RECORD_ASYNC, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE, 'crud_op' : The categorical CRUD operation in which this transaction falls into, ex: CREATE_OR_MODIFY

Labels

app_name,operation_type,status,crud_op

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

• udsf_db_execution_duration

Type

64-bit Histogram

Description

Processing time taken by UDSF client. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE 'operation_type' : Type of CRUD operation/API type involved., ex:CREATE_OR_MODIFY_RECORD_ASYNC, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE, 'crud_op' : The categorical CRUD operation in which this transaction falls into, ex: CREATE_OR_MODIFY

Labels

app_name,operation_type,status,crud_op

Boundaries

[1, 2, 4, 8, 16, 32, 64, 128, 256, 384, 512, 768, 1024, 1536, 2048, 4096, 8192, 16384]

• udsf_hiredis_bufferwrite_duration

Type

64-bit Histogram

Description

Duration observed between hiredis buffer append and write to socket. This is fair approximation for time spent before data is written to socket as the buffer threshold accounted for append and write is dynamic and not fixed size. Per socket, the buffer append start-time is recorded along with size intended to transmit and end-time recorded once data greater than recorded size is written on the socket

Labels

app_name

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

Summary

1. [cores-mgr.alert.rules Alerts](#)

cores-mgr.alert.rules

Alert Name	Description	Alert Info	Diagnostics
application_crash_alert	Applications in namespace {{ "{{ \$labels.app_namespace }}" }} are crashing at a higher rate. This could indicate service impact due to applications unavailability.	1. Summary: High rate of application crashes Severity: critical Realert Time: 2m Expression: increase(num_generated_application_cores_total[15m]) > 10	1. Potential Impact: This might lead to unavailability of the service provided by the application in namespace {{ "{{ \$labels.app_namespace }}" }}. Repair Action: Troubleshoot the crashing application to check if it can be fixed or collect the diagnostic information like process core dump, application logs etc. for further debugging.

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- [Summary](#)
- [Cores Manager Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Cores Manager Metrics	num_generated_application_cores	Counter	Total number of core files generated by applications within a given namespace.
Cores Manager Metrics	num_persisted_application_cores	Counter	Total number of core files persisted by the cores mgr on the destination persistent volume
Cores Manager Metrics	num_deleted_application_cores	Counter	Total number of core files deleted by cores manager because of retention policy

Cores Manager Metrics

- num_generated_application_cores
Type
64-bit Counter
Description
Total number of core files generated by applications within a given namespace.
Labels
app_namespace
- num_persisted_application_cores
Type
64-bit Counter
Description
Total number of core files persisted by the cores manager on the destination persistent volume
Labels
app_namespace
- num_deleted_application_cores
Type
64-bit Counter
Description
Total number of core files deleted by cores manager because of retention policy
Labels
app_namespace

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- [ChargingCmn statistics](#)

Summary

Group	Metric Name	Type	Synopsis
ChargingCmn statistics	svcmgr_service_available_ind_notification	Counter	How many times, Service Available Indication message received
ChargingCmn statistics	svcmgr_service_available_ind_parse_errs	Counter	How many times, Service Available Indication parse error happened
ChargingCmn statistics	svcmgr_service_available_ind_notification_ack	Counter	How many times, Service Available Indication message is acknowledged
ChargingCmn statistics	svcmgr_service_unavailable_ind_notification	Counter	How many times, Service Unavailable Indication message
ChargingCmn statistics	svcmgr_service_unavailable_ind_parse_errs	Counter	How many times, Service Unavailable Indication parse error happened
ChargingCmn statistics	svcmgr_service_unavailable_ind_notification_ack	Counter	How many times, Service Unavailable Indication message is acknowledged
ChargingCmn statistics	svcmgr_publish_msg_sent	Counter	How many times, Publication message sent
ChargingCmn statistics	svcmgr_publish_msg_success	Counter	How many times, Publication message success response received
ChargingCmn statistics	svcmgr_publish_msg_failure	Counter	How many times, failure received for publication message

Group	Metric Name	Type	Synopsis
ChargingCmn statistics	svcmgr_subscribe_msg_sent	Counter	How many times, Subscribe message sent
ChargingCmn statistics	svcmgr_subscribe_msg_success	Counter	How many times, subscribe message success
ChargingCmn statistics	svcmgr_subscribe_msg_failure	Counter	How many times, failure received for subscribe message
ChargingCmn statistics	svcmgr_unsubscribe_msg_sent	Counter	How many times, unsubscribe message sent
ChargingCmn statistics	svcmgr_unsubscribe_msg_failure	Counter	How many times, failure received for unsubscribe message
ChargingCmn statistics	svcmgr_config_req_sent	Counter	How many times, config request sent
ChargingCmn statistics	svcmgr_config_req_failure	Counter	How many times, failure received for config request
ChargingCmn statistics	svcmgr_config_req_success	Counter	How many times, success received for config request
ChargingCmn statistics	svcmgr_config_register_response	Counter	How many times, config register response received
ChargingCmn statistics	svcmgr_config_register_response_ack	Counter	How many times, config register response is acknowledged
ChargingCmn statistics	svcmgr_config_register_response_parse_err	Counter	How many times, parse error happened for config register response
ChargingCmn statistics	svcmgr_config_update_notification	Counter	How many times, config update notification received
ChargingCmn statistics	svcmgr_config_update_notification_response	Counter	How many times, config update response
ChargingCmn statistics	svcmgr_intf_schema_update_ind_notification	Counter	How many times, interface schema update indication message received
ChargingCmn statistics	svcmgr_intf_schema_update_ind_notification_ack	Counter	How many times, interface schema update indication message is acknowledged
ChargingCmn statistics	svcmgr_intf_schema_update_ind_parse_errs	Counter	How many times, interface schema update indication parse error happened

ChargingCmn statistics

- svcmgr_service_available_ind_notification

Type

64-bit Counter

Description

Number of Times Service Available Indication notification received

Labels

servicename

- svcmgr_service_available_ind_parse_errs

Type

64-bit Counter

Description

Number of Times Service Available Indication parse error happened

Labels

• svcmgr_service_available_ind_notification_ack
Type
64-bit Counter
Description
Number of Times Service Available Indication message is acknowledged
Labels

• svcmgr_service_unavailable_ind_notification
Type
64-bit Counter
Description
Number of Times Service Unavailable Indication notification received
Labels
servicename

• svcmgr_service_unavailable_ind_parse_errs
Type
64-bit Counter
Description
Number of Times Service Unavailable Indication parse error happened
Labels

• svcmgr_service_unavailable_ind_notification_ack
Type
64-bit Counter
Description
Number of Times Service Unavailable Indication message is acknowledged
Labels
servicename

• svcmgr_publish_msg_sent
Type
64-bit Counter
Description
Number of Times Publication message sent
Labels
servicename, retry

• svcmgr_publish_msg_success
Type
64-bit Counter
Description
Number of Times Publication message success response received

Labels

servicename, causecode, retry

- **svcmgr_publish_msg_failure**

Type

64-bit Counter

Description

Number of Times failure received for publication message

Labels

servicename, causecode, retry

- **svcmgr_subscribe_msg_sent**

Type

64-bit Counter

Description

Number of Times Subscribe message sent

Labels

servicename, retry

- **svcmgr_subscribe_msg_success**

Type

64-bit Counter

Description

Number of Times Send subscribe message success response received

Labels

servicename, causecode, retry

- **svcmgr_subscribe_msg_failure**

Type

64-bit Counter

Description

Number of time failure received for subscribe message

Labels

servicename, causecode, retry

- **svcmgr_unsubscribe_msg_sent**

Type

64-bit Counter

Description

Number of Times unsubscribe message sent

Labels

servicename

- **svcmgr_unsubscribe_msg_failure**

Type

64-bit Counter

Description

Number of Times failure received for unsubscribe message

Labels

servicename

- **svcmgr_config_req_sent**

Type

64-bit Counter

Description

Number of Times config request sent

Labels

retry

- **svcmgr_config_req_failure**

Type

64-bit Counter

Description

Number of Times failure received for config request

Labels

causecode, retry

- **svcmgr_config_req_success**

Type

64-bit Counter

Description

Number of Times success received for config request

Labels

causecode, retry

- **svcmgr_config_register_response**

Type

64-bit Counter

Description

Number of Times config register response received

Labels

statuscode

- **svcmgr_config_register_response_ack**

Type

64-bit Counter

Description

Number of Times config register response is acknowledged

Labels

• svcmgr_config_register_response_parse_err Type 64-bit Counter Description Number of Times parse error happened for config register response Labels
• svcmgr_config_update_notification Type 64-bit Counter Description Number of Times config update notification received Labels
• svcmgr_config_update_notification_response Type 64-bit Counter Description Number of Times config update response sent Labels
• svcmgr_intf_schema_update_ind_notification Type 64-bit Counter Description Number of Times interface schema update indication message received Labels servicename
• svcmgr_intf_schema_update_ind_notification_ack Type 64-bit Counter Description Number of Times interface schema update indication message is acknowledged Labels servicename
• svcmgr_intf_schema_update_ind_parse_errs Type 64-bit Counter Description Number of Times Service interface schema update indication parse error happened Labels

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- [Diameter Base Etcd Interface Stats](#)
- [Diameter Base Peer Interface Stats](#)
- [Diameter Base DNS Interface Stats](#)

Summary

Group	Metric Name	Type	Synopsis
Diameter Base Application Interface Stats	diameter_application_request	Counter	Total number of application requests received
Diameter Base Application Interface Stats	diameter_application_request_aborted_by_cc	Counter	Total number of application requests aborts received from etcd
Diameter Base Application Interface Stats	diameter_application_response	Counter	Total number of application responses sent
Diameter Base Application Interface Stats	diameter_application_notify_request	Counter	Total number of application notify requests sent
Diameter Base Application Interface Stats	diameter_application_notify_response	Counter	Total number of application notify responses received
Diameter Base Etcd Interface Stats	etcd_watch_request_sent	Counter	Total number of etcd watch request sent
Diameter Base Etcd Interface Stats	etcd_watch_response_received	Counter	Total number of etcd watch responses received
Diameter Base Etcd Interface Stats	etcd_range_request_sent	Counter	Total number of etcd range request sent
Diameter Base Etcd Interface Stats	etcd_range_response_received	Counter	Total number of etcd range responses received
Diameter Base Etcd Interface Stats	etcd_lease_grant_request_sent	Counter	Total number of etcd lease_grant request sent
Diameter Base Etcd Interface Stats	etcd_lease_grant_response_received	Counter	Total number of etcd lease grant responses received
Diameter Base Etcd Interface Stats	etcd_put_request_sent	Counter	Total number of etcd put requests sent
Diameter Base Etcd Interface Stats	etcd_put_response_received	Counter	Total number of etcd put responses received
Diameter Base Etcd Interface Stats	etcd_watch_notify_event_received	Counter	Total number of etcd watch request sent
Diameter Base Etcd Interface Stats	etcd_delete_key_request_sent	Counter	Total number of etcd delete key requests sent
Diameter Base Etcd Interface Stats	etcd_delete_key_response_received	Counter	Total number of etcd delete key responses received
Diameter Base Etcd Interface Stats	etcd_lease_keepalive_request_sent	Counter	Total number of etcd lease keepalive requests sent
Diameter Base Etcd Interface Stats	etcd_lease_keepalive_response_received	Counter	Total number of etcd lease keepalive responses received
Diameter Base Etcd Interface Stats	etcd_request_retry_count	Counter	Total number of etcd retries sent
Diameter Base Etcd Interface Stats	etcd_resp_app_transaction_not_found	Counter	Total number of etcd retries sent
Diameter Base Peer Interface Stats	diameter_peer_request_sent	Counter	Total number of diameter requests sent to a diameter peer
Diameter Base Peer Interface Stats	diameter_peer_response_received	Counter	Total number of diameter responses received from a diameter peer
Diameter Base Peer Interface Stats	diameter_peer_request_received	Counter	Total number of diameter requests sent to a diameter peer
Diameter Base Peer Interface Stats	diameter_peer_response_sent	Counter	Total number of diameter responses received from a diameter peer
Diameter Base Peer Interface Stats	diameter_peer_connection_lost	Counter	Total number of times a connection is lost with a diameter peer
Diameter Base Peer Interface Stats	diameter_peer_connection_established	Counter	Total number of times a connection is established with a diameter peer
Diameter Base DNS Interface Stats	dns_hostname_query_request_sent	Counter	Total number of dns query requests sent for resolving hostname.
Diameter Base DNS Interface Stats	dns_hostname_query_response_received	Counter	Total number of dns query responses for resolving hostname.
Diameter Base DNS Interface Stats	dns_service_query_request_sent	Counter	Total number of dns query requests sent for resolving service.
Diameter Base DNS Interface Stats	dns_service_query_response_received	Counter	Total number of dns query responses for resolving service.

Diameter Base Application Interface Stats

- diameter_application_request
 - Type
 - 64-bit Counter
 - Description
 - Total number of REST API requests received by Diameter-Base from the diameter applications. The application requests are per Destination-Realm, Destination-Host received in the request message and the application ID received in the diameter request message.
 - Labels
 - destination_realm, destination_host, application_id**
- diameter_application_request_aborted_by_cc
 - Type
 - 64-bit Counter
 - Description
 - Total number of ETCD notify request received by Diameter-Base for aborting the diameter application requests.
 - Labels
 - destination_realm, destination_host, application_id**
- diameter_application_response
 - Type
 - 64-bit Counter
 - Description
 - Total number of REST API responses sent by Diameter-Base to the diameter applications. The application responses are per Destination-Realm, Destination-Host received in the request message and the application ID received in the diameter request message.
 - Labels
 - destination_realm, destination_host, application_id, status**
- diameter_application_notify_request
 - Type
 - 64-bit Counter
 - Description
 - Total number of notify requests sent by Diameter-Base to the diameter applications. The application requests are per application name and the application ID received in the diameter request message.
 - Labels
 - application_name, application_id**
- diameter_application_notify_response
 - Type
 - 64-bit Counter
 - Description
 - Total number of notify responses received by Diameter-Base from the diameter applications. The application responses are per Destination-Realm, Destination-Host received in the request message and the application ID received in the diameter request message.
 - Labels
 - application_name, application_id, status**

Diameter Base EtcD Interface Stats

- etcd_watch_request_sent
 - Type
 - 64-bit Counter
 - Description
 - Total number of etcd watch request messages sent. The number of request messages is unique by the etcd table key
 - Labels
 - etcd_table_key**
- etcd_watch_response_received
 - Type
 - 64-bit Counter
 - Description
 - Total number of etcd watch response messages received. The number of response messages is unique by the response code and etcd table key
 - Labels
 - response_code, etcd_table_key**
- etcd_range_request_sent
 - Type
 - 64-bit Counter
 - Description
 - Total number of etcd range request messages sent. The number of request messages is unique by the etcd table key
 - Labels
 - etcd_table_key**
- etcd_range_response_received
 - Type
 - 64-bit Counter
 - Description
 - Total number of etcd range response messages received. The number of reponse messages is unique by the response code and etcd table key
 - Labels
 - response_code, etcd_table_key**
- etcd_lease_grant_request_sent
 - Type
 - 64-bit Counter
 - Description
 - Total number of etcd lease grant request messages sent.
 - Labels
 - etcd_table_key**
- etcd_lease_grant_response_received
 - Type
 - 64-bit Counter

	Description Total number of etcd lease grant response messages received. Labels response_code, etcd_table_key
• etcd_put_request_sent	Type 64-bit Counter Description Total number of etcd put request messages sent. Labels etcd_table_key
• etcd_put_response_received	Type 64-bit Counter Description Total number of etcd put response messages received. Labels response_code, etcd_table_key
• etcd_watch_notify_event_received	Type 64-bit Counter Description Total number of etcd watch notify event messages received. The number of request messages is unique by the etcd table key, eventType Labels etcd_table_key, watch_event_type
• etcd_delete_key_request_sent	Type 64-bit Counter Description Total number of etcd delete key request messages sent. The number of request messages is unique by the etcd table key Labels etcd_table_key
• etcd_delete_key_response_received	Type 64-bit Counter Description Total number of etcd delete key response messages received. The number of response messages is unique by the response code and etcd table key Labels response_code, etcd_table_key

- etcd_lease_keepalive_request_sent

Type

64-bit Counter

Description

Total number of etcd lease keepalive request messages sent. The number of request messages is unique by the etcd table key

Labels

- etcd_lease_keepalive_response_received

Type

64-bit Counter

Description

Total number of etcd lease keepalive response messages received. The number of response messages is unique by the response code

Labels

response_code

- etcd_request_retry_count

Type

64-bit Counter

Description

Total number of etcd watch request messages retries. The number of retries is unique by the message type

Labels

etcd_message_type

- etcd_resp_app_transaction_not_found

Type

64-bit Counter

Description

Total number of etcd transactions not found. The number is unique by the message type

Labels

etcd_message_type

Diameter Base Peer Interface Stats

diameter_peer_request_sent
Type
64-bit Counter
Description
Total number of diameter request messages sent to a configured diameter peer. The number of request messages is unique by the peer's realm/identity, diameter application ID and the diameter command code in the request.
Labels
<i>destination_realm, destination_host, application_id, command_code</i>
diameter_peer_response_received
Type
64-bit Counter

Description

Total number of diameter response messages received from a configured diameter peer. The number of response messages is unique by the peer's realm/identity, diameter application ID, the diameter command code and the result code in the diameter response message.

Labels

destination_realm, destination_host, application_id, command_code, result_code

- diameter_peer_request_received

Type

64-bit Counter

Description

Total number of diameter request messages received from a configured diameter peer. The number of request messages is unique by the peer's realm/identity, diameter application ID and the diameter command code in the request.

Labels

destination_realm, destination_host, application_id, command_code

- diameter_peer_response_sent

Type

64-bit Counter

Description

Total number of diameter response messages sent to a configured diameter peer. The number of response messages is unique by the peer's realm/identity, diameter application ID, the diameter command code and the result code in the diameter response message.

Labels

destination_realm, destination_host, application_id, command_code, result_code

- diameter_peer_connection_lost

Type

64-bit Counter

Description

Total number of times a diameter connection is lost with the configured diameter peer identified by its realm and identity.

Labels

destination_realm, destination_host

- diameter_peer_connection_established

Type

64-bit Counter

Description

Total number of times a diameter connection is established with the configured diameter peer identified by its realm and identity.

Labels

destination_realm, destination_host

Diameter Base DNS Interface Stats

- dns_hostname_query_request_sent

Type

64-bit Counter

Description

Total number of dns query requests sent for resolving hostname.

Labels

hostname

- dns_hostname_query_response_received

Type

64-bit Counter

Description

Total number of dns query responses received for resolving hostname.

Labels

hostname, result

- dns_service_query_request_sent

Type

64-bit Counter

Description

Total number of dns query requests sent for resolving diam peer service.

Labels

service

- dns_service_query_response_received

Type

64-bit Counter

Description

Total number of dns query responses received for resolving diam peer service.

Labels

service, result

Table of Contents

- [Summary](#)
- [ETCD Client Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_put_requests	Counter	The total number of ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_requests_sent	Counter	The total number of ETCD Client PUT Requests sent.
ETCD Client Metrics	etcd_put_successes	Counter	The total number of successful ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_failures	Counter	The total number of failed ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_with_lease_requests	Counter	The total number of ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_requests_sent	Counter	The total number of ETCD Client PUT with Lease Requests sent.
ETCD Client Metrics	etcd_put_with_lease_successes	Counter	The total number of successful ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_failures	Counter	The total number of failed ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_range_requests	Counter	The total number of ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_requests_sent	Counter	The total number of ETCD Client Range Requests sent.
ETCD Client Metrics	etcd_range_successes	Counter	The total number of successful ETCD Client Range Requests.

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_range_failures	Counter	The total number of failed ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_delete_requests	Counter	The total number of ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_requests_sent	Counter	The total number of ETCD Client Range Delete Requests sent.
ETCD Client Metrics	etcd_range_delete_successes	Counter	The total number of successful ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_failures	Counter	The total number of failed ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_txn_requests	Counter	The total number of ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_requests_sent	Counter	The total number of ETCD Client Txn Requests sent.
ETCD Client Metrics	etcd_txn_successes	Counter	The total number of successful ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_failures	Counter	The total number of failed ETCD Client Txn Requests.
ETCD Client Metrics	etcd_watch_requests	Counter	The total number of ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_requests_sent	Counter	The total number of ETCD Client Watch Requests sent.
ETCD Client Metrics	etcd_watch_successes	Counter	The total number of successful ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_failures	Counter	The total number of failed ETCD Client Watch Requests.
ETCD Client Metrics	etcdwatcher_requests	Counter	The total number of ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_requests_sent	Counter	The total number of ETCD Watcher Requests sent.
ETCD Client Metrics	etcdwatcher_successes	Counter	The total number of successful ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_failures	Counter	The total number of failed ETCD Watcher Requests.

ETCD Client Metrics

- etcd_put_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests.

Labels

keyPrefix

- etcd_put_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests sent.

Labels

keyPrefix

- etcd_put_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT Requests.

Labels

keyPrefix

- etcd_put_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT Requests.

Labels

keyPrefix, failureReason

• etcd_put_with_lease_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests.

Labels

keyPrefix

• etcd_put_with_lease_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests sent.

Labels

keyPrefix

• etcd_put_with_lease_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT with Lease Requests.

Labels

keyPrefix

• etcd_put_with_lease_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT with Lease Requests.

Labels

keyPrefix, failureReason

• etcd_range_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests.

Labels

keyPrefix

• etcd_range_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests sent.

Labels

keyPrefix

• etcd_range_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Requests.

Labels

keyPrefix

• etcd_range_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Requests.

Labels

keyPrefix, failureReason

• etcd_range_delete_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests.

Labels

keyPrefix

• etcd_range_delete_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests sent.

Labels

keyPrefix

• etcd_range_delete_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Delete Requests.

Labels

keyPrefix

• etcd_range_delete_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Delete Requests.

Labels

keyPrefix, failureReason

• etcd_txn_requests

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests.

Labels

keyPrefix

• etcd_txn_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests sent.

Labels

keyPrefix

• etcd_txn_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Txn Requests.

Labels

keyPrefix

• etcd_txn_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Txn Requests.

Labels

keyPrefix, failureReason

- etcd_watch_requests

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests sent.

Labels

keyPrefix

- etcd_watch_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Watch Requests.

Labels

keyPrefix, failureReason

- etcdwatcher_requests

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests.

Labels

keyPrefix

- etcdwatcher_requests_sent

Type

64-bit Counter

Description
The total number of ETCD Watcher Requests sent.
Labels
keyPrefix
etcdwatcher_successes
Type
64-bit Counter
Description
The total number of successful ETCD Watcher Requests.
Labels
keyPrefix
etcdwatcher_failures
Type
64-bit Counter
Description
The total number of failed ETCD Watcher Requests.
Labels
keyPrefix, failureReason

Table of Contents

- [Summary](#)
- [HTTP Punting Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_sent	Counter	The total number of HTTP requests successfully punting.
HTTP Punting Metrics	punting_http_requests_failed	Counter	The total number of HTTP requests punting failed.
HTTP Punting Metrics	punting_http_responses_received	Counter	The total number of HTTP responses received for the requests punting.
HTTP Punting Metrics	punting_http_responses_forwarded	Counter	The total number of HTTP responses forwarded after punting transaction completion.
HTTP Punting Metrics	punting_http_requests_aborted	Counter	The total number of HTTP requests aborted.
HTTP Punting Metrics	punting_http_request_duration_seconds	Histogram	Punting operation duration in seconds.

HTTP Punting Metrics

- punting_http_requests_sent
 - Type
 - 64-bit Counter
 - Description
 - The total number of HTTP requests successfully punting.

Labels

message_type

- punting_http_requests_failed

Type

64-bit Counter

Description

The total number of HTTP requests for which the punting failed.

Labels

message_type

- punting_http_responses_received

Type

64-bit Counter

Description

The total number of HTTP responses received for the requests punted.

Labels

message_type, status_code

- punting_http_responses_forwarded

Type

64-bit Counter

Description

The total number of HTTP responses forwarded after punting transaction completion.

Labels

message_type, status_code

- punting_http_requests_aborted

Type

64-bit Counter

Description

The total number of HTTP requests aborted.

Labels

message_type

- punting_http_request_duration_seconds

Type

64-bit Histogram

Description

The punting transaction duration is measured from the time the HTTP request is sent out to the time the response is received for the punting request.

Labels

message_type

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

Table of Contents

- [Summary](#)
- [Packet Streaming Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Packet Streaming Metrics	num_packet_streaming_attempted	Counter	The total number of attempted streaming of packets
Packet Streaming Metrics	num_packet_streaming_success	Counter	The total number of packets streamed successfully
Packet Streaming Metrics	num_packet_streaming_failed	Counter	The total number of failed streaming of packets
Packet Streaming Metrics	num_packet_streaming_bytes	Counter	The total number of bytes successfully streamed

Packet Streaming Metrics

- num_packet_streaming_attempted
 - Type
 - 64-bit Counter
 - Description
 - The total number of attempted streaming of packets
 - Labels
 - interface_name, server_fqdn, kafka_topic*
- num_packet_streaming_success
 - Type
 - 64-bit Counter
 - Description
 - The total number of packets streamed successfully
 - Labels
 - interface_name, server_fqdn, kafka_topic*
- num_packet_streaming_failed
 - Type
 - 64-bit Counter
 - Description
 - The total number of failed streaming of packets
 - Labels
 - interface_name, server_fqdn, kafka_topic*
- num_packet_streaming_bytes
 - Type
 - 64-bit Counter
 - Description
 - The total number of bytes successfully streamed
 - Labels
 - interface_name, server_fqdn, kafka_topic*

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- [Summary](#)
- [Circuit Breaker Metrics](#)
- [Service Endpoint Learner Metrics](#)
- [Service Endpoint Manager Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Circuit Breaker Metrics	sem_cb_endpoints	Gauge	Total number of endpoints for a service in respective states
Service Endpoint Learner Metrics	sem_num_dns_errors	Counter	Total number of DNS resolution errors for a given service
Service Endpoint Learner Metrics	sem_num_endpoints	Gauge	Total number of endpoints learnt for a service
Service Endpoint Manager Metrics	sem_num_endpoint_transactions_sent	Counter	Total number of times an endpoint is given to the application for communication with it
Service Endpoint Manager Metrics	sem_num_endpoint_transaction_status_received	Counter	Total number of transactions for a given endpoint received with the transaction status

Circuit Breaker Metrics

• [sem_cb_endpoints](#)

Type

64-bit Gauge

Description

Total number of endpoints for a service in respective states. There are 3 states viz. Active, Blocked, Restoring. In Active state, the endpoint is responding as expected, i.e. the response code doesn't indicate a service failure. The endpoint will be moved to Blocked state based on response failures which indicate service failure. After a timeout, the endpoint will be moved to Restoring state to check if the endpoint has recovered. In this state, if endpoint responds with response code other than service failure, it will be moved to Active state. Else, it will be moved back to Blocked state.

Labels

service_name, endpoint_state

Service Endpoint Learner Metrics

• [sem_num_dns_errors](#)

Type

64-bit Counter

Description

Total number of DNS resolution errors for a given service, labels are service name for DNS resolution and DNS error codes

Labels

service_name, error_code

• [sem_num_endpoints](#)

Type

64-bit Gauge

Description

Total number of endpoints learnt from the Kubernetes DNS server for a given service

Labels

service

Service Endpoint Manager Metrics

- `sem_num_endpoint_transactions_sent`

Type

64-bit Counter

Description

Total number of times an endpoint is given to the application for communication with it

Labels

service
- `sem_num_endpoint_transaction_status_received`

Type

64-bit Counter

Description

Total number of transactions for a given endpoint received with the transaction status

Labels

service, status

Table of Contents

- [Summary](#)
- [Unified Load Balancer Endpoint Manager Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Unified Load Balancer Endpoint Manager Metrics	ulb_registration_request_sent	Counter	Total number of times an registration request is sent to an unified load balancer pod
Unified Load Balancer Endpoint Manager Metrics	ulb_registration_response_received	Counter	Total number of times registration response is received from unified load balancer pod
Unified Load Balancer Endpoint Manager Metrics	ulb_deregistration_request_sent	Counter	Total number of times an deregistration request is sent to an unified load balancer pod
Unified Load Balancer Endpoint Manager Metrics	ulb_deregistration_response_received	Counter	Total number of times deregistration response is received from unified load balancer pod
Unified Load Balancer Endpoint Manager Metrics	ulb_heartbeat_request_sent	Counter	Total number of times an heartbeat request is sent to an unified load balancer pod
Unified Load Balancer Endpoint Manager Metrics	ulb_heartbeat_response_received	Counter	Total number of times heartbeat response is received from unified load balancer pod

Unified Load Balancer Endpoint Manager Metrics

- `ulb_registration_request_sent`

Type

64-bit Counter

Description

Total number of times registration request is sent to an unified load balancer pod

Labels

ulbName
- `ulb_registration_response_received`

Type

64-bit Counter

Description

Total number of times registration response is received from unified load balancer pod

Labels

UlbName, status

- ulb_deregistration_request_sent

Type

64-bit Counter

Description

Total number of times deregistration request is sent to an unified load balancer pod

Labels

ulbName

- ulb_deregistration_response_received

Type

64-bit Counter

Description

Total number of times deregistration response is received from unified load balancer pod

Labels

UlbName, status

- ulb_heartbeat_request_sent

Type

64-bit Counter

Description

Total number of times heartbeat request is sent to an unified load balancer pod

Labels

ulbName

- ulb_heartbeat_response_received

Type

64-bit Counter

Description

Total number of times heartbeat response is received from unified load balancer pod

Labels

UlbName, status

Table of Contents

- [Summary](#)
- [SCTP statistics](#)

Summary

Group	Metric Name	Type	Synopsis
SCTP statistics	sctpActiveEstabs	Counter	The number of times associations have been established.
SCTP statistics	sctpPassiveEstabs	Counter	The number of times associations have been established.
SCTP statistics	sctpAborted	Counter	The number of times associations have been aborted.
SCTP statistics	sctpShutdowns	Counter	The number of times associations have been gracefully terminated.
SCTP statistics	sctpOutOfBlues	Counter	The number of out of the blue packets received.
SCTP statistics	sctpChecksumErrors	Counter	The number of SCTP packets received with an invalid checksum.
SCTP statistics	sctpOutCtrlChunks	Counter	The number of SCTP control chunks sent.
SCTP statistics	sctpOutOrderChunks	Counter	The number of SCTP ordered data chunks sent.
SCTP statistics	sctpOutUnorderChunks	Counter	The number of SCTP unordered data chunks sent.
SCTP statistics	sctpInCtrlChunks	Counter	The number of SCTP control chunks received.
SCTP statistics	sctpInOrderChunks	Counter	The number of SCTP ordered data chunks received.
SCTP statistics	sctpInUnorderChunks	Counter	The number of SCTP unordered data chunks received.
SCTP statistics	sctpOutSCTPPacks	Counter	The number of SCTP packets sent.
SCTP statistics	sctpInSCTPPacks	Counter	The number of SCTP packets received.
SCTP statistics	sctpCurrEstab	Gauge	The current number of established associations.

SCTP statistics

- [sctpActiveEstabs](#)

Type

64-bit Counter

Description

The number of times that associations have made a direct transition to the ESTABLISHED state from the COOKIE-ECHOED state: COOKIE-ECHOED -> ESTABLISHED. The upper layer initiated the association attempt.

Labels

- [sctpPassiveEstabs](#)

Type

64-bit Counter

Description

The number of times that associations have made a direct transition to the ESTABLISHED state from the CLOSED state: CLOSED -> ESTABLISHED. The remote endpoint initiated the association attempt.

Labels

- [sctpAborted](#)

Type

64-bit Counter

Description

The number of times that associations have made a direct transition to the CLOSED state from any state due to sending or receiving of an ABORT chunk: AnyState --Abort--> CLOSED. Ungraceful termination of the association.

Labels

- [sctpShutdowns](#)

Type

64-bit Counter

Description

The number of times that associations have made a direct transition to the CLOSED state from either the SHUTDOWN-SENT state or the SHUTDOWN-ACK-SENT state. Graceful termination of the association.

Labels

- sctpOutOfBlues

Type

64-bit Counter

Description

The number of out of the blue packets received by the host. An out of the blue packet is an SCTP packet correctly formed, including the proper checksum, but for which the receiver was unable to identify an appropriate association.

Labels

- sctpChecksumErrors

Type

64-bit Counter

Description

The number of SCTP packets received with an invalid checksum.

Labels

- sctpOutCtrlChunks

Type

64-bit Counter

Description

The number of SCTP control chunks sent (retransmissions are not included). Control chunks are those chunks different from DATA.

Labels

- sctpOutOrderChunks

Type

64-bit Counter

Description

The number of SCTP ordered data chunks sent (retransmissions are not included).

Labels

- sctpOutUnorderChunks

Type

64-bit Counter

Description

The number of SCTP unordered chunks (data chunks in which the U bit is set to 1) sent (retransmissions are not included).

Labels

- sctpInCtrlChunks

Type

64-bit Counter

Description

The number of SCTP control chunks received (no duplicate chunks included).

Labels

- sctpInOrderChunks

Type

64-bit Counter

Description

The number of SCTP ordered data chunks received (no duplicate chunks included).

Labels

- sctpInUnorderChunks

Type

64-bit Counter

Description

The number of SCTP unordered chunks (data chunks in which the U bit is set to 1) received (no duplicate chunks included).

Labels

- sctpOutSCTPPacks

Type

64-bit Counter

Description

The number of SCTP packets sent. Retransmitted DATA chunks are included.

Labels

- sctpInSCTPPacks

Type

64-bit Counter

Description

The number of SCTP packets received. Duplicates are included.

Labels

- sctpCurrEstab

Type

64-bit Gauge

Description

The number of associations for which the current state is either ESTABLISHED, SHUTDOWN-RECEIVED or SHUTDOWN-PENDING.

Labels

- [Summary](#)
- [DNS Agent Service statistics](#)
- [BIND statistics from DNS Agent Serice](#)

Summary

Group	Metric Name	Type	Synopsis
DNS Agent Service statistics	dnsagent_static_cache_hits	Counter	The number of requests resolved by static cache.
DNS Agent Service statistics	dnsagent_query_request	Counter	The number of query requests received from other pods.
DNS Agent Service statistics	dnsagent_query_response	Counter	The number of query responses sent.
BIND statistics from DNS Agent Serice	dnsagent_bind_cachestats_cache_hits	Counter	Bind cachestats statistic: CacheHits
BIND statistics from DNS Agent Serice	dnsagent_bind_cachestats_cache_misses	Counter	Bind cachestats statistic: CacheMisses
BIND statistics from DNS Agent Serice	dnsagent_bind_cachestats_delete_lru	Counter	Bind cachestats statistic: DeleteLRU
BIND statistics from DNS Agent Serice	dnsagent_bind_cachestats_delete_ttl	Counter	Bind cachestats statistic: DeleteTTL
BIND statistics from DNS Agent Serice	dnsagent_bind_resolver_queryv4	Counter	Bind resolver statistic: Queryv4
BIND statistics from DNS Agent Serice	dnsagent_bind_resolver_queryv6	Counter	Bind resolver statistic: Queryv6
BIND statistics from DNS Agent Serice	dnsagent_bind_resolver_responsev4	Counter	Bind resolver statistic: Responsev4
BIND statistics from DNS Agent Serice	dnsagent_bind_resolver_responsev6	Counter	Bind resolver statistic: Responsev6
BIND statistics from DNS Agent Serice	dnsagent_bind_resolver_nxdomain	Counter	Bind resolver statistic: NXDOMAIN
BIND statistics from DNS Agent Serice	dnsagent_bind_resolver_servfail	Counter	Bind resolver statistic: SERVFAIL
BIND statistics from DNS Agent Serice	dnsagent_bind_resolver_formerr	Counter	Bind resolver statistic: FORMERR
BIND statistics from DNS Agent Serice	dnsagent_bind_resolver_other_error	Counter	Bind resolver statistic: OtherError
BIND statistics from DNS Agent Serice	dnsagent_bind_resolver_refused	Counter	Bind resolver statistic: REFUSED
BIND statistics from DNS Agent Serice	dnsagent_bind_resolver_bad_edns_version	Counter	Bind resolver statistic: BadEDNSVersion
BIND statistics from DNS Agent Serice	dnsagent_bind_resolver_bad_cookie_rcode	Counter	Bind resolver statistic: BadCookieRcode
BIND statistics from DNS Agent Serice	dnsagent_bind_resolver_query_timeout	Counter	Bind resolver statistic: QueryTimeout
BIND statistics from DNS Agent Serice	dnsagent_bind_resolver_truncated	Counter	Bind resolver statistic: Truncated
BIND statistics from DNS Agent Serice	dnsagent_bind_resolver_retry	Counter	Bind resolver statistic: Retry
BIND statistics from DNS Agent Serice	dnsagent_bind_cachestats_cache_nodes	Gauge	Bind cachestats statistic: CacheNodes
BIND statistics from DNS Agent Serice	dnsagent_bind_cachestats_heap_mem_total	Gauge	Bind cachestats statistic: HeapMemTotal
BIND statistics from DNS Agent Serice	dnsagent_bind_cachestats_heap_mem_in_use	Gauge	Bind cachestats statistic: HeapMemInUse

DNS Agent Service statistics

- [dnsagent_static_cache_hits](#)

Type

64-bit Counter

Description

The number of requests resolved by static cache.

Labels

view, query_type

- [dnsagent_query_request](#)

Type

64-bit Counter

Description

The number of query requests received from other pods.

Labels

view, query_type

- dnsagent_query_response

Type

64-bit Counter

Description

The number of query responses sent.

Labels

view, query_type, status

BIND statistics from DNS Agent Serice

- dnsagent_bind_cachestats_cache_hits

Type

64-bit Counter

Description

The the number of hits in dynamic cache; this includes negative entries.

Labels

view

- dnsagent_bind_cachestats_cache_misses

Type

64-bit Counter

Description

The number of misses in dynamic cache.

Labels

view

- dnsagent_bind_cachestats_delete_lru

Type

64-bit Counter

Description

The number of cache entries deleted because the cache was full.

Labels

view

- dnsagent_bind_cachestats_delete_ttl

Type

64-bit Counter

Description

The number of cache entries deleted because of expired TTL.

Labels

view

- dnsagent_bind_resolver_queryv4

Type

64-bit Counter

Description

The number of IPv4 DNS requests sent.

Labels

view

- dnsagent_bind_resolver_queryv6

Type

64-bit Counter

Description

The number of IPv6 DNS requests sent.

Labels

view

- dnsagent_bind_resolver_responsev4

Type

64-bit Counter

Description

The number of IPv4 DNS responses received.

Labels

view

- dnsagent_bind_resolver_responsev6

Type

64-bit Counter

Description

The number of IPv6 DNS responses received.

Labels

view

- dnsagent_bind_resolver_nxdomain

Type

64-bit Counter

Description

The number of NXDOMAIN responses received.

Labels

view

- dnsagent_bind_resolver_servfail

Type

64-bit Counter

Description

The number of Bind SERVFAIL stats pegged on responses.

Labels

view

- dnsagent_bind_resolver_formerr

Type

64-bit Counter

Description

The number of Bind FORMERR stats pegged on responses.

Labels

view

- dnsagent_bind_resolver_other_error

Type

64-bit Counter

Description

The number of Bind OtherError stats pegged on responses.

Labels

view

- dnsagent_bind_resolver_refused

Type

64-bit Counter

Description

The number of Bind REFUSED stats pegged on responses.

Labels

view

- dnsagent_bind_resolver_bad_edns_version

Type

64-bit Counter

Description

The number of Bind BadEDNSVersion stats pegged on responses.

Labels

view

- dnsagent_bind_resolver_bad_cookie_rcode

Type

64-bit Counter

Description

The number of Bind BadCookieRcode stats pegged on responses.

Labels

view

- dnsagent_bind_resolver_query_timeout

Type

64-bit Counter

Description

The number of outgoing DNS requests that timed out.

Labels

[view](#)

- dnsagent_bind_resolver_truncated

Type

64-bit Counter

Description

The number of truncated DNS responses received.

Labels

[view](#)

- dnsagent_bind_resolver_retry

Type

64-bit Counter

Description

The number of requests that are resent; this could be a fallback to TCP or other trigger.

Labels

[view](#)

- dnsagent_bind_cachestats_cache_nodes

Type

64-bit Gauge

Description

The number of nodes in the dynamic cache; correlates to the number of entries in the cache but usually a little higher.

Labels

[view](#)

- dnsagent_bind_cachestats_heap_mem_total

Type

64-bit Gauge

Description

The number of bytes currently allocated for dynamic cache functionality.

Labels

[view](#)

- dnsagent_bind_cachestats_heap_mem_in_use

Type

64-bit Gauge

Description

The number of bytes currently used for dynamic cache functionality.

Labels

[view](#)

Table of Contents

- [Summary](#)
- [HTTP Punting Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_sent	Counter	The total number of HTTP requests successfully punted.
HTTP Punting Metrics	punting_http_requests_failed	Counter	The total number of HTTP requests punting failed.
HTTP Punting Metrics	punting_http_responses_received	Counter	The total number of HTTP responses received for the requests punted.
HTTP Punting Metrics	punting_http_responses_forwarded	Counter	The total number of HTTP responses forwarded after punting transaction completion.
HTTP Punting Metrics	punting_http_requests_aborted	Counter	The total number of HTTP requests aborted.
HTTP Punting Metrics	punting_http_request_duration_seconds	Histogram	Punting operation duration in seconds.

HTTP Punting Metrics

- [punting_http_requests_sent](#)

Type

64-bit Counter

Description

The total number of HTTP requests successfully punted.

Labels

message_type
- [punting_http_requests_failed](#)

Type

64-bit Counter

Description

The total number of HTTP requests for which the punting failed.

Labels

message_type
- [punting_http_responses_received](#)

Type

64-bit Counter

Description

The total number of HTTP responses received for the requests punted.

Labels

message_type, status_code
- [punting_http_responses_forwarded](#)

Type

64-bit Counter

Description

The total number of HTTP responses forwarded after punting transaction completion.

Labels

message_type, status_code

- [punting_http_requests_aborted](#)

Type

64-bit Counter

Description

The total number of HTTP requests aborted.

Labels

message_type

- [punting_http_request_duration_seconds](#)

Type

64-bit Histogram

Description

The punting transaction duration is measured from the time the HTTP request is sent out to the time the response is received for the punting request.

Labels

message_type

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

Table of Contents

- [Summary](#)
- [Packet Streaming Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Packet Streaming Metrics	num_packet_streaming_attempted	Counter	The total number of attempted streaming of packets
Packet Streaming Metrics	num_packet_streaming_success	Counter	The total number of packets streamed successfully
Packet Streaming Metrics	num_packet_streaming_failed	Counter	The total number of failed streaming of packets
Packet Streaming Metrics	num_packet_streaming_bytes	Counter	The total number of bytes successfully streamed

Packet Streaming Metrics

- [num_packet_streaming_attempted](#)

Type

64-bit Counter

Description

The total number of attempted streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_success

Type

64-bit Counter

Description

The total number of packets streamed successfully

Labels

interface_name, server_fqdn, kafka_topic
- num_packet_streaming_failed

Type

64-bit Counter

Description

The total number of failed streaming of packets

Labels

interface_name, server_fqdn, kafka_topic
- num_packet_streaming_bytes

Type

64-bit Counter

Description

The total number of bytes successfully streamed

Labels

interface_name, server_fqdn, kafka_topic

Table of Contents

- [Summary](#)
- [Elasticsearch Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Elasticsearch Metrics	elasticsearch_breakers_tripped	Counter	tripped for breaker.
Elasticsearch Metrics	elasticsearch_jvm_gc_collection_seconds_count	Counter	Count of JVM GC runs.
Elasticsearch Metrics	elasticsearch_jvm_gc_collection_seconds_sum	Counter	GC run time in seconds.
Elasticsearch Metrics	elasticsearch_indices_translog_operations	Counter	Total translog operations.
Elasticsearch Metrics	elasticsearch_indices_translog_size_in_bytes	Counter	Total translog size in bytes.
Elasticsearch Metrics	elasticsearch_jvm_memory_pool_peak_used_bytes	Counter	JVM memory peak used by pool.
Elasticsearch Metrics	elasticsearch_transport_rx_size_bytes_total	Counter	Total number of bytes received.
Elasticsearch Metrics	elasticsearch_indices_search_query_time_seconds	Counter	Total search query time in seconds.
Elasticsearch Metrics	elasticsearch_indices_store_throttle_time_seconds_total	Counter	Throttle time for index store in seconds.

Group	Metric Name	Type	Synopsis
Elasticsearch Metrics	elasticsearch_indices_indexing_index_time_seconds_total	Counter	Cumulative index time in seconds.
Elasticsearch Metrics	elasticsearch_indices_query_cache_evictions	Counter	Evictions from query cache.
Elasticsearch Metrics	elasticsearch_indices_filter_cache_evictions	Counter	Evictions from filter cache.
Elasticsearch Metrics	elasticsearch_cluster_health_up	Gauge	The elastic-search cluster's end-point status.
Elasticsearch Metrics	elasticsearch_process_cpu_percent	Gauge	Average CPU Usage
Elasticsearch Metrics	elasticsearch_filesystem_data_available_bytes	Gauge	Disk usage Percentage
Elasticsearch Metrics	elasticsearch_jvm_memory_used_bytes	Gauge	Average JVM memory Used
Elasticsearch Metrics	elasticsearch_cluster_health_status	Gauge	Raises when the Elasticsearch cluster status is YELLOW for 2 minutes.
Elasticsearch Metrics	elasticsearch_cluster_health_relocating_shards	Gauge	The number of slices that the current node is migrating to other nodes.
Elasticsearch Metrics	elasticsearch_cluster_health_initializing_shards	Gauge	Initializing fragment.
Elasticsearch Metrics	elasticsearch_cluster_health_unassigned_shards	Gauge	Number of unallocated tiles.
Elasticsearch Metrics	elasticsearch_cluster_health_number_of_pending_tasks	Gauge	Only the master node can handle cluster level metadata changes,
Elasticsearch Metrics	elasticsearch_jvm_memory_max_bytes	Gauge	JVM memory max
Elasticsearch Metrics	elasticsearch_cluster_health_number_of_nodes	Gauge	Number of nodes in the cluster.
Elasticsearch Metrics	elasticsearch_cluster_health_number_of_data_nodes	Gauge	Number of data nodes in the cluster.
Elasticsearch Metrics	elasticsearch_process_open_files_count	Gauge	Open file descriptors
Elasticsearch Metrics	elasticsearch_indices_fielddata_memory_size_bytes	Gauge	Field data cache memory usage in bytes.
Elasticsearch Metrics	elasticsearch_cluster_health_active_primary_shards	Gauge	The number of primary shards in your cluster. This is an aggregate total across all indices.
Elasticsearch Metrics	elasticsearch_cluster_health_active_shards	Gauge	Aggregate total of all shards across all indices, which includes replica shards.
Elasticsearch Metrics	elasticsearch_cluster_health_delayed_unassigned_shards	Gauge	Shards delayed to reduce reallocation overhead.
Elasticsearch Metrics	elasticsearch_breakers_estimated_size_bytes	Gauge	Estimated size in bytes of breaker.
Elasticsearch Metrics	elasticsearch_breakers_limit_size_bytes	Gauge	Limit size in bytes for breaker.
Elasticsearch Metrics	elasticsearch_os_load1	Gauge	Shortterm load average.
Elasticsearch Metrics	elasticsearch_os_load15	Gauge	Longterm load average.
Elasticsearch Metrics	elasticsearch_os_load5	Gauge	Midterm load average
Elasticsearch Metrics	elasticsearch_jvm_memory_committed_bytes	Gauge	JVM memory currently committed by area.
Elasticsearch Metrics	elasticsearch_filesystem_data_size_bytes	Gauge	Size of block device in bytes.

Group	Metric Name	Type	Synopsis
Elasticsearch Metrics	elasticsearch_indices_docs	Gauge	Count of documents on this node.
Elasticsearch Metrics	elasticsearch_indices_query_cache_memory_size_bytes	Gauge	Query cache memory usage in bytes.
Elasticsearch Metrics	elasticsearch_indices_fielddata_evictions	Gauge	Evictions from field data.
Elasticsearch Metrics	elasticsearch_indices_segments_count	Gauge	Count of index segments on this node.
Elasticsearch Metrics	elasticsearch_indices_segments_memory_bytes	Gauge	Current memory size of segments in bytes.
Elasticsearch Metrics	elasticsearch_breakers_tripped	Gauge	tripped for breaker.

Elasticsearch Metrics

- [elasticsearch_breakers_tripped](#)

Type

64-bit Counter

Description

This metrics records tripped for breaker.

Labels

namespaces

- [elasticsearch_jvm_gc_collection_seconds_count](#)

Type

64-bit Counter

Description

Count of JVM GC runs.

Labels

namespaces

- [elasticsearch_jvm_gc_collection_seconds_sum](#)

Type

64-bit Counter

Description

GC run time in seconds.

Labels

namespaces

- [elasticsearch_indices_translog_operations](#)

Type

64-bit Counter

Description

Total translog operations.

Labels

namespaces

elasticsearch_indices_translog_size_in_bytes
Type
64-bit Counter
Description
Total translog size in bytes.
Labels
namespaces
elasticsearch_jvm_memory_pool_peak_used_bytes
Type
64-bit Counter
Description
JVM memory peak used by pool.
Labels
namespaces
elasticsearch_transport_rx_size_bytes_total
Type
64-bit Counter
Description
Total number of bytes received.
Labels
namespaces
elasticsearch_indices_search_query_time_seconds
Type
64-bit Counter
Description
Total search query time in seconds.
Labels
namespaces
elasticsearch_indices_store_throttle_time_seconds_total
Type
64-bit Counter
Description
Throttle time for index store in seconds.
Labels
namespaces
elasticsearch_indices_indexing_index_time_seconds_total
Type
64-bit Counter
Description

Cumulative index time in seconds.

Labels

namespaces

- elasticsearch_indices_query_cache_evictions

Type

64-bit Counter

Description

Evictions from query cache.

Labels

namespaces

- elasticsearch_indices_filter_cache_evictions

Type

64-bit Counter

Description

Evictions from filter cache.

Labels

namespaces

- elasticsearch_cluster_health_up

Type

64-bit Gauge

Description

The elastic-search cluster's end-point status.

Labels

namespace

- elasticsearch_process_cpu_percent

Type

64-bit Gauge

Description

Average CPU Usage

Labels

namespace

- elasticsearch_filesystem_data_available_bytes

Type

64-bit Gauge

Description

Disk usage Percentage

Labels

namespace

- elasticsearch_jvm_memory_used_bytes

Type

64-bit Gauge

Description

Average JVM memory Used

Labels

namespace

- `elasticsearch_cluster_health_status`

Type

64-bit Gauge

Description

Raises when the Elasticsearch cluster status is YELLOW for 2 minutes, meaning that Elasticsearch has allocated all of the primary shards but some or all of the replicas have not been allocated. For the exact reason, inspect the Elasticsearch logs on the log nodes in `/var/log/elasticsearch/elasticsearch.log`. To verify the current status of the cluster, run `curl -XGET '/_cat/health?pretty'`, where `host` is `elasticsearch:client:server:host` and `port` is `elasticsearch:client:server:port` defined in your model.

Labels

namespace

- `elasticsearch_cluster_health_relocating_shards`

Type

64-bit Gauge

Description

The number of slices that the current node is migrating to other nodes , Usually it is 0, This value will increase when a node in the cluster joins or exits.

Labels

namespace

- `elasticsearch_cluster_health_initializing_shards`

Type

64-bit Gauge

Description

Initializing fragment.

Labels

namespaces

- `elasticsearch_cluster_health_unassigned_shards`

Type

64-bit Gauge

Description

Number of unallocated tiles , Usually it is 0, This value increases when the replica fragmentation of a node is lost.

Labels

namespaces

- `elasticsearch_cluster_health_number_of_pending_tasks`

Type

64-bit Gauge

Description

Only the master node can handle cluster level metadata changes (Create index , Update mapping , Distribution, fragmentation, etc), adopt pending-tasks API You can view the tasks waiting in the queue , In most cases, the queue for metadata changes remains basically zero.

Labels

namespaces

- elasticsearch_jvm_memory_max_bytes

Type

64-bit Gauge

Description

JVM memory max

Labels

namespaces

- elasticsearch_cluster_health_number_of_nodes

Type

64-bit Gauge

Description

Number of nodes in the cluster.

Labels

namespaces

- elasticsearch_cluster_health_number_of_data_nodes

Type

64-bit Gauge

Description

Number of data nodes in the cluster.

Labels

namespaces

- elasticsearch_process_open_files_count

Type

64-bit Gauge

Description

Open file descriptors.

Labels

namespaces

- elasticsearch_indices_fielddata_memory_size_bytes

Type

64-bit Gauge

Description

Field data cache memory usage in bytes.

Labels

namespaces

- elasticsearch_cluster_health_active_primary_shards

Type

64-bit Gauge

Description

The number of primary shards in your cluster. This is an aggregate total across all indices.

Labels

namespaces

- elasticsearch_cluster_health_active_shards

Type

64-bit Gauge

Description

Aggregate total of all shards across all indices, which includes replica shards.

Labels

namespaces

- elasticsearch_cluster_health_delayed_unassigned_shards

Type

64-bit Gauge

Description

Shards delayed to reduce reallocation overhead.

Labels

namespaces

- elasticsearch_breakers_estimated_size_bytes

Type

64-bit Gauge

Description

Estimated size in bytes of breaker.

Labels

namespaces

- elasticsearch_breakers_limit_size_bytes

Type

64-bit Gauge

Description

Limit size in bytes for breaker.

Labels

namespaces

- elasticsearch_os_load1

Type

64-bit Gauge

Description

Shortterm load average.

Labels

namespaces

- elasticsearch_os_load15

Type

64-bit Gauge

Description

Longterm load average

Labels

namespaces

- elasticsearch_os_load5

Type

64-bit Gauge

Description

Midterm load average

Labels

namespaces

- elasticsearch_jvm_memory_committed_bytes

Type

64-bit Gauge

Description

JVM memory currently committed by area.

Labels

namespaces

- elasticsearch_filesystem_data_size_bytes

Type

64-bit Gauge

Description

Size of block device in bytes.

Labels

namespaces

- elasticsearch_indices_docs

Type

64-bit Gauge

Description

Count of documents on this node.

Labels

namespaces

- elasticsearch_indices_query_cache_memory_size_bytes

Type

64-bit Gauge

Description

Query cache memory usage in bytes.

Labels

namespaces

- elasticsearch_indices fielddata evictions

Type

64-bit Gauge

Description

Evictions from field data.

Labels

namespaces

- elasticsearch_indices segments count

Type

64-bit Gauge

Description

Count of index segments on this node

Labels

namespaces

- elasticsearch_indices segments memory bytes

Type

64-bit Gauge

Description

Current memory size of segments in bytes.

Labels

namespaces

- elasticsearch_breakers tripped

Type

64-bit Gauge

Description

This metrics records tripped for breaker.

Labels

namespaces

Table of Contents

- [Summary](#)
- [ETCD Server Statistics](#)
- [Disk Statistics](#)
- [Snapshot Statistics](#)

Summary

Group	Metric Name	Type	Synopsis
ETCD Server Statistics	leader_changes_seen_total	Counter	The number of leader changes seen
ETCD Server Statistics	proposals_failed_total	Counter	The total number of failed proposals seen
ETCD Server Statistics	has_leader	Gauge	Whether or not a leader exists. 1 is existence, 0 is not.
ETCD Server Statistics	proposals_committed_total	Gauge	The total number of consensus proposals committed
ETCD Server Statistics	proposals_applied_total	Gauge	The total number of consensus proposals applied
ETCD Server Statistics	proposals_pending	Gauge	The current number of pending proposals
Disk Statistics	wal_fsync_duration_seconds	Histogram	The latency distributions of fsync called by wal
Disk Statistics	backend_commit_duration_seconds	Histogram	The latency distributions of commit called by backend
Snapshot Statistics	snapshot_save_total_duration_seconds	Histogram	The total latency distributions of save called by snapshot

ETCD Server Statistics

- [leader_changes_seen_total](#)

Type

64-bit Counter

Description

Counts the number of leader changes the member has seen since its start.

Labels

- [proposals_failed_total](#)

Type

64-bit Counter

Description

Normally related to two issues: temporary failures related to a leader election or longer downtime caused by a loss of quorum in the cluster.

Labels

- [has_leader](#)

Type

64-bit Gauge

Description

Indicates whether the member has a leader. If all the members in the cluster do not have any leader, the entire cluster is totally unavailable.

Labels

- [proposals_committed_total](#)

Type

64-bit Gauge

Description

Records the total number of consensus proposals committed. This gauge should increase over time if the cluster is healthy.

Labels

- [proposals_applied_total](#)

Type

64-bit Gauge

Description

Records the total number of consensus proposals applied. The etcd server applies every committed proposal asynchronously.

Labels

- proposals_pending

Type

64-bit Gauge

Description

Indicates how many proposals are queued to commit. Rising pending proposals suggests there is a high client load or the member cannot commit proposals.

Labels

Disk Statistics

- wal_fsync_duration_seconds

Type

64-bit Histogram

Description

A wal_fsync is called when etcd persists its log entries to disk before applying them.

Labels

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.32, 0.064, 0.128, 0.256, 0.0512, 1.024, 2.048, 4.096, 8.192]

- backend_commit_duration_seconds

Type

64-bit Histogram

Description

A backend_commit is called when etcd commits an incremental snapshot of its most recent changes to disk.

Labels

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.32, 0.064, 0.128, 0.256, 0.0512, 1.024, 2.048, 4.096, 8.192]

Snapshot Statistics

- snapshot_save_total_duration_seconds

Type

64-bit Histogram

Description

Abnormally high snapshot duration (snapshot_save_total_duration_seconds) indicates disk issues and might cause the cluster to be unstable.

Labels

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.32, 0.064, 0.128, 0.256, 0.0512, 1.024, 2.048, 4.096, 8.192]

Table of Contents

- [Summary](#)
- [ETCD Client Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_put_requests	Counter	The total number of ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_requests_sent	Counter	The total number of ETCD Client PUT Requests sent.
ETCD Client Metrics	etcd_put_successes	Counter	The total number of successful ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_failures	Counter	The total number of failed ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_with_lease_requests	Counter	The total number of ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_requests_sent	Counter	The total number of ETCD Client PUT with Lease Requests sent.
ETCD Client Metrics	etcd_put_with_lease_successes	Counter	The total number of successful ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_failures	Counter	The total number of failed ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_range_requests	Counter	The total number of ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_requests_sent	Counter	The total number of ETCD Client Range Requests sent.
ETCD Client Metrics	etcd_range_successes	Counter	The total number of successful ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_failures	Counter	The total number of failed ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_delete_requests	Counter	The total number of ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_requests_sent	Counter	The total number of ETCD Client Range Delete Requests sent.
ETCD Client Metrics	etcd_range_delete_successes	Counter	The total number of successful ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_failures	Counter	The total number of failed ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_txn_requests	Counter	The total number of ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_requests_sent	Counter	The total number of ETCD Client Txn Requests sent.
ETCD Client Metrics	etcd_txn_successes	Counter	The total number of successful ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_failures	Counter	The total number of failed ETCD Client Txn Requests.
ETCD Client Metrics	etcd_watch_requests	Counter	The total number of ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_requests_sent	Counter	The total number of ETCD Client Watch Requests sent.
ETCD Client Metrics	etcd_watch_successes	Counter	The total number of successful ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_failures	Counter	The total number of failed ETCD Client Watch Requests.
ETCD Client Metrics	etcdwatcher_requests	Counter	The total number of ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_requests_sent	Counter	The total number of ETCD Watcher Requests sent.
ETCD Client Metrics	etcdwatcher_successes	Counter	The total number of successful ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_failures	Counter	The total number of failed ETCD Watcher Requests.

ETCD Client Metrics

- [etcd_put_requests](#)

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests.

Labels

keyPrefix

- [etcd_put_requests_sent](#)

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests sent.

Labels

keyPrefix

• etcd_put_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT Requests.

Labels

keyPrefix

• etcd_put_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT Requests.

Labels

keyPrefix, failureReason

• etcd_put_with_lease_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests.

Labels

keyPrefix

• etcd_put_with_lease_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests sent.

Labels

keyPrefix

• etcd_put_with_lease_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT with Lease Requests.

Labels

keyPrefix, failureReason

- etcd_range_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests sent.

Labels

keyPrefix

- etcd_range_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Requests.

Labels

keyPrefix, failureReason

- etcd_range_delete_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests.

Labels

keyPrefix

- etcd_range_delete_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests sent.

Labels

keyPrefix

- etcd_range_delete_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Delete Requests.

Labels

keyPrefix

- etcd_range_delete_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Delete Requests.

Labels

keyPrefix, failureReason

- etcd_txn_requests

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests.

Labels

keyPrefix

- etcd_txn_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests sent.

Labels

keyPrefix

- etcd_txn_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Txn Requests.

Labels

keyPrefix

- etcd_txn_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Txn Requests.

Labels

keyPrefix, failureReason

- etcd_watch_requests

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests sent.

Labels

keyPrefix

- etcd_watch_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_failures

Type

64-bit Counter

	Description
	The total number of failed ETCD Client Watch Requests.
	Labels
	<i>keyPrefix, failureReason</i>
• etcdwatcher_requests	
	Type
	64-bit Counter
	Description
	The total number of ETCD Watcher Requests.
	Labels
	<i>keyPrefix</i>
• etcdwatcher_requests_sent	
	Type
	64-bit Counter
	Description
	The total number of ETCD Watcher Requests sent.
	Labels
	<i>keyPrefix</i>
• etcdwatcher_successes	
	Type
	64-bit Counter
	Description
	The total number of successful ETCD Watcher Requests.
	Labels
	<i>keyPrefix</i>
• etcdwatcher_failures	
	Type
	64-bit Counter
	Description
	The total number of failed ETCD Watcher Requests.
	Labels
	<i>keyPrefix, failureReason</i>

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- [Summary](#)
- [Event Exposure Interface Statistics](#)
- [Event Exposure UDSF Interface Statistics](#)
- [Event Exposure Procedure Statistics](#)

Summary

Group	Metric Name	Type	Synopsis
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Group	Metric Name	Type	Synopsis
Event Exposure Interface Statistics	evtexp_notification_requests_received_from_smfcc	Counter	Event Notification requests received from Smfcc
Event Exposure Interface Statistics	evtexp_notification_successes_sent_to_smfcc	Counter	Event Notification successes sent to Smfcc
Event Exposure Interface Statistics	evtexp_notification_failures_sent_to_smfcc	Counter	Event Notification failures sent to Smfcc
Event Exposure Interface Statistics	evtexp_notification_requests_sent_to_nef	Counter	Event Notification requests sent to NEF
Event Exposure Interface Statistics	evtexp_notification_successes_received_from_nef	Counter	Event Notification successes received from NEF
Event Exposure Interface Statistics	evtexp_notification_failures_received_from_nef	Counter	Event Notification failures received from NEF
Event Exposure Interface Statistics	evtexp_pfdfetch_requests_received_from_smfcc	Counter	Pfd Fetch requests received from Smfcc
Event Exposure Interface Statistics	evtexp_pfdfetch_successes_sent_to_smfcc	Counter	Pfd Fetch api successes sent to Smfcc
Event Exposure Interface Statistics	evtexp_pfdfetch_failures_sent_to_smfcc	Counter	Pfd Fetch api failures sent to Smfcc
Event Exposure Interface Statistics	evtexp_nef_nrf_discovery_requests_sent	Counter	NEF Nrf Discovery sent from event exposure
Event Exposure Interface Statistics	evtexp_nef_nrf_discovery_successes	Counter	NEF Discovery Success Received from NRF
Event Exposure Interface Statistics	evtexp_nef_nrf_discovery_failures	Counter	NEF Discovery Failures Received from NRF
Event Exposure Interface Statistics	evtexp_nef_nrf_access_token_requests_sent	Counter	NEF Nrf Access Token(Oauth) sent from event exposure
Event Exposure Interface Statistics	evtexp_nef_nrf_access_token_successes	Counter	NEF Access Token(Oauth) Success Received from NRF
Event Exposure Interface Statistics	evtexp_nef_nrf_access_token_failures	Counter	NEF Access Token(Oauth) Failures Received from NRF
Event Exposure Interface Statistics	evtexp_nef_pfd_fetch_requests_sent	Counter	NEF Pfd Fetch Request sent from event exposure
Event Exposure Interface Statistics	evtexp_nef_pfd_fetch_successes	Counter	NEF Pfd Fetch Success Received from NRF
Event Exposure Interface Statistics	evtexp_nef_pfd_fetch_failures	Counter	NEF Pfd Fetch Failures Received from NRF
Event Exposure Interface Statistics	evtexp_nef_pfd_subscribe_requests_sent	Counter	NEF Pfd Subscribe Request sent from event exposure
Event Exposure Interface Statistics	evtexp_nef_pfd_subscribe_successes	Counter	NEF Pfd Subscribe Success Received from NRF
Event Exposure Interface Statistics	evtexp_nef_pfd_subscribe_failures	Counter	NEF Pfd Subscribe Failures Received from NRF
Event Exposure Interface Statistics	evtexp_nef_pfd_delete_subscribe_requests_sent	Counter	NEF Pfd Delete Subscribe Request sent from event exposure
Event Exposure Interface Statistics	evtexp_nef_pfd_delete_subscribe_successes	Counter	NEF Pfd Delete Subscribe Success Received from NRF
Event Exposure Interface Statistics	evtexp_nef_pfd_delete_subscribe_failures	Counter	NEF Pfd Delete Subscribe Failures Received from NRF
Event Exposure Interface Statistics	evtexp_pfcpc_pfdmgmt_requests_sent	Counter	PFD Mgmt message to PCFP Requests
Event Exposure Interface Statistics	evtexp_pfcpc_pfdmgmt_requests_successes	Counter	PFD Mgmt message to PCFP Successes
Event Exposure Interface Statistics	evtexp_pfcpc_pfdmgmt_requests_failures	Counter	PFD Mgmt message to PCFP Failures

Group	Metric Name	Type	Synopsis
Event Exposure Interface Statistics	evtexp_timer_start_requests	Counter	Timer Start Requests
Event Exposure Interface Statistics	evtexp_timer_start_successes	Counter	Timer Start Successes
Event Exposure Interface Statistics	evtexp_timer_start_failures	Counter	Timer Start Failures
Event Exposure Interface Statistics	evtexp_timer_stop_requests	Counter	Timer stop Requests
Event Exposure Interface Statistics	evtexp_timer_stop_successes	Counter	Timer stop Successes
Event Exposure Interface Statistics	evtexp_timer_stop_failures	Counter	Timer stop Failures
Event Exposure Interface Statistics	evtexp_pfcg_assoc_guard_timeout	Counter	Guard Timeout Event
Event Exposure Interface Statistics	evtexp_nef_pfd_subscribe_notify_received	Counter	NEF Pfd Subscribe Notify Received from NEF
Event Exposure Interface Statistics	evtexp_nef_pfd_subscribe_notify_successes	Counter	NEF PFD Subscribe Notify Success Sent from SMF
Event Exposure Interface Statistics	evtexp_nef_pfd_subscribe_notify_failures	Counter	NEF PFD Subscribe Notify Failures Sent from SMF
Event Exposure Interface Statistics	evtexp_pfd_etcd_update_requests_sent	Counter	Number of PFD ETCD Update Requests
Event Exposure Interface Statistics	evtexp_pfd_etcd_update_requests_successes	Counter	PFD ETCD Update Successes
Event Exposure Interface Statistics	evtexp_pfd_etcd_update_requests_failures	Counter	PFD ETCD Update Failures
Event Exposure Interface Statistics	evtexp_pfd_etcd_delete_requests_sent	Counter	Number of PFD ETCD Delete Requests
Event Exposure Interface Statistics	evtexp_pfd_etcd_delete_requests_successes	Counter	PFD ETCD Delete Successes
Event Exposure Interface Statistics	evtexp_pfd_etcd_delete_requests_failures	Counter	PFD ETCD Delete Failures
Event Exposure Interface Statistics	evtexp_pfd_cache_timeout_received	Counter	Count of PFD Cache Timeouts
Event Exposure UDSF Interface Statistics	evtexp_udsf_read_and_lock_record_request_attempts	Counter	Total number of udsf read and lock record request attempts
Event Exposure UDSF Interface Statistics	evtexp_udsf_read_and_lock_record_request_successes	Counter	Total number of udsf read and lock record request successes
Event Exposure UDSF Interface Statistics	evtexp_udsf_read_and_lock_record_request_failures	Counter	Total number of udsf read and lock record request failures
Event Exposure UDSF Interface Statistics	evtexp_udsf_create_record_and_lock_request_attempts	Counter	Total number of udsf create record and lock request attempts
Event Exposure UDSF Interface Statistics	evtexp_udsf_create_record_and_lock_request_successes	Counter	Total number of udsf create record and lock request successes
Event Exposure UDSF Interface Statistics	evtexp_udsf_create_record_and_lock_request_failures	Counter	Total number of udsf create record and lock request failures
Event Exposure UDSF Interface Statistics	evtexp_udsf_modify_record_and_unlock_request_attempts	Counter	Total number of udsf modify record and unlock request attempts
Event Exposure UDSF Interface Statistics	evtexp_udsf_modify_record_and_unlock_request_successes	Counter	Total number of udsf modify record and unlock request successes
Event Exposure UDSF Interface Statistics	evtexp_udsf_modify_record_and_unlock_request_failures	Counter	Total number of udsf modify record and unlock request failures
Event Exposure UDSF Interface Statistics	evtexp_udsf_delete_record_request_attempts	Counter	Total number of udsf delete record request attempts

Group	Metric Name	Type	Synopsis
Event Exposure UDSF Interface Statistics	evtexp_udsf_delete_record_request_successes	Counter	Total number of udsf delete record request successes
Event Exposure UDSF Interface Statistics	evtexp_udsf_delete_record_request_failures	Counter	Total number of udsf delete record requests failures
Event Exposure UDSF Interface Statistics	evtexp_udsf_get_record_and_steal_lock_request_attempts	Counter	Total number of udsf get record and steal lock requests attempts
Event Exposure UDSF Interface Statistics	evtexp_udsf_get_record_and_steal_lock_request_successes	Counter	Total number of udsf get record and steal lock request successes
Event Exposure UDSF Interface Statistics	evtexp_udsf_get_record_and_steal_lock_request_failures	Counter	Total number of udsf get record and steal lock request failures
Event Exposure UDSF Interface Statistics	evtexp_udsf_multi_read_request_attempts	Counter	Total number of udsf Multi read request attempts
Event Exposure UDSF Interface Statistics	evtexp_udsf_multi_read_request_successes	Counter	Total number of udsf Multi read request successes
Event Exposure UDSF Interface Statistics	evtexp_udsf_multi_read_request_failures	Counter	Total number of udsf Multi read request failures
Event Exposure UDSF Interface Statistics	evtexp_udsf_multi_read_exceeded_max_records	Counter	Total number of udsf Multi read request with more than optimal number of records
Event Exposure UDSF Interface Statistics	evtexp_udsf_batch_read_request_attempts	Counter	Total number of udsf Batch read request attempts
Event Exposure UDSF Interface Statistics	evtexp_udsf_batch_read_request_successes	Counter	Total number of udsf Batch read request successes
Event Exposure UDSF Interface Statistics	evtexp_udsf_batch_read_request_failures	Counter	Total number of udsf batch read request failures
Event Exposure UDSF Interface Statistics	evtexp_udsf_connection_available_indication_received	Counter	Total number of udsf available indication received
Event Exposure UDSF Interface Statistics	evtexp_udsf_connection_unavailable_indication_received	Counter	Total number of udsf unavailable indication received
Event Exposure UDSF Interface Statistics	evtexp_udsf_connection_unavailable_transaction_failure	Counter	Total number of udsf unavailable but transaction was attempted
Event Exposure Procedure Statistics	evtexp_nef_event_notification_proc_attempts	Counter	NEF event notification procedures attempted
Event Exposure Procedure Statistics	evtexp_nef_event_notification_proc_successes	Counter	NEF event notification procedures successful
Event Exposure Procedure Statistics	evtexp_nef_event_notification_proc_failures	Counter	NEF event notification procedures failures
Event Exposure Procedure Statistics	evtexp_pfd_fetch_proc_attempts	Counter	PFD Fetch Procedures attempted
Event Exposure Procedure Statistics	evtexp_pfd_fetch_proc_successes	Counter	PFD Fetch Procedures successful
Event Exposure Procedure Statistics	evtexp_pfd_fetch_proc_failures	Counter	PFD Fetch Procedures failed
Event Exposure Procedure Statistics	evtexp_pfd_sub_notify_proc_attempts	Counter	PFD Subscribe Notify Procedures attempted
Event Exposure Procedure Statistics	evtexp_pfd_sub_notify_proc_successes	Counter	PFD Subscribe Notify Procedures successful
Event Exposure Procedure Statistics	evtexp_pfd_sub_notify_proc_failures	Counter	PFD Subscribe Notify Procedures failures
Event Exposure Procedure Statistics	evtexp_pfd_new_pfcf_association_proc_attempts	Counter	PFD New PFCF Association Procedures attempted
Event Exposure Procedure Statistics	evtexp_pfd_new_pfcf_association_proc_successes	Counter	PFD New PFCF Association Procedures successes
Event Exposure Procedure Statistics	evtexp_pfd_new_pfcf_association_proc_failures	Counter	PFD New PFCF Association Procedures failed

Group	Metric Name	Type	Synopsis
Event Exposure Procedure Statistics	evtexp_pfd_cache_timer_proc_attempts	Counter	PFD Cache Timer Expiry Procedures attempted
Event Exposure Procedure Statistics	evtexp_pfd_cache_timer_proc_successes	Counter	PFD Cache Timer Expiry Procedures successful
Event Exposure Procedure Statistics	evtexp_pfd_cache_timer_proc_failures	Counter	PFD Cache Timer Expiry Procedures failures
Event Exposure Procedure Statistics	evtexp_pfd_cache_clear_cli_proc_attempts	Counter	PFD Cache Clear CLI Procedures attempts
Event Exposure Procedure Statistics	evtexp_pfd_cache_clear_cli_proc_successes	Counter	PFD Cache Clear CLI Procedures successes
Event Exposure Procedure Statistics	evtexp_pfd_cache_clear_cli_proc_failures	Counter	PFD Cache Clear CLI Procedures failures
Event Exposure Procedure Statistics	evtexp_pfd_data_show_cli_proc_attempts	Counter	PFD Data Show CLI Procedures attempts
Event Exposure Procedure Statistics	evtexp_pfd_data_show_cli_proc_successes	Counter	PFD Data Show CLI Procedures successes
Event Exposure Procedure Statistics	evtexp_pfd_data_show_cli_proc_failures	Counter	PFD Data Show CLI Procedures failures

Event Exposure Interface Statistics

- [evtexp_notification_requests_received_from_smfcc](#)

Type

64-bit Counter

Description

Number of Event Notification requests received from Smfcc

Labels
- [evtexp_notification_successes_sent_to_smfcc](#)

Type

64-bit Counter

Description

Number of Event Notification successes sent to Smfcc

Labels
- [evtexp_notification_failures_sent_to_smfcc](#)

Type

64-bit Counter

Description

Number of Event Notification failures sent to Smfcc

Labels

causecode
- [evtexp_notification_requests_sent_to_nef](#)

Type

64-bit Counter

Description

Number of Event Notification requests sent to NEF

Labels

subscriptionid

- evtexp_notification_successes_received_from_nef

Type

64-bit Counter

Description

Number of Event Notification successes received from NEF

Labels

subscriptionid

- evtexp_notification_failures_received_from_nef

Type

64-bit Counter

Description

Number of Event Notification failures received from NEF

Labels

subscriptionid, statuscode

- evtexp_pfdfetch_requests_received_from_smfcc

Type

64-bit Counter

Description

Number of Pfd Fetch requests received from Smfcc

Labels

- evtexp_pfdfetch_successes_sent_to_smfcc

Type

64-bit Counter

Description

Number of Pfd Fetch successes sent to Smfcc

Labels

- evtexp_pfdfetch_failures_sent_to_smfcc

Type

64-bit Counter

Description

Number of Pfd failures sent to Smfcc

Labels

causecode

- evtexp_nef_nrf_discovery_requests_sent

Type

64-bit Counter

Description

Number of NEF NRF Discovery Request sent

Labels

• evtexp_nef_nrf_discovery_successes

Type

64-bit Counter

Description

Number of NEF NRF Discovery Success Received

Labels

• evtexp_nef_nrf_discovery_failures

Type

64-bit Counter

Description

Number of NEF NRF Discovery Failure Received

Labels

causecode

• evtexp_nef_nrf_access_token_requests_sent

Type

64-bit Counter

Description

Number of NEF NRF Access Token(Oauth) Request sent

Labels

• evtexp_nef_nrf_access_token_successes

Type

64-bit Counter

Description

Number of NEF NRF Access Token(Oauth) Success Received

Labels

• evtexp_nef_nrf_access_token_failures

Type

64-bit Counter

Description

Number of NEF NRF Access Token(Oauth) Failure Received

Labels

causecode

• evtexp_nef_pfd_fetch_requests_sent

Type

64-bit Counter

Description

Number of NEF PFD Fetch Request sent

Labels

appid

- evtexp_nef_pfd_fetch_successes

Type

64-bit Counter

Description

Number of NEF Pfd Fetch Success Received

Labels

appid

- evtexp_nef_pfd_fetch_failures

Type

64-bit Counter

Description

Number of NEF Pfd Fetch Failure Received

Labels

appid, causecode

- evtexp_nef_pfd_subscribe_requests_sent

Type

64-bit Counter

Description

Number of NEF PFD Subscribe Request sent

Labels

appid

- evtexp_nef_pfd_subscribe_successes

Type

64-bit Counter

Description

Number of NEF Pfd Subscribe Success Received

Labels

appid

- evtexp_nef_pfd_subscribe_failures

Type

64-bit Counter

Description

Number of NEF Pfd Subscribe Failure Received

Labels

appid, causecode

- evtexp_nef_pfd_delete_subscribe_requests_sent

Type

64-bit Counter

Description

Number of NEF PFD Delete Subscribe Request sent

Labels

appid

- evtexp_nef_pfd_delete_subscribe_successes

Type

64-bit Counter

Description

Number of NEF Pfd Delete Subscribe Success Received

Labels

appid

- evtexp_nef_pfd_delete_subscribe_failures

Type

64-bit Counter

Description

Number of NEF Pfd Delete Subscribe Failure Received

Labels

appid, causecode

- evtexp_pfc_pfdmgmt_requests_sent

Type

64-bit Counter

Description

Number of attempts to send PFD Management Messages to PFCP

Labels

peernodeid

- evtexp_pfc_pfdmgmt_requests_successes

Type

64-bit Counter

Description

Number of PFD Management messages successfully sent to PFCP

Labels

peernodeid

- evtexp_pfc_pfdmgmt_requests_failures

Type

64-bit Counter

Description

Number of PFD Management messages failed sent to PFCP

Labels

peernodeid, causecode

• evtexp_timer_start_requests

Type

64-bit Counter

Description

Number of requests to start Timer

Labels

timerType, timerKey

• evtexp_timer_start_successes

Type

64-bit Counter

Description

Number of successes to start Timer

Labels

timerType, timerKey

• evtexp_timer_start_failures

Type

64-bit Counter

Description

Number of failures to start Timer

Labels

timerType, timerKey, causecode

• evtexp_timer_stop_requests

Type

64-bit Counter

Description

Number of requests to stop Timer

Labels

timerType, timerKey

• evtexp_timer_stop_successes

Type

64-bit Counter

Description

Number of successes to stop Timer

Labels

timerType, timerKey

- evtexp_timer_stop_failures

Type

64-bit Counter

Description

Number of failures to stop Timer

Labels

timerType, timerKey, causecode

- evtexp_pfcip_assoc_guard_timeout

Type

64-bit Counter

Description

Number of Guard Timeout Events

Labels

assocPeerNode

- evtexp_nef_pfd_subscribe_notify_received

Type

64-bit Counter

Description

Number of NEF PFD Subscribe Notify Received

Labels

appid

- evtexp_nef_pfd_subscribe_notify_successes

Type

64-bit Counter

Description

Number of NEF PFD Subscribe Notify Success Sent

Labels

appid

- evtexp_nef_pfd_subscribe_notify_failures

Type

64-bit Counter

Description

Number of NEF PFD Subscribe Notify Failures Sent

Labels

appid, causecode

- evtexp_pfd_etcd_update_requests_sent

Type

64-bit Counter

Description

Number of PFD ETCD update requests sent from EE Pod

Labels

appid

- `evtxp_pfd_etcd_update_requests_successes`

Type

64-bit Counter

Description

Number of successes PFD ETCD update requests

Labels

appid

- `evtxp_pfd_etcd_update_requests_failures`

Type

64-bit Counter

Description

Number of failures PFD ETCD update requests

Labels

appid, causecode

- `evtxp_pfd_etcd_delete_requests_sent`

Type

64-bit Counter

Description

Number of PFD ETCD delete requests sent from EE Pod

Labels

appid

- `evtxp_pfd_etcd_delete_requests_successes`

Type

64-bit Counter

Description

Number of successes PFD ETCD delete requests

Labels

appid

- `evtxp_pfd_etcd_delete_requests_failures`

Type

64-bit Counter

Description

Number of failures PFD ETCD delete requests

Labels

appid, causecode

• evtxp_pfd_cache_timeout_received
Type
64-bit Counter
Description
Number of PFD Cache Timeout Events
Labels
appid

Event Exposure UDSF Interface Statistics

• evtxp_udsf_read_and_lock_record_request_attempts
Type
64-bit Counter
Description
Number of udsf read and lock record request attempts
Labels
appid
• evtxp_udsf_read_and_lock_record_request_successes
Type
64-bit Counter
Description
Number of udsf read and lock record request successess
Labels
appid
• evtxp_udsf_read_and_lock_record_request_failures
Type
64-bit Counter
Description
Number of udsf read and lock record request failures
Labels
appid, cause
• evtxp_udsf_create_record_and_lock_request_attempts
Type
64-bit Counter
Description
Number of usdf create record and lock request attempts
Labels
appid
• evtxp_udsf_create_record_and_lock_request_successes
Type
64-bit Counter

Description

Number of usdf create record and lock request successes

Labels

appid

- evtexp_udsf_create_record_and_lock_request_failures

Type

64-bit Counter

Description

Number of usdf create record and lock request failures

Labels

appid, cause

- evtexp_udsf_modify_record_and_unlock_request_attempts

Type

64-bit Counter

Description

Number of usdf modify record and unlock request attempts

Labels

appid

- evtexp_udsf_modify_record_and_unlock_request_successes

Type

64-bit Counter

Description

Number of usdf modify record and unlock request successes

Labels

appid

- evtexp_udsf_modify_record_and_unlock_request_failures

Type

64-bit Counter

Description

Number of usdf modify record and unlock request failures

Labels

appid, cause

- evtexp_udsf_delete_record_request_attempts

Type

64-bit Counter

Description

Number of usdf delete record request attempts

Labels

appid

• evtxp_udsf_delete_record_request_successes
Type
64-bit Counter
Description
Number of udsf delete record request successes
Labels
appid
• evtxp_udsf_delete_record_request_failures
Type
64-bit Counter
Description
Number of udsf delete record request failures
Labels
appid, cause
• evtxp_udsf_get_record_and_steal_lock_request_attempts
Type
64-bit Counter
Description
Number of udsf get record and steal lock request attempts
Labels
appid
• evtxp_udsf_get_record_and_steal_lock_request_successes
Type
64-bit Counter
Description
Number of udsf get record and steal lock request successes
Labels
appid
• evtxp_udsf_get_record_and_steal_lock_request_failures
Type
64-bit Counter
Description
Number of udsf get record and steal lock request failures
Labels
appid, cause
• evtxp_udsf_multi_read_request_attempts
Type
64-bit Counter
Description

Number of udsf Multi Read request attempts

Labels

peer

- evtexp_udsf_multi_read_request_successes

Type

64-bit Counter

Description

Number of udsf Multi Read request successes

Labels

peer

- evtexp_udsf_multi_read_request_failures

Type

64-bit Counter

Description

Number of udsf Multi Read request failures

Labels

peer, cause

- evtexp_udsf_multi_read_exceeded_max_records

Type

64-bit Counter

Description

Number of udsf Multi Read request with more than optimal number of records

Labels

- evtexp_udsf_batch_read_request_attempts

Type

64-bit Counter

Description

Number of udsf Batch Read request attempts

Labels

peer

- evtexp_udsf_batch_read_request_successes

Type

64-bit Counter

Description

Number of udsf Batch Read request successes

Labels

peer

- evtexp_udsf_batch_read_request_failures

Type

64-bit Counter

Description

Number of udsf batch Read request failures

Labels

peer, cause

- evtexp_udsf_connection_available_indication_received

Type

64-bit Counter

Description

Number of udsf available indication received

Labels

- evtexp_udsf_connection_unavailable_indication_received

Type

64-bit Counter

Description

Number of udsf unavailable indication received

Labels

- evtexp_udsf_connection_unavailable_transaction_failure

Type

64-bit Counter

Description

Number of udsf unavailable but transaction was attempted

Labels

procname

Event Exposure Procedure Statistics

- evtexp_nef_event_notification_proc_attempts

Type

64-bit Counter

Description

Number of NEF event notification procedures attempted

Labels

subscriptionid

- evtexp_nef_event_notification_proc_successes

Type

64-bit Counter

Description

Number of NEF event notification procedures successful

Labels

subscriptionid

- evtexp_nef_event_notification_proc_failures

Type

64-bit Counter

Description

Number of NEF event notification procedures failures

Labels

subscriptionid, causecode

- evtexp_pfd_fetch_proc_attempts

Type

64-bit Counter

Description

Number of PFD Fetch procedures attempted

Labels

- evtexp_pfd_fetch_proc_successes

Type

64-bit Counter

Description

Number of PFD Fetch procedures successful

Labels

- evtexp_pfd_fetch_proc_failures

Type

64-bit Counter

Description

Number of PFD Fetch procedures failed

Labels

causecode

- evtexp_pfd_sub_notify_proc_attempts

Type

64-bit Counter

Description

Number of PFD Subscribe Notify procedures attempted

Labels

appid

- evtexp_pfd_sub_notify_proc_successes

Type

64-bit Counter

Description

Number of PFD Subscribe Notify procedures successful

Labels

appid

- evtexp_pfd_sub_notify_proc_failures

Type

64-bit Counter

Description

Number of PFD Subscribe Notify procedures failures

Labels

appid, causecode

- evtexp_pfd_new_pfcip_association_proc_attempts

Type

64-bit Counter

Description

Number of PFD New PFCIP Association Procedure attempted

Labels

peerNodeName

- evtexp_pfd_new_pfcip_association_proc_successes

Type

64-bit Counter

Description

Number of PFD New PFCIP Association Procedure successes

Labels

peerNodeName

- evtexp_pfd_new_pfcip_association_proc_failures

Type

64-bit Counter

Description

Number of PFD New PFCIP Association Procedure failed

Labels

peerNodeName, causecode

- evtexp_pfd_cache_timer_proc_attempts

Type

64-bit Counter

Description

Number of PFD Cache Timer Expiry procedures attempted

Labels

appid

- evtexp_pfd_cache_timer_proc_successes

Type

64-bit Counter

Description

Number of PFD Cache Timer Expiry procedures successful

Labels

appid

- evtexp_pfd_cache_timer_proc_failures

Type

64-bit Counter

Description

Number of PFD Cache Timer Expiry procedures failures

Labels

appid, causecode

- evtexp_pfd_cache_clear_cli_proc_attempts

Type

64-bit Counter

Description

Number of PFD Cache Clear CLI procedures attempts

Labels

appid

- evtexp_pfd_cache_clear_cli_proc_successes

Type

64-bit Counter

Description

Number of PFD Cache Clear CLI procedures successes

Labels

appid

- evtexp_pfd_cache_clear_cli_proc_failures

Type

64-bit Counter

Description

Number of PFD Cache Clear CLI procedures failures

Labels

appid, causecode

- evtexp_pfd_data_show_cli_proc_attempts

Type

64-bit Counter

Description

Number of PFD Data Show CLI procedures attempts

Labels

appid

- evtxp_pfd_data_show_cli_proc_successes

Type

64-bit Counter

Description

Number of PFD Data Show CLI procedures successes

Labels

appid

- evtxp_pfd_data_show_cli_proc_failures

Type

64-bit Counter

Description

Number of PFD Data Show CLI procedures failures

Labels

appid, causecode

Table of Contents

- [Summary](#)
- [HTTP Punting Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_sent	Counter	The total number of HTTP requests successfully punted.
HTTP Punting Metrics	punting_http_requests_failed	Counter	The total number of HTTP requests punting failed.
HTTP Punting Metrics	punting_http_responses_received	Counter	The total number of HTTP responses received for the requests punted.
HTTP Punting Metrics	punting_http_responses_forwarded	Counter	The total number of HTTP responses forwarded after punting transaction completion.
HTTP Punting Metrics	punting_http_requests_aborted	Counter	The total number of HTTP requests aborted.
HTTP Punting Metrics	punting_http_request_duration_seconds	Histogram	Punting operation duration in seconds.

HTTP Punting Metrics

- punting_http_requests_sent

Type

64-bit Counter

Description

The total number of HTTP requests successfully punted.

Labels

message_type

punting_http_requests_failed
Type
64-bit Counter
Description
The total number of HTTP requests for which the punting failed.
Labels
message_type
punting_http_responses_received
Type
64-bit Counter
Description
The total number of HTTP responses received for the requests punted.
Labels
message_type, status_code
punting_http_responses_forwarded
Type
64-bit Counter
Description
The total number of HTTP responses forwarded after punting transaction completion.
Labels
message_type, status_code
punting_http_requests_aborted
Type
64-bit Counter
Description
The total number of HTTP requests aborted.
Labels
message_type
punting_http_request_duration_seconds
Type
64-bit Histogram
Description
The punting transaction duration is measured from the time the HTTP request is sent out to the time the response is received for the punting request.
Labels
message_type
Boundaries
[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

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- [Packet Streaming Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Packet Streaming Metrics	num_packet_streaming_attempted	Counter	The total number of attempted streaming of packets
Packet Streaming Metrics	num_packet_streaming_success	Counter	The total number of packets streamed successfully
Packet Streaming Metrics	num_packet_streaming_failed	Counter	The total number of failed streaming of packets
Packet Streaming Metrics	num_packet_streaming_bytes	Counter	The total number of bytes successfully streamed

Packet Streaming Metrics

- num_packet_streaming_attempted

Type

64-bit Counter

Description

The total number of attempted streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_success

Type

64-bit Counter

Description

The total number of packets streamed successfully

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_failed

Type

64-bit Counter

Description

The total number of failed streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_bytes

Type

64-bit Counter

Description

The total number of bytes successfully streamed

Labels

interface_name, server_fqdn, kafka_topic

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- [Summary](#)

- [Circuit Breaker Metrics](#)
- [Service Endpoint Learner Metrics](#)
- [Service Endpoint Manager Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Circuit Breaker Metrics	sem_cb_endpoints	Gauge	Total number of endpoints for a service in respective states
Service Endpoint Learner Metrics	sem_num_dns_errors	Counter	Total number of DNS resolution errors for a given service
Service Endpoint Learner Metrics	sem_num_endpoints	Gauge	Total number of endpoints learnt for a service
Service Endpoint Manager Metrics	sem_num_endpoint_transactions_sent	Counter	Total number of times an endpoint is given to the application for communication with it
Service Endpoint Manager Metrics	sem_num_endpoint_transaction_status_received	Counter	Total number of transactions for a given endpoint received with the transaction status

Circuit Breaker Metrics

- [sem_cb_endpoints](#)

Type

64-bit Gauge

Description

Total number of endpoints for a service in respective states. There are 3 states viz. Active, Blocked, Restoring. In Active state, the endpoint is responding as expected, i.e. the response code doesn't indicate a service failure. The endpoint will be moved to Blocked state based on response failures which indicate service failure. After a timeout, the endpoint will be moved to Restoring state to check if the endpoint has recovered. In this state, if endpoint responds with response code other than service failure, it will be moved to Active state. Else, it will be moved back to Blocked state.

Labels

service_name, endpoint_state

Service Endpoint Learner Metrics

- [sem_num_dns_errors](#)

Type

64-bit Counter

Description

Total number of DNS resolution errors for a given service, labels are service name for DNS resolution and DNS error codes

Labels

service_name, error_code

- [sem_num_endpoints](#)

Type

64-bit Gauge

Description

Total number of endpoints learnt from the Kubernetes DNS server for a given service

Labels

service

Service Endpoint Manager Metrics

- **sem_num_endpoint_transactions_sent**

Type

64-bit Counter

Description

Total number of times an endpoint is given to the application for communication with it

Labels

service
- **sem_num_endpoint_transaction_status_received**

Type

64-bit Counter

Description

Total number of transactions for a given endpoint received with the transaction status

Labels

service, status

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- [Summary](#)
- [UDSF DBClient Metrics](#)
- [UDSF API latency statistics](#)

Summary

Group	Metric Name	Type	Synopsis
UDSF DBClient Metrics	udsf_db_operation_request	Counter	The total number of udsf dbclient requests received.
UDSF DBClient Metrics	udsf_db_operation_response	Counter	The total number of udsf dbclient responses sent.
UDSF DBClient Metrics	udsf_dbc_total_init_success	Counter	The total number of UDSF INIT successes.
UDSF DBClient Metrics	udsf_dbc_total_init_failure	Counter	The total number of UDSF INIT failures.
UDSF DBClient Metrics	udsf_dbc_total_deinit_success	Counter	The total number of UDSF DEINIT successes.
UDSF DBClient Metrics	udsf_dbc_total_deinit_failure	Counter	The total number of UDSF DEINIT failures.
UDSF DBClient Metrics	udsf_db_transaction_counter	Counter	The total number of CRUD operations.
UDSF API latency statistics	udsf_db_transaction_duration	Histogram	Roundtrip duration taken by transaction in seconds.
UDSF API latency statistics	udsf_db_execution_duration	Histogram	Execution time taken by transaction in microseconds.
UDSF API latency statistics	udsf_hiredis_bufferwrite_duration	Histogram	Duration of time taken in hiredis to write data to socket in seconds

UDSF DBClient Metrics

- **udsf_db_operation_request**

Type

64-bit Counter

Description

The total number of udsf dbclient requests received from worker. Labels- 'app_name' : Originating thread name, ex:smfcc, 'operation_type' : Type of CRUD operation/API type involved, ex:CREATE_OR_MODIFY_RECORD_ASYNC

Labels

app_name, operation_type
- **udsf_db_operation_response**

Type

64-bit Counter

Description

The total number of udsf dbclient responses sent. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE 'operation_type' : Type of CRUD operation/API type involved., ex:CREATE_OR_MODIFY_RECORD_ASYNC

Labels

app_name, status, operation_type

• udsf_dbc_total_init_success

Type

64-bit Counter

Description

The total number of times UDSF API init was success. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: SUCCESS

Labels

app_name, status

• udsf_dbc_total_init_failure

Type

64-bit Counter

Description

The total number of times UDSF API init failed. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: FAILURE

Labels

app_name, status

• udsf_dbc_total_deinit_success

Type

64-bit Counter

Description

The total number of times UDSF API Deinit succeeded. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: SUCCESS

Labels

app_name, status

• udsf_dbc_total_deinit_failure

Type

64-bit Counter

Description

The total number of times UDSF API Deinit failed. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: FAILURE

Labels

app_name, status

• udsf_db_transaction_counter

Type

64-bit Counter

Description

The total number of times CRUD operations are incrementing. Labels- 'app_name' : Originating thread name, ex:smfcc, 'crud_op' : The categorical CRUD operation in which this sub-transation falls into, ex: CREATE_OR_MODIFY

Labels

app_name, crud_op

UDSF API latency statistics

- udsf_db_transaction_duration

Type

64-bit Histogram

Description

Duration observed on total journey/roundtrip taken by a transaction after receiving the request. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE 'operation_type' : Type of CRUD operation/API type involved., ex:CREATE_OR_MODIFY_RECORD_ASYNC, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE, 'crud_op' : The categorical CRUD operation in which this transation falls into, ex: CREATE_OR_MODIFY

Labels

app_name,operation_type,status,crud_op

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

- udsf_db_execution_duration

Type

64-bit Histogram

Description

Processing time taken by UDSF client. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE 'operation_type' : Type of CRUD operation/API type involved., ex:CREATE_OR_MODIFY_RECORD_ASYNC, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE, 'crud_op' : The categorical CRUD operation in which this transation falls into, ex: CREATE_OR_MODIFY

Labels

app_name,operation_type,status,crud_op

Boundaries

[1, 2, 4, 8, 16, 32, 64, 128, 256, 384, 512, 768, 1024, 1536, 2048, 4096, 8192, 16384]

- udsf_hiredis_bufferwrite_duration

Type

64-bit Histogram

Description

Duration observed between hiredis buffer append and write to socket. This is fair approximation for time spent before data is written to socket as the buffer threshold accounted for append and write is dynamic and not fixed size. Per socket, the buffer append start-time is recorded along with size intended to transmit and end-time recorded once data greater than recorded size is written on the socket

Labels

app_name

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

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- [Summary](#)
- [GTPC layer general statistics](#)
- [Path management statistics at GTPC layer](#)
- [Consumer message statistics at GTPC layer](#)

Summary

Group	Metric Name	Type	Synopsis
GTPC layer general statistics	gtpc_agent_in_packet	Counter	The number of packets received from gtpc peer.
GTPC layer general statistics	gtpc_agent_out_packet	Counter	The number of packets sent to gtpc peer.
GTPC layer general statistics	gtpc_agent_in_bytes	Counter	The number of bytes received by gtpc.
GTPC layer general statistics	gtpc_agent_out_bytes	Counter	The number of bytes sent by gtpc.
GTPC layer general statistics	gtpc_agent_ulb_tunneled_in_packet	Counter	The number of udp tunnelled packets received from ulb.
GTPC layer general statistics	gtpc_agent_ulb_tunneled_out_packet	Counter	The number of udp tunnelled packets sent to ulb.
GTPC layer general statistics	gtpc_agent_ulb_packet_decoding_failed	Counter	The number of decoding failed for the packets received from the ulb.
GTPC layer general statistics	gtpc_agent_ulb_packet_dropped_due_no_ulb_instance	Counter	The number of packets dropped due to unavailability of ulb instance.
GTPC layer general statistics	gtpc_agent_k8_watch_request_attempts	Counter	The number of watch requests sent by gtpcagent to k8 api server.
GTPC layer general statistics	gtpc_agent_k8_watch_notifications	Counter	The number of watch notifications by gtpcagent from k8 api server.
GTPC layer general statistics	gtpc_agent_k8_watch_object_notifications	Counter	The number of watch notifications by gtpcagent from k8 api server for a specific k8 resource.
GTPC layer general statistics	gtpc_agent_k8_watch_notification_failure	Counter	The number of watch notifications which could not be processed.
GTPC layer general statistics	gtpc_agent_path	Gauge	The current number of GTP anchor paths.
Path management statistics at GTPC layer	gtpc_agent_out_echo_response	Counter	The number of echo responses sent to peer.
Path management statistics at GTPC layer	gtpc_agent_out_echo_response_failure	Counter	The number of failed echo response attempts sent towards peer.
Path management statistics at GTPC layer	gtpc_agent_out_echo_request	Counter	The number of echo requests sent to peer.
Path management statistics at GTPC layer	gtpc_agent_out_echo_request_failure	Counter	The number of failed echo request attempts towards peer.
Path management statistics at GTPC layer	gtpc_agent_in_echo_request	Counter	The number of echo requests received from peer
Path management statistics at GTPC layer	gtpc_agent_in_echo_response	Counter	The number of echo responses received from peer
Path management statistics at GTPC layer	gtpc_agent_path_failure	Counter	The number of path failure detected for peer.
Path management statistics at GTPC layer	gtpc_agent_in_echo_response_decode_failure	Counter	The number of echo response decode failure.
Path management statistics at GTPC layer	gtpc_agent_out_version_not_supported	Counter	The number of version not supported packet sent.
Consumer message statistics at GTPC layer	gtpc_agent_request_received	Counter	The number of requests received from peer.
Consumer message statistics at GTPC layer	gtpc_agent_response_sent	Counter	The number of responses sent to peer.
Consumer message statistics at GTPC layer	gtpc_agent_response_received	Counter	The number of responses received from peer.
Consumer message statistics at GTPC layer	gtpc_agent_request_sent	Counter	The number of requests sent to peer.

Group	Metric Name	Type	Synopsis
Consumer message statistics at GTPC layer	gtpc_agent_retransmit_messages_received	Counter	The number of retransmitted messages received.
Consumer message statistics at GTPC layer	gtpc_agent_retransmit_messages_sent	Counter	The number of retransmitted messages sent.

GTPC layer general statistics

- [gtpc_agent_in_packet](#)

Type

64-bit Counter

Description

The number of packets received from gtpc peer. This include all packets for which header decoding is successful.

Labels

gtpVersion, interface
- [gtpc_agent_out_packet](#)

Type

64-bit Counter

Description

The number of packets sent to gtpc peer. This is incremented after Gtpc agent write packet on socket.

Labels

gtpVersion, interface
- [gtpc_agent_in_bytes](#)

Type

64-bit Counter

Description

The number of bytes received from gtpc peer. This include bytes of all packets for which header decoding is successful.

Labels

gtpVersion, interface
- [gtpc_agent_out_bytes](#)

Type

64-bit Counter

Description

The number of bytes sent by gtpc. This is incremented after Gtpc agent write packet on socket.

Labels

gtpVersion, interface
- [gtpc_agent_ulb_tunneled_in_packet](#)

Type

64-bit Counter

Description

The number of udp tunnelled packets received from ulb. This include all packets for which outer header decoding is successful.

Labels

gtpVersion, interface, ulb

- gtpc_agent_ulb_tunneled_out_packet

Type

64-bit Counter

Description

The number of udp tunnelled packets sent to ulb. This is incremented after Gtpc Agent write packet on socket.

Labels

gtpVersion, interface, ulb

- gtpc_agent_ulb_packet_decoding_failed

Type

64-bit Counter

Description

The number of decoding failed for the packets received from the ulb. This is incremented if decoding the IP/UDP header fails at Gtpc Agent for the packets received from the ulb

Labels

- gtpc_agent_ulb_packet_dropped_due_no_ulb_instance

Type

64-bit Counter

Description

The number of packets dropped as there is no ulb instance associated with the given NF IP Address.

Labels

interface, nfipaddr

- gtpc_agent_k8_watch_request_attempts

Type

64-bit Counter

Description

The number of watch requests sent by gtpc agent to K8 api server via srclient.

Labels

uri, watchobjecttype

- gtpc_agent_k8_watch_notifications

Type

64-bit Counter

Description

The number of watch notifications received by gtpc agent from k8 api server via srclient.

Labels

watchobjecturlpath, notificationtype

- gtpc_agent_k8_watch_object_notifications

Type

64-bit Counter

Description
The number of watch notifications received by gtpc agent from k8 api server via srclient for a specific k8 resource.
Labels
<i>k8resourcename, k8resourcetype, notificationtype</i>
gtpc_agent_k8_watch_notification_failure
Type
64-bit Counter
Description
The number of watch notifications which could not be processed.
Labels
<i>watchobjecturlpath, reason</i>
gtpc_agent_path
Type
64-bit Gauge
Description
The current number of GTP anchor paths.
Labels
<i>gtpVersion, interface</i>

Path management statistics at GTPC layer

gtpc_agent_out_echo_response
Type
64-bit Counter
Description
The number of echo responses sent to peer. V1 protocol message defined in 3GPP TS 29.060 and V2 protocol message are defined in 3GPP TS 29.274.
Labels
<i>gtpVersion, interface, peer</i>
gtpc_agent_out_echo_response_failure
Type
64-bit Counter
Description
The number of failed echo response attempts towards peer. V1 protocol message defined in 3GPP TS 29.060 and V2 protocol message are defined in 3GPP TS 29.274.
Labels
<i>gtpVersion, interface, peer, reason</i>
gtpc_agent_out_echo_request
Type
64-bit Counter
Description
The number of echo requests sent to peer. V1 protocol message defined in 3GPP TS 29.060 and V2 protocol message are defined in 3GPP TS 29.274.

Labels

gtpVersion, interface, peer

• gtpc_agent_out_echo_request_failure

Type

64-bit Counter

Description

The number of failed echo request attempts towards peer. V1 protocol message defined in 3GPP TS 29.060 and V2 protocol message are defined in 3GPP TS 29.274.

Labels

gtpVersion, interface, peer, reason

• gtpc_agent_in_echo_request

Type

64-bit Counter

Description

The number of echo requests received from peer. V1 protocol message defined in 3GPP TS 29.060 and V2 protocol message are defined in 3GPP TS 29.274.

Labels

gtpVersion, interface, peer

• gtpc_agent_in_echo_response

Type

64-bit Counter

Description

The number of echo responses received from peer. V1 protocol message defined in 3GPP TS 29.060 and V2 protocol message are defined in 3GPP TS 29.274.

Labels

gtpVersion, interface, peer

• gtpc_agent_path_failure

Type

64-bit Counter

Description

The number of path failure detected for peer. This is incremented when restart counter changes in echo message or body of any Gtpc message

Labels

gtpVersion, interface, peer

• gtpc_agent_in_echo_response_decode_failure

Type

64-bit Counter

Description

The number of echo response decode failure.

Labels

gtpVersion, interface, peer

• gtpc_agent_out_version_not_supported

Type

64-bit Counter

Description

The number of version not supported packet sent. This is incremented when interface does not supports V2 protocol.

Labels

gtpVersion, interface, peer

Consumer message statistics at GTPC layer

- gtpc_agent_request_received

Type

64-bit Counter

Description

The number of requests received from peer. V1 protocol message defined in 3GPP TS 29.060 and V2 protocol message are defined in 3GPP TS 29.274.

Labels

gtpVersion, interface, message

- gtpc_agent_response_sent

Type

64-bit Counter

Description

The number of responses sent to peer. V1 protocol message defined in 3GPP TS 29.060 and V2 protocol message are defined in 3GPP TS 29.274.

Labels

gtpVersion, interface, message

- gtpc_agent_response_received

Type

64-bit Counter

Description

The number of responses received from peer. V1 protocol message defined in 3GPP TS 29.060 and V2 protocol message are defined in 3GPP TS 29.274.

Labels

gtpVersion, interface, message

- gtpc_agent_request_sent

Type

64-bit Counter

Description

The number of requests sent to peer. V1 protocol message defined in 3GPP TS 29.060 and V2 protocol message are defined in 3GPP TS 29.274.

Labels

gtpVersion, interface, message

- gtpc_agent_retransmit_messages_received

Type

64-bit Counter

Description

The number of retransmitted messages received. This is incremented if we receive message with same sequence number from same IP address and Port

Labels

gtpVersion, interface, message

- gtpc_agent_retransmit_messages_sent

Type

64-bit Counter

Description

The number of retransmitted messages sent. This is incremented when T3 timer expires and request is sent again

Labels

gtpVersion, interface, message

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- [Summary](#)
- [ChargingCmn statistics](#)

Summary

Group	Metric Name	Type	Synopsis
ChargingCmn statistics	svcmgr_service_available_ind_notification	Counter	How many times, Service Available Indication message received
ChargingCmn statistics	svcmgr_service_available_ind_parse_errs	Counter	How many times, Service Available Indication parse error happened
ChargingCmn statistics	svcmgr_service_available_ind_notification_ack	Counter	How many times, Service Available Indication message is acknowledged
ChargingCmn statistics	svcmgr_service_unavailable_ind_notification	Counter	How many times, Service Unavailable Indication message
ChargingCmn statistics	svcmgr_service_unavailable_ind_parse_errs	Counter	How many times, Service Unavailable Indication parse error happened
ChargingCmn statistics	svcmgr_service_unavailable_ind_notification_ack	Counter	How many times, Service Unavailable Indication message is acknowledged
ChargingCmn statistics	svcmgr_publish_msg_sent	Counter	How many times, Publication message sent
ChargingCmn statistics	svcmgr_publish_msg_success	Counter	How many times, Publication message success response received
ChargingCmn statistics	svcmgr_publish_msg_failure	Counter	How many times, failure received for publication message
ChargingCmn statistics	svcmgr_subscribe_msg_sent	Counter	How many times, Subscribe message sent
ChargingCmn statistics	svcmgr_subscribe_msg_success	Counter	How many times, subscribe message success
ChargingCmn statistics	svcmgr_subscribe_msg_failure	Counter	How many times, failure received for subscribe message
ChargingCmn statistics	svcmgr_unsubscribe_msg_sent	Counter	How many times, unsubscribe message sent
ChargingCmn statistics	svcmgr_unsubscribe_msg_failure	Counter	How many times, failure received for unsubscribe message
ChargingCmn statistics	svcmgr_config_req_sent	Counter	How many times, config request sent
ChargingCmn statistics	svcmgr_config_req_failure	Counter	How many times, failure received for config request
ChargingCmn statistics	svcmgr_config_req_success	Counter	How many times, success received for config request

Group	Metric Name	Type	Synopsis
ChargingCmn statistics	svcmgr_config_register_response	Counter	How many times, config register response received
ChargingCmn statistics	svcmgr_config_register_response_ack	Counter	How many times, config register response is acknowledged
ChargingCmn statistics	svcmgr_config_register_response_parse_err	Counter	How many times, parse error happened for config register response
ChargingCmn statistics	svcmgr_config_update_notification	Counter	How many times, config update notification received
ChargingCmn statistics	svcmgr_config_update_notification_response	Counter	How many times, config update response
ChargingCmn statistics	svcmgr_intf_schema_update_ind_notification	Counter	How many times, interface schema update indication message received
ChargingCmn statistics	svcmgr_intf_schema_update_ind_notification_ack	Counter	How many times, interface schema update indication message is acknowledged
ChargingCmn statistics	svcmgr_intf_schema_update_ind_parse_errs	Counter	How many times, interface schema update indication parse error happened

ChargingCmn statistics

- svcmgr_service_available_ind_notification

Type

64-bit Counter

Description

Number of Times Service Available Indication notification received

Labels

servicename

- svcmgr_service_available_ind_parse_errs

Type

64-bit Counter

Description

Number of Times Service Available Indication parse error happened

Labels

- svcmgr_service_available_ind_notification_ack

Type

64-bit Counter

Description

Number of Times Service Available Indication message is acknowledged

Labels

servicename

- svcmgr_service_unavailable_ind_notification

Type

64-bit Counter

Description

Number of Times Service Unavailable Indication notification received

Labels

servicename

- `svcmgr_service_unavailable_ind_parse_errs`

Type

64-bit Counter

Description

Number of Times Service Unavailable Indication parse error happened

Labels

- `svcmgr_service_unavailable_ind_notification_ack`

Type

64-bit Counter

Description

Number of Times Service Unavailable Indication message is acknowledged

Labels

servicename

- `svcmgr_publish_msg_sent`

Type

64-bit Counter

Description

Number of Times Publication message sent

Labels

servicename, retry

- `svcmgr_publish_msg_success`

Type

64-bit Counter

Description

Number of Times Publication message success response received

Labels

servicename, causecode, retry

- `svcmgr_publish_msg_failure`

Type

64-bit Counter

Description

Number of Times failure received for publication message

Labels

servicename, causecode, retry

- `svcmgr_subscribe_msg_sent`

Type

64-bit Counter

Description
Number of Times Subscribe message sent
Labels
<i>servicename, retry</i>
svcmgr_subscribe_msg_success
Type
64-bit Counter
Description
Number of Times Send subscribe message success response received
Labels
<i>servicename, causecode, retry</i>
svcmgr_subscribe_msg_failure
Type
64-bit Counter
Description
Number of time failure received for subscribe message
Labels
<i>servicename, causecode, retry</i>
svcmgr_unsubscribe_msg_sent
Type
64-bit Counter
Description
Number of Times unsubscribe message sent
Labels
<i>servicename</i>
svcmgr_unsubscribe_msg_failure
Type
64-bit Counter
Description
Number of Times failure received for unsubscribe message
Labels
<i>servicename</i>
svcmgr_config_req_sent
Type
64-bit Counter
Description
Number of Times config request sent
Labels
<i>retry</i>

• svcmgr_config_req_failure Type 64-bit Counter Description Number of Times failure received for config request Labels causecode, retry
• svcmgr_config_req_success Type 64-bit Counter Description Number of Times success received for config request Labels causecode, retry
• svcmgr_config_register_response Type 64-bit Counter Description Number of Times config register response received Labels statuscode
• svcmgr_config_register_response_ack Type 64-bit Counter Description Number of Times config register response is acknowledged Labels
• svcmgr_config_register_response_parse_err Type 64-bit Counter Description Number of Times parse error happened for config register response Labels
• svcmgr_config_update_notification Type 64-bit Counter Description Number of Times config update notification received

Labels

- `svcmgr_config_update_notification_response`

Type

64-bit Counter

Description

Number of Times config update response sent

Labels

- `svcmgr_intf_schema_update_ind_notification`

Type

64-bit Counter

Description

Number of Times interface schema update indication message received

Labels

servicename

- `svcmgr_intf_schema_update_ind_notification_ack`

Type

64-bit Counter

Description

Number of Times interface schema update indication message is acknowledged

Labels

servicename

- `svcmgr_intf_schema_update_ind_parse_errs`

Type

64-bit Counter

Description

Number of Times Service interface schema update indication parse error happened

Labels

Table of Contents

- [Summary](#)
- [ETCD Client Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_put_requests	Counter	The total number of ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_requests_sent	Counter	The total number of ETCD Client PUT Requests sent.
ETCD Client Metrics	etcd_put_successes	Counter	The total number of successful ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_failures	Counter	The total number of failed ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_with_lease_requests	Counter	The total number of ETCD Client PUT with Lease Requests.

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_put_with_lease_requests_sent	Counter	The total number of ETCD Client PUT with Lease Requests sent.
ETCD Client Metrics	etcd_put_with_lease_successes	Counter	The total number of successful ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_failures	Counter	The total number of failed ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_range_requests	Counter	The total number of ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_requests_sent	Counter	The total number of ETCD Client Range Requests sent.
ETCD Client Metrics	etcd_range_successes	Counter	The total number of successful ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_failures	Counter	The total number of failed ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_delete_requests	Counter	The total number of ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_requests_sent	Counter	The total number of ETCD Client Range Delete Requests sent.
ETCD Client Metrics	etcd_range_delete_successes	Counter	The total number of successful ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_failures	Counter	The total number of failed ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_txn_requests	Counter	The total number of ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_requests_sent	Counter	The total number of ETCD Client Txn Requests sent.
ETCD Client Metrics	etcd_txn_successes	Counter	The total number of successful ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_failures	Counter	The total number of failed ETCD Client Txn Requests.
ETCD Client Metrics	etcd_watch_requests	Counter	The total number of ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_requests_sent	Counter	The total number of ETCD Client Watch Requests sent.
ETCD Client Metrics	etcd_watch_successes	Counter	The total number of successful ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_failures	Counter	The total number of failed ETCD Client Watch Requests.
ETCD Client Metrics	etcdwatcher_requests	Counter	The total number of ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_requests_sent	Counter	The total number of ETCD Watcher Requests sent.
ETCD Client Metrics	etcdwatcher_successes	Counter	The total number of successful ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_failures	Counter	The total number of failed ETCD Watcher Requests.

ETCD Client Metrics

- etcd_put_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests.

Labels

keyPrefix

- etcd_put_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests sent.

Labels

keyPrefix

- etcd_put_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT Requests.

Labels

keyPrefix

- etcd_put_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT Requests.

Labels

keyPrefix, failureReason

- etcd_put_with_lease_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests sent.

Labels

keyPrefix

- etcd_put_with_lease_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT with Lease Requests.

Labels

keyPrefix, failureReason

- etcd_range_requests
 - Type
 - 64-bit Counter
 - Description
 - The total number of ETCD Client Range Requests.
 - Labels
 - keyPrefix**
- etcd_range_requests_sent
 - Type
 - 64-bit Counter
 - Description
 - The total number of ETCD Client Range Requests sent.
 - Labels
 - keyPrefix**
- etcd_range_successes
 - Type
 - 64-bit Counter
 - Description
 - The total number of successful ETCD Client Range Requests.
 - Labels
 - keyPrefix**
- etcd_range_failures
 - Type
 - 64-bit Counter
 - Description
 - The total number of failed ETCD Client Range Requests.
 - Labels
 - keyPrefix, failureReason**
- etcd_range_delete_requests
 - Type
 - 64-bit Counter
 - Description
 - The total number of ETCD Client Range Delete Requests.
 - Labels
 - keyPrefix**
- etcd_range_delete_requests_sent
 - Type
 - 64-bit Counter
 - Description

The total number of ETCD Client Range Delete Requests sent.

Labels

keyPrefix

- etcd_range_delete_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Delete Requests.

Labels

keyPrefix

- etcd_range_delete_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Delete Requests.

Labels

keyPrefix, failureReason

- etcd_txn_requests

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests.

Labels

keyPrefix

- etcd_txn_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests sent.

Labels

keyPrefix

- etcd_txn_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Txn Requests.

Labels

keyPrefix

- etcd_txn_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Txn Requests.

Labels

keyPrefix, failureReason

• etcd_watch_requests

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests.

Labels

keyPrefix

• etcd_watch_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests sent.

Labels

keyPrefix

• etcd_watch_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Watch Requests.

Labels

keyPrefix

• etcd_watch_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Watch Requests.

Labels

keyPrefix, failureReason

• etcdwatcher_requests

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests.

Labels

keyPrefix

- etcdwatcher_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests sent.

Labels

keyPrefix

- etcdwatcher_successes

Type

64-bit Counter

Description

The total number of successful ETCD Watcher Requests.

Labels

keyPrefix

- etcdwatcher_failures

Type

64-bit Counter

Description

The total number of failed ETCD Watcher Requests.

Labels

keyPrefix, failureReason

Table of Contents

- [Summary](#)
- [HTTP Punting Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_sent	Counter	The total number of HTTP requests successfully punted.
HTTP Punting Metrics	punting_http_requests_failed	Counter	The total number of HTTP requests punting failed.
HTTP Punting Metrics	punting_http_responses_received	Counter	The total number of HTTP responses received for the requests punted.
HTTP Punting Metrics	punting_http_responses_forwarded	Counter	The total number of HTTP responses forwarded after punting transaction completion.
HTTP Punting Metrics	punting_http_requests_aborted	Counter	The total number of HTTP requests aborted.
HTTP Punting Metrics	punting_http_request_duration_seconds	Histogram	Punting operation duration in seconds.

HTTP Punting Metrics

• punting_http_requests_sent
Type
64-bit Counter
Description
The total number of HTTP requests successfully punted.
Labels
<i>message_type</i>
• punting_http_requests_failed
Type
64-bit Counter
Description
The total number of HTTP requests for which the punting failed.
Labels
<i>message_type</i>
• punting_http_responses_received
Type
64-bit Counter
Description
The total number of HTTP responses received for the requests punted.
Labels
<i>message_type, status_code</i>
• punting_http_responses_forwarded
Type
64-bit Counter
Description
The total number of HTTP responses forwarded after punting transaction completion.
Labels
<i>message_type, status_code</i>
• punting_http_requests_aborted
Type
64-bit Counter
Description
The total number of HTTP requests aborted.
Labels
<i>message_type</i>
• punting_http_request_duration_seconds
Type
64-bit Histogram
Description

The punting transaction duration is measured from the time the HTTP request is sent out to the time the response is received for the punting request.

Labels

message_type

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

Table of Contents

- [Summary](#)
- [Packet Streaming Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Packet Streaming Metrics	num_packet_streaming_attempted	Counter	The total number of attempted streaming of packets
Packet Streaming Metrics	num_packet_streaming_success	Counter	The total number of packets streamed successfully
Packet Streaming Metrics	num_packet_streaming_failed	Counter	The total number of failed streaming of packets
Packet Streaming Metrics	num_packet_streaming_bytes	Counter	The total number of bytes successfully streamed

Packet Streaming Metrics

- num_packet_streaming_attempted
 - Type
 - 64-bit Counter
 - Description
 - The total number of attempted streaming of packets
 - Labels
 - interface_name, server_fqdn, kafka_topic*
- num_packet_streaming_success
 - Type
 - 64-bit Counter
 - Description
 - The total number of packets streamed successfully
 - Labels
 - interface_name, server_fqdn, kafka_topic*
- num_packet_streaming_failed
 - Type
 - 64-bit Counter
 - Description
 - The total number of failed streaming of packets
 - Labels
 - interface_name, server_fqdn, kafka_topic*
- num_packet_streaming_bytes
 - Type

64-bit Counter
Description
The total number of bytes successfully streamed
Labels
<i>interface_name, server_fqdn, kafka_topic</i>

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- [Summary](#)
- [Circuit Breaker Metrics](#)
- [Service Endpoint Learner Metrics](#)
- [Service Endpoint Manager Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Circuit Breaker Metrics	sem_cb_endpoints	Gauge	Total number of endpoints for a service in respective states
Service Endpoint Learner Metrics	sem_num_dns_errors	Counter	Total number of DNS resolution errors for a given service
Service Endpoint Learner Metrics	sem_num_endpoints	Gauge	Total number of endpoints learnt for a service
Service Endpoint Manager Metrics	sem_num_endpoint_transactions_sent	Counter	Total number of times an endpoint is given to the application for communication with it
Service Endpoint Manager Metrics	sem_num_endpoint_transaction_status_received	Counter	Total number of transactions for a given endpoint received with the transaction status

Circuit Breaker Metrics

• sem_cb_endpoints
Type
64-bit Gauge
Description
Total number of endpoints for a service in respective states. There are 3 states viz. Active, Blocked, Restoring. In Active state, the endpoint is responding as expected, i.e. the response code doesn't indicate a service failure. The endpoint will be moved to Blocked state based on response failures which indicate service failure. After a timeout, the endpoint will be moved to Restoring state to check if the endpoint has recovered. In this state, if endpoint responds with response code other than service failure, it will be moved to Active state. Else, it will be moved back to Blocked state.
Labels
<i>service_name, endpoint_state</i>

Service Endpoint Learner Metrics

• sem_num_dns_errors
Type
64-bit Counter
Description
Total number of DNS resolution errors for a given service, labels are service name for DNS resolution and DNS error codes
Labels
<i>service_name, error_code</i>
• sem_num_endpoints
Type

64-bit Gauge

Description

Total number of endpoints learnt from the Kubernetes DNS server for a given service

Labels

service

Service Endpoint Manager Metrics

- `sem_num_endpoint_transactions_sent`

Type

64-bit Counter

Description

Total number of times an endpoint is given to the application for communication with it

Labels

service
- `sem_num_endpoint_transaction_status_received`

Type

64-bit Counter

Description

Total number of transactions for a given endpoint received with the transaction status

Labels

service, status

Table of Contents

- [Summary](#)
- [Unified Load Balancer Endpoint Manager Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Unified Load Balancer Endpoint Manager Metrics	ulb_registration_request_sent	Counter	Total number of times an registration request is sent to an unified load balancer pod
Unified Load Balancer Endpoint Manager Metrics	ulb_registration_response_received	Counter	Total number of times registration response is received from unified load balancer pod
Unified Load Balancer Endpoint Manager Metrics	ulb_deregistration_request_sent	Counter	Total number of times an deregistration request is sent to an unified load balancer pod
Unified Load Balancer Endpoint Manager Metrics	ulb_deregistration_response_received	Counter	Total number of times deregistration response is received from unified load balancer pod
Unified Load Balancer Endpoint Manager Metrics	ulb_heartbeat_request_sent	Counter	Total number of times an heartbeat request is sent to an unified load balancer pod
Unified Load Balancer Endpoint Manager Metrics	ulb_heartbeat_response_received	Counter	Total number of times heartbeat response is received from unified load balancer pod

Unified Load Balancer Endpoint Manager Metrics

- `ulb_registration_request_sent`

Type

64-bit Counter

Description

Total number of times registration request is sent to an unified load balancer pod

Labels

ulbName

- ulb_registration_response_received

Type

64-bit Counter

Description

Total number of times registration response is received from unified load balancer pod

Labels

UlbName, status

- ulb_deregistration_request_sent

Type

64-bit Counter

Description

Total number of times deregistration request is sent to an unified load balancer pod

Labels

ulbName

- ulb_deregistration_response_received

Type

64-bit Counter

Description

Total number of times deregistration response is received from unified load balancer pod

Labels

UlbName, status

- ulb_heartbeat_request_sent

Type

64-bit Counter

Description

Total number of times heartbeat request is sent to an unified load balancer pod

Labels

ulbName

- ulb_heartbeat_response_received

Type

64-bit Counter

Description

Total number of times heartbeat response is received from unified load balancer pod

Labels

UlbName, status

Table of Contents

- [Summary](#)
- [AMF Mobility Management Layer Statistics for Call Control Service](#)
- [Hello World Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
AMF Mobility Management Layer Statistics for Call Control Service	hello_dnn_rattype_active_sessions_app1	Gauge	This is the prometheus builtin help. Keep it short
AMF Mobility Management Layer Statistics for Call Control Service	hello_timer_procedure_duration_app1	Gauge	This is the prometheus builtin help. Keep it short
Hello World Metrics	hello_kafka_callback_hit	Counter	Total number of times kafka subscribe callback hit

AMF Mobility Management Layer Statistics for Call Control Service

- [hello_dnn_rattype_active_sessions_app1](#)

Type

64-bit Gauge

Description

This is used for documentation and can be long, preferably a paragraph

Labels

dnn, ratType

- [hello_timer_procedure_duration_app1](#)

Type

64-bit Gauge

Description

This is used for documentation and can be long, preferably a paragraph

Metrics

Instances	Labels	Description
1	type: default	This is used for documentation and can be long, preferably a paragraph
2	type: create	This is used for documentation and can be long, preferably a paragraph

Hello World Metrics

- [hello_kafka_callback_hit](#)

Type

64-bit Counter

Description

Total number of times kafka subscribe callback hit

Labels

topic_name

Table of Contents

- [Summary](#)

- [AMF Mobility Management Layer Statistics for Call Control Service](#)

Summary

Group	Metric Name	Type	Synopsis
AMF Mobility Management Layer Statistics for Call Control Service	hello_timer_procedure_duration_app2	Histogram	This is the prometheus builtin help. Keep it short
AMF Mobility Management Layer Statistics for Call Control Service	hello_dnn_http_requests_app2	Histogram	This is the prometheus builtin help. Keep it short

AMF Mobility Management Layer Statistics for Call Control Service

- [hello_dnn_http_requests_app2](#)

Type

64-bit Histogram

Description

This is used for documentation and can be long, preferably a paragraph

Labels

dnn

Boundaries

[0, 5, 15, 20, 25, 30]

- [hello_timer_procedure_duration_app2](#)

Type

64-bit Histogram

Description

This is used for documentation and can be long, preferably a paragraph

Metrics

Instances	Labels	Description	Boundaries
1	type: default	This is used for documentation and can be long, preferably a paragraph	[0, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100]
2	type: create	This is used for documentation and can be long, preferably a paragraph	[0, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100]

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- [Summary](#)
- [ETCD Client Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_put_requests	Counter	The total number of ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_requests_sent	Counter	The total number of ETCD Client PUT Requests sent.
ETCD Client Metrics	etcd_put_successes	Counter	The total number of successful ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_failures	Counter	The total number of failed ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_with_lease_requests	Counter	The total number of ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_requests_sent	Counter	The total number of ETCD Client PUT with Lease Requests sent.
ETCD Client Metrics	etcd_put_with_lease_successes	Counter	The total number of successful ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_failures	Counter	The total number of failed ETCD Client PUT with Lease Requests.

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_range_requests	Counter	The total number of ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_requests_sent	Counter	The total number of ETCD Client Range Requests sent.
ETCD Client Metrics	etcd_range_successes	Counter	The total number of successful ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_failures	Counter	The total number of failed ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_delete_requests	Counter	The total number of ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_requests_sent	Counter	The total number of ETCD Client Range Delete Requests sent.
ETCD Client Metrics	etcd_range_delete_successes	Counter	The total number of successful ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_failures	Counter	The total number of failed ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_txn_requests	Counter	The total number of ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_requests_sent	Counter	The total number of ETCD Client Txn Requests sent.
ETCD Client Metrics	etcd_txn_successes	Counter	The total number of successful ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_failures	Counter	The total number of failed ETCD Client Txn Requests.
ETCD Client Metrics	etcd_watch_requests	Counter	The total number of ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_requests_sent	Counter	The total number of ETCD Client Watch Requests sent.
ETCD Client Metrics	etcd_watch_successes	Counter	The total number of successful ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_failures	Counter	The total number of failed ETCD Client Watch Requests.
ETCD Client Metrics	etcdwatcher_requests	Counter	The total number of ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_requests_sent	Counter	The total number of ETCD Watcher Requests sent.
ETCD Client Metrics	etcdwatcher_successes	Counter	The total number of successful ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_failures	Counter	The total number of failed ETCD Watcher Requests.

ETCD Client Metrics

- `etcd_put_requests`

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests.

Labels

keyPrefix

- `etcd_put_requests_sent`

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests sent.

Labels

keyPrefix

- `etcd_put_successes`

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT Requests.

Labels

keyPrefix

- etcd_put_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT Requests.

Labels

keyPrefix, failureReason

- etcd_put_with_lease_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests sent.

Labels

keyPrefix

- etcd_put_with_lease_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT with Lease Requests.

Labels

keyPrefix, failureReason

- etcd_range_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests sent.

Labels

keyPrefix

- etcd_range_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Requests.

Labels

keyPrefix, failureReason

- etcd_range_delete_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests.

Labels

keyPrefix

- etcd_range_delete_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests sent.

Labels

keyPrefix

- etcd_range_delete_successes

Type
64-bit Counter
Description
The total number of successful ETCD Client Range Delete Requests.
Labels
<i>keyPrefix</i>

- etcd_range_delete_failures

Type
64-bit Counter
Description
The total number of failed ETCD Client Range Delete Requests.
Labels
<i>keyPrefix, failureReason</i>

- etcd_txn_requests

Type
64-bit Counter
Description
The total number of ETCD Client Txn Requests.
Labels
<i>keyPrefix</i>

- etcd_txn_requests_sent

Type
64-bit Counter
Description
The total number of ETCD Client Txn Requests sent.
Labels
<i>keyPrefix</i>

- etcd_txn_successes

Type
64-bit Counter
Description
The total number of successful ETCD Client Txn Requests.
Labels
<i>keyPrefix</i>

- etcd_txn_failures

Type
64-bit Counter
Description

The total number of failed ETCD Client Txn Requests.

Labels

keyPrefix, failureReason

• etcd_watch_requests

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests.

Labels

keyPrefix

• etcd_watch_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests sent.

Labels

keyPrefix

• etcd_watch_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Watch Requests.

Labels

keyPrefix

• etcd_watch_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Watch Requests.

Labels

keyPrefix, failureReason

• etcdwatcher_requests

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests.

Labels

keyPrefix

• etcdwatcher_requests_sent

Type
64-bit Counter
Description
The total number of ETCD Watcher Requests sent.
Labels
<i>keyPrefix</i>
• etcdwatcher_successes
Type
64-bit Counter
Description
The total number of successful ETCD Watcher Requests.
Labels
<i>keyPrefix</i>
• etcdwatcher_failures
Type
64-bit Counter
Description
The total number of failed ETCD Watcher Requests.
Labels
<i>keyPrefix, failureReason</i>

Table of Contents

- [Summary](#)
- [HTTP Punting Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_sent	Counter	The total number of HTTP requests successfully punted.
HTTP Punting Metrics	punting_http_requests_failed	Counter	The total number of HTTP requests punting failed.
HTTP Punting Metrics	punting_http_responses_received	Counter	The total number of HTTP responses received for the requests punted.
HTTP Punting Metrics	punting_http_responses_forwarded	Counter	The total number of HTTP responses forwarded after punting transaction completion.
HTTP Punting Metrics	punting_http_requests_aborted	Counter	The total number of HTTP requests aborted.
HTTP Punting Metrics	punting_http_request_duration_seconds	Histogram	Punting operation duration in seconds.

HTTP Punting Metrics

- punting_http_requests_sent
- | |
|----------------|
| Type |
| 64-bit Counter |

Description

The total number of HTTP requests successfully punted.

Labels

message_type

- `punting_http_requests_failed`

Type

64-bit Counter

Description

The total number of HTTP requests for which the punting failed.

Labels

message_type

- `punting_http_responses_received`

Type

64-bit Counter

Description

The total number of HTTP responses received for the requests punted.

Labels

message_type, status_code

- `punting_http_responses_forwarded`

Type

64-bit Counter

Description

The total number of HTTP responses forwarded after punting transaction completion.

Labels

message_type, status_code

- `punting_http_requests_aborted`

Type

64-bit Counter

Description

The total number of HTTP requests aborted.

Labels

message_type

- `punting_http_request_duration_seconds`

Type

64-bit Histogram

Description

The punting transaction duration is measured from the time the HTTP request is sent out to the time the response is received for the punting request.

Labels

message_type

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

Summary

1. [hello-world.alert.rules Alerts](#)

hello-world.alert.rules

Alert Name	Description	Alert Info	Diag
hello_dnn_ratype_registration_request_alert	impact for hello_dnn_ratype_registration_request_alert {{ "{{ \$labels.dnn }}" }}	1. Summary: summarize for hello_dnn_ratype_registration_request_alert	1. N.
		Severity: major	
		Realert Time: 5m	
hello_api_status_http_requests_alert	impact for hello_api_status_http_requests_alert	Expression: rate(hello_dnn_ratype_registration_request_total[1m]) > 100	1. N.
		Severity: major	
		Realert Time: 5m	
hello_api_status_http_requests_alert		Expression: rate(hello_api_status_http_requests_total[1m]) > 100	1. N.
		Severity: major	
		Realert Time: 5m	

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- [Summary](#)
- [Packet Streaming Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Packet Streaming Metrics	num_packet_streaming_attempted	Counter	The total number of attempted streaming of packets
Packet Streaming Metrics	num_packet_streaming_success	Counter	The total number of packets streamed successfully
Packet Streaming Metrics	num_packet_streaming_failed	Counter	The total number of failed streaming of packets
Packet Streaming Metrics	num_packet_streaming_bytes	Counter	The total number of bytes successfully streamed

Packet Streaming Metrics

- num_packet_streaming_attempted
 - Type
 - 64-bit Counter
 - Description
 - The total number of attempted streaming of packets
 - Labels
 - interface_name, server_fqdn, kafka_topic*
- num_packet_streaming_success
 - Type

64-bit Counter

Description

The total number of packets streamed successfully

Labels

interface_name, server_fqdn, kafka_topic

• num_packet_streaming_failed

Type

64-bit Counter

Description

The total number of failed streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

• num_packet_streaming_bytes

Type

64-bit Counter

Description

The total number of bytes successfully streamed

Labels

interface_name, server_fqdn, kafka_topic

Table of Contents

- [Summary](#)
- [AMF Mobility Management Layer Statistics for Call Control Service](#)
- [Session Management Layer Statistics for AMF Call Control Service](#)

Summary

Group	Metric Name	Type	Synopsis
AMF Mobility Management Layer Statistics for Call Control Service	hello_dnn_rattype_registration_request	Counter	This is the prometheus builtin help. Keep it short
AMF Mobility Management Layer Statistics for Call Control Service	hello_slice_registration_request	Counter	This is the prometheus builtin help. Keep it short
AMF Mobility Management Layer Statistics for Call Control Service	hello_api_status_http_requests	Counter	This is the prometheus builtin help. Keep it short
Session Management Layer Statistics for AMF Call Control Service	hello_dnn_rattype_pdu_establishment_request	Counter	This is the prometheus builtin help. Keep it short

AMF Mobility Management Layer Statistics for Call Control Service

• hello_dnn_rattype_registration_request

Type

64-bit Counter

Description

This is used for documentation and can be long, preferably a paragraph

Labels

dnn, ratType

- `hello_slice_registration_request`

Type

64-bit Counter

Description

This is used for documentation and can be long, preferably a paragraph

Labels

slice

- `hello_api_status_http_requests`

Type

64-bit Counter

Description

None

Metrics

Instances	Labels	Description
1	<i>method: POST, statuscode: 200</i>	This is used for documentation and can be long, preferably a paragraph
2	<i>method: POST, statuscode: 400</i>	This is used for documentation and can be long, preferably a paragraph

Session Management Layer Statistics for AMF Call Control Service

- `hello_dnn_rattype_pdu_establishment_request`

Type

64-bit Counter

Description

This is used for documentation and can be long, preferably a paragraph

Labels

dnn, ratType

Table of Contents

- [Summary](#)
- [Circuit Breaker Metrics](#)
- [Service Endpoint Learner Metrics](#)
- [Service Endpoint Manager Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Circuit Breaker Metrics	sem_cb_endpoints	Gauge	Total number of endpoints for a service in respective states
Service Endpoint Learner Metrics	sem_num_dns_errors	Counter	Total number of DNS resolution errors for a given service
Service Endpoint Learner Metrics	sem_num_endpoints	Gauge	Total number of endpoints learnt for a service
Service Endpoint Manager Metrics	sem_num_endpoint_transactions_sent	Counter	Total number of times an endpoint is given to the application for communication with it
Service Endpoint Manager Metrics	sem_num_endpoint_transaction_status_received	Counter	Total number of transactions for a given endpoint received with the transaction status

Circuit Breaker Metrics

- **sem_cb_endpoints**

Type
64-bit Gauge
Description
Total number of endpoints for a service in respective states. There are 3 states viz. Active, Blocked, Restoring. In Active state, the endpoint is responding as expected, i.e. the response code doesn't indicate a service failure. The endpoint will be moved to Blocked state based on response failures which indicate service failure. After a timeout, the endpoint will be moved to Restoring state to check if the endpoint has recovered. In this state, if endpoint responds with response code other than service failure, it will be moved to Active state. Else, it will be moved back to Blocked state.
Labels
service_name, endpoint_state

Service Endpoint Learner Metrics

- **sem_num_dns_errors**

Type
64-bit Counter
Description
Total number of DNS resolution errors for a given service, labels are service name for DNS resolution and DNS error codes
Labels
service_name, error_code

- **sem_num_endpoints**

Type
64-bit Gauge
Description
Total number of endpoints learnt from the Kubernetes DNS server for a given service
Labels
service

Service Endpoint Manager Metrics

- **sem_num_endpoint_transactions_sent**

Type
64-bit Counter
Description
Total number of times an endpoint is given to the application for communication with it
Labels
service

- **sem_num_endpoint_transaction_status_received**

Type
64-bit Counter
Description
Total number of transactions for a given endpoint received with the transaction status
Labels

service, status

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- [Summary](#)
- [IKEv2 message level Statistics](#)
- [IKEv2 payload statistics](#)
- [IKEv2 Notify payload statistics](#)
- [IkeMgr statistics](#)
- [IkeMgr HTTP statistics](#)
- [IkeMgr UDSF statistics](#)
- [IkeMgr Punting statistics](#)
- [IkeMgr Flex Block statistics](#)
- [IkeMgr Tunnel Close statistics](#)
- [IkeMgr miscellaneous statistics](#)

Summary

Group	Metric Name	Type	Synopsis
IKEv2 message level Statistics	ikev2_message_exchange	Counter	Total IKE messages exchanged
IKEv2 payload statistics	ikev2_payload	Counter	Number of IKE payloads
IKEv2 payload statistics	ikev2_payload_decode_error	Counter	Number of IKE payloads with error
IKEv2 Notify payload statistics	ikev2_notify_payload	Counter	Number of IKE notify payloads
IKEv2 Notify payload statistics	ikev2_notify_payload_decode_error	Counter	Number of IKE notify payloads with error
IkeMgr statistics	ikemgr_ike_sa_attempts	Counter	Number of IKE SA connections attempts
IkeMgr statistics	ikemgr_ike_sa_success	Counter	Number of IKE SA connections setup successfully
IkeMgr statistics	ikemgr_ike_sa_failed	Counter	Number of IKE SA connections failed
IkeMgr statistics	ikemgr_ike_sa_failed_half_open_sa_closed	Counter	Number of IKE SA failed due to clean up of Half Open IKE SAs
IkeMgr statistics	ikemgr_ipsec_sa_attempts	Counter	Number of IPsec SA attempts
IkeMgr statistics	ikemgr_ipsec_sa_success	Counter	Number of IPsec SA success
IkeMgr statistics	ikemgr_ipsec_sa_failed	Counter	Number of IPsec SA failed
IkeMgr statistics	ikemgr_ipsec_sa_update_attempts	Counter	Number of IPsec SA update attempts
IkeMgr statistics	ikemgr_ipsec_sa_update_success	Counter	Number of IPsec SA update success
IkeMgr statistics	ikemgr_ipsec_sa_update_failed	Counter	Number of IPsec SA update failed
IkeMgr statistics	ikemgr_ipsec_sa_discarded_temp_sa	Counter	Number of IPsec SA discarded due to temporary SA
IkeMgr statistics	ikemgr_ike_sa_rekey_attempts	Counter	Number of IKE SA rekey attempts
IkeMgr statistics	ikemgr_ike_sa_rekey_success	Counter	Number of IKE SA rekey success
IkeMgr statistics	ikemgr_ike_sa_rekey_fail	Counter	Number of IKE SA rekey fail
IkeMgr statistics	ikemgr_ipsec_sa_rekey_attempts	Counter	Number of IPsec SA rekey attempts
IkeMgr statistics	ikemgr_ipsec_sa_rekey_success	Counter	Number of IPsec SA rekey success
IkeMgr statistics	ikemgr_ipsec_sa_rekey_fail	Counter	Number of IPsec SA rekey fail
IkeMgr statistics	ikev2_eap_attempts	Counter	Number of EAP attempts
IkeMgr statistics	ikev2_eap_success	Counter	Number of EAP success
IkeMgr statistics	ikev2_eap_failure	Counter	Number of EAP fail
IkeMgr statistics	ikev2_dpd_attempts	Counter	Number of DPD attempts
IkeMgr statistics	ikev2_dpd_success	Counter	Number of DPD success
IkeMgr statistics	ikev2_dpd_failure	Counter	Number of DPD fail
IkeMgr statistics	ikemgr_ike_sa_closed	Counter	Number of IKE SA close

Group	Metric Name	Type	Synopsis
IkeMgr statistics	ikemgr_ipsec_sa_closed	Counter	Number of IPsec SA close
IkeMgr statistics	ikemgr_ike_sa_terminated	Counter	Number of IKE SA close due to COOKIE notification
IkeMgr statistics	ikemgr_ike_sa_terminated_cookie	Counter	Number of IKE SA close due to COOKIE notification
IkeMgr statistics	ikemgr_ike_sa_terminated_invalid_ke_payload	Counter	Number of IKE SA close due to INVALID KE payload
IkeMgr statistics	ikev2_active_ipsec_sa	Gauge	Current number of active IPsec SAs
IkeMgr statistics	ikev2_active_ike_sa	Gauge	Current number of active IKEv2 connections
IkeMgr statistics	ikev2_tunnel_setup_duration_msec	Gauge	Time taken to setup an IKE tunnel (msec)
IkeMgr statistics	ikev2_ipsec_sa_setup_duration_msec	Gauge	Time taken to setup a IPsec SA (msec)
IkeMgr statistics	ikev2_ike_half_open_sa	Gauge	Number of half open IKE SA connections
IkeMgr HTTP statistics	ikemgr_http_request	Counter	Number of http request sent or received
IkeMgr HTTP statistics	ikemgr_http_request_send_error	Counter	Number of http requests send error
IkeMgr HTTP statistics	ikemgr_http_response	Counter	Number of http responses received
IkeMgr HTTP statistics	ikemgr_http_response_send_error	Counter	Number of http responses send error
IkeMgr UDSF statistics	ikemgr_udsf_request_send_success	Counter	Number of udsf requests
IkeMgr UDSF statistics	ikemgr_udsf_request_send_error	Counter	Number of udsf requests send error
IkeMgr UDSF statistics	ikemgr_udsf_receive_success_response	Counter	Number of udsf success response received
IkeMgr UDSF statistics	ikemgr_udsf_receive_error_response	Counter	Number of udsf error response received
IkeMgr UDSF statistics	ikemgr_udsf_client_connected	Counter	Number of udsf client connects
IkeMgr UDSF statistics	ikemgr_udsf_client_disconnected	Counter	Number of udsf client disconnects
IkeMgr Punting statistics	ikemgr_punt_request_send_success	Counter	Number of punt requests send successful
IkeMgr Punting statistics	ikemgr_punt_request_send_error	Counter	Number of punt requests send successful error
IkeMgr Punting statistics	ikemgr_punt_error_response	Counter	Number of punt request error response
IkeMgr Punting statistics	ikemgr_punt_error_response_steal_record_lock_send_success	Counter	Number record steal lock sent for punting fail response
IkeMgr Punting statistics	ikemgr_punt_error_response_steal_record_lock_send_failed	Counter	Number record steal lock send failed for punting fail response
IkeMgr Flex Block statistics	flex_block_id_alloc_success	Counter	Number of flex block id alloc success
IkeMgr Flex Block statistics	flex_block_id_alloc_failure	Counter	Number of flex block id alloc failure
IkeMgr Flex Block statistics	flex_block_id_release_success	Counter	Number of flex block id release success
IkeMgr Tunnel Close statistics	ikemgr_tunnel_close	Counter	Number of IKE tunnels closed
IkeMgr Tunnel Close statistics	ikemgr_app_initiated_tunnel_close	Counter	Number of IKE tunnel close initiated by the application
IkeMgr miscellaneous statistics	ikemgr_tunnel_not_found	Counter	Number of tunnel not found errors while processing an operation
IkeMgr miscellaneous statistics	ikemgr_ikesa_config_info_response_parse_error	Counter	Number of IKE SA config info response parse error
IkeMgr miscellaneous statistics	ikemgr_ikesa_config_info_not_received_error	Counter	Number of IKE SA config info not received
IkeMgr miscellaneous statistics	ikemgr_ikesa_internal_ip_address_allocation_error	Counter	Number of IKE SA Internal IP address allocation request failures
IkeMgr miscellaneous statistics	ikemgr_ikesa_nas_ip_address_allocation_error	Counter	Number of IKE SA NAS IP address allocation failures
IkeMgr miscellaneous statistics	ikemgr_ikesa_user_plane_ip_address_allocation_error	Counter	Number of IKE SA USER PLANE IP address allocation failures
IkeMgr miscellaneous statistics	ikemgr_ppe_info_not_received_error	Counter	Number of PPE info not received
IkeMgr miscellaneous statistics	ikemgr_ppe_identifier_not_received_error	Counter	Number of PPE identifier not received

Group	Metric Name	Type	Synopsis
IkeMgr miscellaneous statistics	ikemgr_procedure_timeout	Counter	Number of procedure timeouts
IkeMgr miscellaneous statistics	ikemgr_no_procedure_found	Counter	Number of no procedure found errors
IkeMgr miscellaneous statistics	ikemgr_ppe_pod_direct_url_not_found	Counter	Number of no procedure found errors
IkeMgr miscellaneous statistics	ikemgr_udsf_get_record_local_cache_entry_exists	Counter	Number of UDSF get record responses having local cahce entry in memory
IkeMgr miscellaneous statistics	ikemgr_udsf_record_unlock_and_evict_suppressed	Counter	Number of UDSF unlock and evict record attempts suppressed
IkeMgr miscellaneous statistics	ikemgr_stack_timer_start_success	Counter	Number of stack timer start success
IkeMgr miscellaneous statistics	ikemgr_stack_timer_start_fail	Counter	Number of stack timer start fail
IkeMgr miscellaneous statistics	ikemgr_stack_timer_update_success	Counter	Number of stack timer update success
IkeMgr miscellaneous statistics	ikemgr_stack_timer_update_fail	Counter	Number of stack timer update fail
IkeMgr miscellaneous statistics	ikemgr_stack_timer_stop_success	Counter	Number of stack timer stop success
IkeMgr miscellaneous statistics	ikemgr_stack_timer_stop_fail	Counter	Number of stack timer stop fail
IkeMgr miscellaneous statistics	ikemgr_stack_timer_expired	Counter	Number of stack timer expired
IkeMgr miscellaneous statistics	ikemgr_stack_queued_work_items_count	Gauge	Number of stack work items in queue

IKEv2 message level Statistics

- ikev2_message_exchange

Type

64-bit Counter

Description

Number of IKE messages exchanged

Labels

network_context, ike_local_ip, direction, msg_type, msg_sub_type, ike_message_type

IKEv2 payload statistics

- ikev2_payload

Type

64-bit Counter

Description

Number of IKE payloads

Labels

network_context, ike_local_ip, direction, ike_payload_type

- ikev2_payload_decode_error

Type

64-bit Counter

Description

Number of IKE payloads with error

Labels

network_context, ike_local_ip, direction, ike_payload_type

IKEv2 Notify payload statistics

- ikev2_notify_payload

Type

64-bit Counter

Description

Number of IKE notify payloads

Labels

network_context, ike_local_ip, direction, ike_notify_payload_type

- ikev2_notify_payload_decode_error

Type

64-bit Counter

Description

Number of IKE notify payloads with error

Labels

network_context, ike_local_ip, direction, ike_notify_payload_type

IkeMgr statistics

- ikemgr_ike_sa_attempts

Type

64-bit Counter

Description

Number of IKE SA attempts

Labels

network_context, ike_local_ip, direction

- ikemgr_ike_sa_success

Type

64-bit Counter

Description

Number of IKE SA setup successfully

Labels

network_context, ike_local_ip, direction

- ikemgr_ike_sa_failed

Type

64-bit Counter

Description

Number of IKE SA failed

Labels

network_context, ike_local_ip, direction

- ikemgr_ike_sa_failed_half_open_sa_closed

Type

64-bit Counter

Description

Number of IKE SA failed due to clean up of Half Open IKE SAs

Labels

network_context, ike_local_ip, direction

- ikemgr_ipsec_sa_attempts

Type

64-bit Counter

Description

Number of IPsec SA attempts

Labels

network_context, ike_local_ip, direction

- ikemgr_ipsec_sa_success

Type

64-bit Counter

Description

Number of IPsec SA success

Labels

network_context, ike_local_ip, direction

- ikemgr_ipsec_sa_failed

Type

64-bit Counter

Description

Number of IPsec SA failed

Labels

network_context, ike_local_ip, direction

- ikemgr_ipsec_sa_update_attempts

Type

64-bit Counter

Description

Number of IPsec SA update attempts

Labels

network_context, ike_local_ip, direction

- ikemgr_ipsec_sa_update_success

Type

64-bit Counter

Description

Number of IPsec SA update success

Labels

network_context, ike_local_ip, direction

• ikemgr_ipsec_sa_update_failed

Type

64-bit Counter

Description

Number of IPsec SA update failed

Labels

network_context, ike_local_ip, direction

• ikemgr_ipsec_sa_discarded_temp_sa

Type

64-bit Counter

Description

Number of IPsec SA discarded due to temporary SA

Labels

network_context, ike_local_ip, direction

• ikemgr_ike_sa_rekey_attempts

Type

64-bit Counter

Description

Number of IKE SA rekey attempts

Labels

network_context, ike_local_ip, direction

• ikemgr_ike_sa_rekey_success

Type

64-bit Counter

Description

Number of IKE SA rekey success

Labels

network_context, ike_local_ip, direction

• ikemgr_ike_sa_rekey_fail

Type

64-bit Counter

Description

Number of IKE SA rekey fail

Labels

network_context, ike_local_ip, direction

• ikemgr_ipsec_sa_rekey_attempts

Type

64-bit Counter

Description

Number of IPsec SA rekey attempts

Labels

network_context, ike_local_ip, direction

• ikemgr_ipsec_sa_rekey_success

Type

64-bit Counter

Description

Number of IPsec SA rekey success

Labels

network_context, ike_local_ip, direction

• ikemgr_ipsec_sa_rekey_fail

Type

64-bit Counter

Description

Number of IPsec SA rekey fail

Labels

network_context, ike_local_ip, direction

• ikev2_eap_attempts

Type

64-bit Counter

Description

Number of EAP attempts

Labels

network_context, ike_local_ip, direction

• ikev2_eap_success

Type

64-bit Counter

Description

Number of EAP success

Labels

network_context, ike_local_ip, direction

• ikev2_eap_failure

Type

64-bit Counter

Description

Number of EAP fail

Labels

network_context, ike_local_ip, direction

• ikev2_dpd_attempts

Type

64-bit Counter

Description

Number of DPD attempts

Labels

network_context, ike_local_ip, direction

• ikev2_dpd_success

Type

64-bit Counter

Description

Number of DPD success

Labels

network_context, ike_local_ip, direction

• ikev2_dpd_failure

Type

64-bit Counter

Description

Number of DPD fail

Labels

network_context, ike_local_ip, direction

• ikemgr_ike_sa_closed

Type

64-bit Counter

Description

Number of IKE SA close

Labels

network_context, ike_local_ip, direction

• ikemgr_ipsec_sa_closed

Type

64-bit Counter

Description

Number of IPsec SA close

Labels

network_context, ike_local_ip, direction

• ikemgr_ike_sa_terminated

Type

64-bit Counter

Description

Number of IKE SA terminated due to abandoning of SA (like COOKIE sent, sent or received INVALID_KE payload etc...)

Labels

network_context, ike_local_ip, direction

• ikemgr_ike_sa_terminated_cookie

Type

64-bit Counter

Description

Number of IKE SA close due to COOKIE notification

Labels

network_context, ike_local_ip, direction

• ikemgr_ike_sa_terminated_invalid_ke_payload

Type

64-bit Counter

Description

Number of IKE SA close due to INVALID KE payload

Labels

network_context, ike_local_ip, direction

• ikev2_active_ipsec_sa

Type

64-bit Gauge

Description

Current number of active IPsec SAs

Labels

network_context, ike_local_ip

• ikev2_active_ike_sa

Type

64-bit Gauge

Description

Current number of active IKEv2 connections

Labels

network_context, ike_local_ip

• ikev2_tunnel_setup_duration_msec

Type

64-bit Gauge

Description
Time taken to setup an IKE tunnel (msec)
Labels
<i>network_context, ike_local_ip</i>
• ikev2_ipsec_sa_setup_duration_msec
Type
64-bit Gauge
Description
Time taken to setup a IPsec SA (msec)
Labels
<i>network_context, ike_local_ip</i>
• ikev2_ike_half_open_sa
Type
64-bit Gauge
Description
Number of half open IKE SA
Labels
<i>network_context, ike_local_ip</i>

IkeMgr HTTP statistics

• ikemgr_http_request
Type
64-bit Counter
Description
Number of http requests sent or received
Labels
<i>module, http_msg_type, direction</i>
• ikemgr_http_request_send_error
Type
64-bit Counter
Description
Number of http requests send error
Labels
<i>module, http_msg_type, error</i>
• ikemgr_http_response
Type
64-bit Counter
Description
Number of http responses sent or received
Labels

module, http_msg_type, resp_code, direction

- ikemgr_http_response_send_error

Type

64-bit Counter

Description

Number of http responses send error

Labels

module, http_msg_type, resp_code, error

IkeMgr UDSF statistics

- ikemgr_udsf_request_send_success

Type

64-bit Counter

Description

Number of udsf requests

Labels

udsf_msg_type

- ikemgr_udsf_request_send_error

Type

64-bit Counter

Description

Number of udsf requests send error

Labels

udsf_msg_type, error_msg

- ikemgr_udsf_receive_success_response

Type

64-bit Counter

Description

Number of udsf success response received

Labels

udsf_msg_type

- ikemgr_udsf_receive_error_response

Type

64-bit Counter

Description

Number of udsf error response received

Labels

udsf_msg_type, error_msg

- ikemgr_udsf_client_connected

Type

64-bit Counter

Description

Number of udsf client connects

Labels

- ikemgr_udsf_client_disconnected

Type

64-bit Counter

Description

Number of udsf client disconnects

Labels

IkeMgr Punting statistics

- ikemgr_punt_request_send_success

Type

64-bit Counter

Description

Number of punt request send successful

Labels

sr_entry_exists

- ikemgr_punt_request_send_error

Type

64-bit Counter

Description

Number of punt request send error

Labels

sr_entry_exists

- ikemgr_punt_error_response

Type

64-bit Counter

Description

Number of punt request error response

Labels

sr_entry_exists, status_code, resp_code

- ikemgr_punt_error_response_steal_record_lock_send_success

Type

64-bit Counter

Description

Number record steal lock sent for punting fail response

Labels

• ikemgr_punt_error_response_steal_record_lock_send_failed
Type
64-bit Counter
Description
Number record steal lock send failed for punting fail response
Labels

IkeMgr Flex Block statistics

• flex_block_id_alloc_success
Type
64-bit Counter
Description
Number of flex block id alloc success
Labels
<i>flex_block_pool_name</i>
• flex_block_id_alloc_failure
Type
64-bit Counter
Description
Number of flex block id alloc failure
Labels
<i>flex_block_pool_name</i>
• flex_block_id_release_success
Type
64-bit Counter
Description
Number of flex block id release success
Labels
<i>flex_block_pool_name</i>

IkeMgr Tunnel Close statistics

• ikemgr_tunnel_close
Type
64-bit Counter
Description
Number of IKE tunnels closed
Labels
<i>network_context, ike_local_ip, cause, tunnel_establised</i>
• ikemgr_app_initiated_tunnel_close

Type

64-bit Counter

Description

Number of IKE tunnel close initiated by the application

Labels

network_context, ike_local_ip, cause, tunnel_established

IkeMgr miscellaneous statistics

- ikemgr_tunnel_not_found

Type

64-bit Counter

Description

Number of tunnel not found errors while processing an operation

Labels

operation

- ikemgr_ikesa_config_info_response_parse_error

Type

64-bit Counter

Description

Number of IKE SA config info response parse error

Labels

network_context, ike_local_ip, operation

- ikemgr_ikesa_config_info_not_received_error

Type

64-bit Counter

Description

Number of IKE SA config info not received

Labels

network_context, ike_local_ip, operation

- ikemgr_ikesa_internal_ip_address_allocation_error

Type

64-bit Counter

Description

Number of IKE SA Internal IP address allocation request failures

Labels

network_context, ike_local_ip, operation

- ikemgr_ikesa_nas_ip_address_allocation_error

Type

64-bit Counter

Description

Number of IKE SA NAS IP address allocation failures

Labels

network_context, ike_local_ip, operation

- ikemgr_ikesa_user_plane_ip_address_allocation_error

Type

64-bit Counter

Description

Number of IKE SA USER PLANE IP address allocation failures

Labels

network_context, ike_local_ip, operation

- ikemgr_ppe_info_not_received_error

Type

64-bit Counter

Description

Number of PPE info not received

Labels

network_context, ike_local_ip, operation

- ikemgr_ppe_identifier_not_received_error

Type

64-bit Counter

Description

Number of PPE identifier not received

Labels

network_context, ike_local_ip, operation

- ikemgr_procedure_timeout

Type

64-bit Counter

Description

Number of procedure timeouts

Labels

network_context, ike_local_ip

- ikemgr_no_procedure_found

Type

64-bit Counter

Description

Number of no procedure found errors

Labels

network_context, ike_local_ip

- ikemgr_ppe_pod_direct_url_not_found

Type

64-bit Counter

Description

Number of API requests failed to get POD direct URL for PPE

Labels

network_context, ike_local_ip, operation

- ikemgr_udsf_get_record_local_cache_entry_exists

Type

64-bit Counter

Description

Number of UDSF get record responses having local cahce entry in memory

Labels

network_context, ike_local_ip

- ikemgr_udsf_record_unlock_and_evict_suppressed

Type

64-bit Counter

Description

Number of UDSF unlock and evict record attempts suppressed

Labels

network_context, ike_local_ip

- ikemgr_stack_timer_start_success

Type

64-bit Counter

Description

Number of stack timer start success

Labels

timer_type

- ikemgr_stack_timer_start_fail

Type

64-bit Counter

Description

Number of stack timer start fail

Labels

timer_type, error

- ikemgr_stack_timer_update_success

Type

64-bit Counter

Description

Number of stack timer update success

Labels

timer_type

- ikemgr_stack_timer_update_fail

Type

64-bit Counter

Description

Number of stack timer update fail

Labels

timer_type, error

- ikemgr_stack_timer_stop_success

Type

64-bit Counter

Description

Number of stack timer stop success

Labels

timer_type

- ikemgr_stack_timer_stop_fail

Type

64-bit Counter

Description

Number of stack timer stop fail

Labels

timer_type, error

- ikemgr_stack_timer_expired

Type

64-bit Counter

Description

Number of stack timer expired

Labels

timer_type

- ikemgr_stack_queued_work_items_count

Type

64-bit Gauge

Description

Number of stack work items in queue

Labels

Table of Contents

- [Summary](#)
- [ETCD Client Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_put_requests	Counter	The total number of ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_requests_sent	Counter	The total number of ETCD Client PUT Requests sent.
ETCD Client Metrics	etcd_put_successes	Counter	The total number of successful ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_failures	Counter	The total number of failed ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_with_lease_requests	Counter	The total number of ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_requests_sent	Counter	The total number of ETCD Client PUT with Lease Requests sent.
ETCD Client Metrics	etcd_put_with_lease_successes	Counter	The total number of successful ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_failures	Counter	The total number of failed ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_range_requests	Counter	The total number of ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_requests_sent	Counter	The total number of ETCD Client Range Requests sent.
ETCD Client Metrics	etcd_range_successes	Counter	The total number of successful ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_failures	Counter	The total number of failed ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_delete_requests	Counter	The total number of ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_requests_sent	Counter	The total number of ETCD Client Range Delete Requests sent.
ETCD Client Metrics	etcd_range_delete_successes	Counter	The total number of successful ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_failures	Counter	The total number of failed ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_txn_requests	Counter	The total number of ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_requests_sent	Counter	The total number of ETCD Client Txn Requests sent.
ETCD Client Metrics	etcd_txn_successes	Counter	The total number of successful ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_failures	Counter	The total number of failed ETCD Client Txn Requests.
ETCD Client Metrics	etcd_watch_requests	Counter	The total number of ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_requests_sent	Counter	The total number of ETCD Client Watch Requests sent.
ETCD Client Metrics	etcd_watch_successes	Counter	The total number of successful ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_failures	Counter	The total number of failed ETCD Client Watch Requests.
ETCD Client Metrics	etcdwatcher_requests	Counter	The total number of ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_requests_sent	Counter	The total number of ETCD Watcher Requests sent.
ETCD Client Metrics	etcdwatcher_successes	Counter	The total number of successful ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_failures	Counter	The total number of failed ETCD Watcher Requests.

ETCD Client Metrics

- [etcd_put_requests](#)

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests.

Labels

keyPrefix

- [etcd_put_requests_sent](#)

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests sent.

Labels

keyPrefix

- etcd_put_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT Requests.

Labels

keyPrefix

- etcd_put_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT Requests.

Labels

keyPrefix, failureReason

- etcd_put_with_lease_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests sent.

Labels

keyPrefix

- etcd_put_with_lease_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT with Lease Requests.

Labels

keyPrefix, failureReason

• etcd_range_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests.

Labels

keyPrefix

• etcd_range_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests sent.

Labels

keyPrefix

• etcd_range_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Requests.

Labels

keyPrefix

• etcd_range_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Requests.

Labels

keyPrefix, failureReason

• etcd_range_delete_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests.

Labels

keyPrefix

• etcd_range_delete_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests sent.

Labels

keyPrefix

• etcd_range_delete_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Delete Requests.

Labels

keyPrefix

• etcd_range_delete_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Delete Requests.

Labels

keyPrefix, failureReason

• etcd_txn_requests

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests.

Labels

keyPrefix

• etcd_txn_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests sent.

Labels

keyPrefix

• etcd_txn_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Txn Requests.

Labels

keyPrefix

- etcd_txn_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Txn Requests.

Labels

keyPrefix, failureReason

- etcd_watch_requests

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests sent.

Labels

keyPrefix

- etcd_watch_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Watch Requests.

Labels

keyPrefix, failureReason

• etcdwatcher_requests

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests.

Labels

keyPrefix

• etcdwatcher_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests sent.

Labels

keyPrefix

• etcdwatcher_successes

Type

64-bit Counter

Description

The total number of successful ETCD Watcher Requests.

Labels

keyPrefix

• etcdwatcher_failures

Type

64-bit Counter

Description

The total number of failed ETCD Watcher Requests.

Labels

keyPrefix, failureReason

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- [Summary](#)
- [HTTP Punting Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_sent	Counter	The total number of HTTP requests successfully punted.
HTTP Punting Metrics	punting_http_requests_failed	Counter	The total number of HTTP requests punting failed.
HTTP Punting Metrics	punting_http_responses_received	Counter	The total number of HTTP responses received for the requests punted.

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_responses_forwarded	Counter	The total number of HTTP responses forwarded after punting transaction completion.
HTTP Punting Metrics	punting_http_requests_aborted	Counter	The total number of HTTP requests aborted.
HTTP Punting Metrics	punting_http_request_duration_seconds	Histogram	Punting operation duration in seconds.

HTTP Punting Metrics

- [punting_http_requests_sent](#)

Type

64-bit Counter

Description

The total number of HTTP requests successfully punted.

Labels

message_type

- [punting_http_requests_failed](#)

Type

64-bit Counter

Description

The total number of HTTP requests for which the punting failed.

Labels

message_type

- [punting_http_responses_received](#)

Type

64-bit Counter

Description

The total number of HTTP responses received for the requests punted.

Labels

message_type, status_code

- [punting_http_responses_forwarded](#)

Type

64-bit Counter

Description

The total number of HTTP responses forwarded after punting transaction completion.

Labels

message_type, status_code

- [punting_http_requests_aborted](#)

Type

64-bit Counter

Description

The total number of HTTP requests aborted.

Labels

message_type

- `punting_http_request_duration_seconds`

Type

64-bit Histogram

Description

The punting transaction duration is measured from the time the HTTP request is sent out to the time the response is received for the punting request.

Labels

message_type

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

Table of Contents

- [Summary](#)
- [Packet Streaming Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Packet Streaming Metrics	num_packet_streaming_attempted	Counter	The total number of attempted streaming of packets
Packet Streaming Metrics	num_packet_streaming_success	Counter	The total number of packets streamed successfully
Packet Streaming Metrics	num_packet_streaming_failed	Counter	The total number of failed streaming of packets
Packet Streaming Metrics	num_packet_streaming_bytes	Counter	The total number of bytes successfully streamed

Packet Streaming Metrics

- `num_packet_streaming_attempted`

Type

64-bit Counter

Description

The total number of attempted streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- `num_packet_streaming_success`

Type

64-bit Counter

Description

The total number of packets streamed successfully

Labels

interface_name, server_fqdn, kafka_topic

- `num_packet_streaming_failed`

Type

64-bit Counter

Description

The total number of failed streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_bytes

Type

64-bit Counter

Description

The total number of bytes successfully streamed

Labels

interface_name, server_fqdn, kafka_topic

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- [Summary](#)
- [UDSF DBClient Metrics](#)
- [UDSF API latency statistics](#)

Summary

Group	Metric Name	Type	Synopsis
UDSF DBClient Metrics	udsf_db_operation_request	Counter	The total number of udsf dbclient requests received.
UDSF DBClient Metrics	udsf_db_operation_response	Counter	The total number of udsf dbclient responses sent.
UDSF DBClient Metrics	udsf_dbc_total_init_success	Counter	The total number of UDSF INIT successes.
UDSF DBClient Metrics	udsf_dbc_total_init_failure	Counter	The total number of UDSF INIT failures.
UDSF DBClient Metrics	udsf_dbc_total_deinit_success	Counter	The total number of UDSF DEINIT successes.
UDSF DBClient Metrics	udsf_dbc_total_deinit_failure	Counter	The total number of UDSF DEINIT failures.
UDSF DBClient Metrics	udsf_db_transaction_counter	Counter	The total number of CRUD operations.
UDSF API latency statistics	udsf_db_transaction_duration	Histogram	Roundtrip duration taken by transaction in seconds.
UDSF API latency statistics	udsf_db_execution_duration	Histogram	Execution time taken by transaction in microseconds.
UDSF API latency statistics	udsf_hiredis_bufferwrite_duration	Histogram	Duration of time taken in hiredis to write data to socket in seconds

UDSF DBClient Metrics

- udsf_db_operation_request

Type

64-bit Counter

Description

The total number of udsf dbclient requests received from worker. Labels- 'app_name' : Originating thread name, ex:smfcc, 'operation_type' : Type of CRUD operation/API type involved, ex:CREATE_OR_MODIFY_RECORD_ASYNC

Labels

app_name, operation_type

- udsf_db_operation_response

Type

64-bit Counter

Description

The total number of udsf dbclient responses sent. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE 'operation_type' : Type of CRUD operation/API type involved., ex:CREATE_OR_MODIFY_RECORD_ASYNC

Labels

app_name, status, operation_type

- udsf_dbc_total_init_success

Type

64-bit Counter

Description

The total number of times UDSF API init was success. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: SUCCESS

Labels

app_name, status

- udsf_dbc_total_init_failure

Type

64-bit Counter

Description

The total number of times UDSF API init failed. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: FAILURE

Labels

app_name, status

- udsf_dbc_total_deinit_success

Type

64-bit Counter

Description

The total number of times UDSF API Deinit succeeded. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: SUCCESS

Labels

app_name, status

- udsf_dbc_total_deinit_failure

Type

64-bit Counter

Description

The total number of times UDSF API Deinit failed. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: FAILURE

Labels

app_name, status

- udsf_db_transaction_counter

Type

64-bit Counter

Description

The total number of times CRUD operations are incrementing. Labels- 'app_name' : Originating thread name, ex:smfcc, 'crud_op' : The categorical CRUD operation in which this sub-transaction falls into, ex: CREATE_OR_MODIFY

Labels

app_name, crud_op

UDSF API latency statistics

- udsf_db_transaction_duration

Type

64-bit Histogram

Description

Duration observed on total journey/roundtrip taken by a transaction after receiving the request. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE 'operation_type' : Type of CRUD operation/API type involved., ex:CREATE_OR_MODIFY_RECORD_ASYNC, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE, 'crud_op' : The categorical CRUD operation in which this transaction falls into, ex: CREATE_OR_MODIFY

Labels

app_name,operation_type,status,crud_op

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

- udsf_db_execution_duration

Type

64-bit Histogram

Description

Processing time taken by UDSF client. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE 'operation_type' : Type of CRUD operation/API type involved., ex:CREATE_OR_MODIFY_RECORD_ASYNC, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE, 'crud_op' : The categorical CRUD operation in which this transaction falls into, ex: CREATE_OR_MODIFY

Labels

app_name,operation_type,status,crud_op

Boundaries

[1, 2, 4, 8, 16, 32, 64, 128, 256, 384, 512, 768, 1024, 1536, 2048, 4096, 8192, 16384]

- udsf_hiredis_bufferwrite_duration

Type

64-bit Histogram

Description

Duration observed between hiredis buffer append and write to socket. This is fair approximation for time spent before data is written to socket as the buffer threshold accounted for append and write is dynamic and not fixed size. Per socket, the buffer append start-time is recorded along with size intended to transmit and end-time recorded once data greater than recorded size is written on the socket

Labels

app_name

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

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- [Summary](#)
- [DB Client Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
DB Client Metrics	dbclient_http_requests	Counter	HTTP requests.
DB Client Metrics	dbclient_internal	Counter	Internal counters.
DB Client Metrics	dbclient_thread_pool_ops	Gauge	Number of in-flight CRUD operations in the shared thread pool as of this sampling interval.
DB Client Metrics	dbclient_database_request_duration_seconds	Histogram	Database operation durations in seconds.

DB Client Metrics

- dbclient_http_requests

Type

64-bit Counter

Description

HTTP requests from workers and sidecars, organized based on the requested PaaS database type, the logical database within the PaaS database type, HTTP method, and the HTTP response status code.

Labels

db_type, db, method, http_status_code

- dbclient_internal

Type

64-bit Counter

Description

Internal counters organized based on the PaaS database type.

CONFIGURATION:

- configClusterStatus: cluster status requests
- configClusterCreatedOrConfigChanged: configMap created or configuration change
- configClusterConfigRefreshUnchanged: configMap refreshed and unchanged
- configClusterDeleted: configMap database cluster deleted
- configClusterErroneousEvent: configMap erroneous or incomplete or unexpected configuration data
- configErrorInd: incoming error indications from SrClient

TRACING:

- traceSpanStarted: worker initiated requests sent to jaeger tracing
- traceSpanAddErrorTag: jaeger traced events that were flagged with an Error tag
- traceSpanFinished: worker initiated requests that finished jaeger tracing

SOFTLOCK:

- softLockReadWithLockPkRequests: incoming requests using a primary key
- softLockReadWithLockSkRequests: incoming requests using a secondary key
- softLockReadWithLockOK: successfully executed
- softLockReadWithLockRetry: requests requiring a retry at DB Client
- softLockReadWithLockRetryExhausted: failed requests after retry
- softLockReadWithLockRecordNotUpdated: failures to take softLock since already claimant, another worker is claimant, etc
- softLockReadWithLockFailedReadAfterUpdate: softLock succeeded but subsequent read failed (record deleted?)
- softLockReadWithLockFailedRead: failed to find record to read (may not exist)
- softLockReleaseLockPkRequests: incoming requests using a primary key
- softLockReleaseLockSkRequests: incoming requests using a secondary key
- softLockReleaseLockOK: successfully executed
- softLockReleaseLockRecordNotUpdated: failures to release softLock since record already unassigned, another worker is claimant, etc
- softLockReleaseLockFailedRead: failed to find record to read (may not exist)

- countRegularRequests: regular count requests
- countAggregationArrayRequests: array count requests
- countAggregationFnValuesRequests: field-name field-value count requests

ERRORED OPERATIONS:

- resourceAllocFailure: resource allocation failures preventing the execution of an operation
- serverFailure: errors returned from the server
- malformedAppInput: misbehaving application with erroneous input
- overloadedTryLater: cannot accept operation due to thread pool congestion due to in-flight operations
- notConnectedToDatabase: cannot accept operation since not connected to the database

Labels

db_type, internal_counter

- dbclient_thread_pool_ops

Type

64-bit Gauge

Description

Each database operation consumes a thread from the thread pool while the operation is executed.

Labels

db_type

- dbclient_database_request_duration_seconds

Type

64-bit Histogram

Description

The duration is measured from database submission to receipt of the response, organized based on the requested PaaS database type and the CRUD operation (such as CREATE, READ, UPDATE, DELETE, and COUNT).

Labels

db_type, crud_op

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

Table of Contents

- [Summary](#)
- [ETCD Client Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_put_requests	Counter	The total number of ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_requests_sent	Counter	The total number of ETCD Client PUT Requests sent.
ETCD Client Metrics	etcd_put_successes	Counter	The total number of successful ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_failures	Counter	The total number of failed ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_with_lease_requests	Counter	The total number of ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_requests_sent	Counter	The total number of ETCD Client PUT with Lease Requests sent.
ETCD Client Metrics	etcd_put_with_lease_successes	Counter	The total number of successful ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_failures	Counter	The total number of failed ETCD Client PUT with Lease Requests.

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_range_requests	Counter	The total number of ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_requests_sent	Counter	The total number of ETCD Client Range Requests sent.
ETCD Client Metrics	etcd_range_successes	Counter	The total number of successful ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_failures	Counter	The total number of failed ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_delete_requests	Counter	The total number of ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_requests_sent	Counter	The total number of ETCD Client Range Delete Requests sent.
ETCD Client Metrics	etcd_range_delete_successes	Counter	The total number of successful ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_failures	Counter	The total number of failed ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_txn_requests	Counter	The total number of ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_requests_sent	Counter	The total number of ETCD Client Txn Requests sent.
ETCD Client Metrics	etcd_txn_successes	Counter	The total number of successful ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_failures	Counter	The total number of failed ETCD Client Txn Requests.
ETCD Client Metrics	etcd_watch_requests	Counter	The total number of ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_requests_sent	Counter	The total number of ETCD Client Watch Requests sent.
ETCD Client Metrics	etcd_watch_successes	Counter	The total number of successful ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_failures	Counter	The total number of failed ETCD Client Watch Requests.
ETCD Client Metrics	etcdwatcher_requests	Counter	The total number of ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_requests_sent	Counter	The total number of ETCD Watcher Requests sent.
ETCD Client Metrics	etcdwatcher_successes	Counter	The total number of successful ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_failures	Counter	The total number of failed ETCD Watcher Requests.

ETCD Client Metrics

- etcd_put_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests.

Labels

keyPrefix

- etcd_put_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests sent.

Labels

keyPrefix

- etcd_put_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT Requests.

Labels

keyPrefix

- etcd_put_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT Requests.

Labels

keyPrefix, failureReason

- etcd_put_with_lease_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests sent.

Labels

keyPrefix

- etcd_put_with_lease_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT with Lease Requests.

Labels

keyPrefix, failureReason

- etcd_range_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests sent.

Labels

keyPrefix

- etcd_range_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Requests.

Labels

keyPrefix, failureReason

- etcd_range_delete_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests.

Labels

keyPrefix

- etcd_range_delete_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests sent.

Labels

keyPrefix

- etcd_range_delete_successes

Type
64-bit Counter
Description
The total number of successful ETCD Client Range Delete Requests.
Labels
keyPrefix

- etcd_range_delete_failures

Type
64-bit Counter
Description
The total number of failed ETCD Client Range Delete Requests.
Labels
keyPrefix, failureReason

- etcd_txn_requests

Type
64-bit Counter
Description
The total number of ETCD Client Txn Requests.
Labels
keyPrefix

- etcd_txn_requests_sent

Type
64-bit Counter
Description
The total number of ETCD Client Txn Requests sent.
Labels
keyPrefix

- etcd_txn_successes

Type
64-bit Counter
Description
The total number of successful ETCD Client Txn Requests.
Labels
keyPrefix

- etcd_txn_failures

Type
64-bit Counter
Description

The total number of failed ETCD Client Txn Requests.

Labels

keyPrefix, failureReason

• etcd_watch_requests

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests.

Labels

keyPrefix

• etcd_watch_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests sent.

Labels

keyPrefix

• etcd_watch_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Watch Requests.

Labels

keyPrefix

• etcd_watch_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Watch Requests.

Labels

keyPrefix, failureReason

• etcdwatcher_requests

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests.

Labels

keyPrefix

• etcdwatcher_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests sent.

Labels

keyPrefix

• etcdwatcher_successes

Type

64-bit Counter

Description

The total number of successful ETCD Watcher Requests.

Labels

keyPrefix

• etcdwatcher_failures

Type

64-bit Counter

Description

The total number of failed ETCD Watcher Requests.

Labels

keyPrefix, failureReason

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- [Summary](#)
- [FlexBlock and UnaffinedResource API requests](#)

Summary

Group	Metric Name	Type	Synopsis
FlexBlock and UnaffinedResource API requests	flexblock_api_status_http_requests	Counter	FlexBlockManager API requests

FlexBlock and UnaffinedResource API requests

• flexblock_api_status_http_requests

Type

64-bit Counter

Description

None

Metrics

Instances	Labels	Description
1	operrId: declareFlexBlockType, statuscode: 200	declareFlexBlockType success
2	operrId: declareFlexBlockType, statuscode: 400	declareFlexBlockType bad request
3	operrId: declareFlexBlockType, statuscode: 404	declareFlexBlockType not found
4	operrId: declareFlexBlockType, statuscode: 408	declareFlexBlockType time out

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Instances	Labels	Description
5	operId: declareFlexBlockType, statusCode: 500	declareFlexBlockType service unavailable
6	operId: declareResourceType, statusCode: 200	declareResourceType success
7	operId: declareResourceType, statusCode: 400	declareResourceType bad request
8	operId: declareResourceType, statusCode: 404	declareResourceType not found
9	operId: declareResourceType, statusCode: 408	declareResourceType time out
10	operId: declareResourceType, statusCode: 500	declareResourceType service unavailable
11	operId: declareResourcePool, statusCode: 200	declareResourcePool success
12	operId: declareResourcePool, statusCode: 400	declareResourcePool bad request
13	operId: declareResourcePool, statusCode: 404	declareResourcePool not found
14	operId: declareResourcePool, statusCode: 408	declareResourcePool time out
15	operId: declareResourcePool, statusCode: 423	declareResourcePool resource locked
16	operId: declareResourcePool, statusCode: 500	declareResourcePool service unavailable
17	operId: getResourcePool, statusCode: 200	getResourcePool success
18	operId: getResourcePool, statusCode: 404	getResourcePool not found
19	operId: getResourcePool, statusCode: 500	getResourcePool internal error
20	operId: deleteResourcePool, statusCode: 200	deleteResourcePool success
21	operId: deleteResourcePool, statusCode: 404	deleteResourcePool not found
22	operId: deleteResourcePool, statusCode: 423	deleteResourcePool pool locked
23	operId: deleteResourcePool, statusCode: 500	deleteResourcePool DB
24	operId: getResourceChunk, statusCode: 200	getResourceChunk success
25	operId: getResourceChunk, statusCode: 404	getResourceChunk not found
26	operId: getResourceChunk, statusCode: 500	getResourceChunk internal error
27	operId: declareFlexBlockPool, statusCode: 200	declareFlexBlockPool success
28	operId: declareFlexBlockPool, statusCode: 400	declareFlexBlockPool bad request
29	operId: declareFlexBlockPool, statusCode: 404	declareFlexBlockPool not found
30	operId: declareFlexBlockPool, statusCode: 408	declareFlexBlockPool time out
31	operId: declareFlexBlockPool, statusCode: 500	declareFlexBlockPool service unavailable
32	operId: getFlexBlockPool, statusCode: 200	getFlexBlockPool success
33	operId: getFlexBlockPool, statusCode: 404	getFlexBlockPool not found
34	operId: getFlexBlockPool, statusCode: 500	getFlexBlockPool internal error
35	operId: deleteFlexBlockPool, statusCode: 200	deleteFlexBlockPool success
36	operId: deleteFlexBlockPool, statusCode: 404	deleteFlexBlockPool not found
37	operId: deleteFlexBlockPool, statusCode: 423	deleteFlexBlockPool pool locked
38	operId: deleteFlexBlockPool, statusCode: 500	deleteFlexBlockPool DB
39	operId: getFlexBlock, statusCode: 200	getFlexBlock success
40	operId: getFlexBlock, statusCode: 404	getFlexBlock not found
41	operId: getFlexBlock, statusCode: 500	getFlexBlock internal error
42	operId: getFlexBlockType, statusCode: 200	getFlexBlockType success
43	operId: getFlexBlockType, statusCode: 404	getFlexBlockType not found
44	operId: getFlexBlockType, statusCode: 500	getFlexBlockType internal error
45	operId: getResourceType, statusCode: 200	getResourceType success
46	operId: getResourceType, statusCode: 404	getResourceType not found
47	operId: getResourceType, statusCode: 500	getResourceType internal error
48	operId: addFlexBlock, statusCode: 200	addFlexBlock success
49	operId: addFlexBlock, statusCode: 400	addFlexBlock bad request

Instances	Labels	Description
50	operId: addFlexBlock, statusCode: 404	addFlexBlock not found
51	operId: addFlexBlock, statusCode: 408	addFlexBlock request timed out
52	operId: addFlexBlock, statusCode: 409	addFlexBlock DB conflict
53	operId: addFlexBlock, statusCode: 423	addFlexBlock FlexBlock pool was locked
54	operId: addFlexBlock, statusCode: 500	addFlexBlock DB
55	operId: addFlexBlock, statusCode: 503	addFlexBlock service unavailable
56	operId: cliamFlexBlock, statusCode: 200	cliamFlexBlock success
57	operId: cliamFlexBlock, statusCode: 404	cliamFlexBlock not found
58	operId: cliamFlexBlock, statusCode: 400	claimFlexBlock bad request
59	operId: cliamFlexBlock, statusCode: 408	claimFlexBlock request timed out
60	operId: claimFlexBlock, statusCode: 409	claimFlexBlock DB conflict
61	operId: claimFlexBlock, statusCode: 423	claimFlexBlock resource locked
62	operId: cliamFlexBlock, statusCode: 500	claimFlexBlock DB
63	operId: cliamFlexBlock, statusCode: 503	claimFlexBlock service unavailable
64	operId: deleteFlexBlock, statusCode: 200	deleteFlexBlock success
65	operId: deleteFlexBlock, statusCode: 400	deleteFlexBlock bad request
66	operId: deleteFlexBlock, statusCode: 408	deleteFlexBlock request timed out
67	operId: deleteFlexBlock, statusCode: 409	deleteFlexBlock DB conflict
68	operId: deleteFlexBlock, statusCode: 500	deleteFlexBlock DB
69	operId: deleteFlexBlock, statusCode: 503	deleteFlexBlock service unavailable
70	operId: addResourceChunks, statusCode: 200	addResourceChunks success
71	operId: addResourceChunks, statusCode: 400	addResourceChunks bad request
72	operId: addResourceChunks, statusCode: 404	addResourceChunks not found
73	operId: addResourceChunks, statusCode: 405	addResourceChunks not allowed
74	operId: addResourceChunks, statusCode: 408	addResourceChunks request timed out
75	operId: addResourceChunks, statusCode: 409	addResourceChunks DB conflict
76	operId: addResourceChunks, statusCode: 423	addResourceChunks pool locked
77	operId: addResourceChunks, statusCode: 500	addResourceChunks DB
78	operId: addResourceChunks, statusCode: 503	addResourceChunks service unavailable
79	operId: deleteResourceChunks, statusCode: 200	deleteResourceChunks success
80	operId: deleteResourceChunks, statusCode: 400	deleteResourceChunks bad request
81	operId: deleteResourceChunks, statusCode: 404	deleteResourceChunks not found
82	operId: deleteResourceChunks, statusCode: 408	deleteResourceChunks request timed out
83	operId: deleteResourceChunks, statusCode: 409	deleteResourceChunks DB conflict
84	operId: deleteResourceChunks, statusCode: 423	deleteResourceChunks pool locked
85	operId: deleteResourceChunks, statusCode: 500	deleteResourceChunks DB
86	operId: auditResourceValues, statusCode: 200	auditResourceValues success
87	operId: auditResourceValues, statusCode: 400	auditResourceValues bad request
88	operId: auditResourceValues, statusCode: 404	auditResourceValues not found
89	operId: auditResourceValues, statusCode: 500	auditResourceValues internal error
90	operId: allocateFromFlexBlock, statusCode: 200	allocateFromFlexBlock success
91	operId: allocateFromFlexBlock, statusCode: 400	allocateFromFlexBlock bad request
92	operId: allocateFromFlexBlock, statusCode: 404	allocateFromFlexBlock not found
93	operId: allocateFromFlexBlock, statusCode: 405	allocateFromFlexBlock not allowed
94	operId: allocateFromFlexBlock, statusCode: 423	allocateFromFlexBlock currently locked

Instances	Labels	Description
95	operId: allocateFromFlexBlock, statusCode: 500	allocateFromFlexBlock internal error
96	operId: allocateFromFlexBlock, statusCode: 503	allocateFromFlexBlock resources exhausted
97	operId: releaseToFlexBlock, statusCode: 200	releaseToFlexBlock success
98	operId: releaseToFlexBlock, statusCode: 400	releaseToFlexBlock bad request
99	operId: releaseToFlexBlock, statusCode: 404	releaseToFlexBlock not found
100	operId: releaseToFlexBlock, statusCode: 405	releaseToFlexBlock not allowed
101	operId: releaseToFlexBlock, statusCode: 500	releaseToFlexBlock internal error
102	operId: getValue, statusCode: 200	getValue success
103	operId: getValue, statusCode: 404	getValue not found
104	operId: getValue, statusCode: 500	getValue internal error
105	operId: getClaimant, statusCode: 200	getClaimant success
106	operId: getClaimant, statusCode: 400	getClaimant not found
107	operId: getClaimant, statusCode: 500	getClaimant not found
108	operId: getClaimant, statusCode: 404	getClaimant bad request
109	operId: allocateFromClaimant, statusCode: 200	allocateFromClaimant success
110	operId: allocateFromClaimant, statusCode: 400	allocateFromClaimant bad request
111	operId: allocateFromClaimant, statusCode: 404	allocateFromClaimant not found
112	operId: allocateFromClaimant, statusCode: 423	allocateFromClaimant currently locked
113	operId: allocateFromClaimant, statusCode: 503	allocateFromClaimant resources exhausted
114	operId: releaseToClaimant, statusCode: 200	releaseToClaimant success
115	operId: releaseToClaimant, statusCode: 400	releaseToClaimant bad request
116	operId: releaseToClaimant, statusCode: 404	releaseToClaimant not found
117	operId: allocateFromUnaffined, statusCode: 200	allocateFromUnaffined success
118	operId: allocateFromUnaffined, statusCode: 400	allocateFromUnaffined bad request
119	operId: allocateFromUnaffined, statusCode: 404	allocateFromUnaffined not found
120	operId: allocateFromUnaffined, statusCode: 405	allocateFromUnaffined not allowed
121	operId: allocateFromUnaffined, statusCode: 408	allocateFromUnaffined request timed out
122	operId: allocateFromUnaffined, statusCode: 409	allocateFromUnaffined db conflict
123	operId: allocateFromUnaffined, statusCode: 423	allocateFromUnaffined currently locked
124	operId: allocateFromUnaffined, statusCode: 500	allocateFromUnaffined DB
125	operId: allocateFromUnaffined, statusCode: 503	allocateFromUnaffined resources exhausted
126	operId: releaseToUnaffined, statusCode: 200	releaseToUnaffined success
127	operId: releaseToUnaffined, statusCode: 400	releaseToUnaffined bad request
128	operId: releaseToUnaffined, statusCode: 404	releaseToUnaffined not found
129	operId: releaseToUnaffined, statusCode: 405	releaseToUnaffined not allowed
130	operId: releaseToUnaffined, statusCode: 408	releaseToUnaffined request timed out
131	operId: releaseToUnaffined, statusCode: 409	releaseToUnaffined db conflict
132	operId: releaseToUnaffined, statusCode: 423	releaseToUnaffined currently locked
133	operId: releaseToUnaffined, statusCode: 500	releaseToUnaffined DB
134	operId: deleteFlexBlockType, statusCode: 200	deleteFlexBlockType success
135	operId: deleteFlexBlockType, statusCode: 404	deleteFlexBlockType not found
136	operId: deleteFlexBlockType, statusCode: 408	deleteFlexBlockType request timed out
137	operId: deleteFlexBlockType, statusCode: 500	deleteFlexBlockType DB
138	operId: deleteResourceType, statusCode: 200	deleteResourceType success
139	operId: deleteResourceType, statusCode: 404	deleteResourceType not found

Instances	Labels	Description
140	<i>operId: deleteResourceType, statusCode: 408</i>	deleteResourceType request timed out
141	<i>operId: deleteResourceType, statusCode: 500</i>	deleteResourceType DB
142	<i>operId: releaseFlexBlock, statusCode: 200</i>	releaseFlexBlock success
143	<i>operId: releaseFlexBlock, statusCode: 400</i>	releaseFlexBlock bad request
144	<i>operId: releaseFlexBlock, statusCode: 404</i>	releaseFlexBlock not found
145	<i>operId: releaseFlexBlock, statusCode: 408</i>	releaseFlexBlock request timed out
146	<i>operId: releaseFlexBlock, statusCode: 409</i>	releaseFlexBlock DB conflict
147	<i>operId: releaseFlexBlock, statusCode: 423</i>	releaseFlexBlock resource locked
148	<i>operId: releaseFlexBlock, statusCode: 500</i>	releaseFlexBlock internal error

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- [Summary](#)
- [UDSF DBClient Metrics](#)
- [UDSF API latency statistics](#)

Summary

Group	Metric Name	Type	Synopsis
UDSF DBClient Metrics	udsf_db_operation_request	Counter	The total number of udsf dbclient requests received.
UDSF DBClient Metrics	udsf_db_operation_response	Counter	The total number of udsf dbclient responses sent.
UDSF DBClient Metrics	udsf_dbc_total_init_success	Counter	The total number of UDSF INIT successes.
UDSF DBClient Metrics	udsf_dbc_total_init_failure	Counter	The total number of UDSF INIT failures.
UDSF DBClient Metrics	udsf_dbc_total_deinit_success	Counter	The total number of UDSF DEINIT successes.
UDSF DBClient Metrics	udsf_dbc_total_deinit_failure	Counter	The total number of UDSF DEINIT failures.
UDSF DBClient Metrics	udsf_db_transaction_counter	Counter	The total number of CRUD operations.
UDSF API latency statistics	udsf_db_transaction_duration	Histogram	Roundtrip duration taken by transaction in seconds.
UDSF API latency statistics	udsf_db_execution_duration	Histogram	Execution time taken by transaction in microseconds.
UDSF API latency statistics	udsf_hiredis_bufferwrite_duration	Histogram	Duration of time taken in hiredis to write data to socket in seconds

UDSF DBClient Metrics

- udsf_db_operation_request
 - Type
 - 64-bit Counter
 - Description
 - The total number of udsf dbclient requests received from worker. Labels- 'app_name' : Originating thread name, ex:smfcc, 'operation_type' : Type of CRUD operation/API type involved, ex:CREATE_OR_MODIFY_RECORD_ASYNC
 - Labels
 - app_name, operation_type***
- udsf_db_operation_response
 - Type
 - 64-bit Counter
 - Description
 - The total number of udsf dbclient responses sent. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE 'operation_type' : Type of CRUD operation/API type involved., ex:CREATE_OR_MODIFY_RECORD_ASYNC
 - Labels

app_name, status, operation_type

- udsf_dbc_total_init_success

Type

64-bit Counter

Description

The total number of times UDSF API init was success. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: SUCCESS

Labels

app_name, status

- udsf_dbc_total_init_failure

Type

64-bit Counter

Description

The total number of times UDSF API init failed. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: FAILURE

Labels

app_name, status

- udsf_dbc_total_deinit_success

Type

64-bit Counter

Description

The total number of times UDSF API Deinit succeeded. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: SUCCESS

Labels

app_name, status

- udsf_dbc_total_deinit_failure

Type

64-bit Counter

Description

The total number of times UDSF API Deinit failed. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: FAILURE

Labels

app_name, status

- udsf_db_transaction_counter

Type

64-bit Counter

Description

The total number of times CRUD operations are incrementing. Labels- 'app_name' : Originating thread name, ex:smfcc, 'crud_op' : The categorical CRUD operation in which this sub-transation falls into, ex: CREATE_OR_MODIFY

Labels

app_name, crud_op

UDSF API latency statistics

- udsf_db_transaction_duration

Type

64-bit Histogram

Description

Duration observed on total journey/roundtrip taken by a transaction after receiving the request. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE 'operation_type' : Type of CRUD operation/API type involved., ex:CREATE_OR_MODIFY_RECORD_ASYNC, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE, 'crud_op' : The categorical CRUD operation in which this transation falls into, ex: CREATE_OR_MODIFY

Labels

app_name,operation_type,status,crud_op

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]
- udsf_db_execution_duration

Type

64-bit Histogram

Description

Processing time taken by UDSF client. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE 'operation_type' : Type of CRUD operation/API type involved., ex:CREATE_OR_MODIFY_RECORD_ASYNC, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE, 'crud_op' : The categorical CRUD operation in which this transation falls into, ex: CREATE_OR_MODIFY

Labels

app_name,operation_type,status,crud_op

Boundaries

[1, 2, 4, 8, 16, 32, 64, 128, 256, 384, 512, 768, 1024, 1536, 2048, 4096, 8192, 16384]
- udsf_hiredis_bufferwrite_duration

Type

64-bit Histogram

Description

Duration observed between hiredis buffer append and write to socket. This is fair approximation for time spent before data is written to socket as the buffer threshold accounted for append and write is dynamic and not fixed size. Per socket, the buffer append start-time is recorded along with size intended to transmit and end-time recorded once data greater than recorded size is written on the socket

Labels

app_name

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

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- [Summary](#)
- [ETCD Client Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
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Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_put_requests	Counter	The total number of ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_requests_sent	Counter	The total number of ETCD Client PUT Requests sent.
ETCD Client Metrics	etcd_put_successes	Counter	The total number of successful ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_failures	Counter	The total number of failed ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_with_lease_requests	Counter	The total number of ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_requests_sent	Counter	The total number of ETCD Client PUT with Lease Requests sent.
ETCD Client Metrics	etcd_put_with_lease_successes	Counter	The total number of successful ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_failures	Counter	The total number of failed ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_range_requests	Counter	The total number of ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_requests_sent	Counter	The total number of ETCD Client Range Requests sent.
ETCD Client Metrics	etcd_range_successes	Counter	The total number of successful ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_failures	Counter	The total number of failed ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_delete_requests	Counter	The total number of ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_requests_sent	Counter	The total number of ETCD Client Range Delete Requests sent.
ETCD Client Metrics	etcd_range_delete_successes	Counter	The total number of successful ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_failures	Counter	The total number of failed ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_txn_requests	Counter	The total number of ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_requests_sent	Counter	The total number of ETCD Client Txn Requests sent.
ETCD Client Metrics	etcd_txn_successes	Counter	The total number of successful ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_failures	Counter	The total number of failed ETCD Client Txn Requests.
ETCD Client Metrics	etcd_watch_requests	Counter	The total number of ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_requests_sent	Counter	The total number of ETCD Client Watch Requests sent.
ETCD Client Metrics	etcd_watch_successes	Counter	The total number of successful ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_failures	Counter	The total number of failed ETCD Client Watch Requests.
ETCD Client Metrics	etcdwatcher_requests	Counter	The total number of ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_requests_sent	Counter	The total number of ETCD Watcher Requests sent.
ETCD Client Metrics	etcdwatcher_successes	Counter	The total number of successful ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_failures	Counter	The total number of failed ETCD Watcher Requests.

ETCD Client Metrics

- etcd_put_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests.

Labels

keyPrefix

- etcd_put_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests sent.

Labels

keyPrefix

- etcd_put_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT Requests.

Labels

keyPrefix

- etcd_put_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT Requests.

Labels

keyPrefix, failureReason

- etcd_put_with_lease_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests sent.

Labels

keyPrefix

- etcd_put_with_lease_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT with Lease Requests.

Labels

keyPrefix, failureReason

- etcd_range_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests sent.

Labels

keyPrefix

- etcd_range_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Requests.

Labels

keyPrefix, failureReason

- etcd_range_delete_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests.

Labels

keyPrefix

- etcd_range_delete_requests_sent
 - Type
 - 64-bit Counter
 - Description
 - The total number of ETCD Client Range Delete Requests sent.
 - Labels
 - keyPrefix**
- etcd_range_delete_successes
 - Type
 - 64-bit Counter
 - Description
 - The total number of successful ETCD Client Range Delete Requests.
 - Labels
 - keyPrefix**
- etcd_range_delete_failures
 - Type
 - 64-bit Counter
 - Description
 - The total number of failed ETCD Client Range Delete Requests.
 - Labels
 - keyPrefix, failureReason**
- etcd_txn_requests
 - Type
 - 64-bit Counter
 - Description
 - The total number of ETCD Client Txn Requests.
 - Labels
 - keyPrefix**
- etcd_txn_requests_sent
 - Type
 - 64-bit Counter
 - Description
 - The total number of ETCD Client Txn Requests sent.
 - Labels
 - keyPrefix**
- etcd_txn_successes
 - Type
 - 64-bit Counter
 - Description

The total number of successful ETCD Client Txn Requests.

Labels

keyPrefix

- etcd_txn_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Txn Requests.

Labels

keyPrefix, failureReason

- etcd_watch_requests

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests sent.

Labels

keyPrefix

- etcd_watch_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Watch Requests.

Labels

keyPrefix, failureReason

- etcdwatcher_requests

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests.

Labels

keyPrefix

• etcdwatcher_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests sent.

Labels

keyPrefix

• etcdwatcher_successes

Type

64-bit Counter

Description

The total number of successful ETCD Watcher Requests.

Labels

keyPrefix

• etcdwatcher_failures

Type

64-bit Counter

Description

The total number of failed ETCD Watcher Requests.

Labels

keyPrefix, failureReason

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- [Summary](#)
- [Interface Manager Status](#)

Summary

Group	Metric Name	Type	Synopsis
Interface Manager Status	intfmgr_external_interfaces	Gauge	Number of external interfaces known to Interface Manager
Interface Manager Status	intfmgr_vlan_subinterfaces	Gauge	Number of vlan subinterfaces in a network context.
Interface Manager Status	intfmgr_ipv4_vlan_subnets	Gauge	Number of IPv4 vlan subnets on vlan_id subinterfaces in net_ctxt.
Interface Manager Status	intfmgr_ipv6_vlan_subnets	Gauge	Number of IPv6 vlan subnets on vlan_id subinterfaces in net_ctxt.
Interface Manager Status	intfmgr_ipv4_loopback_subnets	Gauge	Number of IPv4 loopback subnets in net_ctxt.
Interface Manager Status	intfmgr_ipv6_loopback_subnets	Gauge	Number of IPv6 loopback subnets in net_ctxt.

Interface Manager Status

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- `intfmgr_vlan_subinterfaces`

Type
64-bit Gauge
Description
Number of <code>vlan_id</code> vlan subinterfaces in <code>net_ctxt</code> network context.
Labels
<code>net_ctxt, vlan_id</code>

- `intfmgr_ipv4_vlan_subnets`

Type
64-bit Gauge
Description
Number of IPv4 vlan subnets on the <code>vlan_id</code> vlan subinterfaces in <code>net_ctxt</code> network context.
Labels
<code>net_ctxt, vlan_id</code>

- `intfmgr_ipv6_vlan_subnets`

Type
64-bit Gauge
Description
Number of IPv6 vlan subnets on the <code>vlan_id</code> vlan subinterfaces in <code>net_ctxt</code> network context.
Labels
<code>net_ctxt, vlan_id</code>

- `intfmgr_ipv4_loopback_subnets`

Type
64-bit Gauge
Description
Number of IPv4 loopback subnets in <code>net_ctxt</code> network context.
Labels
<code>net_ctxt</code>

- `intfmgr_ipv6_loopback_subnets`

Type
64-bit Gauge
Description
Number of IPv6 loopback subnets in <code>net_ctxt</code> network context.
Labels
<code>net_ctxt</code>

- `intfmgr_external_interfaces`

Type
64-bit Gauge
Description

Number of external interfaces currently reported to Interface Manager.

Metrics

Instances	Labels	Description
1	*****	Number of external interfaces known to Interface Manager

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- [Summary](#)
- [Istio Default Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Istio Default Metrics	istio_requests_total	Counter	This counter is incremented for every request handled by an Istio proxy
Istio Default Metrics	istio_tcp_sent_bytes_total	Counter	This counter measures the size of total bytes sent during response in case of a TCP connection
Istio Default Metrics	istio_tcp_received_bytes_total	Counter	This counter measures the size of total bytes received during request in case of a TCP connection
Istio Default Metrics	istio_tcp_connections_opened_total	Counter	Thiscounter is incremented for every connection opened
Istio Default Metrics	istio_tcp_connections_closed_total	Counter	This counter is incremented for every connection closed
Istio Default Metrics	process_virtual_memory_bytes	Counter	Virtual memory size in bytes.
Istio Default Metrics	process_resident_memory_bytes	Counter	Resident memory size in bytes.
Istio Default Metrics	go_memstats_heap_sys_bytes	Counter	bytes of memory obtained for the heap from OS.
Istio Default Metrics	go_memstats_heap_alloc_bytes	Counter	shows how much bytes of memory is allocated on the heap for the objects
Istio Default Metrics	go_memstats_alloc_bytes	Counter	shows how much bytes of memory is allocated on the heap for the objects
Istio Default Metrics	go_memstats_heap_inuse_bytes	Counter	shows how many bytes in in-use spans.
Istio Default Metrics	go_memstats_stack_inuse_bytes	Counter	shows how many bytes of memory is used by stack memory spans, which have atleast one object in them.
Istio Default Metrics	container_memory_working_set_bytes	Counter	The amount of working set memory, this includes recently accessed memory, dirty memory, and kernel memory
Istio Default Metrics	go_goroutines	Counter	Number of goroutines that currently exist.
Istio Default Metrics	pilot_xds_cds_reject	Counter	Pilot rejected CDS configs.
Istio Default Metrics	pilot_xds_write_timeout	Counter	Pilot XDS response write timeouts.
Istio Default Metrics	pilot_total_xds_internal_errors	Counter	Total number of internal XDS errors in pilot.
Istio Default Metrics	pilot_total_xds_rejects	Counter	Total number of XDS responses from pilot rejected by proxy.
Istio Default Metrics	pilot_xds_push_context_errors	Counter	Number of errors (timeouts) initiating push context.
Istio Default Metrics	pilot_conflict_inbound_listener	Counter	Number of conflicting inbound listeners.
Istio Default Metrics	pilot_conflict_outbound_listener_tcp_over_current_tcp	Counter	Number of conflicting tcp listeners with current tcp listener.

Group	Metric Name	Type	Synopsis
Istio Default Metrics	pilot_virt_services	Counter	Total virtual services known to pilot.
Istio Default Metrics	pilot_services	Counter	Total services known to pilot.
Istio Default Metrics	pilot_xds	Counter	Number of endpoints connected to this pilot using XDS.
Istio Default Metrics	pilot_k8s_cfg_events	Counter	Events from k8s config.
Istio Default Metrics	container_memory_usage_bytes	Counter	Current memory usage in bytes.
Istio Default Metrics	pilot_xds_eds_reject	Counter	Pilot rejected EDS.
Istio Default Metrics	pilot_xds_pushes	Counter	Pilot build and send errors for lds, rds, cds and eds.
Istio Default Metrics	avg_over_time	Counter	To compute a moving average of some metric.
Istio Default Metrics	mixer_runtime_dispatches_total	Counter	Total number of adapter dispatches handled by Mixer.
Istio Default Metrics	istio_request_duration_seconds	Histogram	This distribution measures the duration of requests
Istio Default Metrics	istio_request_bytes	Histogram	This distribution measures the HTTP request body sizes
Istio Default Metrics	istio_response_bytes	Histogram	This distribution measures HTTP response body sizes

Istio Default Metrics

- `istio_requests_total`

Type

64-bit Counter

Description

This counter is incremented for every request handled by an Istio proxy.

Labels

connection_security_policy, destination_app, destination_principal, destination_service, destination_service_name, destination_service_namespace, destination_version, destination_workload, destination_workload_namespace, permissive_response_code, permissive_response_policyid, reporter, request_protocol, response_code, response_flags, service, source_app, source_principal, source_version, source_workload, source_workload_namespace

- `istio_tcp_sent_bytes_total`

Type

64-bit Counter

Description

This counter measures the size of total bytes sent during response in case of a TCP connection.

Labels

connection_security_policy, destination_app, destination_principal, destination_service, destination_service_name, destination_service_namespace, destination_version, destination_workload, destination_workload_namespace, permissive_response_code, permissive_response_policyid, reporter, request_protocol, response_code, response_flags, service, source_app, source_principal, source_version, source_workload, source_workload_namespace

- `istio_tcp_received_bytes_total`

Type

64-bit Counter

Description

This counter measures the size of total bytes received during request in case of a TCP connection.

Labels

connection_security_policy, destination_app, destination_principal, destination_service, destination_service_name, destination_service_namespace, destination_version, destination_workload, destination_workload_namespace, permissive_response_code, permissive_response_policyid, reporter, request_protocol, response_code, response_flags, service, source_app, source_principal, source_version, source_workload, source_workload_namespace

• istio_tcp_connections_opened_total

Type

64-bit Counter

Description

This counter is incremented for every connection opened.

Labels

connection_security_policy, destination_app, destination_principal, destination_service, destination_service_name, destination_service_namespace, destination_version, destination_workload, destination_workload_namespace, permissive_response_code, permissive_response_policyid, reporter, request_protocol, response_code, response_flags, service, source_app, source_principal, source_version, source_workload, source_workload_namespace

• istio_tcp_connections_closed_total

Type

64-bit Counter

Description

This counter is incremented for every connection closed.

Labels

connection_security_policy, destination_app, destination_principal, destination_service, destination_service_name, destination_service_namespace, destination_version, destination_workload, destination_workload_namespace, permissive_response_code, permissive_response_policyid, reporter, request_protocol, response_code, response_flags, service, source_app, source_principal, source_version, source_workload, source_workload_namespace

• process_virtual_memory_bytes

Type

64-bit Counter

Description

Virtual memory size in bytes.

Labels

• process_resident_memory_bytes

Type

64-bit Counter

Description

Resident memory size in bytes.

Labels

• go_memstats_heap_sys_bytes

Type

64-bit Counter

Description

bytes of memory obtained for the heap from OS.

Labels

- `go_memstats_heap_alloc_bytes`

Type

64-bit Counter

Description

shows how much bytes of memory is allocated on the heap for the objects

Labels

- `go_memstats_alloc_bytes`

Type

64-bit Counter

Description

shows how much bytes of memory is allocated on the heap for the objects

Labels

- `go_memstats_heap_inuse_bytes`

Type

64-bit Counter

Description

shows how many bytes in in-use spans.

Labels

- `go_memstats_stack_inuse_bytes`

Type

64-bit Counter

Description

shows how many bytes of memory is used by stack memory spans, which have atleast one object in them.

Labels

- `container_memory_working_set_bytes`

Type

64-bit Counter

Description

The amount of working set memory, this includes recently accessed memory, dirty memory, and kernel memory

Labels

- `go_goroutines`

Type

64-bit Counter

Description

Number of goroutines that currently exist.

Labels

- **pilot_xds_cds_reject**

Type

64-bit Counter

Description

Pilot rejected CDS configs.

Labels

- **pilot_xds_write_timeout**

Type

64-bit Counter

Description

Pilot XDS response write timeouts.

Labels

- **pilot_total_xds_internal_errors**

Type

64-bit Counter

Description

Total number of internal XDS errors in pilot.

Labels

- **pilot_total_xds_rejects**

Type

64-bit Counter

Description

Total number of XDS responses from pilot rejected by proxy.

Labels

- **pilot_xds_push_context_errors**

Type

64-bit Counter

Description

Number of errors (timeouts) initiating push context.

Labels

- **pilot_conflict_inbound_listener**

Type

64-bit Counter

Description

Number of conflicting inbound listeners.

Labels

- **pilot_conflict_outbound_listener_tcp_over_current_tcp**

Type

64-bit Counter

Description

Number of conflicting tcp listeners with current tcp listener.

Labels

- **pilot_virt_services**

Type

64-bit Counter

Description

Total virtual services known to pilot.

Labels

- **pilot_services**

Type

64-bit Counter

Description

Total services known to pilot.

Labels

- **pilot_xds**

Type

64-bit Counter

Description

Number of endpoints connected to this pilot using XDS.

Labels

- **pilot_k8s_cfg_events**

Type

64-bit Counter

Description

Events from k8s config.

Labels

- **container_memory_usage_bytes**

Type

64-bit Counter

Description

Current memory usage in bytes.

Labels

- pilot_xds_eds_reject

Type

64-bit Counter

Description

Pilot rejected EDS.

Labels

- pilot_xds_pushes

Type

64-bit Counter

Description

Pilot build and send errors for lds, rds, cds and eds.

Labels

- avg_over_time

Type

64-bit Counter

Description

To compute a moving average of some metric.

Labels

- mixer_runtime_dispatches_total

Type

64-bit Counter

Description

Total number of adapter dispatches handled by Mixer.

Labels

- istio_request_duration_seconds

Type

64-bit Histogram

Description

This distribution measures the duration of requests.

Labels

connection_security_policy,destination_app,destination_principal,destination_service,destination_service_name,destination_service_namespace,destination_version,destination_workload,destination_workload_namespace,permissive_response_code,permissive_response_policyid,reporter,request_protocol,response_code,response_flags,service,source_app,source_principal,source_version,source_workload,source_workload_namespace

Boundaries

[0.005, 0.01, 0.025, 0.05, 0.1, 0.25, 0.5, 1, 2.5, 5, 10]

- istio_request_bytes

Type

64-bit Histogram

Description

This distribution measures the HTTP request body sizes.

Labels

connection_security_policy,destination_app,destination_principal,destination_service,destination_service_name,destination_service_namespace,destination_version,destination_workload,destination_workload_namespace,permissive_response_code,permissive_response_policyid,reporter,request_protocol,response_code,response_flags,service,source_app,source_principal,source_version,source_workload,source_workload_namespace

Boundaries

[1, 10, 100, 1000, 10000, 100000, 1000000.0, 10000000.0, 100000000.0]

• [istio_response_bytes](#)

Type

64-bit Histogram

Description

This distribution measures HTTP response body sizes.

Labels

connection_security_policy,destination_app,destination_principal,destination_service,destination_service_name,destination_service_namespace,destination_version,destination_workload,destination_workload_namespace,permissive_response_code,permissive_response_policyid,reporter,request_protocol,response_code,response_flags,service,source_app,source_principal,source_version,source_workload,source_workload_namespace

Boundaries

[1, 10, 100, 1000, 10000, 100000, 1000000.0, 10000000.0, 100000000.0]

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- [Summary](#)
- [Istio Config Agent Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Istio Config Agent Metrics	istio_config_agent_requests	Counter	Counter for the number of requests handled by the agent from the client
Istio Config Agent Metrics	istio_config_agent_responses	Counter	Counter for the responses sent by the agent handlers to the client
Istio Config Agent Metrics	istio_config_agent_request_duration_seconds	Histogram	Histogram for the runtime of agent handler methods

Istio Config Agent Metrics

• [istio_config_agent_requests](#)

Type

64-bit Counter

Description

Counter for HTTP requests handled by the agent server, organized based on the HTTP request method, and the HTTP request path

Labels

method, path

• [istio_config_agent_responses](#)

Type

64-bit Counter

Description

Counter for HTTP responses sent by the agent, organized based on the HTTP request method, HTTP request path, HTTP status code and the App Result (if non 2XX status code)

Labels

method, path, http_status_code, app_result

- istio_config_agent_request_duration_seconds

Type

64-bit Histogram

Description

Histogram for the agent handlers (/register, /unregister{/id}), organized based on the HTTP request method, and HTTP request path

Labels

method,path

Boundaries

[0.005, 0.01, 0.025, 0.05, 0.1, 0.25, 0.5, 1, 2.5, 5, 10]

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- [Summary](#)
- [HTTP Punting Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_sent	Counter	The total number of HTTP requests successfully punted.
HTTP Punting Metrics	punting_http_requests_failed	Counter	The total number of HTTP requests punting failed.
HTTP Punting Metrics	punting_http_responses_received	Counter	The total number of HTTP responses received for the requests punted.
HTTP Punting Metrics	punting_http_responses_forwarded	Counter	The total number of HTTP responses forwarded after punting transaction completion.
HTTP Punting Metrics	punting_http_requests_aborted	Counter	The total number of HTTP requests aborted.
HTTP Punting Metrics	punting_http_request_duration_seconds	Histogram	Punting operation duration in seconds.

HTTP Punting Metrics

- punting_http_requests_sent

Type

64-bit Counter

Description

The total number of HTTP requests successfully punted.

Labels

message_type

- punting_http_requests_failed

Type

64-bit Counter

Description

The total number of HTTP requests for which the punting failed.

Labels

message_type

- `punting_http_responses_received`

Type

64-bit Counter

Description

The total number of HTTP responses received for the requests punted.

Labels

message_type, status_code

- `punting_http_responses_forwarded`

Type

64-bit Counter

Description

The total number of HTTP responses forwarded after punting transaction completion.

Labels

message_type, status_code

- `punting_http_requests_aborted`

Type

64-bit Counter

Description

The total number of HTTP requests aborted.

Labels

message_type

- `punting_http_request_duration_seconds`

Type

64-bit Histogram

Description

The punting transaction duration is measured from the time the HTTP request is sent out to the time the response is received for the punting request.

Labels

message_type

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

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- [Summary](#)
- [Kafka Client Producer Metrics](#)
- [Kafka Admin Operations Metrics](#)
- [Kafka Client Consumer Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Kafka Client Producer Metrics	kafka_client_produce_requests	Counter	Total number of message produce requests received.
Kafka Client Producer Metrics	kafka_client_produce_responses	Counter	Total number of message produce responses sent.
Kafka Client Producer Metrics	kafka_client_message_produce_status	Counter	Status of messages produced.
Kafka Admin Operations Metrics	kafka_admin_topic_unknown_op_request_messages	Counter	Number of unknown admin operation requests received for topics.
Kafka Admin Operations Metrics	kafka_admin_topic_unknown_op_response_messages	Counter	Number of unknown admin operation responses sent.
Kafka Admin Operations Metrics	kafka_admin_topic_create_request_messages	Counter	Total number of topic create request messages received.
Kafka Admin Operations Metrics	kafka_admin_topic_create_response_messages	Counter	Total number of topic create response messages sent.
Kafka Admin Operations Metrics	kafka_admin_topics_create_status	Counter	Status of individual topics created.
Kafka Admin Operations Metrics	kafka_admin_topic_modify_request_messages	Counter	Total number of topic modify request messages received.
Kafka Admin Operations Metrics	kafka_admin_topic_modify_response_messages	Counter	Total number of topic modify response messages sent.
Kafka Admin Operations Metrics	kafka_admin_topics_modify_status	Counter	Status of individual topics modified.
Kafka Client Consumer Metrics	kafka_client_consumer_subscribe_requests	Counter	Total number of consumer subscribe requests received.
Kafka Client Consumer Metrics	kafka_client_consumer_subscribe_responses	Counter	Total number of consumer subscribe responses sent.
Kafka Client Consumer Metrics	kafka_client_consumer_modification_requests	Counter	Total number of consumer modification requests received.
Kafka Client Consumer Metrics	kafka_client_consumer_modification_responses	Counter	Total number of consumer modification responses sent.
Kafka Client Consumer Metrics	kafka_client_consumer_unsubscribe_requests	Counter	Total number of consumer unsubscribe requests received.
Kafka Client Consumer Metrics	kafka_client_consumer_unsubscribe_responses	Counter	Total number of consumer unsubscribe responses sent.
Kafka Client Consumer Metrics	kafka_client_consume_callbacks	Counter	Total number of consume callbacks invoked
Kafka Client Consumer Metrics	kafka_client_consume_callback_responses	Counter	Total number of responses to consume callback invocations.
Kafka Client Consumer Metrics	kafka_client_commit_requests	Counter	Total number of commit requests received.
Kafka Client Consumer Metrics	kafka_client_commit_responses	Counter	Total number of commit responses sent.
Kafka Client Consumer Metrics	kafka_client_commits	Counter	Total number of commit responses sent.

Kafka Client Producer Metrics

- `kafka_client_produce_requests`

Type

64-bit Counter

Description

Total number of message produce requests received from the worker for a given topic name.

Labels

`topic_name`

- `kafka_client_produce_responses`

Type

64-bit Counter

Description

Total number of message produce responses sent to worker for a given topic name.

Labels

topic_name, response_code

- kafka_client_message_produce_status

Type

64-bit Counter

Description

Once the request is received by the worker, Kafka client enqueues it to be published to the broker and immediately responds back. The status of messages produced is received at a later point. This metric captures the status of the messages produced. This status label will indicate whether the message delivery was successful or not. The partition label will indicate partition ID the message was written to.

Labels

topic_name, status, partition

Kafka Admin Operations Metrics

- kafka_admin_topic_unknown_op_request_messages

Type

64-bit Counter

Description

Number of unknown admin operation requests received for topics.

Labels

-
- kafka_admin_topic_unknown_op_response_messages

Type

64-bit Counter

Description

Number of unknown admin operation responses sent.

Labels

response_code

- kafka_admin_topic_create_request_messages

Type

64-bit Counter

Description

Total number of topic create request messages received.

Labels

-
- kafka_admin_topic_create_response_messages

Type

64-bit Counter

Description

Total number of topic create response messages sent.

Labels

response_code

- kafka_admin_topics_create_status

Type

64-bit Counter

Description

Status of individual topics created.

Labels

topic_name, status

- kafka_admin_topic_modify_request_messages

Type

64-bit Counter

Description

Total number of topic modify request messages received.

Labels

- kafka_admin_topic_modify_response_messages

Type

64-bit Counter

Description

Total number of topic modify response messages sent.

Labels

response_code

- kafka_admin_topics_modify_status

Type

64-bit Counter

Description

Status of individual topics modified.

Labels

topic_name, status

Kafka Client Consumer Metrics

- kafka_client_consumer_subscribe_requests

Type

64-bit Counter

Description

Total number of consumer subscribe requests received.

Labels

topic_name

- kafka_client_consumer_subscribe_responses

Type

64-bit Counter

Description

Total number of consumer subscribe responses sent.

Labels

topic_name, response_code

• kafka_client_consumer_modification_requests

Type

64-bit Counter

Description

Total number of consumer modification requests received.

Labels

topic_name

• kafka_client_consumer_modification_responses

Type

64-bit Counter

Description

Total number of consumer modification responses sent.

Labels

topic_name, response_code

• kafka_client_consumer_unsubscribe_requests

Type

64-bit Counter

Description

Total number of consumer unsubscribe requests received.

Labels

topic_name

• kafka_client_consumer_unsubscribe_responses

Type

64-bit Counter

Description

Total number of consumer unsubscribe responses sent.

Labels

topic_name, response_code

• kafka_client_consume_callbacks

Type

64-bit Counter

Description

Total number of messages consumed from library and corresponding invocation of respective callbacks.

Labels

topic_name

• kafka_client_consume_callback_responses

Type

64-bit Counter

Description

Total number of responses to consume callback invocations.

Labels

topic_name, response_code

- kafka_client_commit_requests

Type

64-bit Counter

Description

Total number of commit requests received.

Labels

-
- kafka_client_commit_responses

Type

64-bit Counter

Description

Total number of commit responses sent.

Labels

response_code

- kafka_client_commits

Type

64-bit Counter

Description

Total number of commit responses sent.

Labels

topic_name

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- [Summary](#)
- [Packet Streaming Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Packet Streaming Metrics	num_packet_streaming_attempted	Counter	The total number of attempted streaming of packets
Packet Streaming Metrics	num_packet_streaming_success	Counter	The total number of packets streamed successfully
Packet Streaming Metrics	num_packet_streaming_failed	Counter	The total number of failed streaming of packets
Packet Streaming Metrics	num_packet_streaming_bytes	Counter	The total number of bytes successfully streamed

Packet Streaming Metrics

- num_packet_streaming_attempted

Type

64-bit Counter

Description

The total number of attempted streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_success

Type

64-bit Counter

Description

The total number of packets streamed successfully

Labels

interface_name, server_fqdn, kafka_topic
- num_packet_streaming_failed

Type

64-bit Counter

Description

The total number of failed streaming of packets

Labels

interface_name, server_fqdn, kafka_topic
- num_packet_streaming_bytes

Type

64-bit Counter

Description

The total number of bytes successfully streamed

Labels

interface_name, server_fqdn, kafka_topic

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- [Summary](#)
- [ChargingCmn statistics](#)

Summary

Group	Metric Name	Type	Synopsis
ChargingCmn statistics	svcmgr_service_available_ind_notification	Counter	How many times, Service Available Indication message received
ChargingCmn statistics	svcmgr_service_available_ind_parse_errs	Counter	How many times, Service Available Indication parse error happened
ChargingCmn statistics	svcmgr_service_available_ind_notification_ack	Counter	How many times, Service Available Indication message is acknowledged
ChargingCmn statistics	svcmgr_service_unavailable_ind_notification	Counter	How many times, Service Unavailable Indication message
ChargingCmn statistics	svcmgr_service_unavailable_ind_parse_errs	Counter	How many times, Service Unavailable Indication parse error happened

Group	Metric Name	Type	Synopsis
ChargingCmn statistics	svcmgr_service_unavailable_ind_notification_ack	Counter	How many times, Service Unavailable Indication message is acknowledged
ChargingCmn statistics	svcmgr_publish_msg_sent	Counter	How many times, Publication message sent
ChargingCmn statistics	svcmgr_publish_msg_success	Counter	How many times, Publication message success response received
ChargingCmn statistics	svcmgr_publish_msg_failure	Counter	How many times, failure received for publication message
ChargingCmn statistics	svcmgr_subscribe_msg_sent	Counter	How many times, Subscribe message sent
ChargingCmn statistics	svcmgr_subscribe_msg_success	Counter	How many times, subscribe message success
ChargingCmn statistics	svcmgr_subscribe_msg_failure	Counter	How many times, failure received for subscribe message
ChargingCmn statistics	svcmgr_unsubscribe_msg_sent	Counter	How many times, unsubscribe message sent
ChargingCmn statistics	svcmgr_unsubscribe_msg_failure	Counter	How many times, failure received for unsubscribe message
ChargingCmn statistics	svcmgr_config_req_sent	Counter	How many times, config request sent
ChargingCmn statistics	svcmgr_config_req_failure	Counter	How many times, failure received for config request
ChargingCmn statistics	svcmgr_config_req_success	Counter	How many times, success received for config request
ChargingCmn statistics	svcmgr_config_register_response	Counter	How many times, config register response received
ChargingCmn statistics	svcmgr_config_register_response_ack	Counter	How many times, config register response is acknowledged
ChargingCmn statistics	svcmgr_config_register_response_parse_err	Counter	How many times, parse error happened for config register response
ChargingCmn statistics	svcmgr_config_update_notification	Counter	How many times, config update notification received
ChargingCmn statistics	svcmgr_config_update_notification_response	Counter	How many times, config update response
ChargingCmn statistics	svcmgr_intf_schema_update_ind_notification	Counter	How many times, interface schema update indication message received
ChargingCmn statistics	svcmgr_intf_schema_update_ind_notification_ack	Counter	How many times, interface schema update indication message is acknowledged
ChargingCmn statistics	svcmgr_intf_schema_update_ind_parse_errs	Counter	How many times, interface schema update indication parse error happened

ChargingCmn statistics

- svcmgr_service_available_ind_notification

Type

64-bit Counter

Description

Number of Times Service Available Indication notification received

Labels

servicename

- svcmgr_service_available_ind_parse_errs

Type

64-bit Counter

Description

Number of Times Service Available Indication parse error happened

Labels

- `svcmgr_service_available_ind_notification_ack`

Type

64-bit Counter

Description

Number of Times Service Available Indication message is acknowledged

Labels

servicename

- `svcmgr_service_unavailable_ind_notification`

Type

64-bit Counter

Description

Number of Times Service Unavailable Indication notification received

Labels

servicename

- `svcmgr_service_unavailable_ind_parse_errs`

Type

64-bit Counter

Description

Number of Times Service Unavailable Indication parse error happened

Labels

- `svcmgr_service_unavailable_ind_notification_ack`

Type

64-bit Counter

Description

Number of Times Service Unavailable Indication message is acknowledged

Labels

servicename

- `svcmgr_publish_msg_sent`

Type

64-bit Counter

Description

Number of Times Publication message sent

Labels

servicename, retry

- `svcmgr_publish_msg_success`

Type

64-bit Counter

Description

Number of Times Publication message success response received

Labels

servicename, causecode, retry

- `svcmgr_publish_msg_failure`

Type

64-bit Counter

Description

Number of Times failure received for publication message

Labels

servicename, causecode, retry

- `svcmgr_subscribe_msg_sent`

Type

64-bit Counter

Description

Number of Times Subscribe message sent

Labels

servicename, retry

- `svcmgr_subscribe_msg_success`

Type

64-bit Counter

Description

Number of Times Send subscribe message success response received

Labels

servicename, causecode, retry

- `svcmgr_subscribe_msg_failure`

Type

64-bit Counter

Description

Number of time failure received for subscribe message

Labels

servicename, causecode, retry

- `svcmgr_unsubscribe_msg_sent`

Type

64-bit Counter

Description

Number of Times unsubscribe message sent

Labels

servicename

- **svcmgr_unsubscribe_msg_failure**

Type

64-bit Counter

Description

Number of Times failure received for unsubscribe message

Labels

servicename

- **svcmgr_config_req_sent**

Type

64-bit Counter

Description

Number of Times config request sent

Labels

retry

- **svcmgr_config_req_failure**

Type

64-bit Counter

Description

Number of Times failure received for config request

Labels

causecode, retry

- **svcmgr_config_req_success**

Type

64-bit Counter

Description

Number of Times success received for config request

Labels

causecode, retry

- **svcmgr_config_register_response**

Type

64-bit Counter

Description

Number of Times config register response received

Labels

statuscode

• svcmgr_config_register_response_ack Type 64-bit Counter Description Number of Times config register response is acknowledged Labels
• svcmgr_config_register_response_parse_err Type 64-bit Counter Description Number of Times parse error happened for config register response Labels
• svcmgr_config_update_notification Type 64-bit Counter Description Number of Times config update notification received Labels
• svcmgr_config_update_notification_response Type 64-bit Counter Description Number of Times config update response sent Labels
• svcmgr_intf_schema_update_ind_notification Type 64-bit Counter Description Number of Times interface schema update indication message received Labels <i>servicename</i>
• svcmgr_intf_schema_update_ind_notification_ack Type 64-bit Counter Description Number of Times interface schema update indication message is acknowledged Labels

<i>servicename</i>
svcmgr_intf_schema_update_ind_parse_errs
Type
64-bit Counter
Description
Number of Times Service interface schema update indication parse error happened
Labels

Table of Contents

- [Summary](#)
- [ETCD Client Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_put_requests	Counter	The total number of ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_requests_sent	Counter	The total number of ETCD Client PUT Requests sent.
ETCD Client Metrics	etcd_put_successes	Counter	The total number of successful ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_failures	Counter	The total number of failed ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_with_lease_requests	Counter	The total number of ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_requests_sent	Counter	The total number of ETCD Client PUT with Lease Requests sent.
ETCD Client Metrics	etcd_put_with_lease_successes	Counter	The total number of successful ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_failures	Counter	The total number of failed ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_range_requests	Counter	The total number of ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_requests_sent	Counter	The total number of ETCD Client Range Requests sent.
ETCD Client Metrics	etcd_range_successes	Counter	The total number of successful ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_failures	Counter	The total number of failed ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_delete_requests	Counter	The total number of ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_requests_sent	Counter	The total number of ETCD Client Range Delete Requests sent.
ETCD Client Metrics	etcd_range_delete_successes	Counter	The total number of successful ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_failures	Counter	The total number of failed ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_txn_requests	Counter	The total number of ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_requests_sent	Counter	The total number of ETCD Client Txn Requests sent.
ETCD Client Metrics	etcd_txn_successes	Counter	The total number of successful ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_failures	Counter	The total number of failed ETCD Client Txn Requests.
ETCD Client Metrics	etcd_watch_requests	Counter	The total number of ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_requests_sent	Counter	The total number of ETCD Client Watch Requests sent.
ETCD Client Metrics	etcd_watch_successes	Counter	The total number of successful ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_failures	Counter	The total number of failed ETCD Client Watch Requests.
ETCD Client Metrics	etcdwatcher_requests	Counter	The total number of ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_requests_sent	Counter	The total number of ETCD Watcher Requests sent.
ETCD Client Metrics	etcdwatcher_successes	Counter	The total number of successful ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_failures	Counter	The total number of failed ETCD Watcher Requests.

ETCD Client Metrics

etcd_put_requests
Type
64-bit Counter
Description
The total number of ETCD Client PUT Requests.
Labels
keyPrefix
etcd_put_requests_sent
Type
64-bit Counter
Description
The total number of ETCD Client PUT Requests sent.
Labels
keyPrefix
etcd_put_successes
Type
64-bit Counter
Description
The total number of successful ETCD Client PUT Requests.
Labels
keyPrefix
etcd_put_failures
Type
64-bit Counter
Description
The total number of failed ETCD Client PUT Requests.
Labels
keyPrefix, failureReason
etcd_put_with_lease_requests
Type
64-bit Counter
Description
The total number of ETCD Client PUT with Lease Requests.
Labels
keyPrefix
etcd_put_with_lease_requests_sent
Type
64-bit Counter
Description

The total number of ETCD Client PUT with Lease Requests sent.

Labels

keyPrefix

- etcd_put_with_lease_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT with Lease Requests.

Labels

keyPrefix, failureReason

- etcd_range_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests sent.

Labels

keyPrefix

- etcd_range_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Requests.

Labels

keyPrefix, failureReason

• etcd_range_delete_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests.

Labels

keyPrefix

• etcd_range_delete_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests sent.

Labels

keyPrefix

• etcd_range_delete_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Delete Requests.

Labels

keyPrefix

• etcd_range_delete_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Delete Requests.

Labels

keyPrefix, failureReason

• etcd_txn_requests

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests.

Labels

keyPrefix

• etcd_txn_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests sent.

Labels

keyPrefix

• etcd_txn_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Txn Requests.

Labels

keyPrefix

• etcd_txn_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Txn Requests.

Labels

keyPrefix, failureReason

• etcd_watch_requests

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests.

Labels

keyPrefix

• etcd_watch_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests sent.

Labels

keyPrefix

• etcd_watch_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Watch Requests.

Labels

keyPrefix, failureReason

- etcdwatcher_requests

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests.

Labels

keyPrefix

- etcdwatcher_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests sent.

Labels

keyPrefix

- etcdwatcher_successes

Type

64-bit Counter

Description

The total number of successful ETCD Watcher Requests.

Labels

keyPrefix

- etcdwatcher_failures

Type

64-bit Counter

Description

The total number of failed ETCD Watcher Requests.

Labels

keyPrefix, failureReason

Table of Contents

- [Summary](#)
- [HTTP Punting Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_sent	Counter	The total number of HTTP requests successfully punted.
HTTP Punting Metrics	punting_http_requests_failed	Counter	The total number of HTTP requests punting failed.
HTTP Punting Metrics	punting_http_responses_received	Counter	The total number of HTTP responses received for the requests punted.
HTTP Punting Metrics	punting_http_responses_forwarded	Counter	The total number of HTTP responses forwarded after punting transaction completion.
HTTP Punting Metrics	punting_http_requests_aborted	Counter	The total number of HTTP requests aborted.
HTTP Punting Metrics	punting_http_request_duration_seconds	Histogram	Punting operation duration in seconds.

HTTP Punting Metrics

- [punting_http_requests_sent](#)

Type

64-bit Counter

Description

The total number of HTTP requests successfully punted.

Labels

message_type
- [punting_http_requests_failed](#)

Type

64-bit Counter

Description

The total number of HTTP requests for which the punting failed.

Labels

message_type
- [punting_http_responses_received](#)

Type

64-bit Counter

Description

The total number of HTTP responses received for the requests punted.

Labels

message_type, status_code
- [punting_http_responses_forwarded](#)

Type

64-bit Counter

Description

The total number of HTTP responses forwarded after punting transaction completion.

Labels

message_type, status_code

- [punting_http_requests_aborted](#)

Type

64-bit Counter

Description

The total number of HTTP requests aborted.

Labels

message_type

- [punting_http_request_duration_seconds](#)

Type

64-bit Histogram

Description

The punting transaction duration is measured from the time the HTTP request is sent out to the time the response is received for the punting request.

Labels

message_type

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

Table of Contents

- [Summary](#)
- [Packet Streaming Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Packet Streaming Metrics	num_packet_streaming_attempted	Counter	The total number of attempted streaming of packets
Packet Streaming Metrics	num_packet_streaming_success	Counter	The total number of packets streamed successfully
Packet Streaming Metrics	num_packet_streaming_failed	Counter	The total number of failed streaming of packets
Packet Streaming Metrics	num_packet_streaming_bytes	Counter	The total number of bytes successfully streamed

Packet Streaming Metrics

- [num_packet_streaming_attempted](#)

Type

64-bit Counter

Description

The total number of attempted streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

• num_packet_streaming_success

Type

64-bit Counter

Description

The total number of packets streamed successfully

Labels

interface_name, server_fqdn, kafka_topic

• num_packet_streaming_failed

Type

64-bit Counter

Description

The total number of failed streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

• num_packet_streaming_bytes

Type

64-bit Counter

Description

The total number of bytes successfully streamed

Labels

interface_name, server_fqdn, kafka_topic

Table of Contents

- [Summary](#)
- [Kafka Replay Agent Interface Stats](#)

Summary

Group	Metric Name	Type	Synopsis
Kafka Replay Agent Interface Stats	kafka_replay_messages_received	Counter	Total number of replay messages received
Kafka Replay Agent Interface Stats	kafka_replay_messages_commit_request	Counter	Total number of commit requests sent
Kafka Replay Agent Interface Stats	kafka_replay_messages_commit_response	Counter	Total number of commit responses received
Kafka Replay Agent Interface Stats	kafka_replay_messages_committed	Counter	Total number of replay messages committed
Kafka Replay Agent Interface Stats	kafka_replay_message_delivery_request	Counter	Total number of replay message requests sent
Kafka Replay Agent Interface Stats	kafka_replay_message_delivery_response	Counter	Total number of replay message responses received
Kafka Replay Agent Interface Stats	kafka_replay_message_miscellaneous_information	Gauge	Total number of message elements are available in miscellaneous map
Kafka Replay Agent Interface Stats	kafka_replay_outstanding_message_information	Gauge	Total number of message elements are available in outstanding message map

Group	Metric Name	Type	Synopsis
Kafka Replay Agent Interface Stats	kafka_replay_replayed_message_information	Gauge	Total number of already replayed message elements are available in replayed message map

Kafka Replay Agent Interface Stats

kafka_replay_messages_received
Type
64-bit Counter
Description
Total number of replay messages received from the Kafka Message Broker which need to be replayed to the destinations
Labels
topicName, destination
kafka_replay_messages_commit_request
Type
64-bit Counter
Description
Total number of commit messages sent by the Kafka Replay Agent to the Kafka Message Broker which it has successfully replayed to the destinations
Labels
topicName
kafka_replay_messages_commit_response
Type
64-bit Counter
Description
Total number of commit responses received by Kafka Replay Agent in response to commit requests sent.
Labels
topicName, status_code
kafka_replay_messages_committed
Type
64-bit Counter
Description
Total number of replay messages for which the Kafka Replay Agent has committed with the Kafka Message Broker after its successful delivery to the destination
Labels
topicName
kafka_replay_message_delivery_request
Type
64-bit Counter
Description
Total number of replay message requests sent to the destination which is either primary or secondary destination
Labels
destination

kafka_replay_message_delivery_response
Type
64-bit Counter
Description
Total number of replay message responses received from the destination which is either primary or secondary destination
Labels
<i>destination, status_code</i>
kafka_replay_message_miscellaneous_information
Type
64-bit Gauge
Description
Number of message info available in miscellaneous map
Labels
kafka_replay_outstanding_message_information
Type
64-bit Gauge
Description
Number of message info available in outstanding message map
Labels
kafka_replay_replayed_message_information
Type
64-bit Gauge
Description
Number of already replayed message info available in replayed message map
Labels

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- [Summary](#)
- [Circuit Breaker Metrics](#)
- [Service Endpoint Learner Metrics](#)
- [Service Endpoint Manager Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Circuit Breaker Metrics	sem_cb_endpoints	Gauge	Total number of endpoints for a service in respective states
Service Endpoint Learner Metrics	sem_num_dns_errors	Counter	Total number of DNS resolution errors for a given service
Service Endpoint Learner Metrics	sem_num_endpoints	Gauge	Total number of endpoints learnt for a service
Service Endpoint Manager Metrics	sem_num_endpoint_transactions_sent	Counter	Total number of times an endpoint is given to the application for communication with it

Group	Metric Name	Type	Synopsis
Service Endpoint Manager Metrics	sem_num_endpoint_transaction_status_received	Counter	Total number of transactions for a given endpoint received with the transaction status

Circuit Breaker Metrics

- [sem_cb_endpoints](#)

Type

64-bit Gauge

Description

Total number of endpoints for a service in respective states. There are 3 states viz. Active, Blocked, Restoring. In Active state, the endpoint is responding as expected, i.e. the response code doesn't indicate a service failure. The endpoint will be moved to Blocked state based on response failures which indicate service failure. After a timeout, the endpoint will be moved to Restoring state to check if the endpoint has recovered. In this state, if endpoint responds with response code other than service failure, it will be moved to Active state. Else, it will be moved back to Blocked state.

Labels

service_name, endpoint_state

Service Endpoint Learner Metrics

- [sem_num_dns_errors](#)

Type

64-bit Counter

Description

Total number of DNS resolution errors for a given service, labels are service name for DNS resolution and DNS error codes

Labels

service_name, error_code

- [sem_num_endpoints](#)

Type

64-bit Gauge

Description

Total number of endpoints learnt from the Kubernetes DNS server for a given service

Labels

service

Service Endpoint Manager Metrics

- [sem_num_endpoint_transactions_sent](#)

Type

64-bit Counter

Description

Total number of times an endpoint is given to the application for communication with it

Labels

service

- [sem_num_endpoint_transaction_status_received](#)

Type

64-bit Counter
Description
Total number of transactions for a given endpoint received with the transaction status
Labels
<i>service, status</i>

Table of Contents

- [Summary](#)
- [kube-prom Stats](#)

Summary

Group	Metric Name	Type	Synopsis
kube-prom Stats	node_cpu_seconds_total	Counter	This is a counter metric that counts the number of seconds the CPU has been running in a particular mode.
kube-prom Stats	node_disk_io_time_seconds_total	Counter	Number of seconds spent doing I/Os
kube-prom Stats	node_disk_io_time_weighted_seconds_total	Counter	Weighted # of seconds spent doing I/Os.
kube-prom Stats	node_network_receive_bytes_total	Counter	The main metrics to look for bandwidth usage.
kube-prom Stats	node_network_transmit_bytes_total	Counter	The main metrics to look for bandwidth usage.
kube-prom Stats	node_network_receive_drop_total	Counter	Network device statistic receive_drop.
kube-prom Stats	node_network_transmit_drop_total	Counter	Network device statistic transmit_drop.
kube-prom Stats	apiserver_request_total	Counter	The metric apiserver_request_total can be used to monitor the requests to the service.
kube-prom Stats	container_cpu_usage_seconds_total	Counter	Cumulative cpu time consumed per cpu in seconds.
kube-prom Stats	up	Counter	Tells if the instance is healthy.
kube-prom Stats	kube_state_metrics_list_total	Counter	Kube-state-metrics is experiencing errors at an elevated rate in list operations.
kube-prom Stats	node_network_receive_errs_total	Counter	Informs about Network device statistic receive_errs.
kube-prom Stats	node_network_transmit_errs_total	Counter	Informs about Network device statistic transmit_errs.
kube-prom Stats	kube_pod_container_status_restarts_total	Counter	The number of container restarts per container.

Group	Metric Name	Type	Synopsis
kube-prom Stats	container_cpu_cfs_throttled_periods_total	Counter	The metrics targets only container cpu over limit.
kube-prom Stats	container_cpu_cfs_periods_total	Counter	Number of elapsed enforcement period intervals
kube-prom Stats	rest_client_requests_total	Counter	Number of HTTP requests, partitioned by status code, method, and host.
kube-prom Stats	prometheus_rule_evaluation_failures_total	Counter	The total number of rule evaluation failures.
kube-prom Stats	prometheus_rule_group_iterations_missed_total	Counter	The total number of rule group evaluations missed due to slow rule group evaluation.
kube-prom Stats	prometheus_operator_node_address_lookup_errors_total	Counter	Number of times a node IP address could not be determined.
kube-prom Stats	apiserver_client_certificate_expiration_seconds_count	Counter	The remaining lifetimes of client certificates.
kube-prom Stats	apiserver_request_duration_seconds_sum	Counter	Latency can be extracted from the histogram buckets
kube-prom Stats	apiserver_request_duration_seconds_count	Counter	Latency can be extracted from the histogram buckets.
kube-prom Stats	kubelet_pleg_relist_duration_seconds_bucket	Counter	This is the duration in seconds for relisting Pods in PLEG
kube-prom Stats	kube_state_metrics_watch_total	Counter	Number of total resource watches in kube-state-metrics
kube-prom Stats	etcd_server_leader_changes_seen_total	Counter	The number of etcd leader changes seen
kube-prom Stats	grpc_server_handled_total	Counter	Total number of RPCs completed on the server, regardless of success or failure.
kube-prom Stats	grpc_server_handling_seconds_bucket	Counter	Contains the counts of RPCs by status and method in respective handling-time buckets
kube-prom Stats	etcd_disk_backend_commit_duration_seconds_bucket	Counter	The total count of observations for a bucket in the histogram: The latency distributions of commit called by backend
kube-prom Stats	etcd_disk_wal_fsync_duration_seconds_bucket	Counter	The total count of observations for a bucket in the histogram: The latency distributions of fsync called by WAL
kube-prom Stats	prometheus_tsdb_head_samples_appended_total	Counter	Total number of appended samples.
kube-prom Stats	prometheus_target_scrapes_sample_duplicate_timestamp_total	Counter	Total number of samples rejected due to duplicate timestamps but different values.
kube-prom Stats	prometheus_target_scrapes_sample_out_of_order_total	Counter	Total number of samples rejected due to not being out of the expected order.

Group	Metric Name	Type	Synopsis
kube-prom Stats	etcd_server_proposals_failed_total	Counter	The total number of failed proposals seen.
kube-prom Stats	kubelet_pod_worker_duration_seconds_bucket	Counter	The total count of observation for a bucket in the histogram: Duration in seconds to sync a single pod. Broken down by operation type: create, update or sync
kube-prom Stats	prometheus_sd_consul_rpc_failures_total	Counter	The number of consul rpc call failures
kube-prom Stats	container_memory_working_set_bytes	Gauge	The amount of working set memory this includes recently accessed memory.
kube-prom Stats	container_memory_rss	Gauge	The amount of swap cache memory.
kube-prom Stats	container_memory_cache	Gauge	Number of bytes of page cache memory.
kube-prom Stats	container_memory_swap	Gauge	Container swap usage in bytes.
kube-prom Stats	kube_pod_status_phase	Gauge	how many Pods are running at any given time.
kube-prom Stats	kube_pod_owner	Gauge	Kubernetes object is the Pod owner.
kube-prom Stats	node_memory_MemFree_bytes	Gauge	the amount of free memory left on the system.
kube-prom Stats	node_memory_Cached_bytes	Gauge	The cache and buffer data
kube-prom Stats	node_memory_Buffers_bytes	Gauge	The cache and buffer data
kube-prom Stats	node_filesystem_size_bytes	Gauge	How many bytes are total for use by normal users.
kube-prom Stats	node_filesystem_avail_bytes	Gauge	How many bytes are free for use by normal users.
kube-prom Stats	node_filesystem_files_free	Gauge	Total number of permitted inodes.
kube-prom Stats	node_timex_offset_seconds	Gauge	Ensure NTP is configured correctly on this host.
kube-prom Stats	node_network_up	Gauge	If operstate is up targets to 1.
kube-prom Stats	kube_pod_container_resource_requests_memory_bytes	Gauge	Bytes of memory requested by a container.
kube-prom Stats	kube_replicaset_owner	Gauge	Information about the ReplicaSet's owner.

Group	Metric Name	Type	Synopsis
kube-prom Stats	node_memory_Slab_bytes	Gauge	Memory information field Slab_bytes.
kube-prom Stats	node_nf_conntrack_entries	Gauge	Number of currently allocated flow entries for connection tracking.
kube-prom Stats	node_nf_conntrack_entries_limit	Gauge	Maximum size of connection tracking table.
kube-prom Stats	kube_deployment_status_observed_generation	Gauge	Deployment was changed at least once for last 5 minutes.
kube-prom Stats	kube_deployment_metadata_generation	Gauge	Sequence number representing the latest generation of a Deployment.
kube-prom Stats	kube_deployment_status_replicas_updated	Gauge	The number of updated replicas per deployment.
kube-prom Stats	kube_deployment_spec_replicas	Gauge	Number of pods specified for a Deployment or DaemonSet.
kube-prom Stats	kube_deployment_status_replicas_available	Gauge	Number of pods currently available for a Deployment or DaemonSet.
kube-prom Stats	kube_statefulset_status_replicas_ready	Gauge	The number of ready replicas per StatefulSet.
kube-prom Stats	kube_statefulset_status_replicas	Gauge	The number of replicas per StatefulSet.
kube-prom Stats	kube_statefulset_status_replicas_updated	Gauge	The number of updated replicas per StatefulSet.
kube-prom Stats	kube_statefulset_status_observed_generation	Gauge	The generation observed by the StatefulSet controller.
kube-prom Stats	kube_statefulset_metadata_generation	Gauge	Sequence number representing a specific generation of the desired state for the StatefulSet.
kube-prom Stats	kube_statefulset_status_current_revision	Gauge	Indicates the version of the StatefulSet used to generate Pods in the sequence [0,currentReplicas).
kube-prom Stats	kube_statefulset_status_update_revision	Gauge	Indicates the version of the StatefulSet used to generate Pods in the sequence [replicas-updatedReplicas,replicas)
kube-prom Stats	kube_daemonset_status_number_ready	Gauge	The number of nodes that should be running daemon pod.
kube-prom Stats	kube_daemonset_status_desired_number_scheduled	Gauge	The number of nodes that should be running the daemon pod.
kube-prom Stats	kube_pod_container_status_waiting_reason	Gauge	Describes the reason the container is currently in waiting state.
kube-prom Stats	kube_daemonset_status_current_number_scheduled	Gauge	The number of nodes running at least one daemon pod and are supposed to.

Group	Metric Name	Type	Synopsis
kube-prom Stats	kube_daemonset_status_number_misscheduled	Gauge	The number of nodes running a daemon pod but are not supposed to.
kube-prom Stats	kube_cronjob_next_schedule_time	Gauge	Next time the cronjob should be scheduled.
kube-prom Stats	kube_job_spec_completions	Gauge	The desired number of successfully finished pods the job should be run with.
kube-prom Stats	kube_job_status_succeeded	Gauge	The number of pods which reached Phase Succeeded.
kube-prom Stats	kube_job_failed	Gauge	The job has failed its execution.
kube-prom Stats	kube_hpa_status_desired_replicas	Gauge	Desired number of replicas of pods managed by this autoscaler.
kube-prom Stats	kube_hpa_status_current_replicas	Gauge	Current number of replicas of pods managed by this autoscaler.
kube-prom Stats	kube_hpa_spec_max_replicas	Gauge	Upper limit for the number of pods that can be set by the autoscaler; cannot be smaller than MinReplicas.
kube-prom Stats	kube_pod_container_resource_requests_cpu_cores	Gauge	The number of requested cpu cores by a container.
kube-prom Stats	kube_node_status_allocatable_cpu_cores	Gauge	The CPU resources of a node that are available for scheduling.
kube-prom Stats	kube_resourcequota	Gauge	Unix creation timestamp
kube-prom Stats	kubelet_volume_stats_available_bytes	Gauge	Number of available bytes in the volume
kube-prom Stats	kubelet_volume_stats_capacity_bytes	Gauge	[ALPHA] Capacity in bytes of the volume
kube-prom Stats	kube_persistentvolume_status_phase	Gauge	The phase indicates if a volume is available, bound to a claim, or released by a claim.
kube-prom Stats	prometheus_remote_storage_highest_timestamp_in_seconds	Gauge	Highest timestamp that has come into the remote storage via the Appender interface.
kube-prom Stats	prometheus_remote_storage_shards_desired	Gauge	The sharding on the throughput and rate
kube-prom Stats	prometheus_remote_storage_shards_max	Gauge	Max shards based on your current configuration.
kube-prom Stats	alertmanager_config_last_reload_successful	Gauge	Whether the last configuration reload attempt was successful.
kube-prom Stats	node_load1	Gauge	This metric is a gauge and is already averaged for you

Group	Metric Name	Type	Synopsis
kube-prom Stats	node_memory_MemAvailable_bytes	Gauge	The metric will show you the current amount of memory free in bytes
kube-prom Stats	node_memory_MemTotal_bytes	Gauge	node_memory_MemTotal_bytes provides us with the amount of memory on the server as a whole
kube-prom Stats	node_vmstat_pgmajfault	Gauge	This metrics describes major page faults
kube-prom Stats	container_memory_usage_bytes	Gauge	Current memory usage in bytes, including all memory regardless of when it was accessed.
kube-prom Stats	kube_pod_info	Gauge	Information about pod
kube-prom Stats	node_filesystem_free_bytes	Gauge	The total size of the filesystem in bytes.
kube-prom Stats	kube_pod_container_status_ready	Gauge	Describes whether the containers readiness check succeeded.
kube-prom Stats	etcd_server_id	Gauge	Server or member ID in hexadecimal format. 1 for 'server_id' label with current ID. Sampled every 60 seconds.
kube-prom Stats	etcd_server_has_leader	Gauge	This metric indicates whether the etcd nodes have a leader or not
kube-prom Stats	node_filesystem_readonly	Gauge	node_filesystem_readonly indicates if the filesystem is readonly
kube-prom Stats	node_timex_sync_status	Gauge	Is clock synchronized to a reliable server (1 = yes, 0 = no)
kube-prom Stats	kubernetes_build_info	Gauge	A metric with a constant '1' value labeled by major, minor, git version, git commit, git tree state, build date, Go version, and compiler from which Kubernetes was built, and platform on which it is running.
kube-prom Stats	kube_node_status_condition	Gauge	Current health status of the node
kube-prom Stats	kube_node_spec_taint	Gauge	The taint of a cluster node
kube-prom Stats	kubelet_node_name	Gauge	The node's name. The count is always 1.
kube-prom Stats	kube_pod_container_resource_limits_memory_bytes	Gauge	The limit on memory to be used by a container in bytes.
kube-prom Stats	prometheus_config_last_reload_successful	Gauge	Whether the last configuration reload attempt was successful.
kube-prom Stats	prometheus_notifications_queue_capacity	Gauge	The capacity of the alert notifications queue.
kube-prom Stats	node_filesystem_files	Gauge	node_filesystem_files is used for the total number of permitted file nodes

Group	Metric Name	Type	Synopsis
kube-prom Stats	kube_node_status_allocatable_memory_bytes	Gauge	The memory resources of a node that are available for scheduling.
kube-prom Stats	aggregator_unavailable_apiservice	Gauge	Gauge of APIServices which are marked as unavailable broken down by APIService name.
kube-prom Stats	kube_node_status_capacity_pods	Gauge	The total pod resources of the node.
kube-prom Stats	kube_statefulset_replicas	Gauge	Number of desired pods for a statefulset
kube-prom Stats	apiserver_request_duration_seconds_bucket	Histogram	Latency can be extracted from the histogram buckets.
kube-prom Stats	scheduler_e2e_scheduling_duration_seconds_bucket	Histogram	E2e scheduling latency in seconds.
kube-prom Stats	scheduler_scheduling_algorithm_duration_seconds_bucket	Histogram	Scheduling algorithm latency in seconds.
kube-prom Stats	scheduler_binding_duration_seconds_bucket	Histogram	Binding latency in seconds.
kube-prom Stats	apiserver_client_certificate_expiration_seconds_bucket	Histogram	Distribution of the remaining lifetime on the certificate used to authenticate a request.
kube-prom Stats	etcd_network_peer_round_trip_time_seconds_bucket	Histogram	Round-Trip-Time histogram between peers

kube-prom Stats

- node_cpu_seconds_total

Type

64-bit Counter

Description

This is a counter metric that counts the number of seconds the CPU has been running in a particular mode. The CPU has several modes such as iowait, idle, user, and system. Because the objective is to count usage, use a query that excludes idle time Labels: container, cpu, endpoint, instance, job, mode, namespace, pod, service

Labels

- node_disk_io_time_seconds_total

Type

64-bit Counter

Description

Number of seconds spent doing I/Os. This field increases so long as field 9 is nonzero. Labels: container, cpu, endpoint, instance, job, mode, namespace, pod, service

Labels

- node_disk_io_time_weighted_seconds_total

Type

64-bit Counter

Description

Weighted # of seconds spent doing I/Os. This field is incremented at each I/O start, I/O completion, I/O merge, or read of these stats by the number of I/Os in progress (field 9) times the number of seconds spent doing I/O since the last update of this field. This can provide an easy measure of both I/O completion time and the backlog that may be accumulating. Labels: container, device, endpoint, instance, job, namespace, pod, service

Labels

- node_network_receive_bytes_total

Type

64-bit Counter

Description

The main metrics you're probably looking for are bandwidth usage which comes from the netdev module, and is covered by node_network_receive_bytes_total and node_network_transmit_bytes_total. As bytes are the base unit, if you want bits per second you can multiply by 8 in your graphs such as rate(node_network_transmit_bytes_total[5m]) * 8. Labels: container, device, endpoint, instance, job, namespace, pod, service

Labels

- node_network_transmit_bytes_total

Type

64-bit Counter

Description

The main metrics you're probably looking for are bandwidth usage which comes from the netdev module, and is covered by node_network_receive_bytes_total and node_network_transmit_bytes_total. As bytes are the base unit, if you want bits per second you can multiply by 8 in your graphs such as rate(node_network_transmit_bytes_total[5m]) * 8. Labels: container, device, endpoint, instance, job, namespace, pod, service

Labels

- node_network_receive_drop_total

Type

64-bit Counter

Description

node_network_receive_drop_total Network device statistic receive_drop. Labels: container, device, endpoint, instance, job, namespace, pod, service

Labels

- node_network_transmit_drop_total

Type

64-bit Counter

Description

node_network_transmit_drop_total Network device statistic transmit_drop. Labels: container, device, endpoint, instance, job, namespace, pod, service

Labels

- apiserver_request_total

Type

64-bit Counter

Description

The metric apiserver_request_total can be used to monitor the requests to the service, from where they are coming, to which service, which action and whether they were successful. Labels: code, component, endpoint, instance, job, namespace, resource, scope, service, subresource, verb, version

Labels

- container_cpu_usage_seconds_total

Type

64-bit Counter

Description

Cumulative cpu time consumed per cpu in seconds. process_cpu_seconds_total: Total user and system CPU time spent in seconds. Labels: container, cpu, endpoint, id, image, instance, job, name, namespace, node, pod, service.

Labels

- up

Type

64-bit Counter

Description

1 if the instance is healthy, i.e. reachable, or 0 if the scrape failed. Labels: container, endpoint, instance, job, namespace, pod, service

Labels

- kube_state_metrics_list_total

Type

64-bit Counter

Description

Kube-state-metrics is experiencing errors at an elevated rate in list operations. This is likely causing it to not be able to expose metrics about Kubernetes objects correctly or at all. Labels: container, endpoint, instance, job, namespace, pod, resource, result, service

Labels

- node_network_receive_errs_total

Type

64-bit Counter

Description

This metrics node_network_receive_errs_total informs about Network device statistic receive_errs. Labels: container, device, endpoint, instance, job, namespace, pod, service

Labels

- node_network_transmit_errs_total

Type

64-bit Counter

Description

This metrics node_network_transmit_errs_total informs about Network device statistic transmit_errs. Labels: container, device, endpoint, instance, job, namespace, pod, service

Labels

- kube_pod_container_status_restarts_total

Type

64-bit Counter

Description

This metrics will tell the number of container restarts per container. Labels: container, endpoint, instance, job, namespace, pod, service, uid

Labels

- container_cpu_cfs_throttled_periods_total

Type

64-bit Counter

Description

The metrics targets only container cpu over limit, rate(container_cpu_cfs_throttled_seconds_total) > 0. Labels: container, endpoint, id, image, instance, job, name, namespace, node, pod, service

Labels

- container_cpu_cfs_periods_total

Type

64-bit Counter

Description

The metrics targets number of elapsed enforcement period intervals Labels: container, endpoint, id, image, instance, job, name, namespace, node, pod, service

Labels

- rest_client_requests_total

Type

64-bit Counter

Description

The metrics rest_client_requests_total tells the number of HTTP requests, partitioned by status code, method, and host. Labels: code, endpoint, host, instance, job, method, namespace, service

Labels

- prometheus_rule_evaluation_failures_total

Type

64-bit Counter

Description

The metrics prometheus_rule_evaluation_failures_total informs about the total number of rule evaluation failures. Labels: container, endpoint, instance, job, namespace, pod, rule_group, service

Labels

- prometheus_rule_group_iterations_missed_total

Type

64-bit Counter

Description

The metrics prometheus_rule_group_iterations_missed_total informs about the total number of rule group evaluations missed due to slow rule group evaluation. Labels: container, endpoint, instance, job, namespace, pod, rule_group, service

Labels

- prometheus_operator_node_address_lookup_errors_total

Type

64-bit Counter

Description

The metrics prometheus_operator_node_address_lookup_errors_total informs about the Number of times a node IP address could not be determined.

Labels

- apiserver_client_certificate_expiration_seconds_count

Type

64-bit Counter

Description

The apiserver_client_certificate_expiration_seconds_count metric measures the remaining lifetimes of client certificates that were used to authenticate to the apiserver. This does not appear to be a useful metric to monitor, as clients' use of short-expiration credentials do not indicate an impending problem. Labels: endpoint, instance, job, namespace, node, service

Labels

- apiserver_request_duration_seconds_sum

Type

64-bit Counter

Description

Latency can be extracted from the apiserver_request_duration_seconds histogram buckets. Labels: component, dry_run, endpoint, group, instance, job, namespace, resource, scope, service, verb, version

Labels

- apiserver_request_duration_seconds_count

Type

64-bit Counter

Description

Latency can be extracted from the apiserver_request_duration_seconds histogram buckets. Labels: component, dry_run, endpoint, group, instance, job, namespace, resource, scope, service, verb, version

Labels

- kubelet_pleg_relist_duration_seconds_bucket

Type

64-bit Counter

Description

kubelet_pleg_relist_duration_seconds_bucket metrics displays the duration in seconds for relisting Pods in PLEG Labels: endpoint, instance, job, le, namespace, node, service

Labels

- kube_state_metrics_watch_total

Type

64-bit Counter

Description

kube_state_metrics_watch_total metric gives the number of total resource watches in kube-state-metrics Labels: container, endpoint, instance, job, namespace, pod, resource, result, service

Labels

• etcd_server_leader_changes_seen_total

Type

64-bit Counter

Description

The metric etcd_server_leader_changes_seen_total tracks the number of leader transitions within the cluster. Labels: endpoint, instance, job, namespace, pod, service

Labels

• grpc_server_handled_total

Type

64-bit Counter

Description

grpc_server_handled_total gives the count of total number of RPCs completed on the server, regardless of success or failure. Labels: endpoint, grpc_code, grpc_method, grpc_service, grpc_type, instance, job, namespace, pod, service

Labels

• grpc_server_handling_seconds_bucket

Type

64-bit Counter

Description

These buckets can be used by Prometheus to estimate SLAs

Labels

• etcd_disk_backend_commit_duration_seconds_bucket

Type

64-bit Counter

Description

The total count of observations for a bucket in the histogram: The latency distributions of commit called by backend. Labels: endpoint, instance, job, le, namespace, pod, service

Labels

• etcd_disk_wal_fsync_duration_seconds_bucket

Type

64-bit Counter

Description

The total count of observations for a bucket in the histogram: The latency distributions of fsync called by WAL Labels: endpoint, instance, job, le, namespace, pod, service

Labels

• prometheus_tsdb_head_samples_appended_total

Type

64-bit Counter

Description

Total number of appended samples. Labels: endpoint, instance, job, namespace, pod, service

Labels

- prometheus_target_scrapes_sample_duplicate_timestamp_total

Type

64-bit Counter

Description

Total number of samples rejected due to duplicate timestamps but different values. Labels: endpoint, instance, job, namespace, pod, service

Labels

- prometheus_target_scrapes_sample_out_of_order_total

Type

64-bit Counter

Description

Total number of samples rejected due to not being out of the expected order. Labels: endpoint, instance, job, namespace, pod, service

Labels

- etcd_server_proposals_failed_total

Type

64-bit Counter

Description

The total number of failed proposals seen. Labels: endpoint, instance, job, namespace, pod, service

Labels

- kubelet_pod_worker_duration_seconds_bucket

Type

64-bit Counter

Description

The total count of observation for a bucket in the histogram: Duration in seconds to sync a single pod. Broken down by operation type: create, update or sync
Labels: endpoint, instance, job, le, namespace, node, operation_type, service

Labels

- prometheus_sd_consul_rpc_failures_total

Type

64-bit Counter

Description

The number of consul rpc call failures Labels: container, endpoint, fedType, fedUniqueName, instance, job, namespace, nodename, pod, service

Labels

• container_memory_working_set_bytes Type 64-bit Gauge Description The amount of working set memory, this includes recently accessed memory, dirty memory, and kernel memory. Labels: container, endpoint, id, image, instance, job, name, namespace, node, pod, service. Labels
• container_memory_rss Type 64-bit Gauge Description The amount of anonymous and swap cache memory. Labels: container, endpoint, id, image, instance, job, name, namespace, node, pod, service. Labels
• container_memory_cache Type 64-bit Gauge Description This metrics is a output of number of bytes of page cache memory. Labels: container, endpoint, id, image, instance, job, name, namespace, node, pod, service. Labels
• container_memory_swap Type 64-bit Gauge Description This metrics gives info of Container swap usage in bytes. Labels: container, endpoint, id, image, instance, job, name, namespace, node, pod, service. Labels
• kube_pod_status_phase Type 64-bit Gauge Description This metrics will allow you to know how many Pods are running at any given time. Labels: container, endpoint, instance, job, namespace, phase, pod, service, uid Labels
• kube_pod_owner Type 64-bit Gauge Description kube_pod_owner tells us which Kubernetes object is the Pod owner. Labels: container, endpoint, instance, job, namespace, owner_is_controller, owner_kind, owner_name, pod, service, uid

Labels

- node_memory_MemFree_bytes

Type

64-bit Gauge

Description

node_memory_MemFree_bytes denotes the amount of free memory left on the system, not including caches and buffers that can be cleared. To see the amount of available memory, including caches and buffers that can be opened up, we would use node_memory_MemAvailable_bytes. And if we wanted to see the cache and buffer data itself, we would use node_memory_Cached_bytes and node_memory_Buffers_bytes. Labels: container, endpoint, instance, job, namespace, pod, service

Labels

- node_memory_Cached_bytes

Type

64-bit Gauge

Description

node_memory_MemFree_bytes denotes the amount of free memory left on the system, not including caches and buffers that can be cleared. To see the amount of available memory, including caches and buffers that can be opened up, we would use node_memory_MemAvailable_bytes. And if we wanted to see the cache and buffer data itself, we would use node_memory_Cached_bytes and node_memory_Buffers_bytes. Labels: container, endpoint, instance, job, namespace, pod, service

Labels

- node_memory_Buffers_bytes

Type

64-bit Gauge

Description

node_memory_MemFree_bytes denotes the amount of free memory left on the system, not including caches and buffers that can be cleared. To see the amount of available memory, including caches and buffers that can be opened up, we would use node_memory_MemAvailable_bytes. And if we wanted to see the cache and buffer data itself, we would use node_memory_Cached_bytes and node_memory_Buffers_bytes. Labels: container, endpoint, instance, job, namespace, pod, service

Labels

- node_filesystem_size_bytes

Type

64-bit Gauge

Description

The matching metrics from the node exporter for 1K-blocks is node_filesystem_size_bytes. The node exporter uses base units, so node_filesystem_size_bytes / 1024 should produce the same value (or going the other way, use df -B1). Available is node_filesystem_avail_bytes, which is how many bytes are free for use by normal users. Labels: container, device, endpoint, fstype, instance, job, mountpoint, namespace, pod, service

Labels

- node_filesystem_avail_bytes

Type

64-bit Gauge

Description

The matching metrics from the node exporter for 1K-blocks is node_filesystem_size_bytes. The node exporter uses base units, so node_filesystem_size_bytes / 1024 should produce the same value (or going the other way, use df -B1). Available is node_filesystem_avail_bytes, which is how many bytes are free for use by

normal users. Labels: container, device, endpoint, fstype, instance, job, mountpoint, namespace, pod, service

Labels

- **node_filesystem_files_free**

Type

64-bit Gauge

Description

Space is only one resource to manage, Unix filesystems are also usually limited in terms of inodes - basically the number of files there are. You can view these with `df -i`. Similarly the relevant metrics are `node_filesystem_files` for the total number of permitted inodes and `node_filesystem_files_free` for how many more you can have. Labels: container, device, endpoint, fstype, instance, job, mountpoint, namespace, pod, service

Labels

- **node_timex_offset_seconds**

Type

64-bit Gauge

Description

Clock on `NodeClockSkewDetected` is out of sync by more than 300s. Ensure NTP is configured correctly on this host. Labels: container, endpoint, instance, job, namespace, pod, service

Labels

- **node_network_up**

Type

64-bit Gauge

Description

The metrics `node_network_up` is 1 if operstate is up. Labels: container, device, endpoint, instance, job, namespace, pod, service

Labels

- **kube_pod_container_resource_requests_memory_bytes**

Type

64-bit Gauge

Description

`kube_pod_container_resource_requests_memory_bytes` Bytes of memory requested by a container. Labels: container, instance, job, namespace, node, pod

Labels

- **kube_replicaset_owner**

Type

64-bit Gauge

Description

`kube_replicaset_owner` Information about the ReplicaSet's owner. Labels: instance, job, namespace, owner_is_controller, owner_kind, owner_name, replicaset

Labels

- **node_memory_Slab_bytes**

Type

64-bit Gauge

Description

node_memory_Slab_bytes Memory information field Slab_bytes. Labels: endpoint, instance, job, namespace, pod, service

Labels

• node_nf_conntrack_entries

Type

64-bit Gauge

Description

This metrics node_nf_conntrack_entries informs for Number of currently allocated flow entries for connection tracking. Labels: endpoint, instance, job, namespace, pod, service

Labels

• node_nf_conntrack_entries_limit

Type

64-bit Gauge

Description

This metrics node_nf_conntrack_entries_limit informs about Maximum size of connection tracking table. Labels: endpoint, instance, job, namespace, pod, service

Labels

• kube_deployment_status_observed_generation

Type

64-bit Gauge

Description

This metrics extractes changes in that deployment was changed at least once for last 5 minutes. Labels: deployment, instance, job, namespace

Labels

• kube_deployment_metadata_generation

Type

64-bit Gauge

Description

Sequence number representing the latest generation of a Deployment. Labels: deployment, instance, job, namespace

Labels

• kube_deployment_status_replicas_updated

Type

64-bit Gauge

Description

The metrics kube_deployment_status_replicas_updated is the number of updated replicas per deployment. Labels: deployment, instance, job, namespace

Labels

• kube_deployment_spec_replicas

Type

64-bit Gauge

Description

Number of pods specified for a Deployment or DaemonSet. Labels: deployment, instance, job, namespace

Labels

- kube_deployment_status_replicas_available

Type

64-bit Gauge

Description

Number of pods currently available for a Deployment or DaemonSet. Labels: deployment, instance, job, namespace

Labels

- kube_statefulset_status_replicas_ready

Type

64-bit Gauge

Description

The metrics kube_statefulset_status_replicas_ready informs the number of ready replicas per StatefulSet. Labels: instance, job, namespace, statefulset

Labels

- kube_statefulset_status_replicas

Type

64-bit Gauge

Description

The metrics kube_statefulset_status_replicas informs about the number of replicas per StatefulSet. Labels: instance, job, namespace, statefulset

Labels

- kube_statefulset_status_replicas_updated

Type

64-bit Gauge

Description

The metrics kube_statefulset_status_replicas_updated informs about the number of updated replicas per StatefulSet. Labels: instance, job, namespace, statefulset

Labels

- kube_statefulset_status_observed_generation

Type

64-bit Gauge

Description

The metrics kube_statefulset_status_observed_generation informs about the generation observed by the StatefulSet controller. Labels: instance, job, namespace, statefulset

Labels

• kube_statefulset_metadata_generation Type 64-bit Gauge Description The metrics kube_statefulset_metadata_generation informs about Sequence number representing a specific generation of the desired state for the StatefulSet. Labels: instance, job, namespace, statefulset Labels
• kube_statefulset_status_current_revision Type 64-bit Gauge Description The metrics kube_statefulset_status_current_revision Indicates the version of the StatefulSet used to generate Pods in the sequence [0,currentReplicas). Labels: instance, job, namespace, revision, statefulset Labels
• kube_statefulset_status_update_revision Type 64-bit Gauge Description The metrics kube_statefulset_status_update_revision Indicates the version of the StatefulSet used to generate Pods in the sequence [replicas-updatedReplicas,replicas) Labels: instance, job, namespace, revision, statefulset Labels
• kube_daemonset_status_number_ready Type 64-bit Gauge Description The number of nodes that should be running the daemon pod and have one or more of the daemon pod running and ready. Labels: daemonset, instance, job, namespace Labels
• kube_daemonset_status_desired_number_scheduled Type 64-bit Gauge Description The metrics informs about the number of nodes that should be running the daemon pod. Labels: daemonset, instance, job, namespace Labels
• kube_pod_container_status_waiting_reason Type 64-bit Gauge Description

The metrics kube_pod_container_status_waiting_reason describes the reason the container is currently in waiting state. Labels: container, instance, job, namespace, pod, reason

Labels

- kube_daemonset_status_current_number_scheduled

Type

64-bit Gauge

Description

The metrics kube_daemonset_status_current_number_scheduled informs about the number of nodes running at least one daemon pod and are supposed to. Labels: daemonset, instance, job, namespace

Labels

- kube_daemonset_status_number_misscheduled

Type

64-bit Gauge

Description

The metrics kube_daemonset_status_number_misscheduled informs the number of nodes running a daemon pod but are not supposed to. Labels: daemonset, instance, job, namespace

Labels

- kube_cronjob_next_schedule_time

Type

64-bit Gauge

Description

The metrics kube_cronjob_next_schedule_time informs the next time the cronjob should be scheduled. The time after lastScheduleTime, or after the cron job's creation time if it's never been scheduled. Use this to determine if the job is delayed. Labels: cronjob, instance, job, namespace

Labels

- kube_job_spec_completions

Type

64-bit Gauge

Description

The metrics kube_job_spec_completions informs about the desired number of successfully finished pods the job should be run with. Labels: instance, job, job_name, namespace

Labels

- kube_job_status_succeeded

Type

64-bit Gauge

Description

The metrics kube_job_status_succeeded informs about the number of pods which reached Phase Succeeded. Labels: instance, job, job_name, namespace

Labels

- kube_job_failed

Type

64-bit Gauge

Description

The metrics kube_job_failed about the job has failed its execution. Labels: condition, instance, job, job_name, namespace

Labels

- kube_hpa_status_desired_replicas

Type

64-bit Gauge

Description

The metrics kube_hpa_status_desired_replicas tells the desired number of replicas of pods managed by this autoscaler. Labels: hpa, instance, job, namespace

Labels

- kube_hpa_status_current_replicas

Type

64-bit Gauge

Description

The metrics kube_hpa_status_current_replicas tells the current number of replicas of pods managed by this autoscaler. Labels: hpa, instance, job, namespace

Labels

- kube_hpa_spec_max_replicas

Type

64-bit Gauge

Description

The metrics kube_hpa_spec_max_replicas tells the Upper limit for the number of pods that can be set by the autoscaler; cannot be smaller than MinReplicas. Labels: hpa, instance, job, namespace

Labels

- kube_pod_container_resource_requests_cpu_cores

Type

64-bit Gauge

Description

The metrics kube_pod_container_resource_requests_cpu_cores informs the number of requested cpu cores by a container. Labels: container, instance, job, namespace, node, pod

Labels

- kube_node_status_allocatable_cpu_cores

Type

64-bit Gauge

Description

The metrics kube_node_status_allocatable_cpu_cores informs the CPU resources of a node that are available for scheduling. Labels: instance, job, node

Labels

• kube_resourcequota Type 64-bit Gauge Description The metrics kube_resourcequota_created is Unix creation timestamp Labels
• kubelet_volume_stats_available_bytes Type 64-bit Gauge Description The metrics kubelet_volume_stats_available_bytes [ALPHA] is the number of available bytes in the volume Labels: endpoint, instance, job, metrics_path, namespace, node, persistentvolumeclaim, service Labels
• kubelet_volume_stats_capacity_bytes Type 64-bit Gauge Description The metrics kubelet_volume_stats_capacity_bytes informs about the [ALPHA] capacity in bytes of the volume Labels: endpoint, instance, job, metrics_path, namespace, node, persistentvolumeclaim, service Labels
• kube_persistentvolume_status_phase Type 64-bit Gauge Description The metrics kube_persistentvolume_status_phase targets the phase indicates if a volume is available, bound to a claim, or released by a claim. Labels: instance, job, persistentvolume, phase Labels
• prometheus_remote_storage_highest_timestamp_in_seconds Type 64-bit Gauge Description The metrics prometheus_remote_storage_highest_timestamp_in_seconds informs the highest timestamp that has come into the remote storage via the Appender interface, in seconds since epoch. Labels: endpoint, instance, job, namespace, pod, service Labels
• prometheus_remote_storage_shards_desired Type 64-bit Gauge Description

The metrics `prometheus_remote_storage_shards_desired` will show us how many shards the sharding calculation wants to run based on the throughput and rate.

Labels

- `prometheus_remote_storage_shards_max`

Type

64-bit Gauge

Description

`prometheus_remote_storage_shards_max` show us the max shards based on your current configuration.

Labels

- `alertmanager_config_last_reload_successful`

Type

64-bit Gauge

Description

The metrics `alertmanager_config_last_reload_successful` informs whether the last configuration reload attempt was successful. Labels: `endpoint`, `instance`, `job`, `namespace`, `pod`, `service`

Labels

- `node_load1`

Type

64-bit Gauge

Description

To inspect the one-minute load average for a single machine we can use a simple query consisting only of a metric name `node_load1` Labels: `endpoint`, `instance`, `job`, `namespace`, `pod`, `service`

Labels

- `node_memory_MemAvailable_bytes`

Type

64-bit Gauge

Description

To see the amount of available memory, including caches and buffers that can be opened up, we would use `node_memory_MemAvailable_bytes` Labels: `endpoint`, `instance`, `job`, `namespace`, `pod`, `service`

Labels

- `node_memory_MemTotal_bytes`

Type

64-bit Gauge

Description

if we have X GB of memory, then this would always be X GB of memory, until we allocate more. While on its own this is not the most helpful number, it helps us calculate the amount of in-use memory Labels: `endpoint`, `instance`, `job`, `namespace`, `pod`, `service`

Labels

- `node_vmstat_pgmajfault`

Type

64-bit Gauge

Description

High rate of major page fault can be used to determine whether the node is under heavy memory pressure Labels: endpoint, instance, job, namespace, pod, service

Labels

• container_memory_usage_bytes

Type

64-bit Gauge

Description

This metric also includes cached (think filesystem cache) items that can be evicted under memory pressure. Labels: container, endpoint, id, image, instance, job, metrics_path, name, namespace, node, pod, service

Labels

• kube_pod_info

Type

64-bit Gauge

Description

This provides detailed information about the pods, such as the pod name, namespace, and the node name that the pod is running on. Labels: created_by_kind, created_by_name, host_ip, instance, job, namespace, node, pod, pod_ip, priority_class, uid

Labels

• node_filesystem_free_bytes

Type

64-bit Gauge

Description

node_filesystem_free_bytes contains the amount of free space, including the space reserved for root, while node_filesystem_avail_bytes contains only the available space for all users. Labels: device, endpoint, fstype, instance, job, mountpoint, namespace, pod, service

Labels

• kube_pod_container_status_ready

Type

64-bit Gauge

Description

Will be 1 if the container is ready, and 0 if it is in a not ready state. Labels: container, instance, job, namespace, pod

Labels

• etcd_server_id

Type

64-bit Gauge

Description

This metric will give 1 for 'server_id' label with current ID
Labels: endpoint, instance, job, namespace, pod, server_id, service

Labels

• etcd_server_has_leader Type 64-bit Gauge Description If the value is 1, then there is a leader in the cluster, if the value is 0, it means there is no leader elected in the cluster. Labels: endpoint, instance, job, namespace, pod, server_id, service Labels
• node_filesystem_readonly Type 64-bit Gauge Description node_filesystem_readonly is set to 1 if the filesystem is in read-only mode, 0 otherwise. Labels: device, endpoint, fstype, instance, job, mountpoint, namespace, pod, service Labels
• node_timex_sync_status Type 64-bit Gauge Description node_timex_sync_status checks if clock synchronized to a reliable server (1 = yes, 0 = no) Labels: endpoint, instance, job, namespace, pod, service Labels
• kubernetes_build_info Type 64-bit Gauge Description A metric with a constant '1' value labeled by major, minor, git version, git commit, git tree state, build date, Go version, and compiler from which Kubernetes was built, and platform on which it is running. Labels: build_date, compiler, endpoint, git_commit, git_tree_state, git_version, go_version, instance, job, major, minor, namespace, platform, service Labels
• kube_node_status_condition Type 64-bit Gauge Description Current health status of the node. Returns a set of node conditions true, false, or unknown for each Labels: condition, instance, job, node, status Labels
• kube_node_spec_taint Type 64-bit Gauge Description

kube_node_spec_taint describes the taint of a cluster node Labels: condition, instance, job, node, status

Labels

- kubelet_node_name

Type

64-bit Gauge

Description

kubelet_node_name describes the node's name. The count is always 1. Labels: endpoint, instance, job, metrics_path, namespace, node, service

Labels

- kube_pod_container_resource_limits_memory_bytes

Type

64-bit Gauge

Description

kube_pod_container_resource_limits_memory_bytes describes the limit on memory to be used by a container in bytes. Labels: container, instance, job, namespace, node, pod

Labels

- prometheus_config_last_reload_successful

Type

64-bit Gauge

Description

Whether the last configuration reload attempt was successful. Labels: endpoint, instance, job, namespace, pod, service

Labels

- prometheus_notifications_queue_capacity

Type

64-bit Gauge

Description

prometheus_notifications_queue_capacity describes the capacity of the alert notifications queue. Labels: endpoint, instance, job, namespace, pod, service

Labels

- node_filesystem_files

Type

64-bit Gauge

Description

node_filesystem_files is used for the total number of permitted file nodes Labels: device, endpoint, fstype, instance, job, mountpoint, namespace, pod, service

Labels

- kube_node_status_allocatable_memory_bytes

Type

64-bit Gauge

Description

The memory resources of a node that are available for scheduling. Labels: instance, job, node

Labels

- aggregator_unavailable_apiservice

Type

64-bit Gauge

Description

Gauge of APIServices which are marked as unavailable broken down by APIService name. Labels: endpoint, instance, job, name, namespace, service

Labels

- kube_node_status_capacity_pods

Type

64-bit Gauge

Description

The total pod resources of the node. Labels: instance, job, node

Labels

- kube_statefulset_replicas

Type

64-bit Gauge

Description

Number of desired pods for a statefulset. Labels: container, fedType, fedUniqueName, instance, job, namespace, nodename, statefulset

Labels

- apiserver_request_duration_seconds_bucket

Type

64-bit Histogram

Description

Latency can be extracted from the apiserver_request_duration_seconds histogram buckets. Labels: component, dry_run, endpoint, group, instance, job, le, namespace, resource, scope, service, verb, version

Labels

Boundaries

[0.1, 0.5, 1, 5, 30, '+Inf']

- scheduler_e2e_scheduling_duration_seconds_bucket

Type

64-bit Histogram

Description

This metrics is for E2e scheduling latency in seconds (scheduling algorithm + binding) (bucket)

Labels

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384, '+Inf']

• scheduler_scheduling_algorithm_duration_seconds_bucket

Type

64-bit Histogram

Description

This metrics is for Scheduling algorithm latency in seconds (bucket)

Labels

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384, '+Inf']

• scheduler_binding_duration_seconds_bucket

Type

64-bit Histogram

Description

This metrics is for Binding latency in seconds (bucket)

Labels

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384, '+Inf']

• apiserver_client_certificate_expiration_seconds_bucket

Type

64-bit Histogram

Description

Distribution of the remaining lifetime on the certificate used to authenticate a request. Labels: endpoint, instance, job, le, namespace, service

Labels

Boundaries

[0, 1800, 3600, 7200, 21600, 43200, 86400, 172800, 345600, 604800, 2592000.0, 7776000.0, 15552000.0, 31104000.0, '+Inf']

• etcd_network_peer_round_trip_time_seconds_bucket

Type

64-bit Histogram

Description

This metric indicates the RTT (round trip time) latency for etcd to replicate a request between etcd members.

Labels

Boundaries

[0.0001, 0.0002, 0.0004, 0.0008, 0.0016, 0.0032, 0.0064, 0.0128, 0.0256, 0.0512, 0.1024, 0.2048, 0.4096, 0.8192, 1.6384, '+Inf']

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Summary

Group	Metric Name	Type	Synopsis
Prometheus Stats	coredns_dns_request_count_total	Counter	Counter of DNS requests made per zone, protocol and family.
Prometheus Stats	coredns_dns_request_type_count_total	Counter	Counter of DNS requests per type, per zone.
Prometheus Stats	coredns_dns_request_do_count_total	Counter	queries that have the DO bit set
Prometheus Stats	coredns_cache_size	Counter	Size of DNS cache.
Prometheus Stats	coredns_cache_misses_total	Counter	The count of cache misses.
Prometheus Stats	coredns_cache_hits_total	Counter	The count of cache hits.
Prometheus Stats	up	Counter	1 if the instance is healthy, i.e. reachable, or 0 if the scrape failed.
Prometheus Stats	apiserver_request_total	Counter	monitors the requests to the service
Prometheus Stats	etcd_helper_cache_entry_total	Counter	Counter of etcd helper cache entries.
Prometheus Stats	etcd_helper_cache_hit_total	Counter	Counter of etcd helper cache hits.
Prometheus Stats	etcd_helper_cache_miss_total	Counter	Counter of etcd helper cache miss.
Prometheus Stats	process_resident_memory_bytes	Counter	Resident memory size in bytes.
Prometheus Stats	process_cpu_seconds_total	Counter	Total user and system CPU time spent in seconds.
Prometheus Stats	go_goroutines	Counter	Number of goroutines that currently exist.
Prometheus Stats	node_cpu_seconds_total	Counter	node_cpu Seconds the cpus spent in each mode.
Prometheus Stats	kube_pod_container_resource_limits_cpu_cores	Counter	Total CPU limits (cores) of a pod
Prometheus Stats	node_namespace_pod_container:container_cpu_usage_seconds_total:sum_rate	Counter	CPU usage by namespace
Prometheus Stats	mixin_pod_workload	Counter	counter for the number of pods running in a namespace.
Prometheus Stats	kube_pod_container_resource_limits_memory_bytes	Counter	The limit on memory to be used by a container in bytes.
Prometheus Stats	container_network_receive_bytes_total	Counter	Bytes received over the network by the container per second in the last minute
Prometheus Stats	container_network_transmit_bytes_total	Counter	Bytes transmitted over the network by the container per second in the last minute
Prometheus Stats	container_network_receive_packets_total	Counter	Cumulative count of packets received
Prometheus Stats	container_network_transmit_packets_total	Counter	Cumulative count of packets transmitted
Prometheus Stats	container_network_receive_packets_dropped_total	Counter	Cumulative count of packets dropped while receiving
Prometheus Stats	container_network_transmit_packets_dropped_total	Counter	Cumulative count of packets dropped while transmitting
Prometheus Stats	node_vmstat_pgmajfault	Counter	/proc/vmstat information field pgmajfault.
Prometheus Stats	node_disk_io_time_seconds	Counter	total Total seconds spent doing I/Os.

Group	Metric Name	Type	Synopsis
Prometheus Stats	node_disk_io_time_weighted_seconds	Counter	The weighted # of seconds spent doing I/Os.
Prometheus Stats	node_disk_read_bytes_total	Counter	The total number of bytes read successfully.
Prometheus Stats	node_disk_written_bytes_total	Counter	The total number of bytes written successfully.
Prometheus Stats	node_time_seconds	Counter	System time in seconds since epoch (1970).
Prometheus Stats	node_memory_SwapTotal_bytes	Counter	Memory information field SwapTotal_bytes.
Prometheus Stats	node_boot_time_seconds	Counter	Node boot time, in unixtime.
Prometheus Stats	node_network_receive_packets_total	Counter	Network device statistic receive_packets.
Prometheus Stats	node_network_transmit_packets_total	Counter	Network device statistic transmit_packets.
Prometheus Stats	node_network_receive_compressed_total	Counter	Network device statistic receive_compressed.
Prometheus Stats	node_network_transmit_compressed_total	Counter	Network device statistic transmit_compressed.
Prometheus Stats	node_network_receive_multicast_total	Counter	Network device statistic receive_multicast.
Prometheus Stats	node_network_receive_fifo_total	Counter	Network device statistic receive_fifo.
Prometheus Stats	node_network_transmit_fifo_total	Counter	Network device statistic transmit_fifo.
Prometheus Stats	node_network_receive_frame_total	Counter	Network device statistic receive_frame.
Prometheus Stats	node_network_transmit_carrier_total	Counter	Network device statistic transmit_carrier.
Prometheus Stats	node_network_transmit_colls_total	Counter	Network device statistic transmit_colls.
Prometheus Stats	node_arp_entries	Counter	ARP entries by device
Prometheus Stats	node_network_mtu_bytes	Counter	mtu_bytes value of /sys/class/net/.
Prometheus Stats	node_network_speed_bytes	Counter	speed_bytes value of /sys/class/net/.
Prometheus Stats	node_network_transmit_queue_length	Counter	transmit_queue_length value of /sys/class/net/.
Prometheus Stats	node_network_carrier	Counter	carrier value of /sys/class/net/.
Prometheus Stats	kubelet_running_container_count	Counter	Number of containers currently running
Prometheus Stats	volume_manager_total_volumes	Counter	Number of volumes in Volume Manager
Prometheus Stats	kubelet_node_config_error	Counter	This metric is true (1) if the node is experiencing a configuration-related error, false (0) otherwise.
Prometheus Stats	kubelet_runtime_operations_total	Counter	Total count of runtime operations of each type.
Prometheus Stats	kubelet_runtime_operations_errors_total	Counter	Cumulative number of runtime operation errors by operation type.
Prometheus Stats	storage_operation_duration_seconds_count	Counter	Storage operation duration

Group	Metric Name	Type	Synopsis
Prometheus Stats	storage_operation_errors_total	Counter	Storage errors total
Prometheus Stats	kubelet_cgroup_manager_duration_seconds_count	Counter	Duration in seconds for cgroup manager operations. Broken down by method.
Prometheus Stats	node_sockstat_TCP_alloc	Counter	Number of TCP sockets in state alloc.
Prometheus Stats	node_sockstat_TCP_inuse	Counter	Number of TCP sockets in state inuse.
Prometheus Stats	node_sockstat_TCP_mem	Counter	Number of TCP sockets in state mem.
Prometheus Stats	node_sockstat_UDPLITE_inuse	Counter	Number of UDPLITE sockets in state inuse.
Prometheus Stats	node_sockstat_TCP_orphan	Counter	Number of TCP sockets in state orphan.
Prometheus Stats	node_sockstat_UDP_inuse	Counter	Number of UDP sockets in state inuse.
Prometheus Stats	node_sockstat_UDP_mem	Counter	Number of UDP sockets in state mem.
Prometheus Stats	node_sockstat_FRAG_inuse	Counter	Number of FRAG sockets in state inuse.
Prometheus Stats	node_sockstat_RAW_inuse	Counter	Number of RAW sockets in state inuse.
Prometheus Stats	node_sockstat_UDP_mem_bytes	Counter	Number of UDP sockets in state mem_bytes.
Prometheus Stats	node_sockstat_TCP_mem_bytes	Counter	Number of TCP sockets in state mem_bytes.
Prometheus Stats	node_sockstat_FRAG_memory	Counter	Number of FRAG sockets in state memory.
Prometheus Stats	node_sockstat_sockets_used	Counter	Number of IPv4 sockets in use.
Prometheus Stats	node_scrape_collector_duration_seconds	Counter	node_exporter: Duration of a collector scrape.
Prometheus Stats	node_textfile_scrape_error	Counter	1 if there was an error opening or reading a file, 0 otherwise
Prometheus Stats	container_cpu_usage_seconds_total	Counter	Cumulative cpu time consumed per cpu in seconds
Prometheus Stats	container_memory_usage_bytes	Counter	measures the current memory usage, including all memory regardless of when it was accessed.
Prometheus Stats	kube_statefulset_status_observed_generation	Counter	The generation observed by the StatefulSet controller.
Prometheus Stats	kube_statefulset_status_replicas	Counter	The number of replicas per StatefulSet.
Prometheus Stats	kube_statefulset_status_replicas_ready	Counter	The number of ready replicas per StatefulSet.
Prometheus Stats	kube_statefulset_status_replicas_current	Counter	The number of current replicas per StatefulSet.
Prometheus Stats	coredns_dns_response_rcode_count_total	Counter	response per zone and rcode.
Prometheus Stats	kubelet_pod_start_duration_seconds_count	Counter	Number of pod start operations.
Prometheus Stats	kubelet_volume_stats_inodes_used	Gauge	Number of used inodes in the volume.
Prometheus Stats	kubelet_volume_stats_inodes	Gauge	Maximum number of inodes in the volume.

Group	Metric Name	Type	Synopsis
Prometheus Stats	coredns_dns_request_size_bytes_bucket	Histogram	Size of the buffer in bytes
Prometheus Stats	coredns_dns_request_duration_seconds_bucket	Histogram	Histogram of the time (in seconds) each request took.
Prometheus Stats	coredns_dns_response_size_bytes_bucket	Histogram	Size of the returned response in bytes.
Prometheus Stats	etcd_request_cache_get_duration_seconds_bucket	Histogram	Latency in seconds of getting an object from etcd cache
Prometheus Stats	etcd_request_cache_add_duration_seconds_bucket	Histogram	Latency in seconds of adding an object to etcd cache
Prometheus Stats	kubelet_runtime_operations_duration_seconds_bucket	Histogram	Duration of the operations. Useful to calculate percentiles.
Prometheus Stats	storage_operation_duration_seconds_bucket	Histogram	Storage operation duration
Prometheus Stats	kubelet_cgroup_manager_duration_seconds_bucket	Histogram	Duration in seconds for cgroup manager operations. Broken down by method.
Prometheus Stats	kubelet_pleg_relist_interval_seconds_bucket	Histogram	Interval in seconds between relisting in PLEG.
Prometheus Stats	kubelet_pleg_relist_duration_seconds_bucket	Histogram	Duration in seconds for relisting pods in PLEG.
Prometheus Stats	rest_client_request_latency_seconds_bucket	Histogram	Request latency in seconds.

Prometheus Stats

- coredns_dns_request_count_total

Type

64-bit Counter

Description

Counter of DNS requests made per zone, protocol and family.

Labels

- coredns_dns_request_type_count_total

Type

64-bit Counter

Description

Counter of DNS requests per type, per zone.

Labels

- coredns_dns_request_do_count_total

Type

64-bit Counter

Description

queries that have the DO bit set

Labels

- coredns_cache_size

Type

64-bit Counter

Description

Size of DNS cache.

Labels

- `coredns_cache_misses_total`

Type

64-bit Counter

Description

The count of cache misses..

Labels

- `coredns_cache_hits_total`

Type

64-bit Counter

Description

The count of cache hits.

Labels

- `up`

Type

64-bit Counter

Description

1 if the instance is healthy, i.e. reachable, or 0 if the scrape failed.

Labels

- `apiserver_request_total`

Type

64-bit Counter

Description

monitors the requests to the service

Labels

- `etcd_helper_cache_entry_total`

Type

64-bit Counter

Description

Counter of etcd helper cache entries. This can be different from `etcd_helper_cache_miss_count` because two concurrent threads can miss the cache and generate the same entry twice.

Labels

• etcd_helper_cache_hit_total Type 64-bit Counter Description Counter of etcd helper cache hits. Labels
• etcd_helper_cache_miss_total Type 64-bit Counter Description Counter of etcd helper cache miss. Labels
• process_resident_memory_bytes Type 64-bit Counter Description Resident memory size in bytes. Labels
• process_cpu_seconds_total Type 64-bit Counter Description Total user and system CPU time spent in seconds. Labels
• go_goroutines Type 64-bit Counter Description Number of goroutines that currently exist. Labels
• node_cpu_seconds_total Type 64-bit Counter Description node_cpu Seconds the cpus spent in each mode. Labels

• kube_pod_container_resource_limits_cpu_cores
Type
64-bit Counter
Description
Total CPU limits (cores) of a pod
Labels
• node_namespace_pod_container:container_cpu_usage_seconds_total:sum_rate
Type
64-bit Counter
Description
CPU usage by namespace
Labels
• mixin_pod_workload
Type
64-bit Counter
Description
counter for the number of pods running in a namespace.
Labels
• kube_pod_container_resource_limits_memory_bytes
Type
64-bit Counter
Description
The limit on memory to be used by a container in bytes.
Labels
• container_network_receive_bytes_total
Type
64-bit Counter
Description
Bytes received over the network by the container per second in the last minute
Labels
• container_network_transmit_bytes_total
Type
64-bit Counter
Description
Bytes transmitted over the network by the container per second in the last minute
Labels

• container_network_receive_packets_total Type 64-bit Counter Description Cumulative count of packets received Labels
• container_network_transmit_packets_total Type 64-bit Counter Description Cumulative count of packets transmitted Labels
• container_network_receive_packets_dropped_total Type 64-bit Counter Description Cumulative count of packets dropped while receiving Labels
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• node_vmstat_pgmajfault Type 64-bit Counter Description /proc/vmstat information field pgmajfault. Labels
• node_disk_io_time_seconds Type 64-bit Counter Description total Total seconds spent doing I/Os. Labels

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• node_disk_read_bytes_total Type 64-bit Counter Description The total number of bytes read successfully. Labels
• node_disk_written_bytes_total Type 64-bit Counter Description The total number of bytes written successfully. Labels
• node_time_seconds Type 64-bit Counter Description System time in seconds since epoch (1970). Labels
• node_memory_SwapTotal_bytes Type 64-bit Counter Description Memory information field SwapTotal_bytes. Labels
• node_boot_time_seconds Type 64-bit Counter Description Node boot time, in unixtime. Labels

node_network_receive_packets_total
Type
64-bit Counter
Description
Network device statistic receive_packets.
Labels
node_network_transmit_packets_total
Type
64-bit Counter
Description
Network device statistic transmit_packets.
Labels
node_network_receive_compressed_total
Type
64-bit Counter
Description
Network device statistic receive_compressed.
Labels
node_network_transmit_compressed_total
Type
64-bit Counter
Description
Network device statistic transmit_compressed.
Labels
node_network_receive_multicast_total
Type
64-bit Counter
Description
Network device statistic receive_multicast.
Labels
node_network_receive_fifo_total
Type
64-bit Counter
Description
Network device statistic receive_fifo.
Labels

node_network_transmit_fifo_total
Type
64-bit Counter
Description
Network device statistic transmit_fifo.
Labels
node_network_receive_frame_total
Type
64-bit Counter
Description
Network device statistic receive_frame.
Labels
node_network_transmit_carrier_total
Type
64-bit Counter
Description
Network device statistic transmit_carrier.
Labels
node_network_transmit_colls_total
Type
64-bit Counter
Description
Network device statistic transmit_colls.
Labels
node_arp_entries
Type
64-bit Counter
Description
ARP entries by device
Labels
node_network_mtu_bytes
Type
64-bit Counter
Description
mtu_bytes value of /sys/class/net/.
Labels

• node_network_speed_bytes Type 64-bit Counter Description speed_bytes value of /sys/class/net/. Labels
• node_network_transmit_queue_length Type 64-bit Counter Description transmit_queue_length value of /sys/class/net/. Labels
• node_network_carrier Type 64-bit Counter Description carrier value of /sys/class/net/. Labels
• kubelet_running_container_count Type 64-bit Counter Description Number of containers currently running Labels
• volume_manager_total_volumes Type 64-bit Counter Description Number of volumes in Volume Manager Labels
• kubelet_node_config_error Type 64-bit Counter Description This metric is true (1) if the node is experiencing a configuration-related error, false (0) otherwise. Labels

• kubelet_runtime_operations_total Type 64-bit Counter Description Total count of runtime operations of each type. Labels
• kubelet_runtime_operations_errors_total Type 64-bit Counter Description Count of errors in the operations. This can be a good indicator of low level issues in the node, like problems with container runtime. Labels
• storage_operation_duration_seconds_count Type 64-bit Counter Description Storage operation duration Labels
• storage_operation_errors_total Type 64-bit Counter Description Storage errors total Labels
• kubelet_cgroup_manager_duration_seconds_count Type 64-bit Counter Description Duration in seconds for cgroup manager operations. Broken down by method. Labels
• node_sockstat_TCP_alloc Type 64-bit Counter Description Number of TCP sockets in state alloc. Labels

- node_sockstat_TCP_inuse

Type

64-bit Counter

Description

Number of TCP sockets in state inuse.

Labels

- node_sockstat_TCP_mem

Type

64-bit Counter

Description

Number of TCP sockets in state mem.

Labels

- node_sockstat_UDPLITE_inuse

Type

64-bit Counter

Description

Number of UDPLITE sockets in state inuse.

Labels

- node_sockstat_TCP_orphan

Type

64-bit Counter

Description

Number of TCP sockets in state orphan.

Labels

- node_sockstat_UDP_inuse

Type

64-bit Counter

Description

Number of UDP sockets in state inuse.

Labels

- node_sockstat_UDP_mem

Type

64-bit Counter

Description

Number of UDP sockets in state mem.

Labels

• node_sockstat_FRAG_inuse Type 64-bit Counter Description Number of FRAG sockets in state inuse. Labels
• node_sockstat_RAW_inuse Type 64-bit Counter Description Number of RAW sockets in state inuse. Labels
• node_sockstat_UDP_mem_bytes Type 64-bit Counter Description Number of UDP sockets in state mem_bytes. Labels
• node_sockstat_TCP_mem_bytes Type 64-bit Counter Description Number of TCP sockets in state mem_bytes. Labels
• node_sockstat_FRAG_memory Type 64-bit Counter Description Number of FRAG sockets in state memory. Labels
• node_sockstat_sockets_used Type 64-bit Counter Description Number of IPv4 sockets in use. Labels

node_scrape_collector_duration_seconds
Type
64-bit Counter
Description
node_exporter: Duration of a collector scrape.
Labels
node_textfile_scrape_error
Type
64-bit Counter
Description
1 if there was an error opening or reading a file, 0 otherwise
Labels
container_cpu_usage_seconds_total
Type
64-bit Counter
Description
Cumulative cpu time consumed per cpu in seconds
Labels
container_memory_usage_bytes
Type
64-bit Counter
Description
measures the current memory usage, including all memory regardless of when it was accessed.
Labels
kube_statefulset_status_observed_generation
Type
64-bit Counter
Description
The generation observed by the StatefulSet controller.
Labels
kube_statefulset_status_replicas
Type
64-bit Counter
Description
The number of replicas per StatefulSet.
Labels

• kube_statefulset_status_replicas_ready Type 64-bit Counter Description The number of ready replicas per StatefulSet. Labels
• kube_statefulset_status_replicas_current Type 64-bit Counter Description The number of current replicas per StatefulSet. Labels
• coredns_dns_response_rcode_count_total Type 64-bit Counter Description response per zone and rcode. Labels
• kubelet_pod_start_duration_seconds_count Type 64-bit Counter Description Number of pod start operations. Labels
• kubelet_volume_stats_inodes_used Type 64-bit Gauge Description The metrics kubelet_volume_stats_inodes_used informs about the [ALPHA] Number of used inodes in the volume. Labels
• kubelet_volume_stats_inodes Type 64-bit Gauge Description The metrics kubelet_volume_stats_inodes about the [ALPHA] Maximum number of inodes in the volume. Labels

- **coredns_dns_request_size_bytes_bucket**

Type	
64-bit Histogram	
Description	
Size of the buffer in bytes	
Labels	
Boundaries	
[0, 100, 200, 300, 400, 511, 1023, 2047, 4095, 8291, 16000, 32000, 48000, 64000, '+Inf']	

- **coredns_dns_request_duration_seconds_bucket**

Type	
64-bit Histogram	
Description	
Histogram of the time (in seconds) each request took.	
Labels	
Boundaries	
[0.00025, 0.0005, 0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, '+Inf']	

- **coredns_dns_response_size_bytes_bucket**

Type	
64-bit Histogram	
Description	
Size of the returned response in bytes.	
Labels	
Boundaries	
[0, 100, 200, 300, 400, 511, 1023, 2047, 4095, 8291, 16000, 32000, 48000, 64000, '+Inf']	

- **etcd_request_cache_get_duration_seconds_bucket**

Type	
64-bit Histogram	
Description	
Latency in seconds of getting an object from etcd cache	
Labels	
Boundaries	
[0.005, 0.01, 0.025, 0.05, 0.1, 0.25, 0.5, 1, 2.5, 5, 10, '+Inf']	

- **etcd_request_cache_add_duration_seconds_bucket**

Type
64-bit Histogram
Description
Latency in seconds of adding an object to etcd cache

Labels

Boundaries

[0.005, 0.01, 0.025, 0.05, 0.1, 0.25, 0.5, 1, 2.5, 5, 10, '+Inf']

- kubelet_runtime_operations_duration_seconds_bucket

Type

64-bit Histogram

Description

Duration of the operations. Useful to calculate percentiles.

Labels

Boundaries

[0.005, 0.0125, 0.0312, 0.0781, 0.1953, 0.4883, 1.2207, 3.0517, 7.6294, 19.0735, 47.6837, 119.2093, 298.0232, 745.058, '+Inf']

- storage_operation_duration_seconds_bucket

Type

64-bit Histogram

Description

Storage operation duration

Labels

Boundaries

[0.1, 0.25, 0.5, 1, 2.5, 5, 10, 15, 25, 50, 120, 300, 600, '+Inf']

- kubelet_cgroup_manager_duration_seconds_bucket

Type

64-bit Histogram

Description

Duration in seconds for cgroup manager operations. Broken down by method.

Labels

Boundaries

[0.005, 0.01, 0.025, 0.05, 0.1, 0.25, 0.5, 1, 2.5, 5, 10, '+Inf']

- kubelet_pleg_relist_interval_seconds_bucket

Type

64-bit Histogram

Description

Interval in seconds between relisting in PLEG.

Labels

Boundaries

[0.005, 0.01, 0.025, 0.05, 0.1, 0.25, 0.5, 1, 2.5, 5, 10, '+Inf']

- kubelet_pleg_relist_duration_seconds_bucket

Type

64-bit Histogram
Description
Duration in seconds for relisting pods in PLEG.
Labels
Boundaries
[0.005, 0.01, 0.025, 0.05, 0.1, 0.25, 0.5, 1, 2.5, 5, 10, '+Inf']
rest_client_request_latency_seconds_bucket
Type
64-bit Histogram
Description
Request latency in seconds.
Labels
Boundaries
[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, '+Inf']

Table of Contents

- Summary
- ETCD Client Metrics

Summary

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_put_requests	Counter	The total number of ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_requests_sent	Counter	The total number of ETCD Client PUT Requests sent.
ETCD Client Metrics	etcd_put_successes	Counter	The total number of successful ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_failures	Counter	The total number of failed ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_with_lease_requests	Counter	The total number of ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_requests_sent	Counter	The total number of ETCD Client PUT with Lease Requests sent.
ETCD Client Metrics	etcd_put_with_lease_successes	Counter	The total number of successful ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_failures	Counter	The total number of failed ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_range_requests	Counter	The total number of ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_requests_sent	Counter	The total number of ETCD Client Range Requests sent.
ETCD Client Metrics	etcd_range_successes	Counter	The total number of successful ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_failures	Counter	The total number of failed ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_delete_requests	Counter	The total number of ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_requests_sent	Counter	The total number of ETCD Client Range Delete Requests sent.
ETCD Client Metrics	etcd_range_delete_successes	Counter	The total number of successful ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_failures	Counter	The total number of failed ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_txn_requests	Counter	The total number of ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_requests_sent	Counter	The total number of ETCD Client Txn Requests sent.
ETCD Client Metrics	etcd_txn_successes	Counter	The total number of successful ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_failures	Counter	The total number of failed ETCD Client Txn Requests.
ETCD Client Metrics	etcd_watch_requests	Counter	The total number of ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_requests_sent	Counter	The total number of ETCD Client Watch Requests sent.

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_watch_successes	Counter	The total number of successful ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_failures	Counter	The total number of failed ETCD Client Watch Requests.
ETCD Client Metrics	etcdwatcher_requests	Counter	The total number of ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_requests_sent	Counter	The total number of ETCD Watcher Requests sent.
ETCD Client Metrics	etcdwatcher_successes	Counter	The total number of successful ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_failures	Counter	The total number of failed ETCD Watcher Requests.

ETCD Client Metrics

- etcd_put_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests.

Labels

keyPrefix

- etcd_put_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests sent.

Labels

keyPrefix

- etcd_put_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT Requests.

Labels

keyPrefix

- etcd_put_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT Requests.

Labels

keyPrefix, failureReason

- etcd_put_with_lease_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests sent.

Labels

keyPrefix

- etcd_put_with_lease_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT with Lease Requests.

Labels

keyPrefix, failureReason

- etcd_range_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests sent.

Labels

keyPrefix

etcd_range_successes
Type
64-bit Counter
Description
The total number of successful ETCD Client Range Requests.
Labels
keyPrefix
etcd_range_failures
Type
64-bit Counter
Description
The total number of failed ETCD Client Range Requests.
Labels
keyPrefix, failureReason
etcd_range_delete_requests
Type
64-bit Counter
Description
The total number of ETCD Client Range Delete Requests.
Labels
keyPrefix
etcd_range_delete_requests_sent
Type
64-bit Counter
Description
The total number of ETCD Client Range Delete Requests sent.
Labels
keyPrefix
etcd_range_delete_successes
Type
64-bit Counter
Description
The total number of successful ETCD Client Range Delete Requests.
Labels
keyPrefix
etcd_range_delete_failures
Type
64-bit Counter
Description

The total number of failed ETCD Client Range Delete Requests.

Labels

keyPrefix, failureReason

• etcd_txn_requests

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests.

Labels

keyPrefix

• etcd_txn_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests sent.

Labels

keyPrefix

• etcd_txn_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Txn Requests.

Labels

keyPrefix

• etcd_txn_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Txn Requests.

Labels

keyPrefix, failureReason

• etcd_watch_requests

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests.

Labels

keyPrefix

• etcd_watch_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests sent.

Labels

keyPrefix

• etcd_watch_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Watch Requests.

Labels

keyPrefix

• etcd_watch_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Watch Requests.

Labels

keyPrefix, failureReason

• etcdwatcher_requests

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests.

Labels

keyPrefix

• etcdwatcher_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests sent.

Labels

keyPrefix

• etcdwatcher_successes

Type

64-bit Counter

Description

The total number of successful ETCD Watcher Requests.

Labels

keyPrefix

- etcdwatcher_failures

Type

64-bit Counter

Description

The total number of failed ETCD Watcher Requests.

Labels

keyPrefix, failureReason

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- [Summary](#)
- [HTTP Punting Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_sent	Counter	The total number of HTTP requests successfully punted.
HTTP Punting Metrics	punting_http_requests_failed	Counter	The total number of HTTP requests punting failed.
HTTP Punting Metrics	punting_http_responses_received	Counter	The total number of HTTP responses received for the requests punted.
HTTP Punting Metrics	punting_http_responses_forwarded	Counter	The total number of HTTP responses forwarded after punting transaction completion.
HTTP Punting Metrics	punting_http_requests_aborted	Counter	The total number of HTTP requests aborted.
HTTP Punting Metrics	punting_http_request_duration_seconds	Histogram	Punting operation duration in seconds.

HTTP Punting Metrics

- punting_http_requests_sent

Type

64-bit Counter

Description

The total number of HTTP requests successfully punted.

Labels

message_type

- punting_http_requests_failed

Type

64-bit Counter

Description

The total number of HTTP requests for which the punting failed.

Labels

message_type

- **punting_http_responses_received**

Type
64-bit Counter

Description
The total number of HTTP responses received for the requests punted.

Labels
message_type, status_code
- **punting_http_responses_forwarded**

Type
64-bit Counter

Description
The total number of HTTP responses forwarded after punting transaction completion.

Labels
message_type, status_code
- **punting_http_requests_aborted**

Type
64-bit Counter

Description
The total number of HTTP requests aborted.

Labels
message_type
- **punting_http_request_duration_seconds**

Type
64-bit Histogram

Description
The punting transaction duration is measured from the time the HTTP request is sent out to the time the response is received for the punting request.

Labels
message_type

Boundaries
[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

Table of Contents

- [Summary](#)
- [LB Agent VPP Threads Cpu Utilization](#)

Summary

Group	Metric Name	Type	Synopsis
LB Agent VPP Threads Cpu Utilization	lb_vpp_thread_cpu_utilization	Gauge	VPP thread CPU utilization reported as percentage.
LB Agent VPP Threads Cpu Utilization	lb_vpp_thread_cpu_high_water	Gauge	VPP thread CPU high water reported as percentage.

LB Agent VPP Threads Cpu Utilization

- lb_vpp_thread_cpu_utilization
Type
64-bit Gauge
Description
VPP thread CPU utilization reported as percentage.
Labels
thread_id, thread_type
- lb_vpp_thread_cpu_high_water
Type
64-bit Gauge
Description
VPP thread CPU high water reported as percentage.
Labels
thread_id, thread_type

Table of Contents

- [Summary](#)
- [Packet Streaming Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Packet Streaming Metrics	num_packet_streaming_attempted	Counter	The total number of attempted streaming of packets
Packet Streaming Metrics	num_packet_streaming_success	Counter	The total number of packets streamed successfully
Packet Streaming Metrics	num_packet_streaming_failed	Counter	The total number of failed streaming of packets
Packet Streaming Metrics	num_packet_streaming_bytes	Counter	The total number of bytes successfully streamed

Packet Streaming Metrics

- num_packet_streaming_attempted
Type
64-bit Counter
Description
The total number of attempted streaming of packets
Labels
interface_name, server_fqdn, kafka_topic
- num_packet_streaming_success
Type
64-bit Counter
Description
The total number of packets streamed successfully
Labels
interface_name, server_fqdn, kafka_topic

- num_packet_streaming_failed

Type

64-bit Counter

Description

The total number of failed streaming of packets

Labels

interface_name, server_fqdn, kafka_topic
- num_packet_streaming_bytes

Type

64-bit Counter

Description

The total number of bytes successfully streamed

Labels

interface_name, server_fqdn, kafka_topic

Table of Contents

- [Summary](#)
- [ETCD Client Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_put_requests	Counter	The total number of ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_requests_sent	Counter	The total number of ETCD Client PUT Requests sent.
ETCD Client Metrics	etcd_put_successes	Counter	The total number of successful ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_failures	Counter	The total number of failed ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_with_lease_requests	Counter	The total number of ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_requests_sent	Counter	The total number of ETCD Client PUT with Lease Requests sent.
ETCD Client Metrics	etcd_put_with_lease_successes	Counter	The total number of successful ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_failures	Counter	The total number of failed ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_range_requests	Counter	The total number of ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_requests_sent	Counter	The total number of ETCD Client Range Requests sent.
ETCD Client Metrics	etcd_range_successes	Counter	The total number of successful ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_failures	Counter	The total number of failed ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_delete_requests	Counter	The total number of ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_requests_sent	Counter	The total number of ETCD Client Range Delete Requests sent.
ETCD Client Metrics	etcd_range_delete_successes	Counter	The total number of successful ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_failures	Counter	The total number of failed ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_txn_requests	Counter	The total number of ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_requests_sent	Counter	The total number of ETCD Client Txn Requests sent.
ETCD Client Metrics	etcd_txn_successes	Counter	The total number of successful ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_failures	Counter	The total number of failed ETCD Client Txn Requests.
ETCD Client Metrics	etcd_watch_requests	Counter	The total number of ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_requests_sent	Counter	The total number of ETCD Client Watch Requests sent.
ETCD Client Metrics	etcd_watch_successes	Counter	The total number of successful ETCD Client Watch Requests.

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_watch_failures	Counter	The total number of failed ETCD Client Watch Requests.
ETCD Client Metrics	etcdwatcher_requests	Counter	The total number of ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_requests_sent	Counter	The total number of ETCD Watcher Requests sent.
ETCD Client Metrics	etcdwatcher_successes	Counter	The total number of successful ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_failures	Counter	The total number of failed ETCD Watcher Requests.

ETCD Client Metrics

- `etcd_put_requests`

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests.

Labels

keyPrefix

- `etcd_put_requests_sent`

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests sent.

Labels

keyPrefix

- `etcd_put_successes`

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT Requests.

Labels

keyPrefix

- `etcd_put_failures`

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT Requests.

Labels

keyPrefix, failureReason

- `etcd_put_with_lease_requests`

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests sent.

Labels

keyPrefix

- etcd_put_with_lease_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT with Lease Requests.

Labels

keyPrefix, failureReason

- etcd_range_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests sent.

Labels

keyPrefix

- etcd_range_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Requests.

Labels

keyPrefix, failureReason

- etcd_range_delete_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests.

Labels

keyPrefix

- etcd_range_delete_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests sent.

Labels

keyPrefix

- etcd_range_delete_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Delete Requests.

Labels

keyPrefix

- etcd_range_delete_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Delete Requests.

Labels

keyPrefix, failureReason

• etcd_txn_requests

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests.

Labels

keyPrefix

• etcd_txn_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests sent.

Labels

keyPrefix

• etcd_txn_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Txn Requests.

Labels

keyPrefix

• etcd_txn_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Txn Requests.

Labels

keyPrefix, failureReason

• etcd_watch_requests

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests.

Labels

keyPrefix

• etcd_watch_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests sent.

Labels

keyPrefix

- etcd_watch_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Watch Requests.

Labels

keyPrefix, failureReason

- etcdwatcher_requests

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests.

Labels

keyPrefix

- etcdwatcher_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests sent.

Labels

keyPrefix

- etcdwatcher_successes

Type

64-bit Counter

Description

The total number of successful ETCD Watcher Requests.

Labels

keyPrefix

- etcdwatcher_failures

Type

64-bit Counter

Description

The total number of failed ETCD Watcher Requests.

Labels

keyPrefix, failureReason

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- [Summary](#)
- [HTTP Punting Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_sent	Counter	The total number of HTTP requests successfully punting.
HTTP Punting Metrics	punting_http_requests_failed	Counter	The total number of HTTP requests punting failed.
HTTP Punting Metrics	punting_http_responses_received	Counter	The total number of HTTP responses received for the requests punting.
HTTP Punting Metrics	punting_http_responses_forwarded	Counter	The total number of HTTP responses forwarded after punting transaction completion.
HTTP Punting Metrics	punting_http_requests_aborted	Counter	The total number of HTTP requests aborted.
HTTP Punting Metrics	punting_http_request_duration_seconds	Histogram	Punting operation duration in seconds.

HTTP Punting Metrics

- punting_http_requests_sent

Type

64-bit Counter

Description

The total number of HTTP requests successfully punting.

Labels

message_type

- punting_http_requests_failed

Type

64-bit Counter

Description

The total number of HTTP requests for which the punting failed.

Labels

message_type

- punting_http_responses_received

Type

64-bit Counter

Description

The total number of HTTP responses received for the requests punted.

Labels

message_type, status_code

- `punting_http_responses_forwarded`

Type

64-bit Counter

Description

The total number of HTTP responses forwarded after punting transaction completion.

Labels

message_type, status_code

- `punting_http_requests_aborted`

Type

64-bit Counter

Description

The total number of HTTP requests aborted.

Labels

message_type

- `punting_http_request_duration_seconds`

Type

64-bit Histogram

Description

The punting transaction duration is measured from the time the HTTP request is sent out to the time the response is received for the punting request.

Labels

message_type

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

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- [Summary](#)
- [Diameter statistics for MME Call Control Service](#)

Summary

Group	Metric Name	Type	Synopsis
Diameter statistics for MME Call Control Service	<code>mmecc_diameter_failure_response_received</code>	Counter	Total number of diameter responses received with failure result-code
Diameter statistics for MME Call Control Service	<code>mmecc_diameter_failure_response_sent</code>	Counter	Total number of diameter responses sent with failure result-code

Diameter statistics for MME Call Control Service

- mmecc_diameter_failure_response_received

Type

64-bit Counter

Description

Total number of diameter response messages received from a diameter peer node with failure result code.

Labels

command_code, result_code, peer
- mmecc_diameter_failure_response_sent

Type

64-bit Counter

Description

Total number of diameter response messages sent to a diameter peer node with failure result code.

Labels

command_code, result_code, peer

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Mobility Management layer statistics for MME Call Control Service	mmecc_nas_emm_attach_request	Counter	The number of attach request messages received by MME from UE.
Mobility Management layer statistics for MME Call Control Service	mmecc_nas_emm_attach_accept	Counter	The number of attach accept messages sent from MME to UE.
Mobility Management layer statistics for MME Call Control Service	mmecc_nas_emm_attach_complete	Counter	The number of attach complete messages received by MME from UE.
Mobility Management layer statistics for MME Call Control Service	mmecc_emm_attach_eps_attempt	Counter	EPS attach procedure attempts
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Mobility Management layer statistics for MME Call Control Service	mmecc_emm_attach_eps_success_restrict_n1	Counter	EPS attach procedure success with n1 restriction
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Mobility Management layer statistics for MME Call Control Service	mmecc_emm_attach_eps_failure	Counter	EPS attach procedure failures
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Group	Metric Name	Type	Synopsis
Mobility Management layer statistics for MME Call Control Service	mmecc_nas_emm_ue_detach_request	Counter	The number of UE detach request messages received by MME from UE.
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Mobility Management layer statistics for MME Call Control Service	mmecc_emm_eps_detach_ue_attempt	Counter	Detach procedure attempts initiated by UE.
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Mobility Management layer statistics for MME Call Control Service	mmecc_emm_eps_detach_ue_attempt_poweroff	Counter	Detach procedure attempts initiated by UE.
Mobility Management layer statistics for MME Call Control Service	mmecc_emm_eps_detach_mme_attempt_gw	Counter	Detach procedure attempts initiated by MME due to GW initiated detach.
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Mobility Management layer statistics for MME Call Control Service	mmecc_emm_eps_detach_mme_attempt_mrt	Counter	Detach procedure attempts initiated by MME due to MRT expiry.
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Mobility Management layer statistics for MME Call Control Service	mmecc_emm_eps_detach_hss_success	Counter	Detach procedure success initiated by HSS.
Mobility Management layer statistics for MME Call Control Service	mmecc_emm_eps_detach_mme_success	Counter	The number of successful detach procedures initiated by MME.
Mobility Management layer statistics for MME Call Control Service	mmecc_emm_service_request_attempt	Counter	Service request procedure attempts initiated by UE.
Mobility Management layer statistics for MME Call Control Service	mmecc_emm_service_request_success	Counter	Service request procedure success.
Mobility Management layer statistics for MME Call Control Service	mmecc_emm_service_request_failure	Counter	Service request procedure failures
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Group	Metric Name	Type	Synopsis
Mobility Management layer statistics for MME Call Control Service	mmecc_emm_control_plane_service_request_failure	Counter	Control plane service request procedure failures
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Mobility Management layer statistics for MME Call Control Service	mmecc_emm_eps_paging_attempt	Counter	The number of attempted page procedures.
Mobility Management layer statistics for MME Call Control Service	mmecc_emm_eps_paging_failure	Counter	The number of failed paging procedures
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Mobility Management layer statistics for MME Call Control Service	mmecc_emm_eps_paging_third_retry_attempt	Counter	The number of paging attempt third retry.
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Mobility Management layer statistics for MME Call Control Service	mmecc_emm_intra_mme_tau_success	Counter	The number of successful intra-MME normal tracking area update procedures.
Mobility Management layer statistics for MME Call Control Service	mmecc_emm_intra_mme_tau_attempt_intra_sgw	Counter	The number of attempted intra-MME normal tracking area update procedures that do not require SGW relocation.
Mobility Management layer statistics for MME Call Control Service	mmecc_emm_intra_mme_tau_attempt_inter_sgw	Counter	The number of attempted intra-MME normal tracking area update procedures that require SGW relocation.
Mobility Management layer statistics for MME Call Control Service	mmecc_emm_intra_mme_tau_success_intra_sgw	Counter	The number of successful intra-MME normal tracking area update procedures that do not require SGW relocation.
Mobility Management layer statistics for MME Call Control Service	mmecc_emm_intra_mme_tau_success_inter_sgw	Counter	The number of successful intra-MME normal tracking area update procedures that require SGW relocation.
Mobility Management layer statistics for MME Call Control Service	mmecc_emm_periodic_tau_attempt	Counter	The number of attempted periodic tracking area update procedures.

Group	Metric Name	Type	Synopsis
Mobility Management layer statistics for MME Call Control Service	mmecc_emm_periodic_tau_success	Counter	The number of successful periodic tracking area update procedures.
Mobility Management layer statistics for MME Call Control Service	mmecc_emm_periodic_tau_failure	Counter	The number of failed periodic tracking area update procedures.
Mobility Management layer statistics for MME Call Control Service	mmecc_emm_intra_mme_tau_failure	Counter	The number of failed intra-MME normal tracking area update procedures.
Mobility Management layer statistics for MME Call Control Service	mmecc_emm_intra_mme_tau_success_restrict_dcnr	Counter	The number of successful intra-MME normal tracking area update procedures that are restricted for DCNR subscribers.
Mobility Management layer statistics for MME Call Control Service	mmecc_emm_intra_mme_tau_ignore	Counter	The number of intra-MME normal tracking area update procedures ignored by MME.
Mobility Management layer statistics for MME Call Control Service	mmecc_emm_attach_eps_ignore	Counter	The number of attach requests ignored by MME.
Mobility Management layer statistics for MME Call Control Service	mmecc_emm_attach_eps_success_restrict_dcnr	Counter	The number of successful EPS attach procedures for DCNR capable subscribers that are restricted for NR as Secondary RAT in E-UTRAN by HSS or local configuration
Mobility Management layer statistics for MME Call Control Service	mmecc_emm_guti_reallocation_attempt	Counter	The number of attempted GUTI reallocation procedures.
Mobility Management layer statistics for MME Call Control Service	mmecc_emm_guti_reallocation_success	Counter	The number of successful GUTI reallocation procedures.
Mobility Management layer statistics for MME Call Control Service	mmecc_emm_guti_reallocation_failure	Counter	The number of failed GUTI reallocation procedures.
Mobility Management layer statistics for MME Call Control Service	mmecc_emm_standalone_security_attempt	Counter	The number of attempted standalone security procedures.
Mobility Management layer statistics for MME Call Control Service	mmecc_emm_standalone_security_success	Counter	The number of successful standalone security procedures.
Mobility Management layer statistics for MME Call Control Service	mmecc_emm_standalone_security_failure	Counter	The number of failed standalone security procedures.
Mobility Management layer statistics for MME Call Control Service	mmecc_nas_emm_service_request	Counter	The number of service requests received by MME.
Mobility Management layer statistics for MME Call Control Service	mmecc_nas_emm_service_accept	Counter	The number of service accepts sent by MME.
Mobility Management layer statistics for MME Call Control Service	mmecc_nas_emm_service_reject	Counter	The number of service rejects sent by MME.
Mobility Management layer statistics for MME Call Control Service	mmecc_nas_emm_guti_reallocation_command	Counter	The number of GUTI reallocation command sent by MME.
Mobility Management layer statistics for MME Call Control Service	mmecc_nas_emm_guti_reallocation_complete	Counter	The number of GUTI reallocation complete received by MME.
Mobility Management layer statistics for MME Call Control Service	mmecc_emm_inter_new_mme_tau_attempt	Counter	The number of attempted Inter-MME tracking area update procedures on the new MME side.

Group	Metric Name	Type	Synopsis
Mobility Management layer statistics for MME Call Control Service	mmecc_emm_inter_new_mme_tau_success	Counter	The number of successful Inter-MME tracking area update procedures on the new MME side.
Mobility Management layer statistics for MME Call Control Service	mmecc_emm_inter_new_mme_tau_success_restrict_dcnr	Counter	The number of successful Inter-MME tracking area update procedures on the new MME side for the restricted DCNR subscribers.
Mobility Management layer statistics for MME Call Control Service	mmecc_emm_inter_new_mme_tau_success_restrict_n1	Counter	The number of successful Inter-MME tracking area update procedures on the new MME side for the restricted N1 subscribers.
Mobility Management layer statistics for MME Call Control Service	mmecc_emm_inter_new_mme_tau_ignore	Counter	The number of ignored Inter-MME tracking area update procedures on the new MME side.
Mobility Management layer statistics for MME Call Control Service	mmecc_emm_inter_new_mme_tau_failure	Counter	The number of failed Inter-MME tracking area update procedures on the new MME side.
Mobility Management layer statistics for MME Call Control Service	mmecc_nas_emm_tracking_area_update_request	Counter	The number of tracking area update request messages received by MME from UE.
Mobility Management layer statistics for MME Call Control Service	mmecc_nas_emm_tracking_area_update_accept	Counter	The number of tracking area update accept messages sent from MME to UE.
Mobility Management layer statistics for MME Call Control Service	mmecc_nas_emm_tracking_area_update_complete	Counter	The number of tracking area update complete messages received by MME from UE.
Mobility Management layer statistics for MME Call Control Service	mmecc_nas_emm_tracking_area_update_reject	Counter	The number of tracking area update reject messages sent from MME to UE.
Mobility Management layer statistics for MME Call Control Service	mmecc_nas_emm_periodic_tracking_area_update_request	Counter	The number of periodic tracking area update request messages received by MME from UE.
Mobility Management layer statistics for MME Call Control Service	mmecc_nas_emm_periodic_tracking_area_update_accept	Counter	The number of periodic tracking area update accept messages sent from MME to UE.
Mobility Management layer statistics for MME Call Control Service	mmecc_nas_emm_periodic_tracking_area_update_reject	Counter	The number of periodic tracking area update reject messages sent from MME to UE.
Mobility Management layer statistics for MME Call Control Service	mmecc_emm_inter_old_mme_tau_attempt	Counter	The number of attempted Inter-MME tracking area update procedures on the old MME side.
Mobility Management layer statistics for MME Call Control Service	mmecc_emm_inter_old_mme_tau_success	Counter	The number of successful Inter-MME tracking area update procedures on the old MME side.
Mobility Management layer statistics for MME Call Control Service	mmecc_emm_inter_old_mme_tau_failure	Counter	The number of failed Inter-MME tracking area update procedures on the old MME side.
Mobility Management layer statistics for MME Call Control Service	mmecc_emm_inter_rat_new_mme_tau_attempt	Counter	The number of attempted Inter-RAT tracking area update procedures on the new MME side.
Mobility Management layer statistics for MME Call Control Service	mmecc_emm_inter_rat_new_mme_tau_success	Counter	The number of successful Inter-RAT tracking area update procedures on the new MME side.
Mobility Management layer statistics for MME Call Control Service	mmecc_emm_inter_rat_new_mme_tau_failure	Counter	The number of failed Inter-RAT tracking area update procedures on the new MME side.
Mobility Management layer statistics for MME Call Control Service	mmecc_emm_inter_rat_old_mme_idle_mobility_attempt	Counter	The number of attempted Inter-RAT idle mode mobility procedures on the old MME side.

Group	Metric Name	Type	Synopsis
Mobility Management layer statistics for MME Call Control Service	mmecc_emm_inter_rat_old_mme_idle_mobility_success	Counter	The number of successful Inter-RAT idle mode mobility procedures on the old MME side.
Mobility Management layer statistics for MME Call Control Service	mmecc_emm_inter_rat_old_mme_idle_mobility_failure	Counter	The number of failed Inter-RAT idle mode mobility procedures on the old MME side.
Mobility Management layer statistics for MME Call Control Service	mmecc_mm_procedure_duration	Histogram	MMECC MM procedure execution time in milliseconds

Mobility Management layer statistics for MME Call Control Service

mmecc_nas_emm_attach_reject
Type
64-bit Counter
Description
The number of attach reject messages sent from MME to UE. The attach reject message is defined by TS 24.301 section 8.2.3. Label tai.
Labels
tai
mmecc_nas_emm_attach_request
Type
64-bit Counter
Description
The number of attach request messages received by MME from UE. The attach request message is defined by TS 24.301 section 8.2.4. Label tai.
Labels
tai
mmecc_nas_emm_attach_accept
Type
64-bit Counter
Description
The number of attach accept messages sent from MME to UE. The attach accept message is defined by TS 24.301 section 8.2.1. Label tai.
Labels
tai
mmecc_nas_emm_attach_complete
Type
64-bit Counter
Description
The number of attach complete messages received by MME from UE. The attach complete message is defined by TS 24.301 section 8.2.2. Label tai.
Labels
tai
mmecc_emm_attach_eps_attempt
Type

64-bit Counter

Description

The number of attempted attach requests with attach type "eps attach". The counter is pegged on receiving attach request from the UE. Labels include tai, capability (e.g. dcnr, n1, srvc), and ue_service (e.g. combined, attach-without-pdn, emergency, sms-only, regular).

Labels

tai, capability, ue_service

- mmecc_emm_attach_eps_success

Type

64-bit Counter

Description

The number of successful attach requests with attach type "eps attach". The counter is pegged on successfully completing attach procedure and moving the UE to attached state. Labels include tai, class_name(e.g. gold, silver), capability (e.g. dcnr, n1, srvc), and ue_service (e.g. combined or eps-only, attach-without-pdn, emergency, sms-only, regular).

Labels

tai, class_name, capability, ue_service

- mmecc_emm_attach_eps_success_restrict_n1

Type

64-bit Counter

Description

The number of successful attach requests with attach type "eps attach" with n1 restriction. The counter is pegged on successfully completing attach procedure and moving the UE to attached state. Labels include tai, class_name(e.g. gold, silver), capability (e.g. dcnr, n1, srvc), and ue_service (e.g. combined or eps-only, attach-without-pdn, emergency, sms-only, regular).

Labels

tai, class_name, capability, ue_service

- mmecc_emm_attach_eps_success_visiting_user

Type

64-bit Counter

Description

The number of successful attach requests with attach type "eps attach" for visiting users. The counter is pegged on successfully completing attach procedure and moving the UE to attached state. Labels include class_name(e.g. gold, silver), ue_plmn (imsi plmn), and ue_service (e.g. combined or eps-only, attach-without-pdn, emergency, sms-only, regular).

Labels

class_name, ue_plmn, ue_service

- mmecc_emm_attach_eps_failure

Type

64-bit Counter

Description

The number of failed attach requests with attach type "eps attach". The counter is pegged if attach procedure is aborted/rejected. Labels include tai, result (i.e. cause), reason, class_name(e.g. gold, silver), capability (e.g. dcnr, n1, srvc), and ue_service (e.g. combined, attach-without-pdn, emergency, sms-only, regular).

Labels

tai, result, reason, class_name, capability, ue_service

- mmecc_emm_attach_eps_failure_visiting_user

Type

64-bit Counter

Description

The number of failed attach requests for visiting users. The counter is pegged if attach procedure is aborted/rejected. Labels include class_name(e.g. gold, silver), reason, result (cause), ue_plmn (imsi plmn) and ue_service (e.g. combined, attach-without-pdn, emergency, sms-only, regular).

Labels

class_name, reason, result, ue_plmn, ue_service

- mmecc_nas_emm_ue_detach_request

Type

64-bit Counter

Description

The number of UE detach request messages received by MME from UE. The UE detach request message is defined by TS 23.401 section 8.2.11.1. Label tai.

Labels

tai

- mmecc_nas_emm_ue_detach_accept

Type

64-bit Counter

Description

The number of UE detach accept messages received by MME from UE. The UE detach accept message is defined by TS 23.401 section 8.2.10.1. Label tai.

Labels

tai

- mmecc_nas_emm_network_detach_request

Type

64-bit Counter

Description

The number of Network detach request messages sent from MME to UE. The Network detach request message is defined by TS 23.401 section 8.2.11.2.1. Label tai.

Labels

tai

- mmecc_nas_emm_network_detach_accept

Type

64-bit Counter

Description

The number of Network detach accept messages sent from MME to UE. The Network detach accept message is defined by TS 23.401 section 8.2.10.2. Label tai.

Labels

tai

- mmecc_emm_eps_detach_ue_attempt

Type

64-bit Counter

Description

The number of attempted detach requests initiated by UE. The counter is pegged on receiving detach request from the UE

Labels

tai

- mmecc_emm_eps_detach_ue_success

Type

64-bit Counter

Description

The number of successful detach requests initiated by UE. The counter is pegged on successfully completing detach procedure and moving the UE to detached state.

Labels

tai

- mmecc_emm_eps_detach_ue_attempt_normal

Type

64-bit Counter

Description

The number of attempted detach requests initiated by UE with switch off bit not set. The counter is pegged on receiving detach request from the UE

Labels

tai

- mmecc_emm_eps_detach_ue_attempt_poweroff

Type

64-bit Counter

Description

The number of attempted detach requests initiated by UE with switch off bit set. The counter is pegged on receiving detach request from the UE

Labels

tai

- mmecc_emm_eps_detach_mme_attempt_gw

Type

64-bit Counter

Description

The number of attempted MME initiated EPS detach procedures when the GW deactivates the last PDN connection of the UE. The counter is pegged on receiving Delete Bearer Request from SGW

Labels

tai

- mmecc_emm_eps_detach_mme_success_gw

Type

64-bit Counter

Description

The number of successful MME initiated EPS detach procedures when the GW deactivates the last PDN connection of the UE. The counter is pegged on completing detach procedure and moving the UE to detached state.

Labels

tai

- mmecc_emm_eps_detach_mme_attempt_mrt

Type

64-bit Counter

Description

The number of attempted MME initiated EPS detach procedures when the Mobile Reachable Timer of the UE expires. The counter is pegged during Mobile Reachable Timer expiry

Labels

tai

- mmecc_emm_eps_detach_mme_success_mrt

Type

64-bit Counter

Description

The number of successful MME initiated EPS detach procedures when the Mobile Reachable Timer of the UE expires. The counter is pegged on completing detach procedure and moving the UE to detached state.

Labels

tai

- mmecc_emm_eps_detach_hss_success

Type

64-bit Counter

Description

The number of successful detach requests initiated by HSS. The counter is pegged on successfully completing detach procedure initiated by HSS.

Labels

tai, reason

- mmecc_emm_eps_detach_mme_success

Type

64-bit Counter

Description

The number of successful detach procedures initiated by MME. The counter is pegged on successfully completing detach procedure initiated by MME.

Labels

tai, reason

- mmecc_emm_service_request_attempt

Type

64-bit Counter

Description

The number of attempted service requests initiated by UE. The counter is pegged on receiving service request from the UE

Labels

tai, ue_service

- mmecc_emm_service_request_success

Type

64-bit Counter

Description

The number of successful service requests initiated by UE. The counter is pegged on successfully completing service request procedure.

Labels

tai, ue_service

• mmecc_emm_service_request_failure

Type

64-bit Counter

Description

The number of failed service request procedures. The counter is pegged if the service request procedure is aborted/rejected.

Labels

tai, ue_service, result, reason

• mmecc_emm_control_plane_service_request_attempt

Type

64-bit Counter

Description

The number of attempted control plane service requests initiated by UE. The counter is pegged on receiving control plane service request from the UE

Labels

tai, ue_service, service_type

• mmecc_emm_control_plane_service_request_success

Type

64-bit Counter

Description

The number of successful control plane service requests initiated by UE. The counter is pegged on successfully completing control plane service request procedure.

Labels

tai, ue_service, service_type

• mmecc_emm_control_plane_service_request_failure

Type

64-bit Counter

Description

The number of failed control plane service requests procedures. The counter is pegged if the control plane service request procedure is aborted/rejected.

Labels

tai, ue_service, service_type, result, reason

• mmecc_emm_extended_service_request_attempt

Type

64-bit Counter

Description

The number of attempted extended service requests initiated by UE. The counter is pegged on receiving extended service request from the UE

Labels

tai, ue_service, service_type

• mmecc_emm_extended_service_request_success

Type

64-bit Counter

Description

The number of successful extended service requests initiated by UE. The counter is pegged on successfully completing extended service request procedure.

Labels

tai, ue_service, service_type

- mmecc_emm_extended_service_request_failure

Type

64-bit Counter

Description

The number of failed extended service request procedures. The counter is pegged if the extended service request procedure is aborted/rejected.

Labels

tai, ue_service, service_type, result, reason

- mmecc_emm_eps_paging_attempt

Type

64-bit Counter

Description

The number of attempted page procedures. The counter is pegged on initiating page procedure to UE. Labels include initiator(e.g. gw_sig, gw_data, mme and hss), svc_type(e.g. regular, voice, priority), paging_profile_set and paging_profile.

Labels

initiator, svc_type, paging_profile_set, paging_profile

- mmecc_emm_eps_paging_failure

Type

64-bit Counter

Description

The number of failed paging procedures. The counter is pegged if the paging procedure is aborted/rejected. Labels include initiator(e.g. gw_sig, gw_data, mme and hss), svc_type(e.g. regular, voice, priority), paging_profile_set, paging_profile and reason.

Labels

initiator, svc_type, paging_profile_set, paging_profile, reason

- mmecc_emm_eps_paging_first_request_failure

Type

64-bit Counter

Description

The number of first page request failures while using this paging-profile inside of the first paging occasion. Only UE in eDRX mode can have more than one paging occasion.

Labels

initiator, svc_type, paging_profile_set, paging_profile

- mmecc_emm_eps_paging_first_retry_attempt

Type

64-bit Counter

Description

The number of paging attempt first retry. The counter is pegged when paging attempt fails to get response and first retry starts. Labels include initiator(e.g. gw_sig, gw_data, mme and hss), svc_type(e.g. regular, voice, priority), paging_profile_set, paging_profile and paging_occasion_count.

Labels

initiator, svc_type, paging_profile_set, paging_profile, paging_occasion_count

- mmecc_emm_eps_paging_second_retry_attempt

Type

64-bit Counter

Description

The number of paging attempt second retry. The counter is pegged when paging attempt fails to get response and second retry starts. Labels include initiator(e.g. gw_sig, gw_data, mme and hss), svc_type(e.g. regular, voice, priority), paging_profile_set, paging_profile and paging_occasion_count.

Labels

initiator, svc_type, paging_profile_set, paging_profile, paging_occasion_count

- mmecc_emm_eps_paging_third_retry_attempt

Type

64-bit Counter

Description

The number of paging attempt third retry. The counter is pegged when paging attempt fails to get response and third retry starts. Labels include initiator(e.g. gw_sig, gw_data, mme and hss), svc_type(e.g. regular, voice, priority), paging_profile_set, paging_profile and paging_occasion_count.

Labels

initiator, svc_type, paging_profile_set, paging_profile, paging_occasion_count

- mmecc_emm_eps_paging_fourth_retry_attempt

Type

64-bit Counter

Description

The number of paging attempt fourth retry. The counter is pegged when paging attempt fails to get response and fourth retry starts. Labels include initiator(e.g. gw_sig, gw_data, mme and hss), svc_type(e.g. regular, voice, priority), paging_profile_set, paging_profile and paging_occasion_count.

Labels

initiator, svc_type, paging_profile_set, paging_profile, paging_occasion_count

- mmecc_emm_eps_paging_edrx_ptw_expired

Type

64-bit Counter

Description

The number of times all the paging attempts made for an eDRX paging occasion failed to connect the UE before the paging time window expired. Only when the configured amount of extended-drx max-paging-occasions are exhausted, or the paging-profile retries are exhausted within a single occasion, does it count as an overall paging procedure failure as indicated by paging failure metric.

Labels

initiator, svc_type, paging_profile_set, paging_profile

- mmecc_emm_intra_mme_tau_attempt

Type

64-bit Counter

Description

The number of attempted intra-MME normal tracking area update procedures. The counter is pegged when intra-MME tracking area update procedure is attempted by UE. Labels include tai, capability (e.g. dcnr, capability), and ue_service (e.g. combined, emergency, sms-only, regular).

Labels

tai, capability, ue_service

- mmecc_emm_intra_mme_tau_success

Type

64-bit Counter

Description

The number of successful intra-MME normal tracking area update procedures. The counter is pegged when intra-MME tracking area update procedure is processed successfully. Labels include tai, capability (e.g. dcnr, capability), and ue_service (e.g. combined or eps-only, emergency, sms-only, regular).

Labels

tai, capability, ue_service

- mmecc_emm_intra_mme_tau_attempt_intra_sgw

Type

64-bit Counter

Description

The number of attempted intra-MME normal tracking area update procedures that do not require SGW relocation. The counter is pegged when intra-MME tracking area update procedure is attempted by UE with no SGW relocation. Label tai.

Labels

tai

- mmecc_emm_intra_mme_tau_attempt_inter_sgw

Type

64-bit Counter

Description

The number of attempted intra-MME normal tracking area update procedures that require SGW relocation. The counter is pegged when intra-MME tracking area update procedure is attempted by UE with SGW relocation. Label tai.

Labels

tai

- mmecc_emm_intra_mme_tau_success_intra_sgw

Type

64-bit Counter

Description

The number of successful intra-MME normal tracking area update procedures that do not require SGW relocation. The counter is pegged when intra-MME tracking area update procedure with no SGW relocation is processed successfully. Label tai.

Labels

tai

- mmecc_emm_intra_mme_tau_success_inter_sgw

Type

64-bit Counter

Description

The number of successful intra-MME normal tracking area update procedures that require SGW relocation. The counter is pegged when intra-MME tracking area update procedure with SGW relocation is processed successfully. Label tai.

Labels

tai

- mmecc_emm_periodic_tau_attempt

Type

64-bit Counter

Description

The number of attempted periodic tracking area update procedures. The counter is pegged when periodic tracking area update procedure is attempted by UE. Label tai.

Labels

tai

- mmecc_emm_periodic_tau_success

Type

64-bit Counter

Description

The number of successful periodic tracking area update procedures. The counter is pegged when periodic tracking area update procedure is processed successfully. Label tai.

Labels

tai

- mmecc_emm_periodic_tau_failure

Type

64-bit Counter

Description

The number of failed periodic tracking area update procedures. The counter is pegged when intra-MME tracking area update procedure is aborted/rejected. Labels include tai, result and reason

Labels

tai, result, reason

- mmecc_emm_intra_mme_tau_failure

Type

64-bit Counter

Description

The number of failed intra-MME normal tracking area update procedures. The counter is pegged when intra-MME tracking area update procedure is aborted/rejected. Labels include tai, capability (e.g. dcnr, capability), ue_service (e.g. combined, emergency, sms-only, regular), result (i.e. cause), and reason.

Labels

tai, capability, ue_service, result, reason

- mmecc_emm_intra_mme_tau_success_restrict_dcnr

Type

64-bit Counter

Description

The number of successful intra-MME normal tracking area update procedures that are restricted for DCNR subscribers. The counter is pegged when intra-MME tracking area update procedures that are Restricted for DCNR subscribers is processed successfully by UE. Label tai.

Labels

tai

- mmecc_emm_intra_mme_tau_ignore

Type

64-bit Counter

Description

The number of intra-MME normal tracking area update procedures ignored by MME. The counter is pegged when intra-MME tracking area update procedure is ignored by MME. Labels include tai, capability (e.g. dcnr), ue_service (e.g. combined, emergency, sms-only, regular), and reason.

Labels

tai, capability, ue_service, reason

- mmecc_emm_attach_eps_ignore

Type

64-bit Counter

Description

The number of attach requests ignored by MME. Labels include tai, reason, capability (e.g. dcnr, n1, srvcc), and ue_service (e.g. combined, attach-without-pdn, emergency, sms-only, regular).

Labels

tai, reason, capability, ue_service

- mmecc_emm_attach_eps_success_restrict_dcnr

Type

64-bit Counter

Description

The number of successful EPS attach procedures for DCNR capable subscribers that are restricted for "NR as Secondary RAT in E-UTRAN" by HSS or local configuration. "NR as Secondary RAT in E-UTRAN" access restriction by HSS is defined in TS 29.272 section 7.3.31. Label tai.

Labels

tai

- mmecc_emm_guti_reallocation_attempt

Type

64-bit Counter

Description

The number of attempted GUTI reallocation procedures. The counter is pegged when reallocation is attempted by MME on GUTI lifetime expiry. Label tai.

Labels

tai

- mmecc_emm_guti_reallocation_success

Type

64-bit Counter

Description

The number of successful GUTI reallocation procedures. The counter is pegged when reallocation procedure is processed successfully. Label tai.

Labels

tai

- mmecc_emm_guti_reallocation_failure

Type

64-bit Counter

Description

The number of failed GUTI reallocation procedures. The counter is pegged when GUTI reallocation procedure is aborted. Labels include tai, result and reason

Labels

tai, result, reason

- mmecc_emm_standalone_security_attempt

Type

64-bit Counter

Description

The number of attempted standalone security procedures. The counter is pegged when procedure is attempted by MME on security lifetime expiry. Label tai.

Labels

tai

- mmecc_emm_standalone_security_success

Type

64-bit Counter

Description

The number of successful standalone security procedures. The counter is pegged when security procedure is processed successfully. Label tai.

Labels

tai

- mmecc_emm_standalone_security_failure

Type

64-bit Counter

Description

The number of failed standalone security procedures. The counter is pegged when security procedure is aborted. Labels include tai, result and reason

Labels

tai, result, reason

- mmecc_nas_emm_service_request

Type

64-bit Counter

Description

The number of service requests received by MME. The SERVICE REQUEST message is defined in section 8.2.25 of 3GPP TS 24.301.

Labels

tai

- mmecc_nas_emm_service_accept

Type

64-bit Counter

Description

The number of service accepts sent by MME. The SERVICE ACCEPT message is defined in section 8.2.34 of 3GPP TS 24.301.

Labels

tai

- mmecc_nas_emm_service_reject

Type

64-bit Counter

Description

The number of service rejects sent by MME. The SERVICE REJECT message is defined in section 8.2.24 of 3GPP TS 24.301.

Labels

tai, result

- mmecc_nas_emm_guti_reallocation_command

Type

64-bit Counter

Description

The number of GUTI reallocation command sent by MME. The GUTI REALLOCATION COMMAND message is defined in section 8.2.16 of 3GPP TS 24.301.

Labels

tai

- mmecc_nas_emm_guti_reallocation_complete

Type

64-bit Counter

Description

The number of GUTI reallocation complete received by MME. The GUTI REALLOCATION COMPLETE message is defined in section 8.2.17 of 3GPP TS 24.301.

Labels

tai

- mmecc_emm_inter_new_mme_tau_attempt

Type

64-bit Counter

Description

The number of attempted Inter-MME tracking area update procedures on the new MME side. The ITAU procedure is defined in TS 23.401 section 5.3.3.1. The counter is pegged when Inter-MME tracking area update procedure is attempted on the new MME side. Labels include tai, ue_service (e.g. combined, attach-without-pdn, emergency, sms-only, regular) and capability (e.g. dcnr, n1, srvc).

Labels

tai, ue_service, capability

- mmecc_emm_inter_new_mme_tau_success

Type

64-bit Counter

Description

The number of successful Inter-MME tracking area update procedures on the new MME side. The ITAU procedure is defined in TS 23.401 section 5.3.3.1. The counter is pegged when Inter-MME tracking area update procedure is successful on the new MME side. Labels include tai, peer_ip, ue_service (e.g. combined or eps-only, attach-without-pdn, emergency, sms-only, regular), capability (e.g. dcnr, n1, srvc) and sgw_change (yes, no).

Labels

tai, peer_ip, ue_service, capability, sgw_change

- mmecc_emm_inter_new_mme_tau_success_restrict_dcnr

Type

64-bit Counter

Description

The number of successful Inter-MME tracking area update procedures on the new MME side for the restricted DCNR subscribers. The ITAU procedure is defined in TS 23.401 section 5.3.3.1. The counter is pegged when Inter-MME tracking area update procedure is successful on the new MME side. Labels include tai, peer_ip, ue_service (e.g. combined or eps-only, attach-without-pdn, emergency, sms-only, regular), capability (e.g. dcnr, n1, srvc) and sgw_change (yes, no).

Labels

tai, peer_ip, ue_service, capability, sgw_change

- mmecc_emm_inter_new_mme_tau_success_restrict_n1

Type

64-bit Counter

Description

The number of successful Inter-MME tracking area update procedures on the new MME side for the restricted N1 subscribers. The ITAU procedure is defined in TS 23.401 section 5.3.3.1. The counter is pegged when Inter-MME tracking area update procedure is successful on the new MME side. Labels include tai, peer_ip, ue_service (e.g. combined or eps-only, attach-without-pdn, emergency, sms-only, regular), capability (e.g. dcnr, n1, srvc).

Labels

tai, peer_ip, ue_service, capability

- mmecc_emm_inter_new_mme_tau_ignore

Type

64-bit Counter

Description

The number of ignored Inter-MME tracking area update procedures on the new MME side. The ITAU procedure is defined in TS 23.401 section 5.3.3.1. The counter is pegged when Inter-MME tracking area update procedure is ignored on the new MME side. Labels include tai, peer_ip, ue_service (e.g. combined, attach-without-pdn, emergency, sms-only, regular), capability (e.g. dcnr, n1, srvc) and reason.

Labels

tai, peer_ip, ue_service, capability, reason

- mmecc_emm_inter_new_mme_tau_failure

Type

64-bit Counter

Description

The number of failed Inter-MME tracking area update procedures on the new MME side. The ITAU procedure is defined in TS 23.401 section 5.3.3.1. The counter is pegged when Inter-MME tracking area update procedure is aborted/rejected on the new MME side. Labels include tai, result, reason, peer_ip, capability (e.g. dcnr, n1, srvc) and sgw_change (yes, no).

Labels

tai, result, reason, peer_ip, capability, sgw_change

- mmecc_nas_emm_tracking_area_update_request

Type

64-bit Counter

Description

The number of tracking area update request messages received by MME from UE as defined in 3GPP TS 24.301 section 8.2.29. Label tai.

Labels

tai

- mmecc_nas_emm_tracking_area_update_accept

Type

64-bit Counter

Description

The number of tracking area update accept messages sent from MME to UE as defined in 3GPP TS 24.301 section 8.2.26. Label tai.

Labels

tai

• mmecc_nas_emm_tracking_area_update_complete

Type

64-bit Counter

Description

The number of tracking area update complete messages received by MME from UE as defined in 3GPP TS 24.301 section 8.2.27. Label tai.

Labels

tai

• mmecc_nas_emm_tracking_area_update_reject

Type

64-bit Counter

Description

The number of tracking area update reject messages sent from MME to UE as defined in 3GPP TS 24.301 section 8.2.28. Labels include tai, result (i.e. cause).

Labels

tai, result

• mmecc_nas_emm_periodic_tracking_area_update_request

Type

64-bit Counter

Description

The number of periodic tracking area update request messages received by MME from UE as defined in 3GPP TS 24.301 section 8.2.29. Label tai.

Labels

tai

• mmecc_nas_emm_periodic_tracking_area_update_accept

Type

64-bit Counter

Description

The number of periodic tracking area update accept messages sent from MME to UE as defined in 3GPP TS 24.301 section 8.2.26. Label tai.

Labels

tai

• mmecc_nas_emm_periodic_tracking_area_update_reject

Type

64-bit Counter

Description

The number of periodic tracking area update reject messages sent from MME to UE as defined in 3GPP TS 24.301 section 8.2.28. Labels include tai, result (i.e. cause).

Labels

tai, result

- mmecc_emm_inter_old_mme_tau_attempt

Type

64-bit Counter

Description

The number of Inter-MME tracking area update procedures that are attempted on the old MME side. The ITAU procedure is defined in TS 23.401 section 5.3.3.1 & 5.3.3.2. This counter is pegged when the old MME receives a UE Context Request over S10 interface

Labels

peer_ip, capability

- mmecc_emm_inter_old_mme_tau_success

Type

64-bit Counter

Description

The number of Inter-MME tracking area update procedures that are successful on the old MME side. The ITAU procedure is defined in TS 23.401 section 5.3.3.1 & 5.3.3.2.

Labels

peer_ip, capability

- mmecc_emm_inter_old_mme_tau_failure

Type

64-bit Counter

Description

The number of Inter-MME tracking area update procedures that are failed on the old MME side. The ITAU procedure is defined in TS 23.401 section 5.3.3.1 & 5.3.3.2. This counter is pegged when Inter-MME tracking area update procedure is aborted/rejected on the old MME side.

Labels

result, peer_ip, capability

- mmecc_emm_inter_rat_new_mme_tau_attempt

Type

64-bit Counter

Description

The number of attempted Inter-RAT tracking area update procedures on the new MME side. The ITAU procedure is defined in TS 23.401 section 5.3.3.1. The counter is pegged when Inter-RAT tracking area update procedure is attempted on the new MME side. Labels include tai, ue_service (e.g. combined, attach-without-pdn, emergency, sms-only, regular) and capability (e.g. dcnr, n1, srvc) and peer_type (amf).

Labels

tai, ue_service, capability, peer_type

- mmecc_emm_inter_rat_new_mme_tau_success

Type

64-bit Counter

Description

The number of successful Inter-RAT tracking area update procedures on the new MME side. The ITAU procedure is defined in TS 23.401 section 5.3.3.1. The counter is pegged when Inter-RAT tracking area update procedure is successful on the new MME side. Labels include tai, peer_ip, ue_service (e.g. combined or eps-only, attach-without-pdn, emergency, sms-only, regular), capability (e.g. dcnr, n1, srvc), peer_type (amf) and sgw_change (yes or no)

Labels

tai, peer_ip, ue_service, capability, peer_type, sgw_change

- mmecc_emm_inter_rat_new_mme_tau_failure

Type

64-bit Counter

Description

The number of failed Inter-RAT tracking area update procedures on the new MME side. The ITAU procedure is defined in TS 23.401 section 5.3.3.1. The counter is pegged when Inter-RAT tracking area update procedure is aborted/rejected on the new MME side. Labels include tai, result, reason, peer_ip, capability (e.g. dcnr, n1), peer_type (amf) and sgw_change (yes or no).

Labels

tai, result, reason, peer_ip, capability, peer_type, sgw_change

- mmecc_emm_inter_rat_old_mme_idle_mobility_attempt

Type

64-bit Counter

Description

The number of Inter-RAT idle mode mobility procedures that are attempted on the old MME side. The ITAU procedure is defined in TS 23.401 section 5.3.3.1 & 5.3.3.2. This counter is pegged when the old MME receives a UE Context Request over N26 interface. Labels include peer_ip, capability (e.g. dcnr, n1) and peer_type (amf).

Labels

peer_ip, capability, peer_type

- mmecc_emm_inter_rat_old_mme_idle_mobility_success

Type

64-bit Counter

Description

The number of Inter-RAT idle mode mobility procedures that are successful on the old MME side. The ITAU procedure is defined in TS 23.401 section 5.3.3.1 & 5.3.3.2. Labels include peer_ip, capability (e.g. dcnr, n1, srvc) and peer_type (amf).

Labels

peer_ip, capability, peer_type

- mmecc_emm_inter_rat_old_mme_idle_mobility_failure

Type

64-bit Counter

Description

The number of Inter-RAT idle mode mobility procedures that are failed on the old MME side. The ITAU procedure is defined in TS 23.401 section 5.3.3.1 & 5.3.3.2. This counter is pegged when Inter-MME tracking area update procedure is aborted/rejected on the old MME side. Labels include result, peer_ip, capability (e.g. dcnr, n1, srvc) and peer_type (amf).

Labels

result, peer_ip, capability, peer_type

- mmecc_mm_procedure_duration

Type

64-bit Histogram

Description

The time observed for MMECC MM procedure execution in milliseconds

Labels

procedure,status

Boundaries

[10, 25, 50, 100, 150, 200, 500, 1000, 2000, 20000]

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Summary

Group	Metric Name	Type	Synopsis
eNB layer statistics for MME Call Control Service	mmecc_s1_intra_mme_config_transfer_attempt	Counter	S1 intra-MME Configuration Transfer attempts
eNB layer statistics for MME Call Control Service	mmecc_s1_intra_mme_config_transfer_success	Counter	S1 intra-MME Configuration Transfer success
eNB layer statistics for MME Call Control Service	mmecc_s1_intra_mme_config_transfer_failure	Counter	S1 intra-MME Configuration Transfer failures
eNB layer statistics for MME Call Control Service	mmecc_inter_nodal_source_mme_config_transfer_attempt	Counter	Inter-nodal Configuration Transfer attempts on the source MME
eNB layer statistics for MME Call Control Service	mmecc_inter_nodal_source_mme_config_transfer_failure	Counter	Inter-nodal Configuration Transfer failures on the source MME
eNB layer statistics for MME Call Control Service	mmecc_inter_nodal_target_mme_config_transfer_attempt	Counter	Inter-nodal Configuration Transfer attempts on the target MME
eNB layer statistics for MME Call Control Service	mmecc_inter_nodal_target_mme_config_transfer_failure	Counter	Inter-nodal Configuration Transfer failures on the target MME
eNB layer statistics for MME Call Control Service	mmecc_s1_enb_reset_proc_initiated	Counter	eNB Reset procedures initiated at MME
eNB layer statistics for MME Call Control Service	mmecc_s1_enb_reset_proc_completed	Counter	eNB Reset procedures successfully completed at MME
eNB layer statistics for MME Call Control Service	mmecc_s1_enb_reset_partial_ue_not_found	Counter	Partial eNB Reset ignored for a UE because the UE could not be found

eNB layer statistics for MME Call Control Service

• mmecc_s1_intra_mme_config_transfer_attempt
Type
64-bit Counter
Description
The number of attempted intra-MME Configuration Transfer procedures. The counter is pegged on receiving uplink eNB Configuration Transfer message from eNB.
Labels
• mmecc_s1_intra_mme_config_transfer_success
Type
64-bit Counter
Description
The number of successful intra-MME Configuration Transfer procedures. The counter is pegged on completing the intra-MME Configuration Transfer procedure.
Labels
• mmecc_s1_intra_mme_config_transfer_failure

Type

64-bit Counter

Description

The number of failed intra-MME Configuration Transfer procedures. The counter is pegged if there are any failures processing the intra-MME Configuration Transfer procedure.

Labels

reason

- mmecc_inter_nodal_source_mme_config_transfer_attempt

Type

64-bit Counter

Description

The number of attempted inter-nodal Configuration Transfer procedures on the source MME. The counter is pegged on receiving uplink eNB Configuration Transfer message from eNB. The Configuration Transfer procedure is defined in 3GPP TS 23.401 section 5.14 (over S10 interface) and in 3GPP TS 23.501 section 5.17.7 (over N26 interface). Since Configuration Transfer Tunnel message does not have a response message, success is defined as those Configuration Transfer Tunnel messages which are successfully sent to target MME or AMF.

Labels

peer_type

- mmecc_inter_nodal_source_mme_config_transfer_failure

Type

64-bit Counter

Description

The number of failed inter-nodal Configuration Transfer procedures on the source MME. The counter is pegged if there are any failures in processing the inter-nodal Configuration Transfer procedure. The failure reason can be DNS failure or failure to send Configuration Transfer Tunnel message which is indicated as reason label in string format. The Configuration Transfer procedure is defined in 3GPP TS 23.401 section 5.14 (over S10 interface) and in 3GPP TS 23.501 section 5.17.7 (over N26 interface).

Labels

reason, peer_type

- mmecc_inter_nodal_target_mme_config_transfer_attempt

Type

64-bit Counter

Description

The number of attempted inter-nodal Configuration Transfer procedures on the target MME. The counter is pegged on receiving Configuration Transfer Tunnel message from source MME or AMF. The Configuration Transfer procedure is defined in 3GPP TS 23.401 section 5.14 (over S10 interface) and in 3GPP TS 23.501 section 5.17.7 (over N26 interface). The success is defined as those MME Configuration Transfer messages which are successfully sent to target eNB.

Labels

- mmecc_inter_nodal_target_mme_config_transfer_failure

Type

64-bit Counter

Description

The number of failed inter-nodal Configuration Transfer procedures on the target MME. The counter is pegged if there are any failures in processing the inter-nodal Configuration Transfer procedure. The failure reason can be message decode failure or failure to send MME Configuration Transfer message which is indicated as reason label in string format. The Configuration Transfer procedure is defined in 3GPP TS 23.401 section 5.14 (over S10 interface) and in 3GPP TS 23.501 section 5.17.7 (over N26 interface).

Labels

reason

- mmecc_s1_enb_reset_proc_initiated

Type

64-bit Counter

Description

The number of enb reset procedures initiated at MME. The counter is pegged if the MME successfully records the reset event and triggers processing. The eNB Reset procedure is defined in 3GPP TS 36.413, v15.10.0, sections 8.7.1, 9.1.8.1 and 9.1.8.2.

Labels

globalenbid, trigger

- mmecc_s1_enb_reset_proc_completed

Type

64-bit Counter

Description

The number of enb reset procedures successfully completed at MME. The counter is pegged on successfull completion of the processing of the reset event. The eNB Reset procedure is defined in 3GPP TS 36.413, v15.10.0, sections 8.7.1, 9.1.8.1 and 9.1.8.2.

Labels

globalenbid, trigger

- mmecc_s1_enb_reset_partial_ue_not_found

Type

64-bit Counter

Description

The number of times UE was not found to reset the S1 connection. The counter is pegged if the UE is not found for the S1AP ID present in the eNB Reset message. The eNB Reset procedure is defined in 3GPP TS 36.413, v15.10.0, sections 8.7.1, 9.1.8.1 and 9.1.8.2.

Labels

globalenbid

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- [Session Management layers statistics for MME Call Control Service](#)

Summary

Group	Metric Name	Type	Synopsis
Session Management layers statistics for MME Call Control Service	mmecc_esm_activate_default_bearer_attach_initial_attempt	Counter	Default EPS bearer attach procedure attempts
Session Management layers statistics for MME Call Control Service	mmecc_esm_activate_default_bearer_attach_initial_success	Counter	Default EPS bearer attach procedure success
Session Management layers statistics for MME Call Control Service	mmecc_esm_activate_default_bearer_attach_ho_attempt	Counter	Default EPS bearer attach procedure attempts
Session Management layers statistics for MME Call Control Service	mmecc_esm_activate_default_bearer_attach_ho_success	Counter	Default EPS bearer attach procedure success
Session Management layers statistics for MME Call Control Service	mmecc_esm_activate_default_bearer_attach_ho_failure	Counter	Attach failure due to non-3GPP to 3GPP handover
Session Management layers statistics for MME Call Control Service	mmecc_esm_activate_default_bearer_attach_failure	Counter	Default EPS bearer attach procedure failures
Session Management layers statistics for MME Call Control Service	mmecc_esm_activate_default_bearer_attach_attempt_visiting_user	Counter	Default EPS bearer attach procedure attempts for visiting users

Group	Metric Name	Type	Synopsis
Session Management layers statistics for MME Call Control Service	mmecc_esm_activate_default_bearer_attach_success_visiting_user	Counter	Default EPS bearer attach procedure success for visiting users
Session Management layers statistics for MME Call Control Service	mmecc_esm_activate_default_bearer_attach_success_local_breakout	Counter	Default EPS bearer attach procedure success for visiting users with local breakout
Session Management layers statistics for MME Call Control Service	mmecc_esm_activate_default_bearer_attach_failure_visiting_user	Counter	Default EPS bearer attach procedure failures for visiting user
Session Management layers statistics for MME Call Control Service	mmecc_esm_activate_default_bearer_pdncon_initial_attempt	Counter	Default EPS bearer PDN connection procedure attempts
Session Management layers statistics for MME Call Control Service	mmecc_esm_activate_default_bearer_pdncon_initial_success	Counter	Default EPS bearer PDN connection procedure success
Session Management layers statistics for MME Call Control Service	mmecc_esm_activate_default_bearer_pdncon_ho_attempt	Counter	Default EPS bearer PDN connection procedure attempts
Session Management layers statistics for MME Call Control Service	mmecc_esm_activate_default_bearer_pdncon_ho_success	Counter	Default EPS bearer PDN connection procedure success
Session Management layers statistics for MME Call Control Service	mmecc_esm_activate_default_bearer_pdncon_ho_failure	Counter	Stand alone default PDN connection procedure failure due to non-3GPP to 3GPP handover
Session Management layers statistics for MME Call Control Service	mmecc_esm_activate_default_bearer_pdncon_failure	Counter	Default EPS bearer PDN connection procedure failures
Session Management layers statistics for MME Call Control Service	mmecc_esm_activate_default_bearer_pdncon_attempt_visiting_user	Counter	Default EPS bearer PDN connection procedure attempts for visiting users
Session Management layers statistics for MME Call Control Service	mmecc_esm_activate_default_bearer_pdncon_success_visiting_user	Counter	Default EPS bearer PDN connection procedure success for visiting users
Session Management layers statistics for MME Call Control Service	mmecc_esm_activate_default_bearer_pdncon_success_local_breakout	Counter	Default EPS bearer PDN connection procedure success for visiting users with local breakout
Session Management layers statistics for MME Call Control Service	mmecc_esm_activate_default_bearer_pdncon_failure_visiting_user	Counter	Default EPS bearer PDN connection procedure failures for visiting users
Session Management layers statistics for MME Call Control Service	mmecc_esm_bearer_resource_alloc_ue_attempt	Counter	UE initiated bearer resource allocation attempts
Session Management layers statistics for MME Call Control Service	mmecc_esm_bearer_resource_alloc_ue_success	Counter	UE initiated bearer resource allocation success
Session Management layers statistics for MME Call Control Service	mmecc_esm_bearer_resource_alloc_ue_failure	Counter	UE initiated bearer resource allocation failure
Session Management layers statistics for MME Call Control Service	mmecc_esm_activate_dedicated_bearer_attempt	Counter	Dedicated EPS bearer activation attempts
Session Management layers statistics for MME Call Control Service	mmecc_esm_activate_dedicated_bearer_success	Counter	Dedicated EPS bearer activation success
Session Management layers statistics for MME Call Control Service	mmecc_esm_activate_dedicated_bearer_failure	Counter	Dedicated EPS bearer activation failure
Session Management layers statistics for MME Call Control Service	mmecc_esm_activate_dedicated_bearer_success_visiting_user	Counter	Dedicated EPS bearer activation success for visiting users
Session Management layers statistics for MME Call Control Service	mmecc_esm_deactivate_dedicated_bearer_gw_attempt	Counter	SGW initiated dedicated EPS bearer deactivation attempts
Session Management layers statistics for MME Call Control Service	mmecc_esm_deactivate_dedicated_bearer_gw_success	Counter	SGW initiate dedicated EPS bearer deactivation successes
Session Management layers statistics for MME Call Control Service	mmecc_esm_deactivate_dedicated_bearer_mme_attempt	Counter	MME initiated dedicated EPS bearer deactivation attempts
Session Management layers statistics for MME Call Control Service	mmecc_esm_deactivate_dedicated_bearer_mme_success	Counter	MME initiated dedicated EPS bearer deactivation successes
Session Management layers statistics for MME Call Control Service	mmecc_esm_deactivate_dedicated_bearer_ue_attempt	Counter	UE initiated dedicated EPS bearer deactivation attempts

Group	Metric Name	Type	Synopsis
Session Management layers statistics for MME Call Control Service	mmecc_esm_deactivate_dedicated_bearer_ue_success	Counter	UE initiated dedicated EPS bearer deactivation successes
Session Management layers statistics for MME Call Control Service	mmecc_esm_bearer_resource_mod_ue_attempt	Counter	UE initiated bearer resource modification attempts
Session Management layers statistics for MME Call Control Service	mmecc_esm_bearer_resource_mod_ue_success	Counter	UE initiated bearer resource modification success
Session Management layers statistics for MME Call Control Service	mmecc_esm_bearer_resource_mod_ue_failure	Counter	UE initiated bearer resource modification failure
Session Management layers statistics for MME Call Control Service	mmecc_esm_deactivate_default_bearer_pdndiscon_attempt	Counter	Default EPS pdn disconnect procedure attempts
Session Management layers statistics for MME Call Control Service	mmecc_esm_deactivate_default_bearer_pdndiscon_success	Counter	Default EPS pdn disconnect procedure success
Session Management layers statistics for MME Call Control Service	mmecc_esm_deactivate_default_bearer_pdndiscon_failure	Counter	Default EPS pdn disconnect procedure failures
Session Management layers statistics for MME Call Control Service	mmecc_esm_deactivate_default_bearer_mme	Counter	The number of successful default bearer deactivation procedures initiated by MME.
Session Management layers statistics for MME Call Control Service	mmecc_esm_deactivate_default_bearer_hss	Counter	The number of successful default bearer deactivation procedures initiated by HSS.
Session Management layers statistics for MME Call Control Service	mmecc_esm_deactivate_default_bearer_ue_detach	Counter	The number of successful default bearer deactivation procedures initiated by UE.
Session Management layers statistics for MME Call Control Service	mmecc_esm_deactivate_default_bearer_gw_attempts	Counter	GW initiated pdn disconnect procedure attempts
Session Management layers statistics for MME Call Control Service	mmecc_esm_deactivate_default_bearer_gw_success	Counter	GW initiated pdn disconnect procedure success
Session Management layers statistics for MME Call Control Service	mmecc_esm_sgw_relocation_attempt	Counter	SGW relocation attempts
Session Management layers statistics for MME Call Control Service	mmecc_esm_sgw_relocation_success	Counter	SGW relocation success
Session Management layers statistics for MME Call Control Service	mmecc_esm_active_tau_attempt	Counter	Intra-MME Tau attempts with active flag set
Session Management layers statistics for MME Call Control Service	mmecc_esm_active_tau_success	Counter	Intra-MME Tau procedure success with active flag set
Session Management layers statistics for MME Call Control Service	mmecc_esm_service_request_attempt	Counter	Service request procedure attempts
Session Management layers statistics for MME Call Control Service	mmecc_esm_service_request_success	Counter	Service request procedure success
Session Management layers statistics for MME Call Control Service	mmecc_esm_modify_default_bearer_ue_attempt	Counter	The number of attempted default bearer modification procedures initiated by UE
Session Management layers statistics for MME Call Control Service	mmecc_esm_modify_default_bearer_ue_success	Counter	The number of successful default bearer modification procedures initiated by UE
Session Management layers statistics for MME Call Control Service	mmecc_esm_modify_default_bearer_ue_failure	Counter	The number of failed default bearer modification procedures initiated by UE
Session Management layers statistics for MME Call Control Service	mmecc_esm_modify_default_bearer_pgw_attempt	Counter	The number of attempted default bearer modification procedures initiated by PGW
Session Management layers statistics for MME Call Control Service	mmecc_esm_modify_default_bearer_pgw_success	Counter	The number of successful default bearer modification procedures initiated by PGW

Group	Metric Name	Type	Synopsis
Session Management layers statistics for MME Call Control Service	mmecc_esm_modify_default_bearer_pgw_failure	Counter	The number of failed default bearer modification procedures initiated by PGW
Session Management layers statistics for MME Call Control Service	mmecc_esm_modify_default_bearer_hss_attempt	Counter	The number of attempted default bearer modification procedures initiated by HSS
Session Management layers statistics for MME Call Control Service	mmecc_esm_modify_default_bearer_hss_success	Counter	The number of successful default bearer modification procedures initiated by HSS
Session Management layers statistics for MME Call Control Service	mmecc_esm_modify_default_bearer_hss_failure	Counter	The number of failed default bearer modification procedures initiated by HSS
Session Management layers statistics for MME Call Control Service	mmecc_esm_modify_dedicated_bearer_ue_attempt	Counter	The number of attempted dedicated bearer modification procedures initiated by UE
Session Management layers statistics for MME Call Control Service	mmecc_esm_modify_dedicated_bearer_ue_success	Counter	The number of successful dedicated bearer modification procedures initiated by UE
Session Management layers statistics for MME Call Control Service	mmecc_esm_modify_dedicated_bearer_ue_failure	Counter	The number of failed dedicated bearer modification procedures initiated by UE
Session Management layers statistics for MME Call Control Service	mmecc_esm_modify_dedicated_bearer_pgw_attempt	Counter	The number of attempted dedicated bearer modification procedures initiated by PGW
Session Management layers statistics for MME Call Control Service	mmecc_esm_modify_dedicated_bearer_pgw_success	Counter	The number of successful dedicated bearer modification procedures initiated by PGW
Session Management layers statistics for MME Call Control Service	mmecc_esm_modify_dedicated_bearer_pgw_failure	Counter	The number of failed dedicated bearer modification procedures initiated by PGW
Session Management layers statistics for MME Call Control Service	mmecc_esm_modify_dedicated_bearer_hss_attempt	Counter	The number of attempted dedicated bearer modification procedures initiated by HSS
Session Management layers statistics for MME Call Control Service	mmecc_esm_modify_dedicated_bearer_hss_success	Counter	The number of successful dedicated bearer modification procedures initiated by HSS
Session Management layers statistics for MME Call Control Service	mmecc_esm_modify_dedicated_bearer_hss_failure	Counter	The number of failed dedicated bearer modification procedures initiated by HSS
Session Management layers statistics for MME Call Control Service	mmecc_esm_erab_modification_indication_attempt	Counter	The number of attempted e-Rab modification indication procedures initiated by E-UTRAN
Session Management layers statistics for MME Call Control Service	mmecc_esm_erab_modification_indication_success	Counter	The number of successful e-Rab modification indication procedures initiated by E-UTRAN
Session Management layers statistics for MME Call Control Service	mmecc_esm_erab_modification_indication_failure	Counter	The number of failed e-Rab modification indication procedures
Session Management layers statistics for MME Call Control Service	mmecc_esm_deactivate_dedicated_bearer_wlan_ho_attempt	Counter	SGW initiated dedicated EPS bearer deactivation attempts during 3GPP to non-3GPP HO (Wireless LAN Handover)
Session Management layers statistics for MME Call Control Service	mmecc_esm_deactivate_dedicated_bearer_wlan_ho_success	Counter	SGW initiated dedicated EPS bearer deactivation success during 3GPP to non-3GPP HO (Wireless LAN Handover)
Session Management layers statistics for MME Call Control Service	mmecc_esm_deactivate_default_bearer_wlan_ho_attempts	Counter	SGW initiated pdn disconnect procedure attempts during 3GPP to non-3GPP HO (Wireless LAN Handover)

Group	Metric Name	Type	Synopsis
Session Management layers statistics for MME Call Control Service	mmecc_esm_deactivate_default_bearer_wlan_ho_success	Counter	SGW initiated PDN disconnect procedure success during 3GPP to non-3GPP HO (Wireless LAN Handover)
Session Management layers statistics for MME Call Control Service	mmecc_esm_data_transport_downlink_packets_gw	Counter	The number of downlink GTP-U packets count
Session Management layers statistics for MME Call Control Service	mmecc_esm_data_transport_downlink_octets_gw	Counter	The number of downlink GTP-U octets count
Session Management layers statistics for MME Call Control Service	mmecc_esm_data_transport_uplink_packets_gw	Counter	The number of uplink GTP-U packets count
Session Management layers statistics for MME Call Control Service	mmecc_esm_data_transport_uplink_octets_gw	Counter	The number of uplink GTP-U octets count
Session Management layers statistics for MME Call Control Service	mmecc_s11u_incoming_echo_request	Counter	The number of echo request messages received on the S11-U interface
Session Management layers statistics for MME Call Control Service	mmecc_s11u_incoming_error_indication	Counter	The number of error indication messages received on the S11-U interface
Session Management layers statistics for MME Call Control Service	mmecc_s11u_outgoing_echo_response	Counter	The number of echo response messages sent on the S11-U interface
Session Management layers statistics for MME Call Control Service	mmecc_s11u_outgoing_error_indication	Counter	The number of error indication messages sent on the S11-U interface
Session Management layers statistics for MME Call Control Service	mmecc_s11u_incoming_data_octets_gw	Counter	The number of octets received on the S11-U interface
Session Management layers statistics for MME Call Control Service	mmecc_s11u_incoming_data_packets_gw	Counter	The number of packets received on the S11-U interface
Session Management layers statistics for MME Call Control Service	mmecc_s11u_outgoing_data_octets_gw	Counter	The number of octets sent on the S11-U interface
Session Management layers statistics for MME Call Control Service	mmecc_s11u_outgoing_data_packets_gw	Counter	The number of packets sent on the S11-U interface
Session Management layers statistics for MME Call Control Service	mmecc_sm_procedure_duration	Histogram	MMECC SM procedure execution time in milliseconds

Session Management layers statistics for MME Call Control Service

- mmecc_esm_activate_default_bearer_attach_initial_attempt

Type

64-bit Counter

Description

The number of attempted default bearer activation in attach procedure with request-type initial request. The counter is pegged on sending activate default eps bearer context request to UE. Labels include ue_service (e.g. emergency, regular), pdn_type (requested pdn-type) and data_interface (e.g. s1-u, s11-u).

Labels

ue_service, pdn_type, data_interface
- mmecc_esm_activate_default_bearer_attach_initial_success

Type

64-bit Counter

Description

The number of successful default bearer activation in attach procedures with request-type initial request. The counter is pegged on successful activation of default bearer. Labels include ue_service (e.g. emergency, regular), pdn_type (selected pdn-type) and data_interface (e.g. s1-u, s11-u).

Labels

ue_service, pdn_type, data_interface

- mmecc_esm_activate_default_bearer_attach_ho_attempt

Type

64-bit Counter

Description

The number of attempted default bearer activation in attach procedure with request-type handover. The counter is pegged on sending activate default eps bearer context request to UE.

Labels

- mmecc_esm_activate_default_bearer_attach_ho_success

Type

64-bit Counter

Description

The number of successful default bearer activation in attach procedures with request-type handover. The counter is pegged on successful activation of default bearer.

Labels

- mmecc_esm_activate_default_bearer_attach_ho_failure

Type

64-bit Counter

Description

The counter is incremented when request-type is handover and attach procedure is aborted/rejected.

Labels

result, reason

- mmecc_esm_activate_default_bearer_attach_failure

Type

64-bit Counter

Description

The number of failed default bearer activation in attach procedures. The counter is pegged if activation of default bearer is aborted/rejected. Labels include result, reason, pdn_type and data_interface.

Labels

result, reason, pdn_type, data_interface

- mmecc_esm_activate_default_bearer_attach_attempt_visiting_user

Type

64-bit Counter

Description

The number of attempted default bearer activation in attach procedure for visiting users. The counter is pegged on sending activate default eps bearer context request to UE. Labels include data_interface (e.g. s1-u, s11-u), pdn_type (requested pdn-type), ue_plmn (imsi plmn) and ue_service (e.g. emergency, regular).

Labels

data_interface, pdn_type, ue_plmn, ue_service

- mmecc_esm_activate_default_bearer_attach_success_visiting_user

Type

64-bit Counter

Description

The number of successful default bearer activation in attach procedures for visiting users. The counter is pegged on successful activation of default bearer. Labels include `data_interface` (e.g. s1-u, s11-u), `pdn_type` (requested pdn-type), `ue_plmn` (imsi plmn) and `ue_service` (e.g. emergency, regular).

Labels

data_interface, pdn_type, ue_plmn, ue_service

- `mmecc_esm_activate_default_bearer_attach_success_local_breakout`

Type

64-bit Counter

Description

The number of successful default bearer activation with PGW selected in VPLMN network during attach procedures for visiting users. The counter is pegged on successful activation of default bearer. Labels include `data_interface` (e.g. s1-u, s11-u), `pdn_type` (requested pdn-type), `ue_plmn` (imsi plmn) and `ue_service` (e.g. emergency, regular).

Labels

data_interface, pdn_type, ue_plmn, ue_service

- `mmecc_esm_activate_default_bearer_attach_failure_visting_user`

Type

64-bit Counter

Description

The number of failed default bearer activation in attach procedures for visiting users. The counter is pegged if activation of default bearer is aborted/rejected. Labels include `data_interface` (e.g. s1-u, s11-u), `pdn_type` (requested pdn-type), `reason`, `result` (cause) and `ue_plmn` (imsi plmn).

Labels

data_interface, pdn_type, reason, result, ue_plmn

- `mmecc_esm_activate_default_bearer_pdncon_initial_attempt`

Type

64-bit Counter

Description

The number of attempted PDN connection procedures with request-type initial request. The counter is incremented on receipt of a PDN connectivity request message. Labels include `ue_service` (e.g. emergency, regular), `pdn_type` (requested pdn-type) and `data_interface` (e.g. s1-u, s11-u).

Labels

ue_service, pdn_type, data_interface

- `mmecc_esm_activate_default_bearer_pdncon_initial_success`

Type

64-bit Counter

Description

The number of successful PDN connection procedures with request-type initial request. The counter is incremented when a standalone PDN connectivity request is successfully processed. Labels include `ue_service` (e.g. emergency, regular), `pdn_type` (selected pdn-type) and `data_interface` (e.g. s1-u, s11-u).

Labels

ue_service, pdn_type, data_interface

- `mmecc_esm_activate_default_bearer_pdncon_ho_attempt`

Type

64-bit Counter

Description

The number of attempted PDN connection procedures with request-type handover. The counter is incremented on receipt of a PDN connectivity request message.

Labels

- mmecc_esm_activate_default_bearer_pdncon_ho_success

Type

64-bit Counter

Description

The number of successful PDN connection procedures with request-type handover. The counter is incremented when a standalone PDN connectivity request is successfully processed.

Labels

- mmecc_esm_activate_default_bearer_pdncon_ho_failure

Type

64-bit Counter

Description

The counter is incremented when request-type is handover and a standalone PDN connectivity request is aborted/rejected.

Labels

result, reason

- mmecc_esm_activate_default_bearer_pdncon_failure

Type

64-bit Counter

Description

The number of failed PDN connection procedures. The counter is incremented when a standalone PDN connectivity request is aborted/rejected. Labels include result, reason, pdn_type and data_interface.

Labels

result, reason, pdn_type, data_interface

- mmecc_esm_activate_default_bearer_pdncon_attempt_visiting_user

Type

64-bit Counter

Description

The number of attempted PDN connection procedures for visiting users. The counter is incremented on receipt of a PDN connectivity request message. Labels include data_interface (e.g. s1-u, s11-u), pdn_type (requested pdn-type), ue_plmn (imsi plmn) and ue_service (e.g. emergency, regular).

Labels

data_interface, pdn_type, ue_plmn, ue_service

- mmecc_esm_activate_default_bearer_pdncon_success_visiting_user

Type

64-bit Counter

Description

The number of successful PDN connection procedures for visiting users. The counter is incremented when a standalone PDN connectivity request is successfully processed. Labels include `data_interface` (e.g. s1-u, s11-u), `pdn_type` (requested pdn-type), `ue_plmn` (imsi plmn) and `ue_service` (e.g. emergency, regular).

Labels

data_interface, pdn_type, ue_plmn, ue_service

- `mmecc_esm_activate_default_bearer_pdncon_success_local_breakout`

Type

64-bit Counter

Description

The number of successful PDN connection procedures for visiting users with PGW selected in VPLMN network. The counter is incremented when a standalone PDN connectivity request is successfully processed. Labels include `data_interface` (e.g. s1-u, s11-u), `pdn_type` (requested pdn-type), `ue_plmn` (imsi plmn) and `ue_service` (e.g. emergency, regular).

Labels

data_interface, pdn_type, ue_plmn, ue_service

- `mmecc_esm_activate_default_bearer_pdncon_failure_visiting_user`

Type

64-bit Counter

Description

The number of failed PDN connection procedures for visiting users. The counter is incremented when a standalone PDN connectivity request is aborted/rejected. Labels include `data_interface` (e.g. s1-u, s11-u), `pdn_type` (requested pdn-type), `reason`, `result` (cause) and `ue_plmn` (imsi plmn).

Labels

data_interface, pdn_type, reason, result, ue_plmn

- `mmecc_esm_bearer_resource_alloc_ue_attempt`

Type

64-bit Counter

Description

The number of attempted Bearer Resource Allocation procedures. The counter is pegged after successfully sending a Bearer Resource Command to the SGW.

Labels

- `mmecc_esm_bearer_resource_alloc_ue_success`

Type

64-bit Counter

Description

The number of successful Bearer Resource Allocation procedures. The counter is incremented after receiving a Create/Update Bearer Request in response to the UE Initiated Bearer Resource Allocation procedure.

Labels

- `mmecc_esm_bearer_resource_alloc_ue_failure`

Type

64-bit Counter

Description

The number of failed Bearer Resource Allocation procedures. The counter is incremented when bearer resource allocation procedure is aborted/rejected.

Labels

result, reason

- mmecc_esm_activate_dedicated_bearer_attempt

Type

64-bit Counter

Description

The number of attempted dedicated EPS bearer activation procedures. The counter is pegged on receipt of a Create Bearer Request message from GW.

Labels

- mmecc_esm_activate_dedicated_bearer_success

Type

64-bit Counter

Description

The number of successful dedicated EPS bearer activation procedures. The counter is pegged on transmission of a Create Bearer Response message.

Labels

- mmecc_esm_activate_dedicated_bearer_failure

Type

64-bit Counter

Description

The number of failed dedicated EPS bearer activation procedures. The counter is pegged if activation of dedicated bearer is aborted/rejected.

Labels

result, reason

- mmecc_esm_activate_dedicated_bearer_success_visiting_user

Type

64-bit Counter

Description

The number of successful dedicated EPS bearer activation procedures for visiting users. The counter is pegged on transmission of a Create Bearer Response message. Label - ue_plmn (imsi plmn).

Labels

ue_plmn

- mmecc_esm_deactivate_dedicated_bearer_gw_attempt

Type

64-bit Counter

Description

The number of attempted SGW initiated dedicated EPS bearer deactivation procedures. The counter is pegged on initiation of dedicated bearer deactivation procedure due to reception of standalone Delete Bearer Request from the S-GW.

Labels

- mmecc_esm_deactivate_dedicated_bearer_gw_success

Type

64-bit Counter

Description

The number of successful SGW initiated dedicated EPS bearer deactivation procedures. The counter is pegged on sending a Delete Bearer Response to the S-GW on successful completion of dedicated bearer deactivation procedure initiated due to reception of standalone Delete Bearer Request from SGW.

Labels

- mmecc_esm_deactivate_dedicated_bearer_mme_attempt

Type

64-bit Counter

Description

The number of attempted MME initiated dedicated EPS bearer deactivation procedures. The counter is pegged on initiation of dedicated bearer deactivation procedure by sending a Delete Bearer Command to the S-GW.

Labels

- mmecc_esm_deactivate_dedicated_bearer_mme_success

Type

64-bit Counter

Description

The number of successful dedicated EPS bearer deactivation procedures. The counter is pegged on sending a Delete Bearer Response to the S-GW on successful completion of dedicated bearer deactivation procedure initiated by MME by sending a Delete Bearer Command to the S-GW.

Labels

- mmecc_esm_deactivate_dedicated_bearer_ue_attempt

Type

64-bit Counter

Description

The number of attempted UE initiated dedicated EPS bearer deactivation procedures. The counter is pegged on initiation of dedicated bearer deactivation procedure on reception of a Delete Bearer Request from the S-GW on sending a Bearer Resource Command to the S-GW due to UE triggers.

Labels

- mmecc_esm_deactivate_dedicated_bearer_ue_success

Type

64-bit Counter

Description

The number of successful dedicated EPS bearer deactivation procedures. The counter is pegged on sending a Delete Bearer Response to the S-GW on successful completion of dedicated bearer deactivation procedure initiated on reception of a Delete Bearer Request from the S-GW on sending a Bearer Resource Command to the S-GW due to UE triggers..

Labels

- mmecc_esm_bearer_resource_mod_ue_attempt

Type

64-bit Counter

Description

The number of attempted Bearer Resource Modification procedures. The counter is pegged after successfully sending a Bearer Resource Command to the S-GW.

Labels

- mmecc_esm_bearer_resource_mod_ue_success

Type

64-bit Counter

Description

The number of successful Bearer Resource Modification procedures. The counter is incremented when Bearer Resource Modification procedure is successful.

Labels

- mmecc_esm_bearer_resource_mod_ue_failure

Type

64-bit Counter

Description

The number of failed Bearer Resource Modification procedures. The counter is incremented when bearer resource modification procedure is aborted/rejected.

Labels

result, reason

- mmecc_esm_deactivate_default_bearer_pdndiscon_attempt

Type

64-bit Counter

Description

The number of attempted PDN disconnect procedures. The counter is incremented when a PDN Disconnect Request is received from the UE.

Labels

pdn_type, data_interface

- mmecc_esm_deactivate_default_bearer_pdndiscon_success

Type

64-bit Counter

Description

The number of successful PDN disconnect procedures. The counter is incremented when a PDN Disconnect Request is processed successfully.

Labels

pdn_type, data_interface

- mmecc_esm_deactivate_default_bearer_pdndiscon_failure

Type

64-bit Counter

Description

The number of failed PDN disconnect procedures. The counter is pegged if PDN Disconnect Request is aborted/rejected.

Labels

result, reason, pdn type, data interface

- mmecc esm deactivate default bearer mme

Type

64-bit Counter

Description

The number of default bearer deactivation procedures initiated by MME. The counter is incremented on successful completion of MME initiated default bearer deactivation.

Labels

• mmecc_esm_deactivate_default_bearer_hss

Type

64-bit Counter

Description

The number of default bearer deactivation procedures initiated by HSS. The counter is incremented on successful completion of HSS initiated default bearer deactivation, because of Insert Subscriber Data(ISD) or Delete Subscriber Data(DSD).

Labels

• mmecc_esm_deactivate_default_bearer_ue_detach

Type

64-bit Counter

Description

The number of default bearer deactivation procedures initiated by UE initiated detach. The counter is incremented on successful completion of default bearer deactivation as part of detach.

Labels

• mmecc_esm_deactivate_default_bearer_gw_attempts

Type

64-bit Counter

Description

The number of attempted PDN disconnect procedures. The counter is incremented when a Delete Bearer Request is received from the GW.

Labels

pdn_type, data_interface

• mmecc_esm_deactivate_default_bearer_gw_success

Type

64-bit Counter

Description

The number of successful PDN disconnect procedures. The counter is incremented on successful completion of GW initiated PDN disconnect.

Labels

reason, pdn_type, data_interface

• mmecc_esm_sgw_relocation_attempt

Type

64-bit Counter

Description

The number of attempted SGW relocations. The counter is incremented when SGW relocation is attempted by ESM.

Labels

• mmecc_esm_sgw_relocation_success

Type

64-bit Counter

Description

The number of successful SGW relocations. The counter is incremented when SGW relocation is processed successfully.

Labels

- mmecc_esm_active_tau_attempt

Type

64-bit Counter

Description

The number of attempted Intra-MME Tau with active flag. The counter is incremented when active flag is set for Intra-MME Tau.

Labels

- mmecc_esm_active_tau_success

Type

64-bit Counter

Description

The number of succesful Intra-MME Tau procedures with active flag set. The counter is incremented when Intra-MME Tau with active flag set is processed successfully.

Labels

- mmecc_esm_service_request_attempt

Type

64-bit Counter

Description

The number of attempted service request procedures forwarded from EMM to ESM. The counter is incremented when service request procedure is attempted.

Labels

- mmecc_esm_service_request_success

Type

64-bit Counter

Description

The number of successful service request procedures forwarded from EMM to ESM. The counter is incremented when service request procedure is processed successfully.

Labels

- mmecc_esm_modify_default_bearer_ue_attempt

Type

64-bit Counter

Description

The number of attempted default bearer modification procedures initiated by UE. UE initiated bearer modification procedure is defined in TS 23.401 section 5.4.5

Labels

pdn_type, data_interface

- mmecc_esm_modify_default_bearer_ue_success

Type

64-bit Counter

Description

The number of successful default bearer modification procedures initiated by UE. UE initiated bearer modification procedure is defined in TS 23.401 section 5.4.5

Labels

pdn_type, data_interface

- mmecc_esm_modify_default_bearer_ue_failure

Type

64-bit Counter

Description

The number of failed default bearer modification procedures initiated by UE. UE initiated bearer modification procedure is defined in TS 23.401 section 5.4.5

Labels

result, reason, pdn_type, data_interface

- mmecc_esm_modify_default_bearer_pgw_attempt

Type

64-bit Counter

Description

The number of attempted default bearer modification procedures initiated by PGW. PGW initiated bearer modification procedure is defined in TS 23.401 section 5.4.2.1

Labels

pdn_type, data_interface

- mmecc_esm_modify_default_bearer_pgw_success

Type

64-bit Counter

Description

The number of successful default bearer modification procedures initiated by PGW. PGW initiated bearer modification procedure is defined in TS 23.401 section 5.4.2.1

Labels

pdn_type, data_interface

- mmecc_esm_modify_default_bearer_pgw_failure

Type

64-bit Counter

Description

The number of failed default bearer modification procedures initiated by PGW. PGW initiated bearer modification procedure is defined in TS 23.401 section 5.4.2.1

Labels

result, reason, pdn_type, data_interface

- mmecc_esm_modify_default_bearer_hss_attempt

Type

64-bit Counter

Description

The number of attempted default bearer modification procedures initiated by HSS. HSS initiated bearer modification procedure is defined in TS 23.401 section 5.4.2.2

Labels

- mmecc_esm_modify_default_bearer_hss_success

Type

64-bit Counter

Description

The number of successful default bearer modification procedures initiated by HSS. HSS initiated bearer modification procedure is defined in TS 23.401 section 5.4.2.2

Labels

- mmecc_esm_modify_default_bearer_hss_failure

Type

64-bit Counter

Description

The number of failed default bearer modification procedures initiated by HSS. HSS initiated bearer modification procedure is defined in TS 23.401 section 5.4.2.2

Labels

result, reason

- mmecc_esm_modify_dedicated_bearer_ue_attempt

Type

64-bit Counter

Description

The number of attempted dedicated bearer modification procedures initiated by UE. UE initiated bearer modification procedure is defined in TS 23.401 section 5.4.5

Labels

- mmecc_esm_modify_dedicated_bearer_ue_success

Type

64-bit Counter

Description

The number of successful dedicated bearer modification procedures initiated by UE. UE initiated bearer modification procedure is defined in TS 23.401 section 5.4.5

Labels

- mmecc_esm_modify_dedicated_bearer_ue_failure

Type

64-bit Counter

Description

The number of failed dedicated bearer modification procedures initiated by UE. UE initiated bearer modification procedure is defined in TS 23.401 section 5.4.5

Labels

result, reason

• mmecc_esm_modify_dedicated_bearer_pgw_attempt Type 64-bit Counter Description The number of attempted dedicated bearer modification procedures initiated by PGW. PGW initiated bearer modification procedure is defined in TS 23.401 section 5.4.2.1 Labels
• mmecc_esm_modify_dedicated_bearer_pgw_success Type 64-bit Counter Description The number of successful dedicated bearer modification procedures initiated by PGW. PGW initiated bearer modification procedure is defined in TS 23.401 section 5.4.2.1 Labels
• mmecc_esm_modify_dedicated_bearer_pgw_failure Type 64-bit Counter Description The number of failed dedicated bearer modification procedures initiated by PGW. PGW initiated bearer modification procedure is defined in TS 23.401 section 5.4.2.1 Labels result, reason
• mmecc_esm_modify_dedicated_bearer_hss_attempt Type 64-bit Counter Description The number of attempted dedicated bearer modification procedures initiated by HSS. HSS initiated bearer modification procedure is defined in TS 23.401 section 5.4.2.2 Labels
• mmecc_esm_modify_dedicated_bearer_hss_success Type 64-bit Counter Description The number of successful dedicated bearer modification procedures initiated by HSS. HSS initiated bearer modification procedure is defined in TS 23.401 section 5.4.2.2 Labels
• mmecc_esm_modify_dedicated_bearer_hss_failure Type 64-bit Counter

Description

The number of failed dedicated bearer modification procedures initiated by HSS. HSS initiated bearer modification procedure is defined in TS 23.401 section 5.4.2.2

Labels

result, reason

- mmecc_esm_erab_modification_indication_attempt

Type

64-bit Counter

Description

The number of attempted e-Rab modification indication procedures. E-UTRAN initiated E-RAB modification procedure is defined in TS 23.401 section 5.4.7

Labels

qci

- mmecc_esm_erab_modification_indication_success

Type

64-bit Counter

Description

The number of successful e-Rab modification indication procedures. E-UTRAN initiated E-RAB modification procedure is defined in TS 23.401 section 5.4.7

Labels

qci

- mmecc_esm_erab_modification_indication_failure

Type

64-bit Counter

Description

The number of failed e-Rab modification indication procedures. The counter is pegged if e-Rab modification indication procedure is aborted

Labels

qci

- mmecc_esm_deactivate_dedicated_bearer_wlan_ho_attempt

Type

64-bit Counter

Description

The number of attempted SGW initiated dedicated EPS bearer deactivation procedures during 3GPP to non-3GPP HO (Wireless LAN Handover). The counter is pegged on initiation of dedicated bearer deactivation procedure due to reception of standalone Delete Bearer Request from the SGW during 3GPP to non-3GPP HO.

Labels

-
- mmecc_esm_deactivate_dedicated_bearer_wlan_ho_success

Type

64-bit Counter

Description

The number of successful SGW initiated dedicated EPS bearer deactivation procedures during 3GPP to non-3GPP HO (Wireless LAN Handover). The counter is pegged on sending a Delete Bearer Response to the SGW on successful completion of dedicated bearer deactivation procedure initiated due to reception of standalone Delete Bearer Request from SGW during 3GPP to non-3GPP HO.

Labels

- mmecc_esm_deactivate_default_bearer_wlan_ho_attempts

Type

64-bit Counter

Description

The number of attempted PDN disconnect procedures during 3GPP to non-3GPP HO (Wireless LAN Handover). The counter is incremented when a Delete Bearer Request is received from the SGW during 3GPP to non-3GPP HO.

Labels

- mmecc_esm_deactivate_default_bearer_wlan_ho_success

Type

64-bit Counter

Description

The number of successful PDN disconnect procedures during 3GPP to non-3GPP HO (Wireless LAN Handover). The counter is incremented on successful completion of SGW initiated PDN disconnect during 3GPP to non-3GPP HO.

Labels

- mmecc_esm_data_transport_downlink_packets_gw

Type

64-bit Counter

Description

The number of downlink GTP-U packets received from the SGW to be sent over NAS as IP data in an ESM DATA TRANSPORT message, as defined by TS 24.301 section 6.6.4.

Labels

pdn_type

- mmecc_esm_data_transport_downlink_octets_gw

Type

64-bit Counter

Description

The number of downlink GTP-U octets received from the SGW to be sent over NAS as IP data in an ESM DATA TRANSPORT message, as defined by TS 24.301 section 6.6.4.

Labels

pdn_type

- mmecc_esm_data_transport_uplink_packets_gw

Type

64-bit Counter

Description

The number of uplink GTP-U packets received over NAS in an ESM DATA TRANSPORT message to be sent towards the SGW as IP data, as defined by TS 24.301 section 6.6.4.

Labels

pdn_type

- mmecc_esm_data_transport_uplink_octets_gw

Type

64-bit Counter

Description

The number of uplink GTP-U octets received over NAS in an ESM DATA TRANSPORT message to be sent towards the SGW as IP data, as defined by TS 24.301 section 6.6.4.

Labels

pdn_type

- mmecc_s11u_incoming_echo_request

Type

64-bit Counter

Description

The number of echo request messages received on the S11-U interface from the SGW for data over NAS transport

Labels

- mmecc_s11u_incoming_error_indication

Type

64-bit Counter

Description

The number of error indication messages received on the S11-U interface from the SGW for data over NAS transport

Labels

- mmecc_s11u_outgoing_echo_response

Type

64-bit Counter

Description

The number of echo response messages sent on the S11-U interface to the SGW for data over NAS transport

Labels

- mmecc_s11u_outgoing_error_indication

Type

64-bit Counter

Description

The number of error indication messages sent on the S11-U interface to the SGW for data over NAS transport

Labels

- mmecc_s11u_incoming_data_octects_gw

Type

64-bit Counter

Description

The number of octets received on the S11-U interface from the SGW for data over NAS transport

Labels

• mmecc_s11u_incoming_data_packets_gw
Type
64-bit Counter
Description
The number of packets received on the S11-U interface from the SGW for data over NAS transport
Labels
• mmecc_s11u_outgoing_data_octects_gw
Type
64-bit Counter
Description
The number of octets sent on the S11-U interface to the SGW for data over NAS transport
Labels
• mmecc_s11u_outgoing_data_packets_gw
Type
64-bit Counter
Description
The number of packets sent on the S11-U interface to the SGW for data over NAS transport
Labels
• mmecc_sm_procedure_duration
Type
64-bit Histogram
Description
The time observed for MMECC SM procedure execution in milliseconds
Labels
<i>procedure,status</i>
Boundaries
[10, 25, 50, 100, 150, 200, 500, 1000, 2000, 20000]

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Handover procedure statistics for MME Call Control Service	mmecc_s1_intra_handover_success	Counter	S1 Handover procedure success
Handover procedure statistics for MME Call Control Service	mmecc_s1_intra_handover_failure	Counter	S1 Handover procedure failure

Group	Metric Name	Type	Synopsis
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Handover procedure statistics for MME Call Control Service	mmecc_s1_inter_source_mme_handover_success	Counter	Inter Source MME S1 Handover procedure success
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Handover procedure statistics for MME Call Control Service	mmecc_srvc_source_mme_handover_success	Counter	Source MME SRVCC S1 Handover procedure success
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Group	Metric Name	Type	Synopsis
Handover procedure statistics for MME Call Control Service	mmecc_srvcc_source_mme_handover_cancel	Counter	Source MME SRVCC Handover procedure cancel

Handover procedure statistics for MME Call Control Service

- mmecc_s1_intra_handover_attempt

Type

64-bit Counter

Description

The number of intra-MME S1 Handover attempts with or without SGW change.

Labels

tai

- mmecc_s1_intra_handover_success

Type

64-bit Counter

Description

The number of intra-MME S1 Handovers that are successful with or without SGW change. The intra-MME Handover procedure is defined in TS 23.401 section 5.5.1.2. The counter is pegged when intra-MME S1 Handover procedure is successful. Label includes sgw_change(yes/no)

Labels

tai, sgw_change

- mmecc_s1_intra_handover_failure

Type

64-bit Counter

Description

The number of intra-MME S1 Handovers that failed with or without SGW change. The intra-MME Handover procedure is defined in TS 23.401 section 5.5.1.2. The counter is pegged when intra-MME S1 Handover procedure fails. Label includes sgw_change(yes/no)

Labels

tai, result, reason, sgw_change

- mmecc_s1_intra_handover_cancel

Type

64-bit Counter

Description

The number of intra-MME S1 Handovers that are cancelled. This counter is pegged when Handover Cancel message is received and mmecc_s1_intra_handover_failure counter is not pegged when mmecc_s1_intra_handover_cancel is pegged

Labels

tai, result

- mmecc_s1_inter_source_mme_handover_attempt

Type

64-bit Counter

Description

The number of attempted Inter-MME S1 Handover procedures on the source MME side. The Inter MME Handover procedure is defined in TS 23.401 section 5.5.1.2. The counter is pegged when Inter-MME S1 Handover procedure is attempted on the source MME side. Labels include peer_ip, ue_service (emergency/non

emergency)

Labels

peer_ip, ue_service

- mmecc_s1_inter_source_mme_handover_success

Type

64-bit Counter

Description

The number of successful Inter-MME S1 Handover procedures on the source MME side. The Inter MME Handover procedure is defined in TS 23.401 section 5.5.1.2. The counter is pegged when Inter-MME S1 Handover procedure is successful on the source MME side. Labels include peer_ip, ue_service (emergency/non emergency) and sgw_change (yes/no)

Labels

peer_ip, ue_service, sgw_change

- mmecc_s1_inter_source_mme_handover_failure

Type

64-bit Counter

Description

The number of failed Inter-MME S1 Handover procedures on the source MME side. The Inter MME Handover procedure is defined in TS 23.401 section 5.5.1.2. The counter is pegged when Inter-MME S1 Handover fails on the source MME side. Labels include peer_ip, result, reason, ue_service (emergency/non emergency) and sgw_change (yes/no)

Labels

peer_ip, result, reason, ue_service, sgw_change

- mmecc_s1_inter_source_mme_handover_cancel

Type

64-bit Counter

Description

The number of cancelled Inter-MME S1 Handover procedures on the source MME side. The Inter MME Handover procedure is defined in TS 23.401 section 5.5.1.2. The counter is pegged when Inter-MME S1 Handover procedure is cancelled on the source MME side. Labels include peer_ip, ue_service (emergency/non emergency)

Labels

peer_ip, ue_service

- mmecc_s1_inter_target_mme_handover_attempt

Type

64-bit Counter

Description

The number of attempted Inter-MME S1 Handover procedures on the target MME side. The Inter MME Handover procedure is defined in TS 23.401 section 5.5.1.2. The counter is pegged when Inter-MME S1 Handover procedure is attempted on the target MME side. Labels include peer_ip, ue_service (emergency/non emergency)

Labels

peer_ip, ue_service

- mmecc_s1_inter_target_mme_handover_success

Type

64-bit Counter

Description

The number of successful Inter-MME S1 Handover procedures on the target MME side. The Inter MME Handover procedure is defined in TS 23.401 section 5.5.1.2. The counter is pegged when Inter-MME S1 Handover procedure is successful on the target MME side. Labels include peer_ip, ue_service (emergency/non emergency) and sgw_change (yes/no)

Labels

peer_ip, ue_service, sgw_change

- mmecc_s1_inter_target_mme_handover_failure

Type

64-bit Counter

Description

The number of failed Inter-MME S1 Handover on the target MME side. The Inter MME Handover procedure is defined in TS 23.401 section 5.5.1.2. The counter is pegged when Inter-MME S1 Handover fails on the target MME side. Labels include peer_ip, result, reason, ue_service (emergency/non emergency) and sgw_change (yes/no)

Labels

peer_ip, result, reason, ue_service, sgw_change

- mmecc_s1_inter_target_mme_handover_cancel

Type

64-bit Counter

Description

The number of cancelled Inter-MME S1 Handover procedures on the target MME side. The Inter MME Handover procedure is defined in TS 23.401 section 5.5.1.2. The counter is pegged when Inter-MME S1 Handover procedure is cancelled on the target MME side. Labels include peer_ip, ue_service (emergency/non emergency)

Labels

peer_ip, ue_service

- mmecc_inter_rat_source_mme_handover_attempt

Type

64-bit Counter

Description

The number of attempted Inter-RAT Handover procedures on the source MME side. The Inter-RAT Handover procedure is defined in TS 23.401 section 5.5.2. The counter is pegged when Inter-RAT Handover procedure is attempted on the source MME side. Labels include ue_service (emergency/non emergency) and technology (5gs)

Labels

ue_service, technology

- mmecc_inter_rat_source_mme_handover_success

Type

64-bit Counter

Description

The number of successful Inter-RAT Handover on the source MME side. The Inter-RAT Handover procedure is defined in TS 23.401 section 5.5.2. The counter is pegged when Inter-RAT Handover successful on the source MME side. Labels include peer_ip, ue_service (emergency/non emergency), sgw_change (yes/no) and technology (5gs)

Labels

peer_ip, ue_service, sgw_change, technology

- mmecc_inter_rat_source_mme_handover_failure

Type

64-bit Counter

Description

The number of failed Inter-RAT Handover procedures on the source MME side. The Inter-RAT Handover procedure is defined in TS 23.401 section 5.5.2. The counter is pegged when Inter-RAT Handover procedure is failed on the source MME side. Labels include peer_ip, result, reason, ue_service (emergency/non emergency), sgw_change (yes/no) and technology (5gs)

Labels

peer_ip, result, reason, ue_service, sgw_change, technology

- mmecc_inter_rat_source_mme_handover_cancel

Type

64-bit Counter

Description

The number of cancelled Inter-RAT Handover procedures on the source MME side. The Inter-RAT Handover procedure is defined in TS 23.401 section 5.5.2. The counter is pegged when Inter-RAT Handover procedure is cancelled on the source MME side. Labels include peer_ip, ue_service (emergency/non emergency) and technology (5gs)

Labels

peer_ip, ue_service, technology

- mmecc_inter_rat_target_mme_handover_attempt

Type

64-bit Counter

Description

The number of attempted Inter-RAT Handover procedures on the target MME side. The Inter-RAT Handover procedure is defined in TS 23.401 section 5.5.2. The counter is pegged when Inter-RAT Handover procedure is attempted on the target MME side. Labels include peer_ip, ue_service (emergency/non emergency) and technology (5gs)

Labels

peer_ip, ue_service, technology

- mmecc_inter_rat_target_mme_handover_success

Type

64-bit Counter

Description

The number of successful Inter-RAT Handover procedures on the target MME side. The Inter-RAT Handover procedure is defined in TS 23.401 section 5.5.2. The counter is pegged when Inter-RAT Handover successful on the target MME side. Labels include peer_ip, ue_service (emergency/non emergency), sgw_change (yes/no) and technology (5gs)

Labels

peer_ip, ue_service, sgw_change, technology

- mmecc_inter_rat_target_mme_handover_failure

Type

64-bit Counter

Description

The number of failed Inter-RAT Handover procedures on the target MME side. The Inter-RAT Handover procedure is defined in TS 23.401 section 5.5.2. The counter is pegged when Inter-RAT Handover procedure is failed on the target MME side. Labels include peer_ip, result, reason, ue_service (emergency/non emergency), sgw_change (yes/no) and technology (5gs)

Labels

peer_ip, result, reason, ue_service, sgw_change, technology

- mmecc_inter_rat_target_mme_handover_cancel

Type

64-bit Counter

Description

The number of cancelled Inter-RAT Handover procedures on the target MME side. The Inter-RAT Handover procedure is defined in TS 23.401 section 5.5.2. The counter is pegged when Inter-RAT Handover procedure is cancelled on the target MME side. Labels include peer_ip, ue_service (emergency/non emergency) and technology (5gs)

Labels

peer_ip, ue_service, technology

- mmecc_x2_handover_attempt

Type

64-bit Counter

Description

The number of attempts for X2 handovers.

Labels

tai

- mmecc_x2_handover_success

Type

64-bit Counter

Description

The number of X2 Handovers that succeeded, regardless of whether a TAU request message is received or not.

Labels

tai

- mmecc_x2_handover_failure

Type

64-bit Counter

Description

The number of X2 Handovers that failed. Cause values will be given in decimal. Possible values for reason include mme, sgw, dns.

Labels

tai, result, reason

- mmecc_x2_handover_inter_sgw_attempt

Type

64-bit Counter

Description

The number of attempts of X2 Handovers with SGW change.

Labels

tai

- mmecc_x2_handover_inter_sgw_success

Type

64-bit Counter

Description

The number of X2 Handovers with SGW change that succeeded.

Labels

tai

- mmecc_x2_handover_inter_sgw_failure

Type

64-bit Counter

Description

The number of X2 Handovers with SGW change that failed. Cause values will be given in decimal. Possible values for reason include mme, sgw.

Labels

tai, result, reason

- mmecc_srvc_source_mme_handover_attempt

Type

64-bit Counter

Description

The number of attempted SRVCC Handover procedures on the source MME side. The SRVCC Handover procedure is defined in TS 23.216 section 6.2.2. The counter is pegged when srvc Handover procedure is attempted on the source MME side. Labels include peer_ip, ue_service (emergency/non emergency), ho_type (cs-only-no-dtm/cs-only/ps-cs/ps-cs-only) and technology (gprs/umts)

Labels

peer_ip, ue_service, ho_type, technology

- mmecc_srvc_source_mme_handover_success

Type

64-bit Counter

Description

The number of successful SRVCC Handover procedures on the source MME side. The SRVCC Handover procedure is defined in TS 23.216 section 6.2.2. The counter is pegged when SRVCC S1 Handover procedure is successful on the source MME side. Labels include peer_ip, ue_service (emergency/non emergency), ho_type (cs-only-no-dtm/cs-only/ps-cs/ps-cs-only) and technology (gprs/umts)

Labels

peer_ip, ue_service, ho_type, technology

- mmecc_srvc_source_mme_handover_failure

Type

64-bit Counter

Description

The number of failed SRVCC Handover procedures on the source MME side. The SRVCC Handover procedure is defined in TS 23.216 section 6.2.2. The counter is pegged when SRVCC Handover fails on the source MME side. Labels include peer_ip, result, reason, ue_service (emergency/non emergency), ho_type (cs-only-no-dtm/cs-only/ps-cs/ps-cs-only) and technology (gprs/umts)

Labels

peer_ip, result, reason, ue_service, ho_type, technology

- mmecc_srvc_source_mme_handover_cancel

Type

64-bit Counter

Description

The number of cancelled SRVCC Handover procedures on the source MME side. The SRVCC Handover procedure is defined in TS 23.216 section 6.2.2. The counter is pegged when SRVCC Handover procedure is cancelled on the source MME side. Labels include peer_ip, ue_service (emergency/non emergency), ho_type (cs-only-no-dtm/cs-only/ps-cs/ps-cs-only) and technology (gprs/umts)

Labels

peer_ip, ue_service, ho_type, technology

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Summary

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NAS statistics for MME Call Control Service	mmecc_nas_outgoing_message	Counter	The number of outgoing NAS messages sent by MME.

NAS statistics for MME Call Control Service

- [mmecc_nas_incoming_message](#)

Type

64-bit Counter

Description

The number of incoming NAS messages received by MME. The counter is pegged on receiving NAS message from the UE. Labels include tai and msg_type.

Labels

tai, msg_type

- [mmecc_nas_outgoing_message](#)

Type

64-bit Counter

Description

The number of outgoing NAS messages sent by MME. The counter is pegged on sending NAS message to the UE. Labels include tai and msg_type.

Labels

tai, msg_type

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SGSAP statistics for MME Call Control Service	mmecc_sgsap_ta_to_la_mapping_failure	Counter	The number of times TA to LA mapping not found and default LA is not configured.
SGSAP statistics for MME Call Control Service	mmecc_sgsap_paging_request	Counter	The number of Paging request sent by MSC/VLR for CS call.

SGSAP statistics for MME Call Control Service

- [mmecc_sgsap_default_la_selected](#)

Type

64-bit Counter
Description
The number of times default LA is selected. Label tai.
Labels
tai
mmecc_sgsap_ta_to_la_mapping_failure
Type
64-bit Counter
Description
The number of times TA to LA mapping not found and default LA is not configured. Label tai.
Labels
tai
mmecc_sgsap_paging_request
Type
64-bit Counter
Description
The number of Paging request sent by MSC/VLR for CS call. This metric is instrumented for each CS service type. Label CS service type (CSFB or SMS), vlr-name, vlr-association.
Labels
cs_service_type, vlr_name, vlr_association

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- [Summary](#)
- [Packet Streaming Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Packet Streaming Metrics	num_packet_streaming_attempted	Counter	The total number of attempted streaming of packets
Packet Streaming Metrics	num_packet_streaming_success	Counter	The total number of packets streamed successfully
Packet Streaming Metrics	num_packet_streaming_failed	Counter	The total number of failed streaming of packets
Packet Streaming Metrics	num_packet_streaming_bytes	Counter	The total number of bytes successfully streamed

Packet Streaming Metrics

num_packet_streaming_attempted
Type
64-bit Counter
Description
The total number of attempted streaming of packets
Labels
interface_name, server_fqdn, kafka_topic
num_packet_streaming_success
Type

64-bit Counter

Description

The total number of packets streamed successfully

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_failed

Type

64-bit Counter

Description

The total number of failed streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_bytes

Type

64-bit Counter

Description

The total number of bytes successfully streamed

Labels

interface_name, server_fqdn, kafka_topic

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- [Summary](#)
- [Circuit Breaker Metrics](#)
- [Service Endpoint Learner Metrics](#)
- [Service Endpoint Manager Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Circuit Breaker Metrics	sem_cb_endpoints	Gauge	Total number of endpoints for a service in respective states
Service Endpoint Learner Metrics	sem_num_dns_errors	Counter	Total number of DNS resolution errors for a given service
Service Endpoint Learner Metrics	sem_num_endpoints	Gauge	Total number of endpoints learnt for a service
Service Endpoint Manager Metrics	sem_num_endpoint_transactions_sent	Counter	Total number of times an endpoint is given to the application for communication with it
Service Endpoint Manager Metrics	sem_num_endpoint_transaction_status_received	Counter	Total number of transactions for a given endpoint received with the transaction status

Circuit Breaker Metrics

- sem_cb_endpoints

Type

64-bit Gauge

Description

Total number of endpoints for a service in respective states. There are 3 states viz. Active, Blocked, Restoring. In Active state, the endpoint is responding as expected, i.e. the response code doesn't indicate a service failure. The endpoint will be moved to Blocked state based on response failures which indicate service

failure. After a timeout, the endpoint will be moved to Restoring state to check if the endpoint has recovered. In this state, if endpoint responds with response code other than service failure, it will be moved to Active state. Else, it will be moved back to Blocked state.

Labels

service_name, endpoint_state

Service Endpoint Learner Metrics

- sem_num_dns_errors

Type

64-bit Counter

Description

Total number of DNS resolution errors for a given service, labels are service name for DNS resolution and DNS error codes

Labels

service_name, error_code

- sem_num_endpoints

Type

64-bit Gauge

Description

Total number of endpoints learnt from the Kubernetes DNS server for a given service

Labels

service

Service Endpoint Manager Metrics

- sem_num_endpoint_transactions_sent

Type

64-bit Counter

Description

Total number of times an endpoint is given to the application for communication with it

Labels

service

- sem_num_endpoint_transaction_status_received

Type

64-bit Counter

Description

Total number of transactions for a given endpoint received with the transaction status

Labels

service, status

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- [Summary](#)
- [UDSF DBClient Metrics](#)
- [UDSF API latency statistics](#)

Summary

Group	Metric Name	Type	Synopsis
UDSF DBClient Metrics	udsf_db_operation_request	Counter	The total number of udsf dbclient requests received.
UDSF DBClient Metrics	udsf_db_operation_response	Counter	The total number of udsf dbclient responses sent.
UDSF DBClient Metrics	udsf_dbc_total_init_success	Counter	The total number of UDSF INIT successes.
UDSF DBClient Metrics	udsf_dbc_total_init_failure	Counter	The total number of UDSF INIT failures.
UDSF DBClient Metrics	udsf_dbc_total_deinit_success	Counter	The total number of UDSF DEINIT successes.
UDSF DBClient Metrics	udsf_dbc_total_deinit_failure	Counter	The total number of UDSF DEINIT failures.
UDSF DBClient Metrics	udsf_db_transaction_counter	Counter	The total number of CRUD operations.
UDSF API latency statistics	udsf_db_transaction_duration	Histogram	Roundtrip duration taken by transaction in seconds.
UDSF API latency statistics	udsf_db_execution_duration	Histogram	Execution time taken by transaction in microseconds.
UDSF API latency statistics	udsf_hiredis_bufferwrite_duration	Histogram	Duration of time taken in hiredis to write data to socket in seconds

UDSF DBClient Metrics

- [udsf_db_operation_request](#)

Type

64-bit Counter

Description

The total number of udsf dbclient requests received from worker. Labels- 'app_name' : Originating thread name, ex:smfcc, 'operation_type' : Type of CRUD operation/API type involved, ex:CREATE_OR_MODIFY_RECORD_ASYNC

Labels

app_name, operation_type
- [udsf_db_operation_response](#)

Type

64-bit Counter

Description

The total number of udsf dbclient responses sent. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE 'operation_type' : Type of CRUD operation/API type involved., ex:CREATE_OR_MODIFY_RECORD_ASYNC

Labels

app_name, status, operation_type
- [udsf_dbc_total_init_success](#)

Type

64-bit Counter

Description

The total number of times UDSF API init was success. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: SUCCESS

Labels

app_name, status
- [udsf_dbc_total_init_failure](#)

Type

64-bit Counter

Description

The total number of times UDSF API init failed. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: FAILURE

Labels

app_name, status

• udsf_dbc_total_deinit_success

Type

64-bit Counter

Description

The total number of times UDSF API Deinit succeeded. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: SUCCESS

Labels

app_name, status

• udsf_dbc_total_deinit_failure

Type

64-bit Counter

Description

The total number of times UDSF API Deinit failed. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: FAILURE

Labels

app_name, status

• udsf_db_transaction_counter

Type

64-bit Counter

Description

The total number of times CRUD operations are incrementing. Labels- 'app_name' : Originating thread name, ex:smfcc, 'crud_op' : The categorical CRUD operation in which this sub-transation falls into, ex: CREATE_OR_MODIFY

Labels

app_name, crud_op

UDSF API latency statistics

• udsf_db_transaction_duration

Type

64-bit Histogram

Description

Duration observed on total journey/roundtrip taken by a transaction after receiving the request. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE 'operation_type' : Type of CRUD operation/API type involved., ex:CREATE_OR_MODIFY_RECORD_ASYNC, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE, 'crud_op' : The categorical CRUD operation in which this transation falls into, ex: CREATE_OR_MODIFY

Labels

app_name,operation_type,status,crud_op

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

• udsf_db_execution_duration

Type

64-bit Histogram

Description

Processing time taken by UDSF client. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE 'operation_type' : Type of CRUD operation/API type involved., ex:CREATE_OR_MODIFY_RECORD_ASYNC, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE, 'crud_op' : The categorical CRUD operation in which this transation falls into, ex: CREATE_OR_MODIFY

Labels

app_name,operation_type,status,crud_op

Boundaries

[1, 2, 4, 8, 16, 32, 64, 128, 256, 384, 512, 768, 1024, 1536, 2048, 4096, 8192, 16384]

• udsf_hiredis_bufferwrite_duration

Type

64-bit Histogram

Description

Duration observed between hiredis buffer append and write to socket. This is fair approximation for time spent before data is written to socket as the buffer threshold accounted for append and write is dynamic and not fixed size. Per socket, the buffer append start-time is recorded along with size intended to transmit and end-time recorded once data greater than recorded size is written on the socket

Labels

app_name

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

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- [Summary](#)
- [ETCD Client Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_put_requests	Counter	The total number of ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_requests_sent	Counter	The total number of ETCD Client PUT Requests sent.
ETCD Client Metrics	etcd_put_successes	Counter	The total number of successful ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_failures	Counter	The total number of failed ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_with_lease_requests	Counter	The total number of ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_requests_sent	Counter	The total number of ETCD Client PUT with Lease Requests sent.
ETCD Client Metrics	etcd_put_with_lease_successes	Counter	The total number of successful ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_failures	Counter	The total number of failed ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_range_requests	Counter	The total number of ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_requests_sent	Counter	The total number of ETCD Client Range Requests sent.
ETCD Client Metrics	etcd_range_successes	Counter	The total number of successful ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_failures	Counter	The total number of failed ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_delete_requests	Counter	The total number of ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_requests_sent	Counter	The total number of ETCD Client Range Delete Requests sent.
ETCD Client Metrics	etcd_range_delete_successes	Counter	The total number of successful ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_failures	Counter	The total number of failed ETCD Client Range Delete Requests.

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_txn_requests	Counter	The total number of ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_requests_sent	Counter	The total number of ETCD Client Txn Requests sent.
ETCD Client Metrics	etcd_txn_successes	Counter	The total number of successful ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_failures	Counter	The total number of failed ETCD Client Txn Requests.
ETCD Client Metrics	etcd_watch_requests	Counter	The total number of ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_requests_sent	Counter	The total number of ETCD Client Watch Requests sent.
ETCD Client Metrics	etcd_watch_successes	Counter	The total number of successful ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_failures	Counter	The total number of failed ETCD Client Watch Requests.
ETCD Client Metrics	etcdwatcher_requests	Counter	The total number of ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_requests_sent	Counter	The total number of ETCD Watcher Requests sent.
ETCD Client Metrics	etcdwatcher_successes	Counter	The total number of successful ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_failures	Counter	The total number of failed ETCD Watcher Requests.

ETCD Client Metrics

- **etcd_put_requests**

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests.

Labels

keyPrefix

- **etcd_put_requests_sent**

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests sent.

Labels

keyPrefix

- **etcd_put_successes**

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT Requests.

Labels

keyPrefix

- **etcd_put_failures**

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT Requests.

Labels

keyPrefix, failureReason

• etcd_put_with_lease_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests.

Labels

keyPrefix

• etcd_put_with_lease_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests sent.

Labels

keyPrefix

• etcd_put_with_lease_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT with Lease Requests.

Labels

keyPrefix

• etcd_put_with_lease_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT with Lease Requests.

Labels

keyPrefix, failureReason

• etcd_range_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests.

Labels

keyPrefix

• etcd_range_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests sent.

Labels

keyPrefix

- etcd_range_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Requests.

Labels

keyPrefix, failureReason

- etcd_range_delete_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests.

Labels

keyPrefix

- etcd_range_delete_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests sent.

Labels

keyPrefix

- etcd_range_delete_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Delete Requests.

Labels

keyPrefix

- etcd_range_delete_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Delete Requests.

Labels

keyPrefix, failureReason

- etcd_txn_requests

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests.

Labels

keyPrefix

- etcd_txn_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests sent.

Labels

keyPrefix

- etcd_txn_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Txn Requests.

Labels

keyPrefix

- etcd_txn_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Txn Requests.

Labels

keyPrefix, failureReason

- etcd_watch_requests

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests sent.

Labels

keyPrefix

- etcd_watch_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Watch Requests.

Labels

keyPrefix, failureReason

- etcdwatcher_requests

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests.

Labels

keyPrefix

- etcdwatcher_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests sent.

Labels

keyPrefix

- etcdwatcher_successes
 - Type
 - 64-bit Counter
 - Description
 - The total number of successful ETCD Watcher Requests.
 - Labels
 - keyPrefix**
- etcdwatcher_failures
 - Type
 - 64-bit Counter
 - Description
 - The total number of failed ETCD Watcher Requests.
 - Labels
 - keyPrefix, failureReason**

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- [Summary](#)
- [HTTP Punting Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_sent	Counter	The total number of HTTP requests successfully punted.
HTTP Punting Metrics	punting_http_requests_failed	Counter	The total number of HTTP requests punting failed.
HTTP Punting Metrics	punting_http_responses_received	Counter	The total number of HTTP responses received for the requests punted.
HTTP Punting Metrics	punting_http_responses_forwarded	Counter	The total number of HTTP responses forwarded after punting transaction completion.
HTTP Punting Metrics	punting_http_requests_aborted	Counter	The total number of HTTP requests aborted.
HTTP Punting Metrics	punting_http_request_duration_seconds	Histogram	Punting operation duration in seconds.

HTTP Punting Metrics

- punting_http_requests_sent
 - Type
 - 64-bit Counter
 - Description
 - The total number of HTTP requests successfully punted.
 - Labels
 - message_type**
- punting_http_requests_failed
 - Type

64-bit Counter

Description

The total number of HTTP requests for which the punting failed.

Labels

message_type

- punting_http_responses_received

Type

64-bit Counter

Description

The total number of HTTP responses received for the requests punted.

Labels

message_type, status_code

- punting_http_responses_forwarded

Type

64-bit Counter

Description

The total number of HTTP responses forwarded after punting transaction completion.

Labels

message_type, status_code

- punting_http_requests_aborted

Type

64-bit Counter

Description

The total number of HTTP requests aborted.

Labels

message_type

- punting_http_request_duration_seconds

Type

64-bit Histogram

Description

The punting transaction duration is measured from the time the HTTP request is sent out to the time the response is received for the punting request.

Labels

message_type

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

Table of Contents

- [Summary](#)
- [S1 statistics](#)

Summary

Group	Metric Name	Type	Synopsis
S1 statistics	mmes1_s1_incoming_message	Counter	Incoming S1 message.
S1 statistics	mmes1_s1_outgoing_message	Counter	Outgoing S1 message.
S1 statistics	mmes1_s1_erab_setup_request	Counter	E-RAB Setup Request messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 8.2.1.
S1 statistics	mmes1_s1_erab_setup_response	Counter	E-RAB Setup Response messages sent from eNB to MME as defined in the S1AP standard TS 36.413 section 8.2.1.
S1 statistics	mmes1_s1_erab_modify_request	Counter	E-RAB Modify Request messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 8.2.2.
S1 statistics	mmes1_s1_erab_modify_response	Counter	E-RAB Modify Response messages sent from eNB to MME as defined in the S1AP standard TS 36.413 section 8.2.2.
S1 statistics	mmes1_s1_erab_release_indication	Counter	E-RAB Release Indication messages sent from eNB to MME as defined in the S1AP standard TS 36.413 section 8.2.3.
S1 statistics	mmes1_s1_erab_release_command	Counter	E-RAB Release Command messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 8.2.3.
S1 statistics	mmes1_s1_erab_release_response	Counter	E-RAB Release Response messages sent from eNB to MME as defined in the S1AP standard TS 36.413 section 8.2.3.
S1 statistics	mmes1_s1_erab_modification_indication	Counter	E-RAB Modification Indication messages sent from eNB to MME as defined in the S1AP standard TS 36.413 section 8.2.4.
S1 statistics	mmes1_s1_erab_modification_confirm	Counter	E-RAB Modification Confirm messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 8.2.4.
S1 statistics	mmes1_s1_initial_context_setup_request	Counter	Initial Context Setup Request messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 8.3.1.
S1 statistics	mmes1_s1_initial_context_setup_response	Counter	Initial Context Setup Response messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.3.1.
S1 statistics	mmes1_s1_initial_context_setup_failure	Counter	Initial Context Setup Failure messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.3.1.
S1 statistics	mmes1_s1_ue_context_release_request	Counter	UE Context Release Request messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.3.2.
S1 statistics	mmes1_s1_ue_context_release_command	Counter	UE Context Release Command messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 8.3.3.
S1 statistics	mmes1_s1_ue_context_release_complete	Counter	UE Context Release Complete messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.3.3.
S1 statistics	mmes1_s1_ue_context_modification_request	Counter	UE Context Modification Request messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 8.3.4.
S1 statistics	mmes1_s1_ue_context_modification_response	Counter	UE Context Modification Response messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.3.4.
S1 statistics	mmes1_s1_ue_context_modification_failure	Counter	UE Context Modification Failure messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.3.4.
S1 statistics	mmes1_s1_ue_context_modification_indication	Counter	UE Context Modification Indication messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.3.6.
S1 statistics	mmes1_s1_ue_context_modification_confirm	Counter	UE Context Modification Indication Confirm messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 8.3.6.
S1 statistics	mmes1_s1_handover_required	Counter	Handover Required messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.4.1.
S1 statistics	mmes1_s1_handover_command	Counter	Handover Command messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 8.4.1.
S1 statistics	mmes1_s1_handover_preparation_failure	Counter	Handover Preparation Failure messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 8.4.1.
S1 statistics	mmes1_s1_handover_request	Counter	Handover Request messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 8.4.2.

Group	Metric Name	Type	Synopsis
S1 statistics	mmes1_s1_handover_request_ack	Counter	Handover Request Acknowledge messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.4.2.
S1 statistics	mmes1_s1_handover_failure	Counter	Handover Failure messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.4.2.
S1 statistics	mmes1_s1_handover_notify	Counter	Handover Notify messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.4.3.
S1 statistics	mmes1_s1_path_switch_request	Counter	Path Switch Request messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.4.4.
S1 statistics	mmes1_s1_path_switch_request_ack	Counter	Path Switch Request Acknowledge messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 8.4.4.
S1 statistics	mmes1_s1_path_switch_request_failure	Counter	Path Switch Request Failure messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 8.4.4.
S1 statistics	mmes1_s1_handover_cancel	Counter	Handover Cancel messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.4.5.
S1 statistics	mmes1_s1_handover_cancel_ack	Counter	Handover Cancel Acknowledge messages by MME to eNB as defined in the S1AP standard TS 36.413 section 8.4.5.
S1 statistics	mmes1_s1_enb_status_transfer	Counter	eNB Status Transfer messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.4.6.
S1 statistics	mmes1_s1_mme_status_transfer	Counter	MME Status Transfer messages from MME to eNB as defined in the S1AP standard TS 36.413 section 8.4.7.
S1 statistics	mmes1_s1_paging	Counter	Paging messages by MME to eNB as defined in the S1AP standard TS 36.413 section 8.5.
S1 statistics	mmes1_s1_initial_ue_message	Counter	Initial UE messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.6.2.1.
S1 statistics	mmes1_s1_downlink_nas_transport	Counter	Downlink NAS Transport messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 8.6.2.2.
S1 statistics	mmes1_s1_uplink_nas_transport	Counter	Uplink NAS Transport messages sent from eNB to MME as defined in the S1AP standard TS 36.413 section 8.6.2.3.
S1 statistics	mmes1_s1_nas_non_delivery_indication	Counter	NAS Non Delivery Indication messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.6.2.4.
S1 statistics	mmes1_s1_reset_tx	Counter	Reset messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 8.7.1.
S1 statistics	mmes1_s1_reset_rx	Counter	Reset messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.7.1.
S1 statistics	mmes1_s1_reset_ack_tx	Counter	Reset Acknowledge messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 8.7.1.
S1 statistics	mmes1_s1_reset_ack_rx	Counter	Reset Acknowledge messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.7.1.
S1 statistics	mmes1_s1_error_indication_tx	Counter	Error Indication messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 8.7.2.
S1 statistics	mmes1_s1_error_indication_rx	Counter	Error Indication messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.7.2.
S1 statistics	mmes1_s1_setup_request	Counter	S1 Setup Request messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.7.3.
S1 statistics	mmes1_s1_setup_response	Counter	S1 Setup Response messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 8.7.3.
S1 statistics	mmes1_s1_setup_failure	Counter	S1 Setup Failure messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 8.7.3.
S1 statistics	mmes1_s1_enb_configuration_update	Counter	ENB Configuration Update messages from eNB to MME as defined in the S1AP standard TS 36.413 section 9.1.8.7.
S1 statistics	mmes1_s1_enb_configuration_update_acknowledge	Counter	ENB Configuration Update Acknowledge messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 9.1.8.8.

Group	Metric Name	Type	Synopsis
S1 statistics	mmes1_s1_enb_configuration_update_failure	Counter	ENB Configuration Update Failure messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 9.1.8.9.
S1 statistics	mmes1_s1_mme_configuration_update	Counter	MME Configuration Update messages from MME to eNB as defined in the S1AP standard TS 36.413 section 9.1.8.10.
S1 statistics	mmes1_s1_mme_configuration_update_acknowledge	Counter	MME Configuration Update Acknowledge messages sent by eNB to MME as defined in the S1AP standard TS 36.413 section 9.1.8.11.
S1 statistics	mmes1_s1_mme_configuration_update_failure	Counter	MME Configuration Update Failure messages sent by eNB to MME as defined in the S1AP standard TS 36.413 section 9.1.8.12.
S1 statistics	mmes1_s1_enb_configuration_transfer	Counter	eNB Configuration Transfer messages sent by eNB to MME as defined in the S1AP standard TS 36.413 section 8.15.2.1.
S1 statistics	mmes1_s1_mme_configuration_transfer	Counter	MME Configuration Update Failure messages sent by eNB to MME as defined in the S1AP standard TS 36.413 section 8.16.2.1.
S1 statistics	mmes1_s1_ue_capability_info_indication	Counter	UE Capability Info Indication messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.9.
S1 statistics	mmes1_s1_connection_establishment_indication	Counter	Connection Establishment Indication messages from MME to eNB as defined in the S1AP standard TS 36.413 section 8.3.9.
S1 statistics	mmes1_enb_sctp_association_initiate_attempt	Counter	SCTP association initiation attempts by MME towards the eNB.
S1 statistics	mmes1_enb_sctp_association_initiate_success	Counter	SCTP association initiation attempts by MME towards the eNB that were successful.
S1 statistics	mmes1_enb_takeover_attempt	Counter	eNB Takeover procedures attempted.
S1 statistics	mmes1_enb_takeover_success	Counter	eNB Takeover procedures completed successfully.
S1 statistics	mmes1_enb_takeover_failure	Counter	eNB Takeover procedures that encountered failure.
S1 statistics	mmes1_s1_reroute_nas_request	Counter	Reroute NAS Request messages from MME to eNB as defined in the S1AP standard TS 36.413 section 8.6.2.5.

S1 statistics

- mmes1_s1_incoming_message
 - Type
 - 64-bit Counter
 - Description
 - Incoming S1 message, sent by eNB to MME, as defined in the S1AP standard TS 36.413.
 - Labels
 - globalenbid, msg_type**
- mmes1_s1_outgoing_message
 - Type
 - 64-bit Counter
 - Description
 - Outgoing S1 message, sent by MME to eNB, as defined in the S1AP standard TS 36.413.
 - Labels
 - globalenbid, msg_type**
- mmes1_s1_erab_setup_request
 - Type
 - 64-bit Counter

Description

E-RAB Setup Request messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 8.2.1. This message is sent by the MME and is used to request the eNB to assign resources on Uu and S1 for one or several E-RABs.

Labels

globalenbid

- mmes1_s1_erab_setup_response

Type

64-bit Counter

Description

E-RAB Setup Response messages sent from eNB to MME as defined in the S1AP standard TS 36.413 section 8.2.1. This message is sent by the eNB and is used to report the outcome of the request from the E-RAB SETUP REQUEST message.

Labels

globalenbid

- mmes1_s1_erab_modify_request

Type

64-bit Counter

Description

E-RAB Modify Request messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 8.2.2. This message is sent by the MME and is used to request the eNB to modify the Data Radio Bearers and the allocated resources on Uu and S1 for one or several E-RABs or to change the S-GW as defined in TS 23.401.

Labels

globalenbid

- mmes1_s1_erab_modify_response

Type

64-bit Counter

Description

E-RAB Modify Response messages sent from eNB to MME as defined in the S1AP standard TS 36.413 section 8.2.2. This message is sent by the eNB and is used to report the outcome of the request from the E-RAB MODIFY REQUEST message.

Labels

globalenbid

- mmes1_s1_erab_release_indication

Type

64-bit Counter

Description

E-RAB Release Indication messages sent from eNB to MME as defined in the S1AP standard TS 36.413 section 8.2.3. This message is sent by the eNB and is used to indicate the MME to release one or several E-RABs for one UE.

Labels

globalenbid

- mmes1_s1_erab_release_command

Type

64-bit Counter

Description

E-RAB Release Command messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 8.2.3. This message is sent by the MME and is used to request the eNB to release allocated resources on Uu and S1 for one or several E-RABs.

Labels

globalenbid

- mmes1_s1_erab_release_response

Type

64-bit Counter

Description

E-RAB Release Response messages sent from eNB to MME as defined in the S1AP standard TS 36.413 section 8.2.3. This message is sent by the eNB and is used to report the outcome of the request from the E-RAB RELEASE COMMAND message.

Labels

globalenbid

- mmes1_s1_erab_modification_indication

Type

64-bit Counter

Description

E-RAB Modification Indication messages sent from eNB to MME as defined in the S1AP standard TS 36.413 section 8.2.4. This message is sent by the eNB and is used to request the MME to apply the indicated modification for one or several E-RABs.

Labels

globalenbid

- mmes1_s1_erab_modification_confirm

Type

64-bit Counter

Description

E-RAB Modification Confirm messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 8.2.4. This message is sent by the MME and is used to report the outcome of the request from the E-RAB MODIFICATION INDICATION message.

Labels

globalenbid

- mmes1_s1_initial_context_setup_request

Type

64-bit Counter

Description

Initial Context Setup Request messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 8.3.1. This message is sent by the MME to request the setup of a UE context.

Labels

globalenbid

- mmes1_s1_initial_context_setup_response

Type

64-bit Counter

Description

Initial Context Setup Response messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.3.1. This message is sent by the eNB to confirm the setup of a UE context.

Labels

globalenbid

- mmes1_s1_initial_context_setup_failure

Type

64-bit Counter

Description

Initial Context Setup Failure messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.3.1. This message is sent by the eNB to indicate that the setup of the UE context was unsuccessful.

Labels

cause, globalenbid

- mmes1_s1_ue_context_release_request

Type

64-bit Counter

Description

UE Context Release Request messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.3.2. This message is sent by the eNB to request the release of the UE-associated S1-logical connection over the S1 interface.

Labels

globalenbid

- mmes1_s1_ue_context_release_command

Type

64-bit Counter

Description

UE Context Release Command messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 8.3.3. This message is sent by the MME to request the release of the UE-associated S1-logical connection over the S1 interface.

Labels

globalenbid

- mmes1_s1_ue_context_release_complete

Type

64-bit Counter

Description

UE Context Release Complete messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.3.3. This message is sent by the eNB to confirm the release of the UE-associated S1-logical connection over the S1 interface.

Labels

globalenbid

- mmes1_s1_ue_context_modification_request

Type

64-bit Counter

Description

UE Context Modification Request messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 8.3.4. This message is sent by the MME to provide UE Context information changes to the eNB.

Labels

globalenbid

- mmes1_s1_ue_context_modification_response

Type

64-bit Counter

Description

UE Context Modification Response messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.3.4. This message is sent by the eNB to confirm the performed UE context updates.

Labels

globalenbid

- mmes1_s1_ue_context_modification_failure

Type

64-bit Counter

Description

UE Context Modification Failure messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.3.4. This message is sent by the eNB in case the performed UE context update is not successful.

Labels

cause, globalenbid

- mmes1_s1_ue_context_modification_indication

Type

64-bit Counter

Description

UE Context Modification Indication messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.3.6. This message is sent by the eNB to request the MME to modify the UE Context information.

Labels

globalenbid

- mmes1_s1_ue_context_modification_confirm

Type

64-bit Counter

Description

UE Context Modification Indication Confirm messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 8.3.6. This message is sent by the MME to confirm the modification of the UE Context information.

Labels

globalenbid

- mmes1_s1_handover_required

Type

64-bit Counter

Description

Handover Required messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.4.1. This message is sent by the source eNB to the MME to request the preparation of resources at the target.

Labels

globalenbid

- mmes1_s1_handover_command

Type

64-bit Counter

Description

Handover Command messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 8.4.1. This message is sent by the MME to inform the source eNB that resources for the handover have been prepared at the target side.

Labels

globalenbid

- mmes1_s1_handover_preparation_failure

Type

64-bit Counter

Description

Handover Preparation Failure messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 8.4.1. This message is sent by the MME to inform the source eNB that the Handover Preparation has failed.

Labels

cause, globalenbid

- mmes1_s1_handover_request

Type

64-bit Counter

Description

Handover Request messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 8.4.2. This message is sent by the MME to the target eNB to request the preparation of resources.

Labels

globalenbid

- mmes1_s1_handover_request_ack

Type

64-bit Counter

Description

Handover Request Acknowledge messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.4.2. This message is sent by the target eNB to inform the MME about the prepared resources at the target.

Labels

globalenbid

- mmes1_s1_handover_failure

Type

64-bit Counter

Description

Handover Failure messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.4.2. This message is sent by the target eNB to inform the MME that the preparation of resources has failed.

Labels

cause, globalenbid

- mmes1_s1_handover_notify

Type

64-bit Counter

Description

Handover Notify messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.4.3. This message is sent by the target eNB to inform the MME that the UE has been identified in the target cell and the S1 handover has been completed.

Labels

globalenbid

- mmes1_s1_path_switch_request

Type

64-bit Counter

Description

Path Switch Request messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.4.4. This message is sent by the eNB to request the MME to switch DL GTP tunnel termination point(s) from one end-point to another.

Labels

globalenbid

- mmes1_s1_path_switch_request_ack

Type

64-bit Counter

Description

Path Switch Request Acknowledge messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 8.4.4. This message is sent by the MME to inform the eNB that the path switch has been successfully completed in the EPC.

Labels

globalenbid

- mmes1_s1_path_switch_request_failure

Type

64-bit Counter

Description

Path Switch Request Failure messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 8.4.4. This message is sent by the MME to inform the eNB that a failure has occurred in the EPC during the Path switch request procedure.

Labels

cause, globalenbid

- mmes1_s1_handover_cancel

Type

64-bit Counter

Description

Handover Cancel messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.4.5. This message is sent by the source eNB to the MME to request the cancellation of an ongoing handover.

Labels

globalenbid

- mmes1_s1_handover_cancel_ack

Type

64-bit Counter

Description

Handover Cancel Acknowledge messages by MME to eNB as defined in the S1AP standard TS 36.413 section 8.4.5. This message is sent by the MME to the source eNB to confirm that the ongoing handover was cancelled.

Labels

globalenbid

- mmes1_s1_enb_status_transfer

Type

64-bit Counter

Description

eNB Status Transfer messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.4.6. This message is used to transfer the PDCP-SN/HFN information from the source eNB to the target eNB via the MME during an S1 handover

Labels

globalenbid

- mmes1_s1_mme_status_transfer

Type

64-bit Counter

Description

MME Status Transfer messages from MME to eNB as defined in the S1AP standard TS 36.413 section 8.4.7. This message is used to transfer the PDCP-SN/HFN information from the source eNB to the target eNB via the MME during an S1 Handover

Labels

globalenbid

- mmes1_s1_paging

Type

64-bit Counter

Description

Paging messages by MME to eNB as defined in the S1AP standard TS 36.413 section 8.5. This message is sent by the MME and is used to page a UE in one or several tracking areas.

Labels

globalenbid

- mmes1_s1_initial_ue_message

Type

64-bit Counter

Description

Initial UE messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.6.2.1. This message is sent by the eNB to transfer the initial layer 3 message to the MME over the S1 interface.

Labels

globalenbid

- mmes1_s1_downlink_nas_transport

Type

64-bit Counter

Description

Downlink NAS Transport messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 8.6.2.2. This message is sent by the MME and is used to carry NAS information over the S1 interface.

Labels

globalenbid

• mmes1_s1_uplink_nas_transport

Type

64-bit Counter

Description

Uplink NAS Transport messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.6.2.3. This message is sent by the eNB and is used to carry NAS information over the S1 interface.

Labels

globalenbid

• mmes1_s1_nas_non_delivery_indication

Type

64-bit Counter

Description

NAS Non Delivery Indication messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.6.2.4. This message is sent by the eNB and is used for reporting the non-delivery of a NAS PDU previously received within a DOWNLINK NAS TRANSPORT message over the S1 interface.

Labels

globalenbid

• mmes1_s1_reset_tx

Type

64-bit Counter

Description

Reset messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 8.7.1. This message is sent by the MME and is used to request that the S1 interface, or parts of the S1 interface, to be reset.

Labels

globalenbid

• mmes1_s1_reset_rx

Type

64-bit Counter

Description

Reset messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.7.1. This message is sent by the eNB and is used to request that the S1 interface, or parts of the S1 interface, to be reset.

Labels

globalenbid

• mmes1_s1_reset_ack_tx

Type

64-bit Counter

Description

Reset Acknowledge messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 8.7.1. This message is sent by the MME as a response to a RESET message.

Labels

globalenbid

- mmes1_s1_reset_ack_rx

Type

64-bit Counter

Description

Reset Acknowledge messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.7.1. This message is sent by the eNB as a response to a RESET message.

Labels

globalenbid

- mmes1_s1_error_indication_tx

Type

64-bit Counter

Description

Error Indication messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 8.7.2. This message is sent by the MME and is used to indicate that some error has been detected in the node.

Labels

globalenbid

- mmes1_s1_error_indication_rx

Type

64-bit Counter

Description

Error Indication messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.7.2. This message is sent by the eNB and is used to indicate that some error has been detected in the node.

Labels

globalenbid

- mmes1_s1_setup_request

Type

64-bit Counter

Description

S1 Setup Request messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.7.3. This message is sent by the eNB to transfer information for a TNL association.

Labels

globalenbid

- mmes1_s1_setup_response

Type

64-bit Counter

Description

S1 Setup Response messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 8.7.3. This message is sent by the MME to transfer information for a TNL association.

Labels

globalenbid

- mmes1_s1_setup_failure

Type

64-bit Counter

Description

S1 Setup Failure messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 8.7.3. This message is sent by the eNB to indicate that the setup of the UE context was unsuccessful.

Labels

cause, globalenbid, reason

- mmes1_s1_enb_configuration_update

Type

64-bit Counter

Description

ENB Configuration Update messages from eNB to MME as defined in the S1AP standard TS 36.413 section 9.1.8.7. This message is sent by the eNB to transfer updated information for a TNL association.

Labels

globalenbid

- mmes1_s1_enb_configuration_update_acknowledge

Type

64-bit Counter

Description

ENB Configuration Update Acknowledge messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 9.1.8.8. This message is sent by the MME to acknowledge the eNB transfer updated information for a TNL association.

Labels

globalenbid

- mmes1_s1_enb_configuration_update_failure

Type

64-bit Counter

Description

ENB Configuration Update Failure messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 9.1.8.9. This message is sent by the MME to indicate S1 eNB Configuration Update failure

Labels

cause, globalenbid

- mmes1_s1_mme_configuration_update

Type

64-bit Counter

Description

MME Configuration Update messages from MME to eNB as defined in the S1AP standard TS 36.413 section 9.1.8.10. This message is sent by the MME to transfer updated information for a TNL association.

Labels

globalenbid

- mmes1_s1_mme_configuration_update_acknowledge

Type

64-bit Counter

Description

MME Configuration Update Acknowledge messages sent by eNB to MME as defined in the S1AP standard TS 36.413 section 9.1.8.11. This message is sent by the eNB to acknowledge the MME transfer updated information for a TNL association.

Labels

globalenbid

- mmes1_s1_mme_configuration_update_failure

Type

64-bit Counter

Description

MME Configuration Update Failure messages sent by eNB to MME as defined in the S1AP standard TS 36.413 section 9.1.8.12. This message is sent by the MME to indicate S1 MME Configuration Update failure

Labels

globalenbid

- mmes1_s1_enb_configuration_transfer

Type

64-bit Counter

Description

eNB Configuration Transfer messages sent by eNB to MME as defined in the S1AP standard TS 36.413 section 8.15.2.1. This message is sent by the eNB to transfer SON information to another eNB via S1 and Core Network

Labels

globalenbid

- mmes1_s1_mme_configuration_transfer

Type

64-bit Counter

Description

MME Configuration Transfer messages sent by MME to eNB as defined in the S1AP standard TS 36.413 section 8.16.2.1. This message is sent by the MME to eNB to relay the SON information received in eNB Configuration Transfer message

Labels

globalenbid

- mmes1_s1_ue_capability_info_indication

Type

64-bit Counter

Description

UE Capability Info Indication messages from eNB to MME as defined in the S1AP standard TS 36.413 section 8.9. This message is sent by the eNB to provide UE capability-related information to the MME over the S1 interface.

Labels

globalenbid

- mmes1_s1_connection_establishment_indication

Type

64-bit Counter

Description

Connection Establishment Indication messages from MME to eNB as defined in the S1AP standard TS 36.413 section 8.3.9. This message is sent by the MME to complete the establishment of the UE-associated logical S1-connection, and/or trigger the eNB to obtain and report UE Radio Capability.

Labels

globalenbid

- mmes1_enb_sctp_association_initiate_attempt

Type

64-bit Counter

Description

SCTP association initiation attempts by MME towards the eNB.

Labels

globalenbid

- mmes1_enb_sctp_association_initiate_success

Type

64-bit Counter

Description

SCTP association initiation attempts by MME towards the eNB that were successful.

Labels

globalenbid

- mmes1_enb_takeover_attempt

Type

64-bit Counter

Description

eNB Takeover procedures attempted.

Labels

globalenbid

- mmes1_enb_takeover_success

Type

64-bit Counter

Description

eNB Takeover procedures completed successfully.

Labels

globalenbid

- mmes1_enb_takeover_failure

Type

64-bit Counter

Description

eNB Takeover procedures that encountered failure.

Labels

globalenbid, cause

- mmes1_s1_reroute_nas_request

Type

64-bit Counter

Description

Reroute NAS Request messages from MME to eNB as defined in the S1AP standard TS 36.413 section 8.6.2.5. This message is sent by the MME to the eNB for rerouting of the INITIAL UE MESSAGE message to the MME in the indicated Dedicated Core Network.

Labels

globalenbid

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- [Summary](#)
- [Packet Streaming Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Packet Streaming Metrics	num_packet_streaming_attempted	Counter	The total number of attempted streaming of packets
Packet Streaming Metrics	num_packet_streaming_success	Counter	The total number of packets streamed successfully
Packet Streaming Metrics	num_packet_streaming_failed	Counter	The total number of failed streaming of packets
Packet Streaming Metrics	num_packet_streaming_bytes	Counter	The total number of bytes successfully streamed

Packet Streaming Metrics

- num_packet_streaming_attempted

Type

64-bit Counter

Description

The total number of attempted streaming of packets

Labels

interface_name, server_fqdn, kafka_topic
- num_packet_streaming_success

Type

64-bit Counter

Description

The total number of packets streamed successfully

Labels

interface_name, server_fqdn, kafka_topic
- num_packet_streaming_failed

Type

64-bit Counter

Description

The total number of failed streaming of packets

Labels

interface_name, server_fqdn, kafka_topic
- num_packet_streaming_bytes

Type

64-bit Counter

Description

The total number of bytes successfully streamed

Labels

interface_name, server_fqdn, kafka_topic

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- [Summary](#)
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Summary

Group	Metric Name	Type	Synopsis
Circuit Breaker Metrics	sem_cb_endpoints	Gauge	Total number of endpoints for a service in respective states
Service Endpoint Learner Metrics	sem_num_dns_errors	Counter	Total number of DNS resolution errors for a given service
Service Endpoint Learner Metrics	sem_num_endpoints	Gauge	Total number of endpoints learnt for a service
Service Endpoint Manager Metrics	sem_num_endpoint_transactions_sent	Counter	Total number of times an endpoint is given to the application for communication with it
Service Endpoint Manager Metrics	sem_num_endpoint_transaction_status_received	Counter	Total number of transactions for a given endpoint received with the transaction status

Circuit Breaker Metrics

• [sem_cb_endpoints](#)

Type

64-bit Gauge

Description

Total number of endpoints for a service in respective states. There are 3 states viz. Active, Blocked, Restoring. In Active state, the endpoint is responding as expected, i.e. the response code doesn't indicate a service failure. The endpoint will be moved to Blocked state based on response failures which indicate service failure. After a timeout, the endpoint will be moved to Restoring state to check if the endpoint has recovered. In this state, if endpoint responds with response code other than service failure, it will be moved to Active state. Else, it will be moved back to Blocked state.

Labels

service_name, endpoint_state

Service Endpoint Learner Metrics

• [sem_num_dns_errors](#)

Type

64-bit Counter

Description

Total number of DNS resolution errors for a given service, labels are service name for DNS resolution and DNS error codes

Labels

service_name, error_code

• [sem_num_endpoints](#)

Type

64-bit Gauge

Description

Total number of endpoints learnt from the Kubernetes DNS server for a given service

Labels

service

Service Endpoint Manager Metrics

- `sem_num_endpoint_transactions_sent`

Type

64-bit Counter

Description

Total number of times an endpoint is given to the application for communication with it

Labels

service
- `sem_num_endpoint_transaction_status_received`

Type

64-bit Counter

Description

Total number of transactions for a given endpoint received with the transaction status

Labels

service, status

Table of Contents

- [Summary](#)
- [Unified Load Balancer Endpoint Manager Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Unified Load Balancer Endpoint Manager Metrics	ulb_registration_request_sent	Counter	Total number of times an registration request is sent to an unified load balancer pod
Unified Load Balancer Endpoint Manager Metrics	ulb_registration_response_received	Counter	Total number of times registration response is received from unified load balancer pod
Unified Load Balancer Endpoint Manager Metrics	ulb_deregistration_request_sent	Counter	Total number of times an deregistration request is sent to an unified load balancer pod
Unified Load Balancer Endpoint Manager Metrics	ulb_deregistration_response_received	Counter	Total number of times deregistration response is received from unified load balancer pod
Unified Load Balancer Endpoint Manager Metrics	ulb_heartbeat_request_sent	Counter	Total number of times an heartbeat request is sent to an unified load balancer pod
Unified Load Balancer Endpoint Manager Metrics	ulb_heartbeat_response_received	Counter	Total number of times heartbeat response is received from unified load balancer pod

Unified Load Balancer Endpoint Manager Metrics

- `ulb_registration_request_sent`

Type

64-bit Counter

Description

Total number of times registration request is sent to an unified load balancer pod

Labels

ulbName

• ulb_registration_response_received

Type

64-bit Counter

Description

Total number of times registration response is received from unified load balancer pod

Labels

UlbName, status

• ulb_deregistration_request_sent

Type

64-bit Counter

Description

Total number of times deregistration request is sent to an unified load balancer pod

Labels

ulbName

• ulb_deregistration_response_received

Type

64-bit Counter

Description

Total number of times deregistration response is received from unified load balancer pod

Labels

UlbName, status

• ulb_heartbeat_request_sent

Type

64-bit Counter

Description

Total number of times heartbeat request is sent to an unified load balancer pod

Labels

ulbName

• ulb_heartbeat_response_received

Type

64-bit Counter

Description

Total number of times heartbeat response is received from unified load balancer pod

Labels

UlbName, status

Table of Contents

- [Summary](#)
- [SCTP statistics](#)

Summary

Group	Metric Name	Type	Synopsis
SCTP statistics	sctpActiveEstabs	Counter	The number of times associations have been established.
SCTP statistics	sctpPassiveEstabs	Counter	The number of times associations have been established.
SCTP statistics	sctpAbortedds	Counter	The number of times associations have been aborted.
SCTP statistics	sctpShutdowns	Counter	The number of times associations have been gracefully terminated.
SCTP statistics	sctpOutOfBlues	Counter	The number of out of the blue packets received.
SCTP statistics	sctpChecksumErrors	Counter	The number of SCTP packets received with an invalid checksum.
SCTP statistics	sctpOutCtrlChunks	Counter	The number of SCTP control chunks sent.
SCTP statistics	sctpOutOrderChunks	Counter	The number of SCTP ordered data chunks sent.
SCTP statistics	sctpOutUnorderChunks	Counter	The number of SCTP unordered data chunks sent.
SCTP statistics	sctpInCtrlChunks	Counter	The number of SCTP control chunks received.
SCTP statistics	sctpInOrderChunks	Counter	The number of SCTP ordered data chunks received.
SCTP statistics	sctpInUnorderChunks	Counter	The number of SCTP unordered data chunks received.
SCTP statistics	sctpOutSCTPPacks	Counter	The number of SCTP packets sent.
SCTP statistics	sctpInSCTPPacks	Counter	The number of SCTP packets received.
SCTP statistics	sctpCurrEstab	Gauge	The current number of established associations.

SCTP statistics

- [sctpActiveEstabs](#)

Type

64-bit Counter

Description

The number of times that associations have made a direct transition to the ESTABLISHED state from the COOKIE-ECHOED state: COOKIE-ECHOED -> ESTABLISHED. The upper layer initiated the association attempt.

Labels
- [sctpPassiveEstabs](#)

Type

64-bit Counter

Description

The number of times that associations have made a direct transition to the ESTABLISHED state from the CLOSED state: CLOSED -> ESTABLISHED. The remote endpoint initiated the association attempt.

Labels
- [sctpAbortedds](#)

Type

64-bit Counter

Description

The number of times that associations have made a direct transition to the CLOSED state from any state due to sending or receiving of an ABORT chunk: AnyState --Abort--> CLOSED. Ungraceful termination of the association.

Labels

- **sctpShutdowns**

Type

64-bit Counter

Description

The number of times that associations have made a direct transition to the CLOSED state from either the SHUTDOWN-SENT state or the SHUTDOWN-ACK-SENT state. Graceful termination of the association.

Labels

- **sctpOutOfBlues**

Type

64-bit Counter

Description

The number of out of the blue packets received by the host. An out of the blue packet is an SCTP packet correctly formed, including the proper checksum, but for which the receiver was unable to identify an appropriate association.

Labels

- **sctpChecksumErrors**

Type

64-bit Counter

Description

The number of SCTP packets received with an invalid checksum.

Labels

- **sctpOutCtrlChunks**

Type

64-bit Counter

Description

The number of SCTP control chunks sent (retransmissions are not included). Control chunks are those chunks different from DATA.

Labels

- **sctpOutOrderChunks**

Type

64-bit Counter

Description

The number of SCTP ordered data chunks sent (retransmissions are not included).

Labels

- **sctpOutUnorderChunks**

Type

64-bit Counter

Description

The number of SCTP unordered chunks (data chunks in which the U bit is set to 1) sent (retransmissions are not included).

Labels

• sctpInCtrlChunks

Type

64-bit Counter

Description

The number of SCTP control chunks received (no duplicate chunks included).

Labels

• sctpInOrderChunks

Type

64-bit Counter

Description

The number of SCTP ordered data chunks received (no duplicate chunks included).

Labels

• sctpInUnorderChunks

Type

64-bit Counter

Description

The number of SCTP unordered chunks (data chunks in which the U bit is set to 1) received (no duplicate chunks included).

Labels

• sctpOutSCTPPacks

Type

64-bit Counter

Description

The number of SCTP packets sent. Retransmitted DATA chunks are included.

Labels

• sctpInSCTPPacks

Type

64-bit Counter

Description

The number of SCTP packets received. Duplicates are included.

Labels

• sctpCurrEstab

Type

64-bit Gauge
Description
The number of associations for which the current state is either ESTABLISHED, SHUTDOWN-RECEIVED or SHUTDOWN-PENDING.
Labels

Table of Contents

- [Summary](#)
- [HTTP Punting Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_sent	Counter	The total number of HTTP requests successfully punted.
HTTP Punting Metrics	punting_http_requests_failed	Counter	The total number of HTTP requests punting failed.
HTTP Punting Metrics	punting_http_responses_received	Counter	The total number of HTTP responses received for the requests punted.
HTTP Punting Metrics	punting_http_responses_forwarded	Counter	The total number of HTTP responses forwarded after punting transaction completion.
HTTP Punting Metrics	punting_http_requests_aborted	Counter	The total number of HTTP requests aborted.
HTTP Punting Metrics	punting_http_request_duration_seconds	Histogram	Punting operation duration in seconds.

HTTP Punting Metrics

- [punting_http_requests_sent](#)

Type64-bit CounterDescriptionThe total number of HTTP requests successfully punted.Labels**message_type**
- [punting_http_requests_failed](#)

Type64-bit CounterDescriptionThe total number of HTTP requests for which the punting failed.Labels**message_type**
- [punting_http_responses_received](#)

Type64-bit CounterDescriptionThe total number of HTTP responses received for the requests punted.

Labels

message_type, status_code

- `punting_http_responses_forwarded`

Type

64-bit Counter

Description

The total number of HTTP responses forwarded after punting transaction completion.

Labels

message_type, status_code

- `punting_http_requests_aborted`

Type

64-bit Counter

Description

The total number of HTTP requests aborted.

Labels

message_type

- `punting_http_request_duration_seconds`

Type

64-bit Histogram

Description

The punting transaction duration is measured from the time the HTTP request is sent out to the time the response is received for the punting request.

Labels

message_type

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

Table of Contents

- [Summary](#)
- [VLR msg statistics for MME Sgs Service](#)

Summary

Group	Metric Name	Type	Synopsis
VLR msg statistics for MME Sgs Service	mmesgs_sgsap_outgoing_messages	Counter	Outgoing messages to VLR over MME SGS interface
VLR msg statistics for MME Sgs Service	mmesgs_sgsap_incoming_messages	Counter	Incoming messages from VLR over MME SGS interface
VLR msg statistics for MME Sgs Service	mmesgs_sgsap_message_timeout	Counter	Message response timeout from VLR
VLR msg statistics for MME Sgs Service	mmesgs_sgsap_outgoing_octets	Counter	The number of outgoing SGs octets(bytes).
VLR msg statistics for MME Sgs Service	mmesgs_sgsap_outgoing_pdus	Counter	The number of outgoing SGs PDUs.
VLR msg statistics for MME Sgs Service	mmesgs_sgsap_incoming_octets	Counter	The number of incoming SGs octets(bytes).
VLR msg statistics for MME Sgs Service	mmesgs_sgsap_incoming_pdus	Counter	The number of incoming SGs PDUs.

VLR msg statistics for MME Sgs Service

- `mmesgs_sgsap_outgoing_messages`

Type

64-bit Counter

Description

The number of outgoing SGsAP messages sent to the MSC/VLR.

Labels

msg_type, vlr_name, vlr_association

• mmesgs_sgsap_incoming_messages

Type

64-bit Counter

Description

The number of incoming SGsAP messages received from MSC/VLR.

Labels

msg_type, vlr_name, vlr_association

• mmesgs_sgsap_message_timeout

Type

64-bit Counter

Description

The number of timeouts that has occurred due to MSC/VLR not responding to SGsAP message from MME

Labels

msg_type, vlr_name, vlr_association

• mmesgs_sgsap_outgoing_octets

Type

64-bit Counter

Description

The number of outgoing SGs octets(bytes) sent to the MSC/VLR over the SCTP association

Labels

vlr_name, vlr_association

• mmesgs_sgsap_outgoing_pdus

Type

64-bit Counter

Description

The number of outgoing SGs PDU(s) sent to the MSC/VLR over the SCTP association

Labels

vlr_name, vlr_association

• mmesgs_sgsap_incoming_octets

Type

64-bit Counter

Description

The number of incoming SGs octets(bytes) received from the MSC/VLR over the SCTP association

Labels

vlr_name, vlr_association

- mmesgs_sgsap_incoming_pdus

Type

64-bit Counter

Description

The number of incoming SGs PDU(s) received from the MSC/VLR over the SCTP association

Labels

vlr_name, vlr_association

Table of Contents

- [Summary](#)
- [Packet Streaming Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Packet Streaming Metrics	num_packet_streaming_attempted	Counter	The total number of attempted streaming of packets
Packet Streaming Metrics	num_packet_streaming_success	Counter	The total number of packets streamed successfully
Packet Streaming Metrics	num_packet_streaming_failed	Counter	The total number of failed streaming of packets
Packet Streaming Metrics	num_packet_streaming_bytes	Counter	The total number of bytes successfully streamed

Packet Streaming Metrics

- num_packet_streaming_attempted

Type

64-bit Counter

Description

The total number of attempted streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_success

Type

64-bit Counter

Description

The total number of packets streamed successfully

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_failed

Type

64-bit Counter

Description

The total number of failed streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_bytes

Type

64-bit Counter

Description

The total number of bytes successfully streamed

Labels

interface_name, server_fqdn, kafka_topic

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- [Summary](#)
- [Circuit Breaker Metrics](#)
- [Service Endpoint Learner Metrics](#)
- [Service Endpoint Manager Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Circuit Breaker Metrics	sem_cb_endpoints	Gauge	Total number of endpoints for a service in respective states
Service Endpoint Learner Metrics	sem_num_dns_errors	Counter	Total number of DNS resolution errors for a given service
Service Endpoint Learner Metrics	sem_num_endpoints	Gauge	Total number of endpoints learnt for a service
Service Endpoint Manager Metrics	sem_num_endpoint_transactions_sent	Counter	Total number of times an endpoint is given to the application for communication with it
Service Endpoint Manager Metrics	sem_num_endpoint_transaction_status_received	Counter	Total number of transactions for a given endpoint received with the transaction status

Circuit Breaker Metrics

- sem_cb_endpoints

Type

64-bit Gauge

Description

Total number of endpoints for a service in respective states. There are 3 states viz. Active, Blocked, Restoring. In Active state, the endpoint is responding as expected, i.e. the response code doesn't indicate a service failure. The endpoint will be moved to Blocked state based on response failures which indicate service failure. After a timeout, the endpoint will be moved to Restoring state to check if the endpoint has recovered. In this state, if endpoint responds with response code other than service failure, it will be moved to Active state. Else, it will be moved back to Blocked state.

Labels

service_name, endpoint_state

Service Endpoint Learner Metrics

- sem_num_dns_errors

Type

64-bit Counter

Description

Total number of DNS resolution errors for a given service, labels are service name for DNS resolution and DNS error codes

Labels

service_name, error_code

- sem_num_endpoints

Type

64-bit Gauge

Description

Total number of endpoints learnt from the Kubernetes DNS server for a given service

Labels

service

Service Endpoint Manager Metrics

- sem_num_endpoint_transactions_sent

Type

64-bit Counter

Description

Total number of times an endpoint is given to the application for communication with it

Labels

service

- sem_num_endpoint_transaction_status_received

Type

64-bit Counter

Description

Total number of transactions for a given endpoint received with the transaction status

Labels

service, status

Table of Contents

- [Summary](#)
- [Unified Load Balancer Endpoint Manager Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Unified Load Balancer Endpoint Manager Metrics	ulb_registration_request_sent	Counter	Total number of times an registration request is sent to an unified load balancer pod
Unified Load Balancer Endpoint Manager Metrics	ulb_registration_response_received	Counter	Total number of times registration response is received from unified load balancer pod
Unified Load Balancer Endpoint Manager Metrics	ulb_deregistration_request_sent	Counter	Total number of times an deregistration request is sent to an unified load balancer pod
Unified Load Balancer Endpoint Manager Metrics	ulb_deregistration_response_received	Counter	Total number of times deregistration response is received from unified load balancer pod
Unified Load Balancer Endpoint Manager Metrics	ulb_heartbeat_request_sent	Counter	Total number of times an heartbeat request is sent to an unified load balancer pod
Unified Load Balancer Endpoint Manager Metrics	ulb_heartbeat_response_received	Counter	Total number of times heartbeat response is received from unified load balancer pod

Unified Load Balancer Endpoint Manager Metrics

- ulb_registration_request_sent
 - Type
 - 64-bit Counter
 - Description
 - Total number of times registration request is sent to an unified load balancer pod
 - Labels
 - ulbName**
- ulb_registration_response_received
 - Type
 - 64-bit Counter
 - Description
 - Total number of times registration response is received from unified load balancer pod
 - Labels
 - UlbName, status**
- ulb_deregistration_request_sent
 - Type
 - 64-bit Counter
 - Description
 - Total number of times deregistration request is sent to an unified load balancer pod
 - Labels
 - ulbName**
- ulb_deregistration_response_received
 - Type
 - 64-bit Counter
 - Description
 - Total number of times deregistration response is received from unified load balancer pod
 - Labels
 - UlbName, status**
- ulb_heartbeat_request_sent
 - Type
 - 64-bit Counter
 - Description
 - Total number of times heartbeat request is sent to an unified load balancer pod
 - Labels
 - ulbName**
- ulb_heartbeat_response_received
 - Type
 - 64-bit Counter

Description

Total number of times heartbeat response is received from unified load balancer pod

Labels

UlbName, status

Table of Contents

- [Summary](#)
- [SCTP statistics](#)

Summary

Group	Metric Name	Type	Synopsis
SCTP statistics	sctpActiveEstabs	Counter	The number of times associations have been established.
SCTP statistics	sctpPassiveEstabs	Counter	The number of times associations have been established.
SCTP statistics	sctpAborted	Counter	The number of times associations have been aborted.
SCTP statistics	sctpShutdowns	Counter	The number of times associations have been gracefully terminated.
SCTP statistics	sctpOutOfBlues	Counter	The number of out of the blue packets received.
SCTP statistics	sctpChecksumErrors	Counter	The number of SCTP packets received with an invalid checksum.
SCTP statistics	sctpOutCtrlChunks	Counter	The number of SCTP control chunks sent.
SCTP statistics	sctpOutOrderChunks	Counter	The number of SCTP ordered data chunks sent.
SCTP statistics	sctpOutUnorderChunks	Counter	The number of SCTP unordered data chunks sent.
SCTP statistics	sctpInCtrlChunks	Counter	The number of SCTP control chunks received.
SCTP statistics	sctpInOrderChunks	Counter	The number of SCTP ordered data chunks received.
SCTP statistics	sctpInUnorderChunks	Counter	The number of SCTP unordered data chunks received.
SCTP statistics	sctpOutSCTPPacks	Counter	The number of SCTP packets sent.
SCTP statistics	sctpInSCTPPacks	Counter	The number of SCTP packets received.
SCTP statistics	sctpCurrEstab	Gauge	The current number of established associations.

SCTP statistics

- [sctpActiveEstabs](#)

Type

64-bit Counter

Description

The number of times that associations have made a direct transition to the ESTABLISHED state from the COOKIE-ECHOED state: COOKIE-ECHOED -> ESTABLISHED. The upper layer initiated the association attempt.

Labels

- [sctpPassiveEstabs](#)

Type

64-bit Counter

Description

The number of times that associations have made a direct transition to the ESTABLISHED state from the CLOSED state: CLOSED -> ESTABLISHED. The remote endpoint initiated the association attempt.

Labels

- [sctpAborted](#)

Type

64-bit Counter

Description

The number of times that associations have made a direct transition to the CLOSED state from any state due to sending or receiving of an ABORT chunk: AnyState --Abort--> CLOSED. Ungraceful termination of the association.

Labels

- sctpShutdowns

Type

64-bit Counter

Description

The number of times that associations have made a direct transition to the CLOSED state from either the SHUTDOWN-SENT state or the SHUTDOWN-ACK-SENT state. Graceful termination of the association.

Labels

- sctpOutOfBlues

Type

64-bit Counter

Description

The number of out of the blue packets received by the host. An out of the blue packet is an SCTP packet correctly formed, including the proper checksum, but for which the receiver was unable to identify an appropriate association.

Labels

- sctpChecksumErrors

Type

64-bit Counter

Description

The number of SCTP packets received with an invalid checksum.

Labels

- sctpOutCtrlChunks

Type

64-bit Counter

Description

The number of SCTP control chunks sent (retransmissions are not included). Control chunks are those chunks different from DATA.

Labels

- sctpOutOrderChunks

Type

64-bit Counter

Description

The number of SCTP ordered data chunks sent (retransmissions are not included).

Labels

- sctpOutUnorderChunks

Type

64-bit Counter

Description

The number of SCTP unordered chunks (data chunks in which the U bit is set to 1) sent (retransmissions are not included).

Labels

- sctpInCtrlChunks

Type

64-bit Counter

Description

The number of SCTP control chunks received (no duplicate chunks included).

Labels

- sctpInOrderChunks

Type

64-bit Counter

Description

The number of SCTP ordered data chunks received (no duplicate chunks included).

Labels

- sctpInUnorderChunks

Type

64-bit Counter

Description

The number of SCTP unordered chunks (data chunks in which the U bit is set to 1) received (no duplicate chunks included).

Labels

- sctpOutSCTPPacks

Type

64-bit Counter

Description

The number of SCTP packets sent. Retransmitted DATA chunks are included.

Labels

- sctpInSCTPPacks

Type

64-bit Counter

Description

The number of SCTP packets received. Duplicates are included.

Labels

- sctpCurrEstab

Type
64-bit Gauge
Description
The number of associations for which the current state is either ESTABLISHED, SHUTDOWN-RECEIVED or SHUTDOWN-PENDING.
Labels

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- [Summary](#)
- [ETCD Client Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_put_requests	Counter	The total number of ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_requests_sent	Counter	The total number of ETCD Client PUT Requests sent.
ETCD Client Metrics	etcd_put_successes	Counter	The total number of successful ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_failures	Counter	The total number of failed ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_with_lease_requests	Counter	The total number of ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_requests_sent	Counter	The total number of ETCD Client PUT with Lease Requests sent.
ETCD Client Metrics	etcd_put_with_lease_successes	Counter	The total number of successful ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_failures	Counter	The total number of failed ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_range_requests	Counter	The total number of ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_requests_sent	Counter	The total number of ETCD Client Range Requests sent.
ETCD Client Metrics	etcd_range_successes	Counter	The total number of successful ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_failures	Counter	The total number of failed ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_delete_requests	Counter	The total number of ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_requests_sent	Counter	The total number of ETCD Client Range Delete Requests sent.
ETCD Client Metrics	etcd_range_delete_successes	Counter	The total number of successful ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_failures	Counter	The total number of failed ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_txn_requests	Counter	The total number of ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_requests_sent	Counter	The total number of ETCD Client Txn Requests sent.
ETCD Client Metrics	etcd_txn_successes	Counter	The total number of successful ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_failures	Counter	The total number of failed ETCD Client Txn Requests.
ETCD Client Metrics	etcd_watch_requests	Counter	The total number of ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_requests_sent	Counter	The total number of ETCD Client Watch Requests sent.
ETCD Client Metrics	etcd_watch_successes	Counter	The total number of successful ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_failures	Counter	The total number of failed ETCD Client Watch Requests.
ETCD Client Metrics	etcdwatcher_requests	Counter	The total number of ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_requests_sent	Counter	The total number of ETCD Watcher Requests sent.
ETCD Client Metrics	etcdwatcher_successes	Counter	The total number of successful ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_failures	Counter	The total number of failed ETCD Watcher Requests.

ETCD Client Metrics

- [etcd_put_requests](#)

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests.

Labels

keyPrefix

- etcd_put_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests sent.

Labels

keyPrefix

- etcd_put_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT Requests.

Labels

keyPrefix

- etcd_put_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT Requests.

Labels

keyPrefix, failureReason

- etcd_put_with_lease_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests sent.

Labels

keyPrefix

• etcd_put_with_lease_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT with Lease Requests.

Labels

keyPrefix

• etcd_put_with_lease_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT with Lease Requests.

Labels

keyPrefix, failureReason

• etcd_range_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests.

Labels

keyPrefix

• etcd_range_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests sent.

Labels

keyPrefix

• etcd_range_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Requests.

Labels

keyPrefix

• etcd_range_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Requests.

Labels

keyPrefix, failureReason

- etcd_range_delete_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests.

Labels

keyPrefix

- etcd_range_delete_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests sent.

Labels

keyPrefix

- etcd_range_delete_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Delete Requests.

Labels

keyPrefix

- etcd_range_delete_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Delete Requests.

Labels

keyPrefix, failureReason

- etcd_txn_requests

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests.

Labels

keyPrefix

- etcd_txn_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests sent.

Labels

keyPrefix

- etcd_txn_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Txn Requests.

Labels

keyPrefix

- etcd_txn_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Txn Requests.

Labels

keyPrefix, failureReason

- etcd_watch_requests

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests sent.

Labels

keyPrefix

- etcd_watch_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Watch Requests.

Labels

keyPrefix, failureReason

- etcdwatcher_requests

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests.

Labels

keyPrefix

- etcdwatcher_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests sent.

Labels

keyPrefix

- etcdwatcher_successes

Type

64-bit Counter

Description

The total number of successful ETCD Watcher Requests.

Labels

keyPrefix

- etcdwatcher_failures

Type

64-bit Counter

Description

The total number of failed ETCD Watcher Requests.

Labels

keyPrefix, failureReason

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- [Summary](#)
- [HTTP Punting Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_sent	Counter	The total number of HTTP requests successfully punted.
HTTP Punting Metrics	punting_http_requests_failed	Counter	The total number of HTTP requests punting failed.
HTTP Punting Metrics	punting_http_responses_received	Counter	The total number of HTTP responses received for the requests punted.
HTTP Punting Metrics	punting_http_responses_forwarded	Counter	The total number of HTTP responses forwarded after punting transaction completion.
HTTP Punting Metrics	punting_http_requests_aborted	Counter	The total number of HTTP requests aborted.
HTTP Punting Metrics	punting_http_request_duration_seconds	Histogram	Punting operation duration in seconds.

HTTP Punting Metrics

- [punting_http_requests_sent](#)

Type

64-bit Counter

Description

The total number of HTTP requests successfully punted.

Labels

message_type
- [punting_http_requests_failed](#)

Type

64-bit Counter

Description

The total number of HTTP requests for which the punting failed.

Labels

message_type
- [punting_http_responses_received](#)

Type

64-bit Counter

Description

The total number of HTTP responses received for the requests punted.

Labels

message_type, status_code
- [punting_http_responses_forwarded](#)

Type

64-bit Counter

Description

The total number of HTTP responses forwarded after punting transaction completion.

Labels

message_type, status_code

- `punting_http_requests_aborted`

Type

64-bit Counter

Description

The total number of HTTP requests aborted.

Labels

message_type

- `punting_http_request_duration_seconds`

Type

64-bit Histogram

Description

The punting transaction duration is measured from the time the HTTP request is sent out to the time the response is received for the punting request.

Labels

message_type

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

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- [Summary](#)
- [Ta statistics](#)

Summary

Group	Metric Name	Type	Synopsis
Ta statistics	mmeta_enb_relocations_attempted	Counter	eNB Relocations Attempted.
Ta statistics	mmeta_enb_relocations_successful	Counter	eNB Relocation attempts successful.
Ta statistics	mmeta_enb_relocations_failed	Counter	eNB Relocation attempts failed.

Ta statistics

- `mmeta_enb_relocations_attempted`

Type

64-bit Counter

Description

eNB Relocations Attempted.

Labels

globalenbid, trigger

- `mmeta_enb_relocations_successful`

Type

64-bit Counter

Description

eNB Relocation attempts successful.

Labels

globalenbid

- mmeta_enb_relocations_failed

Type

64-bit Counter

Description

eNB Relocation attempts failed.

Labels

globalenbid, cause

Table of Contents

- [Summary](#)
- [Packet Streaming Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Packet Streaming Metrics	num_packet_streaming_attempted	Counter	The total number of attempted streaming of packets
Packet Streaming Metrics	num_packet_streaming_success	Counter	The total number of packets streamed successfully
Packet Streaming Metrics	num_packet_streaming_failed	Counter	The total number of failed streaming of packets
Packet Streaming Metrics	num_packet_streaming_bytes	Counter	The total number of bytes successfully streamed

Packet Streaming Metrics

- num_packet_streaming_attempted

Type

64-bit Counter

Description

The total number of attempted streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_success

Type

64-bit Counter

Description

The total number of packets streamed successfully

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_failed

Type

64-bit Counter

Description

The total number of failed streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_bytes

Type

64-bit Counter

Description

The total number of bytes successfully streamed

Labels

interface_name, server_fqdn, kafka_topic

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- [Summary](#)
- [Circuit Breaker Metrics](#)
- [Service Endpoint Learner Metrics](#)
- [Service Endpoint Manager Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Circuit Breaker Metrics	sem_cb_endpoints	Gauge	Total number of endpoints for a service in respective states
Service Endpoint Learner Metrics	sem_num_dns_errors	Counter	Total number of DNS resolution errors for a given service
Service Endpoint Learner Metrics	sem_num_endpoints	Gauge	Total number of endpoints learnt for a service
Service Endpoint Manager Metrics	sem_num_endpoint_transactions_sent	Counter	Total number of times an endpoint is given to the application for communication with it
Service Endpoint Manager Metrics	sem_num_endpoint_transaction_status_received	Counter	Total number of transactions for a given endpoint received with the transaction status

Circuit Breaker Metrics

- sem_cb_endpoints

Type

64-bit Gauge

Description

Total number of endpoints for a service in respective states. There are 3 states viz. Active, Blocked, Restoring. In Active state, the endpoint is responding as expected, i.e. the response code doesn't indicate a service failure. The endpoint will be moved to Blocked state based on response failures which indicate service failure. After a timeout, the endpoint will be moved to Restoring state to check if the endpoint has recovered. In this state, if endpoint responds with response code other than service failure, it will be moved to Active state. Else, it will be moved back to Blocked state.

Labels

service_name, endpoint_state

Service Endpoint Learner Metrics

- sem_num_dns_errors

Type

64-bit Counter

Description

Total number of DNS resolution errors for a given service, labels are service name for DNS resolution and DNS error codes

Labels

service_name, error_code

- sem_num_endpoints

Type

64-bit Gauge

Description

Total number of endpoints learnt from the Kubernetes DNS server for a given service

Labels

service

Service Endpoint Manager Metrics

- sem_num_endpoint_transactions_sent

Type

64-bit Counter

Description

Total number of times an endpoint is given to the application for communication with it

Labels

service

- sem_num_endpoint_transaction_status_received

Type

64-bit Counter

Description

Total number of transactions for a given endpoint received with the transaction status

Labels

service, status

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- [Summary](#)
- [N3IWF Statistics](#)
- [N3IWF PDU Session Statistics](#)
- [N3IWF Qos Flow Statistics](#)

Summary

Group	Metric Name	Type	Synopsis
N3IWF Statistics	n3iwf_current_pdu_sessions	Gauge	Current Number of PDU Sessions
N3IWF PDU Session Statistics	n3iwf_pdu_session_setup_request	Counter	Number of PDU Sessions requested
N3IWF PDU Session Statistics	n3iwf_pdu_session_setup_success	Counter	Number of PDU Sessions successfully setup
N3IWF PDU Session Statistics	n3iwf_pdu_session_setup_failure	Counter	Number of PDU Sessions failed to setup

Group	Metric Name	Type	Synopsis
N3IWF Qos Flow Statistics	n3iwf_initial_qos_flows_setup_attempt	Counter	Number of Qos Flows requested
N3IWF Qos Flow Statistics	n3iwf_initial_qos_flows_setup_success	Counter	Number of Qos Flows successfully setup
N3IWF Qos Flow Statistics	n3iwf_initial_qos_flows_setup_failure	Counter	Number of Qos Flows failed to setup
N3IWF Qos Flow Statistics	n3iwf_additional_qos_flows_setup_attempt	Counter	Number of Qos Flows requested
N3IWF Qos Flow Statistics	n3iwf_additional_qos_flows_setup_success	Counter	Number of Qos Flows successfully setup
N3IWF Qos Flow Statistics	n3iwf_additional_qos_flows_setup_failure	Counter	Number of Qos Flows failed to setup
N3IWF Qos Flow Statistics	n3iwf_qos_flows_release_attempt	Counter	Number of Qos Flows release requested
N3IWF Qos Flow Statistics	n3iwf_qos_flows_release_success	Counter	Number of Qos Flows released
N3IWF Qos Flow Statistics	n3iwf_active_qos_flows_release	Counter	Number of active Qos Flows released

N3IWF Statistics

- [n3iwf_current_pdu_sessions](#)

Type

64-bit Gauge

Description

Current Number of PDU Sessions

Labels

snssai

N3IWF PDU Session Statistics

- [n3iwf_pdu_session_setup_request](#)

Type

64-bit Counter

Description

Number of PDU Sessions in the PDU Session Resource Setup Requests received

Labels

snssai

- [n3iwf_pdu_session_setup_success](#)

Type

64-bit Counter

Description

Number of PDU Sessions successfully setup in the PDU Session Resource Setup Responses sent

Labels

snssai

- [n3iwf_pdu_session_setup_failure](#)

Type

64-bit Counter

Description

Number of PDU Sessions failed to setup in the PDU Session Resource Setup Responses sent

Labels

snssai, cause_group, cause

N3IWF Qos Flow Statistics

- n3iwf_initial_qos_flows_setup_attempt

Type

64-bit Counter

Description

Number of Qos Flows attempted to setup in the Initial Context Setup Requests received

Labels

snssai, five_qi

- n3iwf_initial_qos_flows_setup_success

Type

64-bit Counter

Description

Number of Qos Flows successfully setup in the Initial Context Setup Responses sent

Labels

snssai, five_qi

- n3iwf_initial_qos_flows_setup_failure

Type

64-bit Counter

Description

Number of Qos Flows failed to setup in the Initial Context Setup Responses sent

Labels

snssai, five_qi, cause_group, cause

- n3iwf_additional_qos_flows_setup_attempt

Type

64-bit Counter

Description

Number of Qos Flows attempted to setup in the PDU Session Resource Setup/Modify Requests received

Labels

snssai, five_qi

- n3iwf_additional_qos_flows_setup_success

Type

64-bit Counter

Description

Number of Qos Flows successfully setup in the PDU Session Resource Setup/Modify Responses sent

Labels

snssai, five_qi

- n3iwf_additional_qos_flows_setup_failure

Type

64-bit Counter

Description

Number of Qos Flows failed to setup in the PDU Session Resource Setup/Modify Responses sent

Labels

snssai, five_qi, cause_group, cause

- n3iwf_qos_flows_release_attempt

Type

64-bit Counter

Description

Number of QoS flows attempted to release due to/in PDU Session Resource Release Command, PDU Session Resource Modify Request or UE Context Release Command messages or any internal event

Labels

snssai, five_qi

- n3iwf_qos_flows_release_success

Type

64-bit Counter

Description

Number of QoS flows released while responding in PDU Session Resource Release Response, PDU Session Resource Modify Response or UE Context Release Complete messages or any internal event

Labels

snssai, five_qi

- n3iwf_active_qos_flows_release

Type

64-bit Counter

Description

Number of released QoS flows that were active at the time of release

Labels

snssai, five_qi

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- [Summary](#)
- [ETCD Client Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_put_requests	Counter	The total number of ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_requests_sent	Counter	The total number of ETCD Client PUT Requests sent.
ETCD Client Metrics	etcd_put_successes	Counter	The total number of successful ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_failures	Counter	The total number of failed ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_with_lease_requests	Counter	The total number of ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_requests_sent	Counter	The total number of ETCD Client PUT with Lease Requests sent.
ETCD Client Metrics	etcd_put_with_lease_successes	Counter	The total number of successful ETCD Client PUT with Lease Requests.

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_put_with_lease_failures	Counter	The total number of failed ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_range_requests	Counter	The total number of ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_requests_sent	Counter	The total number of ETCD Client Range Requests sent.
ETCD Client Metrics	etcd_range_successes	Counter	The total number of successful ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_failures	Counter	The total number of failed ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_delete_requests	Counter	The total number of ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_requests_sent	Counter	The total number of ETCD Client Range Delete Requests sent.
ETCD Client Metrics	etcd_range_delete_successes	Counter	The total number of successful ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_failures	Counter	The total number of failed ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_txn_requests	Counter	The total number of ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_requests_sent	Counter	The total number of ETCD Client Txn Requests sent.
ETCD Client Metrics	etcd_txn_successes	Counter	The total number of successful ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_failures	Counter	The total number of failed ETCD Client Txn Requests.
ETCD Client Metrics	etcd_watch_requests	Counter	The total number of ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_requests_sent	Counter	The total number of ETCD Client Watch Requests sent.
ETCD Client Metrics	etcd_watch_successes	Counter	The total number of successful ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_failures	Counter	The total number of failed ETCD Client Watch Requests.
ETCD Client Metrics	etcdwatcher_requests	Counter	The total number of ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_requests_sent	Counter	The total number of ETCD Watcher Requests sent.
ETCD Client Metrics	etcdwatcher_successes	Counter	The total number of successful ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_failures	Counter	The total number of failed ETCD Watcher Requests.

ETCD Client Metrics

- **etcd_put_requests**

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests.

Labels

keyPrefix

- **etcd_put_requests_sent**

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests sent.

Labels

keyPrefix

- **etcd_put_successes**

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT Requests.

Labels

keyPrefix

- etcd_put_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT Requests.

Labels

keyPrefix, failureReason

- etcd_put_with_lease_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests sent.

Labels

keyPrefix

- etcd_put_with_lease_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT with Lease Requests.

Labels

keyPrefix, failureReason

- etcd_range_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests.

Labels

keyPrefix

• etcd_range_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests sent.

Labels

keyPrefix

• etcd_range_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Requests.

Labels

keyPrefix

• etcd_range_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Requests.

Labels

keyPrefix, failureReason

• etcd_range_delete_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests.

Labels

keyPrefix

• etcd_range_delete_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests sent.

Labels

keyPrefix

- etcd_range_delete_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Delete Requests.

Labels

keyPrefix

- etcd_range_delete_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Delete Requests.

Labels

keyPrefix, failureReason

- etcd_txn_requests

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests.

Labels

keyPrefix

- etcd_txn_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests sent.

Labels

keyPrefix

- etcd_txn_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Txn Requests.

Labels

keyPrefix

- etcd_txn_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Txn Requests.

Labels

keyPrefix, failureReason

- etcd_watch_requests

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests sent.

Labels

keyPrefix

- etcd_watch_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Watch Requests.

Labels

keyPrefix, failureReason

- etcdwatcher_requests

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests.

Labels

keyPrefix

- `etcdwatcher_requests_sent`

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests sent.

Labels

keyPrefix
- `etcdwatcher_successes`

Type

64-bit Counter

Description

The total number of successful ETCD Watcher Requests.

Labels

keyPrefix
- `etcdwatcher_failures`

Type

64-bit Counter

Description

The total number of failed ETCD Watcher Requests.

Labels

keyPrefix, failureReason

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- [Summary](#)
- [HTTP Punting Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_sent	Counter	The total number of HTTP requests successfully punted.
HTTP Punting Metrics	punting_http_requests_failed	Counter	The total number of HTTP requests punting failed.
HTTP Punting Metrics	punting_http_responses_received	Counter	The total number of HTTP responses received for the requests punted.
HTTP Punting Metrics	punting_http_responses_forwarded	Counter	The total number of HTTP responses forwarded after punting transaction completion.
HTTP Punting Metrics	punting_http_requests_aborted	Counter	The total number of HTTP requests aborted.
HTTP Punting Metrics	punting_http_request_duration_seconds	Histogram	Punting operation duration in seconds.

HTTP Punting Metrics

- `punting_http_requests_sent`

Type

64-bit Counter

Description

The total number of HTTP requests successfully punted.

Labels

message_type

- `punting_http_requests_failed`

Type

64-bit Counter

Description

The total number of HTTP requests for which the punting failed.

Labels

message_type

- `punting_http_responses_received`

Type

64-bit Counter

Description

The total number of HTTP responses received for the requests punted.

Labels

message_type, status_code

- `punting_http_responses_forwarded`

Type

64-bit Counter

Description

The total number of HTTP responses forwarded after punting transaction completion.

Labels

message_type, status_code

- `punting_http_requests_aborted`

Type

64-bit Counter

Description

The total number of HTTP requests aborted.

Labels

message_type

- `punting_http_request_duration_seconds`

Type

64-bit Histogram

Description

The punting transaction duration is measured from the time the HTTP request is sent out to the time the response is received for the punting request.

Labels

message_type

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

Table of Contents

- [Summary](#)
- [Packet Streaming Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Packet Streaming Metrics	num_packet_streaming_attempted	Counter	The total number of attempted streaming of packets
Packet Streaming Metrics	num_packet_streaming_success	Counter	The total number of packets streamed successfully
Packet Streaming Metrics	num_packet_streaming_failed	Counter	The total number of failed streaming of packets
Packet Streaming Metrics	num_packet_streaming_bytes	Counter	The total number of bytes successfully streamed

Packet Streaming Metrics

- num_packet_streaming_attempted

Type

64-bit Counter

Description

The total number of attempted streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_success

Type

64-bit Counter

Description

The total number of packets streamed successfully

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_failed

Type

64-bit Counter

Description

The total number of failed streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_bytes

Type

64-bit Counter

Description

The total number of bytes successfully streamed

Labels

interface_name, server_fqdn, kafka_topic

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- [Summary](#)
- [UDSF DBClient Metrics](#)
- [UDSF API latency statistics](#)

Summary

Group	Metric Name	Type	Synopsis
UDSF DBClient Metrics	udsf_db_operation_request	Counter	The total number of udsf dbclient requests received.
UDSF DBClient Metrics	udsf_db_operation_response	Counter	The total number of udsf dbclient responses sent.
UDSF DBClient Metrics	udsf_dbc_total_init_success	Counter	The total number of UDSF INIT successes.
UDSF DBClient Metrics	udsf_dbc_total_init_failure	Counter	The total number of UDSF INIT failures.
UDSF DBClient Metrics	udsf_dbc_total_deinit_success	Counter	The total number of UDSF DEINIT successes.
UDSF DBClient Metrics	udsf_dbc_total_deinit_failure	Counter	The total number of UDSF DEINIT failures.
UDSF DBClient Metrics	udsf_db_transaction_counter	Counter	The total number of CRUD operations.
UDSF API latency statistics	udsf_db_transaction_duration	Histogram	Roundtrip duration taken by transaction in seconds.
UDSF API latency statistics	udsf_db_execution_duration	Histogram	Execution time taken by transaction in microseconds.
UDSF API latency statistics	udsf_hiredis_bufferwrite_duration	Histogram	Duration of time taken in hiredis to write data to socket in seconds

UDSF DBClient Metrics

- [udsf_db_operation_request](#)

Type

64-bit Counter

Description

The total number of udsf dbclient requests received from worker. Labels- 'app_name' : Originating thread name, ex:smfcc, 'operation_type' : Type of CRUD operation/API type involved, ex:CREATE_OR_MODIFY_RECORD_ASYNC

Labels

app_name, operation_type

- [udsf_db_operation_response](#)

Type

64-bit Counter

Description

The total number of udsf dbclient responses sent. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE 'operation_type' : Type of CRUD operation/API type involved., ex:CREATE_OR_MODIFY_RECORD_ASYNC

Labels

app_name, status, operation_type

- [udsf_dbc_total_init_success](#)

Type

64-bit Counter

Description

The total number of times UDSF API init was success. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: SUCCESS

Labels

app_name, status

- udsf_dbc_total_init_failure

Type

64-bit Counter

Description

The total number of times UDSF API init failed. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: FAILURE

Labels

app_name, status

- udsf_dbc_total_deinit_success

Type

64-bit Counter

Description

The total number of times UDSF API Deinit succeeded. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: SUCCESS

Labels

app_name, status

- udsf_dbc_total_deinit_failure

Type

64-bit Counter

Description

The total number of times UDSF API Deinit failed. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: FAILURE

Labels

app_name, status

- udsf_db_transaction_counter

Type

64-bit Counter

Description

The total number of times CRUD operations are incrementing. Labels- 'app_name' : Originating thread name, ex:smfcc, 'crud_op' : The categorical CRUD operation in which this sub-transation falls into, ex: CREATE_OR_MODIFY

Labels

app_name, crud_op

UDSF API latency statistics

- udsf_db_transaction_duration

Type

64-bit Histogram

Description

Duration observed on total journey/roundtrip taken by a transaction after receiving the request. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE 'operation_type' : Type of CRUD operation/API type involved., ex:CREATE_OR_MODIFY_RECORD_ASYNC, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE, 'crud_op' : The categorical CRUD operation in which this transation falls into, ex: CREATE_OR_MODIFY

Labels

app_name,operation_type,status,crud_op

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

• udsf_db_execution_duration

Type

64-bit Histogram

Description

Processing time taken by UDSF client. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE 'operation_type' : Type of CRUD operation/API type involved., ex:CREATE_OR_MODIFY_RECORD_ASYNC, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE, 'crud_op' : The categorical CRUD operation in which this transation falls into, ex: CREATE_OR_MODIFY

Labels

app_name,operation_type,status,crud_op

Boundaries

[1, 2, 4, 8, 16, 32, 64, 128, 256, 384, 512, 768, 1024, 1536, 2048, 4096, 8192, 16384]

• udsf_hiredis_bufferwrite_duration

Type

64-bit Histogram

Description

Duration observed between hiredis buffer append and write to socket. This is fair approximation for time spent before data is written to socket as the buffer threshold accounted for append and write is dynamic and not fixed size. Per socket, the buffer append start-time is recorded along with size intended to transmit and end-time recorded once data greater than recorded size is written on the socket

Labels

app_name

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

Table of Contents

- [Summary](#)
- [ETCD Client Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_put_requests	Counter	The total number of ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_requests_sent	Counter	The total number of ETCD Client PUT Requests sent.
ETCD Client Metrics	etcd_put_successes	Counter	The total number of successful ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_failures	Counter	The total number of failed ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_with_lease_requests	Counter	The total number of ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_requests_sent	Counter	The total number of ETCD Client PUT with Lease Requests sent.
ETCD Client Metrics	etcd_put_with_lease_successes	Counter	The total number of successful ETCD Client PUT with Lease Requests.

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_put_with_lease_failures	Counter	The total number of failed ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_range_requests	Counter	The total number of ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_requests_sent	Counter	The total number of ETCD Client Range Requests sent.
ETCD Client Metrics	etcd_range_successes	Counter	The total number of successful ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_failures	Counter	The total number of failed ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_delete_requests	Counter	The total number of ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_requests_sent	Counter	The total number of ETCD Client Range Delete Requests sent.
ETCD Client Metrics	etcd_range_delete_successes	Counter	The total number of successful ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_failures	Counter	The total number of failed ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_txn_requests	Counter	The total number of ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_requests_sent	Counter	The total number of ETCD Client Txn Requests sent.
ETCD Client Metrics	etcd_txn_successes	Counter	The total number of successful ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_failures	Counter	The total number of failed ETCD Client Txn Requests.
ETCD Client Metrics	etcd_watch_requests	Counter	The total number of ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_requests_sent	Counter	The total number of ETCD Client Watch Requests sent.
ETCD Client Metrics	etcd_watch_successes	Counter	The total number of successful ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_failures	Counter	The total number of failed ETCD Client Watch Requests.
ETCD Client Metrics	etcdwatcher_requests	Counter	The total number of ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_requests_sent	Counter	The total number of ETCD Watcher Requests sent.
ETCD Client Metrics	etcdwatcher_successes	Counter	The total number of successful ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_failures	Counter	The total number of failed ETCD Watcher Requests.

ETCD Client Metrics

- [etcd_put_requests](#)

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests.

Labels

keyPrefix

- [etcd_put_requests_sent](#)

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests sent.

Labels

keyPrefix

- [etcd_put_successes](#)

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT Requests.

Labels

keyPrefix

• etcd_put_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT Requests.

Labels

keyPrefix, failureReason

• etcd_put_with_lease_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests.

Labels

keyPrefix

• etcd_put_with_lease_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests sent.

Labels

keyPrefix

• etcd_put_with_lease_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT with Lease Requests.

Labels

keyPrefix

• etcd_put_with_lease_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT with Lease Requests.

Labels

keyPrefix, failureReason

• etcd_range_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests.

Labels

keyPrefix

• etcd_range_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests sent.

Labels

keyPrefix

• etcd_range_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Requests.

Labels

keyPrefix

• etcd_range_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Requests.

Labels

keyPrefix, failureReason

• etcd_range_delete_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests.

Labels

keyPrefix

• etcd_range_delete_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests sent.

Labels

keyPrefix

- etcd_range_delete_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Delete Requests.

Labels

keyPrefix

- etcd_range_delete_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Delete Requests.

Labels

keyPrefix, failureReason

- etcd_txn_requests

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests.

Labels

keyPrefix

- etcd_txn_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests sent.

Labels

keyPrefix

- etcd_txn_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Txn Requests.

Labels

keyPrefix

- etcd_txn_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Txn Requests.

Labels

keyPrefix, failureReason

- etcd_watch_requests

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests sent.

Labels

keyPrefix

- etcd_watch_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Watch Requests.

Labels

keyPrefix, failureReason

- etcdwatcher_requests

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests.

Labels

keyPrefix

- etcdwatcher_requests_sent
 - Type
 - 64-bit Counter
 - Description
 - The total number of ETCD Watcher Requests sent.
 - Labels
 - keyPrefix**
- etcdwatcher_successes
 - Type
 - 64-bit Counter
 - Description
 - The total number of successful ETCD Watcher Requests.
 - Labels
 - keyPrefix**
- etcdwatcher_failures
 - Type
 - 64-bit Counter
 - Description
 - The total number of failed ETCD Watcher Requests.
 - Labels
 - keyPrefix, failureReason**

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- [Summary](#)
- [HTTP Punting Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_sent	Counter	The total number of HTTP requests successfully punted.
HTTP Punting Metrics	punting_http_requests_failed	Counter	The total number of HTTP requests punting failed.
HTTP Punting Metrics	punting_http_responses_received	Counter	The total number of HTTP responses received for the requests punted.
HTTP Punting Metrics	punting_http_responses_forwarded	Counter	The total number of HTTP responses forwarded after punting transaction completion.
HTTP Punting Metrics	punting_http_requests_aborted	Counter	The total number of HTTP requests aborted.
HTTP Punting Metrics	punting_http_request_duration_seconds	Histogram	Punting operation duration in seconds.

HTTP Punting Metrics

- punting_http_requests_sent
 - Type

64-bit Counter

Description

The total number of HTTP requests successfully punted.

Labels

message_type

- `punting_http_requests_failed`

Type

64-bit Counter

Description

The total number of HTTP requests for which the punting failed.

Labels

message_type

- `punting_http_responses_received`

Type

64-bit Counter

Description

The total number of HTTP responses received for the requests punted.

Labels

message_type, status_code

- `punting_http_responses_forwarded`

Type

64-bit Counter

Description

The total number of HTTP responses forwarded after punting transaction completion.

Labels

message_type, status_code

- `punting_http_requests_aborted`

Type

64-bit Counter

Description

The total number of HTTP requests aborted.

Labels

message_type

- `punting_http_request_duration_seconds`

Type

64-bit Histogram

Description

The punting transaction duration is measured from the time the HTTP request is sent out to the time the response is received for the punting request.

Labels

<i>message_type</i>
Boundaries
[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

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- [Summary](#)
- [NGAP Message Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
NGAP Message Metrics	ngap_ng_setup_request_sent	Counter	NG Setup Request sent
NGAP Message Metrics	ngap_ng_setup_response_rcvd	Counter	NG Setup Response received
NGAP Message Metrics	ngap_ng_setup_failure_rcvd	Counter	NG Setup Failure received
NGAP Message Metrics	ngap_ran_config_update_sent	Counter	RAN configuration update sent
NGAP Message Metrics	ngap_ran_config_update_ack_rcvd	Counter	RAN configuration update acknowledge received
NGAP Message Metrics	ngap_ran_config_update_failure_rcvd	Counter	RAN configuration update failure received
NGAP Message Metrics	ngap_amf_config_update_rcvd	Counter	AMF configuration update received
NGAP Message Metrics	ngap_amf_config_update_ack_sent	Counter	AMF configuration update acknowledge sent
NGAP Message Metrics	ngap_amf_config_update_failure_sent	Counter	AMF configuration update failure sent
NGAP Message Metrics	ngap_ng_reset_rcvd	Counter	NG reset received
NGAP Message Metrics	ngap_ng_reset_sent	Counter	NG reset sent
NGAP Message Metrics	ngap_ng_reset_ack_rcvd	Counter	NG reset acknowledge received
NGAP Message Metrics	ngap_ng_reset_ack_sent	Counter	NG reset acknowledge sent
NGAP Message Metrics	ngap_amf_status_indication_rcvd	Counter	AMF status indication rceived
NGAP Message Metrics	ngap_initial_ue_message_sent	Counter	Initial UE Message sent
NGAP Message Metrics	ngap_uplink_nas_transport_sent	Counter	Uplink NAS Transport sent
NGAP Message Metrics	ngap_downlink_nas_transport_rcvd	Counter	Downlink NAS transport received
NGAP Message Metrics	ngap_nas_non_delivery_indication_sent	Counter	NAS NON Delivery sent
NGAP Message Metrics	ngap_initial_context_setup_request_rcvd	Counter	Initial context setup request received
NGAP Message Metrics	ngap_initial_context_setup_response_sent	Counter	Initial context setup response sent
NGAP Message Metrics	ngap_initial_context_setup_failure_sent	Counter	Initial context setup failure sent
NGAP Message Metrics	ngap_ue_context_release_request_sent	Counter	UE context release request sent
NGAP Message Metrics	ngap_ue_context_release_command_rcvd	Counter	UE context release command received
NGAP Message Metrics	ngap_ue_context_release_complete_sent	Counter	UE context release complete counter sent
NGAP Message Metrics	ngap_ue_context_modification_request_rcvd	Counter	UE context modification request received
NGAP Message Metrics	ngap_ue_context_modification_response_sent	Counter	UE context modification response sent
NGAP Message Metrics	ngap_ue_context_modification_failure_sent	Counter	UE context modification failure sent
NGAP Message Metrics	ngap_pdu_session_resource_setup_request_rcvd	Counter	PDU session resource setup request received
NGAP Message Metrics	ngap_pdu_session_resource_setup_response_sent	Counter	PDU session resource setup response sent
NGAP Message Metrics	ngap_pdu_session_resource_modify_request_rcvd	Counter	PDU session resource modify request received
NGAP Message Metrics	ngap_pdu_session_resource_modify_response_sent	Counter	PDU session resource modify response sent
NGAP Message Metrics	ngap_pdu_session_resource_release_command_rcvd	Counter	PDU session resource release request received
NGAP Message Metrics	ngap_pdu_session_resource_release_response_sent	Counter	PDU session resource release response sent
NGAP Message Metrics	ngap_pdu_session_resource_notify_sent	Counter	PDU session resource notify sent
NGAP Message Metrics	ngap_ue_tnla_binding_release_request_rcvd	Counter	UE TNLA binding release request received
NGAP Message Metrics	ngap_error_indication_rcvd	Counter	Error Indicaiton received

Group	Metric Name	Type	Synopsis
NGAP Message Metrics	ngap_error_indication_sent	Counter	Error Indication sent
NGAP Message Metrics	ngap_message_error_rcvd	Counter	erroneous messages received
NGAP Message Metrics	ngap_message_rcvd_invalid_procedure_code	Counter	Messages received with Invalid Procedure Code
NGAP Message Metrics	ngap_message_rcvd_invalid_message_type	Counter	Messages received with Invalid Message Type

NGAP Message Metrics

- [ngap_ng_setup_request_sent](#)

Type

64-bit Counter

Description

Transmitted NG setup request counter

Labels

tnla

- [ngap_ng_setup_response_rcvd](#)

Type

64-bit Counter

Description

Received NG setup response counter

Labels

tnla, amf_name

- [ngap_ng_setup_failure_rcvd](#)

Type

64-bit Counter

Description

Received NG setup failures counter

Labels

tnla, cause_group, cause

- [ngap_ran_config_update_sent](#)

Type

64-bit Counter

Description

Transmitted RAN configuration update counter

Labels

tnla, amf_name

- [ngap_ran_config_update_ack_rcvd](#)

Type

64-bit Counter

Description

Received RAN configuration update acknowledge counter

Labels

tnla, amf_name

- ngap_ran_config_update_failure_rcvd

Type

64-bit Counter

Description

Received RAN configuration update failure counter

Labels

tnla, amf_name, cause_group, cause

- ngap_amf_config_update_rcvd

Type

64-bit Counter

Description

Received AMF configuration update counter

Labels

tnla, amf_name

- ngap_amf_config_update_ack_sent

Type

64-bit Counter

Description

Transmitted AMF configuration update acknowledge counter

Labels

tnla, amf_name

- ngap_amf_config_update_failure_sent

Type

64-bit Counter

Description

Transmitted AMF configuration update failure counter

Labels

tnla, amf_name, cause_group, cause

- ngap_ng_reset_rcvd

Type

64-bit Counter

Description

Received NG reset counter

Labels

tnla, amf_name

- ngap_ng_reset_sent

Type

64-bit Counter

Description

Transmitted NG reset counter

Labels

tnla, amf_name, cause_group, cause

• ngap_ng_reset_ack_rcvd

Type

64-bit Counter

Description

Received NG reset acknowledge counter

Labels

tnla, amf_name

• ngap_ng_reset_ack_sent

Type

64-bit Counter

Description

Transmitted NG reset acknowledge counter

Labels

tnla, amf_name

• ngap_amf_status_indication_rcvd

Type

64-bit Counter

Description

Received AMF Status Indication message counter

Labels

tnla, amf_name

• ngap_initial_ue_message_sent

Type

64-bit Counter

Description

Transmitted Initial UE message counter

Labels

tnla, amf_name

• ngap_uplink_nas_transport_sent

Type

64-bit Counter

Description

Transmitted uplink Nas transport counter

Labels

tnla, amf_name

- ngap_downlink_nas_transport_rcvd

Type

64-bit Counter

Description

Receveid NAS transport counter

Labels

tnla, amf_name

- ngap_nas_non_delivery_indication_sent

Type

64-bit Counter

Description

Transmitted Nas non delivery indication counter

Labels

tnla, amf_name, cause_group, cause

- ngap_initial_context_setup_request_rcvd

Type

64-bit Counter

Description

Received initial context setup request counter

Labels

tnla, amf_name

- ngap_initial_context_setup_response_sent

Type

64-bit Counter

Description

Transmitted initial context setup response counter

Labels

tnla, amf_name

- ngap_initial_context_setup_failure_sent

Type

64-bit Counter

Description

Transmitted initial context setup failure counter

Labels

tnla, amf_name, cause_group, cause

- ngap_ue_context_release_request_sent

Type

64-bit Counter

Description

Transmitted UE context release request counter

Labels

tnla, amf_name, cause_group, cause

- ngap_ue_context_release_command_rcvd

Type

64-bit Counter

Description

Received UE context release command counter

Labels

tnla, amf_name

- ngap_ue_context_release_complete_sent

Type

64-bit Counter

Description

Transmitted UE context release complete counter

Labels

tnla, amf_name

- ngap_ue_context_modification_request_rcvd

Type

64-bit Counter

Description

Received UE context modification request counter

Labels

tnla, amf_name

- ngap_ue_context_modification_response_sent

Type

64-bit Counter

Description

Transmitted UE context modification response counter

Labels

tnla, amf_name

- ngap_ue_context_modification_failure_sent

Type

64-bit Counter

Description

Transmitted UE context modification failure counter

Labels

tnla, amf_name, cause_group, cause

ngap_pdu_session_resource_setup_request_rcvd
Type
64-bit Counter
Description
Received PDU session resource setup request counter
Labels
tnla, amf_name
ngap_pdu_session_resource_setup_response_sent
Type
64-bit Counter
Description
Transmitted PDU session resource setup response counter
Labels
tnla, amf_name
ngap_pdu_session_resource_modify_request_rcvd
Type
64-bit Counter
Description
Received PDU session resource modify request counter
Labels
tnla, amf_name
ngap_pdu_session_resource_modify_response_sent
Type
64-bit Counter
Description
Transmitted PDU session resource modify response counter
Labels
tnla, amf_name
ngap_pdu_session_resource_release_command_rcvd
Type
64-bit Counter
Description
Received PDU session resource release request counter
Labels
tnla, amf_name
ngap_pdu_session_resource_release_response_sent
Type
64-bit Counter
Description

Transmitted PDU session resource release response counter

Labels

tnla, amf_name

• ngap_pdu_session_resource_notify_sent

Type

64-bit Counter

Description

Transmitted PDU session resource notify counter

Labels

tnla, amf_name

• ngap_ue_tnla_binding_release_request_rcvd

Type

64-bit Counter

Description

UE TNLA binding release request counter

Labels

tnla, amf_name

• ngap_error_indication_rcvd

Type

64-bit Counter

Description

Received error indication counter

Labels

tnla, amf_name

• ngap_error_indication_sent

Type

64-bit Counter

Description

Transmitted error indication counter

Labels

tnla, amf_name, cause_group, cause

• ngap_message_error_rcvd

Type

64-bit Counter

Description

Received erroneous message counter

Labels

tnla, amf_name

• ngap_message_rcvd_invalid_procedure_code

Type

64-bit Counter

Description

Total number of messages received with invalid procedure code

Labels

tnla, amf_name

- ngap_message_rcvd_invalid_message_type

Type

64-bit Counter

Description

Total number of messages received with invalid message type

Labels

tnla, amf_name

Table of Contents

- [Summary](#)
- [Packet Streaming Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Packet Streaming Metrics	num_packet_streaming_attempted	Counter	The total number of attempted streaming of packets
Packet Streaming Metrics	num_packet_streaming_success	Counter	The total number of packets streamed successfully
Packet Streaming Metrics	num_packet_streaming_failed	Counter	The total number of failed streaming of packets
Packet Streaming Metrics	num_packet_streaming_bytes	Counter	The total number of bytes successfully streamed

Packet Streaming Metrics

- num_packet_streaming_attempted

Type

64-bit Counter

Description

The total number of attempted streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_success

Type

64-bit Counter

Description

The total number of packets streamed successfully

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_failed

Type

64-bit Counter

Description

The total number of failed streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_bytes

Type

64-bit Counter

Description

The total number of bytes successfully streamed

Labels

interface_name, server_fqdn, kafka_topic

Table of Contents

- [Summary](#)
- [SCTP connection Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
SCTP connection Metrics	sctp_communication_up	Counter	SCTP communication up event notification
SCTP connection Metrics	sctp_communication_lost	Counter	SCTP communication lost event notification
SCTP connection Metrics	sctp_shutdown_rcvd	Counter	SCTP shutdown event notification
SCTP connection Metrics	sctp_association_setup_failed	Counter	SCTP association failed to setup event notification
SCTP connection Metrics	sctp_association_restarts	Counter	SCTP restart event notification
SCTP connection Metrics	sctp_remote_error	Counter	SCTP remote error event notification
SCTP connection Metrics	sctp_send_failure	Counter	SCTP send failure

SCTP connection Metrics

- sctp_communication_up

Type

64-bit Counter

Description

Total number of SCTP communication up

Labels

tnla

- sctp_communication_lost

Type

64-bit Counter

Description

Total number of SCTP communication lost

Labels

tnla

- sctp_shutdown_rcvd

Type

64-bit Counter

Description

Total number of SCTP shutdown receivd

Labels

tnla

- sctp_association_setup_failed

Type

64-bit Counter

Description

Total number of SCTP association failed to setup

Labels

tnla

- sctp_association_restarts

Type

64-bit Counter

Description

Total number of SCTP association restarts

Labels

tnla

- sctp_remote_error

Type

64-bit Counter

Description

Total number of SCTP remote error

Labels

tnla

- sctp_send_failure

Type

64-bit Counter

Description

Total number of SCTP send failure

Labels

tnla

Table of Contents

- [Summary](#)
- [Circuit Breaker Metrics](#)
- [Service Endpoint Learner Metrics](#)

- [Service Endpoint Manager Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Circuit Breaker Metrics	sem_cb_endpoints	Gauge	Total number of endpoints for a service in respective states
Service Endpoint Learner Metrics	sem_num_dns_errors	Counter	Total number of DNS resolution errors for a given service
Service Endpoint Learner Metrics	sem_num_endpoints	Gauge	Total number of endpoints learnt for a service
Service Endpoint Manager Metrics	sem_num_endpoint_transactions_sent	Counter	Total number of times an endpoint is given to the application for communication with it
Service Endpoint Manager Metrics	sem_num_endpoint_transaction_status_received	Counter	Total number of transactions for a given endpoint received with the transaction status

Circuit Breaker Metrics

- [sem_cb_endpoints](#)

Type

64-bit Gauge

Description

Total number of endpoints for a service in respective states. There are 3 states viz. Active, Blocked, Restoring. In Active state, the endpoint is responding as expected, i.e. the response code doesn't indicate a service failure. The endpoint will be moved to Blocked state based on response failures which indicate service failure. After a timeout, the endpoint will be moved to Restoring state to check if the endpoint has recovered. In this state, if endpoint responds with response code other than service failure, it will be moved to Active state. Else, it will be moved back to Blocked state.

Labels

service_name, endpoint_state

Service Endpoint Learner Metrics

- [sem_num_dns_errors](#)

Type

64-bit Counter

Description

Total number of DNS resolution errors for a given service, labels are service name for DNS resolution and DNS error codes

Labels

service_name, error_code

- [sem_num_endpoints](#)

Type

64-bit Gauge

Description

Total number of endpoints learnt from the Kubernetes DNS server for a given service

Labels

service

Service Endpoint Manager Metrics

- [sem_num_endpoint_transactions_sent](#)

Type

64-bit Counter

Description

Total number of times an endpoint is given to the application for communication with it

Labels

service

• sem_num_endpoint_transaction_status_received

Type

64-bit Counter

Description

Total number of transactions for a given endpoint received with the transaction status

Labels

service, status

Table of Contents

- [Summary](#)
- [Unified Load Balancer Endpoint Manager Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Unified Load Balancer Endpoint Manager Metrics	ulb_registration_request_sent	Counter	Total number of times an registration request is sent to an unified load balancer pod
Unified Load Balancer Endpoint Manager Metrics	ulb_registration_response_received	Counter	Total number of times registration response is received from unified load balancer pod
Unified Load Balancer Endpoint Manager Metrics	ulb_deregistration_request_sent	Counter	Total number of times an deregistration request is sent to an unified load balancer pod
Unified Load Balancer Endpoint Manager Metrics	ulb_deregistration_response_received	Counter	Total number of times deregistration response is received from unified load balancer pod
Unified Load Balancer Endpoint Manager Metrics	ulb_heartbeat_request_sent	Counter	Total number of times an heartbeat request is sent to an unified load balancer pod
Unified Load Balancer Endpoint Manager Metrics	ulb_heartbeat_response_received	Counter	Total number of times heartbeat response is received from unified load balancer pod

Unified Load Balancer Endpoint Manager Metrics

• ulb_registration_request_sent

Type

64-bit Counter

Description

Total number of times registration request is sent to an unified load balancer pod

Labels

ulbName

• ulb_registration_response_received

Type

64-bit Counter

Description

Total number of times registration response is received from unified load balancer pod

Labels

UlbName, status

- `ulb_deregistration_request_sent`

Type

64-bit Counter

Description

Total number of times deregistration request is sent to an unified load balancer pod

Labels

ulbName

- `ulb_deregistration_response_received`

Type

64-bit Counter

Description

Total number of times deregistration response is received from unified load balancer pod

Labels

UlbName, status

- `ulb_heartbeat_request_sent`

Type

64-bit Counter

Description

Total number of times heartbeat request is sent to an unified load balancer pod

Labels

ulbName

- `ulb_heartbeat_response_received`

Type

64-bit Counter

Description

Total number of times heartbeat response is received from unified load balancer pod

Labels

UlbName, status

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- [Summary](#)
- [SCTP statistics](#)

Summary

Group	Metric Name	Type	Synopsis
SCTP statistics	sctpActiveEstabs	Counter	The number of times associations have been established.
SCTP statistics	sctpPassiveEstabs	Counter	The number of times associations have been established.

Group	Metric Name	Type	Synopsis
SCTP statistics	sctpAborted	Counter	The number of times associations have been aborted.
SCTP statistics	sctpShutdowns	Counter	The number of times associations have been gracefully terminated.
SCTP statistics	sctpOutOfBlues	Counter	The number of out of the blue packets received.
SCTP statistics	sctpChecksumErrors	Counter	The number of SCTP packets received with an invalid checksum.
SCTP statistics	sctpOutCtrlChunks	Counter	The number of SCTP control chunks sent.
SCTP statistics	sctpOutOrderChunks	Counter	The number of SCTP ordered data chunks sent.
SCTP statistics	sctpOutUnorderChunks	Counter	The number of SCTP unordered data chunks sent.
SCTP statistics	sctpInCtrlChunks	Counter	The number of SCTP control chunks received.
SCTP statistics	sctpInOrderChunks	Counter	The number of SCTP ordered data chunks received.
SCTP statistics	sctpInUnorderChunks	Counter	The number of SCTP unordered data chunks received.
SCTP statistics	sctpOutSCTPPacks	Counter	The number of SCTP packets sent.
SCTP statistics	sctpInSCTPPacks	Counter	The number of SCTP packets received.
SCTP statistics	sctpCurrEstab	Gauge	The current number of established associations.

SCTP statistics

- **sctpActiveEstabs**

Type

64-bit Counter

Description

The number of times that associations have made a direct transition to the ESTABLISHED state from the COOKIE-ECHOED state: COOKIE-ECHOED -> ESTABLISHED. The upper layer initiated the association attempt.

Labels

- **sctpPassiveEstabs**

Type

64-bit Counter

Description

The number of times that associations have made a direct transition to the ESTABLISHED state from the CLOSED state: CLOSED -> ESTABLISHED. The remote endpoint initiated the association attempt.

Labels

- **sctpAborted**

Type

64-bit Counter

Description

The number of times that associations have made a direct transition to the CLOSED state from any state due to sending or receiving of an ABORT chunk: AnyState --Abort--> CLOSED. Ungraceful termination of the association.

Labels

- **sctpShutdowns**

Type

64-bit Counter

Description

The number of times that associations have made a direct transition to the CLOSED state from either the SHUTDOWN-SENT state or the SHUTDOWN-ACK-SENT state. Graceful termination of the association.

Labels

- **sctpOutOfBlues**

Type

64-bit Counter

Description

The number of out of the blue packets received by the host. An out of the blue packet is an SCTP packet correctly formed, including the proper checksum, but for which the receiver was unable to identify an appropriate association.

Labels

- **sctpChecksumErrors**

Type

64-bit Counter

Description

The number of SCTP packets received with an invalid checksum.

Labels

- **sctpOutCtrlChunks**

Type

64-bit Counter

Description

The number of SCTP control chunks sent (retransmissions are not included). Control chunks are those chunks different from DATA.

Labels

- **sctpOutOrderChunks**

Type

64-bit Counter

Description

The number of SCTP ordered data chunks sent (retransmissions are not included).

Labels

- **sctpOutUnorderChunks**

Type

64-bit Counter

Description

The number of SCTP unordered chunks (data chunks in which the U bit is set to 1) sent (retransmissions are not included).

Labels

- **sctpInCtrlChunks**

Type

64-bit Counter

Description

The number of SCTP control chunks received (no duplicate chunks included).

Labels

- sctpInOrderChunks

Type

64-bit Counter

Description

The number of SCTP ordered data chunks received (no duplicate chunks included).

Labels

- sctpInUnorderChunks

Type

64-bit Counter

Description

The number of SCTP unordered chunks (data chunks in which the U bit is set to 1) received (no duplicate chunks included).

Labels

- sctpOutSCTPPacks

Type

64-bit Counter

Description

The number of SCTP packets sent. Retransmitted DATA chunks are included.

Labels

- sctpInSCTPPacks

Type

64-bit Counter

Description

The number of SCTP packets received. Duplicates are included.

Labels

- sctpCurrEstab

Type

64-bit Gauge

Description

The number of associations for which the current state is either ESTABLISHED, SHUTDOWN-RECEIVED or SHUTDOWN-PENDING.

Labels

Table of Contents

- [Summary](#)
- [ETCD Client Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_put_requests	Counter	The total number of ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_requests_sent	Counter	The total number of ETCD Client PUT Requests sent.
ETCD Client Metrics	etcd_put_successes	Counter	The total number of successful ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_failures	Counter	The total number of failed ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_with_lease_requests	Counter	The total number of ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_requests_sent	Counter	The total number of ETCD Client PUT with Lease Requests sent.
ETCD Client Metrics	etcd_put_with_lease_successes	Counter	The total number of successful ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_failures	Counter	The total number of failed ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_range_requests	Counter	The total number of ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_requests_sent	Counter	The total number of ETCD Client Range Requests sent.
ETCD Client Metrics	etcd_range_successes	Counter	The total number of successful ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_failures	Counter	The total number of failed ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_delete_requests	Counter	The total number of ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_requests_sent	Counter	The total number of ETCD Client Range Delete Requests sent.
ETCD Client Metrics	etcd_range_delete_successes	Counter	The total number of successful ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_failures	Counter	The total number of failed ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_txn_requests	Counter	The total number of ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_requests_sent	Counter	The total number of ETCD Client Txn Requests sent.
ETCD Client Metrics	etcd_txn_successes	Counter	The total number of successful ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_failures	Counter	The total number of failed ETCD Client Txn Requests.
ETCD Client Metrics	etcd_watch_requests	Counter	The total number of ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_requests_sent	Counter	The total number of ETCD Client Watch Requests sent.
ETCD Client Metrics	etcd_watch_successes	Counter	The total number of successful ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_failures	Counter	The total number of failed ETCD Client Watch Requests.
ETCD Client Metrics	etcdwatcher_requests	Counter	The total number of ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_requests_sent	Counter	The total number of ETCD Watcher Requests sent.
ETCD Client Metrics	etcdwatcher_successes	Counter	The total number of successful ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_failures	Counter	The total number of failed ETCD Watcher Requests.

ETCD Client Metrics

- etcd_put_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests.

Labels

keyPrefix

- etcd_put_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests sent.

Labels

keyPrefix

- etcd_put_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT Requests.

Labels

keyPrefix

- etcd_put_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT Requests.

Labels

keyPrefix, failureReason

- etcd_put_with_lease_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests sent.

Labels

keyPrefix

- etcd_put_with_lease_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT with Lease Requests.

Labels

keyPrefix, failureReason

• etcd_range_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests.

Labels

keyPrefix

• etcd_range_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests sent.

Labels

keyPrefix

• etcd_range_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Requests.

Labels

keyPrefix

• etcd_range_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Requests.

Labels

keyPrefix, failureReason

• etcd_range_delete_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests.

Labels

keyPrefix

• etcd_range_delete_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests sent.

Labels

keyPrefix

• etcd_range_delete_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Delete Requests.

Labels

keyPrefix

• etcd_range_delete_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Delete Requests.

Labels

keyPrefix, failureReason

• etcd_txn_requests

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests.

Labels

keyPrefix

• etcd_txn_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests sent.

Labels

keyPrefix

• etcd_txn_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Txn Requests.

Labels

keyPrefix

- etcd_txn_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Txn Requests.

Labels

keyPrefix, failureReason

- etcd_watch_requests

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests sent.

Labels

keyPrefix

- etcd_watch_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Watch Requests.

Labels

keyPrefix, failureReason

• etcdwatcher_requests

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests.

Labels

keyPrefix

• etcdwatcher_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests sent.

Labels

keyPrefix

• etcdwatcher_successes

Type

64-bit Counter

Description

The total number of successful ETCD Watcher Requests.

Labels

keyPrefix

• etcdwatcher_failures

Type

64-bit Counter

Description

The total number of failed ETCD Watcher Requests.

Labels

keyPrefix, failureReason

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- [Summary](#)
- [HTTP Punting Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_sent	Counter	The total number of HTTP requests successfully punted.
HTTP Punting Metrics	punting_http_requests_failed	Counter	The total number of HTTP requests punting failed.
HTTP Punting Metrics	punting_http_responses_received	Counter	The total number of HTTP responses received for the requests punted.

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_responses_forwarded	Counter	The total number of HTTP responses forwarded after punting transaction completion.
HTTP Punting Metrics	punting_http_requests_aborted	Counter	The total number of HTTP requests aborted.
HTTP Punting Metrics	punting_http_request_duration_seconds	Histogram	Punting operation duration in seconds.

HTTP Punting Metrics

- `punting_http_requests_sent`

Type

64-bit Counter

Description

The total number of HTTP requests successfully punted.

Labels

message_type

- `punting_http_requests_failed`

Type

64-bit Counter

Description

The total number of HTTP requests for which the punting failed.

Labels

message_type

- `punting_http_responses_received`

Type

64-bit Counter

Description

The total number of HTTP responses received for the requests punted.

Labels

message_type, status_code

- `punting_http_responses_forwarded`

Type

64-bit Counter

Description

The total number of HTTP responses forwarded after punting transaction completion.

Labels

message_type, status_code

- `punting_http_requests_aborted`

Type

64-bit Counter

Description

The total number of HTTP requests aborted.

Labels

message_type

- `punting_http_request_duration_seconds`

Type

64-bit Histogram

Description

The punting transaction duration is measured from the time the HTTP request is sent out to the time the response is received for the punting request.

Labels

message_type

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

Table of Contents

- [Summary](#)
- [Packet Streaming Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Packet Streaming Metrics	num_packet_streaming_attempted	Counter	The total number of attempted streaming of packets
Packet Streaming Metrics	num_packet_streaming_success	Counter	The total number of packets streamed successfully
Packet Streaming Metrics	num_packet_streaming_failed	Counter	The total number of failed streaming of packets
Packet Streaming Metrics	num_packet_streaming_bytes	Counter	The total number of bytes successfully streamed

Packet Streaming Metrics

- `num_packet_streaming_attempted`

Type

64-bit Counter

Description

The total number of attempted streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- `num_packet_streaming_success`

Type

64-bit Counter

Description

The total number of packets streamed successfully

Labels

interface_name, server_fqdn, kafka_topic

- `num_packet_streaming_failed`

Type

64-bit Counter

Description

The total number of failed streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_bytes

Type

64-bit Counter

Description

The total number of bytes successfully streamed

Labels

interface_name, server_fqdn, kafka_topic

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- [Summary](#)
- [ETCD Client Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_put_requests	Counter	The total number of ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_requests_sent	Counter	The total number of ETCD Client PUT Requests sent.
ETCD Client Metrics	etcd_put_successes	Counter	The total number of successful ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_failures	Counter	The total number of failed ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_with_lease_requests	Counter	The total number of ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_requests_sent	Counter	The total number of ETCD Client PUT with Lease Requests sent.
ETCD Client Metrics	etcd_put_with_lease_successes	Counter	The total number of successful ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_failures	Counter	The total number of failed ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_range_requests	Counter	The total number of ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_requests_sent	Counter	The total number of ETCD Client Range Requests sent.
ETCD Client Metrics	etcd_range_successes	Counter	The total number of successful ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_failures	Counter	The total number of failed ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_delete_requests	Counter	The total number of ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_requests_sent	Counter	The total number of ETCD Client Range Delete Requests sent.
ETCD Client Metrics	etcd_range_delete_successes	Counter	The total number of successful ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_failures	Counter	The total number of failed ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_txn_requests	Counter	The total number of ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_requests_sent	Counter	The total number of ETCD Client Txn Requests sent.
ETCD Client Metrics	etcd_txn_successes	Counter	The total number of successful ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_failures	Counter	The total number of failed ETCD Client Txn Requests.
ETCD Client Metrics	etcd_watch_requests	Counter	The total number of ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_requests_sent	Counter	The total number of ETCD Client Watch Requests sent.
ETCD Client Metrics	etcd_watch_successes	Counter	The total number of successful ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_failures	Counter	The total number of failed ETCD Client Watch Requests.
ETCD Client Metrics	etcdwatcher_requests	Counter	The total number of ETCD Watcher Requests.

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcdwatcher_requests_sent	Counter	The total number of ETCD Watcher Requests sent.
ETCD Client Metrics	etcdwatcher_successes	Counter	The total number of successful ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_failures	Counter	The total number of failed ETCD Watcher Requests.

ETCD Client Metrics

- [etcd_put_requests](#)

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests.

Labels

keyPrefix
- [etcd_put_requests_sent](#)

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests sent.

Labels

keyPrefix
- [etcd_put_successes](#)

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT Requests.

Labels

keyPrefix
- [etcd_put_failures](#)

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT Requests.

Labels

keyPrefix, failureReason
- [etcd_put_with_lease_requests](#)

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests sent.

Labels

keyPrefix

- etcd_put_with_lease_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT with Lease Requests.

Labels

keyPrefix, failureReason

- etcd_range_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests sent.

Labels

keyPrefix

- etcd_range_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Requests.

Labels

keyPrefix, failureReason

- etcd_range_delete_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests.

Labels

keyPrefix

- etcd_range_delete_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests sent.

Labels

keyPrefix

- etcd_range_delete_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Delete Requests.

Labels

keyPrefix

- etcd_range_delete_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Delete Requests.

Labels

keyPrefix, failureReason

- etcd_txn_requests

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests.

Labels

keyPrefix
- etcd_txn_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests sent.

Labels

keyPrefix
- etcd_txn_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Txn Requests.

Labels

keyPrefix
- etcd_txn_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Txn Requests.

Labels

keyPrefix, failureReason
- etcd_watch_requests

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests.

Labels

keyPrefix
- etcd_watch_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests sent.

Labels

keyPrefix

- etcd_watch_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Watch Requests.

Labels

keyPrefix, failureReason

- etcdwatcher_requests

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests.

Labels

keyPrefix

- etcdwatcher_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests sent.

Labels

keyPrefix

- etcdwatcher_successes

Type

64-bit Counter

Description

The total number of successful ETCD Watcher Requests.

Labels

keyPrefix

- etcdwatcher_failures

Type
64-bit Counter
Description
The total number of failed ETCD Watcher Requests.
Labels
<i>keyPrefix, failureReason</i>

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- [Summary](#)
- [HTTP Punting Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_sent	Counter	The total number of HTTP requests successfully punted.
HTTP Punting Metrics	punting_http_requests_failed	Counter	The total number of HTTP requests punting failed.
HTTP Punting Metrics	punting_http_responses_received	Counter	The total number of HTTP responses received for the requests punted.
HTTP Punting Metrics	punting_http_responses_forwarded	Counter	The total number of HTTP responses forwarded after punting transaction completion.
HTTP Punting Metrics	punting_http_requests_aborted	Counter	The total number of HTTP requests aborted.
HTTP Punting Metrics	punting_http_request_duration_seconds	Histogram	Punting operation duration in seconds.

HTTP Punting Metrics

- [punting_http_requests_sent](#)

Type

64-bit Counter

Description

The total number of HTTP requests successfully punted.

Labels

message_type
- [punting_http_requests_failed](#)

Type

64-bit Counter

Description

The total number of HTTP requests for which the punting failed.

Labels

message_type
- [punting_http_responses_received](#)

Type

64-bit Counter

Description

The total number of HTTP responses received for the requests punted.

Labels

message_type, status_code

- `punting_http_responses_forwarded`

Type

64-bit Counter

Description

The total number of HTTP responses forwarded after punting transaction completion.

Labels

message_type, status_code

- `punting_http_requests_aborted`

Type

64-bit Counter

Description

The total number of HTTP requests aborted.

Labels

message_type

- `punting_http_request_duration_seconds`

Type

64-bit Histogram

Description

The punting transaction duration is measured from the time the HTTP request is sent out to the time the response is received for the punting request.

Labels

message_type

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

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- [Summary](#)
- [Slice Inventory Manager statistics](#)

Summary

Group	Metric Name	Type	Synopsis
Slice Inventory Manager statistics	nfim_nnrf_registration_request	Counter	NF registration attempts
Slice Inventory Manager statistics	nfim_nnrf_registration_response	Counter	NF registration responses
Slice Inventory Manager statistics	nfim_nnrf_deregistration_request	Counter	NF deregistration attempts
Slice Inventory Manager statistics	nfim_nnrf_deregistration_response	Counter	NF deregistration responses
Slice Inventory Manager statistics	nfim_nnrf_cli_nf_delete_request	Counter	CLI Initiated NF delete attempts
Slice Inventory Manager statistics	nfim_nnrf_cli_nf_delete_response	Counter	CLI Initiated NF delete responses
Slice Inventory Manager statistics	nfim_nnrf_update_request	Counter	NF update attempts
Slice Inventory Manager statistics	nfim_nnrf_update_response	Counter	NF update responses

Group	Metric Name	Type	Synopsis
Slice Inventory Manager statistics	nfim_nnrf_subscription_request	Counter	NF update attempts
Slice Inventory Manager statistics	nfim_nnrf_subscription_response	Counter	NF update responses
Slice Inventory Manager statistics	nfim_nnrf_subscription_delete	Counter	NF update attempts
Slice Inventory Manager statistics	nfim_nnrf_subscription_delete_response	Counter	NF update responses
Slice Inventory Manager statistics	nfim_nnrf_notification	Counter	NF update attempts
Slice Inventory Manager statistics	nfim_nnrf_notification_response	Counter	NF update responses
Slice Inventory Manager statistics	nfim_nnrf_heartbeat	Counter	NF update responses
Slice Inventory Manager statistics	nfim_nnrf_heartbeat_timeout	Counter	NF update responses
Slice Inventory Manager statistics	nfim_nnrf_sync_initial_data_request_tx	Counter	Initial Sync Requests sent
Slice Inventory Manager statistics	nfim_nnrf_sync_initial_data_request_rx	Counter	Initial Sync Requests received
Slice Inventory Manager statistics	nfim_nnrf_sync_initial_data_response_tx	Counter	Initial Sync Responses sent
Slice Inventory Manager statistics	nfim_nnrf_sync_initial_data_response_rx	Counter	Initial Sync Responses received
Slice Inventory Manager statistics	nfim_nnrf_sync_nf_profile_request_tx	Counter	Sync Nf Profile Requests sent
Slice Inventory Manager statistics	nfim_nnrf_sync_nf_profile_request_rx	Counter	Sync Nf Profile Requests received
Slice Inventory Manager statistics	nfim_nnrf_sync_nf_profile_response_tx	Counter	Sync Nf Profile Responses sent
Slice Inventory Manager statistics	nfim_nnrf_sync_nf_profile_response_rx	Counter	Sync Nf Profile Responses received
Slice Inventory Manager statistics	nfim_nnrf_sync_subscription_data_request_tx	Counter	Sync Subscription Data Requests sent
Slice Inventory Manager statistics	nfim_nnrf_sync_subscription_data_request_rx	Counter	Sync Subscription Data Requests received
Slice Inventory Manager statistics	nfim_nnrf_sync_subscription_data_response_tx	Counter	Sync Subscription Data Responses sent
Slice Inventory Manager statistics	nfim_nnrf_sync_subscription_data_response_rx	Counter	Sync Subscription Data Responses received
Slice Inventory Manager statistics	nfim_nnrf_sync_config_data_request_tx	Counter	Sync Config Data Requests sent
Slice Inventory Manager statistics	nfim_nnrf_sync_config_data_request_rx	Counter	Sync Config Data Requests received
Slice Inventory Manager statistics	nfim_nnrf_sync_config_data_response_tx	Counter	Sync Config Data Responses sent
Slice Inventory Manager statistics	nfim_nnrf_sync_config_data_response_rx	Counter	Sync Config Data Responses received
Slice Inventory Manager statistics	nfim_nnrf_peer_nrf_heartbeat_request_tx	Counter	Peer Nrf Heartbeat Requests sent
Slice Inventory Manager statistics	nfim_nnrf_peer_nrf_heartbeat_request_rx	Counter	Peer Nrf Heartbeat Requests received
Slice Inventory Manager statistics	nfim_nnrf_peer_nrf_heartbeat_response_tx	Counter	Peer Nrf Heartbeat Responses sent
Slice Inventory Manager statistics	nfim_nnrf_peer_nrf_heartbeat_response_rx	Counter	Peer Nrf Heartbeat Responses received
Slice Inventory Manager statistics	nfim_nnrf_options_request	Counter	Options request
Slice Inventory Manager statistics	nfim_nnrf_options_response	Counter	Options responses

Slice Inventory Manager statistics

- nfim_nnrf_registration_request

Type

64-bit Counter

Description

The number of NF registration attempts received by Slice Inventory Manager

Labels

source_nf_type, source_nf_instance_id, slice

- nfim_nnrf_registration_response

Type

64-bit Counter

Description

The number of NF registration responses sent by Slice Inventory Manager

Labels

source_nf_type, source_nf_instance_id, slice, status

• nfim_nnrf_deregistration_request

Type

64-bit Counter

Description

The number of NF deregistration attempts received by Slice Inventory Manager

Labels

source_nf_type, source_nf_instance_id

• nfim_nnrf_deregistration_response

Type

64-bit Counter

Description

The number of NF deregistration responses sent by Slice Inventory Manager

Labels

source_nf_type, source_nf_instance_id, status

• nfim_nnrf_cli_nf_delete_request

Type

64-bit Counter

Description

The number of NF delete attempts received by Slice Inventory Manager from CLI

Labels

source_nf_type, source_nf_instance_id

• nfim_nnrf_cli_nf_delete_response

Type

64-bit Counter

Description

The number of NF delete responses sent by Slice Inventory Manager to CLI

Labels

source_nf_type, source_nf_instance_id, status

• nfim_nnrf_update_request

Type

64-bit Counter

Description

The number of NF update attempts received by Slice Inventory Manager

Labels

source_nf_type, source_nf_instance_id

• nfim_nnrf_update_response

Type

64-bit Counter

Description

The number of NF update responses sent by Slice Inventory Manager

Labels

source_nf_type, source_nf_instance_id, status

• nfim_nnrf_subscription_request

Type

64-bit Counter

Description

The number of NF subscription requests received by Slice Inventory Manager

Labels

requested_nf_type, requested_nf_instance_id

• nfim_nnrf_subscription_response

Type

64-bit Counter

Description

The number of NF subscription responses sent by Slice Inventory Manager

Labels

requested_nf_type, requested_nf_instance_id, status

• nfim_nnrf_subscription_delete

Type

64-bit Counter

Description

The number of NF subscription deletion received by Slice Inventory Manager

Labels

requested_nf_type, requested_nf_instance_id

• nfim_nnrf_subscription_delete_response

Type

64-bit Counter

Description

The number of NF subscription deletion responses sent by Slice Inventory Manager

Labels

requested_nf_type, requested_nf_instance_id, status

• nfim_nnrf_notification

Type

64-bit Counter

Description

The number of NF notifications sent by Slice Inventory Manager

Labels

requested_nf_type, requested_nf_instance_id

• nfim_nnr_notification_response

Type

64-bit Counter

Description

The number of NF notifications responses received by Slice Inventory Manager

Labels

requested_nf_type, requested_nf_instance_id, status

• nfim_nnr_heartbeat

Type

64-bit Counter

Description

The number of NF heartbeat messages received by Slice Inventory Manager

Labels

source_nf_type, source_nf_instance_id

• nfim_nnr_heartbeat_timeout

Type

64-bit Counter

Description

The number of NF heartbeat timeouts determined by Slice Inventory Manager

Labels

source_nf_type, source_nf_instance_id

• nfim_nnr_sync_initial_data_request_tx

Type

64-bit Counter

Description

The number of Initial Sync Request messages sent by NRF

Labels

source_nrf_id, target_nrf_id

• nfim_nnr_sync_initial_data_request_rx

Type

64-bit Counter

Description

The number of Initial Sync Request messages received by NRF

Labels

source_nrf_id, target_nrf_id

• nfim_nnr_sync_initial_data_response_tx

Type

64-bit Counter

Description

The number of Initial Sync Response messages sent by NRF

Labels

source_nrf_id, target_nrf_id, status

• nfim_nnr_sync_initial_data_response_rx

Type

64-bit Counter

Description

The number of Initial Sync Response messages received by NRF

Labels

source_nrf_id, target_nrf_id, status

• nfim_nnr_sync_nf_profile_request_tx

Type

64-bit Counter

Description

The number of Sync Nf Profile Request messages sent by NRF

Labels

source_nrf_id, target_nrf_id

• nfim_nnr_sync_nf_profile_request_rx

Type

64-bit Counter

Description

The number of Sync Nf Profile Request messages received by NRF

Labels

source_nrf_id, target_nrf_id

• nfim_nnr_sync_nf_profile_response_tx

Type

64-bit Counter

Description

The number of Sync Nf Profile Response messages sent by NRF

Labels

source_nrf_id, target_nrf_id, status

• nfim_nnr_sync_nf_profile_response_rx

Type

64-bit Counter

Description

The number of Sync Nf Profile Response messages received by NRF

Labels

source_nrf_id, target_nrf_id, status

• nfim_nnrf_sync_subscription_data_request_tx

Type

64-bit Counter

Description

The number of Sync Subscription Data Request messages sent by NRF

Labels

source_nrf_id, target_nrf_id

• nfim_nnrf_sync_subscription_data_request_rx

Type

64-bit Counter

Description

The number of Sync Subscription Data Request messages received by NRF

Labels

source_nrf_id, target_nrf_id

• nfim_nnrf_sync_subscription_data_response_tx

Type

64-bit Counter

Description

The number of Sync Subscription Data Response messages sent by NRF

Labels

source_nrf_id, target_nrf_id, status

• nfim_nnrf_sync_subscription_data_response_rx

Type

64-bit Counter

Description

The number of Sync Subscription Data Response messages received by NRF

Labels

source_nrf_id, target_nrf_id, status

• nfim_nnrf_sync_config_data_request_tx

Type

64-bit Counter

Description

The number of Sync Config Data Request messages sent by NRF

Labels

source_nrf_id, target_nrf_id

• nfim_nnrf_sync_config_data_request_rx

Type

64-bit Counter

Description

The number of Sync Config Data Request messages received by NRF

Labels

source_nrf_id, target_nrf_id

- `nfim_nnrf_sync_config_data_response_tx`

Type

64-bit Counter

Description

The number of Sync Config Data Response messages sent by NRF

Labels

source_nrf_id, target_nrf_id, status

- `nfim_nnrf_sync_config_data_response_rx`

Type

64-bit Counter

Description

The number of Sync Config Data Response messages received by NRF

Labels

source_nrf_id, target_nrf_id, status

- `nfim_nnrf_peer_nrf_heartbeat_request_tx`

Type

64-bit Counter

Description

The number of Peer Nrf Heartbeat Request messages sent by NRF

Labels

source_nrf_id, target_nrf_id

- `nfim_nnrf_peer_nrf_heartbeat_request_rx`

Type

64-bit Counter

Description

The number of Peer Nrf Heartbeat Request messages received by NRF

Labels

source_nrf_id, target_nrf_id

- `nfim_nnrf_peer_nrf_heartbeat_response_tx`

Type

64-bit Counter

Description

The number of Peer Nrf Heartbeat Response messages sent by NRF

Labels

source_nrf_id, target_nrf_id, status

- `nfim_nnrf_peer_nrf_heartbeat_response_rx`

Type

64-bit Counter

Description

The number of Peer Nrf Heartbeat Response messages received by NRF

Labels

source_nrf_id, target_nrf_id, status

- `nfim_nnrf_options_request`

Type

64-bit Counter

Description

The number of options requests received by NRF

Labels

- `nfim_nnrf_options_response`

Type

64-bit Counter

Description

The number of options responses sent by NRF

Labels

status

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- [Summary](#)
- [Packet Streaming Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Packet Streaming Metrics	num_packet_streaming_attempted	Counter	The total number of attempted streaming of packets
Packet Streaming Metrics	num_packet_streaming_success	Counter	The total number of packets streamed successfully
Packet Streaming Metrics	num_packet_streaming_failed	Counter	The total number of failed streaming of packets
Packet Streaming Metrics	num_packet_streaming_bytes	Counter	The total number of bytes successfully streamed

Packet Streaming Metrics

- `num_packet_streaming_attempted`

Type

64-bit Counter

Description

The total number of attempted streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- `num_packet_streaming_success`

Type

64-bit Counter

Description

The total number of packets streamed successfully

Labels

interface_name, server_fqdn, kafka_topic

• num_packet_streaming_failed

Type

64-bit Counter

Description

The total number of failed streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

• num_packet_streaming_bytes

Type

64-bit Counter

Description

The total number of bytes successfully streamed

Labels

interface_name, server_fqdn, kafka_topic

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- [Summary](#)
- [Circuit Breaker Metrics](#)
- [Service Endpoint Learner Metrics](#)
- [Service Endpoint Manager Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Circuit Breaker Metrics	sem_cb_endpoints	Gauge	Total number of endpoints for a service in respective states
Service Endpoint Learner Metrics	sem_num_dns_errors	Counter	Total number of DNS resolution errors for a given service
Service Endpoint Learner Metrics	sem_num_endpoints	Gauge	Total number of endpoints learnt for a service
Service Endpoint Manager Metrics	sem_num_endpoint_transactions_sent	Counter	Total number of times an endpoint is given to the application for communication with it
Service Endpoint Manager Metrics	sem_num_endpoint_transaction_status_received	Counter	Total number of transactions for a given endpoint received with the transaction status

Circuit Breaker Metrics

- sem_cb_endpoints

Type

64-bit Gauge

Description

Total number of endpoints for a service in respective states. There are 3 states viz. Active, Blocked, Restoring. In Active state, the endpoint is responding as expected, i.e. the response code doesn't indicate a service failure. The endpoint will be moved to Blocked state based on response failures which indicate service failure. After a timeout, the endpoint will be moved to Restoring state to check if the endpoint has recovered. In this state, if endpoint responds with response code other than service failure, it will be moved to Active state. Else, it will be moved back to Blocked state.

Labels

service_name, endpoint_state

Service Endpoint Learner Metrics

- sem_num_dns_errors

Type

64-bit Counter

Description

Total number of DNS resolution errors for a given service, labels are service name for DNS resolution and DNS error codes

Labels

service_name, error_code

- sem_num_endpoints

Type

64-bit Gauge

Description

Total number of endpoints learnt from the Kubernetes DNS server for a given service

Labels

service

Service Endpoint Manager Metrics

- sem_num_endpoint_transactions_sent

Type

64-bit Counter

Description

Total number of times an endpoint is given to the application for communication with it

Labels

service

- sem_num_endpoint_transaction_status_received

Type

64-bit Counter

Description

Total number of transactions for a given endpoint received with the transaction status

Labels

service, status

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- [Summary](#)
- [ETCD Client Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_put_requests	Counter	The total number of ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_requests_sent	Counter	The total number of ETCD Client PUT Requests sent.
ETCD Client Metrics	etcd_put_successes	Counter	The total number of successful ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_failures	Counter	The total number of failed ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_with_lease_requests	Counter	The total number of ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_requests_sent	Counter	The total number of ETCD Client PUT with Lease Requests sent.
ETCD Client Metrics	etcd_put_with_lease_successes	Counter	The total number of successful ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_failures	Counter	The total number of failed ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_range_requests	Counter	The total number of ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_requests_sent	Counter	The total number of ETCD Client Range Requests sent.
ETCD Client Metrics	etcd_range_successes	Counter	The total number of successful ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_failures	Counter	The total number of failed ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_delete_requests	Counter	The total number of ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_requests_sent	Counter	The total number of ETCD Client Range Delete Requests sent.
ETCD Client Metrics	etcd_range_delete_successes	Counter	The total number of successful ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_failures	Counter	The total number of failed ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_txn_requests	Counter	The total number of ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_requests_sent	Counter	The total number of ETCD Client Txn Requests sent.
ETCD Client Metrics	etcd_txn_successes	Counter	The total number of successful ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_failures	Counter	The total number of failed ETCD Client Txn Requests.
ETCD Client Metrics	etcd_watch_requests	Counter	The total number of ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_requests_sent	Counter	The total number of ETCD Client Watch Requests sent.
ETCD Client Metrics	etcd_watch_successes	Counter	The total number of successful ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_failures	Counter	The total number of failed ETCD Client Watch Requests.
ETCD Client Metrics	etcdwatcher_requests	Counter	The total number of ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_requests_sent	Counter	The total number of ETCD Watcher Requests sent.
ETCD Client Metrics	etcdwatcher_successes	Counter	The total number of successful ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_failures	Counter	The total number of failed ETCD Watcher Requests.

ETCD Client Metrics

- [etcd_put_requests](#)

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests.

Labels

keyPrefix

- [etcd_put_requests_sent](#)

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests sent.

Labels

keyPrefix

- etcd_put_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT Requests.

Labels

keyPrefix

- etcd_put_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT Requests.

Labels

keyPrefix, failureReason

- etcd_put_with_lease_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests sent.

Labels

keyPrefix

- etcd_put_with_lease_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT with Lease Requests.

Labels

keyPrefix, failureReason

• etcd_range_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests.

Labels

keyPrefix

• etcd_range_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests sent.

Labels

keyPrefix

• etcd_range_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Requests.

Labels

keyPrefix

• etcd_range_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Requests.

Labels

keyPrefix, failureReason

• etcd_range_delete_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests.

Labels

keyPrefix

• etcd_range_delete_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests sent.

Labels

keyPrefix

• etcd_range_delete_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Delete Requests.

Labels

keyPrefix

• etcd_range_delete_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Delete Requests.

Labels

keyPrefix, failureReason

• etcd_txn_requests

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests.

Labels

keyPrefix

• etcd_txn_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests sent.

Labels

keyPrefix

• etcd_txn_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Txn Requests.

Labels

keyPrefix

• etcd_txn_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Txn Requests.

Labels

keyPrefix, failureReason

• etcd_watch_requests

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests.

Labels

keyPrefix

• etcd_watch_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests sent.

Labels

keyPrefix

• etcd_watch_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Watch Requests.

Labels

keyPrefix

• etcd_watch_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Watch Requests.

Labels

keyPrefix, failureReason

• etcdwatcher_requests

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests.

Labels

keyPrefix

• etcdwatcher_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests sent.

Labels

keyPrefix

• etcdwatcher_successes

Type

64-bit Counter

Description

The total number of successful ETCD Watcher Requests.

Labels

keyPrefix

• etcdwatcher_failures

Type

64-bit Counter

Description

The total number of failed ETCD Watcher Requests.

Labels

keyPrefix, failureReason

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- [Summary](#)
- [HTTP Punting Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_sent	Counter	The total number of HTTP requests successfully punted.
HTTP Punting Metrics	punting_http_requests_failed	Counter	The total number of HTTP requests punting failed.
HTTP Punting Metrics	punting_http_responses_received	Counter	The total number of HTTP responses received for the requests punted.

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_responses_forwarded	Counter	The total number of HTTP responses forwarded after punting transaction completion.
HTTP Punting Metrics	punting_http_requests_aborted	Counter	The total number of HTTP requests aborted.
HTTP Punting Metrics	punting_http_request_duration_seconds	Histogram	Punting operation duration in seconds.

HTTP Punting Metrics

- punting_http_requests_sent

Type

64-bit Counter

Description

The total number of HTTP requests successfully punted.

Labels

message_type
- punting_http_requests_failed

Type

64-bit Counter

Description

The total number of HTTP requests for which the punting failed.

Labels

message_type
- punting_http_responses_received

Type

64-bit Counter

Description

The total number of HTTP responses received for the requests punted.

Labels

message_type, status_code
- punting_http_responses_forwarded

Type

64-bit Counter

Description

The total number of HTTP responses forwarded after punting transaction completion.

Labels

message_type, status_code
- punting_http_requests_aborted

Type

64-bit Counter

Description

The total number of HTTP requests aborted.

Labels

message_type

- `punting_http_request_duration_seconds`

Type

64-bit Histogram

Description

The punting transaction duration is measured from the time the HTTP request is sent out to the time the response is received for the punting request.

Labels

message_type

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

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- [Summary](#)
- [NF Selector statistics](#)

Summary

Group	Metric Name	Type	Synopsis
NF Selector statistics	nfsel_nnrf_discovery_request	Counter	NF Discovery Requests
NF Selector statistics	nfsel_nnrf_discovery_response	Counter	NF Discovery responses
NF Selector statistics	nfsel_nnrf_discovery_af_cache_response	Counter	NF Discovery cache hit for af
NF Selector statistics	nfsel_nnrf_discovery_amf_cache_response	Counter	NF Discovery cache hit for amf
NF Selector statistics	nfsel_nnrf_discovery_ausf_cache_response	Counter	NF Discovery cache hit for ausf
NF Selector statistics	nfsel_nnrf_discovery_bsf_cache_response	Counter	NF Discovery cache hit for bsf
NF Selector statistics	nfsel_nnrf_discovery_cbcf_cache_response	Counter	NF Discovery cache hit for cbcf
NF Selector statistics	nfsel_nnrf_discovery_chf_cache_response	Counter	NF Discovery cache hit for chf
NF Selector statistics	nfsel_nnrf_discovery_eir_cache_response	Counter	NF Discovery cache hit for eir
NF Selector statistics	nfsel_nnrf_discovery_gmlc_cache_response	Counter	NF Discovery cache hit for gmlc
NF Selector statistics	nfsel_nnrf_discovery_lmf_cache_response	Counter	NF Discovery cache hit for lmf
NF Selector statistics	nfsel_nnrf_discovery_n3iwf_cache_response	Counter	NF Discovery cache hit for n3iwf
NF Selector statistics	nfsel_nnrf_discovery_nef_cache_response	Counter	NF Discovery cache hit for nef
NF Selector statistics	nfsel_nnrf_discovery_nssf_cache_response	Counter	NF Discovery cache hit for nssf
NF Selector statistics	nfsel_nnrf_discovery_nwdaf_cache_response	Counter	NF Discovery cache hit for nwdaf
NF Selector statistics	nfsel_nnrf_discovery_pcf_cache_response	Counter	NF Discovery cache hit for pcf
NF Selector statistics	nfsel_nnrf_discovery_pcscf_cache_response	Counter	NF Discovery cache hit for pcscf
NF Selector statistics	nfsel_nnrf_discovery_smf_cache_response	Counter	NF Discovery cache hit for smf
NF Selector statistics	nfsel_nnrf_discovery_smsf_cache_response	Counter	NF Discovery cache hit for smsf
NF Selector statistics	nfsel_nnrf_discovery_udm_cache_response	Counter	NF Discovery cache hit for udm
NF Selector statistics	nfsel_nnrf_discovery_udr_cache_response	Counter	NF Discovery cache hit for udr
NF Selector statistics	nfsel_nnrf_discovery_upf_cache_response	Counter	NF Discovery cache hit for upf

NF Selector statistics

- `nfsel_nnrf_discovery_request`

Type

64-bit Counter

Description

The number of NF discovery requests received by NF Selector

Labels

source_nf_type, source_nf_instance_id, slice, target_nf_type

- nfsel_nnrf_discovery_response

Type

64-bit Counter

Description

The number of NF discovery responses sent by NF Selector

Labels

source_nf_type, source_nf_instance_id, slice, target_nf_type, status

- nfsel_nnrf_discovery_af_cache_response

Type

64-bit Counter

Description

The number of NF discovery cache hit for af sent by NF Selector

Labels

- nfsel_nnrf_discovery_amf_cache_response

Type

64-bit Counter

Description

The number of NF discovery cache hit for amf sent by NF Selector

Labels

- nfsel_nnrf_discovery_ausf_cache_response

Type

64-bit Counter

Description

The number of NF discovery cache hit for ausf sent by NF Selector

Labels

- nfsel_nnrf_discovery_bsf_cache_response

Type

64-bit Counter

Description

The number of NF discovery cache hit for bsf sent by NF Selector

Labels

- nfssel_nnrf_discovery_cbcf_cache_response

Type
64-bit Counter
Description
The number of NF discovery cache hit for cbcf sent by NF Selector
Labels

- nfssel_nnrf_discovery_chf_cache_response

Type
64-bit Counter
Description
The number of NF discovery cache hit for chf sent by NF Selector
Labels

- nfssel_nnrf_discovery_eir_cache_response

Type
64-bit Counter
Description
The number of NF discovery cache hit for eir sent by NF Selector
Labels

- nfssel_nnrf_discovery_gmlc_cache_response

Type
64-bit Counter
Description
The number of NF discovery cache hit for gmlc sent by NF Selector
Labels

- nfssel_nnrf_discovery_lmf_cache_response

Type
64-bit Counter
Description
The number of NF discovery cache hit for lmf sent by NF Selector
Labels

- nfssel_nnrf_discovery_n3iwf_cache_response

Type
64-bit Counter
Description
The number of NF discovery cache hit for n3iwf sent by NF Selector
Labels

- nysel_nnrf_discovery_nef_cache_response

Type

64-bit Counter

Description

The number of NF discovery cache hit for nef sent by NF Selector

Labels

- ntsel_nnrf_discovery_nssf_cache_response

Type

64-bit Counter

Description

The number of NF discovery cache hit for nssf sent by NF Selector

Labels

- nfmf_nmrp_discovery_nwdaf_cache_response

Type

64-bit Counter

Description

The number of NF discovery cache hit for nwdaf sent by NF Selector

Labels

- ntsel_nnrf_discovery_pcf_cache_response

Type

64-bit Counter

Description

The number of NF discovery cache hit for pcf sent by NF Selector

Labels

- nysel_nnrf_discovery_pcscf_cache_response

Type

64-bit Counter

Description

The number of NF discovery cache hit for pcscf sent by NF Selector

Labels

- nsel_nnrf_discovery_smf_cache_response

Type

64-bit Counter

Description

The number of NF discovery cache hit for smf sent by NF Selector

Labels

• nfsel_nnrf_discovery_smsf_cache_response
Type
64-bit Counter
Description
The number of NF discovery cache hit for smsf sent by NF Selector
Labels

• nfsel_nnrf_discovery_udm_cache_response
Type
64-bit Counter
Description
The number of NF discovery cache hit for udm sent by NF Selector
Labels

• nfsel_nnrf_discovery_udr_cache_response
Type
64-bit Counter
Description
The number of NF discovery cache hit for udr sent by NF Selector
Labels

• nfsel_nnrf_discovery_upf_cache_response
Type
64-bit Counter
Description
The number of NF discovery cache hit for upf sent by NF Selector
Labels

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- [Summary](#)
- [Packet Streaming Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Packet Streaming Metrics	num_packet_streaming_attempted	Counter	The total number of attempted streaming of packets
Packet Streaming Metrics	num_packet_streaming_success	Counter	The total number of packets streamed successfully
Packet Streaming Metrics	num_packet_streaming_failed	Counter	The total number of failed streaming of packets
Packet Streaming Metrics	num_packet_streaming_bytes	Counter	The total number of bytes successfully streamed

Packet Streaming Metrics

- [num_packet_streaming_attempted](#)
- Type

64-bit Counter

Description

The total number of attempted streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

• num_packet_streaming_success

Type

64-bit Counter

Description

The total number of packets streamed successfully

Labels

interface_name, server_fqdn, kafka_topic

• num_packet_streaming_failed

Type

64-bit Counter

Description

The total number of failed streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

• num_packet_streaming_bytes

Type

64-bit Counter

Description

The total number of bytes successfully streamed

Labels

interface_name, server_fqdn, kafka_topic

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- [Summary](#)
- [ETCD Client Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_put_requests	Counter	The total number of ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_requests_sent	Counter	The total number of ETCD Client PUT Requests sent.
ETCD Client Metrics	etcd_put_successes	Counter	The total number of successful ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_failures	Counter	The total number of failed ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_with_lease_requests	Counter	The total number of ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_requests_sent	Counter	The total number of ETCD Client PUT with Lease Requests sent.
ETCD Client Metrics	etcd_put_with_lease_successes	Counter	The total number of successful ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_failures	Counter	The total number of failed ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_range_requests	Counter	The total number of ETCD Client Range Requests.

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_range_requests_sent	Counter	The total number of ETCD Client Range Requests sent.
ETCD Client Metrics	etcd_range_successes	Counter	The total number of successful ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_failures	Counter	The total number of failed ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_delete_requests	Counter	The total number of ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_requests_sent	Counter	The total number of ETCD Client Range Delete Requests sent.
ETCD Client Metrics	etcd_range_delete_successes	Counter	The total number of successful ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_failures	Counter	The total number of failed ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_txn_requests	Counter	The total number of ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_requests_sent	Counter	The total number of ETCD Client Txn Requests sent.
ETCD Client Metrics	etcd_txn_successes	Counter	The total number of successful ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_failures	Counter	The total number of failed ETCD Client Txn Requests.
ETCD Client Metrics	etcd_watch_requests	Counter	The total number of ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_requests_sent	Counter	The total number of ETCD Client Watch Requests sent.
ETCD Client Metrics	etcd_watch_successes	Counter	The total number of successful ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_failures	Counter	The total number of failed ETCD Client Watch Requests.
ETCD Client Metrics	etcdwatcher_requests	Counter	The total number of ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_requests_sent	Counter	The total number of ETCD Watcher Requests sent.
ETCD Client Metrics	etcdwatcher_successes	Counter	The total number of successful ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_failures	Counter	The total number of failed ETCD Watcher Requests.

ETCD Client Metrics

- [etcd_put_requests](#)

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests.

Labels

keyPrefix

- [etcd_put_requests_sent](#)

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests sent.

Labels

keyPrefix

- [etcd_put_successes](#)

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT Requests.

Labels

keyPrefix

- etcd_put_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT Requests.

Labels

keyPrefix, failureReason

- etcd_put_with_lease_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests sent.

Labels

keyPrefix

- etcd_put_with_lease_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT with Lease Requests.

Labels

keyPrefix, failureReason

- etcd_range_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests.

Labels

keyPrefix

- `etcd_range_requests_sent`

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests sent.

Labels

keyPrefix

- `etcd_range_successes`

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Requests.

Labels

keyPrefix

- `etcd_range_failures`

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Requests.

Labels

keyPrefix, failureReason

- `etcd_range_delete_requests`

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests.

Labels

keyPrefix

- `etcd_range_delete_requests_sent`

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests sent.

Labels

keyPrefix

- etcd_range_delete_successes

Type
64-bit Counter
Description
The total number of successful ETCD Client Range Delete Requests.
Labels
<i>keyPrefix</i>

- etcd_range_delete_failures

Type
64-bit Counter
Description
The total number of failed ETCD Client Range Delete Requests.
Labels
<i>keyPrefix, failureReason</i>

- etcd_txn_requests

Type
64-bit Counter
Description
The total number of ETCD Client Txn Requests.
Labels
<i>keyPrefix</i>

- etcd_txn_requests_sent

Type
64-bit Counter
Description
The total number of ETCD Client Txn Requests sent.
Labels
<i>keyPrefix</i>

- etcd_txn_successes

Type
64-bit Counter
Description
The total number of successful ETCD Client Txn Requests.
Labels
<i>keyPrefix</i>

- etcd_txn_failures

Type
64-bit Counter
Description

The total number of failed ETCD Client Txn Requests.

Labels

keyPrefix, failureReason

• etcd_watch_requests

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests.

Labels

keyPrefix

• etcd_watch_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests sent.

Labels

keyPrefix

• etcd_watch_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Watch Requests.

Labels

keyPrefix

• etcd_watch_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Watch Requests.

Labels

keyPrefix, failureReason

• etcdwatcher_requests

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests.

Labels

keyPrefix

• etcdwatcher_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests sent.

Labels

keyPrefix

- etcdwatcher_successes

Type

64-bit Counter

Description

The total number of successful ETCD Watcher Requests.

Labels

keyPrefix
- etcdwatcher_failures

Type

64-bit Counter

Description

The total number of failed ETCD Watcher Requests.

Labels

keyPrefix, failureReason

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- [Summary](#)
- [HTTP Punting Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_sent	Counter	The total number of HTTP requests successfully punted.
HTTP Punting Metrics	punting_http_requests_failed	Counter	The total number of HTTP requests punting failed.
HTTP Punting Metrics	punting_http_responses_received	Counter	The total number of HTTP responses received for the requests punted.
HTTP Punting Metrics	punting_http_responses_forwarded	Counter	The total number of HTTP responses forwarded after punting transaction completion.
HTTP Punting Metrics	punting_http_requests_aborted	Counter	The total number of HTTP requests aborted.
HTTP Punting Metrics	punting_http_request_duration_seconds	Histogram	Punting operation duration in seconds.

HTTP Punting Metrics

- punting_http_requests_sent

Type

64-bit Counter

Description

The total number of HTTP requests successfully punted.

Labels

message_type

- `punting_http_requests_failed`

Type

64-bit Counter

Description

The total number of HTTP requests for which the punting failed.

Labels

message_type

- `punting_http_responses_received`

Type

64-bit Counter

Description

The total number of HTTP responses received for the requests punted.

Labels

message_type, status_code

- `punting_http_responses_forwarded`

Type

64-bit Counter

Description

The total number of HTTP responses forwarded after punting transaction completion.

Labels

message_type, status_code

- `punting_http_requests_aborted`

Type

64-bit Counter

Description

The total number of HTTP requests aborted.

Labels

message_type

- `punting_http_request_duration_seconds`

Type

64-bit Histogram

Description

The punting transaction duration is measured from the time the HTTP request is sent out to the time the response is received for the punting request.

Labels

message_type

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

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- [Summary](#)
- [NRF Agent statistics](#)

Summary

Group	Metric Name	Type	Synopsis
NRF Agent statistics	nrfa_nnrf_disc_request	Counter	NF discovery attempts
NRF Agent statistics	nrfa_nnrf_disc_response	Counter	NF discovery responses
NRF Agent statistics	nrfa_nnrf_nfm_registration_request	Counter	NF registration attempts
NRF Agent statistics	nrfa_nnrf_nfm_registration_response	Counter	NF registration responses
NRF Agent statistics	nrfa_nnrf_nfm_deregistration_request	Counter	NF deregistration attempts
NRF Agent statistics	nrfa_nnrf_nfm_deregistration_response	Counter	NF deregistration responses
NRF Agent statistics	nrfa_nnrf_nfm_update_request	Counter	NF update attempts
NRF Agent statistics	nrfa_nnrf_nfm_update_response	Counter	NF update responses
NRF Agent statistics	nrfa_nnrf_nfm_heartbeat_request	Counter	Heartbeat messages sent to the NRF
NRF Agent statistics	nrfa_nnrf_nfm_heartbeat_response	Counter	Heartbeat responses received from the NRF
NRF Agent statistics	nrfa_nnrf_nfm_heartbeat_failure	Counter	The number of occurrences NF transitioned to suspended due to heartbeat exhaustion
NRF Agent statistics	nrfa_nnrf_nfm_subscription_request	Counter	NF subscription request
NRF Agent statistics	nrfa_nnrf_nfm_subscription_response	Counter	NF subscription responses
NRF Agent statistics	nrfa_nnrf_nfm_notification	Counter	NF notifications
NRF Agent statistics	nrfa_nnrf_nfm_notification_response	Counter	NF notification responses
NRF Agent statistics	nrfa_nf_discovery_request	Counter	Internal discovery requests
NRF Agent statistics	nrfa_nf_discovery_cache_hit	Counter	Discovery cache hits
NRF Agent statistics	nrfa_nf_discovery_cache_miss	Counter	Discovery cache misses
NRF Agent statistics	nrfa_nf_discovery_configured_hit	Counter	Configured cache hits
NRF Agent statistics	nrfa_nnrf_service_heartbeat_request	Counter	Keep alive messages sent to the NRF for the nnrf-nfm service
NRF Agent statistics	nrfa_nnrf_service_heartbeat_response	Counter	Keep alive responses received from the NRF for the nnrf-nfm service
NRF Agent statistics	nrfa_nnrf_discovery_heartbeat_request	Counter	Keep alive messages sent to the NRF for the nnrf-disc service
NRF Agent statistics	nrfa_nnrf_discovery_heartbeat_response	Counter	Keep alive responses received from the NRF for the nnrf-disc service
NRF Agent statistics	nrfa_nnrf_failover_count	Counter	Number of times Primary NRF Failed for the nnrf-nfm service
NRF Agent statistics	nrfa_nnrf_recovery_count	Counter	Number of times Primary NRF recovered from nnrf-nfm service
NRF Agent statistics	nrfa_nnrf_discovery_failover_count	Counter	Number of times Primary NRF Failed for the nnrf-disc service
NRF Agent statistics	nrfa_nnrf_discovery_recovery_count	Counter	Number of times Primary NRF recovered from nnrf-disc service
NRF Agent statistics	nrfa_nf_entries_discovered	Gauge	Current number of NF discovered entries cached
NRF Agent statistics	nrfa_nf_entries_configured	Gauge	Current number of NF configured entries cached

NRF Agent statistics

- [nrfa_nnrf_disc_request](#)

Type

64-bit Counter

Description

The number of NF discovery attempts sent by NRF Agent. Refer to 29.510 for message definition.

Labels

requester_nf_type, target_nf_type, nrf

- nrfa_nnrf_disc_response

Type

64-bit Counter

Description

The number of NF discovery responses received by NRF Agent. Refer to 29.510 for message definition.

Labels

requester_nf_type, target_nf_type, status, nrf

- nrfa_nnrf_nfm_registration_request

Type

64-bit Counter

Description

The number of NF registration attempts sent by NRF Agent. Refer to 29.510 for message definition.

Labels

requester_nf_type, nrf

- nrfa_nnrf_nfm_registration_response

Type

64-bit Counter

Description

The number of NF registration responses received by NRF Agent. Refer to 29.510 for message definition.

Labels

requester_nf_type, status, nrf

- nrfa_nnrf_nfm_deregistration_request

Type

64-bit Counter

Description

The number of NF deregistration attempts sent by NRF Agent. Refer to 29.510 for message definition.

Labels

requester_nf_type, nrf

- nrfa_nnrf_nfm_deregistration_response

Type

64-bit Counter

Description

The number of NF deregistration responses received by NRF Agent. Refer to 29.510 for message definition.

Labels

requester_nf_type, status, nrf

- nrfa_nnrf_nfm_update_request

Type

64-bit Counter

Description

The number of NF update attempts sent by NRF Agent. Refer to 29.510 for message definition.

Labels

requester_nf_type, nrf

- **nrfa_nnrf_nfm_update_response**

Type

64-bit Counter

Description

The number of NF update responses received by NRF Agent. Refer to 29.510 for message definition.

Labels

requester_nf_type, status, nrf

- **nrfa_nnrf_nfm_heartbeat_request**

Type

64-bit Counter

Description

The number of heartbeat messages sent to NRF. Refer to 29.510 for message definition.

Labels

requester_nf_type, nrf

- **nrfa_nnrf_nfm_heartbeat_response**

Type

64-bit Counter

Description

The number of heartbeat responses received from NRF. Refer to 29.510 for message definition.

Labels

requester_nf_type, status, nrf

- **nrfa_nnrf_nfm_heartbeat_failure**

Type

64-bit Counter

Description

The number of occurrences heartbeats retries were exhausted causing the NF to transition from registered to suspended

Labels

requester_nf_type, nrf

- **nrfa_nnrf_nfm_subscription_request**

Type

64-bit Counter

Description

The number of subscription messages sent to NRF. Refer to 29.510 for message definition.

Labels

requester_nf_type, target_nf_type, nrf

- nrfa_nnrf_nfm_subscription_response

Type

64-bit Counter

Description

The number of subscription responses received from NRF. Refer to 29.510 for message definition.

Labels

requester_nf_type, target_nf_type, status, nrf

- nrfa_nnrf_nfm_notification

Type

64-bit Counter

Description

The number of notification messages received from NRF. Refer to 30.510 for message definition.

Labels

requester_nf_type, target_nf_type, nrf

- nrfa_nnrf_nfm_notification_response

Type

64-bit Counter

Description

The number of notification messages received from NRF. Refer to 29.510 for message definition.

Labels

requester_nf_type, target_nf_type, status, nrf

- nrfa_nf_discovery_request

Type

64-bit Counter

Description

The number of internal discovery requests received from our NF

Labels

requester_nf_type, target_nf_type

- nrfa_nf_discovery_cache_hit

Type

64-bit Counter

Description

The number of discovery cache hit

Labels

requester_nf_type, target_nf_type

- nrfa_nf_discovery_cache_miss

Type

64-bit Counter

Description

The number of discovery cache misses

Labels

requester_nf_type, target_nf_type

- nrfa_nf_discovery_configured_hit

Type

64-bit Counter

Description

The number of configured cache hit

Labels

requester_nf_type, target_nf_type

- nrfa_nnrf_service_heartbeat_request

Type

64-bit Counter

Description

The number of NF management service keep alive messages sent to NRF.

Labels

requester_nf_type, nrf, service_name

- nrfa_nnrf_service_heartbeat_response

Type

64-bit Counter

Description

The number of NF management service keep alive responses received from NRF.

Labels

requester_nf_type, status, nrf, service_name

- nrfa_nnrf_discovery_heartbeat_request

Type

64-bit Counter

Description

The number of NF discovery service keep alive messages sent to NRF.

Labels

requester_nf_type, nrf, service_name

- nrfa_nnrf_discovery_heartbeat_response

Type

64-bit Counter

Description

The number of NF discovery service keep alive responses received from NRF.

Labels

requester_nf_type, status, nrf, service_name

nrfa_nnrf_failover_count
Type
64-bit Counter
Description
The Number of times Primary NRF Failed for the NF management service.
Labels
<i>nr</i> , <i>service_name</i>
nrfa_nnrf_recovery_count
Type
64-bit Counter
Description
The number of times Primary NRF recovered from the NF management service.
Labels
<i>nr</i> , <i>service_name</i>
nrfa_nnrf_discovery_failover_count
Type
64-bit Counter
Description
The Number of times Primary NRF Failed the NF discovery service.
Labels
<i>nr</i> , <i>service_name</i>
nrfa_nnrf_discovery_recovery_count
Type
64-bit Counter
Description
The number of times Primary NRF recovered from NF discovery service.
Labels
<i>nr</i> , <i>service_name</i>
nrfa_nf_entries_discovered
Type
64-bit Gauge
Description
The number of NF discovered entries currently cached
Labels
<i>target_nf_type</i>
nrfa_nf_entries_configured
Type
64-bit Gauge
Description

The number of NF configured entries currently cached

Labels

target_nf_type

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- [Summary](#)
- [Packet Streaming Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Packet Streaming Metrics	num_packet_streaming_attempted	Counter	The total number of attempted streaming of packets
Packet Streaming Metrics	num_packet_streaming_success	Counter	The total number of packets streamed successfully
Packet Streaming Metrics	num_packet_streaming_failed	Counter	The total number of failed streaming of packets
Packet Streaming Metrics	num_packet_streaming_bytes	Counter	The total number of bytes successfully streamed

Packet Streaming Metrics

- num_packet_streaming_attempted

Type

64-bit Counter

Description

The total number of attempted streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_success

Type

64-bit Counter

Description

The total number of packets streamed successfully

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_failed

Type

64-bit Counter

Description

The total number of failed streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_bytes

Type

64-bit Counter

Description

The total number of bytes successfully streamed

Labels

interface_name, server_fqdn, kafka_topic

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- [Summary](#)
- [Circuit Breaker Metrics](#)
- [Service Endpoint Learner Metrics](#)
- [Service Endpoint Manager Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Circuit Breaker Metrics	sem_cb_endpoints	Gauge	Total number of endpoints for a service in respective states
Service Endpoint Learner Metrics	sem_num_dns_errors	Counter	Total number of DNS resolution errors for a given service
Service Endpoint Learner Metrics	sem_num_endpoints	Gauge	Total number of endpoints learnt for a service
Service Endpoint Manager Metrics	sem_num_endpoint_transactions_sent	Counter	Total number of times an endpoint is given to the application for communication with it
Service Endpoint Manager Metrics	sem_num_endpoint_transaction_status_received	Counter	Total number of transactions for a given endpoint received with the transaction status

Circuit Breaker Metrics

- [sem_cb_endpoints](#)

Type

64-bit Gauge

Description

Total number of endpoints for a service in respective states. There are 3 states viz. Active, Blocked, Restoring. In Active state, the endpoint is responding as expected, i.e. the response code doesn't indicate a service failure. The endpoint will be moved to Blocked state based on response failures which indicate service failure. After a timeout, the endpoint will be moved to Restoring state to check if the endpoint has recovered. In this state, if endpoint responds with response code other than service failure, it will be moved to Active state. Else, it will be moved back to Blocked state.

Labels

service_name, endpoint_state

Service Endpoint Learner Metrics

- [sem_num_dns_errors](#)

Type

64-bit Counter

Description

Total number of DNS resolution errors for a given service, labels are service name for DNS resolution and DNS error codes

Labels

service_name, error_code

- [sem_num_endpoints](#)

Type

64-bit Gauge

Description

Total number of endpoints learnt from the Kubernetes DNS server for a given service

Labels

service

Service Endpoint Manager Metrics

- sem_num_endpoint_transactions_sent

Type

64-bit Counter

Description

Total number of times an endpoint is given to the application for communication with it

Labels

service

- sem_num_endpoint_transaction_status_received

Type

64-bit Counter

Description

Total number of transactions for a given endpoint received with the transaction status

Labels

service, status

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- [Summary](#)
- [ChargingCmn statistics](#)

Summary

Group	Metric Name	Type	Synopsis
ChargingCmn statistics	svcmgr_service_available_ind_notification	Counter	How many times, Service Available Indication message received
ChargingCmn statistics	svcmgr_service_available_ind_parse_errs	Counter	How many times, Service Available Indication parse error happened
ChargingCmn statistics	svcmgr_service_available_ind_notification_ack	Counter	How many times, Service Available Indication message is acknowledged
ChargingCmn statistics	svcmgr_service_unavailable_ind_notification	Counter	How many times, Service Unavailable Indication message
ChargingCmn statistics	svcmgr_service_unavailable_ind_parse_errs	Counter	How many times, Service Unavailable Indication parse error happened
ChargingCmn statistics	svcmgr_service_unavailable_ind_notification_ack	Counter	How many times, Service Unavailable Indication message is acknowledged
ChargingCmn statistics	svcmgr_publish_msg_sent	Counter	How many times, Publication message sent
ChargingCmn statistics	svcmgr_publish_msg_success	Counter	How many times, Publication message success response received
ChargingCmn statistics	svcmgr_publish_msg_failure	Counter	How many times, failure received for publication message
ChargingCmn statistics	svcmgr_subscribe_msg_sent	Counter	How many times, Subscribe message sent

Group	Metric Name	Type	Synopsis
ChargingCmn statistics	svcmgr_subscribe_msg_success	Counter	How many times, subscribe message success
ChargingCmn statistics	svcmgr_subscribe_msg_failure	Counter	How many times, failure received for subscribe message
ChargingCmn statistics	svcmgr_unsubscribe_msg_sent	Counter	How many times, unsubscribe message sent
ChargingCmn statistics	svcmgr_unsubscribe_msg_failure	Counter	How many times, failure received for unsubscribe message
ChargingCmn statistics	svcmgr_config_req_sent	Counter	How many times, config request sent
ChargingCmn statistics	svcmgr_config_req_failure	Counter	How many times, failure received for config request
ChargingCmn statistics	svcmgr_config_req_success	Counter	How many times, success received for config request
ChargingCmn statistics	svcmgr_config_register_response	Counter	How many times, config register response received
ChargingCmn statistics	svcmgr_config_register_response_ack	Counter	How many times, config register response is acknowledged
ChargingCmn statistics	svcmgr_config_register_response_parse_err	Counter	How many times, parse error happened for config register response
ChargingCmn statistics	svcmgr_config_update_notification	Counter	How many times, config update notification received
ChargingCmn statistics	svcmgr_config_update_notification_response	Counter	How many times, config update response
ChargingCmn statistics	svcmgr_intf_schema_update_ind_notification	Counter	How many times, interface schema update indication message received
ChargingCmn statistics	svcmgr_intf_schema_update_ind_notification_ack	Counter	How many times, interface schema update indication message is acknowledged
ChargingCmn statistics	svcmgr_intf_schema_update_ind_parse_errs	Counter	How many times, interface schema update indication parse error happened

ChargingCmn statistics

- svcmgr_service_available_ind_notification

Type

64-bit Counter

Description

Number of Times Service Available Indication notification received

Labels

servicename

- svcmgr_service_available_ind_parse_errs

Type

64-bit Counter

Description

Number of Times Service Available Indication parse error happened

Labels

- svcmgr_service_available_ind_notification_ack

Type

64-bit Counter

Description

Number of Times Service Available Indication message is acknowledged

Labels

servicename

- **svcmgr_service_unavailable_ind_notification**

Type

64-bit Counter

Description

Number of Times Service Unavailable Indication notification received

Labels

servicename

- **svcmgr_service_unavailable_ind_parse_errs**

Type

64-bit Counter

Description

Number of Times Service Unavailable Indication parse error happened

Labels

- **svcmgr_service_unavailable_ind_notification_ack**

Type

64-bit Counter

Description

Number of Times Service Unavailable Indication message is acknowledged

Labels

servicename

- **svcmgr_publish_msg_sent**

Type

64-bit Counter

Description

Number of Times Publication message sent

Labels

servicename, retry

- **svcmgr_publish_msg_success**

Type

64-bit Counter

Description

Number of Times Publication message success response received

Labels

servicename, causecode, retry

• svcmgr_publish_msg_failure
Type
64-bit Counter
Description
Number of Times failure received for publication message
Labels
<i>servicename, causecode, retry</i>
• svcmgr_subscribe_msg_sent
Type
64-bit Counter
Description
Number of Times Subscribe message sent
Labels
<i>servicename, retry</i>
• svcmgr_subscribe_msg_success
Type
64-bit Counter
Description
Number of Times Send subscribe message success response received
Labels
<i>servicename, causecode, retry</i>
• svcmgr_subscribe_msg_failure
Type
64-bit Counter
Description
Number of time failure received for subscribe message
Labels
<i>servicename, causecode, retry</i>
• svcmgr_unsubscribe_msg_sent
Type
64-bit Counter
Description
Number of Times unsubscribe message sent
Labels
<i>servicename</i>
• svcmgr_unsubscribe_msg_failure
Type
64-bit Counter
Description

Number of Times failure received for unsubscribe message

Labels

servicename

- **svcmgr_config_req_sent**

Type

64-bit Counter

Description

Number of Times config request sent

Labels

retry

- **svcmgr_config_req_failure**

Type

64-bit Counter

Description

Number of Times failure received for config request

Labels

causecode, retry

- **svcmgr_config_req_success**

Type

64-bit Counter

Description

Number of Times success received for config request

Labels

causecode, retry

- **svcmgr_config_register_response**

Type

64-bit Counter

Description

Number of Times config register response received

Labels

statuscode

- **svcmgr_config_register_response_ack**

Type

64-bit Counter

Description

Number of Times config register response is acknowledged

Labels

- **svcmgr_config_register_response_parse_err**

Type

64-bit Counter

Description

Number of Times parse error happened for config register response

Labels

• svcmgr_config_update_notification

Type

64-bit Counter

Description

Number of Times config update notification received

Labels

• svcmgr_config_update_notification_response

Type

64-bit Counter

Description

Number of Times config update response sent

Labels

• svcmgr_intf_schema_update_ind_notification

Type

64-bit Counter

Description

Number of Times interface schema update indication message received

Labels

servicename

• svcmgr_intf_schema_update_ind_notification_ack

Type

64-bit Counter

Description

Number of Times interface schema update indication message is acknowledged

Labels

servicename

• svcmgr_intf_schema_update_ind_parse_errs

Type

64-bit Counter

Description

Number of Times Service interface schema update indication parse error happened

Labels

- [Summary](#)
- [NRF Monitor common statistics](#)

Summary

Group	Metric Name	Type	Synopsis
NRF Monitor common statistics	ready_pods	Gauge	The number of ready pods in the federation

NRF Monitor common statistics

- [ready_pods](#)

Type

64-bit Gauge

Description

The number of ready pods in the federation

Labels

pod_type

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- [Summary](#)
- [ETCD Client Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_put_requests	Counter	The total number of ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_requests_sent	Counter	The total number of ETCD Client PUT Requests sent.
ETCD Client Metrics	etcd_put_successes	Counter	The total number of successful ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_failures	Counter	The total number of failed ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_with_lease_requests	Counter	The total number of ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_requests_sent	Counter	The total number of ETCD Client PUT with Lease Requests sent.
ETCD Client Metrics	etcd_put_with_lease_successes	Counter	The total number of successful ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_failures	Counter	The total number of failed ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_range_requests	Counter	The total number of ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_requests_sent	Counter	The total number of ETCD Client Range Requests sent.
ETCD Client Metrics	etcd_range_successes	Counter	The total number of successful ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_failures	Counter	The total number of failed ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_delete_requests	Counter	The total number of ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_requests_sent	Counter	The total number of ETCD Client Range Delete Requests sent.
ETCD Client Metrics	etcd_range_delete_successes	Counter	The total number of successful ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_failures	Counter	The total number of failed ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_txn_requests	Counter	The total number of ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_requests_sent	Counter	The total number of ETCD Client Txn Requests sent.
ETCD Client Metrics	etcd_txn_successes	Counter	The total number of successful ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_failures	Counter	The total number of failed ETCD Client Txn Requests.
ETCD Client Metrics	etcd_watch_requests	Counter	The total number of ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_requests_sent	Counter	The total number of ETCD Client Watch Requests sent.
ETCD Client Metrics	etcd_watch_successes	Counter	The total number of successful ETCD Client Watch Requests.

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_watch_failures	Counter	The total number of failed ETCD Client Watch Requests.
ETCD Client Metrics	etcdwatcher_requests	Counter	The total number of ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_requests_sent	Counter	The total number of ETCD Watcher Requests sent.
ETCD Client Metrics	etcdwatcher_successes	Counter	The total number of successful ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_failures	Counter	The total number of failed ETCD Watcher Requests.

ETCD Client Metrics

- etcd_put_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests.

Labels

keyPrefix

- etcd_put_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests sent.

Labels

keyPrefix

- etcd_put_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT Requests.

Labels

keyPrefix

- etcd_put_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT Requests.

Labels

keyPrefix, failureReason

- etcd_put_with_lease_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests sent.

Labels

keyPrefix

- etcd_put_with_lease_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT with Lease Requests.

Labels

keyPrefix, failureReason

- etcd_range_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests sent.

Labels

keyPrefix

- etcd_range_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Requests.

Labels

keyPrefix, failureReason

- etcd_range_delete_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests.

Labels

keyPrefix

- etcd_range_delete_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests sent.

Labels

keyPrefix

- etcd_range_delete_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Delete Requests.

Labels

keyPrefix

- etcd_range_delete_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Delete Requests.

Labels

keyPrefix, failureReason

• etcd_txn_requests

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests.

Labels

keyPrefix

• etcd_txn_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests sent.

Labels

keyPrefix

• etcd_txn_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Txn Requests.

Labels

keyPrefix

• etcd_txn_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Txn Requests.

Labels

keyPrefix, failureReason

• etcd_watch_requests

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests.

Labels

keyPrefix

• etcd_watch_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests sent.

Labels

keyPrefix

- etcd_watch_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Watch Requests.

Labels

keyPrefix, failureReason

- etcdwatcher_requests

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests.

Labels

keyPrefix

- etcdwatcher_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests sent.

Labels

keyPrefix

- etcdwatcher_successes

Type

64-bit Counter

Description

The total number of successful ETCD Watcher Requests.

Labels

keyPrefix

- etcdwatcher_failures

Type

64-bit Counter

Description

The total number of failed ETCD Watcher Requests.

Labels

keyPrefix, failureReason

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- [Summary](#)
- [HTTP Punting Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_sent	Counter	The total number of HTTP requests successfully punted.
HTTP Punting Metrics	punting_http_requests_failed	Counter	The total number of HTTP requests punting failed.
HTTP Punting Metrics	punting_http_responses_received	Counter	The total number of HTTP responses received for the requests punted.
HTTP Punting Metrics	punting_http_responses_forwarded	Counter	The total number of HTTP responses forwarded after punting transaction completion.
HTTP Punting Metrics	punting_http_requests_aborted	Counter	The total number of HTTP requests aborted.
HTTP Punting Metrics	punting_http_request_duration_seconds	Histogram	Punting operation duration in seconds.

HTTP Punting Metrics

- punting_http_requests_sent

Type

64-bit Counter

Description

The total number of HTTP requests successfully punted.

Labels

message_type

- punting_http_requests_failed

Type

64-bit Counter

Description

The total number of HTTP requests for which the punting failed.

Labels

message_type

- punting_http_responses_received

Type

64-bit Counter

Description

The total number of HTTP responses received for the requests punted.

Labels

message_type, status_code

- `punting_http_responses_forwarded`

Type

64-bit Counter

Description

The total number of HTTP responses forwarded after punting transaction completion.

Labels

message_type, status_code

- `punting_http_requests_aborted`

Type

64-bit Counter

Description

The total number of HTTP requests aborted.

Labels

message_type

- `punting_http_request_duration_seconds`

Type

64-bit Histogram

Description

The punting transaction duration is measured from the time the HTTP request is sent out to the time the response is received for the punting request.

Labels

message_type

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

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- [Summary](#)
- [NfMonitor statistics](#)
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- [Feature Functionality statistics](#)
- [Consul statistics](#)

Summary

Group	Metric Name	Type	Synopsis
NfMonitor statistics	nfmonitor_unstable_nf_counter	Counter	Howmany times, the Nf is found to be Unstable

Group	Metric Name	Type	Synopsis
NfMonitor statistics	nfmonitor_force_restart_due_to_nf_unstability	Counter	Numbe of times the VNFRestart triggered due to Unstability of the NF
NfMonitor statistics	nfmonitor_cumulative_readiness_failure_count	Counter	Numbe of times Service sound to have insuffucient count due to Readiness probe
NfMonitor statistics	nfmonitor_cumulative_readiness_failure_alerts	Counter	Numbe of times Service sound to have insuffucient count due to Readiness probe
NfMonitor statistics	nfmonitor_cumulative_k8api_get_query_failures	Counter	Numbe of times the k8 get Api query failures
DebugNfMonitorCmn	nfmonitor_force_restart_counter	Counter	Number of times force-restart triggered
NfManagement statistics	nfmonitor_nfmgmt_service_request_attempts	Counter	Howmany times, the Nf is attempting to send NRF Requests
NfManagement statistics	nfmonitor_nfmgmt_service_request_success	Counter	Numbe of times Success response is received for the requests to NRF
NfManagement statistics	nfmonitor_nfmgmt_service_request_failures	Counter	Numbe of times Failure response is received for the requests to NRF
NfManagement statistics	nfmonitor_nfmgmt_aborted_request_msgs	Counter	Numbe of times the requests to NRF is Aborted/Cancelled
Debug statistics	nfmonitor_vnf_restart_trigger	Counter	Howmany times, the Force Restart is received
Debug statistics	nfmonitor_fsm_state_changes	Counter	The statistics for the the FSM state changes
Debug statistics	nfmonitor_deactivation_triggers	Counter	Numbe of times the NF deactivation trigger is received
Debug statistics	nfmonitor_critical_service_up_notification	Counter	Numbe of times the Critical Service Up Notification received
Debug statistics	nfmonitor_critical_service_down_notification	Counter	Numbe of times the Critical Service Down Notification received
Debug statistics	nfmonitor_critical_services_overall_up_notification	Counter	Numbe of times the Up notification received when all monitored Critical Services are available
Debug statistics	nfmonitor_critical_services_overall_down_notification	Counter	Numbe of times the Down notification received when at least one Critical Service was unavailable
Debug statistics	nfmonitor_hysteresis_timer_started	Counter	Numbe of times the Hysteresis Timer is started due to missing Critical Service
Debug statistics	nfmonitor_hysteresis_timer_stopped	Counter	Numbe of times the Hysteresis Timer is stopped due to Critical Service become available
Debug statistics	nfmonitor_hysteresis_timer_expired	Counter	Numbe of times the Hysteresis Timer is expired due to missing Critical Service
Debug statistics	nfmonitor_personality_update_triggers	Counter	Numbe of times the Nf Personality update is received
Debug statistics	nfmonitor_maintenance_trigger_received	Counter	Numbe of times the Maintenance Mode is activated
Debug statistics	nfmonitor_profile_update_triggers	Counter	Numbe of times the Nf Profile Update happened
Debug statistics	nfmonitor_critical_service_down_on_register_entry	Counter	Numbe of times the Critical Services were down while starting with RegisterFsmState
ApiServer statistics	nfmonitor_deployment_create_watch_request	Counter	Howmany times, the deployment watch requests are created
ApiServer statistics	nfmonitor_deployment_delete_watch_request	Counter	Howmany times, the deployment watch requests are deleted
ApiServer statistics	nfmonitor_deployment_watch_notification_success	Counter	Howmany times, the watch notifications are successful
ApiServer statistics	nfmonitor_deployment_watch_notification_error	Counter	Howmany times, the watch notifications are failed
ApiServer statistics	nfmonitor_limit_range_query_attempts	Counter	Howmany times, the limit-range query attempts are made
ApiServer statistics	nfmonitor_limit_range_query_failures	Counter	Howmany times, the limit-range query attempts are failed
ApiServer statistics	nfmonitor_limit_range_query_success	Counter	Howmany times, the limit-range query attempts are succeeded
ApiServer statistics	nfmonitor_hpa_query_attempts	Counter	Howmany times, the Hpa query attempts are made
ApiServer statistics	nfmonitor_hpa_query_failures	Counter	Howmany times, the Hpa query attempts are failed
ApiServer statistics	nfmonitor_hpa_query_success	Counter	Howmany times, the Hpa query attempts are succeeded
ApiServer statistics	nfmonitor_readiness_probe_query_attempts	Counter	Howmany times, the Readiness Probe query attempts are made
ApiServer statistics	nfmonitor_readiness_probe_query_failures	Counter	Howmany times, the Readiness Probe query attempts are failed
ApiServer statistics	nfmonitor_readiness_probe_query_success	Counter	Howmany times, the Readiness Probe query attempts are succeeded

Group	Metric Name	Type	Synopsis
Feature Functionality statistics	nfmonitor_critical_service_frequent_restart	Counter	Howmany times, minimum count is not met for the stable Critical Services
Feature Functionality statistics	nfmonitor_alarm_due_to_frequent_restart	Counter	Howmany times alarm raised, due to minimum count is not met for the stable Critical Services
Feature Functionality statistics	nfmonitor_force_restart_due_to_frequent_restart	Counter	Howmany times, force restart triggerd because of the unstable Critical Services
Feature Functionality statistics	nfmonitor_critical_service_readiness_failure	Counter	Howmany times, minimum count is not met for the readiness succeeded Critical Services
Feature Functionality statistics	nfmonitor_alarm_due_to_readiness_failure	Counter	Alarm raised, due to minimum count is not met for the readiness succeeded Critical Services
Feature Functionality statistics	nfmonitor_force_restart_due_to_readiness_failure	Counter	Force restart triggerd because of Critical Service Readiness failure
Consul statistics	nfmonitor_publish_status_into_consul	Counter	Number of times Publish message is sent to Consul
Consul statistics	nfmonitor_consul_read_attempts	Counter	Number of times Consul read attempts
Consul statistics	nfmonitor_consul_read_failures	Counter	Number of times Consul read attempts failed
Consul statistics	nfmonitor_consul_read_success	Counter	Number of times Consul read attempts succeeded

NfMonitor statistics

- [nfmonitor_unstable_nf_counter](#)

Type

64-bit Counter

Description

Number of Times Nf is found Unstable

Labels

- [nfmonitor_force_restart_due_to_nf_unstability](#)

Type

64-bit Counter

Description

The number of times the Force Restart triggered due to NF-Unstability

Labels

- [nfmonitor_cumulative_readiness_failure_count](#)

Type

64-bit Counter

Description

The number of times the service is found to be failing the Readiness probe

Labels

critical_service_name, pod_name

- [nfmonitor_cumulative_readiness_failure_alerts](#)

Type

64-bit Counter

Description

Total number of EventAlerts generated due to Readiness probe failure

Labels
<i>critical_service_name, pod_name</i>
nfmonitor_cumulative_k8api_get_query_failures
Type
64-bit Counter
Description
Total number of k8 api get pod query failures
Labels
<i>critical_service_name, pod_name</i>

DebugNfMonitorCmn

nfmonitor_force_restart_counter
Type
64-bit Counter
Description
Number of times force-restart triggered
Labels
<i>agent, name</i>

NfManagement statistics

nfmonitor_nfmgmt_service_request_attempts
Type
64-bit Counter
Description
Number of Requests sent to NRF
Labels
<i>instance_id, msg_type, method, retry</i>
nfmonitor_nfmgmt_service_request_success
Type
64-bit Counter
Description
The number of times the Request to NRF is Successful
Labels
<i>instance_id, msg_type, method, retry, status_code</i>
nfmonitor_nfmgmt_service_request_failures
Type
64-bit Counter
Description
The number of times the Request to NRF is Failure
Labels

instance_id, msg_type, method, retry, status_code

- nfmonitor_nfmgmt_aborted_request_msgs

Type

64-bit Counter

Description

The number of times the Request to NRF is Aborted

Labels

instance_id, msg_type

Debug statistics

- nfmonitor_vnf_restart_trigger

Type

64-bit Counter

Description

Number of VNF Restart triggers received

Labels

curr_nf_status, nfmgmt_state, nfmonitor_fsm_state, up_count, timestamp, trigger, type

- nfmonitor_fsm_state_changes

Type

64-bit Counter

Description

The FSM State Change events stats

Labels

curr_nf_status, nfmgmt_state, up_count, from_state, to_state

- nfmonitor_deactivation_triggers

Type

64-bit Counter

Description

The number of times the deactivation trigger received

Labels

nf_status, nfmgmt_state, up_count, nfmonitor_fsm_state

- nfmonitor_critical_service_up_notification

Type

64-bit Counter

Description

The number of times the CriticalService Up notification received

Labels

service, criticality, nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_critical_service_down_notification

Type

64-bit Counter

Description

The number of times the CriticalService Down notification received

Labels

service, criticality, nf_status, nfmgmt_state, up_count, fsm_state

• nfmonitor_critical_services_overall_up_notification

Type

64-bit Counter

Description

The number of times the Overall CriticalServices Up notification received

Labels

nf_status, nfmgmt_state, up_count, fsm_state

• nfmonitor_critical_services_overall_down_notification

Type

64-bit Counter

Description

The number of times the Overall CriticalServices Down notification received

Labels

nf_status, nfmgmt_state, up_count, fsm_state

• nfmonitor_hysteresis_timer_started

Type

64-bit Counter

Description

The number of times the Hysteresis Timer is started

Labels

nf_status, nfmgmt_state, up_count, fsm_state

• nfmonitor_hysteresis_timer_stopped

Type

64-bit Counter

Description

The number of times the Hysteresis Timer is stopped

Labels

nf_status, nfmgmt_state, up_count, fsm_state

• nfmonitor_hysteresis_timer_expired

Type

64-bit Counter

Description

The number of times the Hysteresis Timer is expired

Labels

nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_personality_update_triggers

Type

64-bit Counter

Description

The number of times the Nf Personality Update is received

Labels

nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_maintenance_trigger_received

Type

64-bit Counter

Description

The number of times the Maintenance Mode is activated

Labels

nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_profile_update_triggers

Type

64-bit Counter

Description

The number of times the Nf Profile Update happened

Labels

nf_status, nfmgmt_state, up_count, fsm_state, update_operation

- nfmonitor_critical_service_down_on_register_entry

Type

64-bit Counter

Description

The number of times the Critical Services were down while starting with RegisterFsmState

Labels

nf_status, up_count

ApiServer statistics

- nfmonitor_deployment_create_watch_request

Type

64-bit Counter

Description

The number of Watch Reqs created for deployment type

Labels

resource

- nfmonitor_deployment_delete_watch_request

Type

64-bit Counter

Description

The number of Watch Reqs deleted for deployment type

Labels

resource

- nfmonitor_deployment_watch_notification_success

Type

64-bit Counter

Description

The number of times Watch notifications are successful

Labels

resource

- nfmonitor_deployment_watch_notification_error

Type

64-bit Counter

Description

The number of times Watch notifications are failed

Labels

resource, reason

- nfmonitor_limit_range_query_attempts

Type

64-bit Counter

Description

The number of Limit Range query attempts

Labels

-
- nfmonitor_limit_range_query_failures

Type

64-bit Counter

Description

The number of Limit Range query failures

Labels

status_code

- nfmonitor_limit_range_query_success

Type

64-bit Counter

Description

The number of Limit Range query succeeded

Labels

status_code

nfmonitor_hpa_query_attempts
Type
64-bit Counter
Description
The number of Hpa query attempts
Labels
<i>resource, continue_param</i>
nfmonitor_hpa_query_failures
Type
64-bit Counter
Description
The number of Hpa query failures
Labels
<i>resource, continue_param, status_code</i>
nfmonitor_hpa_query_success
Type
64-bit Counter
Description
The number of Hpa query succeeded
Labels
<i>resource, continue_param, status_code</i>
nfmonitor_readiness_probe_query_attempts
Type
64-bit Counter
Description
The number of Readiness Probe query attempts
Labels
<i>resource, continue_param</i>
nfmonitor_readiness_probe_query_failures
Type
64-bit Counter
Description
The number of Readiness Probe query failures
Labels
<i>resource, continue_param, status_code</i>
nfmonitor_readiness_probe_query_success
Type
64-bit Counter
Description

The number of Readiness Probe query succeeded

Labels

resource, continue_param, status_code

Feature Functionality statistics

- nfmonitor_critical_service_frequent_restart

Type

64-bit Counter

Description

Howmany times the minimum required stable CriticalService count is not met

Labels

service

- nfmonitor_alarm_due_to_frequent_restart

Type

64-bit Counter

Description

Howmany times Alarm is raised due to minimum required stable CriticalService count is not met

Labels

service, alarm_status

- nfmonitor_force_restart_due_to_frequent_restart

Type

64-bit Counter

Description

Howmany times Force Restart initiated due to missing stable CriticalService count

Labels

service

- nfmonitor_critical_service_readiness_failure

Type

64-bit Counter

Description

Howmany times the minimum required readiness succeeded CriticalService count is not met

Labels

service

- nfmonitor_alarm_due_to_readiness_failure

Type

64-bit Counter

Description

Alarm raised due to minimum count is not met for the readiness succeeded CriticalService

Labels

service, alarm_status

• <code>nfmonitor_force_restart_due_to_readiness_failure</code>
Type
64-bit Counter
Description
Howmany times Force Restart initiated due to readiness failure of CriticalService
Labels
<i>service</i>

Consul statistics

• <code>nfmonitor_publish_status_into_consul</code>
Type
64-bit Counter
Description
Howmany times Publish is sent to Consul
Labels
<i>nf_status, nf_state, up_count, fsm_state</i>
• <code>nfmonitor_consul_read_attempts</code>
Type
64-bit Counter
Description
Howmany times Consul read attempt is made
Labels
<i>fsm_state, retry</i>
• <code>nfmonitor_consul_read_failures</code>
Type
64-bit Counter
Description
Howmany times Consul read attempt are failed
Labels
<i>fsm_state, retry, status_code</i>
• <code>nfmonitor_consul_read_success</code>
Type
64-bit Counter
Description
Howmany times Consul read attempt are succeeded
Labels
<i>fsm_state, retry, status_code</i>

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- [Summary](#)
- [NRF Monitor statistics](#)

Summary

Group	Metric Name	Type	Synopsis
NRF Monitor statistics	nrf_nf_registered	Gauge	Current number of NFs in Registered State
NRF Monitor statistics	nrf_nf_suspended	Gauge	Current number of NFs in Suspended State
NRF Monitor statistics	nrf_nf_type	Gauge	Current number of subscriptions per NF type
NRF Monitor statistics	nrf_nf_instance	Gauge	Current number of subscriptions per NF instance

NRF Monitor statistics

- [nrf_nf_registered](#)

Type

64-bit Gauge

Description

The number of NFs currently in Registered State

Labels

source_nf_type

- [nrf_nf_suspended](#)

Type

64-bit Gauge

Description

The number of NFs currently in Suspended State

Labels

source_nf_type

- [nrf_nf_type](#)

Type

64-bit Gauge

Description

The number of subscriptions per NF type

Labels

source_nf_type

- [nrf_nf_instance](#)

Type

64-bit Gauge

Description

The number of subscriptions per NF instance

Labels

source_nf_type

Table of Contents

- [Summary](#)
- [Packet Streaming Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Packet Streaming Metrics	num_packet_streaming_attempted	Counter	The total number of attempted streaming of packets
Packet Streaming Metrics	num_packet_streaming_success	Counter	The total number of packets streamed successfully
Packet Streaming Metrics	num_packet_streaming_failed	Counter	The total number of failed streaming of packets
Packet Streaming Metrics	num_packet_streaming_bytes	Counter	The total number of bytes successfully streamed

Packet Streaming Metrics

- num_packet_streaming_attempted

Type

64-bit Counter

Description

The total number of attempted streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_success

Type

64-bit Counter

Description

The total number of packets streamed successfully

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_failed

Type

64-bit Counter

Description

The total number of failed streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_bytes

Type

64-bit Counter

Description

The total number of bytes successfully streamed

Labels

interface_name, server_fqdn, kafka_topic

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- [Summary](#)
- [Circuit Breaker Metrics](#)
- [Service Endpoint Learner Metrics](#)

- [Service Endpoint Manager Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Circuit Breaker Metrics	sem_cb_endpoints	Gauge	Total number of endpoints for a service in respective states
Service Endpoint Learner Metrics	sem_num_dns_errors	Counter	Total number of DNS resolution errors for a given service
Service Endpoint Learner Metrics	sem_num_endpoints	Gauge	Total number of endpoints learnt for a service
Service Endpoint Manager Metrics	sem_num_endpoint_transactions_sent	Counter	Total number of times an endpoint is given to the application for communication with it
Service Endpoint Manager Metrics	sem_num_endpoint_transaction_status_received	Counter	Total number of transactions for a given endpoint received with the transaction status

Circuit Breaker Metrics

- [sem_cb_endpoints](#)

Type

64-bit Gauge

Description

Total number of endpoints for a service in respective states. There are 3 states viz. Active, Blocked, Restoring. In Active state, the endpoint is responding as expected, i.e. the response code doesn't indicate a service failure. The endpoint will be moved to Blocked state based on response failures which indicate service failure. After a timeout, the endpoint will be moved to Restoring state to check if the endpoint has recovered. In this state, if endpoint responds with response code other than service failure, it will be moved to Active state. Else, it will be moved back to Blocked state.

Labels

service_name, endpoint_state

Service Endpoint Learner Metrics

- [sem_num_dns_errors](#)

Type

64-bit Counter

Description

Total number of DNS resolution errors for a given service, labels are service name for DNS resolution and DNS error codes

Labels

service_name, error_code

- [sem_num_endpoints](#)

Type

64-bit Gauge

Description

Total number of endpoints learnt from the Kubernetes DNS server for a given service

Labels

service

Service Endpoint Manager Metrics

- [sem_num_endpoint_transactions_sent](#)

Type
64-bit Counter
Description
Total number of times an endpoint is given to the application for communication with it
Labels
service
sem_num_endpoint_transaction_status_received
Type
64-bit Counter
Description
Total number of transactions for a given endpoint received with the transaction status
Labels
service, status

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- [Summary](#)
- [ChargingCmn statistics](#)

Summary

Group	Metric Name	Type	Synopsis
ChargingCmn statistics	svcmgr_service_available_ind_notification	Counter	How many times, Service Available Indication message received
ChargingCmn statistics	svcmgr_service_available_ind_parse_errs	Counter	How many times, Service Available Indication parse error happened
ChargingCmn statistics	svcmgr_service_available_ind_notification_ack	Counter	How many times, Service Available Indication message is acknowledged
ChargingCmn statistics	svcmgr_service_unavailable_ind_notification	Counter	How many times, Service Unavailable Indication message
ChargingCmn statistics	svcmgr_service_unavailable_ind_parse_errs	Counter	How many times, Service Unavailable Indication parse error happened
ChargingCmn statistics	svcmgr_service_unavailable_ind_notification_ack	Counter	How many times, Service Unavailable Indication message is acknowledged
ChargingCmn statistics	svcmgr_publish_msg_sent	Counter	How many times, Publication message sent
ChargingCmn statistics	svcmgr_publish_msg_success	Counter	How many times, Publication message success response received
ChargingCmn statistics	svcmgr_publish_msg_failure	Counter	How many times, failure received for publication message
ChargingCmn statistics	svcmgr_subscribe_msg_sent	Counter	How many times, Subscribe message sent
ChargingCmn statistics	svcmgr_subscribe_msg_success	Counter	How many times, subscribe message success
ChargingCmn statistics	svcmgr_subscribe_msg_failure	Counter	How many times, failure received for subscribe message
ChargingCmn statistics	svcmgr_unsubscribe_msg_sent	Counter	How many times, unsubscribe message sent
ChargingCmn statistics	svcmgr_unsubscribe_msg_failure	Counter	How many times, failure received for unsubscribe message

Group	Metric Name	Type	Synopsis
ChargingCmn statistics	svcmgr_config_req_sent	Counter	How many times, config request sent
ChargingCmn statistics	svcmgr_config_req_failure	Counter	How many times, failure received for config request
ChargingCmn statistics	svcmgr_config_req_success	Counter	How many times, success received for config request
ChargingCmn statistics	svcmgr_config_register_response	Counter	How many times, config register response received
ChargingCmn statistics	svcmgr_config_register_response_ack	Counter	How many times, config register response is acknowledged
ChargingCmn statistics	svcmgr_config_register_response_parse_err	Counter	How many times, parse error happened for config register response
ChargingCmn statistics	svcmgr_config_update_notification	Counter	How many times, config update notification received
ChargingCmn statistics	svcmgr_config_update_notification_response	Counter	How many times, config update response
ChargingCmn statistics	svcmgr_intf_schema_update_ind_notification	Counter	How many times, interface schema update indication message received
ChargingCmn statistics	svcmgr_intf_schema_update_ind_notification_ack	Counter	How many times, interface schema update indication message is acknowledged
ChargingCmn statistics	svcmgr_intf_schema_update_ind_parse_errs	Counter	How many times, interface schema update indication parse error happened

ChargingCmn statistics

- svcmgr_service_available_ind_notification

Type

64-bit Counter

Description

Number of Times Service Available Indication notification received

Labels

servicename

- svcmgr_service_available_ind_parse_errs

Type

64-bit Counter

Description

Number of Times Service Available Indication parse error happened

Labels

- svcmgr_service_available_ind_notification_ack

Type

64-bit Counter

Description

Number of Times Service Available Indication message is acknowledged

Labels

servicename

- svcmgr_service_unavailable_ind_notification

Type
64-bit Counter
Description
Number of Times Service Unavailable Indication notification received
Labels
<i>servicename</i>

- svcmgr_service_unavailable_ind_parse_errs

Type
64-bit Counter
Description
Number of Times Service Unavailable Indication parse error happened
Labels

- svcmgr_service_unavailable_ind_notification_ack

Type
64-bit Counter
Description
Number of Times Service Unavailable Indication message is acknowledged
Labels
<i>servicename</i>

- svcmgr_publish_msg_sent

Type
64-bit Counter
Description
Number of Times Publication message sent
Labels
<i>servicename, retry</i>

- svcmgr_publish_msg_success

Type
64-bit Counter
Description
Number of Times Publication message success response received
Labels
<i>servicename, causecode, retry</i>

- svcmgr_publish_msg_failure

Type
64-bit Counter
Description
Number of Times failure received for publication message

Labels

servicename, causecode, retry

- **svcmgr_subscribe_msg_sent**

Type

64-bit Counter

Description

Number of Times Subscribe message sent

Labels

servicename, retry

- **svcmgr_subscribe_msg_success**

Type

64-bit Counter

Description

Number of Times Send subscribe message success response received

Labels

servicename, causecode, retry

- **svcmgr_subscribe_msg_failure**

Type

64-bit Counter

Description

Number of time failure received for subscribe message

Labels

servicename, causecode, retry

- **svcmgr_unsubscribe_msg_sent**

Type

64-bit Counter

Description

Number of Times unsubscribe message sent

Labels

servicename

- **svcmgr_unsubscribe_msg_failure**

Type

64-bit Counter

Description

Number of Times failure received for unsubscribe message

Labels

servicename

- **svcmgr_config_req_sent**

Type

64-bit Counter

Description

Number of Times config request sent

Labels

retry

• `svcmgr_config_req_failure`

Type

64-bit Counter

Description

Number of Times failure received for config request

Labels

causecode, retry

• `svcmgr_config_req_success`

Type

64-bit Counter

Description

Number of Times success received for config request

Labels

causecode, retry

• `svcmgr_config_register_response`

Type

64-bit Counter

Description

Number of Times config register response received

Labels

statuscode

• `svcmgr_config_register_response_ack`

Type

64-bit Counter

Description

Number of Times config register response is acknowledged

Labels

• `svcmgr_config_register_response_parse_err`

Type

64-bit Counter

Description

Number of Times parse error happened for config register response

Labels

• svcmgr_config_update_notification	Type
	64-bit Counter
	Description
	Number of Times config update notification received
	Labels
• svcmgr_config_update_notification_response	Type
	64-bit Counter
	Description
	Number of Times config update response sent
	Labels
• svcmgr_intf_schema_update_ind_notification	Type
	64-bit Counter
	Description
	Number of Times interface schema update indication message received
	Labels
servicename	
• svcmgr_intf_schema_update_ind_notification_ack	Type
	64-bit Counter
	Description
	Number of Times interface schema update indication message is acknowledged
	Labels
servicename	
• svcmgr_intf_schema_update_ind_parse_errs	Type
	64-bit Counter
	Description
	Number of Times Service interface schema update indication parse error happened
	Labels

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- [Summary](#)
- [Nssf Monitor common statistics](#)

Summary

Group	Metric Name	Type	Synopsis
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Group	Metric Name	Type	Synopsis
Nssf Monitor common statistics	ready_pods	Gauge	The number of ready pods in the federation

Nssf Monitor common statistics

- ready_pods

Type

64-bit Gauge

Description

The number of ready pods in the federation

Labels

pod_type

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- [Summary](#)
- [ETCD Client Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_put_requests	Counter	The total number of ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_requests_sent	Counter	The total number of ETCD Client PUT Requests sent.
ETCD Client Metrics	etcd_put_successes	Counter	The total number of successful ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_failures	Counter	The total number of failed ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_with_lease_requests	Counter	The total number of ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_requests_sent	Counter	The total number of ETCD Client PUT with Lease Requests sent.
ETCD Client Metrics	etcd_put_with_lease_successes	Counter	The total number of successful ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_failures	Counter	The total number of failed ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_range_requests	Counter	The total number of ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_requests_sent	Counter	The total number of ETCD Client Range Requests sent.
ETCD Client Metrics	etcd_range_successes	Counter	The total number of successful ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_failures	Counter	The total number of failed ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_delete_requests	Counter	The total number of ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_requests_sent	Counter	The total number of ETCD Client Range Delete Requests sent.
ETCD Client Metrics	etcd_range_delete_successes	Counter	The total number of successful ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_failures	Counter	The total number of failed ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_txn_requests	Counter	The total number of ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_requests_sent	Counter	The total number of ETCD Client Txn Requests sent.
ETCD Client Metrics	etcd_txn_successes	Counter	The total number of successful ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_failures	Counter	The total number of failed ETCD Client Txn Requests.
ETCD Client Metrics	etcd_watch_requests	Counter	The total number of ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_requests_sent	Counter	The total number of ETCD Client Watch Requests sent.
ETCD Client Metrics	etcd_watch_successes	Counter	The total number of successful ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_failures	Counter	The total number of failed ETCD Client Watch Requests.
ETCD Client Metrics	etcdwatcher_requests	Counter	The total number of ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_requests_sent	Counter	The total number of ETCD Watcher Requests sent.
ETCD Client Metrics	etcdwatcher_successes	Counter	The total number of successful ETCD Watcher Requests.

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcdwatcher_failures	Counter	The total number of failed ETCD Watcher Requests.

ETCD Client Metrics

- etcd_put_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests.

Labels

keyPrefix

- etcd_put_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests sent.

Labels

keyPrefix

- etcd_put_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT Requests.

Labels

keyPrefix

- etcd_put_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT Requests.

Labels

keyPrefix, failureReason

- etcd_put_with_lease_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests sent.

Labels

keyPrefix
- etcd_put_with_lease_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT with Lease Requests.

Labels

keyPrefix
- etcd_put_with_lease_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT with Lease Requests.

Labels

keyPrefix, failureReason
- etcd_range_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests.

Labels

keyPrefix
- etcd_range_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests sent.

Labels

keyPrefix
- etcd_range_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Requests.

Labels

keyPrefix

• etcd_range_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Requests.

Labels

keyPrefix, failureReason

• etcd_range_delete_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests.

Labels

keyPrefix

• etcd_range_delete_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests sent.

Labels

keyPrefix

• etcd_range_delete_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Delete Requests.

Labels

keyPrefix

• etcd_range_delete_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Delete Requests.

Labels

keyPrefix, failureReason

• etcd_txn_requests

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests.

Labels

keyPrefix

- etcd_txn_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests sent.

Labels

keyPrefix

- etcd_txn_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Txn Requests.

Labels

keyPrefix

- etcd_txn_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Txn Requests.

Labels

keyPrefix, failureReason

- etcd_watch_requests

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests sent.

Labels

keyPrefix

• etcd_watch_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Watch Requests.

Labels

keyPrefix

• etcd_watch_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Watch Requests.

Labels

keyPrefix, failureReason

• etcdwatcher_requests

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests.

Labels

keyPrefix

• etcdwatcher_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests sent.

Labels

keyPrefix

• etcdwatcher_successes

Type

64-bit Counter

Description

The total number of successful ETCD Watcher Requests.

Labels

keyPrefix

• etcdwatcher_failures

Type

64-bit Counter
Description
The total number of failed ETCD Watcher Requests.
Labels
keyPrefix, failureReason

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- [Summary](#)
- [HTTP Punting Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_sent	Counter	The total number of HTTP requests successfully punted.
HTTP Punting Metrics	punting_http_requests_failed	Counter	The total number of HTTP requests punting failed.
HTTP Punting Metrics	punting_http_responses_received	Counter	The total number of HTTP responses received for the requests punted.
HTTP Punting Metrics	punting_http_responses_forwarded	Counter	The total number of HTTP responses forwarded after punting transaction completion.
HTTP Punting Metrics	punting_http_requests_aborted	Counter	The total number of HTTP requests aborted.
HTTP Punting Metrics	punting_http_request_duration_seconds	Histogram	Punting operation duration in seconds.

HTTP Punting Metrics

- [punting_http_requests_sent](#)

Type

64-bit Counter

Description

The total number of HTTP requests successfully punted.

Labels

message_type
- [punting_http_requests_failed](#)

Type

64-bit Counter

Description

The total number of HTTP requests for which the punting failed.

Labels

message_type
- [punting_http_responses_received](#)

Type

64-bit Counter

Description

The total number of HTTP responses received for the requests punted.

Labels

message_type, status_code

- `punting_http_responses_forwarded`

Type

64-bit Counter

Description

The total number of HTTP responses forwarded after punting transaction completion.

Labels

message_type, status_code

- `punting_http_requests_aborted`

Type

64-bit Counter

Description

The total number of HTTP requests aborted.

Labels

message_type

- `punting_http_request_duration_seconds`

Type

64-bit Histogram

Description

The punting transaction duration is measured from the time the HTTP request is sent out to the time the response is received for the punting request.

Labels

message_type

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

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- [Summary](#)
- [NfMonitor statistics](#)
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- [NfManagement statistics](#)
- [Debug statistics](#)
- [ApiServer statistics](#)
- [Feature Functionality statistics](#)
- [Consul statistics](#)

Summary

Group	Metric Name	Type	Synopsis
NfMonitor statistics	nfmonitor_unstable_nf_counter	Counter	Howmany times, the Nf is found to be Unstable
NfMonitor statistics	nfmonitor_force_restart_due_to_nf_unstability	Counter	Numbe of times the VNFRestart triggered due to Unstability of the NF
NfMonitor statistics	nfmonitor_cumulative_readiness_failure_count	Counter	Numbe of times Service sound to have insuffucient count due to Readiness probe

Group	Metric Name	Type	Synopsis
NfMonitor statistics	nfmonitor_cumulative_readiness_failure_alerts	Counter	Numbe of times Service sound to have insuffucient count due to Readiness probe
NfMonitor statistics	nfmonitor_cumulative_k8api_get_query_failures	Counter	Numbe of times the k8 get Api query failures
DebugNfMonitorCmn	nfmonitor_force_restart_counter	Counter	Number of times force-restart triggered
NfManagement statistics	nfmonitor_nfmgmt_service_request_attempts	Counter	Howmany times, the Nf is attempting to send NRF Requests
NfManagement statistics	nfmonitor_nfmgmt_service_request_success	Counter	Numbe of times Success response is received for the requests to NRF
NfManagement statistics	nfmonitor_nfmgmt_service_request_failures	Counter	Numbe of times Failure response is received for the requests to NRF
NfManagement statistics	nfmonitor_nfmgmt_aborted_request_msgs	Counter	Numbe of times the requests to NRF is Aborted/Cancelled
Debug statistics	nfmonitor_vnf_restart_trigger	Counter	Howmany times, the Force Restart is received
Debug statistics	nfmonitor_fsm_state_changes	Counter	The statistics for the the FSM state changes
Debug statistics	nfmonitor_deactivation_triggers	Counter	Numbe of times the NF deactivation trigger is received
Debug statistics	nfmonitor_critical_service_up_notification	Counter	Numbe of times the Critical Service Up Notification received
Debug statistics	nfmonitor_critical_service_down_notification	Counter	Numbe of times the Critical Service Down Notification received
Debug statistics	nfmonitor_critical_services_overall_up_notification	Counter	Numbe of times the Up notification received when all monitored Critical Services are available
Debug statistics	nfmonitor_critical_services_overall_down_notification	Counter	Numbe of times the Down notification received when at least one Critical Service was unavailable
Debug statistics	nfmonitor_hysterisis_timer_started	Counter	Numbe of times the Hysterisis Timer is started due to missing Critical Service
Debug statistics	nfmonitor_hysterisis_timer_stopped	Counter	Numbe of times the Hysterisis Timer is stopped due to Critical Service become available
Debug statistics	nfmonitor_hysterisis_timer_expired	Counter	Numbe of times the Hysterisis Timer is expired due to missing Critical Service
Debug statistics	nfmonitor_personality_update_triggers	Counter	Numbe of times the Nf Personality update is received
Debug statistics	nfmonitor_maintenance_trigger_received	Counter	Numbe of times the Maintenance Mode is activated
Debug statistics	nfmonitor_profile_update_triggers	Counter	Numbe of times the Nf Profile Update happened
Debug statistics	nfmonitor_critical_service_down_on_register_entry	Counter	Numbe of times the Critical Services were down while starting with RegisterFsmState
ApiServer statistics	nfmonitor_deployment_create_watch_request	Counter	Howmany times, the deployment watch requests are created
ApiServer statistics	nfmonitor_deployment_delete_watch_request	Counter	Howmany times, the deployment watch requests are deleted
ApiServer statistics	nfmonitor_deployment_watch_notification_success	Counter	Howmany times, the watch notifications are successful
ApiServer statistics	nfmonitor_deployment_watch_notification_error	Counter	Howmany times, the watch notifications are failed
ApiServer statistics	nfmonitor_limit_range_query_attempts	Counter	Howmany times, the limit-range query attempts are made
ApiServer statistics	nfmonitor_limit_range_query_failures	Counter	Howmany times, the limit-range query attempts are failed
ApiServer statistics	nfmonitor_limit_range_query_success	Counter	Howmany times, the limit-range query attempts are succeeded
ApiServer statistics	nfmonitor_hpa_query_attempts	Counter	Howmany times, the Hpa query attempts are made
ApiServer statistics	nfmonitor_hpa_query_failures	Counter	Howmany times, the Hpa query attempts are failed
ApiServer statistics	nfmonitor_hpa_query_success	Counter	Howmany times, the Hpa query attempts are succeeded
ApiServer statistics	nfmonitor_readiness_probe_query_attempts	Counter	Howmany times, the Readiness Probe query attempts are made
ApiServer statistics	nfmonitor_readiness_probe_query_failures	Counter	Howmany times, the Readiness Probe query attempts are failed
ApiServer statistics	nfmonitor_readiness_probe_query_success	Counter	Howmany times, the Readiness Probe query attempts are succeeded
Feature Functionality statistics	nfmonitor_critical_service_frequent_restart	Counter	Howmany times, minimum count is not met for the stable Critical Services
Feature Functionality statistics	nfmonitor_alarm_due_to_frequent_restart	Counter	Howmany times alarm raised, due to minimum count is not met for the stable Critical Services

Group	Metric Name	Type	Synopsis
Feature Functionality statistics	nfmonitor_force_restart_due_to_frequent_restart	Counter	Howmany times, force restart triggerd because of the unstable Critical Services
Feature Functionality statistics	nfmonitor_critical_service_readiness_failure	Counter	Howmany times, minimum count is not met for the readiness succeeded Critical Services
Feature Functionality statistics	nfmonitor_alarm_due_to_readiness_failure	Counter	Alarm raised, due to minimum count is not met for the readiness succeeded Critical Services
Feature Functionality statistics	nfmonitor_force_restart_due_to_readiness_failure	Counter	Force restart triggerd because of Critical Service Readiness failure
Consul statistics	nfmonitor_publish_status_into_consul	Counter	Number of times Publish message is sent to Consul
Consul statistics	nfmonitor_consul_read_attempts	Counter	Number of times Consul read attempts
Consul statistics	nfmonitor_consul_read_failures	Counter	Number of times Consul read attempts failed
Consul statistics	nfmonitor_consul_read_success	Counter	Number of times Consul read attempts succeeded

NfMonitor statistics

- [nfmonitor_unstable_nf_counter](#)

Type

64-bit Counter

Description

Number of Times Nf is found Unstable

Labels

- [nfmonitor_force_restart_due_to_nf_unstability](#)

Type

64-bit Counter

Description

The number of times the Force Restart triggered due to NF-Unstability

Labels

- [nfmonitor_cumulative_readiness_failure_count](#)

Type

64-bit Counter

Description

The number of times the service is found to be failing the Readiness probe

Labels

critical_service_name, pod_name

- [nfmonitor_cumulative_readiness_failure_alerts](#)

Type

64-bit Counter

Description

Total number of EventAlerts generated due to Readiness probe failure

Labels

critical_service_name, pod_name

nfmonitor_cumulative_k8api_get_query_failures
Type
64-bit Counter
Description
Total number of k8 api get pod query failures
Labels
<i>critical_service_name, pod_name</i>

DebugNfMonitorCmn

nfmonitor_force_restart_counter
Type
64-bit Counter
Description
Number of times force-restart triggered
Labels
<i>agent, name</i>

NfManagement statistics

nfmonitor_nfmgmt_service_request_attempts
Type
64-bit Counter
Description
Number of Requests sent to NRF
Labels
<i>instance_id, msg_type, method, retry</i>
nfmonitor_nfmgmt_service_request_success
Type
64-bit Counter
Description
The number of times the Request to NRF is Successful
Labels
<i>instance_id, msg_type, method, retry, status_code</i>
nfmonitor_nfmgmt_service_request_failures
Type
64-bit Counter
Description
The number of times the Request to NRF is Failure
Labels
<i>instance_id, msg_type, method, retry, status_code</i>
nfmonitor_nfmgmt_aborted_request_msgs

Type

64-bit Counter

Description

The number of times the Request to NRF is Aborted

Labels

instance_id, msg_type

Debug statistics

- nfmonitor_vnf_restart_trigger

Type

64-bit Counter

Description

Number of VNF Restart triggers received

Labels

curr_nf_status, nfmgmt_state, nfmonitor_fsm_state, up_count, timestamp, trigger, type

- nfmonitor_fsm_state_changes

Type

64-bit Counter

Description

The FSM State Change events stats

Labels

curr_nf_status, nfmgmt_state, up_count, from_state, to_state

- nfmonitor_deactivation_triggers

Type

64-bit Counter

Description

The number of times the deactivation trigger received

Labels

nf_status, nfmgmt_state, up_count, nfmonitor_fsm_state

- nfmonitor_critical_service_up_notification

Type

64-bit Counter

Description

The number of times the CriticalService Up notification received

Labels

service, criticality, nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_critical_service_down_notification

Type

64-bit Counter

Description

The number of times the CriticalService Down notification received

Labels

service, criticality, nf_status, nfmgmt_state, up_count, fsm_state

• nfmonitor_critical_services_overall_up_notification

Type

64-bit Counter

Description

The number of times the Overall CriticalServices Up notification received

Labels

nf_status, nfmgmt_state, up_count, fsm_state

• nfmonitor_critical_services_overall_down_notification

Type

64-bit Counter

Description

The number of times the Overall CriticalServices Down notification received

Labels

nf_status, nfmgmt_state, up_count, fsm_state

• nfmonitor_hysteresis_timer_started

Type

64-bit Counter

Description

The number of times the Hysteresis Timer is started

Labels

nf_status, nfmgmt_state, up_count, fsm_state

• nfmonitor_hysteresis_timer_stopped

Type

64-bit Counter

Description

The number of times the Hysteresis Timer is stopped

Labels

nf_status, nfmgmt_state, up_count, fsm_state

• nfmonitor_hysteresis_timer_expired

Type

64-bit Counter

Description

The number of times the Hysteresis Timer is expired

Labels

nf_status, nfmgmt_state, up_count, fsm_state

• nfmonitor_personality_update_triggers

Type

64-bit Counter

Description

The number of times the Nf Personality Update is received

Labels

nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_maintenance_trigger_received

Type

64-bit Counter

Description

The number of times the Maintenance Mode is activated

Labels

nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_profile_update_triggers

Type

64-bit Counter

Description

The number of times the Nf Profile Update happened

Labels

nf_status, nfmgmt_state, up_count, fsm_state, update_operation

- nfmonitor_critical_service_down_on_register_entry

Type

64-bit Counter

Description

The number of times the Critical Services were down while starting with RegisterFsmState

Labels

nf_status, up_count

ApiServer statistics

- nfmonitor_deployment_create_watch_request

Type

64-bit Counter

Description

The number of Watch Reqs created for deployment type

Labels

resource

- nfmonitor_deployment_delete_watch_request

Type

64-bit Counter

Description

The number of Watch Reqs deleted for deployment type

Labels

resource

- nfmonitor_deployment_watch_notification_success

Type

64-bit Counter

Description

The number of times Watch notifications are successful

Labels

resource

- nfmonitor_deployment_watch_notification_error

Type

64-bit Counter

Description

The number of times Watch notifications are failed

Labels

resource, reason

- nfmonitor_limit_range_query_attempts

Type

64-bit Counter

Description

The number of Limit Range query attempts

Labels

- nfmonitor_limit_range_query_failures

Type

64-bit Counter

Description

The number of Limit Range query failures

Labels

status_code

- nfmonitor_limit_range_query_success

Type

64-bit Counter

Description

The number of Limit Range query succeeded

Labels

status_code

- nfmonitor_hpa_query_attempts

Type

64-bit Counter

Description

The number of Hpa query attempts

Labels

resource, continue_param

• nfmonitor_hpa_query_failures

Type

64-bit Counter

Description

The number of Hpa query failures

Labels

resource, continue_param, status_code

• nfmonitor_hpa_query_success

Type

64-bit Counter

Description

The number of Hpa query succeeded

Labels

resource, continue_param, status_code

• nfmonitor_readiness_probe_query_attempts

Type

64-bit Counter

Description

The number of Readiness Probe query attempts

Labels

resource, continue_param

• nfmonitor_readiness_probe_query_failures

Type

64-bit Counter

Description

The number of Readiness Probe query failures

Labels

resource, continue_param, status_code

• nfmonitor_readiness_probe_query_success

Type

64-bit Counter

Description

The number of Readiness Probe query succeeded

Labels

resource, continue_param, status_code

Feature Functionality statistics

- nfmonitor_critical_service_frequent_restart

Type

64-bit Counter

Description

Howmany times the minimum required stable CriticalService count is not met

Labels

service

- nfmonitor_alarm_due_to_frequent_restart

Type

64-bit Counter

Description

Howmany times Alarm is raised due to minimum required stable CriticalService count is not met

Labels

service, alarm_status

- nfmonitor_force_restart_due_to_frequent_restart

Type

64-bit Counter

Description

Howmany times Force Restart initiated due to missing stable CriticalService count

Labels

service

- nfmonitor_critical_service_readiness_failure

Type

64-bit Counter

Description

Howmany times the minimum required readiness succeeded CriticalService count is not met

Labels

service

- nfmonitor_alarm_due_to_readiness_failure

Type

64-bit Counter

Description

Alarm raised due to minimum count is not met for the readiness succeeded CriticalService

Labels

service, alarm_status

- nfmonitor_force_restart_due_to_readiness_failure

Type

64-bit Counter
Description
Howmany times Force Restart initiated due to readiness failure of CriticalService
Labels
service

Consul statistics

- nfmonitor_publish_status_into_consul

Type
64-bit Counter
Description
Howmany times Publish is sent to Consul
Labels
nf_status, nf_state, up_count, fsm_state

- nfmonitor_consul_read_attempts

Type
64-bit Counter
Description
Howmany times Consul read attempt is made
Labels
fsm_state, retry

- nfmonitor_consul_read_failures

Type
64-bit Counter
Description
Howmany times Consul read attempt are failed
Labels
fsm_state, retry, status_code

- nfmonitor_consul_read_success

Type
64-bit Counter
Description
Howmany times Consul read attempt are succeeded
Labels
fsm_state, retry, status_code

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- [Summary](#)
- [Packet Streaming Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Packet Streaming Metrics	num_packet_streaming_attempted	Counter	The total number of attempted streaming of packets
Packet Streaming Metrics	num_packet_streaming_success	Counter	The total number of packets streamed successfully
Packet Streaming Metrics	num_packet_streaming_failed	Counter	The total number of failed streaming of packets
Packet Streaming Metrics	num_packet_streaming_bytes	Counter	The total number of bytes successfully streamed

Packet Streaming Metrics

- [num_packet_streaming_attempted](#)

Type

64-bit Counter

Description

The total number of attempted streaming of packets

Labels

interface_name, server_fqdn, kafka_topic
- [num_packet_streaming_success](#)

Type

64-bit Counter

Description

The total number of packets streamed successfully

Labels

interface_name, server_fqdn, kafka_topic
- [num_packet_streaming_failed](#)

Type

64-bit Counter

Description

The total number of failed streaming of packets

Labels

interface_name, server_fqdn, kafka_topic
- [num_packet_streaming_bytes](#)

Type

64-bit Counter

Description

The total number of bytes successfully streamed

Labels

interface_name, server_fqdn, kafka_topic

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- [Summary](#)
- [Circuit Breaker Metrics](#)
- [Service Endpoint Learner Metrics](#)
- [Service Endpoint Manager Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Circuit Breaker Metrics	sem_cb_endpoints	Gauge	Total number of endpoints for a service in respective states
Service Endpoint Learner Metrics	sem_num_dns_errors	Counter	Total number of DNS resolution errors for a given service
Service Endpoint Learner Metrics	sem_num_endpoints	Gauge	Total number of endpoints learnt for a service
Service Endpoint Manager Metrics	sem_num_endpoint_transactions_sent	Counter	Total number of times an endpoint is given to the application for communication with it
Service Endpoint Manager Metrics	sem_num_endpoint_transaction_status_received	Counter	Total number of transactions for a given endpoint received with the transaction status

Circuit Breaker Metrics

- [sem_cb_endpoints](#)

Type

64-bit Gauge

Description

Total number of endpoints for a service in respective states. There are 3 states viz. Active, Blocked, Restoring. In Active state, the endpoint is responding as expected, i.e. the response code doesn't indicate a service failure. The endpoint will be moved to Blocked state based on response failures which indicate service failure. After a timeout, the endpoint will be moved to Restoring state to check if the endpoint has recovered. In this state, if endpoint responds with response code other than service failure, it will be moved to Active state. Else, it will be moved back to Blocked state.

Labels

service_name, endpoint_state

Service Endpoint Learner Metrics

- [sem_num_dns_errors](#)

Type

64-bit Counter

Description

Total number of DNS resolution errors for a given service, labels are service name for DNS resolution and DNS error codes

Labels

service_name, error_code

- [sem_num_endpoints](#)

Type

64-bit Gauge

Description

Total number of endpoints learnt from the Kubernetes DNS server for a given service

Labels

service

Service Endpoint Manager Metrics

- [sem_num_endpoint_transactions_sent](#)

Type

64-bit Counter

Description

Total number of times an endpoint is given to the application for communication with it

Labels

service

- sem_num_endpoint_transaction_status_received

Type

64-bit Counter

Description

Total number of transactions for a given endpoint received with the transaction status

Labels

service, status

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- [Summary](#)
- [ETCD Client Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_put_requests	Counter	The total number of ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_requests_sent	Counter	The total number of ETCD Client PUT Requests sent.
ETCD Client Metrics	etcd_put_successes	Counter	The total number of successful ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_failures	Counter	The total number of failed ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_with_lease_requests	Counter	The total number of ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_requests_sent	Counter	The total number of ETCD Client PUT with Lease Requests sent.
ETCD Client Metrics	etcd_put_with_lease_successes	Counter	The total number of successful ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_failures	Counter	The total number of failed ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_range_requests	Counter	The total number of ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_requests_sent	Counter	The total number of ETCD Client Range Requests sent.
ETCD Client Metrics	etcd_range_successes	Counter	The total number of successful ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_failures	Counter	The total number of failed ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_delete_requests	Counter	The total number of ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_requests_sent	Counter	The total number of ETCD Client Range Delete Requests sent.
ETCD Client Metrics	etcd_range_delete_successes	Counter	The total number of successful ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_failures	Counter	The total number of failed ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_txn_requests	Counter	The total number of ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_requests_sent	Counter	The total number of ETCD Client Txn Requests sent.
ETCD Client Metrics	etcd_txn_successes	Counter	The total number of successful ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_failures	Counter	The total number of failed ETCD Client Txn Requests.
ETCD Client Metrics	etcd_watch_requests	Counter	The total number of ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_requests_sent	Counter	The total number of ETCD Client Watch Requests sent.
ETCD Client Metrics	etcd_watch_successes	Counter	The total number of successful ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_failures	Counter	The total number of failed ETCD Client Watch Requests.
ETCD Client Metrics	etcdwatcher_requests	Counter	The total number of ETCD Watcher Requests.

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcdwatcher_requests_sent	Counter	The total number of ETCD Watcher Requests sent.
ETCD Client Metrics	etcdwatcher_successes	Counter	The total number of successful ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_failures	Counter	The total number of failed ETCD Watcher Requests.

ETCD Client Metrics

- [etcd_put_requests](#)

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests.

Labels

keyPrefix
- [etcd_put_requests_sent](#)

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests sent.

Labels

keyPrefix
- [etcd_put_successes](#)

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT Requests.

Labels

keyPrefix
- [etcd_put_failures](#)

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT Requests.

Labels

keyPrefix, failureReason
- [etcd_put_with_lease_requests](#)

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests sent.

Labels

keyPrefix

- etcd_put_with_lease_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT with Lease Requests.

Labels

keyPrefix, failureReason

- etcd_range_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests sent.

Labels

keyPrefix

- etcd_range_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Requests.

Labels

keyPrefix, failureReason

- etcd_range_delete_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests.

Labels

keyPrefix

- etcd_range_delete_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests sent.

Labels

keyPrefix

- etcd_range_delete_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Delete Requests.

Labels

keyPrefix

- etcd_range_delete_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Delete Requests.

Labels

keyPrefix, failureReason

- `etcd_txn_requests`

Type
64-bit Counter

Description
The total number of ETCD Client Txn Requests.

Labels
keyPrefix
- `etcd_txn_requests_sent`

Type
64-bit Counter

Description
The total number of ETCD Client Txn Requests sent.

Labels
keyPrefix
- `etcd_txn_successes`

Type
64-bit Counter

Description
The total number of successful ETCD Client Txn Requests.

Labels
keyPrefix
- `etcd_txn_failures`

Type
64-bit Counter

Description
The total number of failed ETCD Client Txn Requests.

Labels
keyPrefix, failureReason
- `etcd_watch_requests`

Type
64-bit Counter

Description
The total number of ETCD Client Watch Requests.

Labels
keyPrefix
- `etcd_watch_requests_sent`

Type
64-bit Counter

Description

The total number of ETCD Client Watch Requests sent.

Labels

keyPrefix

- etcd_watch_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Watch Requests.

Labels

keyPrefix, failureReason

- etcdwatcher_requests

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests.

Labels

keyPrefix

- etcdwatcher_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests sent.

Labels

keyPrefix

- etcdwatcher_successes

Type

64-bit Counter

Description

The total number of successful ETCD Watcher Requests.

Labels

keyPrefix

- etcdwatcher_failures

Type
64-bit Counter
Description
The total number of failed ETCD Watcher Requests.
Labels
<i>keyPrefix, failureReason</i>

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- [Summary](#)
- [HTTP Punting Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_sent	Counter	The total number of HTTP requests successfully punted.
HTTP Punting Metrics	punting_http_requests_failed	Counter	The total number of HTTP requests punting failed.
HTTP Punting Metrics	punting_http_responses_received	Counter	The total number of HTTP responses received for the requests punted.
HTTP Punting Metrics	punting_http_responses_forwarded	Counter	The total number of HTTP responses forwarded after punting transaction completion.
HTTP Punting Metrics	punting_http_requests_aborted	Counter	The total number of HTTP requests aborted.
HTTP Punting Metrics	punting_http_request_duration_seconds	Histogram	Punting operation duration in seconds.

HTTP Punting Metrics

- [punting_http_requests_sent](#)

Type

64-bit Counter

Description

The total number of HTTP requests successfully punted.

Labels

message_type
- [punting_http_requests_failed](#)

Type

64-bit Counter

Description

The total number of HTTP requests for which the punting failed.

Labels

message_type
- [punting_http_responses_received](#)

Type

64-bit Counter

Description

The total number of HTTP responses received for the requests punting.

Labels

message_type, status_code

- `punting_http_responses_forwarded`

Type

64-bit Counter

Description

The total number of HTTP responses forwarded after punting transaction completion.

Labels

message_type, status_code

- `punting_http_requests_aborted`

Type

64-bit Counter

Description

The total number of HTTP requests aborted.

Labels

message_type

- `punting_http_request_duration_seconds`

Type

64-bit Histogram

Description

The punting transaction duration is measured from the time the HTTP request is sent out to the time the response is received for the punting request.

Labels

message_type

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

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- [Summary](#)
- [Oauth Token statistics](#)

Summary

Group	Metric Name	Type	Synopsis
Oauth Token statistics	oauth_nnrf_access_token_request	Counter	Number of NF Access Token Request Messages Received
Oauth Token statistics	oauth_nnrf_access_token_response	Counter	Number of NF Access Token Response Messages sent

Oauth Token statistics

- `oauth_nnrf_access_token_request`

Type

64-bit Counter

Description

Number of NF Access Token Request Messages received. The access token service, as defined by TS 29.510, provided by the NRF may be invoked by many NF types or NF instance id, as indicated by the source-nf-type label or nf-instance-id label.

Labels

source_nf_type, nf_instance_id

- [oauth_nnrf_access_token_response](#)

Type

64-bit Counter

Description

Number of NF Access Token Response Messages sent. The access token service, as defined by TS 29.510, provided by the NRF may be invoked by many NF types or NF instance id, as indicated by the source-nf-type or nf-instance-id label. The status label is populated with an HTTP cause followed by an application cause per failure type, separated by a dot.

Labels

source_nf_type, nf_instance_id, status

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- [Summary](#)
- [Packet Streaming Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Packet Streaming Metrics	num_packet_streaming_attempted	Counter	The total number of attempted streaming of packets
Packet Streaming Metrics	num_packet_streaming_success	Counter	The total number of packets streamed successfully
Packet Streaming Metrics	num_packet_streaming_failed	Counter	The total number of failed streaming of packets
Packet Streaming Metrics	num_packet_streaming_bytes	Counter	The total number of bytes successfully streamed

Packet Streaming Metrics

- [num_packet_streaming_attempted](#)

Type

64-bit Counter

Description

The total number of attempted streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- [num_packet_streaming_success](#)

Type

64-bit Counter

Description

The total number of packets streamed successfully

Labels

interface_name, server_fqdn, kafka_topic

- [num_packet_streaming_failed](#)

Type

64-bit Counter

Description

The total number of failed streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

• num_packet_streaming_bytes

Type

64-bit Counter

Description

The total number of bytes successfully streamed

Labels

interface_name, server_fqdn, kafka_topic

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- [Summary](#)
- [ETCD Client Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_put_requests	Counter	The total number of ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_requests_sent	Counter	The total number of ETCD Client PUT Requests sent.
ETCD Client Metrics	etcd_put_successes	Counter	The total number of successful ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_failures	Counter	The total number of failed ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_with_lease_requests	Counter	The total number of ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_requests_sent	Counter	The total number of ETCD Client PUT with Lease Requests sent.
ETCD Client Metrics	etcd_put_with_lease_successes	Counter	The total number of successful ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_failures	Counter	The total number of failed ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_range_requests	Counter	The total number of ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_requests_sent	Counter	The total number of ETCD Client Range Requests sent.
ETCD Client Metrics	etcd_range_successes	Counter	The total number of successful ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_failures	Counter	The total number of failed ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_delete_requests	Counter	The total number of ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_requests_sent	Counter	The total number of ETCD Client Range Delete Requests sent.
ETCD Client Metrics	etcd_range_delete_successes	Counter	The total number of successful ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_failures	Counter	The total number of failed ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_txn_requests	Counter	The total number of ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_requests_sent	Counter	The total number of ETCD Client Txn Requests sent.
ETCD Client Metrics	etcd_txn_successes	Counter	The total number of successful ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_failures	Counter	The total number of failed ETCD Client Txn Requests.
ETCD Client Metrics	etcd_watch_requests	Counter	The total number of ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_requests_sent	Counter	The total number of ETCD Client Watch Requests sent.
ETCD Client Metrics	etcd_watch_successes	Counter	The total number of successful ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_failures	Counter	The total number of failed ETCD Client Watch Requests.
ETCD Client Metrics	etcdwatcher_requests	Counter	The total number of ETCD Watcher Requests.

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcdwatcher_requests_sent	Counter	The total number of ETCD Watcher Requests sent.
ETCD Client Metrics	etcdwatcher_successes	Counter	The total number of successful ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_failures	Counter	The total number of failed ETCD Watcher Requests.

ETCD Client Metrics

- [etcd_put_requests](#)

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests.

Labels

keyPrefix
- [etcd_put_requests_sent](#)

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests sent.

Labels

keyPrefix
- [etcd_put_successes](#)

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT Requests.

Labels

keyPrefix
- [etcd_put_failures](#)

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT Requests.

Labels

keyPrefix, failureReason
- [etcd_put_with_lease_requests](#)

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests sent.

Labels

keyPrefix

- etcd_put_with_lease_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT with Lease Requests.

Labels

keyPrefix, failureReason

- etcd_range_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests sent.

Labels

keyPrefix

- etcd_range_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Requests.

Labels

keyPrefix, failureReason

- etcd_range_delete_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests.

Labels

keyPrefix

- etcd_range_delete_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests sent.

Labels

keyPrefix

- etcd_range_delete_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Delete Requests.

Labels

keyPrefix

- etcd_range_delete_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Delete Requests.

Labels

keyPrefix, failureReason

etcd_txn_requests
Type
64-bit Counter
Description
The total number of ETCD Client Txn Requests.
Labels
keyPrefix
etcd_txn_requests_sent
Type
64-bit Counter
Description
The total number of ETCD Client Txn Requests sent.
Labels
keyPrefix
etcd_txn_successes
Type
64-bit Counter
Description
The total number of successful ETCD Client Txn Requests.
Labels
keyPrefix
etcd_txn_failures
Type
64-bit Counter
Description
The total number of failed ETCD Client Txn Requests.
Labels
keyPrefix, failureReason
etcd_watch_requests
Type
64-bit Counter
Description
The total number of ETCD Client Watch Requests.
Labels
keyPrefix
etcd_watch_requests_sent
Type
64-bit Counter
Description

The total number of ETCD Client Watch Requests sent.

Labels

keyPrefix

- etcd_watch_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Watch Requests.

Labels

keyPrefix, failureReason

- etcdwatcher_requests

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests.

Labels

keyPrefix

- etcdwatcher_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests sent.

Labels

keyPrefix

- etcdwatcher_successes

Type

64-bit Counter

Description

The total number of successful ETCD Watcher Requests.

Labels

keyPrefix

- etcdwatcher_failures

Type
64-bit Counter
Description
The total number of failed ETCD Watcher Requests.
Labels
<i>keyPrefix, failureReason</i>

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- [Summary](#)
- [HTTP Punting Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_sent	Counter	The total number of HTTP requests successfully punted.
HTTP Punting Metrics	punting_http_requests_failed	Counter	The total number of HTTP requests punting failed.
HTTP Punting Metrics	punting_http_responses_received	Counter	The total number of HTTP responses received for the requests punted.
HTTP Punting Metrics	punting_http_responses_forwarded	Counter	The total number of HTTP responses forwarded after punting transaction completion.
HTTP Punting Metrics	punting_http_requests_aborted	Counter	The total number of HTTP requests aborted.
HTTP Punting Metrics	punting_http_request_duration_seconds	Histogram	Punting operation duration in seconds.

HTTP Punting Metrics

- [punting_http_requests_sent](#)

Type

64-bit Counter

Description

The total number of HTTP requests successfully punted.

Labels

message_type
- [punting_http_requests_failed](#)

Type

64-bit Counter

Description

The total number of HTTP requests for which the punting failed.

Labels

message_type
- [punting_http_responses_received](#)

Type

64-bit Counter

Description

The total number of HTTP responses received for the requests punted.

Labels

message_type, status_code

- `punting_http_responses_forwarded`

Type

64-bit Counter

Description

The total number of HTTP responses forwarded after punting transaction completion.

Labels

message_type, status_code

- `punting_http_requests_aborted`

Type

64-bit Counter

Description

The total number of HTTP requests aborted.

Labels

message_type

- `punting_http_request_duration_seconds`

Type

64-bit Histogram

Description

The punting transaction duration is measured from the time the HTTP request is sent out to the time the response is received for the punting request.

Labels

message_type

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

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- [Summary](#)
- [Packet Streaming Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Packet Streaming Metrics	num_packet_streaming_attempted	Counter	The total number of attempted streaming of packets
Packet Streaming Metrics	num_packet_streaming_success	Counter	The total number of packets streamed successfully
Packet Streaming Metrics	num_packet_streaming_failed	Counter	The total number of failed streaming of packets
Packet Streaming Metrics	num_packet_streaming_bytes	Counter	The total number of bytes successfully streamed

Packet Streaming Metrics

- `num_packet_streaming_attempted`

Type

64-bit Counter

Description

The total number of attempted streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_success

Type

64-bit Counter

Description

The total number of packets streamed successfully

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_failed

Type

64-bit Counter

Description

The total number of failed streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_bytes

Type

64-bit Counter

Description

The total number of bytes successfully streamed

Labels

interface_name, server_fqdn, kafka_topic

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- [Summary](#)
- [PFCP etcd interface message statistics](#)

Summary

Group	Metric Name	Type	Synopsis
PFCP etcd interface message statistics	pfc_p_etcd_record_request_attempts	Counter	total number of etcd requests sent out from LiX1
PFCP etcd interface message statistics	pfc_p_etcd_record_request_success	Counter	total number of successful etcd requests from LiX1
PFCP etcd interface message statistics	pfc_p_etcd_record_request_failures	Counter	total number of failed etcd requests from LiX1

PFCP etcd interface message statistics

- pfc_p_etcd_record_request_attempts

Type

64-bit Counter
Description
Total number of requests sent out to ETCD for performing any operation on the record
Labels
key, optype
• pfc_p_etcd_record_request_success
Type
64-bit Counter
Description
Total number of successful ETCD requests
Labels
key, optype
• pfc_p_etcd_record_request_failures
Type
64-bit Counter
Description
Total number of failed ETCD requests
Labels
key, optype, reason

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- [Summary](#)
- [PFCP-U PPE Message Statistics](#)
- [PFCP-U PPE Session Establishment Statistics](#)

Summary

Group	Metric Name	Type	Synopsis
PFCP-U PPE Message Statistics	pfc_p_ppe_session_establishment_request_sent	Counter	Session establishment request messages sent from pfc_p-u to ppe.
PFCP-U PPE Message Statistics	pfc_p_ppe_session_establishment_response_received	Counter	Session establishment response messages sent from ppe to pfc_p-u.
PFCP-U PPE Message Statistics	pfc_p_ppe_session_establishment_response_success	Counter	Session establishment success response messages sent from ppe to pfc_p-u.
PFCP-U PPE Message Statistics	pfc_p_ppe_session_establishment_response_failure	Counter	Session establishment failure response messages sent from ppe to pfc_p-u.
PFCP-U PPE Message Statistics	pfc_p_ppe_session_establishment_retransmitted_request_sent	Counter	Session establishment request retransmitted messages sent from pfc_p-u to ppe.
PFCP-U PPE Message Statistics	pfc_p_ppe_session_establishment_request_timeout	Counter	Session establishment request timed out due no response from ppe.
PFCP-U PPE Message Statistics	pfc_p_ppe_session_establishment_request_send_failure_due_to_ppe_not_found	Counter	Session establishment request messages sent from pfc_p-u due to ppe not found.
PFCP-U PPE Message Statistics	pfc_p_ppe_session_modification_request_sent	Counter	Session modification response messages sent from pfc_p-u to ppe.
PFCP-U PPE Message Statistics	pfc_p_ppe_session_modification_response_received	Counter	Session modification response messages sent from ppe to pfc_p-u.
PFCP-U PPE Message Statistics	pfc_p_ppe_session_modification_response_success	Counter	Session modification success response messages sent from ppe to pfc_p-u.

Group	Metric Name	Type	Synopsis
PFCP-U PPE Message Statistics	pfc_ppe_session_modification_response_failure	Counter	Session modification failure response messages sent from ppe to pfc-p-u.
PFCP-U PPE Message Statistics	pfc_ppe_session_modification_retransmitted_request_sent	Counter	Session modification request retransmitted messages sent from pfc-p-u to ppe.
PFCP-U PPE Message Statistics	pfc_ppe_session_modification_request_timeout	Counter	Session modification request timed out due no response from ppe.
PFCP-U PPE Message Statistics	pfc_ppe_session_modification_request_send_failure_due_to_ppe_not_found	Counter	Session modification request messages sent from pfc-p-u due to ppe not found.
PFCP-U PPE Message Statistics	pfc_ppe_session_deletion_request_sent	Counter	Session deletion response messages sent from pfc-p-u to ppe.
PFCP-U PPE Message Statistics	pfc_ppe_session_deletion_response_received	Counter	Session deletion response messages sent from ppe to pfc-p-u.
PFCP-U PPE Message Statistics	pfc_ppe_session_deletion_response_success	Counter	Session modification success response messages sent from ppe to pfc-p-u.
PFCP-U PPE Message Statistics	pfc_ppe_session_deletion_response_failure	Counter	Session modification failure response messages sent from ppe to pfc-p-u.
PFCP-U PPE Message Statistics	pfc_ppe_session_deletion_retransmitted_request_sent	Counter	Session deletion request retransmitted messages sent from pfc-p-u to ppe.
PFCP-U PPE Message Statistics	pfc_ppe_session_deletion_request_timeout	Counter	Session deletion request timed out due no response from ppe.
PFCP-U PPE Message Statistics	pfc_ppe_session_deletion_request_send_failure_due_to_ppe_not_found	Counter	Session deletion request messages sent from pfc-p-u due to ppe not found.
PFCP-U PPE Message Statistics	pfc_ppe_session_report_request_received	Counter	Session report request messages sent from ppe to pfc-p-u.
PFCP-U PPE Message Statistics	pfc_ppe_session_report_response_sent	Counter	Session report response messages response from pfc-p-u to ppe.
PFCP-U PPE Message Statistics	pfc_ppe_session_report_response_send_failure_due_to_ppe_not_found	Counter	Session report response messages response send failed from pfc-p-u due to ppe not found.
PFCP-U PPE Message Statistics	pfc_ppe_session_response_proc_context_not_found	Counter	Session response messages recieved from ppe to pfc-p-u but no proc context found.
PFCP-U PPE Message Statistics	pfc_ppe_session_establishment_total_duration_milliseconds	Histogram	Histogram of the duration, in milliseconds, of pfc-p establishment request transaction between pfc-p-u and ppe.
PFCP-U PPE Message Statistics	pfc_ppe_session_modification_total_duration_milliseconds	Histogram	Histogram of the duration, in milliseconds, of pfc-p modification request transaction between pfc-p-u and ppe.
PFCP-U PPE Message Statistics	pfc_ppe_session_deletion_total_duration_milliseconds	Histogram	Histogram of the duration, in milliseconds, of pfc-p deletion request transaction between pfc-p-u and ppe.
PFCP-U PPE Session Establishment Statistics	pfc_ppe_session_establishment_request_initial_success	Counter	Number of non-crankback, initial session establishment requests successfully sent to target SM Instances.
PFCP-U PPE Session Establishment Statistics	pfc_ppe_session_establishment_request_retry_success	Counter	Number of non-crankback, retry session establishment requests successfully sent to target SM Instances.
PFCP-U PPE Session Establishment Statistics	pfc_ppe_session_establishment_request_failure	Counter	Number of non-crankback, initial and retry session establishment requests that failed to be sent.
PFCP-U PPE Session Establishment Statistics	pfc_ppe_session_establishment_request_crankback_success	Counter	Number of crankback initiated session establishment requests successfully sent to an alternative SM Instance.
PFCP-U PPE Session Establishment Statistics	pfc_ppe_session_establishment_request_crankback_failure	Counter	Number of crankback initiated session establishment requests that failed to be sent.

Group	Metric Name	Type	Synopsis
PFCP-U PPE Session Establishment Statistics	pfc_ppe_session_establishment_response_succ	Counter	Number of crankback initiated session establishment requests successfully sent to an alternative SM Instance.
PFCP-U PPE Session Establishment Statistics	pfc_ppe_session_establishment_response_fail	Counter	Number of session establishment responses reporting failed session establishment. Includes responses for initial, retry and crankback session establishment requests.

PFCP-U PPE Message Statistics

- [pfc_ppe_session_establishment_request_sent](#)

Type

64-bit Counter

Description

The number of session establishment request messages sent from pfc-p-u to ppe.

Labels

ppe_id
- [pfc_ppe_session_establishment_response_received](#)

Type

64-bit Counter

Description

The number of session establishment response messages received from ppe to pfc-p-u.

Labels

ppe_id
- [pfc_ppe_session_establishment_response_success](#)

Type

64-bit Counter

Description

The number of session establishment success response messages received from ppe to pfc-p-u.

Labels

ppe_id
- [pfc_ppe_session_establishment_response_failure](#)

Type

64-bit Counter

Description

The number of session establishment failure response messages received from ppe to pfc-p-u.

Labels

ppe_id
- [pfc_ppe_session_establishment_retransmitted_request_sent](#)

Type

64-bit Counter

Description

The number of session establishment request retransmitted due to no response from ppe.

Labels

ppe_id

- pfc_ppe_session_establishment_request_timeout

Type

64-bit Counter

Description

The number of session establishment request timed out due no response even after retransmission.

Labels

ppe_id

- pfc_ppe_session_establishment_request_send_failure_due_to_ppe_not_found

Type

64-bit Counter

Description

The number of session establishment request message sent failed from pfc-u due to ppe not found.

Labels

ppe_id

- pfc_ppe_session_modification_request_sent

Type

64-bit Counter

Description

The number of session modification request messages sent from pfc-u to ppe.

Labels

ppe_id

- pfc_ppe_session_modification_response_received

Type

64-bit Counter

Description

The number of session modification response messages received from ppe to pfc-u.

Labels

ppe_id

- pfc_ppe_session_modification_response_success

Type

64-bit Counter

Description

The number of session modification success response messages received from ppe to pfc-u.

Labels

ppe_id

- pfc_ppe_session_modification_response_failure

Type

64-bit Counter

Description

The number of session modification failure response messages received from ppe to pfcp-u.

Labels

ppe_id

- pfcp_ppe_session_modification_retransmitted_request_sent

Type

64-bit Counter

Description

The number of session modification request retransmitted due to no response from ppe.

Labels

ppe_id

- pfcp_ppe_session_modification_request_timeout

Type

64-bit Counter

Description

The number of session modification request timed out due no response even after retransmission.

Labels

ppe_id

- pfcp_ppe_session_modification_request_send_failure_due_to_ppe_not_found

Type

64-bit Counter

Description

The number of session modification request message sent failed from pfcp-u due to ppe not found.

Labels

ppe_id

- pfcp_ppe_session_deletion_request_sent

Type

64-bit Counter

Description

The number of session deletion request messages sent from pfcp-u to ppe.

Labels

ppe_id

- pfcp_ppe_session_deletion_response_received

Type

64-bit Counter

Description

The number of session deletion response messages received from ppe to pfcp-u.

Labels

ppe_id

- pfc_ppe_session_deletion_response_success

Type

64-bit Counter

Description

The number of session deletion success response messages received from ppe to pfc-u.

Labels

ppe_id

- pfc_ppe_session_deletion_response_failure

Type

64-bit Counter

Description

The number of session deletion failure response messages received from ppe to pfc-u.

Labels

ppe_id

- pfc_ppe_session_deletion_retransmitted_request_sent

Type

64-bit Counter

Description

The number of session deletion request retransmitted due to no response from ppe.

Labels

ppe_id

- pfc_ppe_session_deletion_request_timeout

Type

64-bit Counter

Description

The number of session deletion request timed out due no response even after retransmission.

Labels

ppe_id

- pfc_ppe_session_deletion_request_send_failure_due_to_ppe_not_found

Type

64-bit Counter

Description

The number of session deletion request message sent failed from pfc-u due to ppe not found.

Labels

ppe_id

- pfc_ppe_session_report_request_received

Type

64-bit Counter

Description

The number of session report request messages received from ppe to pfcu-u.

Labels

ppe_id

• pfcu_ppe_session_report_response_sent

Type

64-bit Counter

Description

The number of session report response messages sent from pfcu-u to ppe.

Labels

ppe_id

• pfcu_ppe_session_report_response_send_failure_due_to_ppe_not_found

Type

64-bit Counter

Description

The number of session report response messages send failed from pfcu-u due to ppe not found.

Labels

ppe_id

• pfcu_ppe_session_response_proc_context_not_found

Type

64-bit Counter

Description

The number of session response messages recieved from ppe to pfcu-u but no proc context found.

Labels

ppe_id, msg_type

• pfcu_ppe_session_establishment_total_duration_milliseconds

Type

64-bit Histogram

Description

Histogram of the duration, in milliseconds, of pfcu establishment request transaction between pfcu-u and ppe.

Labels

ppe_id

Boundaries

[5, 10, 15, 20, 30, 50, 100, 500, 1000, 2500]

• pfcu_ppe_session_modification_total_duration_milliseconds

Type

64-bit Histogram

Description

Histogram of the duration, in milliseconds, of pfcP modification request transaction between pfcP-u and ppe.

Labels

ppe_id

Boundaries

[5, 10, 15, 20, 30, 50, 100, 500, 1000, 2500]

- pfcP_ppe_session_deletion_total_duration_milliseconds

Type

64-bit Histogram

Description

Histogram of the duration, in milliseconds, of pfcP deletion request transaction between pfcP-u and ppe.

Labels

ppe_id

Boundaries

[5, 10, 15, 20, 30, 50, 100, 500, 1000, 2500]

PFCE-U PPE Session Establishment Statistics

- pfcP_ppe_session_establishment_request_initial_success

Type

64-bit Counter

Description

Number of non-crankback, initial session establishment requests successfully sent to target SM Instances.

Labels

ppe_id, sm_instance_id, apn_dnn

- pfcP_ppe_session_establishment_request_retry_success

Type

64-bit Counter

Description

Number of non-crankback, retry session establishment requests successfully sent to target SM Instances.

Labels

ppe_id, sm_instance_id, apn_dnn

- pfcP_ppe_session_establishment_request_failure

Type

64-bit Counter

Description

Number of non-crankback, initial and retry session establishment requests that failed to be sent. Failure reason is identified in the 'cause' label.

Labels

ppe_id, sm_instance_id, apn_dnn, cause

- pfcP_ppe_session_establishment_request_crankback_success

Type

64-bit Counter

Description

Number of crankback initiated session establishment requests successfully sent to an alternative SM Instance.

Labels

ppe_id, sm_instance_id, apn_dnn

- pfc_ppe_session_establishment_request_crankback_failure

Type

64-bit Counter

Description

Number of crankback initiated session establishment requests that failed to be sent. Failure reason is identified in the 'cause' label.

Labels

ppe_id, sm_instance_id, apn_dnn, cause

- pfc_ppe_session_establishment_response_succ

Type

64-bit Counter

Description

Number of session establishment responses reporting successful session establishment. Includes responses for initial, retry and crankback session establishment requests.

Labels

ppe_id, sm_instance_id, apn_dnn

- pfc_ppe_session_establishment_response_fail

Type

64-bit Counter

Description

Number of session establishment responses reporting failed session establishment. Includes responses for initial, retry and crankback session establishment requests. Failure reason is identified in the 'cause' label.

Labels

ppe_id, sm_instance_id, apn_dnn, cause

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- [Summary](#)
- [PFCP Procedure Statistics](#)

Summary

Group	Metric Name	Type	Synopsis
PFCP Procedure Statistics	pfc_p association_setup_attempt	Counter	Association setup attempts.
PFCP Procedure Statistics	pfc_p association_setup_success	Counter	Association setup successes.
PFCP Procedure Statistics	pfc_p association_setup_failure	Counter	Association setup failures.
PFCP Procedure Statistics	pfc_p peer_association_setup_attempt	Counter	Peer association setup attempts.
PFCP Procedure Statistics	pfc_p peer_association_setup_success	Counter	Peer association setup successes.

Group	Metric Name	Type	Synopsis
PFCP Procedure Statistics	pfcpeer_association_setup_failure	Counter	Peer association setup failures.
PFCP Procedure Statistics	pfcpeer_association_update_attempt	Counter	Association update attempts.
PFCP Procedure Statistics	pfcpeer_association_update_success	Counter	Association update successes.
PFCP Procedure Statistics	pfcpeer_association_update_failure	Counter	Association update failures.
PFCP Procedure Statistics	pfcpeer_peer_association_update_attempt	Counter	Peer association update attempts.
PFCP Procedure Statistics	pfcpeer_peer_association_update_success	Counter	Peer association update successes.
PFCP Procedure Statistics	pfcpeer_peer_association_update_failure	Counter	Peer association update failures.
PFCP Procedure Statistics	pfcpeer_association_release_attempt	Counter	Association release attempts.
PFCP Procedure Statistics	pfcpeer_association_release_success	Counter	Association release successes.
PFCP Procedure Statistics	pfcpeer_association_release_failure	Counter	Association release failures.
PFCP Procedure Statistics	pfcpeer_peer_association_release_attempt	Counter	Peer association release attempts.
PFCP Procedure Statistics	pfcpeer_peer_association_release_success	Counter	Peer association release successes.
PFCP Procedure Statistics	pfcpeer_peer_association_release_failure	Counter	Peer association release failures.
PFCP Procedure Statistics	pfcpeer_heartbeat_attempt	Counter	HeartBeat attempts.
PFCP Procedure Statistics	pfcpeer_heartbeat_success	Counter	HeartBeat successes.
PFCP Procedure Statistics	pfcpeer_heartbeat_failure	Counter	HeartBeat failures.
PFCP Procedure Statistics	pfcpeer_peer_heartbeat_attempt	Counter	Peer HeartBeat attempts.
PFCP Procedure Statistics	pfcpeer_peer_heartbeat_success	Counter	Peer HeartBeat successes.
PFCP Procedure Statistics	pfcpeer_peer_heartbeat_failure	Counter	Peer HeartBeat failures.
PFCP Procedure Statistics	pfcpeer_peer_heartbeat_received_without_association	Counter	Peer HeartBeat attempts without association.
PFCP Procedure Statistics	pfcpeer_peer_heartbeat_peer_association_added_or_updated_during_etcd_sync_wait_timer	Counter	Peer info added or updated during etcd sync wait time.
PFCP Procedure Statistics	pfcpeer_session_establishment_attempt	Counter	Session establishment attempts.
PFCP Procedure Statistics	pfcpeer_session_establishment_success	Counter	Session establishment successes.
PFCP Procedure Statistics	pfcpeer_session_establishment_failure	Counter	Session establishment failures.
PFCP Procedure Statistics	pfcpeer_session_modification_attempt	Counter	Session modification attempts.
PFCP Procedure Statistics	pfcpeer_session_modification_success	Counter	Session modification successes.
PFCP Procedure Statistics	pfcpeer_session_modification_failure	Counter	Session modification failures.

Group	Metric Name	Type	Synopsis
PFCP Procedure Statistics	pfc_p_session_deletion_attempt	Counter	Session deletion attempts.
PFCP Procedure Statistics	pfc_p_session_deletion_success	Counter	Session deletion successes.
PFCP Procedure Statistics	pfc_p_session_deletion_failure	Counter	Session deletion failures.
PFCP Procedure Statistics	pfc_p_session_report_attempt	Counter	Session report attempts.
PFCP Procedure Statistics	pfc_p_session_report_success	Counter	Session report successes.
PFCP Procedure Statistics	pfc_p_session_report_failure	Counter	Session report failures.
PFCP Procedure Statistics	pfc_p_pfdm_attempt	Counter	Pfdm attempts.
PFCP Procedure Statistics	pfc_p_pfdm_success	Counter	Pfdm successes.
PFCP Procedure Statistics	pfc_p_pfdm_failure	Counter	Pfdm failures.
PFCP Procedure Statistics	pfc_p_nodes_dnn_report_attempts	Counter	Nodes dnn report attempts.
PFCP Procedure Statistics	pfc_p_nodes_dnn_report_success	Counter	Node dnn report successes.
PFCP Procedure Statistics	pfc_p_nodes_dnn_report_failed	Counter	Node dnn report failures.
PFCP Procedure Statistics	pfc_p_nodes_dnn_report_sent_to_peer	Counter	Node dnn report sent to peer
PFCP Procedure Statistics	pfc_p_nodes_dnn_report_sent_to_peer_failed	Counter	Node dnn report sent to peer
PFCP Procedure Statistics	pfc_p_master_elections_count	Counter	pfc_p master election count.
PFCP Procedure Statistics	pfc_p_master_changed_count	Counter	pfc_p node master changed count.
PFCP Procedure Statistics	pfc_p_node_promoted_to_master_count	Counter	dnn report election node promoted count.
PFCP Procedure Statistics	pfc_p_node_demoted_from_master_count	Counter	dnn report election node demoted count.

PFCP Procedure Statistics

- [pfc_p_association_setup_attempt](#)

Type

64-bit Counter

Description

The number of association setup attempts.

Labels

peer_id

- [pfc_p_association_setup_success](#)

Type

64-bit Counter

Description

The number of association setup successes.

Labels

peer_id

- pfc_p_association_setup_failure

Type

64-bit Counter

Description

The number of association setup failures.

Labels

peer_id, cause

- pfc_p_peer_association_setup_attempt

Type

64-bit Counter

Description

The number of peer association setup attempts.

Labels

peer_id

- pfc_p_peer_association_setup_success

Type

64-bit Counter

Description

The number of peer association setup successes.

Labels

peer_id

- pfc_p_peer_association_setup_failure

Type

64-bit Counter

Description

The number of peer association setup failures.

Labels

peer_id, cause

- pfc_p_association_update_attempt

Type

64-bit Counter

Description

The number of association update attempts.

Labels

peer_id, release_reason

- pfc_p_association_update_success

Type

64-bit Counter

Description

The number of association update successes.

Labels

peer_id, release_reason

- pfc_p_association_update_failure

Type

64-bit Counter

Description

The number of association update failures.

Labels

peer_id, release_reason, cause

- pfc_p_peer_association_update_attempt

Type

64-bit Counter

Description

The number of association update peer attempts.

Labels

peer_id, release_reason

- pfc_p_peer_association_update_success

Type

64-bit Counter

Description

The number of association update peer successes.

Labels

peer_id, release_reason

- pfc_p_peer_association_update_failure

Type

64-bit Counter

Description

The number of association update peer failures.

Labels

peer_id, release_reason, cause

- pfc_p_association_release_attempt

Type

64-bit Counter

Description

The number of association release attempts.

Labels

peer_id, release_reason

- pfcip_association_release_success

Type

64-bit Counter

Description

The number of association release successes.

Labels

peer_id, release_reason

- pfcip_association_release_failure

Type

64-bit Counter

Description

The number of association release failures.

Labels

peer_id, cause, release_reason

- pfcip_peer_association_release_attempt

Type

64-bit Counter

Description

The number of association release peer attempts.

Labels

peer_id, release_reason

- pfcip_peer_association_release_success

Type

64-bit Counter

Description

The number of association release peer successes.

Labels

peer_id, release_reason

- pfcip_peer_association_release_failure

Type

64-bit Counter

Description

The number of association release peer failures.

Labels

peer_id, cause, release_reason

- pfcip_heartbeat_attempt

Type

64-bit Counter

Description

The number of heartbeat attempts.

Labels

peer_id

• pfc_p_heartbeat_success

Type

64-bit Counter

Description

The number of heartbeat successes.

Labels

peer_id

• pfc_p_heartbeat_failure

Type

64-bit Counter

Description

The number of heartbeat failures.

Labels

peer_id, cause

• pfc_p_peer_heartbeat_attempt

Type

64-bit Counter

Description

The number of heartbeat peer attempts.

Labels

peer_id

• pfc_p_peer_heartbeat_success

Type

64-bit Counter

Description

The number of heartbeat peer successes.

Labels

peer_id

• pfc_p_peer_heartbeat_failure

Type

64-bit Counter

Description

The number of heartbeat peer failures.

Labels

peer_id, cause

- pfcpeer_heartbeat_received_without_association

Type

64-bit Counter

Description

The number of heartbeat peer attempts without association.

Labels

peer_id

- pfcpeer_heartbeat_peer_association_added_or_updated_during_etcd_sync_wait_timer

Type

64-bit Counter

Description

The number of peer info added or updated during etcd sync wait time.

Labels

peer_id

- pfcpeer_session_establishment_attempt

Type

64-bit Counter

Description

The number of session establishment attempts.

Labels

interface, dnn, nssai, peer_id

- pfcpeer_session_establishment_success

Type

64-bit Counter

Description

The number of session establishment successes.

Labels

interface, dnn, nssai, peer_id

- pfcpeer_session_establishment_failure

Type

64-bit Counter

Description

The number of session establishment failures.

Labels

interface, dnn, nssai, peer_id, cause

- pfcpeer_session_modification_attempt

Type

64-bit Counter

Description

The number of session modification attempts.

Labels

interface, dnn, nssai, peer_id

- pfcpl_session_modification_success

Type

64-bit Counter

Description

The number of session modification successes.

Labels

interface, dnn, nssai, peer_id

- pfcpl_session_modification_failure

Type

64-bit Counter

Description

The number of session modification failures.

Labels

interface, dnn, nssai, peer_id, cause

- pfcpl_session_deletion_attempt

Type

64-bit Counter

Description

The number of session deletion attempts.

Labels

interface, dnn, nssai, peer_id

- pfcpl_session_deletion_success

Type

64-bit Counter

Description

The number of session deletion successes.

Labels

interface, dnn, nssai, peer_id

- pfcpl_session_deletion_failure

Type

64-bit Counter

Description

The number of session deletion failures.

Labels

interface, dnn, nssai, peer_id, cause

- pfc_session_report_attempt
 - Type
 - 64-bit Counter
 - Description
 - The number of session report attempts.
 - Labels
 - interface, dnn, nssai, peer_id**
- pfc_session_report_success
 - Type
 - 64-bit Counter
 - Description
 - The number of session report successes.
 - Labels
 - interface, dnn, nssai, peer_id**
- pfc_session_report_failure
 - Type
 - 64-bit Counter
 - Description
 - The number of session report failures.
 - Labels
 - interface, dnn, nssai, peer_id, cause**
- pfc_pfdm_attempt
 - Type
 - 64-bit Counter
 - Description
 - The number of pfdm attempts.
 - Labels
 - peer_id**
- pfc_pfdm_success
 - Type
 - 64-bit Counter
 - Description
 - The number of pfdm successes.
 - Labels
 - peer_id**
- pfc_pfdm_failure
 - Type
 - 64-bit Counter
 - Description

The number of pfdm failures.

Labels

peer_id, cause

• pfcpl_nodes_dnn_report_attempts

Type

64-bit Counter

Description

The number of node dnn report attempts.

Labels

• pfcpl_nodes_dnn_report_success

Type

64-bit Counter

Description

The number of node dnn report successes.

Labels

• pfcpl_nodes_dnn_report_failed

Type

64-bit Counter

Description

The number of node dnn report failures.

Labels

• pfcpl_nodes_dnn_report_sent_to_peer

Type

64-bit Counter

Description

The number of node dnn reports sent to peer

Labels

peer_ip, peer_port, message

• pfcpl_nodes_dnn_report_sent_to_peer_failed

Type

64-bit Counter

Description

The number of node dnn reports sent to peer failed

Labels

peer_ip, peer_port, message

• pfcpl_master_elections_count

Type

64-bit Counter

Description

The number of elections done count

Labels

- pfc_p_master_changed_count

Type

64-bit Counter

Description

The number of elections changed to promoted or demoted count

Labels

- pfc_p_node_promoted_to_master_count

Type

64-bit Counter

Description

The number of node promoted done count

Labels

- pfc_p_node_demoted_from_master_count

Type

64-bit Counter

Description

The number of node demoted done counted

Labels

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- [PFCP Protocol Message Statistics](#)

Summary

Group	Metric Name	Type	Synopsis
PFCP Protocol Message Statistics	pfc_p_association_setup_request_sent	Counter	Association setup sent request messages.
PFCP Protocol Message Statistics	pfc_p_association_setup_retransmitted_request_sent	Counter	Association setup retransmitted sent request messages.
PFCP Protocol Message Statistics	pfc_p_association_setup_request_received	Counter	Association setup received request messages.
PFCP Protocol Message Statistics	pfc_p_association_setup_retransmitted_request_received	Counter	Association setup received retransmitted request messages.
PFCP Protocol Message Statistics	pfc_p_association_update_request_sent	Counter	Association update sent request messages.

Group	Metric Name	Type	Synopsis
PFCP Protocol Message Statistics	pfc_p_association_update_retransmitted_request_sent	Counter	Association update retransmitted sent request messages.
PFCP Protocol Message Statistics	pfc_p_association_update_request_received	Counter	Association update received request messages.
PFCP Protocol Message Statistics	pfc_p_association_update_retransmitted_request_received	Counter	Association update received retransmitted request messages.
PFCP Protocol Message Statistics	pfc_p_association_release_request_sent	Counter	Association release sent request messages.
PFCP Protocol Message Statistics	pfc_p_association_release_retransmitted_request_sent	Counter	Association release retransmitted sent request messages.
PFCP Protocol Message Statistics	pfc_p_association_release_request_received	Counter	Association release received request messages.
PFCP Protocol Message Statistics	pfc_p_association_release_retransmitted_request_received	Counter	Association release received retransmitted request messages.
PFCP Protocol Message Statistics	pfc_p_association_setup_response_sent	Counter	Association setup sent response messages.
PFCP Protocol Message Statistics	pfc_p_association_setup_response_received	Counter	Association setup received response messages.
PFCP Protocol Message Statistics	pfc_p_association_update_response_sent	Counter	Association update sent response messages.
PFCP Protocol Message Statistics	pfc_p_association_update_response_received	Counter	Association update received response messages.
PFCP Protocol Message Statistics	pfc_p_association_release_response_sent	Counter	Association release sent response messages.
PFCP Protocol Message Statistics	pfc_p_association_release_response_received	Counter	Association release received response messages.
PFCP Protocol Message Statistics	pfc_p_version_not_supported_response_sent	Counter	Version not supported sent response messages.
PFCP Protocol Message Statistics	pfc_p_version_not_supported_response_received	Counter	Version not supported received response messages.
PFCP Protocol Message Statistics	pfc_p_heartbeat_request_sent	Counter	HeartBeat sent request messages.
PFCP Protocol Message Statistics	pfc_p_heartbeat_retransmitted_request_sent	Counter	HeartBeat retransmitted sent request messages.
PFCP Protocol Message Statistics	pfc_p_heartbeat_request_received	Counter	HeartBeat received request messages.
PFCP Protocol Message Statistics	pfc_p_heartbeat_retransmitted_request_received	Counter	HeartBeat received retransmitted request messages.

Group	Metric Name	Type	Synopsis
PFCP Protocol Message Statistics	pfc_p_heartbeat_response_sent	Counter	HeartBeat sent response messages.
PFCP Protocol Message Statistics	pfc_p_heartbeat_response_received	Counter	HeartBeat received response messages.
PFCP Protocol Message Statistics	pfc_p_session_establishment_request_sent	Counter	Session establishment sent request messages.
PFCP Protocol Message Statistics	pfc_p_session_establishment_retransmitted_request_sent	Counter	Session establishment retransmitted sent request messages.
PFCP Protocol Message Statistics	pfc_p_session_establishment_request_received	Counter	Session establishment received request messages.
PFCP Protocol Message Statistics	pfc_p_session_establishment_retransmitted_request_received	Counter	Session establishment received retransmitted request messages.
PFCP Protocol Message Statistics	pfc_p_session_modification_request_sent	Counter	Session modification sent request messages.
PFCP Protocol Message Statistics	pfc_p_session_modification_retransmitted_request_sent	Counter	Session modification retransmitted sent request messages.
PFCP Protocol Message Statistics	pfc_p_session_modification_request_received	Counter	Session modification received request messages.
PFCP Protocol Message Statistics	pfc_p_session_modification_retransmitted_request_received	Counter	Session modification received retransmitted request messages.
PFCP Protocol Message Statistics	pfc_p_session_deletion_request_sent	Counter	Session deletion sent request messages.
PFCP Protocol Message Statistics	pfc_p_session_deletion_retransmitted_request_sent	Counter	Session deletion retransmitted sent request messages.
PFCP Protocol Message Statistics	pfc_p_session_deletion_request_received	Counter	Session deletion received request messages.
PFCP Protocol Message Statistics	pfc_p_session_deletion_retransmitted_request_received	Counter	Session deletion received retransmitted request messages.
PFCP Protocol Message Statistics	pfc_p_session_report_request_sent	Counter	Session report sent request messages.
PFCP Protocol Message Statistics	pfc_p_session_report_retransmitted_request_sent	Counter	Session report retransmitted sent request messages.
PFCP Protocol Message Statistics	pfc_p_session_report_request_received	Counter	Session report received request messages.
PFCP Protocol Message Statistics	pfc_p_session_report_retransmitted_request_received	Counter	Session report received retransmitted request messages.
PFCP Protocol Message Statistics	pfc_p_session_establishment_response_sent	Counter	Session establishment sent response messages.

Group	Metric Name	Type	Synopsis
PFCP Protocol Message Statistics	pfcpl_session_establishment_response_received	Counter	Session establishment received response messages.
PFCP Protocol Message Statistics	pfcpl_session_modification_response_sent	Counter	Session modification sent response messages.
PFCP Protocol Message Statistics	pfcpl_session_modification_response_received	Counter	Session modification received response messages.
PFCP Protocol Message Statistics	pfcpl_session_deletion_response_sent	Counter	Session deletion sent response messages.
PFCP Protocol Message Statistics	pfcpl_session_deletion_response_received	Counter	Session deletion received response messages.
PFCP Protocol Message Statistics	pfcpl_node_report_request_sent	Counter	Node report sent request messages.
PFCP Protocol Message Statistics	pfcpl_node_report_type_request_received	Counter	Node report received request messages.
PFCP Protocol Message Statistics	pfcpl_node_report_type_retransmitted_request_received	Counter	Node report received retransmitted request messages.
PFCP Protocol Message Statistics	pfcpl_node_report_response_received	Counter	Node report received response messages.
PFCP Protocol Message Statistics	pfcpl_node_report_retransmitted_request_sent	Counter	Node report retransmitted request messages sent.
PFCP Protocol Message Statistics	pfcpl_session_report_response_sent	Counter	Session report sent response messages.
PFCP Protocol Message Statistics	pfcpl_session_report_response_received	Counter	Session report received response messages.
PFCP Protocol Message Statistics	pfcpl_pfdrm_request_sent	Counter	Pfdrm sent request messages.
PFCP Protocol Message Statistics	pfcpl_pfdrm_retransmitted_request_sent	Counter	Pfdrm retransmitted sent request messages.
PFCP Protocol Message Statistics	pfcpl_pfdrm_request_received	Counter	Pfdrm received request messages.
PFCP Protocol Message Statistics	pfcpl_pfdrm_retransmitted_request_received	Counter	Pfdrm received retransmitted request messages.
PFCP Protocol Message Statistics	pfcpl_pfdrm_response_sent	Counter	Pfdrm sent response messages.
PFCP Protocol Message Statistics	pfcpl_pfdrm_response_received	Counter	Pfdrm received response messages.
PFCP Protocol Message Statistics	pfcpl_bundle_message_sent	Counter	Number of PFCP bundle messages sent to a particular peer(represented by peer_ip and peer_port) due to a particular trigger(timeout or max buffer size or max number of PFCP messages in a single bundle).

Group	Metric Name	Type	Synopsis
PFCP Protocol Message Statistics	pfc_p_bundle_message_sent_failure	Counter	Number of PFCP bundle messages sent failures to a particular peer(represented by peer_ip and peer_port), cause represents failure reason(like udp send socket failure).
PFCP Protocol Message Statistics	pfc_p_bundle_message_received	Counter	Number of PFCP bundle messages received from a particular peer(represented by peer_ip and peer_port).
PFCP Protocol Message Statistics	pfc_p_bundle_message_received_failure	Counter	Number of PFCP bundle messages decode failures, cause represents failure reason(like invalid packet).
PFCP Protocol Message Statistics	pfc_p_number_of_messages_in_bundled_request_sent	Histogram	Number of messages present in bundled request sent
PFCP Protocol Message Statistics	pfc_p_duration_of_bundling_in_bundled_request_sent	Histogram	Duration of bundling in bundled request sent in milli second
PFCP Protocol Message Statistics	pfc_p_size_of_message_of_bundled_request_sent	Histogram	Size of message in bytes of bundled request sent
PFCP Protocol Message Statistics	pfc_p_number_of_messages_in_bundled_request_received	Histogram	Number of messages present in bundled request received
PFCP Protocol Message Statistics	pfc_p_size_of_message_of_bundled_request_received	Histogram	Size of message in bytes of bundled request received

PFCP Protocol Message Statistics

- pfc_p_association_setup_request_sent

Type

64-bit Counter

Description

The number of association setup request messages sent.

Labels

peer_ip
- pfc_p_association_setup_retransmitted_request_sent

Type

64-bit Counter

Description

The number of association setup retransmitted request messages sent.

Labels

peer_ip
- pfc_p_association_setup_request_received

Type

64-bit Counter

Description

The number of association setup request messages received.

Labels

peer_ip

- pfcpl_association_setup_retransmitted_request_received

Type

64-bit Counter

Description

The number of association setup retransmitted request messages received.

Labels

peer_ip

- pfcpl_association_update_request_sent

Type

64-bit Counter

Description

The number of association update request messages sent.

Labels

peer_ip

- pfcpl_association_update_retransmitted_request_sent

Type

64-bit Counter

Description

The number of association update retransmitted request messages sent.

Labels

peer_ip

- pfcpl_association_update_request_received

Type

64-bit Counter

Description

The number of association update request messages received.

Labels

peer_ip

- pfcpl_association_update_retransmitted_request_received

Type

64-bit Counter

Description

The number of association update retransmitted request messages received.

Labels

peer_ip

- pfcpl_association_release_request_sent

Type

64-bit Counter

Description

The number of association release request messages sent.

Labels

peer_ip

- pfcf_association_release_retransmitted_request_sent

Type

64-bit Counter

Description

The number of association release retransmitted request messages sent.

Labels

peer_ip

- pfcf_association_release_request_received

Type

64-bit Counter

Description

The number of association release request messages received.

Labels

peer_ip

- pfcf_association_release_retransmitted_request_received

Type

64-bit Counter

Description

The number of association release retransmitted request messages received.

Labels

peer_ip

- pfcf_association_setup_response_sent

Type

64-bit Counter

Description

The number of association setup response messages sent.

Labels

peer_ip, pfcf_cause

- pfcf_association_setup_response_received

Type

64-bit Counter

Description

The number of association setup response messages received.

Labels

peer_ip, pfcf_cause

- pfcf_association_update_response_sent

Type

64-bit Counter

Description

The number of association update response messages sent.

Labels

peer_ip, pfc_p_cause

- pfc_p_association_update_response_received

Type

64-bit Counter

Description

The number of association update response messages received.

Labels

peer_ip, pfc_p_cause

- pfc_p_association_release_response_sent

Type

64-bit Counter

Description

The number of association release response messages sent.

Labels

peer_ip, pfc_p_cause

- pfc_p_association_release_response_received

Type

64-bit Counter

Description

The number of association release response messages received.

Labels

peer_ip, pfc_p_cause

- pfc_p_version_not_supported_response_sent

Type

64-bit Counter

Description

The number of version not supported response messages sent.

Labels

peer_ip, pfc_p_cause

- pfc_p_version_not_supported_response_received

Type

64-bit Counter

Description

The number of version not supported response messages received.

Labels

peer_ip, pfc_p_cause

- pfc_p_heartbeat_request_sent

Type

64-bit Counter

Description

The number of heartbeat request messages sent.

Labels

peer_ip

- pfc_p_heartbeat_retransmitted_request_sent

Type

64-bit Counter

Description

The number of heartbeat retransmitted request messages sent.

Labels

peer_ip

- pfc_p_heartbeat_request_received

Type

64-bit Counter

Description

The number of heartbeat request messages received.

Labels

peer_ip

- pfc_p_heartbeat_retransmitted_request_received

Type

64-bit Counter

Description

The number of heartbeat retransmitted request messages received.

Labels

peer_ip

- pfc_p_heartbeat_response_sent

Type

64-bit Counter

Description

The number of heartbeat response messages sent.

Labels

peer_ip

- pfc_p_heartbeat_response_received

Type

64-bit Counter

Description

The number of heartbeat response messages received.

Labels

peer_ip

• pfc_session_establishment_request_sent

Type

64-bit Counter

Description

The number of session establishment request messages sent.

Labels

interface, peer_ip, nssai

• pfc_session_establishment_retransmitted_request_sent

Type

64-bit Counter

Description

The number of session establishment retransmitted request messages sent.

Labels

interface, peer_ip, nssai

• pfc_session_establishment_request_received

Type

64-bit Counter

Description

The number of session establishment request messages received.

Labels

peer_ip

• pfc_session_establishment_retransmitted_request_received

Type

64-bit Counter

Description

The number of session establishment retransmitted request messages received.

Labels

peer_ip

• pfc_session_modification_request_sent

Type

64-bit Counter

Description

The number of session modification request messages sent.

Labels

interface, peer_ip, nssai

- pfcf_session_modification_retransmitted_request_sent

Type

64-bit Counter

Description

The number of session modification retransmitted request messages sent.

Labels

interface, peer_ip, nssai

- pfcf_session_modification_request_received

Type

64-bit Counter

Description

The number of session modification request messages received.

Labels

peer_ip

- pfcf_session_modification_retransmitted_request_received

Type

64-bit Counter

Description

The number of session modification retransmitted request messages received.

Labels

peer_ip

- pfcf_session_deletion_request_sent

Type

64-bit Counter

Description

The number of session deletion request messages sent.

Labels

interface, peer_ip, nssai

- pfcf_session_deletion_retransmitted_request_sent

Type

64-bit Counter

Description

The number of session deletion retransmitted request messages sent.

Labels

interface, peer_ip, nssai

- pfcf_session_deletion_request_received

Type

64-bit Counter

Description

The number of session deletion request messages received.

Labels

peer_ip

- pfc_session_deletion_retransmitted_request_received

Type

64-bit Counter

Description

The number of session deletion retransmitted request messages received.

Labels

peer_ip

- pfc_session_report_request_sent

Type

64-bit Counter

Description

The number of session report request messages sent.

Labels

peer_ip

- pfc_session_report_retransmitted_request_sent

Type

64-bit Counter

Description

The number of session report retransmitted request messages sent.

Labels

peer_ip

- pfc_session_report_request_received

Type

64-bit Counter

Description

The number of session report request messages received.

Labels

peer_ip

- pfc_session_report_retransmitted_request_received

Type

64-bit Counter

Description

The number of session report retransmitted request messages received.

Labels

peer_ip

- pfcpl_session_establishment_response_sent
Type
64-bit Counter
Description
The number of session establishment response messages sent.
Labels
peer_ip, pfcpl_cause
- pfcpl_session_establishment_response_received
Type
64-bit Counter
Description
The number of session establishment response messages received.
Labels
interface, peer_ip, pfcpl_cause, nssai
- pfcpl_session_modification_response_sent
Type
64-bit Counter
Description
The number of session modification response messages sent.
Labels
peer_ip, pfcpl_cause
- pfcpl_session_modification_response_received
Type
64-bit Counter
Description
The number of session modification response messages received.
Labels
interface, peer_ip, pfcpl_cause, nssai
- pfcpl_session_deletion_response_sent
Type
64-bit Counter
Description
The number of session deletion response messages sent.
Labels
peer_ip, pfcpl_cause
- pfcpl_session_deletion_response_received
Type
64-bit Counter
Description

The number of session deletion response messages received.

Labels

interface, peer_ip, pfc_p_cause, nssai

• pfc_p_node_report_request_sent

Type

64-bit Counter

Description

The number of node report request messages sent.

Labels

peer_ip

• pfc_p_node_report_type_request_received

Type

64-bit Counter

Description

The number of node report request messages received.

Labels

peer_ip

• pfc_p_node_report_type_retransmitted_request_received

Type

64-bit Counter

Description

The number of node report retransmitted request messages received.

Labels

peer_ip

• pfc_p_node_report_response_received

Type

64-bit Counter

Description

The number of node report response messages received.

Labels

peer_ip, pfc_p_cause

• pfc_p_node_report_retransmitted_request_sent

Type

64-bit Counter

Description

The number of node report retransmitted request messages sent.

Labels

peer_ip

• pfc_p_session_report_response_sent

Type

64-bit Counter

Description

The number of session report response messages sent.

Labels

interface, peer_ip, pfc_p_cause

- pfc_session_report_response_received

Type

64-bit Counter

Description

The number of session report response messages received.

Labels

peer_ip, pfc_p_cause

- pfc_pfdm_request_sent

Type

64-bit Counter

Description

The number of pfdm request messages sent.

Labels

peer_ip

- pfc_pfdm_retransmitted_request_sent

Type

64-bit Counter

Description

The number of pfdm retransmitted request messages sent.

Labels

peer_ip

- pfc_pfdm_request_received

Type

64-bit Counter

Description

The number of pfdm request messages received.

Labels

peer_ip

- pfc_pfdm_retransmitted_request_received

Type

64-bit Counter

Description

The number of pfdm retransmitted request messages received.

Labels

peer_ip

- pfc_p_pfdm_response_sent

Type

64-bit Counter

Description

The number of pfdm response messages sent.

Labels

peer_ip, pfc_p_cause

- pfc_p_pfdm_response_received

Type

64-bit Counter

Description

The number of pfdm response messages received.

Labels

peer_ip, pfc_p_cause

- pfc_p_bundle_message_sent

Type

64-bit Counter

Description

The number of pfc bundle messages sent.

Labels

peer_ip, bundle_trigger

- pfc_p_bundle_message_sent_failure

Type

64-bit Counter

Description

The number of pfc bundle messages sent failures.

Labels

peer_ip, cause

- pfc_p_bundle_message_received

Type

64-bit Counter

Description

The number of pfc bundle messages received.

Labels

peer_ip

- pfc_p_bundle_message_received_failure

Type

64-bit Counter

Description

The number of pfcP bundle messages decode failure.

Labels

peer_ip, cause

• pfcP_number_of_messages_in_bundled_request_sent

Type

64-bit Histogram

Description

The number of messages in bundled request sent

Labels

peer_ip

Boundaries

[1, 2, 4, 6, 8, 10, 12, 14, 16, 20, 24, 28, 32, 40, 48, 56, 64]

• pfcP_duration_of_bundling_in_bundled_request_sent

Type

64-bit Histogram

Description

The duration of bundling in bundled request sent in milli second

Labels

peer_ip

Boundaries

[1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 14, 16, 20, 24, 28, 32, 40, 48, 56, 64, 80, 96, 112, 128, 160, 192, 224, 256]

• pfcP_size_of_message_of_bundled_request_sent

Type

64-bit Histogram

Description

The size of message in bytes of bundled request sent

Labels

peer_ip

Boundaries

[100, 200, 400, 600, 800, 1000, 1500, 2000, 2500, 3500, 4000, 5000, 6000, 7000, 8000, 9000, 10000, 16000, 20000, 24000, 28000, 32000]

• pfcP_number_of_messages_in_bundled_request_received

Type

64-bit Histogram

Description

The number of messages in bundled request received

Labels

peer_ip

Boundaries

[1, 2, 4, 6, 8, 10, 12, 14, 16, 20, 24, 28, 32, 40, 48, 56, 64]

• pfc_p_size_of_message_of_bundled_request_received

Type

64-bit Histogram

Description

The size of message in bytes of bundled request received

Labels

peer_ip

Boundaries

[100, 200, 400, 600, 800, 1000, 1500, 2000, 2500, 3500, 4000, 5000, 6000, 7000, 8000, 9000, 10000, 16000, 20000, 24000, 28000, 32000]

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- [PFCP UDP Message Statistics](#)

Summary

Group	Metric Name	Type	Synopsis
PFCP UDP Message Statistics	pfc_p_udp_message_received	Counter	PFCP UDP received messages.
PFCP UDP Message Statistics	pfc_p_udp_message_sent	Counter	PFCP UDP sent messages.

PFCP UDP Message Statistics

• pfc_p_udp_message_received

Type

64-bit Counter

Description

The number of PFCP UDP messages received.

Labels

traffic, src_ip

• pfc_p_udp_message_sent

Type

64-bit Counter

Description

The number of PFCP UDP messages sent.

Labels

traffic, dst_ip

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- [Summary](#)
- [PFCP udsf interface message statistics](#)

Summary

Group	Metric Name	Type	Synopsis
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Group	Metric Name	Type	Synopsis
PFCP udsf interface message statistics	pfcf_udsf_read_and_lock_record_using_secondary_key_request_attempts	Counter	Total number of udsf read and lock record request using secondary key attempts
PFCP udsf interface message statistics	pfcf_udsf_read_and_lock_record_using_secondary_key_request_successes	Counter	Total number of udsf read and lock record request using secondary key successes
PFCP udsf interface message statistics	pfcf_udsf_read_and_lock_record_using_secondary_key_request_failures	Counter	Total number of udsf read and lock record request using secondary key failures
PFCP udsf interface message statistics	pfcf_udsf_read_and_lock_record_request_attempts	Counter	Total number of udsf read and lock record request attempts
PFCP udsf interface message statistics	pfcf_udsf_read_and_lock_record_request_successes	Counter	Total number of udsf read and lock record request successes
PFCP udsf interface message statistics	pfcf_udsf_read_and_lock_record_request_failures	Counter	Total number of udsf read and lock record request failures
PFCP udsf interface message statistics	pfcf_udsf_create_record_and_lock_request_attempts	Counter	Total number of udsf create record and lock request attempts
PFCP udsf interface message statistics	pfcf_udsf_create_record_and_lock_request_successes	Counter	Total number of udsf create record and lock request successes
PFCP udsf interface message statistics	pfcf_udsf_create_record_and_lock_request_failures	Counter	Total number of udsf create record and lock request failures
PFCP udsf interface message statistics	pfcf_udsf_create_or_modify_record_request_attempts	Counter	Total number of udsf create or modify record request attempts
PFCP udsf interface message statistics	pfcf_udsf_create_or_modify_record_request_successes	Counter	Total number of udsf create or modify record request successes
PFCP udsf interface message statistics	pfcf_udsf_create_or_modify_record_request_failures	Counter	Total number of create or modify record request failures
PFCP udsf interface message statistics	pfcf_udsf_modify_record_and_lock_request_attempts	Counter	Total number of udsf modify record and lock request attempts
PFCP udsf interface message statistics	pfcf_udsf_modify_record_and_lock_request_successes	Counter	Total number of udsf modify record and lock request successes
PFCP udsf interface message statistics	pfcf_udsf_modify_record_and_lock_request_failures	Counter	Total number of udsf modify record and lock request failures
PFCP udsf interface message statistics	pfcf_udsf_modify_record_and_unlock_request_attempts	Counter	Total number of udsf modify record and unlock request attempts
PFCP udsf interface message statistics	pfcf_udsf_modify_record_and_unlock_request_successes	Counter	Total number of udsf modify record and unlock request successes
PFCP udsf interface message statistics	pfcf_udsf_modify_record_and_unlock_request_failures	Counter	Total number of udsf modify record and unlock request failures
PFCP udsf interface message statistics	pfcf_udsf_delete_record_request_attempts	Counter	Total number of udsf delete record request attempts
PFCP udsf interface message statistics	pfcf_udsf_delete_record_request_successes	Counter	Total number of udsf delete record request successes
PFCP udsf interface message statistics	pfcf_udsf_delete_record_request_failures	Counter	Total number of udsf delete record requests failures
PFCP udsf interface message statistics	pfcf_udsf_get_record_and_steal_lock_request_attempts	Counter	Total number of udsf get record and steal lock requests attempts
PFCP udsf interface message statistics	pfcf_udsf_get_record_and_steal_lock_request_successes	Counter	Total number of udsf get record and steal lock request successes
PFCP udsf interface message statistics	pfcf_udsf_get_record_and_steal_lock_request_failures	Counter	Total number of udsf get record and steal lock request failures
PFCP udsf interface message statistics	pfcf_udsf_search_record_request_attempts	Counter	Total number of udsf search record request attempts
PFCP udsf interface message statistics	pfcf_udsf_search_record_request_successes	Counter	Total number of udsf search record request successes

Group	Metric Name	Type	Synopsis
PFCP udsf interface message statistics	pfcpl_udsfl_search_record_request_failures	Counter	Total number of udsf search record request failures
PFCP udsf interface message statistics	pfcpl_udsfl_get_record_request_attempts	Counter	Total number of udsf get record request attempts
PFCP udsf interface message statistics	pfcpl_udsfl_get_record_request_successes	Counter	Total number of udsf get record request successes
PFCP udsf interface message statistics	pfcpl_udsfl_get_record_request_failures	Counter	Total number of udsf get record request failures
PFCP udsf interface message statistics	pfcpl_udsfl_connection_available_indication_received	Counter	Total number of UDSF API Connection Available Indication received
PFCP udsf interface message statistics	pfcpl_udsfl_connection_unavailable_indication_received	Counter	Total number of UDSF API Connection Unavailable Indication received
PFCP udsf interface message statistics	pfcpl_db_transaction_failure_udsfl_connection_unavailable	Counter	Total number of DB transactions failed due to UDSF Connection Unavailable

PFCP udsf interface message statistics

- [pfcpl_udsfl_read_and_lock_record_using_secondary_key_request_attempts](#)

Type

64-bit Counter

Description

Number of udsf read and lock record request using secondary key attempts

Labels

key

- [pfcpl_udsfl_read_and_lock_record_using_secondary_key_request_successes](#)

Type

64-bit Counter

Description

Number of udsf read and lock record request using secondary key successes

Labels

key

- [pfcpl_udsfl_read_and_lock_record_using_secondary_key_request_failures](#)

Type

64-bit Counter

Description

Number of udsf read and lock record request using secondary key failures

Labels

key, cause

- [pfcpl_udsfl_read_and_lock_record_request_attempts](#)

Type

64-bit Counter

Description

Number of udsf read and lock record request attempts

Labels

key

- pfc_p_uds_f_read_and_lock_record_request_successes

Type

64-bit Counter

Description

Number of udsf read and lock record request successess

Labels

key

- pfc_p_uds_f_read_and_lock_record_request_failures

Type

64-bit Counter

Description

Number of udsf read and lock record request failures

Labels

key, cause

- pfc_p_uds_f_create_record_and_lock_request_attempts

Type

64-bit Counter

Description

Number of usdf create record and lock request attempts

Labels

key

- pfc_p_uds_f_create_record_and_lock_request_successes

Type

64-bit Counter

Description

Number of usdf create record and lock request successes

Labels

key

- pfc_p_uds_f_create_record_and_lock_request_failures

Type

64-bit Counter

Description

Number of udsf create record and lock request failures

Labels

key, cause

- pfc_p_uds_f_create_or_modify_record_request_attempts

Type

64-bit Counter

Description

Number of create or modify record request attempts

Labels

key

- pfc_p_uds_f_create_or_modify_record_request_successes

Type

64-bit Counter

Description

Number of create or modify record request successes

Labels

key

- pfc_p_uds_f_create_or_modify_record_request_failures

Type

64-bit Counter

Description

Number of uds_f create or modify record request failures

Labels

key, cause

- pfc_p_uds_f_modify_record_and_lock_request_attempts

Type

64-bit Counter

Description

Number of uds_f modify record and lock request attempts

Labels

key

- pfc_p_uds_f_modify_record_and_lock_request_successes

Type

64-bit Counter

Description

Number of uds_f modify record and lock request successes

Labels

key

- pfc_p_uds_f_modify_record_and_lock_request_failures

Type

64-bit Counter

Description

Number of uds_f modify record and lock request failures

Labels

key, cause

- pfc_p_uds_f_modify_record_and_unlock_request_attempts

Type
64-bit Counter
Description
Number of udsf modify record and unlock request attempts
Labels

key

- pfc_p_uds_f_modify_record_and_unlock_request_successes

Type
64-bit Counter
Description
Number of udsf modify record and unlock request successes
Labels

key

- pfc_p_uds_f_modify_record_and_unlock_request_failures

Type
64-bit Counter
Description
Number of udsf modify record and unlock request failures
Labels

key, cause

- pfc_p_uds_f_delete_record_request_attempts

Type
64-bit Counter
Description
Number of udsf delete record request attempts
Labels

key

- pfc_p_uds_f_delete_record_request_successes

Type
64-bit Counter
Description
Number of udsf delete record request successes
Labels

key

- pfc_p_uds_f_delete_record_request_failures

Type
64-bit Counter
Description

Number of udsf delete record request failures

Labels

key, cause

- pfc_p_udsf_get_record_and_steal_lock_request_attempts

Type

64-bit Counter

Description

Number of udsf get record and steal lock request attempts

Labels

key

- pfc_p_udsf_get_record_and_steal_lock_request_successes

Type

64-bit Counter

Description

Number of udsf get record and steal lock request successes

Labels

key

- pfc_p_udsf_get_record_and_steal_lock_request_failures

Type

64-bit Counter

Description

Number of udsf get record and steal lock request failures

Labels

key, cause

- pfc_p_udsf_search_record_request_attempts

Type

64-bit Counter

Description

Number of udsf search record request attempts

Labels

key

- pfc_p_udsf_search_record_request_successes

Type

64-bit Counter

Description

Number of udsf search record request successes

Labels

key

- pfc_p_udsf_search_record_request_failures

Type

64-bit Counter

Description

Number of udsf search record request failures

Labels

key, cause

- pfc_p_udsf_get_record_request_attempts

Type

64-bit Counter

Description

Number of udsf get record request attempts

Labels

key

- pfc_p_udsf_get_record_request_successes

Type

64-bit Counter

Description

Number of udsf get record request successes

Labels

key

- pfc_p_udsf_get_record_request_failures

Type

64-bit Counter

Description

Number of udsf get record request failures

Labels

key, cause

- pfc_p_udsf_connection_available_indication_received

Type

64-bit Counter

Description

Number of UDSF API Connection Available Indication received

Labels

- pfc_p_udsf_connection_unavailable_indication_received

Type

64-bit Counter

Description

Number of UDSF API Connection Unavailable Indication received

Labels

- pfc_p_db_transaction_failure_udsf_connection_unavailable

Type

64-bit Counter

Description

Number of DB transactions failed due to UDSF Connection Unavailable

Labels

transaction, event_trigger

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- [Circuit Breaker Metrics](#)
- [Service Endpoint Learner Metrics](#)
- [Service Endpoint Manager Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Circuit Breaker Metrics	sem_cb_endpoints	Gauge	Total number of endpoints for a service in respective states
Service Endpoint Learner Metrics	sem_num_dns_errors	Counter	Total number of DNS resolution errors for a given service
Service Endpoint Learner Metrics	sem_num_endpoints	Gauge	Total number of endpoints learnt for a service
Service Endpoint Manager Metrics	sem_num_endpoint_transactions_sent	Counter	Total number of times an endpoint is given to the application for communication with it
Service Endpoint Manager Metrics	sem_num_endpoint_transaction_status_received	Counter	Total number of transactions for a given endpoint received with the transaction status

Circuit Breaker Metrics

- sem_cb_endpoints

Type

64-bit Gauge

Description

Total number of endpoints for a service in respective states. There are 3 states viz. Active, Blocked, Restoring. In Active state, the endpoint is responding as expected, i.e. the response code doesn't indicate a service failure. The endpoint will be moved to Blocked state based on response failures which indicate service failure. After a timeout, the endpoint will be moved to Restoring state to check if the endpoint has recovered. In this state, if endpoint responds with response code other than service failure, it will be moved to Active state. Else, it will be moved back to Blocked state.

Labels

service_name, endpoint_state

Service Endpoint Learner Metrics

- sem_num_dns_errors

Type

64-bit Counter

Description

Total number of DNS resolution errors for a given service, labels are service name for DNS resolution and DNS error codes

Labels

- sem_num_endpoints

64-bit Gauge

Description

Total number of endpoints learnt from the Kubernetes DNS server for a given service

Labels

service

Service Endpoint Manager Metrics

- sem_num_endpoint_transactions_sent

64-bit Counter

Description

Total number of times an endpoint is given to the application for communication with it

Labels

service

- sem_num_endpoint_transaction_status_received

Type

64-bit Counter

Description

Total number of transactions for a given endpoint received with the transaction status

Labels

service, status

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- Summary
- UDSF DBClient Metrics
- UDSF API latency statistics

Summary

Group	Metric Name	Type	Synopsis
UDSF DBClient Metrics	udsf_db_operation_request	Counter	The total number of udsf dbclient requests received.
UDSF DBClient Metrics	udsf_db_operation_response	Counter	The total number of udsf dbclient responses sent.
UDSF DBClient Metrics	udsf_dbc_total_init_success	Counter	The total number of UDSF INIT successes.
UDSF DBClient Metrics	udsf_dbc_total_init_failure	Counter	The total number of UDSF INIT failures.
UDSF DBClient Metrics	udsf_dbc_total_deinit_success	Counter	The total number of UDSF DEINIT successes.
UDSF DBClient Metrics	udsf_dbc_total_deinit_failure	Counter	The total number of UDSF DEINIT failures.
UDSF DBClient Metrics	udsf_db_transaction_counter	Counter	The total number of CRUD operations.
UDSF API latency statistics	udsf_db_transaction_duration	Histogram	Roundtrip duration taken by transaction in seconds.
UDSF API latency statistics	udsf_db_execution_duration	Histogram	Execution time taken by transaction in microseconds.
UDSF API latency statistics	udsf_hiredis_bufferwrite_duration	Histogram	Duration of time taken in hiredis to write data to socket in seconds

UDSF DBClient Metrics

- **udsf_db_operation_request**

Type

64-bit Counter

Description

The total number of udsf dbclient requests received from worker. Labels- 'app_name' : Originating thread name, ex:smfcc, 'operation_type' : Type of CRUD operation/API type involved, ex:CREATE_OR_MODIFY_RECORD_ASYNC

Labels

app_name, operation_type
- **udsf_db_operation_response**

Type

64-bit Counter

Description

The total number of udsf dbclient responses sent. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE 'operation_type' : Type of CRUD operation/API type involved., ex:CREATE_OR_MODIFY_RECORD_ASYNC

Labels

app_name, status, operation_type
- **udsf_dbc_total_init_success**

Type

64-bit Counter

Description

The total number of times UDSF API init was success. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: SUCCESS

Labels

app_name, status
- **udsf_dbc_total_init_failure**

Type

64-bit Counter

Description

The total number of times UDSF API init failed. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: FAILURE

Labels

app_name, status
- **udsf_dbc_total_deinit_success**

Type

64-bit Counter

Description

The total number of times UDSF API Deinit succeeded. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: SUCCESS

Labels

app_name, status

- udsf_dbc_total_deinit_failure
 - Type
 - 64-bit Counter
 - Description
 - The total number of times UDSF API Deinit failed. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: FAILURE
 - Labels
 - app_name, status**
- udsf_db_transaction_counter
 - Type
 - 64-bit Counter
 - Description
 - The total number of times CRUD operations are incrementing. Labels- 'app_name' : Originating thread name, ex:smfcc, 'crud_op' : The categorical CRUD operation in which this sub-transation falls into, ex: CREATE_OR_MODIFY
 - Labels
 - app_name, crud_op**

UDSF API latency statistics

- udsf_db_transaction_duration
 - Type
 - 64-bit Histogram
 - Description
 - Duration observed on total journey/roundtrip taken by a transaction after receiving the request. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE 'operation_type' : Type of CRUD operation/API type involved., ex:CREATE_OR_MODIFY_RECORD_ASYNC, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE, 'crud_op' : The categorical CRUD operation in which this transation falls into, ex: CREATE_OR_MODIFY
 - Labels
 - app_name,operation_type,status,crud_op**
 - Boundaries
 - [0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]
- udsf_db_execution_duration
 - Type
 - 64-bit Histogram
 - Description
 - Processing time taken by UDSF client. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE 'operation_type' : Type of CRUD operation/API type involved., ex:CREATE_OR_MODIFY_RECORD_ASYNC, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE, 'crud_op' : The categorical CRUD operation in which this transation falls into, ex: CREATE_OR_MODIFY
 - Labels
 - app_name,operation_type,status,crud_op**
 - Boundaries
 - [1, 2, 4, 8, 16, 32, 64, 128, 256, 384, 512, 768, 1024, 1536, 2048, 4096, 8192, 16384]
- udsf_hiredis_bufferwrite_duration

Type

64-bit Histogram

Description

Duration observed between hiredis buffer append and write to socket. This is fair approximation for time spent before data is written to socket as the buffer threshold accounted for append and write is dynamic and not fixed size. Per socket, the buffer append start-time is recorded along with size intended to transmit and end-time recorded once data greater than recorded size is written on the socket

Labels

app_name

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

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- [TDEF Proxy CPU Statistics](#)
- [TDEF Proxy Overload Control State Transition Statistics](#)
- [TDEF Proxy Overload Control Connection Statistics](#)

Summary

Group	Metric Name	Type	Synopsis
TDEF Proxy CPU Statistics	num_cpu_busy_percentage	Gauge	Percentage - %
TDEF Proxy Overload Control State Transition Statistics	num_enter_non_overload_state	Counter	# of times a proxy exited overload state
TDEF Proxy Overload Control State Transition Statistics	num_enter_minor_overload_state	Counter	# of times a proxy entered overload state
TDEF Proxy Overload Control State Transition Statistics	num_enter_major_overload_state	Counter	# of times a proxy entered major overload state
TDEF Proxy Overload Control State Transition Statistics	num_enter_critical_overload_state	Counter	# of times a proxy entered critical overload state
TDEF Proxy Overload Control Connection Statistics	num_http_connections_bypassed_proxy_when_minor_overloaded	Gauge	# of HTTP connections bypassed proxy when minor overloaded
TDEF Proxy Overload Control Connection Statistics	num_http_connections_reached_proxy_when_minor_overloaded	Gauge	# of HTTP connections reached proxy when minor overloaded
TDEF Proxy Overload Control Connection Statistics	num_tcp_connections_bypassed_proxy_when_minor_overloaded	Gauge	# of TCP connections bypassed proxy when minor overloaded
TDEF Proxy Overload Control Connection Statistics	num_tcp_connections_reached_proxy_when_minor_overloaded	Gauge	# of TCP connections reached proxy when minor overloaded
TDEF Proxy Overload Control Connection Statistics	num_udp_connections_bypassed_proxy_when_minor_overloaded	Gauge	# of UDP connections bypassed proxy when minor overloaded
TDEF Proxy Overload Control Connection Statistics	num_udp_connections_reached_proxy_when_minor_overloaded	Gauge	# of UDP connections reached proxy when minor overloaded
TDEF Proxy Overload Control Connection Statistics	num_http_connections_bypassed_proxy_when_major_overloaded	Gauge	# of HTTP connections bypassed proxy when major overloaded
TDEF Proxy Overload Control Connection Statistics	num_http_connections_reached_proxy_when_major_overloaded	Gauge	# of HTTP connections reached proxy when major overloaded
TDEF Proxy Overload Control Connection Statistics	num_tcp_connections_bypassed_proxy_when_major_overloaded	Gauge	# of TCP connections bypassed proxy when major overloaded
TDEF Proxy Overload Control Connection Statistics	num_tcp_connections_reached_proxy_when_major_overloaded	Gauge	# of TCP connections reached proxy when major overloaded
TDEF Proxy Overload Control Connection Statistics	num_udp_connections_bypassed_proxy_when_major_overloaded	Gauge	# of UDP connections bypassed proxy when major overloaded

Group	Metric Name	Type	Synopsis
TDEF Proxy Overload Control Connection Statistics	num_udp_connections_reached_proxy_when_major_overloaded	Gauge	# of UDP connections reached proxy when major overloaded
TDEF Proxy Overload Control Connection Statistics	num_http_connections_bypassed_proxy_when_critical_overloaded	Gauge	# of HTTP connections bypassed proxy when critically overloaded
TDEF Proxy Overload Control Connection Statistics	num_http_connections_reached_proxy_when_critical_overloaded	Gauge	# of HTTP connections reached proxy when critically overloaded
TDEF Proxy Overload Control Connection Statistics	num_tcp_connections_bypassed_proxy_when_critical_overloaded	Gauge	# of TCP connections bypassed proxy when critically overloaded
TDEF Proxy Overload Control Connection Statistics	num_tcp_connections_reached_proxy_when_critical_overloaded	Gauge	# of TCP connections reached proxy when critically overloaded
TDEF Proxy Overload Control Connection Statistics	num_udp_connections_bypassed_proxy_when_critical_overloaded	Gauge	# of UDP connections bypassed proxy when critically overloaded
TDEF Proxy Overload Control Connection Statistics	num_udp_connections_reached_proxy_when_critical_overloaded	Gauge	# of UDP connections reached proxy when critically overloaded

TDEF Proxy CPU Statistics

- num_cpu_busy_percentage

Type

64-bit Gauge

Description

CPU busy percentage

Labels

pod_id, proxy_id

TDEF Proxy Overload Control State Transition Statistics

- num_enter_non_overload_state

Type

64-bit Counter

Description

The number of times that a proxy exited overload state

Labels

pod_id, proxy_id, proxy_service, table

- num_enter_minor_overload_state

Type

64-bit Counter

Description

The number of times that a proxy entered minor overload state

Labels

pod_id, proxy_id, proxy_service, table

- num_enter_major_overload_state

Type

64-bit Counter

Description

The number of times that a proxy entered major overload state

Labels

pod_id, proxy_id, proxy_service, table

- num_enter_critical_overload_state

Type

64-bit Counter

Description

The number of times that a proxy entered critical overload state

Labels

pod_id, proxy_id, proxy_service, table

TDEF Proxy Overload Control Connection Statistics

- num_http_connections_bypassed_proxy_when_minor_overloaded

Type

64-bit Gauge

Description

The number of HTTP connections bypassed proxy when minor overloaded

Labels

pod_id, proxy_id, proxy_service, table

- num_http_connections_reached_proxy_when_minor_overloaded

Type

64-bit Gauge

Description

The number of HTTP connections processed by proxy when minor overloaded

Labels

pod_id, proxy_id, proxy_service, table

- num_tcp_connections_bypassed_proxy_when_minor_overloaded

Type

64-bit Gauge

Description

The number of TCP connections bypassed proxy when minor overloaded

Labels

pod_id, proxy_id, proxy_service, table

- num_tcp_connections_reached_proxy_when_minor_overloaded

Type

64-bit Gauge

Description

The number of TCP connections processed by proxy when minor overloaded

Labels

pod_id, proxy_id, proxy_service, table

- num_udp_connections_bypassed_proxy_when_minor_overloaded
Type
64-bit Gauge
Description
The number of UDP connections bypassed proxy when minor overloaded
Labels
pod_id, proxy_id, proxy_service, table
- num_udp_connections_reached_proxy_when_minor_overloaded
Type
64-bit Gauge
Description
The number of UDP connections processed by proxy when minor overloaded
Labels
pod_id, proxy_id, proxy_service, table
- num_http_connections_bypassed_proxy_when_major_overloaded
Type
64-bit Gauge
Description
The number of HTTP connections bypassed proxy when major overloaded
Labels
pod_id, proxy_id, proxy_service, table
- num_http_connections_reached_proxy_when_major_overloaded
Type
64-bit Gauge
Description
The number of HTTP connections processed by proxy when major overloaded
Labels
pod_id, proxy_id, proxy_service, table
- num_tcp_connections_bypassed_proxy_when_major_overloaded
Type
64-bit Gauge
Description
The number of TCP connections bypassed proxy when major overloaded
Labels
pod_id, proxy_id, proxy_service, table
- num_tcp_connections_reached_proxy_when_major_overloaded
Type
64-bit Gauge
Description

The number of TCP connections processed by proxy when major overloaded

Labels

pod_id, proxy_id, proxy_service, table

- num_udp_connections_bypassed_proxy_when_major_overloaded

Type

64-bit Gauge

Description

The number of UDP connections bypassed proxy when major overloaded

Labels

pod_id, proxy_id, proxy_service, table

- num_udp_connections_reached_proxy_when_major_overloaded

Type

64-bit Gauge

Description

The number of UDP connections processed by proxy when major overloaded

Labels

pod_id, proxy_id, proxy_service, table

- num_http_connections_bypassed_proxy_when_critical_overloaded

Type

64-bit Gauge

Description

The number of HTTP connections bypassed proxy when critically overloaded

Labels

pod_id, proxy_id, proxy_service, table

- num_http_connections_reached_proxy_when_critical_overloaded

Type

64-bit Gauge

Description

The number of HTTP connections processed by proxy when critically overloaded

Labels

pod_id, proxy_id, proxy_service, table

- num_tcp_connections_bypassed_proxy_when_critical_overloaded

Type

64-bit Gauge

Description

The number of TCP connections bypassed proxy when critically overloaded

Labels

pod_id, proxy_id, proxy_service, table

- num_tcp_connections_reached_proxy_when_critical_overloaded

- Type

64-bit Gauge

Description

The number of TCP connections processed by proxy when critically overloaded

Labels

pod_id, proxy_id, proxy_service, table
- num_udp_connections_bypassed_proxy_when_critical_overloaded
- Type

64-bit Gauge

Description

The number of UDP connections bypassed proxy when critically overloaded

Labels

pod_id, proxy_id, proxy_service, table
- num_udp_connections_reached_proxy_when_critical_overloaded
- Type

64-bit Gauge

Description

The number of UDP connections processed by proxy when critically overloaded

Labels

pod_id, proxy_id, proxy_service, table

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Summary

Group	Metric Name	Type	Synopsis
Summary of DNS Client Statistics	tdef_dns_client_query_attempts	Counter	Asynchronous DNS - Total Attempts
Summary of DNS Client Statistics	tdef_dns_client_query_success	Counter	Asynchronous DNS - Successes
Summary of DNS Client Statistics	tdef_dns_client_query_failure	Counter	Asynchronous DNS - Failures
Summary of DNS Client Statistics	tdef_dns_client_query_current	Gauge	Asynchronous DNS - Current
DNS Client Request Statistics	tdef_dns_client_request_ipv4	Counter	Asynchronous DNS Requests - IPv4
DNS Client Request Statistics	tdef_dns_client_request_ipv6	Counter	Asynchronous DNS Requests - IPv6
DNS Client Request Statistics	tdef_dns_client_request_ipv4v6	Counter	Asynchronous DNS Requests - IPv4v6
DNS Client Result Statistics	tdef_dns_client_result_cache_hit	Counter	Asynchronous DNS Results - Cache Hit
DNS Client Result Statistics	tdef_dns_client_result_ipv4	Counter	Asynchronous DNS Results - IPv4
DNS Client Result Statistics	tdef_dns_client_result_ipv6	Counter	Asynchronous DNS Results - IPv6
DNS Client Result Statistics	tdef_dns_client_result_ipv4v6	Counter	Asynchronous DNS Results - IPv4v6
Dns Client Failure Statistics	tdef_dns_client_failure_dropped	Counter	Asynchronous DNS Failures - Query Dropped
Dns Client Failure Statistics	tdef_dns_client_failure_canceled	Counter	Asynchronous DNS Failures - Query Canceled

Group	Metric Name	Type	Synopsis
Dns Client Failure Statistics	tdef_dns_client_failure_not_reachable	Counter	Asynchronous DNS Failures - Server Not Reachable
Dns Client Failure Statistics	tdef_dns_client_failure_timeout	Counter	Asynchronous DNS Failures - Response Timeout
Dns Client Failure Statistics	tdef_dns_client_failure_no_data	Counter	Asynchronous DNS Failures - No Data
Dns Client Failure Statistics	tdef_dns_client_failure_not_found	Counter	Asynchronous DNS Failures - Name Not Found
Dns Client Failure Statistics	tdef_dns_client_failure_server_refused	Counter	Asynchronous DNS Failures - Server Refused
Dns Client Failure Statistics	tdef_dns_client_failure_other	Counter	Asynchronous DNS Failures - Others

Summary of DNS Client Statistics

- tdef_dns_client_query_attempts

Type

64-bit Counter

Description

Total number of Asynchronous DNS Query Attempts

Labels

pod_id, proxy_id

- tdef_dns_client_query_success

Type

64-bit Counter

Description

Total number of Asynchronous DNS Query Success

Labels

pod_id, proxy_id

- tdef_dns_client_query_failure

Type

64-bit Counter

Description

Total number of Asynchronous DNS Query Failure

Labels

pod_id, proxy_id

- tdef_dns_client_query_current

Type

64-bit Gauge

Description

Current number of Asynchronous DNS Query IPv4 Attempts

Labels

pod_id, proxy_id

DNS Client Request Statistics

- tdef_dns_client_request_ipv4

Type

64-bit Counter

Description

Total number of Asynchronous DNS Query IPv4 Requests

Labels

pod_id, proxy_id

- tdef_dns_client_request_ipv6

Type

64-bit Counter

Description

Total number of Asynchronous DNS Query IPv6 Requests

Labels

pod_id, proxy_id

- tdef_dns_client_request_ipv4v6

Type

64-bit Counter

Description

Total number of Asynchronous DNS Query IPv4v6 Requests

Labels

pod_id, proxy_id

DNS Client Result Statistics

- tdef_dns_client_result_cache_hit

Type

64-bit Counter

Description

Total number of Asynchronous DNS Query Cache Hit Results

Labels

pod_id, proxy_id

- tdef_dns_client_result_ipv4

Type

64-bit Counter

Description

Total number of Asynchronous DNS Query IPv4 Results

Labels

pod_id, proxy_id

- tdef_dns_client_result_ipv6

Type

64-bit Counter

Description

Total number of Asynchronous DNS Query IPv6 Results

Labels
<i>pod_id, proxy_id</i>
tdef_dns_client_result_ipv4v6
Type
64-bit Counter
Description
Total number of Asynchronous DNS Query IPv4v6 Results
Labels
<i>pod_id, proxy_id</i>

Dns Client Failure Statistics

tdef_dns_client_failure_dropped
Type
64-bit Counter
Description
Total number of Asynchronous DNS Query Dropped, Not Sent
Labels
<i>pod_id, proxy_id</i>
tdef_dns_client_failure_canceled
Type
64-bit Counter
Description
Total number of Asynchronous DNS Query Canceled
Labels
<i>pod_id, proxy_id</i>
tdef_dns_client_failure_not_reachable
Type
64-bit Counter
Description
Total number of Asynchronous DNS Server Not-Reachable
Labels
<i>pod_id, proxy_id</i>
tdef_dns_client_failure_timeout
Type
64-bit Counter
Description
Total number of Asynchronous DNS Response Timeout
Labels
<i>pod_id, proxy_id</i>
tdef_dns_client_failure_no_data

Type

64-bit Counter

Description

Total number of Asynchronous DNS Response No Data

Labels

pod_id, proxy_id

- tdef_dns_client_failure_not_found

Type

64-bit Counter

Description

Total number of Asynchronous DNS Response Name Not Found

Labels

pod_id, proxy_id

- tdef_dns_client_failure_server_refused

Type

64-bit Counter

Description

Total number of Asynchronous DNS Response Server Refused

Labels

pod_id, proxy_id

- tdef_dns_client_failure_other

Type

64-bit Counter

Description

Total number of Asynchronous DNS Response Dailure With Other Reason

Labels

pod_id, proxy_id

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Summary

Group	Metric Name	Type	Synopsis
Summary of TDEF HTTP Proxy Statistics	tdef_http_proxy_service_attempts	Counter	HTTP-Proxy Service - Attempts
Summary of TDEF HTTP Proxy Statistics	tdef_http_proxy_service_success	Counter	HTTP-Proxy Service - Successes
Summary of TDEF HTTP Proxy Statistics	tdef_http_proxy_service_aborted_by_client	Counter	HTTP-Proxy Service - Aborts

Group	Metric Name	Type	Synopsis
Summary of TDEF HTTP Proxy Statistics	tdef_http_proxy_service_failed	Counter	HTTP-Proxy Service - Failures
Summary of TDEF HTTP Proxy Statistics	tdef_http_proxy_client_requests_total	Counter	Client Requests - Total
Summary of TDEF HTTP Proxy Statistics	tdef_http_proxy_client_requests_parsed_successfully	Counter	Client Requests - Parsed Successfully
Summary of TDEF HTTP Proxy Statistics	tdef_http_proxy_client_requests_aborted_by_client	Counter	Client Requests - Incomplete
Summary of TDEF HTTP Proxy Statistics	tdef_http_proxy_client_requests_processing_failed	Counter	Client Requests - Processing Failed
Summary of TDEF HTTP Proxy Statistics	tdef_http_proxy_client_requests_pipeline_dropped	Counter	Client Requests - Pipeline Dropped
Summary of TDEF HTTP Proxy Statistics	tdef_http_proxy_client_requests_invalid	Counter	Client Requests - Invalid
Summary of TDEF HTTP Proxy Statistics	tdef_http_proxy_client_request_bytes_total	Counter	Client Requests - Total Bytes
Summary of TDEF HTTP Proxy Statistics	tdef_http_proxy_client_response_bytes_total	Counter	Client Responses - Total Bytes
Summary of TDEF HTTP Proxy Statistics	tdef_http_proxy_server_request_bytes_total	Counter	Server Requests - Total Bytes
Summary of TDEF HTTP Proxy Statistics	tdef_http_proxy_server_response_bytes_total	Counter	Server Responses - Total Bytes
Summary of TDEF HTTP Proxy Statistics	tdef_http_proxy_server_response_total	Counter	Server Responses - Total
Summary of TDEF HTTP Proxy Statistics	tdef_http_proxy_transaction_edr_creates	Counter	HTTP-Proxy - EDR creates
Summary of TDEF HTTP Proxy Statistics	tdef_http_proxy_transaction_edr_attempts	Counter	HTTP-Proxy - EDR attempts
Summary of TDEF HTTP Proxy Statistics	tdef_http_proxy_transaction_edr_successes	Counter	HTTP-Proxy - EDR successes
Summary of TDEF HTTP Proxy Statistics	tdef_http_proxy_transaction_edr_failures	Counter	HTTP-Proxy - EDR failures
Summary of TDEF HTTP Proxy Statistics	tdef_http_proxy_client_http_service_time_min	Gauge	Service Time - Min (ms)
Summary of TDEF HTTP Proxy Statistics	tdef_http_proxy_client_http_service_time_avg	Gauge	Service Time - Avg (ms)
Summary of TDEF HTTP Proxy Statistics	tdef_http_proxy_client_http_service_time_max	Gauge	Service Time - Max (ms)
Summary of TDEF HTTP Proxy Statistics	tdef_http_proxy_client_requests_currently_in_service	Gauge	Client Requests - Current
Summary of TDEF HTTP Proxy Statistics	tdef_http_proxy_client_session_current	Gauge	Client Sessions - Current
Summary of TDEF HTTP Proxy Statistics	tdef_http_proxy_http_response_time_min	Gauge	Server Responses - Min (ms)
Summary of TDEF HTTP Proxy Statistics	tdef_http_proxy_http_response_time_avg	Gauge	Server Responses - Avg (ms)
Summary of TDEF HTTP Proxy Statistics	tdef_http_proxy_http_response_time_max	Gauge	Server Responses - Max (ms)
TDEF HTTP Proxy Detail Statistics	tdef_http_proxy_client_requests_http_v09	Counter	Client Requests - HTTP v0.9
TDEF HTTP Proxy Detail Statistics	tdef_http_proxy_client_requests_http_v10	Counter	Client Requests - HTTP v1.0
TDEF HTTP Proxy Detail Statistics	tdef_http_proxy_client_requests_http_v11	Counter	Client Requests - HTTP v1.1
TDEF HTTP Proxy Detail Statistics	tdef_http_proxy_persistent_client_requests	Counter	Client Requests - Persistent
TDEF HTTP Proxy Detail Statistics	tdef_http_proxy_pipelined_client_requests	Counter	Client Requests - Pipelined
TDEF HTTP Proxy Detail Statistics	tdef_http_proxy_ipv4_client_requests	Counter	Client Requests - IPv4
TDEF HTTP Proxy Detail Statistics	tdef_http_proxy_ipv6_client_requests	Counter	Client Requests - IPv6
TDEF HTTP Proxy Detail Statistics	tdef_http_proxy_transparent_proxy_requests	Counter	Client Requests - Transparent Proxy
TDEF HTTP Proxy Detail Statistics	tdef_http_proxy_explicit_proxy_requests	Counter	Client Requests - Explicit Proxy
TDEF HTTP Proxy Detail Statistics	tdef_http_proxy_client_http_response_status_information	Counter	Client Response Status - Information (1xx)
TDEF HTTP Proxy Detail Statistics	tdef_http_proxy_client_http_response_status_success_ok	Counter	Client Response Status - Success OK (200)
TDEF HTTP Proxy Detail Statistics	tdef_http_proxy_client_http_response_status_success_other	Counter	Client Response Status - Success Other (2xx)
TDEF HTTP Proxy Detail Statistics	tdef_http_proxy_client_http_response_status_redirection	Counter	Client Response Status - Redirection (3xx)
TDEF HTTP Proxy Detail Statistics	tdef_http_proxy_client_http_response_status_client_error	Counter	Client Response Status - Client Error (4xx)
TDEF HTTP Proxy Detail Statistics	tdef_http_proxy_client_http_response_status_server_error	Counter	Client Response Status - Server Error (5xx)
TDEF HTTP Proxy Detail Statistics	tdef_http_proxy_server_http_responses_ipv4	Counter	Server Responses - IPv4
TDEF HTTP Proxy Detail Statistics	tdef_http_proxy_server_http_responses_ipv6	Counter	Server Responses - IPv6
TDEF HTTP Proxy Detail Statistics	tdef_http_proxy_dpi_update_failures	Counter	DPI update failure - HTTP
TDEF HTTP Proxy Detail Statistics	tdef_http_proxy_dpi_update_for_deleted_request	Counter	DPI updates for deleted request - HTTP
TDEF HTTP Proxy Detail Statistics	tdef_http_proxy_dpi_updates	Counter	DPI updates - HTTP

Group	Metric Name	Type	Synopsis
TDEF HTTP Proxy Detail Statistics	tdef_http_proxy_transaction_edr_callback_successes	Counter	EDR callback - successes
TDEF HTTP Proxy Detail Statistics	tdef_http_proxy_transaction_edr_callback_failures	Counter	EDR callback - failures
TDEF HTTP Proxy Detail Statistics	tdef_http_proxy_client_http_request_size_min	Gauge	Client Request Size - Min (bytes)
TDEF HTTP Proxy Detail Statistics	tdef_http_proxy_client_http_request_size_avg	Gauge	Client Request Size - Avg (bytes)
TDEF HTTP Proxy Detail Statistics	tdef_http_proxy_client_http_request_size_max	Gauge	Client Request Size - Max (bytes)
TDEF HTTP Proxy Detail Statistics	tdef_http_proxy_client_http_response_size_min	Gauge	Client Response Size - Min (bytes)
TDEF HTTP Proxy Detail Statistics	tdef_http_proxy_client_http_response_size_avg	Gauge	Client Response Size - Avg (bytes)
TDEF HTTP Proxy Detail Statistics	tdef_http_proxy_client_http_response_size_max	Gauge	Client Response Size - Max (bytes)
TDEF HTTP Proxy Failure Statistics	tdef_http_proxy_service_failed_request_compile_error	Counter	HTTP-Proxy Service Failed - Request Compile Error
TDEF HTTP Proxy Failure Statistics	tdef_http_proxy_service_failed_connection_init_failure	Counter	HTTP-Proxy Service Failed - Connection Init Failure
TDEF HTTP Proxy Failure Statistics	tdef_http_proxy_service_failed_over_sized_response_header	Counter	HTTP-Proxy Service Failed - Oversized Response Header
TDEF HTTP Proxy Failure Statistics	tdef_http_proxy_service_failed_bad_response_status_line	Counter	HTTP-Proxy Service Failed - Bad Response Status Line
TDEF HTTP Proxy Failure Statistics	tdef_http_proxy_service_failed_bad_response_header	Counter	HTTP-Proxy Service Failed - Bad Response Header
TDEF HTTP Proxy Failure Statistics	tdef_http_proxy_service_failed_response_processing_failure	Counter	HTTP-Proxy Service Failed - Response Processing Failure
TDEF HTTP Proxy Failure Statistics	tdef_http_proxy_service_failed_server_connection_close	Counter	HTTP-Proxy Service Failed - Server Connection Close
TDEF HTTP Proxy Failure Statistics	tdef_http_proxy_service_failed_client_connection_timeout	Counter	HTTP-Proxy Service Failed - Client Connection Timeout
TDEF HTTP Proxy Failure Statistics	tdef_http_proxy_service_failed_resource_unavailable	Counter	HTTP-Proxy Service Failed - Resources unavailable
TDEF HTTP Proxy Failure Statistics	tdef_http_proxy_service_redirected	Counter	HTTP-Proxy Service Failed - Requests redirected
TDEF HTTP Proxy Failure Statistics	tdef_http_proxy_service_blocked	Counter	HTTP-Proxy Service Failed - Requests blocked
TDEF HTTP Proxy Failure Statistics	tdef_http_proxy_service_failed_maintenance_connection_close	Counter	HTTP-Proxy Service Failed - Maintenance Connection Close
TDEF HTTP Proxy Failure Statistics	tdef_http_proxy_service_out_of_buffer	Counter	HTTP-Proxy Service Failed - Out of Buffer
TDEF HTTP Proxy Failure Statistics	tdef_http_proxy_service_failed_server_connection_timeout	Counter	HTTP-Proxy Service Aborted - Server Connection Timeout
TDEF HTTP Proxy Failure Statistics	tdef_http_proxy_service_aborted_by_client	Counter	HTTP-Proxy Service - Aborts
TDEF HTTP Proxy Failure Statistics	tdef_http_proxy_service_aborted_no_resp_from_server	Counter	HTTP-Proxy Service Aborted - No Response from Server
TDEF HTTP Proxy Failure Statistics	tdef_http_proxy_service_aborted_request_to_local_addr	Counter	HTTP-Proxy Service Aborted - Request to Local Address
TDEF HTTP Proxy Failure Statistics	tdef_http_proxy_connection_failures_overload	Counter	Connection Failures - Terminated due to Overload
TDEF HTTP Proxy Failure Statistics	tdef_http_proxy_connection_failures_invalid_metadata	Counter	Connection Failures - Invalid Metadata
TDEF HTTP Proxy Failure Statistics	tdef_http_proxy_connection_failures_proxy_is_not_ready	Counter	Connection Failures - Proxy is not Ready
TDEF HTTP Proxy Failure Statistics	tdef_http_proxy_connection_failures_fd_limit_exceeded	Counter	Connection Failures - FD limit exceeded
TDEF HTTP Proxy Failure Statistics	tdef_http_proxy_connection_failures_out_of_resources	Counter	Connection Failures - Out of Resources
TDEF HTTP Proxy Connection Statistics	tdef_http_proxy_total_client_connections	Counter	Client Connections - Total
TDEF HTTP Proxy Connection Statistics	tdef_http_proxy_keep_alive_requested	Counter	Client Connections - Keep-Alive Requested
TDEF HTTP Proxy Connection Statistics	tdef_http_proxy_persistent_client_connections	Counter	Client Connections - Persistent

Group	Metric Name	Type	Synopsis
TDEF HTTP Proxy Connection Statistics	tdef_http_proxy_reused_client_connections	Counter	Client Connections - Reused
TDEF HTTP Proxy Connection Statistics	tdef_http_proxy_pipelined_client_connections	Counter	Client Connections - Pipelined
TDEF HTTP Proxy Connection Statistics	tdef_http_proxy_ipv4_client_connections	Counter	Client Connections - IPv4
TDEF HTTP Proxy Connection Statistics	tdef_http_proxy_ipv6_client_connections	Counter	Client Connections - IPv6
TDEF HTTP Proxy Connection Statistics	tdef_http_proxy_non_http_client_connections	Counter	Client Connections - Non-HTTP Message Received
TDEF HTTP Proxy Connection Statistics	tdef_http_proxy_no_request_client_connections	Counter	Client Connections - Request Receive Timeouts
TDEF HTTP Proxy Connection Statistics	tdef_http_proxy_reverse_handoff_client_connections	Counter	Client Connections - Reverse Handoffs
TDEF HTTP Proxy Connection Statistics	tdef_http_proxy_handoff_client_connections	Counter	Client Connections - Handoffs
TDEF HTTP Proxy Connection Statistics	tdef_http_proxy_failures_client_connections	Counter	Client Connections - Failures
TDEF HTTP Proxy Connection Statistics	tdef_http_proxy_idle_timeout_updated_client_connections	Counter	Client Connections - Idle timeout updated
TDEF HTTP Proxy Connection Statistics	tdef_http_proxy_total_server_connections	Counter	Server Connections - Total
TDEF HTTP Proxy Connection Statistics	tdef_http_proxy_ipv4_server_connections	Counter	Server Connections - IPv4
TDEF HTTP Proxy Connection Statistics	tdef_http_proxy_ipv6_server_connections	Counter	Server Connections - IPv6
TDEF HTTP Proxy Connection Statistics	tdef_http_proxy_reused_same_client_server_connections	Counter	Server Connections - Reused by Same Client
TDEF HTTP Proxy Connection Statistics	tdef_http_proxy_reused_server_connections	Counter	Server Connections - Reused from Pool
TDEF HTTP Proxy Connection Statistics	tdef_http_proxy_prefip_family_used_server_connections	Counter	Server Connections - Preferred IP Family Used
TDEF HTTP Proxy Connection Statistics	tdef_http_proxy_prefip_family_unavailable_server_connections	Counter	Server Connections - Preferred IP Family Not Available
TDEF HTTP Proxy Connection Statistics	tdef_http_proxy_ip_transparent_server_connections	Counter	Server Connections - IP Transparent
TDEF HTTP Proxy Connection Statistics	tdef_http_proxy_idle_timeout_updated_server_connections	Counter	Server Connections - Idle timeout updated
TDEF HTTP Proxy Connection Statistics	tdef_http_proxy_fastopen_attempts	Counter	Fast Open - Attempts
TDEF HTTP Proxy Connection Statistics	tdef_http_proxy_fastopen_successes	Counter	Fast Open - Successes
TDEF HTTP Proxy Connection Statistics	tdef_http_proxy_fastopen_failure_init	Counter	Fast Open - Connection Initialization Failures
TDEF HTTP Proxy Connection Statistics	tdef_http_proxy_fastopen_failure_duplicate	Counter	Fast Open - Connection Already in Pending List
TDEF HTTP Proxy Connection Statistics	tdef_http_proxy_fastopen_failure_already_in_session	Counter	Fast Open - Connection Already in Session
TDEF HTTP Proxy Connection Statistics	tdef_http_proxy_fastopen_failure_policy	Counter	Fast Open - Policy Denied
TDEF HTTP Proxy Connection Statistics	tdef_http_proxy_fastopen_failure_other	Counter	Fast Open - Not Ready
TDEF HTTP Proxy Connection Statistics	tdef_http_proxy_fastopen_pending_list_full	Counter	Fast Open - Pending Connection List Full
TDEF HTTP Proxy Connection Statistics	tdef_http_proxy_fastopen_ageout	Counter	Fast Open - Pending Connection Idle Timeout
TDEF HTTP Proxy Connection Statistics	tdef_http_proxy_current_client_connections	Gauge	Client Connections - Currently in Use
TDEF HTTP Proxy Connection Statistics	tdef_http_proxy_current_server_connections	Gauge	Server Connections - Currently in Use
TDEF HTTP Proxy Connection Statistics	tdef_http_proxy_currently_idle_server_connection	Gauge	Server Connections - Currently Idle
TDEF HTTP Proxy Connection Statistics	tdef_http_proxy_fastopen_current_pending_connection	Gauge	Fast Open - Current Pending Connections
TDEF HTTP Proxy Compression Statistics	tdef_http_proxy_compression_rsp_not_compressed_by_server	Counter	Compression - Responses Not Compressed by Server
TDEF HTTP Proxy Compression Statistics	tdef_http_proxy_compression_rsp_compressed_by_server	Counter	Compression - Responses Compressed by Server
TDEF HTTP Proxy Compression Statistics	tdef_http_proxy_compression_rsp_compressed_by_proxy	Counter	Compression - Responses Compressed by Proxy
TDEF HTTP Proxy Compression Statistics	tdef_http_proxy_compression_rsp_force_compressed_by_proxy	Counter	Compression - Responses Forcefully Compressed by Proxy

Group	Metric Name	Type	Synopsis
TDEF HTTP Proxy Compression Statistics	tdef_http_proxy_compression_bytes_not_compressed_by_server	Counter	Compression - Bytes Not Compressed by Server
TDEF HTTP Proxy Compression Statistics	tdef_http_proxy_compression_bytes_compressed_by_server	Counter	Compression - Bytes Compressed by Server
TDEF HTTP Proxy Compression Statistics	tdef_http_proxy_compression_bytes_compressed_by_proxy	Counter	Compression - Bytes Compressed by Proxy
TDEF HTTP Proxy Compression Statistics	tdef_http_proxy_compression_bytes_force_compressed_by_proxy	Counter	Compression - Bytes Forcefully Compressed by Proxy
TDEF HTTP Proxy Compression Statistics	tdef_http_proxy_compression_cancel_content_type_mismatch	Counter	Compression Cancel - Content-Type Mismatches
TDEF HTTP Proxy URL Rewrite Statistics	tdef_http_proxy_url_rewrite_evaluated	Counter	URL-Rewrite - Evaluated
TDEF HTTP Proxy URL Rewrite Statistics	tdef_http_proxy_url_rewrite_attempts	Counter	URL-Rewrite - Attempts
TDEF HTTP Proxy URL Rewrite Statistics	tdef_http_proxy_url_rewrite_success	Counter	URL-Rewrite - Successes
TDEF HTTP Proxy URL Rewrite Statistics	tdef_http_proxy_url_rewrite_dnslkup_v4attempts	Counter	URL-Rewrite - DNS IPv4 Attempts
TDEF HTTP Proxy URL Rewrite Statistics	tdef_http_proxy_url_rewrite_dnslkup_v6attempts	Counter	URL-Rewrite - DNS IPv6 Attempts
TDEF HTTP Proxy URL Rewrite Statistics	tdef_http_proxy_url_rewrite_dnslkup_v4v6attempts	Counter	URL-Rewrite - DNS IPV4V6 Attempts
TDEF HTTP Proxy URL Rewrite Statistics	tdef_http_proxy_url_rewrite_dnslkup_v4success	Counter	URL-Rewrite - DNS IPv4 Successes
TDEF HTTP Proxy URL Rewrite Statistics	tdef_http_proxy_url_rewrite_dnslkup_v6success	Counter	URL-Rewrite - DNS IPv6 Successes
TDEF HTTP Proxy URL Rewrite Statistics	tdef_http_proxy_url_rewrite_dnslkup_v4v6success	Counter	URL-Rewrite - DNS IPV4V6 Successes
TDEF HTTP Proxy Header Enrichment Statistics	tdef_http_proxy_header_enrichment_supi	Counter	HTTP Header Enrichment - SUPI
TDEF HTTP Proxy Header Enrichment Statistics	tdef_http_proxy_header_enrichment_imei	Counter	HTTP Header Enrichment - IMEI
TDEF HTTP Proxy Header Enrichment Statistics	tdef_http_proxy_header_enrichment_gpsi	Counter	HTTP Header Enrichment - GPSI
TDEF HTTP Proxy Header Enrichment Statistics	tdef_http_proxy_header_enrichment_plmid	Counter	HTTP Header Enrichment - PLMID
TDEF HTTP Proxy Header Enrichment Statistics	tdef_http_proxy_header_enrichment_http_version	Counter	HTTP Header Enrichment - HTTP Version
TDEF HTTP Proxy Header Enrichment Statistics	tdef_http_proxy_header_enrichment_ue_ip_address	Counter	HTTP Header Enrichment - UE IP address
TDEF HTTP Proxy Header Enrichment Statistics	tdef_http_proxy_header_enrichment_ue_timezone	Counter	HTTP Header Enrichment - UE TimeZone
TDEF HTTP Proxy Header Enrichment Statistics	tdef_http_proxy_header_enrichment_gw_ip_address	Counter	HTTP Header Enrichment - GW IP Address
TDEF HTTP Proxy Header Enrichment Statistics	tdef_http_proxy_header_enrichment_gw_ipv6_address	Counter	HTTP Header Enrichment - GW IPv6 Address
TDEF HTTP Proxy Header Enrichment Statistics	tdef_http_proxy_header_enrichment_subscriber_id	Counter	HTTP Header Enrichment - Subscriber ID
TDEF HTTP Proxy Header Enrichment Statistics	tdef_http_proxy_header_enrichment_node_name	Counter	HTTP Header Enrichment - Node Name
TDEF HTTP Proxy Header Enrichment Statistics	tdef_http_proxy_header_enrichment_other	Counter	HTTP Header Enrichment - Other
TDEF HTTP Proxy Header Enrichment Statistics	tdef_http_proxy_header_enrichment_supi_encrypted	Counter	HTTP Header Enrichment - SUPI Encrypted
TDEF HTTP Proxy Header Enrichment Statistics	tdef_http_proxy_header_enrichment_imei_encrypted	Counter	HTTP Header Enrichment - IMEI Encrypted
TDEF HTTP Proxy Header Enrichment Statistics	tdef_http_proxy_header_enrichment_gpsi_encrypted	Counter	HTTP Header Enrichment - GPSI Encrypted
TDEF HTTP Proxy Header Enrichment Statistics	tdef_http_proxy_header_enrichment_plmid_encrypted	Counter	HTTP Header Enrichment - PLMID Encrypted

Group	Metric Name	Type	Synopsis
TDEF HTTP Proxy Header Enrichment Statistics	tdef_http_proxy_header_enrichment_http_version_encrypted	Counter	HTTP Header Enrichment - HTTP Version Encrypted
TDEF HTTP Proxy Header Enrichment Statistics	tdef_http_proxy_header_enrichment_ue_ip_address_encrypted	Counter	HTTP Header Enrichment - UE IP address Encrypted
TDEF HTTP Proxy Header Enrichment Statistics	tdef_http_proxy_header_enrichment_ue_timezone_encrypted	Counter	HTTP Header Enrichment - UE TimeZone Encrypted
TDEF HTTP Proxy Header Enrichment Statistics	tdef_http_proxy_header_enrichment_gw_ip_address_encrypted	Counter	HTTP Header Enrichment - GW IP Address Encrypted
TDEF HTTP Proxy Header Enrichment Statistics	tdef_http_proxy_header_enrichment_gw_ipv6_address_encrypted	Counter	HTTP Header Enrichment - GW IPv6 Address Encrypted
TDEF HTTP Proxy Header Enrichment Statistics	tdef_http_proxy_header_enrichment_subscriber_id_encrypted	Counter	HTTP Header Enrichment - Subscriber ID Encrypted
TDEF HTTP Proxy Header Enrichment Statistics	tdef_http_proxy_header_enrichment_node_name_encrypted	Counter	HTTP Header Enrichment - Node Name Encrypted
TDEF HTTP Proxy Header Enrichment Statistics	tdef_http_proxy_header_enrichment_other_encrypted	Counter	HTTP Header Enrichment - Other Encrypted
TDEF HTTP Proxy Header Enrichment Statistics	tdef_http_proxy_header_enrichment_vsa_reseller_id	Counter	HTTP Header Enrichment - vsa_reseller_id
TDEF HTTP Proxy Header Enrichment Statistics	tdef_http_proxy_header_enrichment_vsa_subclass_id	Counter	HTTP Header Enrichment - vsa_subclass_id
TDEF HTTP Proxy Header Enrichment Statistics	tdef_http_proxy_header_enrichment_rat_type	Counter	HTTP Header Enrichment - rat_type
TDEF HTTP Proxy Header Enrichment Statistics	tdef_http_proxy_header_enrichment_pei	Counter	HTTP Header Enrichment - pei
TDEF HTTP Proxy Header Enrichment Statistics	tdef_http_proxy_header_enrichment_incoming_dnn	Counter	HTTP Header Enrichment - incoming_dnn
TDEF HTTP Proxy Header Enrichment Statistics	tdef_http_proxy_header_enrichment_current_dnn	Counter	HTTP Header Enrichment - current_dnn
TDEF HTTP Proxy Header Enrichment Statistics	tdef_http_proxy_header_enrichment_matched_dnn	Counter	HTTP Header Enrichment - matched_dnn
TDEF HTTP Proxy Header Enrichment Statistics	tdef_http_proxy_header_enrichment_vsa_reseller_id_encrypted	Counter	HTTP Header Enrichment - vsa_reseller_id Encrypted
TDEF HTTP Proxy Header Enrichment Statistics	tdef_http_proxy_header_enrichment_vsa_subclass_id_encrypted	Counter	HTTP Header Enrichment - vsa_subclass_id Encrypted
TDEF HTTP Proxy Header Enrichment Statistics	tdef_http_proxy_header_enrichment_rat_type_encrypted	Counter	HTTP Header Enrichment - rat_type Encrypted
TDEF HTTP Proxy Header Enrichment Statistics	tdef_http_proxy_header_enrichment_pei_encrypted	Counter	HTTP Header Enrichment - pei Encrypted
TDEF HTTP Proxy Header Enrichment Statistics	tdef_http_proxy_header_enrichment_incoming_dnn_encrypted	Counter	HTTP Header Enrichment - incoming_dnn Encrypted
TDEF HTTP Proxy Header Enrichment Statistics	tdef_http_proxy_header_enrichment_current_dnn_encrypted	Counter	HTTP Header Enrichment - current_dnn Encrypted
TDEF HTTP Proxy Header Enrichment Statistics	tdef_http_proxy_header_enrichment_matched_dnn_encrypted	Counter	HTTP Header Enrichment - matched_dnn Encrypted

Summary of TDEF HTTP Proxy Statistics

- tdef_http_proxy_service_attempts

Type

64-bit Counter

Description

Total number of proxy service attempts

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_service_success

Type

64-bit Counter

Description

Total number of proxy service success

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_service_aborted_by_client

Type

64-bit Counter

Description

Total number of proxy service aborted

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_service_failed

Type

64-bit Counter

Description

Total number of proxy service failed

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_client_requests_total

Type

64-bit Counter

Description

Total number of client requests received

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_client_requests_parsed_successfully

Type

64-bit Counter

Description

Total number of client requests parsed successfully

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_client_requests_aborted_by_client

Type

64-bit Counter

Description

Total number of incomplete client requests

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_client_requests_processing_failed

Type

64-bit Counter

Description

Total number of client requests processing failed

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_client_requests_pipeline_dropped

Type

64-bit Counter

Description

Total number of client requests pipeline dropped

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_client_requests_invalid

Type

64-bit Counter

Description

Total number of invalid client requests

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_client_request_bytes_total

Type

64-bit Counter

Description

Total number of Bytes Client Request

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_client_response_bytes_total

Type

64-bit Counter

Description

Total number of Bytes Client Response

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_server_request_bytes_total

Type

64-bit Counter

Description

Total number of Bytes Server Request

Labels

pod_id, proxy_id, proxy_service, table
- tdef_http_proxy_server_response_bytes_total

Type

64-bit Counter

Description

Total number of Bytes Server Response

Labels

pod_id, proxy_id, proxy_service, table
- tdef_http_proxy_server_response_total

Type

64-bit Counter

Description

Total number of HTTP server responses

Labels

pod_id, proxy_id, proxy_service, table
- tdef_http_proxy_transaction_edr_creates

Type

64-bit Counter

Description

Total number of HTTP transaction EDR creates

Labels

pod_id, proxy_id, proxy_service, table
- tdef_http_proxy_transaction_edr_attempts

Type

64-bit Counter

Description

Total number of HTTP transaction EDR attempts

Labels

pod_id, proxy_id, proxy_service, table
- tdef_http_proxy_transaction_edr_successes

Type

64-bit Counter

Description

Total number of HTTP transaction EDR successes

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_transaction_edr_failures

Type

64-bit Counter

Description

Total number of HTTP transaction EDR failures

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_client_http_service_time_min

Type

64-bit Gauge

Description

Client HTTP Service Time Min (ms)

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_client_http_service_time_avg

Type

64-bit Gauge

Description

Client HTTP Service Time Avg (ms)

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_client_http_service_time_max

Type

64-bit Gauge

Description

Client HTTP Service Time Max (ms)

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_client_requests_currently_in_service

Type

64-bit Gauge

Description

Total number of client requests currently in service

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_client_session_current

Type

64-bit Gauge

Description

Total number of current Proxy Sessions

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_http_response_time_min

Type

64-bit Gauge

Description

Server HTTP Response Time Min (ms)

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_http_response_time_avg

Type

64-bit Gauge

Description

Server HTTP Response Time Avg (ms)

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_http_response_time_max

Type

64-bit Gauge

Description

Server HTTP Response Time Max (ms)

Labels

pod_id, proxy_id, proxy_service, table

TDEF HTTP Proxy Detail Statistics

- tdef_http_proxy_client_requests_http_v09

Type

64-bit Counter

Description

Total number of Http V0.9 client requests received

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_client_requests_http_v10

Type

64-bit Counter

Description

Total number of Http V1.0 client requests received

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_client_requests_http_v11

Type

64-bit Counter

Description

Total number of Http V1.1 client requests received

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_persistent_client_requests

Type

64-bit Counter

Description

Total number of client requests persistent

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_pipelined_client_requests

Type

64-bit Counter

Description

Total number of pipelined client requests

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_ipv4_client_requests

Type

64-bit Counter

Description

Total number of IPv4 client requests

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_ipv6_client_requests

Type

64-bit Counter

Description

Total number of IPv6 client requests

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_transparent_proxy_requests

Type

64-bit Counter

Description

Total number of client requests destined to transparent proxy

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_explicit_proxy_requests

Type

64-bit Counter

Description

Total number of client requests destined to an explicit proxy

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_client_http_response_status_information

Type

64-bit Counter

Description

Client HTTP Response Status Information (1xx)

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_client_http_response_status_success_ok

Type

64-bit Counter

Description

Client HTTP Response Status Success OK (200)

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_client_http_response_status_success_other

Type

64-bit Counter

Description

Client HTTP Response Status Success Other (2xx)

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_client_http_response_status_redirection

Type

64-bit Counter

Description

Client HTTP Response Status Redirection (3xx)

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_client_http_response_status_client_error

Type

64-bit Counter

Description

Client HTTP Response Status Client Error (4xx)

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_client_http_response_status_server_error

Type

64-bit Counter

Description

Client HTTP Response Status Server Error (5xx)

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_server_http_responses_ipv4

Type

64-bit Counter

Description

Total number of IPv4 server responses

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_server_http_responses_ipv6

Type

64-bit Counter

Description

Total number of IPv6 server responses

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_dpi_update_failures

Type

64-bit Counter

Description

Total number of DPI update failures

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_dpi_update_for_deleted_request

Type

64-bit Counter

Description

Total number of DPI updates for deleted requests

Labels

pod_id, proxy_id, proxy_service, table

• tdef_http_proxy_dpi_updates

Type

64-bit Counter

Description

Total number of DPI updates

Labels

pod_id, proxy_id, proxy_service, table

• tdef_http_proxy_transaction_edr_callback_successes

Type

64-bit Counter

Description

Total number of transaction EDR callback successes

Labels

pod_id, proxy_id, proxy_service, table

• tdef_http_proxy_transaction_edr_callback_failures

Type

64-bit Counter

Description

Total number of transaction EDR callback failures

Labels

pod_id, proxy_id, proxy_service, table

• tdef_http_proxy_client_http_request_size_min

Type

64-bit Gauge

Description

Client HTTP Request Size Min (Bytes)

Labels

pod_id, proxy_id, proxy_service, table

• tdef_http_proxy_client_http_request_size_avg

Type

64-bit Gauge

Description

Client HTTP Request Size Avg (Bytes)

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_client_http_request_size_max

Type

64-bit Gauge

Description

Client HTTP Request Size Max (Bytes)

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_client_http_response_size_min

Type

64-bit Gauge

Description

Client HTTP Response Size Min (Bytes)

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_client_http_response_size_avg

Type

64-bit Gauge

Description

Client HTTP Response Size Avg (Bytes)

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_client_http_response_size_max

Type

64-bit Gauge

Description

Client HTTP Response Size Max (Bytes)

Labels

pod_id, proxy_id, proxy_service, table

TDEF HTTP Proxy Failure Statistics

- tdef_http_proxy_service_failed_request_compile_error

Type

64-bit Counter

Description

Total number of proxy service failed - Request Compile Error

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_service_failed_connection_init_failure

Type

64-bit Counter

Description

Total number of proxy service failed - Connection Init Failure

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_service_failed_over_sized_response_header

Type

64-bit Counter

Description

Total number of proxy service failed- Oversized Response Header

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_service_failed_bad_response_status_line

Type

64-bit Counter

Description

Total number of proxy service failed - Bad Response Status Line

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_service_failed_bad_response_header

Type

64-bit Counter

Description

Total number of proxy service failed - Bad Response Header

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_service_failed_response_processing_failure

Type

64-bit Counter

Description

Total number of proxy service failed - Response Processing Failure

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_service_failed_server_connection_close

Type

64-bit Counter

Description

Total number of proxy service failed - Server Connection Close

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_service_failed_client_connection_timeout

Type

64-bit Counter

Description

Total number of proxy service failed - Client Connection Timeout

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_service_failed_resource_unavailable

Type

64-bit Counter

Description

Total number of proxy service failed - Resources unavailable

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_service_redirected

Type

64-bit Counter

Description

Total number of proxy service failed - Requests redirected

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_service_blocked

Type

64-bit Counter

Description

Total number of proxy service failed - Requests blocked

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_service_failed_maintenance_connection_close

Type

64-bit Counter

Description

Total number of proxy service failed - Maintenance Connection Close

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_service_out_of_buffer

Type

64-bit Counter

Description

Total number of proxy service failed - Out of buffer

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_service_failed_server_connection_timeout

Type

64-bit Counter

Description

Total number of proxy service failed - Server Connection Timeout

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_service_aborted_by_client

Type

64-bit Counter

Description

Total number of proxy service aborted

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_service_aborted_no_resp_from_server

Type

64-bit Counter

Description

Total number of proxy service aborted - No Response from Server

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_service_aborted_request_to_local_addr

Type

64-bit Counter

Description

Total number of proxy service aborted - Request to Local Address

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_connection_failures_overload

Type

64-bit Counter

Description

Total number of client connection failures - Terminated due to Overload

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_connection_failures_invalid_metadata

Type

64-bit Counter

Description

Total number of client connection failures - Invalid Metadata

Labels

pod_id, proxy_id, proxy_service, table
- tdef_http_proxy_connection_failures_proxy_is_not_ready

Type

64-bit Counter

Description

Total number of client connection failures - Proxy is not Ready

Labels

pod_id, proxy_id, proxy_service, table
- tdef_http_proxy_connection_failures_fd_limit_exceeded

Type

64-bit Counter

Description

Total number of client connection failures - FD limit exceeded

Labels

pod_id, proxy_id, proxy_service, table
- tdef_http_proxy_connection_failures_out_of_resources

Type

64-bit Counter

Description

Total number of client connection failures - Out of Resources

Labels

pod_id, proxy_id, proxy_service, table

TDEF HTTP Proxy Connection Statistics

- tdef_http_proxy_total_client_connections

Type

64-bit Counter

Description

Total number of client connections

Labels

pod_id, proxy_id, proxy_service, table
- tdef_http_proxy_keep_alive_requested

Type

64-bit Counter

Description

Total number of keep-alive requested from client

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_persistent_client_connections

Type

64-bit Counter

Description

Total number of persistent client connections

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_reused_client_connections

Type

64-bit Counter

Description

Total number of reused client connections

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_pipelined_client_connections

Type

64-bit Counter

Description

Total number of pipelined client connections

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_ipv4_client_connections

Type

64-bit Counter

Description

Total number of IPv4 client connections

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_ipv6_client_connections

Type

64-bit Counter

Description

Total number of IPv6 client connections

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_non_http_client_connections

Type

64-bit Counter

Description

Total number of client connections with non-http request

Labels

pod_id, proxy_id, proxy_service, table
- tdef_http_proxy_no_request_client_connnections

Type

64-bit Counter

Description

Total number of client connections with request receive timeout

Labels

pod_id, proxy_id, proxy_service, table
- tdef_http_proxy_reverse_handoff_client_connnections

Type

64-bit Counter

Description

Total number of client connections from reverse handoff

Labels

pod_id, proxy_id, proxy_service, table
- tdef_http_proxy_handoff_client_connnections

Type

64-bit Counter

Description

Total number of client connections handoffs

Labels

pod_id, proxy_id, proxy_service, table
- tdef_http_proxy_failures_client_connnections

Type

64-bit Counter

Description

Total number of client connection failures

Labels

pod_id, proxy_id, proxy_service, table
- tdef_http_proxy_idle_timeout_updated_client_connections

Type

64-bit Counter

Description

Total number of client connection idle timeout updated from PPE

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_total_server_connections

Type

64-bit Counter

Description

Total number of server connections

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_ipv4_server_connections

Type

64-bit Counter

Description

Total number of IPv4 server connections

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_ipv6_server_connections

Type

64-bit Counter

Description

Total number of IPv6 server connections

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_reused_same_client_server_connections

Type

64-bit Counter

Description

The number of times server connections were reused by same client connections

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_reused_server_connections

Type

64-bit Counter

Description

The number of times idle server connections were reused

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_prefip_family_used_server_connections

Type
64-bit Counter
Description
Total number of preferred ip family used server connections
Labels
<i>pod_id, proxy_id, proxy_service, table</i>
• tdef_http_proxy_prefip_family_unavailable_server_connections
Type
64-bit Counter
Description
Total number of preferred ip family unavailable server connections
Labels
<i>pod_id, proxy_id, proxy_service, table</i>
• tdef_http_proxy_ip_transparent_server_connections
Type
64-bit Counter
Description
Total number of IP transparent server connections
Labels
<i>pod_id, proxy_id, proxy_service, table</i>
• tdef_http_proxy_idle_timeout_updated_server_connections
Type
64-bit Counter
Description
Total number of server connection idle timeout updated from PPE
Labels
<i>pod_id, proxy_id, proxy_service, table</i>
• tdef_http_proxy_fastopen_attempts
Type
64-bit Counter
Description
Total number of HTTP server fast open connection attempts
Labels
<i>pod_id, proxy_id, proxy_service, table</i>
• tdef_http_proxy_fastopen_successes
Type
64-bit Counter
Description
Total number of HTTP server fast open connection successes

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_fastopen_failure_init

Type

64-bit Counter

Description

Total number of HTTP server fast open initialization failures

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_fastopen_failure_duplicate

Type

64-bit Counter

Description

Total number of HTTP server fast open duplicated connections

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_fastopen_failure_already_in_session

Type

64-bit Counter

Description

Total number of HTTP server fast open connections already in the session

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_fastopen_failure_policy

Type

64-bit Counter

Description

Total number of HTTP server fast open connections rejected by policy

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_fastopen_failure_other

Type

64-bit Counter

Description

Total number of HTTP server fast open connections other failures

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_fastopen_pending_list_full

Type

64-bit Counter

Description

Total number of HTTP server fast open connections pending list full

Labels

pod_id, proxy_id, proxy_service, table

• tdef_http_proxy_fastopen_ageout

Type

64-bit Counter

Description

Total number of HTTP server fast open connections closed due to aging

Labels

pod_id, proxy_id, proxy_service, table

• tdef_http_proxy_current_client_connections

Type

64-bit Gauge

Description

Total number of client connections currently in use

Labels

pod_id, proxy_id, proxy_service, table

• tdef_http_proxy_current_server_connections

Type

64-bit Gauge

Description

Total number of server connections currently in use

Labels

pod_id, proxy_id, proxy_service, table

• tdef_http_proxy_currently_idle_server_connection

Type

64-bit Gauge

Description

The number of idle server connections in the server connection pool

Labels

pod_id, proxy_id, proxy_service, table

• tdef_http_proxy_fastopen_current_pending_connection

Type

64-bit Gauge

Description

Total number of HTTP server fast open connections in pending list

Labels

pod_id, proxy_id, proxy_service, table

TDEF HTTP Proxy Compression Statistics

- tdef_http_proxy_compression_rsp_not_compressed_by_server

Type

64-bit Counter

Description

Total number of responses that were not compressed by the server

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_compression_rsp_compressed_by_server

Type

64-bit Counter

Description

Total number of responses that were compressed by the server

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_compression_rsp_compressed_by_proxy

Type

64-bit Counter

Description

Total number of responses that were compressed by proxy

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_compression_rsp_force_compressed_by_proxy

Type

64-bit Counter

Description

Total number of responses that were forced to compress by proxy

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_compression_bytes_not_compressed_by_server

Type

64-bit Counter

Description

Total number of bytes that were not compressed by the server

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_compression_bytes_compressed_by_server

Type

64-bit Counter

Description

Total number of bytes that were compressed by the server

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_compression_bytes_compressed_by_proxy

Type

64-bit Counter

Description

Total number of bytes that were compressed by proxy

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_compression_bytes_force_compressed_by_proxy

Type

64-bit Counter

Description

Total number of bytes that were forced to compress by proxy

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_compression_cancel_content_type_mismatch

Type

64-bit Counter

Description

Total number of responses that were cancel because content-type mismatches

Labels

pod_id, proxy_id, proxy_service, table

TDEF HTTP Proxy URL Rewrite Statistics

- tdef_http_proxy_url_rewrite_evaluated

Type

64-bit Counter

Description

Total number of URL Rewrite Evaluated

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_url_rewrite_attempts

Type

64-bit Counter

Description

Total number of URL Rewrite Attempts

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_url_rewrite_success

Type

64-bit Counter

Description

Total number of URL Rewrite Successes

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_url_rewrite_dnslkup_v4attempts

Type

64-bit Counter

Description

Total number of URL Rewrite DNS Lookup V4 Attempts

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_url_rewrite_dnslkup_v6attempts

Type

64-bit Counter

Description

Total number of URL Rewrite DNS Lookup V6 Attempts

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_url_rewrite_dnslkup_v4v6attempts

Type

64-bit Counter

Description

Total number of URL Rewrite DNS Lookup V4V6 Attempts

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_url_rewrite_dnslkup_v4success

Type

64-bit Counter

Description

Total number of URL Rewrite DNS Lookup V4 Successes

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_url_rewrite_dnslkup_v6success

Type

64-bit Counter

Description

Total number of URL Rewrite DNS Lookup V6 Successes

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_url_rewrite_dnslkup_v4v6success

Type

64-bit Counter

Description

Total number of URL Rewrite DNS Lookup V4V6 Successes

Labels

pod_id, proxy_id, proxy_service, table

TDEF HTTP Proxy Header Enrichment Statistics

- tdef_http_proxy_header_enrichment_supi

Type

64-bit Counter

Description

Total number of HTTP Header Enrichment - SUPI

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_header_enrichment_imei

Type

64-bit Counter

Description

Total number of HTTP Header Enrichment - IMEI

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_header_enrichment_gps

Type

64-bit Counter

Description

Total number of HTTP Header Enrichment - GPSI

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_header_enrichment_plmid

Type

64-bit Counter

Description

Total number of HTTP Header Enrichment - PLMID

Labels
<i>pod_id, proxy_id, proxy_service, table</i>
tdef_http_proxy_header_enrichment_http_version
Type
64-bit Counter
Description
Total number of HTTP Header Enrichment - HTTP Version
Labels
<i>pod_id, proxy_id, proxy_service, table</i>
tdef_http_proxy_header_enrichment_ue_ip_address
Type
64-bit Counter
Description
Total number of HTTP Header Enrichment - UE IP Address
Labels
<i>pod_id, proxy_id, proxy_service, table</i>
tdef_http_proxy_header_enrichment_ue_timezone
Type
64-bit Counter
Description
Total number of HTTP Header Enrichment - UE TimeZone
Labels
<i>pod_id, proxy_id, proxy_service, table</i>
tdef_http_proxy_header_enrichment_gw_ip_address
Type
64-bit Counter
Description
Total number of HTTP Header Enrichment - GW IP Address
Labels
<i>pod_id, proxy_id, proxy_service, table</i>
tdef_http_proxy_header_enrichment_gw_ipv6_address
Type
64-bit Counter
Description
Total number of HTTP Header Enrichment - GW IPv6 Address
Labels
<i>pod_id, proxy_id, proxy_service, table</i>
tdef_http_proxy_header_enrichment_subscriber_id
Type

64-bit Counter

Description

Total number of HTTP Header Enrichment - Subscriber ID

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_header_enrichment_node_name

Type

64-bit Counter

Description

Total number of HTTP Header Enrichment - Node Name

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_header_enrichment_other

Type

64-bit Counter

Description

Total number of HTTP Header Enrichment - Other

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_header_enrichment_supi_encrypted

Type

64-bit Counter

Description

Total number of HTTP Header Enrichment - SUPI Encrypted

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_header_enrichment_imei_encrypted

Type

64-bit Counter

Description

Total number of HTTP Header Enrichment - IMEI Encrypted

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_header_enrichment_gpsi_encrypted

Type

64-bit Counter

Description

Total number of HTTP Header Enrichment - GPSI Encrypted

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_header_enrichment_plmid_encrypted

Type

64-bit Counter

Description

Total number of HTTP Header Enrichment - PLMID Encrypted

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_header_enrichment_http_version_encrypted

Type

64-bit Counter

Description

Total number of HTTP Header Enrichment - HTTP Version Encrypted

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_header_enrichment_ue_ip_address_encrypted

Type

64-bit Counter

Description

Total number of HTTP Header Enrichment - UE IP Address Encrypted

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_header_enrichment_ue_timezone_encrypted

Type

64-bit Counter

Description

Total number of HTTP Header Enrichment - UE TimeZone Encrypted

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_header_enrichment_gw_ip_address_encrypted

Type

64-bit Counter

Description

Total number of HTTP Header Enrichment - GW IP Address Encrypted

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_header_enrichment_gw_ipv6_address_encrypted

Type

64-bit Counter

Description

Total number of HTTP Header Enrichment - GW IPv6 Address Encrypted

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_header_enrichment_subscriber_id_encrypted

Type

64-bit Counter

Description

Total number of HTTP Header Enrichment - Subscriber ID Encrypted

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_header_enrichment_node_name_encrypted

Type

64-bit Counter

Description

Total number of HTTP Header Enrichment - Node Name Encrypted

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_header_enrichment_other_encrypted

Type

64-bit Counter

Description

Total number of HTTP Header Enrichment - Other Encrypted

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_header_enrichment_vsa_reseller_id

Type

64-bit Counter

Description

Total number of HTTP Header Enrichment - vsa_reseller_id

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_header_enrichment_vsa_subclass_id

Type

64-bit Counter

Description

Total number of HTTP Header Enrichment - vsa_subclass_id

Labels

pod_id, proxy_id, proxy_service, table

Type

Description

Labels

- tdef_http_proxy_header_enrichment_pei

Type

Description

Labels

- tdef_http_proxy_header_enrichment_incoming_dnn

Type

Description

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_header_enrichment_current_dnn

Type

Description

Total number of HTTP Header Enrichment - current_dnn

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_header_enrichment_matched_dnn

Type

64-bit Counter

Description

Total number of HTTP Header Enrichment - matched_dnn

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_header_enrichment_vsa_reseller_id_encrypted

Type

64-bit Counter

Description

Total number of HTTP Header Enrichment - vsa_reseller_id Encrypted

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_header_enrichment_vsa_subclass_id_encrypted

Type

64-bit Counter

Description

Total number of HTTP Header Enrichment - vsa_subclass_id Encrypted

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_header_enrichment_rat_type_encrypted

Type

64-bit Counter

Description

Total number of HTTP Header Enrichment - rat_type Encrypted

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_header_enrichment_pei_encrypted

Type

64-bit Counter

Description

Total number of HTTP Header Enrichment - pei Encrypted

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_header_enrichment_incoming_dnn_encrypted

Type

64-bit Counter

Description

Total number of HTTP Header Enrichment - incoming_dnn Encrypted

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_header_enrichment_current_dnn_encrypted

Type

64-bit Counter

Description

Total number of HTTP Header Enrichment - current_dnn Encrypted

Labels

pod_id, proxy_id, proxy_service, table

- tdef_http_proxy_header_enrichment_matched_dnn_encrypted

Type

64-bit Counter

Description

Total number of HTTP Header Enrichment - matched_dnn Encrypted

Labels

pod_id, proxy_id, proxy_service, table

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- [Summary](#)
- [TDEF Proxy Packet Statistics](#)

Summary

Group	Metric Name	Type	Synopsis
TDEF Proxy Packet Statistics	tdef_proxy_rx_packets	Gauge	# of RX packets
TDEF Proxy Packet Statistics	tdef_proxy_tx_packets	Gauge	# of TX packet
TDEF Proxy Packet Statistics	tdef_proxy_tx_packet_drop_buffer_depletion	Gauge	# of TX packet drops due to buffer depletion
TDEF Proxy Packet Statistics	tdef_proxy_tx_packet_drop_copy_data_error	Gauge	# of TX packet drops due to data copy error
TDEF Proxy Packet Statistics	tdef_proxy_rx_packet_drop_buffer_depletion	Gauge	# of RX packet drops due to buffer depletion
TDEF Proxy Packet Statistics	tdef_proxy_rx_packet_drop_invalid_checksum	Gauge	# of RX packet drops due to invalid checksum

TDEF Proxy Packet Statistics

- [tdef_proxy_rx_packets](#)

Type

64-bit Gauge

Description

The number of RX packets

Labels

pod_id, proxy_id, proxy_service, table

- [tdef_proxy_tx_packets](#)

Type

64-bit Gauge

Description

The number of TX packets

Labels

pod_id, proxy_id, proxy_service, table

- [tdef_proxy_tx_packet_drop_buffer_depletion](#)

Type

64-bit Gauge

Description

The number of TX packet drops caused by packet buffer depletion

Labels

pod_id, proxy_id, proxy_service, table

- [tdef_proxy_tx_packet_drop_copy_data_error](#)

Type

64-bit Gauge

Description

The number of TX packet drops caused by data copy error

Labels

pod_id, proxy_id, proxy_service, table

- [tdef_proxy_rx_packet_drop_buffer_depletion](#)

Type

64-bit Gauge

Description

The number of RX packet drops caused by packet buffer depletion

Labels

pod_id, proxy_id, proxy_service, table

- [tdef_proxy_rx_packet_drop_invalid_checksum](#)

Type

64-bit Gauge

Description

The number of TX packet drops caused by failed packet checksum check

Labels

pod_id, proxy_id, proxy_service, table

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- [Summary](#)
- [HTTP Server Plugin Summary Statistics](#)

Summary

Group	Metric Name	Type	Synopsis
HTTP Server Plugin Summary Statistics	tdef_secure_entitlement_server_num_summary_attempts	Counter	Secure Entitlement Server - Total number of attempts.
HTTP Server Plugin Summary Statistics	tdef_secure_entitlement_server_num_summary_successes	Counter	Secure Entitlement Server - Total number of successes.
HTTP Server Plugin Summary Statistics	tdef_secure_entitlement_server_num_summary_failures	Counter	Secure Entitlement Server - Total number of failures.
HTTP Server Plugin Summary Statistics	tdef_secure_entitlement_server_num_detail_http_error_responses	Counter	Secure Entitlement Server - Total number of HTTP error responses.
HTTP Server Plugin Summary Statistics	tdef_secure_entitlement_server_num_detail_http_ok_responses	Counter	Secure Entitlement Server - Total number of HTTP OK responses.
HTTP Server Plugin Summary Statistics	tdef_secure_entitlement_server_num_failures_internal_failures	Counter	Secure Entitlement Server - Total number of internal failures.
HTTP Server Plugin Summary Statistics	tdef_secure_entitlement_server_num_failures_invalid_requests	Counter	Secure Entitlement Server - Total number of invalid requests sent to server.
HTTP Server Plugin Summary Statistics	tdef_secure_entitlement_server_num_failures_no_consent	Counter	Secure Entitlement Server - Total number of requests with invalid/missing consent ID.

Group	Metric Name	Type	Synopsis
HTTP Server Plugin Summary Statistics	tdef_simple_entitlement_server_num_summary_attempts	Counter	Simple Entitlement Server - Total number of attempts.
HTTP Server Plugin Summary Statistics	tdef_simple_entitlement_server_num_summary_successes	Counter	Simple Entitlement Server - Total number of successes.
HTTP Server Plugin Summary Statistics	tdef_simple_entitlement_server_num_summary_failures	Counter	Simple Entitlement Server - Total number of failures.
HTTP Server Plugin Summary Statistics	tdef_simple_entitlement_server_num_detail_http_error_responses	Counter	Simple Entitlement Server - Total number of HTTP error responses.
HTTP Server Plugin Summary Statistics	tdef_simple_entitlement_server_num_detail_http_ok_responses	Counter	Simple Entitlement Server - Total number of HTTP OK responses.
HTTP Server Plugin Summary Statistics	tdef_simple_entitlement_server_num_failures_internal_failures	Counter	Simple Entitlement Server - Total number of internal failures.
HTTP Server Plugin Summary Statistics	tdef_simple_entitlement_server_num_failures_invalid_requests	Counter	Simple Entitlement Server - Total number of invalid requests sent to server.
HTTP Server Plugin Summary Statistics	tdef_simple_entitlement_server_num_failures_no_consent	Counter	Simple Entitlement Server - Total number of requests with invalid/missing consent ID.
HTTP Server Plugin Summary Statistics	tdef_ship_api_server_num_summary_attempts	Counter	SHIP API Server - Total number of attempts.
HTTP Server Plugin Summary Statistics	tdef_ship_api_server_num_summary_successes	Counter	SHIP API Server - Total number of successes.
HTTP Server Plugin Summary Statistics	tdef_ship_api_server_num_summary_failures	Counter	SHIP API Server - Total number of failures.
HTTP Server Plugin Summary Statistics	tdef_ship_api_server_num_detail_http_error_responses	Counter	SHIP API Server - Total number of HTTP error responses.
HTTP Server Plugin Summary Statistics	tdef_ship_api_server_num_detail_http_ok_responses	Counter	SHIP API Server - Total number of HTTP OK responses.
HTTP Server Plugin Summary Statistics	tdef_ship_api_server_num_failures_internal_failures	Counter	SHIP API Server - Total number of internal failures.
HTTP Server Plugin Summary Statistics	tdef_ship_api_server_num_failures_invalid_requests	Counter	SHIP API Server - Total number of invalid requests sent to server.
HTTP Server Plugin Summary Statistics	tdef_ship_api_server_num_failures_no_consent	Counter	SHIP API Server - Total number of requests with invalid/missing consent ID.
HTTP Server Plugin Summary Statistics	tdef_ship_redirect_server_num_summary_attempts	Counter	Redirect SHIP Server - Total number of attempts.
HTTP Server Plugin Summary Statistics	tdef_ship_redirect_server_num_summary_successes	Counter	Redirect SHIP Server - Total number of successes.
HTTP Server Plugin Summary Statistics	tdef_ship_redirect_server_num_summary_failures	Counter	Redirect SHIP Server - Total number of failures.
HTTP Server Plugin Summary Statistics	tdef_ship_redirect_server_num_detail_http_error_responses	Counter	Redirect SHIP Server - Total number of HTTP error responses.
HTTP Server Plugin Summary Statistics	tdef_ship_redirect_server_num_detail_http_ok_responses	Counter	Redirect SHIP Server - Total number of HTTP OK responses.
HTTP Server Plugin Summary Statistics	tdef_ship_redirect_server_num_failures_internal_failures	Counter	Redirect SHIP Server - Total number of internal failures.
HTTP Server Plugin Summary Statistics	tdef_ship_redirect_server_num_failures_invalid_requests	Counter	Redirect SHIP Server - Total number of invalid requests sent to server.
HTTP Server Plugin Summary Statistics	tdef_ship_redirect_server_num_failures_no_consent	Counter	Redirect SHIP Server - Total number of requests with invalid/missing consent ID.

HTTP Server Plugin Summary Statistics

- tdef_secure_entitlement_server_num_summary_attempts

Type

64-bit Counter

	Description Secure Entitlement Server - Total number of attempts. Labels pod_id, proxy_id
• tdef_secure_entitlement_server_num_summary_successes	Type 64-bit Counter Description Secure Entitlement Server - Total number of successes. Labels pod_id, proxy_id
• tdef_secure_entitlement_server_num_summary_failures	Type 64-bit Counter Description Secure Entitlement Server - Total number of failures. Labels pod_id, proxy_id
• tdef_secure_entitlement_server_num_detail_http_error_responses	Type 64-bit Counter Description Secure Entitlement Server - Total number of HTTP error responses. Labels pod_id, proxy_id
• tdef_secure_entitlement_server_num_detail_http_ok_responses	Type 64-bit Counter Description Secure Entitlement Server - Total number of HTTP OK responses. Labels pod_id, proxy_id
• tdef_secure_entitlement_server_num_failures_internal_failures	Type 64-bit Counter Description Secure Entitlement Server - Total number of internal failures. Labels pod_id, proxy_id

tdef_secure_entitlement_server_num_failures_invalid_requests
Type
64-bit Counter
Description
Secure Entitlement Server - Total number of invalid requests sent to server.
Labels
pod_id, proxy_id
tdef_secure_entitlement_server_num_failures_no_consent
Type
64-bit Counter
Description
Secure Entitlement Server - Total number of requests with invalid/missing consent ID.
Labels
pod_id, proxy_id
tdef_simple_entitlement_server_num_summary_attempts
Type
64-bit Counter
Description
Simple Entitlement Server - Total number of attempts.
Labels
pod_id, proxy_id
tdef_simple_entitlement_server_num_summary_successes
Type
64-bit Counter
Description
Simple Entitlement Server - Total number of successes.
Labels
pod_id, proxy_id
tdef_simple_entitlement_server_num_summary_failures
Type
64-bit Counter
Description
Simple Entitlement Server - Total number of failures.
Labels
pod_id, proxy_id
tdef_simple_entitlement_server_num_detail_http_error_responses
Type
64-bit Counter
Description

Simple Entitlement Server - Total number of HTTP error responses.

Labels

pod_id, proxy_id

- tdef_simple_entitlement_server_num_detail_http_ok_responses

Type

64-bit Counter

Description

Simple Entitlement Server - Total number of HTTP OK responses.

Labels

pod_id, proxy_id

- tdef_simple_entitlement_server_num_failures_internal_failures

Type

64-bit Counter

Description

Simple Entitlement Server - Total number of internal failures.

Labels

pod_id, proxy_id

- tdef_simple_entitlement_server_num_failures_invalid_requests

Type

64-bit Counter

Description

Simple Entitlement Server - Total number of invalid requests sent to server.

Labels

pod_id, proxy_id

- tdef_simple_entitlement_server_num_failures_no_consent

Type

64-bit Counter

Description

Simple Entitlement Server - Total number of requests with invalid/missing consent ID.

Labels

pod_id, proxy_id

- tdef_ship_api_server_num_summary_attempts

Type

64-bit Counter

Description

SHIP API Server - Total number of attempts.

Labels

pod_id, proxy_id

- tdef_ship_api_server_num_summary_successes

Type

64-bit Counter

Description

SHIP API Server - Total number of successes.

Labels

pod_id, proxy_id

- tdef_ship_api_server_num_summary_failures

Type

64-bit Counter

Description

SHIP API Server - Total number of failures.

Labels

pod_id, proxy_id

- tdef_ship_api_server_num_detail_http_error_responses

Type

64-bit Counter

Description

SHIP API Server - Total number of HTTP error responses.

Labels

pod_id, proxy_id

- tdef_ship_api_server_num_detail_http_ok_responses

Type

64-bit Counter

Description

SHIP API Server - Total number of HTTP OK responses.

Labels

pod_id, proxy_id

- tdef_ship_api_server_num_failures_internal_failures

Type

64-bit Counter

Description

SHIP API Server - Total number of internal failures.

Labels

pod_id, proxy_id

- tdef_ship_api_server_num_failures_invalid_requests

Type

64-bit Counter

Description

SHIP API Server - Total number of invalid requests sent to server.

Labels

pod_id, proxy_id

- tdef_ship_api_server_num_failures_no_consent

Type

64-bit Counter

Description

SHIP API Server - Total number of requests with invalid/missing consent ID.

Labels

pod_id, proxy_id

- tdef_ship_redirect_server_num_summary_attempts

Type

64-bit Counter

Description

Redirect SHIP Server - Total number of attempts.

Labels

pod_id, proxy_id

- tdef_ship_redirect_server_num_summary_successes

Type

64-bit Counter

Description

Redirect SHIP Server - Total number of successes.

Labels

pod_id, proxy_id

- tdef_ship_redirect_server_num_summary_failures

Type

64-bit Counter

Description

Redirect SHIP Server - Total number of failures.

Labels

pod_id, proxy_id

- tdef_ship_redirect_server_num_detail_http_error_responses

Type

64-bit Counter

Description

Redirect SHIP Server - Total number of HTTP error responses.

Labels

pod_id, proxy_id

- tdef_ship_redirect_server_num_detail_http_ok_responses

Type

64-bit Counter

Description

Redirect SHIP Server - Total number of HTTP OK responses.

Labels

pod_id, proxy_id

- tdef_ship_redirect_server_num_failures_internal_failures

Type

64-bit Counter

Description

Redirect SHIP Server - Total number of internal failures.

Labels

pod_id, proxy_id

- tdef_ship_redirect_server_num_failures_invalid_requests

Type

64-bit Counter

Description

Redirect SHIP Server - Total number of invalid requests sent to server.

Labels

pod_id, proxy_id

- tdef_ship_redirect_server_num_failures_no_consent

Type

64-bit Counter

Description

Redirect SHIP Server - Total number of requests with invalid/missing consent ID.

Labels

pod_id, proxy_id

Table of Contents

- [Summary](#)
- [TCP Optimization Statistics](#)

Summary

Group	Metric Name	Type	Synopsis
TCP Optimization Statistics	tdef_tcp_opt_total_client_connections	Counter	TCP Client Connections - Total
TCP Optimization Statistics	tdef_tcp_opt_served_client_connections	Counter	TCP Client Connections - Service Applied
TCP Optimization Statistics	tdef_tcp_opt_failed_served_client_connections	Counter	TCP Client Connections - Service Failed
TCP Optimization Statistics	tdef_tcp_opt_queued_syn_current_count	Counter	TCP Client Connections - Queued SYN Current Count
TCP Optimization Statistics	tdef_tcp_opt_queued_syn_packets	Counter	TCP Client Connections - Queued SYN Packets
TCP Optimization Statistics	tdef_tcp_opt_queued_syn_failed_connections	Counter	TCP Client Connections - Queued SYN Failed Connections
TCP Optimization Statistics	tdef_tcp_opt_queued_syn_forwarded	Counter	TCP Client Connections - Queued SYN Forwarded
TCP Optimization Statistics	tdef_tcp_opt_queued_syn_retransmit_received	Counter	TCP Client Connections - Queued SYN Received Retransmit

Group	Metric Name	Type	Synopsis
TCP Optimization Statistics	tdef_tcp_opt_queued_syn_deleted_timeout	Counter	TCP Client Connections - Queued SYN Deleted Timeout
TCP Optimization Statistics	tdef_tcp_opt_queued_syn_reset_sent	Counter	TCP Client Connections - Queued SYN Reset Sent
TCP Optimization Statistics	tdef_tcp_opt_queued_syn_not_saved_queue_full	Counter	TCP Client Connections - Queued SYN Not Saved Queue Full
TCP Optimization Statistics	tdef_tcp_opt_queued_syn_not_found	Counter	TCP Client Connections - Queued SYN Not Found
TCP Optimization Statistics	tdef_tcp_opt_total_server_connections	Counter	TCP Server Connections - Total
TCP Optimization Statistics	tdef_tcp_opt_served_server_connections	Counter	TCP Server Connections - Service Applied
TCP Optimization Statistics	tdef_tcp_opt_failed_served_server_connections	Counter	TCP Server Connections - Service Failed
TCP Optimization Statistics	tdef_tcp_opt_intelligent_buffering_applied	Counter	TCP Server Connections - Intelligent Buffering Applied
TCP Optimization Statistics	tdef_tcp_opt_total_icmpv4_packets_rcd	Counter	TCP Optimization - Total ICMPv4 packets received
TCP Optimization Statistics	tdef_tcp_opt_icmpv4_packets_too_big_rcd	Counter	TCP Optimization - Total ICMPv4 Packets Too Big received
TCP Optimization Statistics	tdef_tcp_opt_v4_server_connection_mtu_changed	Counter	TCP Optimization - Total v4 Server Connection MTU changed
TCP Optimization Statistics	tdef_tcp_opt_total_icmpv6_packets_rcd	Counter	TCP Optimization - Total ICMPv6 packets received
TCP Optimization Statistics	tdef_tcp_opt_icmpv6_packets_too_big_rcd	Counter	TCP Optimization - Total ICMPv6 Packets Too Big received
TCP Optimization Statistics	tdef_tcp_opt_v6_server_connection_mtu_changed	Counter	TCP Optimization - Total v6 Server Connection MTU changed
TCP Optimization Statistics	tdef_tcp_opt_active_client_connections	Gauge	TCP Client Connections - Active
TCP Optimization Statistics	tdef_tcp_opt_active_server_connections	Gauge	TCP Server Connections - Active

TCP Optimization Statistics

- tdef_tcp_opt_total_client_connections

Type

64-bit Counter

Description

None

Labels

pod_id, proxy_id, proxy_service

- tdef_tcp_opt_served_client_connections

Type

64-bit Counter

Description

None

Labels

pod_id, proxy_id, proxy_service

- tdef_tcp_opt_failed_served_client_connections

Type

64-bit Counter

Description

None

Labels

pod_id, proxy_id, proxy_service

- tdef_tcp_opt_queued_syn_current_count

Type

64-bit Counter

Description

Number of currently queued SYN packets for delayed ACK

Labels

pod_id, proxy_id, proxy_service

• tdef_tcp_opt_queued_syn_packets

Type

64-bit Counter

Description

Number of SYN packets that were queued for delayed ACK

Labels

pod_id, proxy_id, proxy_service

• tdef_tcp_opt_queued_syn_failed_connections

Type

64-bit Counter

Description

Number of SYN packets that were queued for delayed ACK and server connection failed

Labels

pod_id, proxy_id, proxy_service

• tdef_tcp_opt_queued_syn_forwarded

Type

64-bit Counter

Description

Number of SYN packets that were delivered to the TCP stack

Labels

pod_id, proxy_id, proxy_service

• tdef_tcp_opt_queued_syn_retransmit_received

Type

64-bit Counter

Description

Number of SYN Retramits received while SYN was in delayed queue

Labels

pod_id, proxy_id, proxy_service

• tdef_tcp_opt_queued_syn_deleted_timeout

Type

64-bit Counter

Description

Number of SYN packets deleted from queue because of timeout

Labels

pod_id, proxy_id, proxy_service

- tdef_tcp_opt_queued_syn_reset_sent

Type

64-bit Counter

Description

Number of Queued SYNs that were responded with TCP Resets

Labels

pod_id, proxy_id, proxy_service

- tdef_tcp_opt_queued_syn_not_saved_queue_full

Type

64-bit Counter

Description

Number of Queued SYNs that could not be saved because queue full

Labels

pod_id, proxy_id, proxy_service

- tdef_tcp_opt_queued_syn_not_found

Type

64-bit Counter

Description

Number of Queued SYNs that could not be forwarded to TCP stack because SYN was not found in queue

Labels

pod_id, proxy_id, proxy_service

- tdef_tcp_opt_total_server_connections

Type

64-bit Counter

Description

Total number of TCP server connections

Labels

pod_id, proxy_id, proxy_service

- tdef_tcp_opt_serviced_server_connections

Type

64-bit Counter

Description

Number of server connections that TCP Optimization Service applied

Labels

pod_id, proxy_id, proxy_service

- tdef_tcp_opt_failed_serviced_server_connections

Type

64-bit Counter

Description

Number of tcp server connections that failed applying TCP Optimization service

Labels

pod_id, proxy_id, proxy_service

- tdef_tcp_opt_intelligent_buffering_applied

Type

64-bit Counter

Description

Number of tcp server connections that applied intelligent buffering service

Labels

pod_id, proxy_id, proxy_service

- tdef_tcp_opt_total_icmpv4_packets_recd

Type

64-bit Counter

Description

Number of total ICMPv4 packets received

Labels

pod_id, proxy_id, proxy_service

- tdef_tcp_opt_icmpv4_packets_too_big_recd

Type

64-bit Counter

Description

Number of total ICMPv4 Packets Too Big received

Labels

pod_id, proxy_id, proxy_service

- tdef_tcp_opt_v4_server_connection_mtu_changed

Type

64-bit Counter

Description

Number of total v4 Server Connection MTU changed

Labels

pod_id, proxy_id, proxy_service

- tdef_tcp_opt_total_icmpv6_packets_recd

Type

64-bit Counter

Description

Number of total ICMPv6 packets received

Labels

pod_id, proxy_id, proxy_service

- tdef_tcp_opt_icmpv6_packets_too_big_recd

Type

64-bit Counter

Description

Number of total ICMPv6 Packets Too Big received

Labels

pod_id, proxy_id, proxy_service
- tdef_tcp_opt_v6_server_connection_mtu_changed

Type

64-bit Counter

Description

Number of total v6 Server Connection MTU changed

Labels

pod_id, proxy_id, proxy_service
- tdef_tcp_opt_active_client_connections

Type

64-bit Gauge

Description

None

Labels

pod_id, proxy_id, proxy_service
- tdef_tcp_opt_active_server_connections

Type

64-bit Gauge

Description

Number of active TCP server connections

Labels

pod_id, proxy_id, proxy_service

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- [TCP Perf Statistics](#)

Summary

Group	Metric Name	Type	Synopsis
TCP Perf Statistics	tdef_tcp_perf_sync_cookies	Counter	TCP Event - Number of SYN-Cookies Sent
TCP Perf Statistics	tdef_tcp_perf_listen_overflows	Counter	TCP Event - Listen Overflows
TCP Perf Statistics	tdef_tcp_perf_listen_drops	Counter	TCP Event - Listen Drops
TCP Perf Statistics	tdef_tcp_perf_active_connection_openings	Counter	TCP Event - Active Connection Openings
TCP Perf Statistics	tdef_tcp_perf_passive_connection_openings	Counter	TCP Event - Passive Connection Openings
TCP Perf Statistics	tdef_tcp_perf_packet_drops	Counter	TCP Event - number of dropped packets

Group	Metric Name	Type	Synopsis
TCP Perf Statistics	tdef_tcp_perf_resets_received_for_embryonic_syn_rcv_sockets	Counter	TCP Event - Resets Received for Embryonic SYN_RECV Sockets
TCP Perf Statistics	tdef_tcp_perf_tcp_data_packets_sent	Counter	TCP Event - Data packets sent
TCP Perf Statistics	tdef_tcp_perf_tcp_data_packets_retransmitted	Counter	TCP Event - Data packets retransmitted
TCP Perf Statistics	tdef_tcp_perf_tcp_sack_scoreboard_overflow	Counter	TCP Event - SACK scoreboard overflows
TCP Perf Statistics	tdef_tcp_perf_tcp_socket_inuse	Gauge	TCP Socket - In Use
TCP Perf Statistics	tdef_tcp_perf_tcp_socket_orphan	Gauge	TCP Socket - Orphan
TCP Perf Statistics	tdef_tcp_perf_tcp_socket_est	Gauge	TCP Socket - Established
TCP Perf Statistics	tdef_tcp_perf_tcp_socket_tw	Gauge	TCP Socket - Time Wait

TCP Perf Statistics

- [tdef_tcp_perf_sync_cookies](#)

Type

64-bit Counter

Description

None

Labels

pod_id, proxy_id

- [tdef_tcp_perf_listen_overflows](#)

Type

64-bit Counter

Description

None

Labels

pod_id, proxy_id

- [tdef_tcp_perf_listen_drops](#)

Type

64-bit Counter

Description

None

Labels

pod_id, proxy_id

- [tdef_tcp_perf_active_connection_openings](#)

Type

64-bit Counter

Description

TCP event active socket connection openings

Labels

pod_id, proxy_id

- [tdef_tcp_perf_passive_connection_openings](#)

Type

64-bit Counter

Description

TCP event passive socket connection openings

Labels

pod_id, proxy_id

• tdef_tcp_perf_packet_drops

Type

64-bit Counter

Description

TCP event number of dropped packets

Labels

pod_id, proxy_id

• tdef_tcp_perf_resets_received_for_embryonic_syn_rcv_sockets

Type

64-bit Counter

Description

TCP event resets received for embryonic SYN_RECV sockets

Labels

pod_id, proxy_id

• tdef_tcp_perf_tcp_data_packets_sent

Type

64-bit Counter

Description

TCP event data packets sent

Labels

pod_id, proxy_id

• tdef_tcp_perf_tcp_data_packets_retransmitted

Type

64-bit Counter

Description

TCP event data packets retransmitted

Labels

pod_id, proxy_id

• tdef_tcp_perf_tcp_sack_scoreboard_overflow

Type

64-bit Counter

Description

TCP event SACK scoreboard overflows

Labels

pod_id, proxy_id

- tdef_tcp_perf_tcp_socket_inuse

Type

64-bit Gauge

Description

None

Labels

pod_id, proxy_id

- tdef_tcp_perf_tcp_socket_orphan

Type

64-bit Gauge

Description

None

Labels

pod_id, proxy_id

- tdef_tcp_perf_tcp_socket_est

Type

64-bit Gauge

Description

None

Labels

pod_id, proxy_id

- tdef_tcp_perf_tcp_socket_tw

Type

64-bit Gauge

Description

Number of Time Wait TCP sockets

Labels

pod_id, proxy_id

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- [TCP Proxy Detail Statistics](#)
- [TCP Proxy Connection Statistics](#)
- [TCP Proxy Failure Statistics](#)
- [TCP Proxy DNS Statistics](#)

Summary

Group	Metric Name	Type	Synopsis
Summary of TCP Proxy Statistics	num_tcp_splicing_summary_attempts	Counter	Summary - Attempts
Summary of TCP Proxy Statistics	num_tcp_splicing_summary_successes	Counter	Summary - Successes

Group	Metric Name	Type	Synopsis
Summary of TCP Proxy Statistics	num_tcp_splicing_summary_aborts	Counter	Summary - Aborts
Summary of TCP Proxy Statistics	num_tcp_splicing_summary_failures	Counter	Summary - Failures
Summary of TCP Proxy Statistics	num_tcp_splicing_transaction_edr_creates	Counter	Summary - EDRs Creates
Summary of TCP Proxy Statistics	num_tcp_splicing_transaction_edr_attempts	Counter	Summary - EDRs Attempts
Summary of TCP Proxy Statistics	num_tcp_splicing_transaction_edr_successes	Counter	Summary - EDRs Successes
Summary of TCP Proxy Statistics	num_tcp_splicing_transaction_edr_failures	Counter	Summary - EDRs Failures
Summary of TCP Proxy Statistics	num_tcp_service_duration_min	Gauge	Duration - Min (seconds)
Summary of TCP Proxy Statistics	num_tcp_service_duration_max	Gauge	Duration - Max (seconds)
Summary of TCP Proxy Statistics	num_tcp_service_duration_avg	Gauge	Duration - Average (seconds)
TCP Proxy Detail Statistics	num_tcp_service_handoff_reverse_handoffs	Counter	Handoff - Reverse Handoffs
TCP Proxy Detail Statistics	num_tcp_splicing_protocol_http1x_over_tls	Counter	Protocol - HTTP/1.x over TLS
TCP Proxy Detail Statistics	num_tcp_splicing_protocol_http20	Counter	Protocol - HTTP/2.0
TCP Proxy Detail Statistics	num_tcp_splicing_protocol_http20_over_tls	Counter	Protocol - HTTP/2.0 over TLS
TCP Proxy Detail Statistics	num_tcp_splicing_protocol_websocket	Counter	Protocol - WebSocket
TCP Proxy Detail Statistics	num_tcp_splicing_protocol_others	Counter	Protocol - Others
TCP Proxy Detail Statistics	num_tcp_splicing_protocol_http	Counter	Protocol - HTTP
TCP Proxy Detail Statistics	num_tcp_splicing_protocol_webrtc	Counter	Protocol - WebRTC
TCP Proxy Detail Statistics	num_tcp_splicing_protocol_facebookzero	Counter	Protocol - FacebookZero
TCP Proxy Detail Statistics	num_tcp_splicing_reason_https_credentials_not_available	Counter	Reason - HTTPS Credentials Not Available
TCP Proxy Detail Statistics	num_tcp_splicing_reason_http_connection_upgrade	Counter	Reason - HTTP Connection Upgrade
TCP Proxy Detail Statistics	num_tcp_splicing_reason_tcp_ip_flow	Counter	Reason - TCP IP Flow
TCP Proxy Detail Statistics	num_tcp_splicing_reason_nonhttp	Counter	Reason - Non-HTTP
TCP Proxy Detail Statistics	num_tcp_splicing_reason_http_request_receive_timeout	Counter	Reason - HTTP Request Receive Timeout
TCP Proxy Detail Statistics	num_tcp_splicing_reason_http_connect	Counter	Reason - HTTP CONNECT
TCP Proxy Detail Statistics	num_tcp_splicing_reason_bypass_http_proxy	Counter	Reason - Bypass HTTP Proxy
TCP Proxy Detail Statistics	num_tcp_splicing_dpi_updates	Counter	Dpi Updates
TCP Proxy Detail Statistics	num_tcp_splicing_dpi_update_failures	Counter	DPI update failures
TCP Proxy Detail Statistics	num_tcp_splicing_dpi_update_for_deleted_flow	Counter	DPI updates for deleted flows
TCP Proxy Detail Statistics	num_tcp_splicing_flow_idle_timeout_updated	Counter	Flow Idle Timeout Updated
TCP Proxy Connection Statistics	num_tcp_splicing_client_connections_total	Counter	Client Connections - Total
TCP Proxy Connection Statistics	num_tcp_splicing_client_connections_ipv4	Counter	Client Connections - IPv4
TCP Proxy Connection Statistics	num_tcp_splicing_client_connections_ipv6	Counter	Client Connections - IPv6
TCP Proxy Connection Statistics	num_tcp_splicing_client_connections_bytes_received	Counter	Client Connections - Bytes Received
TCP Proxy Connection Statistics	num_tcp_splicing_client_connections_bytes_sent	Counter	Client Connections - Bytes Sent
TCP Proxy Connection Statistics	num_tcp_splicing_client_connections_read_failure	Counter	Client Connections - Read Failure
TCP Proxy Connection Statistics	num_tcp_splicing_client_connections_write_failure	Counter	Client Connections - Write Failure
TCP Proxy Connection Statistics	num_tcp_splicing_client_connections_write_incomplete	Counter	Client Connections - Write Incomplete
TCP Proxy Connection Statistics	num_tcp_splicing_server_connections_total	Counter	Server Connections - Total
TCP Proxy Connection Statistics	num_tcp_splicing_server_connections_ipv4	Counter	Server Connections - IPv4
TCP Proxy Connection Statistics	num_tcp_splicing_server_connections_ipv6	Counter	Server Connections - IPv6
TCP Proxy Connection Statistics	num_tcp_splicing_server_connections_ip_transparent	Counter	Server Connections - IP Transparent
TCP Proxy Connection Statistics	num_tcp_splicing_server_connections_bytes_received	Counter	Server Connections - Bytes Received
TCP Proxy Connection Statistics	num_tcp_splicing_server_connections_bytes_sent	Counter	Server Connections - Bytes Sent
TCP Proxy Connection Statistics	num_tcp_splicing_server_connections_read_failure	Counter	Server Connections - Read Failure
TCP Proxy Connection Statistics	num_tcp_splicing_server_connections_write_failure	Counter	Server Connections - Write Failure

Group	Metric Name	Type	Synopsis
TCP Proxy Connection Statistics	num_tcp_splicing_server_connections_write_incomplete	Counter	Server Connections - Write Incomplete
TCP Proxy Connection Statistics	num_tcp_splicing_server_connections_connect_failed	Counter	Server Connections - Connect Failed
TCP Proxy Connection Statistics	num_fastopen_attempts	Counter	Fast Open - Attempts
TCP Proxy Connection Statistics	num_fastopen_successes	Counter	Fast Open - Successes
TCP Proxy Connection Statistics	num_fastopen_failure_init_conn	Counter	Fast Open - Connection Initialization Failures
TCP Proxy Connection Statistics	num_fastopen_failure_duplicate	Counter	Fast Open - Connection Already in Pending List
TCP Proxy Connection Statistics	num_fastopen_failure_already_created	Counter	Fast Open - Connection Already in Session
TCP Proxy Connection Statistics	num_fastopen_failure_policy	Counter	Fast Open - Policy Denied
TCP Proxy Connection Statistics	num_fastopen_failure_other	Counter	Fast Open - Not Ready
TCP Proxy Connection Statistics	num_fastopen_list_full	Counter	Fast Open - Pending Connection List Full
TCP Proxy Connection Statistics	num_fastopen_aging	Counter	Fast Open - Pending Connection Idle Timeout
TCP Proxy Connection Statistics	num_tcp_splicing_client_connections_current_in_use	Gauge	Client Connections - Currently in Use
TCP Proxy Connection Statistics	num_tcp_splicing_server_connections_current_in_use	Gauge	Server Connections - Currently in Use
TCP Proxy Connection Statistics	num_fastopen_current_pending_conn	Gauge	Fast Open - Current Pending Connections
TCP Proxy Failure Statistics	num_tcp_splicing_aborts_client_abort	Counter	Aborts - Client Abort
TCP Proxy Failure Statistics	num_tcp_splicing_aborts_serverabort	Counter	Aborts - Server Abort
TCP Proxy Failure Statistics	num_tcp_splicing_aborts_idle_timeout	Counter	Aborts - Idle Timeout
TCP Proxy Failure Statistics	num_tcp_splicing_aborts_client_socket_error	Counter	Aborts - Client Socket Error
TCP Proxy Failure Statistics	num_tcp_splicing_aborts_server_socket_error	Counter	Aborts - Server Socket Error
TCP Proxy Failure Statistics	num_tcp_splicing_aborts_client_read_error	Counter	Aborts - Client Read Error
TCP Proxy Failure Statistics	num_tcp_splicing_aborts_client_write_error	Counter	Aborts - Client Write Error
TCP Proxy Failure Statistics	num_tcp_splicing_aborts_server_read_error	Counter	Aborts - Server Read Error
TCP Proxy Failure Statistics	num_tcp_splicing_aborts_server_write_error	Counter	Aborts - Server Write Error
TCP Proxy Failure Statistics	num_tcp_splicing_aborts_no_data_in_connection	Counter	Aborts - No Data in connection
TCP Proxy Failure Statistics	num_tcp_splicing_aborts_client_error	Counter	Aborts - Client Error
TCP Proxy Failure Statistics	num_tcp_splicing_aborts_server_error	Counter	Aborts - Server Error
TCP Proxy Failure Statistics	num_tcp_splicing_failures_out_of_buffers	Counter	Failures - Out of buffers
TCP Proxy Failure Statistics	num_tcp_splicing_failures_server_connection_failure	Counter	Failures - Server connection failure
TCP Proxy Failure Statistics	num_tcp_splicing_failures_dns_request_failure	Counter	Failures - DNS request failure
TCP Proxy Failure Statistics	num_tcp_splicing_failures_dns_response_failure	Counter	Failures - DNS response failure
TCP Proxy Failure Statistics	num_tcp_splicing_failures_out_of_socket	Counter	Failures - Out of socket
TCP Proxy DNS Statistics	num_tcp_splicing_dns_queries_sent	Counter	DNS Queries - Sent
TCP Proxy DNS Statistics	num_tcp_splicing_dns_queries_failed	Counter	DNS Queries - Failed
TCP Proxy DNS Statistics	num_tcp_splicing_dns_queries_canceled	Counter	DNS Queries - Canceled
TCP Proxy DNS Statistics	num_tcp_splicing_dns_replies_success	Counter	DNS Replies - Success
TCP Proxy DNS Statistics	num_tcp_splicing_dns_replies_failure	Counter	DNS Replies - Failure
TCP Proxy DNS Statistics	num_tcp_splicing_dns_replies_stale	Counter	DNS Replies - Stale
TCP Proxy DNS Statistics	num_tcp_splicing_dns_replies_from_cache	Counter	DNS Replies - From Cache

Summary of TCP Proxy Statistics

- [num_tcp_splicing_summary_attempts](#)

Type
64-bit Counter
Description

Number of tcp splicing summary attempts

Labels

pod_id, proxy_id, table, proxy_service

• num_tcp_splicing_summary_successes

Type

64-bit Counter

Description

Number of tcp splicing summary successes

Labels

pod_id, proxy_id, table, proxy_service

• num_tcp_splicing_summary_aborts

Type

64-bit Counter

Description

Number of tcp splicing summary aborts

Labels

pod_id, proxy_id, table, proxy_service

• num_tcp_splicing_summary_failures

Type

64-bit Counter

Description

Number of tcp splicing summary failures

Labels

pod_id, proxy_id, table, proxy_service

• num_tcp_splicing_transaction_edr_creates

Type

64-bit Counter

Description

Number of tcp splicing transaction EDRs creates

Labels

pod_id, proxy_id, table, proxy_service

• num_tcp_splicing_transaction_edr_attempts

Type

64-bit Counter

Description

Number of tcp splicing transaction EDRs attempts

Labels

pod_id, proxy_id, table, proxy_service

• num_tcp_splicing_transaction_edr_successes

Type

64-bit Counter

Description

Number of tcp splicing transaction EDRs successes

Labels

pod_id, proxy_id, table, proxy_service

• num_tcp_splicing_transaction_edr_failures

Type

64-bit Counter

Description

Number of tcp splicing transaction EDRs failures

Labels

pod_id, proxy_id, table, proxy_service

• num_tcp_service_duration_min

Type

64-bit Gauge

Description

Number of tcp service duration min seconds

Labels

pod_id, proxy_id, table, proxy_service

• num_tcp_service_duration_max

Type

64-bit Gauge

Description

Number of tcp service duration max seconds

Labels

pod_id, proxy_id, table, proxy_service

• num_tcp_service_duration_avg

Type

64-bit Gauge

Description

Number of tcp service duration average seconds

Labels

pod_id, proxy_id, table, proxy_service

TCP Proxy Detail Statistics

• num_tcp_service_handoff_reverse_handoffs

Type

64-bit Counter

Description

Number of tcp service handoff reverse handoffs

Labels

pod_id, proxy_id, table, proxy_service

- num_tcp_splicing_protocol_http1x_over_tls

Type

64-bit Counter

Description

Number of tcp splicing protocol http1x over tls

Labels

pod_id, proxy_id, table, proxy_service

- num_tcp_splicing_protocol_http20

Type

64-bit Counter

Description

Number of tcp splicing protocol http20

Labels

pod_id, proxy_id, table, proxy_service

- num_tcp_splicing_protocol_http20_over_tls

Type

64-bit Counter

Description

Number of tcp splicing protocol http20 over tls

Labels

pod_id, proxy_id, table, proxy_service

- num_tcp_splicing_protocol_websocket

Type

64-bit Counter

Description

Number of tcp splicing protocol websocket

Labels

pod_id, proxy_id, table, proxy_service

- num_tcp_splicing_protocol_others

Type

64-bit Counter

Description

Number of tcp splicing protocol others

Labels

pod_id, proxy_id, table, proxy_service

- num_tcp_splicing_protocol_http

Type

64-bit Counter

Description

Number of tcp splicing protocol http

Labels

pod_id, proxy_id, table, proxy_service

- num_tcp_splicing_protocol_webrtc

Type

64-bit Counter

Description

Number of tcp splicing protocol webRTC

Labels

pod_id, proxy_id, table, proxy_service

- num_tcp_splicing_protocol_facebookzero

Type

64-bit Counter

Description

Number of tcp splicing protocol FacebookZero

Labels

pod_id, proxy_id, table, proxy_service

- num_tcp_splicing_reason_https_credentials_not_available

Type

64-bit Counter

Description

Number of tcp splicing reason https credentials not available

Labels

pod_id, proxy_id, table, proxy_service

- num_tcp_splicing_reason_http_connection_upgrade

Type

64-bit Counter

Description

Number of tcp splicing reason http connection upgrade

Labels

pod_id, proxy_id, table, proxy_service

- num_tcp_splicing_reason_tcp_ip_flow

Type

64-bit Counter

Description

Number of tcp splicing reason tcp ip flow

Labels

pod_id, proxy_id, table, proxy_service

- num_tcp_splicing_reason_nonhttp

Type

64-bit Counter

Description

Number of tcp splicing reason nonhttp

Labels

pod_id, proxy_id, table, proxy_service

- num_tcp_splicing_reason_http_request_receive_timeout

Type

64-bit Counter

Description

Number of tcp splicing reason http request receive timeout

Labels

pod_id, proxy_id, table, proxy_service

- num_tcp_splicing_reason_http_connect

Type

64-bit Counter

Description

Number of tcp splicing reason http connect

Labels

pod_id, proxy_id, table, proxy_service

- num_tcp_splicing_reason_bypass_http_proxy

Type

64-bit Counter

Description

Number of tcp splicing reason bypass http proxy

Labels

pod_id, proxy_id, table, proxy_service

- num_tcp_splicing_dpi_updates

Type

64-bit Counter

Description

Number of tcp splicing Dpi Updates

Labels

pod_id, proxy_id, table, proxy_service

- num_tcp_splicing_dpi_update_failures

Type

64-bit Counter

Description

Number of tcp splicing Dpi Update failures

Labels

pod_id, proxy_id, table, proxy_service

- num_tcp_splicing_dpi_update_for_deleted_flow

Type

64-bit Counter

Description

Number of tcp splicing Dpi Updates for deleted flows

Labels

pod_id, proxy_id, table, proxy_service

- num_tcp_splicing_flow_idle_timeout_updated

Type

64-bit Counter

Description

Number of tcp splicing flow idle timeout updated from PPE

Labels

pod_id, proxy_id, table, proxy_service

TCP Proxy Connection Statistics

- num_tcp_splicing_client_connections_total

Type

64-bit Counter

Description

Number of tcp splicing client connections total

Labels

pod_id, proxy_id, table, proxy_service

- num_tcp_splicing_client_connections_ipv4

Type

64-bit Counter

Description

Number of tcp splicing client connections ipv4

Labels

pod_id, proxy_id, table, proxy_service

- num_tcp_splicing_client_connections_ipv6

Type

64-bit Counter

Description

Number of tcp splicing client connections ipv6

Labels

pod_id, proxy_id, table, proxy_service

- num_tcp_splicing_client_connections_bytes_received

Type

64-bit Counter

Description

Number of tcp splicing client connections bytes received

Labels

pod_id, proxy_id, table, proxy_service

- num_tcp_splicing_client_connections_bytes_sent

Type

64-bit Counter

Description

Number of tcp splicing client connections bytes sent

Labels

pod_id, proxy_id, table, proxy_service

- num_tcp_splicing_client_connections_read_failure

Type

64-bit Counter

Description

Number of tcp splicing client connections read failure

Labels

pod_id, proxy_id, table, proxy_service

- num_tcp_splicing_client_connections_write_failure

Type

64-bit Counter

Description

Number of tcp splicing client connections write failure

Labels

pod_id, proxy_id, table, proxy_service

- num_tcp_splicing_client_connections_write_incomplete

Type

64-bit Counter

Description

Number of tcp splicing client connections write incomplete

Labels

pod_id, proxy_id, table, proxy_service

- num_tcp_splicing_server_connections_total

Type

64-bit Counter

Description

Number of tcp splicing server connections total

Labels

pod_id, proxy_id, table, proxy_service

• num_tcp_splicing_server_connections_ipv4

Type

64-bit Counter

Description

Number of tcp splicing server connections ipv4

Labels

pod_id, proxy_id, table, proxy_service

• num_tcp_splicing_server_connections_ipv6

Type

64-bit Counter

Description

Number of tcp splicing server connections ipv6

Labels

pod_id, proxy_id, table, proxy_service

• num_tcp_splicing_server_connections_ip_transparent

Type

64-bit Counter

Description

Number of tcp splicing server connections ip transparent

Labels

pod_id, proxy_id, table, proxy_service

• num_tcp_splicing_server_connections_bytes_received

Type

64-bit Counter

Description

Number of tcp splicing server connections bytes received

Labels

pod_id, proxy_id, table, proxy_service

• num_tcp_splicing_server_connections_bytes_sent

Type

64-bit Counter

Description

Number of tcp splicing server connections bytes sent

Labels

pod_id, proxy_id, table, proxy_service

• num_tcp_splicing_server_connections_read_failure

Type

64-bit Counter

Description

Number of tcp splicing server connections read failure

Labels

pod_id, proxy_id, table, proxy_service

• num_tcp_splicing_server_connections_write_failure

Type

64-bit Counter

Description

Number of tcp splicing server connections write failure

Labels

pod_id, proxy_id, table, proxy_service

• num_tcp_splicing_server_connections_write_incomplete

Type

64-bit Counter

Description

Number of tcp splicing server connections write incomplete

Labels

pod_id, proxy_id, table, proxy_service

• num_tcp_splicing_server_connections_connect_failed

Type

64-bit Counter

Description

Number of tcp splicing server connections connect failed

Labels

pod_id, proxy_id, table, proxy_service

• num_fastopen_attempts

Type

64-bit Counter

Description

Total number of HTTPS server connection fast open attempts

Labels

pod_id, proxy_id, table, proxy_service

• num_fastopen_successes

Type

64-bit Counter

Description

Total number of HTTPS server connection fast open successes

Labels

pod_id, proxy_id, table, proxy_service

• num_fastopen_failure_init_conn

Type

64-bit Counter

Description

Total number of HTTPS server connection fast open initialization failure

Labels

pod_id, proxy_id, table, proxy_service

• num_fastopen_failure_duplicate

Type

64-bit Counter

Description

Total number of HTTPS server connection fast open duplicated connection

Labels

pod_id, proxy_id, table, proxy_service

• num_fastopen_failure_already_created

Type

64-bit Counter

Description

Total number of HTTPS server connection fast open already in the session

Labels

pod_id, proxy_id, table, proxy_service

• num_fastopen_failure_policy

Type

64-bit Counter

Description

Total number of HTTPS server connection fast open policy reject

Labels

pod_id, proxy_id, table, proxy_service

• num_fastopen_failure_other

Type

64-bit Counter

Description

Total number of HTTPS server connection fast open failure others

Labels

pod_id, proxy_id, table, proxy_service

- num_fastopen_list_full
 - Type
 - 64-bit Counter
 - Description
 - Total number of HTTPS server connection fast open pending list full
 - Labels
 - pod_id, proxy_id, table, proxy_service**
- num_fastopen_aging
 - Type
 - 64-bit Counter
 - Description
 - Total number of HTTPS server connection fast open connection closed due to aging
 - Labels
 - pod_id, proxy_id, table, proxy_service**
- num_tcp_splicing_client_connections_current_in_use
 - Type
 - 64-bit Gauge
 - Description
 - Number of tcp splicing client connections currently in use
 - Labels
 - pod_id, proxy_id, table, proxy_service**
- num_tcp_splicing_server_connections_current_in_use
 - Type
 - 64-bit Gauge
 - Description
 - Number of tcp splicing server connections currently in use
 - Labels
 - pod_id, proxy_id, table, proxy_service**
- num_fastopen_current_pending_conn
 - Type
 - 64-bit Gauge
 - Description
 - Total number of HTTPS server connection fast open in pending list
 - Labels
 - pod_id, proxy_id, table, proxy_service**

TCP Proxy Failure Statistics

- num_tcp_splicing_aborts_client_abort
 - Type
 - 64-bit Counter

Description

Number of tcp splicing aborts client abort

Labels

pod_id, proxy_id, table, proxy_service

- num_tcp_splicing_aborts_serverabort

Type

64-bit Counter

Description

Number of tcp splicing aborts serverabort

Labels

pod_id, proxy_id, table, proxy_service

- num_tcp_splicing_aborts_idle_timeout

Type

64-bit Counter

Description

Number of tcp splicing aborts idle timeout

Labels

pod_id, proxy_id, table, proxy_service

- num_tcp_splicing_aborts_client_socket_error

Type

64-bit Counter

Description

Number of tcp splicing aborts client socket error

Labels

pod_id, proxy_id, table, proxy_service

- num_tcp_splicing_aborts_server_socket_error

Type

64-bit Counter

Description

Number of tcp splicing aborts server socket error

Labels

pod_id, proxy_id, table, proxy_service

- num_tcp_splicing_aborts_client_read_error

Type

64-bit Counter

Description

Number of tcp splicing aborts client read error

Labels

pod_id, proxy_id, table, proxy_service

• num_tcp_splicing_aborts_client_write_error
Type
64-bit Counter
Description
Number of tcp splicing aborts client write error
Labels
pod_id, proxy_id, table, proxy_service
• num_tcp_splicing_aborts_server_read_error
Type
64-bit Counter
Description
Number of tcp splicing aborts server read error
Labels
pod_id, proxy_id, table, proxy_service
• num_tcp_splicing_aborts_server_write_error
Type
64-bit Counter
Description
Number of tcp splicing aborts server write error
Labels
pod_id, proxy_id, table, proxy_service
• num_tcp_splicing_aborts_no_data_in_connection
Type
64-bit Counter
Description
Number of tcp splicing aborts no data in connection
Labels
pod_id, proxy_id, table, proxy_service
• num_tcp_splicing_aborts_client_error
Type
64-bit Counter
Description
Number of tcp splicing aborts client error
Labels
pod_id, proxy_id, table, proxy_service
• num_tcp_splicing_aborts_server_error
Type
64-bit Counter
Description

Number of tcp splicing aborts server error

Labels

pod_id, proxy_id, table, proxy_service

- num_tcp_splicing_failures_out_of_buffers

Type

64-bit Counter

Description

Number of tcp splicing failures out of buffers

Labels

pod_id, proxy_id, table, proxy_service

- num_tcp_splicing_failures_server_connection_failure

Type

64-bit Counter

Description

Number of tcp splicing failures server connection failure

Labels

pod_id, proxy_id, table, proxy_service

- num_tcp_splicing_failures_dns_request_failure

Type

64-bit Counter

Description

Number of tcp splicing failures dns request failure

Labels

pod_id, proxy_id, table, proxy_service

- num_tcp_splicing_failures_dns_response_failure

Type

64-bit Counter

Description

Number of tcp splicing failures dns response failure

Labels

pod_id, proxy_id, table, proxy_service

- num_tcp_splicing_failures_out_of_socket

Type

64-bit Counter

Description

Number of tcp splicing failures out of socket

Labels

pod_id, proxy_id, table, proxy_service

TCP Proxy DNS Statistics

• num_tcp_splicing_dns_queries_sent
Type
64-bit Counter
Description
Number of tcp splicing dns queries sent
Labels
pod_id, proxy_id, table, proxy_service
• num_tcp_splicing_dns_queries_failed
Type
64-bit Counter
Description
Number of tcp splicing dns queries failed
Labels
pod_id, proxy_id, table, proxy_service
• num_tcp_splicing_dns_queries_canceled
Type
64-bit Counter
Description
Number of tcp splicing dns queries canceled
Labels
pod_id, proxy_id, table, proxy_service
• num_tcp_splicing_dns_replies_success
Type
64-bit Counter
Description
Number of tcp splicing dns replies success
Labels
pod_id, proxy_id, table, proxy_service
• num_tcp_splicing_dns_replies_failure
Type
64-bit Counter
Description
Number of tcp splicing dns replies failure
Labels
pod_id, proxy_id, table, proxy_service
• num_tcp_splicing_dns_replies_stale
Type
64-bit Counter
Description

Number of tcp splicing dns replies stale

Labels

pod_id, proxy_id, table, proxy_service

- num_tcp_splicing_dns_replies_from_cache

Type

64-bit Counter

Description

Number of tcp splicing dns replies from cache

Labels

pod_id, proxy_id, table, proxy_service

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Summary

Group	Metric Name	Type	Synopsis
Summary of TLS Statistics	tls_num_client_connections_total	Counter	TLS total number of client connections.
Summary of TLS Statistics	tls_num_client_connections_term	Counter	TLS total number of terminated client connections.
Summary of TLS Statistics	tls_num_client_connections_ipv4	Counter	TLS total number of IPv4 client connections.
Summary of TLS Statistics	tls_num_client_connections_ipv6	Counter	TLS total number of IPv6 client connections.
Summary of TLS Statistics	tls_num_client_bytes_recvd	Counter	TLS total number of bytes received from client.
Summary of TLS Statistics	tls_num_client_bytes_sent	Counter	TLS total number of bytes sent to client.
Summary of TLS Statistics	tls_num_client_tls_errors	Counter	TLS total number of client TLS errors.
Summary of TLS Statistics	tls_num_client_fd_errors	Counter	TLS total number of client set FD errors.
Summary of TLS Statistics	tls_num_client_handshake_errors	Counter	TLS total number of client handshake errors.
Summary of TLS Statistics	tls_num_client_read_errors	Counter	TLS total number of client read errors.
Summary of TLS Statistics	tls_num_client_write_errors	Counter	TLS total number of client write errors.
Summary of TLS Statistics	tls_num_client_requests_tls_v10	Counter	TLS total number of TLS1.0 client requests received.
Summary of TLS Statistics	tls_num_client_requests_tls_v11	Counter	TLS total number of TLS1.1 client requests received.
Summary of TLS Statistics	tls_num_client_requests_tls_v12	Counter	TLS total number of TLS1.2 client requests received.
Summary of TLS Statistics	tls_num_server_connections_total	Counter	TLS total number of server connections.
Summary of TLS Statistics	tls_num_server_connections_orig	Counter	TLS total number of originated server connections.
Summary of TLS Statistics	tls_num_server_connections_ipv4	Counter	TLS total number of IPv4 server connections.
Summary of TLS Statistics	tls_num_server_connections_ipv6	Counter	TLS total number of IPv6 server connections.
Summary of TLS Statistics	tls_num_server_bytes_recvd	Counter	TLS total number of bytes received from server.
Summary of TLS Statistics	tls_num_server_bytes_sent	Counter	TLS total number of bytes sent to server.
Summary of TLS Statistics	tls_num_server_tls_errors	Counter	TLS total number of server TLS errors.
Summary of TLS Statistics	tls_num_server_fd_errors	Counter	TLS total number of server set FD errors.
Summary of TLS Statistics	tls_num_server_handshake_errors	Counter	TLS total number of server handshake errors.
Summary of TLS Statistics	tls_num_server_authentication_errors	Counter	TLS total number of server authentication errors.
Summary of TLS Statistics	tls_num_server_read_errors	Counter	TLS total number of server read errors.
Summary of TLS Statistics	tls_num_server_write_errors	Counter	TLS total number of server write errors.
Summary of TLS Statistics	tls_num_server_requests_tls_v10	Counter	TLS total number of TLS1.0 server requests sent.

Group	Metric Name	Type	Synopsis
Summary of TLS Statistics	tls_num_server_requests_tls_v11	Counter	TLS total number of TLS1.1 server requests sent.
Summary of TLS Statistics	tls_num_server_requests_tls_v12	Counter	TLS total number of TLS1.2 server requests sent.
Summary of TLS Statistics	tls_num_client_connections_currently_in_use	Gauge	TLS number of client connections currently in use.
Summary of TLS Statistics	tls_num_server_connections_currently_in_use	Gauge	TLS number of server connections currently in use.

Summary of TLS Statistics

- [tls_num_client_connections_total](#)

Type

64-bit Counter

Description

TLS total number of client connections.

Labels

pod_id, proxy_id

- [tls_num_client_connections_term](#)

Type

64-bit Counter

Description

TLS total number of terminated client connections.

Labels

pod_id, proxy_id

- [tls_num_client_connections_ipv4](#)

Type

64-bit Counter

Description

TLS total number of IPv4 client connections.

Labels

pod_id, proxy_id

- [tls_num_client_connections_ipv6](#)

Type

64-bit Counter

Description

TLS total number of IPv6 client connections.

Labels

pod_id, proxy_id

- [tls_num_client_bytes_recvd](#)

Type

64-bit Counter

Description

TLS total number of bytes received from client.

Labels

pod_id, proxy_id

- `tls_num_client_bytes_sent`

Type

64-bit Counter

Description

TLS total number of bytes sent to client.

Labels

pod_id, proxy_id

- `tls_num_client_tls_errors`

Type

64-bit Counter

Description

TLS total number of client TLS errors.

Labels

pod_id, proxy_id

- `tls_num_client_fd_errors`

Type

64-bit Counter

Description

TLS total number of client set FD errors.

Labels

pod_id, proxy_id

- `tls_num_client_handshake_errors`

Type

64-bit Counter

Description

TLS total number of client handshake errors.

Labels

pod_id, proxy_id

- `tls_num_client_read_errors`

Type

64-bit Counter

Description

TLS total number of client read errors.

Labels

pod_id, proxy_id

- `tls_num_client_write_errors`

Type

64-bit Counter

Description

TLS total number of client write errors.

Labels

pod_id, proxy_id

• `tls_num_client_requests_tls_v10`

Type

64-bit Counter

Description

TLS total number of TLS1.0 client requests received.

Labels

pod_id, proxy_id

• `tls_num_client_requests_tls_v11`

Type

64-bit Counter

Description

TLS total number of TLS1.1 client requests received.

Labels

pod_id, proxy_id

• `tls_num_client_requests_tls_v12`

Type

64-bit Counter

Description

TLS total number of TLS1.2 client requests received.

Labels

pod_id, proxy_id

• `tls_num_server_connections_total`

Type

64-bit Counter

Description

TLS total number of server connections.

Labels

pod_id, proxy_id

• `tls_num_server_connections_orig`

Type

64-bit Counter

Description

TLS total number of originated server connections.

Labels

pod_id, proxy_id

- `tls_num_server_connections_ipv4`

Type

64-bit Counter

Description

TLS total number of IPv4 server connections.

Labels

pod_id, proxy_id

- `tls_num_server_connections_ipv6`

Type

64-bit Counter

Description

TLS total number of IPv6 server connections.

Labels

pod_id, proxy_id

- `tls_num_server_bytes_recvd`

Type

64-bit Counter

Description

TLS total number of bytes received from server.

Labels

pod_id, proxy_id

- `tls_num_server_bytes_sent`

Type

64-bit Counter

Description

TLS total number of bytes sent to server.

Labels

pod_id, proxy_id

- `tls_num_server_tls_errors`

Type

64-bit Counter

Description

TLS total number of server TLS errors.

Labels

pod_id, proxy_id

- `tls_num_server_fd_errors`

Type

64-bit Counter

Description

TLS total number of server set FD errors.

Labels

pod_id, proxy_id

- `tls_num_server_handshake_errors`

Type

64-bit Counter

Description

TLS total number of server handshake errors.

Labels

pod_id, proxy_id

- `tls_num_server_authentication_errors`

Type

64-bit Counter

Description

TLS total number of server authentication errors.

Labels

pod_id, proxy_id

- `tls_num_server_read_errors`

Type

64-bit Counter

Description

TLS total number of server read errors.

Labels

pod_id, proxy_id

- `tls_num_server_write_errors`

Type

64-bit Counter

Description

TLS total number of server write errors.

Labels

pod_id, proxy_id

- `tls_num_server_requests_tls_v10`

Type

64-bit Counter

Description

TLS total number of TLS1.0 server requests sent.

Labels

pod_id, proxy_id

- `tls_num_server_requests_tls_v11`

Type

64-bit Counter

Description

TLS total number of TLS1.1 server requests sent.

Labels

pod_id, proxy_id
- `tls_num_server_requests_tls_v12`

Type

64-bit Counter

Description

TLS total number of TLS1.2 server requests sent.

Labels

pod_id, proxy_id
- `tls_num_client_connections_currently_in_use`

Type

64-bit Gauge

Description

TLS number of client connections currently in use.

Labels

pod_id, proxy_id
- `tls_num_server_connections_currently_in_use`

Type

64-bit Gauge

Description

TLS number of server connections currently in use.

Labels

pod_id, proxy_id

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Summary

Group	Metric Name	Type	Synopsis
Tdef UDP Proxy Connection Statistics	tdef_udp_proxy_num_client_connections_total	Counter	Number of UDP Proxy client connections
Tdef UDP Proxy Connection Statistics	tdef_udp_proxy_num_client_connections_ipv4	Counter	Number of UDP Proxy IPv4 client connections

Group	Metric Name	Type	Synopsis
Tdef UDP Proxy Connection Statistics	tdef_udp_proxy_num_client_connections_ipv6	Counter	Number of UDP Proxy IPv6 client connections
Tdef UDP Proxy Connection Statistics	tdef_udp_proxy_num_client_connections_bytes_received	Counter	Number of UDP Proxy client connection bytes received
Tdef UDP Proxy Connection Statistics	tdef_udp_proxy_num_client_connections_bytes_sent	Counter	Number of UDP Proxy client connection bytes sent
Tdef UDP Proxy Connection Statistics	tdef_udp_proxy_num_client_connections_packets_received	Counter	Number of UDP Proxy client connection packets received
Tdef UDP Proxy Connection Statistics	tdef_udp_proxy_num_client_connections_packets_sent	Counter	Number of UDP Proxy client connection packets sent
Tdef UDP Proxy Connection Statistics	tdef_udp_proxy_num_server_connections_total	Counter	Number of UDP Proxy server connections
Tdef UDP Proxy Connection Statistics	tdef_udp_proxy_num_server_connections_ipv4	Counter	Number of UDP Proxy IPv4 server connections
Tdef UDP Proxy Connection Statistics	tdef_udp_proxy_num_server_connections_ipv6	Counter	Number of UDP Proxy IPv6 server connections
Tdef UDP Proxy Connection Statistics	tdef_udp_proxy_num_server_connections_ip_transparent	Counter	Number of UDP Proxy IP transparent server connections
Tdef UDP Proxy Connection Statistics	tdef_udp_proxy_num_server_connections_bytes_received	Counter	Number of UDP Proxy server connection bytes received
Tdef UDP Proxy Connection Statistics	tdef_udp_proxy_num_server_connections_bytes_sent	Counter	Number of UDP Proxy server connection bytes sent
Tdef UDP Proxy Connection Statistics	tdef_udp_proxy_num_server_connections_packets_received	Counter	Number of UDP Proxy server connection packets received
Tdef UDP Proxy Connection Statistics	tdef_udp_proxy_num_server_connections_packets_sent	Counter	Number of UDP Proxy server connection packets sent
Tdef UDP Proxy Connection Statistics	tdef_udp_proxy_num_dpi_updates	Counter	Number of UDP Proxy DPI updates
Tdef UDP Proxy Connection Statistics	tdef_udp_proxy_num_dpi_update_failures	Counter	Number of UDP Proxy DPI update failures
Tdef UDP Proxy Connection Statistics	tdef_udp_proxy_num_dpi_update_for_deleted_flow	Counter	Number of UDP Proxy DPI updates for deleted flows
Tdef UDP Proxy Connection Statistics	tdef_udp_proxy_num_flow_idle_timeout_updates	Counter	Number of UDP Proxy Flow Idle Timeout updated from PPE
Tdef UDP Proxy Connection Statistics	tdef_udp_proxy_num_client_connections_current	Gauge	Current number of UDP Proxy client connections
Tdef UDP Proxy Connection Statistics	tdef_udp_proxy_num_server_connections_current	Gauge	Current number of UDP Proxy server connections
Tdef UDP Proxy Detail Statistics	tdef_udp_proxy_policy_num_evaluated	Counter	Number of UDP Proxy policies evaluated
Tdef UDP Proxy Detail Statistics	tdef_udp_proxy_policy_num_matched	Counter	Number of UDP Proxy policies matched
Tdef UDP Proxy Detail Statistics	tdef_udp_proxy_policy_num_ignored	Counter	Number of UDP Proxy policies ignored
Tdef UDP Proxy Detail Statistics	tdef_udp_proxy_policy_num_not_available	Counter	Number of UDP Proxy policies not available
Tdef UDP Proxy Detail Statistics	tdef_udp_proxy_policy_num_reevaluations	Counter	Number of UDP Proxy policies reevaluated
Tdef UDP Proxy Detail Statistics	tdef_udp_proxy_policy_num_matches_after_reevaluation	Counter	Number of UDP Proxy policies matched after reevaluation
Tdef UDP Proxy Detail Statistics	tdef_udp_proxy_policy_num_not_available_after_reevaluation	Counter	Number of UDP Proxy policies not available after reevaluation
Tdef UDP Proxy Detail Statistics	tdef_udp_proxy_traffic_control_profile_num_action_allowed	Counter	Number of UDP Proxy traffic control profile actions allowed
Tdef UDP Proxy Detail Statistics	tdef_udp_proxy_traffic_control_profile_num_steered_to_ip	Counter	Number of UDP Proxy traffic control profile steering to IP applied
Tdef UDP Proxy Detail Statistics	tdef_udp_proxy_traffic_control_profile_num_steered_to_host	Counter	Number of UDP Proxy traffic control profile steering to host applied

Group	Metric Name	Type	Synopsis
Tdef UDP Proxy DNS Statistics	tdef_udp_proxy_dns_queries_num_sent	Counter	Number of UDP Proxy DNS queries sent
Tdef UDP Proxy DNS Statistics	tdef_udp_proxy_dns_queries_num_failed	Counter	Number of UDP Proxy DNS queries failed
Tdef UDP Proxy DNS Statistics	tdef_udp_proxy_dns_queries_num_canceled	Counter	Number of UDP Proxy DNS queries canceled
Tdef UDP Proxy DNS Statistics	tdef_udp_proxy_dns_replies_num_success	Counter	Number of UDP Proxy DNS replies succeeded
Tdef UDP Proxy DNS Statistics	tdef_udp_proxy_dns_replies_num_failure	Counter	Number of UDP Proxy DNS replies failed
Tdef UDP Proxy DNS Statistics	tdef_udp_proxy_dns_replies_num_stale	Counter	Number of UDP Proxy DNS stale replies
Tdef UDP Proxy DNS Statistics	tdef_udp_proxy_dns_replies_num_from_cache	Counter	Number of UDP Proxy DNS replies from cache
Tdef UDP Proxy Failure Statistics	tdef_udp_proxy_aborts_num_no_response_from_server	Counter	Number of UDP Proxy connection no response from server aborts
Tdef UDP Proxy Failure Statistics	tdef_udp_proxy_failures_num_server_connection_init_failures	Counter	Number of UDP Proxy server connection initialization failures
Tdef UDP Proxy Failure Statistics	tdef_udp_proxy_failures_num_out_of_socket	Counter	Number of UDP Proxy connection out of socket failures
Tdef UDP Proxy Failure Statistics	tdef_udp_proxy_aborts_num_client_session_deleted	Counter	Number of UDP Proxy client session deleted aborts
Tdef UDP Proxy Failure Statistics	tdef_udp_proxy_aborts_num_client_socket_error	Counter	Number of UDP Proxy client connection socket error aborts
Tdef UDP Proxy Failure Statistics	tdef_udp_proxy_aborts_num_server_socket_error	Counter	Number of UDP Proxy server connection socket error aborts
Tdef UDP Proxy QUIC Statistics	tdef_udp_proxy_num_quic_client_hello_packets	Counter	Number of UDP Proxy QUIC client hello packets
Tdef UDP Proxy QUIC Statistics	tdef_udp_proxy_num_quic_server_hello_packets	Counter	Number of UDP Proxy QUIC server hello packets
Tdef UDP Proxy QUIC Statistics	tdef_udp_proxy_num_quic_other_packets	Counter	Number of UDP Proxy QUIC other packets
Tdef UDP Proxy QUIC Statistics	tdef_udp_proxy_num_quic_client_resets	Counter	Number of UDP Proxy QUIC client resets
Tdef UDP Proxy QUIC Statistics	tdef_udp_proxy_num_quic_server_resets	Counter	Number of UDP Proxy QUIC server resets
Tdef UDP Proxy QUIC Statistics	tdef_udp_proxy_num_quic_server_rejects	Counter	Number of UDP Proxy QUIC server rejects
Tdef UDP Proxy QUIC Statistics	tdef_udp_proxy_num_quic_bytes_upstream	Counter	Number of UDP Proxy QUIC bytes upstream
Tdef UDP Proxy QUIC Statistics	tdef_udp_proxy_num_quic_bytes_downstream	Counter	Number of UDP Proxy QUIC bytes downstream
Tdef UDP Proxy QUIC Statistics	tdef_udp_proxy_num_quic_connections	Counter	Number of UDP Proxy QUIC Flows
Tdef UDP Proxy QUIC Statistics	tdef_udp_proxy_num_quic_version_43	Counter	Number of UDP Proxy QUIC Flows Version 43
Tdef UDP Proxy QUIC Statistics	tdef_udp_proxy_num_quic_version_44	Counter	Number of UDP Proxy QUIC Flows Version 44
Tdef UDP Proxy QUIC Statistics	tdef_udp_proxy_num_quic_version_46	Counter	Number of UDP Proxy QUIC Flows Version 46
Tdef UDP Proxy QUIC Statistics	tdef_udp_proxy_num_quic_version_50	Counter	Number of UDP Proxy QUIC Flows Version 50
Tdef UDP Proxy QUIC Statistics	tdef_udp_proxy_num_quic_version_other	Counter	Number of UDP Proxy QUIC Flows Version Other
Tdef UDP Proxy QUIC Statistics	tdef_udp_proxy_num_quic_connections_current	Gauge	Current number of UDP Proxy QUIC connections
Tdef Summary of UDP Proxy Statistics	tdef_udp_proxy_summary_num_attempts	Counter	Number of UDP Proxy summary attempts
Tdef Summary of UDP Proxy Statistics	tdef_udp_proxy_summary_num_successes	Counter	Number of UDP Proxy summary successes
Tdef Summary of UDP Proxy Statistics	tdef_udp_proxy_summary_num_aborts	Counter	Number of UDP Proxy summary aborts
Tdef Summary of UDP Proxy Statistics	tdef_udp_proxy_summary_num_failures	Counter	Number of UDP Proxy summary failures
Tdef Summary of UDP Proxy Statistics	tdef_udp_proxy_transaction_edr_creates	Counter	Number of UDP Proxy transaction EDR Creates
Tdef Summary of UDP Proxy Statistics	tdef_udp_proxy_transaction_edr_attempts	Counter	Number of UDP Proxy transaction EDR Attempts
Tdef Summary of UDP Proxy Statistics	tdef_udp_proxy_transaction_edr_successes	Counter	Number of UDP Proxy transaction EDR Successes
Tdef Summary of UDP Proxy Statistics	tdef_udp_proxy_transaction_edr_failures	Counter	Number of UDP Proxy transaction EDR Failures

Group	Metric Name	Type	Synopsis
Tdef Summary of UDP Proxy Statistics	tdef_udp_proxy_duration_min_msecs	Gauge	Minimum UDP Proxy duration in milliseconds
Tdef Summary of UDP Proxy Statistics	tdef_udp_proxy_duration_max_msecs	Gauge	Maximum UDP Proxy duration in milliseconds
Tdef Summary of UDP Proxy Statistics	tdef_udp_proxy_duration_avg_msecs	Gauge	Average UDP Proxy duration in milliseconds

Tdef UDP Proxy Connection Statistics

- [tdef_udp_proxy_num_client_connections_total](#)

Type
64-bit Counter

Description
Number of UDP Proxy client connections

Labels
pod_id, proxy_id, table, proxy_service
- [tdef_udp_proxy_num_client_connections_ipv4](#)

Type
64-bit Counter

Description
Number of UDP Proxy IPv4 client connections

Labels
pod_id, proxy_id, table, proxy_service
- [tdef_udp_proxy_num_client_connections_ipv6](#)

Type
64-bit Counter

Description
Number of UDP Proxy IPv6 client connections

Labels
pod_id, proxy_id, table, proxy_service
- [tdef_udp_proxy_num_client_connections_bytes_received](#)

Type
64-bit Counter

Description
Number of UDP Proxy client connection bytes received

Labels
pod_id, proxy_id, table, proxy_service
- [tdef_udp_proxy_num_client_connections_bytes_sent](#)

Type
64-bit Counter

Description

Number of UDP Proxy client connection bytes sent

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_num_client_connections_packets_received

Type

64-bit Counter

Description

Number of UDP Proxy client connection packets received

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_num_client_connections_packets_sent

Type

64-bit Counter

Description

Number of UDP Proxy client connection packets sent

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_num_server_connections_total

Type

64-bit Counter

Description

Number of UDP Proxy server connections

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_num_server_connections_ipv4

Type

64-bit Counter

Description

Number of UDP Proxy IPv4 server connections

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_num_server_connections_ipv6

Type

64-bit Counter

Description

Number of UDP Proxy IPv6 server connections

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_num_server_connections_ip_transparent

Type

64-bit Counter

Description

Number of UDP Proxy IP transparent server connections

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_num_server_connections_bytes_received

Type

64-bit Counter

Description

Number of UDP Proxy server connection bytes received

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_num_server_connections_bytes_sent

Type

64-bit Counter

Description

Number of UDP Proxy server connection bytes sent

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_num_server_connections_packets_received

Type

64-bit Counter

Description

Number of UDP Proxy server connection packets received

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_num_server_connections_packets_sent

Type

64-bit Counter

Description

Number of UDP Proxy server connection packets sent

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_num_dpi_updates

Type

64-bit Counter

Description

Number of UDP Proxy DPI updates

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_num_dpi_update_failures

Type

64-bit Counter

Description

Number of UDP Proxy DPI update failures

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_num_dpi_update_for_deleted_flow

Type

64-bit Counter

Description

Number of UDP Proxy DPI updates for deleted flows

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_num_flow_idle_timeout_updates

Type

64-bit Counter

Description

Number of UDP Proxy Flow Idle Timeout updated from PPE

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_num_client_connections_current

Type

64-bit Gauge

Description

Current number of UDP Proxy client connections

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_num_server_connections_current

Type

64-bit Gauge

Description

Current number of UDP Proxy server connections

Labels

pod_id, proxy_id, table, proxy_service

Tdef UDP Proxy Detail Statistics

- tdef_udp_proxy_policy_num_evaluated

Type

64-bit Counter

Description

Number of UDP Proxy policies evaluated

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_policy_num_matched

Type

64-bit Counter

Description

Number of UDP Proxy policies matched

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_policy_num_ignored

Type

64-bit Counter

Description

Number of UDP Proxy policies ignored

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_policy_num_not_available

Type

64-bit Counter

Description

Number of UDP Proxy policies not available

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_policy_num_reevaluations

Type

64-bit Counter

Description

Number of UDP Proxy policies reevaluated

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_policy_num_matches_after_reevaluation

Type

64-bit Counter

Description

Number of UDP Proxy policies matched after reevaluation

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_policy_num_not_available_after_reevaluation

Type

64-bit Counter

Description

Number of UDP Proxy policies not available after reevaluation

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_traffic_control_profile_num_action_allowed

Type

64-bit Counter

Description

Number of UDP Proxy traffic control profile actions allowed

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_traffic_control_profile_num_steered_to_ip

Type

64-bit Counter

Description

Number of UDP Proxy traffic control profile steering to IP applied

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_traffic_control_profile_num_steered_to_host

Type

64-bit Counter

Description

Number of UDP Proxy traffic control profile steering to host applied

Labels

pod_id, proxy_id, table, proxy_service

Tdef UDP Proxy DNS Statistics

- tdef_udp_proxy_dns_queries_num_sent

Type

64-bit Counter

Description

Number of UDP Proxy DNS queries sent

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_dns_queries_num_failed

Type

64-bit Counter

Description

Number of UDP Proxy DNS queries failed

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_dns_queries_num_canceled

Type

64-bit Counter

Description

Number of UDP Proxy DNS queries canceled

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_dns_replies_num_success

Type

64-bit Counter

Description

Number of UDP Proxy DNS replies succeeded

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_dns_replies_num_failure

Type

64-bit Counter

Description

Number of UDP Proxy DNS replies failed

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_dns_replies_num_stale

Type

64-bit Counter

Description

Number of UDP Proxy DNS stale replies

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_dns_replies_num_from_cache

Type

64-bit Counter

Description

Number of UDP Proxy DNS replies from cache

Labels

pod_id, proxy_id, table, proxy_service

Tdef UDP Proxy Failure Statistics

- tdef_udp_proxy_aborts_num_no_response_from_server

Type

64-bit Counter

Description

Number of UDP Proxy connection no response from server aborts

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_failures_num_server_connection_init_failures

Type

64-bit Counter

Description

Number of UDP Proxy server connection initialization failures

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_failures_num_out_of_socket

Type

64-bit Counter

Description

Number of UDP Proxy connection out of socket failures

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_aborts_num_client_session_deleted

Type

64-bit Counter

Description

Number of UDP Proxy client session deleted aborts

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_aborts_num_client_socket_error

Type

64-bit Counter

Description

Number of UDP Proxy client connection socket error aborts

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_aborts_num_server_socket_error

Type

64-bit Counter

Description

Number of UDP Proxy server connection socket error aborts

Labels

pod_id, proxy_id, table, proxy_service

Tdef UDP Proxy QUIC Statistics

- tdef_udp_proxy_num_quic_client_hello_packets

Type

64-bit Counter

Description

Number of UDP Proxy QUIC client hello packets

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_num_quic_server_hello_packets

Type

64-bit Counter

Description

Number of UDP Proxy QUIC server hello packets

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_num_quic_other_packets

Type

64-bit Counter

Description

Number of UDP Proxy QUIC other packets

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_num_quic_client_resets

Type

64-bit Counter

Description

Number of UDP Proxy QUIC client resets

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_num_quic_server_resets

Type

64-bit Counter

Description

Number of UDP Proxy QUIC server resets

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_num_quic_server_rejects

Type

64-bit Counter

Description

Number of UDP Proxy QUIC server rejects

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_num_quic_bytes_upstream

Type

64-bit Counter

Description

Number of UDP Proxy QUIC bytes upstream

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_num_quic_bytes_downstream

Type

64-bit Counter

Description

Number of UDP Proxy QUIC bytes downstream

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_num_quic_connections

Type

64-bit Counter

Description

Number of UDP Proxy QUIC Flows

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_num_quic_version_43

Type

64-bit Counter

Description

Number of UDP Proxy QUIC Flows Version 43

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_num_quic_version_44

Type

64-bit Counter

Description

Number of UDP Proxy QUIC Flows Version 44

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_num_quic_version_46

Type

64-bit Counter

Description

Number of UDP Proxy QUIC Flows Version 46

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_num_quic_version_50

Type

64-bit Counter

Description

Number of UDP Proxy QUIC Flows Version 50

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_num_quic_version_other

Type

64-bit Counter

Description

Number of UDP Proxy QUIC Flows Version Other

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_num_quic_connections_current

Type

64-bit Gauge

Description

Current number of UDP Proxy QUIC connections

Labels

pod_id, proxy_id, table, proxy_service

Tdef Summary of UDP Proxy Statistics

- tdef_udp_proxy_summary_num_attempts

Type

64-bit Counter

Description

Number of UDP Proxy summary attempts

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_summary_num_successes

Type

64-bit Counter

Description

Number of UDP Proxy summary successes

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_summary_num_aborts

Type

64-bit Counter

Description

Number of UDP Proxy summary aborts

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_summary_num_failures

Type

64-bit Counter

Description

Number of UDP Proxy summary failures

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_transaction_edr_creates

Type

64-bit Counter

Description

Number of UDP Proxy transaction EDR Creates

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_transaction_edr_attempts

Type

64-bit Counter

Description

Number of UDP Proxy transaction EDR Attempts

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_transaction_edr_successes

Type

64-bit Counter

Description

Number of UDP Proxy transaction EDR Successes

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_transaction_edr_failures

Type

64-bit Counter

Description

Number of UDP Proxy transaction EDR Failures

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_duration_min_msecs

Type

64-bit Gauge

Description

Minimum UDP Proxy duration in milliseconds

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_duration_max_msecs

Type

64-bit Gauge

Description

Maximum UDP Proxy duration in milliseconds

Labels

pod_id, proxy_id, table, proxy_service

- tdef_udp_proxy_duration_avg_msecs

Type

64-bit Gauge

Description

Average UDP Proxy duration in milliseconds

Labels

pod_id, proxy_id, table, proxy_service

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- [Tdef Video Optimization Http Shaping Statistics](#)

Summary

Group	Metric Name	Type	Synopsis
Tdef Video Optimization Udp Shaping Statistics	tdef_video_opt_udp_shaping_summary_num_attempts	Counter	Number of Tdef Video Optimization UDP Shaping summary attempts
Tdef Video Optimization Udp Shaping Statistics	tdef_video_opt_udp_shaping_summary_num_successes	Counter	Number of Tdef Video Optimization UDP Shaping summary successes
Tdef Video Optimization Udp Shaping Statistics	tdef_video_opt_udp_shaping_summary_num_cancels	Counter	Number of Tdef Video Optimization UDP Shaping summary cancels
Tdef Video Optimization Udp Shaping Statistics	tdef_video_opt_udp_shaping_total_num_bytes	Counter	Tdef Video Optimization UDP Shaping total number of bytes
Tdef Video Optimization Udp Shaping Statistics	tdef_video_opt_udp_shaping_bypasses_num_opted_out_by_subscriber	Counter	Number of Tdef Video Optimization UDP Shaping subscriber opt-out bypasses
Tdef Video Optimization Udp Shaping Statistics	tdef_video_opt_udp_shaping_bypasses_num_disabled_flows	Counter	Number of Tdef Video Optimization UDP Shaping disabled flows bypasses
Tdef Video Optimization Udp Shaping Statistics	tdef_video_opt_udp_shaping_bypasses_num_subscriber_opted_in_reqs_not_met	Counter	Number of Tdef Video Optimization UDP Shaping subscriber opt-in requirements not met bypasses
Tdef Video Optimization Udp Shaping Statistics	tdef_video_opt_udp_shaping_bypasses_num_not_quic	Counter	Number of Tdef Video Optimization UDP Shaping not QUIC bypasses
Tdef Video Optimization Udp Shaping Statistics	tdef_video_opt_udp_shaping_bypasses_num_timer_limit_hit	Counter	Number of Tdef Video Optimization UDP Shaping timer limit hit bypasses
Tdef Video Optimization Udp Shaping Statistics	tdef_video_opt_udp_shaping_bypasses_num_canceled_too_short	Counter	Number of Tdef Video Optimization UDP Shaping canceled too short bypasses
Tdef Video Optimization Udp Shaping Statistics	tdef_video_opt_udp_shaping_num_aggregation_bypasses	Counter	Number of Tdef Video Optimization UDP Shaping aggregation bypasses
Tdef Video Optimization Udp Shaping Statistics	tdef_video_opt_udp_shaping_bypasses_num_savings_estimation	Counter	Number of Tdef Video Optimization UDP Shaping savings estimation bypasses
Tdef Video Optimization Udp Shaping Statistics	tdef_video_opt_udp_shaping_bypasses_num_bytes_savings_estimation	Counter	Tdef Video Optimization UDP Shaping bypass savings estimation number of bytes
Tdef Video Optimization Udp Shaping Statistics	tdef_video_opt_udp_shaping_buffering_bytes_dropped	Counter	Number of Tdef Video Optimization UDP Shaping Buffering bytes dropped
Tdef Video Optimization Udp Shaping Statistics	tdef_video_opt_udp_shaping_buffering_packets_dropped	Counter	Number of Tdef Video Optimization UDP Shaping Buffering packets dropped
Tdef Video Optimization Udp Shaping Statistics	tdef_video_opt_udp_shaping_num_connections_current	Gauge	Current number of UDP proxy rate limited connections
Tdef Video Optimization Tcp Shaping Statistics	tdef_video_opt_tcp_shaping_num_summary_attempts	Counter	Number of Tdef Video Optimization TCP Shaping summary attempts
Tdef Video Optimization Tcp Shaping Statistics	tdef_video_opt_tcp_shaping_num_summary_successes	Counter	Number of Tdef Video Optimization TCP Shaping summary successes
Tdef Video Optimization Tcp Shaping Statistics	tdef_video_opt_tcp_shaping_num_summary_cancels	Counter	Number of Tdef Video Optimization TCP Shaping summary cancels
Tdef Video Optimization Tcp Shaping Statistics	tdef_video_opt_tcp_shaping_num_bytes	Counter	Tdef Video Optimization TCP Shaping total number of bytes
Tdef Video Optimization Tcp Shaping Statistics	tdef_video_opt_tcp_shaping_num_cancels_rsp_too_short	Counter	Number of Tdef Video Optimization TCP Shaping cancels - response too short
Tdef Video Optimization Tcp Shaping Statistics	tdef_video_opt_tcp_shaping_num_optedout_by_subscriber	Counter	Number of Tdef Video Optimization TCP Shaping Opted Out by Subscriber
Tdef Video Optimization Tcp Shaping Statistics	tdef_video_opt_tcp_shaping_num_disabled	Counter	Number of Tdef Video Optimization TCP Shaping Disabled

Group	Metric Name	Type	Synopsis
Tdef Video Optimization Tcp Shaping Statistics	tdef_video_opt_tcp_shaping_num_optedin_req_not_met	Counter	Number of Tdef Video Optimization TCP Shaping Opted by Subscriber but Requirements not met
Tdef Video Optimization Tcp Shaping Statistics	tdef_video_opt_tcp_shaping_num_bypass_savings_estimation	Counter	Number of Tdef Video Optimization TCP Shaping Bypass due to Savings Estimation
Tdef Video Optimization Tcp Shaping Statistics	tdef_video_opt_tcp_shaping_num_bypass_overload	Counter	Number of Tdef Video Optimization TCP Shaping Bypass due to Overload
Tdef Video Optimization Tcp Shaping Statistics	tdef_video_opt_tcp_shaping_num_session_count_total	Counter	Number of Tdef Video Optimization TCP Shaping sessions - Total
Tdef Video Optimization Tcp Shaping Statistics	tdef_video_opt_tcp_shaping_num_session_aggregation_bypass	Counter	Number of Tdef Video Optimization TCP Shaping sessions aggregation bypass
Tdef Video Optimization Tcp Shaping Statistics	tdef_video_opt_tcp_shaping_num_timer_limit_hit	Counter	Tdef Video Optimization TCP Shaping Timer Limit Hits
Tdef Video Optimization Tcp Shaping Statistics	tdef_video_opt_tcp_shaping_num_sessions_250_500	Counter	Number of Tdef Video Optimization TCP Shaping sessions from 250 to 500kbps (LD)
Tdef Video Optimization Tcp Shaping Statistics	tdef_video_opt_tcp_shaping_num_sessions_500_1500	Counter	Number of Tdef Video Optimization TCP Shaping sessions from 500 to 1500kbps (SD)
Tdef Video Optimization Tcp Shaping Statistics	tdef_video_opt_tcp_shaping_num_sessions_1500_3000	Counter	Number of Tdef Video Optimization TCP Shaping sessions from 1500 to 3000kbps (HD)
Tdef Video Optimization Tcp Shaping Statistics	tdef_video_opt_tcp_shaping_num_sessions_3000_6000	Counter	Number of Tdef Video Optimization TCP Shaping sessions from 3000 to 6000kbps (Full HD)
Tdef Video Optimization Tcp Shaping Statistics	tdef_video_opt_tcp_shaping_num_sessions_6000_more	Counter	Number of Tdef Video Optimization TCP Shaping sessions greater than 6000kbps (UHD)
Tdef Video Optimization Tcp Shaping Statistics	tdef_video_opt_tcp_shaping_num_bytes_ld	Counter	Number of Tdef Video Optimization TCP Shaping Bytes - LD
Tdef Video Optimization Tcp Shaping Statistics	tdef_video_opt_tcp_shaping_num_bytes_sd	Counter	Number of Tdef Video Optimization TCP Shaping Bytes - SD
Tdef Video Optimization Tcp Shaping Statistics	tdef_video_opt_tcp_shaping_num_bytes_hd	Counter	Number of Tdef Video Optimization TCP Shaping Bytes - HD
Tdef Video Optimization Tcp Shaping Statistics	tdef_video_opt_tcp_shaping_num_bytes_full_hd	Counter	Number of Tdef Video Optimization TCP Shaping Bytes - Full HD
Tdef Video Optimization Tcp Shaping Statistics	tdef_video_opt_tcp_shaping_num_bytes_uhd	Counter	Number of Tdef Video Optimization TCP Shaping Bytes - UHD
Tdef Video Optimization Tcp Shaping Statistics	tdef_video_opt_tcp_shaping_num_connections_current	Gauge	Current number of Tdef Video Optimization Tcp shaped connections
Tdef Video Optimization Tcp Shaping Statistics	tdef_video_opt_tcp_shaping_num_session_average_flow_count	Gauge	Tdef Video Optimization TCP Shaping Flows/Session - Average
Tdef Video Optimization Tcp Shaping Statistics	tdef_video_opt_tcp_shaping_num_session_average_duration	Gauge	Tdef Video Optimization TCP Shaping sessions - Average duration
Tdef Video Optimization Http Shaping Statistics	tdef_video_opt_http_shaping_num_session_total	Counter	Number of Tdef Video Optimization HTTP Shaping total number of sessions
Tdef Video Optimization Http Shaping Statistics	tdef_video_opt_http_shaping_num_session_total_hls	Counter	Number of Tdef Video Optimization HTTP Shaping Total HLS ABR Sessions
Tdef Video Optimization Http Shaping Statistics	tdef_video_opt_http_shaping_num_session_total_dash	Counter	Number of Tdef Video Optimization HTTP Shaping Total DASH ABR Sessions

Group	Metric Name	Type	Synopsis
Tdef Video Optimization Http Shaping Statistics	tdef_video_opt_http_shaping_num_session_total_hds	Counter	Number of Tdef Video Optimization HTTP Shaping Total HDS ABR Sessions
Tdef Video Optimization Http Shaping Statistics	tdef_video_opt_http_shaping_num_session_total_other	Counter	Number of Tdef Video Optimization HTTP Shaping Total Other ABR Sessions
Tdef Video Optimization Http Shaping Statistics	tdef_video_opt_http_shaping_num_session_duration_min	Counter	Tdef Video Optimization HTTP Shaping Session - Duration Min (seconds)
Tdef Video Optimization Http Shaping Statistics	tdef_video_opt_http_shaping_num_session_duration_max	Counter	Tdef Video Optimization HTTP Shaping Session - Duration Max (seconds)
Tdef Video Optimization Http Shaping Statistics	tdef_video_opt_http_shaping_num_session_duration_total	Counter	Tdef Video Optimization HTTP Shaping Session - Duration Total (seconds)
Tdef Video Optimization Http Shaping Statistics	tdef_video_opt_http_shaping_num_session_bytes_total	Counter	Number of Tdef Video Optimization HTTP Shaping Session - Total Bytes
Tdef Video Optimization Http Shaping Statistics	tdef_video_opt_http_shaping_num_session_bytes_hls	Counter	Number of Tdef Video Optimization HTTP Shaping Session - HLS Bytes
Tdef Video Optimization Http Shaping Statistics	tdef_video_opt_http_shaping_num_session_bytes_dash	Counter	Number of Tdef Video Optimization HTTP Shaping Session - DASH Bytes
Tdef Video Optimization Http Shaping Statistics	tdef_video_opt_http_shaping_num_session_bytes_hds	Counter	Number of Tdef Video Optimization HTTP Shaping Session - HDS Bytes
Tdef Video Optimization Http Shaping Statistics	tdef_video_opt_http_shaping_num_session_bytes_other	Counter	Number of Tdef Video Optimization HTTP Shaping Session - Other Bytes
Tdef Video Optimization Http Shaping Statistics	tdef_video_opt_http_shaping_num_session_bitrate_0_500	Counter	Number of Tdef Video Optimization HTTP Shaping Session Bitrate 0-500 kbps
Tdef Video Optimization Http Shaping Statistics	tdef_video_opt_http_shaping_num_session_bitrate_500_1000	Counter	Number of Tdef Video Optimization HTTP Shaping Session Bitrate 500-1000 kbps
Tdef Video Optimization Http Shaping Statistics	tdef_video_opt_http_shaping_num_session_bitrate_1000_2500	Counter	Number of Tdef Video Optimization HTTP Shaping Session Bitrate 1000-2500 kbps
Tdef Video Optimization Http Shaping Statistics	tdef_video_opt_http_shaping_num_session_bitrate_2500_more	Counter	Number of Tdef Video Optimization HTTP Shaping Session Bitrate 2500-more kbps
Tdef Video Optimization Http Shaping Statistics	tdef_video_opt_http_shaping_num_session_bypass_savings_estimation	Counter	Number of Tdef Video Optimization HTTP Shaping Session Bypass - Savings Estimation
Tdef Video Optimization Http Shaping Statistics	tdef_video_opt_http_shaping_num_segment_total_hls	Counter	Number of Tdef Video Optimization HTTP Shaping Total HLS ABR Segment
Tdef Video Optimization Http Shaping Statistics	tdef_video_opt_http_shaping_num_segment_total_dash	Counter	Number of Tdef Video Optimization HTTP Shaping Total DASH ABR Segment
Tdef Video Optimization Http Shaping Statistics	tdef_video_opt_http_shaping_num_segment_total_hds	Counter	Number of Tdef Video Optimization HTTP Shaping Total HDS ABR Segment
Tdef Video Optimization Http Shaping Statistics	tdef_video_opt_http_shaping_num_segment_total_other	Counter	Number of Tdef Video Optimization HTTP Shaping Total HDS Other ABR Segment
Tdef Video Optimization Http Shaping Statistics	tdef_video_opt_http_shaping_num_segment_bitrate_0_500	Counter	Number of Tdef Video Optimization HTTP Shaping Segment Bitrate 0-500 kbps
Tdef Video Optimization Http Shaping Statistics	tdef_video_opt_http_shaping_num_segment_bitrate_500_1000	Counter	Number of Tdef Video Optimization HTTP Shaping Segment Bitrate 500-1000 kbps

Group	Metric Name	Type	Synopsis
Tdef Video Optimization Http Shaping Statistics	tdef_video_opt_http_shaping_num_segment_bitrate_1000_2500	Counter	Number of Tdef Video Optimization HTTP Shaping Segment Bitrate 1000-2500 kbps
Tdef Video Optimization Http Shaping Statistics	tdef_video_opt_http_shaping_num_segment_bitrate_2500_more	Counter	Number of Tdef Video Optimization HTTP Shaping Segment Bitrate 2500-more kbps
Tdef Video Optimization Http Shaping Statistics	tdef_video_opt_http_shaping_num_segment_bitrate_enforced	Counter	Number of Tdef Video Optimization HTTP Shaping Segment - Enforced Bitrate Lower Than Media Bitrate
Tdef Video Optimization Http Shaping Statistics	tdef_video_opt_http_shaping_num_segment_subscriber_opt_out	Counter	Number of Tdef Video Optimization HTTP Shaping Segment Subscriber Opt Out
Tdef Video Optimization Http Shaping Statistics	tdef_video_opt_http_shaping_num_segment_subscriber_opt_in_req_not_met	Counter	Number of Tdef Video Optimization HTTP Shaping Segment Subscriber Opt In Requirement Not Met
Tdef Video Optimization Http Shaping Statistics	tdef_video_opt_http_shaping_num_segment_no_maxrate_cp_manifest	Counter	Number of Tdef Video Optimization HTTP Shaping Segment Bypass - CP Manifest with no Max-Rate
Tdef Video Optimization Http Shaping Statistics	tdef_video_opt_http_shaping_num_segment_invalid_maxrate_cp_manifest	Counter	Number of Tdef Video Optimization HTTP Shaping Segment Bypass - CP Manifest with Invalid Max-Rate
Tdef Video Optimization Http Shaping Statistics	tdef_video_opt_http_shaping_num_segment_bypass_savings_estimation	Counter	Number of Tdef Video Optimization HTTP Shaping Segment Bypass - Savings Estimation
Tdef Video Optimization Http Shaping Statistics	tdef_video_opt_http_shaping_num_segment_bypass_short_response	Counter	Number of Tdef Video Optimization HTTP Shaping Segment Bypass - Response Too Short
Tdef Video Optimization Http Shaping Statistics	tdef_video_opt_http_shaping_num_segment_cp_manifest_with_maxrate	Counter	Number of Tdef Video Optimization HTTP Shaping Segment - CP Manifest with Max-Rate
Tdef Video Optimization Http Shaping Statistics	tdef_video_opt_http_shaping_num_segment_aggregation_bypass	Counter	Number of Tdef Video Optimization HTTP Shaping Segment - Aggregation Bypass
Tdef Video Optimization Http Shaping Statistics	tdef_video_opt_http_shaping_num_session_current	Gauge	Current Number of Tdef Video Optimization HTTP Shaping Session - Current
Tdef Video Optimization Http Shaping Statistics	tdef_video_opt_http_shaping_num_segment_current	Gauge	Current Number of Tdef Video Optimization HTTP Shaping Segment - Current

Tdef Video Optimization Udp Shaping Statistics

- tdef_video_opt_udp_shaping_summary_num_attempts

Type

64-bit Counter

Description

Number of Tdef Video Optimization UDP Shaping summary attempts

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_udp_shaping_summary_num_successes

Type

64-bit Counter

Description

Number of Tdef Video Optimization UDP Shaping summary successes

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_udp_shaping_summary_num_cancels

Type

64-bit Counter

Description

Number of Tdef Video Optimization UDP Shaping summary cancels

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_udp_shaping_total_num_bytes

Type

64-bit Counter

Description

Tdef Video Optimization UDP Shaping total number of bytes

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_udp_shaping_bypasses_num_opted_out_by_subscriber

Type

64-bit Counter

Description

Number of Tdef Video Optimization UDP Shaping subscriber opt-out bypasses

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_udp_shaping_bypasses_num_disabled_flows

Type

64-bit Counter

Description

Number of Tdef Video Optimization UDP Shaping disabled flows bypasses

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_udp_shaping_bypasses_num_subscriber_opted_in_reqs_not_met

Type

64-bit Counter

Description

Number of Tdef Video Optimization UDP Shaping subscriber opt-in requirements not met bypasses

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_udp_shaping_bypasses_num_not_quic

Type

64-bit Counter

Description

Number of Tdef Video Optimization UDP Shaping not QUIC bypasses

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_udp_shaping_bypasses_num_timer_limit_hit

Type

64-bit Counter

Description

Number of Tdef Video Optimization UDP Shaping timer limit hit bypasses

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_udp_shaping_bypasses_num_canceled_too_short

Type

64-bit Counter

Description

Number of Tdef Video Optimization UDP Shaping canceled too short bypasses

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_udp_shaping_num_aggregation_bypasses

Type

64-bit Counter

Description

Number of Tdef Video Optimization UDP Shaping aggregation bypasses

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_udp_shaping_bypasses_num_savings_estimation

Type

64-bit Counter

Description

Number of Tdef Video Optimization UDP Shaping savings estimation bypasses

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_udp_shaping_bypasses_num_bytes_savings_estimation

Type

64-bit Counter

Description

Tdef Video Optimization UDP Shaping bypass savings estimation number of bytes

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_udp_shaping_buffering_bytes_dropped

Type

64-bit Counter

Description

Number of Tdef Video Optimization UDP Shaping Buffering bytes dropped

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_udp_shaping_buffering_packets_dropped

Type

64-bit Counter

Description

Number of Tdef Video Optimization UDP Shaping Buffering packets dropped

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_udp_shaping_num_connections_current

Type

64-bit Gauge

Description

Current number of UDP proxy rate limited connections

Labels

pod_id, proxy_id, video_service

Tdef Video Optimization Tcp Shaping Statistics

- tdef_video_opt_tcp_shaping_num_summary_attempts

Type

64-bit Counter

Description

Number of Tdef Video Optimization TCP Shaping summary attempts

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_tcp_shaping_num_summary_successes

Type

64-bit Counter

Description

Number of Tdef Video Optimization TCP Shaping summary successes

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_tcp_shaping_num_summary_cancels

Type

64-bit Counter

Description

Number of Tdef Video Optimization TCP Shaping summary cancels

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_tcp_shaping_num_bytes

Type

64-bit Counter

Description

Tdef Video Optimization TCP Shaping total number of bytes

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_tcp_shaping_num_cancels_rsp_too_short

Type

64-bit Counter

Description

Number of Tdef Video Optimization TCP Shaping cancels - response too short

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_tcp_shaping_num_optedout_by_subscriber

Type

64-bit Counter

Description

Number of Tdef Video Optimization TCP Shaping Opted Out by Subscriber

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_tcp_shaping_num_disabled

Type

64-bit Counter

Description

Number of Tdef Video Optimization TCP Shaping Disabled

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_tcp_shaping_num_optedin_req_not_met

Type

64-bit Counter

Description

Number of Tdef Video Optimization TCP Shaping Opted by Subscriber but Requirements not met

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_tcp_shaping_num_bypass_savings_estimation

Type

64-bit Counter

Description

Number of Tdef Video Optimization TCP Shaping Bypass due to Savings Estimation

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_tcp_shaping_num_bypass_overload

Type

64-bit Counter

Description

Number of Tdef Video Optimization TCP Shaping Bypass due to Overload

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_tcp_shaping_num_session_count_total

Type

64-bit Counter

Description

Number of Tdef Video Optimization TCP Shaping sessions - Total

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_tcp_shaping_num_session_aggregation_bypass

Type

64-bit Counter

Description

Number of Tdef Video Optimization TCP Shaping sessions aggregation bypass

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_tcp_shaping_num_timer_limit_hit

Type

64-bit Counter

Description

Tdef Video Optimization TCP Shaping Timer Limit Hits

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_tcp_shaping_num_sessions_250_500

Type

64-bit Counter

Description

Number of Tdef Video Optimization TCP Shaping sessions from 250 to 500kbps (LD)

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_tcp_shaping_num_sessions_500_1500

Type

64-bit Counter

Description

Number of Tdef Video Optimization TCP Shaping sessions from 500 to 1500kbps (SD)

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_tcp_shaping_num_sessions_1500_3000

Type

64-bit Counter

Description

Number of Tdef Video Optimization TCP Shaping sessions from 1500 to 3000kbps (HD)

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_tcp_shaping_num_sessions_3000_6000

Type

64-bit Counter

Description

Number of Tdef Video Optimization TCP Shaping sessions from 3000 to 6000kbps (Full HD)

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_tcp_shaping_num_sessions_6000_more

Type

64-bit Counter

Description

Number of Tdef Video Optimization TCP Shaping sessions greater than 6000kbps (UHD)

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_tcp_shaping_num_bytes_ld

Type

64-bit Counter

Description

Number of Tdef Video Optimization TCP Shaping Bytes - LD

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_tcp_shaping_num_bytes_sd

Type

64-bit Counter

Description

Number of Tdef Video Optimization TCP Shaping Bytes - SD

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_tcp_shaping_num_bytes_hd

Type

64-bit Counter

Description

Number of Tdef Video Optimization TCP Shaping Bytes - HD

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_tcp_shaping_num_bytes_full_hd

Type

64-bit Counter

Description

Number of Tdef Video Optimization TCP Shaping Bytes - Full HD

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_tcp_shaping_num_bytes_uhd

Type

64-bit Counter

Description

Number of Tdef Video Optimization TCP Shaping Bytes - UHD

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_tcp_shaping_num_connections_current

Type

64-bit Gauge

Description

Current number of Tdef Video Optimization Tcp shaped connections

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_tcp_shaping_num_session_average_flow_count

Type

64-bit Gauge

Description
Tdef Video Optimization TCP Shaping Flows/Session - Average
Labels
<i>pod_id, proxy_id, video_service</i>
• tdef_video_opt_tcp_shaping_num_session_average_duration
Type
64-bit Gauge
Description
Tdef Video Optimization TCP Shaping sessions - Average duration
Labels
<i>pod_id, proxy_id, video_service</i>

Tdef Video Optimization Http Shaping Statistics

• tdef_video_opt_http_shaping_num_session_total
Type
64-bit Counter
Description
Number of Tdef Video Optimization HTTP Shaping total number of sessions
Labels
<i>pod_id, proxy_id, video_service</i>
• tdef_video_opt_http_shaping_num_session_total_hls
Type
64-bit Counter
Description
Number of Tdef Video Optimization HTTP Shaping Total HLS ABR Sessions
Labels
<i>pod_id, proxy_id, video_service</i>
• tdef_video_opt_http_shaping_num_session_total_dash
Type
64-bit Counter
Description
Number of Tdef Video Optimization HTTP Shaping Total DASH ABR Sessions
Labels
<i>pod_id, proxy_id, video_service</i>
• tdef_video_opt_http_shaping_num_session_total_hds
Type
64-bit Counter
Description
Number of Tdef Video Optimization HTTP Shaping Total HDS ABR Sessions
Labels

pod_id, proxy_id, video_service

- tdef_video_opt_http_shaping_num_session_total_other

Type

64-bit Counter

Description

Number of Tdef Video Optimization HTTP Shaping Total Other ABR Sessions

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_http_shaping_num_session_duration_min

Type

64-bit Counter

Description

Tdef Video Optimization HTTP Shaping Session - Duration Min (seconds)

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_http_shaping_num_session_duration_max

Type

64-bit Counter

Description

Tdef Video Optimization HTTP Shaping Session - Duration Max (seconds)

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_http_shaping_num_session_duration_total

Type

64-bit Counter

Description

Tdef Video Optimization HTTP Shaping Session - Duration Total (seconds)

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_http_shaping_num_session_bytes_total

Type

64-bit Counter

Description

Number of Tdef Video Optimization HTTP Shaping Session - Total Bytes

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_http_shaping_num_session_bytes_hls

Type

64-bit Counter

Description

Number of Tdef Video Optimization HTTP Shaping Session - HLS Bytes

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_http_shaping_num_session_bytes_dash

Type

64-bit Counter

Description

Number of Tdef Video Optimization HTTP Shaping Session - DASH Bytes

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_http_shaping_num_session_bytes_hds

Type

64-bit Counter

Description

Number of Tdef Video Optimization HTTP Shaping Session - HDS Bytes

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_http_shaping_num_session_bytes_other

Type

64-bit Counter

Description

Number of Tdef Video Optimization HTTP Shaping Session - Other Bytes

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_http_shaping_num_session_bitrate_0_500

Type

64-bit Counter

Description

Number of Tdef Video Optimization HTTP Shaping Session Bitrate 0-500 kbps

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_http_shaping_num_session_bitrate_500_1000

Type

64-bit Counter

Description

Number of Tdef Video Optimization HTTP Shaping Session Bitrate 500-1000 kbps

Labels

pod_id, proxy_id, video_service

tdef_video_opt_http_shaping_num_session_bitrate_1000_2500
Type
64-bit Counter
Description
Number of Tdef Video Optimization HTTP Shaping Session Bitrate 1000-2500 kbps
Labels
pod_id, proxy_id, video_service
tdef_video_opt_http_shaping_num_session_bitrate_2500_more
Type
64-bit Counter
Description
Number of Tdef Video Optimization HTTP Shaping Session Bitrate 2500-more kbps
Labels
pod_id, proxy_id, video_service
tdef_video_opt_http_shaping_num_session_bypass_savings_estimation
Type
64-bit Counter
Description
Number of Tdef Video Optimization HTTP Shaping Session Bypass - Savings Estimation
Labels
pod_id, proxy_id, video_service
tdef_video_opt_http_shaping_num_segment_total_hls
Type
64-bit Counter
Description
Number of Tdef Video Optimization HTTP Shaping Total HLS ABR Segment
Labels
pod_id, proxy_id, video_service
tdef_video_opt_http_shaping_num_segment_total_dash
Type
64-bit Counter
Description
Number of Tdef Video Optimization HTTP Shaping Total DASH ABR Segment
Labels
pod_id, proxy_id, video_service
tdef_video_opt_http_shaping_num_segment_total_hds
Type
64-bit Counter
Description

Number of Tdef Video Optimization HTTP Shaping Total HDS ABR Segment

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_http_shaping_num_segment_total_other

Type

64-bit Counter

Description

Number of Tdef Video Optimization HTTP Shaping Total Other ABR Segment

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_http_shaping_num_segment_bitrate_0_500

Type

64-bit Counter

Description

Number of Tdef Video Optimization HTTP Shaping Segment Bitrate 0-500 kbps

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_http_shaping_num_segment_bitrate_500_1000

Type

64-bit Counter

Description

Number of Tdef Video Optimization HTTP Shaping Segment Bitrate 500-1000 kbps

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_http_shaping_num_segment_bitrate_1000_2500

Type

64-bit Counter

Description

Number of Tdef Video Optimization HTTP Shaping Segment Bitrate 1000-2500 kbps

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_http_shaping_num_segment_bitrate_2500_more

Type

64-bit Counter

Description

Number of Tdef Video Optimization HTTP Shaping Segment Bitrate 2500-more kbps

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_http_shaping_num_segment_bitrate_enforced

Type

64-bit Counter

Description

Number of Tdef Video Optimization HTTP Shaping Segment - Enforced Bitrate Lower Than Media Bitrate

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_http_shaping_num_segment_subscriber_opt_out

Type

64-bit Counter

Description

Number of Tdef Video Optimization HTTP Shaping Segment Subscriber Opt Out

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_http_shaping_num_segment_subscriber_opt_in_req_not_met

Type

64-bit Counter

Description

Number of Tdef Video Optimization HTTP Shaping Segment Subscriber Opt In Requirement Not Met

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_http_shaping_num_segment_no_maxrate_cp_manifest

Type

64-bit Counter

Description

Number of Tdef Video Optimization HTTP Shaping Segment Bypass - CP Manifest with no Max-Rate

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_http_shaping_num_segment_invalid_maxrate_cp_manifest

Type

64-bit Counter

Description

Number of Tdef Video Optimization HTTP Shaping Segment Bypass - CP Manifest with Invalid Max-Rate

Labels

pod_id, proxy_id, video_service

- tdef_video_opt_http_shaping_num_segment_bypass_savings_estimation

Type

64-bit Counter

Description

Number of Tdef Video Optimization HTTP Shaping Segment Bypass - Savings Estimation

Labels
<i>pod_id, proxy_id, video_service</i>
tdef_video_opt_http_shaping_num_segment_bypass_short_response
Type
64-bit Counter
Description
Number of Tdef Video Optimization HTTP Shaping Segment Bypass - Response Too Short
Labels
<i>pod_id, proxy_id, video_service</i>
tdef_video_opt_http_shaping_num_segment_cp_manifest_with_maxrate
Type
64-bit Counter
Description
Number of Tdef Video Optimization HTTP Shaping Segment - CP Manifest with Max-Rate
Labels
<i>pod_id, proxy_id, video_service</i>
tdef_video_opt_http_shaping_num_segment_aggregation_bypass
Type
64-bit Counter
Description
Number of Tdef Video Optimization HTTP Shaping Segment - Aggregation Bypass
Labels
<i>pod_id, proxy_id, video_service</i>
tdef_video_opt_http_shaping_num_session_current
Type
64-bit Gauge
Description
Current Number of Tdef Video Optimization HTTP Shaping Session - Current
Labels
<i>pod_id, proxy_id, video_service</i>
tdef_video_opt_http_shaping_num_segment_current
Type
64-bit Gauge
Description
Current Number of Tdef Video Optimization HTTP Shaping Segment - Current
Labels
<i>pod_id, proxy_id, video_service</i>

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- ETCD Client Metrics

Summary

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_put_requests	Counter	The total number of ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_requests_sent	Counter	The total number of ETCD Client PUT Requests sent.
ETCD Client Metrics	etcd_put_successes	Counter	The total number of successful ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_failures	Counter	The total number of failed ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_with_lease_requests	Counter	The total number of ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_requests_sent	Counter	The total number of ETCD Client PUT with Lease Requests sent.
ETCD Client Metrics	etcd_put_with_lease_successes	Counter	The total number of successful ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_failures	Counter	The total number of failed ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_range_requests	Counter	The total number of ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_requests_sent	Counter	The total number of ETCD Client Range Requests sent.
ETCD Client Metrics	etcd_range_successes	Counter	The total number of successful ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_failures	Counter	The total number of failed ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_delete_requests	Counter	The total number of ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_requests_sent	Counter	The total number of ETCD Client Range Delete Requests sent.
ETCD Client Metrics	etcd_range_delete_successes	Counter	The total number of successful ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_failures	Counter	The total number of failed ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_txn_requests	Counter	The total number of ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_requests_sent	Counter	The total number of ETCD Client Txn Requests sent.
ETCD Client Metrics	etcd_txn_successes	Counter	The total number of successful ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_failures	Counter	The total number of failed ETCD Client Txn Requests.
ETCD Client Metrics	etcd_watch_requests	Counter	The total number of ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_requests_sent	Counter	The total number of ETCD Client Watch Requests sent.
ETCD Client Metrics	etcd_watch_successes	Counter	The total number of successful ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_failures	Counter	The total number of failed ETCD Client Watch Requests.
ETCD Client Metrics	etcdwatcher_requests	Counter	The total number of ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_requests_sent	Counter	The total number of ETCD Watcher Requests sent.
ETCD Client Metrics	etcdwatcher_successes	Counter	The total number of successful ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_failures	Counter	The total number of failed ETCD Watcher Requests.

ETCD Client Metrics

- etcd_put_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests.

Labels

keyPrefix

- etcd_put_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests sent.

Labels

keyPrefix

- etcd_put_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT Requests.

Labels

keyPrefix

- etcd_put_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT Requests.

Labels

keyPrefix, failureReason

- etcd_put_with_lease_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests sent.

Labels

keyPrefix

- etcd_put_with_lease_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_failures
 - Type
 - 64-bit Counter
 - Description
 - The total number of failed ETCD Client PUT with Lease Requests.
 - Labels
 - keyPrefix, failureReason**

- etcd_range_requests
 - Type
 - 64-bit Counter
 - Description
 - The total number of ETCD Client Range Requests.
 - Labels
 - keyPrefix**

- etcd_range_requests_sent
 - Type
 - 64-bit Counter
 - Description
 - The total number of ETCD Client Range Requests sent.
 - Labels
 - keyPrefix**

- etcd_range_successes
 - Type
 - 64-bit Counter
 - Description
 - The total number of successful ETCD Client Range Requests.
 - Labels
 - keyPrefix**

- etcd_range_failures
 - Type
 - 64-bit Counter
 - Description
 - The total number of failed ETCD Client Range Requests.
 - Labels
 - keyPrefix, failureReason**

- etcd_range_delete_requests
 - Type
 - 64-bit Counter
 - Description

The total number of ETCD Client Range Delete Requests.

Labels

keyPrefix

- etcd_range_delete_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests sent.

Labels

keyPrefix

- etcd_range_delete_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Delete Requests.

Labels

keyPrefix

- etcd_range_delete_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Delete Requests.

Labels

keyPrefix, failureReason

- etcd_txn_requests

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests.

Labels

keyPrefix

- etcd_txn_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests sent.

Labels

keyPrefix

- etcd_txn_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Txn Requests.

Labels

keyPrefix

- etcd_txn_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Txn Requests.

Labels

keyPrefix, failureReason

- etcd_watch_requests

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests sent.

Labels

keyPrefix

- etcd_watch_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Watch Requests.

Labels

keyPrefix, failureReason

- etcdwatcher_requests

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests.

Labels

keyPrefix
- etcdwatcher_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests sent.

Labels

keyPrefix
- etcdwatcher_successes

Type

64-bit Counter

Description

The total number of successful ETCD Watcher Requests.

Labels

keyPrefix
- etcdwatcher_failures

Type

64-bit Counter

Description

The total number of failed ETCD Watcher Requests.

Labels

keyPrefix, failureReason

Table of Contents

- [Summary](#)
- [HTTP Punting Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_sent	Counter	The total number of HTTP requests successfully punted.
HTTP Punting Metrics	punting_http_requests_failed	Counter	The total number of HTTP requests punting failed.

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_responses_received	Counter	The total number of HTTP responses received for the requests punted.
HTTP Punting Metrics	punting_http_responses_forwarded	Counter	The total number of HTTP responses forwarded after punting transaction completion.
HTTP Punting Metrics	punting_http_requests_aborted	Counter	The total number of HTTP requests aborted.
HTTP Punting Metrics	punting_http_request_duration_seconds	Histogram	Punting operation duration in seconds.

HTTP Punting Metrics

- [punting_http_requests_sent](#)

Type

64-bit Counter

Description

The total number of HTTP requests successfully punted.

Labels

message_type

- [punting_http_requests_failed](#)

Type

64-bit Counter

Description

The total number of HTTP requests for which the punting failed.

Labels

message_type

- [punting_http_responses_received](#)

Type

64-bit Counter

Description

The total number of HTTP responses received for the requests punted.

Labels

message_type, status_code

- [punting_http_responses_forwarded](#)

Type

64-bit Counter

Description

The total number of HTTP responses forwarded after punting transaction completion.

Labels

message_type, status_code

- [punting_http_requests_aborted](#)

Type

64-bit Counter

Description

The total number of HTTP requests aborted.

Labels

message_type

- punting_http_request_duration_seconds

Type

64-bit Histogram

Description

The punting transaction duration is measured from the time the HTTP request is sent out to the time the response is received for the punting request.

Labels

message_type

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

Table of Contents

- [Summary](#)
- [Packet Streaming Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Packet Streaming Metrics	num_packet_streaming_attempted	Counter	The total number of attempted streaming of packets
Packet Streaming Metrics	num_packet_streaming_success	Counter	The total number of packets streamed successfully
Packet Streaming Metrics	num_packet_streaming_failed	Counter	The total number of failed streaming of packets
Packet Streaming Metrics	num_packet_streaming_bytes	Counter	The total number of bytes successfully streamed

Packet Streaming Metrics

- num_packet_streaming_attempted

Type

64-bit Counter

Description

The total number of attempted streaming of packets

Labels

interface_name, server_fqdn, kafka_topic
- num_packet_streaming_success

Type

64-bit Counter

Description

The total number of packets streamed successfully

Labels

interface_name, server_fqdn, kafka_topic
- num_packet_streaming_failed

Type

64-bit Counter

Description

The total number of failed streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

• num_packet_streaming_bytes

Type

64-bit Counter

Description

The total number of bytes successfully streamed

Labels

interface_name, server_fqdn, kafka_topic

Table of Contents

- [Summary](#)
- [ChargingCmn statistics](#)

Summary

Group	Metric Name	Type	Synopsis
ChargingCmn statistics	svcmgr_service_available_ind_notification	Counter	How many times, Service Available Indication message received
ChargingCmn statistics	svcmgr_service_available_ind_parse_errs	Counter	How many times, Service Available Indication parse error happened
ChargingCmn statistics	svcmgr_service_available_ind_notification_ack	Counter	How many times, Service Available Indication message is acknowledged
ChargingCmn statistics	svcmgr_service_unavailable_ind_notification	Counter	How many times, Service Unavailable Indication message
ChargingCmn statistics	svcmgr_service_unavailable_ind_parse_errs	Counter	How many times, Service Unavailable Indication parse error happened
ChargingCmn statistics	svcmgr_service_unavailable_ind_notification_ack	Counter	How many times, Service Unavailable Indication message is acknowledged
ChargingCmn statistics	svcmgr_publish_msg_sent	Counter	How many times, Publication message sent
ChargingCmn statistics	svcmgr_publish_msg_success	Counter	How many times, Publication message success response received
ChargingCmn statistics	svcmgr_publish_msg_failure	Counter	How many times, failure received for publication message
ChargingCmn statistics	svcmgr_subscribe_msg_sent	Counter	How many times, Subscribe message sent
ChargingCmn statistics	svcmgr_subscribe_msg_success	Counter	How many times, subscribe message success
ChargingCmn statistics	svcmgr_subscribe_msg_failure	Counter	How many times, failure received for subscribe message
ChargingCmn statistics	svcmgr_unsubscribe_msg_sent	Counter	How many times, unsubscribe message sent
ChargingCmn statistics	svcmgr_unsubscribe_msg_failure	Counter	How many times, failure received for unsubscribe message

Group	Metric Name	Type	Synopsis
ChargingCmn statistics	svcmgr_config_req_sent	Counter	How many times, config request sent
ChargingCmn statistics	svcmgr_config_req_failure	Counter	How many times, failure received for config request
ChargingCmn statistics	svcmgr_config_req_success	Counter	How many times, success received for config request
ChargingCmn statistics	svcmgr_config_register_response	Counter	How many times, config register response received
ChargingCmn statistics	svcmgr_config_register_response_ack	Counter	How many times, config register response is acknowledged
ChargingCmn statistics	svcmgr_config_register_response_parse_err	Counter	How many times, parse error happened for config register response
ChargingCmn statistics	svcmgr_config_update_notification	Counter	How many times, config update notification received
ChargingCmn statistics	svcmgr_config_update_notification_response	Counter	How many times, config update response
ChargingCmn statistics	svcmgr_intf_schema_update_ind_notification	Counter	How many times, interface schema update indication message received
ChargingCmn statistics	svcmgr_intf_schema_update_ind_notification_ack	Counter	How many times, interface schema update indication message is acknowledged
ChargingCmn statistics	svcmgr_intf_schema_update_ind_parse_errs	Counter	How many times, interface schema update indication parse error happened

ChargingCmn statistics

- svcmgr_service_available_ind_notification

Type

64-bit Counter

Description

Number of Times Service Available Indication notification received

Labels

servicename

- svcmgr_service_available_ind_parse_errs

Type

64-bit Counter

Description

Number of Times Service Available Indication parse error happened

Labels

- svcmgr_service_available_ind_notification_ack

Type

64-bit Counter

Description

Number of Times Service Available Indication message is acknowledged

Labels

servicename

- `svcmgr_service_unavailable_ind_notification`

Type
64-bit Counter
Description
Number of Times Service Unavailable Indication notification received
Labels
<i>servicename</i>

- `svcmgr_service_unavailable_ind_parse_errs`

Type
64-bit Counter
Description
Number of Times Service Unavailable Indication parse error happened
Labels

- `svcmgr_service_unavailable_ind_notification_ack`

Type
64-bit Counter
Description
Number of Times Service Unavailable Indication message is acknowledged
Labels
<i>servicename</i>

- `svcmgr_publish_msg_sent`

Type
64-bit Counter
Description
Number of Times Publication message sent
Labels
<i>servicename, retry</i>

- `svcmgr_publish_msg_success`

Type
64-bit Counter
Description
Number of Times Publication message success response received
Labels
<i>servicename, causecode, retry</i>

- `svcmgr_publish_msg_failure`

Type
64-bit Counter
Description
Number of Times failure received for publication message

Labels

servicename, causecode, retry

• **svcmgr_subscribe_msg_sent**

Type

64-bit Counter

Description

Number of Times Subscribe message sent

Labels

servicename, retry

• **svcmgr_subscribe_msg_success**

Type

64-bit Counter

Description

Number of Times Send subscribe message success response received

Labels

servicename, causecode, retry

• **svcmgr_subscribe_msg_failure**

Type

64-bit Counter

Description

Number of time failure received for subscribe message

Labels

servicename, causecode, retry

• **svcmgr_unsubscribe_msg_sent**

Type

64-bit Counter

Description

Number of Times unsubscribe message sent

Labels

servicename

• **svcmgr_unsubscribe_msg_failure**

Type

64-bit Counter

Description

Number of Times failure received for unsubscribe message

Labels

servicename

• **svcmgr_config_req_sent**

Type

64-bit Counter

Description

Number of Times config request sent

Labels

retry

- `svcmgr_config_req_failure`

Type

64-bit Counter

Description

Number of Times failure received for config request

Labels

causecode, retry

- `svcmgr_config_req_success`

Type

64-bit Counter

Description

Number of Times success received for config request

Labels

causecode, retry

- `svcmgr_config_register_response`

Type

64-bit Counter

Description

Number of Times config register response received

Labels

statuscode

- `svcmgr_config_register_response_ack`

Type

64-bit Counter

Description

Number of Times config register response is acknowledged

Labels

- `svcmgr_config_register_response_parse_err`

Type

64-bit Counter

Description

Number of Times parse error happened for config register response

Labels

• svcmgr_config_update_notification
Type
64-bit Counter
Description
Number of Times config update notification received
Labels
• svcmgr_config_update_notification_response
Type
64-bit Counter
Description
Number of Times config update response sent
Labels
• svcmgr_intf_schema_update_ind_notification
Type
64-bit Counter
Description
Number of Times interface schema update indication message received
Labels
servicename
• svcmgr_intf_schema_update_ind_notification_ack
Type
64-bit Counter
Description
Number of Times interface schema update indication message is acknowledged
Labels
servicename
• svcmgr_intf_schema_update_ind_parse_errs
Type
64-bit Counter
Description
Number of Times Service interface schema update indication parse error happened
Labels

Table of Contents

- [Summary](#)
- [ETCD Client Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
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Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_put_requests	Counter	The total number of ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_requests_sent	Counter	The total number of ETCD Client PUT Requests sent.
ETCD Client Metrics	etcd_put_successes	Counter	The total number of successful ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_failures	Counter	The total number of failed ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_with_lease_requests	Counter	The total number of ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_requests_sent	Counter	The total number of ETCD Client PUT with Lease Requests sent.
ETCD Client Metrics	etcd_put_with_lease_successes	Counter	The total number of successful ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_failures	Counter	The total number of failed ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_range_requests	Counter	The total number of ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_requests_sent	Counter	The total number of ETCD Client Range Requests sent.
ETCD Client Metrics	etcd_range_successes	Counter	The total number of successful ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_failures	Counter	The total number of failed ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_delete_requests	Counter	The total number of ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_requests_sent	Counter	The total number of ETCD Client Range Delete Requests sent.
ETCD Client Metrics	etcd_range_delete_successes	Counter	The total number of successful ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_failures	Counter	The total number of failed ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_txn_requests	Counter	The total number of ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_requests_sent	Counter	The total number of ETCD Client Txn Requests sent.
ETCD Client Metrics	etcd_txn_successes	Counter	The total number of successful ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_failures	Counter	The total number of failed ETCD Client Txn Requests.
ETCD Client Metrics	etcd_watch_requests	Counter	The total number of ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_requests_sent	Counter	The total number of ETCD Client Watch Requests sent.
ETCD Client Metrics	etcd_watch_successes	Counter	The total number of successful ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_failures	Counter	The total number of failed ETCD Client Watch Requests.
ETCD Client Metrics	etcdwatcher_requests	Counter	The total number of ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_requests_sent	Counter	The total number of ETCD Watcher Requests sent.
ETCD Client Metrics	etcdwatcher_successes	Counter	The total number of successful ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_failures	Counter	The total number of failed ETCD Watcher Requests.

ETCD Client Metrics

- etcd_put_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests.

Labels

keyPrefix

- etcd_put_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests sent.

Labels

keyPrefix

- etcd_put_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT Requests.

Labels

keyPrefix

- etcd_put_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT Requests.

Labels

keyPrefix, failureReason

- etcd_put_with_lease_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests sent.

Labels

keyPrefix

- etcd_put_with_lease_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT with Lease Requests.

Labels

keyPrefix, failureReason

- etcd_range_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests sent.

Labels

keyPrefix

- etcd_range_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Requests.

Labels

keyPrefix, failureReason

- etcd_range_delete_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests.

Labels

keyPrefix

- etcd_range_delete_requests_sent
 - Type
 - 64-bit Counter
 - Description
 - The total number of ETCD Client Range Delete Requests sent.
 - Labels
 - keyPrefix**
- etcd_range_delete_successes
 - Type
 - 64-bit Counter
 - Description
 - The total number of successful ETCD Client Range Delete Requests.
 - Labels
 - keyPrefix**
- etcd_range_delete_failures
 - Type
 - 64-bit Counter
 - Description
 - The total number of failed ETCD Client Range Delete Requests.
 - Labels
 - keyPrefix, failureReason**
- etcd_txn_requests
 - Type
 - 64-bit Counter
 - Description
 - The total number of ETCD Client Txn Requests.
 - Labels
 - keyPrefix**
- etcd_txn_requests_sent
 - Type
 - 64-bit Counter
 - Description
 - The total number of ETCD Client Txn Requests sent.
 - Labels
 - keyPrefix**
- etcd_txn_successes
 - Type
 - 64-bit Counter
 - Description

The total number of successful ETCD Client Txn Requests.

Labels

keyPrefix

- etcd_txn_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Txn Requests.

Labels

keyPrefix, failureReason

- etcd_watch_requests

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests sent.

Labels

keyPrefix

- etcd_watch_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Watch Requests.

Labels

keyPrefix, failureReason

- etcdwatcher_requests

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests.

Labels

keyPrefix

• etcdwatcher_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests sent.

Labels

keyPrefix

• etcdwatcher_successes

Type

64-bit Counter

Description

The total number of successful ETCD Watcher Requests.

Labels

keyPrefix

• etcdwatcher_failures

Type

64-bit Counter

Description

The total number of failed ETCD Watcher Requests.

Labels

keyPrefix, failureReason

Table of Contents

- [Summary](#)
- [HTTP Punting Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_sent	Counter	The total number of HTTP requests successfully punted.
HTTP Punting Metrics	punting_http_requests_failed	Counter	The total number of HTTP requests punting failed.
HTTP Punting Metrics	punting_http_responses_received	Counter	The total number of HTTP responses received for the requests punted.
HTTP Punting Metrics	punting_http_responses_forwarded	Counter	The total number of HTTP responses forwarded after punting transaction completion.

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Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_aborted	Counter	The total number of HTTP requests aborted.
HTTP Punting Metrics	punting_http_request_duration_seconds	Histogram	Punting operation duration in seconds.

HTTP Punting Metrics

- [punting_http_requests_sent](#)

Type

64-bit Counter

Description

The total number of HTTP requests successfully punted.

Labels

message_type
- [punting_http_requests_failed](#)

Type

64-bit Counter

Description

The total number of HTTP requests for which the punting failed.

Labels

message_type
- [punting_http_responses_received](#)

Type

64-bit Counter

Description

The total number of HTTP responses received for the requests punted.

Labels

message_type, status_code
- [punting_http_responses_forwarded](#)

Type

64-bit Counter

Description

The total number of HTTP responses forwarded after punting transaction completion.

Labels

message_type, status_code
- [punting_http_requests_aborted](#)

Type

64-bit Counter

Description

The total number of HTTP requests aborted.

Labels

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

Summary

Group	Metric Name	Type	Synopsis
Packet Streaming Metrics	num_packet_streaming_attempted	Counter	The total number of attempted streaming of packets
Packet Streaming Metrics	num_packet_streaming_success	Counter	The total number of packets streamed successfully
Packet Streaming Metrics	num_packet_streaming_failed	Counter	The total number of failed streaming of packets
Packet Streaming Metrics	num_packet_streaming_bytes	Counter	The total number of bytes successfully streamed

Packet Streaming Metrics

- ### Description

The total number of failed streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_bytes

Type

64-bit Counter

Description

The total number of bytes successfully streamed

Labels

interface_name, server_fqdn, kafka_topic

Table of Contents

- Summary
- SMF Radius Server Group Statistics
- SMF Radius Server Group Peer Statistics
- SMF Radius Server Statistics
- SMF Radius Server Group Peer Port Statistics
- SMF Radius Inband Gtpu Statistics

Summary

Group	Metric Name	Type	Synopsis
SMF Radius Server Group Statistics	smfcc_radius_server_group_access_req_rej_no_client_intf	Counter	SMF Radius Server Group Request Reject
SMF Radius Server Group Statistics	smfcc_radius_server_group_access_req_rej_client_intf_admin_down	Counter	SMF Radius Server Group Request Reject
SMF Radius Server Group Statistics	smfcc_radius_server_group_tx_send_msg_failure	Counter	SMF Radius Server Group send message failure
SMF Radius Server Group Statistics	smfcc_radius_server_group_tx_send_msg_success	Counter	SMF Radius Server Group send message Success
SMF Radius Server Group Statistics	smfcc_radius_server_group_rx_msg_failure	Counter	SMF Radius Server Group receive messages Failed
SMF Radius Server Group Statistics	smfcc_radius_server_group_rx_msg_success	Counter	SMF Radius Server Group receive message Success
SMF Radius Server Group Statistics	smfcc_radius_server_group_rx_server_timeout	Counter	SMF Radius Server Group receive message timeout
SMF Radius Server Group Peer Statistics	smfcc_radius_server_group_peer_tx_buffer_unavailable	Counter	Radius Server Group Peer tx buffer unavailable
SMF Radius Server Group Peer Statistics	smfcc_radius_server_group_peer_tx_encode_error	Counter	Radius Server Group Peer tx encode error
SMF Radius Server Group Peer Statistics	smfcc_radius_server_group_peer_tx_start_timer_failure	Counter	Radius Server Group Peer tx start timer failure
SMF Radius Server Group Peer Statistics	smfcc_radius_server_group_peer_tx_send_msg_failure	Counter	Radius Server Group Peer tx send message failure
SMF Radius Server Group Peer Statistics	smfcc_radius_server_group_peer_tx_send_msg_success	Counter	Radius Server Group Peer tx send message success
SMF Radius Server Group Peer Statistics	smfcc_radius_server_group_peer_re_tx_send_msg_failure	Counter	Radius Server Group Peer tx send message failure
SMF Radius Server Group Peer Statistics	smfcc_radius_server_group_peer_re_tx_send_msg_success	Counter	Radius Server Group Peer tx send message success
SMF Radius Server Group Peer Statistics	smfcc_radius_server_group_peer_rx_no_port_found	Counter	Radius Server Group Peer tx buffer unavailable

Group	Metric Name	Type	Synopsis
SMF Radius Server Group Peer Statistics	smfcc_radius_server_group_peer_rx_invalid_msg_len	Counter	Radius Server Group Peer tx buffer unavailable
SMF Radius Server Group Peer Statistics	smfcc_radius_server_group_peer_rx_verify_reply_failure	Counter	Radius Server Group Peer tx encode error
SMF Radius Server Group Peer Statistics	smfcc_radius_server_group_peer_rx_buffer_unavailable	Counter	Radius Server Group Peer tx start timer failure
SMF Radius Server Group Peer Statistics	smfcc_radius_server_group_peer_rx_decode_error	Counter	Radius Server Group Peer tx send message failure
SMF Radius Server Group Peer Statistics	smfcc_radius_server_group_peer_rx_server_timeout	Counter	Radius Server Group Peer tx send message success
SMF Radius Server Group Peer Statistics	smfcc_radius_server_group_peer_rx_failure	Counter	Radius Server Group Peer tx send message failure
SMF Radius Server Group Peer Statistics	smfcc_radius_server_group_peer_rx_success	Counter	Radius Server Group Peer tx send message success
SMF Radius Server Group Peer Statistics	smfcc_radius_server_group_peer_re_rx_failure	Counter	Radius Server Group Peer tx send message failure
SMF Radius Server Group Peer Statistics	smfcc_radius_server_group_peer_re_rx_success	Counter	Radius Server Group Peer tx send message success
SMF Radius Server Statistics	smfcc_radius_module_resp_tx_get_handle_failure	Counter	Radius Server Group Peer
SMF Radius Server Statistics	smfcc_radius_module_resp_tx_send_failure	Counter	Radius Server Group Peer
SMF Radius Server Statistics	smfcc_radius_module_access_req_tx_invalid_service_id	Counter	Radius Server Group Peer
SMF Radius Server Statistics	smfcc_radius_module_access_req_tx_invalid_group_id	Counter	Radius Server Group Peer
SMF Radius Server Statistics	smfcc_radius_module_access_req_tx_success	Counter	Radius Server Group Peer
SMF Radius Server Statistics	smfcc_radius_module_access_req_tx_failure	Counter	Radius Server Group Peer
SMF Radius Server Statistics	smfcc_radius_module_access_resp_rx_failure	Counter	Radius Server Group Peer
SMF Radius Server Statistics	smfcc_radius_module_access_accept_resp_rx_success	Counter	Radius Server Group Peer
SMF Radius Server Statistics	smfcc_radius_module_access_accept_resp_rx_failure	Counter	Radius Server Group Peer
SMF Radius Server Statistics	smfcc_radius_module_access_challenge_resp_rx_failure	Counter	Radius Server Group Peer
SMF Radius Server Statistics	smfcc_radius_module_access_challenge_resp_rx_success	Counter	Radius Server Group Peer
SMF Radius Server Statistics	smfcc_radius_module_access_reject_resp_rx_failure	Counter	Radius Server Group Peer
SMF Radius Server Statistics	smfcc_radius_module_access_reject_resp_rx_success	Counter	Radius Server Group Peer
SMF Radius Server Statistics	smfcc_radius_module_acct_record_req_tx_invalid_group_id	Counter	Radius Server Group Peer
SMF Radius Server Statistics	smfcc_radius_module_acct_record_req_tx_success	Counter	Radius Server Group Peer
SMF Radius Server Statistics	smfcc_radius_module_acct_record_req_tx_failure	Counter	Radius Server Group Peer
SMF Radius Server Group Peer Port Statistics	smfcc_radius_client_port_tx_socket_inactive	Counter	Radius Server Group Peer Port tx sequence unavailable
SMF Radius Server Group Peer Port Statistics	smfcc_radius_client_port_tx_seq_num_unavailable	Counter	Radius Server Group Peer Port tx sequence unavailable
SMF Radius Server Group Peer Port Statistics	smfcc_radius_client_port_tx_send_message_failure	Counter	Radius Server Group Peer Port tx send send message failure
SMF Radius Server Group Peer Port Statistics	smfcc_radius_client_port_tx_send_message_success	Counter	Radius Server Group Peer Port
SMF Radius Server Group Peer Port Statistics	smfcc_radius_client_port_rx_message_node_not_found	Counter	Radius Server Group Peer Port
SMF Radius Server Group Peer Port Statistics	smfcc_radius_client_port_rx_failure	Counter	Radius Server Group Peer Port
SMF Radius Server Group Peer Port Statistics	smfcc_radius_client_port_rx_success	Counter	Radius Server Group Peer tx buffer unavailable
SMF Radius Inband Gtpu Statistics	smfcc_radius_inband_punting_rx_failure	Counter	no of rx failure while processing gtpu packet. with no port mapping

Group	Metric Name	Type	Synopsis
SMF Radius Inband Gtpu Statistics	smfcc_radius_inband_punting_rx_success	Counter	no of rx success while processing gtpu packet. with port mapping
SMF Radius Inband Gtpu Statistics	smfcc_radius_inband_punting_self	Counter	no of rx pkts punted to self
SMF Radius Inband Gtpu Statistics	smfcc_radius_inband_gtpu_send_success	Counter	success no of send pkts over gtpu tunnel
SMF Radius Inband Gtpu Statistics	smfcc_radius_inband_gtpu_send_failure	Counter	failure no of send pkts over gtpu tunnel
SMF Radius Inband Gtpu Statistics	smfcc_radius_inband_gtpu_receive_success	Counter	success no of rx pkts over gtpu tunnel
SMF Radius Inband Gtpu Statistics	smfcc_radius_inband_gtpu_receive_failure	Counter	failure no of rx pkts over gtpu tunnel
SMF Radius Inband Gtpu Statistics	smfcc_radius_inband_gtpu_missing_port_map	Counter	no of missing port map for pkts received over gtpu tunnel
SMF Radius Inband Gtpu Statistics	smfcc_radius_inband_gtpu_rx_error	Counter	no of gtpu error rx pkts

SMF Radius Server Group Statistics

- smfcc_radius_server_group_access_req_rej_no_client_intf

Type

64-bit Counter

Description

Number of Access Request Reject with no client interface

Labels

name

- smfcc_radius_server_group_access_req_rej_client_intf_admin_down

Type

64-bit Counter

Description

Number of Access Request Reject with client interface admin down

Labels

name

- smfcc_radius_server_group_tx_send_msg_failure

Type

64-bit Counter

Description

Number of send messages failed

Labels

name

- smfcc_radius_server_group_tx_send_msg_success

Type

64-bit Counter

Description

Number of send messages success

Labels

name

• smfcc_radius_server_group_rx_msg_failure
Type
64-bit Counter
Description
Number of receive messages Failed
Labels
<i>name</i>
• smfcc_radius_server_group_rx_msg_success
Type
64-bit Counter
Description
Number of receive messages success
Labels
<i>name</i>
• smfcc_radius_server_group_rx_server_timeout
Type
64-bit Counter
Description
Number of receive server timeout
Labels
<i>name</i>

SMF Radius Server Group Peer Statistics

• smfcc_radius_server_group_peer_tx_buffer_unavailable
Type
64-bit Counter
Description
no of tx buffer unavailable
Labels
<i>group, peer</i>
• smfcc_radius_server_group_peer_tx_encode_error
Type
64-bit Counter
Description
no of tx encode error
Labels
<i>group, peer</i>
• smfcc_radius_server_group_peer_tx_start_timer_failure
Type
64-bit Counter

Description

no of tx start timer failure

Labels

group, peer

- smfcc_radius_server_group_peer_tx_send_msg_failure

Type

64-bit Counter

Description

Number of send messages failed

Labels

group, peer

- smfcc_radius_server_group_peer_tx_send_msg_success

Type

64-bit Counter

Description

Number of send message success

Labels

group, peer

- smfcc_radius_server_group_peer_re_tx_send_msg_failure

Type

64-bit Counter

Description

Number of send messages failed

Labels

group, peer

- smfcc_radius_server_group_peer_re_tx_send_msg_success

Type

64-bit Counter

Description

Number of send message success

Labels

group, peer

- smfcc_radius_server_group_peer_rx_no_port_found

Type

64-bit Counter

Description

no of rx no port found

Labels

group, peer

smfcc_radius_server_group_peer_rx_invalid_msg_len
Type
64-bit Counter
Description
no of rx invalid msg length
Labels
group, peer
smfcc_radius_server_group_peer_rx_verify_reply_failure
Type
64-bit Counter
Description
no of tx encode error
Labels
group, peer
smfcc_radius_server_group_peer_rx_buffer_unavailable
Type
64-bit Counter
Description
no of tx start timer failure
Labels
group, peer
smfcc_radius_server_group_peer_rx_decode_error
Type
64-bit Counter
Description
Number of send messages failed
Labels
group, peer
smfcc_radius_server_group_peer_rx_server_timeout
Type
64-bit Counter
Description
Number of send message success
Labels
group, peer
smfcc_radius_server_group_peer_rx_failure
Type
64-bit Counter
Description

Number of send messages failed

Labels

group, peer

- smfcc_radius_server_group_peer_rx_success

Type

64-bit Counter

Description

Number of send message success

Labels

group, peer

- smfcc_radius_server_group_peer_re_rx_failure

Type

64-bit Counter

Description

Number of send messages failed

Labels

group, peer

- smfcc_radius_server_group_peer_re_rx_success

Type

64-bit Counter

Description

Number of send message success

Labels

group, peer

SMF Radius Server Statistics

- smfcc_radius_module_resp_tx_get_handle_failure

Type

64-bit Counter

Description

no of tx buffer unavailable

Labels

name

- smfcc_radius_module_resp_tx_send_failure

Type

64-bit Counter

Description

Number of send message failure

Labels

name

smfcc_radius_module_access_req_tx_invalid_service_id
Type
64-bit Counter
Description
Number of receive invalid service id
Labels
name
smfcc_radius_module_access_req_tx_invalid_group_id
Type
64-bit Counter
Description
Number of receive invalid group id
Labels
name
smfcc_radius_module_access_req_tx_success
Type
64-bit Counter
Description
Number of receive success
Labels
name
smfcc_radius_module_access_req_tx_failure
Type
64-bit Counter
Description
Number of receive failure
Labels
name
smfcc_radius_module_access_resp_rx_failure
Type
64-bit Counter
Description
Number of send failure
Labels
name
smfcc_radius_module_access_accept_resp_rx_success
Type
64-bit Counter
Description

Number of accept response send success

Labels

name

- smfcc_radius_module_access_accept_resp_rx_failure

Type

64-bit Counter

Description

Number of accept response send failure

Labels

name

- smfcc_radius_module_access_challenge_resp_rx_failure

Type

64-bit Counter

Description

Number of challenge response send failure

Labels

name

- smfcc_radius_module_access_challenge_resp_rx_success

Type

64-bit Counter

Description

Number of challenge response send success

Labels

name

- smfcc_radius_module_access_reject_resp_rx_failure

Type

64-bit Counter

Description

Number of reject response send failure

Labels

name

- smfcc_radius_module_access_reject_resp_rx_success

Type

64-bit Counter

Description

Number of reject response send success

Labels

name

- smfcc_radius_module_acct_record_req_tx_invalid_group_id

Type

64-bit Counter

Description

Number of acct request received with invalid group id

Labels

name

- smfcc_radius_module_acct_record_req_tx_success

Type

64-bit Counter

Description

Number of acct request received success

Labels

name

- smfcc_radius_module_acct_record_req_tx_failure

Type

64-bit Counter

Description

Number of acct request received failure

Labels

name

SMF Radius Server Group Peer Port Statistics

- smfcc_radius_client_port_tx_socket_inactive

Type

64-bit Counter

Description

no of tx socket inactive

Labels

name, port

- smfcc_radius_client_port_tx_seq_num_unavailable

Type

64-bit Counter

Description

no of tx sequence number unavailable

Labels

name, port

- smfcc_radius_client_port_tx_send_message_failure

Type

64-bit Counter

Description

no of tx send message failure

Labels

name, port

- smfcc_radius_client_port_tx_send_message_success

Type

64-bit Counter

Description

no of tx send message success

Labels

name, port

- smfcc_radius_client_port_rx_message_node_not_found

Type

64-bit Counter

Description

no of rx message node not found

Labels

name, port

- smfcc_radius_client_port_rx_failure

Type

64-bit Counter

Description

no of rx failure

Labels

name, port

- smfcc_radius_client_port_rx_success

Type

64-bit Counter

Description

no of rx success

Labels

name, port

SMF Radius Inband Gtpu Statistics

- smfcc_radius_inband_punting_rx_failure

Type

64-bit Counter

Description

no of rx failure while processing gtpu packet. with no port mapping

Labels

name, port

smfcc_radius_inband_punting_rx_success
Type
64-bit Counter
Description
no of rx success while processing gtpu packet. with port mapping
Labels
name
smfcc_radius_inband_punting_self
Type
64-bit Counter
Description
no of rx pkts punted to self
Labels
name, port
smfcc_radius_inband_gtpu_send_success
Type
64-bit Counter
Description
success no of send pkts over gtpu tunnel
Labels
name, role
smfcc_radius_inband_gtpu_send_failure
Type
64-bit Counter
Description
failure no of send pkts over gtpu tunnel
Labels
name
smfcc_radius_inband_gtpu_receive_success
Type
64-bit Counter
Description
success no of rx pkts over gtpu tunnel
Labels
name, role
smfcc_radius_inband_gtpu_receive_failure
Type
64-bit Counter
Description

failure no of rx pkts over gtpu tunnel

Labels

name

- smfcc_radius_inband_gtpu_missing_port_map

Type

64-bit Counter

Description

no of missing port map for pkts received over gtpu tunnel

Labels

name, role, key

- smfcc_radius_inband_gtpu_rx_error

Type

64-bit Counter

Description

no of gtpu error rx pkts

Labels

name, role

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- [Summary](#)
- [Circuit Breaker Metrics](#)
- [Service Endpoint Learner Metrics](#)
- [Service Endpoint Manager Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Circuit Breaker Metrics	sem_cb_endpoints	Gauge	Total number of endpoints for a service in respective states
Service Endpoint Learner Metrics	sem_num_dns_errors	Counter	Total number of DNS resolution errors for a given service
Service Endpoint Learner Metrics	sem_num_endpoints	Gauge	Total number of endpoints learnt for a service
Service Endpoint Manager Metrics	sem_num_endpoint_transactions_sent	Counter	Total number of times an endpoint is given to the application for communication with it
Service Endpoint Manager Metrics	sem_num_endpoint_transaction_status_received	Counter	Total number of transactions for a given endpoint received with the transaction status

Circuit Breaker Metrics

- sem_cb_endpoints

Type

64-bit Gauge

Description

Total number of endpoints for a service in respective states. There are 3 states viz. Active, Blocked, Restoring. In Active state, the endpoint is responding as expected, i.e. the response code doesn't indicate a service failure. The endpoint will be moved to Blocked state based on response failures which indicate service failure. After a timeout, the endpoint will be moved to Restoring state to check if the endpoint has recovered. In this state, if endpoint responds with response code other than service failure, it will be moved to Active state. Else, it will be moved back to Blocked state.

Labels

service_name, endpoint_state

Service Endpoint Learner Metrics

- sem_num_dns_errors

Type

64-bit Counter

Description

Total number of DNS resolution errors for a given service, labels are service name for DNS resolution and DNS error codes

Labels

service_name, error_code

- sem_num_endpoints

Type

64-bit Gauge

Description

Total number of endpoints learnt from the Kubernetes DNS server for a given service

Labels

service

Service Endpoint Manager Metrics

- sem_num_endpoint_transactions_sent

Type

64-bit Counter

Description

Total number of times an endpoint is given to the application for communication with it

Labels

service

- sem_num_endpoint_transaction_status_received

Type

64-bit Counter

Description

Total number of transactions for a given endpoint received with the transaction status

Labels

service, status

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- [Summary](#)
- [REDIS Cluster Stats](#)
- [REDIS Replication Stats](#)
- [REDIS Cluster Persistence Stats](#)
- [REDIS Memory Stats](#)
- [REDIS CPU Stats](#)
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Summary

Group	Metric Name	Type	Synopsis
REDIS Cluster Stats	redis_uptime_in_seconds	Counter	Redis server uptime
REDIS Cluster Stats	redis_commands_processed_total	Counter	Total number of commands processed by the server
REDIS Cluster Stats	redis_rejected_connections_total	Counter	Total number of connections rejected.
REDIS Cluster Stats	redis_replica_resyncs_full	Counter	Total number of full resyncs
REDIS Cluster Stats	redis_expired_keys_total	Counter	Total number of expired keys.
REDIS Cluster Stats	redis_evicted_keys_total	Counter	Total number of evicted keys.
REDIS Cluster Stats	redis_keyspace_hits_total	Counter	Total number of successful lookup in db
REDIS Cluster Stats	redis_keyspace_misses_total	Counter	Total number of failed lookup in db
REDIS Cluster Stats	redis_total_error_replies	Counter	Total number of issued error replies
REDIS Cluster Stats	redis_tracking_total_keys	Counter	Total number of keys being tracked by the server
REDIS Cluster Stats	redis_defrag_hits	Counter	Total number of keys that were actively defragmented
REDIS Cluster Stats	redis_defrag_misses	Counter	Total number of keys skipped by the active defragmentation process
REDIS Replication Stats	redis_connected_slaves	Counter	Total number of connected replicas
REDIS Replication Stats	redis_repl_backlog_is_active	Counter	Flag indicating replication backlog is active
REDIS Replication Stats	redis_master_sync_in_progress	Counter	Indicate the master is syncing to the replica
REDIS Cluster Persistence Stats	redis_rdb_bgsave_in_progress	Counter	Redis RDB save status
REDIS Cluster Persistence Stats	rdb_changes_since_last_save	Counter	Total number of changes since the last dump
REDIS Memory Stats	redis_memory_used_bytes	Counter	Total number of bytes allocated by Redis.
REDIS Memory Stats	redis_memory_max_bytes	Counter	Total value of max_memory allocated to redis.
REDIS Memory Stats	redis_mem_fragmentation_ratio	Counter	Ratio between used_memory_rss and used_memory.
REDIS Memory Stats	redis_allocator_rss_ratio	Counter	Ratio between allocator_resident and allocator_active
REDIS CPU Stats	redis_cpu_sys_seconds_total	Counter	System CPU consumed by the Redis server
REDIS CPU Stats	redis_cpu_user_seconds_total	Counter	User CPU consumed by the Redis server
REDIS Clients Stats	redis_connected_clients	Counter	Total number of Redis client connections
REDIS Clients Stats	redis_cluster_connections	Counter	Total number of sockets opened.
REDIS Clients Stats	redis_config_maxclients	Counter	Total sum of connected_clients, connected_slaves and cluster_connections.
REDIS Clients Stats	redis_blocked_clients	Counter	Total number of clients pending on a blocking call.
REDIS Clients Stats	clients_in_timeout_table	Counter	Total number of clients in the clients timeout table.

REDIS Cluster Stats

- redis_uptime_in_seconds

Type

64-bit Counter

Description

Number of seconds since Redis server start

Labels

server

- redis_commands_processed_total

Type

64-bit Counter

Description

Total number of commands processed by the server

Labels

commands

- redis_rejected_connections_total

Type

64-bit Counter

Description

Number of connections rejected because of maxclients limit

Labels

rejected_connections

- redis_replica_resyncs_full

Type

64-bit Counter

Description

The number of full resyncs with replicas

Labels

replica_resync

- redis_expired_keys_total

Type

64-bit Counter

Description

Total number of key expiration events

Labels

expired_keys

- redis_evicted_keys_total

Type

64-bit Counter

Description

The number of evicted keys due to maxmemory limit

Labels

evicted_keys

- redis_keyspace_hits_total

Type

64-bit Counter

Description

Number of successful lookup of keys in the main dictionary

Labels

keyspace_hits

- redis_keyspace_misses_total

Type

64-bit Counter

Description

Number of failed lookup of keys in the main dictionary

Labels

keyspace_misses

- redis_total_error_replies

Type

64-bit Counter

Description

Total number of issued error replies, that is the sum of rejected commands (errors prior command execution) and failed commands (errors within the command execution)

Labels

error_replies

- redis_tracking_total_keys

Type

64-bit Counter

Description

Number of keys being tracked by the server

Labels

tracked_keys

- redis_defrag_hits

Type

64-bit Counter

Description

Number of keys that were actively defragmented

Labels

defrag_hits

- redis_defrag_misses

Type

64-bit Counter

Description

Number of keys that were skipped by the active defragmentation process

Labels

defrag_misses

REDIS Replication Stats

- redis_connected_slaves

Type

64-bit Counter

Description

Number of connected replicas

Labels

connected_replicas

- redis_repl_backlog_is_active

Type

64-bit Counter

Description

Flag indicating replication backlog is active

Labels

repl_backlog

- redis_master_sync_in_progress

Type

64-bit Counter

Description

Indicate the master is syncing to the replica

Labels

master_sync

REDIS Cluster Persistence Stats

- redis_rdb_bgsave_in_progress

Type

64-bit Counter

Description

Flag indicating a RDB save is on-going

Labels

rdb_bgsave

- rdb_changes_since_last_save

Type

64-bit Counter

Description

Total number of changes since the last dump

Labels

rdb_changes

REDIS Memory Stats

- redis_memory_used_bytes

Type

64-bit Counter

Description

Total number of bytes allocated by Redis using its allocator (either standard libc, jemalloc, or an alternative allocator such as tcmalloc)

Labels

used_memory

- redis_memory_max_bytes

Type

64-bit Counter

Description

The value of the maxmemory configuration directive

Labels

max_memory

- redis_mem_fragmentation_ratio

Type

64-bit Counter

Description

Ratio between used_memory_rss and used_memory. Note that this doesn't only includes fragmentation, but also other process overheads (see the allocator_* metrics), and also overheads like code, shared libraries, stack, etc.

Labels

used_memory_rss, used_memory

- redis_allocator_rss_ratio

Type

64-bit Counter

Description

Ratio between allocator_resident and allocator_active. This usually indicates pages that the allocator can and probably will soon release back to the OS.

Labels

allocator_resident, allocator_active

REDIS CPU Stats

- redis_cpu_sys_seconds_total

Type

64-bit Counter

Description

System CPU consumed by the Redis server, which is the sum of system CPU consumed by all threads of the server process (main thread and background threads)

Labels

cpu_sys_seconds

- redis_cpu_user_seconds_total

Type

64-bit Counter

Description

User CPU consumed by the Redis server, which is the sum of user CPU consumed by all threads of the server process (main thread and background threads)

Labels

cpu_user_seconds

REDIS Clients Stats

- redis_connected_clients

Type

64-bit Counter

Description

Number of client connections (excluding connections from replicas)

Labels

clients

- redis_cluster_connections

Type

64-bit Counter

Description

An approximation of the number of sockets used by the cluster's bus

Labels

socket

- redis_config_maxclients

Type

64-bit Counter

Description

The value of the maxclients configuration directive. This is the upper limit for the sum of connected_clients, connected_slaves and cluster_connections

Labels

max_clients, connected_slaves, connected_clients, cluster_connections

- redis_blocked_clients

Type

64-bit Counter

Description

Number of clients pending on a blocking call (BLPOP, BRPOP, BRPOPLPUSH, BLMOVE, BZPOPMIN, BZPOPMAX)

Labels

blocked_clients

- clients_in_timeout_table

Type

64-bit Counter

Description

Number of clients in the clients timeout table

Labels

clients_timeout_table

Summary

1. [redis-cluster.alert.rules Alerts](#)

redis-cluster.alert.rules

Alert Name	Description	Alert Info	Diagnostics
RedisInstanceDown	Redis instance is down.	1. Summary: redis instance is down Severity: critical Realert Time: 5m Expression: <pre>redis_up == 0</pre>	1. Potential Impact: Applications using redis for stroing data will face issues doing redis operations as server is down. Repair Action: Verify why redis cluster is down using container logs, check cluster state.
RedisClusterRestartingFrequently	redis server is unavailable.	1. Summary: redis server is unavailable Severity: critical Realert Time: 5m Expression: <pre>redis_uptime_in_seconds < 300</pre>	1. Potential Impact: Applications using redis clusters might face some problem doing any operations as server is unavailable. Repair Action: Verify why redis cluster is unavailable using container logs, can check memory issues by describing pods.
RedisUsageOfNetworkConnections	Usage of network connections in redis is over 85 percent.	1. Summary: Usage of network connections in redis is over 85 percent Severity: critical Realert Time: 5m Expression: <pre>((redis_cluster_connections + redis_connected_clients) / 10000) > 0.85</pre>	1. Potential Impact: Applications using redis might face issues while making new connections. Repair Action: usage of network connections in redis is over 85 percent, reduce connections.
RedisMemoryFragmentationRatio	Redis mem fragmentation ratio is over 1.5	1. Summary: Redis mem fragmentation ratio is over 1.5 Severity: major Realert Time: 15m Expression: <pre>redis_allocator_frag_ratio > 1.5</pre>	1. Potential Impact: Applications using redis cluster shall restart the redis server to ensure optimal use of memory among all instances/pods. Repair Action: Restart the Redis server to ensure optimal use of the memory or distribute the memory optimally.
RedisMemoryUsage	Redis memory usage > 75	1. Summary: Redis memory usage > 75 Severity: critical Realert Time: 10m Expression: <pre>(100 * (redis_memory_used_bytes / redis_memory_max_bytes)) > 75</pre>	1. Potential Impact: Applications using redis server should check memory leakage issue, its degrading and data cannot be stored further. Repair Action: Check the memory leakage in redis server

Alert Name	Description	Alert Info	Diagnostics
RedisCacheHitRatiosLow	Redis is experiencing low hit ratio	1. Summary: Redis is experiencing low hit ratio < 0.9 Severity: major Realert Time: 5m Expression: <pre>(rate(redis_keyspace_hits_total[5m]) / (rate(redis_keyspace_misses_total[5m]) + rate(redis_keyspace_hits_total[5m]))) < 0.9</pre>	1. Potential Impact: Applications using redis cluster might face delay in response time. Repair Action: A low hit ratio can cause an increase in the response time of your Redis database
RedisIsEvictingKey	High number of evicted keys from redis	1. Summary: High number of evicted keys from redis Severity: critical Realert Time: 5m Expression: <pre>redis_evicted_keys_total > 10</pre>	1. Potential Impact: Applications using redis cluster please check if there is any issue with memory usage stats as high number of keys are getting evicted from server. Repair Action: Having a high cache hit ratio or a high number of evicted keys can be a symptom of low available memory, check memory stats.
RedisAverageResponseTime	Average response time is over 250ms for redis queries	1. Summary: Average response time is over 250ms for redis queries Severity: critical Realert Time: 5m Expression: <pre>(avg(irate(redis_commands_duration_seconds_total[5m])) / avg(irate(redis_commands_total[5m])) * 1000) > 0.25</pre>	1. Potential Impact: Applications using redis server should check memory stats, any specific load causing more response time. Repair Action: Check possibility why redis server is having latency issues, check memory, load and long running queries.
RedisDisconnectedSlaves	Redis is having disconnected slaves with master	1. Summary: Redis is having disconnected slaves with master Severity: critical Realert Time: 5m Expression: <pre>sum(redis_instance_info{role='slave'}) - sum(redis_instance_info{role='master'}) < 0</pre>	1. Potential Impact: Applications using redis server might face data replication issues and replicas might be unavailable as there are some disconnected slaves from master. Repair Action: Redis is having disconnected slaves with master, check replicas cluster state.
RedisMissingSlots	Redis cluster has missing slots	1. Summary: Redis cluster has missing slots Severity: major Realert Time: 1m Expression: <pre>((16384 - redis_cluster_slots_ok) + (16384 - redis_cluster_slots_assigned) + redis_cluster_slots_fail) > 0</pre>	1. Potential Impact: Applications using redis cluster might face data inconsistency and some keys might be unavailable due to missing slots. Repair Action: Check the redis cluster configuration and ensure all slots are assigned and available. Follow the Redis missing slot troubleshooting guide.

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- [Summary](#)
- [Metric based alerts for redis](#)

Summary

Group	Metric Name	Type	Synopsis
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Group	Metric Name	Type	Synopsis
Metric based alerts for redis	redis_down	Counter	None

Metric based alerts for redis

- redis_down

Type

64-bit Counter

Description

The redis instance is down

Labels

instance

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- [Summary](#)
- [REDIS Operator Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
REDIS Operator Metrics	controller_runtime_reconcile_total	Counter	Total number of reconciliations per controller
REDIS Operator Metrics	controller_runtime_reconcile_errors_total	Counter	Total number of reconciliation errors per controller
REDIS Operator Metrics	controller_runtime_reconcile_time_seconds	Counter	Length of time per reconciliation per controller
REDIS Operator Metrics	controller_runtime_max_concurrent_reconciles	Counter	Maximum number of concurrent reconciles per controller
REDIS Operator Metrics	controller_runtime_active_workers	Counter	Number of currently used workers per controller

REDIS Operator Metrics

- controller_runtime_reconcile_total

Type

64-bit Counter

Description

ReconcileTotal is a prometheus counter metrics which holds the total number of reconciliations per controller. It has two labels. controller label refers to the controller name and result label refers to the reconcile result i.e. success, error, requeue, requeue_after.

Labels

reconciliations, result, controller

- controller_runtime_reconcile_errors_total

Type

64-bit Counter

Description

ReconcileErrors is a prometheus counter metrics which holds the total number of errors from the Reconciler.

Labels

controller, reconciliation_errors

- controller_runtime_reconcile_time_seconds

Type

64-bit Counter

Description

ReconcileTime is a prometheus metric which keeps track of the duration of reconciliations.

Labels

controller, reconcile_time

- controller_runtime_max_concurrent_reconciles

Type

64-bit Counter

Description

WorkerCount is a prometheus metric which holds the number of concurrent reconciles per controller.

Labels

controller, max_concurrent_reconciles

- controller_runtime_active_workers

Type

64-bit Counter

Description

ActiveWorkers is a prometheus metric which holds the number of active workers per controller.

Labels

controller, runtime_active_workers

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- [Syncer Metrics for Redis](#)
- [Syncer Metrics for RedisConnection](#)
- [Metrics for Sites](#)
- [Metrics for RedisParser](#)
- [Metrics for Redis Context](#)
- [Metrics for InitialSync](#)

Summary

Group	Metric Name	Type	Synopsis
Syncer Metrics for Redis	local_redis_attempts	Counter	Total number of connection attempts towards local redis node.
Syncer Metrics for Redis	local_redis_connect_failures	Counter	Total number of connection failures towards local redis node.
Syncer Metrics for Redis	local_redis_connects	Counter	Total number of successful connects towards local redis node.
Syncer Metrics for Redis	local_redis_connect_timeouts	Counter	Total number of connect timeouts towards local redis node.
Syncer Metrics for Redis	local_redis_connect_retries	Counter	Total number of connect retries towards local redis node.
Syncer Metrics for Redis	local_redis_read_operations	Counter	Total number of read operations happened for every site towards local redis node.
Syncer Metrics for Redis	local_redis_write_operations	Counter	Total number of write operations for every site towards local redis node.
Syncer Metrics for Redis	local_redis_initial_sync_opns_errors	Counter	Total number of errors encountered while reading/writing bytes towards local redis node during initial sync.
Syncer Metrics for Redis	local_redis_connection_status	Gauge	Determine status that tcp connection is up/down.
Syncer Metrics for Redis	local_redis_read_bytes	Gauge	Total number of bytes read by every site towards local redis node.
Syncer Metrics for Redis	local_redis_writes_bytes	Gauge	Total number of bytes written by every site towards local redis node.
Syncer Metrics for Redis	local_redis_initial_sync_read_bytes	Gauge	Total number of bytes read towards local redis node during initial sync.

Group	Metric Name	Type	Synopsis
Syncer Metrics for Redis	local_redis_initial_sync_write_bytes	Gauge	Total number of bytes sent towards local redis node during initial sync.
Syncer Metrics for RedisConnection	redis_requests	Counter	Total number of requests sent to redis.
Syncer Metrics for RedisConnection	redis_replies	Counter	Total number of replies received by redis which can replyEmpty/ParseFailure or success.
Syncer Metrics for RedisConnection	redis_state	Counter	Total number of states changed by redis, where states can be checked by currentState/newState.
Metrics for Sites	site_consuming_paused	Gauge	Total number of sites paused for consuming data.
Metrics for Sites	site_state	Gauge	Current state of the remote site
Metrics for Sites	site_events	Gauge	Total number of event for a remote site.
Metrics for Sites	site_event_discards	Gauge	Total number of events discarded for a remote site.
Metrics for Sites	site_cmds_sent	Gauge	Total number of commands sent by every site.
Metrics for Sites	site_parse_callbacks	Gauge	Total number of callbacks for parsing.
Metrics for Sites	site_commands_discards	Gauge	Total number of commands discarded by every site.
Metrics for RedisParser	parser_parsed_cmds	Gauge	Total number of cmds parsed.
Metrics for RedisParser	parser_cmds_sent	Gauge	Total number of cmds sent.
Metrics for RedisParser	parser_cmds_discards	Gauge	Total number of cmds discarded while parsing.
Metrics for Redis Context	redis_context_connection_attempts	Counter	Total number of connection attempts towards remote redis node.
Metrics for Redis Context	redis_context_connection_connects	Counter	Total number of successful connects towards remote redis node.
Metrics for Redis Context	redis_context_connection_failures	Counter	Total number of connection failures towards remote redis node.
Metrics for Redis Context	redis_context_connection_disconnects	Counter	Total number of connection disconnects towards remote redis node.
Metrics for Redis Context	redis_context_commands_buffer_full	Counter	Number of command buffer full instances that has happened.
Metrics for Redis Context	redis_context_connection_status	Gauge	Determine status that redis connection is up/down.
Metrics for Redis Context	redis_context_commands_buffered	Gauge	Total number of commands in buffer.
Metrics for Redis Context	redis_context_commands_sent	Gauge	Total number of commands sent to the remote site.
Metrics for Redis Context	redis_context_command_send_failures	Gauge	Total number of command send failures.
Metrics for Redis Context	redis_context_command_responses	Gauge	Total number of command send failures.
Metrics for Redis Context	redis_context_cmds_latency	Histogram	Roundtrip duration taken by commands sent from syncer to redis in seconds.
Metrics for InitialSync	initial_sync_requests	Counter	Total number of psync commands to local redis during initial sync.
Metrics for InitialSync	initial_sync_commands_decoding_errors	Counter	Total number of type of errors occured for parsing during initial sync
Metrics for InitialSync	last_initial_sync_start_time	Gauge	It refers to the time when the previous initial sync occurred.
Metrics for InitialSync	last_initial_sync_duration	Gauge	Duration of initial sync when the previous initial sync occurred.
Metrics for InitialSync	last_initial_sync_status	Gauge	Status of the last initial sync whether success/failure.
Metrics for InitialSync	last_initial_sync_type	Gauge	Initial sync's last synced type..which can be full/partial
Metrics for InitialSync	initial_sync_commands_discards	Gauge	Total number of commands discarded by every site.
Metrics for InitialSync	initial_sync_commands_parsed	Gauge	Total number of commands parsed by every site during initial sync.

Syncer Metrics for Redis

- local_redis_attempts

Type

64-bit Counter

Description

Total number of connection attempts towards local redis node, where siteName will indicate site's name.

Labels

siteName

- local_redis_connect_failures

Type

64-bit Counter

Description

Total number of connection failures towards local redis node, where siteName will indicate site's name & errorMsg indicate the failure.

Labels

siteName, errorMsg

- local_redis_connects

Type

64-bit Counter

Description

Total number of successful connects towards local redis node, where siteName will indicate site's name.

Labels

siteName

- local_redis_connect_timeouts

Type

64-bit Counter

Description

Total number connect timeouts towards local redis node, where siteName will indicate site's name.

Labels

siteName

- local_redis_connect_retries

Type

64-bit Counter

Description

Total number of connect retries towards local redis node, where siteName will indicate site's name.

Labels

siteName

- local_redis_read_operations

Type

64-bit Counter

Description

Total number of read operations happened for every site towards local redis node, where siteName will indicate site's name.

Labels

siteName

- local_redis_write_operations

Type

64-bit Counter

Description

Total number of write operations happened for every site towards local redis node, where siteName will indicate site's name.

Labels

siteName

- local_redis_initial_sync_opns_errors

Type

64-bit Counter

Description

Total number of errors encountered while reading/writing bytes towards local redis node during initial sync, where siteName will indicate site's name, errorMsg will display the relevant msg.

Labels

siteName, opnType, errorMsg

- local_redis_connection_status

Type

64-bit Gauge

Description

Determine status that local redis connection is up/down, where siteName will indicate site's name.

Labels

siteName

- local_redis_read_bytes

Type

64-bit Gauge

Description

Total number of bytes read by every site towards local redis node, where siteName will indicate site's name.

Labels

siteName

- local_redis_writes_bytes

Type

64-bit Gauge

Description

Total number of bytes written by every site towards local redis node, where siteName will indicate site's name.

Labels

siteName

- local_redis_initial_sync_read_bytes

Type

64-bit Gauge

Description

Total number of bytes read by every site towards local redis node during initial sync, where siteName will indicate site's name.

Labels

siteName

- local_redis_initial_sync_write_bytes

Type
64-bit Gauge
Description
Total number of bytes sent by every site towards local redis node during initial sync, where siteName will indicate site's name.
Labels
siteName

Syncer Metrics for RedisConnection

- redis_requests

Type
64-bit Counter
Description
Total number of requests sent to redis.
Labels

- redis_replies

Type
64-bit Counter
Description
Total number of replies received by redis which can replyEmpty/ParseFailure or success.
Labels
status

- redis_state

Type
64-bit Counter
Description
Total number of states changed by redis, where states can be checked by currentState/newState.
Labels
currentState, newState

Metrics for Sites

- site_consuming_paused

Type
64-bit Gauge
Description
Total number of sites paused for consuming data, where siteName will indicate site's name.
Labels
siteName

- site_state

Type

64-bit Gauge

Description

Current state of the remote site. The states can be 1-Connecting, 2-Synchronizing or 3-Synchronized

Labels

siteName

• site_events

Type

64-bit Gauge

Description

Total number of events for a remote site, where siteName will indicate site's name, destination can be local/remote, and event can be connected/disconnecte/read.

Labels

siteName, destination, event

• site_event_discards

Type

64-bit Gauge

Description

Total number of events discarded for a given remote site labelled by reason of discard, siteName and event

Labels

siteName, destination, event, reason

• site_cmds_sent

Type

64-bit Gauge

Description

Total number of commands sent,where siteName will indicate site's name, destination can be local/remote.

Labels

siteName, destination

• site_parse_callbacks

Type

64-bit Gauge

Description

Total number of ping callbacks for parsing, where siteName will indicate site's name, callbackType can cmd/Error/Ping.

Labels

siteName, callbackType

• site_commands_discards

Type

64-bit Gauge

Description

Total number of commands discarded by every site,where siteName will indicate site's name, and cmdName is command discarded.

Labels

siteName, cmdName

Metrics for RedisParser

- parser_parsed_cmds

Type

64-bit Gauge

Description

Total number of cmds parsed/processed.

Labels

siteName, cmdName

- parser_cmds_sent

Type

64-bit Gauge

Description

Total number of cmds sent for further processing.

Labels

siteName

- parser_cmds_discards

Type

64-bit Gauge

Description

Total number of cmds discarded while parsing.

Labels

siteName

Metrics for Redis Context

- redis_context_connection_attempts

Type

64-bit Counter

Description

Total number of connection attempts towards remote redis node, where siteName will indicate site's name.

Labels

siteName

- redis_context_connection_connects

Type

64-bit Counter

Description

Total number of successful connects towards remote redis node, where siteName will indicate site's name.

Labels

siteName

- redis_context_connection_failures

Type

64-bit Counter

Description

Total number of connection failures towards remote redis node, where siteName will indicate site's name.

Labels

siteName

- redis_context_connection_disconnects

Type

64-bit Counter

Description

Total number of connection disconnects towards remote redis node, where siteName will indicate site's name.

Labels

siteName

- redis_context_commands_buffer_full

Type

64-bit Counter

Description

Number of command buffer full instances that has happened.

Labels

siteName

- redis_context_connection_status

Type

64-bit Gauge

Description

Determine status that redis connection is up/down, where siteName will indicate site's name.

Labels

siteName

- redis_context_commands_buffered

Type

64-bit Gauge

Description

Total number of commands in buffer.

Labels

siteName

- redis_context_commands_sent

Type

64-bit Gauge

Description

Total number of commands sent to the remote site.

Labels

siteName

- redis_context_command_send_failures

Type

64-bit Gauge

Description

Total number of command send failures.

Labels

siteName

- redis_context_command_responses

Type

64-bit Gauge

Description

Total number of command send failures.

Labels

siteName, status

- redis_context_cmds_latency

Type

64-bit Histogram

Description

Duration observed on total journey/roundtrip taken by a command sent from syncer to redis where cmdName will indicate the command and siteName be the site's name.

Labels

siteName

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

Metrics for InitialSync

- initial_sync_requests

Type

64-bit Counter

Description

Total number of psync commands requests to local redis,where replicald indicates -1/replicald sent.

Labels

siteName, replicald

- initial_sync_commands_decoding_errors

Type

64-bit Counter

Description

Total number of type of errors occurred for parsing during initial sync,where siteName will indicate site's name, and errorType can be StringType/HashType/Zset2/Zset etc.

Labels

siteName, errorType

- last_initial_sync_start_time

Type

64-bit Gauge

Description

It refers to the time when the previous initial sync occurred.

Labels

siteName

- last_initial_sync_duration

Type

64-bit Gauge

Description

It refers to the duration when the previous initial sync occurred.

Labels

siteName

- last_initial_sync_status

Type

64-bit Gauge

Description

Status of the last initial sync whether success/failure.

Labels

siteName

- last_initial_sync_type

Type

64-bit Gauge

Description

It refers to Initial sync's synced type which can be full/partial.

Labels

siteName

- initial_sync_commands_discards

Type

64-bit Gauge

Description

Total number of commands discarded by every site during initial sync,where siteName will indicate site's name, and cmdName is command discarded.

Labels

siteName, cmdName, reason

- initial_sync_commands_parsed

Type

64-bit Gauge

Description

Total number of commands parsed by every site during initial sync,where siteName will indicate site's name, and cmdType can be totalCmds/moduleDataCmds.

Labels

siteName, cmdType

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- [Summary](#)
- [ETCD Client Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_put_requests	Counter	The total number of ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_requests_sent	Counter	The total number of ETCD Client PUT Requests sent.
ETCD Client Metrics	etcd_put_successes	Counter	The total number of successful ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_failures	Counter	The total number of failed ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_with_lease_requests	Counter	The total number of ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_requests_sent	Counter	The total number of ETCD Client PUT with Lease Requests sent.
ETCD Client Metrics	etcd_put_with_lease_successes	Counter	The total number of successful ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_failures	Counter	The total number of failed ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_range_requests	Counter	The total number of ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_requests_sent	Counter	The total number of ETCD Client Range Requests sent.
ETCD Client Metrics	etcd_range_successes	Counter	The total number of successful ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_failures	Counter	The total number of failed ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_delete_requests	Counter	The total number of ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_requests_sent	Counter	The total number of ETCD Client Range Delete Requests sent.
ETCD Client Metrics	etcd_range_delete_successes	Counter	The total number of successful ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_failures	Counter	The total number of failed ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_txn_requests	Counter	The total number of ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_requests_sent	Counter	The total number of ETCD Client Txn Requests sent.
ETCD Client Metrics	etcd_txn_successes	Counter	The total number of successful ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_failures	Counter	The total number of failed ETCD Client Txn Requests.
ETCD Client Metrics	etcd_watch_requests	Counter	The total number of ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_requests_sent	Counter	The total number of ETCD Client Watch Requests sent.
ETCD Client Metrics	etcd_watch_successes	Counter	The total number of successful ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_failures	Counter	The total number of failed ETCD Client Watch Requests.
ETCD Client Metrics	etcdwatcher_requests	Counter	The total number of ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_requests_sent	Counter	The total number of ETCD Watcher Requests sent.
ETCD Client Metrics	etcdwatcher_successes	Counter	The total number of successful ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_failures	Counter	The total number of failed ETCD Watcher Requests.

ETCD Client Metrics

- etcd_put_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests.

Labels

keyPrefix

- etcd_put_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests sent.

Labels

keyPrefix

- etcd_put_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT Requests.

Labels

keyPrefix

- etcd_put_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT Requests.

Labels

keyPrefix, failureReason

- etcd_put_with_lease_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests sent.

Labels

keyPrefix

- etcd_put_with_lease_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT with Lease Requests.

Labels

keyPrefix, failureReason

- etcd_range_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests sent.

Labels

keyPrefix

- etcd_range_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Requests.

Labels

keyPrefix, failureReason

• etcd_range_delete_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests.

Labels

keyPrefix

• etcd_range_delete_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests sent.

Labels

keyPrefix

• etcd_range_delete_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Delete Requests.

Labels

keyPrefix

• etcd_range_delete_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Delete Requests.

Labels

keyPrefix, failureReason

• etcd_txn_requests

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests.

Labels

keyPrefix

- etcd_txn_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests sent.

Labels

keyPrefix

- etcd_txn_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Txn Requests.

Labels

keyPrefix

- etcd_txn_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Txn Requests.

Labels

keyPrefix, failureReason

- etcd_watch_requests

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests sent.

Labels

keyPrefix

- etcd_watch_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Watch Requests.

Labels

keyPrefix, failureReason

- etcdwatcher_requests

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests.

Labels

keyPrefix

- etcdwatcher_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests sent.

Labels

keyPrefix

- etcdwatcher_successes

Type

64-bit Counter

Description

The total number of successful ETCD Watcher Requests.

Labels

keyPrefix

- etcdwatcher_failures

Type

64-bit Counter

Description

The total number of failed ETCD Watcher Requests.

Labels

keyPrefix, failureReason

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- [Summary](#)
- [ETCD Client Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_put_requests	Counter	The total number of ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_requests_sent	Counter	The total number of ETCD Client PUT Requests sent.
ETCD Client Metrics	etcd_put_successes	Counter	The total number of successful ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_failures	Counter	The total number of failed ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_with_lease_requests	Counter	The total number of ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_requests_sent	Counter	The total number of ETCD Client PUT with Lease Requests sent.
ETCD Client Metrics	etcd_put_with_lease_successes	Counter	The total number of successful ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_failures	Counter	The total number of failed ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_range_requests	Counter	The total number of ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_requests_sent	Counter	The total number of ETCD Client Range Requests sent.
ETCD Client Metrics	etcd_range_successes	Counter	The total number of successful ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_failures	Counter	The total number of failed ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_delete_requests	Counter	The total number of ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_requests_sent	Counter	The total number of ETCD Client Range Delete Requests sent.
ETCD Client Metrics	etcd_range_delete_successes	Counter	The total number of successful ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_failures	Counter	The total number of failed ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_txn_requests	Counter	The total number of ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_requests_sent	Counter	The total number of ETCD Client Txn Requests sent.
ETCD Client Metrics	etcd_txn_successes	Counter	The total number of successful ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_failures	Counter	The total number of failed ETCD Client Txn Requests.
ETCD Client Metrics	etcd_watch_requests	Counter	The total number of ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_requests_sent	Counter	The total number of ETCD Client Watch Requests sent.
ETCD Client Metrics	etcd_watch_successes	Counter	The total number of successful ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_failures	Counter	The total number of failed ETCD Client Watch Requests.
ETCD Client Metrics	etcdwatcher_requests	Counter	The total number of ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_requests_sent	Counter	The total number of ETCD Watcher Requests sent.
ETCD Client Metrics	etcdwatcher_successes	Counter	The total number of successful ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_failures	Counter	The total number of failed ETCD Watcher Requests.

ETCD Client Metrics

- [etcd_put_requests](#)

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests.

Labels

keyPrefix

- [etcd_put_requests_sent](#)

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests sent.

Labels

keyPrefix

• etcd_put_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT Requests.

Labels

keyPrefix

• etcd_put_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT Requests.

Labels

keyPrefix, failureReason

• etcd_put_with_lease_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests.

Labels

keyPrefix

• etcd_put_with_lease_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests sent.

Labels

keyPrefix

• etcd_put_with_lease_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT with Lease Requests.

Labels

keyPrefix

• etcd_put_with_lease_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT with Lease Requests.

Labels

keyPrefix, failureReason

• etcd_range_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests.

Labels

keyPrefix

• etcd_range_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests sent.

Labels

keyPrefix

• etcd_range_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Requests.

Labels

keyPrefix

• etcd_range_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Requests.

Labels

keyPrefix, failureReason

• etcd_range_delete_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests.

Labels

keyPrefix

• etcd_range_delete_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests sent.

Labels

keyPrefix

• etcd_range_delete_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Delete Requests.

Labels

keyPrefix

• etcd_range_delete_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Delete Requests.

Labels

keyPrefix, failureReason

• etcd_txn_requests

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests.

Labels

keyPrefix

• etcd_txn_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests sent.

Labels

keyPrefix

- etcd_txn_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Txn Requests.

Labels

keyPrefix

- etcd_txn_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Txn Requests.

Labels

keyPrefix, failureReason

- etcd_watch_requests

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests sent.

Labels

keyPrefix

- etcd_watch_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_failures

Type

64-bit Counter

Description			
The total number of failed ETCD Client Watch Requests.			
Labels			
<i>keyPrefix, failureReason</i>			
• etcdwatcher_requests			
Type			
64-bit Counter			
Description			
The total number of ETCD Watcher Requests.			
Labels			
<i>keyPrefix</i>			
• etcdwatcher_requests_sent			
Type			
64-bit Counter			
Description			
The total number of ETCD Watcher Requests sent.			
Labels			
<i>keyPrefix</i>			
• etcdwatcher_successes			
Type			
64-bit Counter			
Description			
The total number of successful ETCD Watcher Requests.			
Labels			
<i>keyPrefix</i>			
• etcdwatcher_failures			
Type			
64-bit Counter			
Description			
The total number of failed ETCD Watcher Requests.			
Labels			
<i>keyPrefix, failureReason</i>			

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- [Summary](#)
- [HTTP Punting Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_sent	Counter	The total number of HTTP requests successfully punted.

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_failed	Counter	The total number of HTTP requests punting failed.
HTTP Punting Metrics	punting_http_responses_received	Counter	The total number of HTTP responses received for the requests punted.
HTTP Punting Metrics	punting_http_responses_forwarded	Counter	The total number of HTTP responses forwarded after punting transaction completion.
HTTP Punting Metrics	punting_http_requests_aborted	Counter	The total number of HTTP requests aborted.
HTTP Punting Metrics	punting_http_request_duration_seconds	Histogram	Punting operation duration in seconds.

HTTP Punting Metrics

- [punting_http_requests_sent](#)

Type

64-bit Counter

Description

The total number of HTTP requests successfully punted.

Labels

message_type

- [punting_http_requests_failed](#)

Type

64-bit Counter

Description

The total number of HTTP requests for which the punting failed.

Labels

message_type

- [punting_http_responses_received](#)

Type

64-bit Counter

Description

The total number of HTTP responses received for the requests punted.

Labels

message_type, status_code

- [punting_http_responses_forwarded](#)

Type

64-bit Counter

Description

The total number of HTTP responses forwarded after punting transaction completion.

Labels

message_type, status_code

- [punting_http_requests_aborted](#)

Type

64-bit Counter

Description

The total number of HTTP requests aborted.

Labels

message_type

- punting_http_request_duration_seconds

Type

64-bit Histogram

Description

The punting transaction duration is measured from the time the HTTP request is sent out to the time the response is received for the punting request.

Labels

message_type

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

Table of Contents

- [Summary](#)
- [Packet Streaming Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Packet Streaming Metrics	num_packet_streaming_attempted	Counter	The total number of attempted streaming of packets
Packet Streaming Metrics	num_packet_streaming_success	Counter	The total number of packets streamed successfully
Packet Streaming Metrics	num_packet_streaming_failed	Counter	The total number of failed streaming of packets
Packet Streaming Metrics	num_packet_streaming_bytes	Counter	The total number of bytes successfully streamed

Packet Streaming Metrics

- num_packet_streaming_attempted

Type

64-bit Counter

Description

The total number of attempted streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_success

Type

64-bit Counter

Description

The total number of packets streamed successfully

Labels

interface_name, server_fqdn, kafka_topic

• num_packet_streaming_failed

Type

64-bit Counter

Description

The total number of failed streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

• num_packet_streaming_bytes

Type

64-bit Counter

Description

The total number of bytes successfully streamed

Labels

interface_name, server_fqdn, kafka_topic

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- [Summary](#)
- [ETCD Client Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_put_requests	Counter	The total number of ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_requests_sent	Counter	The total number of ETCD Client PUT Requests sent.
ETCD Client Metrics	etcd_put_successes	Counter	The total number of successful ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_failures	Counter	The total number of failed ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_with_lease_requests	Counter	The total number of ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_requests_sent	Counter	The total number of ETCD Client PUT with Lease Requests sent.
ETCD Client Metrics	etcd_put_with_lease_successes	Counter	The total number of successful ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_failures	Counter	The total number of failed ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_range_requests	Counter	The total number of ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_requests_sent	Counter	The total number of ETCD Client Range Requests sent.
ETCD Client Metrics	etcd_range_successes	Counter	The total number of successful ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_failures	Counter	The total number of failed ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_delete_requests	Counter	The total number of ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_requests_sent	Counter	The total number of ETCD Client Range Delete Requests sent.
ETCD Client Metrics	etcd_range_delete_successes	Counter	The total number of successful ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_failures	Counter	The total number of failed ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_txn_requests	Counter	The total number of ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_requests_sent	Counter	The total number of ETCD Client Txn Requests sent.
ETCD Client Metrics	etcd_txn_successes	Counter	The total number of successful ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_failures	Counter	The total number of failed ETCD Client Txn Requests.
ETCD Client Metrics	etcd_watch_requests	Counter	The total number of ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_requests_sent	Counter	The total number of ETCD Client Watch Requests sent.

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_watch_successes	Counter	The total number of successful ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_failures	Counter	The total number of failed ETCD Client Watch Requests.
ETCD Client Metrics	etcdwatcher_requests	Counter	The total number of ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_requests_sent	Counter	The total number of ETCD Watcher Requests sent.
ETCD Client Metrics	etcdwatcher_successes	Counter	The total number of successful ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_failures	Counter	The total number of failed ETCD Watcher Requests.

ETCD Client Metrics

- etcd_put_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests.

Labels

keyPrefix

- etcd_put_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests sent.

Labels

keyPrefix

- etcd_put_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT Requests.

Labels

keyPrefix

- etcd_put_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT Requests.

Labels

keyPrefix, failureReason

- etcd_put_with_lease_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests sent.

Labels

keyPrefix

- etcd_put_with_lease_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT with Lease Requests.

Labels

keyPrefix, failureReason

- etcd_range_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests sent.

Labels

keyPrefix

etcd_range_successes
Type
64-bit Counter
Description
The total number of successful ETCD Client Range Requests.
Labels
keyPrefix
etcd_range_failures
Type
64-bit Counter
Description
The total number of failed ETCD Client Range Requests.
Labels
keyPrefix, failureReason
etcd_range_delete_requests
Type
64-bit Counter
Description
The total number of ETCD Client Range Delete Requests.
Labels
keyPrefix
etcd_range_delete_requests_sent
Type
64-bit Counter
Description
The total number of ETCD Client Range Delete Requests sent.
Labels
keyPrefix
etcd_range_delete_successes
Type
64-bit Counter
Description
The total number of successful ETCD Client Range Delete Requests.
Labels
keyPrefix
etcd_range_delete_failures
Type
64-bit Counter
Description

The total number of failed ETCD Client Range Delete Requests.

Labels

keyPrefix, failureReason

• etcd_txn_requests

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests.

Labels

keyPrefix

• etcd_txn_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests sent.

Labels

keyPrefix

• etcd_txn_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Txn Requests.

Labels

keyPrefix

• etcd_txn_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Txn Requests.

Labels

keyPrefix, failureReason

• etcd_watch_requests

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests.

Labels

keyPrefix

• etcd_watch_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests sent.

Labels

keyPrefix

• etcd_watch_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Watch Requests.

Labels

keyPrefix

• etcd_watch_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Watch Requests.

Labels

keyPrefix, failureReason

• etcdwatcher_requests

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests.

Labels

keyPrefix

• etcdwatcher_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests sent.

Labels

keyPrefix

• etcdwatcher_successes

Type

64-bit Counter

Description

The total number of successful ETCD Watcher Requests.

Labels

keyPrefix

- etcdwatcher_failures

Type

64-bit Counter

Description

The total number of failed ETCD Watcher Requests.

Labels

keyPrefix, failureReason

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- [Summary](#)
- [HTTP Punting Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_sent	Counter	The total number of HTTP requests successfully punted.
HTTP Punting Metrics	punting_http_requests_failed	Counter	The total number of HTTP requests punting failed.
HTTP Punting Metrics	punting_http_responses_received	Counter	The total number of HTTP responses received for the requests punted.
HTTP Punting Metrics	punting_http_responses_forwarded	Counter	The total number of HTTP responses forwarded after punting transaction completion.
HTTP Punting Metrics	punting_http_requests_aborted	Counter	The total number of HTTP requests aborted.

HTTP Punting Metrics	punting_http_request_duration_seconds	Histogram	Punting operation duration in seconds.
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HTTP Punting Metrics

- punting_http_requests_sent

Type

64-bit Counter

Description

The total number of HTTP requests successfully punted.

Labels

message_type

- punting_http_requests_failed

Type

64-bit Counter

Description

The total number of HTTP requests for which the punting failed.

Labels

message_type

- **punting_http_responses_received**

Type
64-bit Counter

Description
The total number of HTTP responses received for the requests punted.

Labels
message_type, status_code
- **punting_http_responses_forwarded**

Type
64-bit Counter

Description
The total number of HTTP responses forwarded after punting transaction completion.

Labels
message_type, status_code
- **punting_http_requests_aborted**

Type
64-bit Counter

Description
The total number of HTTP requests aborted.

Labels
message_type
- **punting_http_request_duration_seconds**

Type
64-bit Histogram

Description
The punting transaction duration is measured from the time the HTTP request is sent out to the time the response is received for the punting request.

Labels
message_type

Boundaries
[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

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- [Summary](#)
- [Slice Selector statistics](#)

Summary

Group	Metric Name	Type	Synopsis
Slice Selector statistics	ss_nssf_nsselection_request_for_registration	Counter	Slice selection request attempts for UE registration
Slice Selector statistics	ss_nssf_nsselection_response_for_registration	Counter	Slice selection responses for UE registration
Slice Selector statistics	ss_nssf_nsselection_request_for_pdu_session	Counter	Slice selection request attempts for PDU session establishment
Slice Selector statistics	ss_nssf_nsselection_response_for_pdu_session	Counter	Slice selection responses for PDU session establishment

Group	Metric Name	Type	Synopsis
Slice Selector statistics	ss_nnssf_nsselection_request_for_ue_config_update	Counter	Slice selection request attempts for UE Configuration Update
Slice Selector statistics	ss_nnssf_nsselection_response_for_ue_config_update	Counter	Slice selection responses for UE Configuration Update
Slice Selector statistics	ss_nnssf_nssaiavailability_update_put_request	Counter	NSSAI Availability Update PUT request attempts
Slice Selector statistics	ss_nnssf_nssaiavailability_update_put_response	Counter	NSSAI Availability Update PUT responses returned
Slice Selector statistics	ss_nnssf_nssaiavailability_update_patch_request	Counter	NSSAI Availability Update PATCH request attempts
Slice Selector statistics	ss_nnssf_nssaiavailability_update_patch_response	Counter	NSSAI Availability Update PATCH responses returned
Slice Selector statistics	ss_nnssf_nssaiavailability_delete_request	Counter	NSSAI Availability DELETE request attempts
Slice Selector statistics	ss_nnssf_nssaiavailability_delete_response	Counter	NSSAI Availability DELETE responses returned
Slice Selector statistics	ss_nnssf_nssaiavailability_subscribe_post_request	Counter	NSSAI Availability Subscribe POST request attempts
Slice Selector statistics	ss_nnssf_nssaiavailability_subscribe_post_response	Counter	NSSAI Availability Subscribe POST responses returned
Slice Selector statistics	ss_nnssf_nssaiavailability_subscribe_patch_request	Counter	NSSAI Availability Unsubscribe PATCH request attempts
Slice Selector statistics	ss_nnssf_nssaiavailability_subscribe_patch_response	Counter	NSSAI Availability Unsubscribe PATCH responses returned
Slice Selector statistics	ss_nnssf_nssaiavailability_unsubscribe_request	Counter	NSSAI Availability Unsubscribe DELETE request attempts
Slice Selector statistics	ss_nnssf_nssaiavailability_unsubscribe_response	Counter	NSSAI Availability Unsubscribe DELETE responses returned
Slice Selector statistics	ss_nnssf_nssaiavailability_notify_request	Counter	NSSAI Availability Notify POST request sent
Slice Selector statistics	ss_nnssf_nssaiavailability_notify_response	Counter	NSSAI Availability Notify POST responses returned
Slice Selector statistics	ss_nnssf_nssaiavailability_options_request	Counter	NSSAI Availability Options request attempts
Slice Selector statistics	ss_nnssf_nssaiavailability_options_response	Counter	NSSAI Availability Options responses returned

Slice Selector statistics

- ss_nnssf_nsselection_request_for_registration
 - Type
 - 64-bit Counter
 - Description
 - Number of Network Slice Selection Request messages associated with the UE registration event received by Slice Selector, as defined by TS 29.531 section 6.1.
 - Labels
 - source_nf_type, source_nf_instance_id**
- ss_nnssf_nsselection_response_for_registration
 - Type
 - 64-bit Counter
 - Description
 - Number of Network Slice Selection Response messages associated with the UE registration event returned by Slice Selector, as defined by TS 29.531 section 6.1.
 - Labels
 - source_nf_type, source_nf_instance_id, status**
- ss_nnssf_nsselection_request_for_pdu_session
 - Type
 - 64-bit Counter
 - Description
 - Number of Network Slice Selection Request messages associated with the PDU session establishment event received by Slice Selector, as defined by TS 29.531 section 6.1.
 - Labels
 - source_nf_type, source_nf_instance_id, requested_snssai**

• ss_nssf_nsselection_response_for_pdu_session Type 64-bit Counter Description Number of Network Slice Selection Response messages associated with the PDU session establishment event returned by Slice Selector, as defined by TS 29.531 section 6.1. Labels source_nf_type, source_nf_instance_id, requested_snssai, status
• ss_nssf_nsselection_request_for_ue_config_update Type 64-bit Counter Description Number of Network Slice Selection Request messages associated with the UE Configuration Update event received by Slice Selector, as defined by TS 29.531 section 6.1. Labels source_nf_type, source_nf_instance_id
• ss_nssf_nsselection_response_for_ue_config_update Type 64-bit Counter Description Number of Network Slice Selection Response messages associated with the UE Configuration Update event returned by Slice Selector, as defined by TS 29.531 section 6.1. Labels source_nf_type, source_nf_instance_id, status
• ss_nssf_nssaiavailability_update_put_request Type 64-bit Counter Description Number of NSSAI Availability Update PUT Request messages received by Slice Selector, as defined by TS 29.531 section 6.2. Labels source_nf_instance_id
• ss_nssf_nssaiavailability_update_put_response Type 64-bit Counter Description Number of NSSAI Availability Update PUT Response messages returned by Slice Selector, as defined by TS 29.531 section 6.2. Labels source_nf_instance_id, status
• ss_nssf_nssaiavailability_update_patch_request Type 64-bit Counter

Description

Number of NSSAI Availability Update PATCH Request messages received by Slice Selector, as defined by TS 29.531 section 6.2.

Labels

source_nf_instance_id

- ss_nnssf_nssaiavailability_update_patch_response

Type

64-bit Counter

Description

Number of NSSAI Availability Update PATCH Response messages returned by Slice Selector, as defined by TS 29.531 section 6.2.

Labels

source_nf_instance_id, status

- ss_nnssf_nssaiavailability_delete_request

Type

64-bit Counter

Description

Number of NSSAI Availability DELETE Request messages received by Slice Selector, as defined by TS 29.531 section 6.2.

Labels

source_nf_instance_id

- ss_nnssf_nssaiavailability_delete_response

Type

64-bit Counter

Description

Number of NSSAI Availability DELETE Response messages returned by Slice Selector, as defined by TS 29.531 section 6.2.

Labels

source_nf_instance_id, status

- ss_nnssf_nssaiavailability_subscribe_post_request

Type

64-bit Counter

Description

Number of NSSAI Availability Subscribe POST Request messages received by Slice Selector, as defined by TS 29.531 section 6.2.

Labels

- ss_nnssf_nssaiavailability_subscribe_post_response

Type

64-bit Counter

Description

Number of NSSAI Availability Subscribe POST Response messages returned by Slice Selector, as defined by TS 29.531 section 6.2.

Labels

status

- ss_nnssf_nssaiavailability_subscribe_patch_request

Type

64-bit Counter

Description

Number of NSSAI Availability Subscribe PATCH Request messages received by Slice Selector, as defined by TS 29.531 section 6.2.

Labels

- `ss_nnssf_nssaiavailability_subscribe_patch_response`

Type

64-bit Counter

Description

Number of NSSAI Availability Subscribe PATCH Response messages returned by Slice Selector, as defined by TS 29.531 section 6.2.

Labels

status

- `ss_nnssf_nssaiavailability_unsubscribe_request`

Type

64-bit Counter

Description

Number of NSSAI Availability Unsubscribe DELETE Request messages received by Slice Selector, as defined by TS 29.531 section 6.2.

Labels

- `ss_nnssf_nssaiavailability_unsubscribe_response`

Type

64-bit Counter

Description

Number of NSSAI Availability Unsubscribe DELETE Response messages returned by Slice Selector, as defined by TS 29.531 section 6.2.

Labels

status

- `ss_nnssf_nssaiavailability_notify_request`

Type

64-bit Counter

Description

Number of NSSAI Availability Notify POST Request messages sent by Slice Selector, as defined by TS 29.531 section 6.2.

Labels

- `ss_nnssf_nssaiavailability_notify_response`

Type

64-bit Counter

Description

Number of NSSAI Availability Notify POST Response messages received by Slice Selector, as defined by TS 29.531 section 6.2.

Labels

status

- `ss_nssf_nssaiavailability_options_request`

Type

64-bit Counter

Description

Number of NSSAI Availability Options Request messages received by Slice Selector, as defined by TS 29.531 section 6.2.

Labels

- `ss_nssf_nssaiavailability_options_response`

Type

64-bit Counter

Description

Number of NSSAI Availability Options Response messages returned by Slice Selector, as defined by TS 29.531 section 6.2.

Labels

status

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- [Summary](#)
- [Packet Streaming Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Packet Streaming Metrics	num_packet_streaming_attempted	Counter	The total number of attempted streaming of packets
Packet Streaming Metrics	num_packet_streaming_success	Counter	The total number of packets streamed successfully
Packet Streaming Metrics	num_packet_streaming_failed	Counter	The total number of failed streaming of packets
Packet Streaming Metrics	num_packet_streaming_bytes	Counter	The total number of bytes successfully streamed

Packet Streaming Metrics

- `num_packet_streaming_attempted`

Type

64-bit Counter

Description

The total number of attempted streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- `num_packet_streaming_success`

Type

64-bit Counter

Description

The total number of packets streamed successfully

Labels

interface_name, server_fqdn, kafka_topic

- `num_packet_streaming_failed`

Type

64-bit Counter

Description

The total number of failed streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_bytes

Type

64-bit Counter

Description

The total number of bytes successfully streamed

Labels

interface_name, server_fqdn, kafka_topic

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- [Summary](#)
- [Circuit Breaker Metrics](#)
- [Service Endpoint Learner Metrics](#)
- [Service Endpoint Manager Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Circuit Breaker Metrics	sem_cb_endpoints	Gauge	Total number of endpoints for a service in respective states
Service Endpoint Learner Metrics	sem_num_dns_errors	Counter	Total number of DNS resolution errors for a given service
Service Endpoint Learner Metrics	sem_num_endpoints	Gauge	Total number of endpoints learnt for a service
Service Endpoint Manager Metrics	sem_num_endpoint_transactions_sent	Counter	Total number of times an endpoint is given to the application for communication with it
Service Endpoint Manager Metrics	sem_num_endpoint_transaction_status_received	Counter	Total number of transactions for a given endpoint received with the transaction status

Circuit Breaker Metrics

- sem_cb_endpoints

Type

64-bit Gauge

Description

Total number of endpoints for a service in respective states. There are 3 states viz. Active, Blocked, Restoring. In Active state, the endpoint is responding as expected, i.e. the response code doesn't indicate a service failure. The endpoint will be moved to Blocked state based on response failures which indicate service failure. After a timeout, the endpoint will be moved to Restoring state to check if the endpoint has recovered. In this state, if endpoint responds with response code other than service failure, it will be moved to Active state. Else, it will be moved back to Blocked state.

Labels

service_name, endpoint_state

Service Endpoint Learner Metrics

- sem_num_dns_errors

Type

64-bit Counter

Description

Total number of DNS resolution errors for a given service, labels are service name for DNS resolution and DNS error codes

Labels

service_name, error_code

- sem_num_endpoints

Type

64-bit Gauge

Description

Total number of endpoints learnt from the Kubernetes DNS server for a given service

Labels

service

Service Endpoint Manager Metrics

- sem_num_endpoint_transactions_sent

Type

64-bit Counter

Description

Total number of times an endpoint is given to the application for communication with it

Labels

service

- sem_num_endpoint_transaction_status_received

Type

64-bit Counter

Description

Total number of transactions for a given endpoint received with the transaction status

Labels

service, status

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- [Summary](#)
- [ChargingCmn statistics](#)

Summary

Group	Metric Name	Type	Synopsis
ChargingCmn statistics	svcmgr_service_available_ind_notification	Counter	How many times, Service Available Indication message received
ChargingCmn statistics	svcmgr_service_available_ind_parse_errs	Counter	How many times, Service Available Indication parse error happened
ChargingCmn statistics	svcmgr_service_available_ind_notification_ack	Counter	How many times, Service Available Indication message is acknowledged

Group	Metric Name	Type	Synopsis
ChargingCmn statistics	svcmgr_service_unavailable_ind_notification	Counter	How many times, Service Unavailable Indication message
ChargingCmn statistics	svcmgr_service_unavailable_ind_parse_errs	Counter	How many times, Service Unavailable Indication parse error happened
ChargingCmn statistics	svcmgr_service_unavailable_ind_notification_ack	Counter	How many times, Service Unavailable Indication message is acknowledged
ChargingCmn statistics	svcmgr_publish_msg_sent	Counter	How many times, Publication message sent
ChargingCmn statistics	svcmgr_publish_msg_success	Counter	How many times, Publication message success response received
ChargingCmn statistics	svcmgr_publish_msg_failure	Counter	How many times, failure received for publication message
ChargingCmn statistics	svcmgr_subscribe_msg_sent	Counter	How many times, Subscribe message sent
ChargingCmn statistics	svcmgr_subscribe_msg_success	Counter	How many times, subscribe message success
ChargingCmn statistics	svcmgr_subscribe_msg_failure	Counter	How many times, failure received for subscribe message
ChargingCmn statistics	svcmgr_unsubscribe_msg_sent	Counter	How many times, unsubscribe message sent
ChargingCmn statistics	svcmgr_unsubscribe_msg_failure	Counter	How many times, failure received for unsubscribe message
ChargingCmn statistics	svcmgr_config_req_sent	Counter	How many times, config request sent
ChargingCmn statistics	svcmgr_config_req_failure	Counter	How many times, failure received for config request
ChargingCmn statistics	svcmgr_config_req_success	Counter	How many times, success received for config request
ChargingCmn statistics	svcmgr_config_register_response	Counter	How many times, config register response received
ChargingCmn statistics	svcmgr_config_register_response_ack	Counter	How many times, config register response is acknowledged
ChargingCmn statistics	svcmgr_config_register_response_parse_err	Counter	How many times, parse error happened for config register response
ChargingCmn statistics	svcmgr_config_update_notification	Counter	How many times, config update notification received
ChargingCmn statistics	svcmgr_config_update_notification_response	Counter	How many times, config update response
ChargingCmn statistics	svcmgr_intf_schema_update_ind_notification	Counter	How many times, interface schema update indication message received
ChargingCmn statistics	svcmgr_intf_schema_update_ind_notification_ack	Counter	How many times, interface schema update indication message is acknowledged
ChargingCmn statistics	svcmgr_intf_schema_update_ind_parse_errs	Counter	How many times, interface schema update indication parse error happened

ChargingCmn statistics

- svcmgr_service_available_ind_notification

Type

64-bit Counter

Description

Number of Times Service Available Indication notification received

Labels

servicename

- svcmgr_service_available_ind_parse_errs

Type

64-bit Counter

Description

Number of Times Service Available Indication parse error happened

Labels

- svcmgr_service_available_ind_notification_ack

Type

64-bit Counter

Description

Number of Times Service Available Indication message is acknowledged

Labels

servicename

- svcmgr_service_unavailable_ind_notification

Type

64-bit Counter

Description

Number of Times Service Unavailable Indication notification received

Labels

servicename

- svcmgr_service_unavailable_ind_parse_errs

Type

64-bit Counter

Description

Number of Times Service Unavailable Indication parse error happened

Labels

- svcmgr_service_unavailable_ind_notification_ack

Type

64-bit Counter

Description

Number of Times Service Unavailable Indication message is acknowledged

Labels

servicename

- svcmgr_publish_msg_sent

Type

64-bit Counter

Description

Number of Times Publication message sent

Labels

servicename, retry

• **svcmgr_publish_msg_success**

Type

64-bit Counter

Description

Number of Times Publication message success response received

Labels

servicename, causecode, retry

• **svcmgr_publish_msg_failure**

Type

64-bit Counter

Description

Number of Times failure received for publication message

Labels

servicename, causecode, retry

• **svcmgr_subscribe_msg_sent**

Type

64-bit Counter

Description

Number of Times Subscribe message sent

Labels

servicename, retry

• **svcmgr_subscribe_msg_success**

Type

64-bit Counter

Description

Number of Times Send subscribe message success response received

Labels

servicename, causecode, retry

• **svcmgr_subscribe_msg_failure**

Type

64-bit Counter

Description

Number of time failure received for subscribe message

Labels

servicename, causecode, retry

• **svcmgr_unsubscribe_msg_sent**

Type
64-bit Counter
Description
Number of Times unsubscribe message sent
Labels
<i>servicename</i>
• svcmgr_unsubscribe_msg_failure
Type
64-bit Counter
Description
Number of Times failure received for unsubscribe message
Labels
<i>servicename</i>
• svcmgr_config_req_sent
Type
64-bit Counter
Description
Number of Times config request sent
Labels
<i>retry</i>
• svcmgr_config_req_failure
Type
64-bit Counter
Description
Number of Times failure received for config request
Labels
<i>causecode, retry</i>
• svcmgr_config_req_success
Type
64-bit Counter
Description
Number of Times success received for config request
Labels
<i>causecode, retry</i>
• svcmgr_config_register_response
Type
64-bit Counter
Description
Number of Times config register response received

Labels

statuscode

- svcmgr_config_register_response_ack

Type

64-bit Counter

Description

Number of Times config register response is acknowledged

Labels

- svcmgr_config_register_response_parse_err

Type

64-bit Counter

Description

Number of Times parse error happened for config register response

Labels

- svcmgr_config_update_notification

Type

64-bit Counter

Description

Number of Times config update notification received

Labels

- svcmgr_config_update_notification_response

Type

64-bit Counter

Description

Number of Times config update response sent

Labels

- svcmgr_intf_schema_update_ind_notification

Type

64-bit Counter

Description

Number of Times interface schema update indication message received

Labels

servicename

- svcmgr_intf_schema_update_ind_notification_ack

Type

64-bit Counter

Description

Number of Times interface schema update indication message is acknowledged
Labels
<i>servicename</i>
svcmgr_intf_schema_update_ind_parse_errs
Type
64-bit Counter
Description
Number of Times Service interface schema update indication parse error happened
Labels

Table of Contents

- [Summary](#)
- [ETCD Client Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_put_requests	Counter	The total number of ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_requests_sent	Counter	The total number of ETCD Client PUT Requests sent.
ETCD Client Metrics	etcd_put_successes	Counter	The total number of successful ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_failures	Counter	The total number of failed ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_with_lease_requests	Counter	The total number of ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_requests_sent	Counter	The total number of ETCD Client PUT with Lease Requests sent.
ETCD Client Metrics	etcd_put_with_lease_successes	Counter	The total number of successful ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_failures	Counter	The total number of failed ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_range_requests	Counter	The total number of ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_requests_sent	Counter	The total number of ETCD Client Range Requests sent.
ETCD Client Metrics	etcd_range_successes	Counter	The total number of successful ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_failures	Counter	The total number of failed ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_delete_requests	Counter	The total number of ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_requests_sent	Counter	The total number of ETCD Client Range Delete Requests sent.
ETCD Client Metrics	etcd_range_delete_successes	Counter	The total number of successful ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_failures	Counter	The total number of failed ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_txn_requests	Counter	The total number of ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_requests_sent	Counter	The total number of ETCD Client Txn Requests sent.
ETCD Client Metrics	etcd_txn_successes	Counter	The total number of successful ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_failures	Counter	The total number of failed ETCD Client Txn Requests.
ETCD Client Metrics	etcd_watch_requests	Counter	The total number of ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_requests_sent	Counter	The total number of ETCD Client Watch Requests sent.
ETCD Client Metrics	etcd_watch_successes	Counter	The total number of successful ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_failures	Counter	The total number of failed ETCD Client Watch Requests.
ETCD Client Metrics	etcdwatcher_requests	Counter	The total number of ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_requests_sent	Counter	The total number of ETCD Watcher Requests sent.
ETCD Client Metrics	etcdwatcher_successes	Counter	The total number of successful ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_failures	Counter	The total number of failed ETCD Watcher Requests.

ETCD Client Metrics

- etcd_put_requests
 - Type
 - 64-bit Counter
 - Description
 - The total number of ETCD Client PUT Requests.
 - Labels
 - keyPrefix**
- etcd_put_requests_sent
 - Type
 - 64-bit Counter
 - Description
 - The total number of ETCD Client PUT Requests sent.
 - Labels
 - keyPrefix**
- etcd_put_successes
 - Type
 - 64-bit Counter
 - Description
 - The total number of successful ETCD Client PUT Requests.
 - Labels
 - keyPrefix**
- etcd_put_failures
 - Type
 - 64-bit Counter
 - Description
 - The total number of failed ETCD Client PUT Requests.
 - Labels
 - keyPrefix, failureReason**
- etcd_put_with_lease_requests
 - Type
 - 64-bit Counter
 - Description
 - The total number of ETCD Client PUT with Lease Requests.
 - Labels
 - keyPrefix**
- etcd_put_with_lease_requests_sent
 - Type
 - 64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests sent.

Labels

keyPrefix

- `etcd_put_with_lease_successes`

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- `etcd_put_with_lease_failures`

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT with Lease Requests.

Labels

keyPrefix, failureReason

- `etcd_range_requests`

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests.

Labels

keyPrefix

- `etcd_range_requests_sent`

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests sent.

Labels

keyPrefix

- `etcd_range_successes`

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_failures
 - Type
 - 64-bit Counter
 - Description
 - The total number of failed ETCD Client Range Requests.
 - Labels
 - keyPrefix, failureReason**
- etcd_range_delete_requests
 - Type
 - 64-bit Counter
 - Description
 - The total number of ETCD Client Range Delete Requests.
 - Labels
 - keyPrefix**
- etcd_range_delete_requests_sent
 - Type
 - 64-bit Counter
 - Description
 - The total number of ETCD Client Range Delete Requests sent.
 - Labels
 - keyPrefix**
- etcd_range_delete_successes
 - Type
 - 64-bit Counter
 - Description
 - The total number of successful ETCD Client Range Delete Requests.
 - Labels
 - keyPrefix**
- etcd_range_delete_failures
 - Type
 - 64-bit Counter
 - Description
 - The total number of failed ETCD Client Range Delete Requests.
 - Labels
 - keyPrefix, failureReason**
- etcd_txn_requests
 - Type
 - 64-bit Counter
 - Description

The total number of ETCD Client Txn Requests.

Labels

keyPrefix

• etcd_txn_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests sent.

Labels

keyPrefix

• etcd_txn_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Txn Requests.

Labels

keyPrefix

• etcd_txn_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Txn Requests.

Labels

keyPrefix, failureReason

• etcd_watch_requests

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests.

Labels

keyPrefix

• etcd_watch_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests sent.

Labels

keyPrefix

• etcd_watch_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Watch Requests.

Labels

keyPrefix

• etcd_watch_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Watch Requests.

Labels

keyPrefix, failureReason

• etcdwatcher_requests

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests.

Labels

keyPrefix

• etcdwatcher_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests sent.

Labels

keyPrefix

• etcdwatcher_successes

Type

64-bit Counter

Description

The total number of successful ETCD Watcher Requests.

Labels

keyPrefix

• etcdwatcher_failures

Type

64-bit Counter

Description

The total number of failed ETCD Watcher Requests.

Labels

keyPrefix, failureReason

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- [Summary](#)
- [HTTP Punting Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_sent	Counter	The total number of HTTP requests successfully punted.
HTTP Punting Metrics	punting_http_requests_failed	Counter	The total number of HTTP requests punting failed.
HTTP Punting Metrics	punting_http_responses_received	Counter	The total number of HTTP responses received for the requests punted.
HTTP Punting Metrics	punting_http_responses_forwarded	Counter	The total number of HTTP responses forwarded after punting transaction completion.
HTTP Punting Metrics	punting_http_requests_aborted	Counter	The total number of HTTP requests aborted.
HTTP Punting Metrics	punting_http_request_duration_seconds	Histogram	Punting operation duration in seconds.

HTTP Punting Metrics

- [punting_http_requests_sent](#)

Type

64-bit Counter

Description

The total number of HTTP requests successfully punted.

Labels

message_type
- [punting_http_requests_failed](#)

Type

64-bit Counter

Description

The total number of HTTP requests for which the punting failed.

Labels

message_type
- [punting_http_responses_received](#)

Type

64-bit Counter

Description

The total number of HTTP responses received for the requests punted.

Labels

message_type, status_code

- [punting_http_responses_forwarded](#)

Type

64-bit Counter

Description

The total number of HTTP responses forwarded after punting transaction completion.

Labels

message_type, status_code
- [punting_http_requests_aborted](#)

Type

64-bit Counter

Description

The total number of HTTP requests aborted.

Labels

message_type
- [punting_http_request_duration_seconds](#)

Type

64-bit Histogram

Description

The punting transaction duration is measured from the time the HTTP request is sent out to the time the response is received for the punting request.

Labels

message_type

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

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- [Summary](#)
- [SMF Monitor Interface Stats](#)
- [SMF Monitor Service Stats](#)
- [SMF Call-Control Service Stats](#)

Summary

Group	Metric Name	Type	Synopsis
SMF Monitor Interface Stats	monitor_smf_nfmanagement_service_events	Counter	Total number of NFManagement Service events triggered by SMF Monitor
SMF Monitor Service Stats	monitor_smf_critical_service_up	Counter	Number of critical services required by SMF which are up
SMF Monitor Service Stats	monitor_smf_critical_service_down	Counter	Number of critical services required by SMF which are down
SMF Monitor Service Stats	monitor_smf_minor_load_level_entered	Counter	Number of times SMF has entered into the minor load level
SMF Monitor Service Stats	monitor_smf_minor_load_level_exited	Counter	Number of times SMF has exited out of the minor load level
SMF Monitor Service Stats	monitor_smf_major_load_level_entered	Counter	Number of times SMF has entered into the major load level

Group	Metric Name	Type	Synopsis
SMF Monitor Service Stats	monitor_smf_major_load_level_exited	Counter	Number of times SMF has exited out of the major load level
SMF Monitor Service Stats	monitor_smf_critical_load_level_entered	Counter	Number of times SMF has entered into the critical load level
SMF Monitor Service Stats	monitor_smf_critical_load_level_exited	Counter	Number of times SMF has exited out of the critical load level
SMF Monitor Service Stats	monitor_smf_pfcpc_associations_become_unavailable	Counter	Number of times the SMF has found all of the PFCPC Associations went Down
SMF Monitor Service Stats	monitor_smf_pfcpc_associations_become_available	Counter	Number of times the SMF has found at least one of the PFCPC Associations came UP
SMF Monitor Service Stats	monitor_smf_registration_update_on_etcd_dnnlist_update	Counter	Number of SMF Registration Updates triggered due to the Update on the UEIP Pool Exhaustion of DNN List
SMF Call-Control Service Stats	smf_simultaneous_attached_users	Gauge	Total number of unique subscribers attached to the network
SMF Call-Control Service Stats	smf_simultaneous_attached_active_users	Gauge	Total number of unique subscribers attached to the network and active
SMF Call-Control Service Stats	smf_sgw_simultaneous_attached_active_users	Gauge	Total number of unique sgw subscribers attached to the network and active
SMF Call-Control Service Stats	smf_current_pdu_sessions	Gauge	Total number of subscriber sessions present in SMF
SMF Call-Control Service Stats	smf_current_cumulative_qos_flows	Gauge	Total number of qos flows present in SMF
SMF Call-Control Service Stats	smf_pgw_current_cumulative_bearers	Gauge	Current number of bearers present in SMF+PGW-C
SMF Call-Control Service Stats	smf_pgw_current_bearers	Gauge	Current number of bearers present in SMF+PGW-C with labels
SMF Call-Control Service Stats	smf_current_qos_flows	Gauge	Total number of qos flows present for a fiveQi arp_pl combination in SMF
SMF Call-Control Service Stats	smf_current_qos_flows_per_snssai	Gauge	Total number of qos flows present for a snssai in SMF
SMF Call-Control Service Stats	smf_current_num_ue_per_qos_flow	Gauge	Total number of distinct ues present for a fiveQi arp_pl combination in SMF
SMF Call-Control Service Stats	smf_current_pdu_sessions_using_default_quota	Gauge	Total number of pdu sessions using locally configured default quota
SMF Call-Control Service Stats	smf_current_pdu_sessions_using_default_policy	Gauge	Total number of pdu sessions using locally configured default policy
SMF Call-Control Service Stats	smf_current_pdu_sessions_per_upf	Gauge	Total number of pdu sessions per upf basis
SMF Call-Control Service Stats	smf_current_pdu_sessions_per_ip_anchor_upf	Gauge	Total number of pdu sessions per ip anchor upf basis
SMF Call-Control Service Stats	smf_current_pdu_sessions_without_charging	Gauge	Total number of pdu sessions without charging in SMF
SMF Call-Control Service Stats	smf_current_pdu_sessions_using_udm_allocated_static_ue_ip_address	Gauge	Total number of pdu sessions with UDM allocated static ue IP in SMF
SMF Call-Control Service Stats	smf_current_pdu_sessions_using_radius_allocated_ue_ip_address	Gauge	Total number of pdu sessions with Radius allocated ue IP in SMF
SMF Call-Control Service Stats	smf_current_pdu_sessions_per_dnn	Gauge	Total number of pdu sessions per dnn
SMF Call-Control Service Stats	smf_current_pdu_sessions_per_rat_type	Gauge	Total number of pdu sessions per rat type
SMF Call-Control Service Stats	smf_current_outbound_roaming_sessions	Gauge	Total number of outbound roamers in SMF
SMF Call-Control Service Stats	smf_current_inbound_roaming_home_routed_sessions	Gauge	Total number of inbound roamers home routed in SMF

Group	Metric Name	Type	Synopsis
SMF Call-Control Service Stats	smf_current_pdu_sessions_per_mapped_dnn	Gauge	Total number of pdu sessions per mapped dnn
SMF Call-Control Service Stats	smf_current_idle_pdu_sessions	Gauge	Total number of idle pdu sessions present in SMF
SMF Call-Control Service Stats	smf_current_pdu_sessions_using_gx_policy	Gauge	Current number of pdu sessions using Gx policy
SMF Call-Control Service Stats	smf_sgw_current_idle_pdu_sessions	Gauge	Total number of idle pdu sessions present in SMF SGW
SMF Call-Control Service Stats	smf_sgw_idle_subscribers	Gauge	Total number of idle subscribers present in SMF SGW

SMF Monitor Interface Stats

- monitor_smf_nfmanagement_service_events

Type

64-bit Counter

Description

Total number of NFManagement Service events triggered by SMF Monitor. The event types include registration, deregistration and update of NF Profile for the given instance of SMF NF.

Labels

event_type

SMF Monitor Service Stats

- monitor_smf_critical_service_up

Type

64-bit Counter

Description

Number of critical services required by SMF which have become available for use

Labels

critical_service_name

- monitor_smf_critical_service_down

Type

64-bit Counter

Description

Number of critical services required by SMF which have become unavailable for use

Labels

critical_service_name

- monitor_smf_minor_load_level_entered

Type

64-bit Counter

Description

Number of times SMF has entered into the minor load level

Labels

• monitor_smf_minor_load_level_exited Type 64-bit Counter Description Number of times SMF has exited out of the minor load level Labels
• monitor_smf_major_load_level_entered Type 64-bit Counter Description Number of times SMF has entered into the major load level Labels
• monitor_smf_major_load_level_exited Type 64-bit Counter Description Number of times SMF has exited out of the major load level Labels
• monitor_smf_critical_load_level_entered Type 64-bit Counter Description Number of times SMF has entered into the critical load level Labels
• monitor_smf_critical_load_level_exited Type 64-bit Counter Description Number of times SMF has exited out of the critical load level Labels
• monitor_smf_pfcps_associations_become_unavailable Type 64-bit Counter Description Number of times the SMF has found none of it's PFCP connections are available Labels

- monitor_smf_pfcps_associations_become_available
Type
64-bit Counter
Description
Number of times the SMF has found at least one of its PFCP connection has become available
Labels
FED_NAME
- monitor_smf_registration_update_on_etcd_dnnlist_update
Type
64-bit Counter
Description
Number of SMF Registration Updates triggered for the UEIP Pool Exhaustion status of the DNN List
Labels
FED_NAME

SMF Call-Control Service Stats

- smf_simultaneous_attached_users
Type
64-bit Gauge
Description
Number of unique attached subscribers in SMF
Labels
FED_NAME
- smf_simultaneous_attached_active_users
Type
64-bit Gauge
Description
Number of unique attached subscribers in Active state in SMF
Labels
FED_NAME
- smf_sgw_simultaneous_attached_active_users
Type
64-bit Gauge
Description
Number of unique attached sgw subscribers in Active state in SMF
Labels
FED_NAME
- smf_current_pdu_sessions
Type
64-bit Gauge

Description

Total number of pdu sessions in SMF

Labels

FED_NAME

- smf_current_cumulative_qos_flows

Type

64-bit Gauge

Description

Total number of Qos Flows in SMF

Labels

FED_NAME

- smf_pgw_current_cumulative_bearers

Type

64-bit Gauge

Description

Current number of bearers in SMF+PGW-C

Labels

FED_NAME

- smf_pgw_current_bearers

Type

64-bit Gauge

Description

Current number of bearers in SMF+PGW-C with labels

Labels

fiveQi, arpPI

- smf_current_qos_flows

Type

64-bit Gauge

Description

Total number of qos flows per fiveQi and arp pl

Labels

fiveQi, arpPI

- smf_current_qos_flows_per_snssai

Type

64-bit Gauge

Description

Total number of qos flows per snssai

Labels

nssai

smf_current_num_ue_per_qos_flow
Type
64-bit Gauge
Description
Total number of distinct ues per fiveQi and arp pl
Labels
fiveQi, arpPl, nssai
smf_current_pdu_sessions_using_default_quota
Type
64-bit Gauge
Description
Total number of pdu sessions with default quota
Labels
FED_NAME
smf_current_pdu_sessions_using_default_policy
Type
64-bit Gauge
Description
Total number of pdu sessions with default policy
Labels
FED_NAME
smf_current_pdu_sessions_per_upf
Type
64-bit Gauge
Description
Total number of pdu sessions per upf
Labels
upf_node_id
smf_current_pdu_sessions_per_ip_anchor_upf
Type
64-bit Gauge
Description
Total number of pdu sessions per ip anchor upf
Labels
upf_node_id
smf_current_pdu_sessions_without_charging
Type
64-bit Gauge
Description

Total number of pdu sessions without charging

Labels

FED_NAME

- smf_current_pdu_sessions_using_udm_allocated_static_ue_ip_address

Type

64-bit Gauge

Description

Total number of pdu sessions with UDM allocated static ue IP

Labels

FED_NAME

- smf_current_pdu_sessions_using_radius_allocated_ue_ip_address

Type

64-bit Gauge

Description

Total number of pdu sessions with Radius allocated ue IP

Labels

FED_NAME

- smf_current_pdu_sessions_per_dnn

Type

64-bit Gauge

Description

Total number of pdu sessions per dnn

Labels

dnn

- smf_current_pdu_sessions_per_rat_type

Type

64-bit Gauge

Description

Total number of pdu sessions per rat type

Labels

ratType

- smf_current_outbound_roaming_sessions

Type

64-bit Gauge

Description

Total number of outbound roamers in SMF

Labels

FED_NAME

- smf_current_inbound_roaming_home_routed_sessions

Type

64-bit Gauge

Description

Total number of inbound roamers home routed in SMF

Labels

FED_NAME

- smf_current_pdu_sessions_per_mapped_dnn

Type

64-bit Gauge

Description

Total number of pdu sessions per mapped dnn

Labels

mappedDnn

- smf_current_idle_pdu_sessions

Type

64-bit Gauge

Description

Total number of idle pdu sessions in SMF

Labels

FED_NAME

- smf_current_pdu_sessions_using_gx_policy

Type

64-bit Gauge

Description

Current number of pdu sessions using Gx policy

Labels

FED_NAME

- smf_sgw_current_idle_pdu_sessions

Type

64-bit Gauge

Description

Total number of idle pdu sessions in SMF SGW

Labels

FED_NAME

- smf_sgw_idle_subscribers

Type

64-bit Gauge

Description

Total number of idle subscribers in SMF SGW

Labels

FED_NAME

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- [Summary](#)
- [NfMonitor statistics](#)
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Summary

Group	Metric Name	Type	Synopsis
NfMonitor statistics	nfmonitor_unstable_nf_counter	Counter	Howmany times, the Nf is found to be Unstable
NfMonitor statistics	nfmonitor_force_restart_due_to_nf_unstability	Counter	Numbe of times the VNFRestart triggered due to Unstability of the NF
NfMonitor statistics	nfmonitor_cumulative_readiness_failure_count	Counter	Numbe of times Service sound to have insuffucient count due to Readiness probe
NfMonitor statistics	nfmonitor_cumulative_readiness_failure_alerts	Counter	Numbe of times Service sound to have insuffucient count due to Readiness probe
NfMonitor statistics	nfmonitor_cumulative_k8api_get_query_failures	Counter	Numbe of times the k8 get Api query failures
DebugNfMonitorCmn	nfmonitor_force_restart_counter	Counter	Number of times force-restart triggered
NfManagement statistics	nfmonitor_nfmgmt_service_request_attempts	Counter	Howmany times, the Nf is attempting to send NRF Requests
NfManagement statistics	nfmonitor_nfmgmt_service_request_success	Counter	Numbe of times Success response is received for the requests to NRF
NfManagement statistics	nfmonitor_nfmgmt_service_request_failures	Counter	Numbe of times Failure response is received for the requests to NRF
NfManagement statistics	nfmonitor_nfmgmt_aborted_request_msgs	Counter	Numbe of times the requests to NRF is Aborted/Cancelled
Debug statistics	nfmonitor_vnf_restart_trigger	Counter	Howmany times, the Force Restart is received
Debug statistics	nfmonitor_fsm_state_changes	Counter	The statistics for the the FSM state changes
Debug statistics	nfmonitor_deactivation_triggers	Counter	Numbe of times the NF deactivation trigger is received
Debug statistics	nfmonitor_critical_service_up_notification	Counter	Numbe of times the Critical Service Up Notification received
Debug statistics	nfmonitor_critical_service_down_notification	Counter	Numbe of times the Critical Service Down Notification received
Debug statistics	nfmonitor_critical_services_overall_up_notification	Counter	Numbe of times the Up notification received when all monitored Critical Services are available
Debug statistics	nfmonitor_critical_services_overall_down_notification	Counter	Numbe of times the Down notification received when at least one Critical Service was unavailable
Debug statistics	nfmonitor_hysteresis_timer_started	Counter	Numbe of times the Hysteresis Timer is started due to missing Critical Service
Debug statistics	nfmonitor_hysteresis_timer_stopped	Counter	Numbe of times the Hysteresis Timer is stopped due to Critical Service become available
Debug statistics	nfmonitor_hysteresis_timer_expired	Counter	Numbe of times the Hysteresis Timer is expired due to missing Critical Service
Debug statistics	nfmonitor_personality_update_triggers	Counter	Numbe of times the Nf Personality update is received
Debug statistics	nfmonitor_maintenance_trigger_received	Counter	Numbe of times the Maintenance Mode is activated
Debug statistics	nfmonitor_profile_update_triggers	Counter	Numbe of times the Nf Profile Update happened
Debug statistics	nfmonitor_critical_service_down_on_register_entry	Counter	Numbe of times the Critical Services were down while starting with RegisterFsmState

Group	Metric Name	Type	Synopsis
ApiServer statistics	nfmonitor_deployment_create_watch_request	Counter	Howmany times, the deployment watch requests are created
ApiServer statistics	nfmonitor_deployment_delete_watch_request	Counter	Howmany times, the deployment watch requests are deleted
ApiServer statistics	nfmonitor_deployment_watch_notification_success	Counter	Howmany times, the watch notifications are successful
ApiServer statistics	nfmonitor_deployment_watch_notification_error	Counter	Howmany times, the watch notifications are failed
ApiServer statistics	nfmonitor_limit_range_query_attempts	Counter	Howmany times, the limit-range query attempts are made
ApiServer statistics	nfmonitor_limit_range_query_failures	Counter	Howmany times, the limit-range query attempts are failed
ApiServer statistics	nfmonitor_limit_range_query_success	Counter	Howmany times, the limit-range query attempts are succeeded
ApiServer statistics	nfmonitor_hpa_query_attempts	Counter	Howmany times, the Hpa query attempts are made
ApiServer statistics	nfmonitor_hpa_query_failures	Counter	Howmany times, the Hpa query attempts are failed
ApiServer statistics	nfmonitor_hpa_query_success	Counter	Howmany times, the Hpa query attempts are succeeded
ApiServer statistics	nfmonitor_readiness_probe_query_attempts	Counter	Howmany times, the Readiness Probe query attempts are made
ApiServer statistics	nfmonitor_readiness_probe_query_failures	Counter	Howmany times, the Readiness Probe query attempts are failed
ApiServer statistics	nfmonitor_readiness_probe_query_success	Counter	Howmany times, the Readiness Probe query attempts are succeeded
Feature Functionality statistics	nfmonitor_critical_service_frequent_restart	Counter	Howmany times, minimum count is not met for the stable Critical Services
Feature Functionality statistics	nfmonitor_alarm_due_to_frequent_restart	Counter	Howmany times alarm raised, due to minimum count is not met for the stable Critical Services
Feature Functionality statistics	nfmonitor_force_restart_due_to_frequent_restart	Counter	Howmany times, force restart triggerd because of the unstable Critical Services
Feature Functionality statistics	nfmonitor_critical_service_readiness_failure	Counter	Howmany times, minimum count is not met for the readiness succeeded Critical Services
Feature Functionality statistics	nfmonitor_alarm_due_to_readiness_failure	Counter	Alarm raised, due to minimum count is not met for the readiness succeeded Critical Services
Feature Functionality statistics	nfmonitor_force_restart_due_to_readiness_failure	Counter	Force restart triggerd because of Critical Service Readiness failure
Consul statistics	nfmonitor_publish_status_into_consul	Counter	Number of times Publish message is sent to Consul
Consul statistics	nfmonitor_consul_read_attempts	Counter	Number of times Consul read attempts
Consul statistics	nfmonitor_consul_read_failures	Counter	Number of times Consul read attempts failed
Consul statistics	nfmonitor_consul_read_success	Counter	Number of times Consul read attempts succeeded

NfMonitor statistics

- [nfmonitor_unstable_nf_counter](#)

Type

64-bit Counter

Description

Number of Times Nf is found Unstable

Labels

- [nfmonitor_force_restart_due_to_nf_unstability](#)

Type

64-bit Counter

Description

The number of times the Force Restart triggered due to NF-Unstability

Labels

- [nfmonitor_cumulative_readiness_failure_count](#)

Type

64-bit Counter

Description

The number of times the service is found to be failing the Readiness probe

Labels

critical_service_name, pod_name

- nfmonitor_cumulative_readiness_failure_alerts

Type

64-bit Counter

Description

Total number of EventAlerts generated due to Readiness probe failure

Labels

critical_service_name, pod_name

- nfmonitor_cumulative_k8api_get_query_failures

Type

64-bit Counter

Description

Total number of k8 api get pod query failures

Labels

critical_service_name, pod_name

DebugNfMonitorCmn

- nfmonitor_force_restart_counter

Type

64-bit Counter

Description

Number of times force-restart triggered

Labels

agent, name

NfManagement statistics

- nfmonitor_nfmgmt_service_request_attempts

Type

64-bit Counter

Description

Number of Requests sent to NRF

Labels

instance_id, msg_type, method, retry

- nfmonitor_nfmgmt_service_request_success

Type

64-bit Counter

Description

The number of times the Request to NRF is Successful

Labels

instance_id, msg_type, method, retry, status_code

- nfmonitor_nfmgmt_service_request_failures

Type

64-bit Counter

Description

The number of times the Request to NRF is Failure

Labels

instance_id, msg_type, method, retry, status_code

- nfmonitor_nfmgmt_aborted_request_msgs

Type

64-bit Counter

Description

The number of times the Request to NRF is Aborted

Labels

instance_id, msg_type

Debug statistics

- nfmonitor_vnf_restart_trigger

Type

64-bit Counter

Description

Number of VNF Restart triggers received

Labels

curr_nf_status, nfmgmt_state, nfmonitor_fsm_state, up_count, timestamp, trigger, type

- nfmonitor_fsm_state_changes

Type

64-bit Counter

Description

The FSM State Change events stats

Labels

curr_nf_status, nfmgmt_state, up_count, from_state, to_state

- nfmonitor_deactivation_triggers

Type

64-bit Counter

Description

The number of times the deactivation trigger received

Labels

nf_status, nfmgmt_state, up_count, nfmonitor_fsm_state

- nfmonitor_critical_service_up_notification

Type

64-bit Counter

Description

The number of times the CriticalService Up notification received

Labels

service, criticality, nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_critical_service_down_notification

Type

64-bit Counter

Description

The number of times the CriticalService Down notification received

Labels

service, criticality, nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_critical_services_overall_up_notification

Type

64-bit Counter

Description

The number of times the Overall CriticalServices Up notification received

Labels

nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_critical_services_overall_down_notification

Type

64-bit Counter

Description

The number of times the Overall CriticalServices Down notification received

Labels

nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_hysteresis_timer_started

Type

64-bit Counter

Description

The number of times the Hysteresis Timer is started

Labels

nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_hysteresis_timer_stopped

Type

64-bit Counter

Description

The number of times the Hysterisis Timer is stopped

Labels

nf_status, nfmgmt_state, up_count, fsm_state

• nfmonitor_hysterisis_timer_expired

Type

64-bit Counter

Description

The number of times the Hysterisis Timer is expired

Labels

nf_status, nfmgmt_state, up_count, fsm_state

• nfmonitor_personality_update_triggers

Type

64-bit Counter

Description

The number of times the Nf Personality Update is received

Labels

nf_status, nfmgmt_state, up_count, fsm_state

• nfmonitor_maintenance_trigger_received

Type

64-bit Counter

Description

The number of times the Maintenance Mode is activated

Labels

nf_status, nfmgmt_state, up_count, fsm_state

• nfmonitor_profile_update_triggers

Type

64-bit Counter

Description

The number of times the Nf Profile Update happened

Labels

nf_status, nfmgmt_state, up_count, fsm_state, update_operation

• nfmonitor_critical_service_down_on_register_entry

Type

64-bit Counter

Description

The number of times the Critical Services were down while starting with RegisterFsmState

Labels

nf_status, up_count

ApiServer statistics

- nfmonitor_deployment_create_watch_request

Type

64-bit Counter

Description

The number of Watch Reqs created for deployment type

Labels

resource

- nfmonitor_deployment_delete_watch_request

Type

64-bit Counter

Description

The number of Watch Reqs deleted for deployment type

Labels

resource

- nfmonitor_deployment_watch_notification_success

Type

64-bit Counter

Description

The number of times Watch notifications are successful

Labels

resource

- nfmonitor_deployment_watch_notification_error

Type

64-bit Counter

Description

The number of times Watch notifications are failed

Labels

resource, reason

- nfmonitor_limit_range_query_attempts

Type

64-bit Counter

Description

The number of Limit Range query attempts

Labels

-
- nfmonitor_limit_range_query_failures

Type

64-bit Counter

Description

The number of Limit Range query failures

Labels

status_code

• nfmonitor_limit_range_query_success

Type

64-bit Counter

Description

The number of Limit Range query succeeded

Labels

status_code

• nfmonitor_hpa_query_attempts

Type

64-bit Counter

Description

The number of Hpa query attempts

Labels

resource, continue_param

• nfmonitor_hpa_query_failures

Type

64-bit Counter

Description

The number of Hpa query failures

Labels

resource, continue_param, status_code

• nfmonitor_hpa_query_success

Type

64-bit Counter

Description

The number of Hpa query succeeded

Labels

resource, continue_param, status_code

• nfmonitor_readiness_probe_query_attempts

Type

64-bit Counter

Description

The number of Readiness Probe query attempts

Labels

resource, continue_param

- nfmonitor_readiness_probe_query_failures

Type

64-bit Counter

Description

The number of Readiness Probe query failures

Labels

resource, continue_param, status_code

- nfmonitor_readiness_probe_query_success

Type

64-bit Counter

Description

The number of Readiness Probe query succeeded

Labels

resource, continue_param, status_code

Feature Functionality statistics

- nfmonitor_critical_service_frequent_restart

Type

64-bit Counter

Description

Howmany times the minimum required stable CriticalService count is not met

Labels

service

- nfmonitor_alarm_due_to_frequent_restart

Type

64-bit Counter

Description

Howmany times Alarm is raised due to minimum required stable CriticalService count is not met

Labels

service, alarm_status

- nfmonitor_force_restart_due_to_frequent_restart

Type

64-bit Counter

Description

Howmany times Force Restart initiated due to missing stable CriticalService count

Labels

service

- nfmonitor_critical_service_readiness_failure

Type

64-bit Counter

Description

Howmany times the minimum required readiness succeeded CriticalService count is not met

Labels

service

- nfmonitor_alarm_due_to_readiness_failure

Type

64-bit Counter

Description

Alarm raised due to minimum count is not met for the readiness succeeded CriticalService

Labels

service, alarm_status

- nfmonitor_force_restart_due_to_readiness_failure

Type

64-bit Counter

Description

Howmany times Force Restart initiated due to readiness failure of CriticalService

Labels

service

Consul statistics

- nfmonitor_publish_status_into_consul

Type

64-bit Counter

Description

Howmany times Publish is sent to Consul

Labels

nf_status, nf_state, up_count, fsm_state

- nfmonitor_consul_read_attempts

Type

64-bit Counter

Description

Howmany times Consul read attempt is made

Labels

fsm_state, retry

- nfmonitor_consul_read_failures

Type

64-bit Counter

Description

Howmany times Consul read attempt are failed

Labels

fsm_state, retry, status_code

- nfmonitor_consul_read_success

Type

64-bit Counter

Description

Howmany times Consul read attempt are succeeded

Labels

fsm_state, retry, status_code

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- [Summary](#)
- [Packet Streaming Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Packet Streaming Metrics	num_packet_streaming_attempted	Counter	The total number of attempted streaming of packets
Packet Streaming Metrics	num_packet_streaming_success	Counter	The total number of packets streamed successfully
Packet Streaming Metrics	num_packet_streaming_failed	Counter	The total number of failed streaming of packets
Packet Streaming Metrics	num_packet_streaming_bytes	Counter	The total number of bytes successfully streamed

Packet Streaming Metrics

- num_packet_streaming_attempted

Type

64-bit Counter

Description

The total number of attempted streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_success

Type

64-bit Counter

Description

The total number of packets streamed successfully

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_failed

Type

64-bit Counter

Description

The total number of failed streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_bytes

Type

64-bit Counter

Description

The total number of bytes successfully streamed

Labels

interface_name, server_fqdn, kafka_topic

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- [Summary](#)
- [Circuit Breaker Metrics](#)
- [Service Endpoint Learner Metrics](#)
- [Service Endpoint Manager Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Circuit Breaker Metrics	sem_cb_endpoints	Gauge	Total number of endpoints for a service in respective states
Service Endpoint Learner Metrics	sem_num_dns_errors	Counter	Total number of DNS resolution errors for a given service
Service Endpoint Learner Metrics	sem_num_endpoints	Gauge	Total number of endpoints learnt for a service
Service Endpoint Manager Metrics	sem_num_endpoint_transactions_sent	Counter	Total number of times an endpoint is given to the application for communication with it
Service Endpoint Manager Metrics	sem_num_endpoint_transaction_status_received	Counter	Total number of transactions for a given endpoint received with the transaction status

Circuit Breaker Metrics

- sem_cb_endpoints

Type

64-bit Gauge

Description

Total number of endpoints for a service in respective states. There are 3 states viz. Active, Blocked, Restoring. In Active state, the endpoint is responding as expected, i.e. the response code doesn't indicate a service failure. The endpoint will be moved to Blocked state based on response failures which indicate service failure. After a timeout, the endpoint will be moved to Restoring state to check if the endpoint has recovered. In this state, if endpoint responds with response code other than service failure, it will be moved to Active state. Else, it will be moved back to Blocked state.

Labels

service_name, endpoint_state

Service Endpoint Learner Metrics

- sem_num_dns_errors

Type

64-bit Counter

Description

Total number of DNS resolution errors for a given service, labels are service name for DNS resolution and DNS error codes

Labels

service_name, error_code

- sem_num_endpoints

Type

64-bit Gauge

Description

Total number of endpoints learnt from the Kubernetes DNS server for a given service

Labels

service

Service Endpoint Manager Metrics

- sem_num_endpoint_transactions_sent

Type

64-bit Counter

Description

Total number of times an endpoint is given to the application for communication with it

Labels

service

- sem_num_endpoint_transaction_status_received

Type

64-bit Counter

Description

Total number of transactions for a given endpoint received with the transaction status

Labels

service, status

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- [Summary](#)
- [ETCD Client Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_put_requests	Counter	The total number of ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_requests_sent	Counter	The total number of ETCD Client PUT Requests sent.
ETCD Client Metrics	etcd_put_successes	Counter	The total number of successful ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_failures	Counter	The total number of failed ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_with_lease_requests	Counter	The total number of ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_requests_sent	Counter	The total number of ETCD Client PUT with Lease Requests sent.
ETCD Client Metrics	etcd_put_with_lease_successes	Counter	The total number of successful ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_failures	Counter	The total number of failed ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_range_requests	Counter	The total number of ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_requests_sent	Counter	The total number of ETCD Client Range Requests sent.

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_range_successes	Counter	The total number of successful ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_failures	Counter	The total number of failed ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_delete_requests	Counter	The total number of ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_requests_sent	Counter	The total number of ETCD Client Range Delete Requests sent.
ETCD Client Metrics	etcd_range_delete_successes	Counter	The total number of successful ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_failures	Counter	The total number of failed ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_txn_requests	Counter	The total number of ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_requests_sent	Counter	The total number of ETCD Client Txn Requests sent.
ETCD Client Metrics	etcd_txn_successes	Counter	The total number of successful ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_failures	Counter	The total number of failed ETCD Client Txn Requests.
ETCD Client Metrics	etcd_watch_requests	Counter	The total number of ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_requests_sent	Counter	The total number of ETCD Client Watch Requests sent.
ETCD Client Metrics	etcd_watch_successes	Counter	The total number of successful ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_failures	Counter	The total number of failed ETCD Client Watch Requests.
ETCD Client Metrics	etcdwatcher_requests	Counter	The total number of ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_requests_sent	Counter	The total number of ETCD Watcher Requests sent.
ETCD Client Metrics	etcdwatcher_successes	Counter	The total number of successful ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_failures	Counter	The total number of failed ETCD Watcher Requests.

ETCD Client Metrics

- etcd_put_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests.

Labels

keyPrefix

- etcd_put_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests sent.

Labels

keyPrefix

- etcd_put_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT Requests.

Labels

keyPrefix

- etcd_put_failures
 - Type
 - 64-bit Counter
 - Description
 - The total number of failed ETCD Client PUT Requests.
 - Labels
 - keyPrefix, failureReason**
- etcd_put_with_lease_requests
 - Type
 - 64-bit Counter
 - Description
 - The total number of ETCD Client PUT with Lease Requests.
 - Labels
 - keyPrefix**
- etcd_put_with_lease_requests_sent
 - Type
 - 64-bit Counter
 - Description
 - The total number of ETCD Client PUT with Lease Requests sent.
 - Labels
 - keyPrefix**
- etcd_put_with_lease_successes
 - Type
 - 64-bit Counter
 - Description
 - The total number of successful ETCD Client PUT with Lease Requests.
 - Labels
 - keyPrefix**
- etcd_put_with_lease_failures
 - Type
 - 64-bit Counter
 - Description
 - The total number of failed ETCD Client PUT with Lease Requests.
 - Labels
 - keyPrefix, failureReason**
- etcd_range_requests
 - Type
 - 64-bit Counter
 - Description

The total number of ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests sent.

Labels

keyPrefix

- etcd_range_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Requests.

Labels

keyPrefix, failureReason

- etcd_range_delete_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests.

Labels

keyPrefix

- etcd_range_delete_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests sent.

Labels

keyPrefix

- etcd_range_delete_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Delete Requests.

Labels

keyPrefix

- etcd_range_delete_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Delete Requests.

Labels

keyPrefix, failureReason

- etcd_txn_requests

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests.

Labels

keyPrefix

- etcd_txn_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests sent.

Labels

keyPrefix

- etcd_txn_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Txn Requests.

Labels

keyPrefix

- etcd_txn_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Txn Requests.

Labels

keyPrefix, failureReason

• etcd_watch_requests

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests.

Labels

keyPrefix

• etcd_watch_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests sent.

Labels

keyPrefix

• etcd_watch_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Watch Requests.

Labels

keyPrefix

• etcd_watch_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Watch Requests.

Labels

keyPrefix, failureReason

• etcdwatcher_requests

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests.

Labels

keyPrefix

• etcdwatcher_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests sent.

Labels

keyPrefix

- etcdwatcher_successes

Type

64-bit Counter

Description

The total number of successful ETCD Watcher Requests.

Labels

keyPrefix

- etcdwatcher_failures

Type

64-bit Counter

Description

The total number of failed ETCD Watcher Requests.

Labels

keyPrefix, failureReason

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- [Summary](#)
- [HTTP Punting Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_sent	Counter	The total number of HTTP requests successfully punted.
HTTP Punting Metrics	punting_http_requests_failed	Counter	The total number of HTTP requests punting failed.
HTTP Punting Metrics	punting_http_responses_received	Counter	The total number of HTTP responses received for the requests punted.
HTTP Punting Metrics	punting_http_responses_forwarded	Counter	The total number of HTTP responses forwarded after punting transaction completion.
HTTP Punting Metrics	punting_http_requests_aborted	Counter	The total number of HTTP requests aborted.
HTTP Punting Metrics	punting_http_request_duration_seconds	Histogram	Punting operation duration in seconds.

HTTP Punting Metrics

- punting_http_requests_sent

Type

64-bit Counter

Description

The total number of HTTP requests successfully punted.

Labels

message_type

- `punting_http_requests_failed`

Type

64-bit Counter

Description

The total number of HTTP requests for which the punting failed.

Labels

message_type

- `punting_http_responses_received`

Type

64-bit Counter

Description

The total number of HTTP responses received for the requests punted.

Labels

message_type, status_code

- `punting_http_responses_forwarded`

Type

64-bit Counter

Description

The total number of HTTP responses forwarded after punting transaction completion.

Labels

message_type, status_code

- `punting_http_requests_aborted`

Type

64-bit Counter

Description

The total number of HTTP requests aborted.

Labels

message_type

- `punting_http_request_duration_seconds`

Type

64-bit Histogram

Description

The punting transaction duration is measured from the time the HTTP request is sent out to the time the response is received for the punting request.

Labels

message_type

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

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- [Summary](#)
- [In memory bufferstatistics](#)

Summary

Group	Metric Name	Type	Synopsis
In memory bufferstatistics	inmemory_buffer_pdu_added	Counter	Count of the PDUs added to the inmemorybuffer
In memory bufferstatistics	inmemory_buffer_pdu_removed	Counter	Count of the PDUs removed from the inmemorybuffer
In memory bufferstatistics	inmemory_buffer_pdu_dropped	Counter	Count of the PDUs dropped from the inmemorybuffer
In memory bufferstatistics	inmemory_buffer_pdu_remove_all	Counter	Count of times removeAllPdus is called for the inmemorybuffer

In memory bufferstatistics

- [inmemory_buffer_pdu_added](#)

Type

64-bit Counter

Description

Count of the PDUs added to the inmemorybuffer

Labels

did, position
- [inmemory_buffer_pdu_removed](#)

Type

64-bit Counter

Description

Count of the PDUs removed from the inmemorybuffer

Labels

did, position
- [inmemory_buffer_pdu_dropped](#)

Type

64-bit Counter

Description

Count of the PDUs dropped from the inmemorybuffer

Labels

did, position
- [inmemory_buffer_pdu_remove_all](#)

Type

64-bit Counter

Description

Count of times removeAllPdus is called for the inmemorybuffer

Labels

did, reason

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- [Summary](#)
- [LiCmn statistics](#)
- [Key Ciphering Statistics](#)

Summary

Group	Metric Name	Type	Synopsis
LiCmn statistics	svcmgr_service_available_ind_notification	Counter	How many times, Service Available Indication message received
LiCmn statistics	svcmgr_service_available_ind_parse_errs	Counter	How many times, Service Available Indication parse error happened
LiCmn statistics	svcmgr_service_available_ind_notification_ack	Counter	How many times, Service Available Indication message is acknowledged
LiCmn statistics	svcmgr_service_unavailable_ind_notification	Counter	How many times, Service Unavailable Indication message
LiCmn statistics	svcmgr_service_unavailable_ind_parse_errs	Counter	How many times, Service Unavailable Indication parse error happened
LiCmn statistics	svcmgr_service_unavailable_ind_notification_ack	Counter	How many times, Service Unavailable Indication message is acknowledged
LiCmn statistics	svcmgr_publish_msg_sent	Counter	How many times, Publication message sent
LiCmn statistics	svcmgr_publish_msg_success	Counter	How many times, Publication message success response received
LiCmn statistics	svcmgr_publish_msg_failure	Counter	How many times, failure received for publication message
LiCmn statistics	svcmgr_subscribe_msg_sent	Counter	How many times, Subscribe message sent
LiCmn statistics	svcmgr_subscribe_msg_success	Counter	How many times, subscribe message success
LiCmn statistics	svcmgr_subscribe_msg_failure	Counter	How many times, failure received for subscribe message
LiCmn statistics	svcmgr_unsubscribe_msg_sent	Counter	How many times, unsubscribe message sent
LiCmn statistics	svcmgr_unsubscribe_msg_failure	Counter	How many times, failure received for unsubscribe message
LiCmn statistics	svcmgr_config_req_sent	Counter	How many times, config request sent
LiCmn statistics	svcmgr_config_req_failure	Counter	How many times, failure received for config request
LiCmn statistics	svcmgr_config_req_success	Counter	How many times, success received for config request
LiCmn statistics	svcmgr_config_register_response	Counter	How many times, config register response received
LiCmn statistics	svcmgr_config_register_response_ack	Counter	How many times, config register response is acknowledged
LiCmn statistics	svcmgr_config_register_response_parse_err	Counter	How many times, parse error happened for config register response
LiCmn statistics	svcmgr_config_update_notification	Counter	How many times, config update notification received
LiCmn statistics	svcmgr_config_update_notification_response	Counter	How many times, config update response
LiCmn statistics	svcmgr_intf_schema_update_ind_notification	Counter	How many times, interface schema update indication message received
LiCmn statistics	svcmgr_intf_schema_update_ind_notification_ack	Counter	How many times, interface schema update indication message is acknowledged
LiCmn statistics	svcmgr_intf_schema_update_ind_parse_errs	Counter	How many times, interface schema update indication parse error happened
Key Ciphering Statistics	li_cmn_cipher_object_registered	Counter	Report number of cipher objects registered
Key Ciphering Statistics	li_cmn_secret_key_rotated	Counter	Report number of times secret key has been changed
Key Ciphering Statistics	li_cmn_enciphering_failed	Counter	Report number of times enciphering has failed
Key Ciphering Statistics	li_cmn_deciphering_failed	Counter	Report number of times deciphering has failed
Key Ciphering Statistics	li_cmn_key_version_not_available	Counter	Report number of times when the key corresponding to the key version is not available.

LiCmn statistics

- [svcmgr_service_available_ind_notification](#)

Type

64-bit Counter

Description

Number of Times Service Available Indication notification received

Labels

servicename

- **svcmgr_service_available_ind_parse_errs**

Type

64-bit Counter

Description

Number of Times Service Available Indication parse error happened

Labels

- **svcmgr_service_available_ind_notification_ack**

Type

64-bit Counter

Description

Number of Times Service Available Indication message is acknowledged

Labels

servicename

- **svcmgr_service_unavailable_ind_notification**

Type

64-bit Counter

Description

Number of Times Service Unavailable Indication notification received

Labels

servicename

- **svcmgr_service_unavailable_ind_parse_errs**

Type

64-bit Counter

Description

Number of Times Service Unavailable Indication parse error happened

Labels

- **svcmgr_service_unavailable_ind_notification_ack**

Type

64-bit Counter

Description

Number of Times Service Unavailable Indication message is acknowledged

Labels

servicename

- **svcmgr_publish_msg_sent**

Type

64-bit Counter

Description

Number of Times Publication message sent

Labels

servicename, retry

- **svcmgr_publish_msg_success**

Type

64-bit Counter

Description

Number of Times Publication message success response received

Labels

servicename, causecode, retry

- **svcmgr_publish_msg_failure**

Type

64-bit Counter

Description

Number of Times failure received for publication message

Labels

servicename, causecode, retry

- **svcmgr_subscribe_msg_sent**

Type

64-bit Counter

Description

Number of Times Subscribe message sent

Labels

servicename, retry

- **svcmgr_subscribe_msg_success**

Type

64-bit Counter

Description

Number of Times Send subscribe message success response received

Labels

servicename, causecode, retry

- **svcmgr_subscribe_msg_failure**

Type

64-bit Counter

Description

Number of time failure received for subscribe message

Labels

servicename, causecode, retry

- **svcmgr_unsubscribe_msg_sent**

Type

64-bit Counter

Description

Number of Times unsubscribe message sent

Labels

servicename

- **svcmgr_unsubscribe_msg_failure**

Type

64-bit Counter

Description

Number of Times failure received for unsubscribe message

Labels

servicename

- **svcmgr_config_req_sent**

Type

64-bit Counter

Description

Number of Times config request sent

Labels

retry

- **svcmgr_config_req_failure**

Type

64-bit Counter

Description

Number of Times failure received for config request

Labels

causecode, retry

- **svcmgr_config_req_success**

Type

64-bit Counter

Description

Number of Times success received for config request

Labels

causecode, retry

• svcmgr_config_register_response Type 64-bit Counter Description Number of Times config register response received Labels statuscode
• svcmgr_config_register_response_ack Type 64-bit Counter Description Number of Times config register response is acknowledged Labels
• svcmgr_config_register_response_parse_err Type 64-bit Counter Description Number of Times parse error happened for config register response Labels
• svcmgr_config_update_notification Type 64-bit Counter Description Number of Times config update notification received Labels
• svcmgr_config_update_notification_response Type 64-bit Counter Description Number of Times config update response sent Labels
• svcmgr_intf_schema_update_ind_notification Type 64-bit Counter Description Number of Times interface schema update indication message received Labels

servicename

- svcmgr_intf_schema_update_ind_notification_ack

Type

64-bit Counter

Description

Number of Times interface schema update indication message is acknowledged

Labels

servicename

- svcmgr_intf_schema_update_ind_parse_errs

Type

64-bit Counter

Description

Number of Times Service interface schema update indication parse error happened

Labels

Key Ciphering Statistics

- li_cm_n_cipher_object_registered

Type

64-bit Counter

Description

Total number of cipher objects registered

Labels

key_version

- li_cm_n_secret_key_rotated

Type

64-bit Counter

Description

Total number of times secret keys has been rotated

Labels

- li_cm_n_enciphering_failed

Type

64-bit Counter

Description

Total number of times enciphering has failed

Labels

key_version

- li_cm_n_deciphering_failed

Type

64-bit Counter

Description

Total number of times deciphering has failed

Labels

key_version

- li_cmn_key_version_not_available

Type

64-bit Counter

Description

Total number of times key corresponding to the key version is not available.

Labels

key_version

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- Summary
- NfMonitor statistics
- DebugNfMonitorCmn
- NfManagement statistics
- Debug statistics
- ApiServer statistics
- Feature Functionality statistics
- Consul statistics

Summary

Group	Metric Name	Type	Synopsis
NfMonitor statistics	nfmonitor_unstable_nf_counter	Counter	Howmany times, the Nf is found to be Unstable
NfMonitor statistics	nfmonitor_force_restart_due_to_nf_unstability	Counter	Numbe of times the VNFRestart triggered due to Unstability of the NF
NfMonitor statistics	nfmonitor_cumulative_readiness_failure_count	Counter	Numbe of times Service sound to have insuffucient count due to Readiness probe
NfMonitor statistics	nfmonitor_cumulative_readiness_failure_alerts	Counter	Numbe of times Service sound to have insuffucient count due to Readiness probe
NfMonitor statistics	nfmonitor_cumulative_k8api_get_query_failures	Counter	Numbe of times the k8 get Api query failures
DebugNfMonitorCmn	nfmonitor_force_restart_counter	Counter	Number of times force-restart triggered
NfManagement statistics	nfmonitor_nfmgmt_service_request_attempts	Counter	Howmany times, the Nf is attempting to send NRF Requests
NfManagement statistics	nfmonitor_nfmgmt_service_request_success	Counter	Numbe of times Success response is received for the requests to NRF
NfManagement statistics	nfmonitor_nfmgmt_service_request_failures	Counter	Numbe of times Failure response is received for the requests to NRF
NfManagement statistics	nfmonitor_nfmgmt_aborted_request_msgs	Counter	Numbe of times the requests to NRF is Aborted/Cancelled
Debug statistics	nfmonitor_vnf_restart_trigger	Counter	Howmany times, the Force Restart is received
Debug statistics	nfmonitor_fsm_state_changes	Counter	The statistics for the the FSM state changes
Debug statistics	nfmonitor_deactivation_triggers	Counter	Numbe of times the NF deactivation trigger is received
Debug statistics	nfmonitor_critical_service_up_notification	Counter	Numbe of times the Critical Service Up Notification received
Debug statistics	nfmonitor_critical_service_down_notification	Counter	Numbe of times the Critical Service Down Notification received
Debug statistics	nfmonitor_critical_services_overall_up_notification	Counter	Numbe of times the Up notification received when all monitored Critical Services are available
Debug statistics	nfmonitor_critical_services_overall_down_notification	Counter	Numbe of times the Down notification received when at least one Critical Service was unavailable

Group	Metric Name	Type	Synopsis
Debug statistics	nfmonitor_hysteresis_timer_started	Counter	Numbe of times the Hysteresis Timer is started due to missing Critical Service
Debug statistics	nfmonitor_hysteresis_timer_stopped	Counter	Numbe of times the Hysteresis Timer is stopped due to Critical Service become available
Debug statistics	nfmonitor_hysteresis_timer_expired	Counter	Numbe of times the Hysteresis Timer is expired due to missing Critical Service
Debug statistics	nfmonitor_personality_update_triggers	Counter	Numbe of times the Nf Personality update is received
Debug statistics	nfmonitor_maintenance_trigger_received	Counter	Numbe of times the Maintenance Mode is activated
Debug statistics	nfmonitor_profile_update_triggers	Counter	Numbe of times the Nf Profile Update happened
Debug statistics	nfmonitor_critical_service_down_on_register_entry	Counter	Numbe of times the Critical Services were down while starting with RegisterFsmState
ApiServer statistics	nfmonitor_deployment_create_watch_request	Counter	Howmany times, the deployment watch requests are created
ApiServer statistics	nfmonitor_deployment_delete_watch_request	Counter	Howmany times, the deployment watch requests are deleted
ApiServer statistics	nfmonitor_deployment_watch_notification_success	Counter	Howmany times, the watch notifications are successful
ApiServer statistics	nfmonitor_deployment_watch_notification_error	Counter	Howmany times, the watch notifications are failed
ApiServer statistics	nfmonitor_limit_range_query_attempts	Counter	Howmany times, the limit-range query attempts are made
ApiServer statistics	nfmonitor_limit_range_query_failures	Counter	Howmany times, the limit-range query attempts are failed
ApiServer statistics	nfmonitor_limit_range_query_success	Counter	Howmany times, the limit-range query attempts are succeeded
ApiServer statistics	nfmonitor_hpa_query_attempts	Counter	Howmany times, the Hpa query attempts are made
ApiServer statistics	nfmonitor_hpa_query_failures	Counter	Howmany times, the Hpa query attempts are failed
ApiServer statistics	nfmonitor_hpa_query_success	Counter	Howmany times, the Hpa query attempts are succeeded
ApiServer statistics	nfmonitor_readiness_probe_query_attempts	Counter	Howmany times, the Readiness Probe query attempts are made
ApiServer statistics	nfmonitor_readiness_probe_query_failures	Counter	Howmany times, the Readiness Probe query attempts are failed
ApiServer statistics	nfmonitor_readiness_probe_query_success	Counter	Howmany times, the Readiness Probe query attempts are succeeded
Feature Functionality statistics	nfmonitor_critical_service_frequent_restart	Counter	Howmany times, minimum count is not met for the stable Critical Services
Feature Functionality statistics	nfmonitor_alarm_due_to_frequent_restart	Counter	Howmany times alarm raised, due to minimum count is not met for the stable Critical Services
Feature Functionality statistics	nfmonitor_force_restart_due_to_frequent_restart	Counter	Howmany times, force restart triggerd because of the unstable Critical Services
Feature Functionality statistics	nfmonitor_critical_service_readiness_failure	Counter	Howmany times, minimum count is not met for the readiness succeeded Critical Services
Feature Functionality statistics	nfmonitor_alarm_due_to_readiness_failure	Counter	Alarm raised, due to minimum count is not met for the readiness succeeded Critical Services
Feature Functionality statistics	nfmonitor_force_restart_due_to_readiness_failure	Counter	Force restart triggerd because of Critical Service Readiness failure
Consul statistics	nfmonitor_publish_status_into_consul	Counter	Number of times Publish message is sent to Consul
Consul statistics	nfmonitor_consul_read_attempts	Counter	Number of times Consul read attempts
Consul statistics	nfmonitor_consul_read_failures	Counter	Number of times Consul read attempts failed
Consul statistics	nfmonitor_consul_read_success	Counter	Number of times Consul read attempts succeeded

NfMonitor statistics

- [nfmonitor_unstable_nf_counter](#)

Type

64-bit Counter

Description

Number of Times Nf is found Unstable

Labels

nfmonitor_force_restart_due_to_nf_unstability	Type
	64-bit Counter
	Description
	The number of times the Force Restart triggered due to NF-Unstability
	Labels
nfmonitor_cumulative_readiness_failure_count	Type
	64-bit Counter
	Description
	The number of times the service is found to be failing the Readiness probe
	Labels
critical_service_name, pod_name	
nfmonitor_cumulative_readiness_failure_alerts	Type
	64-bit Counter
	Description
	Total number of EventAlerts generated due to Readiness probe failure
	Labels
critical_service_name, pod_name	
nfmonitor_cumulative_k8api_get_query_failures	Type
	64-bit Counter
	Description
	Total number of k8 api get pod query failures
	Labels
critical_service_name, pod_name	

DebugNfMonitorCmn

nfmonitor_force_restart_counter	Type
	64-bit Counter
	Description
	Number of times force-restart triggered
	Labels
agent, name	

NfManagement statistics

nfmonitor_nfmgmt_service_request_attempts

Type

64-bit Counter

Description

Number of Requests sent to NRF

Labels

instance_id, msg_type, method, retry

- nfmonitor_nfmgmt_service_request_success

Type

64-bit Counter

Description

The number of times the Request to NRF is Successful

Labels

instance_id, msg_type, method, retry, status_code

- nfmonitor_nfmgmt_service_request_failures

Type

64-bit Counter

Description

The number of times the Request to NRF is Failure

Labels

instance_id, msg_type, method, retry, status_code

- nfmonitor_nfmgmt_aborted_request_msgs

Type

64-bit Counter

Description

The number of times the Request to NRF is Aborted

Labels

instance_id, msg_type

Debug statistics

- nfmonitor_vnf_restart_trigger

Type

64-bit Counter

Description

Number of VNF Restart triggers received

Labels

curr_nf_status, nfmgmt_state, nfmonitor_fsm_state, up_count, timestamp, trigger, type

- nfmonitor_fsm_state_changes

Type

64-bit Counter

Description

The FSM State Change events stats

Labels

curr_nf_status, nfmgmt_state, up_count, from_state, to_state

- nfmonitor_deactivation_triggers

Type

64-bit Counter

Description

The number of times the deactivation trigger received

Labels

nf_status, nfmgmt_state, up_count, nfmonitor_fsm_state

- nfmonitor_critical_service_up_notification

Type

64-bit Counter

Description

The number of times the CriticalService Up notification received

Labels

service, criticality, nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_critical_service_down_notification

Type

64-bit Counter

Description

The number of times the CriticalService Down notification received

Labels

service, criticality, nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_critical_services_overall_up_notification

Type

64-bit Counter

Description

The number of times the Overall CriticalServices Up notification received

Labels

nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_critical_services_overall_down_notification

Type

64-bit Counter

Description

The number of times the Overall CriticalServices Down notification received

Labels

nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_hysterisis_timer_started

Type

64-bit Counter

Description

The number of times the Hysterisis Timer is started

Labels

nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_hysterisis_timer_stopped

Type

64-bit Counter

Description

The number of times the Hysterisis Timer is stopped

Labels

nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_hysterisis_timer_expired

Type

64-bit Counter

Description

The number of times the Hysterisis Timer is expired

Labels

nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_personality_update_triggers

Type

64-bit Counter

Description

The number of times the Nf Personality Update is received

Labels

nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_maintenance_trigger_received

Type

64-bit Counter

Description

The number of times the Maintenance Mode is activated

Labels

nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_profile_update_triggers

Type

64-bit Counter

Description

The number of times the Nf Profile Update happened

Labels

nf_status, nfmgmt_state, up_count, fsm_state, update_operation

- nfmonitor_critical_service_down_on_register_entry

Type

64-bit Counter

Description

The number of times the Critical Services were down while starting with RegisterFsmState

Labels

nf_status, up_count

ApiServer statistics

- nfmonitor_deployment_create_watch_request

Type

64-bit Counter

Description

The number of Watch Reqs created for deployment type

Labels

resource

- nfmonitor_deployment_delete_watch_request

Type

64-bit Counter

Description

The number of Watch Reqs deleted for deployment type

Labels

resource

- nfmonitor_deployment_watch_notification_success

Type

64-bit Counter

Description

The number of times Watch notifications are successful

Labels

resource

- nfmonitor_deployment_watch_notification_error

Type

64-bit Counter

Description

The number of times Watch notifications are failed

Labels

resource, reason

- nfmonitor_limit_range_query_attempts

Type

64-bit Counter

Description

The number of Limit Range query attempts

Labels

- `nfmonitor_limit_range_query_failures`

Type

64-bit Counter

Description

The number of Limit Range query failures

Labels

status_code

- `nfmonitor_limit_range_query_success`

Type

64-bit Counter

Description

The number of Limit Range query succeeded

Labels

status_code

- `nfmonitor_hpa_query_attempts`

Type

64-bit Counter

Description

The number of Hpa query attempts

Labels

resource, continue_param

- `nfmonitor_hpa_query_failures`

Type

64-bit Counter

Description

The number of Hpa query failures

Labels

resource, continue_param, status_code

- `nfmonitor_hpa_query_success`

Type

64-bit Counter

Description

The number of Hpa query succeeded

Labels

resource, continue_param, status_code

- nfmonitor_readiness_probe_query_attempts

Type

64-bit Counter

Description

The number of Readiness Probe query attempts

Labels

resource, continue_param

- nfmonitor_readiness_probe_query_failures

Type

64-bit Counter

Description

The number of Readiness Probe query failures

Labels

resource, continue_param, status_code

- nfmonitor_readiness_probe_query_success

Type

64-bit Counter

Description

The number of Readiness Probe query succeeded

Labels

resource, continue_param, status_code

Feature Functionality statistics

- nfmonitor_critical_service_frequent_restart

Type

64-bit Counter

Description

Howmany times the minimum required stable CriticalService count is not met

Labels

service

- nfmonitor_alarm_due_to_frequent_restart

Type

64-bit Counter

Description

Howmany times Alarm is raised due to minimum required stable CriticalService count is not met

Labels

service, alarm_status

- nfmonitor_force_restart_due_to_frequent_restart

Type

64-bit Counter

Description

Howmany times Force Restart initiated due to missing stable CriticalService count

Labels

service

- nfmonitor_critical_service_readiness_failure

Type

64-bit Counter

Description

Howmany times the minimum required readiness succeeded CriticalService count is not met

Labels

service

- nfmonitor_alarm_due_to_readiness_failure

Type

64-bit Counter

Description

Alarm raised due to minimum count is not met for the readiness succeeded CriticalService

Labels

service, alarm_status

- nfmonitor_force_restart_due_to_readiness_failure

Type

64-bit Counter

Description

Howmany times Force Restart initiated due to readiness failure of CriticalService

Labels

service

Consul statistics

- nfmonitor_publish_status_into_consul

Type

64-bit Counter

Description

Howmany times Publish is sent to Consul

Labels

nf_status, nf_state, up_count, fsm_state

- nfmonitor_consul_read_attempts

Type

64-bit Counter

Description

Howmany times Consul read attempt is made

Labels

fsm_state, retry

- nfmonitor_consul_read_failures

Type

64-bit Counter

Description

Howmany times Consul read attempt are failed

Labels

fsm_state, retry, status_code

- nfmonitor_consul_read_success

Type

64-bit Counter

Description

Howmany times Consul read attempt are succeeded

Labels

fsm_state, retry, status_code

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- [Packet Streaming Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Packet Streaming Metrics	num_packet_streaming_attempted	Counter	The total number of attempted streaming of packets
Packet Streaming Metrics	num_packet_streaming_success	Counter	The total number of packets streamed successfully
Packet Streaming Metrics	num_packet_streaming_failed	Counter	The total number of failed streaming of packets
Packet Streaming Metrics	num_packet_streaming_bytes	Counter	The total number of bytes successfully streamed

Packet Streaming Metrics

- num_packet_streaming_attempted

Type

64-bit Counter

Description

The total number of attempted streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_success

Type

64-bit Counter

Description

The total number of packets streamed successfully

Labels

interface_name, server_fqdn, kafka_topic

• num_packet_streaming_failed

Type

64-bit Counter

Description

The total number of failed streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

• num_packet_streaming_bytes

Type

64-bit Counter

Description

The total number of bytes successfully streamed

Labels

interface_name, server_fqdn, kafka_topic

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- [Summary](#)
- [PCFP Procedure Statistics](#)

Summary

Group	Metric Name	Type	Synopsis
PCFP Procedure Statistics	pfcf_association_setup_attempt	Counter	Association setup attempts.
PCFP Procedure Statistics	pfcf_association_setup_success	Counter	Association setup successes.
PCFP Procedure Statistics	pfcf_association_setup_failure	Counter	Association setup failures.
PCFP Procedure Statistics	pfcf_peer_association_setup_attempt	Counter	Peer association setup attempts.
PCFP Procedure Statistics	pfcf_peer_association_setup_success	Counter	Peer association setup successes.
PCFP Procedure Statistics	pfcf_peer_association_setup_failure	Counter	Peer association setup failures.
PCFP Procedure Statistics	pfcf_association_update_attempt	Counter	Association update attempts.
PCFP Procedure Statistics	pfcf_association_update_success	Counter	Association update successes.
PCFP Procedure Statistics	pfcf_association_update_failure	Counter	Association update failures.
PCFP Procedure Statistics	pfcf_peer_association_update_attempt	Counter	Peer association update attempts.
PCFP Procedure Statistics	pfcf_peer_association_update_success	Counter	Peer association update successes.
PCFP Procedure Statistics	pfcf_peer_association_update_failure	Counter	Peer association update failures.

Group	Metric Name	Type	Synopsis
PFCP Procedure Statistics	pfcf_association_release_attempt	Counter	Association release attempts.
PFCP Procedure Statistics	pfcf_association_release_success	Counter	Association release successes.
PFCP Procedure Statistics	pfcf_association_release_failure	Counter	Association release failures.
PFCP Procedure Statistics	pfcf_peer_association_release_attempt	Counter	Peer association release attempts.
PFCP Procedure Statistics	pfcf_peer_association_release_success	Counter	Peer association release successes.
PFCP Procedure Statistics	pfcf_peer_association_release_failure	Counter	Peer association release failures.
PFCP Procedure Statistics	pfcf_heartbeat_attempt	Counter	HeartBeat attempts.
PFCP Procedure Statistics	pfcf_heartbeat_success	Counter	HeartBeat successes.
PFCP Procedure Statistics	pfcf_heartbeat_failure	Counter	HeartBeat failures.
PFCP Procedure Statistics	pfcf_peer_heartbeat_attempt	Counter	Peer HeartBeat attempts.
PFCP Procedure Statistics	pfcf_peer_heartbeat_success	Counter	Peer HeartBeat successes.
PFCP Procedure Statistics	pfcf_peer_heartbeat_failure	Counter	Peer HeartBeat failures.
PFCP Procedure Statistics	pfcf_peer_heartbeat_received_without_association	Counter	Peer HeartBeat attempts without association.
PFCP Procedure Statistics	pfcf_peer_heartbeat_peer_association_added_or_updated_during_etcd_sync_wait_timer	Counter	Peer info added or updated during etcd sync wait time.
PFCP Procedure Statistics	pfcf_session_establishment_attempt	Counter	Session establishment attempts.
PFCP Procedure Statistics	pfcf_session_establishment_success	Counter	Session establishment successes.
PFCP Procedure Statistics	pfcf_session_establishment_failure	Counter	Session establishment failures.
PFCP Procedure Statistics	pfcf_session_modification_attempt	Counter	Session modification attempts.
PFCP Procedure Statistics	pfcf_session_modification_success	Counter	Session modification successes.
PFCP Procedure Statistics	pfcf_session_modification_failure	Counter	Session modification failures.
PFCP Procedure Statistics	pfcf_session_deletion_attempt	Counter	Session deletion attempts.
PFCP Procedure Statistics	pfcf_session_deletion_success	Counter	Session deletion successes.
PFCP Procedure Statistics	pfcf_session_deletion_failure	Counter	Session deletion failures.
PFCP Procedure Statistics	pfcf_session_report_attempt	Counter	Session report attempts.
PFCP Procedure Statistics	pfcf_session_report_success	Counter	Session report successes.
PFCP Procedure Statistics	pfcf_session_report_failure	Counter	Session report failures.
PFCP Procedure Statistics	pfcf_pfdm_attempt	Counter	Pfdm attempts.

Group	Metric Name	Type	Synopsis
PFCP Procedure Statistics	pfc_pfdm_success	Counter	Pfdm successes.
PFCP Procedure Statistics	pfc_pfdm_failure	Counter	Pfdm failures.
PFCP Procedure Statistics	pfc_nodes_dnn_report_attempts	Counter	Nodes dnn report attempts.
PFCP Procedure Statistics	pfc_nodes_dnn_report_success	Counter	Node dnn report successes.
PFCP Procedure Statistics	pfc_nodes_dnn_report_failed	Counter	Node dnn report failures.
PFCP Procedure Statistics	pfc_nodes_dnn_report_sent_to_peer	Counter	Node dnn report sent to peer
PFCP Procedure Statistics	pfc_nodes_dnn_report_sent_to_peer_failed	Counter	Node dnn report sent to peer
PFCP Procedure Statistics	pfc_master_elections_count	Counter	pfc master election count.
PFCP Procedure Statistics	pfc_master_changed_count	Counter	pfc node master changed count.
PFCP Procedure Statistics	pfc_node_promoted_to_master_count	Counter	dnn report election node promoted count.
PFCP Procedure Statistics	pfc_node_demoted_from_master_count	Counter	dnn report election node demoted count.

PFCP Procedure Statistics

- [pfc_association_setup_attempt](#)

Type

64-bit Counter

Description

The number of association setup attempts.

Labels

peer_id

- [pfc_association_setup_success](#)

Type

64-bit Counter

Description

The number of association setup successes.

Labels

peer_id

- [pfc_association_setup_failure](#)

Type

64-bit Counter

Description

The number of association setup failures.

Labels

peer_id, cause

• pfcpeer_association_setup_attempt
Type
64-bit Counter
Description
The number of peer association setup attempts.
Labels
<i>peer_id</i>
• pfcpeer_association_setup_success
Type
64-bit Counter
Description
The number of peer association setup successes.
Labels
<i>peer_id</i>
• pfcpeer_association_setup_failure
Type
64-bit Counter
Description
The number of peer association setup failures.
Labels
<i>peer_id, cause</i>
• pfcpeer_association_update_attempt
Type
64-bit Counter
Description
The number of association update attempts.
Labels
<i>peer_id, release_reason</i>
• pfcpeer_association_update_success
Type
64-bit Counter
Description
The number of association update successes.
Labels
<i>peer_id, release_reason</i>
• pfcpeer_association_update_failure
Type
64-bit Counter
Description

The number of association update failures.

Labels

peer_id, release_reason, cause

- pfcpeer_association_update_attempt

Type

64-bit Counter

Description

The number of association update peer attempts.

Labels

peer_id, release_reason

- pfcpeer_association_update_success

Type

64-bit Counter

Description

The number of association update peer successes.

Labels

peer_id, release_reason

- pfcpeer_association_update_failure

Type

64-bit Counter

Description

The number of association update peer failures.

Labels

peer_id, release_reason, cause

- pfcpeer_association_release_attempt

Type

64-bit Counter

Description

The number of association release attempts.

Labels

peer_id, release_reason

- pfcpeer_association_release_success

Type

64-bit Counter

Description

The number of association release successes.

Labels

peer_id, release_reason

- pfcpeer_association_release_failure

Type

64-bit Counter

Description

The number of association release failures.

Labels

peer_id, cause, release_reason

• pfcpeer_association_release_attempt

Type

64-bit Counter

Description

The number of association release peer attempts.

Labels

peer_id, release_reason

• pfcpeer_association_release_success

Type

64-bit Counter

Description

The number of association release peer successes.

Labels

peer_id, release_reason

• pfcpeer_association_release_failure

Type

64-bit Counter

Description

The number of association release peer failures.

Labels

peer_id, cause, release_reason

• pfcpeer_heartbeat_attempt

Type

64-bit Counter

Description

The number of heartbeat attempts.

Labels

peer_id

• pfcpeer_heartbeat_success

Type

64-bit Counter

Description

The number of heartbeat successes.

Labels

peer_id

- pfcpeer_heartbeat_failure

Type

64-bit Counter

Description

The number of heartbeat failures.

Labels

peer_id, cause

- pfcpeer_peer_heartbeat_attempt

Type

64-bit Counter

Description

The number of heartbeat peer attempts.

Labels

peer_id

- pfcpeer_peer_heartbeat_success

Type

64-bit Counter

Description

The number of heartbeat peer successes.

Labels

peer_id

- pfcpeer_peer_heartbeat_failure

Type

64-bit Counter

Description

The number of heartbeat peer failures.

Labels

peer_id, cause

- pfcpeer_peer_heartbeat_received_without_association

Type

64-bit Counter

Description

The number of heartbeat peer attempts without association.

Labels

peer_id

- pfcpeer_peer_heartbeat_peer_association_added_or_updated_during_etcd_sync_wait_timer

Type

64-bit Counter

Description

The number of peer info added or updated during etcd sync wait time.

Labels

peer_id

• pfc_session_establishment_attempt

Type

64-bit Counter

Description

The number of session establishment attempts.

Labels

interface, dnn, nssai, peer_id

• pfc_session_establishment_success

Type

64-bit Counter

Description

The number of session establishment successes.

Labels

interface, dnn, nssai, peer_id

• pfc_session_establishment_failure

Type

64-bit Counter

Description

The number of session establishment failures.

Labels

interface, dnn, nssai, peer_id, cause

• pfc_session_modification_attempt

Type

64-bit Counter

Description

The number of session modification attempts.

Labels

interface, dnn, nssai, peer_id

• pfc_session_modification_success

Type

64-bit Counter

Description

The number of session modification successes.

Labels

interface, dnn, nssai, peer_id

• pfcpsessionmodificationfailure

Type

64-bit Counter

Description

The number of session modification failures.

Labels

interface, dnn, nssai, peer_id, cause

• pfcpsessiondeletionattempt

Type

64-bit Counter

Description

The number of session deletion attempts.

Labels

interface, dnn, nssai, peer_id

• pfcpsessiondeletionsuccess

Type

64-bit Counter

Description

The number of session deletion successes.

Labels

interface, dnn, nssai, peer_id

• pfcpsessiondeletionfailure

Type

64-bit Counter

Description

The number of session deletion failures.

Labels

interface, dnn, nssai, peer_id, cause

• pfcpsessionreportattempt

Type

64-bit Counter

Description

The number of session report attempts.

Labels

interface, dnn, nssai, peer_id

• pfcpsessionreportsuccess

Type

64-bit Counter

Description

The number of session report successes.

Labels

interface, dnn, nssai, peer_id

- pfcpl_session_report_failure

Type

64-bit Counter

Description

The number of session report failures.

Labels

interface, dnn, nssai, peer_id, cause

- pfcpl_pfdm_attempt

Type

64-bit Counter

Description

The number of pfdm attempts.

Labels

peer_id

- pfcpl_pfdm_success

Type

64-bit Counter

Description

The number of pfdm successes.

Labels

peer_id

- pfcpl_pfdm_failure

Type

64-bit Counter

Description

The number of pfdm failures.

Labels

peer_id, cause

- pfcpl_nodes_dnn_report_attempts

Type

64-bit Counter

Description

The number of node dnn report attempts.

Labels

- pfcpl_nodes_dnn_report_success

Type

64-bit Counter

Description

The number of node dnn report successes.

Labels

- pfcpxnodes_dnn_report_failed

Type

64-bit Counter

Description

The number of node dnn report failures.

Labels

- pfcpxnodes_dnn_report_sent_to_peer

Type

64-bit Counter

Description

The number of node dnn reports sent to peer

Labels

peer_ip, peer_port, message

- pfcpxnodes_dnn_report_sent_to_peer_failed

Type

64-bit Counter

Description

The number of node dnn reports sent to peer failed

Labels

peer_ip, peer_port, message

- pfcpxmaster_elections_count

Type

64-bit Counter

Description

The number of elections done count

Labels

- pfcpxmaster_changed_count

Type

64-bit Counter

Description

The number of elections changed to promoted or demoted count

Labels

- pfcf_node_promoted_to_master_count

Type
64-bit Counter
Description
The number of node promoted done count
Labels

- pfcf_node_demoted_from_master_count

Type
64-bit Counter
Description
The number of node demoted done counted
Labels

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- [PFCP Protocol Message Statistics](#)

Summary

Group	Metric Name	Type	Synopsis
PFCP Protocol Message Statistics	pfcf_association_setup_request_sent	Counter	Association setup sent request messages.
PFCP Protocol Message Statistics	pfcf_association_setup_retransmitted_request_sent	Counter	Association setup retransmitted sent request messages.
PFCP Protocol Message Statistics	pfcf_association_setup_request_received	Counter	Association setup received request messages.
PFCP Protocol Message Statistics	pfcf_association_setup_retransmitted_request_received	Counter	Association setup received retransmitted request messages.
PFCP Protocol Message Statistics	pfcf_association_update_request_sent	Counter	Association update sent request messages.
PFCP Protocol Message Statistics	pfcf_association_update_retransmitted_request_sent	Counter	Association update retransmitted sent request messages.
PFCP Protocol Message Statistics	pfcf_association_update_request_received	Counter	Association update received request messages.
PFCP Protocol Message Statistics	pfcf_association_update_retransmitted_request_received	Counter	Association update received retransmitted request messages.
PFCP Protocol Message Statistics	pfcf_association_release_request_sent	Counter	Association release sent request messages.
PFCP Protocol Message Statistics	pfcf_association_release_retransmitted_request_sent	Counter	Association release retransmitted sent request messages.

Group	Metric Name	Type	Synopsis
PFCP Protocol Message Statistics	pfc_p_association_release_request_received	Counter	Association release received request messages.
PFCP Protocol Message Statistics	pfc_p_association_release_retransmitted_request_received	Counter	Association release received retransmitted request messages.
PFCP Protocol Message Statistics	pfc_p_association_setup_response_sent	Counter	Association setup sent response messages.
PFCP Protocol Message Statistics	pfc_p_association_setup_response_received	Counter	Association setup received response messages.
PFCP Protocol Message Statistics	pfc_p_association_update_response_sent	Counter	Association update sent response messages.
PFCP Protocol Message Statistics	pfc_p_association_update_response_received	Counter	Association update received response messages.
PFCP Protocol Message Statistics	pfc_p_association_release_response_sent	Counter	Association release sent response messages.
PFCP Protocol Message Statistics	pfc_p_association_release_response_received	Counter	Association release received response messages.
PFCP Protocol Message Statistics	pfc_p_version_not_supported_response_sent	Counter	Version not supported sent response messages.
PFCP Protocol Message Statistics	pfc_p_version_not_supported_response_received	Counter	Version not supported received response messages.
PFCP Protocol Message Statistics	pfc_p_heartbeat_request_sent	Counter	HeartBeat sent request messages.
PFCP Protocol Message Statistics	pfc_p_heartbeat_retransmitted_request_sent	Counter	HeartBeat retransmitted sent request messages.
PFCP Protocol Message Statistics	pfc_p_heartbeat_request_received	Counter	HeartBeat received request messages.
PFCP Protocol Message Statistics	pfc_p_heartbeat_retransmitted_request_received	Counter	HeartBeat received retransmitted request messages.
PFCP Protocol Message Statistics	pfc_p_heartbeat_response_sent	Counter	HeartBeat sent response messages.
PFCP Protocol Message Statistics	pfc_p_heartbeat_response_received	Counter	HeartBeat received response messages.
PFCP Protocol Message Statistics	pfc_p_session_establishment_request_sent	Counter	Session establishment sent request messages.
PFCP Protocol Message Statistics	pfc_p_session_establishment_retransmitted_request_sent	Counter	Session establishment retransmitted sent request messages.
PFCP Protocol Message Statistics	pfc_p_session_establishment_request_received	Counter	Session establishment received request messages.

Group	Metric Name	Type	Synopsis
PFCP Protocol Message Statistics	pfcplib_session_establishment_retransmitted_request_received	Counter	Session establishment received retransmitted request messages.
PFCP Protocol Message Statistics	pfcplib_session_modification_request_sent	Counter	Session modification sent request messages.
PFCP Protocol Message Statistics	pfcplib_session_modification_retransmitted_request_sent	Counter	Session modification retransmitted sent request messages.
PFCP Protocol Message Statistics	pfcplib_session_modification_request_received	Counter	Session modification received request messages.
PFCP Protocol Message Statistics	pfcplib_session_modification_retransmitted_request_received	Counter	Session modification received retransmitted request messages.
PFCP Protocol Message Statistics	pfcplib_session_deletion_request_sent	Counter	Session deletion sent request messages.
PFCP Protocol Message Statistics	pfcplib_session_deletion_retransmitted_request_sent	Counter	Session deletion retransmitted sent request messages.
PFCP Protocol Message Statistics	pfcplib_session_deletion_request_received	Counter	Session deletion received request messages.
PFCP Protocol Message Statistics	pfcplib_session_deletion_retransmitted_request_received	Counter	Session deletion received retransmitted request messages.
PFCP Protocol Message Statistics	pfcplib_session_report_request_sent	Counter	Session report sent request messages.
PFCP Protocol Message Statistics	pfcplib_session_report_retransmitted_request_sent	Counter	Session report retransmitted sent request messages.
PFCP Protocol Message Statistics	pfcplib_session_report_request_received	Counter	Session report received request messages.
PFCP Protocol Message Statistics	pfcplib_session_report_retransmitted_request_received	Counter	Session report received retransmitted request messages.
PFCP Protocol Message Statistics	pfcplib_session_establishment_response_sent	Counter	Session establishment sent response messages.
PFCP Protocol Message Statistics	pfcplib_session_establishment_response_received	Counter	Session establishment received response messages.
PFCP Protocol Message Statistics	pfcplib_session_modification_response_sent	Counter	Session modification sent response messages.
PFCP Protocol Message Statistics	pfcplib_session_modification_response_received	Counter	Session modification received response messages.
PFCP Protocol Message Statistics	pfcplib_session_deletion_response_sent	Counter	Session deletion sent response messages.
PFCP Protocol Message Statistics	pfcplib_session_deletion_response_received	Counter	Session deletion received response messages.

Group	Metric Name	Type	Synopsis
PFCP Protocol Message Statistics	pfcf_node_report_request_sent	Counter	Node report sent request messages.
PFCP Protocol Message Statistics	pfcf_node_report_type_request_received	Counter	Node report received request messages.
PFCP Protocol Message Statistics	pfcf_node_report_type_retransmitted_request_received	Counter	Node report received retransmitted request messages.
PFCP Protocol Message Statistics	pfcf_node_report_response_received	Counter	Node report received response messages.
PFCP Protocol Message Statistics	pfcf_node_report_retransmitted_request_sent	Counter	Node report retransmitted request messages sent.
PFCP Protocol Message Statistics	pfcf_session_report_response_sent	Counter	Session report sent response messages.
PFCP Protocol Message Statistics	pfcf_session_report_response_received	Counter	Session report received response messages.
PFCP Protocol Message Statistics	pfcf_pfdm_request_sent	Counter	Pfdm sent request messages.
PFCP Protocol Message Statistics	pfcf_pfdm_retransmitted_request_sent	Counter	Pfdm retransmitted sent request messages.
PFCP Protocol Message Statistics	pfcf_pfdm_request_received	Counter	Pfdm received request messages.
PFCP Protocol Message Statistics	pfcf_pfdm_retransmitted_request_received	Counter	Pfdm received retransmitted request messages.
PFCP Protocol Message Statistics	pfcf_pfdm_response_sent	Counter	Pfdm sent response messages.
PFCP Protocol Message Statistics	pfcf_pfdm_response_received	Counter	Pfdm received response messages.
PFCP Protocol Message Statistics	pfcf_bundle_message_sent	Counter	Number of PFCP bundle messages sent to a particular peer(represented by peer_ip and peer_port) due to a particular trigger(timeout or max buffer size or max number of PFCP messages in a single bundle).
PFCP Protocol Message Statistics	pfcf_bundle_message_sent_failure	Counter	Number of PFCP bundle messages sent failures to a particular peer(represented by peer_ip and peer_port), cause represents failure reason(like udp send socket failure).
PFCP Protocol Message Statistics	pfcf_bundle_message_received	Counter	Number of PFCP bundle messages received from a particular peer(represented by peer_ip and peer_port).
PFCP Protocol Message Statistics	pfcf_bundle_message_received_failure	Counter	Number of PFCP bundle messages decode failures, cause represents failure reason(like invalid packet).
PFCP Protocol Message Statistics	pfcf_number_of_messages_in_bundled_request_sent	Histogram	Number of messages present in bundled request sent
PFCP Protocol Message Statistics	pfcf_duration_of_bundling_in_bundled_request_sent	Histogram	Duration of bundling in bundled request sent in milli second

Group	Metric Name	Type	Synopsis
PFCP Protocol Message Statistics	pfcpl_size_of_message_of_bundled_request_sent	Histogram	Size of message in bytes of bundled request sent
PFCP Protocol Message Statistics	pfcpl_number_of_messages_in_bundled_request_received	Histogram	Number of messages present in bundled request received
PFCP Protocol Message Statistics	pfcpl_size_of_message_of_bundled_request_received	Histogram	Size of message in bytes of bundled request received

PFCP Protocol Message Statistics

- pfcpl_association_setup_request_sent
 - Type
 - 64-bit Counter
 - Description
 - The number of association setup request messages sent.
 - Labels
 - peer_ip**
- pfcpl_association_setup_retransmitted_request_sent
 - Type
 - 64-bit Counter
 - Description
 - The number of association setup retransmitted request messages sent.
 - Labels
 - peer_ip**
- pfcpl_association_setup_request_received
 - Type
 - 64-bit Counter
 - Description
 - The number of association setup request messages received.
 - Labels
 - peer_ip**
- pfcpl_association_setup_retransmitted_request_received
 - Type
 - 64-bit Counter
 - Description
 - The number of association setup retransmitted request messages received.
 - Labels
 - peer_ip**
- pfcpl_association_update_request_sent
 - Type

64-bit Counter

Description

The number of association update request messages sent.

Labels

peer_ip

• pfc_p_association_update_retransmitted_request_sent

Type

64-bit Counter

Description

The number of association update retransmitted request messages sent.

Labels

peer_ip

• pfc_p_association_update_request_received

Type

64-bit Counter

Description

The number of association update request messages received.

Labels

peer_ip

• pfc_p_association_update_retransmitted_request_received

Type

64-bit Counter

Description

The number of association update retransmitted request messages received.

Labels

peer_ip

• pfc_p_association_release_request_sent

Type

64-bit Counter

Description

The number of association release request messages sent.

Labels

peer_ip

• pfc_p_association_release_retransmitted_request_sent

Type

64-bit Counter

Description

The number of association release retransmitted request messages sent.

Labels

peer_ip

- pfcfp_association_release_request_received

Type

64-bit Counter

Description

The number of association release request messages received.

Labels

peer_ip

- pfcfp_association_release_retransmitted_request_received

Type

64-bit Counter

Description

The number of association release retransmitted request messages received.

Labels

peer_ip

- pfcfp_association_setup_response_sent

Type

64-bit Counter

Description

The number of association setup response messages sent.

Labels

peer_ip, pfcfp_cause

- pfcfp_association_setup_response_received

Type

64-bit Counter

Description

The number of association setup response messages received.

Labels

peer_ip, pfcfp_cause

- pfcfp_association_update_response_sent

Type

64-bit Counter

Description

The number of association update response messages sent.

Labels

peer_ip, pfcfp_cause

- pfcfp_association_update_response_received

Type

64-bit Counter

Description

The number of association update response messages received.

Labels

peer_ip, pfc_p_cause

- pfc_p_association_release_response_sent

Type

64-bit Counter

Description

The number of association release response messages sent.

Labels

peer_ip, pfc_p_cause

- pfc_p_association_release_response_received

Type

64-bit Counter

Description

The number of association release response messages received.

Labels

peer_ip, pfc_p_cause

- pfc_p_version_not_supported_response_sent

Type

64-bit Counter

Description

The number of version not supported response messages sent.

Labels

peer_ip, pfc_p_cause

- pfc_p_version_not_supported_response_received

Type

64-bit Counter

Description

The number of version not supported response messages received.

Labels

peer_ip, pfc_p_cause

- pfc_p_heartbeat_request_sent

Type

64-bit Counter

Description

The number of heartbeat request messages sent.

Labels

peer_ip

- pfc_p_heartbeat_retransmitted_request_sent

Type
64-bit Counter
Description
The number of heartbeat retransmitted request messages sent.
Labels

peer_ip

- pfc_p_heartbeat_request_received

Type
64-bit Counter
Description
The number of heartbeat request messages received.
Labels

peer_ip

- pfc_p_heartbeat_retransmitted_request_received

Type
64-bit Counter
Description
The number of heartbeat retransmitted request messages received.
Labels

peer_ip

- pfc_p_heartbeat_response_sent

Type
64-bit Counter
Description
The number of heartbeat response messages sent.
Labels

peer_ip

- pfc_p_heartbeat_response_received

Type
64-bit Counter
Description
The number of heartbeat response messages received.
Labels

peer_ip

- pfc_p_session_establishment_request_sent

Type
64-bit Counter
Description

The number of session establishment request messages sent.

Labels

interface, peer_ip, nssai

- pfcpsessionestablishmentretransmittedrequestsent

Type

64-bit Counter

Description

The number of session establishment retransmitted request messages sent.

Labels

interface, peer_ip, nssai

- pfcpsessionestablishmentrequestreceived

Type

64-bit Counter

Description

The number of session establishment request messages received.

Labels

peer_ip

- pfcpsessionestablishmentretransmittedrequestreceived

Type

64-bit Counter

Description

The number of session establishment retransmitted request messages received.

Labels

peer_ip

- pfcpsessionmodificationrequestsent

Type

64-bit Counter

Description

The number of session modification request messages sent.

Labels

interface, peer_ip, nssai

- pfcpsessionmodificationretransmittedrequestsent

Type

64-bit Counter

Description

The number of session modification retransmitted request messages sent.

Labels

interface, peer_ip, nssai

- pfcpsessionmodificationrequestreceived

Type

64-bit Counter

Description

The number of session modification request messages received.

Labels

peer_ip

- pfcpl_session_modification_retransmitted_request_received

Type

64-bit Counter

Description

The number of session modification retransmitted request messages received.

Labels

peer_ip

- pfcpl_session_deletion_request_sent

Type

64-bit Counter

Description

The number of session deletion request messages sent.

Labels

interface, peer_ip, nssai

- pfcpl_session_deletion_retransmitted_request_sent

Type

64-bit Counter

Description

The number of session deletion retransmitted request messages sent.

Labels

interface, peer_ip, nssai

- pfcpl_session_deletion_request_received

Type

64-bit Counter

Description

The number of session deletion request messages received.

Labels

peer_ip

- pfcpl_session_deletion_retransmitted_request_received

Type

64-bit Counter

Description

The number of session deletion retransmitted request messages received.

Labels

peer_ip

- pfc_session_report_request_sent

Type

64-bit Counter

Description

The number of session report request messages sent.

Labels

peer_ip

- pfc_session_report_retransmitted_request_sent

Type

64-bit Counter

Description

The number of session report retransmitted request messages sent.

Labels

peer_ip

- pfc_session_report_request_received

Type

64-bit Counter

Description

The number of session report request messages received.

Labels

peer_ip

- pfc_session_report_retransmitted_request_received

Type

64-bit Counter

Description

The number of session report retransmitted request messages received.

Labels

peer_ip

- pfc_session_establishment_response_sent

Type

64-bit Counter

Description

The number of session establishment response messages sent.

Labels

peer_ip, pfc_cause

- pfc_session_establishment_response_received

Type

64-bit Counter

Description

The number of session establishment response messages received.

Labels

interface, peer_ip, pfc_p_cause, nssai

• pfc_p_session_modification_response_sent

Type

64-bit Counter

Description

The number of session modification response messages sent.

Labels

peer_ip, pfc_p_cause

• pfc_p_session_modification_response_received

Type

64-bit Counter

Description

The number of session modification response messages received.

Labels

interface, peer_ip, pfc_p_cause, nssai

• pfc_p_session_deletion_response_sent

Type

64-bit Counter

Description

The number of session deletion response messages sent.

Labels

peer_ip, pfc_p_cause

• pfc_p_session_deletion_response_received

Type

64-bit Counter

Description

The number of session deletion response messages received.

Labels

interface, peer_ip, pfc_p_cause, nssai

• pfc_p_node_report_request_sent

Type

64-bit Counter

Description

The number of node report request messages sent.

Labels

peer_ip

- pfcpl_node_report_type_request_received

Type

64-bit Counter

Description

The number of node report request messages received.

Labels

peer_ip

- pfcpl_node_report_type_retransmitted_request_received

Type

64-bit Counter

Description

The number of node report retransmitted request messages received.

Labels

peer_ip

- pfcpl_node_report_response_received

Type

64-bit Counter

Description

The number of node report response messages received.

Labels

peer_ip, pfcpl_cause

- pfcpl_node_report_retransmitted_request_sent

Type

64-bit Counter

Description

The number of node report retransmitted request messages sent.

Labels

peer_ip

- pfcpl_session_report_response_sent

Type

64-bit Counter

Description

The number of session report response messages sent.

Labels

interface, peer_ip, pfcpl_cause

- pfcpl_session_report_response_received

Type

64-bit Counter

Description
The number of session report response messages received.
Labels
<i>peer_ip, pfc_p_cause</i>
• pfc_p_pdm_request_sent
Type
64-bit Counter
Description
The number of pdm request messages sent.
Labels
<i>peer_ip</i>
• pfc_p_pdm_retransmitted_request_sent
Type
64-bit Counter
Description
The number of pdm retransmitted request messages sent.
Labels
<i>peer_ip</i>
• pfc_p_pdm_request_received
Type
64-bit Counter
Description
The number of pdm request messages received.
Labels
<i>peer_ip</i>
• pfc_p_pdm_retransmitted_request_received
Type
64-bit Counter
Description
The number of pdm retransmitted request messages received.
Labels
<i>peer_ip</i>
• pfc_p_pdm_response_sent
Type
64-bit Counter
Description
The number of pdm response messages sent.
Labels
<i>peer_ip, pfc_p_cause</i>

• pfc_p_pfdm_response_received Type 64-bit Counter Description The number of pfdm response messages received. Labels peer_ip, pfc_p_cause
• pfc_p_bundle_message_sent Type 64-bit Counter Description The number of pfc_p bundle messages sent. Labels peer_ip, bundle_trigger
• pfc_p_bundle_message_sent_failure Type 64-bit Counter Description The number of pfc_p bundle messages sent failures. Labels peer_ip, cause
• pfc_p_bundle_message_received Type 64-bit Counter Description The number of pfc_p bundle messages received. Labels peer_ip
• pfc_p_bundle_message_received_failure Type 64-bit Counter Description The number of pfc_p bundle messages decode failure. Labels peer_ip, cause
• pfc_p_number_of_messages_in_bundled_request_sent Type 64-bit Histogram Description

The number of messages in bundled request sent

Labels

peer_ip

Boundaries

[1, 2, 4, 6, 8, 10, 12, 14, 16, 20, 24, 28, 32, 40, 48, 56, 64]

• pfc_p_duration_of_bundling_in_bundled_request_sent

Type

64-bit Histogram

Description

The duration of bundling in bundled request sent in milli second

Labels

peer_ip

Boundaries

[1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 14, 16, 20, 24, 28, 32, 40, 48, 56, 64, 80, 96, 112, 128, 160, 192, 224, 256]

• pfc_p_size_of_message_of_bundled_request_sent

Type

64-bit Histogram

Description

The size of message in bytes of bundled request sent

Labels

peer_ip

Boundaries

[100, 200, 400, 600, 800, 1000, 1500, 2000, 2500, 3500, 4000, 5000, 6000, 7000, 8000, 9000, 10000, 16000, 20000, 24000, 28000, 32000]

• pfc_p_number_of_messages_in_bundled_request_received

Type

64-bit Histogram

Description

The number of messages in bundled request received

Labels

peer_ip

Boundaries

[1, 2, 4, 6, 8, 10, 12, 14, 16, 20, 24, 28, 32, 40, 48, 56, 64]

• pfc_p_size_of_message_of_bundled_request_received

Type

64-bit Histogram

Description

The size of message in bytes of bundled request received

Labels

peer_ip

Boundaries

[100, 200, 400, 600, 800, 1000, 1500, 2000, 2500, 3500, 4000, 5000, 6000, 7000, 8000, 9000, 10000, 16000, 20000, 24000, 28000, 32000]

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- [PFCP UDP Message Statistics](#)

Summary

Group	Metric Name	Type	Synopsis
PFCP UDP Message Statistics	pfcip_udp_message_received	Counter	PFCP UDP received messages.
PFCP UDP Message Statistics	pfcip_udp_message_sent	Counter	PFCP UDP sent messages.

PFCP UDP Message Statistics

- [pfcip_udp_message_received](#)

Type

64-bit Counter

Description

The number of PFCP UDP messages received.

Labels

traffic, src_ip

- [pfcip_udp_message_sent](#)

Type

64-bit Counter

Description

The number of PFCP UDP messages sent.

Labels

traffic, dst_ip

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- [PGW Call Control Procedure Statistics](#)
- [PGW Call Performance Statistics](#)

Summary

Group	Metric Name	Type	Synopsis
PGW Call Control Procedure Statistics	pgwcc_initial_attach_attempts	Counter	Initial Attach Procedures attempted
PGW Call Control Procedure Statistics	pgwcc_initial_attach_successes	Counter	Initial Attach Procedures successful
PGW Call Control Procedure Statistics	pgwcc_initial_attach_failures	Counter	Initial Attach Procedures failed
PGW Call Control Procedure Statistics	pgwcc_detach_attempts	Counter	Detach Procedures attempted
PGW Call Control Procedure Statistics	pgwcc_detach_successes	Counter	Detach Procedures successful
PGW Call Control Procedure Statistics	pgwcc_detach_failures	Counter	Detach Procedures failed
PGW Call Performance Statistics	pgwcc_pscsf_sessions_activated	Counter	Sessions activated using P-CSCF address
PGW Call Performance Statistics	pgwcc_pscsf_sessions_deactivated	Counter	Sessions deactivated using P-CSCF address

PGW Call Control Procedure Statistics

- pgwcc_initial_attach_attempts
Type
64-bit Counter
Description
Number of Initial Attach Procedures attempted
Labels
dnn
- pgwcc_initial_attach_successes
Type
64-bit Counter
Description
Number of Initial Attach Procedures successful
Labels
dnn, snssai
- pgwcc_initial_attach_failures
Type
64-bit Counter
Description
Number of Initial Attach Procedures failed
Labels
dnn, snssai, causecode
- pgwcc_detach_attempts
Type
64-bit Counter
Description
Number of Detach Procedures attempted
Labels
dnn
- pgwcc_detach_successes
Type
64-bit Counter
Description
Number of Detach Procedures successful
Labels
dnn, snssai
- pgwcc_detach_failures
Type
64-bit Counter
Description

Number of Detach Procedures failed

Labels

dnn, snssai, causecode

PGW Call Performance Statistics

pgwcc_pcsf_sessions_activated

Type

64-bit Counter

Description

Number of session activated using P-CSCF address

Labels

dnn, snssai, ratType, ipaddress

pgwcc_pcsf_sessions_deactivated

Type

64-bit Counter

Description

Number of session deactivated using P-CSCF address

Labels

dnn, snssai, ratType, ipaddress

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- SMF Radius Server Group Statistics
- SMF Radius Server Group Peer Statistics
- SMF Radius Server Statistics
- SMF Radius Server Group Peer Port Statistics
- SMF Radius Inband Gtpu Statistics

Summary

Group	Metric Name	Type	Synopsis
SMF Radius Server Group Statistics	smfcc_radius_server_group_access_req_rej_no_client_intf	Counter	SMF Radius Server Group Request Reject
SMF Radius Server Group Statistics	smfcc_radius_server_group_access_req_rej_client_intf_admin_down	Counter	SMF Radius Server Group Request Reject
SMF Radius Server Group Statistics	smfcc_radius_server_group_tx_send_msg_failure	Counter	SMF Radius Server Group send message failure
SMF Radius Server Group Statistics	smfcc_radius_server_group_tx_send_msg_success	Counter	SMF Radius Server Group send message Success
SMF Radius Server Group Statistics	smfcc_radius_server_group_rx_msg_failure	Counter	SMF Radius Server Group receive messages Failed
SMF Radius Server Group Statistics	smfcc_radius_server_group_rx_msg_success	Counter	SMF Radius Server Group receive message Success
SMF Radius Server Group Statistics	smfcc_radius_server_group_rx_server_timeout	Counter	SMF Radius Server Group receive message timeout
SMF Radius Server Group Peer Statistics	smfcc_radius_server_group_peer_tx_buffer_unavailable	Counter	Radius Server Group Peer tx buffer unavailable
SMF Radius Server Group Peer Statistics	smfcc_radius_server_group_peer_tx_encode_error	Counter	Radius Server Group Peer tx encode error
SMF Radius Server Group Peer Statistics	smfcc_radius_server_group_peer_tx_start_timer_failure	Counter	Radius Server Group Peer tx start timer failure

Group	Metric Name	Type	Synopsis
SMF Radius Server Group Peer Statistics	smfcc_radius_server_group_peer_tx_send_msg_failure	Counter	Radius Server Group Peer tx send message failure
SMF Radius Server Group Peer Statistics	smfcc_radius_server_group_peer_tx_send_msg_success	Counter	Radius Server Group Peer tx send message success
SMF Radius Server Group Peer Statistics	smfcc_radius_server_group_peer_re_tx_send_msg_failure	Counter	Radius Server Group Peer tx send message failure
SMF Radius Server Group Peer Statistics	smfcc_radius_server_group_peer_re_tx_send_msg_success	Counter	Radius Server Group Peer tx send message success
SMF Radius Server Group Peer Statistics	smfcc_radius_server_group_peer_rx_no_port_found	Counter	Radius Server Group Peer tx buffer unavailable
SMF Radius Server Group Peer Statistics	smfcc_radius_server_group_peer_rx_invalid_msg_len	Counter	Radius Server Group Peer tx buffer unavailable
SMF Radius Server Group Peer Statistics	smfcc_radius_server_group_peer_rx_verify_reply_failure	Counter	Radius Server Group Peer tx encode error
SMF Radius Server Group Peer Statistics	smfcc_radius_server_group_peer_rx_buffer_unavailable	Counter	Radius Server Group Peer tx start timer failure
SMF Radius Server Group Peer Statistics	smfcc_radius_server_group_peer_rx_decode_error	Counter	Radius Server Group Peer tx send message failure
SMF Radius Server Group Peer Statistics	smfcc_radius_server_group_peer_rx_server_timeout	Counter	Radius Server Group Peer tx send message success
SMF Radius Server Group Peer Statistics	smfcc_radius_server_group_peer_rx_failure	Counter	Radius Server Group Peer tx send message failure
SMF Radius Server Group Peer Statistics	smfcc_radius_server_group_peer_rx_success	Counter	Radius Server Group Peer tx send message success
SMF Radius Server Group Peer Statistics	smfcc_radius_server_group_peer_re_rx_failure	Counter	Radius Server Group Peer tx send message failure
SMF Radius Server Group Peer Statistics	smfcc_radius_server_group_peer_re_rx_success	Counter	Radius Server Group Peer tx send message success
SMF Radius Server Statistics	smfcc_radius_module_resp_tx_get_handle_failure	Counter	Radius Server Group Peer
SMF Radius Server Statistics	smfcc_radius_module_resp_tx_send_failure	Counter	Radius Server Group Peer
SMF Radius Server Statistics	smfcc_radius_module_access_req_tx_invalid_service_id	Counter	Radius Server Group Peer
SMF Radius Server Statistics	smfcc_radius_module_access_req_tx_invalid_group_id	Counter	Radius Server Group Peer
SMF Radius Server Statistics	smfcc_radius_module_access_req_tx_success	Counter	Radius Server Group Peer
SMF Radius Server Statistics	smfcc_radius_module_access_req_tx_failure	Counter	Radius Server Group Peer
SMF Radius Server Statistics	smfcc_radius_module_access_resp_rx_failure	Counter	Radius Server Group Peer
SMF Radius Server Statistics	smfcc_radius_module_access_accept_resp_rx_success	Counter	Radius Server Group Peer
SMF Radius Server Statistics	smfcc_radius_module_access_accept_resp_rx_failure	Counter	Radius Server Group Peer
SMF Radius Server Statistics	smfcc_radius_module_access_challenge_resp_rx_failure	Counter	Radius Server Group Peer
SMF Radius Server Statistics	smfcc_radius_module_access_challenge_resp_rx_success	Counter	Radius Server Group Peer
SMF Radius Server Statistics	smfcc_radius_module_access_reject_resp_rx_failure	Counter	Radius Server Group Peer
SMF Radius Server Statistics	smfcc_radius_module_access_reject_resp_rx_success	Counter	Radius Server Group Peer
SMF Radius Server Statistics	smfcc_radius_module_acct_record_req_tx_invalid_group_id	Counter	Radius Server Group Peer
SMF Radius Server Statistics	smfcc_radius_module_acct_record_req_tx_success	Counter	Radius Server Group Peer
SMF Radius Server Statistics	smfcc_radius_module_acct_record_req_tx_failure	Counter	Radius Server Group Peer
SMF Radius Server Group Peer Port Statistics	smfcc_radius_client_port_tx_socket_inactive	Counter	Radius Server Group Peer Port tx sequence unavailable
SMF Radius Server Group Peer Port Statistics	smfcc_radius_client_port_tx_seq_num_unavailable	Counter	Radius Server Group Peer Port tx sequence unavailable
SMF Radius Server Group Peer Port Statistics	smfcc_radius_client_port_tx_send_message_failure	Counter	Radius Server Group Peer Port tx send send message failure

Group	Metric Name	Type	Synopsis
SMF Radius Server Group Peer Port Statistics	smfcc_radius_client_port_tx_send_message_success	Counter	Radius Server Group Peer Port
SMF Radius Server Group Peer Port Statistics	smfcc_radius_client_port_rx_message_node_not_found	Counter	Radius Server Group Peer Port
SMF Radius Server Group Peer Port Statistics	smfcc_radius_client_port_rx_failure	Counter	Radius Server Group Peer Port
SMF Radius Server Group Peer Port Statistics	smfcc_radius_client_port_rx_success	Counter	Radius Server Group Peer tx buffer unavailable
SMF Radius Inband Gtpu Statistics	smfcc_radius_inband_punting_rx_failure	Counter	no of rx failure while processing gtpu packet. with no port mapping
SMF Radius Inband Gtpu Statistics	smfcc_radius_inband_punting_rx_success	Counter	no of rx success while processing gtpu packet. with port mapping
SMF Radius Inband Gtpu Statistics	smfcc_radius_inband_punting_self	Counter	no of rx pkts punted to self
SMF Radius Inband Gtpu Statistics	smfcc_radius_inband_gtpu_send_success	Counter	success no of send pkts over gtpu tunnel
SMF Radius Inband Gtpu Statistics	smfcc_radius_inband_gtpu_send_failure	Counter	failure no of send pkts over gtpu tunnel
SMF Radius Inband Gtpu Statistics	smfcc_radius_inband_gtpu_receive_success	Counter	success no of rx pkts over gtpu tunnel
SMF Radius Inband Gtpu Statistics	smfcc_radius_inband_gtpu_receive_failure	Counter	failure no of rx pkts over gtpu tunnel
SMF Radius Inband Gtpu Statistics	smfcc_radius_inband_gtpu_missing_port_map	Counter	no of missing port map for pkts received over gtpu tunnel
SMF Radius Inband Gtpu Statistics	smfcc_radius_inband_gtpu_rx_error	Counter	no of gtpu error rx pkts

SMF Radius Server Group Statistics

- smfcc_radius_server_group_access_req_rej_no_client_intf
 - Type
 - 64-bit Counter
 - Description
 - Number of Access Request Reject with no client interface
 - Labels
 - name**
- smfcc_radius_server_group_access_req_rej_client_intf_admin_down
 - Type
 - 64-bit Counter
 - Description
 - Number of Access Request Reject with client interface admin down
 - Labels
 - name**
- smfcc_radius_server_group_tx_send_msg_failure
 - Type
 - 64-bit Counter
 - Description
 - Number of send messages failed
 - Labels
 - name**

- smfcc_radius_server_group_tx_send_msg_success

Type
64-bit Counter
Description
Number of send messages success
Labels
<i>name</i>

- smfcc_radius_server_group_rx_msg_failure

Type
64-bit Counter
Description
Number of receive messages Failed
Labels
<i>name</i>

- smfcc_radius_server_group_rx_msg_success

Type
64-bit Counter
Description
Number of receive messages success
Labels
<i>name</i>

- smfcc_radius_server_group_rx_server_timeout

Type
64-bit Counter
Description
Number of receive server timeout
Labels
<i>name</i>

SMF Radius Server Group Peer Statistics

- smfcc_radius_server_group_peer_tx_buffer_unavailable

Type
64-bit Counter
Description
no of tx buffer unavailable
Labels
<i>group, peer</i>

- smfcc_radius_server_group_peer_tx_encode_error

Type
64-bit Counter

Description

no of tx encode error

Labels

group, peer

- smfcc_radius_server_group_peer_tx_start_timer_failure

Type

64-bit Counter

Description

no of tx start timer failure

Labels

group, peer

- smfcc_radius_server_group_peer_tx_send_msg_failure

Type

64-bit Counter

Description

Number of send messages failed

Labels

group, peer

- smfcc_radius_server_group_peer_tx_send_msg_success

Type

64-bit Counter

Description

Number of send message success

Labels

group, peer

- smfcc_radius_server_group_peer_re_tx_send_msg_failure

Type

64-bit Counter

Description

Number of send messages failed

Labels

group, peer

- smfcc_radius_server_group_peer_re_tx_send_msg_success

Type

64-bit Counter

Description

Number of send message success

Labels

group, peer

smfcc_radius_server_group_peer_rx_no_port_found
Type
64-bit Counter
Description
no of rx no port found
Labels
group, peer
smfcc_radius_server_group_peer_rx_invalid_msg_len
Type
64-bit Counter
Description
no of rx invalid msg length
Labels
group, peer
smfcc_radius_server_group_peer_rx_verify_reply_failure
Type
64-bit Counter
Description
no of tx encode error
Labels
group, peer
smfcc_radius_server_group_peer_rx_buffer_unavailable
Type
64-bit Counter
Description
no of tx start timer failure
Labels
group, peer
smfcc_radius_server_group_peer_rx_decode_error
Type
64-bit Counter
Description
Number of send messages failed
Labels
group, peer
smfcc_radius_server_group_peer_rx_server_timeout
Type
64-bit Counter
Description

Number of send message success

Labels

group, peer

- smfcc_radius_server_group_peer_rx_failure

Type

64-bit Counter

Description

Number of send messages failed

Labels

group, peer

- smfcc_radius_server_group_peer_rx_success

Type

64-bit Counter

Description

Number of send message success

Labels

group, peer

- smfcc_radius_server_group_peer_re_rx_failure

Type

64-bit Counter

Description

Number of send messages failed

Labels

group, peer

- smfcc_radius_server_group_peer_re_rx_success

Type

64-bit Counter

Description

Number of send message success

Labels

group, peer

SMF Radius Server Statistics

- smfcc_radius_module_resp_tx_get_handle_failure

Type

64-bit Counter

Description

no of tx buffer unavailable

Labels

name

smfcc_radius_module_resp_tx_send_failure
Type
64-bit Counter
Description
Number of send message failure
Labels
name
smfcc_radius_module_access_req_tx_invalid_service_id
Type
64-bit Counter
Description
Number of receive invalid service id
Labels
name
smfcc_radius_module_access_req_tx_invalid_group_id
Type
64-bit Counter
Description
Number of receive invalid group id
Labels
name
smfcc_radius_module_access_req_tx_success
Type
64-bit Counter
Description
Number of receive success
Labels
name
smfcc_radius_module_access_req_tx_failure
Type
64-bit Counter
Description
Number of receive failure
Labels
name
smfcc_radius_module_access_resp_rx_failure
Type
64-bit Counter
Description

Number of send failure

Labels

name

- smfcc_radius_module_access_accept_resp_rx_success

Type

64-bit Counter

Description

Number of accept response send success

Labels

name

- smfcc_radius_module_access_accept_resp_rx_failure

Type

64-bit Counter

Description

Number of accept response send failure

Labels

name

- smfcc_radius_module_access_challenge_resp_rx_failure

Type

64-bit Counter

Description

Number of challenge response send failure

Labels

name

- smfcc_radius_module_access_challenge_resp_rx_success

Type

64-bit Counter

Description

Number of challenge response send success

Labels

name

- smfcc_radius_module_access_reject_resp_rx_failure

Type

64-bit Counter

Description

Number of reject response send failure

Labels

name

- smfcc_radius_module_access_reject_resp_rx_success

Type

64-bit Counter

Description

Number of reject response send success

Labels

name

- smfcc_radius_module_acct_record_req_tx_invalid_group_id

Type

64-bit Counter

Description

Number of acct request received with invalid group id

Labels

name

- smfcc_radius_module_acct_record_req_tx_success

Type

64-bit Counter

Description

Number of acct request received success

Labels

name

- smfcc_radius_module_acct_record_req_tx_failure

Type

64-bit Counter

Description

Number of acct request received failure

Labels

name

SMF Radius Server Group Peer Port Statistics

- smfcc_radius_client_port_tx_socket_inactive

Type

64-bit Counter

Description

no of tx socket inactive

Labels

name, port

- smfcc_radius_client_port_tx_seq_num_unavailable

Type

64-bit Counter

Description

no of tx sequence number unavailable

Labels

name, port

- smfcc_radius_client_port_tx_send_message_failure

Type

64-bit Counter

Description

no of tx send message failure

Labels

name, port

- smfcc_radius_client_port_tx_send_message_success

Type

64-bit Counter

Description

no of tx send message success

Labels

name, port

- smfcc_radius_client_port_rx_message_node_not_found

Type

64-bit Counter

Description

no of rx message node not found

Labels

name, port

- smfcc_radius_client_port_rx_failure

Type

64-bit Counter

Description

no of rx failure

Labels

name, port

- smfcc_radius_client_port_rx_success

Type

64-bit Counter

Description

no of rx success

Labels

name, port

SMF Radius Inband Gtpu Statistics

smfcc_radius_inband_punting_rx_failure
Type
64-bit Counter
Description
no of rx failure while processing gtpu packet. with no port mapping
Labels
name, port
smfcc_radius_inband_punting_rx_success
Type
64-bit Counter
Description
no of rx success while processing gtpu packet. with port mapping
Labels
name
smfcc_radius_inband_punting_self
Type
64-bit Counter
Description
no of rx pkts punted to self
Labels
name, port
smfcc_radius_inband_gtpu_send_success
Type
64-bit Counter
Description
success no of send pkts over gtpu tunnel
Labels
name, role
smfcc_radius_inband_gtpu_send_failure
Type
64-bit Counter
Description
failure no of send pkts over gtpu tunnel
Labels
name
smfcc_radius_inband_gtpu_receive_success
Type
64-bit Counter
Description

success no of rx pkts over gtpu tunnel

Labels

name, role

- smfcc_radius_inband_gtpu_receive_failure

Type

64-bit Counter

Description

failure no of rx pkts over gtpu tunnel

Labels

name

- smfcc_radius_inband_gtpu_missing_port_map

Type

64-bit Counter

Description

no of missing port map for pkts received over gtpu tunnel

Labels

name, role, key

- smfcc_radius_inband_gtpu_rx_error

Type

64-bit Counter

Description

no of gtpu error rx pkts

Labels

name, role

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Summary

Group	Metric Name	Type	Synopsis
Circuit Breaker Metrics	sem_cb_endpoints	Gauge	Total number of endpoints for a service in respective states
Service Endpoint Learner Metrics	sem_num_dns_errors	Counter	Total number of DNS resolution errors for a given service
Service Endpoint Learner Metrics	sem_num_endpoints	Gauge	Total number of endpoints learnt for a service
Service Endpoint Manager Metrics	sem_num_endpoint_transactions_sent	Counter	Total number of times an endpoint is given to the application for communication with it
Service Endpoint Manager Metrics	sem_num_endpoint_transaction_status_received	Counter	Total number of transactions for a given endpoint received with the transaction status

Circuit Breaker Metrics

sem_cb_endpoints
Type
64-bit Gauge
Description
Total number of endpoints for a service in respective states. There are 3 states viz. Active, Blocked, Restoring. In Active state, the endpoint is responding as expected, i.e. the response code doesn't indicate a service failure. The endpoint will be moved to Blocked state based on response failures which indicate service failure. After a timeout, the endpoint will be moved to Restoring state to check if the endpoint has recovered. In this state, if endpoint responds with response code other than service failure, it will be moved to Active state. Else, it will be moved back to Blocked state.
Labels
service_name, endpoint_state

Service Endpoint Learner Metrics

sem_num_dns_errors
Type
64-bit Counter
Description
Total number of DNS resolution errors for a given service, labels are service name for DNS resolution and DNS error codes
Labels
service_name, error_code
sem_num_endpoints
Type
64-bit Gauge
Description
Total number of endpoints learnt from the Kubernetes DNS server for a given service
Labels
service

Service Endpoint Manager Metrics

sem_num_endpoint_transactions_sent
Type
64-bit Counter
Description
Total number of times an endpoint is given to the application for communication with it
Labels
service
sem_num_endpoint_transaction_status_received
Type
64-bit Counter
Description
Total number of transactions for a given endpoint received with the transaction status
Labels

service, status

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- [SMF NF Set Statistics](#)
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Summary

Group	Metric Name	Type	Synopsis
SMF Call Control Interface Statistics	smfcc_nsmf_pdu_session_create_sm_context_attempts	Counter	Create SM Ctxt Requests received
SMF Call Control Interface Statistics	smfcc_nsmf_pdu_session_create_sm_context_successes	Counter	Create SM Ctxt Requests successful
SMF Call Control Interface Statistics	smfcc_nsmf_pdu_session_create_sm_context_failures	Counter	Create SM Ctxt Requests failed
SMF Call Control Interface Statistics	smfcc_nsmf_pdu_session_create_sm_context_punted	Counter	Create SM Ctxt Requests punted
SMF Call Control Interface Statistics	smfcc_nsmf_pdu_session_create_sm_context_duplicate_attempts	Counter	Create SM Ctxt Requests received
SMF Call Control Interface Statistics	smfcc_vsmf_pdu_session_create_sm_context_duplicate_attempts	Counter	Create SM Ctxt Requests received by vSMF
SMF Call Control Interface Statistics	smfcc_nsmf_pdu_session_update_sm_context_attempts	Counter	Update SM Ctxt Requests received
SMF Call Control Interface Statistics	smfcc_nsmf_pdu_session_update_sm_context_successes	Counter	Update SM Ctxt Requests successful

Group	Metric Name	Type	Synopsis
SMF Call Control Interface Statistics	smfcc_nsmf_pdu_session_update_sm_context_failures	Counter	Update SM Ctxt Requests failed
SMF Call Control Interface Statistics	smfcc_nsmf_pdu_session_update_sm_context_punted	Counter	Update SM Ctxt Requests punted
SMF Call Control Interface Statistics	smfcc_nsmf_pdu_session_release_sm_context_attempts	Counter	Release SM Ctxt Requests received
SMF Call Control Interface Statistics	smfcc_nsmf_pdu_session_release_sm_context_successes	Counter	Release SM Ctxt Requests successful
SMF Call Control Interface Statistics	smfcc_nsmf_pdu_session_release_sm_context_failures	Counter	Release SM Ctxt Requests failed
SMF Call Control Interface Statistics	smfcc_nsmf_pdu_session_release_sm_context_punted	Counter	Release SM Ctxt Requests punted
SMF Call Control Interface Statistics	smfcc_nsmf_pdu_session_create_punted	Counter	Create Requests punted
SMF Call Control Interface Statistics	smfcc_nsmf_pdu_session_create_client_attempts	Counter	Create Requests received at the client
SMF Call Control Interface Statistics	smfcc_nsmf_pdu_session_create_client_successes	Counter	Create Requests successful at the client
SMF Call Control Interface Statistics	smfcc_nsmf_pdu_session_create_client_failures	Counter	Create Requests failed at the client
SMF Call Control Interface Statistics	smfcc_nsmf_pdu_session_create_client_punted	Counter	Create Requests punted at the client
SMF Call Control Interface Statistics	smfcc_nsmf_pdu_session_update_attempts	Counter	Update Requests received
SMF Call Control Interface Statistics	smfcc_nsmf_pdu_session_update_successes	Counter	Update Requests successful
SMF Call Control Interface Statistics	smfcc_nsmf_pdu_session_update_failures	Counter	Update Requests failed
SMF Call Control Interface Statistics	smfcc_nsmf_pdu_session_update_punted	Counter	Update Requests punted
SMF Call Control Interface Statistics	smfcc_nsmf_pdu_session_update_client_attempts	Counter	Update Requests received at the client
SMF Call Control Interface Statistics	smfcc_nsmf_pdu_session_update_client_successes	Counter	Update Requests successful at the client
SMF Call Control Interface Statistics	smfcc_nsmf_pdu_session_update_client_failures	Counter	Update Requests failed at the client
SMF Call Control Interface Statistics	smfcc_nsmf_pdu_session_update_client_punted	Counter	Update Requests punted at the client
SMF Call Control Interface Statistics	smfcc_nsmf_pdu_session_release_attempts	Counter	Release Requests received
SMF Call Control Interface Statistics	smfcc_nsmf_pdu_session_release_successes	Counter	Release Requests successful
SMF Call Control Interface Statistics	smfcc_nsmf_pdu_session_release_failures	Counter	Release Requests failed
SMF Call Control Interface Statistics	smfcc_nsmf_pdu_session_release_punted	Counter	Release Requests punted
SMF Call Control Interface Statistics	smfcc_nsmf_pdu_session_release_client_attempts	Counter	Release Requests received at the client
SMF Call Control Interface Statistics	smfcc_nsmf_pdu_session_release_client_successes	Counter	Release Requests successful at the client
SMF Call Control Interface Statistics	smfcc_nsmf_pdu_session_release_client_failures	Counter	Release Requests failed at the client
SMF Call Control Interface Statistics	smfcc_nsmf_pdu_session_release_client_punted	Counter	Release Requests punted at the client

Group	Metric Name	Type	Synopsis
SMF Call Control Interface Statistics	smfcc_namf_comm_n1n2_message_transfer_attempts	Counter	N1N2 Message Transfer Requests sent
SMF Call Control Interface Statistics	smfcc_namf_comm_n1n2_message_transfer_successes	Counter	N1N2 Message Transfer Requests successful
SMF Call Control Interface Statistics	smfcc_namf_comm_n1n2_message_transfer_failures	Counter	N1N2 Message Transfer Requests failed
SMF Call Control Interface Statistics	smfcc_start_durable_timer_requests	Counter	Durable Timer Start Requests sent
SMF Call Control Interface Statistics	smfcc_start_durable_timer_failures	Counter	Durable Timer Start Requests failed
SMF Call Control Interface Statistics	smfcc_modify_durable_timer_requests	Counter	Durable Timer Modify Requests sent
SMF Call Control Interface Statistics	smfcc_modify_durable_timer_failures	Counter	Durable Timer Modify Requests failed
SMF Call Control Interface Statistics	smfcc_durable_timer_modify_changed_to_start	Counter	Durable Timer Modify Attempts changed to Start
SMF Call Control Interface Statistics	smfcc_stop_durable_timer_requests	Counter	Durable Timer Stop Requests sent
SMF Call Control Interface Statistics	smfcc_stop_durable_timer_failures	Counter	Durable Timer Stop Requests failed
SMF Call Control Interface Statistics	smfcc_durable_timer_fired	Counter	Durable Timer Fired Notifications
SMF Call Control Interface Statistics	smfcc_durable_timer_fired_responses	Counter	Durable Timer Fired Notification Response
SMF Call Control Interface Statistics	smfcc_durable_timer_fired_stale	Counter	Stale Durable Timer Fired Notifications
SMF Call Control Interface Statistics	smfcc_durable_timer_fired_error	Counter	Erroneous Durable Timer Fired Notifications
SMF Call Control Interface Statistics	smfcc_namf_comm_assign_ebi_message_attempts	Counter	Assign Ebi Requests sent
SMF Call Control Interface Statistics	smfcc_namf_comm_assign_ebi_message_successes	Counter	Assign Ebi Requests successful
SMF Call Control Interface Statistics	smfcc_namf_comm_assign_ebi_message_failures	Counter	Assign Ebi Requests failed
SMF Call Control Interface Statistics	smfcc_nsmf_pdu_session_retrieve_sm_context_attempts	Counter	Retrieve Requests received
SMF Call Control Interface Statistics	smfcc_nsmf_pdu_session_retrieve_sm_context_successes	Counter	Retrieve Requests successful
SMF Call Control Interface Statistics	smfcc_nsmf_pdu_session_retrieve_sm_context_failures	Counter	Retrieve Requests failed
SMF Call Control Interface Statistics	smfcc_nsmf_pdu_session_retrieve_sm_context_punted	Counter	Retrieve Requests punted
SMF Call Control Interface Statistics	smfcc_nsmf_pdu_session_sm_context_status_notify_attempts	Counter	SM Ctxt Status Notify sent
SMF Call Control Interface Statistics	smfcc_nsmf_pdu_session_sm_context_status_notify_successes	Counter	SM Ctxt Status Notify successful
SMF Call Control Interface Statistics	smfcc_nsmf_pdu_session_sm_context_status_notify_failures	Counter	SM Ctxt Status notify failed
SMF Call Control Interface Statistics	smfcc_namf_evts_subscr_create_attempts	Counter	Subscription Create Requests sent
SMF Call Control Interface Statistics	smfcc_namf_evts_subscr_create_successes	Counter	Subscription Create Requests successful
SMF Call Control Interface Statistics	smfcc_namf_evts_subscr_create_failures	Counter	Subscription Create Requests failed

Group	Metric Name	Type	Synopsis
SMF Call Control Interface Statistics	smfcc_namf_evts_event_notification_attempts	Counter	AMF Event Notification Request received
SMF Call Control Interface Statistics	smfcc_namf_evts_event_notification_successes	Counter	AMF Event Notification Request successful
SMF Call Control Interface Statistics	smfcc_namf_evts_event_notification_failures	Counter	AMF Event Notification Request failed
SMF Call Control Interface Statistics	smfcc_namf_evts_subscr_modify_attempts	Counter	Subscription Modify Requests sent
SMF Call Control Interface Statistics	smfcc_namf_evts_subscr_modify_successes	Counter	Subscription Modify Requests successful
SMF Call Control Interface Statistics	smfcc_namf_evts_subscr_modify_failures	Counter	Subscription Modify Requests failed
SMF Call Control Interface Statistics	smfcc_namf_evts_subscr_delete_attempts	Counter	Subscription Delete Requests sent
SMF Call Control Interface Statistics	smfcc_namf_evts_subscr_delete_successes	Counter	Subscription Delete Requests successful
SMF Call Control Interface Statistics	smfcc_namf_evts_subscr_delete_failures	Counter	Subscription Delete Requests failed
SMF Call Control Interface Statistics	smfcc_inband_radius_tunnel_timer_start_attempts	Counter	Number of Inband Radius Tunnel Create/Delete Timer Start Request messages sent
SMF Call Control Interface Statistics	smfcc_inband_radius_tunnel_timer_start_successes	Counter	Number of Inband Radius Tunnel Create/Delete Timer Start Request Create successful
SMF Call Control Interface Statistics	smfcc_inband_radius_tunnel_timer_start_failures	Counter	Number of Inband Radius Tunnel Create/Delete Timer Start Request messages failed
SMF Call Control Interface Statistics	smfcc_inband_radius_tunnel_timer_fires	Counter	Number of Inband Radius Tunnel Create/Delete Timer fired
SMF Call Control Interface Statistics	smfcc_inband_radius_tunnel_timer_restarts	Counter	Number of Inband Radius Tunnel Create/Delete Timer Re-Start Request messages sent
SMF PCF Interface Call Control Procedure Statistics	smfcc_nsmf_smpolicy_control_create_service_attempts	Counter	SmPolicy Control Create Service Requests attempted
SMF PCF Interface Call Control Procedure Statistics	smfcc_nsmf_smpolicy_control_create_service_success	Counter	SmPolicy Control Create Service requests successful
SMF PCF Interface Call Control Procedure Statistics	smfcc_nsmf_smpolicy_control_create_service_failures	Counter	SmPolicy Control Create Service Requests failed
SMF PCF Interface Call Control Procedure Statistics	smfcc_nsmf_smpolicy_control_create_service_retry_attempts	Counter	SmPolicy Control Create Service Requests attempted to alternate PCF
SMF PCF Interface Call Control Procedure Statistics	smfcc_nsmf_smpolicy_control_create_service_failures_causing_session_continue	Counter	SmPolicy Control Create Service failures causing session continue
SMF PCF Interface Call Control Procedure Statistics	smfcc_nsmf_smpolicy_control_create_service_failures_causing_session_deactivate	Counter	SmPolicy Control Create Service failures causing session deactivate
SMF PCF Interface Call Control Procedure Statistics	smfcc_nsmf_smpolicy_control_update_service_attempts	Counter	SmPolicy Control Update Service Requests attempted
SMF PCF Interface Call Control Procedure Statistics	smfcc_nsmf_smpolicy_control_update_service_success	Counter	SmPolicy Control Update Service requests successful

Group	Metric Name	Type	Synopsis
SMF PCF Interface Call Control Procedure Statistics	smfcc_nsmf_smpolicy_control_update_service_failures	Counter	SmPolicy Control Update Service Requests failed
SMF PCF Interface Call Control Procedure Statistics	smfcc_nsmf_smpolicy_control_delete_service_attempts	Counter	SmPolicy Control Delete Service Requests attempted
SMF PCF Interface Call Control Procedure Statistics	smfcc_nsmf_smpolicy_control_delete_service_success	Counter	SmPolicy Control Delete Service requests successful
SMF PCF Interface Call Control Procedure Statistics	smfcc_nsmf_smpolicy_control_delete_service_failures	Counter	SmPolicy Control Delete Service Requests failed
SMF PCF Interface Call Control Procedure Statistics	smfcc_nsmf_smpolicy_control_update_notify_attempts	Counter	SmPolicy Control Update Notify Requests attempted
SMF PCF Interface Call Control Procedure Statistics	smfcc_nsmf_smpolicy_control_update_notify_success	Counter	SmPolicy Control Update Notify requests successful
SMF PCF Interface Call Control Procedure Statistics	smfcc_nsmf_smpolicy_control_update_notify_failures	Counter	SmPolicy Control Update Notify Requests failed
SMF PCF Interface Call Control Procedure Statistics	smfcc_nsmf_smpolicy_control_update_notify_punted	Counter	SmPolicy Control Update Notify Requests punted
SMF PCF Interface Call Control Procedure Statistics	smfcc_nsmf_smpolicy_control_terminate_notify_attempts	Counter	SmPolicy Control Terminate Notify Requests attempted
SMF PCF Interface Call Control Procedure Statistics	smfcc_nsmf_smpolicy_control_terminate_notify_success	Counter	SmPolicy Control Terminate Notify requests successful
SMF PCF Interface Call Control Procedure Statistics	smfcc_nsmf_smpolicy_control_terminate_notify_failures	Counter	SmPolicy Control Terminate Notify Requests failed
SMF PCF Interface Call Control Procedure Statistics	smfcc_nsmf_smpolicy_control_terminate_notify_punted	Counter	SmPolicy Control Terminate Notify Requests punted
SMF GX Interface Statistics	smfcc_gx_ccr_init_attempts	Counter	CCR-I attempted
SMF GX Interface Statistics	smfcc_gx_ccr_init_success	Counter	CCR-I successful
SMF GX Interface Statistics	smfcc_gx_ccr_init_failures	Counter	CCR-I failed
SMF GX Interface Statistics	smfcc_gx_ccr_update_attempts	Counter	CCR-U attempted
SMF GX Interface Statistics	smfcc_gx_ccr_update_success	Counter	CCR-U successful
SMF GX Interface Statistics	smfcc_gx_ccr_update_failures	Counter	CCR-U failed
SMF GX Interface Statistics	smfcc_gx_ccr_terminate_attempts	Counter	CCR-T attempted
SMF GX Interface Statistics	smfcc_gx_ccr_terminate_success	Counter	CCR-T successful
SMF GX Interface Statistics	smfcc_gx_ccr_terminate_failures	Counter	CCR-T failed
SMF GX Interface Statistics	smfcc_gx_rar_received	Counter	RAR received from PCRF
SMF GX Interface Statistics	smfcc_gx_rar_rsp_success	Counter	RAA successful towards PCRF
SMF GX Interface Statistics	smfcc_gx_rar_rsp_failures	Counter	RAA failed towards PCRF
SMF GX Interface Statistics	smfcc_gx_rar_terminate_received	Counter	RAR-terminate received
SMF GX Interface Statistics	smfcc_gx_rar_terminate_success	Counter	RAR-terminate successful
SMF GX Interface Statistics	smfcc_gx_rar_terminate_failures	Counter	RAR-terminate failed
SMF Call Control Procedure Statistics	smfcc_pdu_session_establishment_attempts	Counter	PDU Session Establishment Procedures attempted
SMF Call Control Procedure Statistics	smfcc_pdu_session_establishment_successes	Counter	PDU Session Establishment Procedures successful
SMF Call Control Procedure Statistics	smfcc_pdu_session_establishment_failures	Counter	PDU Session Establishment Procedures failed
SMF Call Control Procedure Statistics	smfcc_ue_deregistration_attempts	Counter	UE Deregistration Procedures attempted
SMF Call Control Procedure Statistics	smfcc_ue_deregistration_successes	Counter	UE Deregistration Procedures successful

Group	Metric Name	Type	Synopsis
SMF Call Control Procedure Statistics	smfcc_ue_deregistration_failures	Counter	UE Deregistration Procedures failed
SMF Call Control Procedure Statistics	smfcc_vsmf_ue_deregistration_attempts	Counter	VSMF UE Deregistration Procedures attempted
SMF Call Control Procedure Statistics	smfcc_vsmf_ue_deregistration_successes	Counter	VSMF UE Deregistration Procedures successful
SMF Call Control Procedure Statistics	smfcc_vsmf_ue_deregistration_failures	Counter	VSMF UE Deregistration Procedures failed
SMF Call Control Procedure Statistics	smfcc_an_release_attempts	Counter	AN Release Procedures attempted
SMF Call Control Procedure Statistics	smfcc_an_release_successes	Counter	AN Release Procedures successful
SMF Call Control Procedure Statistics	smfcc_an_release_failures	Counter	AN Release Procedures failed
SMF Call Control Procedure Statistics	smfcc_service_request_attempts	Counter	Service Request Procedures attempted
SMF Call Control Procedure Statistics	smfcc_service_request_successes	Counter	Service Request Procedures successful
SMF Call Control Procedure Statistics	smfcc_service_request_failures	Counter	Service Request Procedures failed
SMF Call Control Procedure Statistics	smfcc_nw_initiated_release_attempts	Counter	NW Initiated Release Procedures attempted
SMF Call Control Procedure Statistics	smfcc_nw_initiated_release_successes	Counter	NW Initiated Release Procedures successful
SMF Call Control Procedure Statistics	smfcc_nw_initiated_release_failures	Counter	NW Initiated Release Procedures failed
SMF Call Control Procedure Statistics	smfcc_xn_handover_initiated_release_attempts	Counter	Xn Handover Initiated Release Procedures attempted
SMF Call Control Procedure Statistics	smfcc_xn_handover_initiated_release_successes	Counter	Xn Handover Initiated Release Procedures successful
SMF Call Control Procedure Statistics	smfcc_xn_handover_initiated_release_failures	Counter	Xn Handover Initiated Release Procedures failed
SMF Call Control Procedure Statistics	smfcc_xn_handover_attempts	Counter	Xn Handover Procedures attempted
SMF Call Control Procedure Statistics	smfcc_xn_handover_successes	Counter	Xn Handover Procedures succeeded
SMF Call Control Procedure Statistics	smfcc_xn_handover_failures	Counter	Xn Handover Procedures failed
SMF Call Control Procedure Statistics	smfcc_xn_handover_idle_mode_successes	Counter	Xn Handover Procedures succeeded for idle mode
SMF Call Control Procedure Statistics	smfcc_xn_handover_idle_mode_failures	Counter	Xn Handover Procedures failed for idle mode
SMF Call Control Procedure Statistics	smfcc_n2_handover_attempts	Counter	N2 Handover Procedures attempted
SMF Call Control Procedure Statistics	smfcc_n2_handover_successes	Counter	N2 Handover Procedures successful
SMF Call Control Procedure Statistics	smfcc_n2_handover_failures	Counter	N2 Handover Procedures failed
SMF Call Control Procedure Statistics	smfcc_n2_handover_cancels	Counter	N2 Handover Procedures canceled
SMF Call Control Procedure Statistics	smfcc_vsmf_n2_handover_attempts	Counter	VSMF N2 Handover Procedures attempted

Group	Metric Name	Type	Synopsis
SMF Call Control Procedure Statistics	smfcc_vsmf_n2_handover_successes	Counter	VSMF N2 Handover Procedures successful
SMF Call Control Procedure Statistics	smfcc_vsmf_n2_handover_failures	Counter	VSMF N2 Handover Procedures failed
SMF Call Control Procedure Statistics	smfcc_vsmf_n2_handover_cancels	Counter	VSMF N2 Handover Procedures cancelled
SMF Call Control Procedure Statistics	smfcc_ue_initiated_modify_qos_attempts	Counter	UE Initiated Modify Qos Procedures attempted
SMF Call Control Procedure Statistics	smfcc_ue_initiated_modify_qos_successes	Counter	UE Initiated Modify Qos Procedures successful
SMF Call Control Procedure Statistics	smfcc_ue_initiated_modify_qos_failures	Counter	UE Initiated Modify Qos failed
SMF Call Control Procedure Statistics	smfcc_vsmf_ue_initiated_modify_attempts	Counter	VSMF UE Initiated Modify Procedures attempted
SMF Call Control Procedure Statistics	smfcc_vsmf_ue_initiated_modify_successes	Counter	VSMF UE Initiated Modify Procedures successful
SMF Call Control Procedure Statistics	smfcc_vsmf_ue_initiated_modify_failures	Counter	VSMF UE Initiated Modify failed
SMF Call Control Procedure Statistics	smfcc_hsmf_ue_initiated_modify_attempts	Counter	HSMF UE Initiated Modify Procedures attempted
SMF Call Control Procedure Statistics	smfcc_hsmf_ue_initiated_modify_successes	Counter	HSMF UE Initiated Modify Procedures successful
SMF Call Control Procedure Statistics	smfcc_hsmf_ue_initiated_modify_failures	Counter	HSMF UE Initiated Modify failed
SMF Call Control Procedure Statistics	smfcc_pcf_initiated_modify_qos_attempts	Counter	PCF Initiated Modify Qos Procedures attempted
SMF Call Control Procedure Statistics	smfcc_pcf_initiated_modify_qos_successes	Counter	PCF Initiated Modify Qos Procedures successful
SMF Call Control Procedure Statistics	smfcc_pcf_initiated_modify_qos_failures	Counter	PCF Initiated Modify Qos failed
SMF Call Control Procedure Statistics	smfcc_pcf_initiated_modify_qos_successes_with_n2_partial_failure	Counter	PCF Initiated Modify Qos Procedures successful with n2 partial failure
SMF Call Control Procedure Statistics	smfcc_ue_initiated_pdu_session_release_attempts	Counter	UE Initiated PDU Session Release Procedures attempted
SMF Call Control Procedure Statistics	smfcc_ue_initiated_pdu_session_release_successes	Counter	UE Initiated PDU Session Release Procedures successful
SMF Call Control Procedure Statistics	smfcc_ue_initiated_pdu_session_release_failures	Counter	UE Initiated PDU Session Release Procedures failed
SMF Call Control Procedure Statistics	smfcc_vsmf_ue_initiated_pdu_session_release_attempts	Counter	VSMF UE Initiated PDU Session Release Procedures attempted for in-bound roamers
SMF Call Control Procedure Statistics	smfcc_vsmf_ue_initiated_pdu_session_release_successes	Counter	VSMF UE Initiated PDU Session Release Procedures successful for in-bound roamers
SMF Call Control Procedure Statistics	smfcc_vsmf_ue_initiated_pdu_session_release_failures	Counter	VSMF UE Initiated PDU Session Release Procedures failed for in-bound roamers
SMF Call Control Procedure Statistics	smfcc_hsmf_ue_initiated_pdu_session_release_attempts	Counter	HSMF UE Initiated PDU Session Release Procedures attempted for out-bound roamers

Group	Metric Name	Type	Synopsis
SMF Call Control Procedure Statistics	smfcc_hsmf_ue_initiated_pdu_session_release_successes	Counter	HSMF UE Initiated PDU Session Release Procedures successful for out-bound roamers
SMF Call Control Procedure Statistics	smfcc_hsmf_ue_initiated_pdu_session_release_failures	Counter	HSMF UE Initiated PDU Session Release Procedures failed for out-bound roamers
SMF Call Control Procedure Statistics	smfcc_vsmf_initiated_release_attempts	Counter	VSMF Initiated Release Procedures attempted
SMF Call Control Procedure Statistics	smfcc_vsmf_initiated_release_successes	Counter	VSMF Initiated Release Procedures successful
SMF Call Control Procedure Statistics	smfcc_vsmf_initiated_release_failures	Counter	VSMF Initiated Release Procedures failed
SMF Call Control Procedure Statistics	smfcc_amf_initiated_pdu_session_release_attempts	Counter	AMF Initiated PDU Session Release Procedures attempted
SMF Call Control Procedure Statistics	smfcc_amf_initiated_pdu_session_release_successes	Counter	AMF Initiated PDU Session Release Procedures successful
SMF Call Control Procedure Statistics	smfcc_amf_initiated_pdu_session_release_failures	Counter	AMF Initiated PDU Session Release Procedures failed
SMF Call Control Procedure Statistics	smfcc_network_initiated_service_request_attempts	Counter	Network Initiated Service Request Procedures attempted
SMF Call Control Procedure Statistics	smfcc_network_initiated_service_request_successes	Counter	Network Initiated Service Request Procedures successful
SMF Call Control Procedure Statistics	smfcc_network_initiated_service_request_failures	Counter	Network Initiated Service Request Procedures failed
SMF Call Control Procedure Statistics	smfcc_pcf_initiated_terminate_attempts	Counter	PCF Initiated Terminate Procedures attempted
SMF Call Control Procedure Statistics	smfcc_pcf_initiated_terminate_successes	Counter	PCF Initiated Terminate Procedures successful
SMF Call Control Procedure Statistics	smfcc_pcf_initiated_terminate_failures	Counter	PCF Initiated Terminate failed
SMF Call Control Procedure Statistics	smfcc_hsmf_pcf_initiated_terminate_attempts	Counter	HSMF PCF Initiated Terminate Procedures attempted
SMF Call Control Procedure Statistics	smfcc_hsmf_pcf_initiated_terminate_successes	Counter	HSMF PCF Initiated Terminate Procedures successful
SMF Call Control Procedure Statistics	smfcc_hsmf_pcf_initiated_terminate_failures	Counter	HSMF PCF Initiated Terminate failed
SMF Call Control Procedure Statistics	smfcc_upf_usage_report_procedure_attempts	Counter	Network of Usage Report Processing Procedure attempted
SMF Call Control Procedure Statistics	smfcc_upf_usage_report_procedure_successes	Counter	Network of Usage Report Processing Procedure successful
SMF Call Control Procedure Statistics	smfcc_upf_usage_report_procedure_failures	Counter	Network of Usage Report Processing Procedure failed
SMF Call Control Procedure Statistics	smfcc_upf_usage_report_punted	Counter	Network of Usage Report Processing Procedure punted
SMF Call Control Procedure Statistics	smfcc_amf_mobility_attempts	Counter	Amf Mobility Procedures attempted

Group	Metric Name	Type	Synopsis
SMF Call Control Procedure Statistics	smfcc_amf_mobility_successes	Counter	Amf Mobility Procedures successful
SMF Call Control Procedure Statistics	smfcc_amf_mobility_failures	Counter	Amf Mobility Procedures failed
SMF Call Control Procedure Statistics	smfcc_vsmf_error_indication_procedure_attempts	Counter	VSMF Error Indication Procedures attempted
SMF Call Control Procedure Statistics	smfcc_vsmf_error_indication_procedure_successes	Counter	VSMF Error Indication Procedures successful
SMF Call Control Procedure Statistics	smfcc_vsmf_error_indication_procedure_failures	Counter	VSMF Error Indication Procedures failed
SMF Call Control Procedure Statistics	smfcc_hsmf_error_indication_procedure_attempts	Counter	HSMF Error Indication Procedures attempted
SMF Call Control Procedure Statistics	smfcc_hsmf_error_indication_procedure_successes	Counter	HSMF Error Indication Procedures successful
SMF Call Control Procedure Statistics	smfcc_hsmf_error_indication_procedure_failures	Counter	HSMF Error Indication Procedures failed
SMF Call Control Procedure Statistics	smfcc_vsmf_up_inactivity_procedure_attempts	Counter	VSMF UP Inactivity Procedures attempted
SMF Call Control Procedure Statistics	smfcc_vsmf_up_inactivity_procedure_successes	Counter	VSMF UP Inactivity Procedures successful
SMF Call Control Procedure Statistics	smfcc_vsmf_up_inactivity_procedure_failures	Counter	VSMF UP Inactivity Procedures failed
SMF Call Control Procedure Statistics	smfcc_hsmf_up_inactivity_procedure_attempts	Counter	HSMF UP Inactivity Procedures attempted
SMF Call Control Procedure Statistics	smfcc_hsmf_up_inactivity_procedure_successes	Counter	HSMF UP Inactivity Procedures successful
SMF Call Control Procedure Statistics	smfcc_hsmf_up_inactivity_procedure_failures	Counter	HSMF UP Inactivity Procedures failed
SMF Call Control Procedure Statistics	smfcc_error_indication_procedure_attempts	Counter	Error Indication Procedures attempted
SMF Call Control Procedure Statistics	smfcc_error_indication_procedure_successes	Counter	Error Indication Procedures successful
SMF Call Control Procedure Statistics	smfcc_error_indication_procedure_failures	Counter	Error Indication Procedures failed
SMF Call Control Procedure Statistics	smfcc_up_inactivity_procedure_attempts	Counter	UP Inactivity Procedures attempted
SMF Call Control Procedure Statistics	smfcc_up_inactivity_procedure_successes	Counter	UP Inactivity Procedures successful
SMF Call Control Procedure Statistics	smfcc_up_inactivity_procedure_failures	Counter	UP Inactivity Procedures failed
SMF Call Control Procedure Statistics	smfcc_process_pcf_update_policy_rules_attempts	Counter	Process Pcf Update Policy Rules Procedures attempted
SMF Call Control Procedure Statistics	smfcc_process_pcf_update_policy_rules_successes	Counter	Process Pcf Update Policy Rules Procedures successful
SMF Call Control Procedure Statistics	smfcc_process_pcf_update_policy_rules_failures	Counter	Process Pcf Update Policy Rules Procedures failed
SMF Call Control Procedure Statistics	smfcc_process_hsmf_pcf_update_policy_rules_attempts	Counter	Process HSMF Pcf Update Policy Rules Procedures attempted
SMF Call Control Procedure Statistics	smfcc_process_hsmf_pcf_update_policy_rules_successes	Counter	Process HSMF Pcf Update Policy Rules Procedures successful

Group	Metric Name	Type	Synopsis
SMF Call Control Procedure Statistics	smfcc_process_hsmf_pcf_update_policy_rules_failures	Counter	Process HSMF Pcf Update Policy Rules Procedures failed
SMF Call Control Procedure Statistics	smfcc_process_pcf_update_policy_rules_successes_with_n2_partial_failure	Counter	Process Pcf Update Policy Rules Procedures successful with n2 partial failure
SMF Call Control Procedure Statistics	smfcc_5gs_to_eps_handover_attempts	Counter	5GS to EPS Handover Procedures attempted
SMF Call Control Procedure Statistics	smfcc_5gs_to_eps_handover_successes	Counter	5GS to EPS Handover Procedures successful
SMF Call Control Procedure Statistics	smfcc_5gs_to_eps_handover_failures	Counter	5GS to EPS Handover Procedures failed
SMF Call Control Procedure Statistics	smfcc_hsmf_5gs_to_eps_handover_attempts	Counter	HSMF 5GS to EPS Handover Procedures attempted
SMF Call Control Procedure Statistics	smfcc_hsmf_5gs_to_eps_handover_successes	Counter	HSMF 5GS to EPS Handover Procedures successful
SMF Call Control Procedure Statistics	smfcc_hsmf_5gs_to_eps_handover_failures	Counter	HSMF 5GS to EPS Handover Procedures failed
SMF Call Control Procedure Statistics	smfcc_5gs_to_eps_idle_mobility_attempts	Counter	5GS to EPS Idle Mode Mobility Procedures attempted
SMF Call Control Procedure Statistics	smfcc_5gs_to_eps_idle_mobility_successes	Counter	5GS to EPS Idle Mode Mobility Procedures successful
SMF Call Control Procedure Statistics	smfcc_5gs_to_eps_idle_mobility_failures	Counter	5GS to EPS Idle Mode Mobility Procedures failed
SMF Call Control Procedure Statistics	smfcc_eps_to_5gs_handover_attempts	Counter	EPS to 5GS Handover Procedures attempted
SMF Call Control Procedure Statistics	smfcc_eps_to_5gs_handover_successes	Counter	EPS to 5GS Handover Procedures successful
SMF Call Control Procedure Statistics	smfcc_eps_to_5gs_handover_failures	Counter	EPS to 5GS Handover Procedures failed
SMF Call Control Procedure Statistics	smfcc_eps_to_5gs_handover_cancels	Counter	EPS to 5GS Handover Procedures canceled
SMF Call Control Procedure Statistics	smfcc_eps_to_5gs_idle_mobility_attempts	Counter	EPS to 5GS in Idle Mobility Procedures attempted
SMF Call Control Procedure Statistics	smfcc_eps_to_5gs_idle_mobility_successes	Counter	EPS to 5GS in Idle Mode Mobility Procedures successful
SMF Call Control Procedure Statistics	smfcc_eps_to_5gs_idle_mobility_failures	Counter	EPS to 5GS in Idle Mode Mobility Procedures failed
SMF Call Control Procedure Statistics	smfcc_vsmf_eps_to_5gs_idle_mobility_attempts	Counter	VSMF EPS to 5GS in Idle Mobility Procedures attempted
SMF Call Control Procedure Statistics	smfcc_vsmf_eps_to_5gs_idle_mobility_successes	Counter	VSMF EPS to 5GS in Idle Mode Mobility Procedures successful
SMF Call Control Procedure Statistics	smfcc_vsmf_eps_to_5gs_idle_mobility_failures	Counter	VSMF EPS to 5GS in Idle Mode Mobility Procedures failed
SMF Call Control Procedure Statistics	smfcc_hsmf_eps_to_5gs_idle_mobility_attempts	Counter	HSMF EPS to 5GS in Idle Mobility Procedures attempted
SMF Call Control Procedure Statistics	smfcc_hsmf_eps_to_5gs_idle_mobility_successes	Counter	HSMF EPS to 5GS in Idle Mode Mobility Procedures successful

Group	Metric Name	Type	Synopsis
SMF Call Control Procedure Statistics	smfcc_hsmf_eps_to_5gs_idle_mobility_failures	Counter	HSMF EPS to 5GS in Idle Mode Mobility Procedures failed
SMF Call Control Procedure Statistics	smfcc_chf_server_notification_procedure_attempts	Counter	CHF Server Notification Processing Procedure attempted
SMF Call Control Procedure Statistics	smfcc_chf_server_notification_procedure_successes	Counter	CHF Server Notification Processing Procedure successful
SMF Call Control Procedure Statistics	smfcc_chf_server_notification_procedure_failures	Counter	CHF Server Notification Processing Procedure failed
SMF Call Control Procedure Statistics	smfcc_hsmf_chf_server_notification_procedure_attempts	Counter	HSMF CHF Server Notification Processing Procedure attempted
SMF Call Control Procedure Statistics	smfcc_hsmf_chf_server_notification_procedure_successes	Counter	HSMF CHF Server Notification Processing Procedure successful
SMF Call Control Procedure Statistics	smfcc_hsmf_chf_server_notification_procedure_failures	Counter	HSMF CHF Server Notification Processing Procedure failed
SMF Call Control Procedure Statistics	smfcc_4g_to_epdg_handover_attempts	Counter	4G to 5EPDG Handover Procedures attempted
SMF Call Control Procedure Statistics	smfcc_4g_to_epdg_handover_successes	Counter	4G to EPDG Handover Procedures successful
SMF Call Control Procedure Statistics	smfcc_4g_to_epdg_handover_failures	Counter	4G to EPDG Handover Procedures failed
SMF Call Control Procedure Statistics	smfcc_5gs_to_epdg_handover_attempts	Counter	5GS to EPDG Handover Procedures attempted
SMF Call Control Procedure Statistics	smfcc_5gs_to_epdg_handover_successes	Counter	5GS to EPDG Handover Procedures successful
SMF Call Control Procedure Statistics	smfcc_5gs_to_epdg_handover_failures	Counter	5GS to EPDG Handover Procedures failed
SMF Call Control Procedure Statistics	smfcc_5gs_to_epdg_idle_handover_attempts	Counter	5GS to EPDG Idle Handover Procedures attempted
SMF Call Control Procedure Statistics	smfcc_5gs_to_epdg_idle_handover_successes	Counter	5GS to EPDG Idle Handover Procedures successful
SMF Call Control Procedure Statistics	smfcc_5gs_to_epdg_idle_handover_failures	Counter	5GS to EPDG Idle Handover Procedures failed
SMF Call Control Procedure Statistics	smfcc_non3gpp_to_5gs_handover_attempts	Counter	Non-3gpp to 5GS Handover Procedures attempted
SMF Call Control Procedure Statistics	smfcc_non3gpp_to_5gs_handover_successes	Counter	Non-3gpp to 5GS Handover Procedures successful
SMF Call Control Procedure Statistics	smfcc_non3gpp_to_5gs_handover_failures	Counter	Non-3gpp to 5GS Handover Procedures failed
SMF Call Control Procedure Statistics	smfcc_non3gpp_to_5gs_idle_mobility_attempts	Counter	Non-3gpp to 5GS Idle Mode Mobility Procedures attempted
SMF Call Control Procedure Statistics	smfcc_non3gpp_to_5gs_idle_mobility_successes	Counter	Non-3gpp to 5GS Idle Mode Mobility Procedures successful
SMF Call Control Procedure Statistics	smfcc_non3gpp_to_5gs_idle_mobility_failures	Counter	Non-3gpp to 5GS Idle Mode Mobility Procedures failed
SMF Call Control Procedure Statistics	smfcc_epdg_to_eps_handover_attempts	Counter	ePDG to EPS Handover Procedures attempted

Group	Metric Name	Type	Synopsis
SMF Call Control Procedure Statistics	smfcc_epdg_to_eps_handover_successes	Counter	ePDG to EPS Handover Procedures successful
SMF Call Control Procedure Statistics	smfcc_epdg_to_eps_handover_failures	Counter	ePDG to EPS Handover Procedures failed
SMF Call Control Procedure Statistics	smfcc_sdm_update_notification_procedure_attempts	Counter	SDM Update Notification Processing Procedure attempted
SMF Call Control Procedure Statistics	smfcc_sdm_update_notification_procedure_successes	Counter	SDM Update Notification Processing Procedure successful
SMF Call Control Procedure Statistics	smfcc_sdm_update_notification_procedure_failures	Counter	SDM Update Notification Processing Procedure failed
SMF Call Control Procedure Statistics	smfcc_rejects_overload	Counter	Procedures rejected due to overload
SMF Call Control Procedure Statistics	smfcc_qos_flow_inactivity_procedure_attempts	Counter	QosFlow Inactivity Procedures attempted
SMF Call Control Procedure Statistics	smfcc_qos_flow_inactivity_procedure_successes	Counter	QosFlow Inactivity Procedures successful
SMF Call Control Procedure Statistics	smfcc_qos_flow_inactivity_procedure_failures	Counter	QosFlow Inactivity Procedures failed
SMF Call Control Procedure Statistics	smfcc_session_idle_procedure_attempts	Counter	Session Idle Procedures attempted
SMF Call Control Procedure Statistics	smfcc_session_idle_procedure_successes	Counter	Session Idle Procedures successful
SMF Call Control Procedure Statistics	smfcc_session_idle_procedure_failures	Counter	Session Idle Procedures failed
SMF Call Control Procedure Statistics	smfcc_vsmf_session_idle_procedure_attempts	Counter	VSMF Session Idle Procedures attempted
SMF Call Control Procedure Statistics	smfcc_vsmf_session_idle_procedure_successes	Counter	VSMF Session Idle Procedures successful
SMF Call Control Procedure Statistics	smfcc_vsmf_session_idle_procedure_failures	Counter	VSMF Session Idle Procedures failed
SMF Call Control Procedure Statistics	smfcc_hsmf_session_idle_procedure_attempts	Counter	HSMF Session Idle Procedures attempted
SMF Call Control Procedure Statistics	smfcc_hsmf_session_idle_procedure_successes	Counter	HSMF Session Idle Procedures successful
SMF Call Control Procedure Statistics	smfcc_hsmf_session_idle_procedure_failures	Counter	HSMF Session Idle Procedures failed
SMF Call Control Procedure Statistics	smfcc_ue_initiated_modify_attempts	Counter	UE Initiated Modify Procedures attempted
SMF Call Control Procedure Statistics	smfcc_ue_initiated_modify_successes	Counter	UE Initiated Modify Procedures successful
SMF Call Control Procedure Statistics	smfcc_ue_initiated_modify_failures	Counter	UE Initiated Modify failed
SMF Call Control Procedure Statistics	smfcc_ssc_mode_3_pdu_session_establishment_attempts	Counter	PDU Session Establishment Procedures attempted due to ssc mode 3 reactivation
SMF Call Control Procedure Statistics	smfcc_ssc_mode_3_pdu_session_establishment_successes	Counter	PDU Session Establishment Procedures due to ssc mode 3 reactivation succeeded
SMF Call Control Procedure Statistics	smfcc_ssc_mode_3_pdu_session_establishment_failures	Counter	PDU Session Establishment Procedures due to ssc mode 3 reactivation failed

Group	Metric Name	Type	Synopsis
SMF Call Control Procedure Statistics	smfcc_ssc_mode3_pdu_session_request_reactivation_attempts	Counter	Ssc mode 3 request reactivation procedures attempted
SMF Call Control Procedure Statistics	smfcc_ssc_mode3_pdu_session_request_reactivation_successes	Counter	Ssc mode 3 request reactivation procedures succeeded
SMF Call Control Procedure Statistics	smfcc_ssc_mode3_pdu_session_request_reactivation_failures	Counter	Ssc mode 3 request reactivation procedures failed
SMF Call Control Procedure Statistics	smfcc_ssc_mode2_nw_initiated_release_attempts	Counter	SSC-Mode-2 NW Initiated Release Procedures attempted
SMF Call Control Procedure Statistics	smfcc_ssc_mode2_nw_initiated_release_successes	Counter	SSC-Mode-2 NW Initiated Release Procedures successful
SMF Call Control Procedure Statistics	smfcc_ssc_mode2_nw_initiated_release_failures	Counter	SSC-Mode-2 NW Initiated Release Procedures failed
SMF Call Control Procedure Statistics	smfcc_tft_update_after_5g_handover_attempts	Counter	Number of SMF Initiated TFT update after 5G Handover complete Procedures attempted
SMF Call Control Procedure Statistics	smfcc_tft_update_after_5g_handover_successes	Counter	Number of SMF Initiated TFT update after 5G Handover complete Procedures successful
SMF Call Control Procedure Statistics	smfcc_tft_update_after_5g_handover_failures	Counter	Number of SMF Initiated TFT update after 5G Handover complete Procedures failed
SMF Call Control Procedure Statistics	smfcc_upf_selection_under_overload	Counter	Number of UPF selections by SMF when all the NRF discovered UPFs are overloaded
SMF Call Control Procedure Statistics	smfcc_pfd_fetch_attempts	Counter	Pfd Fetch Procedures attempted
SMF Call Control Procedure Statistics	smfcc_pfd_fetch_successes	Counter	Pfd Fetch Procedures successful
SMF Call Control Procedure Statistics	smfcc_pfd_fetch_failures	Counter	Pfd Fetch Procedures failed
SMF Call Control Procedure Statistics	smfcc_amf_event_notification_attempts	Counter	AMF Event Notification Procedures attempted
SMF Call Control Procedure Statistics	smfcc_amf_event_notification_successes	Counter	AMF Event Notification Procedures successful
SMF Call Control Procedure Statistics	smfcc_amf_event_notification_failures	Counter	AMF Event Notification Procedures failed
SMF Call Control Procedure Statistics	smfcc_5gs_to_eps_handover_direct_data_forwarding_attempts	Counter	5GS to EPS Handover Direct Data Forwarding Procedures attempted
SMF Call Control Procedure Statistics	smfcc_5gs_to_eps_handover_direct_data_forwarding_successes	Counter	5GS to EPS Handover Direct Data Forwarding Procedures successful
SMF Call Control Procedure Statistics	smfcc_5gs_to_eps_handover_direct_data_forwarding_failures	Counter	5GS to EPS Handover Direct Data Forwarding Procedures failed
SMF Call Control Procedure Statistics	smfcc_eps_to_5gs_handover_without_n26_attempts	Counter	EPS to 5GS Handover Without N26 Interface Procedures attempted

Group	Metric Name	Type	Synopsis
SMF Call Control Procedure Statistics	smfcc_eps_to_5gs_handover_without_n26_successes	Counter	EPS to 5GS Handover Without N26 Interface Procedures successful
SMF Call Control Procedure Statistics	smfcc_eps_to_5gs_handover_without_n26_failures	Counter	EPS to 5GS Handover Without N26 Interface Procedures failed
SMF Call Control Procedure Statistics	smfcc_5gs_to_eps_handover_without_n26_attempts	Counter	EPS to 5GS Handover Without N26 Interface Procedures attempted
SMF Call Control Procedure Statistics	smfcc_5gs_to_eps_handover_without_n26_successes	Counter	EPS to 5GS Handover Without N26 Interface Procedures successful
SMF Call Control Procedure Statistics	smfcc_5gs_to_eps_handover_without_n26_failures	Counter	EPS to 5GS Handover Without N26 Interface Procedures failed
SMF Call Control Procedure Statistics	smfcc_5gs_to_eps_handover_without_n26_rsrc_rel_attempts	Counter	EPS to 5GS Handover Without N26 Interface Procedures attempted
SMF Call Control Procedure Statistics	smfcc_5gs_to_eps_handover_without_n26_rsrc_rel_successes	Counter	EPS to 5GS Handover Without N26 Interface Resource Release Procedures successful
SMF Call Control Procedure Statistics	smfcc_5gs_to_eps_handover_without_n26_rsrc_rel_failures	Counter	EPS to 5GS Handover Without N26 Interface Resource Release Procedures failed
SMF Call Control Procedure Statistics	smfcc_5gs_to_eps_handover_without_n26_brr_crt_attempts	Counter	EPS to 5GS Handover Without N26 Interface Bearer Create Procedures attempted
SMF Call Control Procedure Statistics	smfcc_5gs_to_eps_handover_without_n26_brr_crt_successes	Counter	EPS to 5GS Handover Without N26 Interface Bearer Create Procedures successful
SMF Call Control Procedure Statistics	smfcc_5gs_to_eps_handover_without_n26_brr_crt_failures	Counter	EPS to 5GS Handover Without N26 Interface Bearer Create Procedures failed
SMF Call Control Procedure Statistics	smfcc_5gs_to_eps_idle_handover_without_n26_attempts	Counter	EPS to 5GS Handover Without N26 Interface Procedures attempted
SMF Call Control Procedure Statistics	smfcc_5gs_to_eps_idle_handover_without_n26_successes	Counter	EPS to 5GS Handover Without N26 Interface Procedures successful
SMF Call Control Procedure Statistics	smfcc_5gs_to_eps_idle_handover_without_n26_failures	Counter	EPS to 5GS Handover Without N26 Interface Procedures failed
SMF Call Control Procedure Statistics	smfcc_ran_initiated_qosflow_mobility_attempts	Counter	RAN Initiated Qos Flow Mobility Procedures attempted
SMF Call Control Procedure Statistics	smfcc_ran_initiated_qosflow_mobility_successes	Counter	RAN Initiated Qos Flow Mobility Procedures successful
SMF Call Control Procedure Statistics	smfcc_ran_initiated_qosflow_mobility_partial_successes	Counter	RAN Initiated Qos Flow Mobility Procedures partially successful
SMF Call Control Procedure Statistics	smfcc_ran_initiated_qosflow_mobility_failures	Counter	RAN Initiated Qos Flow Mobility Procedures failed

Group	Metric Name	Type	Synopsis
SMF Call Control Procedure Statistics	smfcc_vsmf_ran_initiated_qosflow_mobility_attempts	Counter	vSMF RAN Initiated Qos Flow Mobility Procedures attempted
SMF Call Control Procedure Statistics	smfcc_vsmf_ran_initiated_qosflow_mobility_successes	Counter	vSMF RAN Initiated Qos Flow Mobility Procedures successful
SMF Call Control Procedure Statistics	smfcc_vsmf_ran_initiated_qosflow_mobility_partial_successes	Counter	vSMF RAN Initiated Qos Flow Mobility Procedures partially successful
SMF Call Control Procedure Statistics	smfcc_vsmf_ran_initiated_qosflow_mobility_failures	Counter	vSMF RAN Initiated Qos Flow Mobility Procedures failed
SMF Call Control Procedure Statistics	smfcc_policy_timeout_handling_attempts	Counter	Policy Timer Timeout Handling Procedures attempted
SMF Call Control Procedure Statistics	smfcc_policy_timeout_handling_successes	Counter	Policy Timer Timeout Handling Procedures successful
SMF Call Control Procedure Statistics	smfcc_policy_timeout_handling_failures	Counter	Policy Timer Timeout Handling Procedures failed
SMF Call Control Procedure Statistics	smfcc_vsmf_hsmf_initiated_pdu_session_assignebi_attempts	Counter	HSmf requested ebi assignment procedures attempted for VSMF sessions
SMF Call Control Procedure Statistics	smfcc_vsmf_hsmf_initiated_pdu_session_assignebi_successes	Counter	HSmf requested ebi assignment procedures successful for VSMF sessions
SMF Call Control Procedure Statistics	smfcc_vsmf_hsmf_initiated_pdu_session_assignebi_failures	Counter	HSmf requested ebi assignment procedures failed for VSMF sessions
SMF Call Control Procedure Statistics	smfcc_vsmf_hsmf_initiated_pdu_session_modification_attempts	Counter	HSmf requested qos modification procedures attempted for VSMF sessions
SMF Call Control Procedure Statistics	smfcc_vsmf_hsmf_initiated_pdu_session_modification_successes	Counter	HSmf requested qos modification procedures successful for VSMF sessions
SMF Call Control Procedure Statistics	smfcc_vsmf_hsmf_initiated_pdu_session_modification_failures	Counter	HSmf requested qos modification procedures failed for VSMF sessions
SMF Call Control Procedure Statistics	smfcc_vsmf_hsmf_initiated_pdu_session_release_attempts	Counter	HSMF Initiated PDU Session Release Procedure attempted for VSMF sessions
SMF Call Control Procedure Statistics	smfcc_vsmf_hsmf_initiated_pdu_session_release_successes	Counter	HSMF Initiated PDU Session Release Procedure successful for VSMF sessions
SMF Call Control Procedure Statistics	smfcc_vsmf_hsmf_initiated_pdu_session_release_failures	Counter	HSMF Initiated PDU Session Release Procedure Failed for VSMF sessions
SMF Call Control Procedure Statistics	smfcc_vsmf_pdu_session_establishment_attempts	Counter	VSMF PDU Session Establishment Procedures attempted
SMF Call Control Procedure Statistics	smfcc_vsmf_pdu_session_establishment_successes	Counter	VSMF PDU Session Establishment Procedures successful

Group	Metric Name	Type	Synopsis
SMF Call Control Procedure Statistics	smfcc_vsmf_pdu_session_establishment_failures	Counter	VSMF PDU Session Establishment Procedures failed
SMF Call Control Procedure Statistics	smfcc_vsmf_an_release_attempts	Counter	VSMF AN Release Procedures for inbound roamers attempted
SMF Call Control Procedure Statistics	smfcc_vsmf_an_release_successes	Counter	VSMF AN Release Procedures for inbound roamers successful
SMF Call Control Procedure Statistics	smfcc_vsmf_an_release_failures	Counter	VSMF AN Release Procedures for inbound roamers failed
SMF Call Control Procedure Statistics	smfcc_vsmf_service_request_attempts	Counter	VSMF Service Request Procedures for inbound roamers attempted
SMF Call Control Procedure Statistics	smfcc_vsmf_service_request_successes	Counter	VSMF Service Request Procedures for inbound roamers successful
SMF Call Control Procedure Statistics	smfcc_vsmf_service_request_failures	Counter	VSMF Service Request Procedures for inbound roamers failed
SMF Call Control Procedure Statistics	smfcc_vsmf_xn_handover_attempts	Counter	VSMF Xn Handover Procedures for inbound roamers attempted
SMF Call Control Procedure Statistics	smfcc_vsmf_xn_handover_successes	Counter	VSMF Xn Handover Procedures for inbound roamers succeeded
SMF Call Control Procedure Statistics	smfcc_vsmf_xn_handover_failures	Counter	VSMF Xn Handover Procedures for inbound roamers failed
SMF Call Control Procedure Statistics	smfcc_vsmf_xn_handover_idle_mode_successes	Counter	VSMF Xn Handover Procedures for inbound roamers succeeded for idle mode
SMF Call Control Procedure Statistics	smfcc_vsmf_xn_handover_idle_mode_failures	Counter	VSMF Xn Handover Procedures for inbound roamers failed for idle mode
SMF Call Control Procedure Statistics	smfcc_vsmf_network_initiated_service_request_attempts	Counter	VSMF Network Initiated Service Request Procedures attempted
SMF Call Control Procedure Statistics	smfcc_vsmf_network_initiated_service_request_successes	Counter	VSMF Network Initiated Service Request Procedures successful
SMF Call Control Procedure Statistics	smfcc_vsmf_network_initiated_service_request_failures	Counter	VSMF Network Initiated Service Request Procedures failed
SMF Call Control Procedure Statistics	smfcc_vsmf_pdu_session_retrieve_attempts	Counter	VSMF Retrieve Procedures attempted
SMF Call Control Procedure Statistics	smfcc_vsmf_pdu_session_retrieve_successes	Counter	VSMF Retrieve Procedures successful
SMF Call Control Procedure Statistics	smfcc_vsmf_pdu_session_retrieve_failures	Counter	VSMF Retrieve Procedures failed
SMF Call Control Procedure Statistics	smfcc_etcd_add_inband_radius_tunnel_attempts	Counter	add inband radius tunnel Procedures attempted
SMF Call Control Procedure Statistics	smfcc_etcd_add_inband_radius_tunnel_successes	Counter	add inband radius tunnel Procedures successful

Group	Metric Name	Type	Synopsis
SMF Call Control Procedure Statistics	smfcc_etcd_add_inband_radius_tunnel_failures	Counter	add inband radius tunnel Procedures failed
SMF Call Control Procedure Statistics	smfcc_etcd_del_inband_radius_tunnel_attempts	Counter	del inband radius tunnel Procedures attempted
SMF Call Control Procedure Statistics	smfcc_etcd_del_inband_radius_tunnel_successes	Counter	del inband radius tunnel Procedures successful
SMF Call Control Procedure Statistics	smfcc_etcd_del_inband_radius_tunnel_failures	Counter	del inband radius tunnel Procedures failed
SMF Call Control Procedure Statistics	smfcc_inband_radius_tunnel_create_attempts	Counter	Number of pdu session create procedures to add inband radius tunnels attempted
SMF Call Control Procedure Statistics	smfcc_inband_radius_tunnel_create_successes	Counter	Number of pdu session create procedures to add inband radius tunnels successful
SMF Call Control Procedure Statistics	smfcc_inband_radius_tunnel_create_failures	Counter	Number of pdu session create procedures to add inband radius tunnels failed
SMF Call Control Procedure Statistics	smfcc_inband_radius_tunnel_delete_attempts	Counter	Number of pdu session delete procedures to cleanup inband radius tunnels attempted
SMF Call Control Procedure Statistics	smfcc_inband_radius_tunnel_delete_successes	Counter	Number of pdu session delete procedures to cleanup inband radius tunnels successful
SMF Call Control Procedure Statistics	smfcc_inband_radius_tunnel_delete_failures	Counter	Number of pdu session delete procedures to cleanup inband radius tunnels failed
SMF Call Control Procedure Statistics	smfcc_hsmf_pdu_session_establishment_attempts	Counter	HSMF PDU Session Establishment Procedures attempted
SMF Call Control Procedure Statistics	smfcc_hsmf_pdu_session_establishment_successes	Counter	HSMF PDU Session Establishment Procedures successful
SMF Call Control Procedure Statistics	smfcc_hsmf_pdu_session_establishment_failures	Counter	HSMF PDU Session Establishment Procedures failed
SMF Call Control Procedure Statistics	smfcc_hsmf_pdu_session_modification_attempts	Counter	HSMF Pdu Session Modification Procedures attempted
SMF Call Control Procedure Statistics	smfcc_hsmf_pdu_session_modification_successes	Counter	HSMF Pdu Session Modification Procedures Successful
SMF Call Control Procedure Statistics	smfcc_hsmf_pdu_session_modification_failures	Counter	HSMF Pdu Session Modification Procedures failed
SMF Call Control Procedure Statistics	smfcc_hsmf_vsmf_initiated_pdu_session_release_attempts	Counter	HSMF Pdu Session release Procedures attempted
SMF Call Control Procedure Statistics	smfcc_hsmf_vsmf_initiated_pdu_session_release_successes	Counter	HSMF Pdu Session Release Procedures successful
SMF Call Control Procedure Statistics	smfcc_hsmf_vsmf_initiated_pdu_session_release_failures	Counter	HSMF Pdu Session release Procedures failed
SMF Call Control Procedure Statistics	smfcc_hsmf_initiated_pdu_session_release_attempts	Counter	HSMF Initiated PDU Session Release Procedure attempted

Group	Metric Name	Type	Synopsis
SMF Call Control Procedure Statistics	smfcc_hsmf_initiated_pdu_session_release_successes	Counter	HSMF Initiated PDU Session Release Procedure successful
SMF Call Control Procedure Statistics	smfcc_hsmf_initiated_pdu_session_release_failures	Counter	HSMF Initiated PDU Session Release Procedure Failed
SMF Call Control Procedure Statistics	smfcc_evt_exp_udm_create_subscription_attempts	Counter	Event Exposure UDM Create Subscription Procedure attempted
SMF Call Control Procedure Statistics	smfcc_evt_exp_udm_create_subscription_successes	Counter	Event Exposure UDM Create Subscription Procedure successful
SMF Call Control Procedure Statistics	smfcc_evt_exp_udm_create_subscription_failures	Counter	Event Exposure UDM Create Subscription Procedure failed
SMF Call Control Procedure Statistics	smfcc_evt_exp_udm_modify_subscription_attempts	Counter	Event Exposure UDM Modify Subscription Procedure attempted
SMF Call Control Procedure Statistics	smfcc_evt_exp_udm_modify_subscription_successes	Counter	Event Exposure UDM Modify Subscription Procedure successful
SMF Call Control Procedure Statistics	smfcc_evt_exp_udm_modify_subscription_failures	Counter	Event Exposure UDM Modify Subscription Procedure failed
SMF Call Control Procedure Statistics	smfcc_evt_exp_udm_delete_subscription_attempts	Counter	Event Exposure UDM Delete Subscription Procedure attempted
SMF Call Control Procedure Statistics	smfcc_evt_exp_udm_delete_subscription_successes	Counter	Event Exposure UDM Delete Subscription Procedure successful
SMF Call Control Procedure Statistics	smfcc_evt_exp_udm_delete_subscription_failures	Counter	Event Exposure UDM Delete Subscription Procedure failed
SMF Call Control Procedure Statistics	smfcc_proc_pdu_session_level_limit_exhaustions	Counter	Number of Limit Exhaustions For Smfcc Procedure Per Pdu Session
SMF Call Control Procedure Statistics	smfcc_proc_process_level_limit_exhaustions	Counter	Number of Limit Exhaustions For Smfcc Procedure Per Process
SMF UDM Interface Statistics	smfcc_udm_registration_attempts	Counter	UDM Registration Processing attempted
SMF UDM Interface Statistics	smfcc_udm_registration_successes	Counter	UDM Registration Processing successful
SMF UDM Interface Statistics	smfcc_udm_registration_failures	Counter	UDM Registration Processing failed
SMF UDM Interface Statistics	smfcc_sdm_get_sm_data_attempts	Counter	SDM Get Sm-Data Processing attempted
SMF UDM Interface Statistics	smfcc_sdm_get_sm_data_successes	Counter	SDM Get Sm-Data Processing successful
SMF UDM Interface Statistics	smfcc_sdm_get_sm_data_failures	Counter	SDM Get Sm-Data Processing failed
SMF UDM Interface Statistics	smfcc_sdm_subscription_attempts	Counter	SDM Subscription Processing attempted
SMF UDM Interface Statistics	smfcc_sdm_subscription_successes	Counter	SDM Subscription Processing successful
SMF UDM Interface Statistics	smfcc_sdm_subscription_failures	Counter	SDM Subscription Processing failed

Group	Metric Name	Type	Synopsis
SMF UDM Interface Statistics	smfcc_cancel_sdm_subscription_attempts	Counter	Cancel SDM Subscription Processing attempted
SMF UDM Interface Statistics	smfcc_cancel_sdm_subscription_successes	Counter	Cancel SDM Subscription Processing successful
SMF UDM Interface Statistics	smfcc_cancel_sdm_subscription_failures	Counter	Cancel SDM Subscription Processing failed
SMF UDM Interface Statistics	smfcc_udm_deregistration_attempts	Counter	UDM Deregistration Processing attempted
SMF UDM Interface Statistics	smfcc_udm_deregistration_successes	Counter	UDM Deregistration Processing successful
SMF UDM Interface Statistics	smfcc_udm_deregistration_failures	Counter	UDM Deregistration Processing failed
SMF UDM Interface Statistics	smfcc_sdm_update_notification_punted	Counter	SDM Update Notification Processing Procedure punted
SMF UDM Interface Statistics	smfcc_udm_static_ue_ip_allocation	Counter	UDM static UE IP allocation
SMF UDM Interface Statistics	smfcc_udm_evt_exp_create_subscription_attempts	Counter	UDM Nsmf Event Exposure Create Subscription attempted
SMF UDM Interface Statistics	smfcc_udm_evt_exp_create_subscription_successes	Counter	UDM Nsmf Event Exposure Create Subscription successes
SMF UDM Interface Statistics	smfcc_udm_evt_exp_create_subscription_failures	Counter	UDM Nsmf Event Exposure Create Subscription failed
SMF UDM Interface Statistics	smfcc_udm_evt_exp_modify_subscription_attempts	Counter	UDM Nsmf Event Exposure Modify Subscription attempted
SMF UDM Interface Statistics	smfcc_udm_evt_exp_modify_subscription_successes	Counter	UDM Nsmf Event Exposure Modify Subscription successes
SMF UDM Interface Statistics	smfcc_udm_evt_exp_modify_subscription_failures	Counter	UDM Nsmf Event Exposure Modify Subscription failed
SMF UDM Interface Statistics	smfcc_udm_evt_exp_delete_subscription_attempts	Counter	UDM Nsmf Event Exposure Delete Subscription attempted
SMF UDM Interface Statistics	smfcc_udm_evt_exp_delete_subscription_successes	Counter	UDM Nsmf Event Exposure Delete Subscription successes
SMF UDM Interface Statistics	smfcc_udm_evt_exp_delete_subscription_failures	Counter	UDM Nsmf Event Exposure Delete Subscription failed
SMF CHF Interface Procedure Statistics	smfcc_nchf_create_session_attempts	Counter	CHF Create Session attempted
SMF CHF Interface Procedure Statistics	smfcc_nchf_create_session_success	Counter	CHF Create Session successful
SMF CHF Interface Procedure Statistics	smfcc_nchf_create_session_failures	Counter	CHF Create Session failed
SMF CHF Interface Procedure Statistics	smfcc_nchf_update_session_attempts	Counter	CHF Update Session attempted
SMF CHF Interface Procedure Statistics	smfcc_nchf_update_session_success	Counter	CHF Update Session successful
SMF CHF Interface Procedure Statistics	smfcc_nchf_update_session_failures	Counter	CHF Update Session failed
SMF CHF Interface Procedure Statistics	smfcc_nchf_release_session_attempts	Counter	CHF Create Release attempted

Group	Metric Name	Type	Synopsis
SMF CHF Interface Procedure Statistics	smfcc_nchf_release_session_success	Counter	CHF Release Session successful
SMF CHF Interface Procedure Statistics	smfcc_nchf_release_session_failures	Counter	CHF Release Session failed
SMF CHF Interface Procedure Statistics	smfcc_nchf_one_time_report_attempts	Counter	CHF One Time Report attempted
SMF CHF Interface Procedure Statistics	smfcc_nchf_one_time_report_success	Counter	CHF One Time Report Successful
SMF CHF Interface Procedure Statistics	smfcc_nchf_one_time_report_failures	Counter	CHF One Time Report Failed
SMF CHF Interface Procedure Statistics	smfcc_nchf_notification_received	Counter	CHF Notification Received
SMF CHF Interface Procedure Statistics	smfcc_nchf_notification_success	Counter	CHF Notification Success sent
SMF CHF Interface Procedure Statistics	smfcc_nchf_notification_failures	Counter	CHF Notification Fail Sent
SMF CHF Interface Procedure Statistics	smfcc_nchf_notification_punted	Counter	CHF Notification punted
SMFCC Monitor Service Interface Statistics	smfcc_get_smfcc_gauge_attempts	Counter	Get SMFCC gauge request received
SMFCC Monitor Service Interface Statistics	smfcc_get_smfcc_gauge_successes	Counter	Get SMFCC gauge requests succeeded
SMFCC Monitor Service Interface Statistics	smfcc_get_smfcc_gauge_failures	Counter	Get SMFCC gauge requests failed
SMF GTPC Interface Statistics	smfcc_gtpc_create_session_attempts	Counter	GTPC Create Session attempts
SMF GTPC Interface Statistics	smfcc_gtpc_create_session_accepts	Counter	GTPC Create Session attempts accepted
SMF GTPC Interface Statistics	smfcc_gtpc_create_session_rejects	Counter	GTPC Create Session attempts rejected
SMF GTPC Interface Statistics	smfcc_gtpc_saegw_create_session_attempts	Counter	GTPC SAEGW Create Session attempts
SMF GTPC Interface Statistics	smfcc_gtpc_saegw_create_session_accepts	Counter	GTPC SAEGW Create Session attempts accepted
SMF GTPC Interface Statistics	smfcc_gtpc_saegw_create_session_rejects	Counter	GTPC SAEGW Create Session attempts rejected
SMF GTPC Interface Statistics	smfcc_gtpc_create_session_punted	Counter	GTPC Create Session punted
SMF GTPC Interface Statistics	smfcc_rate_limit_exceed_create_session_rejects	Counter	Number of GTPC Create Session attempts rejected due to exceeding rate limit
SMF GTPC Interface Statistics	smfcc_gtpc_delete_session_attempts	Counter	GTPC Delete Session attempts
SMF GTPC Interface Statistics	smfcc_gtpc_delete_session_successes	Counter	GTPC Delete Session attempts successful
SMF GTPC Interface Statistics	smfcc_gtpc_delete_session_failures	Counter	GTPC Delete Session attempts failed
SMF GTPC Interface Statistics	smfcc_gtpc_saegw_delete_session_attempts	Counter	GTPC SAEGW Delete Session attempts
SMF GTPC Interface Statistics	smfcc_gtpc_saegw_delete_session_successes	Counter	GTPC SAEGW Delete Session attempts successful
SMF GTPC Interface Statistics	smfcc_gtpc_saegw_delete_session_failures	Counter	GTPC SAEGW Delete Session attempts failed
SMF GTPC Interface Statistics	smfcc_gtpc_delete_bearer_attempts	Counter	GTPC Delete Bearer attempts

Group	Metric Name	Type	Synopsis
SMF GTPC Interface Statistics	smfcc_gtpc_delete_session_punted	Counter	GTPC Delete Session punted
SMF GTPC Interface Statistics	smfcc_gtpc_delete_bearer_successes	Counter	GTPC Delete Bearer attempts successful
SMF GTPC Interface Statistics	smfcc_gtpc_delete_bearer_failures	Counter	GTPC Delete Bearer attempts failed
SMF GTPC Interface Statistics	smfcc_gtpc_saegw_delete_bearer_attempts	Counter	GTPC Delete Bearer attempts for SAEGW
SMF GTPC Interface Statistics	smfcc_gtpc_saegw_delete_bearer_successes	Counter	GTPC Delete Bearer attempts successful for SAEGW
SMF GTPC Interface Statistics	smfcc_gtpc_saegw_delete_bearer_failures	Counter	GTPC Delete Bearer attempts failed for SAEGW
SMF GTPC Interface Statistics	smfcc_gtpc_create_bearer_attempts	Counter	GTPC Create Bearer attempts
SMF GTPC Interface Statistics	smfcc_gtpc_create_bearer_successes	Counter	GTPC Create Bearer attempts successful
SMF GTPC Interface Statistics	smfcc_gtpc_create_bearer_failures	Counter	GTPC Create Bearer attempts failed
SMF GTPC Interface Statistics	smfcc_gtpc_saegw_create_bearer_attempts	Counter	GTPC Create Bearer attempts for SAEGW
SMF GTPC Interface Statistics	smfcc_gtpc_saegw_create_bearer_successes	Counter	GTPC Create Bearer attempts successful for SAEGW
SMF GTPC Interface Statistics	smfcc_gtpc_saegw_create_bearer_failures	Counter	GTPC Create Bearer attempts failed for SAEGW
SMF GTPC Interface Statistics	smfcc_gtpc_epco_5gsm_cause_received	Counter	GTPC peer response received with EPCO having 5GSM cause
SMF GTPC Interface Statistics	smfcc_gtpc_update_bearer_attempts	Counter	GTPC Update Bearer attempts
SMF GTPC Interface Statistics	smfcc_gtpc_update_bearer_successes	Counter	GTPC Update Bearer attempts successful
SMF GTPC Interface Statistics	smfcc_gtpc_update_bearer_failures	Counter	GTPC Update Bearer attempts failed
SMF GTPC Interface Statistics	smfcc_gtpc_saegw_update_bearer_attempts	Counter	GTPC Update Bearer attempts for SAEGW
SMF GTPC Interface Statistics	smfcc_gtpc_saegw_update_bearer_successes	Counter	GTPC Update Bearer attempts successful for SAEGW
SMF GTPC Interface Statistics	smfcc_gtpc_saegw_update_bearer_failures	Counter	GTPC Update Bearer attempts failed for SAEGW
SMF GTPC Interface Statistics	smfcc_gtpc_modify_bearer_request_attempts	Counter	GTPC Modify Bearer Request attempts
SMF GTPC Interface Statistics	smfcc_gtpc_modify_bearer_request_successes	Counter	GTPC Modify Bearer Request attempts successful
SMF GTPC Interface Statistics	smfcc_gtpc_modify_bearer_request_failures	Counter	GTPC Modify Bearer Request attempts failed
SMF GTPC Interface Statistics	smfcc_gtpc_saegw_modify_bearer_request_attempts	Counter	GTPC Modify Bearer Request attempts for SAEGW
SMF GTPC Interface Statistics	smfcc_gtpc_saegw_modify_bearer_request_successes	Counter	GTPC Modify Bearer Request attempts successful for SAEGW

Group	Metric Name	Type	Synopsis
SMF GTPC Interface Statistics	smfcc_gtpc_saegw_modify_bearer_request_failures	Counter	GTPC Modify Bearer Request attempts failed for SAEGW
SMF GTPC Interface Statistics	smfcc_gtpc_modify_bearer_request_punted	Counter	GTPC Modify Bearer Request punted
SMF GTPC Interface Statistics	smfcc_gtpc_modify_bearer_command_attempts	Counter	GTPC Modify Bearer Command attempts
SMF GTPC Interface Statistics	smfcc_gtpc_modify_bearer_command_failures	Counter	GTPC Modify Bearer Command attempts failed
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_modify_bearer_command_attempts	Counter	GTPC Modify Bearer Command attempts for SGW
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_modify_bearer_command_failures	Counter	GTPC Modify Bearer Command attempts failed for SGW
SMF GTPC Interface Statistics	smfcc_gtpc_saegw_modify_bearer_command_attempts	Counter	GTPC Modify Bearer Command attempts for SAEGW
SMF GTPC Interface Statistics	smfcc_gtpc_saegw_modify_bearer_command_failures	Counter	GTPC Modify Bearer Command attempts failed for SAEGW
SMF GTPC Interface Statistics	smfcc_gtpc_modify_bearer_command_punted	Counter	GTPC Modify Bearer Command punted
SMF GTPC Interface Statistics	smfcc_gtpc_bearer_resource_command_attempts	Counter	GTPC Bearer Resource Command attempts
SMF GTPC Interface Statistics	smfcc_gtpc_bearer_resource_command_failures	Counter	GTPC Bearer Resource Command attempts failed
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_bearer_resource_command_attempts	Counter	GTPC Bearer Resource Command attempts for SGW
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_bearer_resource_command_failures	Counter	GTPC Bearer Resource Command attempts failed for SAEGW
SMF GTPC Interface Statistics	smfcc_gtpc_saegw_bearer_resource_command_attempts	Counter	GTPC Bearer Resource Command attempts
SMF GTPC Interface Statistics	smfcc_gtpc_saegw_bearer_resource_command_failures	Counter	GTPC Bearer Resource Command attempts failed
SMF GTPC Interface Statistics	smfcc_gtpc_bearer_resource_command_punted	Counter	GTPC Bearer Resource Command punted
SMF GTPC Interface Statistics	smfcc_gtpc_delete_bearer_command_attempts	Counter	GTPC Delete Bearer Command attempts
SMF GTPC Interface Statistics	smfcc_gtpc_delete_bearer_command_failures	Counter	GTPC Delete Bearer Command attempts failed
SMF GTPC Interface Statistics	smfcc_gtpc_delete_bearer_command_punted	Counter	GTPC Delete Bearer Command punted
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_create_session_attempts	Counter	GTPC Sgw Create Session attempts
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_create_session_accepts	Counter	GTPC Sgw Create Session attempts accepted
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_create_session_rejects	Counter	GTPC Sgw Create Session attempts rejected
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_delete_session_attempts	Counter	GTPC Sgw Delete Session attempts
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_delete_session_successes	Counter	GTPC Sgw Delete Session attempts successful

Group	Metric Name	Type	Synopsis
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_delete_session_failures	Counter	GTPC Sgw Delete Session attempts failed
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_delete_session_sent_to_pgw_attempts	Counter	GTPC Sgw Delete Session attempts
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_delete_session_sent_to_pgw_successes	Counter	GTPC Sgw Delete Session attempts successful
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_delete_session_sent_to_pgw_failures	Counter	GTPC Sgw Delete Session attempts failed
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_create_bearer_attempts	Counter	GTPC Sgw Create Bearer attempts
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_create_bearer_accepts	Counter	GTPC Sgw Create Bearer attempts accepted
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_create_bearer_failures	Counter	GTPC Sgw Create Bearer attempts rejected
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_create_bearer_request_sent_to_mme_attempts	Counter	GTPC Mme Create Bearer attempts
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_create_bearer_request_sent_to_mme_accepts	Counter	GTPC Mme Create Bearer accepted
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_create_bearer_request_sent_to_mme_failures	Counter	GTPC Mme Create Bearer rejected
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_update_bearer_attempts	Counter	GTPC Sgw Update Bearer attempts
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_update_bearer_accepts	Counter	GTPC Sgw Update Bearer attempts accepted
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_update_bearer_failures	Counter	GTPC Sgw Update Bearer attempts rejected
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_update_bearer_request_sent_to_mme_attempts	Counter	GTPC Mme Update Bearer attempts
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_update_bearer_request_sent_to_mme_accepts	Counter	GTPC Mme Update Bearer accepted
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_update_bearer_request_sent_to_mme_failures	Counter	GTPC Mme Update Bearer rejected
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_modify_bearer_request_attempts	Counter	GTPC Sgw Modify Bearer Request attempts
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_modify_bearer_request_successes	Counter	GTPC Sgw Modify Bearer Request attempts successful
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_modify_bearer_request_failures	Counter	GTPC Sgw Modify Bearer Request attempts failed
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_delete_bearer_request_attempts	Counter	GTPC Sgw Delete Bearer Request attempts
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_delete_bearer_request_successes	Counter	GTPC Sgw Delete Bearer Request attempts successful
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_delete_bearer_request_failures	Counter	GTPC Sgw Delete Bearer Request attempts failed
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_delete_bearer_request_sent_to_mme_attempts	Counter	GTPC Sgw Delete Bearer Request attempts sent to MME
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_delete_bearer_request_sent_to_mme_successes	Counter	GTPC Sgw Delete Bearer Request attempts sent to MME successful
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_delete_bearer_request_sent_to_mme_failures	Counter	GTPC Sgw Delete Bearer Request attempts sent to MME failed
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_create_session_duplicate_attempts	Counter	GTPC sgw Create Session attempts

Group	Metric Name	Type	Synopsis
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_create_session_duplicate_drops	Counter	GTPC sgw Create Session duplicate requests dropped
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_modify_bearer_duplicate_attempts	Counter	GTPC sgw Modify Bearer attempts
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_modify_bearer_duplicate_drops	Counter	GTPC sgw Modify Bearer duplicate requests dropped
SMF GTPC Interface Statistics	smfcc_gtpc_pgw_create_session_duplicate_attempts	Counter	GTPC Pgw Create Session attempts
SMF GTPC Interface Statistics	smfcc_gtpc_pgw_create_session_duplicate_drops	Counter	GTPC Pgw Create Session duplicate requests dropped
SMF GTPC Interface Statistics	smfcc_gtpc_pgw_modify_bearer_duplicate_attempts	Counter	GTPC Pgw Modify Bearer attempts
SMF GTPC Interface Statistics	smfcc_gtpc_pgw_modify_bearer_duplicate_drops	Counter	GTPC Pgw Modify Bearer duplicate requests dropped
SMF GTPC Interface Statistics	smfcc_gtpc_saegw_create_session_duplicate_attempts	Counter	GTPC SAEGW Create Session attempts
SMF GTPC Interface Statistics	smfcc_gtpc_saegw_create_session_duplicate_drops	Counter	GTPC SAEGW Create Session duplicate requests dropped
SMF GTPC Interface Statistics	smfcc_gtpc_saegw_modify_bearer_duplicate_attempts	Counter	GTPC SAEGW Modify Bearer attempts
SMF GTPC Interface Statistics	smfcc_gtpc_saegw_modify_bearer_duplicate_drops	Counter	GTPC SAEGW Modify Bearer duplicate requests dropped
SMF GTPC Interface Statistics	smfcc_gtpc_pgw_create_session_context_replacement_count	Counter	GTPC Pgw Create Session context replacement count
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_create_session_context_replacement_count	Counter	GTPC Sgw Create Session context replacement count
SMF GTPC Interface Statistics	smfcc_gtpc_release_access_bearer_request_attempts	Counter	GTPC Sgw Release Access Bearer Request attempts
SMF GTPC Interface Statistics	smfcc_gtpc_release_access_bearer_request_successes	Counter	GTPC Sgw Release Access Bearer Request attempts successful
SMF GTPC Interface Statistics	smfcc_gtpc_release_access_bearer_request_failures	Counter	GTPC Sgw Release Access Bearer Request attempts failed
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_modify_bearer_request_sent_to_pgw_attempts	Counter	GTPC Sgw Modify Bearer Request attempts
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_modify_bearer_request_sent_to_pgw_successes	Counter	GTPC Sgw Modify Bearer Request attempts successful
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_modify_bearer_request_sent_to_pgw_failures	Counter	GTPC Sgw Modify Bearer Request attempts failed
SMF GTPC Interface Statistics	smfcc_gtpc_unsupported_msg_received	Counter	GTPC unsupported messages received that got dropped
SMF GTPC Interface Statistics	smfcc_gtpc_down_link_data_notification_attempts	Counter	GTPC Sgw Down link data notification attempts
SMF GTPC Interface Statistics	smfcc_gtpc_down_link_data_notification_accepts	Counter	GTPC Sgw Down link data notification accepted
SMF GTPC Interface Statistics	smfcc_gtpc_down_link_data_notification_failures	Counter	GTPC Sgw down link data notification attempts rejected
SMF GTPC Interface Statistics	smfcc_gtpc_down_link_notification_failure_indication	Counter	GTPC Sgw Down link data notification failure indication

Group	Metric Name	Type	Synopsis
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_delete_bearer_command_from_mme_attempts	Counter	GTPC SGW Delete Bearer Command attempts from MME
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_delete_bearer_command_to_mme_failures	Counter	GTPC SGW Delete Bearer Command failed from MME
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_delete_bearer_command_to_pgw_attempts	Counter	GTPC SGW Delete Bearer Command attempts to PGW
SMF GTPC Interface Statistics	smfcc_gtpc_sgw_delete_bearer_command_from_pgw_failures	Counter	GTPC SGW Delete Bearer Command failures to PGW
SMF GTPC Interface Statistics	smfcc_gtpc_delete_pdp_context_rx_attempts	Counter	GTPC Delete PDP Context received attempts
SMF GTPC Interface Statistics	smfcc_gtpc_delete_pdp_context_rx_successes	Counter	GTPC Delete PDP Context received attempts successful
SMF GTPC Interface Statistics	smfcc_gtpc_delete_pdp_context_rx_failures	Counter	GTPC Delete PDP Context received attempts failed
SMF GTPC Interface Statistics	smfcc_gtpc_delete_pdp_context_tx_attempts	Counter	GGSN init GTPC Delete PDP Context attempts
SMF GTPC Interface Statistics	smfcc_gtpc_delete_pdp_context_tx_successes	Counter	GGSN init GTPC Delete PDP Context attempts successful
SMF GTPC Interface Statistics	smfcc_gtpc_delete_pdp_context_tx_failures	Counter	GGSN init GTPC Delete PDP Context attempts failed
SMF GTPC Interface Statistics	smfcc_gtpc_ggsn_create_pdp_context_duplicate_attempts	Counter	GTPC Ggsn Create PDP Context attempts
SMF GTPC Interface Statistics	smfcc_gtpc_ggsn_create_pdp_context_duplicate_drops	Counter	GTPC Ggsn Create PDP Context duplicate requests dropped
SMF GTPC Interface Statistics	smfcc_gtpc_ggsn_create_pdp_context_context_replacement_count	Counter	GTPC Ggsn Create PDP Context context replacement count
SMF GTPC Interface Statistics	smfcc_gtpc_ggsn_update_pdp_context_duplicate_attempts	Counter	GTPC Ggsn Update PDP Context attempts
SMF GTPC Interface Statistics	smfcc_gtpc_ggsn_update_pdp_context_duplicate_drops	Counter	GTPC Ggsn Update PDP Context duplicate requests dropped
SMF GTPC Interface Statistics	smfcc_gtpc_create_pdp_context_attempts	Counter	GTPC Create PDP Context attempts
SMF GTPC Interface Statistics	smfcc_gtpc_create_pdp_context_accepts	Counter	GTPC Create PDP Context attempts accepted
SMF GTPC Interface Statistics	smfcc_gtpc_create_pdp_context_rejects	Counter	GTPC Create PDP Context attempts rejected
SMF GTPC Interface Statistics	smfcc_gtpc_update_pdp_context_rx_attempts	Counter	GTPC Update PDP Context attempts
SMF GTPC Interface Statistics	smfcc_gtpc_update_pdp_context_rx_success	Counter	GTPC Update PDP Context attempts accepted
SMF GTPC Interface Statistics	smfcc_gtpc_update_pdp_context_rx_failures	Counter	GTPC Update PDP Context attempts rejected
SMF GTPC Interface Statistics	smfcc_gtpc_update_pdp_context_tx_attempts	Counter	GTPC Update PDP Context attempts
SMF GTPC Interface Statistics	smfcc_gtpc_update_pdp_context_tx_accepts	Counter	GTPC Update PDP Context attempts accepted
SMF GTPC Interface Statistics	smfcc_gtpc_update_pdp_context_tx_failures	Counter	GTPC Update PDP Context attempts failed
SMF GTPC Interface Statistics	smfcc_gtpc_saegw_create_session_context_replacement_count	Counter	GTPC SAEGW Create Session context replacement count

Group	Metric Name	Type	Synopsis
SMFCC PGW Procedures Statistics	smfcc_pgw_initial_attach_attempts	Counter	PGW Initial Attach Procedures attempted
SMFCC PGW Procedures Statistics	smfcc_pgw_initial_attach_successes	Counter	PGW Initial Attach Procedures successful
SMFCC PGW Procedures Statistics	smfcc_pgw_initial_attach_failures	Counter	PGW Initial Attach Procedures failed
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_initial_attach_attempts	Counter	PGW Initial Roaming Attach Procedures attempted
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_initial_attach_successes	Counter	PGW Initial Roaming Attach Procedures successful
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_initial_attach_failures	Counter	PGW Initial Roaming Attach Procedures failed
SMFCC PGW Procedures Statistics	smfcc_pgw_ue_initiated_detach_attempts	Counter	PGW Delete Session Procedures attempted
SMFCC PGW Procedures Statistics	smfcc_pgw_ue_initiated_detach_successes	Counter	PGW Delete Session Procedures successful
SMFCC PGW Procedures Statistics	smfcc_pgw_ue_initiated_detach_failures	Counter	PGW Delete Session Procedures failed
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_ue_initiated_detach_attempts	Counter	PGW Delete Roaming Session Procedures attempted
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_ue_initiated_detach_successes	Counter	PGW Delete Roaming Session Procedures successful
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_ue_initiated_detach_failures	Counter	PGW Delete Roaming Session Procedures failed
SMFCC PGW Procedures Statistics	smfcc_pgw_pcf_initiated_terminate_attempts	Counter	PCF Initiated Terminate Procedures attempted
SMFCC PGW Procedures Statistics	smfcc_pgw_pcf_initiated_terminate_successes	Counter	PCF Initiated Terminate Procedures successful
SMFCC PGW Procedures Statistics	smfcc_pgw_pcf_initiated_terminate_failures	Counter	PCF Initiated Terminate failed
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_pcf_initiated_terminate_attempts	Counter	PCF Initiated Terminate Procedures attempted for roaming sessions
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_pcf_initiated_terminate_successes	Counter	PCF Initiated Terminate Procedures successful
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_pcf_initiated_terminate_failures	Counter	PCF Initiated Terminate failed for roaming sessions
SMFCC PGW Procedures Statistics	smfcc_pgw_nw_initiated_release_attempts	Counter	NW Initiated Release Procedures attempted
SMFCC PGW Procedures Statistics	smfcc_pgw_nw_initiated_release_successes	Counter	NW Initiated Release Procedures successful
SMFCC PGW Procedures Statistics	smfcc_pgw_nw_initiated_release_failures	Counter	NW Initiated Release Procedures failed
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_nw_initiated_release_attempts	Counter	NW Initiated Release Procedures attempted for roaming sessions
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_nw_initiated_release_successes	Counter	NW Initiated Release Procedures successful for roaming sessions
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_nw_initiated_release_failures	Counter	NW Initiated Release Procedures failed for roaming sessions

Group	Metric Name	Type	Synopsis
SMFCC PGW Procedures Statistics	smfcc_pgw_pcf_initiated_policy_modification_attempts	Counter	PCF Initiated Policy Modification Procedures attempted
SMFCC PGW Procedures Statistics	smfcc_pgw_pcf_initiated_policy_modification_successes	Counter	PCF Initiated Policy Modification Procedures successful
SMFCC PGW Procedures Statistics	smfcc_pgw_pcf_initiated_policy_modification_failures	Counter	PCF Initiated Policy Modification Procedures failed
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_pcf_initiated_policy_modification_attempts	Counter	PCF Initiated Policy Modification Procedures attempted for roaming sessions
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_pcf_initiated_policy_modification_successes	Counter	PCF Initiated Policy Modification Procedures successful
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_pcf_initiated_policy_modification_failures	Counter	PCF Initiated Policy Modification Procedures failed for roaming sessions
SMFCC PGW Procedures Statistics	smfcc_pgw_pcf_initiated_dedicated_bearer_activation_attempts	Counter	PCF Initiated Dedicated Bearer Activation Procedures attempted
SMFCC PGW Procedures Statistics	smfcc_pgw_pcf_initiated_dedicated_bearer_activation_successes	Counter	PCF Initiated Bearer Activation Procedures successful
SMFCC PGW Procedures Statistics	smfcc_pgw_pcf_initiated_dedicated_bearer_activation_failures	Counter	PCF Initiated Dedicated Bearer Activation Procedures failed
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_pcf_initiated_dedicated_bearer_activation_attempts	Counter	PCF Initiated Dedicated Bearer Activation Procedures attempted for roaming sessions
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_pcf_initiated_dedicated_bearer_activation_successes	Counter	PCF Initiated Bearer Activation Procedures successful for roaming sessions
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_pcf_initiated_dedicated_bearer_activation_failures	Counter	PCF Initiated Dedicated Bearer Activation Procedures failed for roaming sessions
SMFCC PGW Procedures Statistics	smfcc_pgw_initiated_legacy_dedicated_bearer_activation_attempts_ho	Counter	PGW Initiated Legacy Dedicated Bearer Activation Procedures attempted During HO
SMFCC PGW Procedures Statistics	smfcc_pgw_initiated_legacy_dedicated_bearer_activation_successes_ho	Counter	PGW Initiated Legacy Dedicated Bearer Activation Procedures successful During HO
SMFCC PGW Procedures Statistics	smfcc_pgw_initiated_legacy_dedicated_bearer_activation_failures_ho	Counter	PGW Initiated Legacy Dedicated Bearer Activation Procedures failed
SMFCC PGW Procedures Statistics	smfcc_pgw_initiated_legacy_dedicated_bearer_activation_roaming_session_attempts_ho	Counter	PGW Initiated Legacy Dedicated Bearer Activation Procedures attempted During HO
SMFCC PGW Procedures Statistics	smfcc_pgw_initiated_legacy_dedicated_bearer_activation_roaming_session_successes_ho	Counter	PGW Initiated Legacy Dedicated Bearer Activation Procedures successful During HO

Group	Metric Name	Type	Synopsis
SMFCC PGW Procedures Statistics	smfcc_pgw_initiated_legacy_dedicated_bearer_activation_roaming_session_failures_ho	Counter	PGW Initiated Legacy Dedicated Bearer Activation Procedures failed
SMFCC PGW Procedures Statistics	smfcc_pgw_modify_bearer_request_attempts	Counter	Modify Bearer Request Procedures attempted
SMFCC PGW Procedures Statistics	smfcc_pgw_modify_bearer_request_successes	Counter	Modify Bearer Request Procedures successful
SMFCC PGW Procedures Statistics	smfcc_pgw_modify_bearer_request_failures	Counter	Modify Bearer Request Procedures failed
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_modify_bearer_request_attempts	Counter	Modify Bearer Request Procedures attempted for roaming sessions
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_modify_bearer_request_successes	Counter	Modify Bearer Request Procedures successful for roaming sessions
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_modify_bearer_request_failures	Counter	Modify Bearer Request Procedures failed for roaming sessions
SMFCC PGW Procedures Statistics	smfcc_pgw_mbr_tau_rau_x2_sgw_change_handover_attempts	Counter	Modify Bearer Request Procedures with SGW change attempted
SMFCC PGW Procedures Statistics	smfcc_pgw_mbr_tau_rau_x2_sgw_change_handover_successes	Counter	Modify Bearer Request Procedures with SGW change successful
SMFCC PGW Procedures Statistics	smfcc_pgw_mbr_tau_rau_x2_sgw_change_handover_failures	Counter	Modify Bearer Request Procedures with SGW change failed
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_mbr_tau_rau_x2_sgw_change_handover_attempts	Counter	Modify Bearer Request Procedures with SGW change attempted for roaming sessions
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_mbr_tau_rau_x2_sgw_change_handover_successes	Counter	Modify Bearer Request Procedures with SGW change successful for roaming sessions
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_mbr_tau_rau_x2_sgw_change_handover_failures	Counter	Modify Bearer Request Procedures with SGW change failed for roaming sessions
SMFCC PGW Procedures Statistics	smfcc_pgw_hss_initiated_modify_bearer_command_attempts	Counter	HSS Initiated Modify Bearer Command Procedures attempted
SMFCC PGW Procedures Statistics	smfcc_pgw_hss_initiated_modify_bearer_command_successes	Counter	HSS Initiated Modify Bearer Command Procedures successful
SMFCC PGW Procedures Statistics	smfcc_pgw_hss_initiated_modify_bearer_command_failures	Counter	HSS Initiated Modify Bearer Command Procedures failed
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_hss_initiated_modify_bearer_command_attempts	Counter	HSS Initiated Modify Bearer Command Procedures attempted for roaming sessions
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_hss_initiated_modify_bearer_command_successes	Counter	HSS Initiated Modify Bearer Command Procedures successful for roaming sessions
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_hss_initiated_modify_bearer_command_failures	Counter	HSS Initiated Modify Bearer Command Procedures failed for roaming sessions

Group	Metric Name	Type	Synopsis
SMFCC PGW Procedures Statistics	smfcc_pgw_pcf_initiated_dedicated_bearer_delete_attempts	Counter	PCF Initiated Dedicated Bearer Delete Procedures attempted
SMFCC PGW Procedures Statistics	smfcc_pgw_pcf_initiated_dedicated_bearer_delete_successes	Counter	PCF Initiated Bearer Delete Procedures successful
SMFCC PGW Procedures Statistics	smfcc_pgw_pcf_initiated_dedicated_bearer_delete_failures	Counter	PCF Initiated Dedicated Bearer Delete Procedures failed
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_pcf_initiated_dedicated_bearer_delete_attempts	Counter	PCF Initiated Dedicated Bearer Delete Procedures attempted for roaming sessions
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_pcf_initiated_dedicated_bearer_delete_successes	Counter	PCF Initiated Bearer Delete Procedures successful for roaming sessions
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_pcf_initiated_dedicated_bearer_delete_failures	Counter	PCF Initiated Dedicated Bearer Delete Procedures failed for roaming sessions
SMFCC PGW Procedures Statistics	smfcc_pgw_pcf_initiated_dedicated_bearer_update_attempts	Counter	PCF Initiated Dedicated Bearer Update Procedures attempted
SMFCC PGW Procedures Statistics	smfcc_pgw_pcf_initiated_dedicated_bearer_update_successes	Counter	PCF Initiated Bearer Update Procedures successful
SMFCC PGW Procedures Statistics	smfcc_pgw_pcf_initiated_dedicated_bearer_update_failures	Counter	PCF Initiated Dedicated Bearer Update Procedures failed
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_pcf_initiated_dedicated_bearer_update_attempts	Counter	PCF Initiated Dedicated Bearer Update Procedures attempted for roaming sessions
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_pcf_initiated_dedicated_bearer_update_successes	Counter	PCF Initiated Bearer Update Procedures successful for roaming sessions
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_pcf_initiated_dedicated_bearer_update_failures	Counter	PCF Initiated Dedicated Bearer Update Procedures failed for roaming sessions
SMFCC PGW Procedures Statistics	smfcc_pgw_process_pcf_update_policy_rules_attempts	Counter	Process Pcf Update Policy Rules Procedures attempted
SMFCC PGW Procedures Statistics	smfcc_pgw_process_pcf_update_policy_rules_successes	Counter	Process Pcf Update Policy Rules Procedures successful
SMFCC PGW Procedures Statistics	smfcc_pgw_process_pcf_update_policy_rules_failures	Counter	Process Pcf Update Policy Rules Procedures failed
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_process_pcf_update_policy_rules_attempts	Counter	Process Pcf Update Policy Rules Procedures attempted for roaming sessions
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_process_pcf_update_policy_rules_successes	Counter	Process Pcf Update Policy Rules Procedures successful for roaming sessions
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_process_pcf_update_policy_rules_failures	Counter	Process Pcf Update Policy Rules Procedures failed for roaming sessions
SMFCC PGW Procedures Statistics	smfcc_pgw_bearer_resource_command_unknown_ebi_attempts	Counter	Bearer Resource Command Procedures attempted for Unknown ebi

Group	Metric Name	Type	Synopsis
SMFCC PGW Procedures Statistics	smfcc_pgw_bearer_resource_command_unknown_ebi_successes	Counter	Bearer Resource Command Procedures successful for Unknown ebi
SMFCC PGW Procedures Statistics	smfcc_pgw_bearer_resource_command_unknown_ebi_failures	Counter	Bearer Resource Command Procedures failed for Unknown ebi
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_bearer_resource_command_unknown_ebi_attempts	Counter	Bearer Resource Command Procedures attempted for Unknown ebi in roaming sessions
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_bearer_resource_command_unknown_ebi_successes	Counter	Bearer Resource Command Procedures successful for Unknown ebi
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_bearer_resource_command_unknown_ebi_failures	Counter	Bearer Resource Command Procedures failed for Unknown ebi in roaming sessions
SMFCC PGW Procedures Statistics	smfcc_pgw_bearer_resource_command_attempts	Counter	Bearer Resource Command Procedures attempted
SMFCC PGW Procedures Statistics	smfcc_pgw_bearer_resource_command_successes	Counter	Bearer Resource Command Procedures successful
SMFCC PGW Procedures Statistics	smfcc_pgw_bearer_resource_command_failures	Counter	Bearer Resource Command Procedures failed
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_bearer_resource_command_attempts	Counter	Bearer Resource Command Procedures attempted for roaming sessions
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_bearer_resource_command_successes	Counter	Bearer Resource Command Procedures successful for roaming sessions
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_bearer_resource_command_failures	Counter	Bearer Resource Command Procedures failed for roaming sessions
SMFCC PGW Procedures Statistics	smfcc_pgw_mme_initiated_delete_bearer_command_attempts	Counter	MME Initiated Delete Bearer Command Procedures attempted
SMFCC PGW Procedures Statistics	smfcc_pgw_mme_initiated_delete_bearer_command_successes	Counter	MME Initiated Delete Bearer Command Procedures successful
SMFCC PGW Procedures Statistics	smfcc_pgw_mme_initiated_delete_bearer_command_failures	Counter	MME Initiate Delete Bearer Command Procedures failed
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_mme_initiated_delete_bearer_command_attempts	Counter	MME Initiated Delete Bearer Command Procedures attempted for roaming sessions
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_mme_initiated_delete_bearer_command_successes	Counter	MME Initiated Delete Bearer Command Procedures successful for roaming sessions
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_mme_initiated_delete_bearer_command_failures	Counter	MME Initiate Delete Bearer Command Procedures failed for roaming sessions
SMFCC PGW Procedures Statistics	smfcc_pgw_error_indication_procedure_attempts	Counter	Error Indication Procedures attempted
SMFCC PGW Procedures Statistics	smfcc_pgw_error_indication_procedure_successes	Counter	Error Indication Procedures successful
SMFCC PGW Procedures Statistics	smfcc_pgw_error_indication_procedure_failures	Counter	Error Indication Procedures failed

Group	Metric Name	Type	Synopsis
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_error_indication_procedure_attempts	Counter	Error Indication Procedures attempted for roaming sessions
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_error_indication_procedure_successes	Counter	Error Indication Procedures successful for roaming sessions
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_error_indication_procedure_failures	Counter	Error Indication Procedures failed for roaming sessions
SMFCC PGW Procedures Statistics	smfcc_pgw_qos_flow_inactivity_procedure_attempts	Counter	Pgw QosFlow Inactivity Procedures attempted
SMFCC PGW Procedures Statistics	smfcc_pgw_qos_flow_inactivity_procedure_successes	Counter	Pgw QosFlow Inactivity Procedures successful
SMFCC PGW Procedures Statistics	smfcc_pgw_qos_flow_inactivity_procedure_failures	Counter	Pgw QosFlow Inactivity Procedures failed
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_qos_flow_inactivity_procedure_attempts	Counter	Pgw QosFlow Inactivity Procedures attempted for roaming sessions
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_qos_flow_inactivity_procedure_successes	Counter	Pgw QosFlow Inactivity Procedures successful for roaming sessions
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_qos_flow_inactivity_procedure_failures	Counter	Pgw QosFlow Inactivity Procedures failed for roaming sessions
SMFCC PGW Procedures Statistics	smfcc_pgw_session_idle_procedure_attempts	Counter	Pgw Session Idle Procedures attempted
SMFCC PGW Procedures Statistics	smfcc_pgw_session_idle_procedure_successes	Counter	Pgw Session Idle Procedures successful
SMFCC PGW Procedures Statistics	smfcc_pgw_session_idle_procedure_failures	Counter	Pgw Session Idle Procedures failed
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_idle_procedure_attempts	Counter	Pgw Session Idle Procedures attempted for roaming sessions
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_idle_procedure_successes	Counter	Pgw Session Idle Procedures successful for roaming sessions
SMFCC PGW Procedures Statistics	smfcc_pgw_roaming_session_idle_procedure_failures	Counter	Pgw Session Idle Procedures failed for roaming sessions
SMFCC PGW Procedures Statistics	smfcc_abort_gtpc_retransmission_during_session_deletion_attempts	Counter	Pgw Abort Gtpc Procedures attempted
SMFCC PGW Procedures Statistics	smfcc_abort_gtpc_retransmission_during_session_deletion_successes	Counter	Pgw Abort Gtpc Procedures successful
SMFCC PGW Procedures Statistics	smfcc_abort_gtpc_retransmission_during_session_deletion_failures	Counter	Pgw Abort Gtpc Procedures failed
SMFCC PGW Procedures Statistics	smfcc_pgw_restore_pdn_connection_attempts	Counter	Pgw restore Pdn Connection Procedures attempted
SMFCC PGW Procedures Statistics	smfcc_pgw_restore_pdn_connection_successes	Counter	Pgw restore Pdn Connection Procedures successful
SMFCC PGW Procedures Statistics	smfcc_pgw_restore_pdn_connection_failures	Counter	Pgw restore Pdn Connection Procedures failed
SMFCC PGW Procedures Statistics	smfcc_pgw_interplmn_handover_with_subscriber_type_change_attempts	Counter	Pgw inter plmn handover with subscriber type change attempts

Group	Metric Name	Type	Synopsis
SMFCC PGW Procedures Statistics	smfcc_pgw_interplmn_handover_with_subscriber_type_change_successes	Counter	Pgw inter plmn handover with subscriber type change successes
SMFCC PGW Procedures Statistics	smfcc_pgw_interplmn_handover_with_subscriber_type_change_failures	Counter	Pgw inter plmn handover with subscriber type change failures
SMFCC SGW Procedures Statistics	smfcc_sgw_initial_attach_attempts	Counter	PGW Initial Attach Procedures attempted
SMFCC SGW Procedures Statistics	smfcc_sgw_initial_attach_successes	Counter	SGW Initial Attach Procedures successful
SMFCC SGW Procedures Statistics	smfcc_sgw_initial_attach_failures	Counter	SGW Initial Attach Procedures failed
SMFCC SGW Procedures Statistics	smfcc_sgw_ue_initiated_detach_attempts	Counter	SGW Delete Session Procedures attempted
SMFCC SGW Procedures Statistics	smfcc_sgw_ue_initiated_detach_successes	Counter	SGW Delete Session Procedures successful
SMFCC SGW Procedures Statistics	smfcc_sgw_ue_initiated_detach_failures	Counter	SGW Delete Session Procedures failed
SMFCC SGW Procedures Statistics	smfcc_sgw_nw_initiated_release_attempts	Counter	NW Initiated Release Procedures attempted
SMFCC SGW Procedures Statistics	smfcc_sgw_nw_initiated_release_successes	Counter	NW Initiated Release Procedures successful
SMFCC SGW Procedures Statistics	smfcc_sgw_nw_initiated_release_failures	Counter	NW Initiated Release Procedures failed
SMFCC SGW Procedures Statistics	smfcc_sgw_nw_initiated_dedicated_bearer_activation_attempts	Counter	NW Initiated Dedicated Bearer Activation Procedures attempted
SMFCC SGW Procedures Statistics	smfcc_sgw_nw_initiated_dedicated_bearer_activation_successes	Counter	NW Initiated Bearer Activation Procedures successful
SMFCC SGW Procedures Statistics	smfcc_sgw_nw_initiated_dedicated_bearer_activation_failures	Counter	NW Initiated Dedicated Bearer Activation Procedures failed
SMFCC SGW Procedures Statistics	smfcc_sgw_modify_bearer_request_attempts	Counter	Modify Bearer Request Procedures attempted
SMFCC SGW Procedures Statistics	smfcc_sgw_modify_bearer_request_successes	Counter	Modify Bearer Request Procedures successful
SMFCC SGW Procedures Statistics	smfcc_sgw_modify_bearer_request_failures	Counter	Modify Bearer Request Procedures failed
SMFCC SGW Procedures Statistics	smfcc_sgw_modify_bearer_ho_attempts	Counter	Modify Bearer Request Handover Procedures attempted
SMFCC SGW Procedures Statistics	smfcc_sgw_modify_bearer_ho_success	Counter	Modify Bearer Request Handover Procedures successful
SMFCC SGW Procedures Statistics	smfcc_sgw_modify_bearer_ho_failure	Counter	Modify Bearer Request Handover Procedures failed
SMFCC SGW Procedures Statistics	smfcc_sgw_delete_bearer_request_attempts	Counter	Delete Bearer Request Procedures attempted
SMFCC SGW Procedures Statistics	smfcc_sgw_delete_bearer_request_successes	Counter	Delete Bearer Request Procedures successful
SMFCC SGW Procedures Statistics	smfcc_sgw_delete_bearer_request_failures	Counter	Delete Bearer Request Procedures failed
SMFCC SGW Procedures Statistics	smfcc_sgw_create_bearer_request_attempts	Counter	Create Bearer Request Procedures attempted

Group	Metric Name	Type	Synopsis
SMFCC SGW Procedures Statistics	smfcc_sgw_create_bearer_request_successes	Counter	Create Bearer Request Procedures successful
SMFCC SGW Procedures Statistics	smfcc_sgw_create_bearer_request_failures	Counter	Create Bearer Request Procedures failed
SMFCC SGW Procedures Statistics	smfcc_sgw_update_bearer_request_attempts	Counter	Update Bearer Request Procedures attempted
SMFCC SGW Procedures Statistics	smfcc_sgw_update_bearer_request_successes	Counter	Update Bearer Request Procedures successful
SMFCC SGW Procedures Statistics	smfcc_sgw_update_bearer_request_failures	Counter	Update Bearer Request Procedures failed
SMFCC SGW Procedures Statistics	smfcc_sgw_hss_initiated_modify_bearer_command_attempts	Counter	HSS Initiated Modify Bearer Command Procedures attempted
SMFCC SGW Procedures Statistics	smfcc_sgw_hss_initiated_modify_bearer_command_successes	Counter	HSS Initiated Modify Bearer Command Procedures successful
SMFCC SGW Procedures Statistics	smfcc_sgw_hss_initiated_modify_bearer_command_failures	Counter	HSS Initiated Modify Bearer Command Procedures failed
SMFCC SGW Procedures Statistics	smfcc_sgw_nw_initiated_dedicated_bearer_delete_attempts	Counter	NW Initiated Dedicated Bearer Delete Procedures attempted
SMFCC SGW Procedures Statistics	smfcc_sgw_nw_initiated_dedicated_bearer_delete_successes	Counter	NW Initiated Bearer Delete Procedures successful
SMFCC SGW Procedures Statistics	smfcc_sgw_nw_initiated_dedicated_bearer_delete_failures	Counter	NW Initiated Dedicated Bearer Delete Procedures failed
SMFCC SGW Procedures Statistics	smfcc_sgw_nw_initiated_dedicated_bearer_update_attempts	Counter	NW Initiated Dedicated Bearer Update Procedures attempted
SMFCC SGW Procedures Statistics	smfcc_sgw_nw_initiated_dedicated_bearer_update_successes	Counter	NW Initiated Bearer Update Procedures successful
SMFCC SGW Procedures Statistics	smfcc_sgw_nw_initiated_dedicated_bearer_update_failures	Counter	NW Initiated Dedicated Bearer Update Procedures failed
SMFCC SGW Procedures Statistics	smfcc_sgw_delete_bearer_command_attempts	Counter	Delete Bearer Command Procedures attempted
SMFCC SGW Procedures Statistics	smfcc_sgw_delete_bearer_command_failures	Counter	Delete Bearer Command Procedures failed
SMFCC SGW Procedures Statistics	smfcc_sgw_release_access_bearer_request_attempts	Counter	Release Access Bearer Request Procedures attempted
SMFCC SGW Procedures Statistics	smfcc_sgw_release_access_bearer_request_successes	Counter	Release Access Bearer Request Procedures successful
SMFCC SGW Procedures Statistics	smfcc_sgw_release_access_bearer_request_failures	Counter	Release Access Bearer Request Procedures failed
SMFCC SGW Procedures Statistics	smfcc_sgw_ue_initiated_service_request_attempts	Counter	SGW UE initiated service request Procedures attempted
SMFCC SGW Procedures Statistics	smfcc_sgw_ue_initiated_service_request_successes	Counter	SGW UE initiated service request Procedures successful
SMFCC SGW Procedures Statistics	smfcc_sgw_ue_initiated_service_request_failures	Counter	SGW UE initiated service request Procedures failed

Group	Metric Name	Type	Synopsis
SMFCC SGW Procedures Statistics	smfcc_sgw_nw_initiated_service_request_attempts	Counter	SGW UE initiated service request Procedures attempted
SMFCC SGW Procedures Statistics	smfcc_sgw_nw_initiated_service_request_successes	Counter	SGW Network initiated service request Procedures successful
SMFCC SGW Procedures Statistics	smfcc_sgw_nw_initiated_service_request_failures	Counter	SGW Network initiated service request Procedures failed
SMFCC SGW Procedures Statistics	smfcc_sgw_nw_network_initiated_service_request_timeouts	Counter	SGW Network Initiated Service Request Procedures timeout
SMFCC SGW Procedures Statistics	smfcc_sgw_nw_initiated_service_request_aborted	Counter	SGW Network initiated service request Procedures aborted due to higher arp pl DDN received from upf
SMFCC SGW Procedures Statistics	smfcc_sgw_nw_initiated_service_request_dropped	Counter	SGW Network initiated service request Procedures dropped due to higher arp pl DDN already sent
SMFCC SGW Procedures Statistics	smfcc_sgw_csr_tau_rau_x2ho_attempts	Counter	SGW TAU/RAU/X2-HO with SGW Change Procedures attempted
SMFCC SGW Procedures Statistics	smfcc_sgw_csr_tau_rau_x2ho_success	Counter	SGW TAU/RAU/X2-HO with SGW Change Procedures successful
SMFCC SGW Procedures Statistics	smfcc_sgw_csr_tau_rau_x2ho_failures	Counter	SGW TAU/RAU/X2-HO with SGW Change Procedures failed
SMFCC SGW Procedures Statistics	smfcc_sgw_csr_s1ho_attempts	Counter	SGW S1-HO with SGW Change Procedures attempted
SMFCC SGW Procedures Statistics	smfcc_sgw_csr_s1ho_failures	Counter	SGW S1-HO with SGW Change Procedures failed
SMFCC SGW Procedures Statistics	smfcc_sgw_dedicated_delete_bearer_request_attempts	Counter	Dedicated Delete Bearer Request Procedures attempted
SMFCC SGW Procedures Statistics	smfcc_sgw_dedicated_delete_bearer_request_failures	Counter	Dedicated Delete Bearer Request Procedures failed
SMFCC SGW Procedures Statistics	smfcc_sgw_error_indication_attempts	Counter	SGW Error Indication Procedures attempted
SMFCC SGW Procedures Statistics	smfcc_sgw_error_indication_failures	Counter	SGW Error Indication Procedures failed
SMFCC SGW Procedures Statistics	smfcc_sgw_s1u_error_indication_attempts	Counter	S1U SGW Error Indication Procedures attempted
SMFCC SGW Procedures Statistics	smfcc_sgw_s1u_error_indication_failures	Counter	S1U SGW Error Indication Procedures failed
SMFCC SGW Procedures Statistics	smfcc_sgw_s5s8u_default_bearer_error_indication_attempts	Counter	S5S8U Default Bearer SGW Error Indication Procedures attempted
SMFCC SGW Procedures Statistics	smfcc_sgw_s5s8u_default_bearer_error_indication_failures	Counter	S5S8U Default Bearer SGW Error Indication Procedures failed
SMFCC SGW Procedures Statistics	smfcc_sgw_s5s8u_dedicated_bearer_error_indication_attempts	Counter	S5S8U Dedicated Bearer SGW Error Indication Procedures attempted

Group	Metric Name	Type	Synopsis
SMFCC SGW Procedures Statistics	smfcc_sgw_s5s8u_dedicated_bearer_error_indication_failures	Counter	S5S8U Dedicated Bearer SGW Error Indication Procedures failed
SMFCC SAEGW Procedures Statistics	smfcc_saegw_initial_attach_attempts	Counter	SAEGW Initial Attach Procedures attempted
SMFCC SAEGW Procedures Statistics	smfcc_saegw_initial_attach_successes	Counter	SAEGW Initial Attach Procedures successful
SMFCC SAEGW Procedures Statistics	smfcc_saegw_initial_attach_failures	Counter	SAEGW Initial Attach Procedures failed
SMFCC SAEGW Procedures Statistics	smfcc_saegw_ue_initiated_detach_attempts	Counter	SAEGW Delete Session Procedures attempted
SMFCC SAEGW Procedures Statistics	smfcc_saegw_ue_initiated_detach_successes	Counter	SAEGW Delete Session Procedures successful
SMFCC SAEGW Procedures Statistics	smfcc_saegw_ue_initiated_detach_failures	Counter	SAEGW Delete Session Procedures failed
SMFCC SAEGW Procedures Statistics	smfcc_saegw_modify_bearer_request_attempts	Counter	Modify Bearer Request Procedures attempted
SMFCC SAEGW Procedures Statistics	smfcc_saegw_modify_bearer_request_successes	Counter	Modify Bearer Request Procedures successful
SMFCC SAEGW Procedures Statistics	smfcc_saegw_modify_bearer_request_failures	Counter	Modify Bearer Request Procedures failed
SMFCC SAEGW Procedures Statistics	smfcc_saegw_pcf_initiated_terminate_attempts	Counter	SAEGW PCF Initiated Terminate Procedures attempted
SMFCC SAEGW Procedures Statistics	smfcc_saegw_pcf_initiated_terminate_successes	Counter	SAEGW PCF Initiated Terminate Procedures successful
SMFCC SAEGW Procedures Statistics	smfcc_saegw_pcf_initiated_terminate_failures	Counter	SAEGW PCF Initiated Terminate failed
SMFCC SAEGW Procedures Statistics	smfcc_saegw_nw_initiated_release_attempts	Counter	SAEGW NW Initiated Release Procedures attempted
SMFCC SAEGW Procedures Statistics	smfcc_saegw_nw_initiated_release_successes	Counter	SAEGW NW Initiated Release Procedures successful
SMFCC SAEGW Procedures Statistics	smfcc_saegw_nw_initiated_release_failures	Counter	SAEGW NW Initiated Release Procedures failed
SMFCC SAEGW Procedures Statistics	smfcc_saegw_pcf_initiated_policy_modification_attempts	Counter	PCF Initiated Policy Modification Procedures attempted for SAEGW
SMFCC SAEGW Procedures Statistics	smfcc_saegw_pcf_initiated_policy_modification_successes	Counter	PCF Initiated Policy Modification Procedures successful for SAEGW
SMFCC SAEGW Procedures Statistics	smfcc_saegw_pcf_initiated_policy_modification_failures	Counter	PCF Initiated Policy Modification Procedures failed for SAEGW
SMFCC SAEGW Procedures Statistics	smfcc_saegw_pcf_initiated_dedicated_bearer_activation_attempts	Counter	PCF Initiated Dedicated Bearer Activation Procedures attempted for SAEGW
SMFCC SAEGW Procedures Statistics	smfcc_saegw_pcf_initiated_dedicated_bearer_activation_successes	Counter	PCF Initiated Bearer Activation Procedures successful for SAEGW
SMFCC SAEGW Procedures Statistics	smfcc_saegw_pcf_initiated_dedicated_bearer_activation_failures	Counter	PCF Initiated Dedicated Bearer Activation Procedures failed for SAEGW

Group	Metric Name	Type	Synopsis
SMFCC SAEGW Procedures Statistics	smfcc_saegw_pcf_initiated_dedicated_bearer_delete_attempts	Counter	PCF Initiated Dedicated Bearer Delete Procedures attempted for SAEGW
SMFCC SAEGW Procedures Statistics	smfcc_saegw_pcf_initiated_dedicated_bearer_delete_successes	Counter	PCF Initiated Bearer Delete Procedures successful for SAEGW
SMFCC SAEGW Procedures Statistics	smfcc_saegw_pcf_initiated_dedicated_bearer_delete_failures	Counter	PCF Initiated Dedicated Bearer Delete Procedures failed for SAEGW
SMFCC SAEGW Procedures Statistics	smfcc_saegw_pcf_initiated_dedicated_bearer_update_attempts	Counter	PCF Initiated Dedicated Bearer Update Procedures attempted for SAEGW
SMFCC SAEGW Procedures Statistics	smfcc_saegw_pcf_initiated_dedicated_bearer_update_successes	Counter	PCF Initiated Bearer Update Procedures successful for SAEGW
SMFCC SAEGW Procedures Statistics	smfcc_saegw_pcf_initiated_dedicated_bearer_update_failures	Counter	PCF Initiated Dedicated Bearer Update Procedures failed for SAEGW
SMFCC SAEGW Procedures Statistics	smfcc_saegw_process_pcf_update_policy_rules_attempts	Counter	Process Pcf Update Policy Rules Procedures attempted for SAEGW
SMFCC SAEGW Procedures Statistics	smfcc_saegw_process_pcf_update_policy_rules_successes	Counter	Process Pcf Update Policy Rules Procedures successful for SAEGW
SMFCC SAEGW Procedures Statistics	smfcc_saegw_process_pcf_update_policy_rules_failures	Counter	Process Pcf Update Policy Rules Procedures failed for SAEGW
SMFCC SAEGW Procedures Statistics	smfcc_saegw_policy_timeout_handling_attempts	Counter	Policy Timer Timeout Handling Procedures attempted for SAEGW
SMFCC SAEGW Procedures Statistics	smfcc_saegw_policy_timeout_handling_successes	Counter	Policy Timer Timeout Handling Procedures successful for SAEGW
SMFCC SAEGW Procedures Statistics	smfcc_saegw_policy_timeout_handling_failures	Counter	Policy Timer Timeout Handling Procedures failed for SAEGW
SMFCC SAEGW Procedures Statistics	smfcc_saegw_hss_initiated_modify_bearer_command_attempts	Counter	HSS Initiated Modify Bearer Command Procedures attempted for SAEGW
SMFCC SAEGW Procedures Statistics	smfcc_saegw_hss_initiated_modify_bearer_command_successes	Counter	HSS Initiated Modify Bearer Command Procedures successful for SAEGW
SMFCC SAEGW Procedures Statistics	smfcc_saegw_hss_initiated_modify_bearer_command_failures	Counter	HSS Initiated Modify Bearer Command Procedures failed for SAEGW
SMFCC SAEGW Procedures Statistics	smfcc_saegw_error_indication_procedure_attempts	Counter	Error Indication Procedures attempted for SAEGW
SMFCC SAEGW Procedures Statistics	smfcc_saegw_error_indication_procedure_successes	Counter	Error Indication Procedures successful for SAEGW
SMFCC SAEGW Procedures Statistics	smfcc_saegw_error_indication_procedure_failures	Counter	Error Indication Procedures failed for SAEGW
SMFCC SAEGW Procedures Statistics	smfcc_saegw_session_idle_procedure_attempts	Counter	Session Idle Procedures attempted for SAEGW
SMFCC SAEGW Procedures Statistics	smfcc_saegw_session_idle_procedure_successes	Counter	Session Idle Procedures successful for SAEGW
SMFCC SAEGW Procedures Statistics	smfcc_saegw_session_idle_procedure_failures	Counter	Session Idle Procedures failed for SAEGW

Group	Metric Name	Type	Synopsis
SMFCC SAEGW Procedures Statistics	smfcc_saegw_qos_flow_inactivity_procedure_attempts	Counter	QoSFlow Inactivity Procedures attempted for SAEGW
SMFCC SAEGW Procedures Statistics	smfcc_saegw_qos_flow_inactivity_procedure_successes	Counter	QoSFlow Inactivity Procedures successful for SAEGW
SMFCC SAEGW Procedures Statistics	smfcc_saegw_qos_flow_inactivity_procedure_failures	Counter	QoSFlow Inactivity Procedures failed for SAEGW
SMFCC SAEGW Procedures Statistics	smfcc_saegw_bearer_resource_command_attempts	Counter	Bearer Resource Command Procedures attempted for SAEGW
SMFCC SAEGW Procedures Statistics	smfcc_saegw_bearer_resource_command_successes	Counter	Bearer Resource Command Procedures successful for SAEGW
SMFCC SAEGW Procedures Statistics	smfcc_saegw_bearer_resource_command_failures	Counter	Bearer Resource Command Procedures failed for SAEGW
SMFCC SAEGW Procedures Statistics	smfcc_saegw_bearer_resource_command_unknown_ebi_attempts	Counter	Bearer Resource Command Procedures attempted for Unknown ebi for SAEGW
SMFCC SAEGW Procedures Statistics	smfcc_saegw_bearer_resource_command_unknown_ebi_successes	Counter	Bearer Resource Command Procedures successful for Unknown ebi for SAEGW
SMFCC SAEGW Procedures Statistics	smfcc_saegw_bearer_resource_command_unknown_ebi_failures	Counter	Bearer Resource Command Procedures failed for Unknown ebi for SAEGW
SMFCC SAEGW Procedures Statistics	smfcc_saegw_modify_bearer_ho_attempts	Counter	Modify Bearer Request Handover Procedures attempted for SAEGW
SMFCC SAEGW Procedures Statistics	smfcc_saegw_modify_bearer_ho_successes	Counter	Modify Bearer Request X2 Handover Procedures successful for SAEGW
SMFCC SAEGW Procedures Statistics	smfcc_saegw_modify_bearer_ho_failures	Counter	Modify Bearer Request X2 Handover Procedures failed for SAEGW
SMFCC SAEGW Procedures Statistics	smfcc_saegw_csr_tau_rau_x2ho_attempts	Counter	SAEGW TAU/RAU/X2-HO with SGW Change Procedures attempted
SMFCC SAEGW Procedures Statistics	smfcc_saegw_csr_tau_rau_x2ho_success	Counter	SAEGW TAU/RAU/X2-HO with SGW Change Procedures successful
SMFCC SAEGW Procedures Statistics	smfcc_saegw_csr_tau_rau_x2ho_failures	Counter	SAEGW TAU/RAU/X2-HO with SGW Change Procedures failed
SMFCC SAEGW Procedures Statistics	smfcc_saegw_csr_s1ho_attempts	Counter	SAEGW S1-HO with SGW Change Procedures attempted
SMFCC SAEGW Procedures Statistics	smfcc_saegw_csr_s1ho_failures	Counter	SAEGW S1-HO with SGW Change Procedures failed
SMFCC SAEGW Procedures Statistics	smfcc_saegw_to_pgw_modify_bearer_request_ho_attempts	Counter	SAEGW to PGW Modify Bearer Request Procedures attempted
SMFCC SAEGW Procedures Statistics	smfcc_saegw_to_pgw_modify_bearer_request_ho_failures	Counter	SAEGW to PGW Modify Bearer Request Procedures failed

Group	Metric Name	Type	Synopsis
SMFCC SAEGW Procedures Statistics	smfcc_saegw_to_pgw_ho_delete_session_request_attempts	Counter	SAEGW to PGW HO Delete Session Request Procedures attempted
SMFCC SAEGW Procedures Statistics	smfcc_saegw_to_pgw_ho_delete_session_request_failures	Counter	SAEGW to PGW HO Delete Session Request Procedures failed
SMFCC SAEGW Procedures Statistics	smfcc_saegw_release_access_bearer_request_attempts	Counter	SAEGW Release Access Bearer Request Procedures attempted
SMFCC SAEGW Procedures Statistics	smfcc_saegw_release_access_bearer_request_failures	Counter	SAEGW Release Access Bearer Request Procedures failed
SMFCC SAEGW Procedures Statistics	smfcc_saegw_nw_initiated_service_request_attempts	Counter	SAEGW network initiated service request Procedures attempted
SMFCC SAEGW Procedures Statistics	smfcc_saegw_nw_initiated_service_request_failures	Counter	SAEGW Network initiated service request Procedures failed
SMFCC SAEGW Procedures Statistics	smfcc_saegw_nw_initiated_service_request_aborted	Counter	SAEGW Network initiated service request Procedures aborted due to higher arp pl DDN received from upf
SMFCC SAEGW Procedures Statistics	smfcc_saegw_nw_initiated_service_request_dropped	Counter	SAEGW Network initiated service request Procedures dropped due to higher arp pl DDN already sent/max ddns reached
SMFCC SAEGW Procedures Statistics	smfcc_saegw_ue_initiated_service_request_attempts	Counter	SAEGW UE initiated service request Procedures attempted
SMFCC SAEGW Procedures Statistics	smfcc_saegw_ue_initiated_service_request_failures	Counter	SAEGW UE initiated service request Procedures failed
SMFCC SAEGW Procedures Statistics	smfcc_saegw_nw_network_initiated_service_request_timeouts	Counter	SAEGW Network Initiated Service Request Procedures timeout
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_create_pdp_context_attempts	Counter	GGSN Create PDP Context Procedures attempted
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_create_pdp_context_successes	Counter	GGSN Create PDP Context Procedures successful
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_create_pdp_context_failures	Counter	GGSN Create PDP Context Procedures failed
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_roaming_session_create_pdp_context_attempts	Counter	GGSN Roaming Create PDP Context Procedures attempted
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_roaming_session_create_pdp_context_successes	Counter	GGSN Roaming Create PDP Context Procedures successful
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_roaming_session_create_pdp_context_failures	Counter	GGSN Roaming Create PDP Context Procedures failed
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_sgsn_initiated_update_pdp_context_attempts	Counter	GGSN Update PDP Context Procedures attempted
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_sgsn_initiated_update_pdp_context_successes	Counter	GGSN Update PDP Context Procedures successful
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_sgsn_initiated_update_pdp_context_failures	Counter	GGSN Update PDP Context Procedures failed

Group	Metric Name	Type	Synopsis
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_roaming_session_sgsn_initiated_update_pdp_context_attempts	Counter	GGSN Roaming Update PDP Context Procedures attempted
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_roaming_session_sgsn_initiated_update_pdp_context_successes	Counter	GGSN Roaming Update PDP Context Procedures successful
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_roaming_session_sgsn_initiated_update_pdp_context_failures	Counter	GGSN Roaming Update PDP Context Procedures failed
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_sgsn_initiated_delete_pdp_context_attempts	Counter	GGSN delete PDP Context Procedures attempted
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_sgsn_initiated_delete_pdp_context_successes	Counter	GGSN delete PDP Context Procedures successful
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_sgsn_initiated_delete_pdp_context_failures	Counter	GGSN delete PDP Context Procedures failed
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_sgsn_initiated_roaming_session_delete_pdp_context_attempts	Counter	GGSN delete roaming PDP Context Procedures attempted
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_sgsn_initiated_roaming_session_delete_pdp_context_successes	Counter	GGSN delete roaming PDP Context Procedures successful
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_sgsn_initiated_roaming_session_delete_pdp_context_failures	Counter	GGSN delete roaming PDP Context Procedures failed
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_nw_initiated_release_attempts	Counter	GGSN NW Initiated Release Procedures attempted
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_nw_initiated_release_successes	Counter	GGSN NW Initiated Release Procedures successful
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_nw_initiated_release_failures	Counter	GGSN NW Initiated Release Procedures failed
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_roaming_session_nw_initiated_release_attempts	Counter	GGSN NW Initiated Release Procedures attempted for roaming sessions
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_roaming_session_nw_initiated_release_successes	Counter	GGSN NW Initiated Release Procedures successful for roaming sessions
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_roaming_session_nw_initiated_release_failures	Counter	GGSN NW Initiated Release Procedures failed for roaming sessions
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_pcf_initiated_terminate_attempts	Counter	GGSN PCF Initiated Terminate Procedures attempted
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_pcf_initiated_terminate_successes	Counter	GGSN PCF Initiated Terminate Procedures successful
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_pcf_initiated_terminate_failures	Counter	GGSN PCF Initiated Terminate failed
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_roaming_session_pcf_initiated_terminate_attempts	Counter	GGSN PCF Initiated Terminate Procedures attempted for roaming sessions
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_roaming_session_pcf_initiated_terminate_successes	Counter	GGSN PCF Initiated Terminate Procedures successful
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_roaming_session_pcf_initiated_terminate_failures	Counter	GGSN PCF Initiated Terminate failed for roaming sessions

Group	Metric Name	Type	Synopsis
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_pcf_initiated_policy_modification_attempts	Counter	PCF Initiated (GGSN Initiated) Policy Modification Procedures attempted
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_pcf_initiated_policy_modification_failures	Counter	PCF Initiated (GGSN Initiated) Policy Modification Procedures failed
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_pcf_initiated_policy_modification_successes	Counter	PCF Initiated (GGSN Initiated) Policy Modification Procedures successful
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_roaming_session_pcf_initiated_policy_modification_attempts	Counter	PCF Initiated (GGSN Initiated) Policy Modification Procedures attempted for roaming sessions
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_roaming_session_pcf_initiated_policy_modification_successes	Counter	PCF Initiated (GGSN Initiated) Policy Modification Procedures successful
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_roaming_session_pcf_initiated_policy_modification_failures	Counter	PCF Initiated (GGSN Initiated) Policy Modification Procedures failed for roaming sessions
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_3g_to_4g_handover_attempts	Counter	2G/3G to 4G Handover Procedures attempted
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_3g_to_4g_handover_successes	Counter	2G/3G to 4G Handover Procedures successful
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_3g_to_4g_handover_failures	Counter	2G/3G to 4G Handover Procedures failed
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_4g_to_3g_handover_attempts	Counter	4G to 2G/3G Handover Procedures attempted
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_4g_to_3g_handover_successes	Counter	4G to 2G/3G Handover Procedures successful
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_4g_to_3g_handover_failures	Counter	4G to 2G/3G Handover Procedures failed
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_roaming_session_3g_to_4g_handover_attempts	Counter	2G/3G to 4G Handover Procedures attempted
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_roaming_session_3g_to_4g_handover_successes	Counter	2G/3G to 4G Handover Procedures successful
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_roaming_session_3g_to_4g_handover_failures	Counter	2G/3G to 4G Handover Procedures failed
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_roaming_session_4g_to_3g_handover_attempts	Counter	4G to 2G/3G Handover Procedures attempted
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_roaming_session_4g_to_3g_handover_successes	Counter	4G to 2G/3G Handover Procedures successful
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_roaming_session_4g_to_3g_handover_failures	Counter	4G to 2G/3G Handover Procedures failed
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_inter_sgsn_handover_attempts	Counter	Update PDP Context Request Procedures with SGSN change attempted
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_inter_sgsn_handover_successes	Counter	Update PDP Context Request Procedures with SGSN change successful

Group	Metric Name	Type	Synopsis
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_inter_sgsn_handover_failures	Counter	Update PDP Context Request Procedures with SGSN change failed
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_roaming_session_inter_sgsn_handover_attempts	Counter	Update PDP Context Request Procedures with SGSN change attempted for roaming sessions
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_roaming_session_inter_sgsn_handover_successes	Counter	Update PDP Context Request Procedures with SGSN change successful for roaming sessions
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_roaming_session_inter_sgsn_handover_failures	Counter	Update PDP Context Request Procedures with SGSN change failed for roaming sessions
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_error_indication_attempts	Counter	GGSN-C Error Indication Procedures attempted
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_error_indication_successes	Counter	GGSN-C Error Indication Procedures successful
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_error_indication_failures	Counter	GGSN-C Error Indication Procedures failed
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_roaming_session_error_indication_attempts	Counter	GGSN-C Error Indication Procedures attempted for roaming sessions
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_roaming_session_error_indication_successes	Counter	GGSN-C Error Indication Procedures successful for roaming sessions
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_roaming_session_error_indication_failures	Counter	GGSN-C Error Indication Procedures failed for roaming sessions
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_pcf_initiated_bearer_update_attempts	Counter	PCF Initiated Bearer Update Procedures attempted
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_pcf_initiated_bearer_update_successes	Counter	PCF Initiated Bearer Update Procedures successful
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_pcf_initiated_bearer_update_failures	Counter	PCF Initiated Bearer Update Procedures failed
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_roaming_session_pcf_initiated_bearer_update_attempts	Counter	PCF Initiated Bearer Update Procedures attempted for roaming sessions
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_roaming_session_pcf_initiated_bearer_update_successes	Counter	PCF Initiated Bearer Update Procedures successful for roaming sessions
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_roaming_session_pcf_initiated_bearer_update_failures	Counter	PCF Initiated Bearer Update Procedures failed for roaming sessions
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_process_pcf_update_policy_rules_attempts	Counter	Process Pcf Update Policy Rules Procedures attempted
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_process_pcf_update_policy_rules_successes	Counter	Process Pcf Update Policy Rules Procedures successful
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_process_pcf_update_policy_rules_failures	Counter	Process Pcf Update Policy Rules Procedures failed
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_roaming_session_process_pcf_update_policy_rules_attempts	Counter	Process Pcf Update Policy Rules Procedures attempted for roaming sessions

Group	Metric Name	Type	Synopsis
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_roaming_session_process_pcf_update_policy_rules_successes	Counter	Process Pcf Update Policy Rules Procedures successful for roaming sessions
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_roaming_session_process_pcf_update_policy_rules_failures	Counter	Process Pcf Update Policy Rules Procedures failed for roaming sessions
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_session_idle_procedure_attempts	Counter	Ggsn Session Idle Procedures attempted
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_session_idle_procedure_successes	Counter	Ggsn Session Idle Procedures successful
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_session_idle_procedure_failures	Counter	Ggsn Session Idle Procedures failed
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_roaming_session_idle_procedure_attempts	Counter	Ggsn Session Idle Procedures attempted for roaming sessions
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_roaming_session_idle_procedure_successes	Counter	Ggsn Session Idle Procedures successful for roaming sessions
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_roaming_session_idle_procedure_failures	Counter	Ggsn Session Idle Procedures failed for roaming sessions
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_qos_flow_inactivity_procedure_attempts	Counter	Ggsn QosFlow Inactivity Procedures attempted
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_qos_flow_inactivity_procedure_successes	Counter	Ggsn QosFlow Inactivity Procedures successful
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_qos_flow_inactivity_procedure_failures	Counter	Ggsn QosFlow Inactivity Procedures failed
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_roaming_session_qos_flow_inactivity_procedure_attempts	Counter	Ggsn QosFlow Inactivity Procedures attempted for roaming sessions
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_roaming_session_qos_flow_inactivity_procedure_successes	Counter	Ggsn QosFlow Inactivity Procedures successful for roaming sessions
SMFCC GGSN-C Procedure Statistics	smfcc_ggsn_roaming_session_qos_flow_inactivity_procedure_failures	Counter	Ggsn QosFlow Inactivity Procedures failed for roaming sessions
SMF Call Performance Statistics	smfcc_pcscf_sessions_activated	Counter	Sessions activated using P-CSCF address
SMF Call Performance Statistics	smfcc_pcscf_sessions_deactivated	Counter	Sessions deactivated using P-CSCF address
SMF Call Performance Statistics	smfcc_qos_flows_setup_success	Counter	Qos Flows succssfully setup
SMF Call Performance Statistics	smfcc_qos_flows_setup_failure	Counter	Qos Flows failed to setup
SMF Call Performance Statistics	smfcc_nw_init_qos_flows_released	Counter	Network initiated Qos flows released
SMF Call Performance Statistics	smfcc_access_init_qos_flows_released	Counter	Access initiated Qos flows released
SMF Call Performance Statistics	smfcc_abnormal_qos_flows_released	Counter	Qos flows released abnormally
SMF Call Performance Statistics	smfcc_pcc_rule_add_attempts	Counter	Total number of pcc rule add attempsts
SMF Call Performance Statistics	smfcc_pcc_rule_add_successes	Counter	Total number of pcc rule add attempsts successes
SMF Call Performance Statistics	smfcc_pcc_rule_add_failures	Counter	Total number of pcc rule add attempt failures

Group	Metric Name	Type	Synopsis
SMF Call Performance Statistics	smfcc_pcc_rule_modify_attempts	Counter	Total number of pcc rule modify attempts
SMF Call Performance Statistics	smfcc_pcc_rule_modify_successes	Counter	Total number of pcc rule modify attempts successes
SMF Call Performance Statistics	smfcc_pcc_rule_modify_failures	Counter	Total number of pcc rule modify attempt failures
SMF Call Performance Statistics	smfcc_pcc_rule_delete_attempts	Counter	Total number of pcc rule delete attempts
SMF Call Performance Statistics	smfcc_pcc_rule_delete_successes	Counter	Total number of pcc rule delete attempt successes
SMF Call Performance Statistics	smfcc_pcc_rule_delete_failures	Counter	Total number of pcc rule delete attempt failures
SMF Call Performance Statistics	smfcc_pcc_rule_install_drops	Counter	Total number of pcc rule installs dropped
SMF Call Performance Statistics	smfcc_session_rule_add_failures	Counter	Total number of session rule add attempt failures
SMF Call Performance Statistics	smfcc_session_rule_modify_failures	Counter	Total number of session rule modify attempt failures
SMF Call Performance Statistics	smfcc_session_rule_failures_reported_to_pcf	Counter	Total number of session rule failures reported to PCF
SMF Call Performance Statistics	smfcc_default_policy_used_pdu_session_count	Counter	Total Number of Sessions Created with Configured Default Policy
SMF Call Performance Statistics	smfcc_default_quota_used_pdu_session_count	Counter	Total Number of Sessions which Used Locally Configured Online Quota
SMF Call Performance Statistics	smfcc_dnn_matching_exact_match	Counter	Total Number of Sessions which used exact matched DNN that was configured
SMF Call Performance Statistics	smfcc_dnn_matching_ni_only	Counter	Total Number of Sessions which used NI Only DNN that was configured
SMF Call Performance Statistics	smfcc_dnn_matching_ni_and_malformed_oi	Counter	Total Number of Sessions which used NI and Malformed OI DNN that was configured
SMF Call Performance Statistics	smfcc_dnn_mapping_success	Counter	Total Number of successful DNN Mapping operations
SMF Call Performance Statistics	smfcc_dnn_mapping_failures	Counter	Total Number of failed DNN Mapping operations
SMF Call Performance Statistics	smfcc_pcf_response_reseller_subclass_info	Counter	Total Number of PCF Create Responses with Reseller and Subclass Information
SMF Call Performance Statistics	smfcc_db_read_lock_conflict_record_failures	Counter	Total Number of DB Read Lock failures with conflicted record error
SMF Call Performance Statistics	smfcc_db_write_conflict_record_failures	Counter	Total Number of DB Write Lock/Unlock failures with conflicted record error
SMF Call Performance Statistics	smfcc_db_create_conflict_record_failures	Counter	Total Number of DB Create failures with conflicted record error
SMF Call Performance Statistics	smfcc_db_get_conflict_record_failures	Counter	Total Number of DB Get record failures with conflicted record error

Group	Metric Name	Type	Synopsis
SMF Call Performance Statistics	smfcc_pcc_rule_base_install_attempts	Counter	Total number of pcc rule base install attempts
SMF Call Performance Statistics	smfcc_pcc_rule_base_install_failures	Counter	Total number of pcc rule base install attempt failures
SMFCC PfcP Interface Statistics	smfcc_error_indication_reports	Counter	Total Error Indication Report Received from UPF
SMFCC PfcP Interface Statistics	smfcc_error_indication_report_successes	Counter	Total Success Response to UPF for Error Indication Report
SMFCC PfcP Interface Statistics	smfcc_error_indication_report_failures	Counter	Total Failure Response to UPF for Error Indication Report
SMFCC PfcP Interface Statistics	smfcc_usage_reports	Counter	Total Usage Report Received from UPF
SMFCC PfcP Interface Statistics	smfcc_usage_report_successes	Counter	Total Success Response to UPF for Usage Report
SMFCC PfcP Interface Statistics	smfcc_usage_report_failures	Counter	Total Failure Response to UPF for Usage Report
SMFCC PfcP Interface Statistics	smfcc_up_inactivity_reports	Counter	Total Up Inactivity Report Received from UPF for all sessions
SMFCC PfcP Interface Statistics	smfcc_up_inactivity_reports_5G_sessions	Counter	Total Up Inactivity Report Received from UP for 5G sessionsF
SMFCC PfcP Interface Statistics	smfcc_up_inactivity_reports_5G_always_on_sessions	Counter	Total Up Inactivity Report Received from UPF for 5G always on sessions
SMFCC PfcP Interface Statistics	smfcc_up_inactivity_reports_4G_sessions	Counter	Total Up Inactivity Report Received from UPF for 4G sessions
SMFCC PfcP Interface Statistics	smfcc_up_inactivity_reports_GGSN_sessions	Counter	Total Up Inactivity Report Received from UPF for 3G/2G sessions
SMFCC PfcP Interface Statistics	smfcc_up_activity_reports_5G_sessions	Counter	Total Up activity Report Received from UP for 5G sessionsF
SMFCC PfcP Interface Statistics	smfcc_up_activity_reports_5G_always_on_sessions	Counter	Total Up activity Report Received from UPF for 5G always on sessions
SMFCC PfcP Interface Statistics	smfcc_up_activity_reports_4G_sessions	Counter	Total Up activity Report Received from UPF for 4G sessions
SMFCC PfcP Interface Statistics	smfcc_up_activity_reports_GGSN_sessions	Counter	Total Up activity Report Received from UPF for 3G/2G sessions
SMFCC PfcP Interface Statistics	smfcc_up_inactivity_report_successes_5G_sessions	Counter	Total Success Response to UPF for 5G sessions Up Inactivity Report
SMFCC PfcP Interface Statistics	smfcc_up_inactivity_report_successes_5G_always_on_sessions	Counter	Total Success Response to UPF for 5G always on sessions Up Inactivity Report
SMFCC PfcP Interface Statistics	smfcc_up_inactivity_report_successes_4G_sessions	Counter	Total Success Response to UPF for 4G sessions Up Inactivity Report

Group	Metric Name	Type	Synopsis
SMFCC PfcP Interface Statistics	smfcc_up_inactivity_report_successes_GGSN_sessions	Counter	Total Success Response to UPF for 3G/2G sessions Up Inactivity Report
SMFCC PfcP Interface Statistics	smfcc_up_inactivity_report_failures_5G_sessions	Counter	Total failure Response to UPF for 5G sessions Up Inactivity Report
SMFCC PfcP Interface Statistics	smfcc_up_inactivity_report_failures_5G_always_on_sessions	Counter	Total failure Response to UPF for 5G always on sessions Up Inactivity Report
SMFCC PfcP Interface Statistics	smfcc_up_inactivity_report_failures_4G_sessions	Counter	Total failure Response to UPF for 4G sessions Up Inactivity Report
SMFCC PfcP Interface Statistics	smfcc_up_inactivity_report_failures_GGSN_sessions	Counter	Total failure Response to UPF for 3G/2G sessions Up Inactivity Report
SMFCC PfcP Interface Statistics	smfcc_up_activity_report_successes_5G_sessions	Counter	Total Success Response to UPF for 5G sessions Up activity Report
SMFCC PfcP Interface Statistics	smfcc_up_activity_report_successes_5G_always_on_sessions	Counter	Total Success Response to UPF for 5G always on sessions Up activity Report
SMFCC PfcP Interface Statistics	smfcc_up_activity_report_successes_4G_sessions	Counter	Total Success Response to UPF for 4G sessions Up activity Report
SMFCC PfcP Interface Statistics	smfcc_up_activity_report_successes_GGSN_sessions	Counter	Total Success Response to UPF for 3G/2G sessions Up activity Report
SMFCC PfcP Interface Statistics	smfcc_up_activity_report_failures_5G_sessions	Counter	Total failure Response to UPF for 5G sessions Up activity Report
SMFCC PfcP Interface Statistics	smfcc_up_activity_report_failures_5G_always_on_sessions	Counter	Total failure Response to UPF for 5G always on sessions Up activity Report
SMFCC PfcP Interface Statistics	smfcc_up_activity_report_failures_4G_sessions	Counter	Total failure Response to UPF for 4G sessions Up activity Report
SMFCC PfcP Interface Statistics	smfcc_up_activity_report_failures_GGSN_sessions	Counter	Total failure Response to UPF for 3G/2G sessions Up activity Report
SMFCC PfcP Interface Statistics	smfcc_qos_flow_inactivity_reports	Counter	Total Qos Flow Inactivity Report Received from UPF
SMFCC PfcP Interface Statistics	smfcc_qos_flow_inactivity_report_successes	Counter	Total Success Response to UPF for Qos Flow Inactivity Report
SMFCC PfcP Interface Statistics	smfcc_qos_flow_inactivity_report_failures	Counter	Total Failure Response to UPF for Qos Flow Inactivity Report
SMFCC PfcP Interface Statistics	smfcc_session_idle_inactivity_reports	Counter	Total Session Idle Inactivity Report Received from UPF
SMFCC PfcP Interface Statistics	smfcc_session_idle_inactivity_report_successes	Counter	Total Success Response to UPF for Session Idle Inactivity Report
SMFCC PfcP Interface Statistics	smfcc_session_idle_inactivity_report_failures	Counter	Total Failure Response to UPF for Session Idle Inactivity Report
SMFCC PfcP Interface Statistics	smfcc_down_link_data_reports	Counter	Total Usage Report Received from UPF

Group	Metric Name	Type	Synopsis
SMFCC PfcP Interface Statistics	smfcc_down_link_data_report_successes	Counter	Total Success Response to UPF for down link data Report
SMFCC PfcP Interface Statistics	smfcc_down_link_data_report_failures	Counter	Total Failure Response to UPF for down link data Report
SMFCC UPF Usage Report Statistics	smfcc_usage_report_total_volume	Counter	Total volume report received from UPF in usage report
SMFCC UPF Usage Report Statistics	smfcc_usage_report_downlink_volume	Counter	Downlink volume report received from UPF in usage report
SMFCC UPF Usage Report Statistics	smfcc_usage_report_uplink_volume	Counter	Uplink volume report received from UPF in usage report
SMFCC UPF Usage Report Statistics	smfcc_usage_report_drop_count	Counter	Dropped Duplicate Invalid UPF Usage Report count
SMFCC UPF Usage Report Statistics	smfcc_usage_report_adc_start_count	Counter	Number of ADC Start Trigger Report count
SMFCC UPF Usage Report Statistics	smfcc_usage_report_adc_stop_count	Counter	Number of ADC Stop Trigger Report count
SMF PGW Call Performance Statistics	smfcc_pgw_bearer_setup_success	Counter	Number of bearer setup success on PGW
SMF PGW Call Performance Statistics	smfcc_pgw_bearer_setup_failure	Counter	Number of bearer setup failure on PGW
SMF PGW Call Performance Statistics	smfcc_pgw_nw_init_bearer_release	Counter	Number of Network Initiated PGW bearer release
SMF PGW Call Performance Statistics	smfcc_pgw_access_init_bearer_release	Counter	Number of Access Initiated PGW bearer release
SMF PGW Call Performance Statistics	smfcc_pgw_abnormal_bearer_release	Counter	Number of PGW bearer released abnormally
SMF PGW Call Performance Statistics	smfcc_pgw_bearer_setup_duration	Histogram	PGW bearer setup time in seconds
SMF SAEGW Call Performance Statistics	smfcc_saegw_bearer_setup_success	Counter	Number of bearer setup success on SAEGW
SMF SAEGW Call Performance Statistics	smfcc_saegw_bearer_setup_failure	Counter	Number of bearer setup failure on SAEGW
SMF SAEGW Call Performance Statistics	smfcc_saegw_nw_init_bearer_release	Counter	Number of Network Initiated SAEGW bearer release
SMF SAEGW Call Performance Statistics	smfcc_saegw_access_init_bearer_release	Counter	Number of Access Initiated SAEGW bearer release
SMF SAEGW Call Performance Statistics	smfcc_saegw_abnormal_bearer_release	Counter	Number of SAEGW bearer released abnormally
SMF SAEGW Call Performance Statistics	smfcc_saegw_bearer_setup_duration	Histogram	SAEGW bearer setup time in seconds
SMFCC EDR Statistics	smfcc_edr_generated	Counter	Total number of EDRs generated by SMF
SMFCC EDR Statistics	smfcc_edr_callback_rcvd	Counter	Total number of EDR callbacks received from Kafka Client
SMFCC 3GPP Procedure Duration Statistics	smfcc_3gpp_procedure_duration	Histogram	SMFCC 3GPP procedure execution time in seconds

Group	Metric Name	Type	Synopsis
SMFCC Interface Round Trip Duration Statistics	smfcc_interface_round_trip_duration	Histogram	SMFCC Interface Round Trip time in seconds
SMFCC Event-Exposure Statistics	smfcc_evtexp_notifications_sent	Counter	Total number of Event Exposure Notifications sent by SMF
SMFCC Event-Exposure Statistics	smfcc_evtexp_notification_responses_rcvd	Counter	Total number of Notification responses from Event Exposure pod
SMFCC Miscellaneous Statistics	smfcc_misc_invalid_session_db_records_cleared	Counter	Total number of invalid session DB Records cleared
SMFCC Miscellaneous Statistics	smfcc_misc_db_record_serialization_failures	Counter	Total number of DB record serialization failures
SMFCC Miscellaneous Statistics	smfcc_misc_db_record_compression_failures	Counter	Total number of DB record compression failures
SMFCC Miscellaneous Statistics	smfcc_misc_db_record_deserialization_failures	Counter	Total number of DB record deserialization failures
SMFCC Miscellaneous Statistics	smfcc_misc_db_record_decompression_failures	Counter	Total number of DB record decompression failures
SMFCC Miscellaneous Statistics	smfcc_misc_db_record_delete_failures	Counter	Total number of DB record delete failures
SMFCC Miscellaneous Statistics	smfcc_misc_db_record_delete_on_create_conflict_attempts	Counter	Total number of DB record delete attempts on create conflict
SMFCC Miscellaneous Statistics	smfcc_misc_db_record_deleted_on_create_conflict	Counter	Total number of DB record deleted on create conflict
SMFCC Miscellaneous Statistics	smfcc_misc_db_record_delete_failures_on_create_conflict	Counter	Total number of DB record delete failed on create conflict
SMFCC Miscellaneous Statistics	smfcc_misc_upf_chosen_available_capacity	Counter	Total Number of UPF chosen based on available capacity
SMFCC Miscellaneous Statistics	smfcc_misc_upf_chosen_nf_priority	Counter	Total Number of UPF chosen based on Network Function Priority
SMFCC Miscellaneous Statistics	smfcc_misc_upf_chosen_tai_priority	Counter	Total Number of UPF chosen based on TAI Priority
SMFCC Miscellaneous Statistics	smfcc_misc_upf_dnn_list_wo_upf	Counter	Total Number of UPF dnn usage entries received with no corresponding UPF association
SMFCC Miscellaneous Statistics	smfcc_misc_upf_selection_change_on_ue_ip_exhaust	Counter	Total Number of Selected UPF change based on UE IP pool exhaust detection
SMF ETCD Interface Statistics	smfcc_etcd_gtpc_abort_lease_grant_request	Counter	Lease grant request sent to ETCD from GTPC for abort transaction
SMF ETCD Interface Statistics	smfcc_etcd_gtpc_abort_lease_grant_response	Counter	Lease grant response received from ETCD to GTPC for abort transaction
SMF ETCD Interface Statistics	smfcc_etcd_gtpc_abort_put_with_lease_grant_request	Counter	Put with lease grant request sent to ETCD from GTPC for abort transaction
SMF ETCD Interface Statistics	smfcc_etcd_gtpc_abort_put_with_lease_grant_response	Counter	Put with lease grant response from ETCD to GTPC for abort transaction

Group	Metric Name	Type	Synopsis
SMF ETCD Interface Statistics	smfcc_etcd_inband_radius_tunnel_put_request	Counter	Inband Radius Tunnel info put request sent to ETCD
SMF ETCD Interface Statistics	smfcc_etcd_inband_radius_tunnel_put_response	Counter	Inband Radius Tunnel info put response from ETCD
SMF ETCD Interface Statistics	smfcc_etcd_inband_radius_tunnel_delete_range_request	Counter	Inband Radius Tunnel info delete range request sent to ETCD
SMF ETCD Interface Statistics	smfcc_etcd_inband_radius_tunnel_delete_range_response	Counter	Inband Radius Tunnel info delete range response from ETCD
SMFCC Debug Statistics	smfcc_session_cache_hits	Counter	Number of times the session is found in cache for incoming requests
SMFCC Debug Statistics	smfcc_procedures_dropped	Counter	Total number of 3gpp procedures dropped
SMFCC Debug Statistics	smfcc_clear_bulk_pdu_sessions_operator_request	Counter	Number of times bulk pdu session clear request is initiated by the operator
SMFCC Debug Statistics	smfcc_cookie_session_id_map_delete	Counter	Number of times the session is found in cookie session id map for deletion
SMFCC Debug Statistics	smfcc_force_nf_discovery	Counter	Number of times smfcc sends nf-discovery request with force-discovery
SMFCC Debug Statistics	smfcc_instanceid_change_no_nfset	Counter	Number of times NF Instance ID change detected when no nf set is configured
SMFCC Debug Statistics	smfcc_tcmalloc_memory_used_by_application	Gauge	Total number of bytes used by application
SMFCC Debug Statistics	smfcc_tcmalloc_page_heap_free_list	Gauge	Total number of bytes in free, mapped pages in page heap
SMFCC Debug Statistics	smfcc_tcmalloc_central_cache_free_list	Gauge	Total number of bytes in free pages in central cache
SMFCC Debug Statistics	smfcc_tcmalloc_transfer_cache_free_list	Gauge	Total number of bytes in free pages in transfer cache
SMFCC Debug Statistics	smfcc_tcmalloc_thread_cache_free_list	Gauge	Total number of bytes in free pages in thread cache
SMFCC Debug Statistics	smfcc_tcmalloc_metadata	Gauge	Total number of bytes used by tcmalloc metadata
SMFCC Debug Statistics	smfcc_tcmalloc_actual_memory_used	Gauge	Actual memory used by application
SMFCC Debug Statistics	smfcc_tcmalloc_virtual_address_space_used	Gauge	Virtual address space used by application
SMFCC Debug Statistics	smfcc_cached_session_count	Gauge	Number of cached sessions on smfcc pod
SMF N1 Interface Message Statistics	smfcc_n1_msg_pdu_session_establishment_request_received	Counter	N1 Pdu Session Establishment request messages received
SMF N1 Interface Message Statistics	smfcc_n1_msg_pdu_session_establishment_request_failures	Counter	N1 Pdu Session Establishment request messages failed
SMF N1 Interface Message Statistics	smfcc_n1_msg_pdu_session_establishment_accept_sent	Counter	N1 Pdu Session Establishment accept messages sent

Group	Metric Name	Type	Synopsis
SMF N1 Interface Message Statistics	smfcc_n1_msg_pdu_session_establishment_reject_sent	Counter	N1 Pdu Session Establishment reject messages sent
SMF N1 Interface Message Statistics	smfcc_n1_msg_pdu_session_modification_request_received	Counter	N1 Pdu Session Modification request messages received
SMF N1 Interface Message Statistics	smfcc_n1_msg_pdu_session_modification_request_failures	Counter	N1 Pdu Session modification request failures
SMF N1 Interface Message Statistics	smfcc_n1_msg_pdu_session_modification_reject_sent	Counter	N1 Pdu Session Modification rejects sent
SMF N1 Interface Message Statistics	smfcc_n1_msg_pdu_session_modification_command_sent	Counter	N1 Pdu Session Modification rejects sent
SMF N1 Interface Message Statistics	smfcc_n1_msg_pdu_session_modification_command_response_failures	Counter	N1 Pdu Session Modification command response failures
SMF N1 Interface Message Statistics	smfcc_n1_msg_pdu_session_modification_command_reject_received	Counter	N1 Pdu Session Modification command reject received
SMF N1 Interface Message Statistics	smfcc_n1_msg_pdu_session_modification_complete_received	Counter	N1 Pdu Session Modification Command
SMF N1 Interface Message Statistics	smfcc_n1_msg_pdu_session_release_request_received	Counter	N1 Pdu Session Release request received
SMF N1 Interface Message Statistics	smfcc_n1_msg_pdu_session_release_request_failures	Counter	Number of n1 pdu session release request message failed
SMF N1 Interface Message Statistics	smfcc_n1_msg_pdu_session_release_reject_sent	Counter	Numner of n1 pdu session release reject message sent
SMF N1 Interface Message Statistics	smfcc_n1_msg_pdu_session_release_command_sent	Counter	Number of n1 pdu session release command message sent
SMF N1 Interface Message Statistics	smfcc_n1_msg_pdu_session_release_command_response_failures	Counter	Number of n1 pdu session release command response failures
SMF N1 Interface Message Statistics	smfcc_n1_msg_pdu_session_release_complete_received	Counter	Number of n1 pdu session release complete received
SMF N1 Interface Message Statistics	smfcc_n1_msg_pdu_session_modification_complete_dropped	Counter	Number of n1 pdu session modification complete messages dropped
SMF N1 Interface Message Statistics	smfcc_n1_msg_pdu_session_modification_command_reject_dropped	Counter	Number of n1 pdu session modification command reject messages dropped
SMF N1 Interface Message Statistics	smfcc_n1_msg_pdu_session_release_complete_dropped	Counter	Number of n1 pdu session release complete messages dropped
SMF N2 Interface Message Statistics	smfcc_n2_msg_pdu_session_resource_setup_request_transfer_sent	Counter	N2 pdu session resource setup request messages sent
SMF N2 Interface Message Statistics	smfcc_n2_msg_pdu_session_resource_setup_request_transfer_failures	Counter	N2 pdu session resource setup requests failed to receive response
SMF N2 Interface Message Statistics	smfcc_n2_msg_pdu_session_resource_setup_response_transfer_received	Counter	N2 pdu session resource setup response message received
SMF N2 Interface Message Statistics	smfcc_n2_msg_pdu_session_resource_setup_unsuccessful_transfer_received	Counter	N2 dpu session resource setup failed message received

Group	Metric Name	Type	Synopsis
SMF N2 Interface Message Statistics	smfcc_n2_msg_pdu_session_resource_modify_request_transfer_sent	Counter	N2 pdu session resource modify request messages sent
SMF N2 Interface Message Statistics	smfcc_n2_msg_pdu_session_resource_modify_request_transfer_failures	Counter	N2 pdu session resource modify request messages failed to receive response
SMF N2 Interface Message Statistics	smfcc_n2_msg_pdu_session_resource_modify_response_transfer_received	Counter	N2 pdu session resource modify response received
SMF N2 Interface Message Statistics	smfcc_n2_msg_pdu_session_resource_modify_unsuccessful_transfer_received	Counter	N2 pdu session resource modify failure messages received
SMF N2 Interface Message Statistics	smfcc_n2_msg_pdu_session_resource_release_command_transfer_sent	Counter	N2 pdu session resource release command transfer sent
SMF N2 Interface Message Statistics	smfcc_n2_msg_pdu_session_resource_release_command_transfer_failures	Counter	N2 pdu session resource release requests failed to receive response
SMF N2 Interface Message Statistics	smfcc_n2_msg_pdu_session_resource_release_response_transfer_received	Counter	N2 pdu session resource release response transfer message received
SMF N2 Interface Message Statistics	smfcc_n2_msg_path_switch_request_transfer_message_received	Counter	N2 path switch request transfer message received
SMF N2 Interface Message Statistics	smfcc_n2_msg_path_swith_request_transfer_failed	Counter	N2 path switch request transfer messages failed to process
SMF N2 Interface Message Statistics	smfcc_n2_msg_path_switch_request_setup_failed_transfer_received	Counter	N2 path switch request setup failed transfer messages received
SMF N2 Interface Message Statistics	smfcc_n2_msg_path_switch_request_setup_failed_transfer_failures	Counter	N2 path switch request setup failed tarnser messages failed to process
SMF N2 Interface Message Statistics	smfcc_n2_msg_path_switch_request_acknowledge_transfer_sent	Counter	N2 path switch request acknowledge transfer messages sent
SMF N2 Interface Message Statistics	smfcc_n2_msg_path_switch_request_unsuccessful_transfer_sent	Counter	N2 path switch request unsuccessful transfer messages sent
SMF N2 Interface Message Statistics	smfcc_n2_msg_handover_required_transfer_received	Counter	N2 handover required transfer messages received
SMF N2 Interface Message Statistics	smfcc_n2_msg_handover_required_transfer_failures	Counter	N2 handover required transfer messages failed to process
SMF N2 Interface Message Statistics	smfcc_n2_msg_handover_command_transfer_sent	Counter	N2 handover command transfer messages sent
SMF N2 Interface Message Statistics	smfcc_n2_msg_handover_request_ack_transfer_received	Counter	N2 handover request acknowledge transfer messages received
SMF N2 Interface Message Statistics	smfcc_n2_msg_handover_request_ack_transfer_failures	Counter	N2 handover request acknowledge transfer messages failed process
SMF N2 Interface Message Statistics	smfcc_n2_msg_handover_prep_unsuccessful_transfer_sent	Counter	N2 handover preparation unsuccessful transfer message sent
SMF N2 Interface Message Statistics	smfcc_n2_msg_handover_resource_allocation_unsuccessful_transfer_received	Counter	N2 handover resource allocation unsuccessful transfer messages received

Group	Metric Name	Type	Synopsis
SMF N2 Interface Message Statistics	smfcc_n2_msg_pdu_session_resource_setup_response_transfer_dropped	Counter	Pdu session resource setup response transfer messages dropped
SMF N2 Interface Message Statistics	smfcc_n2_msg_pdu_session_resource_setup_unsuccessful_transfer_dropped	Counter	Pdu session resource setup unsuccessful transfer messages dropped
SMF N2 Interface Message Statistics	smfcc_n2_msg_pdu_session_resource_modify_response_transfer_dropped	Counter	Pdu session resource modify response transfer messages dropped
SMF N2 Interface Message Statistics	smfcc_n2_msg_pdu_session_resource_modify_unsuccessful_transfer_dropped	Counter	Pdu session resource modify unsuccessful transfer messages dropped
SMF N2 Interface Message Statistics	smfcc_n2_msg_pdu_session_resource_release_response_transfer_dropped	Counter	Pdu session resource release response transfer messages dropped
SMF N2 Interface Message Statistics	smfcc_n2_msg_handover_resource_allocation_unsuccessful_transfer_dropped	Counter	Handover resource allocation unsuccessful transfer messages dropped
SMF N2 Interface Message Statistics	smfcc_n2_msg_pdu_session_resource_modify_indication_transfer_received	Counter	N2 pdu session resource modify indication transfer message received
SMF N2 Interface Message Statistics	smfcc_n2_msg_pdu_session_resource_modify_indication_transfer_failed	Counter	N2 pdu session resource modify indication transfer messages failed to process
SMF N2 Interface Message Statistics	smfcc_n2_msg_pdu_session_resource_modify_confirm_transfer_sent	Counter	N2 pdu session resource modify confirm transfer messages sent
SMF N2 Interface Message Statistics	smfcc_n2_msg_pdu_session_resource_modify_indication_unsuccessful_transfer_sent	Counter	N2 pdu session resource modify indication unsuccessful transfer messages sent
SMF udsf interface message statistics	smfcc_udsf_read_and_lock_record_using_secondary_key_request_attempts	Counter	Total number of udsf read and lock record request using secondary key attempts
SMF udsf interface message statistics	smfcc_udsf_read_and_lock_record_using_secondary_key_request_successes	Counter	Total number of udsf read and lock record request using secondary key successes
SMF udsf interface message statistics	smfcc_udsf_read_and_lock_record_using_secondary_key_request_failures	Counter	Total number of udsf read and lock record request using secondary key failures
SMF udsf interface message statistics	smfcc_udsf_read_and_lock_record_request_attempts	Counter	Total number of udsf read and lock record request attempts
SMF udsf interface message statistics	smfcc_udsf_read_and_lock_record_request_successes	Counter	Total number of udsf read and lock record request successes
SMF udsf interface message statistics	smfcc_udsf_read_and_lock_record_request_failures	Counter	Total number of udsf read and lock record request failures
SMF udsf interface message statistics	smfcc_udsf_create_record_and_lock_request_attempts	Counter	Total number of udsf create record and lock request attempts
SMF udsf interface message statistics	smfcc_udsf_create_record_and_lock_request_successes	Counter	Total number of udsf create record and lock request successes

Group	Metric Name	Type	Synopsis
SMF udsf interface message statistics	smfcc_udsf_create_record_and_lock_request_failures	Counter	Total number of udsf create record and lock request failures
SMF udsf interface message statistics	smfcc_udsf_create_or_modify_block_request_attempts	Counter	Total number of udsf create or modify block request attempts
SMF udsf interface message statistics	smfcc_udsf_create_or_modify_block_request_successes	Counter	Total number of udsf create or modify block request successes
SMF udsf interface message statistics	smfcc_udsf_create_or_modify_block_request_failures	Counter	Total number of udsf create or modify block request failures
SMF udsf interface message statistics	smfcc_udsf_modify_record_and_lock_request_attempts	Counter	Total number of udsf modify record and lock request attempts
SMF udsf interface message statistics	smfcc_udsf_modify_record_and_lock_request_successes	Counter	Total number of udsf modify record and lock request successes
SMF udsf interface message statistics	smfcc_udsf_modify_record_and_lock_request_failures	Counter	Total number of modify record and lock request failures
SMF udsf interface message statistics	smfcc_udsf_modify_record_and_unlock_request_attempts	Counter	Total number of udsf modify record and unlock request attempts
SMF udsf interface message statistics	smfcc_udsf_modify_record_and_unlock_request_successes	Counter	Total number of udsf modify record and unlock request successes
SMF udsf interface message statistics	smfcc_udsf_modify_record_and_unlock_request_failures	Counter	Total number of udsf modify record and unlock request failures
SMF udsf interface message statistics	smfcc_udsf_delete_record_request_attempts	Counter	Total number of udsf delete record request attempts
SMF udsf interface message statistics	smfcc_udsf_delete_record_request_successes	Counter	Total number of udsf delete record request successes
SMF udsf interface message statistics	smfcc_udsf_delete_record_request_failures	Counter	Total number of udsf delete record requests failures
SMF udsf interface message statistics	smfcc_udsf_get_record_and_steal_lock_request_attempts	Counter	Total number of udsf get record and steal lock requests attempts
SMF udsf interface message statistics	smfcc_udsf_get_record_and_steal_lock_request_successes	Counter	Total number of udsf get record and steal lock request successes
SMF udsf interface message statistics	smfcc_udsf_get_record_and_steal_lock_request_failures	Counter	Total number of udsf get record and steal lock request failures
SMF udsf interface message statistics	smfcc_udsf_search_record_request_attempts	Counter	Total number of udsf search record request attempts
SMF udsf interface message statistics	smfcc_udsf_search_record_request_successes	Counter	Total number of udsf search record request successes
SMF udsf interface message statistics	smfcc_udsf_search_record_request_failures	Counter	Total number of udsf search record request failures
SMF udsf interface message statistics	smfcc_udsf_get_record_request_attempts	Counter	Total number of udsf get record request attempts
SMF udsf interface message statistics	smfcc_udsf_get_record_request_successes	Counter	Total number of udsf get record request successes
SMF udsf interface message statistics	smfcc_udsf_get_record_request_failures	Counter	Total number of udsf get record request failures

Group	Metric Name	Type	Synopsis
SMF udsf interface message statistics	smfcc_udsf_modify_block_and_unlock_request_attempts	Counter	Total number of udsf modify block and unlock request attempts
SMF udsf interface message statistics	smfcc_udsf_modify_block_and_unlock_request_successes	Counter	Total number of udsf modify block and unlock requests successes
SMF udsf interface message statistics	smfcc_udsf_modify_block_and_unlock_request_failures	Counter	Total number of udsf modify block and unlock request failures
SMF udsf interface message statistics	smfcc_udsf_get_tag_count_request_attempts	Counter	Total number of udsf get tag count request attempts
SMF udsf interface message statistics	smfcc_udsf_get_tag_count_request_successes	Counter	Total number of udsf get tag count request successes
SMF udsf interface message statistics	smfcc_udsf_get_tag_count_request_failures	Counter	Total number of udsf get tag count request failures
SMF udsf interface message statistics	smfcc_udsf_unlock_only_attempts	Counter	Total number of Unlock only request attempts
SMF udsf interface message statistics	smfcc_udsf_unlock_only_successes	Counter	Total number of Unlock only request successes
SMF udsf interface message statistics	smfcc_udsf_unlock_only_request_failures	Counter	Total number of Unlock only request failures
SMF udsf interface message statistics	smfcc_udsf_patch_record_request_attempts	Counter	Total Number of patch record request attempts
SMF udsf interface message statistics	smfcc_udsf_patch_record_request_successes	Counter	Total Number of patch record request successes
SMF udsf interface message statistics	smfcc_udsf_patch_record_request_failures	Counter	Total Number of patch record request failures
SMF udsf interface message statistics	smfcc_udsf_update_record_meta_request_attempts	Counter	Total number of update record meta request attempts
SMF udsf interface message statistics	smfcc_udsf_update_record_meta_request_successes	Counter	Total number of update record meta request successes
SMF udsf interface message statistics	smfcc_udsf_update_record_meta_request_failures	Counter	Total number of update record meta request failures
SMF udsf interface message statistics	smfcc_udsf_connection_available_indication_received	Counter	Total number of UDSF API Connection Available Indication received
SMF udsf interface message statistics	smfcc_udsf_connection_unavailable_indication_received	Counter	Total number of UDSF API Connection Unavailable Indication received
SMF udsf interface message statistics	smfcc_db_transaction_failure_udsf_connection_unavailable	Counter	Total number of DB transactions failed due to UDSF Connection Unavailable
SMF Agent Interface Timeout Statistics	smfcc_chf_transaction_internal_timeout	Counter	Number of internal transaction timeouts happened in chf interface
SMF Agent Interface Timeout Statistics	smfcc_amf_transaction_internal_timeout	Counter	Number of internal transaction timeouts happened in amf interface
SMF Agent Interface Timeout Statistics	smfcc_pcf_transaction_internal_timeout	Counter	Number of internal transaction timeouts happened in pcf interface
SMF Agent Interface Timeout Statistics	smfcc_udm_transaction_internal_timeout	Counter	Number of internal transaction timeouts happened in udm interface

Group	Metric Name	Type	Synopsis
SMF Agent Interface Timeout Statistics	smfcc_subscr analyzer_transaction_internal_timeout	Counter	Number of internal transaction timeouts happened in subscriber analyzer interface
SMF Agent Interface Timeout Statistics	smfcc_urm_transaction_internal_timeout	Counter	Number of internal transaction timeouts happened in urm interface
SMF Agent Interface Timeout Statistics	smfcc_evtexp_transaction_internal_timeout	Counter	Number of internal transaction timeouts happened in evtexp interface
SMF Agent Interface Timeout Statistics	smfcc_nrf_transaction_internal_timeout	Counter	Number of internal transaction timeouts happened in nrf interface
SMF Agent Interface Timeout Statistics	smfcc_pfcp_transaction_internal_timeout	Counter	Number of internal transaction timeouts happened in pfcp interface
SMF Agent Interface Timeout Statistics	smfcc_gtpc_transaction_internal_timeout	Counter	Number of internal transaction timeouts happened in gtpc interface
SMF Agent Interface Timeout Statistics	smfcc_gx_transaction_internal_timeout	Counter	Number of internal transaction timeouts happened in gx interface
SMF Agent Interface Timeout Statistics	smfcc_dtm_transaction_internal_timeout	Counter	Number of internal transaction timeouts happened in durable timer interface
SMF Radius Interface Message Statistics	smfcc_radius_access_requests_sent	Counter	Number of radius access-requests sent
SMF Radius Interface Message Statistics	smfcc_radius_access_requests_send_failed	Counter	Number of radius access-requests send failed
SMF Radius Interface Message Statistics	smfcc_radius_access_accepts	Counter	Number of radius access-accepts received
SMF Radius Interface Message Statistics	smfcc_radius_access_rejects	Counter	Number of radius access-rejects received
SMF Radius Interface Message Statistics	smfcc_radius_authentication_request_timeouts	Counter	Number of radius access-request timeouts
SMF Radius Interface Message Statistics	smfcc_radius_accounting_start_requests_sent	Counter	Number of radius accounting-start requests sent
SMF Radius Interface Message Statistics	smfcc_radius_accounting_start_requests_send_failed	Counter	Number of radius accounting-start requests send failed
SMF Radius Interface Message Statistics	smfcc_radius_accounting_start_response_successes	Counter	Number of radius accounting-start success responses received
SMF Radius Interface Message Statistics	smfcc_radius_accounting_start_response_failures	Counter	Number of radius accounting-start fail responses received
SMF Radius Interface Message Statistics	smfcc_radius_accounting_start_request_timeouts	Counter	Number of radius accounting-start request timeouts
SMF Radius Interface Message Statistics	smfcc_radius_accounting_interim_requests_sent	Counter	Number of radius accounting-interim requests sent
SMF Radius Interface Message Statistics	smfcc_radius_accounting_interim_requests_send_failed	Counter	Number of radius accounting-interim requests send failed

Group	Metric Name	Type	Synopsis
SMF Radius Interface Message Statistics	smfcc_radius_accounting_interim_response_successes	Counter	Number of radius accounting-interim success responses received
SMF Radius Interface Message Statistics	smfcc_radius_accounting_interim_response_failures	Counter	Number of radius accounting-interim fail responses received
SMF Radius Interface Message Statistics	smfcc_radius_accounting_interim_request_timeouts	Counter	Number of radius accounting-interim request timeouts
SMF Radius Interface Message Statistics	smfcc_radius_accounting_stop_requests_sent	Counter	Number of radius accounting-stop requests sent
SMF Radius Interface Message Statistics	smfcc_radius_accounting_stop_requests_send_failed	Counter	Number of radius accounting-stop requests send failed
SMF Radius Interface Message Statistics	smfcc_radius_accounting_stop_response_successes	Counter	Number of radius accounting-stop success responses received
SMF Radius Interface Message Statistics	smfcc_radius_accounting_stop_response_failures	Counter	Number of radius accounting-stop fail responses received
SMF Radius Interface Message Statistics	smfcc_radius_accounting_stop_request_timeouts	Counter	Number of radius accounting-stop request timeouts
SMF Radius Interface Message Statistics	smfcc_inband_radius_gtpu_error_indications_received	Counter	Number of inband radius gtpu error indication request received
SMF Radius Interface Message Statistics	smfcc_inband_radius_gtpu_error_indications_response_sent	Counter	Number of inband radius gtpu error indication response success sent
SMF Radius Interface Message Statistics	smfcc_radius_ip_allocation_requested	Counter	Number of ip allocation requests sent to radius
SMF Radius Interface Message Statistics	smfcc_radius_ip_allocated	Counter	Number of ip allocated response from radius
SMF Radius Interface Message Statistics	smfcc_radius_ip_downgraded	Counter	Number of ip allocation response from radius with downgrade
SMF Radius Interface Message Statistics	smfcc_radius_ip_ignored	Counter	Number of ip allocation response from radius ignored
SMF DB Interface Statistics	smfcc_bulk_delete_timers_start_attempts	Counter	Bulk delete durable timer attempted
SMF DB Interface Statistics	smfcc_bulk_delete_timers_start_failures	Counter	Bulk delete durable timer failures
SMF DB Interface Statistics	smfcc_bulk_delete_timers_start_successes	Counter	Bulk delete durable timer started successfully
SMF DB Interface Statistics	smfcc_bulk_delete_timers_stop_attempts	Counter	Bulk delete durable timer stops attempted
SMF DB Interface Statistics	smfcc_bulk_delete_timers_stop_failures	Counter	Bulk delete durable timer failures
SMF DB Interface Statistics	smfcc_bulk_delete_timers_stop_successes	Counter	Bulk delete durable timer started successfully
SMF DB Interface Statistics	smfcc_bulk_delete_timers_expiry	Counter	Bulk delete durable timer expired
SMF DB Interface Statistics	smfcc_multi_read_requests_sent_to_db	Counter	Multi Read requests sent to Db

Group	Metric Name	Type	Synopsis
SMF DB Interface Statistics	smfcc_multi_read_response_received_from_db_success	Counter	Multi Read success responses received from db
SMF DB Interface Statistics	smfcc_multi_read_response_received_from_db_failure	Counter	Multi Read failure responses received from db
SMF DB Interface Statistics	smfcc_bulk_delete_requests_sent	Counter	Bulk delete requests sent out
SMF DB Interface Statistics	smfcc_bulk_delete_requests_received	Counter	Bulk delete requests received
SMF SMF Interface Statistics	smfcc_nsmf_pdu_session_create_attempts	Counter	PDU Session Create attempts
SMF SMF Interface Statistics	smfcc_nsmf_pdu_session_create_successes	Counter	PDU Session Create successes
SMF SMF Interface Statistics	smfcc_nsmf_pdu_session_create_failures	Counter	PDU Session Create failures
SMF SMF Interface Statistics	smfcc_hsmf_pdu_session_create_duplicate_attempts	Counter	Duplicate Pdu Session Creation Attempted
SMF SMF Interface Statistics	smfcc_nsmf_pdu_session_update_tx_attempts	Counter	PDU Session Update attempts
SMF SMF Interface Statistics	smfcc_nsmf_pdu_session_update_tx_successes	Counter	PDU Session Update successes
SMF SMF Interface Statistics	smfcc_nsmf_pdu_session_update_tx_failures	Counter	PDU Session Update failures
SMF SMF Interface Statistics	smfcc_nsmf_pdu_session_release_request_attempts	Counter	PDU Session Release attempts
SMF SMF Interface Statistics	smfcc_nsmf_pdu_session_release_request_successes	Counter	PDU Session Release successes
SMF SMF Interface Statistics	smfcc_nsmf_pdu_session_release_request_failures	Counter	PDU Session Release failures
SMF SMF Interface Statistics	smfcc_nsmf_pdu_session_update_rx_attempts	Counter	Incoming PDU Session Update attempts
SMF SMF Interface Statistics	smfcc_nsmf_pdu_session_update_rx_successes	Counter	Incoming PDU Session Update successes
SMF SMF Interface Statistics	smfcc_nsmf_pdu_session_update_rx_failures	Counter	Incoming PDU Session Update failures
SMF SMF Interface Statistics	smfcc_nsmf_pdu_session_status_notify_attempts	Counter	PDU Session Notify attempts
SMF SMF Interface Statistics	smfcc_nsmf_pdu_session_status_notify_successes	Counter	PDU Session Status Notify successes
SMF SMF Interface Statistics	smfcc_nsmf_pdu_session_status_notify_failures	Counter	PDU Session Status Notify failures
SMF SMF Interface Statistics	smfcc_nsmf_evt_exp_create_subscription_tx_attempts	Counter	Nsmf Event Exposure Create Subscription attempts
SMF SMF Interface Statistics	smfcc_nsmf_evt_exp_create_subscription_tx_successes	Counter	Nsmf Event Exposure Create Subscription successes
SMF SMF Interface Statistics	smfcc_nsmf_evt_exp_create_subscription_tx_failures	Counter	Nsmf Event Exposure Create Subscription failures
SMF SMF Interface Statistics	smfcc_nsmf_evt_exp_modify_subscription_tx_attempts	Counter	Nsmf Event Exposure Modify Subscription attempts
SMF SMF Interface Statistics	smfcc_nsmf_evt_exp_modify_subscription_tx_successes	Counter	Nsmf Event Exposure Modify Subscription successes

Group	Metric Name	Type	Synopsis
SMF SMF Interface Statistics	smfcc_nsmf_evt_exp_modify_subscription_tx_failures	Counter	Nsmf Event Exposure Modify Subscription failures
SMF SMF Interface Statistics	smfcc_nsmf_evt_exp_delete_subscription_tx_attempts	Counter	Nsmf Event Exposure Delete Subscription attempts
SMF SMF Interface Statistics	smfcc_nsmf_evt_exp_delete_subscription_tx_successes	Counter	Nsmf Event Exposure Delete Subscription successes
SMF SMF Interface Statistics	smfcc_nsmf_evt_exp_delete_subscription_tx_failures	Counter	Nsmf Event Exposure Delete Subscription failures
SMF SMF Interface Statistics	smfcc_nsmf_pdu_session_create_attempts_alternate_sepp	Counter	PDU Session Create attempts
SMF SMF Interface Statistics	smfcc_nsmf_pdu_session_create_successes_alternate_sepp	Counter	PDU Session Create successes
SMF SMF Interface Statistics	smfcc_nsmf_pdu_session_create_failures_alternate_sepp	Counter	PDU Session Create failures
SMF SMF Interface Statistics	smfcc_nsmf_pdu_session_update_tx_attempts_alternate_sepp	Counter	PDU Session Update attempts
SMF SMF Interface Statistics	smfcc_nsmf_pdu_session_update_tx_successes_alternate_sepp	Counter	PDU Session Update successes
SMF SMF Interface Statistics	smfcc_nsmf_pdu_session_update_tx_failures_alternate_sepp	Counter	PDU Session Update failures
SMF SMF Interface Statistics	smfcc_nsmf_pdu_session_release_request_attempts_alternate_sepp	Counter	PDU Session Release attempts
SMF SMF Interface Statistics	smfcc_nsmf_pdu_session_release_request_successes_alternate_sepp	Counter	PDU Session Release successes
SMF SMF Interface Statistics	smfcc_nsmf_pdu_session_release_request_failures_alternate_sepp	Counter	PDU Session Release failures
SMF IntraNF Punting Statistics	smfcc_intra_nf_punting_sent	Counter	Number of intra-nf-punting requests
SMF IntraNF Punting Statistics	smfcc_intra_nf_punting_successes	Counter	Number of intra-nf-punting requests success
SMF IntraNF Punting Statistics	smfcc_intra_nf_punting_failures	Counter	Number of intra-nf-punting requests failed
SMF IntraNF Punting Statistics	smfcc_intra_nf_punting_ip_collision	Counter	Number of intra-nf-punting cases IP addresse collision which resulted force lock
SMF IntraNF Punting Statistics	smfcc_intra_nf_punting_received	Counter	Number of intra-nf-punting requests received
SMF IntraNF Punting Statistics	smfcc_intra_nf_punting_timeout	Counter	Number of intra-nf-punting requests timed out at application
SMF IntraNF Punting Statistics	smfcc_intra_nf_punting_delayed_response	Counter	Number of intra-nf-punting requests getting delayed response after timer expiry
SMF InterNF Punting Statistics	smfcc_inter_nf_punting_sent	Counter	Number of inter-nf-punting requests
SMF InterNF Punting Statistics	smfcc_inter_nf_punting_successes	Counter	Number of inter-nf-punting requests success
SMF InterNF Punting Statistics	smfcc_inter_nf_punting_failures	Counter	Number of inter-nf-punting requests failed
SMF InterNF Punting Statistics	smfcc_inter_nf_punting_received	Counter	Number of inter-nf-punting requests received
SMF InterNF Punting Statistics	smfcc_inter_nf_punting_ignored_for_internal_session_release	Counter	Number of internal session release events ignored for inter-nf-punting

Group	Metric Name	Type	Synopsis
SMF Peer NF Set Statistics	smfcc_selected_nf_instance_status_down	Counter	Number of peer nf instance status down in NRF cache
SMF Peer NF Set Statistics	smfcc_msg_rcvd_from_new_peer_nf_instance	Counter	Number of messages received from new peer nf instance
SMF Peer NF Set Statistics	smfcc_msg_sent_to_new_peer_nf_instance	Counter	Number of messages sent to from new peer nf instance
SMF Peer NF Set Statistics	smfcc_alternative_peer_nf_retry_attempt_success	Counter	Number of times alternative peer NF was successfully selected for retrying message towards NF
SMF Peer NF Set Statistics	smfcc_alternative_peer_nf_retry_attempt_failures	Counter	Number of times alternative peer NF couldn't be selected for retrying message towards NF
SMF NF Set Statistics	smfcc_nf_set_smf_owner_instance_changed	Counter	Number of sessions for which current SMF instance took ownership from other SMF instance within the set
SMF NF Set Statistics	smfcc_nf_set_smf_procedures_failed_during_smf_owner_instance_change	Counter	Number of procedures failed during owner SMF instance change in NF Set
SMF NRF Interface Statistics	smfcc_nnrf_nfm_nfstatus_subscribe_for_nfset_attempts	Counter	Number of NF Status Subscribe requests for NF set
SMF NRF Interface Statistics	smfcc_nnrf_nfm_nfstatus_subscribe_for_nfset_failures	Counter	Number of NF Status Subscribe requests for NF set failed
SMF NRF Interface Statistics	smfcc_nnrf_nfm_nfstatus_subscribe_for_nfset_retries	Counter	Number of NF Status Subscribe requests retried for NF set
SMF NRF Interface Statistics	smfcc_nnrf_nfm_nfstatus_notification_for_nfset_received	Counter	Number of NF Status Notification requests received for NF set
SMF NRF Interface Statistics	smfcc_nnrf_nfm_nfstatus_notification_for_nfset_successes	Counter	Number of NF Status Notification successes for NF set sent
SMF NRF Interface Statistics	smfcc_nnrf_disc_nf_discover_for_nfset_attempts	Counter	Number of NF Discover requests for NF set
SMF NRF Interface Statistics	smfcc_nnrf_disc_nf_discover_for_nfset_failures	Counter	Number of NF Discover requests for NF set failed
SMF NRF Interface Statistics	smfcc_nnrf_disc_nf_discover_for_nfset_retries	Counter	Number of NF Discover requests retried for NF set
SMF NRF Interface Statistics	smfcc_nnrf_nfm_nfstatus_unsubscribe_for_nfset_attempts	Counter	Number of NF Status UnSubscribe requests for NF set
SMF NRF Interface Statistics	smfcc_nnrf_nfm_nfstatus_unsubscribe_for_nfset_failures	Counter	Number of NF Status UnSubscribe requests for NF set failed
SMF NRF Interface Statistics	smfcc_nnrf_nfm_nfstatus_unsubscribe_for_nfset_retries	Counter	Number of NF Status UnSubscribe requests retried for NF set
SMF Subscriber DB Interface Statistics	smfcc_gw_subscriber_db_create_record_attempts	Counter	Number of subscriber record create attempts
SMF Subscriber DB Interface Statistics	smfcc_gw_subscriber_db_create_record_failures	Counter	Number of subscriber record create failures

Group	Metric Name	Type	Synopsis
SMF Subscriber DB Interface Statistics	smfcc_gw_subscriber_db_readlock_record_attempts	Counter	Number of subscriber record readlock attempts
SMF Subscriber DB Interface Statistics	smfcc_gw_subscriber_db_readlock_record_failures	Counter	Number of subscriber record readlock failures
SMF Subscriber DB Interface Statistics	smfcc_gw_subscriber_db_update_record_attempts	Counter	Number of subscriber record update attempts
SMF Subscriber DB Interface Statistics	smfcc_gw_subscriber_db_update_record_failures	Counter	Number of subscriber record update failures
SMF Subscriber DB Interface Statistics	smfcc_gw_subscriber_db_delete_record_attempts	Counter	Number of subscriber record delete attempts
SMF Subscriber DB Interface Statistics	smfcc_gw_subscriber_db_delete_record_failures	Counter	Number of subscriber record delete failures
SMF Cdr Interface Statistics	smfcc_cdr_sent	Counter	Number of cdr sent
SMF Cdr Interface Statistics	smfcc_cdr_fail	Counter	Number of send failure
SMF Cdr Interface Statistics	smfcc_cdr_response_rcvd	Counter	Number of send failure
SMF Gw Subscriber Cache Statistics	smfcc_gw_subscriber_cache_release_request_attempts	Counter	Number of gw subscriber record release attempts
SMF Gw Subscriber Cache Statistics	smfcc_gw_subscriber_cache_release_request_failures	Counter	Number of gw subscriber record release request failures
SMF Gw Subscriber Cache Statistics	smfcc_gw_subscriber_cache_record_steal_lock_attempted	Counter	Number of subscriber record steal lock attempted
SMF Gw Subscriber Cache Statistics	smfcc_gw_subscriber_cache_release_db_read_retry_wait_timer_expiry	Counter	Number of DB read and lock retry when negavtive response received from owner pod
SMF Gw Subscriber Cache Statistics	smfcc_gw_subscriber_cache_release_request_received	Counter	Number of subscriber record release request received
SMF Gw Subscriber Cache Statistics	smfcc_gw_subscriber_cache_release_request_acknowledged	Counter	Number of subscriber record release request responded

SMF Call Control Interface Statistics

- smfcc_nsmf_pdu_session_create_sm_context_attempts

Type

64-bit Counter

Description

Number of Create SM Context request messages received

Labels

dnn, snssai

- smfcc_nsmf_pdu_session_create_sm_context_successes

Type

64-bit Counter

Description

Number of Create SM Context request messages successful

Labels

dnn, snssai

smfcc_nsmf_pdu_session_create_sm_context_failures
Type
64-bit Counter
Description
Number of Create SM Context request messages failed
Labels
dnn, snssai, causecode
smfcc_nsmf_pdu_session_create_sm_context_duplicate_attempts
Type
64-bit Counter
Description
Number of Duplicate Create SM Context request messages received
Labels
dnn, snssai
smfcc_vsmf_pdu_session_create_sm_context_duplicate_attempts
Type
64-bit Counter
Description
Number of Duplicate Create SM Context request messages received by vSMF
Labels
dnn, snssai
smfcc_nsmf_pdu_session_update_sm_context_attempts
Type
64-bit Counter
Description
Number of Update SM Context request messages received
Labels
dnn, snssai
smfcc_nsmf_pdu_session_update_sm_context_successes
Type
64-bit Counter
Description
Number of Update SM Context request messages successful
Labels
dnn, snssai
smfcc_nsmf_pdu_session_update_sm_context_failures
Type
64-bit Counter
Description

Number of Update SM Context request messages failed

Labels

dnn, snssai, causecode

• smfcc_nsmf_pdu_session_release_sm_context_attempts

Type

64-bit Counter

Description

Number of Release SM Context request messages received

Labels

dnn, snssai

• smfcc_nsmf_pdu_session_release_sm_context_successes

Type

64-bit Counter

Description

Number of Release SM Context request messages successful

Labels

dnn, snssai

• smfcc_nsmf_pdu_session_release_sm_context_failures

Type

64-bit Counter

Description

Number of Release SM Context request messages failed

Labels

dnn, snssai, causecode

• smfcc_nsmf_pdu_session_create_punted

Type

64-bit Counter

Description

Number of Create request messages punted

Labels

dnn, snssai

• smfcc_nsmf_pdu_session_create_client_attempts

Type

64-bit Counter

Description

Number of Create request messages received at the client

Labels

dnn, snssai

• smfcc_nsmf_pdu_session_create_client_successes

Type

64-bit Counter

Description

Number of Create request messages successful at the client

Labels

dnn, snssai

• smfcc_nsmf_pdu_session_create_client_failures

Type

64-bit Counter

Description

Number of Create request messages failed at the client

Labels

dnn, snssai, causecode

• smfcc_nsmf_pdu_session_create_client_punted

Type

64-bit Counter

Description

Number of Create request messages punted at the client

Labels

dnn, snssai

• smfcc_nsmf_pdu_session_update_attempts

Type

64-bit Counter

Description

Number of Update request messages received

Labels

dnn, snssai

• smfcc_nsmf_pdu_session_update_successes

Type

64-bit Counter

Description

Number of Update request messages successful

Labels

dnn, snssai

• smfcc_nsmf_pdu_session_update_failures

Type

64-bit Counter

Description

Number of Update request messages failed

Labels

dnn, snssai, causecode

- smfcc_nsmf_pdu_session_update_punted

Type

64-bit Counter

Description

Number of Update request messages punted

Labels

dnn, snssai

- smfcc_nsmf_pdu_session_update_client_attempts

Type

64-bit Counter

Description

Number of Update request messages received at the client

Labels

dnn, snssai

- smfcc_nsmf_pdu_session_update_client_successes

Type

64-bit Counter

Description

Number of Update request messages successful at the client

Labels

dnn, snssai

- smfcc_nsmf_pdu_session_update_client_failures

Type

64-bit Counter

Description

Number of Update request messages failed at the client

Labels

dnn, snssai, causecode

- smfcc_nsmf_pdu_session_update_client_punted

Type

64-bit Counter

Description

Number of Update request messages punted at the client

Labels

dnn, snssai

- smfcc_nsmf_pdu_session_release_attempts

Type

64-bit Counter

Description

Number of Release request messages received

Labels

dnn, snssai

• smfcc_nsmf_pdu_session_release_successes

Type

64-bit Counter

Description

Number of Release request messages successful

Labels

dnn, snssai

• smfcc_nsmf_pdu_session_release_failures

Type

64-bit Counter

Description

Number of Release request messages failed

Labels

dnn, snssai, causecode

• smfcc_nsmf_pdu_session_release_punted

Type

64-bit Counter

Description

Number of Release request messages punted

Labels

dnn, snssai

• smfcc_nsmf_pdu_session_release_client_attempts

Type

64-bit Counter

Description

Number of Release request messages received at the client

Labels

dnn, snssai

• smfcc_nsmf_pdu_session_release_client_successes

Type

64-bit Counter

Description

Number of Release request messages successful at the client

Labels

dnn, snssai

- smfcc_nsmf_pdu_session_release_client_failures

Type

64-bit Counter

Description

Number of Release request messages failed at the client

Labels

dnn, snssai, causecode

- smfcc_nsmf_pdu_session_release_client_punted

Type

64-bit Counter

Description

Number of Release request messages punted at the client

Labels

dnn, snssai

- smfcc_namf_comm_n1n2_message_transfer_attempts

Type

64-bit Counter

Description

Number of N1N2 Message Transfer Request messages sent

Labels

peerid, dnn, snssai

- smfcc_namf_comm_n1n2_message_transfer_successes

Type

64-bit Counter

Description

Number of N1N2 Message Transfer Request messages successful

Labels

peerid, dnn, snssai

- smfcc_namf_comm_n1n2_message_transfer_failures

Type

64-bit Counter

Description

Number of N1N2 Message Transfer Request messages failed

Labels

peerid, dnn, snssai, causecode

- smfcc_start_durable_timer_requests

Type

64-bit Counter

	Description
	Number of Durable Timer Start Request messages sent
	Labels
	<i>timerType, dnn, snssai</i>
• smfcc_start_durable_timer_failures	
	Type
	64-bit Counter
	Description
	Number of Durable Timer Start Request messages failed
	Labels
	<i>timerType, dnn, snssai, causecode</i>
• smfcc_modify_durable_timer_requests	
	Type
	64-bit Counter
	Description
	Number of Durable Timer Modify Request messages sent
	Labels
	<i>timerType, dnn, snssai</i>
• smfcc_modify_durable_timer_failures	
	Type
	64-bit Counter
	Description
	Number of Durable Timer Modify Request messages failed
	Labels
	<i>timerType, dnn, snssai, causecode</i>
• smfcc_durable_timer_modify_changed_to_start	
	Type
	64-bit Counter
	Description
	Number of Durable Timer Modify Attempts changes to Start Timer
	Labels
	<i>timerType, dnn, snssai, reason</i>
• smfcc_stop_durable_timer_requests	
	Type
	64-bit Counter
	Description
	Number of Durable Timer Stop Request messages sent
	Labels
	<i>timerType, dnn, snssai</i>

smfcc_stop_durable_timer_failures
Type
64-bit Counter
Description
Number of Durable Timer Stop Request messages failed
Labels
<i>timerType, dnn, snssai, causecode</i>
smfcc_durable_timer_fired
Type
64-bit Counter
Description
Number of Durable Timer Fired Notifications
Labels
<i>timerType, dnn, snssai</i>
smfcc_durable_timer_fired_responses
Type
64-bit Counter
Description
Number of Durable Timer Fired Notification responses sent
Labels
<i>timerType</i>
smfcc_durable_timer_fired_stale
Type
64-bit Counter
Description
Number of Stale Durable Timers Fired
Labels
<i>timerType, dnn, snssai</i>
smfcc_durable_timer_fired_error
Type
64-bit Counter
Description
Number of Stale Durable Timers Fired With Error
Labels
<i>timerType, dnn, snssai, errvalue</i>
smfcc_namf_comm_assign_ebi_message_attempts
Type
64-bit Counter
Description

Number of Assign Ebi Request messages sent

Labels

peerid, dnn, snssai

- smfcc_namf_comm_assign_ebi_message_successes

Type

64-bit Counter

Description

Number of Assign Ebi Request messages successful

Labels

peerid, dnn, snssai

- smfcc_namf_comm_assign_ebi_message_failures

Type

64-bit Counter

Description

Number of Assign Ebi Request messages failed

Labels

peerid, dnn, snssai, causecode

- smfcc_nsmf_pdu_session_retrieve_sm_context_attempts

Type

64-bit Counter

Description

Number of Retrieve request messages received

Labels

dnn, snssai

- smfcc_nsmf_pdu_session_retrieve_sm_context_successes

Type

64-bit Counter

Description

Number of Retrieve request messages successful

Labels

dnn, snssai

- smfcc_nsmf_pdu_session_retrieve_sm_context_failures

Type

64-bit Counter

Description

Number of Retrieve request messages failed

Labels

dnn, snssai, causecode

- smfcc_nsmf_pdu_session_sm_context_status_notify_attempts

Type

64-bit Counter

Description

Number of SM Context Status Notify messages sent

Labels

peerid, dnn, snssai

- smfcc_nsmf_pdu_session_sm_context_status_notify_successes

Type

64-bit Counter

Description

Number of SM Context Status notify messages successful

Labels

peerid, dnn, snssai

- smfcc_nsmf_pdu_session_sm_context_status_notify_failures

Type

64-bit Counter

Description

Number of SM Context Status notify messages failed

Labels

peerid, dnn, snssai, causecode

- smfcc_namf_evts_subscr_create_attempts

Type

64-bit Counter

Description

Number of Subscription Create Request messages sent

Labels

peerid, dnn, snssai

- smfcc_namf_evts_subscr_create_successes

Type

64-bit Counter

Description

Number of Subscription Create Request messages successful

Labels

peerid, dnn, snssai

- smfcc_namf_evts_subscr_create_failures

Type

64-bit Counter

Description

Number of Subscription Create Request messages failed

Labels
<i>peerid, dnn, snssai, causecode</i>
• smfcc_namf_evts_event_notification_attempts
Type
64-bit Counter
Description
Number of AMF Event Notification Request messages received
Labels
<i>dnn, snssai</i>
• smfcc_namf_evts_event_notification_successes
Type
64-bit Counter
Description
Number of AMF Event Notification Request messages successful
Labels
<i>dnn, snssai</i>
• smfcc_namf_evts_event_notification_failures
Type
64-bit Counter
Description
Number of AMF Event Notification Request messages failed
Labels
<i>dnn, snssai, causecode</i>
• smfcc_namf_evts_subscr_modify_attempts
Type
64-bit Counter
Description
Number of Subscription Modify Request messages sent
Labels
<i>peerid, dnn, snssai</i>
• smfcc_namf_evts_subscr_modify_successes
Type
64-bit Counter
Description
Number of Subscription Modify Request messages successful
Labels
<i>peerid, dnn, snssai</i>
• smfcc_namf_evts_subscr_modify_failures
Type

64-bit Counter

Description

Number of Subscription Modify Request messages failed

Labels

peerid, dnn, snssai, causecode

• smfcc_namf_evts_subscr_delete_attempts

Type

64-bit Counter

Description

Number of Subscription Delete Request messages sent

Labels

peerid, dnn, snssai

• smfcc_namf_evts_subscr_delete_successes

Type

64-bit Counter

Description

Number of Subscription Delete Request messages successful

Labels

peerid, dnn, snssai

• smfcc_namf_evts_subscr_delete_failures

Type

64-bit Counter

Description

Number of Subscription Delete Request messages failed

Labels

peerid, dnn, snssai, causecode

• smfcc_inband_radius_tunnel_timer_start_attempts

Type

64-bit Counter

Description

Number of Inband Radius Tunnel Create/Delete Timer Start Request messages sent

Labels

• smfcc_inband_radius_tunnel_timer_start_successes

Type

64-bit Counter

Description

Number of Inband Radius Tunnel Create/Delete Timer Start Request Create successful

Labels

- smfcc_inband_radius_tunnel_timer_start_failures

Type

64-bit Counter

Description

Number of Inband Radius Tunnel Create/Delete Timer Start Request messages failed

Labels

causecode
- smfcc_inband_radius_tunnel_timer_fires

Type

64-bit Counter

Description

Number of Inband Radius Tunnel Create/Delete Timer fired

Labels

- smfcc_inband_radius_tunnel_timer_restarts

Type

64-bit Counter

Description

Number of Inband Radius Tunnel Create/Delete Timer Re-Start Request messages sent

Labels

- smfcc_nsmf_pdu_session_create_sm_context_punted

Type

64-bit Counter

Description

None

Metrics

Instances	Labels	Description
1	*****	Number of Create SM Context request messages punted
- smfcc_nsmf_pdu_session_update_sm_context_punted

Type

64-bit Counter

Description

None

Metrics

Instances	Labels	Description
1	*****	Number of Update SM Context request messages punted
- smfcc_nsmf_pdu_session_release_sm_context_punted

Type

64-bit Counter

Description		
None		
Metrics		
Instances	Labels	Description
1	*****	Number of Release SM Context request messages punted
• smfcc_nsmf_pdu_session_retrieve_sm_context_punted		
Type		
64-bit Counter		
Description		
None		
Metrics		
Instances	Labels	Description
1	*****	Number of Retrieve request messages punted

SMF PCF Interface Call Control Procedure Statistics

• smfcc_nsmf_smpolicy_control_create_service_attempts		
Type		
64-bit Counter		
Description		
Number of Create SmPolicy Control Service requests attempted		
Labels		
peerid		
• smfcc_nsmf_smpolicy_control_create_service_success		
Type		
64-bit Counter		
Description		
Number of Create SmPolicy Control Service requests successful		
Labels		
peerid		
• smfcc_nsmf_smpolicy_control_create_service_failures		
Type		
64-bit Counter		
Description		
Number of Create SmPolicyControl Service requests failed		
Labels		
peerid, causecode		
• smfcc_nsmf_smpolicy_control_create_service_retry_attempts		
Type		
64-bit Counter		

Description

Number of Create SmPolicy Control Service requests attempted to alternate PCF

Labels

- smfcc_nsmf_smpolicy_control_create_service_failures_causing_session_continue

Type

64-bit Counter

Description

Number of Create SmPolicy Control Service failures causing session continue

Labels

- smfcc_nsmf_smpolicy_control_create_service_failures_causing_session_deactivate

Type

64-bit Counter

Description

Number of Create SmPolicy Control Service failures causing session deactivate

Labels

- smfcc_nsmf_smpolicy_control_update_service_attempts

Type

64-bit Counter

Description

Number of Update SmPolicy Control Service requests attempted

Labels

peerid

- smfcc_nsmf_smpolicy_control_update_service_success

Type

64-bit Counter

Description

Number of Update SmPolicy Control Service requests successful

Labels

peerid

- smfcc_nsmf_smpolicy_control_update_service_failures

Type

64-bit Counter

Description

Number of Update SmPolicyControl Service requests failed

Labels

peerid, causecode

- smfcc_nsmf_smpolicy_control_delete_service_attempts

Type

64-bit Counter

Description

Number of Delete SmPolicy Control Service requests attempted

Labels

peerid

- smfcc_nsmf_smpolicy_control_delete_service_success

Type

64-bit Counter

Description

Number of Delete SmPolicy Control Service requests successful

Labels

peerid

- smfcc_nsmf_smpolicy_control_delete_service_failures

Type

64-bit Counter

Description

Number of Delete SmPolicyControl Service requests failed

Labels

peerid, causecode

- smfcc_nsmf_smpolicy_control_update_notify_attempts

Type

64-bit Counter

Description

Number of SmPolicy Control Update Notify requests attempted

Labels

peerid

- smfcc_nsmf_smpolicy_control_update_notify_success

Type

64-bit Counter

Description

Number of SmPolicy Control Update Notify requests successful

Labels

peerid

- smfcc_nsmf_smpolicy_control_update_notify_failures

Type

64-bit Counter

Description

Number of SmPolicyControl Update Notify requests failed

Labels

peerid, causecode

• smfcc_nsmf_smpolicy_control_terminate_notify_attempts

Type

64-bit Counter

Description

Number of SmPolicy Control Terminate Notify requests attempted

Labels

peerid

• smfcc_nsmf_smpolicy_control_terminate_notify_success

Type

64-bit Counter

Description

Number of SmPolicy Control Terminate Notify requests successful

Labels

peerid

• smfcc_nsmf_smpolicy_control_terminate_notify_failures

Type

64-bit Counter

Description

Number of SmPolicyControl Terminate Notify requests failed

Labels

peerid, causecode

• smfcc_nsmf_smpolicy_control_update_notify_punted

Type

64-bit Counter

Description

None

Metrics

Instances	Labels	Description
1	*****	Number of SmPolicy Control Update Notify requests punted

• smfcc_nsmf_smpolicy_control_terminate_notify_punted

Type

64-bit Counter

Description

None

Metrics

Instances	Labels	Description
1	*****	Number of SmPolicy Control Terminate Notify requests punted

- smfcc_gx_ccr_init_attempts

Type
64-bit Counter
Description
Number of CCR-I attempted
Labels

peerid

- smfcc_gx_ccr_init_success

Type
64-bit Counter
Description
Number of CCR-I successful
Labels

peerid

- smfcc_gx_ccr_init_failures

Type
64-bit Counter
Description
Number of CCR-I failed
Labels

peerid, causecode

- smfcc_gx_ccr_update_attempts

Type
64-bit Counter
Description
Number of CCR-U attempted
Labels

peerid

- smfcc_gx_ccr_update_success

Type
64-bit Counter
Description
Number of CCR-U successful
Labels

peerid

- smfcc_gx_ccr_update_failures

Type
64-bit Counter
Description

Number of CCR-U failed

Labels

peerid, causecode

• smfcc_gx_ccr_terminate_attempts

Type

64-bit Counter

Description

Number of CCR-T attempted

Labels

peerid

• smfcc_gx_ccr_terminate_success

Type

64-bit Counter

Description

Number of CCR-T successful

Labels

peerid

• smfcc_gx_ccr_terminate_failures

Type

64-bit Counter

Description

Number of CCR-T failed

Labels

peerid, causecode

• smfcc_gx_rar_received

Type

64-bit Counter

Description

Number of RAR received from diameter-base

Labels

peerid

• smfcc_gx_rar_rsp_success

Type

64-bit Counter

Description

Number of RAR response/RAA successfully sent to diameter-base

Labels

peerid

• smfcc_gx_rar_rsp_failures

Type

64-bit Counter

Description

Number of RAR response/RAA failed to sent to diameter-base

Labels

peerid, causecode

• smfcc_gx_rar_terminate_received

Type

64-bit Counter

Description

Number of RAR-terminate request received

Labels

peerid

• smfcc_gx_rar_terminate_success

Type

64-bit Counter

Description

Number of RAR-terminate successful

Labels

peerid

• smfcc_gx_rar_terminate_failures

Type

64-bit Counter

Description

Number of RAR-terminate failed

Labels

peerid, causecode

SMF Call Control Procedure Statistics

• smfcc_pdu_session_establishment_attempts

Type

64-bit Counter

Description

Number of PDU Session Establishment Procedures attempted

Labels

dnn, snssai

• smfcc_pdu_session_establishment_successes

Type

64-bit Counter

Description

Number of PDU Session Establishment Procedures successful

Labels

dnn, snssai

• smfcc_pdu_session_establishment_failures

Type

64-bit Counter

Description

Number of PDU Session Establishment Procedures failed

Labels

dnn, snssai, causecode

• smfcc_ue_deregistration_attempts

Type

64-bit Counter

Description

Number of UE Deregistration Procedures attempted

Labels

dnn, snssai

• smfcc_ue_deregistration_successes

Type

64-bit Counter

Description

Number of UE Deregistration Procedures successful

Labels

dnn, snssai

• smfcc_ue_deregistration_failures

Type

64-bit Counter

Description

Number of UE Deregistration Procedures failed

Labels

dnn, snssai, causecode

• smfcc_vsmf_ue_deregistration_attempts

Type

64-bit Counter

Description

Number of VSMF UE Deregistration Procedures attempted

Labels

dnn, snssai

• smfcc_vsmf_ue_deregistration_successes

Type

64-bit Counter

Description

Number of VSMF UE Deregistration Procedures successful

Labels

dnn, snssai

• smfcc_vsmf_ue_deregistration_failures

Type

64-bit Counter

Description

Number of VSMF UE Deregistration Procedures failed

Labels

dnn, snssai, causecode

• smfcc_an_release_attempts

Type

64-bit Counter

Description

Number of AN Release Procedures attempted

Labels

dnn, snssai

• smfcc_an_release_successes

Type

64-bit Counter

Description

Number of AN Release Procedures successful

Labels

dnn, snssai

• smfcc_an_release_failures

Type

64-bit Counter

Description

Number of AN Release Procedures failed

Labels

dnn, snssai, causecode

• smfcc_service_request_attempts

Type

64-bit Counter

Description

Number of Service Request Procedures attempted

Labels

dnn, snssai, amfmobility, iupfrelocation

• smfcc_service_request_successes

Type

64-bit Counter

Description

Number of Service Request Procedures successful

Labels

dnn, snssai, amfmobility, iupfrelocation

• smfcc_service_request_failures

Type

64-bit Counter

Description

Number of Service Request Procedures failed

Labels

dnn, snssai, causecode, amfmobility, iupfrelocation

• smfcc_nw_initiated_release_attempts

Type

64-bit Counter

Description

Number of NW Initiated Release Procedures attempted

Labels

dnn, snssai, causecode

• smfcc_nw_initiated_release_successes

Type

64-bit Counter

Description

Number of NW Initiated Release Procedures successful

Labels

dnn, snssai

• smfcc_nw_initiated_release_failures

Type

64-bit Counter

Description

Number of NW Initiated Release Procedures failed

Labels

dnn, snssai, causecode

• smfcc_xn_handover_initiated_release_attempts

Type

64-bit Counter

Description

Number of Xn Handover Initiated Release Procedures attempted

Labels

dnn, snssai

• smfcc_xn_handover_initiated_release_successes

Type

64-bit Counter

Description

Number of Xn Handover Initiated Release Procedures successful

Labels

dnn, snssai

• smfcc_xn_handover_initiated_release_failures

Type

64-bit Counter

Description

Number of Xn Handover Initiated Release Procedures failed

Labels

dnn, snssai, causecode

• smfcc_xn_handover_attempts

Type

64-bit Counter

Description

Number of Xn Handover Procedures attempted

Labels

dnn, snssai, iupfrelocation

• smfcc_xn_handover_successes

Type

64-bit Counter

Description

Number of Xn Handover Procedures succeeded

Labels

dnn, snssai, iupfrelocation

• smfcc_xn_handover_failures

Type

64-bit Counter

Description

Number of Xn Handover Procedures failed

Labels

dnn, snssai, causecode, iupfrelocation

- smfcc_xn_handover_idle_mode_successes

Type

64-bit Counter

Description

Number of Xn Handover Procedures succeeded for idle mode

Labels

dnn, snssai, iupfrelocation

- smfcc_xn_handover_idle_mode_failures

Type

64-bit Counter

Description

Number of Xn Handover Procedures failed for idle mode

Labels

dnn, snssai, causecode, iupfrelocation

- smfcc_n2_handover_attempts

Type

64-bit Counter

Description

Number of N2 Handover Procedures attempted

Labels

dnn, snssai, amfmobility, iupfrelocation

- smfcc_n2_handover_successes

Type

64-bit Counter

Description

Number of N2 Handover Procedures successful

Labels

dnn, snssai, amfmobility, iupfrelocation

- smfcc_n2_handover_failures

Type

64-bit Counter

Description

Number of N2 Handover Procedures failed

Labels

dnn, snssai, causecode, amfmobility, iupfrelocation

- smfcc_n2_handover_cancels

Type

64-bit Counter

Description
Number of N2 Handover Procedures canceled
Labels
dnn, snssai, amfmobility, iupfrelocation
• smfcc_vsmf_n2_handover_attempts
Type
64-bit Counter
Description
Number of VSMF N2 Handover Procedures attempted for inbound roamers
Labels
dnn, snssai, amfmobility, iupfrelocation
• smfcc_vsmf_n2_handover_successes
Type
64-bit Counter
Description
Number of VSMF N2 Handover Procedures succeeded for inbound roamers
Labels
dnn, snssai, amfmobility, iupfrelocation
• smfcc_vsmf_n2_handover_failures
Type
64-bit Counter
Description
Number of VSMF N2 Handover Procedures failed for inbound roamers
Labels
dnn, snssai, causecode, amfmobility, iupfrelocation
• smfcc_vsmf_n2_handover_cancels
Type
64-bit Counter
Description
Number of VSMF N2 Handover Procedures cancelled for inbound roamers
Labels
dnn, snssai, amfmobility, iupfrelocation
• smfcc_ue_initiated_modify_qos_attempts
Type
64-bit Counter
Description
Number of UE Initiated Modify Qos Procedures attempted
Labels
dnn, snssai

smfcc_ue_initiated_modify_qos_successes
Type
64-bit Counter
Description
Number of UE Initiated Modify Qos Procedures successful
Labels
dnn, snssai
smfcc_ue_initiated_modify_qos_failures
Type
64-bit Counter
Description
Number of UE Initiated Modify Qos Procedures failed
Labels
dnn, snssai, causecode
smfcc_vsmf_ue_initiated_modify_attempts
Type
64-bit Counter
Description
Number of UE Initiated Modify Procedures attempted
Labels
dnn, snssai
smfcc_vsmf_ue_initiated_modify_successes
Type
64-bit Counter
Description
Number of UE Initiated Modify Procedures successful
Labels
dnn, snssai
smfcc_vsmf_ue_initiated_modify_failures
Type
64-bit Counter
Description
Number of UE Initiated Modify Procedures failed
Labels
dnn, snssai, causecode
smfcc_hsmf_ue_initiated_modify_attempts
Type
64-bit Counter
Description

Number of UE Initiated Modify Procedures attempted

Labels

dnn, snssai

- smfcc_hsmf_ue_initiated_modify_successes

Type

64-bit Counter

Description

Number of UE Initiated Modify Procedures successful

Labels

dnn, snssai

- smfcc_hsmf_ue_initiated_modify_failures

Type

64-bit Counter

Description

Number of UE Initiated Modify Procedures failed

Labels

dnn, snssai, causecode

- smfcc_pcf_initiated_modify_qos_attempts

Type

64-bit Counter

Description

Number of PCF Initiated Modify Qos Procedures attempted

Labels

dnn, snssai

- smfcc_pcf_initiated_modify_qos_successes

Type

64-bit Counter

Description

Number of PCF Initiated Modify Qos Procedures successful

Labels

dnn, snssai

- smfcc_pcf_initiated_modify_qos_failures

Type

64-bit Counter

Description

Number of PCF Initiated Modify Qos Procedures failed

Labels

dnn, snssai, causecode

- smfcc_pcf_initiated_modify_qos_successes_with_n2_partial_failure

Type

64-bit Counter

Description

Number of PCF Initiated Modify Qos Procedures successful with n2 partial failure

Labels

dnn, snssai

• smfcc_ue_initiated_pdu_session_release_attempts

Type

64-bit Counter

Description

Number of UE Initiated PDU Session Release Procedures attempted

Labels

dnn, snssai

• smfcc_ue_initiated_pdu_session_release_successes

Type

64-bit Counter

Description

Number of UE Initiated PDU Session Release Procedures successful

Labels

dnn, snssai

• smfcc_ue_initiated_pdu_session_release_failures

Type

64-bit Counter

Description

Number of UE Initiated PDU Session Release Procedures failed

Labels

dnn, snssai, causecode

• smfcc_vsmf_ue_initiated_pdu_session_release_attempts

Type

64-bit Counter

Description

Number of VSMF UE Initiated PDU Session Release Procedures attempted for in-bound roamers

Labels

dnn, snssai

• smfcc_vsmf_ue_initiated_pdu_session_release_successes

Type

64-bit Counter

Description

Number of VSMF UE Initiated PDU Session Release Procedures successful for in-bound roamers

Labels

dnn, snssai

• smfcc_vsmf_ue_initiated_pdu_session_release_failures

Type

64-bit Counter

Description

Number of VSMF UE Initiated PDU Session Release Procedures failed for in-bound roamers

Labels

dnn, snssai, causecode

• smfcc_hsmf_ue_initiated_pdu_session_release_attempts

Type

64-bit Counter

Description

Number of HSMF UE Initiated PDU Session Release Procedures attempted for out-bound roamers

Labels

dnn, snssai

• smfcc_hsmf_ue_initiated_pdu_session_release_successes

Type

64-bit Counter

Description

Number of HSMF UE Initiated PDU Session Release Procedures successful for out-bound roamers

Labels

dnn, snssai

• smfcc_hsmf_ue_initiated_pdu_session_release_failures

Type

64-bit Counter

Description

Number of HSMF UE Initiated PDU Session Release Procedures failed for out-bound roamers

Labels

dnn, snssai, causecode

• smfcc_vsmf_initiated_release_attempts

Type

64-bit Counter

Description

Number of VSMF Initiated Release Procedures attempted

Labels

dnn, snssai, causecode

• smfcc_vsmf_initiated_release_successes

Type

64-bit Counter

Description

Number of VSMF Initiated Release Procedures successful

Labels

dnn, snssai

• smfcc_vsmf_initiated_release_failures

Type

64-bit Counter

Description

Number of VSMF Initiated Release Procedures failed

Labels

dnn, snssai, causecode

• smfcc_amf_initiated_pdu_session_release_attempts

Type

64-bit Counter

Description

Number of AMF Initiated PDU Session Release Procedures attempted

Labels

dnn, snssai

• smfcc_amf_initiated_pdu_session_release_successes

Type

64-bit Counter

Description

Number of AMF Initiated PDU Session Release Procedures successful

Labels

dnn, snssai

• smfcc_amf_initiated_pdu_session_release_failures

Type

64-bit Counter

Description

Number of AMF Initiated PDU Session Release Procedures failed

Labels

dnn, snssai, causecode

• smfcc_network_initiated_service_request_attempts

Type

64-bit Counter

Description

Number of Network Initiated Service Request Procedures attempted

Labels

dnn, snssai

- smfcc_network_initiated_service_request_successes

Type

64-bit Counter

Description

Number of Network Initiated Service Request Procedures successful

Labels

dnn, snssai

- smfcc_network_initiated_service_request_failures

Type

64-bit Counter

Description

Number of Network Initiated Service Request Procedures failed

Labels

dnn, snssai, causecode

- smfcc_pcf_initiated_terminate_attempts

Type

64-bit Counter

Description

Number of PCF Initiated Terminate Procedures attempted

Labels

dnn, snssai

- smfcc_pcf_initiated_terminate_successes

Type

64-bit Counter

Description

Number of PCF Initiated Terminate Procedures successful

Labels

dnn, snssai

- smfcc_pcf_initiated_terminate_failures

Type

64-bit Counter

Description

Number of PCF Initiated Terminate Procedures failed

Labels

dnn, snssai, causecode

- smfcc_hsmf_pcf_initiated_terminate_attempts

Type

64-bit Counter

Description

Number of HSMF PCF Initiated Terminate Procedures attempted

Labels

dnn, snssai

• smfcc_hsmf_pcf_initiated_terminate_successes

Type

64-bit Counter

Description

Number of HSMF PCF Initiated Terminate Procedures successful

Labels

dnn, snssai

• smfcc_hsmf_pcf_initiated_terminate_failures

Type

64-bit Counter

Description

Number of HSMF PCF Initiated Terminate Procedures failed

Labels

dnn, snssai, causecode

• smfcc_upf_usage_report_procedure_attempts

Type

64-bit Counter

Description

Number of Usage Report Processing Procedure attempted

Labels

dnn, snssai

• smfcc_upf_usage_report_procedure_successes

Type

64-bit Counter

Description

Number of Usage Report Processing Procedure successful

Labels

dnn, snssai

• smfcc_upf_usage_report_procedure_failures

Type

64-bit Counter

Description

Number of Usage Report Processing Procedures failed

Labels

dnn, snssai, causecode

smfcc_amf_mobility_attempts
Type
64-bit Counter
Description
Number of Amf Mobility Procedures attempted
Labels
dnn, snssai
smfcc_amf_mobility_successes
Type
64-bit Counter
Description
Number of Amf Mobility Procedures successful
Labels
dnn, snssai
smfcc_amf_mobility_failures
Type
64-bit Counter
Description
Number of Amf Mobility Procedures failed
Labels
dnn, snssai, causecode
smfcc_vsmf_error_indication_procedure_attempts
Type
64-bit Counter
Description
Number of VSMF Error Indication(from RAN) Procedures attempted
Labels
dnn, snssai
smfcc_vsmf_error_indication_procedure_successes
Type
64-bit Counter
Description
Number of VSMF Error Indication(from RAN) Procedures successful
Labels
dnn, snssai
smfcc_vsmf_error_indication_procedure_failures
Type
64-bit Counter
Description

Number of VSMF Error Indication(from RAN) Procedures failed

Labels

dnn, snssai, causecode

• smfcc_hsmf_error_indication_procedure_attempts

Type

64-bit Counter

Description

Number of HSMF Error Indication(from VSMF) Procedures attempted

Labels

dnn, snssai

• smfcc_hsmf_error_indication_procedure_successes

Type

64-bit Counter

Description

Number of HSMF Error Indication(from VSMF) Procedures successful

Labels

dnn, snssai

• smfcc_hsmf_error_indication_procedure_failures

Type

64-bit Counter

Description

Number of HSMF Error Indication(from HSMF) Procedures failed

Labels

dnn, snssai, causecode

• smfcc_vsmf_up_inactivity_procedure_attempts

Type

64-bit Counter

Description

Number of VSMF UP Inactivity Procedures attempted

Labels

dnn, snssai

• smfcc_vsmf_up_inactivity_procedure_successes

Type

64-bit Counter

Description

Number of VSMF UP Inactivity Procedures successful

Labels

dnn, snssai

• smfcc_vsmf_up_inactivity_procedure_failures

Type

64-bit Counter

Description

Number of VSMF UP Inactivity Procedures failed

Labels

dnn, snssai, causecode

• smfcc_hsmf_up_inactivity_procedure_attempts

Type

64-bit Counter

Description

Number of HSMF UP Inactivity Procedures attempted

Labels

dnn, snssai

• smfcc_hsmf_up_inactivity_procedure_successes

Type

64-bit Counter

Description

Number of HSMF UP Inactivity Procedures successful

Labels

dnn, snssai

• smfcc_hsmf_up_inactivity_procedure_failures

Type

64-bit Counter

Description

Number of HSMF UP Inactivity Procedures failed

Labels

dnn, snssai, causecode

• smfcc_error_indication_procedure_attempts

Type

64-bit Counter

Description

Number of Error Indication Procedures attempted

Labels

dnn, snssai

• smfcc_error_indication_procedure_successes

Type

64-bit Counter

Description

Number of Error Indication Procedures successful

Labels

dnn, snssai

• smfcc_error_indication_procedure_failures

Type

64-bit Counter

Description

Number of Error Indication Procedures failed

Labels

dnn, snssai, causecode

• smfcc_up_inactivity_procedure_attempts

Type

64-bit Counter

Description

Number of UP Inactivity Procedures attempted

Labels

dnn, snssai

• smfcc_up_inactivity_procedure_successes

Type

64-bit Counter

Description

Number of UP Inactivity Procedures successful

Labels

dnn, snssai

• smfcc_up_inactivity_procedure_failures

Type

64-bit Counter

Description

Number of UP Inactivity Procedures failed

Labels

dnn, snssai, causecode

• smfcc_process_pcf_update_policy_rules_attempts

Type

64-bit Counter

Description

Number of Process Pcf Update Policy Rules Procedures attempted

Labels

dnn, snssai

• smfcc_process_pcf_update_policy_rules_successes

Type

64-bit Counter

Description

Number of Process Pcf Update Policy Rules Procedures successful

Labels

dnn, snssai

- smfcc_process_pcf_update_policy_rules_failures

Type

64-bit Counter

Description

Number of Process Pcf Update Policy Rules Procedures failed

Labels

dnn, snssai, causecode

- smfcc_process_hsmf_pcf_update_policy_rules_attempts

Type

64-bit Counter

Description

Number of Process HSMF Pcf Update Policy Rules Procedures attempted

Labels

dnn, snssai

- smfcc_process_hsmf_pcf_update_policy_rules_successes

Type

64-bit Counter

Description

Number of Process HSMF Pcf Update Policy Rules Procedures successful

Labels

dnn, snssai

- smfcc_process_hsmf_pcf_update_policy_rules_failures

Type

64-bit Counter

Description

Number of Process HSMF Pcf Update Policy Rules Procedures failed

Labels

dnn, snssai, causecode

- smfcc_process_pcf_update_policy_rules_successes_with_n2_partial_failure

Type

64-bit Counter

Description

Number of Process Pcf Update Policy Rules Procedures successful with n2 partial failure

Labels

dnn, snssai

- smfcc_5gs_to_eps_handover_attempts

Type

64-bit Counter

Description

Number of 5GS to EPS Handover Procedures attempted

Labels

dnn, snssai

- smfcc_5gs_to_eps_handover_successes

Type

64-bit Counter

Description

Number of 5GS to EPS Handover Procedures successful

Labels

dnn, snssai

- smfcc_5gs_to_eps_handover_failures

Type

64-bit Counter

Description

Number of 5GS to EPS Handover Procedures failed

Labels

dnn, snssai, causecode

- smfcc_hsmf_5gs_to_eps_handover_attempts

Type

64-bit Counter

Description

Number of HSMF 5GS to EPS Handover Procedures attempted

Labels

dnn, snssai

- smfcc_hsmf_5gs_to_eps_handover_successes

Type

64-bit Counter

Description

Number of HSMF 5GS to EPS Handover Procedures successful

Labels

dnn, snssai

- smfcc_hsmf_5gs_to_eps_handover_failures

Type

64-bit Counter

Description
Number of HSMF 5GS to EPS Handover Procedures failed
Labels
<i>dnn, snssai, causecode</i>
• smfcc_5gs_to_eps_idle_mobility_attempts
Type
64-bit Counter
Description
Number of 5GS to EPS Idle Mode Mobility Procedures attempted
Labels
<i>dnn, snssai</i>
• smfcc_5gs_to_eps_idle_mobility_successes
Type
64-bit Counter
Description
Number of 5GS to EPS Idle Mode Mobility Procedures successful
Labels
<i>dnn, snssai</i>
• smfcc_5gs_to_eps_idle_mobility_failures
Type
64-bit Counter
Description
Number of 5GS to EPS Idle Mode Mobility Procedures failed
Labels
<i>dnn, snssai, causecode</i>
• smfcc_eps_to_5gs_handover_attempts
Type
64-bit Counter
Description
Number of EPS to 5GS Handover Procedures attempted
Labels
<i>dnn, snssai, iupfrelocation</i>
• smfcc_eps_to_5gs_handover_successes
Type
64-bit Counter
Description
Number of EPS to 5GS Handover Procedures successful
Labels
<i>dnn, snssai, iupfrelocation</i>

- smfcc_eps_to_5gs_handover_failures
 - Type
 - 64-bit Counter
 - Description
 - Number of EPS to 5GS Handover Procedures failed
 - Labels
 - dnn, snssai, causecode, iupfrelocation**
- smfcc_eps_to_5gs_handover_cancels
 - Type
 - 64-bit Counter
 - Description
 - Number of EPS to 5GS Handover Procedures canceled
 - Labels
 - dnn, snssai, iupfrelocation**
- smfcc_eps_to_5gs_idle_mobility_attempts
 - Type
 - 64-bit Counter
 - Description
 - Number of EPS to 5GS in Idle Mode Mobility Procedures attempted
 - Labels
 - dnn, snssai**
- smfcc_eps_to_5gs_idle_mobility_successes
 - Type
 - 64-bit Counter
 - Description
 - Number of EPS to 5GS in Idle Mode Mobility Procedures successful
 - Labels
 - dnn, snssai**
- smfcc_eps_to_5gs_idle_mobility_failures
 - Type
 - 64-bit Counter
 - Description
 - Number of EPS to 5GS in Idle Mode Mobility Procedures failed
 - Labels
 - dnn, snssai, causecode**
- smfcc_vsmf_eps_to_5gs_idle_mobility_attempts
 - Type
 - 64-bit Counter
 - Description

Number of VSMF EPS to 5GS in Idle Mode Mobility Procedures attempted

Labels

dnn, snssai

• smfcc_vsmf_eps_to_5gs_idle_mobility_successes

Type

64-bit Counter

Description

Number of VSMF EPS to 5GS in Idle Mode Mobility Procedures successful

Labels

dnn, snssai

• smfcc_vsmf_eps_to_5gs_idle_mobility_failures

Type

64-bit Counter

Description

Number of VSMF EPS to 5GS in Idle Mode Mobility Procedures failed

Labels

dnn, snssai, causecode

• smfcc_hsmf_eps_to_5gs_idle_mobility_attempts

Type

64-bit Counter

Description

Number of HSMF EPS to 5GS in Idle Mode Mobility Procedures attempted

Labels

dnn, snssai

• smfcc_hsmf_eps_to_5gs_idle_mobility_successes

Type

64-bit Counter

Description

Number of HSMF EPS to 5GS in Idle Mode Mobility Procedures successful

Labels

dnn, snssai

• smfcc_hsmf_eps_to_5gs_idle_mobility_failures

Type

64-bit Counter

Description

Number of HSMF EPS to 5GS in Idle Mode Mobility Procedures failed

Labels

dnn, snssai, causecode

• smfcc_chf_server_notification_procedure_attempts

Type

64-bit Counter

Description

Number of CHF Server Notification Procedure attempted

Labels

dnn, snssai

- smfcc_chf_server_notification_procedure_successes

Type

64-bit Counter

Description

Number of CHF Server Notification Processing Procedure successful

Labels

dnn, snssai

- smfcc_chf_server_notification_procedure_failures

Type

64-bit Counter

Description

Number of CHF Server Notification Processing Procedures failed

Labels

dnn, snssai, causecode

- smfcc_hsmf_chf_server_notification_procedure_attempts

Type

64-bit Counter

Description

Number of HSMF CHF Server Notification Procedure attempted

Labels

dnn, snssai

- smfcc_hsmf_chf_server_notification_procedure_successes

Type

64-bit Counter

Description

Number of HSMF CHF Server Notification Processing Procedure successful

Labels

dnn, snssai

- smfcc_hsmf_chf_server_notification_procedure_failures

Type

64-bit Counter

Description

Number of HSMF CHF Server Notification Processing Procedures failed

Labels

dnn, snssai, causecode

- smfcc_4g_to_epdg_handover_attempts

Type

64-bit Counter

Description

Number of 4G to EPDG Handover Procedures attempted

Labels

dnn, snssai

- smfcc_4g_to_epdg_handover_successes

Type

64-bit Counter

Description

Number of 4G to EPDG Handover Procedures successful

Labels

dnn, snssai

- smfcc_4g_to_epdg_handover_failures

Type

64-bit Counter

Description

Number of 4G to EPDG Handover Procedures failed

Labels

dnn, snssai, causecode

- smfcc_5gs_to_epdg_handover_attempts

Type

64-bit Counter

Description

Number of 5GS to EPDG Handover Procedures attempted

Labels

dnn, snssai

- smfcc_5gs_to_epdg_handover_successes

Type

64-bit Counter

Description

Number of 5GS to EPDG Handover Procedures successful

Labels

dnn, snssai

- smfcc_5gs_to_epdg_handover_failures

Type

64-bit Counter

Description

Number of 5GS to EPDG Handover Procedures failed

Labels

dnn, snssai, causecode

• smfcc_5gs_to_epdg_idle_handover_attempts

Type

64-bit Counter

Description

Number of 5GS to EPDG Idle Handover Procedures attempted

Labels

dnn, snssai

• smfcc_5gs_to_epdg_idle_handover_successes

Type

64-bit Counter

Description

Number of 5GS to EPDG Idle Handover Procedures successful

Labels

dnn, snssai

• smfcc_5gs_to_epdg_idle_handover_failures

Type

64-bit Counter

Description

Number of 5GS to EPDG Idle Handover Procedures failed

Labels

dnn, snssai, causecode

• smfcc_non3gpp_to_5gs_handover_attempts

Type

64-bit Counter

Description

Number of Non-3gpp to 5GS Handover Procedures attempted

Labels

dnn, snssai

• smfcc_non3gpp_to_5gs_handover_successes

Type

64-bit Counter

Description

Number of Non-3gpp to 5GS Handover Procedures successful

Labels

dnn, snssai

- smfcc_non3gpp_to_5gs_handover_failures

Type

64-bit Counter

Description

Number of Non-3gpp to 5GS Handover Procedures failed

Labels

dnn, snssai, causecode

- smfcc_non3gpp_to_5gs_idle_mobility_attempts

Type

64-bit Counter

Description

Number of Non-3gpp to 5GS Idle Mode Mobility Procedures attempted

Labels

dnn, snssai

- smfcc_non3gpp_to_5gs_idle_mobility_successes

Type

64-bit Counter

Description

Number of Non-3gpp to 5GS Idle Mode Mobility Procedures successful

Labels

dnn, snssai

- smfcc_non3gpp_to_5gs_idle_mobility_failures

Type

64-bit Counter

Description

Number of Non-3gpp to 5GS Idle Mode Mobility Procedures failed

Labels

dnn, snssai, causecode

- smfcc_epdg_to_eps_handover_attempts

Type

64-bit Counter

Description

Number of ePDG to EPS Handover Procedures attempted

Labels

dnn, snssai

- smfcc_epdg_to_eps_handover_successes

Type

64-bit Counter

Description

Number of ePDG to EPS Handover Procedures successful

Labels

dnn, snssai

• smfcc_epdg_to_eps_handover_failures

Type

64-bit Counter

Description

Number of ePDG to EPS Handover Procedures failed

Labels

dnn, snssai, causecode

• smfcc_sdm_update_notification_procedure_attempts

Type

64-bit Counter

Description

Number of SDM Update Notification Procedure attempted

Labels

dnn, snssai

• smfcc_sdm_update_notification_procedure_successes

Type

64-bit Counter

Description

Number of SDM Update Notification Processing Procedure successful

Labels

dnn, snssai

• smfcc_sdm_update_notification_procedure_failures

Type

64-bit Counter

Description

Number of SDM Update Notification Processing Procedures failed

Labels

dnn, snssai, causecode

• smfcc_rejects_overload

Type

64-bit Counter

Description

Number of Procedures rejected due to overload

Labels

dnn, snssai, proceduretype, overloadlevel

- smfcc_qos_flow_inactivity_procedure_attempts

Type
64-bit Counter
Description
Number of QosFlow Inactivity Procedures attempted
Labels
dnn, snssai

- smfcc_qos_flow_inactivity_procedure_successes

Type
64-bit Counter
Description
Number of QosFlow Inactivity Procedures successful
Labels
dnn, snssai

- smfcc_qos_flow_inactivity_procedure_failures

Type
64-bit Counter
Description
Number of QosFlow Inactivity Procedures failed
Labels
dnn, snssai, causecode

- smfcc_session_idle_procedure_attempts

Type
64-bit Counter
Description
Number of Session Idle Procedures attempted
Labels
dnn, snssai

- smfcc_session_idle_procedure_successes

Type
64-bit Counter
Description
Number of Session Idle Procedures successful
Labels
dnn, snssai

- smfcc_session_idle_procedure_failures

Type
64-bit Counter
Description

Number of Session Idle Procedures failed

Labels

dnn, snssai, causecode

• smfcc_vsmf_session_idle_procedure_attempts

Type

64-bit Counter

Description

Number of VSMF Session Idle Procedures attempted

Labels

dnn, snssai

• smfcc_vsmf_session_idle_procedure_successes

Type

64-bit Counter

Description

Number of VSMF Session Idle Procedures successful

Labels

dnn, snssai

• smfcc_vsmf_session_idle_procedure_failures

Type

64-bit Counter

Description

Number of VSMF Session Idle Procedures failed

Labels

dnn, snssai, causecode

• smfcc_hsmf_session_idle_procedure_attempts

Type

64-bit Counter

Description

Number of HSMF Session Idle Procedures attempted

Labels

dnn, snssai

• smfcc_hsmf_session_idle_procedure_successes

Type

64-bit Counter

Description

Number of HSMF Session Idle Procedures successful

Labels

dnn, snssai

• smfcc_hsmf_session_idle_procedure_failures

Type

64-bit Counter

Description

Number of HSMF Session Idle Procedures failed

Labels

dnn, snssai, causecode

• smfcc_ue_initiated_modify_attempts

Type

64-bit Counter

Description

Number of UE Initiated Modify Procedures attempted

Labels

dnn, snssai

• smfcc_ue_initiated_modify_successes

Type

64-bit Counter

Description

Number of UE Initiated Modify Procedures successful

Labels

dnn, snssai

• smfcc_ue_initiated_modify_failures

Type

64-bit Counter

Description

Number of UE Initiated Modify Procedures failed

Labels

dnn, snssai, causecode

• smfcc_ssc_mode_3_pdu_session_establishment_attempts

Type

64-bit Counter

Description

Number of PDU Session Establishment Procedures due to ssc mode 3 reactivation attempted

Labels

dnn, snssai

• smfcc_ssc_mode_3_pdu_session_establishment_successes

Type

64-bit Counter

Description

Number of PDU Session Establishment Procedures due to ssc mode 3 reactivation succeeded

Labels

dnn, snssai

- smfcc_ssc_mode_3_pdu_session_establishment_failures

Type

64-bit Counter

Description

Number of PDU Session Establishment Procedures due to ssc mode 3 reactivation failed

Labels

dnn, snssai, causecode

- smfcc_ssc_mode_3_pdu_session_request_reactivation_attempts

Type

64-bit Counter

Description

Number of ssc mode 3 request reactivation procedures attempted

Labels

dnn, snssai

- smfcc_ssc_mode_3_pdu_session_request_reactivation_successes

Type

64-bit Counter

Description

Number of ssc mode 3 request reactivation procedures succeeded

Labels

dnn, snssai

- smfcc_ssc_mode_3_pdu_session_request_reactivation_failures

Type

64-bit Counter

Description

Number of ssc mode 3 request reactivation procedures failed

Labels

dnn, snssai, causecode

- smfcc_ssc_mode2_nw_initiated_release_attempts

Type

64-bit Counter

Description

Number of SSC-Mode-2 NW Initiated Release Procedures attempted

Labels

dnn, snssai

- smfcc_ssc_mode2_nw_initiated_release_successes

Type

64-bit Counter

Description

Number of SSC-Mode-2 NW Initiated Release Procedures successful

Labels

dnn, snssai

• smfcc_ssc_mode2_nw_initiated_release_failures

Type

64-bit Counter

Description

Number of SSC-Mode-2 NW Initiated Release Procedures failed

Labels

dnn, snssai, causecode

• smfcc_tft_update_after_5g_handover_attempts

Type

64-bit Counter

Description

Number of SMF Initiated TFT update after 5G Handover complete during collision of PCF update and Handover request Procedures attempted

Labels

dnn, snssai

• smfcc_tft_update_after_5g_handover_successes

Type

64-bit Counter

Description

Number of SMF Initiated TFT update after 5G Handover complete during collision of PCF update and Handover request Procedures successful

Labels

dnn, snssai

• smfcc_tft_update_after_5g_handover_failures

Type

64-bit Counter

Description

Number of SMF Initiated TFT update after 5G Handover complete during collision of PCF update and Handover request Procedures failed

Labels

dnn, snssai, causecode

• smfcc_upf_selection_under_overload

Type

64-bit Counter

Description

Number of UPF selections by SMF when all the NRF discovered UPFs are overloaded

Labels

dnn, snssai, upftype

• smfcc_pfd_fetch_attempts

Type

64-bit Counter

Description

Number of Pfd Fetch Procedures attempted

Labels

dnn, snssai

• smfcc_pfd_fetch_successes

Type

64-bit Counter

Description

Number of Pfd Fetch Procedures successful

Labels

dnn, snssai

• smfcc_pfd_fetch_failures

Type

64-bit Counter

Description

Number of Pfd Fetch Procedures failed

Labels

dnn, snssai, causecode

• smfcc_amf_event_notification_attempts

Type

64-bit Counter

Description

Number of AMF Event Notification Procedures attempted

Labels

dnn, snssai

• smfcc_amf_event_notification_successes

Type

64-bit Counter

Description

Number of AMF Event Notification Procedures successful

Labels

dnn, snssai

• smfcc_amf_event_notification_failures

Type

64-bit Counter

Description

Number of AMF Event Notification Procedures failed

Labels

dnn, snssai, causecode

- smfcc_5gs_to_eps_handover_direct_data_forwarding_attempts

Type

64-bit Counter

Description

Number of 5GS to EPS Handover Direct Data Forwarding Procedures attempted

Labels

dnn, snssai

- smfcc_5gs_to_eps_handover_direct_data_forwarding_successes

Type

64-bit Counter

Description

Number of 5GS to EPS Handover Direct Data Forwarding Procedures successful

Labels

dnn, snssai

- smfcc_5gs_to_eps_handover_direct_data_forwarding_failures

Type

64-bit Counter

Description

Number of 5GS to EPS Handover Direct Data Forwarding Procedures failed

Labels

dnn, snssai, causecode

- smfcc_eps_to_5gs_handover_without_n26_attempts

Type

64-bit Counter

Description

Number of EPS to 5GS Handover Without N26 Interface Procedures attempted

Labels

dnn, snssai, iupfrelocation

- smfcc_eps_to_5gs_handover_without_n26_successes

Type

64-bit Counter

Description

Number of EPS to 5GS Handover Without N26 Interface Procedures successful

Labels

dnn, snssai, iupfrelocation

smfcc_eps_to_5gs_handover_without_n26_failures
Type
64-bit Counter
Description
Number of EPS to 5GS Handover Procedures failed
Labels
<i>dnn, snssai, causecode, iupfrelocation</i>
smfcc_5gs_to_eps_handover_without_n26_attempts
Type
64-bit Counter
Description
Number of 5GS to EPSHandover Without N26 Interface Procedures attempted
Labels
<i>dnn, snssai</i>
smfcc_5gs_to_eps_handover_without_n26_successes
Type
64-bit Counter
Description
Number of 5GS to EPS Handover Without N26 Interface Procedures successful
Labels
<i>dnn, snssai</i>
smfcc_5gs_to_eps_handover_without_n26_failures
Type
64-bit Counter
Description
Number of 5GS to EPS Handover Procedures failed
Labels
<i>dnn, snssai, causecode</i>
smfcc_5gs_to_eps_handover_without_n26_rsrc_rel_attempts
Type
64-bit Counter
Description
Number of 5GS to EPS Handover Without N26 Interface Resource Release Procedures attempted
Labels
<i>dnn, snssai</i>
smfcc_5gs_to_eps_handover_without_n26_rsrc_rel_successes
Type
64-bit Counter
Description

Number of 5GS to EPS Handover Without N26 Interface Resource Release Procedures successful

Labels

dnn, snssai

- smfcc_5gs_to_eps_handover_without_n26_rsrc_rel_failures

Type

64-bit Counter

Description

Number of 5GS to EPS Handover Procedures Resource Release failed

Labels

dnn, snssai, causecode

- smfcc_5gs_to_eps_handover_without_n26_brr crt_attempts

Type

64-bit Counter

Description

Number of 5GS to EPS Handover Without N26 Interface Bearer Create Procedures attempted

Labels

dnn, snssai

- smfcc_5gs_to_eps_handover_without_n26_brr crt_successes

Type

64-bit Counter

Description

Number of 5GS to EPS Handover Without N26 Interface Bearer Create Procedures successful

Labels

dnn, snssai

- smfcc_5gs_to_eps_handover_without_n26_brr crt_failures

Type

64-bit Counter

Description

Number of 5GS to EPS Handover Procedures Bearer Create failed

Labels

dnn, snssai, causecode

- smfcc_5gs_to_eps_idle_handover_without_n26_attempts

Type

64-bit Counter

Description

Number of 5GS to EPS Idle Handover Without N26 Interface Procedures attempted

Labels

dnn, snssai

- smfcc_5gs_to_eps_idle_handover_without_n26_successes

Type

64-bit Counter

Description

Number of 5GS to EPS Idle Handover Without N26 Interface Procedures successful

Labels

dnn, snssai

- smfcc_5gs_to_eps_idle_handover_without_n26_failures

Type

64-bit Counter

Description

Number of 5GS to EPS Idle Handover Without N26 Interface Procedures failed

Labels

dnn, snssai, causecode

- smfcc_ran_initiated_qosflow_mobility_attempts

Type

64-bit Counter

Description

Number of RAN Initiated Qos Flow Mobility Procedures attempted

Labels

dnn, snssai

- smfcc_ran_initiated_qosflow_mobility_successes

Type

64-bit Counter

Description

Number of RAN Initiated Qos Flow Mobility Procedures successful

Labels

dnn, snssai

- smfcc_ran_initiated_qosflow_mobility_partial_successes

Type

64-bit Counter

Description

Number of RAN Initiated Qos Flow Mobility Procedures partially successful

Labels

dnn, snssai

- smfcc_ran_initiated_qosflow_mobility_failures

Type

64-bit Counter

Description

Number of RAN Initiated Qos Flow Mobility Procedures failed

Labels
<i>dnn, snssai, causecode</i>
• smfcc_vsmf_ran_initiated_qosflow_mobility_attempts
Type
64-bit Counter
Description
Number of vSMF RAN Initiated Qos Flow Mobility Procedures attempted
Labels
<i>dnn, snssai</i>
• smfcc_vsmf_ran_initiated_qosflow_mobility_successes
Type
64-bit Counter
Description
Number of vSMF RAN Initiated Qos Flow Mobility Procedures successful
Labels
<i>dnn, snssai</i>
• smfcc_vsmf_ran_initiated_qosflow_mobility_partial_successes
Type
64-bit Counter
Description
Number of vSMF RAN Initiated Qos Flow Mobility Procedures partially successful
Labels
<i>dnn, snssai</i>
• smfcc_vsmf_ran_initiated_qosflow_mobility_failures
Type
64-bit Counter
Description
Number of vSMF RAN Initiated Qos Flow Mobility Procedures failed
Labels
<i>dnn, snssai, causecode</i>
• smfcc_policy_timeout_handling_attempts
Type
64-bit Counter
Description
Number of Policy Timer Timeout Handling Procedures attempted
Labels
<i>timerType, dnn, snssai</i>
• smfcc_policy_timeout_handling_successes
Type

64-bit Counter

Description

Number of Policy Timer Timeout Handling Procedures successful

Labels

timerType, dnn, snssai

• smfcc_policy_timeout_handling_failures

Type

64-bit Counter

Description

Number of Policy Timer Timeout Handling Procedures failed

Labels

timerType, dnn, snssai, causecode

• smfcc_vsmf_hsmf_initiated_pdu_session_assignebi_attempts

Type

64-bit Counter

Description

Number of HSmf requested ebi assignment procedures attempted for VSMF sessions

Labels

dnn, snssai

• smfcc_vsmf_hsmf_initiated_pdu_session_assignebi_successes

Type

64-bit Counter

Description

Number of HSmf requested ebi assignment procedures successful for VSMF sessions

Labels

dnn, snssai

• smfcc_vsmf_hsmf_initiated_pdu_session_assignebi_failures

Type

64-bit Counter

Description

Number of HSmf requested ebi assignment procedures failed for VSMF sessions

Labels

dnn, snssai, causecode

• smfcc_vsmf_hsmf_initiated_pdu_session_modification_attempts

Type

64-bit Counter

Description

Number of HSmf requested qos modification procedures attempted for VSMF sessions

Labels

dnn, snssai

- smfcc_vsmf_hsmf_initiated_pdu_session_modification_successes

Type

64-bit Counter

Description

Number of HSmf requested qos modification procedures successful for VSMF sessions

Labels

dnn, snssai

- smfcc_vsmf_hsmf_initiated_pdu_session_modification_failures

Type

64-bit Counter

Description

Number of HSmf requested qos modification procedures failed for VSMF sessions

Labels

dnn, snssai, causecode

- smfcc_vsmf_hsmf_initiated_pdu_session_release_attempts

Type

64-bit Counter

Description

Number of HSMF Initiated PDU Session Release Procedure attempted for VSMF sessions

Labels

dnn, snssai, causecode

- smfcc_vsmf_hsmf_initiated_pdu_session_release_successes

Type

64-bit Counter

Description

Number of HSMF Initiated PDU Session Release Procedures successful for VSMF sessions

Labels

dnn, snssai

- smfcc_vsmf_hsmf_initiated_pdu_session_release_failures

Type

64-bit Counter

Description

Number of HSMF Initiated PDU Session Release Procedure Failed for VSMF sessions

Labels

dnn, snssai, causecode

- smfcc_vsmf_pdu_session_establishment_attempts

Type

64-bit Counter

Description
Number of VSMF PDU Session Establishment Procedures attempted
Labels
<i>dnn, snssai</i>
• smfcc_vsmf_pdu_session_establishment_successes
Type
64-bit Counter
Description
Number of VSMF PDU Session Establishment Procedures successful
Labels
<i>dnn, snssai</i>
• smfcc_vsmf_pdu_session_establishment_failures
Type
64-bit Counter
Description
Number of VSMF PDU Session Establishment Procedures failed
Labels
<i>dnn, snssai, causecode</i>
• smfcc_vsmf_an_release_attempts
Type
64-bit Counter
Description
Number of VSMF AN Release Procedures for inbound roamers attempted
Labels
<i>dnn, snssai</i>
• smfcc_vsmf_an_release_successes
Type
64-bit Counter
Description
Number of VSMF AN Release Procedures for inbound roamers successful
Labels
<i>dnn, snssai</i>
• smfcc_vsmf_an_release_failures
Type
64-bit Counter
Description
Number of VSMF AN Release Procedures for inbound roamers failed
Labels
<i>dnn, snssai, causecode</i>

smfcc_vsmf_service_request_attempts
Type
64-bit Counter
Description
Number of VSMF Service Request Procedures for inbound roamers attempted
Labels
dnn, snssai, amfmobility, iupfrelocation
smfcc_vsmf_service_request_successes
Type
64-bit Counter
Description
Number of VSMF Service Request Procedures for inbound roamers successful
Labels
dnn, snssai, amfmobility, iupfrelocation
smfcc_vsmf_service_request_failures
Type
64-bit Counter
Description
Number of VSMF Service Request Procedures for inbound roamers failed
Labels
dnn, snssai, causecode, amfmobility, iupfrelocation
smfcc_vsmf_xn_handover_attempts
Type
64-bit Counter
Description
Number of VSMF Xn Handover Procedures for inbound roamers attempted
Labels
dnn, snssai, iupfrelocation
smfcc_vsmf_xn_handover_successes
Type
64-bit Counter
Description
Number of VSMF Xn Handover Procedures for inbound roamers succeeded
Labels
dnn, snssai, iupfrelocation
smfcc_vsmf_xn_handover_failures
Type
64-bit Counter
Description

Number of VSMF Xn Handover Procedures for inbound roamers failed

Labels

dnn, snssai, causecode, iupfrelocation

• smfcc_vsmf_xn_handover_idle_mode_successes

Type

64-bit Counter

Description

Number of VSMF Xn Handover Procedures for inbound roamers succeeded for idle mode

Labels

dnn, snssai, iupfrelocation

• smfcc_vsmf_xn_handover_idle_mode_failures

Type

64-bit Counter

Description

Number of VSMF Xn Handover Procedures for inbound roamers failed for idle mode

Labels

dnn, snssai, causecode, iupfrelocation

• smfcc_vsmf_network_initiated_service_request_attempts

Type

64-bit Counter

Description

Number of VSMF Network Initiated Service Request Procedures attempted

Labels

dnn, snssai

• smfcc_vsmf_network_initiated_service_request_successes

Type

64-bit Counter

Description

Number of VSMF Network Initiated Service Request Procedures successful

Labels

dnn, snssai

• smfcc_vsmf_network_initiated_service_request_failures

Type

64-bit Counter

Description

Number of VSMF Network Initiated Service Request Procedures failed

Labels

dnn, snssai, causecode

• smfcc_vsmf_pdu_session_retrieve_attempts

Type

64-bit Counter

Description

Number of VSMF Retrieve Proceduresattempted

Labels

dnn, snssai

- smfcc_vsmf_pdu_session_retrieve_successes

Type

64-bit Counter

Description

Number of VSMF Retrieve Procedures successful

Labels

dnn, snssai

- smfcc_vsmf_pdu_session_retrieve_failures

Type

64-bit Counter

Description

Number of VSMF Retrieve Procedures failed

Labels

dnn, snssai, causecode

- smfcc_etcd_add_inband_radius_tunnel_attempts

Type

64-bit Counter

Description

Number of add inband radius tunnel Procedures attempted

Labels

inbandProfile, peerUpfRole, peerUpflp

- smfcc_etcd_add_inband_radius_tunnel_successes

Type

64-bit Counter

Description

Number of add inband radius tunnel Procedures successful

Labels

inbandProfile, peerUpfRole, peerUpflp

- smfcc_etcd_add_inband_radius_tunnel_failures

Type

64-bit Counter

Description

Number of add inband radius tunnel Procedures failed

Labels

inbandProfile, peerUpfRole, peerUpflp

- smfcc_etcd_del_inband_radius_tunnel_attempts

Type

64-bit Counter

Description

Number of del inband radius tunnel Procedures attempted

Labels

inbandProfile, peerUpfRole

- smfcc_etcd_del_inband_radius_tunnel_successes

Type

64-bit Counter

Description

Number of del inband radius tunnel Procedures successful

Labels

inbandProfile, peerUpfRole

- smfcc_etcd_del_inband_radius_tunnel_failures

Type

64-bit Counter

Description

Number of del inband radius tunnel Procedures failed

Labels

inbandProfile, peerUpfRole

- smfcc_inband_radius_tunnel_create_attempts

Type

64-bit Counter

Description

Number of pdu session create procedures to add inband radius tunnels attempted

Labels

inbandProfile, peerUpfName, peerUpfRole, peerUpflp

- smfcc_inband_radius_tunnel_create_successes

Type

64-bit Counter

Description

Number of pdu session create procedures to add inband radius tunnels seccuess

Labels

inbandProfile, peerUpfName, peerUpfRole, peerUpflp

- smfcc_inband_radius_tunnel_create_failures

Type

64-bit Counter

Description

Number of pdu session create procedures to add inband radius tunnels failure

Labels

inbandProfile, peerUpfName, peerUpfRole, peerUpflp

• smfcc_inband_radius_tunnel_delete_attempts

Type

64-bit Counter

Description

Number of pdu session delete procedures to cleanup inband radius tunnels attempted

Labels

inbandProfile, peerUpfName, peerUpfRole, peerUpflp

• smfcc_inband_radius_tunnel_delete_successes

Type

64-bit Counter

Description

Number of pdu session delete procedures to cleanup inband radius tunnels seccuess

Labels

inbandProfile, peerUpfName, peerUpfRole, peerUpflp

• smfcc_inband_radius_tunnel_delete_failures

Type

64-bit Counter

Description

Number of pdu session delete procedures to cleanup inband radius tunnels failure

Labels

inbandProfile, peerUpfName, peerUpfRole, peerUpflp

• smfcc_hsmf_pdu_session_establishment_attempts

Type

64-bit Counter

Description

Number of HSMF PDU Session Establishment Procedures attempted

Labels

dnn, snssai

• smfcc_hsmf_pdu_session_establishment_successes

Type

64-bit Counter

Description

Number of HSMF PDU Session Establishment Procedures successful

Labels

dnn, snssai

- smfcc_hsmf_pdu_session_establishment_failures

Type

64-bit Counter

Description

Number of HSMF PDU Session Establishment Procedures failed

Labels

dnn, snssai, causecode

- smfcc_hsmf_pdu_session_modification_attempts

Type

64-bit Counter

Description

Number of HSMF Pdu Session Modifications attempted

Labels

dnn, snssai

- smfcc_hsmf_pdu_session_modification_successes

Type

64-bit Counter

Description

Number of HSMF Pdu Session Modification Successful

Labels

dnn, snssai

- smfcc_hsmf_pdu_session_modification_failures

Type

64-bit Counter

Description

Number of HSMF Pdu Session Modification failed

Labels

dnn, snssai, causecode

- smfcc_hsmf_vsmf_initiated_pdu_session_release_attempts

Type

64-bit Counter

Description

Number of HSMF Pdu Session Release Procedures initiated by VSMF/AMF attempted

Labels

dnn, snssai

- smfcc_hsmf_vsmf_initiated_pdu_session_release_successes

Type

64-bit Counter

Description

Number of HSMF Pdu Session Release Procedures initiated by VSMF/AMF successful

Labels

dnn, snssai

- smfcc_hsmf_vsmf_initiated_pdu_session_release_failures

Type

64-bit Counter

Description

Number of HSMF Pdu Session Release Procedures initiated by VSMF/AMF failed

Labels

dnn, snssai, causecode

- smfcc_hsmf_initiated_pdu_session_release_attempts

Type

64-bit Counter

Description

Number of HSMF Initiated PDU Session Release Procedure attempted

Labels

dnn, snssai, causecode

- smfcc_hsmf_initiated_pdu_session_release_successes

Type

64-bit Counter

Description

Number of HSMF Initiated PDU Session Release Procedures successful

Labels

dnn, snssai

- smfcc_hsmf_initiated_pdu_session_release_failures

Type

64-bit Counter

Description

Number of HSMF Initiated PDU Session Release Procedure Failed

Labels

dnn, snssai, causecode

- smfcc_evt_exp_udm_create_subscription_attempts

Type

64-bit Counter

Description

Number of Event Exposure UDM Create Subscription Procedures attempted

Labels

-
- smfcc_evt_exp_udm_create_subscription_successes

Type

64-bit Counter

Description

Number of Event Exposure UDM Create Subscription Procedures successful

Labels

- smfcc_evt_exp_udm_create_subscription_failures

Type

64-bit Counter

Description

Number of Event Exposure UDM Create Subscription Procedures failed

Labels

- smfcc_evt_exp_udm_modify_subscription_attempts

Type

64-bit Counter

Description

Number of Event Exposure UDM Moidfy Subscription Procedures attempted

Labels

- smfcc_evt_exp_udm_modify_subscription_successes

Type

64-bit Counter

Description

Number of Event Exposure UDM Modify Subscription Procedures successful

Labels

- smfcc_evt_exp_udm_modify_subscription_failures

Type

64-bit Counter

Description

Number of Event Exposure UDM Modify Subscription Procedures failed

Labels

- smfcc_evt_exp_udm_delete_subscription_attempts

Type

64-bit Counter

Description

Number of Event Exposure UDM Delete Subscription Procedures attempted

Labels

- smfcc_evt_exp_udm_delete_subscription_successes

Type
64-bit Counter
Description
Number of Event Exposure UDM Delete Subscription Procedures successful
Labels
• smfcc_evt_exp_udm_delete_subscription_failures
Type
64-bit Counter
Description
Number of Event Exposure UDM Delete Subscription Procedures failed
Labels
• smfcc_proc_pdu_session_level_limit_exhaustions
Type
64-bit Counter
Description
Number of Limit Exhaustions For Smfcc Procedure Per Pdu Session
Labels
dnn, snssai, proceduretype
• smfcc_proc_process_level_limit_exhaustions
Type
64-bit Counter
Description
Number of Limit Exhaustions For Smfcc Procedure Per Process
Labels
dnn, snssai, proceduretype
• smfcc_upf_usage_report_punted
Type
64-bit Counter
Description
None
Metrics

Instances	Labels	Description
1	*****	Number of Usage Report Processing Procedure punted

SMF UDM Interface Statistics

• smfcc_udm_registration_attempts
Type
64-bit Counter
Description

Number of UDM Registration Processing attempted

Labels

peerid, dnn, snssai

• smfcc_udm_registration_successes

Type

64-bit Counter

Description

Number of UDM Registration Processing successful

Labels

peerid, dnn, snssai

• smfcc_udm_registration_failures

Type

64-bit Counter

Description

Number of UDM Registration Processing failed

Labels

peerid, dnn, snssai, causecode

• smfcc_sdm_get_sm_data_attempts

Type

64-bit Counter

Description

Number of SDM Get Sm-Data Processing attempted

Labels

peerid, dnn, snssai

• smfcc_sdm_get_sm_data_successes

Type

64-bit Counter

Description

Number of SDM Get Sm-Data Processing successful

Labels

peerid, dnn, snssai

• smfcc_sdm_get_sm_data_failures

Type

64-bit Counter

Description

Number of SDM Get Sm-Data Processing failed

Labels

peerid, dnn, snssai, causecode

• smfcc_sdm_subscription_attempts

Type

64-bit Counter

Description

Number of SDM Subscription Processing attempted

Labels

peerid, dnn, snssai

• smfcc_sdm_subscription_successes

Type

64-bit Counter

Description

Number of SDM Subscription Processing successful

Labels

peerid, dnn, snssai

• smfcc_sdm_subscription_failures

Type

64-bit Counter

Description

Number of SDM Subscription Processing failed

Labels

peerid, dnn, snssai, causecode

• smfcc_cancel_sdm_subscription_attempts

Type

64-bit Counter

Description

Number of Cancel SDM Subscription Processing attempted

Labels

peerid, dnn, snssai

• smfcc_cancel_sdm_subscription_successes

Type

64-bit Counter

Description

Number of Cancel SDM Subscription Processing successful

Labels

peerid, dnn, snssai

• smfcc_cancel_sdm_subscription_failures

Type

64-bit Counter

Description

Number of Cancel SDM Subscription Processing failed

Labels

peerid, dnn, snssai, causecode

• smfcc_udm_deregistration_attempts

Type

64-bit Counter

Description

Number of UDM Deregistration Processing attempted

Labels

peerid, dnn, snssai

• smfcc_udm_deregistration_successes

Type

64-bit Counter

Description

Number of UDM Deregistration Processing successful

Labels

peerid, dnn, snssai

• smfcc_udm_deregistration_failures

Type

64-bit Counter

Description

Number of UDM Deregistration Processing failed

Labels

peerid, dnn, snssai, causecode

• smfcc_udm_static_ue_ip_allocation

Type

64-bit Counter

Description

Number of UDM static ue IP allocation

Labels

dnn

• smfcc_udm_evt_exp_create_subscription_attempts

Type

64-bit Counter

Description

Number of UDM Nsmf Event Exposure Create Subscription attempted

Labels

• smfcc_udm_evt_exp_create_subscription_successes

Type

64-bit Counter

Description

Number of UDM Nsmf Event Exposure Create Subscription successes

Labels

- smfcc_udm_evt_exp_create_subscription_failures

Type

64-bit Counter

Description

Number of UDM Nsmf Event Exposure Create Subscription failed

Labels

- smfcc_udm_evt_exp_modify_subscription_attempts

Type

64-bit Counter

Description

Number of UDM Nsmf Event Exposure Modify Subscription attempted

Labels

- smfcc_udm_evt_exp_modify_subscription_successes

Type

64-bit Counter

Description

Number of UDM Nsmf Event Exposure Modify Subscription successes

Labels

- smfcc_udm_evt_exp_modify_subscription_failures

Type

64-bit Counter

Description

Number of UDM Nsmf Event Exposure Modify Subscription failed

Labels

- smfcc_udm_evt_exp_delete_subscription_attempts

Type

64-bit Counter

Description

Number of UDM Nsmf Event Exposure Delete Subscription attempted

Labels

- smfcc_udm_evt_exp_delete_subscription_successes

Type

64-bit Counter

Description		
Number of UDM Nsmf Event Exposure Delete Subscription successes		
Labels		
• smfcc_udm_evt_exp_delete_subscription_failures		
Type		
64-bit Counter		
Description		
Number of UDM Nsmf Event Exposure Delete Subscription failed		
Labels		
• smfcc_sdm_update_notification_punted		
Type		
64-bit Counter		
Description		
None		
Metrics		
Instances	Labels	Description
1	*****	Number of SDM Update Notification Procedure punted

SMF CHF Interface Procedure Statistics

• smfcc_nchf_create_session_attempts		
Type		
64-bit Counter		
Description		
Number of CHF Create Session attempted		
Labels		
peerid, dnn, snssai		
• smfcc_nchf_create_session_success		
Type		
64-bit Counter		
Description		
Number of CHF Create Session successful		
Labels		
peerid, dnn, snssai		
• smfcc_nchf_create_session_failures		
Type		
64-bit Counter		
Description		
Number of CHF Create Session failed		
Labels		

peerid, dnn, snssai, causecode

• smfcc_nchf_update_session_attempts

Type

64-bit Counter

Description

Number of CHF Update Session attempted

Labels

peerid, dnn, snssai

• smfcc_nchf_update_session_success

Type

64-bit Counter

Description

Number of CHF Update Session successful

Labels

peerid, dnn, snssai

• smfcc_nchf_update_session_failures

Type

64-bit Counter

Description

Number of CHF Update Session failed

Labels

peerid, dnn, snssai, causecode

• smfcc_nchf_release_session_attempts

Type

64-bit Counter

Description

Number of CHF Release Session attempted

Labels

peerid, dnn, snssai

• smfcc_nchf_release_session_success

Type

64-bit Counter

Description

Number of CHF Release Session successful

Labels

peerid, dnn, snssai

• smfcc_nchf_release_session_failures

Type

64-bit Counter

Description

Number of CHF Release Session failed

Labels

peerid, dnn, snssai, causecode

• smfcc_nchf_one_time_report_attempts

Type

64-bit Counter

Description

Number of CHF One Time Report attempted

Labels

peerid, dnn, snssai

• smfcc_nchf_one_time_report_success

Type

64-bit Counter

Description

Number of CHF One Time Report successful

Labels

peerid, dnn, snssai

• smfcc_nchf_one_time_report_failures

Type

64-bit Counter

Description

Number of CHF One Time Report Failed

Labels

peerid, dnn, snssai, causecode

• smfcc_nchf_notification_received

Type

64-bit Counter

Description

Number of CHF Notification Received

Labels

peerid, dnn, snssai

• smfcc_nchf_notification_success

Type

64-bit Counter

Description

Number of CHF Notification Success

Labels

peerid, dnn, snssai

- smfcc_nchf_notification_failures

Type

64-bit Counter

Description

Number of CHF Notification Failed

Labels

peerid, dnn, snssai, causecode

- smfcc_nchf_notification_punted

Type

64-bit Counter

Description

None

Metrics

Instances	Labels	Description
1	*****	Number of CHF Notification punted

SMFCC Monitor Service Interface Statistics

- smfcc_get_smfcc_gauge_attempts

Type

64-bit Counter

Description

Number of get smfcc gauge request messages received

Labels

gauge

- smfcc_get_smfcc_gauge_successes

Type

64-bit Counter

Description

Number of get smfcc gauge request messages responded successfully

Labels

gauge

- smfcc_get_smfcc_gauge_failures

Type

64-bit Counter

Description

Number of get smfcc gauge request message failed to process

Labels

gauge, cause

SMF GTPC Interface Statistics

smfcc_gtpc_create_session_attempts
Type
64-bit Counter
Description
Number of GTPC Create Session attempts
Labels
dnn, snssai
smfcc_gtpc_create_session_accepts
Type
64-bit Counter
Description
Number of GTPC Create Session attempts accepted
Labels
dnn, snssai
smfcc_gtpc_create_session_rejects
Type
64-bit Counter
Description
Number of GTPC Create Session attempts rejected
Labels
dnn, snssai, causecode
smfcc_gtpc_saegw_create_session_attempts
Type
64-bit Counter
Description
Number of GTPC SAEGW Create Session attempts
Labels
dnn, snssai
smfcc_gtpc_saegw_create_session_accepts
Type
64-bit Counter
Description
Number of GTPC SAEGW Create Session attempts accepted
Labels
dnn, snssai
smfcc_gtpc_saegw_create_session_rejects
Type
64-bit Counter
Description

Number of GTPC SAEGW Create Session attempts rejected

Labels

dnn, snssai, causecode

• smfcc_gtpc_delete_session_attempts

Type

64-bit Counter

Description

Number of GTPC Delete Session attempts

Labels

dnn, snssai

• smfcc_gtpc_delete_session_successes

Type

64-bit Counter

Description

Number of GTPC Delete Session attempts successful

Labels

dnn, snssai

• smfcc_gtpc_delete_session_failures

Type

64-bit Counter

Description

Number of GTPC Release Session attempts failed

Labels

dnn, snssai, causecode

• smfcc_gtpc_saegw_delete_session_attempts

Type

64-bit Counter

Description

Number of GTPC SAEGW Delete Session attempts

Labels

dnn, snssai, operationIndication

• smfcc_gtpc_saegw_delete_session_successes

Type

64-bit Counter

Description

Number of GTPC SAEGW Delete Session attempts successful

Labels

dnn, snssai, operationIndication

• smfcc_gtpc_saegw_delete_session_failures

Type

64-bit Counter

Description

Number of GTPC SAEGW Release Session attempts failed

Labels

dnn, snssai, operationIndication, causecode

• smfcc_gtpc_delete_bearer_attempts

Type

64-bit Counter

Description

Number of GTPC Delete Bearer attempts

Labels

dnn, snssai

• smfcc_gtpc_delete_bearer_successes

Type

64-bit Counter

Description

Number of GTPC Delete Bearer attempts successful

Labels

dnn, snssai

• smfcc_gtpc_delete_bearer_failures

Type

64-bit Counter

Description

Number of GTPC Release Bearer attempts failed

Labels

dnn, snssai, causecode

• smfcc_gtpc_saegw_delete_bearer_attempts

Type

64-bit Counter

Description

Number of GTPC Delete Bearer attempts for SAEGW

Labels

dnn, snssai

• smfcc_gtpc_saegw_delete_bearer_successes

Type

64-bit Counter

Description

Number of GTPC Delete Bearer attempts successful for SAEGW

Labels
<i>dnn, snssai</i>
smfcc_gtpc_saegw_delete_bearer_failures
Type
64-bit Counter
Description
Number of GTPC Release Bearer attempts failed for SAEGW
Labels
<i>dnn, snssai, causecode</i>
smfcc_gtpc_create_bearer_attempts
Type
64-bit Counter
Description
Number of GTPC Create Bearer attempts
Labels
<i>dnn, snssai</i>
smfcc_gtpc_create_bearer_successes
Type
64-bit Counter
Description
Number of GTPC Create Bearer attempts successful
Labels
<i>dnn, snssai</i>
smfcc_gtpc_create_bearer_failures
Type
64-bit Counter
Description
Number of GTPC Release Bearer attempts failed
Labels
<i>dnn, snssai, causecode</i>
smfcc_gtpc_saegw_create_bearer_attempts
Type
64-bit Counter
Description
Number of GTPC Create Bearer attempts for SAEGW
Labels
<i>dnn, snssai</i>
smfcc_gtpc_saegw_create_bearer_successes
Type

64-bit Counter

Description

Number of GTPC Create Bearer attempts successful for SAEGW

Labels

dnn, snssai

• smfcc_gtpc_saegw_create_bearer_failures

Type

64-bit Counter

Description

Number of GTPC Release Bearer attempts failed for SAEGW

Labels

dnn, snssai, causecode

• smfcc_gtpc_epco_5gsm_cause_received

Type

64-bit Counter

Description

Number of GTPC peer response received with EPCO having 5GSM cause

Labels

dnn, msgType, causecode

• smfcc_gtpc_update_bearer_attempts

Type

64-bit Counter

Description

Number of GTPC Update Bearer attempts

Labels

dnn, snssai

• smfcc_gtpc_update_bearer_successes

Type

64-bit Counter

Description

Number of GTPC Update Bearer attempts successful

Labels

dnn, snssai

• smfcc_gtpc_update_bearer_failures

Type

64-bit Counter

Description

Number of GTPC Update Bearer attempts failed

Labels

dnn, snssai, causecode

• smfcc_gtpc_saegw_update_bearer_attempts

Type

64-bit Counter

Description

Number of GTPC Update Bearer attempts for SAEGW

Labels

dnn, snssai

• smfcc_gtpc_saegw_update_bearer_successes

Type

64-bit Counter

Description

Number of GTPC Update Bearer attempts successful for SAEGW

Labels

dnn, snssai

• smfcc_gtpc_saegw_update_bearer_failures

Type

64-bit Counter

Description

Number of GTPC Update Bearer attempts failed for SAEGW

Labels

dnn, snssai, causecode

• smfcc_gtpc_modify_bearer_request_attempts

Type

64-bit Counter

Description

Number of GTPC Modify Bearer Request attempts

Labels

dnn, snssai

• smfcc_gtpc_modify_bearer_request_successes

Type

64-bit Counter

Description

Number of GTPC Modify Bearer Request attempts successful

Labels

dnn, snssai

• smfcc_gtpc_modify_bearer_request_failures

Type

64-bit Counter

Description
Number of GTPC Modify Bearer Request attempts failed
Labels
<i>dnn, snssai, causecode</i>
smfcc_gtpc_saegw_modify_bearer_request_attempts
Type
64-bit Counter
Description
Number of GTPC Modify Bearer Request attempts for SAEGW
Labels
<i>dnn, snssai</i>
smfcc_gtpc_saegw_modify_bearer_request_successes
Type
64-bit Counter
Description
Number of GTPC Modify Bearer Request attempts successful for SAEGW
Labels
<i>dnn, snssai</i>
smfcc_gtpc_saegw_modify_bearer_request_failures
Type
64-bit Counter
Description
Number of GTPC Modify Bearer Request attempts failed for SAEGW
Labels
<i>dnn, snssai, causecode</i>
smfcc_gtpc_modify_bearer_command_attempts
Type
64-bit Counter
Description
Number of GTPC Modify Bearer Command attempts
Labels
<i>dnn, snssai</i>
smfcc_gtpc_modify_bearer_command_failures
Type
64-bit Counter
Description
Number of GTPC Modify Bearer Command attempts failed
Labels
<i>dnn, snssai, causecode</i>

- smfcc_gtpc_sgw_modify_bearer_command_attempts

Type
64-bit Counter
Description
Number of GTPC Modify Bearer Command attempts for SGW
Labels

dnn

- smfcc_gtpc_sgw_modify_bearer_command_failures

Type
64-bit Counter
Description
Number of GTPC Modify Bearer Command attempts failed for SGW
Labels

dnn, causecode

- smfcc_gtpc_saegw_modify_bearer_command_attempts

Type
64-bit Counter
Description
Number of GTPC Modify Bearer Command attempts for SAEGW
Labels

dnn, snssai

- smfcc_gtpc_saegw_modify_bearer_command_failures

Type
64-bit Counter
Description
Number of GTPC Modify Bearer Command attempts failed for SAEGW
Labels

dnn, snssai, causecode

- smfcc_gtpc_bearer_resource_command_attempts

Type
64-bit Counter
Description
Number of GTPC Bearer Resource Command attempts
Labels

dnn, snssai, brctype

- smfcc_gtpc_bearer_resource_command_failures

Type
64-bit Counter
Description

Number of GTPC Bearer Resource Command attempts failed

Labels

dnn, snssai, causecode, brctype

• smfcc_gtpc_sgw_bearer_resource_command_attempts

Type

64-bit Counter

Description

Number of GTPC Bearer Resource Command attempts for SGW

Labels

dnn, brctype

• smfcc_gtpc_sgw_bearer_resource_command_failures

Type

64-bit Counter

Description

Number of GTPC Bearer Resource Command attempts failed for SAEGW

Labels

dnn, causecode, brctype

• smfcc_gtpc_saegw_bearer_resource_command_attempts

Type

64-bit Counter

Description

Number of GTPC Bearer Resource Command attempts

Labels

dnn, snssai, brctype

• smfcc_gtpc_saegw_bearer_resource_command_failures

Type

64-bit Counter

Description

Number of GTPC Bearer Resource Command attempts failed

Labels

dnn, snssai, causecode, brctype

• smfcc_gtpc_delete_bearer_command_attempts

Type

64-bit Counter

Description

Number of GTPC Delete Bearer Command attempts

Labels

dnn, snssai

• smfcc_gtpc_delete_bearer_command_failures

Type

64-bit Counter

Description

Number of GTPC Delete Bearer Command attempts failed

Labels

dnn, snssai, causecode

• smfcc_gtpc_sgw_create_session_attempts

Type

64-bit Counter

Description

Number of GTPC SGW Create Session attempts

Labels

dnn

• smfcc_gtpc_sgw_create_session_accepts

Type

64-bit Counter

Description

Number of GTPC SGW Create Session attempts accepted

Labels

dnn

• smfcc_gtpc_sgw_create_session_rejects

Type

64-bit Counter

Description

Number of GTPC SGW Create Session attempts rejected

Labels

dnn, causecode

• smfcc_gtpc_sgw_delete_session_attempts

Type

64-bit Counter

Description

Number of GTPC SGW Delete Session attempts

Labels

dnn

• smfcc_gtpc_sgw_delete_session_successes

Type

64-bit Counter

Description

Number of GTPC SGW Delete Session attempts successful

Labels

dnn

- smfcc_gtpc_sgw_delete_session_failures

Type

64-bit Counter

Description

Number of GTPC SGW Release Session attempts failed

Labels

dnn, causecode

- smfcc_gtpc_sgw_delete_session_sent_to_pgw_attempts

Type

64-bit Counter

Description

Number of GTPC SGW Delete Session sent towards PGW attempts

Labels

dnn

- smfcc_gtpc_sgw_delete_session_sent_to_pgw_successes

Type

64-bit Counter

Description

Number of GTPC SGW Delete Session sent towards PGW attempts successful

Labels

dnn

- smfcc_gtpc_sgw_delete_session_sent_to_pgw_failures

Type

64-bit Counter

Description

Number of GTPC SGW Release Session sent towards PGW attempts failed

Labels

dnn, causecode

- smfcc_gtpc_sgw_create_bearer_attempts

Type

64-bit Counter

Description

Number of GTPC SGW Create Bearer attempts

Labels

dnn, ratType

- smfcc_gtpc_sgw_create_bearer_accepts

Type

64-bit Counter
Description
Number of GTPC SGW Create Bearer attempts accepted
Labels
<i>dnn, ratType</i>
• smfcc_gtpc_sgw_create_bearer_failures
Type
64-bit Counter
Description
Number of GTPC SGW Create Bearer attempts rejected
Labels
<i>dnn, ratType, causecode</i>
• smfcc_gtpc_sgw_create_bearer_request_sent_to_mme_attempts
Type
64-bit Counter
Description
Number of GTPC Create Bearer forwarded by SGW to MME attempts
Labels
<i>dnn, ratType</i>
• smfcc_gtpc_sgw_create_bearer_request_sent_to_mme_accepts
Type
64-bit Counter
Description
Number of GTPC Create Bearer forwarded by SGW to MME got accepted
Labels
<i>dnn, ratType</i>
• smfcc_gtpc_sgw_create_bearer_request_sent_to_mme_failures
Type
64-bit Counter
Description
Number of GTPC Create Bearer forwarded by SGW to MME got rejected
Labels
<i>dnn, ratType, causecode</i>
• smfcc_gtpc_sgw_update_bearer_attempts
Type
64-bit Counter
Description
Number of GTPC SGW Update Bearer attempts
Labels

dnn, ratType

- smfcc_gtpc_sgw_update_bearer_accepts

Type

64-bit Counter

Description

Number of GTPC SGW Update Bearer attempts accepted

Labels

dnn, ratType

- smfcc_gtpc_sgw_update_bearer_failures

Type

64-bit Counter

Description

Number of GTPC SGW Update Bearer attempts rejected

Labels

dnn, ratType, causecode

- smfcc_gtpc_sgw_update_bearer_request_sent_to_mme_attempts

Type

64-bit Counter

Description

Number of GTPC Update Bearer forwarded by SGW to MME attempts

Labels

dnn, ratType

- smfcc_gtpc_sgw_update_bearer_request_sent_to_mme_accepts

Type

64-bit Counter

Description

Number of GTPC Update Bearer forwarded by SGW to MME got accepted

Labels

dnn, ratType

- smfcc_gtpc_sgw_update_bearer_request_sent_to_mme_failures

Type

64-bit Counter

Description

Number of GTPC Update Bearer forwarded by SGW to MME got rejected

Labels

dnn, ratType, causecode

- smfcc_gtpc_sgw_modify_bearer_request_attempts

Type

64-bit Counter

	Description Number of GTPC SGW Modify Bearer Request attempts Labels dnn
• smfcc_gtpc_sgw_modify_bearer_request_successes	Type 64-bit Counter Description Number of GTPC SGW Modify Bearer Request attempts successful Labels dnn
• smfcc_gtpc_sgw_modify_bearer_request_failures	Type 64-bit Counter Description Number of GTPC SGW Modify Bearer Request attempts failed Labels dnn, causecode
• smfcc_gtpc_sgw_delete_bearer_request_attempts	Type 64-bit Counter Description Number of GTPC SGW Delete Bearer Request attempts Labels dnn
• smfcc_gtpc_sgw_delete_bearer_request_successes	Type 64-bit Counter Description Number of GTPC SGW Delete Bearer Request attempts successful Labels dnn
• smfcc_gtpc_sgw_delete_bearer_request_failures	Type 64-bit Counter Description Number of GTPC SGW Delete Bearer Request attempts failed Labels dnn, causecode

- smfcc_gtpc_sgw_delete_bearer_request_sent_to_mme_attempts

Type
64-bit Counter
Description
Number of GTPC SGW Delete Bearer Request attempts sent to MME
Labels
dnn

- smfcc_gtpc_sgw_delete_bearer_request_sent_to_mme_successes

Type
64-bit Counter
Description
Number of GTPC SGW Delete Bearer Request attempts sent to MME successful
Labels
dnn

- smfcc_gtpc_sgw_delete_bearer_request_sent_to_mme_failures

Type
64-bit Counter
Description
Number of GTPC SGW Delete Bearer Request attempts sent to MME failed
Labels
dnn, causecode

- smfcc_gtpc_sgw_create_session_duplicate_attempts

Type
64-bit Counter
Description
Number of GTPC sgw Create Session duplicate requests
Labels
dnn, ratType

- smfcc_gtpc_sgw_create_session_duplicate_drops

Type
64-bit Counter
Description
Number of GTPC sgw Create Session duplicate requests dropped
Labels
dnn, ratType

- smfcc_gtpc_sgw_modify_bearer_duplicate_attempts

Type
64-bit Counter
Description

Number of GTPC sgw Modify Bearer duplicate requests
Labels
<i>dnn, ratType</i>
smfcc_gtpc_sgw_modify_bearer_duplicate_drops
Type
64-bit Counter
Description
Number of GTPC sgw Modify Bearer duplicate requests dropped
Labels
<i>dnn, ratType</i>
smfcc_gtpc_pgw_create_session_duplicate_attempts
Type
64-bit Counter
Description
Number of GTPC PGW Create Session duplicate requests
Labels
<i>dnn, snssai, ratType</i>
smfcc_gtpc_pgw_create_session_duplicate_drops
Type
64-bit Counter
Description
Number of GTPC PGW Create Session duplicate requests dropped
Labels
<i>dnn, snssai, ratType</i>
smfcc_gtpc_pgw_modify_bearer_duplicate_attempts
Type
64-bit Counter
Description
Number of GTPC PGW Modify Bearer duplicate requests
Labels
<i>dnn, snssai, ratType</i>
smfcc_gtpc_pgw_modify_bearer_duplicate_drops
Type
64-bit Counter
Description
Number of GTPC PGW Modify Bearer duplicate requests dropped
Labels
<i>dnn, snssai, ratType</i>
smfcc_gtpc_saegw_create_session_duplicate_attempts

Type

64-bit Counter

Description

Number of GTPC SAEGW Create Session duplicate requests

Labels

dnn, snssai, ratType

• smfcc_gtpc_saegw_create_session_duplicate_drops

Type

64-bit Counter

Description

Number of GTPC PGW Create Session duplicate requests dropped

Labels

dnn, snssai, ratType

• smfcc_gtpc_saegw_modify_bearer_duplicate_attempts

Type

64-bit Counter

Description

Number of GTPC SAEGW Modify Bearer duplicate requests

Labels

dnn, snssai, ratType

• smfcc_gtpc_saegw_modify_bearer_duplicate_drops

Type

64-bit Counter

Description

Number of GTPC SAEGW Modify Bearer duplicate requests dropped

Labels

dnn, snssai, ratType

• smfcc_gtpc_pgw_create_session_context_replacement_count

Type

64-bit Counter

Description

Number of GTPC PGW Create Session context replacement count

Labels

dnn, snssai, ratType

• smfcc_gtpc_sgw_create_session_context_replacement_count

Type

64-bit Counter

Description

Number of GTPC SGW Create Session context replacement count

Labels

dnn, ratType

- smfcc_gtpc_release_access_bearer_request_attempts

Type

64-bit Counter

Description

Number of GTPC SGW Release Access Bearer Request attempts

Labels

dnn, ratType

- smfcc_gtpc_release_access_bearer_request_successes

Type

64-bit Counter

Description

Number of GTPC SGW Release Access Bearer Request attempts successful

Labels

dnn, ratType

- smfcc_gtpc_release_access_bearer_request_failures

Type

64-bit Counter

Description

Number of GTPC SGW Release Access Bearer Request attempts failed

Labels

dnn, ratType, causecode

- smfcc_gtpc_sgw_modify_bearer_request_sent_to_pgw_attempts

Type

64-bit Counter

Description

Number of GTPC SGW Modify Bearer Request attempts to pgw

Labels

dnn

- smfcc_gtpc_sgw_modify_bearer_request_sent_to_pgw_successes

Type

64-bit Counter

Description

Number of GTPC SGW Modify Bearer Request attempts to pgw successful

Labels

dnn

- smfcc_gtpc_sgw_modify_bearer_request_sent_to_pgw_failures

Type

64-bit Counter

Description

Number of GTPC SGW Modify Bearer Request attempts to pgw failed

Labels

dnn, causecode

• smfcc_gtpc_unsupported_msg_received

Type

64-bit Counter

Description

Number of unsupported messages received that got dropped

Labels

MessageType

• smfcc_gtpc_down_link_data_notification_attempts

Type

64-bit Counter

Description

Number of GTPC SGW/SAEGW down link data notification attempts

Labels

dnn, ratType

• smfcc_gtpc_down_link_data_notification_accepts

Type

64-bit Counter

Description

Number of GTPC SGW/SAEGW Down link data notification accepted

Labels

dnn, ratType

• smfcc_gtpc_down_link_data_notification_failures

Type

64-bit Counter

Description

Number of GTPC SGW/SAEGW Down link data notification attempts rejected

Labels

dnn, ratType, causecode

• smfcc_gtpc_down_link_notification_failure_indication

Type

64-bit Counter

Description

Number of GTPC SGW/SAEGW Down link data notification failure indication

Labels

dnn, causecode

- smfcc_gtpc_sgw_delete_bearer_command_from_mme_attempts

Type

64-bit Counter

Description

Number of GTPC SGW Delete Bearer Command attempts from MME

Labels

dnn

- smfcc_gtpc_sgw_delete_bearer_command_to_mme_failures

Type

64-bit Counter

Description

Number of GTPC SGW Delete Bearer Command failures to MME (Timeout/Failure Indication)

Labels

dnn, causecode

- smfcc_gtpc_sgw_delete_bearer_command_to_pgw_attempts

Type

64-bit Counter

Description

Number of GTPC SGW Delete Bearer Command attempts to PGW

Labels

dnn

- smfcc_gtpc_sgw_delete_bearer_command_from_pgw_failures

Type

64-bit Counter

Description

Number of GTPC SGW Delete Bearer Command failures from PGW (Timeout/Failure Indication)

Labels

dnn, causecode

- smfcc_gtpc_delete_pdp_context_rx_attempts

Type

64-bit Counter

Description

Number of GTPC Delete PDP Context received attempts

Labels

dnn, snssai

- smfcc_gtpc_delete_pdp_context_rx_successes

Type

64-bit Counter

	Description
	Number of GTPC Delete PDP Context received attempts successful
	Labels
	<i>dnn, snssai</i>
• smfcc_gtpc_delete_pdp_context_rx_failures	
	Type
	64-bit Counter
	Description
	Number of GTPC Release PDP Context received attempts failed
	Labels
	<i>dnn, snssai, causecode</i>
• smfcc_gtpc_delete_pdp_context_tx_attempts	
	Type
	64-bit Counter
	Description
	Number of GGSN initiated GTPC Delete PDP Context attempts
	Labels
	<i>dnn, snssai</i>
• smfcc_gtpc_delete_pdp_context_tx_successes	
	Type
	64-bit Counter
	Description
	Number of GGSN initiated GTPC Delete PDP Context attempts successful
	Labels
	<i>dnn, snssai</i>
• smfcc_gtpc_delete_pdp_context_tx_failures	
	Type
	64-bit Counter
	Description
	Number of GGSN initiated GTPC Release Session attempts failed
	Labels
	<i>dnn, snssai, causecode</i>
• smfcc_gtpc_ggsn_create_pdp_context_duplicate_attempts	
	Type
	64-bit Counter
	Description
	Number of GTPC GGSN Create PDP Context duplicate requests
	Labels
	<i>dnn, snssai, ratType</i>

smfcc_gtpc_ggsn_create_pdp_context_duplicate_drops
Type
64-bit Counter
Description
Number of GTPC GGSN Create PDP Context duplicate requests dropped
Labels
dnn, snssai, ratType
smfcc_gtpc_ggsn_create_pdp_context_context_replacement_count
Type
64-bit Counter
Description
Number of GTPC GGSN Create PDP Context context replacement count
Labels
dnn, snssai, ratType
smfcc_gtpc_ggsn_update_pdp_context_duplicate_attempts
Type
64-bit Counter
Description
Number of GTPC GGSN Update PDP Context duplicate requests
Labels
dnn, snssai, ratType
smfcc_gtpc_ggsn_update_pdp_context_duplicate_drops
Type
64-bit Counter
Description
Number of GTPC GGSN Update PDP Context duplicate requests dropped
Labels
dnn, snssai, ratType
smfcc_gtpc_create_pdp_context_attempts
Type
64-bit Counter
Description
Number of GTPC Create PDP Context attempts
Labels
dnn, snssai
smfcc_gtpc_create_pdp_context_accepts
Type
64-bit Counter
Description

Number of GTPC Create PDP Context attempts accepted

Labels

dnn, snssai

• smfcc_gtpc_create_pdp_context_rejects

Type

64-bit Counter

Description

Number of GTPC Create PDP Context attempts rejected

Labels

dnn, snssai, causecode

• smfcc_gtpc_update_pdp_context_rx_attempts

Type

64-bit Counter

Description

Number of GTPC Create PDP Context attempts

Labels

dnn, snssai

• smfcc_gtpc_update_pdp_context_rx_success

Type

64-bit Counter

Description

Number of GTPC Create PDP Context attempts successful

Labels

dnn, snssai

• smfcc_gtpc_update_pdp_context_rx_failures

Type

64-bit Counter

Description

Number of GTPC Create PDP Context attempts failed

Labels

dnn, snssai, causecode

• smfcc_gtpc_update_pdp_context_tx_attempts

Type

64-bit Counter

Description

Number of GTPC Update PDP Context attempts

Labels

dnn, snssai

• smfcc_gtpc_update_pdp_context_tx_accepts

Type

64-bit Counter

Description

Number of GTPC Update PDP Context attempts accepted

Labels

dnn, snssai

• smfcc_gtpc_update_pdp_context_tx_failures

Type

64-bit Counter

Description

Number of GTPC Update PDP Context attempts failed

Labels

dnn, snssai, causecode

• smfcc_gtpc_saegw_create_session_context_replacement_count

Type

64-bit Counter

Description

Number of GTPC SAEGW Create Session context replacement count

Labels

dnn, snssai, ratType

• smfcc_gtpc_create_session_punted

Type

64-bit Counter

Description

None

Metrics

Instances	Labels	Description
1	*****	Number of GTPC Create Session punted

• smfcc_rate_limit_exceed_create_session_rejects

Type

64-bit Counter

Description

None

Metrics

Instances	Labels	Description
1	*****	Number of GTPC Create Session attempts rejected due to exceeding rate limit

• smfcc_gtpc_delete_session_punted

Type

64-bit Counter

Description		
None		
Metrics		
Instances	Labels	Description
1	*****	Number of GTPC Delete Session punted
• smfcc_gtpc_modify_bearer_request_punted		
Type		
64-bit Counter		
Description		
None		
Metrics		
Instances	Labels	Description
1	*****	Number of GTPC Modify Bearer Request punted
• smfcc_gtpc_modify_bearer_command_punted		
Type		
64-bit Counter		
Description		
None		
Metrics		
Instances	Labels	Description
1	*****	Number of GTPC Modify Bearer Command punted
• smfcc_gtpc_bearer_resource_command_punted		
Type		
64-bit Counter		
Description		
Number of GTPC Bearer Resource Command punted		
Metrics		
Instances	Labels	Description
1	*****	Number of GTPC Bearer Resource Command punted
• smfcc_gtpc_delete_bearer_command_punted		
Type		
64-bit Counter		
Description		
Number of GTPC Delete Bearer Command punted		
Metrics		
Instances	Labels	Description
1	*****	Number of GTPC Delete Bearer Command punted

smfcc_pgw_initial_attach_attempts
Type
64-bit Counter
Description
Number of PGW Initial Attach Procedures attempted
Labels
dnn, snssai, ratType
smfcc_pgw_initial_attach_successes
Type
64-bit Counter
Description
Number of PGW Initial Attach Procedures successful
Labels
dnn, snssai, ratType
smfcc_pgw_initial_attach_failures
Type
64-bit Counter
Description
Number of Initial Attach Procedures failed
Labels
dnn, snssai, ratType, causecode
smfcc_pgw_roaming_session_initial_attach_attempts
Type
64-bit Counter
Description
Number of PGW Roaming Initial Attach Procedures attempted
Labels
dnn, snssai, ratType
smfcc_pgw_roaming_session_initial_attach_successes
Type
64-bit Counter
Description
Number of PGW Initial Roaming Attach Procedures successful
Labels
dnn, snssai, ratType
smfcc_pgw_roaming_session_initial_attach_failures
Type
64-bit Counter
Description

Number of Initial Roaming Attach Procedures failed

Labels

dnn, snssai, ratType, causecode

• smfcc_pgw_ue_initiated_detach_attempts

Type

64-bit Counter

Description

Number of PGW Delete Session Procedures attempted

Labels

dnn, snssai, ratType

• smfcc_pgw_ue_initiated_detach_successes

Type

64-bit Counter

Description

Number of PGW Delete Session Procedures successful

Labels

dnn, snssai, ratType

• smfcc_pgw_ue_initiated_detach_failures

Type

64-bit Counter

Description

Number of PGW Delete Session Procedures failed

Labels

dnn, snssai, ratType, causecode

• smfcc_pgw_roaming_session_ue_initiated_detach_attempts

Type

64-bit Counter

Description

Number of PGW Delete Roaming Session Procedures attempted

Labels

dnn, snssai, ratType

• smfcc_pgw_roaming_session_ue_initiated_detach_successes

Type

64-bit Counter

Description

Number of PGW Delete Roaming Session Procedures successful

Labels

dnn, snssai, ratType

• smfcc_pgw_roaming_session_ue_initiated_detach_failures

Type

64-bit Counter

Description

Number of PGW Delete Roaming Session Procedures failed

Labels

dnn, snssai, ratType, causecode

• smfcc_pgw_pcf_initiated_terminate_attempts

Type

64-bit Counter

Description

Number of PCF Initiated Terminate Procedures attempted

Labels

dnn, snssai, ratType

• smfcc_pgw_pcf_initiated_terminate_successes

Type

64-bit Counter

Description

Number of PCF Initiated Terminate Procedures successful

Labels

dnn, snssai, ratType

• smfcc_pgw_pcf_initiated_terminate_failures

Type

64-bit Counter

Description

Number of PCF Initiated Terminate Procedures failed

Labels

dnn, snssai, ratType, causecode

• smfcc_pgw_roaming_session_pcf_initiated_terminate_attempts

Type

64-bit Counter

Description

Number of PCF Initiated Terminate Procedures attempted for roaming sessions

Labels

dnn, snssai, ratType

• smfcc_pgw_roaming_session_pcf_initiated_terminate_successes

Type

64-bit Counter

Description

Number of PCF Initiated Terminate Procedures successful for roaming sessions

Labels

dnn, snssai, ratType

- smfcc_pgw_roaming_session_pcf_initiated_terminate_failures

Type

64-bit Counter

Description

Number of PCF Initiated Terminate Procedures failed for roaming sessions

Labels

dnn, snssai, ratType, causecode

- smfcc_pgw_nw_initiated_release_attempts

Type

64-bit Counter

Description

Number of NW Initiated Release Procedures attempted

Labels

dnn, snssai, ratType, causecode

- smfcc_pgw_nw_initiated_release_successes

Type

64-bit Counter

Description

Number of NW Initiated Release Procedures successful

Labels

dnn, snssai, ratType

- smfcc_pgw_nw_initiated_release_failures

Type

64-bit Counter

Description

Number of NW Initiated Release Procedures failed

Labels

dnn, snssai, ratType, causecode

- smfcc_pgw_roaming_session_nw_initiated_release_attempts

Type

64-bit Counter

Description

Number of NW Initiated Release Procedures attempted for roaming sessions

Labels

dnn, snssai, ratType, causecode

- smfcc_pgw_roaming_session_nw_initiated_release_successes

Type

64-bit Counter

Description

Number of NW Initiated Release Procedures successful for roaming sessions

Labels

dnn, snssai, ratType

- smfcc_pgw_roaming_session_nw_initiated_release_failures

Type

64-bit Counter

Description

Number of NW Initiated Release Procedures failed for roaming sessions

Labels

dnn, snssai, ratType, causecode

- smfcc_pgw_pcf_initiated_policy_modification_attempts

Type

64-bit Counter

Description

Number of PCF Initiated Policy Modification Procedures attempted

Labels

dnn, snssai, ratType

- smfcc_pgw_pcf_initiated_policy_modification_successes

Type

64-bit Counter

Description

Number of PCF Initiated Policy Modification Procedures successful

Labels

dnn, snssai, ratType

- smfcc_pgw_pcf_initiated_policy_modification_failures

Type

64-bit Counter

Description

Number of PCF Initiated Policy Modification Procedures failed

Labels

dnn, snssai, ratType, causecode

- smfcc_pgw_roaming_session_pcf_initiated_policy_modification_attempts

Type

64-bit Counter

Description

Number of PCF Initiated Policy Modification Procedures attempted for roaming sessions

Labels

dnn, snssai, ratType

- smfcc_pgw_roaming_session_pcf_initiated_policy_modification_successes

Type

64-bit Counter

Description

Number of PCF Initiated Policy Modification Procedures successful for roaming sessions

Labels

dnn, snssai, ratType

- smfcc_pgw_roaming_session_pcf_initiated_policy_modification_failures

Type

64-bit Counter

Description

Number of PCF Initiated Policy Modification Procedures failed for roaming sessions

Labels

dnn, snssai, ratType, causecode

- smfcc_pgw_pcf_initiated_dedicated_bearer_activation_attempts

Type

64-bit Counter

Description

Number of PCF Initiated Dedicated Bearer Activation Procedures attempted

Labels

dnn, snssai, ratType

- smfcc_pgw_pcf_initiated_dedicated_bearer_activation_successes

Type

64-bit Counter

Description

Number of PCF Initiated Dedicated Bearer Activation Procedures successful

Labels

dnn, snssai, ratType

- smfcc_pgw_pcf_initiated_dedicated_bearer_activation_failures

Type

64-bit Counter

Description

Number of PCF Initiated Dedicated Bearer Activation Procedures failed

Labels

dnn, snssai, ratType, causecode

- smfcc_pgw_roaming_session_pcf_initiated_dedicated_bearer_activation_attempts

Type

64-bit Counter

Description

Number of PCF Initiated Dedicated Bearer Activation Procedures attempted for roaming sessions

Labels

dnn, snssai, ratType

- smfcc_pgw_roaming_session_pcf_initiated_dedicated_bearer_activation_successes

Type

64-bit Counter

Description

Number of PCF Initiated Dedicated Bearer Activation Procedures successful for roaming sessions

Labels

dnn, snssai, ratType

- smfcc_pgw_roaming_session_pcf_initiated_dedicated_bearer_activation_failures

Type

64-bit Counter

Description

Number of PCF Initiated Dedicated Bearer Activation Procedures failed for roaming sessions

Labels

dnn, snssai, ratType, causecode

- smfcc_pgw_initiated_legacy_dedicated_bearer_activation_attempts_ho

Type

64-bit Counter

Description

Number of PGW Initiated Legacy Dedicated Bearer Activation Procedures attempted During HO

Labels

dnn, snssai, ratType

- smfcc_pgw_initiated_legacy_dedicated_bearer_activation_successes_ho

Type

64-bit Counter

Description

Number of PGW Initiated Legacy Dedicated Bearer Activation Procedures successful During HO

Labels

dnn, snssai, ratType

- smfcc_pgw_initiated_legacy_dedicated_bearer_activation_failures_ho

Type

64-bit Counter

Description

Number of PGW Initiated Legacy Dedicated Bearer Activation Procedures failed

Labels

dnn, snssai, ratType, causecode

- smfcc_pgw_initiated_legacy_dedicated_bearer_activation_roaming_session_attempts_ho

Type
64-bit Counter
Description
Number of PGW Initiated Legacy Dedicated Bearer Activation Procedures attempted During HO
Labels
dnn, snssai, ratType

- smfcc_pgw_initiated_legacy_dedicated_bearer_activation_roaming_session_successes_ho

Type
64-bit Counter
Description
Number of PGW Initiated Legacy Dedicated Bearer Activation Procedures successful During HO
Labels
dnn, snssai, ratType

- smfcc_pgw_initiated_legacy_dedicated_bearer_activation_roaming_session_failures_ho

Type
64-bit Counter
Description
Number of PGW Initiated Legacy Dedicated Bearer Activation Procedures failed
Labels
dnn, snssai, ratType, causecode

- smfcc_pgw_modify_bearer_request_attempts

Type
64-bit Counter
Description
Number of Modify Bearer Request Procedures attempted
Labels
dnn, snssai, ratType

- smfcc_pgw_modify_bearer_request_successes

Type
64-bit Counter
Description
Number of Modify Bearer Request Procedures successful
Labels
dnn, snssai, ratType

- smfcc_pgw_modify_bearer_request_failures

Type
64-bit Counter
Description

Number of Modify Bearer Request Procedures failed

Labels

dnn, snssai, ratType, causecode

- smfcc_pgw_roaming_session_modify_bearer_request_attempts

Type

64-bit Counter

Description

Number of Modify Bearer Request Procedures attempted for roaming sessions

Labels

dnn, snssai, ratType

- smfcc_pgw_roaming_session_modify_bearer_request_successes

Type

64-bit Counter

Description

Number of Modify Bearer Request Procedures successful for roaming sessions

Labels

dnn, snssai, ratType

- smfcc_pgw_roaming_session_modify_bearer_request_failures

Type

64-bit Counter

Description

Number of Modify Bearer Request Procedures failed for roaming sessions

Labels

dnn, snssai, ratType, causecode

- smfcc_pgw_mbr_tau_rau_x2_sgw_change_handover_attempts

Type

64-bit Counter

Description

Number of Modify Bearer Request with SGW change Procedures attempted

Labels

dnn, snssai, ratType

- smfcc_pgw_mbr_tau_rau_x2_sgw_change_handover_successes

Type

64-bit Counter

Description

Number of Modify Bearer Request with SGW change Procedures successful

Labels

dnn, snssai, ratType

- smfcc_pgw_mbr_tau_rau_x2_sgw_change_handover_failures

Type

64-bit Counter

Description

Number of Modify Bearer Request with SGW change Procedures failed

Labels

dnn, snssai, ratType, causecode

- smfcc_pgw_roaming_session_mbr_tau_rau_x2_sgw_change_handover_attempts

Type

64-bit Counter

Description

Number of Modify Bearer Request with SGW change Procedures attempted for roaming sessions

Labels

dnn, snssai, ratType

- smfcc_pgw_roaming_session_mbr_tau_rau_x2_sgw_change_handover_successes

Type

64-bit Counter

Description

Number of Modify Bearer Request with SGW change Procedures successful for roaming sessions

Labels

dnn, snssai, ratType

- smfcc_pgw_roaming_session_mbr_tau_rau_x2_sgw_change_handover_failures

Type

64-bit Counter

Description

Number of Modify Bearer Request with SGW change Procedures failed for roaming sessions

Labels

dnn, snssai, ratType, causecode

- smfcc_pgw_hss_initiated_modify_bearer_command_attempts

Type

64-bit Counter

Description

Number of HSS Initiated Modify Bearer Command Procedures attempted

Labels

dnn, snssai, ratType

- smfcc_pgw_hss_initiated_modify_bearer_command_successes

Type

64-bit Counter

Description

Number of HSS Initiated Modify Bearer Command Procedures successful

Labels

dnn, snssai, ratType

- smfcc_pgw_hss_initiated_modify_bearer_command_failures

Type

64-bit Counter

Description

Number of HSS Initiated Modify Bearer Command Procedures failed

Labels

dnn, snssai, ratType, causecode

- smfcc_pgw_roaming_session_hss_initiated_modify_bearer_command_attempts

Type

64-bit Counter

Description

Number of HSS Initiated Modify Bearer Command Procedures attempted for roaming sessions

Labels

dnn, snssai, ratType

- smfcc_pgw_roaming_session_hss_initiated_modify_bearer_command_successes

Type

64-bit Counter

Description

Number of HSS Initiated Modify Bearer Command Procedures successful for roaming sessions

Labels

dnn, snssai, ratType

- smfcc_pgw_roaming_session_hss_initiated_modify_bearer_command_failures

Type

64-bit Counter

Description

Number of HSS Initiated Modify Bearer Command Procedures failed for roaming sessions

Labels

dnn, snssai, ratType, causecode

- smfcc_pgw_pcf_initiated_dedicated_bearer_delete_attempts

Type

64-bit Counter

Description

Number of PCF Initiated Dedicated Bearer Delete Procedures attempted

Labels

dnn, snssai, ratType

- smfcc_pgw_pcf_initiated_dedicated_bearer_delete_successes

Type

64-bit Counter

Description

Number of PCF Initiated Dedicated Bearer Delete Procedures successful

Labels

dnn, snssai, ratType

- smfcc_pgw_pcf_initiated_dedicated_bearer_delete_failures

Type

64-bit Counter

Description

Number of PCF Initiated Dedicated Bearer Delete Procedures failed

Labels

dnn, snssai, ratType, causecode

- smfcc_pgw_roaming_session_pcf_initiated_dedicated_bearer_delete_attempts

Type

64-bit Counter

Description

Number of PCF Initiated Dedicated Bearer Delete Procedures attempted for roaming sessions

Labels

dnn, snssai, ratType

- smfcc_pgw_roaming_session_pcf_initiated_dedicated_bearer_delete_successes

Type

64-bit Counter

Description

Number of PCF Initiated Dedicated Bearer Delete Procedures successful for roaming sessions

Labels

dnn, snssai, ratType

- smfcc_pgw_roaming_session_pcf_initiated_dedicated_bearer_delete_failures

Type

64-bit Counter

Description

Number of PCF Initiated Dedicated Bearer Delete Procedures failed for roaming sessions

Labels

dnn, snssai, ratType, causecode

- smfcc_pgw_pcf_initiated_dedicated_bearer_update_attempts

Type

64-bit Counter

Description

Number of PCF Initiated Dedicated Bearer Update Procedures attempted

Labels

dnn, snssai, ratType

- smfcc_pgw_pcf_initiated_dedicated_bearer_update_successes

Type

64-bit Counter

Description

Number of PCF Initiated Dedicated Bearer Update Procedures successful

Labels

dnn, snssai, ratType

- smfcc_pgw_pcf_initiated_dedicated_bearer_update_failures

Type

64-bit Counter

Description

Number of PCF Initiated Dedicated Bearer Update Procedures failed

Labels

dnn, snssai, ratType, causecode

- smfcc_pgw_roaming_session_pcf_initiated_dedicated_bearer_update_attempts

Type

64-bit Counter

Description

Number of PCF Initiated Dedicated Bearer Update Procedures attempted for roaming sessions

Labels

dnn, snssai, ratType

- smfcc_pgw_roaming_session_pcf_initiated_dedicated_bearer_update_successes

Type

64-bit Counter

Description

Number of PCF Initiated Dedicated Bearer Update Procedures successful for roaming sessions

Labels

dnn, snssai, ratType

- smfcc_pgw_roaming_session_pcf_initiated_dedicated_bearer_update_failures

Type

64-bit Counter

Description

Number of PCF Initiated Dedicated Bearer Update Procedures failed for roaming sessions

Labels

dnn, snssai, ratType, causecode

- smfcc_pgw_process_pcf_update_policy_rules_attempts

Type

64-bit Counter

Description
Number of Process Pcf Update Policy Rules Procedures attempted
Labels
<i>dnn, snssai, ratType</i>
• smfcc_pgw_process_pcf_update_policy_rules_successes
Type
64-bit Counter
Description
Number of Process Pcf Update Policy Rules Procedures successful
Labels
<i>dnn, snssai, ratType</i>
• smfcc_pgw_process_pcf_update_policy_rules_failures
Type
64-bit Counter
Description
Number of Process Pcf Update Policy Rules Procedures failed
Labels
<i>dnn, snssai, ratType, causecode</i>
• smfcc_pgw_roaming_session_process_pcf_update_policy_rules_attempts
Type
64-bit Counter
Description
Number of Process Pcf Update Policy Rules Procedures attempted for roaming sessions
Labels
<i>dnn, snssai, ratType</i>
• smfcc_pgw_roaming_session_process_pcf_update_policy_rules_successes
Type
64-bit Counter
Description
Number of Process Pcf Update Policy Rules Procedures successful for roaming sessions
Labels
<i>dnn, snssai, ratType</i>
• smfcc_pgw_roaming_session_process_pcf_update_policy_rules_failures
Type
64-bit Counter
Description
Number of Process Pcf Update Policy Rules Procedures failed for roaming sessions
Labels
<i>dnn, snssai, ratType, causecode</i>

smfcc_pgw_bearer_resource_command_unknown_ebi_attempts
Type
64-bit Counter
Description
Number of Bearer Resource Command Procedures attempted for Unknown ebi
Labels
dnn, snssai, ratType
smfcc_pgw_bearer_resource_command_unknown_ebi_successes
Type
64-bit Counter
Description
Number of Bearer Resource Command Procedures successful for Unknown ebi
Labels
dnn, snssai, ratType
smfcc_pgw_bearer_resource_command_unknown_ebi_failures
Type
64-bit Counter
Description
Number of Bearer Resource Command Procedures failed for Unknown ebi
Labels
dnn, snssai, ratType, causecode
smfcc_pgw_roaming_session_bearer_resource_command_unknown_ebi_attempts
Type
64-bit Counter
Description
Number of Bearer Resource Command Procedures attempted for Unknown ebi in roaming sessions
Labels
dnn, snssai, ratType
smfcc_pgw_roaming_session_bearer_resource_command_unknown_ebi_successes
Type
64-bit Counter
Description
Number of Bearer Resource Command Procedures successful for Unknown ebi in roaming sessions
Labels
dnn, snssai, ratType
smfcc_pgw_roaming_session_bearer_resource_command_unknown_ebi_failures
Type
64-bit Counter
Description

Number of Bearer Resource Command Procedures failed for Unknown ebi in roaming sessions

Labels

dnn, snssai, ratType, causecode

• smfcc_pgw_bearer_resource_command_attempts

Type

64-bit Counter

Description

Number of Bearer Resource Command Procedures attempted

Labels

dnn, snssai, ratType

• smfcc_pgw_bearer_resource_command_successes

Type

64-bit Counter

Description

Number of Bearer Resource Command Procedures successful

Labels

dnn, snssai, ratType

• smfcc_pgw_bearer_resource_command_failures

Type

64-bit Counter

Description

Number of Bearer Resource Command Procedures failed

Labels

dnn, snssai, ratType, causecode

• smfcc_pgw_roaming_session_bearer_resource_command_attempts

Type

64-bit Counter

Description

Number of Bearer Resource Command Procedures attempted for roaming sessions

Labels

dnn, snssai, ratType

• smfcc_pgw_roaming_session_bearer_resource_command_successes

Type

64-bit Counter

Description

Number of Bearer Resource Command Procedures successful for roaming sessions

Labels

dnn, snssai, ratType

• smfcc_pgw_roaming_session_bearer_resource_command_failures

Type

64-bit Counter

Description

Number of Bearer Resource Command Procedures failed for roaming sessions

Labels

dnn, snssai, ratType, causecode

• smfcc_pgw_mme_initiated_delete_bearer_command_attempts

Type

64-bit Counter

Description

Number of MME Initiated Delete Bearer Command Procedures attempted

Labels

dnn, snssai, ratType

• smfcc_pgw_mme_initiated_delete_bearer_command_successes

Type

64-bit Counter

Description

Number of MME Initiated Delete Bearer Command Procedures successful

Labels

dnn, snssai, ratType

• smfcc_pgw_mme_initiated_delete_bearer_command_failures

Type

64-bit Counter

Description

Number of MME Initiated Delete Bearer Command Procedures failed

Labels

dnn, snssai, ratType, causecode

• smfcc_pgw_roaming_session_mme_initiated_delete_bearer_command_attempts

Type

64-bit Counter

Description

Number of MME Initiated Delete Bearer Command Procedures attempted for roaming sessions

Labels

dnn, snssai, ratType

• smfcc_pgw_roaming_session_mme_initiated_delete_bearer_command_successes

Type

64-bit Counter

Description

Number of MME Initiated Delete Bearer Command Procedures successful for roaming sessions

Labels

dnn, snssai, ratType

- smfcc_pgw_roaming_session_mme_initiated_delete_bearer_command_failures

Type

64-bit Counter

Description

Number of MME Initiated Delete Bearer Command Procedures failed for roaming sessions

Labels

dnn, snssai, ratType, causecode

- smfcc_pgw_error_indication_procedure_attempts

Type

64-bit Counter

Description

Number of Error Indication Procedures attempted

Labels

dnn, snssai, ratType

- smfcc_pgw_error_indication_procedure_successes

Type

64-bit Counter

Description

Number of Error Indication Procedures successful

Labels

dnn, snssai, ratType

- smfcc_pgw_error_indication_procedure_failures

Type

64-bit Counter

Description

Number of Error Indication Procedures failed

Labels

dnn, snssai, ratType, causecode

- smfcc_pgw_roaming_session_error_indication_procedure_attempts

Type

64-bit Counter

Description

Number of Error Indication Procedures attempted for roaming sessions

Labels

dnn, snssai, ratType

- smfcc_pgw_roaming_session_error_indication_procedure_successes

Type

64-bit Counter

Description

Number of Error Indication Procedures successful for roaming sessions

Labels

dnn, snssai, ratType

- smfcc_pgw_roaming_session_error_indication_procedure_failures

Type

64-bit Counter

Description

Number of Error Indication Procedures failed for roaming sessions

Labels

dnn, snssai, ratType, causecode

- smfcc_pgw_qos_flow_inactivity_procedure_attempts

Type

64-bit Counter

Description

Number of Pgw QosFlow Inactivity Procedures attempted

Labels

dnn, snssai, ratType

- smfcc_pgw_qos_flow_inactivity_procedure_successes

Type

64-bit Counter

Description

Number of Pgw QosFlow Inactivity Procedures successful

Labels

dnn, snssai, ratType

- smfcc_pgw_qos_flow_inactivity_procedure_failures

Type

64-bit Counter

Description

Number of Pgw QosFlow Inactivity Procedures failed

Labels

dnn, snssai, ratType, causecode

- smfcc_pgw_roaming_session_qos_flow_inactivity_procedure_attempts

Type

64-bit Counter

Description

Number of Pgw QosFlow Inactivity Procedures attempted for roaming sessions

Labels

dnn, snssai, ratType

- smfcc_pgw_roaming_session_qos_flow_inactivity_procedure_successes

Type

64-bit Counter

Description

Number of Pgw QosFlow Inactivity Procedures successful for roaming sessions

Labels

dnn, snssai, ratType

- smfcc_pgw_roaming_session_qos_flow_inactivity_procedure_failures

Type

64-bit Counter

Description

Number of Pgw QosFlow Inactivity Procedures failed for roaming sessions

Labels

dnn, snssai, ratType, causecode

- smfcc_pgw_session_idle_procedure_attempts

Type

64-bit Counter

Description

Number of Pgw Session Idle Procedures attempted

Labels

dnn, snssai, ratType

- smfcc_pgw_session_idle_procedure_successes

Type

64-bit Counter

Description

Number of Pgw Session Idle Procedures successful

Labels

dnn, snssai, ratType

- smfcc_pgw_session_idle_procedure_failures

Type

64-bit Counter

Description

Number of Pgw Session Idle Procedures failed

Labels

dnn, snssai, ratType, causecode

- smfcc_pgw_roaming_session_idle_procedure_attempts

Type

64-bit Counter

Description

Number of Pgw Session Idle Procedures attempted for roaming sessions

Labels

dnn, snssai, ratType

- smfcc_pgw_roaming_session_idle_procedure_successes

Type

64-bit Counter

Description

Number of Pgw Session Idle Procedures successful for roaming sessions

Labels

dnn, snssai, ratType

- smfcc_pgw_roaming_session_idle_procedure_failures

Type

64-bit Counter

Description

Number of Pgw Session Idle Procedures failed for roaming sessions

Labels

dnn, snssai, ratType, causecode

- smfcc_abort_gtpc_retransmission_during_session_deletion_attempts

Type

64-bit Counter

Description

Number of Pgw Abort Gtpc Procedures during Session Delete attempted

Labels

dnn, snssai, messagetype

- smfcc_abort_gtpc_retransmission_during_session_deletion_successes

Type

64-bit Counter

Description

Number of Pgw Abort Gtpc Procedures during Session Delete successful

Labels

dnn, snssai, messagetype

- smfcc_abort_gtpc_retransmission_during_session_deletion_failures

Type

64-bit Counter

Description

Number of Pgw Abort Gtpc Procedures during Session Delete failed

Labels

dnn, snssai, causecode, messagetype

- smfcc_pgw_restore_pdn_connection_attempts

Type

64-bit Counter

Description

Number of Pgw Restore Pdn Connection Procedures attempted

Labels

dnn, snssai
- smfcc_pgw_restore_pdn_connection_successes

Type

64-bit Counter

Description

Number of Pgw Restore Pdn Connection Procedures successful

Labels

dnn, snssai
- smfcc_pgw_restore_pdn_connection_failures

Type

64-bit Counter

Description

Number of Pgw Restore Pdn Connection Procedures failed

Labels

dnn, snssai
- smfcc_pgw_interplmn_handover_with_subscriber_type_change_attempts

Type

64-bit Counter

Description

Number of pgw inter plmn handovers with subscriber type(Homer/Roamer) change attempts

Labels

dnn, snssai, newstatus
- smfcc_pgw_interplmn_handover_with_subscriber_type_change_successes

Type

64-bit Counter

Description

Number of pgw inter plmn handovers with subscriber type(Homer/Roamer) change successes

Labels

dnn, snssai, newstatus
- smfcc_pgw_interplmn_handover_with_subscriber_type_change_failures

Type

64-bit Counter

Description

Number of pgw inter plmn handovers with subscriber type(Homer/Roamer) change failures

Labels

dnn, snssai, newstatus, cause

SMFCC SGW Procedures Statistics

- smfcc_sgw_initial_attach_attempts

Type

64-bit Counter

Description

Number of SGW Initial Attach Procedures attempted

Labels

dnn, ratType

- smfcc_sgw_initial_attach_successes

Type

64-bit Counter

Description

Number of SGW Initial Attach Procedures successful

Labels

dnn, ratType

- smfcc_sgw_initial_attach_failures

Type

64-bit Counter

Description

Number of Initial Attach Procedures failed

Labels

dnn, ratType, causecode

- smfcc_sgw_ue_initiated_detach_attempts

Type

64-bit Counter

Description

Number of SGW Delete Session Procedures attempted

Labels

dnn, ratType

- smfcc_sgw_ue_initiated_detach_successes

Type

64-bit Counter

Description

Number of SGW Delete Session Procedures successful

Labels

dnn, ratType

smfcc_sgw_ue_initiated_detach_failures
Type
64-bit Counter
Description
Number of SGW Delete Session Procedures failed
Labels
<i>dnn, ratType, causecode</i>
smfcc_sgw_nw_initiated_release_attempts
Type
64-bit Counter
Description
Number of NW Initiated Release Procedures attempted
Labels
<i>dnn, ratType</i>
smfcc_sgw_nw_initiated_release_successes
Type
64-bit Counter
Description
Number of NW Initiated Release Procedures successful
Labels
<i>dnn, ratType</i>
smfcc_sgw_nw_initiated_release_failures
Type
64-bit Counter
Description
Number of NW Initiated Release Procedures failed
Labels
<i>dnn, ratType, causecode</i>
smfcc_sgw_nw_initiated_dedicated_bearer_activation_attempts
Type
64-bit Counter
Description
Number of NW Initiated Dedicated Bearer Activation Procedures attempted
Labels
<i>dnn, ratType</i>
smfcc_sgw_nw_initiated_dedicated_bearer_activation_successes
Type
64-bit Counter
Description

Number of NW Initiated Dedicated Bearer Activation Procedures successful

Labels

dnn, ratType

- smfcc_sgw_nw_initiated_dedicated_bearer_activation_failures

Type

64-bit Counter

Description

Number of NW Initiated Dedicated Bearer Activation Procedures failed

Labels

dnn, ratType, causecode

- smfcc_sgw_modify_bearer_request_attempts

Type

64-bit Counter

Description

Number of Modify Bearer Request Procedures attempted

Labels

dnn, ratType

- smfcc_sgw_modify_bearer_request_successes

Type

64-bit Counter

Description

Number of Modify Bearer Request Procedures successful

Labels

dnn, ratType

- smfcc_sgw_modify_bearer_request_failures

Type

64-bit Counter

Description

Number of Modify Bearer Request Procedures failed

Labels

dnn, ratType, causecode

- smfcc_sgw_modify_bearer_ho_attempts

Type

64-bit Counter

Description

Number of Modify Bearer Request Handover Procedures attempted

Labels

dnn, ratType

- smfcc_sgw_modify_bearer_ho_success

Type

64-bit Counter

Description

Number of Modify Bearer Request Handover Procedures successful

Labels

dnn, ratType

- smfcc_sgw_modify_bearer_ho_failure

Type

64-bit Counter

Description

Number of Modify Bearer Request Handover Procedures failed

Labels

dnn, ratType, causecode

- smfcc_sgw_delete_bearer_request_attempts

Type

64-bit Counter

Description

Number of Delete Bearer Request Procedures attempted

Labels

dnn, ratType

- smfcc_sgw_delete_bearer_request_successes

Type

64-bit Counter

Description

Number of Delete Bearer Request Procedures successful

Labels

dnn, ratType

- smfcc_sgw_delete_bearer_request_failures

Type

64-bit Counter

Description

Number of Delete Bearer Request Procedures failed

Labels

dnn, ratType, causecode

- smfcc_sgw_create_bearer_request_attempts

Type

64-bit Counter

Description

Number of Create Bearer Request Procedures attempted

Labels

dnn, ratType

- smfcc_sgw_create_bearer_request_successes

Type

64-bit Counter

Description

Number of Create Bearer Request Procedures successful

Labels

dnn, ratType

- smfcc_sgw_create_bearer_request_failures

Type

64-bit Counter

Description

Number of Create Bearer Request Procedures failed

Labels

dnn, ratType, causecode

- smfcc_sgw_update_bearer_request_attempts

Type

64-bit Counter

Description

Number of Update Bearer Request Procedures attempted

Labels

dnn, ratType

- smfcc_sgw_update_bearer_request_successes

Type

64-bit Counter

Description

Number of Update Bearer Request Procedures successful

Labels

dnn, ratType

- smfcc_sgw_update_bearer_request_failures

Type

64-bit Counter

Description

Number of Update Bearer Request Procedures failed

Labels

dnn, ratType, causecode

- smfcc_sgw_hss_initiated_modify_bearer_command_attempts

Type

64-bit Counter

Description

Number of HSS Initiated Modify Bearer Command Procedures attempted

Labels

dnn, ratType

• smfcc_sgw_hss_initiated_modify_bearer_command_successes

Type

64-bit Counter

Description

Number of HSS Initiated Modify Bearer Command Procedures successful

Labels

dnn, ratType

• smfcc_sgw_hss_initiated_modify_bearer_command_failures

Type

64-bit Counter

Description

Number of HSS Initiated Modify Bearer Command Procedures failed

Labels

dnn, ratType, causecode

• smfcc_sgw_nw_initiated_dedicated_bearer_delete_attempts

Type

64-bit Counter

Description

Number of NW Initiated Dedicated Bearer Delete Procedures attempted

Labels

dnn, ratType

• smfcc_sgw_nw_initiated_dedicated_bearer_delete_successes

Type

64-bit Counter

Description

Number of NW Initiated Dedicated Bearer Delete Procedures successful

Labels

dnn, ratType

• smfcc_sgw_nw_initiated_dedicated_bearer_delete_failures

Type

64-bit Counter

Description

Number of NW Initiated Dedicated Bearer Delete Procedures failed

Labels

dnn, ratType, causecode

- smfcc_sgw_nw_initiated_dedicated_bearer_update_attempts

Type

64-bit Counter

Description

Number of NW Initiated Dedicated Bearer Update Procedures attempted

Labels

dnn, ratType

- smfcc_sgw_nw_initiated_dedicated_bearer_update_successes

Type

64-bit Counter

Description

Number of NW Initiated Dedicated Bearer Update Procedures successful

Labels

dnn, ratType

- smfcc_sgw_nw_initiated_dedicated_bearer_update_failures

Type

64-bit Counter

Description

Number of NW Initiated Dedicated Bearer Update Procedures failed

Labels

dnn, ratType, causecode

- smfcc_sgw_delete_bearer_command_attempts

Type

64-bit Counter

Description

Number of Delete Bearer Command Procedures attempted

Labels

dnn, ratType

- smfcc_sgw_delete_bearer_command_failures

Type

64-bit Counter

Description

Number of Bearer Command Procedures failed

Labels

dnn, ratType, causecode

- smfcc_sgw_release_access_bearer_request_attempts

Type

64-bit Counter

Description
Number of Release Access Bearer Request Procedures attempted
Labels
<i>dnn, ratType</i>
• smfcc_sgw_release_access_bearer_request_successes
Type
64-bit Counter
Description
Number of Release Access Bearer Request Procedures successful
Labels
<i>dnn, ratType</i>
• smfcc_sgw_release_access_bearer_request_failures
Type
64-bit Counter
Description
Number of Release Access Bearer Request Procedures failed
Labels
<i>dnn, ratType, causecode</i>
• smfcc_sgw_ue_initiated_service_request_attempts
Type
64-bit Counter
Description
Number of SGW UE service request Procedures attempted
Labels
<i>dnn, ratType</i>
• smfcc_sgw_ue_initiated_service_request_successes
Type
64-bit Counter
Description
Number of SGW UE initiated service request Procedures successful
Labels
<i>dnn, ratType</i>
• smfcc_sgw_ue_initiated_service_request_failures
Type
64-bit Counter
Description
Number of SGW UE initiated service request Procedures failed
Labels
<i>dnn, ratType, causecode</i>

smfcc_sgw_nw_initiated_service_request_attempts
Type
64-bit Counter
Description
Number of SGW Network service request Procedures attempted
Labels
dnn, ratType
smfcc_sgw_nw_initiated_service_request_successes
Type
64-bit Counter
Description
Number of SGW Network initiated service request Procedures successful
Labels
dnn, ratType
smfcc_sgw_nw_initiated_service_request_failures
Type
64-bit Counter
Description
Number of SGW Network initiated service request Procedures failed
Labels
dnn, ratType, causecode
smfcc_sgw_nw_network_initiated_service_request_timeouts
Type
64-bit Counter
Description
Number of SGW Network Initiated Service Request Procedures Timeouts
Labels
dnn, ratType
smfcc_sgw_nw_initiated_service_request_aborted
Type
64-bit Counter
Description
Number of SGW Network initiated service request Procedures aborted due to higher arp pl DDN received from upf
Labels
dnn, ratType
smfcc_sgw_nw_initiated_service_request_dropped
Type
64-bit Counter
Description

Number of SGW Network initiated service request Procedures dropped due to higher arp pl DDN already sent

Labels

dnn, ratType, cause

• smfcc_sgw_csr_tau_rau_x2ho_attempts

Type

64-bit Counter

Description

Number of SGW TAU/RAU/X2-HO with SGW Change Procedures attempted

Labels

dnn, ratType

• smfcc_sgw_csr_tau_rau_x2ho_success

Type

64-bit Counter

Description

Number of SGW TAU/RAU/X2-HO with SGW Change Procedures successful

Labels

dnn, ratType

• smfcc_sgw_csr_tau_rau_x2ho_failures

Type

64-bit Counter

Description

Number of SGW TAU/RAU/X2-HO with SGW Change Procedures failed

Labels

dnn, ratType, causecode

• smfcc_sgw_csr_s1ho_attempts

Type

64-bit Counter

Description

Number of SGW S1-HO with SGW Change Procedures attempted

Labels

dnn, ratType

• smfcc_sgw_csr_s1ho_failures

Type

64-bit Counter

Description

Number of SGW S1-HO with SGW Change Procedures failed

Labels

dnn, ratType, causecode

• smfcc_sgw_dedicated_delete_bearer_request_attempts

Type

64-bit Counter

Description

Number of Dedicated Delete Bearer Request Procedures attempted

Labels

dnn

- smfcc_sgw_dedicated_delete_bearer_request_failures

Type

64-bit Counter

Description

Number of Dedicated Delete Bearer Request Procedures failed

Labels

dnn, causecode

- smfcc_sgw_error_indication_attempts

Type

64-bit Counter

Description

Number of SGW Error Indication Procedures attempted

Labels

dnn, ratType

- smfcc_sgw_error_indication_failures

Type

64-bit Counter

Description

Number of SGW Error Indication Procedures failed

Labels

dnn, ratType, causecode

- smfcc_sgw_s1u_error_indication_attempts

Type

64-bit Counter

Description

Number of S1U SGW Error Indication Procedures attempted

Labels

dnn, ratType

- smfcc_sgw_s1u_error_indication_failures

Type

64-bit Counter

Description

Number of S1U SGW Error Indication Procedures failed

Labels
<i>dnn, ratType, causecode</i>
smfcc_sgw_s5s8u_default_bearer_error_indication_attempts
Type
64-bit Counter
Description
Number of S5S8U Default Bearer SGW Error Indication Procedures attempted
Labels
<i>dnn, ratType</i>
smfcc_sgw_s5s8u_default_bearer_error_indication_failures
Type
64-bit Counter
Description
Number of S5S8U Default Bearer SGW Error Indication Procedures failed
Labels
<i>dnn, ratType, causecode</i>
smfcc_sgw_s5s8u_dedicated_bearer_error_indication_attempts
Type
64-bit Counter
Description
Number of S5S8U Dedicated Bearer SGW Error Indication Procedures attempted
Labels
<i>dnn, ratType</i>
smfcc_sgw_s5s8u_dedicated_bearer_error_indication_failures
Type
64-bit Counter
Description
Number of S5S8U Dedicated Bearer SGW Error Indication Procedures failed
Labels
<i>dnn, ratType, causecode</i>

SMFCC SAEGW Procedures Statistics

smfcc_saegw_initial_attach_attempts
Type
64-bit Counter
Description
Number of SAEGW Initial Attach Procedures attempted
Labels
<i>dnn, snssai, ratType</i>
smfcc_saegw_initial_attach_successes

Type

64-bit Counter

Description

Number of SAEGW Initial Attach Procedures successful

Labels

dnn, snssai, ratType

• smfcc_saegw_initial_attach_failures

Type

64-bit Counter

Description

Number of SAEGW Initial Attach Procedures failed

Labels

dnn, snssai, ratType, causecode

• smfcc_saegw_ue_initiated_detach_attempts

Type

64-bit Counter

Description

Number of SAEGW Delete Session Procedures attempted

Labels

dnn, snssai, ratType

• smfcc_saegw_ue_initiated_detach_successes

Type

64-bit Counter

Description

Number of SAEGW Delete Session Procedures successful

Labels

dnn, snssai, ratType

• smfcc_saegw_ue_initiated_detach_failures

Type

64-bit Counter

Description

Number of SAEGW Delete Session Procedures failed

Labels

dnn, snssai, ratType, causecode

• smfcc_saegw_modify_bearer_request_attempts

Type

64-bit Counter

Description

Number of Modify Bearer Request Procedures attempted

Labels

dnn, snssai, ratType

- smfcc_saegw_modify_bearer_request_successes

Type

64-bit Counter

Description

Number of Modify Bearer Request Procedures successful

Labels

dnn, snssai, ratType

- smfcc_saegw_modify_bearer_request_failures

Type

64-bit Counter

Description

Number of Modify Bearer Request Procedures failed

Labels

dnn, snssai, ratType, causecode

- smfcc_saegw_pcf_initiated_terminate_attempts

Type

64-bit Counter

Description

Number of SAEGW PCF Initiated Terminate Procedures attempted

Labels

dnn, snssai, ratType

- smfcc_saegw_pcf_initiated_terminate_successes

Type

64-bit Counter

Description

Number of SAEGW PCF Initiated Terminate Procedures successful

Labels

dnn, snssai, ratType

- smfcc_saegw_pcf_initiated_terminate_failures

Type

64-bit Counter

Description

Number of SAEGW PCF Initiated Terminate Procedures failed

Labels

dnn, snssai, ratType, causecode

- smfcc_saegw_nw_initiated_release_attempts

Type

64-bit Counter

Description

Number of SAEGW NW Initiated Release Procedures attempted

Labels

dnn, snssai, ratType, causecode

• smfcc_saegw_nw_initiated_release_successes

Type

64-bit Counter

Description

Number of SAEGW NW Initiated Release Procedures successful

Labels

dnn, snssai, ratType

• smfcc_saegw_nw_initiated_release_failures

Type

64-bit Counter

Description

Number of SAEGW NW Initiated Release Procedures failed

Labels

dnn, snssai, ratType, causecode

• smfcc_saegw_pcf_initiated_policy_modification_attempts

Type

64-bit Counter

Description

Number of PCF Initiated Policy Modification Procedures attempted for SAEGW

Labels

dnn, snssai, ratType

• smfcc_saegw_pcf_initiated_policy_modification_successes

Type

64-bit Counter

Description

Number of PCF Initiated Policy Modification Procedures successfulfor SAEGW

Labels

dnn, snssai, ratType

• smfcc_saegw_pcf_initiated_policy_modification_failures

Type

64-bit Counter

Description

Number of PCF Initiated Policy Modification Procedures failedfor SAEGW

Labels

dnn, snssai, ratType, causecode

• smfcc_saegw_pcf_initiated_dedicated_bearer_activation_attempts

Type

64-bit Counter

Description

Number of PCF Initiated Dedicated Bearer Activation Procedures attempted for SAEGW

Labels

dnn, snssai, ratType

• smfcc_saegw_pcf_initiated_dedicated_bearer_activation_successes

Type

64-bit Counter

Description

Number of PCF Initiated Dedicated Bearer Activation Procedures successful for SAEGW

Labels

dnn, snssai, ratType

• smfcc_saegw_pcf_initiated_dedicated_bearer_activation_failures

Type

64-bit Counter

Description

Number of PCF Initiated Dedicated Bearer Activation Procedures failed for SAEGW

Labels

dnn, snssai, ratType, causecode

• smfcc_saegw_pcf_initiated_dedicated_bearer_delete_attempts

Type

64-bit Counter

Description

Number of PCF Initiated Dedicated Bearer Delete Procedures attempted for SAEGW

Labels

dnn, snssai, ratType

• smfcc_saegw_pcf_initiated_dedicated_bearer_delete_successes

Type

64-bit Counter

Description

Number of PCF Initiated Dedicated Bearer Delete Procedures successful for SAEGW

Labels

dnn, snssai, ratType

• smfcc_saegw_pcf_initiated_dedicated_bearer_delete_failures

Type

64-bit Counter

Description
Number of PCF Initiated Dedicated Bearer Delete Procedures failed for SAEGW
Labels
<i>dnn, snssai, ratType, causecode</i>
• smfcc_saegw_pcf_initiated_dedicated_bearer_update_attempts
Type
64-bit Counter
Description
Number of PCF Initiated Dedicated Bearer Update Procedures attempted for SAEGW
Labels
<i>dnn, snssai, ratType</i>
• smfcc_saegw_pcf_initiated_dedicated_bearer_update_successes
Type
64-bit Counter
Description
Number of PCF Initiated Dedicated Bearer Update Procedures successful for SAEGW
Labels
<i>dnn, snssai, ratType</i>
• smfcc_saegw_pcf_initiated_dedicated_bearer_update_failures
Type
64-bit Counter
Description
Number of PCF Initiated Dedicated Bearer Update Procedures failed for SAEGW
Labels
<i>dnn, snssai, ratType, causecode</i>
• smfcc_saegw_process_pcf_update_policy_rules_attempts
Type
64-bit Counter
Description
Number of Process Pcf Update Policy Rules Procedures attempted for SAEGW
Labels
<i>dnn, snssai, ratType</i>
• smfcc_saegw_process_pcf_update_policy_rules_successes
Type
64-bit Counter
Description
Number of Process Pcf Update Policy Rules Procedures successful for SAEGW
Labels
<i>dnn, snssai, ratType</i>

smfcc_saegw_process_pcf_update_policy_rules_failures
Type
64-bit Counter
Description
Number of Process Pcf Update Policy Rules Procedures failed for SAEGW
Labels
<i>dnn, snssai, ratType, causecode</i>
smfcc_saegw_policy_timeout_handling_attempts
Type
64-bit Counter
Description
Number of Policy Timer Timeout Handling Procedures attempted for SAEGW
Labels
<i>timerType, dnn, snssai</i>
smfcc_saegw_policy_timeout_handling_successes
Type
64-bit Counter
Description
Number of Policy Timer Timeout Handling Procedures successful for SAEGW
Labels
<i>timerType, dnn, snssai</i>
smfcc_saegw_policy_timeout_handling_failures
Type
64-bit Counter
Description
Number of Policy Timer Timeout Handling Procedures failed for SAEGW
Labels
<i>timerType, dnn, snssai, causecode</i>
smfcc_saegw_hss_initiated_modify_bearer_command_attempts
Type
64-bit Counter
Description
Number of HSS Initiated Modify Bearer Command Procedures attempted for SAEGW
Labels
<i>dnn, snssai, ratType</i>
smfcc_saegw_hss_initiated_modify_bearer_command_successes
Type
64-bit Counter
Description

Number of HSS Initiated Modify Bearer Command Procedures successful for SAEGW

Labels

dnn, snssai, ratType

- smfcc_saegw_hss_initiated_modify_bearer_command_failures

Type

64-bit Counter

Description

Number of HSS Initiated Modify Bearer Command Procedures failed for SAEGW

Labels

dnn, snssai, ratType, causecode

- smfcc_saegw_error_indication_procedure_attempts

Type

64-bit Counter

Description

Number of Error Indication Procedures attempted for SAEGW

Labels

dnn, snssai, ratType

- smfcc_saegw_error_indication_procedure_successes

Type

64-bit Counter

Description

Number of Error Indication Procedures successful for SAEGW

Labels

dnn, snssai, ratType

- smfcc_saegw_error_indication_procedure_failures

Type

64-bit Counter

Description

Number of Error Indication Procedures failed for SAEGW

Labels

dnn, snssai, ratType, causecode

- smfcc_saegw_session_idle_procedure_attempts

Type

64-bit Counter

Description

Number of Session Idle Procedures attempted for SAEGW

Labels

dnn, snssai, ratType

- smfcc_saegw_session_idle_procedure_successes

Type

64-bit Counter

Description

Number of Session Idle Procedures successful for SAEGW

Labels

dnn, snssai, ratType

- smfcc_saegw_session_idle_procedure_failures

Type

64-bit Counter

Description

Number of Session Idle Procedures failed for SAEGW

Labels

dnn, snssai, ratType, causecode

- smfcc_saegw_qos_flow_inactivity_procedure_attempts

Type

64-bit Counter

Description

Number of QosFlow Inactivity Procedures attempted for SAEGW

Labels

dnn, snssai, ratType

- smfcc_saegw_qos_flow_inactivity_procedure_successes

Type

64-bit Counter

Description

Number of QosFlow Inactivity Procedures successful for SAEGW

Labels

dnn, snssai, ratType

- smfcc_saegw_qos_flow_inactivity_procedure_failures

Type

64-bit Counter

Description

Number of QosFlow Inactivity Procedures failed for SAEGW

Labels

dnn, snssai, ratType, causecode

- smfcc_saegw_bearer_resource_command_attempts

Type

64-bit Counter

Description

Number of Bearer Resource Command Procedures attempted for SAEGW

Labels

dnn, snssai, ratType

- smfcc_saegw_bearer_resource_command_successes

Type

64-bit Counter

Description

Number of Bearer Resource Command Procedures successful for SAEGW

Labels

dnn, snssai, ratType

- smfcc_saegw_bearer_resource_command_failures

Type

64-bit Counter

Description

Number of Bearer Resource Command Procedures failed for SAEGW

Labels

dnn, snssai, ratType, causecode

- smfcc_saegw_bearer_resource_command_unknown_ebi_attempts

Type

64-bit Counter

Description

Number of Bearer Resource Command Procedures attempted for Unknown ebi for SAEGW

Labels

dnn, snssai, ratType

- smfcc_saegw_bearer_resource_command_unknown_ebi_successes

Type

64-bit Counter

Description

Number of Bearer Resource Command Procedures successful for Unknown ebi for SAEGW

Labels

dnn, snssai, ratType

- smfcc_saegw_bearer_resource_command_unknown_ebi_failures

Type

64-bit Counter

Description

Number of Bearer Resource Command Procedures failed for Unknown ebi for SAEGW

Labels

dnn, snssai, ratType, causecode

- smfcc_saegw_modify_bearer_ho_attempts

Type

64-bit Counter
Description
Number of Modify Bearer Request Handover Procedures attempted for SAEGW
Labels
<i>dnn, snssai, ratType</i>
• smfcc_saegw_modify_bearer_ho_successes
Type
64-bit Counter
Description
Number of Modify Bearer Request X2 Handover Procedures successful for SAEGW
Labels
<i>dnn, snssai, ratType</i>
• smfcc_saegw_modify_bearer_ho_failures
Type
64-bit Counter
Description
Number of Modify Bearer Request X2 Handover Procedures failed for SAEGW
Labels
<i>dnn, snssai, ratType, causecode</i>
• smfcc_saegw_csr_tau_rau_x2ho_attempts
Type
64-bit Counter
Description
Number of SAEGW TAU/RAU/X2-HO with SGW Change Procedures attempted
Labels
<i>dnn, snssai, ratType</i>
• smfcc_saegw_csr_tau_rau_x2ho_success
Type
64-bit Counter
Description
Number of SAEGW TAU/RAU/X2-HO with SGW Change Procedures successful
Labels
<i>dnn, snssai, ratType</i>
• smfcc_saegw_csr_tau_rau_x2ho_failures
Type
64-bit Counter
Description
Number of SAEGW TAU/RAU/X2-HO with SGW Change Procedures failed
Labels

dnn, snssai, ratType, causecode

• smfcc_saegw_csr_s1ho_attempts

Type

64-bit Counter

Description

Number of SAEGW S1-HO with SGW Change Procedures attempted

Labels

dnn, snssai, ratType

• smfcc_saegw_csr_s1ho_failures

Type

64-bit Counter

Description

Number of SAEGW S1-HO with SGW Change Procedures failed

Labels

dnn, snssai, ratType, causecode

• smfcc_saegw_to_pgw_modify_bearer_request_ho_attempts

Type

64-bit Counter

Description

Number of SAEGW to PGW Modify Bearer Request Procedures attempted

Labels

dnn, ratType

• smfcc_saegw_to_pgw_modify_bearer_request_ho_failures

Type

64-bit Counter

Description

Number of SAEGW to PGW Modify Bearer Request Procedures failed

Labels

dnn, ratType, causecode

• smfcc_saegw_to_pgw_ho_delete_session_request_attempts

Type

64-bit Counter

Description

Number of SAEGW to PGW HO Delete Session Request Procedures attempted

Labels

dnn, ratType

• smfcc_saegw_to_pgw_ho_delete_session_request_failures

Type

64-bit Counter

Description

Number of SAEGW to PGW HO Delete Session Request Procedures failed

Labels

dnn, ratType, causecode

- smfcc_saegw_release_access_bearer_request_attempts

Type

64-bit Counter

Description

Number of SAEGW Release Access Bearer Request Procedures attempted

Labels

dnn, snssai, ratType

- smfcc_saegw_release_access_bearer_request_failures

Type

64-bit Counter

Description

Number of SAEGW Release Access Bearer Request Procedures failed

Labels

dnn, snssai, ratType, causecode

- smfcc_saegw_nw_initiated_service_request_attempts

Type

64-bit Counter

Description

Number of SAEGW Network service request Procedures attempted

Labels

dnn, snsaai, ratType

- smfcc_saegw_nw_initiated_service_request_failures

Type

64-bit Counter

Description

Number of SAEGW Network initiated service request Procedures failed

Labels

dnn, snssai, ratType, causecode

- smfcc_saegw_nw_initiated_service_request_aborted

Type

64-bit Counter

Description

Number of SAEGW Network initiated service request Procedures aborted due to higher arp pl DDN received from upf

Labels

dnn, snssai, ratType

- smfcc_saegw_nw_initiated_service_request_dropped
 - Type
 - 64-bit Counter
 - Description
 - Number of SAEGW Network initiated service request Procedures dropped due to higher arp pl DDN already sent/max dnns reached
 - Labels
 - dnn, snssai, ratType, cause**
- smfcc_saegw_ue_initiated_service_request_attempts
 - Type
 - 64-bit Counter
 - Description
 - Number of SAEGW Ue Init service request Procedures attempted
 - Labels
 - dnn, snssai, ratType**
- smfcc_saegw_ue_initiated_service_request_failures
 - Type
 - 64-bit Counter
 - Description
 - Number of SAEGW UE initiated service request Procedures failed
 - Labels
 - dnn, snssai, ratType, causecode**
- smfcc_saegw_nw_network_initiated_service_request_timeouts
 - Type
 - 64-bit Counter
 - Description
 - Number of SAEGW Network Initiated Service Request Procedures Timeouts
 - Labels
 - dnn, snssai, ratType**

SMFCC GGSN-C Procedure Statistics

- smfcc_ggsn_create_pdp_context_attempts
 - Type
 - 64-bit Counter
 - Description
 - Number of GGSN Create PDP Context Procedures attempted
 - Labels
 - dnn, snssai, ratType**
- smfcc_ggsn_create_pdp_context_successes
 - Type
 - 64-bit Counter

	Description Number of GGSN Create PDP Context Procedures successful Labels <i>dnn, snssai, ratType</i>
• smfcc_ggsn_create_pdp_context_failures	Type 64-bit Counter Description Number of GGSN Create PDP Context Procedures failed Labels <i>dnn, snssai, ratType, causecode</i>
• smfcc_ggsn_roaming_session_create_pdp_context_attempts	Type 64-bit Counter Description Number of GGSN Roaming Create PDP Context Procedures attempted Labels <i>dnn, snssai, ratType</i>
• smfcc_ggsn_roaming_session_create_pdp_context_successes	Type 64-bit Counter Description Number of GGSN Roaming Create PDP Context Procedures successful Labels <i>dnn, snssai, ratType</i>
• smfcc_ggsn_roaming_session_create_pdp_context_failures	Type 64-bit Counter Description Number of GGSN Roaming Create PDP Context Procedures failed Labels <i>dnn, snssai, ratType, causecode</i>
• smfcc_ggsn_sgsn_initiated_update_pdp_context_attempts	Type 64-bit Counter Description Number of GGSN Update PDP Context Procedures attempted Labels <i>dnn, snssai, ratType</i>

smfcc_ggsn_sgsn_initiated_update_pdp_context_successes
Type
64-bit Counter
Description
Number of GGSN Update PDP Context Procedures successful
Labels
dnn, snssai, ratType
smfcc_ggsn_sgsn_initiated_update_pdp_context_failures
Type
64-bit Counter
Description
Number of GGSN Update PDP Context Procedures failed
Labels
dnn, snssai, ratType, causecode
smfcc_ggsn_roaming_session_sgsn_initiated_update_pdp_context_attempts
Type
64-bit Counter
Description
Number of GGSN Roaming Update PDP Context Procedures attempted
Labels
dnn, snssai, ratType
smfcc_ggsn_roaming_session_sgsn_initiated_update_pdp_context_successes
Type
64-bit Counter
Description
Number of GGSN Roaming Update PDP Context Procedures successful
Labels
dnn, snssai, ratType
smfcc_ggsn_roaming_session_sgsn_initiated_update_pdp_context_failures
Type
64-bit Counter
Description
Number of GGSN Roaming Update PDP Context Procedures failed
Labels
dnn, snssai, ratType, causecode
smfcc_ggsn_sgsn_initiated_delete_pdp_context_attempts
Type
64-bit Counter
Description

Number of GGSN delete PDP Context Procedures attempted

Labels

dnn, snssai, ratType

- smfcc_ggsn_sgsn_initiated_delete_pdp_context_successes

Type

64-bit Counter

Description

Number of GGSN delete PDP Context Procedures successful

Labels

dnn, snssai, ratType

- smfcc_ggsn_sgsn_initiated_delete_pdp_context_failures

Type

64-bit Counter

Description

Number of GGSN delete PDP Context Procedures failed

Labels

dnn, snssai, ratType, causecode

- smfcc_ggsn_sgsn_initiated_roaming_session_delete_pdp_context_attempts

Type

64-bit Counter

Description

Number of GGSN delete roaming PDP Context Procedures attempted

Labels

dnn, snssai, ratType

- smfcc_ggsn_sgsn_initiated_roaming_session_delete_pdp_context_successes

Type

64-bit Counter

Description

Number of GGSN delete roaming PDP Context Procedures successful

Labels

dnn, snssai, ratType

- smfcc_ggsn_sgsn_initiated_roaming_session_delete_pdp_context_failures

Type

64-bit Counter

Description

Number of GGSN delete roaming PDP Context Procedures failed

Labels

dnn, snssai, ratType, causecode

- smfcc_ggsn_nw_initiated_release_attempts

Type

64-bit Counter

Description

Number of GGSN NW Initiated Release Procedures attempted

Labels

dnn, snssai, ratType, causecode

• smfcc_ggsn_nw_initiated_release_successes

Type

64-bit Counter

Description

Number of GGSN NW Initiated Release Procedures successful

Labels

dnn, snssai, ratType

• smfcc_ggsn_nw_initiated_release_failures

Type

64-bit Counter

Description

Number of GGSN NW Initiated Release Procedures failed

Labels

dnn, snssai, ratType, causecode

• smfcc_ggsn_roaming_session_nw_initiated_release_attempts

Type

64-bit Counter

Description

Number of GGSN NW Initiated Release Procedures attempted for roaming sessions

Labels

dnn, snssai, ratType, causecode

• smfcc_ggsn_roaming_session_nw_initiated_release_successes

Type

64-bit Counter

Description

Number of GGSN NW Initiated Release Procedures successful for roaming sessions

Labels

dnn, snssai, ratType

• smfcc_ggsn_roaming_session_nw_initiated_release_failures

Type

64-bit Counter

Description

Number of GGSN NW Initiated Release Procedures failed for roaming sessions

Labels

dnn, snssai, ratType, causecode

• smfcc_ggsn_pcf_initiated_terminate_attempts

Type

64-bit Counter

Description

Number of GGSN PCF Initiated Terminate Procedures attempted

Labels

dnn, snssai, ratType

• smfcc_ggsn_pcf_initiated_terminate_successes

Type

64-bit Counter

Description

Number of GGSN PCF Initiated Terminate Procedures successful

Labels

dnn, snssai, ratType

• smfcc_ggsn_pcf_initiated_terminate_failures

Type

64-bit Counter

Description

Number of GGSN PCF Initiated Terminate Procedures failed

Labels

dnn, snssai, ratType, causecode

• smfcc_ggsn_roaming_session_pcf_initiated_terminate_attempts

Type

64-bit Counter

Description

Number of GGSN PCF Initiated Terminate Procedures attempted for roaming sessions

Labels

dnn, snssai, ratType

• smfcc_ggsn_roaming_session_pcf_initiated_terminate_successes

Type

64-bit Counter

Description

Number of GGSN PCF Initiated Terminate Procedures successful for roaming sessions

Labels

dnn, snssai, ratType

• smfcc_ggsn_roaming_session_pcf_initiated_terminate_failures

Type

64-bit Counter

Description

Number of GGSN PCF Initiated Terminate Procedures failed for roaming sessions

Labels

dnn, snssai, ratType, causecode

• smfcc_ggsn_pcf_initiated_policy_modification_attempts

Type

64-bit Counter

Description

Number of PCF Initiated (GGSN Initiated) Policy Modification Procedures attempted

Labels

dnn, snssai, ratType

• smfcc_ggsn_pcf_initiated_policy_modification_failures

Type

64-bit Counter

Description

Number of PCF Initiated (GGSN Initiated) Policy Modification Procedures failed

Labels

dnn, snssai, ratType, causecode

• smfcc_ggsn_pcf_initiated_policy_modification_successes

Type

64-bit Counter

Description

Number of PCF Initiated (GGSN Initiated) Policy Modification Procedures successful

Labels

dnn, snssai, ratType

• smfcc_ggsn_roaming_session_pcf_initiated_policy_modification_attempts

Type

64-bit Counter

Description

Number of PCF Initiated (GGSN Initiated) Policy Modification Procedures attempted for roaming sessions

Labels

dnn, snssai, ratType

• smfcc_ggsn_roaming_session_pcf_initiated_policy_modification_successes

Type

64-bit Counter

Description

Number of PCF Initiated (GGSN Initiated) Policy Modification Procedures successful for roaming sessions

Labels

dnn, snssai, ratType

- smfcc_ggsn_roaming_session_pcf_initiated_policy_modification_failures

Type

64-bit Counter

Description

Number of PCF Initiated (GGSN Initiated) Policy Modification Procedures failed for roaming sessions

Labels

dnn, snssai, ratType, causecode

- smfcc_ggsn_3g_to_4g_handover_attempts

Type

64-bit Counter

Description

Number of 2G/3G to 4G Handover Procedures attempted

Labels

dnn, snssai, ratType

- smfcc_ggsn_3g_to_4g_handover_successes

Type

64-bit Counter

Description

Number of 2G/3G to 4G Handover Procedures successful

Labels

dnn, snssai, ratType

- smfcc_ggsn_3g_to_4g_handover_failures

Type

64-bit Counter

Description

Number of 2G/3G to 4G Handover Procedures failed

Labels

dnn, snssai, ratType, causecode

- smfcc_ggsn_4g_to_3g_handover_attempts

Type

64-bit Counter

Description

Number of 4G to 2G/3G Handover Procedures attempted

Labels

dnn, snssai, ratType

- smfcc_ggsn_4g_to_3g_handover_successes

Type

64-bit Counter

Description
Number of 4G to 2G/3G Handover Procedures successful
Labels
<i>dnn, snssai, ratType</i>
• smfcc_ggsn_4g_to_3g_handover_failures
Type
64-bit Counter
Description
Number of 4G to 2G/3G Handover Procedures failed
Labels
<i>dnn, snssai, ratType, causecode</i>
• smfcc_ggsn_roaming_session_3g_to_4g_handover_attempts
Type
64-bit Counter
Description
Number of 2G/3G to 4G Handover Procedures attempted for roaming sessions
Labels
<i>dnn, snssai, ratType</i>
• smfcc_ggsn_roaming_session_3g_to_4g_handover_successes
Type
64-bit Counter
Description
Number of 2G/3G to 4G Handover Procedures successful roaming sessions
Labels
<i>dnn, snssai, ratType</i>
• smfcc_ggsn_roaming_session_3g_to_4g_handover_failures
Type
64-bit Counter
Description
Number of 2G/3G to 4G Handover Procedures failed roaming sessions
Labels
<i>dnn, snssai, ratType, causecode</i>
• smfcc_ggsn_roaming_session_4g_to_3g_handover_attempts
Type
64-bit Counter
Description
Number of 4G to 2G/3G Handover Procedures attempted roaming sessions
Labels
<i>dnn, snssai, ratType</i>

- smfcc_ggsn_roaming_session_4g_to_3g_handover_successes

Type
64-bit Counter
Description
Number of 4G to 2G/3G Handover Procedures successful roaming sessions
Labels
<i>dnn, snssai, ratType</i>

- smfcc_ggsn_roaming_session_4g_to_3g_handover_failures

Type
64-bit Counter
Description
Number of 4G to 2G/3G Handover Procedures failed roaming sessions
Labels
<i>dnn, snssai, ratType, causecode</i>

- smfcc_ggsn_inter_sgsn_handover_attempts

Type
64-bit Counter
Description
Number of Update PDP Context Request with SGSN change Procedures attempted
Labels
<i>dnn, snssai, ratType</i>

- smfcc_ggsn_inter_sgsn_handover_successes

Type
64-bit Counter
Description
Number of Update PDP Context Request with SGSN change Procedures successful
Labels
<i>dnn, snssai, ratType</i>

- smfcc_ggsn_inter_sgsn_handover_failures

Type
64-bit Counter
Description
Number of Update PDP Context Request with SGSN change Procedures failed
Labels
<i>dnn, snssai, ratType, causecode</i>

- smfcc_ggsn_roaming_session_inter_sgsn_handover_attempts

Type
64-bit Counter
Description

Number of Update PDP Context Request with SGSN change Procedures attempted for roaming sessions

Labels

dnn, snssai, ratType

- smfcc_ggsn_roaming_session_inter_sgsn_handover_successes

Type

64-bit Counter

Description

Number of Update PDP Context Request with SGSN change Procedures successful for roaming sessions

Labels

dnn, snssai, ratType

- smfcc_ggsn_roaming_session_inter_sgsn_handover_failures

Type

64-bit Counter

Description

Number of Update PDP Context Request with SGSN change Procedures failed for roaming sessions

Labels

dnn, snssai, ratType, causecode

- smfcc_ggsn_error_indication_attempts

Type

64-bit Counter

Description

Number of GGSN-C Error Indication Procedures attempted

Labels

dnn, snssai

- smfcc_ggsn_error_indication_successes

Type

64-bit Counter

Description

Number of GGSN-C Error Indication Procedures successful

Labels

dnn, snssai

- smfcc_ggsn_error_indication_failures

Type

64-bit Counter

Description

Number of GGSN-C Error Indication Procedures failed

Labels

dnn, snssai, causecode

- smfcc_ggsn_roaming_session_error_indication_attempts

Type

64-bit Counter

Description

Number of GGSN-C Error Indication Procedures attempted for roaming sessions

Labels

dnn, snssai

- smfcc_ggsn_roaming_session_error_indication_successes

Type

64-bit Counter

Description

Number of GGSN-C Error Indication Procedures successful for roaming sessions

Labels

dnn, snssai

- smfcc_ggsn_roaming_session_error_indication_failures

Type

64-bit Counter

Description

Number of GGSN-C Error Indication Procedures failed for roaming sessions

Labels

dnn, snssai, causecode

- smfcc_ggsn_pcf_initiated_bearer_update_attempts

Type

64-bit Counter

Description

Number of PCF Initiated Bearer Update Procedures attempted

Labels

dnn, snssai, ratType

- smfcc_ggsn_pcf_initiated_bearer_update_successes

Type

64-bit Counter

Description

Number of PCF Initiated Bearer Update Procedures successful

Labels

dnn, snssai, ratType

- smfcc_ggsn_pcf_initiated_bearer_update_failures

Type

64-bit Counter

Description

Number of PCF Initiated Bearer Update Procedures failed

Labels

dnn, snssai, ratType, causecode

- smfcc_ggsn_roaming_session_pcf_initiated_bearer_update_attempts

Type

64-bit Counter

Description

Number of PCF Initiated Bearer Update Procedures attempted for roaming sessions

Labels

dnn, snssai, ratType

- smfcc_ggsn_roaming_session_pcf_initiated_bearer_update_successes

Type

64-bit Counter

Description

Number of PCF Initiated Bearer Update Procedures successful for roaming sessions

Labels

dnn, snssai, ratType

- smfcc_ggsn_roaming_session_pcf_initiated_bearer_update_failures

Type

64-bit Counter

Description

Number of PCF Initiated Bearer Update Procedures failed for roaming sessions

Labels

dnn, snssai, ratType, causecode

- smfcc_ggsn_process_pcf_update_policy_rules_attempts

Type

64-bit Counter

Description

Number of Process Pcf Update Policy Rules Procedures attempted

Labels

dnn, snssai, ratType

- smfcc_ggsn_process_pcf_update_policy_rules_successes

Type

64-bit Counter

Description

Number of Process Pcf Update Policy Rules Procedures successful

Labels

dnn, snssai, ratType

- smfcc_ggsn_process_pcf_update_policy_rules_failures

Type

64-bit Counter

Description

Number of Process Pcf Update Policy Rules Procedures failed

Labels

dnn, snssai, ratType, causecode

- smfcc_ggsn_roaming_session_process_pcf_update_policy_rules_attempts

Type

64-bit Counter

Description

Number of Process Pcf Update Policy Rules Procedures attempted for roaming sessions

Labels

dnn, snssai, ratType

- smfcc_ggsn_roaming_session_process_pcf_update_policy_rules_successes

Type

64-bit Counter

Description

Number of Process Pcf Update Policy Rules Procedures successful for roaming sessions

Labels

dnn, snssai, ratType

- smfcc_ggsn_roaming_session_process_pcf_update_policy_rules_failures

Type

64-bit Counter

Description

Number of Process Pcf Update Policy Rules Procedures failed for roaming sessions

Labels

dnn, snssai, ratType, causecode

- smfcc_ggsn_session_idle_procedure_attempts

Type

64-bit Counter

Description

Number of Ggsn Session Idle Procedures attempted

Labels

dnn, snssai, ratType

- smfcc_ggsn_session_idle_procedure_successes

Type

64-bit Counter

Description

Number of Ggsn Session Idle Procedures successful

Labels

dnn, snssai, ratType

- smfcc_ggsn_session_idle_procedure_failures

Type

64-bit Counter

Description

Number of Ggsn Session Idle Procedures failed

Labels

dnn, snssai, ratType, causecode

- smfcc_ggsn_roaming_session_idle_procedure_attempts

Type

64-bit Counter

Description

Number of Ggsn Session Idle Procedures attempted for roaming sessions

Labels

dnn, snssai, ratType

- smfcc_ggsn_roaming_session_idle_procedure_successes

Type

64-bit Counter

Description

Number of Ggsn Session Idle Procedures successful for roaming sessions

Labels

dnn, snssai, ratType

- smfcc_ggsn_roaming_session_idle_procedure_failures

Type

64-bit Counter

Description

Number of Ggsn Session Idle Procedures failed for roaming sessions

Labels

dnn, snssai, ratType, causecode

- smfcc_ggsn_qos_flow_inactivity_procedure_attempts

Type

64-bit Counter

Description

Number of Ggsn QosFlow Inactivity Procedures attempted

Labels

dnn, snssai, ratType

- smfcc_ggsn_qos_flow_inactivity_procedure_successes

Type

64-bit Counter

Description

Number of Ggsn QosFlow Inactivity Procedures successful

Labels

dnn, snssai, ratType

- smfcc_ggsn_qos_flow_inactivity_procedure_failures

Type

64-bit Counter

Description

Number of Ggsn QosFlow Inactivity Procedures failed

Labels

dnn, snssai, ratType, causecode

- smfcc_ggsn_roaming_session_qos_flow_inactivity_procedure_attempts

Type

64-bit Counter

Description

Number of Ggsn QosFlow Inactivity Procedures attempted for roaming sessions

Labels

dnn, snssai, ratType

- smfcc_ggsn_roaming_session_qos_flow_inactivity_procedure_successes

Type

64-bit Counter

Description

Number of Ggsn QosFlow Inactivity Procedures successful for roaming sessions

Labels

dnn, snssai, ratType

- smfcc_ggsn_roaming_session_qos_flow_inactivity_procedure_failures

Type

64-bit Counter

Description

Number of Ggsn QosFlow Inactivity Procedures failed for roaming sessions

Labels

dnn, snssai, ratType, causecode

SMF Call Performance Statistics

- smfcc_pcscf_sessions_activated

Type

64-bit Counter

Description

Number of session activated using P-CSCF address

Labels

dnn, snssai, ipaddress

- smfcc_pcscf_sessions_deactivated

Type

64-bit Counter

Description

Number of session deactivated using P-CSCF address

Labels

dnn, snssai, ipaddress

- smfcc_qos_flows_setup_success

Type

64-bit Counter

Description

Number of Qos flows successfully setup

Labels

snssai, fiveQi, arp

- smfcc_qos_flows_setup_failure

Type

64-bit Counter

Description

Number of Qos flows failed to setup

Labels

snssai, fiveQi, arp, cause

- smfcc_nw_init_qos_flows_released

Type

64-bit Counter

Description

Number of Network initiated Qos flows released

Labels

snssai, fiveQi, arp

- smfcc_access_init_qos_flows_released

Type

64-bit Counter

Description

Number of Access initiated Qos flows released

Labels

snssai, fiveQi, arp

- smfcc_abnormal_qos_flows_released

Type

64-bit Counter

	Description Number of Qos flows released abnormally Labels <i>snssai, fiveQi, arp, cause</i>
• smfcc_pcc_rule_add_attempts	Type 64-bit Counter Description Number of pcc rule add attempts Labels <i>dnn, snssai</i>
• smfcc_pcc_rule_add_successes	Type 64-bit Counter Description Number of pcc rule add attempt successes Labels <i>dnn, snssai</i>
• smfcc_pcc_rule_add_failures	Type 64-bit Counter Description Number of pcc rule add attempt failures Labels <i>dnn, snssai, causecode</i>
• smfcc_pcc_rule_modify_attempts	Type 64-bit Counter Description Number of pcc rule modify attempts Labels <i>dnn, snssai</i>
• smfcc_pcc_rule_modify_successes	Type 64-bit Counter Description Number of pcc rule modify attempt successes Labels <i>dnn, snssai</i>

- smfcc_pcc_rule_modify_failures

Type
64-bit Counter
Description
Number of pcc rule modify attempt failures
Labels
<i>dnn, snssai, causecode</i>

- smfcc_pcc_rule_delete_attempts

Type
64-bit Counter
Description
Number of pcc rule delete attempts
Labels
<i>dnn, snssai</i>

- smfcc_pcc_rule_delete_successes

Type
64-bit Counter
Description
Number of pcc rule delete attempt successes
Labels
<i>dnn, snssai</i>

- smfcc_pcc_rule_delete_failures

Type
64-bit Counter
Description
Number of pcc rule delete attempt failures
Labels
<i>dnn, snssai, causecode</i>

- smfcc_pcc_rule_install_drops

Type
64-bit Counter
Description
Number of pcc rule installs dropped
Labels
<i>dnn, snssai</i>

- smfcc_session_rule_add_failures

Type
64-bit Counter
Description

Number of session rule add attempt failures

Labels

failurecode

- smfcc_session_rule_modify_failures

Type

64-bit Counter

Description

Number of session rule modify attempt failures

Labels

failurecode

- smfcc_session_rule_failures_reported_to_pcf

Type

64-bit Counter

Description

Number of session rule failures reported to PCF

Labels

failurecode

- smfcc_default_policy_used_pdu_session_count

Type

64-bit Counter

Description

Number of sessions which used default policy

Labels

dnn, snssai

- smfcc_default_quota_used_pdu_session_count

Type

64-bit Counter

Description

Number of sessions which used default quota

Labels

dnn, snssai

- smfcc_dnn_matching_exact_match

Type

64-bit Counter

Description

Number of Session setup with exact matching DNN

Labels

dnn

- smfcc_dnn_matching_ni_only

Type
64-bit Counter
Description
Number of Session setup with NI Only matching DNN
Labels
dnn
• smfcc_dnn_matching_ni_and_malformed_oι
Type
64-bit Counter
Description
Number of Session setup with NI and Malformed OI matching DNN
Labels
dnn
• smfcc_dnn_mapping_success
Type
64-bit Counter
Description
Number of successful DNN mapping operations
Labels
source_dnn, target_dnn
• smfcc_dnn_mapping_failures
Type
64-bit Counter
Description
Number of failed DNN mapping operations
Labels
source_dnn, target_dnn
• smfcc_pcf_response_reseller_subclass_info
Type
64-bit Counter
Description
Number of PCF Create Responses with Reseller and Subclass Information
Labels
resellerId, subclassId
• smfcc_db_read_lock_conflict_record_failures
Type
64-bit Counter
Description
Number of DB Read Lock failures with conflicted record error

Labels

dnn, snssai

- smfcc_db_write_conflict_record_failures

Type

64-bit Counter

Description

Number of DB Write Lock/Unlock failures with conflicted record error

Labels

dnn, snssai

- smfcc_db_create_conflict_record_failures

Type

64-bit Counter

Description

Number of DB Create failures with conflicted record error

Labels

dnn, snssai

- smfcc_db_get_conflict_record_failures

Type

64-bit Counter

Description

Number of DB Get record failures with conflicted record error

Labels

dnn, snssai

- smfcc_pcc_rule_base_install_attempts

Type

64-bit Counter

Description

Number of pcc rule base install attempts

Labels

dnn, snssai

- smfcc_pcc_rule_base_install_failures

Type

64-bit Counter

Description

Number of pcc rule base install attempt failures

Labels

dnn, snssai, causecode

SMFCC PfcP Interface Statistics

- smfcc_error_indication_reports

Type

64-bit Counter

Description

Total Error Indication Report Received from UPF

Labels

dnn, snssai

• smfcc_error_indication_report_successes

Type

64-bit Counter

Description

Total Success Response to UPF for Error Indication Report

Labels

dnn, snssai

• smfcc_error_indication_report_failures

Type

64-bit Counter

Description

Total Failure Response to UPF for Error Indication Report

Labels

dnn, snssai, causecode

• smfcc_usage_reports

Type

64-bit Counter

Description

Total Usage Report Received from UPF

Labels

dnn, snssai

• smfcc_usage_report_successes

Type

64-bit Counter

Description

Total Success Response to UPF for Usage Report

Labels

dnn, snssai

• smfcc_usage_report_failures

Type

64-bit Counter

Description

Total Failure Response to UPF for Usage Report

Labels

dnn, snssai, causecode

- smfcc_up_inactivity_reports

Type

64-bit Counter

Description

Total Up Inactivity Report Received from UPF for all sessions

Labels

dnn, snssai

- smfcc_up_inactivity_reports_5G_sessions

Type

64-bit Counter

Description

Total Up Inactivity Report Received from UPF for 5G sessions

Labels

dnn, snssai

- smfcc_up_inactivity_reports_5G_always_on_sessions

Type

64-bit Counter

Description

Total Up Inactivity Report Received from UPF for 5G always on sessions

Labels

dnn, snssai

- smfcc_up_inactivity_reports_4G_sessions

Type

64-bit Counter

Description

Total Up Inactivity Report Received from UPF for 4G sessions

Labels

dnn, snssai

- smfcc_up_inactivity_reports_GGSN_sessions

Type

64-bit Counter

Description

Total Up Inactivity Report Received from UPF for 3G/2G sessions

Labels

dnn, snssai

- smfcc_up_activity_reports_5G_sessions

Type

64-bit Counter

Description

Total Up activity Report Received from UPF for 5G sessions

Labels

dnn, snssai

- smfcc_up_activity_reports_5G_always_on_sessions

Type

64-bit Counter

Description

Total Up activity Report Received from UPF for 5G always on sessions

Labels

dnn, snssai

- smfcc_up_activity_reports_4G_sessions

Type

64-bit Counter

Description

Total Up activity Report Received from UPF for 4G sessions

Labels

dnn, snssai

- smfcc_up_activity_reports_GGSN_sessions

Type

64-bit Counter

Description

Total Up activity Report Received from UPF for 3G/2G sessions

Labels

dnn, snssai

- smfcc_up_inactivity_report_successes_5G_sessions

Type

64-bit Counter

Description

Total Success Response to UPF for 5G sessions Up Inactivity Report

Labels

dnn, snssai

- smfcc_up_inactivity_report_successes_5G_always_on_sessions

Type

64-bit Counter

Description

Total Success Response to UPF for 5G always on sessions Up Inactivity Report

Labels

dnn, snssai

- smfcc_up_inactivity_report_successes_4G_sessions

Type

64-bit Counter

Description

Total Success Response to UPF for 4G sessions Up Inactivity Report

Labels

dnn, snssai

- smfcc_up_inactivity_report_successes_GGSN_sessions

Type

64-bit Counter

Description

Total Success Response to UPF for 3G/2G sessions Up Inactivity Report

Labels

dnn, snssai

- smfcc_up_inactivity_report_failures_5G_sessions

Type

64-bit Counter

Description

Total failure Response to UPF for 5G sessions Up Inactivity Report

Labels

dnn, snssai, causecode

- smfcc_up_inactivity_report_failures_5G_always_on_sessions

Type

64-bit Counter

Description

Total failure Response to UPF for 5G always on sessions Up Inactivity Report

Labels

dnn, snssai, causecode

- smfcc_up_inactivity_report_failures_4G_sessions

Type

64-bit Counter

Description

Total failure Response to UPF for 4G sessions Up Inactivity Report

Labels

dnn, snssai, causecode

- smfcc_up_inactivity_report_failures_GGSN_sessions

Type

64-bit Counter

Description

Total failure Response to UPF for 3G/2G sessions Up Inactivity Report

Labels

dnn, snssai, causecode

- smfcc_up_activity_report_successes_5G_sessions

Type

64-bit Counter

Description

Total Success Response to UPF for 5G sessions Up activity Report

Labels

dnn, snssai

- smfcc_up_activity_report_successes_5G_always_on_sessions

Type

64-bit Counter

Description

Total Success Response to UPF for 5G always on sessions Up activity Report

Labels

dnn, snssai

- smfcc_up_activity_report_successes_4G_sessions

Type

64-bit Counter

Description

Total Success Response to UPF for 4G sessions Up activity Report

Labels

dnn, snssai

- smfcc_up_activity_report_successes_GGSN_sessions

Type

64-bit Counter

Description

Total Success Response to UPF for 3G/2G sessions Up activity Report

Labels

dnn, snssai

- smfcc_up_activity_report_failures_5G_sessions

Type

64-bit Counter

Description

Total failure Response to UPF for 5G sessions Up activity Report

Labels

dnn, snssai, causecode

- smfcc_up_activity_report_failures_5G_always_on_sessions

Type
64-bit Counter
Description
Total failure Response to UPF for 5G always on sessions Up activity Report
Labels
<i>dnn, snssai, causecode</i>

- smfcc_up_activity_report_failures_4G_sessions

Type
64-bit Counter
Description
Total failure Response to UPF for 4G sessions Up activity Report
Labels
<i>dnn, snssai, causecode</i>

- smfcc_up_activity_report_failures_GGSN_sessions

Type
64-bit Counter
Description
Total failure Response to UPF for 3G/2G sessions Up activity Report
Labels
<i>dnn, snssai, causecode</i>

- smfcc_qos_flow_inactivity_reports

Type
64-bit Counter
Description
Total Qos Flow Inactivity Report Received from UPF
Labels
<i>dnn, snssai</i>

- smfcc_qos_flow_inactivity_report_successes

Type
64-bit Counter
Description
Total Success Response to UPF for Qos Flow Inactivity Report
Labels
<i>dnn, snssai</i>

- smfcc_qos_flow_inactivity_report_failures

Type
64-bit Counter
Description

Total Failure Response to UPF for Qos Flow Inactivity Report

Labels

dnn, snssai, causecode

• smfcc_session_idle_inactivity_reports

Type

64-bit Counter

Description

Total Session Idle Inactivity Report Received from UPF

Labels

dnn, snssai

• smfcc_session_idle_inactivity_report_successes

Type

64-bit Counter

Description

Total Success Response to UPF for Session Idle Inactivity Report

Labels

dnn, snssai

• smfcc_session_idle_inactivity_report_failures

Type

64-bit Counter

Description

Total Failure Response to UPF for Session Idle Inactivity Report

Labels

dnn, snssai, causecode

• smfcc_down_link_data_reports

Type

64-bit Counter

Description

Total Down link data Reports Received from UPF

Labels

dnn, rattype

• smfcc_down_link_data_report_successes

Type

64-bit Counter

Description

Total down link data report Success Response to UPF

Labels

dnn, rattype

• smfcc_down_link_data_report_failures

Type

64-bit Counter

Description

Total Failure Response to UPF for down link data Report

Labels

dnn, rattype, causecode

SMFCC UPF Usage Report Statistics

- smfcc_usage_report_total_volume

Type

64-bit Counter

Description

Total Volume Report Received from UPF

Labels

snssai, ratinggroup, serviceid, quotatype

- smfcc_usage_report_downlink_volume

Type

64-bit Counter

Description

Downlink Volume Report Received from UPF

Labels

snssai, ratinggroup, serviceid, quotatype

- smfcc_usage_report_uplink_volume

Type

64-bit Counter

Description

Uplink Volume Report Received from UPF

Labels

snssai, ratinggroup, serviceid, quotatype

- smfcc_usage_report_drop_count

Type

64-bit Counter

Description

Dropped UPF Usage Report Count

Labels

snssai, ratinggroup, serviceid

- smfcc_usage_report_adc_start_count

Type

64-bit Counter

Description

ADC Start Trigger Report Count

Labels

snssai, applicationid

- smfcc_usage_report_adc_stop_count

Type

64-bit Counter

Description

ADC Stop Trigger Report Count

Labels

snssai, applicationid

SMF PGW Call Performance Statistics

- smfcc_pgw_bearer_setup_success

Type

64-bit Counter

Description

Number of 4G bearer setup success on PGW

Labels

snssai, qci, arp

- smfcc_pgw_bearer_setup_failure

Type

64-bit Counter

Description

Number of 4G bearer setup failure on PGW

Labels

snssai, qci, arp

- smfcc_pgw_nw_init_bearer_release

Type

64-bit Counter

Description

Number of Network Initiated 4G bearer release on PGW

Labels

snssai, qci, arp

- smfcc_pgw_access_init_bearer_release

Type

64-bit Counter

Description

Number of Access Initiated 4G bearer release on PGW

Labels

snssai, qci, arp

- smfcc_pgw_abnormal_bearer_release
Type
64-bit Counter
Description
Number of 4G bearer released abnormally on PGW
Labels
snssai, qci, arp, cause
- smfcc_pgw_bearer_setup_duration
Type
64-bit Histogram
Description
Time observed for successfull 4G bearer setup on PGW in seconds
Labels
snssai,qci,arp
Boundaries
[0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0, 2.5, 3, 4, 5]

SMF SAEGW Call Performance Statistics

- smfcc_saegw_bearer_setup_success
Type
64-bit Counter
Description
Number of 4G bearer setup success on SAEGW
Labels
snssai, qci, arp
- smfcc_saegw_bearer_setup_failure
Type
64-bit Counter
Description
Number of 4G bearer setup failure on SAEGW
Labels
snssai, qci, arp
- smfcc_saegw_nw_init_bearer_release
Type
64-bit Counter
Description
Number of Network Initiated 4G bearer release on SAEGW
Labels
snssai, qci, arp
- smfcc_saegw_access_init_bearer_release

Type

64-bit Counter

Description

Number of Access Initiated 4G bearer release on SAEGW

Labels

snssai, qci, arp

• smfcc_saegw_abnormal_bearer_release

Type

64-bit Counter

Description

Number of 4G bearer released abnormally on SAEGW

Labels

snssai, qci, arp, cause

• smfcc_saegw_bearer_setup_duration

Type

64-bit Histogram

Description

Time observed for successfull 4G bearer setup on SAEGW in seconds

Labels

snssai,qci,arp

Boundaries

[0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0, 2.5, 3, 4, 5]

SMFCC EDR Statistics

• smfcc_edr_generated

Type

64-bit Counter

Description

Number of EDRs generated by SMF

Labels

send_status, eventtype, eventoutcome

• smfcc_edr_callback_rcvd

Type

64-bit Counter

Description

Number of EDR callbacks received from Kafka Client

Labels

cbk_status

SMFCC 3GPP Procedure Duration Statistics

smfcc_3gpp_procedure_duration
Type
64-bit Histogram
Description
Time observed for SMFCC 3GPP procedure execution in seconds
Labels
procedure,status
Boundaries
[0.5, 1, 1.5, 2, 3, 5, 10, 20]

SMFCC Interface Round Trip Duration Statistics

smfcc_interface_round_trip_duration
Type
64-bit Histogram
Description
Time observed for SMFCC Interface Round Trip Duration in seconds
Labels
interface
Boundaries
[0.1, 0.2, 0.3, 0.4, 0.5, 0.7, 1, 1.5, 2, 3, 5]

SMFCC Event-Exposure Statistics

smfcc_evtexp_notifications_sent
Type
64-bit Counter
Description
Number of Event Exposure Notifications sent by SMF
Labels
send_status
smfcc_evtexp_notification_responses_rcvd
Type
64-bit Counter
Description
Number of Notification responses from Event Exposure pod
Labels
response_status

SMFCC Miscellaneous Statistics

smfcc_misc_db_record_compression_failures
Type
64-bit Counter

Description

None

Labels

blocktype

- smfcc_misc_db_record_deserialization_failures

Type

64-bit Counter

Description

None

Labels

blocktype

- smfcc_misc_db_record_decompression_failures

Type

64-bit Counter

Description

None

Labels

blocktype

- smfcc_misc_db_record_delete_failures

Type

64-bit Counter

Description

None

Labels

cause

- smfcc_misc_db_record_delete_on_create_conflict_attempts

Type

64-bit Counter

Description

None

Labels

- smfcc_misc_db_record_deleted_on_create_conflict

Type

64-bit Counter

Description

None

Labels

- smfcc_misc_db_record_delete_failures_on_create_conflict

Type

64-bit Counter

Description

None

Labels

cause

- smfcc_misc_upf_chosen_available_capacity

Type

64-bit Counter

Description

Total Number of UPF chosen based on available capacity

Labels

upf_node_id

- smfcc_misc_upf_chosen_nf_priority

Type

64-bit Counter

Description

Total Number of UPF chosen based on Network Function Priority

Labels

upf_node_id

- smfcc_misc_upf_chosen_tai_priority

Type

64-bit Counter

Description

Total Number of UPF chosen based on TAI Priority

Labels

upf_node_id

- smfcc_misc_upf_dnn_list_wo_upf

Type

64-bit Counter

Description

Total Number of UPF dnn usage entries received with no corresponding UPF association

Labels

upf_node_id

- smfcc_misc_upf_selection_change_on_ue_ip_exhaust

Type

64-bit Counter

Description

Total Number of Selected UPF change based on UE IP pool exhaust detection

Labels

upf_node_id

- smfcc_misc_invalid_session_db_records_cleared

Type

64-bit Counter

Description

None

Metrics

Instances	Labels	Description
1	*****	Number of invalid session DB Records cleared

- smfcc_misc_db_record_serialization_failures

Type

64-bit Counter

Description

None

Metrics

Instances	Labels	Description
1	*****	Number of DB record serialization failures

SMF ETCD Interface Statistics

- smfcc_etcd_gtpc_abort_lease_grant_response

Type

64-bit Counter

Description

Lease grant response

Labels

responsecode

- smfcc_etcd_gtpc_abort_put_with_lease_grant_response

Type

64-bit Counter

Description

Put with lease grant response

Labels

responsecode

- smfcc_etcd_inband_radius_tunnel_put_response

Type

64-bit Counter

Description

Inband Radius Tunnel info put response

Labels

responsecode

- smfcc_etcd_inband_radius_tunnel_delete_range_response

Type

64-bit Counter

Description

Inband Radius Tunnel info delete range response

Labels

responsecode

- smfcc_etcd_gtpc_abort_lease_grant_request

Type

64-bit Counter

Description

None

Metrics

Instances	Labels	Description
1	*****	request

- smfcc_etcd_gtpc_abort_put_with_lease_grant_request

Type

64-bit Counter

Description

None

Metrics

Instances	Labels	Description
1	*****	request

- smfcc_etcd_inband_radius_tunnel_put_request

Type

64-bit Counter

Description

None

Metrics

Instances	Labels	Description
1	*****	request

- smfcc_etcd_inband_radius_tunnel_delete_range_request

Type

64-bit Counter

Description

None

Metrics

Instances	Labels	Description
1	*****	request

SMFCC Debug Statistics

- smfcc_session_cache_hits
 - Type
 - 64-bit Counter
 - Description
 - Number of times the session is found in cache for incoming requests
 - Labels
 - interface**
- smfcc_procedures_dropped
 - Type
 - 64-bit Counter
 - Description
 - Number of 3gpp procedures dropped
 - Labels
 - causecode**
- smfcc_clear_bulk_pdu_sessions_operator_request
 - Type
 - 64-bit Counter
 - Description
 - Number of times bulk pdu session clear request is initiated by the operator
 - Labels

- smfcc_cookie_session_id_map_delete
 - Type
 - 64-bit Counter
 - Description
 - Number of times the session is found in cookie session id map for deletion
 - Labels
 - agentProcedureName**
- smfcc_force_nf_discovery
 - Type
 - 64-bit Counter
 - Description
 - Number of times smfcc sends nf-discovery request with force-discovery
 - Labels
 - interface**
- smfcc_instanceid_change_no_nfset

Type

64-bit Counter

Description

Number of times NF Instance ID change detected when no nf set is configured

Labels

oldInstanceid, curlInstanceid

- smfcc_tcmalloc_memory_used_by_application

Type

64-bit Gauge

Description

Number of bytes used by application

Labels

- smfcc_tcmalloc_page_heap_free_list

Type

64-bit Gauge

Description

Number of bytes in free, mapped pages in page heap

Labels

- smfcc_tcmalloc_central_cache_free_list

Type

64-bit Gauge

Description

Number of bytes in free pages in central cache

Labels

- smfcc_tcmalloc_transfer_cache_free_list

Type

64-bit Gauge

Description

Number of bytes in free pages in transfer cache

Labels

- smfcc_tcmalloc_thread_cache_free_list

Type

64-bit Gauge

Description

Number of bytes in free pages in thread cache

Labels

- smfcc_tcmalloc_metadata

Type

64-bit Gauge

Description

Number of bytes used by tcmalloc metadata

Labels

- smfcc_tcmalloc_actual_memory_used

Type

64-bit Gauge

Description

Actual memory used by application

Labels

- smfcc_tcmalloc_virtual_address_space_used

Type

64-bit Gauge

Description

Virtual address space used by application

Labels

- smfcc_cached_session_count

Type

64-bit Gauge

Description

Number of cached sessions on smfcc pod

Labels

SMF N1 Interface Message Statistics

- smfcc_n1_msg_pdu_session_establishment_request_received

Type

64-bit Counter

Description

Number of n1 pdu session establishment request messages received

Labels

dnn, snssai

- smfcc_n1_msg_pdu_session_establishment_request_failures

Type

64-bit Counter

Description

Number of n1 pdu session establishment request messages failed

Labels

dnn, snssai, cause

- smfcc_n1_msg_pdu_session_establishment_accept_sent

Type

64-bit Counter

Description

Number of n1 pdu session establishment accept messages sent

Labels

dnn, snssai

- smfcc_n1_msg_pdu_session_establishment_reject_sent

Type

64-bit Counter

Description

Number of n1 pdu session establishment reject messages sent

Labels

dnn, snssai, nas5gSmCause

- smfcc_n1_msg_pdu_session_modification_request_received

Type

64-bit Counter

Description

Number of n1 pdu session modification request messages received

Labels

dnn, snssai, nas5gSmCause

- smfcc_n1_msg_pdu_session_modification_request_failures

Type

64-bit Counter

Description

Number of n1 pdu session modification request failures

Labels

dnn, snssai, cause

- smfcc_n1_msg_pdu_session_modification_reject_sent

Type

64-bit Counter

Description

Number of n1 pdu session modification reject messages sent

Labels

dnn, snssai, nas5gSmCause

- smfcc_n1_msg_pdu_session_modification_command_sent

Type

64-bit Counter

Description

Number of n1 pdu session modification command messages sent

Labels

dnn, snssai, nas5gSmCause

- smfcc_n1_msg_pdu_session_modification_command_response_failures

Type

64-bit Counter

Description

Number of n1 pdu session modification command messages failed to receive response

Labels

dnn, snssai, cause

- smfcc_n1_msg_pdu_session_modification_command_reject_received

Type

64-bit Counter

Description

Number of n1 pdu session modification command message recieved

Labels

dnn, snssai, nas5gSmCause

- smfcc_n1_msg_pdu_session_modification_complete_received

Type

64-bit Counter

Description

Number of n1 pdu session modification complete received

Labels

dnn, snssai

- smfcc_n1_msg_pdu_session_release_request_received

Type

64-bit Counter

Description

Number of n1 pdu session release request received

Labels

dnn, snssai, nas5gSmCause

- smfcc_n1_msg_pdu_session_release_request_failures

Type

64-bit Counter

Description

Number of n1 pdu session release request message failed

Labels

dnn, snssai, cause

- smfcc_n1_msg_pdu_session_release_reject_sent

Type

64-bit Counter

Description

Number of n1 pdu session release reject message sent

Labels

dnn, snssai, nas5gSmcause
- smfcc_n1_msg_pdu_session_release_command_sent

Type

64-bit Counter

Description

Number of n1 pdu session release command sent

Labels

dnn, snssai, nas5gSmCause
- smfcc_n1_msg_pdu_session_release_command_response_failures

Type

64-bit Counter

Description

Number of n1 pdu session release command messages failed to receive response

Labels

dnn, snssai, cause
- smfcc_n1_msg_pdu_session_release_complete_received

Type

64-bit Counter

Description

Number of n1 pdu session release complete received

Labels

dnn, snssai
- smfcc_n1_msg_pdu_session_modification_complete_dropped

Type

64-bit Counter

Description

Number of n1 pdu session modification complete messages dropped

Labels

dnn, snssai, cause
- smfcc_n1_msg_pdu_session_modification_command_reject_dropped

Type

64-bit Counter

Description

Number of n1 pdu session modification command reject dropped

Labels

dnn, snssai, cause

- smfcc_n1_msg_pdu_session_release_complete_dropped

Type

64-bit Counter

Description

Number of n1 pdu session release complete messages dropped

Labels

dnn, snssai, cause

SMF N2 Interface Message Statistics

- smfcc_n2_msg_pdu_session_resource_setup_request_transfer_sent

Type

64-bit Counter

Description

Number of n2 pdu session resource setup requests sent

Labels

dnn, snssai

- smfcc_n2_msg_pdu_session_resource_setup_request_transfer_failures

Type

64-bit Counter

Description

Number of n2 pdu session resource setup requests failed to receive response

Labels

dnn, snssai, cause

- smfcc_n2_msg_pdu_session_resource_setup_response_transfer_received

Type

64-bit Counter

Description

Number of n2 pdu session resource setup response transfer messages received

Labels

dnn, snssai

- smfcc_n2_msg_pdu_session_resource_setup_unsuccessful_transfer_received

Type

64-bit Counter

Description

Number of n2 pdu session resource setup unsuccessful transfer messages received

Labels

dnn, snssai

smfcc_n2_msg_pdu_session_resource_modify_request_transfer_sent
Type
64-bit Counter
Description
Number of n2 pdu session resource modify requests sent
Labels
dnn, snssai
smfcc_n2_msg_pdu_session_resource_modify_request_transfer_failures
Type
64-bit Counter
Description
Number of n2 pdu session modify requests failed to receive response
Labels
dnn, snssai, cause
smfcc_n2_msg_pdu_session_resource_modify_response_transfer_received
Type
64-bit Counter
Description
Number of n2 pdu session modify response transfer messages received
Labels
dnn, snssai
smfcc_n2_msg_pdu_session_resource_modify_unsuccessful_transfer_received
Type
64-bit Counter
Description
Number of n2 pdu session modify unsuccessful transfer messages received
Labels
dnn, snssai
smfcc_n2_msg_pdu_session_resource_release_command_transfer_sent
Type
64-bit Counter
Description
Number of n2 pdu session resource release command transfer sent
Labels
dnn, snssai
smfcc_n2_msg_pdu_session_resource_release_command_transfer_failures
Type
64-bit Counter
Description

Number of n2 pdu session resource release requests failed to receive response

Labels

dnn, snssai, cause

- smfcc_n2_msg_pdu_session_resource_release_response_transfer_received

Type

64-bit Counter

Description

Number of n2 pdu session resource release response transfer message received

Labels

dnn, snssai

- smfcc_n2_msg_path_switch_request_transfer_message_received

Type

64-bit Counter

Description

Number of n2 path switch request transfer message received

Labels

dnn, snssai

- smfcc_n2_msg_path_switch_request_transfer_failed

Type

64-bit Counter

Description

Number of n2 path switch request transfer messages failed to process

Labels

dnn, snssai, cause

- smfcc_n2_msg_path_switch_request_setup_failed_transfer_received

Type

64-bit Counter

Description

Number of n2 path switch request setup failed transfer received

Labels

dnn, snssai

- smfcc_n2_msg_path_switch_request_setup_failed_transfer_failures

Type

64-bit Counter

Description

Number of n2 path switch request setup failed transfer message failed to process

Labels

dnn, snssai, cause

- smfcc_n2_msg_path_switch_request_acknowledge_transfer_sent

Type

64-bit Counter

Description

Number of n2 path switch request acknowledge transfer messages sent

Labels

dnn, snssai

- smfcc_n2_msg_path_switch_request_unsuccessful_transfer_sent

Type

64-bit Counter

Description

Number of n2 path switch request unsuccessful transfer messages sent

Labels

dnn, snssai

- smfcc_n2_msg_handover_required_transfer_received

Type

64-bit Counter

Description

Number of n2 handover required transfer messages received

Labels

dnn, snssai

- smfcc_n2_msg_handover_required_transfer_failures

Type

64-bit Counter

Description

Number of n2 handover required transfer messages failed process

Labels

dnn, snssai, cause

- smfcc_n2_msg_handover_command_transfer_sent

Type

64-bit Counter

Description

Number of n2 handover command transfer messages sent

Labels

dnn, snssai

- smfcc_n2_msg_handover_request_ack_transfer_received

Type

64-bit Counter

Description

Number of n2 handover request acknowledge transfer messages received

Labels

dnn, snssai

- smfcc_n2_msg_handover_request_ack_transfer_failures

Type

64-bit Counter

Description

Number of n2 handover request ack transfer messages failed to process

Labels

dnn, snssai, cause

- smfcc_n2_msg_handover_prep_unsuccessful_transfer_sent

Type

64-bit Counter

Description

Number of n2 handover preparation unsuccessful transfer messages sent

Labels

dnn, snssai

- smfcc_n2_msg_handover_resource_allocation_unsuccessful_transfer_received

Type

64-bit Counter

Description

Number of n2 handover resource allocation unsuccessful transfer messages received

Labels

dnn, snssai

- smfcc_n2_msg_pdu_session_resource_setup_response_transfer_dropped

Type

64-bit Counter

Description

Number of N2 pdu session resource setup response transfer messages dropped

Labels

dnn, snssai, cause

- smfcc_n2_msg_pdu_session_resource_setup_unsuccessful_transfer_dropped

Type

64-bit Counter

Description

Pdu session resource setup unsuccessful transfer messages dropped

Labels

dnn, snssai, cause

- smfcc_n2_msg_pdu_session_resource_modify_response_transfer_dropped

Type

64-bit Counter

Description

Number of n2 pdu session resource modify response transfer messages dropped

Labels

dnn, snssai, cause

- smfcc_n2_msg_pdu_session_resource_modify_unsuccessful_transfer_dropped

Type

64-bit Counter

Description

Number of n2 pdu session resource modify unsuccessful transfer messages dropped

Labels

dnn, snssai, cause

- smfcc_n2_msg_pdu_session_resource_release_response_transfer_dropped

Type

64-bit Counter

Description

Number of n2 pdu session resource release response transfer messages dropped

Labels

dnn, snssai, cause

- smfcc_n2_msg_handover_resource_allocation_unsuccessful_transfer_dropped

Type

64-bit Counter

Description

Number of n2 handover resource allocation unsuccessful transfer messages dropped

Labels

dnn, snssai, cause

- smfcc_n2_msg_pdu_session_resource_modify_indication_transfer_received

Type

64-bit Counter

Description

Number of pdu session resource modify indication transfer message received

Labels

dnn, snssai

- smfcc_n2_msg_pdu_session_resource_modify_indication_transfer_failed

Type

64-bit Counter

Description

Number of n2 pdu session resource modify indication transfer messages failed to process

Labels

dnn, snssai, cause

- smfcc_n2_msg_pdu_session_resource_modify_confirm_transfer_sent

Type

64-bit Counter

Description

Number of pdu session resource modify confirm transfer messages sent

Labels

dnn, snssai

- smfcc_n2_msg_pdu_session_resource_modify_indication_unsuccessful_transfer_sent

Type

64-bit Counter

Description

Number of n2 pdu session resource modify indication unsuccessful transfer messages sent

Labels

dnn, snssai

SMF udsf interface message statistics

- smfcc_udsf_read_and_lock_record_using_secondary_key_request_attempts

Type

64-bit Counter

Description

Number of udsf read and lock record request using secondary key attempts

Labels

- smfcc_udsf_read_and_lock_record_using_secondary_key_request_successes

Type

64-bit Counter

Description

Number of udsf read and lock record request using secondary key successes

Labels

- smfcc_udsf_read_and_lock_record_using_secondary_key_request_failures

Type

64-bit Counter

Description

Number of udsf read and lock record request using secondary key failures

Labels

cause

- smfcc_udsf_read_and_lock_record_request_attempts

Type

64-bit Counter

Description

Number of udsf read and lock record request attempts

Labels

- smfcc_udsf_read_and_lock_record_request_successes

Type

64-bit Counter

Description

Number of udsf read and lock record request successess

Labels

- smfcc_udsf_read_and_lock_record_request_failures

Type

64-bit Counter

Description

Number of udsf read and lock record request failures

Labels

cause

- smfcc_udsf_create_record_and_lock_request_attempts

Type

64-bit Counter

Description

Number of usdf create record and lock request attempts

Labels

- smfcc_udsf_create_record_and_lock_request_successes

Type

64-bit Counter

Description

Number of usdf create record and lock request successes

Labels

- smfcc_udsf_create_record_and_lock_request_failures

Type

64-bit Counter

Description

Number of udsf create record and lock request failures

Labels

cause

- smfcc_udsf_create_or_modify_block_request_attempts

Type

64-bit Counter

Description

Number of udsf create or modify block request attempts

Labels

• smfcc_udsf_create_or_modify_block_request_successes

Type

64-bit Counter

Description

Number of udsf create or modify block request successes

Labels

• smfcc_udsf_create_or_modify_block_request_failures

Type

64-bit Counter

Description

Number of udsf create or modify block request failures

Labels

cause

• smfcc_udsf_modify_record_and_lock_request_attempts

Type

64-bit Counter

Description

Number of modify record and lock request attempts

Labels

• smfcc_udsf_modify_record_and_lock_request_successes

Type

64-bit Counter

Description

Number of modify record and lock request successes

Labels

• smfcc_udsf_modify_record_and_lock_request_failures

Type

64-bit Counter

Description

Number of modify record and lock request failures

Labels

cause

• smfcc_udsf_modify_record_and_unlock_request_attempts

Type

64-bit Counter

Description

Number of udsf modify record and unlock request attempts

Labels

- smfcc_udsf_modify_record_and_unlock_request_successes

Type

64-bit Counter

Description

Number of udsf modify record and unlock request successes

Labels

- smfcc_udsf_modify_record_and_unlock_request_failures

Type

64-bit Counter

Description

Number of udsf modify record and unlock request failures

Labels

cause

- smfcc_udsf_delete_record_request_attempts

Type

64-bit Counter

Description

Number of udsf delete record request attempts

Labels

- smfcc_udsf_delete_record_request_successes

Type

64-bit Counter

Description

Number of udsf delete record request successes

Labels

- smfcc_udsf_delete_record_request_failures

Type

64-bit Counter

Description

Number of udsf delete record request failures

Labels

cause

- smfcc_udsf_get_record_and_steal_lock_request_attempts

Type
64-bit Counter
Description
Number of udsf get record and steal lock request attempts
Labels

- smfcc_udsf_get_record_and_steal_lock_request_successes

Type
64-bit Counter
Description
Number of udsf get record and steal lock request successes
Labels

- smfcc_udsf_get_record_and_steal_lock_request_failures

Type
64-bit Counter
Description
Number of udsf get record and steal lock request failures
Labels

- *cause*

- smfcc_udsf_search_record_request_attempts

Type
64-bit Counter
Description
Number of udsf search record request attempts
Labels

- smfcc_udsf_search_record_request_successes

Type
64-bit Counter
Description
Number of udsf search record request successes
Labels

- smfcc_udsf_search_record_request_failures

Type
64-bit Counter
Description
Number of udsf search record request failures
Labels

cause

- smfcc_udsf_get_record_request_attempts

Type

64-bit Counter

Description

Number of udsf get record request attempts

Labels

- smfcc_udsf_get_record_request_successes

Type

64-bit Counter

Description

Number of udsf get record request successes

Labels

- smfcc_udsf_get_record_request_failures

Type

64-bit Counter

Description

Number of udsf get record request failures

Labels

cause

- smfcc_udsf_modify_block_and_unlock_request_attempts

Type

64-bit Counter

Description

Number of udsf modify block and unlock request attempts

Labels

- smfcc_udsf_modify_block_and_unlock_request_successes

Type

64-bit Counter

Description

Number of udsf modify block and unlock request successes

Labels

- smfcc_udsf_modify_block_and_unlock_request_failures

Type

64-bit Counter

Description

Number of udsf modify block and unlock request failures

Labels

cause

- smfcc_udsf_get_tag_count_request_attempts

Type

64-bit Counter

Description

Number of udsf get tag count request attempts

Labels

- smfcc_udsf_get_tag_count_request_successes

Type

64-bit Counter

Description

Number of udsf get tag count request successes

Labels

- smfcc_udsf_get_tag_count_request_failures

Type

64-bit Counter

Description

Number of udsf get tag count request failures

Labels

cause

- smfcc_udsf_unlock_only_attempts

Type

64-bit Counter

Description

Number of Unlock only request attempts

Labels

- smfcc_udsf_unlock_only_successes

Type

64-bit Counter

Description

Number of Unlock only request successes

Labels

- smfcc_udsf_unlock_only_request_failures

Type

64-bit Counter

Description

Number of Unlock only request failures

Labels

cause

- smfcc_udsf_patch_record_request_attempts

Type

64-bit Counter

Description

Number of patch record request attempts

Labels

- smfcc_udsf_patch_record_request_successes

Type

64-bit Counter

Description

Number of patch record request successes

Labels

- smfcc_udsf_patch_record_request_failures

Type

64-bit Counter

Description

Number of patch record request failures

Labels

cause

- smfcc_udsf_update_record_meta_request_attempts

Type

64-bit Counter

Description

Number of update record meta request attempts

Labels

- smfcc_udsf_update_record_meta_request_successes

Type

64-bit Counter

Description

Number of update record meta request successes

Labels

- smfcc_udsf_update_record_meta_request_failures

Type

64-bit Counter

Description

Number of update record meta request failures

Labels

cause

- smfcc_udsf_connection_available_indication_received

Type

64-bit Counter

Description

Number of UDSF API Connection Available Indication received

Labels

- smfcc_udsf_connection_unavailable_indication_received

Type

64-bit Counter

Description

Number of UDSF API Connection Unavailable Indication received

Labels

- smfcc_db_transaction_failure_udsf_connection_unavailable

Type

64-bit Counter

Description

Number of DB transactions failed due to UDSF Connection Unavailable

Labels

transaction

SMF Agent Interface Timeout Statistics

- smfcc_chf_transaction_internal_timeout

Type

64-bit Counter

Description

Number of internal transaction timeouts happend in chf interface

Labels

MessageType

- smfcc_amf_transaction_internal_timeout

Type

64-bit Counter

Description

Number of internal transaction timeouts happend in amf interface

Labels

MessageType

smfcc_pcf_transaction_internal_timeout
Type
64-bit Counter
Description
Number of internal transaction timeouts happend in pcf interface
Labels
MessageType
smfcc_udm_transaction_internal_timeout
Type
64-bit Counter
Description
Number of internal transaction timeouts happend in udm interface
Labels
MessageType
smfcc_subscr analyzer_transaction_internal_timeout
Type
64-bit Counter
Description
Number of internal transaction timeouts happend in subscriber analyzer interface
Labels
MessageType
smfcc_urm_transaction_internal_timeout
Type
64-bit Counter
Description
Number of internal transaction timeouts happend in urm interface
Labels
MessageType
smfcc_evtxp_transaction_internal_timeout
Type
64-bit Counter
Description
Number of internal transaction timeouts happend in evtxp interface
Labels
MessageType
smfcc_nrf_transaction_internal_timeout
Type
64-bit Counter
Description

Number of internal transaction timeouts happend in nrf interface

Labels

MessageType

- smfcc_pfcpc_transaction_internal_timeout

Type

64-bit Counter

Description

Number of internal transaction timeouts happend in pfcpc interface

Labels

MessageType

- smfcc_gtpc_transaction_internal_timeout

Type

64-bit Counter

Description

Number of internal transaction timeouts happend in gtpc interface

Labels

MessageType

- smfcc_gx_transaction_internal_timeout

Type

64-bit Counter

Description

Number of internal transaction timeouts happend in gx interface

Labels

MessageType

- smfcc_dtm_transaction_internal_timeout

Type

64-bit Counter

Description

Number of internal transaction timeouts happend in durable timer interface

Labels

MessageType

SMF Radius Interface Message Statistics

- smfcc_radius_access_requests_sent

Type

64-bit Counter

Description

Number of radius access-requests sent

Labels

dnn, rattype

- smfcc_radius_access_requests_send_failed

Type
64-bit Counter
Description
Number of radius access-requests send failed
Labels
dnn, ratype

- smfcc_radius_access_accepts

Type
64-bit Counter
Description
Number of radius access-accepts received
Labels
dnn, ratype

- smfcc_radius_access_rejects

Type
64-bit Counter
Description
Number of radius access-rejects received
Labels
dnn, ratype, cause

- smfcc_radius_authentication_request_timeouts

Type
64-bit Counter
Description
Number of radius access-request timeouts
Labels
dnn, ratype

- smfcc_radius_accounting_start_requests_sent

Type
64-bit Counter
Description
Number of radius accounting-start requests sent
Labels
dnn, ratype

- smfcc_radius_accounting_start_requests_send_failed

Type
64-bit Counter
Description

Number of radius accounting-start requests send failed

Labels

dnn, ratype

• smfcc_radius_accounting_start_response_successes

Type

64-bit Counter

Description

Number of radius accounting-start success responses received

Labels

dnn, ratype

• smfcc_radius_accounting_start_response_failures

Type

64-bit Counter

Description

Number of radius accounting-start fail responses received

Labels

dnn, ratype

• smfcc_radius_accounting_start_request_timeouts

Type

64-bit Counter

Description

Number of radius accounting-start request timeouts

Labels

dnn, ratype

• smfcc_radius_accounting_interim_requests_sent

Type

64-bit Counter

Description

Number of radius accounting-interim requests sent

Labels

dnn, ratype

• smfcc_radius_accounting_interim_requests_send_failed

Type

64-bit Counter

Description

Number of radius accounting-interim requests send failed

Labels

dnn, ratype

• smfcc_radius_accounting_interim_response_successes

Type

64-bit Counter

Description

Number of radius accounting-interim success responses received

Labels

dnn, ratype

- smfcc_radius_accounting_interim_response_failures

Type

64-bit Counter

Description

Number of radius accounting-interim fail responses received

Labels

dnn, ratype

- smfcc_radius_accounting_interim_request_timeouts

Type

64-bit Counter

Description

Number of radius accounting-interim request timeouts

Labels

dnn, ratype

- smfcc_radius_accounting_stop_requests_sent

Type

64-bit Counter

Description

Number of radius accounting-stop requests sent

Labels

dnn, ratype

- smfcc_radius_accounting_stop_requests_send_failed

Type

64-bit Counter

Description

Number of radius accounting-stop requests send failed

Labels

dnn, ratype

- smfcc_radius_accounting_stop_response_successes

Type

64-bit Counter

Description

Number of radius accounting-stop success responses received

Labels

dnn, ratype

- smfcc_radius_accounting_stop_response_failures

Type

64-bit Counter

Description

Number of radius accounting-stop fail responses received

Labels

dnn, ratype

- smfcc_radius_accounting_stop_request_timeouts

Type

64-bit Counter

Description

Number of radius accounting-stop request timeouts

Labels

dnn, ratype

- smfcc_inband_radius_gtpu_error_indications_received

Type

64-bit Counter

Description

Number of inband radius gtpu error indication request received

Labels

inbandProfile, peerUpfName, peerUpfRole, peerUpfIp

- smfcc_inband_radius_gtpu_error_indications_response_sent

Type

64-bit Counter

Description

Number of inband radius gtpu error indication response success sent

Labels

inbandProfile, peerUpfName, peerUpfRole, peerUpfIp

- smfcc_radius_ip_allocation_requested

Type

64-bit Counter

Description

Number of ip allocation requests sent to radius

Labels

type

- smfcc_radius_ip_allocated

Type

64-bit Counter

Description

Number of ip allocated response from radius

Labels

type

- smfcc_radius_ip_downgraded

Type

64-bit Counter

Description

Number of ip allocation response from radius with downgrade

Labels

type

- smfcc_radius_ip_ignored

Type

64-bit Counter

Description

Number of ip allocation response from radius ignored

Labels

type

SMF DB Interface Statistics

- smfcc_bulk_delete_timers_start_attempts

Type

64-bit Counter

Description

Number of Durable timer starts are sent for bulk deactivation procedure

Labels

timerType

- smfcc_bulk_delete_timers_start_failures

Type

64-bit Counter

Description

Number of Durable timer starts failure for bulk deactivation procedure

Labels

timerType

- smfcc_bulk_delete_timers_start_successes

Type

64-bit Counter

Description

Number of Durable timers started successfully for bulk deactivation procedure

Labels

timerType

- smfcc_bulk_delete_timers_stop_attempts

Type

64-bit Counter

Description

Number of Durable timer stops are sent for bulk deactivation procedure

Labels

timerType

- smfcc_bulk_delete_timers_stop_failures

Type

64-bit Counter

Description

Number of Durable timer stops failure for bulk deactivation procedure

Labels

timerType

- smfcc_bulk_delete_timers_stop_successes

Type

64-bit Counter

Description

Number of Durable timers stopped successfully for bulk deactivation procedure

Labels

timerType

- smfcc_bulk_delete_timers_expiry

Type

64-bit Counter

Description

Number of Durable timers expired

Labels

timerType

- smfcc_multi_read_requests_sent_to_db

Type

64-bit Counter

Description

None

Metrics

Instances	Labels	Description
1	*****	Number of Multi read requests sent to Db

- smfcc_multi_read_response_received_from_db_success

Type

64-bit Counter

Description

None

Metrics

Instances	Labels	Description
1	*****	Number of success multi read responses received

- smfcc_multi_read_response_received_from_db_failure

Type
64-bit Counter
Description
None
Metrics

Instances	Labels	Description
1	*****	Number of failed multi read responses received
• smfcc_bulk_delete_requests_sent		

Type
64-bit Counter
Description
None
Metrics

Instances	Labels	Description
1	*****	Number of bulk delete requests sent out by lead smfcc
• smfcc_bulk_delete_requests_received		

Type
64-bit Counter
Description
None
Metrics

Instances	Labels	Description
1	*****	Number of bulk delete requests received from lead smfcc

SMF SMF Interface Statistics

• smfcc_nsmf_pdu_session_create_attempts
Type
64-bit Counter
Description
Number of Pdu Session Create attempts
Labels

dnn, snssai, smfSessionType, smfHost

- smfcc_nsmf_pdu_session_create_successes

Type

64-bit Counter

Description

Number of Pdu Session Create Successes

Labels

dnn, snssai, smfSessionType, smfHost

- smfcc_nsmf_pdu_session_create_failures

Type

64-bit Counter

Description

Number of Pdu Session Create Failures

Labels

dnn, snssai, smfSessionType, smfHost, statusCode

- smfcc_hsmf_pdu_session_create_duplicate_attempts

Type

64-bit Counter

Description

Number of duplicate Pdu Session Create Attempted

Labels

dnn, snssai

- smfcc_nsmf_pdu_session_update_tx_attempts

Type

64-bit Counter

Description

Number of Pdu Session Update attempts

Labels

dnn, snssai, smfSessionType, smfHost

- smfcc_nsmf_pdu_session_update_tx_successes

Type

64-bit Counter

Description

Number of Pdu Session Update Success

Labels

dnn, snssai, smfSessionType, smfHost

- smfcc_nsmf_pdu_session_update_tx_failures

Type

64-bit Counter

Description
Number of Pdu Session Update Failures
Labels
<i>dnn, snssai, smfSessionType, smfHost, statusCode</i>
• smfcc_nsmf_pdu_session_release_request_attempts
Type
64-bit Counter
Description
Number of Pdu Session Release attempts
Labels
<i>dnn, snssai, smfSessionType, smfHost</i>
• smfcc_nsmf_pdu_session_release_request_successes
Type
64-bit Counter
Description
Number of Pdu Session Release Success
Labels
<i>dnn, snssai, smfSessionType, smfHost</i>
• smfcc_nsmf_pdu_session_release_request_failures
Type
64-bit Counter
Description
Number of Pdu Session Release Failure
Labels
<i>dnn, snssai, smfSessionType, smfHost, statusCode</i>
• smfcc_nsmf_pdu_session_update_rx_attempts
Type
64-bit Counter
Description
Number of incoming Pdu Session Update attempts
Labels
<i>dnn, snssai, smfSessionType, smfHost</i>
• smfcc_nsmf_pdu_session_update_rx_successes
Type
64-bit Counter
Description
Number of incoming Pdu Session Update successes
Labels
<i>dnn, snssai, smfSessionType, smfHost</i>

smfcc_nsmf_pdu_session_update_rx_failures
Type
64-bit Counter
Description
Number of incoming Pdu Session Update failures
Labels
dnn, snssai, smfSessionType, statusCode
smfcc_nsmf_pdu_session_status_notify_attempts
Type
64-bit Counter
Description
Number of Pdu Session Status Notify attempts
Labels
dnn, snssai, smfSessionType, smfHost
smfcc_nsmf_pdu_session_status_notify_successes
Type
64-bit Counter
Description
Number of Pdu Session Status Notify successes
Labels
dnn, snssai, smfSessionType, smfHost
smfcc_nsmf_pdu_session_status_notify_failures
Type
64-bit Counter
Description
Number of Pdu Session Status Notify failures
Labels
dnn, snssai, smfSessionType, statusCode, smfHost
smfcc_nsmf_evt_exp_create_subscription_tx_attempts
Type
64-bit Counter
Description
Number of Nsmf Event Exposure Create Subscription attempts
Labels
smfHost
smfcc_nsmf_evt_exp_create_subscription_tx_successes
Type
64-bit Counter
Description

Number of Nsmf Event Exposure Create Subscription successes

Labels

smfHost

- smfcc_nsmf_evt_exp_create_subscription_tx_failures

Type

64-bit Counter

Description

Number of Nsmf Event Exposure Create Subscription failures

Labels

smfHost, statusCode

- smfcc_nsmf_evt_exp_modify_subscription_tx_attempts

Type

64-bit Counter

Description

Number of Nsmf Event Exposure Modify Subscription attempts

Labels

smfHost

- smfcc_nsmf_evt_exp_modify_subscription_tx_successes

Type

64-bit Counter

Description

Number of Nsmf Event Exposure Modify Subscription successes

Labels

smfHost

- smfcc_nsmf_evt_exp_modify_subscription_tx_failures

Type

64-bit Counter

Description

Number of Nsmf Event Exposure Modify Subscription failures

Labels

smfHost, statusCode

- smfcc_nsmf_evt_exp_delete_subscription_tx_attempts

Type

64-bit Counter

Description

Number of Nsmf Event Exposure Delete Subscription attempts

Labels

smfHost

- smfcc_nsmf_evt_exp_delete_subscription_tx_successes

Type

64-bit Counter

Description

Number of Nsmf Event Exposure Delete Subscription successes

Labels

smfHost

- smfcc_nsmf_evt_exp_delete_subscription_tx_failures

Type

64-bit Counter

Description

Number of Nsmf Event Exposure Delete Subscription failures

Labels

smfHost, statusCode

- smfcc_nsmf_pdu_session_create_attempts_alternate_sepp

Type

64-bit Counter

Description

Number of Pdu Session Create attempts using alternate SEPP

Labels

dnn, snssai, smfSessionType, smfHost

- smfcc_nsmf_pdu_session_create_successes_alternate_sepp

Type

64-bit Counter

Description

Number of Pdu Session Create Successes using alternate SEPP

Labels

dnn, snssai, smfSessionType, smfHost

- smfcc_nsmf_pdu_session_create_failures_alternate_sepp

Type

64-bit Counter

Description

Number of Pdu Session Create Failures using alternate SEPP

Labels

dnn, snssai, smfSessionType, smfHost, statusCode

- smfcc_nsmf_pdu_session_update_tx_attempts_alternate_sepp

Type

64-bit Counter

Description

Number of Pdu Session Update attempts using alternate SEPP

Labels

dnn, snssai, smfSessionType, smfHost

- smfcc_nsmf_pdu_session_update_tx_successes_alternate_sepp

Type

64-bit Counter

Description

Number of Pdu Session Update Success using alternate SEPP

Labels

dnn, snssai, smfSessionType, smfHost

- smfcc_nsmf_pdu_session_update_tx_failures_alternate_sepp

Type

64-bit Counter

Description

Number of Pdu Session Update Failures using alternate SEPP

Labels

dnn, snssai, smfSessionType, smfHost, statusCode

- smfcc_nsmf_pdu_session_release_request_attempts_alternate_sepp

Type

64-bit Counter

Description

Number of Pdu Session Release attempts using alternate SEPP

Labels

dnn, snssai, smfSessionType, smfHost

- smfcc_nsmf_pdu_session_release_request_successes_alternate_sepp

Type

64-bit Counter

Description

Number of Pdu Session Release Success using alternate SEPP

Labels

dnn, snssai, smfSessionType, smfHost

- smfcc_nsmf_pdu_session_release_request_failures_alternate_sepp

Type

64-bit Counter

Description

Number of Pdu Session Release Failure using alternate SEPP

Labels

dnn, snssai, smfSessionType, smfHost, statusCode

SMF IntraNF Punting Statistics

- smfcc_intra_nf_punting_sent

Type

64-bit Counter

Description

Number of intra-nf-punting requests sent

Labels

smfPodInstanceld, msgType, isSentPodTerminatingPod

• smfcc_intra_nf_punting_successes

Type

64-bit Counter

Description

Number of intra-nf-punting requests successful

Labels

smfPodInstanceld, msgType, httpRespCode, isSentPodTerminatingPod

• smfcc_intra_nf_punting_failures

Type

64-bit Counter

Description

Number of intra-nf-punting requests failed

Labels

smfPodInstanceld, msgType, httpRespCode, isSentPodTerminatingPod

• smfcc_intra_nf_punting_ip_collision

Type

64-bit Counter

Description

Number of intra-nf-punting cases IP address collision which resulted force lock

Labels

LockString, podServiceId, podIp

• smfcc_intra_nf_punting_received

Type

64-bit Counter

Description

Number of intra-nf-punting requests received

Labels

smfPodInstanceld, msgType, isRecvPodTerminatingPod

• smfcc_intra_nf_punting_timeout

Type

64-bit Counter

Description

Number of intra-nf-punting requests timed out

Labels
<i>smfPodInstanceld, msgType, isSentPodTerminatingPod</i>
smfcc_intra_nf_punting_delayed_response
Type
64-bit Counter
Description
Number of intra-nf-punting requests getting delayed response after timer expiry
Labels
<i>smfPodInstanceld, httpRespCode</i>

SMF InterNF Punting Statistics

smfcc_inter_nf_punting_sent
Type
64-bit Counter
Description
Number of inter-nf-punting requests sent
Labels
<i>smfInstanceld, msgType</i>
smfcc_inter_nf_punting_successes
Type
64-bit Counter
Description
Number of inter-nf-punting requests successful
Labels
<i>smfInstanceld, msgType</i>
smfcc_inter_nf_punting_failures
Type
64-bit Counter
Description
Number of inter-nf-punting requests failed
Labels
<i>smfInstanceld, msgType</i>
smfcc_inter_nf_punting_received
Type
64-bit Counter
Description
Number of inter-nf-punting requests received
Labels
<i>smfInstanceld, msgType</i>
smfcc_inter_nf_punting_ignored_for_internal_session_release

Type

64-bit Counter

Description

Number of internal session release events ignored for inter-nf-punting

Labels

smfInstanceId

SMF Peer NF Set Statistics

- smfcc_selected_nf_instance_status_down

Type

64-bit Counter

Description

Number of peer nf instance status down in NRF cache

Labels

peerNfType, peerNfInstanceId

- smfcc_msg_rcvd_from_new_peer_nf_instance

Type

64-bit Counter

Description

Number of messages received from new peer nf instance

Labels

peerNfType, peerNfInstanceId

- smfcc_msg_sent_to_new_peer_nf_instance

Type

64-bit Counter

Description

Number of messages sent to new peer nf instance

Labels

peerNfType, peerNfInstanceId

- smfcc_alternative_peer_nf_retry_attempt_success

Type

64-bit Counter

Description

Number of retry requests attempted towards the NF

Labels

peerNfServiceType

- smfcc_alternative_peer_nf_retry_attempt_failures

Type

64-bit Counter

Description

Number of retry requests attempt failures towards the NF

Labels

peerNfServiceType, cause

SMF NF Set Statistics

- smfcc_nf_set_smf_owner_instance_changed

Type

64-bit Counter

Description

Number of sessions for which SMF instance changed

Labels

oldSmfInstanceld

- smfcc_nf_set_smf_procedures_failed_during_smf_owner_instance_change

Type

64-bit Counter

Description

Number of procedures failed during owner SMF instance change in NF Set

Labels

procedureName

SMF NRF Interface Statistics

- smfcc_nnrf_nfm_nfstatus_subscribe_for_nfset_attempts

Type

64-bit Counter

Description

Number of NF Status Subscribe requests for NF set attempted

Labels

nfType, nfSetId

- smfcc_nnrf_nfm_nfstatus_subscribe_for_nfset_failures

Type

64-bit Counter

Description

Number of NF Status Subscribe requests for NF set failed

Labels

nfType, nfSetId

- smfcc_nnrf_nfm_nfstatus_subscribe_for_nfset_retries

Type

64-bit Counter

Description

Number of NF Status Subscribe requests for NF set retried

Labels

nfType, nfSetId

- smfcc_nnrf_nfm_nfstatus_notification_for_nfset_received

Type

64-bit Counter

Description

Number of NF Status Notifications for NF set received

Labels

nfType, nfSetId

- smfcc_nnrf_nfm_nfstatus_notification_for_nfset_successes

Type

64-bit Counter

Description

Number of NF Status Notifications for NF set successful

Labels

nfType, nfSetId

- smfcc_nnrf_disc_nf_discover_for_nfset_attempts

Type

64-bit Counter

Description

Number of NF Discover requests for NF set attempted

Labels

nfType, nfSetId

- smfcc_nnrf_disc_nf_discover_for_nfset_failures

Type

64-bit Counter

Description

Number of NF Discover requests for NF set failed

Labels

nfType, nfSetId

- smfcc_nnrf_disc_nf_discover_for_nfset_retries

Type

64-bit Counter

Description

Number of NF Discover requests for NF set retried

Labels

nfType, nfSetId

- smfcc_nnrf_nfm_nfstatus_unsubscribe_for_nfset_attempts

Type

64-bit Counter

Description

Number of NF Status UnSubscribe requests for NF set attempted

Labels

nfType, nfSetId

- smfcc_nnrf_nfm_nfstatus_unsubscribe_for_nfset_failures

Type

64-bit Counter

Description

Number of NF Status UnSubscribe requests for NF set failed

Labels

nfType, nfSetId

- smfcc_nnrf_nfm_nfstatus_unsubscribe_for_nfset_retries

Type

64-bit Counter

Description

Number of NF Status UnSubscribe requests for NF set retried

Labels

nfType, nfSetId

SMF Subscriber DB Interface Statistics

- smfcc_gw_subscriber_db_create_record_attempts

Type

64-bit Counter

Description

Number of subscriber record create attempts

Labels

dnn

- smfcc_gw_subscriber_db_create_record_failures

Type

64-bit Counter

Description

Number of subscriber record create failures

Labels

dnn, cause

- smfcc_gw_subscriber_db_readlock_record_attempts

Type

64-bit Counter

Description

Number of subscriber record readlock attempts

Labels

dnn, lock_req_type

- smfcc_gw_subscriber_db_readlock_record_failures

Type

64-bit Counter

Description

Number of subscriber record readlock failures

Labels

dnn, lock_req_type, cause

- smfcc_gw_subscriber_db_update_record_attempts

Type

64-bit Counter

Description

Number of subscriber record update attempts

Labels

dnn

- smfcc_gw_subscriber_db_update_record_failures

Type

64-bit Counter

Description

Number of subscriber record update failures

Labels

dnn, cause

- smfcc_gw_subscriber_db_delete_record_attempts

Type

64-bit Counter

Description

Number of subscriber record delete attempts

Labels

dnn

- smfcc_gw_subscriber_db_delete_record_failures

Type

64-bit Counter

Description

Number of subscriber record delete failures

Labels

dnn, cause

SMF Cdr Interface Statistics

- smfcc_cdr_sent

Type

64-bit Counter

Description

Number of cdr sent

Labels

gwType, cdrType, retry

- smfcc_cdr_fail

Type

64-bit Counter

Description

Number of cdr failure

Labels

gwType, cdrType, causecode

- smfcc_cdr_response_rcvd

Type

64-bit Counter

Description

Number of cdr failure

Labels

gwType, cdrType, causecode

SMF Gw Subscriber Cache Statistics

- smfcc_gw_subscriber_cache_release_request_attempts

Type

64-bit Counter

Description

Number of gw subscriber record release attempts

Labels

owner_pod

- smfcc_gw_subscriber_cache_release_request_failures

Type

64-bit Counter

Description

Number of gw subscriber record release request failures

Labels

owner_pod, cause

- smfcc_gw_subscriber_cache_record_steal_lock_attempted

Type

64-bit Counter

Description

Number of subscriber cache record steal lock attempted

Labels

owner_pod

- smfcc_gw_subscriber_cache_release_db_read_retry_wait_timer_expiry

Type

64-bit Counter

Description

Number of DB read and lock retry timer expiry when lock owner is different

Labels

owner_pod

- smfcc_gw_subscriber_cache_release_request_received

Type

64-bit Counter

Description

Number of subscriber cache record release request received

Labels

owner_pod

- smfcc_gw_subscriber_cache_release_request_acknowledged

Type

64-bit Counter

Description

Number of subscriber cache record release request responded

Labels

owner_pod, statusCode

Table of Contents

- [Summary](#)
- [UDSF DBClient Metrics](#)
- [UDSF API latency statistics](#)

Summary

Group	Metric Name	Type	Synopsis
UDSF DBClient Metrics	udsf_db_operation_request	Counter	The total number of udsf dbclient requests received.
UDSF DBClient Metrics	udsf_db_operation_response	Counter	The total number of udsf dbclient responses sent.
UDSF DBClient Metrics	udsf_dbc_total_init_success	Counter	The total number of UDSF INIT successes.
UDSF DBClient Metrics	udsf_dbc_total_init_failure	Counter	The total number of UDSF INIT failures.
UDSF DBClient Metrics	udsf_dbc_total_deinit_success	Counter	The total number of UDSF DEINIT successes.
UDSF DBClient Metrics	udsf_dbc_total_deinit_failure	Counter	The total number of UDSF DEINIT failures.
UDSF DBClient Metrics	udsf_db_transaction_counter	Counter	The total number of CRUD operations.
UDSF API latency statistics	udsf_db_transaction_duration	Histogram	Roundtrip duration taken by transaction in seconds.
UDSF API latency statistics	udsf_db_execution_duration	Histogram	Execution time taken by transaction in microseconds.
UDSF API latency statistics	udsf_hiredis_bufferwrite_duration	Histogram	Duration of time taken in hiredis to write data to socket in seconds

UDSF DBClient Metrics

- **udsf_db_operation_request**

Type

64-bit Counter

Description

The total number of udsf dbclient requests received from worker. Labels- 'app_name' : Originating thread name, ex:smfcc, 'operation_type' : Type of CRUD operation/API type involved, ex:CREATE_OR_MODIFY_RECORD_ASYNC

Labels

app_name, operation_type
- **udsf_db_operation_response**

Type

64-bit Counter

Description

The total number of udsf dbclient responses sent. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE 'operation_type' : Type of CRUD operation/API type involved., ex:CREATE_OR_MODIFY_RECORD_ASYNC

Labels

app_name, status, operation_type
- **udsf_dbc_total_init_success**

Type

64-bit Counter

Description

The total number of times UDSF API init was success. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: SUCCESS

Labels

app_name, status
- **udsf_dbc_total_init_failure**

Type

64-bit Counter

Description

The total number of times UDSF API init failed. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: FAILURE

Labels

app_name, status
- **udsf_dbc_total_deinit_success**

Type

64-bit Counter

Description

The total number of times UDSF API Deinit succeeded. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: SUCCESS

Labels

app_name, status

- **udsf_dbc_total_deinit_failure**

Type

64-bit Counter

Description

The total number of times UDSF API Deinit failed. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: FAILURE

Labels

app_name, status
- **udsf_db_transaction_counter**

Type

64-bit Counter

Description

The total number of times CRUD operations are incrementing. Labels- 'app_name' : Originating thread name, ex:smfcc, 'crud_op' : The categorical CRUD operation in which this sub-transation falls into, ex: CREATE_OR_MODIFY

Labels

app_name, crud_op

UDSF API latency statistics

- **udsf_db_transaction_duration**

Type

64-bit Histogram

Description

Duration observed on total journey/roundtrip taken by a transaction after receiving the request. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE 'operation_type' : Type of CRUD operation/API type involved., ex:CREATE_OR_MODIFY_RECORD_ASYNC, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE, 'crud_op' : The categorical CRUD operation in which this transation falls into, ex: CREATE_OR_MODIFY

Labels

app_name,operation_type,status,crud_op

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]
- **udsf_db_execution_duration**

Type

64-bit Histogram

Description

Processing time taken by UDSF client. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE 'operation_type' : Type of CRUD operation/API type involved., ex:CREATE_OR_MODIFY_RECORD_ASYNC, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE, 'crud_op' : The categorical CRUD operation in which this transation falls into, ex: CREATE_OR_MODIFY

Labels

app_name,operation_type,status,crud_op

Boundaries

[1, 2, 4, 8, 16, 32, 64, 128, 256, 384, 512, 768, 1024, 1536, 2048, 4096, 8192, 16384]
- **udsf_hiredis_bufferwrite_duration**

Type

64-bit Histogram

Description

Duration observed between hiredis buffer append and write to socket. This is fair approximation for time spent before data is written to socket as the buffer threshold accounted for append and write is dynamic and not fixed size. Per socket, the buffer append start-time is recorded along with size intended to transmit and end-time recorded once data greater than recorded size is written on the socket

Labels

app_name

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

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- [Summary](#)
- [Classifier query for analyzer micro service](#)

Summary

Group	Metric Name	Type	Synopsis
Classifier query for analyzer micro service	analyzer_classifier_request	Counter	Classifier query requests received by Analyzer.
Classifier query for analyzer micro service	analyzer_classifier_response	Counter	Classifier query response sent by Analyzer.
Classifier query for analyzer micro service	analyzer_instance_query_attempt	Counter	Analyzer query attempts for current instance.
Classifier query for analyzer micro service	analyzer_instance_query_success	Counter	Analyzer query attempt success for current instance.
Classifier query for analyzer micro service	analyzer_instance_query_failure	Counter	Analyzer query attempt failure for current instance.

Classifier query for analyzer micro service

• analyzer_classifier_request

Type

64-bit Counter

Description

The number of classifier query requests received by the analyzer micro service. A single query could contain one or more analyzer instance name in the request body.

Labels

analyzer_instance

• analyzer_classifier_response

Type

64-bit Counter

Description

The number of classifier query response(success/failure) sent by the analyzer micro service. Successful response will contain analyzer instance name along with the class name in the response body.

Labels

analyzer_instance, http_status_code, app_error

• analyzer_instance_query_attempt

Type

64-bit Counter

Description

The number of attempted analyzer query for the current instance. The counter is pegged if the received analyzer instance in query is present in the database.

Labels

analyzer_instance

- analyzer_instance_query_success

Type

64-bit Counter

Description

The number of successful analyzer query for the current instance. The counter is pegged if analyzer instance, class and filterset name are retrieved successfully.

Labels

analyzer_instance, class_name, filter_set

- analyzer_instance_query_failure

Type

64-bit Counter

Description

The number of failed analyzer query for the current instance. The counter is pegged if it failed to retrieve the analyzer instance, class or filterset name.

Labels

analyzer_instance, app_error

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- [Summary](#)
- [HTTP Punting Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_sent	Counter	The total number of HTTP requests successfully punted.
HTTP Punting Metrics	punting_http_requests_failed	Counter	The total number of HTTP requests punting failed.
HTTP Punting Metrics	punting_http_responses_received	Counter	The total number of HTTP responses received for the requests punted.
HTTP Punting Metrics	punting_http_responses_forwarded	Counter	The total number of HTTP responses forwarded after punting transaction completion.
HTTP Punting Metrics	punting_http_requests_aborted	Counter	The total number of HTTP requests aborted.
HTTP Punting Metrics	punting_http_request_duration_seconds	Histogram	Punting operation duration in seconds.

HTTP Punting Metrics

- punting_http_requests_sent

Type

64-bit Counter

Description

The total number of HTTP requests successfully punted.

Labels

message_type

- `punting_http_requests_failed`

Type

64-bit Counter

Description

The total number of HTTP requests for which the punting failed.

Labels

message_type

- `punting_http_responses_received`

Type

64-bit Counter

Description

The total number of HTTP responses received for the requests punted.

Labels

message_type, status_code

- `punting_http_responses_forwarded`

Type

64-bit Counter

Description

The total number of HTTP responses forwarded after punting transaction completion.

Labels

message_type, status_code

- `punting_http_requests_aborted`

Type

64-bit Counter

Description

The total number of HTTP requests aborted.

Labels

message_type

- `punting_http_request_duration_seconds`

Type

64-bit Histogram

Description

The punting transaction duration is measured from the time the HTTP request is sent out to the time the response is received for the punting request.

Labels

message_type

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

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- [Summary](#)
- [Packet Streaming Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Packet Streaming Metrics	num_packet_streaming_attempted	Counter	The total number of attempted streaming of packets
Packet Streaming Metrics	num_packet_streaming_success	Counter	The total number of packets streamed successfully
Packet Streaming Metrics	num_packet_streaming_failed	Counter	The total number of failed streaming of packets
Packet Streaming Metrics	num_packet_streaming_bytes	Counter	The total number of bytes successfully streamed

Packet Streaming Metrics

- num_packet_streaming_attempted

Type

64-bit Counter

Description

The total number of attempted streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_success

Type

64-bit Counter

Description

The total number of packets streamed successfully

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_failed

Type

64-bit Counter

Description

The total number of failed streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_bytes

Type

64-bit Counter

Description

The total number of bytes successfully streamed

Labels

interface_name, server_fqdn, kafka_topic

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- [Summary](#)
- [Unified LB](#)

Summary

Group	Metric Name	Type	Synopsis
Unified LB	numRxSctpV4PacketsFromApp	Counter	The number of SCTP V4 packets that have been received from all the applications by this ULB pod
Unified LB	numRxSctpV6PacketsFromApp	Counter	The number of SCTP V6 packets that have been received from all the applications by this ULB pod
Unified LB	numRxUdpV4PacketsFromApp	Counter	The number of UDP V4 packets that have been received from all the applications by this ULB pod
Unified LB	numRxUdpV6PacketsFromApp	Counter	The number of UDP V6 packets that have been received from all the applications by this ULB pod
Unified LB	numTxSctpV4PacketsToApp	Counter	The number of SCTP V4 packets that have been sent to applications by this ULB pod
Unified LB	numTxSctpV6PacketsToApp	Counter	The number of SCTP V6 packets that have been sent to applications by this ULB pod
Unified LB	numTxUdpV4PacketsToApp	Counter	The number of UDP V4 packets that have been sent to applications by this ULB pod
Unified LB	numTxUdpV6PacketsToApp	Counter	The number of UDP V6 packets that have been sent to applications by this ULB pod
Unified LB	numRxSctpV4PacketsFromPeer	Counter	The number of SCTP V4 packets that have been received from the peers by this ULB pod
Unified LB	numRxSctpV6PacketsFromPeer	Counter	The number of SCTP V6 packets that have been received from the peers by this ULB pod
Unified LB	numRxUdpV4PacketsFromPeer	Counter	The number of UDP V4 packets that have been received from the peers by this ULB pod
Unified LB	numRxUdpV6PacketsFromPeer	Counter	The number of UDP V6 packets that have been received from the peers by this ULB pod
Unified LB	numTxSctpV4PacketsToPeer	Counter	The number of SCTP V4 packets that have been sent to peers by this ULB pod
Unified LB	numTxSctpV6PacketsToPeer	Counter	The number of SCTP V6 packets that have been sent to peers by this ULB pod
Unified LB	numTxUdpV4PacketsToPeer	Counter	The number of UDP V4 packets that have been sent to peers by this ULB pod
Unified LB	numTxUdpV6PacketsToPeer	Counter	The number of UDP V6 packets that have been sent to peers by this ULB pod
Unified LB	numSctpFlowsAdded	Counter	Total number of SCTP flows added to the flow table
Unified LB	numSctpFlowsRemoved	Counter	Total number of SCTP flows removed from the flow table
Unified LB	numUdpFlowsAdded	Counter	Total number of UDP flows added to the flow table
Unified LB	numUdpFlowsRemoved	Counter	Total number of UDP flows removed from the flow table

Unified LB

- [numRxSctpV4PacketsFromApp](#)

Type

64-bit Counter

Description

The number of SCTP V4 packets that have been received from all the applications by this ULB pod

Labels

none
- [numRxSctpV6PacketsFromApp](#)

Type

64-bit Counter

Description

The number of SCTP V6 packets that have been received from all the applications by this ULB pod

Labels

none
- [numRxUdpV4PacketsFromApp](#)

Type

64-bit Counter

Description

The number of UDP V4 packets that have been received from all the applications by this ULB pod

Labels

none

• numRxUdpV6PacketsFromApp

Type

64-bit Counter

Description

The number of UDP V6 packets that have been received from all the applications by this ULB pod

Labels

none

• numTxSctpV4PacketsToApp

Type

64-bit Counter

Description

The number of SCTP V4 packets that have been sent to applications by this ULB pod

Labels

none

• numTxSctpV6PacketsToApp

Type

64-bit Counter

Description

The number of SCTP V6 packets that have been sent to applications by this ULB pod

Labels

none

• numTxUdpV4PacketsToApp

Type

64-bit Counter

Description

The number of UDP V4 packets that have been sent to applications by this ULB pod

Labels

none

• numTxUdpV6PacketsToApp

Type

64-bit Counter

Description

The number of UDP V6 packets that have been sent to applications by this ULB pod

Labels
<i>none</i>
numRxSctpV4PacketsFromPeer
Type
64-bit Counter
Description
The number of SCTP V4 packets that have been received from the peers by this ULB pod
Labels
<i>none</i>
numRxSctpV6PacketsFromPeer
Type
64-bit Counter
Description
The number of SCTP V6 packets that have been received from the peers by this ULB pod
Labels
<i>none</i>
numRxUdpV4PacketsFromPeer
Type
64-bit Counter
Description
The number of UDP V4 packets that have been received from the peers by this ULB pod
Labels
<i>none</i>
numRxUdpV6PacketsFromPeer
Type
64-bit Counter
Description
The number of UDP V6 packets that have been received from the peers by this ULB pod
Labels
<i>none</i>
numTxSctpV4PacketsToPeer
Type
64-bit Counter
Description
The number of SCTP V4 packets that have been sent to peers by this ULB pod
Labels
<i>none</i>
numTxSctpV6PacketsToPeer
Type

64-bit Counter

Description

The number of SCTP V6 packets that have been sent to peers by this ULB pod

Labels

none

• numTxUdpV4PacketsToPeer

Type

64-bit Counter

Description

The number of UDP V4 packets that have been sent to peers by this ULB pod

Labels

none

• numTxUdpV6PacketsToPeer

Type

64-bit Counter

Description

The number of UDP V6 packets that have been sent to peers by this ULB pod

Labels

none

• numSctpFlowsAdded

Type

64-bit Counter

Description

Total number of SCTP flows added to the flow table

Labels

none

• numSctpFlowsRemoved

Type

64-bit Counter

Description

Total number of SCTP flows removed from the flow table

Labels

none

• numUdpFlowsAdded

Type

64-bit Counter

Description

Total number of UDP flows added to the flow table

Labels

none

- numUdpFlowsRemoved

Type

64-bit Counter

Description

Total number of UDP flows removed from the flow table

Labels

none

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- Summary
- ChargingCmn statistics

Summary

Group	Metric Name	Type	Synopsis
ChargingCmn statistics	svcmgr_service_available_ind_notification	Counter	How many times, Service Available Indication message received
ChargingCmn statistics	svcmgr_service_available_ind_parse_errs	Counter	How many times, Service Available Indication parse error happened
ChargingCmn statistics	svcmgr_service_available_ind_notification_ack	Counter	How many times, Service Available Indication message is acknowledged
ChargingCmn statistics	svcmgr_service_unavailable_ind_notification	Counter	How many times, Service Unavailable Indication message
ChargingCmn statistics	svcmgr_service_unavailable_ind_parse_errs	Counter	How many times, Service Unavailable Indication parse error happened
ChargingCmn statistics	svcmgr_service_unavailable_ind_notification_ack	Counter	How many times, Service Unavailable Indication message is acknowledged
ChargingCmn statistics	svcmgr_publish_msg_sent	Counter	How many times, Publication message sent
ChargingCmn statistics	svcmgr_publish_msg_success	Counter	How many times, Publication message success response received
ChargingCmn statistics	svcmgr_publish_msg_failure	Counter	How many times, failure received for publication message
ChargingCmn statistics	svcmgr_subscribe_msg_sent	Counter	How many times, Subscribe message sent
ChargingCmn statistics	svcmgr_subscribe_msg_success	Counter	How many times, subscribe message success
ChargingCmn statistics	svcmgr_subscribe_msg_failure	Counter	How many times, failure received for subscribe message
ChargingCmn statistics	svcmgr_unsubscribe_msg_sent	Counter	How many times, unsubscribe message sent
ChargingCmn statistics	svcmgr_unsubscribe_msg_failure	Counter	How many times, failure received for unsubscribe message
ChargingCmn statistics	svcmgr_config_req_sent	Counter	How many times, config request sent
ChargingCmn statistics	svcmgr_config_req_failure	Counter	How many times, failure received for config request
ChargingCmn statistics	svcmgr_config_req_success	Counter	How many times, success received for config request
ChargingCmn statistics	svcmgr_config_register_response	Counter	How many times, config register response received

Group	Metric Name	Type	Synopsis
ChargingCmn statistics	svcmgr_config_register_response_ack	Counter	How many times, config register response is acknowledged
ChargingCmn statistics	svcmgr_config_register_response_parse_err	Counter	How many times, parse error happened for config register response
ChargingCmn statistics	svcmgr_config_update_notification	Counter	How many times, config update notification received
ChargingCmn statistics	svcmgr_config_update_notification_response	Counter	How many times, config update response
ChargingCmn statistics	svcmgr_intf_schema_update_ind_notification	Counter	How many times, interface schema update indication message received
ChargingCmn statistics	svcmgr_intf_schema_update_ind_notification_ack	Counter	How many times, interface schema update indication message is acknowledged
ChargingCmn statistics	svcmgr_intf_schema_update_ind_parse_errs	Counter	How many times, interface schema update indication parse error happened

ChargingCmn statistics

- svcmgr_service_available_ind_notification

Type

64-bit Counter

Description

Number of Times Service Available Indication notification received

Labels

servicename

- svcmgr_service_available_ind_parse_errs

Type

64-bit Counter

Description

Number of Times Service Available Indication parse error happened

Labels

- svcmgr_service_available_ind_notification_ack

Type

64-bit Counter

Description

Number of Times Service Available Indication message is acknowledged

Labels

servicename

- svcmgr_service_unavailable_ind_notification

Type

64-bit Counter

Description

Number of Times Service Unavailable Indication notification received

Labels

servicename

- svcmgr_service_unavailable_ind_parse_errs

Type

64-bit Counter

Description

Number of Times Service Unavailable Indication parse error happened

Labels

- svcmgr_service_unavailable_ind_notification_ack

Type

64-bit Counter

Description

Number of Times Service Unavailable Indication message is acknowledged

Labels

servicename

- svcmgr_publish_msg_sent

Type

64-bit Counter

Description

Number of Times Publication message sent

Labels

servicename, retry

- svcmgr_publish_msg_success

Type

64-bit Counter

Description

Number of Times Publication message success response received

Labels

servicename, causecode, retry

- svcmgr_publish_msg_failure

Type

64-bit Counter

Description

Number of Times failure received for publication message

Labels

servicename, causecode, retry

- svcmgr_subscribe_msg_sent

Type

64-bit Counter

Description

Number of Times Subscribe message sent

Labels

servicename, retry

- **svcmgr_subscribe_msg_success**

Type

64-bit Counter

Description

Number of Times Send subscribe message success response received

Labels

servicename, causecode, retry

- **svcmgr_subscribe_msg_failure**

Type

64-bit Counter

Description

Number of time failure received for subscribe message

Labels

servicename, causecode, retry

- **svcmgr_unsubscribe_msg_sent**

Type

64-bit Counter

Description

Number of Times unsubscribe message sent

Labels

servicename

- **svcmgr_unsubscribe_msg_failure**

Type

64-bit Counter

Description

Number of Times failure received for unsubscribe message

Labels

servicename

- **svcmgr_config_req_sent**

Type

64-bit Counter

Description

Number of Times config request sent

Labels

retry

- **svcmgr_config_req_failure**

Type

64-bit Counter

Description

Number of Times failure received for config request

Labels

causecode, retry

- `svcmgr_config_req_success`

Type

64-bit Counter

Description

Number of Times success received for config request

Labels

causecode, retry

- `svcmgr_config_register_response`

Type

64-bit Counter

Description

Number of Times config register response received

Labels

statuscode

- `svcmgr_config_register_response_ack`

Type

64-bit Counter

Description

Number of Times config register response is acknowledged

Labels

-
- `svcmgr_config_register_response_parse_err`

Type

64-bit Counter

Description

Number of Times parse error happened for config register response

Labels

-
- `svcmgr_config_update_notification`

Type

64-bit Counter

Description

Number of Times config update notification received

Labels

• svcmgr_config_update_notification_response
Type
64-bit Counter
Description
Number of Times config update response sent
Labels

• svcmgr_intf_schema_update_ind_notification
Type
64-bit Counter
Description
Number of Times interface schema update indication message received
Labels
<i>servicename</i>

• svcmgr_intf_schema_update_ind_notification_ack
Type
64-bit Counter
Description
Number of Times interface schema update indication message is acknowledged
Labels
<i>servicename</i>

• svcmgr_intf_schema_update_ind_parse_errs
Type
64-bit Counter
Description
Number of Times Service interface schema update indication parse error happened
Labels

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• Summary
• ETCD Client Metrics

Summary

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_put_requests	Counter	The total number of ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_requests_sent	Counter	The total number of ETCD Client PUT Requests sent.
ETCD Client Metrics	etcd_put_successes	Counter	The total number of successful ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_failures	Counter	The total number of failed ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_with_lease_requests	Counter	The total number of ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_requests_sent	Counter	The total number of ETCD Client PUT with Lease Requests sent.
ETCD Client Metrics	etcd_put_with_lease_successes	Counter	The total number of successful ETCD Client PUT with Lease Requests.

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_put_with_lease_failures	Counter	The total number of failed ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_range_requests	Counter	The total number of ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_requests_sent	Counter	The total number of ETCD Client Range Requests sent.
ETCD Client Metrics	etcd_range_successes	Counter	The total number of successful ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_failures	Counter	The total number of failed ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_delete_requests	Counter	The total number of ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_requests_sent	Counter	The total number of ETCD Client Range Delete Requests sent.
ETCD Client Metrics	etcd_range_delete_successes	Counter	The total number of successful ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_failures	Counter	The total number of failed ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_txn_requests	Counter	The total number of ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_requests_sent	Counter	The total number of ETCD Client Txn Requests sent.
ETCD Client Metrics	etcd_txn_successes	Counter	The total number of successful ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_failures	Counter	The total number of failed ETCD Client Txn Requests.
ETCD Client Metrics	etcd_watch_requests	Counter	The total number of ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_requests_sent	Counter	The total number of ETCD Client Watch Requests sent.
ETCD Client Metrics	etcd_watch_successes	Counter	The total number of successful ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_failures	Counter	The total number of failed ETCD Client Watch Requests.
ETCD Client Metrics	etcdwatcher_requests	Counter	The total number of ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_requests_sent	Counter	The total number of ETCD Watcher Requests sent.
ETCD Client Metrics	etcdwatcher_successes	Counter	The total number of successful ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_failures	Counter	The total number of failed ETCD Watcher Requests.

ETCD Client Metrics

- [etcd_put_requests](#)

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests.

Labels

keyPrefix

- [etcd_put_requests_sent](#)

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests sent.

Labels

keyPrefix

- [etcd_put_successes](#)

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT Requests.

Labels

keyPrefix

• etcd_put_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT Requests.

Labels

keyPrefix, failureReason

• etcd_put_with_lease_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests.

Labels

keyPrefix

• etcd_put_with_lease_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests sent.

Labels

keyPrefix

• etcd_put_with_lease_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT with Lease Requests.

Labels

keyPrefix

• etcd_put_with_lease_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT with Lease Requests.

Labels

keyPrefix, failureReason

• etcd_range_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests.

Labels

keyPrefix

• etcd_range_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests sent.

Labels

keyPrefix

• etcd_range_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Requests.

Labels

keyPrefix

• etcd_range_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Requests.

Labels

keyPrefix, failureReason

• etcd_range_delete_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests.

Labels

keyPrefix

• etcd_range_delete_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests sent.

Labels

keyPrefix

- etcd_range_delete_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Delete Requests.

Labels

keyPrefix

- etcd_range_delete_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Delete Requests.

Labels

keyPrefix, failureReason

- etcd_txn_requests

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests.

Labels

keyPrefix

- etcd_txn_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Txn Requests sent.

Labels

keyPrefix

- etcd_txn_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Txn Requests.

Labels

keyPrefix

- etcd_txn_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Txn Requests.

Labels

keyPrefix, failureReason

- etcd_watch_requests

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests sent.

Labels

keyPrefix

- etcd_watch_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Watch Requests.

Labels

keyPrefix, failureReason

- etcdwatcher_requests

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests.

Labels

keyPrefix

- `etcdwatcher_requests_sent`

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests sent.

Labels

keyPrefix
- `etcdwatcher_successes`

Type

64-bit Counter

Description

The total number of successful ETCD Watcher Requests.

Labels

keyPrefix
- `etcdwatcher_failures`

Type

64-bit Counter

Description

The total number of failed ETCD Watcher Requests.

Labels

keyPrefix, failureReason

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- [HTTP Punting Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_sent	Counter	The total number of HTTP requests successfully punted.
HTTP Punting Metrics	punting_http_requests_failed	Counter	The total number of HTTP requests punting failed.
HTTP Punting Metrics	punting_http_responses_received	Counter	The total number of HTTP responses received for the requests punted.
HTTP Punting Metrics	punting_http_responses_forwarded	Counter	The total number of HTTP responses forwarded after punting transaction completion.
HTTP Punting Metrics	punting_http_requests_aborted	Counter	The total number of HTTP requests aborted.
HTTP Punting Metrics	punting_http_request_duration_seconds	Histogram	Punting operation duration in seconds.

HTTP Punting Metrics

- `punting_http_requests_sent`

Type

64-bit Counter

Description

The total number of HTTP requests successfully punted.

Labels

message_type

- `punting_http_requests_failed`

Type

64-bit Counter

Description

The total number of HTTP requests for which the punting failed.

Labels

message_type

- `punting_http_responses_received`

Type

64-bit Counter

Description

The total number of HTTP responses received for the requests punted.

Labels

message_type, status_code

- `punting_http_responses_forwarded`

Type

64-bit Counter

Description

The total number of HTTP responses forwarded after punting transaction completion.

Labels

message_type, status_code

- `punting_http_requests_aborted`

Type

64-bit Counter

Description

The total number of HTTP requests aborted.

Labels

message_type

- `punting_http_request_duration_seconds`

Type

64-bit Histogram

Description

The punting transaction duration is measured from the time the HTTP request is sent out to the time the response is received for the punting request.

Labels

message_type

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

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- [UPF Monitor Interface Stats](#)
- [UPF Monitor Service Stats](#)
- [UPF Monitor Stats](#)

Summary

Group	Metric Name	Type	Synopsis
UPF Monitor Interface Stats	monitor_upf_nfmanagement_service_events	Counter	Total number of NFManagement Service events triggered by UPF Monitor
UPF Monitor Service Stats	monitor_upf_critical_service_up	Counter	Number of critical services required by UPF which are up
UPF Monitor Service Stats	monitor_upf_critical_service_down	Counter	Number of critical services required by UPF which are down
UPF Monitor Stats	monitor_upf_total_load_reported	Gauge	Average UPF load reported across all PPEs to NRF
UPF Monitor Stats	monitor_upf_average_session_load	Gauge	Average Session Load Computed across all PPEs in UPF
UPF Monitor Stats	monitor_upf_average_cpu_load	Gauge	Average CPU Load Computed across all PPEs in UPF
UPF Monitor Stats	monitor_upf_ppe_session_load	Gauge	Average Session Load reported by PPE
UPF Monitor Stats	monitor_upf_ppe_cpu_load	Gauge	Average CPU Load reported by PPE

UPF Monitor Interface Stats

- [monitor_upf_nfmanagement_service_events](#)

Type

64-bit Counter

Description

Total number of NFManagement Service events triggered by UPF Monitor. The event types include registration, deregistration and update of NF Profile for the given instance of UPF NF.

Labels

event_type

UPF Monitor Service Stats

- [monitor_upf_critical_service_up](#)

Type

64-bit Counter

Description

Number of critical services required by UPF which have become available for use

Labels

critical_service_name

- [monitor_upf_critical_service_down](#)

Type

64-bit Counter

Description

Number of critical services required by UPF which have become unavailable for use

Labels
critical_service_name

UPF Monitor Stats

- monitor_upf_total_load_reported

Type
64-bit Gauge
Description
Average UPF load reported across all PPEs to NRF
Labels

- monitor_upf_average_session_load

Type
64-bit Gauge
Description
Average Session Load Computed across all PPEs in UPF
Labels

- monitor_upf_average_cpu_load

Type
64-bit Gauge
Description
Average CPU Load Computed across all PPEs in UPF
Labels

- monitor_upf_ppe_session_load

Type
64-bit Gauge
Description
Average Session Load reported by PPE
Labels

ppe_id

- monitor_upf_ppe_cpu_load

Type
64-bit Gauge
Description
Average CPU Load reported by PPE
Labels

ppe_id

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- [NfManagement statistics](#)
- [Debug statistics](#)
- [ApiServer statistics](#)
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- [Consul statistics](#)

Summary

Group	Metric Name	Type	Synopsis
NfMonitor statistics	nfmonitor_unstable_nf_counter	Counter	Howmany times, the Nf is found to be Unstable
NfMonitor statistics	nfmonitor_force_restart_due_to_nf_unstability	Counter	Numbe of times the VNFRestart triggered due to Unstability of the NF
NfMonitor statistics	nfmonitor_cumulative_readiness_failure_count	Counter	Numbe of times Service sound to have insuffucient count due to Readiness probe
NfMonitor statistics	nfmonitor_cumulative_readiness_failure_alerts	Counter	Numbe of times Service sound to have insuffucient count due to Readiness probe
NfMonitor statistics	nfmonitor_cumulative_k8api_get_query_failures	Counter	Numbe of times the k8 get Api query failures
DebugNfMonitorCmn	nfmonitor_force_restart_counter	Counter	Number of times force-restart triggered
NfManagement statistics	nfmonitor_nfmgmt_service_request_attempts	Counter	Howmany times, the Nf is attempting to send NRF Requests
NfManagement statistics	nfmonitor_nfmgmt_service_request_success	Counter	Numbe of times Success response is received for the requests to NRF
NfManagement statistics	nfmonitor_nfmgmt_service_request_failures	Counter	Numbe of times Failure response is received for the requests to NRF
NfManagement statistics	nfmonitor_nfmgmt_aborted_request_msgs	Counter	Numbe of times the requests to NRF is Aborted/Cancelled
Debug statistics	nfmonitor_vnf_restart_trigger	Counter	Howmany times, the Force Restart is received
Debug statistics	nfmonitor_fsm_state_changes	Counter	The statistics for the the FSM state changes
Debug statistics	nfmonitor_deactivation_triggers	Counter	Numbe of times the NF deactivation trigger is received
Debug statistics	nfmonitor_critical_service_up_notification	Counter	Numbe of times the Critical Service Up Notification received
Debug statistics	nfmonitor_critical_service_down_notification	Counter	Numbe of times the Critical Service Down Notification received
Debug statistics	nfmonitor_critical_services_overall_up_notification	Counter	Numbe of times the Up notification received when all monitored Critical Services are available
Debug statistics	nfmonitor_critical_services_overall_down_notification	Counter	Numbe of times the Down notification received when at least one Critical Service was unavailable
Debug statistics	nfmonitor_hysterisis_timer_started	Counter	Numbe of times the Hysterisis Timer is started due to missing Critical Service
Debug statistics	nfmonitor_hysterisis_timer_stopped	Counter	Numbe of times the Hysterisis Timer is stopped due to Critical Service become available
Debug statistics	nfmonitor_hysterisis_timer_expired	Counter	Numbe of times the Hysterisis Timer is expired due to missing Critical Service
Debug statistics	nfmonitor_personality_update_triggers	Counter	Numbe of times the Nf Personality update is received
Debug statistics	nfmonitor_maintenance_trigger_received	Counter	Numbe of times the Maintenance Mode is activated
Debug statistics	nfmonitor_profile_update_triggers	Counter	Numbe of times the Nf Profile Update happened
Debug statistics	nfmonitor_critical_service_down_on_register_entry	Counter	Numbe of times the Critical Services were down while starting with RegisterFsmState
ApiServer statistics	nfmonitor_deployment_create_watch_request	Counter	Howmany times, the deployment watch requests are created
ApiServer statistics	nfmonitor_deployment_delete_watch_request	Counter	Howmany times, the deployment watch requests are deleted
ApiServer statistics	nfmonitor_deployment_watch_notification_success	Counter	Howmany times, the watch notifications are successful
ApiServer statistics	nfmonitor_deployment_watch_notification_error	Counter	Howmany times, the watch notifications are failed
ApiServer statistics	nfmonitor_limit_range_query_attempts	Counter	Howmany times, the limit-range query attempts are made
ApiServer statistics	nfmonitor_limit_range_query_failures	Counter	Howmany times, the limit-range query attempts are failed

Group	Metric Name	Type	Synopsis
ApiServer statistics	nfmonitor_limit_range_query_success	Counter	Howmany times, the limit-range query attempts are succeeded
ApiServer statistics	nfmonitor_hpa_query_attempts	Counter	Howmany times, the Hpa query attempts are made
ApiServer statistics	nfmonitor_hpa_query_failures	Counter	Howmany times, the Hpa query attempts are failed
ApiServer statistics	nfmonitor_hpa_query_success	Counter	Howmany times, the Hpa query attempts are succeeded
ApiServer statistics	nfmonitor_readiness_probe_query_attempts	Counter	Howmany times, the Readiness Probe query attempts are made
ApiServer statistics	nfmonitor_readiness_probe_query_failures	Counter	Howmany times, the Readiness Probe query attempts are failed
ApiServer statistics	nfmonitor_readiness_probe_query_success	Counter	Howmany times, the Readiness Probe query attempts are succeeded
Feature Functionality statistics	nfmonitor_critical_service_frequent_restart	Counter	Howmany times, minimum count is not met for the stable Critical Services
Feature Functionality statistics	nfmonitor_alarm_due_to_frequent_restart	Counter	Howmany times alarm raised, due to minimum count is not met for the stable Critical Services
Feature Functionality statistics	nfmonitor_force_restart_due_to_frequent_restart	Counter	Howmany times, force restart triggerd because of the unstable Critical Services
Feature Functionality statistics	nfmonitor_critical_service_readiness_failure	Counter	Howmany times, minimum count is not met for the readiness succeeded Critical Services
Feature Functionality statistics	nfmonitor_alarm_due_to_readiness_failure	Counter	Alarm raised, due to minimum count is not met for the readiness succeeded Critical Services
Feature Functionality statistics	nfmonitor_force_restart_due_to_readiness_failure	Counter	Force restart triggerd because of Critical Service Readiness failure
Consul statistics	nfmonitor_publish_status_into_consul	Counter	Number of times Publish message is sent to Consul
Consul statistics	nfmonitor_consul_read_attempts	Counter	Number of times Consul read attempts
Consul statistics	nfmonitor_consul_read_failures	Counter	Number of times Consul read attempts failed
Consul statistics	nfmonitor_consul_read_success	Counter	Number of times Consul read attempts succeeded

NfMonitor statistics

- [nfmonitor_unstable_nf_counter](#)

Type

64-bit Counter

Description

Number of Times Nf is found Unstable

Labels

- [nfmonitor_force_restart_due_to_nf_unstability](#)

Type

64-bit Counter

Description

The number of times the Force Restart triggered due to NF-Unstability

Labels

- [nfmonitor_cumulative_readiness_failure_count](#)

Type

64-bit Counter

Description

The number of times the service is found to be failing the Readiness probe

Labels

critical_service_name, pod_name

- nfmonitor_cumulative_readiness_failure_alerts

Type

64-bit Counter

Description

Total number of EventAlerts generated due to Readiness probe failure

Labels

critical_service_name, pod_name

- nfmonitor_cumulative_k8api_get_query_failures

Type

64-bit Counter

Description

Total number of k8 api get pod query failures

Labels

critical_service_name, pod_name

DebugNfMonitorCmn

- nfmonitor_force_restart_counter

Type

64-bit Counter

Description

Number of times force-restart triggered

Labels

agent, name

NfManagement statistics

- nfmonitor_nfmgmt_service_request_attempts

Type

64-bit Counter

Description

Number of Requests sent to NRF

Labels

instance_id, msg_type, method, retry

- nfmonitor_nfmgmt_service_request_success

Type

64-bit Counter

Description

The number of times the Request to NRF is Successful

Labels

instance_id, msg_type, method, retry, status_code

- nfmonitor_nfmgmt_service_request_failures

Type

64-bit Counter

Description

The number of times the Request to NRF is Failure

Labels

instance_id, msg_type, method, retry, status_code

- nfmonitor_nfmgmt_aborted_request_msgs

Type

64-bit Counter

Description

The number of times the Request to NRF is Aborted

Labels

instance_id, msg_type

Debug statistics

- nfmonitor_vnf_restart_trigger

Type

64-bit Counter

Description

Number of VNF Restart triggers received

Labels

curr_nf_status, nfmgmt_state, nfmonitor_fsm_state, up_count, timestamp, trigger, type

- nfmonitor_fsm_state_changes

Type

64-bit Counter

Description

The FSM State Change events stats

Labels

curr_nf_status, nfmgmt_state, up_count, from_state, to_state

- nfmonitor_deactivation_triggers

Type

64-bit Counter

Description

The number of times the deactivation trigger received

Labels

nf_status, nfmgmt_state, up_count, nfmonitor_fsm_state

- nfmonitor_critical_service_up_notification

Type

64-bit Counter

Description

The number of times the CriticalService Up notification received

Labels

service, criticality, nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_critical_service_down_notification

Type

64-bit Counter

Description

The number of times the CriticalService Down notification received

Labels

service, criticality, nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_critical_services_overall_up_notification

Type

64-bit Counter

Description

The number of times the Overall CriticalServices Up notification received

Labels

nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_critical_services_overall_down_notification

Type

64-bit Counter

Description

The number of times the Overall CriticalServices Down notification received

Labels

nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_hysteresis_timer_started

Type

64-bit Counter

Description

The number of times the Hysteresis Timer is started

Labels

nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_hysteresis_timer_stopped

Type

64-bit Counter

Description

The number of times the Hysteresis Timer is stopped

Labels

nf_status, nfmgmt_state, up_count, fsm_state

- nfmonitor_hysterisis_timer_expired

Type
64-bit Counter

Description
The number of times the Hysterisis Timer is expired

Labels

nf_status, nfmgmt_state, up_count, fsm_state
- nfmonitor_personality_update_triggers

Type
64-bit Counter

Description
The number of times the Nf Personality Update is received

Labels

nf_status, nfmgmt_state, up_count, fsm_state
- nfmonitor_maintenance_trigger_received

Type
64-bit Counter

Description
The number of times the Maintenance Mode is activated

Labels

nf_status, nfmgmt_state, up_count, fsm_state
- nfmonitor_profile_update_triggers

Type
64-bit Counter

Description
The number of times the Nf Profile Update happened

Labels

nf_status, nfmgmt_state, up_count, fsm_state, update_operation
- nfmonitor_critical_service_down_on_register_entry

Type
64-bit Counter

Description
The number of times the Critical Services were down while starting with RegisterFsmState

Labels

nf_status, up_count

ApiServer statistics

- nfmonitor_deployment_create_watch_request

Type
64-bit Counter

Description

The number of Watch Reqs created for deployment type

Labels

resource

- nfmonitor_deployment_delete_watch_request

Type

64-bit Counter

Description

The number of Watch Reqs deleted for deployment type

Labels

resource

- nfmonitor_deployment_watch_notification_success

Type

64-bit Counter

Description

The number of times Watch notifications are successful

Labels

resource

- nfmonitor_deployment_watch_notification_error

Type

64-bit Counter

Description

The number of times Watch notifications are failed

Labels

resource, reason

- nfmonitor_limit_range_query_attempts

Type

64-bit Counter

Description

The number of Limit Range query attempts

Labels

-
- nfmonitor_limit_range_query_failures

Type

64-bit Counter

Description

The number of Limit Range query failures

Labels

status_code

- nfmonitor_limit_range_query_success

Type

64-bit Counter

Description

The number of Limit Range query succeeded

Labels

status_code

• nfmonitor_hpa_query_attempts

Type

64-bit Counter

Description

The number of Hpa query attempts

Labels

resource, continue_param

• nfmonitor_hpa_query_failures

Type

64-bit Counter

Description

The number of Hpa query failures

Labels

resource, continue_param, status_code

• nfmonitor_hpa_query_success

Type

64-bit Counter

Description

The number of Hpa query succeeded

Labels

resource, continue_param, status_code

• nfmonitor_readiness_probe_query_attempts

Type

64-bit Counter

Description

The number of Readiness Probe query attempts

Labels

resource, continue_param

• nfmonitor_readiness_probe_query_failures

Type

64-bit Counter

Description

The number of Readiness Probe query failures

Labels
<i>resource, continue_param, status_code</i>
nfmonitor_readiness_probe_query_success
Type
64-bit Counter
Description
The number of Readiness Probe query succeeded
Labels
<i>resource, continue_param, status_code</i>

Feature Functionality statistics

nfmonitor_critical_service_frequent_restart
Type
64-bit Counter
Description
Howmany times the minimum required stable CriticalService count is not met
Labels
<i>service</i>
nfmonitor_alarm_due_to_frequent_restart
Type
64-bit Counter
Description
Howmany times Alarm is raised due to minimum required stable CriticalService count is not met
Labels
<i>service, alarm_status</i>
nfmonitor_force_restart_due_to_frequent_restart
Type
64-bit Counter
Description
Howmany times Force Restart initiated due to missing stable CriticalService count
Labels
<i>service</i>
nfmonitor_critical_service_readiness_failure
Type
64-bit Counter
Description
Howmany times the minimum required readiness succeeded CriticalService count is not met
Labels
<i>service</i>
nfmonitor_alarm_due_to_readiness_failure

Type

64-bit Counter

Description

Alarm raised due to minimum count is not met for the readiness succeeded CriticalService

Labels

service, alarm_status

- nfmonitor_force_restart_due_to_readiness_failure

Type

64-bit Counter

Description

Howmany times Force Restart initiated due to readiness failure of CriticalService

Labels

service

Consul statistics

- nfmonitor_publish_status_into_consul

Type

64-bit Counter

Description

Howmany times Publish is sent to Consul

Labels

nf_status, nf_state, up_count, fsm_state

- nfmonitor_consul_read_attempts

Type

64-bit Counter

Description

Howmany times Consul read attempt is made

Labels

fsm_state, retry

- nfmonitor_consul_read_failures

Type

64-bit Counter

Description

Howmany times Consul read attempt are failed

Labels

fsm_state, retry, status_code

- nfmonitor_consul_read_success

Type

64-bit Counter

Description

Howmany times Consul read attempt are succeeded

Labels

fsm_state, retry, status_code

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- [Packet Streaming Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Packet Streaming Metrics	num_packet_streaming_attempted	Counter	The total number of attempted streaming of packets
Packet Streaming Metrics	num_packet_streaming_success	Counter	The total number of packets streamed successfully
Packet Streaming Metrics	num_packet_streaming_failed	Counter	The total number of failed streaming of packets
Packet Streaming Metrics	num_packet_streaming_bytes	Counter	The total number of bytes successfully streamed

Packet Streaming Metrics

- num_packet_streaming_attempted

Type

64-bit Counter

Description

The total number of attempted streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_success

Type

64-bit Counter

Description

The total number of packets streamed successfully

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_failed

Type

64-bit Counter

Description

The total number of failed streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- num_packet_streaming_bytes

Type

64-bit Counter

Description

The total number of bytes successfully streamed

Labels

interface_name, server_fqdn, kafka_topic

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- [Circuit Breaker Metrics](#)
- [Service Endpoint Learner Metrics](#)
- [Service Endpoint Manager Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Circuit Breaker Metrics	sem_cb_endpoints	Gauge	Total number of endpoints for a service in respective states
Service Endpoint Learner Metrics	sem_num_dns_errors	Counter	Total number of DNS resolution errors for a given service
Service Endpoint Learner Metrics	sem_num_endpoints	Gauge	Total number of endpoints learnt for a service
Service Endpoint Manager Metrics	sem_num_endpoint_transactions_sent	Counter	Total number of times an endpoint is given to the application for communication with it
Service Endpoint Manager Metrics	sem_num_endpoint_transaction_status_received	Counter	Total number of transactions for a given endpoint received with the transaction status

Circuit Breaker Metrics

- [sem_cb_endpoints](#)

Type

64-bit Gauge

Description

Total number of endpoints for a service in respective states. There are 3 states viz. Active, Blocked, Restoring. In Active state, the endpoint is responding as expected, i.e. the response code doesn't indicate a service failure. The endpoint will be moved to Blocked state based on response failures which indicate service failure. After a timeout, the endpoint will be moved to Restoring state to check if the endpoint has recovered. In this state, if endpoint responds with response code other than service failure, it will be moved to Active state. Else, it will be moved back to Blocked state.

Labels

service_name, endpoint_state

Service Endpoint Learner Metrics

- [sem_num_dns_errors](#)

Type

64-bit Counter

Description

Total number of DNS resolution errors for a given service, labels are service name for DNS resolution and DNS error codes

Labels

service_name, error_code

- [sem_num_endpoints](#)

Type

64-bit Gauge

Description

Total number of endpoints learnt from the Kubernetes DNS server for a given service

Labels

service

Service Endpoint Manager Metrics

- sem_num_endpoint_transactions_sent

Type

64-bit Counter

Description

Total number of times an endpoint is given to the application for communication with it

Labels

service

- sem_num_endpoint_transaction_status_received

Type

64-bit Counter

Description

Total number of transactions for a given endpoint received with the transaction status

Labels

service, status

Table of Contents

- [Summary](#)
- [ETCD Client Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_put_requests	Counter	The total number of ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_requests_sent	Counter	The total number of ETCD Client PUT Requests sent.
ETCD Client Metrics	etcd_put_successes	Counter	The total number of successful ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_failures	Counter	The total number of failed ETCD Client PUT Requests.
ETCD Client Metrics	etcd_put_with_lease_requests	Counter	The total number of ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_requests_sent	Counter	The total number of ETCD Client PUT with Lease Requests sent.
ETCD Client Metrics	etcd_put_with_lease_successes	Counter	The total number of successful ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_put_with_lease_failures	Counter	The total number of failed ETCD Client PUT with Lease Requests.
ETCD Client Metrics	etcd_range_requests	Counter	The total number of ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_requests_sent	Counter	The total number of ETCD Client Range Requests sent.
ETCD Client Metrics	etcd_range_successes	Counter	The total number of successful ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_failures	Counter	The total number of failed ETCD Client Range Requests.
ETCD Client Metrics	etcd_range_delete_requests	Counter	The total number of ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_requests_sent	Counter	The total number of ETCD Client Range Delete Requests sent.
ETCD Client Metrics	etcd_range_delete_successes	Counter	The total number of successful ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_range_delete_failures	Counter	The total number of failed ETCD Client Range Delete Requests.
ETCD Client Metrics	etcd_txn_requests	Counter	The total number of ETCD Client Txn Requests.

Group	Metric Name	Type	Synopsis
ETCD Client Metrics	etcd_txn_requests_sent	Counter	The total number of ETCD Client Txn Requests sent.
ETCD Client Metrics	etcd_txn_successes	Counter	The total number of successful ETCD Client Txn Requests.
ETCD Client Metrics	etcd_txn_failures	Counter	The total number of failed ETCD Client Txn Requests.
ETCD Client Metrics	etcd_watch_requests	Counter	The total number of ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_requests_sent	Counter	The total number of ETCD Client Watch Requests sent.
ETCD Client Metrics	etcd_watch_successes	Counter	The total number of successful ETCD Client Watch Requests.
ETCD Client Metrics	etcd_watch_failures	Counter	The total number of failed ETCD Client Watch Requests.
ETCD Client Metrics	etcdwatcher_requests	Counter	The total number of ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_requests_sent	Counter	The total number of ETCD Watcher Requests sent.
ETCD Client Metrics	etcdwatcher_successes	Counter	The total number of successful ETCD Watcher Requests.
ETCD Client Metrics	etcdwatcher_failures	Counter	The total number of failed ETCD Watcher Requests.

ETCD Client Metrics

- [etcd_put_requests](#)

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests.

Labels

keyPrefix

- [etcd_put_requests_sent](#)

Type

64-bit Counter

Description

The total number of ETCD Client PUT Requests sent.

Labels

keyPrefix

- [etcd_put_successes](#)

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT Requests.

Labels

keyPrefix

- [etcd_put_failures](#)

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT Requests.

Labels

keyPrefix, failureReason

- etcd_put_with_lease_requests

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client PUT with Lease Requests sent.

Labels

keyPrefix

- etcd_put_with_lease_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client PUT with Lease Requests.

Labels

keyPrefix

- etcd_put_with_lease_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client PUT with Lease Requests.

Labels

keyPrefix, failureReason

- etcd_range_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Requests sent.

Labels

keyPrefix

- etcd_range_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Requests.

Labels

keyPrefix

- etcd_range_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Range Requests.

Labels

keyPrefix, failureReason

- etcd_range_delete_requests

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests.

Labels

keyPrefix

- etcd_range_delete_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Range Delete Requests sent.

Labels

keyPrefix

- etcd_range_delete_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Range Delete Requests.

Labels

keyPrefix

- etcd_range_delete_failures
 - Type
 - 64-bit Counter
 - Description
 - The total number of failed ETCD Client Range Delete Requests.
 - Labels
 - keyPrefix, failureReason**

- etcd_txn_requests
 - Type
 - 64-bit Counter
 - Description
 - The total number of ETCD Client Txn Requests.
 - Labels
 - keyPrefix**

- etcd_txn_requests_sent
 - Type
 - 64-bit Counter
 - Description
 - The total number of ETCD Client Txn Requests sent.
 - Labels
 - keyPrefix**

- etcd_txn_successes
 - Type
 - 64-bit Counter
 - Description
 - The total number of successful ETCD Client Txn Requests.
 - Labels
 - keyPrefix**

- etcd_txn_failures
 - Type
 - 64-bit Counter
 - Description
 - The total number of failed ETCD Client Txn Requests.
 - Labels
 - keyPrefix, failureReason**

- etcd_watch_requests
 - Type
 - 64-bit Counter
 - Description

The total number of ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Client Watch Requests sent.

Labels

keyPrefix

- etcd_watch_successes

Type

64-bit Counter

Description

The total number of successful ETCD Client Watch Requests.

Labels

keyPrefix

- etcd_watch_failures

Type

64-bit Counter

Description

The total number of failed ETCD Client Watch Requests.

Labels

keyPrefix, failureReason

- etcdwatcher_requests

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests.

Labels

keyPrefix

- etcdwatcher_requests_sent

Type

64-bit Counter

Description

The total number of ETCD Watcher Requests sent.

Labels

keyPrefix

- etcdwatcher_successes

Type

64-bit Counter

Description

The total number of successful ETCD Watcher Requests.

Labels

keyPrefix

• etcdwatcher_failures

Type

64-bit Counter

Description

The total number of failed ETCD Watcher Requests.

Labels

keyPrefix, failureReason

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- [Summary](#)
- [HTTP Punting Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
HTTP Punting Metrics	punting_http_requests_sent	Counter	The total number of HTTP requests successfully punted.
HTTP Punting Metrics	punting_http_requests_failed	Counter	The total number of HTTP requests punting failed.
HTTP Punting Metrics	punting_http_responses_received	Counter	The total number of HTTP responses received for the requests punted.
HTTP Punting Metrics	punting_http_responses_forwarded	Counter	The total number of HTTP responses forwarded after punting transaction completion.
HTTP Punting Metrics	punting_http_requests_aborted	Counter	The total number of HTTP requests aborted.
HTTP Punting Metrics	punting_http_request_duration_seconds	Histogram	Punting operation duration in seconds.

HTTP Punting Metrics

• [punting_http_requests_sent](#)

Type

64-bit Counter

Description

The total number of HTTP requests successfully punted.

Labels

message_type

• [punting_http_requests_failed](#)

Type

64-bit Counter

Description

The total number of HTTP requests for which the punting failed.

Labels

message_type

- punting_http_responses_received

Type

64-bit Counter

Description

The total number of HTTP responses received for the requests punted.

Labels

message_type, status_code

- punting_http_responses_forwarded

Type

64-bit Counter

Description

The total number of HTTP responses forwarded after punting transaction completion.

Labels

message_type, status_code

- punting_http_requests_aborted

Type

64-bit Counter

Description

The total number of HTTP requests aborted.

Labels

message_type

- punting_http_request_duration_seconds

Type

64-bit Histogram

Description

The punting transaction duration is measured from the time the HTTP request is sent out to the time the response is received for the punting request.

Labels

message_type

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

Table of Contents

- [Summary](#)
- [Packet Streaming Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Packet Streaming Metrics	num_packet_streaming_attempted	Counter	The total number of attempted streaming of packets
Packet Streaming Metrics	num_packet_streaming_success	Counter	The total number of packets streamed successfully
Packet Streaming Metrics	num_packet_streaming_failed	Counter	The total number of failed streaming of packets
Packet Streaming Metrics	num_packet_streaming_bytes	Counter	The total number of bytes successfully streamed

Packet Streaming Metrics

- [num_packet_streaming_attempted](#)

Type

64-bit Counter

Description

The total number of attempted streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- [num_packet_streaming_success](#)

Type

64-bit Counter

Description

The total number of packets streamed successfully

Labels

interface_name, server_fqdn, kafka_topic

- [num_packet_streaming_failed](#)

Type

64-bit Counter

Description

The total number of failed streaming of packets

Labels

interface_name, server_fqdn, kafka_topic

- [num_packet_streaming_bytes](#)

Type

64-bit Counter

Description

The total number of bytes successfully streamed

Labels

interface_name, server_fqdn, kafka_topic

Table of Contents

- [Summary](#)
- [Circuit Breaker Metrics](#)
- [Service Endpoint Learner Metrics](#)
- [Service Endpoint Manager Metrics](#)

Summary

Group	Metric Name	Type	Synopsis
Circuit Breaker Metrics	sem_cb_endpoints	Gauge	Total number of endpoints for a service in respective states
Service Endpoint Learner Metrics	sem_num_dns_errors	Counter	Total number of DNS resolution errors for a given service
Service Endpoint Learner Metrics	sem_num_endpoints	Gauge	Total number of endpoints learnt for a service
Service Endpoint Manager Metrics	sem_num_endpoint_transactions_sent	Counter	Total number of times an endpoint is given to the application for communication with it
Service Endpoint Manager Metrics	sem_num_endpoint_transaction_status_received	Counter	Total number of transactions for a given endpoint received with the transaction status

Circuit Breaker Metrics

- [sem_cb_endpoints](#)

Type

64-bit Gauge

Description

Total number of endpoints for a service in respective states. There are 3 states viz. Active, Blocked, Restoring. In Active state, the endpoint is responding as expected, i.e. the response code doesn't indicate a service failure. The endpoint will be moved to Blocked state based on response failures which indicate service failure. After a timeout, the endpoint will be moved to Restoring state to check if the endpoint has recovered. In this state, if endpoint responds with response code other than service failure, it will be moved to Active state. Else, it will be moved back to Blocked state.

Labels

service_name, endpoint_state

Service Endpoint Learner Metrics

- [sem_num_dns_errors](#)

Type

64-bit Counter

Description

Total number of DNS resolution errors for a given service, labels are service name for DNS resolution and DNS error codes

Labels

service_name, error_code

- [sem_num_endpoints](#)

Type

64-bit Gauge

Description

Total number of endpoints learnt from the Kubernetes DNS server for a given service

Labels

service

Service Endpoint Manager Metrics

- [sem_num_endpoint_transactions_sent](#)

Type

64-bit Counter

Description

Total number of times an endpoint is given to the application for communication with it

Labels

service

- sem_num_endpoint_transaction_status_received

Type

64-bit Counter

Description

Total number of transactions for a given endpoint received with the transaction status

Labels

service, status

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- [Summary](#)
- [UDSF DBClient Metrics](#)
- [UDSF API latency statistics](#)

Summary

Group	Metric Name	Type	Synopsis
UDSF DBClient Metrics	udsf_db_operation_request	Counter	The total number of udsf dbclient requests received.
UDSF DBClient Metrics	udsf_db_operation_response	Counter	The total number of udsf dbclient responses sent.
UDSF DBClient Metrics	udsf_dbc_total_init_success	Counter	The total number of UDSF INIT successes.
UDSF DBClient Metrics	udsf_dbc_total_init_failure	Counter	The total number of UDSF INIT failures.
UDSF DBClient Metrics	udsf_dbc_total_deinit_success	Counter	The total number of UDSF DEINIT successes.
UDSF DBClient Metrics	udsf_dbc_total_deinit_failure	Counter	The total number of UDSF DEINIT failures.
UDSF DBClient Metrics	udsf_db_transaction_counter	Counter	The total number of CRUD operations.
UDSF API latency statistics	udsf_db_transaction_duration	Histogram	Roundtrip duration taken by transaction in seconds.
UDSF API latency statistics	udsf_db_execution_duration	Histogram	Execution time taken by transaction in microseconds.
UDSF API latency statistics	udsf_hiredis_bufferwrite_duration	Histogram	Duration of time taken in hiredis to write data to socket in seconds

UDSF DBClient Metrics

- udsf_db_operation_request

Type

64-bit Counter

Description

The total number of udsf dbclient requests received from worker. Labels- 'app_name' : Originating thread name, ex:smfcc, 'operation_type' : Type of CRUD operation/API type involved, ex:CREATE_OR_MODIFY_RECORD_ASYNC

Labels

app_name, operation_type

- udsf_db_operation_response

Type

64-bit Counter

Description

The total number of udsf dbclient responses sent. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE 'operation_type' : Type of CRUD operation/API type involved,, ex:CREATE_OR_MODIFY_RECORD_ASYNC

Labels

app_name, status, operation_type

- udsf_dbc_total_init_success

Type

64-bit Counter

Description

The total number of times UDSF API init was success. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: SUCCESS

Labels

app_name, status

- udsf_dbc_total_init_failure

Type

64-bit Counter

Description

The total number of times UDSF API init failed. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: FAILURE

Labels

app_name, status

- udsf_dbc_total_deinit_success

Type

64-bit Counter

Description

The total number of times UDSF API Deinit succeeded. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: SUCCESS

Labels

app_name, status

- udsf_dbc_total_deinit_failure

Type

64-bit Counter

Description

The total number of times UDSF API Deinit failed. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: FAILURE

Labels

app_name, status

- udsf_db_transaction_counter

Type

64-bit Counter

Description

The total number of times CRUD operations are incrementing. Labels- 'app_name' : Originating thread name, ex:smfcc, 'crud_op' : The categorical CRUD operation in which this sub-transaction falls into, ex: CREATE_OR_MODIFY

Labels

app_name, crud_op

UDSF API latency statistics

• udsf_db_transaction_duration

Type

64-bit Histogram

Description

Duration observed on total journey/roundtrip taken by a transaction after receiving the request. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE 'operation_type' : Type of CRUD operation/API type involved., ex:CREATE_OR_MODIFY_RECORD_ASYNC, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE, 'crud_op' : The categorical CRUD operation in which this transaction falls into, ex: CREATE_OR_MODIFY

Labels

app_name,operation_type,status,crud_op

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

• udsf_db_execution_duration

Type

64-bit Histogram

Description

Processing time taken by UDSF client. Labels- 'app_name' : Originating thread name, ex:smfcc, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE 'operation_type' : Type of CRUD operation/API type involved., ex:CREATE_OR_MODIFY_RECORD_ASYNC, 'status' : SUCCESS or error string in case of FAILURE, ex: PARSING_PARAMETER_FAILURE, 'crud_op' : The categorical CRUD operation in which this transaction falls into, ex: CREATE_OR_MODIFY

Labels

app_name,operation_type,status,crud_op

Boundaries

[1, 2, 4, 8, 16, 32, 64, 128, 256, 384, 512, 768, 1024, 1536, 2048, 4096, 8192, 16384]

• udsf_hiredis_bufferwrite_duration

Type

64-bit Histogram

Description

Duration observed between hiredis buffer append and write to socket. This is fair approximation for time spent before data is written to socket as the buffer threshold accounted for append and write is dynamic and not fixed size. Per socket, the buffer append start-time is recorded along with size intended to transmit and end-time recorded once data greater than recorded size is written on the socket

Labels

app_name

Boundaries

[0.001, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192, 16.384]

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- [Summary](#)
- [URM UDSF Interface Message Statistics](#)

Summary

Group	Metric Name	Type	Synopsis
URM UDSF Interface Message Statistics	urm_udsf_get_record_request_attempts	Counter	Total number of udsf get record request attempts
URM UDSF Interface Message Statistics	urm_udsf_get_record_request_successes	Counter	Total number of udsf get record request successes
URM UDSF Interface Message Statistics	urm_udsf_get_record_request_failures	Counter	Total number of udsf get record request failures
URM UDSF Interface Message Statistics	urm_udsf_read_and_lock_record_request_attempts	Counter	Total number of udsf read and lock record request attempts
URM UDSF Interface Message Statistics	urm_udsf_read_and_lock_record_request_successes	Counter	Total number of udsf read and lock record request successes
URM UDSF Interface Message Statistics	urm_udsf_read_and_lock_record_request_failures	Counter	Total number of udsf read and lock record request failures
URM UDSF Interface Message Statistics	urm_udsf_read_and_lock_record_using_secondary_key_request_attempts	Counter	Total number of udsf read and lock record request using secondary key attempts
URM UDSF Interface Message Statistics	urm_udsf_read_and_lock_record_using_secondary_key_request_successes	Counter	Total number of udsf read and lock record request using secondary key successes
URM UDSF Interface Message Statistics	urm_udsf_read_and_lock_record_using_secondary_key_request_failures	Counter	Total number of udsf read and lock record request using secondary key failures
URM UDSF Interface Message Statistics	urm_udsf_create_or_modify_record_request_attempts	Counter	Total number of udsf create or modify record request attempts
URM UDSF Interface Message Statistics	urm_udsf_create_or_modify_record_request_successes	Counter	Total number of udsf create or modify record request successes
URM UDSF Interface Message Statistics	urm_udsf_create_or_modify_record_request_failures	Counter	Total number of create or modify record request failures
URM UDSF Interface Message Statistics	urm_udsf_modify_record_and_unlock_request_attempts	Counter	Total number of udsf modify record and unlock request attempts
URM UDSF Interface Message Statistics	urm_udsf_modify_record_and_unlock_request_successes	Counter	Total number of udsf modify record and unlock request successes
URM UDSF Interface Message Statistics	urm_udsf_modify_record_and_unlock_request_failures	Counter	Total number of udsf modify record and unlock request failures
URM UDSF Interface Message Statistics	urm_udsf_delete_record_request_attempts	Counter	Total number of udsf delete record request attempts
URM UDSF Interface Message Statistics	urm_udsf_delete_record_request_successes	Counter	Total number of udsf delete record request successes
URM UDSF Interface Message Statistics	urm_udsf_delete_record_request_failures	Counter	Total number of udsf delete record requests failures
URM UDSF Interface Message Statistics	urm_udsf_search_record_request_attempts	Counter	Total number of udsf search record request attempts
URM UDSF Interface Message Statistics	urm_udsf_search_record_request_successes	Counter	Total number of udsf search record request successes
URM UDSF Interface Message Statistics	urm_udsf_search_record_request_failures	Counter	Total number of udsf search record request failures
URM UDSF Interface Message Statistics	urm_udsf_connection_available_indication_received	Counter	Total number of UDSF API Connection Available Indication received
URM UDSF Interface Message Statistics	urm_udsf_connection_unavailable_indication_received	Counter	Total number of UDSF API Connection Unavailable Indication received
URM UDSF Interface Message Statistics	urm_db_transaction_failure_udsf_connection_unavailable	Counter	Total number of DB transactions failed due to UDSF Connection Unavailable
URM UDSF Interface Message Statistics	urm_db_audit_session_record_req	Counter	Audit session record request sent to CRUD DB
URM UDSF Interface Message Statistics	urm_db_audit_session_record_rsp	Counter	Audit session record response received from CRUD DB

URM UDSF Interface Message Statistics

- `urm_udsf_get_record_request_attempts`

Type
64-bit Counter
Description
Number of udsf get record request attempts
Labels

- `urm_udsf_get_record_request_successes`

Type
64-bit Counter
Description
Number of udsf get record request successes
Labels

- `urm_udsf_get_record_request_failures`

Type
64-bit Counter
Description
Number of udsf get record request failures
Labels

cause

- `urm_udsf_read_and_lock_record_request_attempts`

Type
64-bit Counter
Description
Number of udsf read and lock record request attempts
Labels

- `urm_udsf_read_and_lock_record_request_successes`

Type
64-bit Counter
Description
Number of udsf read and lock record request successess
Labels

- `urm_udsf_read_and_lock_record_request_failures`

Type
64-bit Counter
Description
Number of udsf read and lock record request failures

Labels

cause

- urm_udsf_read_and_lock_record_using_secondary_key_request_attempts

Type

64-bit Counter

Description

Number of udsf read and lock record request using secondary key attempts

Labels

- urm_udsf_read_and_lock_record_using_secondary_key_request_successes

Type

64-bit Counter

Description

Number of udsf read and lock record request using secondary key successes

Labels

- urm_udsf_read_and_lock_record_using_secondary_key_request_failures

Type

64-bit Counter

Description

Number of udsf read and lock record request using secondary key failures

Labels

cause

- urm_udsf_create_or_modify_record_request_attempts

Type

64-bit Counter

Description

Number of create or modify record request attempts

Labels

- urm_udsf_create_or_modify_record_request_successes

Type

64-bit Counter

Description

Number of create or modify record request successes

Labels

- urm_udsf_create_or_modify_record_request_failures

Type

64-bit Counter

Description

Number of udsf create or modify record request failures

Labels

cause

- urm_udsf_modify_record_and_unlock_request_attempts

Type

64-bit Counter

Description

Number of udsf modify record and unlock request attempts

Labels

- urm_udsf_modify_record_and_unlock_request_successes

Type

64-bit Counter

Description

Number of udsf modify record and unlock request successes

Labels

- urm_udsf_modify_record_and_unlock_request_failures

Type

64-bit Counter

Description

Number of udsf modify record and unlock request failures

Labels

cause

- urm_udsf_delete_record_request_attempts

Type

64-bit Counter

Description

Number of udsf delete record request attempts

Labels

- urm_udsf_delete_record_request_successes

Type

64-bit Counter

Description

Number of udsf delete record request successes

Labels

- urm_udsf_delete_record_request_failures

Type

64-bit Counter

Description

Number of udsf delete record request failures

Labels

cause

- `urm_udsf_search_record_request_attempts`

Type

64-bit Counter

Description

Number of udsf search record request attempts

Labels

- `urm_udsf_search_record_request_successes`

Type

64-bit Counter

Description

Number of udsf search record request successes

Labels

- `urm_udsf_search_record_request_failures`

Type

64-bit Counter

Description

Number of udsf search record request failures

Labels

cause

- `urm_udsf_connection_available_indication_received`

Type

64-bit Counter

Description

Number of UDSF API Connection Available Indication received

Labels

- `urm_udsf_connection_unavailable_indication_received`

Type

64-bit Counter

Description

Number of UDSF API Connection Unavailable Indication received

Labels

- `urm_db_transaction_failure_udsf_connection_unavailable`

Type

64-bit Counter

Description

Number of DB transactions failed due to UDSF Connection Unavailable

Labels

- transaction**
- urm_db_audit_session_record_rsp

Type

64-bit Counter

Description

Audit session record response received

Labels

- statusCode**
- urm_db_audit_session_record_req

Type

64-bit Counter

Description

None

Metrics

Instances	Labels	Description
1	method: GET	request

Table of Contents

- [Summary](#)
- [URM Dtm Msg Statistics](#)

Summary

Group	Metric Name	Type	Synopsis
URM Dtm Msg Statistics	urm_dtm_start_timer_req	Counter	Start timer request sent to DTM
URM Dtm Msg Statistics	urm_dtm_start_timer_rsp	Counter	Start timer response received from DTM
URM Dtm Msg Statistics	urm_dtm_stop_timer_req	Counter	Stop timer request sent to DTM
URM Dtm Msg Statistics	urm_dtm_stop_timer_rsp	Counter	Stop timer response received from DTM
URM Dtm Msg Statistics	urm_dtm_update_timer_req	Counter	Update timer request sent to DTM
URM Dtm Msg Statistics	urm_dtm_update_timer_rsp	Counter	Update timer response received from DTM
URM Dtm Msg Statistics	urm_dtm_delete_pool_timer_expiry_notification	Counter	Delete Pool timer expiry notification from DTM
URM Dtm Msg Statistics	urm_dtm_delete_pool_timer_expiry_notification_ack	Counter	Delete Pool timer notification ack sent to DTM
URM Dtm Msg Statistics	urm_dtm_unsolicited_ra_timer_expiry_notification	Counter	UnSolicited RA timer expiry notification from DTM
URM Dtm Msg Statistics	urm_dtm_unsolicited_ra_timer_expiry_notification_ack	Counter	UnSolicited RA timer notification ack sent to DTM

URM Dtm Msg Statistics

- urm_dtm_start_timer_rsp

Type

64-bit Counter

Description		
Start timer response received		
Labels		
statusCode		
• urm_dtm_stop_timer_rsp		
Type		
64-bit Counter		
Description		
Stop timer response received		
Labels		
statusCode		
• urm_dtm_update_timer_rsp		
Type		
64-bit Counter		
Description		
Update timer response received		
Labels		
statusCode		
• urm_dtm_delete_pool_timer_expiry_notification_ack		
Type		
64-bit Counter		
Description		
Delete Pool timer notification ack		
Labels		
statusCode		
• urm_dtm_unsolicited_ra_timer_expiry_notification_ack		
Type		
64-bit Counter		
Description		
UnSolicited RA timer notification ack		
Labels		
statusCode		
• urm_dtm_start_timer_req		
Type		
64-bit Counter		
Description		
None		
Metrics		

Instances	Labels	Description
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Instances	Labels	Description
1	method: POST	request
urm_dtm_stop_timer_req		
Type		
64-bit Counter		
Description		
None		
Metrics		

Instances	Labels	Description
1	method: DELETE	request
urm_dtm_update_timer_req		
Type		
64-bit Counter		
Description		
None		
Metrics		

Instances	Labels	Description
1	method: PUT	request
urm_dtm_delete_pool_timer_expiry_notification		
Type		
64-bit Counter		
Description		
None		
Metrics		

Instances	Labels	Description
1	method: POST	request
urm_dtm_unsolicited_ra_timer_expiry_notification		
Type		
64-bit Counter		
Description		
None		
Metrics		

Instances	Labels	Description
1	method: POST	request

Table of Contents

- [Summary](#)
- [URM Etcd Msg Statistics](#)

Summary

Group	Metric Name	Type	Synopsis
URM Etcd Msg Statistics	urm_etcd_ip_pool_table_watch_req	Counter	ip_pool_watch request sent to ETCD DB
URM Etcd Msg Statistics	urm_etcd_ip_pool_table_watch_rsp	Counter	IpPool Table Watch Response received from ETCD DB
URM Etcd Msg Statistics	urm_etcd_up_rsc_info_table_watch_req	Counter	UP Resource Info Table watch request sent to ETCD DB
URM Etcd Msg Statistics	urm_etcd_up_rsc_info_table_watch_rsp	Counter	UP Resource Info Table Watch Response received from ETCD DB
URM Etcd Msg Statistics	urm_etcd_pfcg_association_table_watch_req	Counter	Pfcg Association Table watch request sent to ETCD DB
URM Etcd Msg Statistics	urm_etcd_pfcg_association_table_watch_rsp	Counter	Pfcg Association Table Watch Response received from ETCD DB
URM Etcd Msg Statistics	urm_etcd_ip_pool_table_range_req	Counter	IP Pool Range request sent to ETCD DB
URM Etcd Msg Statistics	urm_etcd_ip_pool_table_range_rsp	Counter	IpPool Table Range Response received from ETCD DB
URM Etcd Msg Statistics	urm_etcd_up_rsc_info_table_range_req	Counter	UP Resource Info Table Range request sent to ETCD DB
URM Etcd Msg Statistics	urm_etcd_up_rsc_info_table_range_rsp	Counter	UP Resource Info Table Range Response received from ETCD DB
URM Etcd Msg Statistics	urm_etcd_ip_pool_table_watch_event	Counter	IP Pool Table Watch Event received from ETCD DB
URM Etcd Msg Statistics	urm_etcd_up_rsc_info_table_watch_event	Counter	UP Resource Info Table Watch Event received from ETCD DB
URM Etcd Msg Statistics	urm_etcd_pfcg_association_table_watch_event	Counter	Pfcg Association Table Watch Event received from ETCD DB

URM Etcd Msg Statistics

- [urm_etcd_ip_pool_table_watch_rsp](#)

Type

64-bit Counter

Description

IP Pool Table Watch Response

Labels

statusCode

- [urm_etcd_up_rsc_info_table_watch_rsp](#)

Type

64-bit Counter

Description

UP Resource Info Table Watch Response

Labels

statusCode

- [urm_etcd_pfcg_association_table_watch_rsp](#)

Type

64-bit Counter

Description

Pfcg Association Table Watch Response

Labels

statusCode

- [urm_etcd_ip_pool_table_range_rsp](#)

Type

64-bit Counter

Description

IP Pool Table Range Response

Labels

statusCode

- urm_etcd_up_rsc_info_table_range_rsp

Type

64-bit Counter

Description

UP Resource Info Table Range Response

Labels

statusCode

- urm_etcd_ip_pool_table_watch_event

Type

64-bit Counter

Description

IP Pool Table Table Watch Event

Labels

event, upfld

- urm_etcd_up_rsc_info_table_watch_event

Type

64-bit Counter

Description

UP Resource Info Table Watch Event

Labels

event, upfld

- urm_etcd_pfc_p_association_table_watch_event

Type

64-bit Counter

Description

Pfc_p Association Table Watch Event

Labels

event, upfld

- urm_etcd_ip_pool_table_watch_req

Type

64-bit Counter

Description

None

Metrics

Instances	Labels	Description
1	poolType: IP-POOL	request

• urm_etcd_up_rsc_info_table_watch_req

Type

64-bit Counter

Description

None

Metrics

Instances	Labels	Description
1	poolType: UP-RSC-INFO	request

• urm_etcd_pfc_p_association_table_watch_req

Type

64-bit Counter

Description

None

Metrics

Instances	Labels	Description
1	*****	request

• urm_etcd_ip_pool_table_range_req

Type

64-bit Counter

Description

None

Metrics

Instances	Labels	Description
1	poolType: IP-POOL	request

• urm_etcd_up_rsc_info_table_range_req

Type

64-bit Counter

Description

None

Metrics

Instances	Labels	Description
1	poolType: UP-RSC-INFO	request

Table of Contents

- [Summary](#)
- [URM Flexblock msg statistics](#)

Summary

Group	Metric Name	Type	Synopsis
URM Flexblock msg statistics	urm_fb_resource_allocation_req	Counter	resource request sent to flexblock by urm

Group	Metric Name	Type	Synopsis
URM Flexblock msg statistics	urm_fb_resource_allocation_rsp	Counter	resource respone received by urm
URM Flexblock msg statistics	urm_fb_resource_release_req	Counter	resource release request sent to flexblock by urm
URM Flexblock msg statistics	urm_fb_resource_release_rsp	Counter	resource release response received by urm
URM Flexblock msg statistics	urm_fb_get_resource_pool_info_req	Counter	get resource pool info request sent to flexblock by urm
URM Flexblock msg statistics	urm_fb_get_resource_pool_info_rsp	Counter	get resource pool info response received by urm
URM Flexblock msg statistics	urm_fb_unaffined_resource_pool_create_req	Counter	resource pool create request sent to flexblock by urm
URM Flexblock msg statistics	urm_fb_unaffined_resource_pool_create_rsp	Counter	resource pool create response received by urm
URM Flexblock msg statistics	urm_fb_unaffined_resource_type_create_req	Counter	resource type create request sent to flexblock by urm
URM Flexblock msg statistics	urm_fb_unaffined_resource_type_create_rsp	Counter	resource type response received by urm
URM Flexblock msg statistics	urm_fb_unaffined_resource_type_delete_req	Counter	unaffined resource type delete request sent to flexblock by urm
URM Flexblock msg statistics	urm_fb_unaffined_resource_type_delete_rsp	Counter	unaffined resource type delete response received by urm
URM Flexblock msg statistics	urm_fb_bulk_resource_allocation_req	Counter	bulk resource request sent to flexblock by urm
URM Flexblock msg statistics	urm_fb_bulk_resource_allocation_rsp	Counter	bulk resource respone received by urm
URM Flexblock msg statistics	urm_fb_bulk_resource_release_req	Counter	bulk resource release request sent to flexblock by urm
URM Flexblock msg statistics	urm_fb_bulk_resource_release_rsp	Counter	bulk resource release response received by urm
URM Flexblock msg statistics	urm_fb_audit_notify_count	Counter	number of audit request notification received from flexblock

URM Flexblock msg statistics

- [urm_fb_resource_allocation_req](#)

Type

64-bit Counter

Description

urm flexblock resource allocation request

Labels

poolId
- [urm_fb_resource_allocation_rsp](#)

Type

64-bit Counter

Description

urm flexblock resource allocation rsp

Labels

poolId, statusCode
- [urm_fb_resource_release_req](#)

Type

64-bit Counter

Description

urm flexblock resource release request

Labels

poolId
- [urm_fb_resource_release_rsp](#)

Type

64-bit Counter

Description

urm flexblock resource release rsp

Labels

poolId, statusCode

- urm_fb_get_resource_pool_info_req

Type

64-bit Counter

Description

urm flexblock get resource pool info request

Labels

poolId

- urm_fb_get_resource_pool_info_rsp

Type

64-bit Counter

Description

urm flexblock get resource pool info rsp

Labels

poolId, statusCode

- urm_fb_unaffined_resource_pool_create_req

Type

64-bit Counter

Description

urm flexblock unaffined resource pool create request

Labels

poolId

- urm_fb_unaffined_resource_pool_create_rsp

Type

64-bit Counter

Description

urm flexblock unaffined resource pool create response

Labels

poolId, statusCode

- urm_fb_unaffined_resource_type_create_req

Type

64-bit Counter

Description

urm flexblock unaffined resource type create request

Labels

poolId

- urm_fb_unaffined_resource_type_create_rsp

Type

64-bit Counter

Description

urm flexblock unaffined resource type create response

Labels

poolId, statusCode

- urm_fb_unaffined_resource_type_delete_req

Type

64-bit Counter

Description

urm flexblock unaffined resource type delete request

Labels

poolId

- urm_fb_unaffined_resource_type_delete_rsp

Type

64-bit Counter

Description

urm flexblock unaffined resource type delete response

Labels

poolId, statusCode

- urm_fb_bulk_resource_allocation_req

Type

64-bit Counter

Description

urm flexblock bulk/cache resource allocation request

Labels

poolId

- urm_fb_bulk_resource_allocation_rsp

Type

64-bit Counter

Description

urm flexblock bulk/cache resource allocation rsp

Labels

poolId, statusCode

- urm_fb_bulk_resource_release_req

Type

64-bit Counter

Description

urm flexblock bulk/cache resource release request

Labels

poolId

• urm_fb_bulk_resource_release_rsp

Type

64-bit Counter

Description

urm flexblock bulk/cache resource release rsp

Labels

poolId, statusCode

• urm_fb_audit_notify_count

Type

64-bit Counter

Description

urm audit request notification from flexblock

Labels

poolId

Table of Contents

- [Summary](#)
- [URM Rest API Statistics](#)

Summary

Group	Metric Name	Type	Synopsis
URM Rest API Statistics	urm_resources_req	Counter	resource request received by urm
URM Rest API Statistics	urm_resources_rsp	Counter	resource response sent by urm
URM Rest API Statistics	urm_status_req	Counter	Status request statistics
URM Rest API Statistics	urm_status_rsp	Counter	Status response statistics
URM Rest API Statistics	urm_user_planes_req	Counter	configured user planes statistics
URM Rest API Statistics	urm_user_planes_rsp	Counter	Configured userplane response statistics
URM Rest API Statistics	urm_smfcc_release_session_notify	Counter	session release notification from urm to smfcc
URM Rest API Statistics	urm_smfcc_release_session_notify_ack	Counter	acknowledgement for session release notification
URM Rest API Statistics	urm_inband_data_ind	Counter	inband gtpu data indication statistics
URM Rest API Statistics	urm_inband_data_ind_rsp	Counter	inband gtpu data indication response statistics
URM Rest API Statistics	urm_router_solicit_received	Counter	router solicit received statistics
URM Rest API Statistics	urm_router_advertisement_sent	Counter	router advertisement sent statistics
URM Rest API Statistics	urm_unsolicited_ra_trigger_ind	Counter	unsolicited ra trigger indication statistics
URM Rest API Statistics	urm_unsolicited_ra_trigger_ind_rsp	Counter	unsolicited ra trigger indication response statistics

URM Rest API Statistics

- urm_resources_req

Type

64-bit Counter

Description

urm resource request

Labels

method, slice, dnn

• urm_resources_rsp

Type

64-bit Counter

Description

urm resource response

Labels

method, slice, dnn, upfld, statusCode, cause

• urm_smfcc_release_session_notify

Type

64-bit Counter

Description

urm to smfcc session release notification

Labels

cause

• urm_smfcc_release_session_notify_ack

Type

64-bit Counter

Description

acknowledgement from smfcc for session release notification

Labels

statusCode

• urm_status_req

Type

64-bit Counter

Description

None

Metrics

Instances	Labels	Description
1	method: GET	urm status get request

• urm_status_rsp

Type

64-bit Counter

Description

None

Metrics

Instances	Labels	Description
1	method: GET, statusCode: 200	urm status get response
2	method: GET, statusCode: 500	urm status get response

• urm_user_planes_req

Type

64-bit Counter

Description

None

Metrics

Instances	Labels	Description
1	method: GET	get configured user planes request

• urm_user_planes_rsp

Type

64-bit Counter

Description

None

Metrics

Instances	Labels	Description
1	method: GET, statusCode: 200	configured user planes response
2	method: GET, statusCode: 400	configured user planes response
3	method: GET, statusCode: 404	configured user planes response
4	method: GET, statusCode: 500	configured user planes response

• urm_inband_data_ind

Type

64-bit Counter

Description

None

Metrics

Instances	Labels	Description
1	method: PUT	number of inband gtpu data received

• urm_inband_data_ind_rsp

Type

64-bit Counter

Description

None

Metrics

Instances	Labels	Description
1	statusCode: 200	inband success response sent
2	statusCode: 404	inband failure respone sent

Instances	Labels	Description
3	statusCode: 500	inband failure response sent
urm_router_solicit_received		
Type		
64-bit Counter		
Description		
None		
Metrics		

Instances	Labels	Description
1	method: PUT	number of router solicit ip packets received
urm_router_advertisement_sent		
Type		
64-bit Counter		
Description		
None		
Metrics		

Instances	Labels	Description
1	method: PUT	number of router advertisement sent
urm_unsolicited_ra_trigger_ind		
Type		
64-bit Counter		
Description		
None		
Metrics		

Instances	Labels	Description
1	method: POST	number of unsolicited ra trigger received
urm_unsolicited_ra_trigger_ind_rsp		
Type		
64-bit Counter		
Description		
None		
Metrics		

Instances	Labels	Description
1	statusCode: 204	success response sent
2	statusCode: 500	failure response sent

Table of Contents

- [Summary](#)
- [URM Resource Pool statistics](#)

Summary

Group	Metric Name	Type	Synopsis
URM Resource Pool statistics	urm_ipv4_resource_allocation_success	Counter	ipv4 resource allocation success by urm
URM Resource Pool statistics	urm_ipv4_resource_allocation_failure	Counter	ipv4 resource allocation failure by urm
URM Resource Pool statistics	urm_ipv6_resource_allocation_success	Counter	ipv6 resource allocation success by urm
URM Resource Pool statistics	urm_ipv6_resource_allocation_failure	Counter	ipv6 resource allocation failure by urm
URM Resource Pool statistics	urm_teid_resource_allocation_success	Counter	teid resource allocation success by urm
URM Resource Pool statistics	urm_teid_resource_allocation_failure	Counter	teid resource allocation failure by urm
URM Resource Pool statistics	urm_ipv4_resource_release_success	Counter	ipv4 resource release success by urm
URM Resource Pool statistics	urm_ipv4_resource_release_failure	Counter	ipv4 resource release failure by urm
URM Resource Pool statistics	urm_ipv6_resource_release_success	Counter	ipv6 resource release success by urm
URM Resource Pool statistics	urm_ipv6_resource_release_failure	Counter	ipv6 resource release failure by urm
URM Resource Pool statistics	urm_teid_resource_release_success	Counter	teid resource release success by urm
URM Resource Pool statistics	urm_teid_resource_release_failure	Counter	teid resource release failure by urm

URM Resource Pool statistics

- [urm_ipv4_resource_allocation_success](#)

Type

64-bit Counter

Description

urm ipv4 resource allocation success

Labels

upfId, poolId

- [urm_ipv4_resource_allocation_failure](#)

Type

64-bit Counter

Description

urm ipv4 resource allocation failure

Labels

upfId, poolId

- [urm_ipv6_resource_allocation_success](#)

Type

64-bit Counter

Description

urm ipv6 resource allocation success

Labels

upfId, poolId

- [urm_ipv6_resource_allocation_failure](#)

Type

64-bit Counter

Description

urm ipv6 resource allocation failure

Labels

upfId, poolId

- urm_teid_resource_allocation_success

Type

64-bit Counter

Description

urm teid resource allocation success

Labels

upfId, poolId

- urm_teid_resource_allocation_failure

Type

64-bit Counter

Description

urm teid resource allocation failure

Labels

upfId, poolId

- urm_ipv4_resource_release_success

Type

64-bit Counter

Description

urm ipv4 resource release success

Labels

upfId, poolId

- urm_ipv4_resource_release_failure

Type

64-bit Counter

Description

urm ipv4 resource release failure

Labels

upfId, poolId

- urm_ipv6_resource_release_success

Type

64-bit Counter

Description

urm ipv6 resource release success

Labels

upfId, poolId

- urm_ipv6_resource_release_failure

Type

64-bit Counter

Description

urm ipv6 resource release failure

Labels

upfId, poolId

• urm_teid_resource_release_success

Type

64-bit Counter

Description

urm teid resource release success

Labels

upfId, poolId

• urm_teid_resource_release_failure

Type

64-bit Counter

Description

urm teid resource release failure

Labels

upfId, poolId

Table of Contents

- [Summary](#)
- [hello_api_status_http_requests_req_rate_per_second](#)
- [hello_api_status_http_requests_success_percentage](#)
- [hello_dnn_ratype_registration_request_eutran_percentage](#)
- [hello_dnn_ratype_registration_request_req_rate_per_second](#)

Summary

Metric Name	Type	Synopsis
hello_api_status_http_requests_req_rate_per_second	counter	http request api success rate per second 10 mins ago
hello_api_status_http_requests_success_percentage	counter	http hello api Success Percentage
hello_dnn_ratype_registration_request_eutran_percentage	counter	ratype eutran registration percentage
hello_dnn_ratype_registration_request_req_rate_per_second	counter	dnn api success rate per second 10 mins ago

hello_api_status_http_requests_req_rate_per_second

Description

This counter indicates the success request rate per second 10 mins ago.

Type

counter

Prometheus Expression

rate(hello_api_status_http_requests{statuscode='200'}[5m] offset 10m)

hello_api_status_http_requests_success_percentage

Description

percentage of total requests success in the last 30 mins

Type

counter

Prometheus Expression

sum(rate(hello_api_status_http_requests{statuscode="200"}[30m])) / sum(rate(hello_api_status_http_requests[30m])) * 100

hello_dnn_rattype_registration_request_eutran_percentage

Description

percentage of dnn rattyp registration in the last 30 mins

Type

counter

Prometheus Expression

sum(rate(hello_dnn_rattype_registration_request{ratType="eutran"}[30m])) / sum(rate(hello_dnn_rattype_registration_request[30m])) * 100

hello_dnn_rattype_registration_request_req_rate_per_second

Description

This counter indicates rattype registration request rate per second 10 mins ago.

Type

counter

Prometheus Expression

rate(hello_dnn_rattype_registration_request{ratType="eutran"}[5m] offset 10m)

Table of Contents

- [Summary](#)
- [mmecc_emm_attach_attempt_rate](#)
- [mmecc_emm_attach_success_rate](#)

Summary

Metric Name	Type	Synopsis
mmecc_emm_attach_attempt_rate	counter	LTE Attach attempt rate
mmecc_emm_attach_success_rate	counter	LTE Attach success rate

mmecc_emm_attach_attempt_rate

Description

The rate of LTE attach attempts per second. This counter indicates the average LTE attach attempts per second received by the MME in the last 5 minutes

Type

counter

Prometheus Expression

(sum(rate(mmecc_emm_attach_eps_attempt[5m])) >= 0 or on() vector(0))

mmecc_emm_attach_success_rate

Description

The LTE attach success rate. This counter indicates the average LTE attach success rate on the MME in the last 5 minutes

Type

counter

Prometheus Expression

((sum(increase(mmecc_emm_attach_eps_success[5m]))) >= 0 or on() vector(0)) / (((sum(increase(mmecc_emm_attach_eps_success[5m]))) >= 0 or on() vector(0)) + ((sum(increase(mmecc_emm_attach_eps_failure[5m]))) >= 0 or on() vector(0)))) * 100

Table of Contents

- [Summary](#)
- [mmes1_s1_incoming_message_rate](#)

Summary

Metric Name	Type	Synopsis
mmes1_s1_incoming_message_rate	counter	Rate of incoming S1AP messages.

mmes1_s1_incoming_message_rate

Description

The rate of S1AP messages per second. This counter indicates the S1AP messages per second received by the MME, averaged over the last 5 minutes

Type

counter

Prometheus Expression

(sum(rate(mmes1_s1_incoming_message[5m])) >= 0 or on() vector(0))

Table of Contents

- [Summary](#)
- [nfim_nnrf_update_request_rate](#)
- [nfim_nnrf_update_success_rate](#)

Summary

Metric Name	Type	Synopsis
nfim_nnrf_update_request_rate	counter	NF Update request rate
nfim_nnrf_update_success_rate	counter	NF Update procedure success rate in last 5m

nfim_nnrf_update_request_rate

Description

The rate of NF update request per second. This counter indicates the average update requests per second received by the NRF in the last 5 minutes

Type

counter

Prometheus Expression

sum(rate(nfim_nnrf_update_request[5m]))>= 0 or on() vector(0)

nfim_nnrf_update_success_rate

Description

The NF update procedure success rate. This counter indicates the average update responses successfully responded by the NRF in the last 5 minutes

Type

counter

Prometheus Expression

((sum(increase(nfim_nnrf_update_response[5m]))- sum(increase(nfim_nnrf_update_response(status=~"5.*")[5m])))/
(sum(increase(nfim_nnrf_update_response[5m])))*100 >= 0 or on() vector(100)

Table of Contents

- [Summary](#)
- [nfsel_nnrf_discovery_success_rate](#)

Summary

Metric Name	Type	Synopsis
nfsel_nnrf_discovery_success_rate	counter	Discovery procedure success rate in last 5m

nfsel_nnrf_discovery_success_rate

Description

The NF discovery procedure success rate. This counter indicates the average discovery responses successfully responded by the NRF in the last 5 minutes

Type

counter

Prometheus Expression

((sum(increase(nfsel_nnrf_discovery_response[5m]))- sum(increase(nfsel_nnrf_discovery_response(status=~"5.*")[5m])))/
(sum(increase(nfsel_nnrf_discovery_response[5m])))*100 >= 0 or on() vector(100)

Table of Contents

- [Summary](#)
- [oauth_nnrf_access_token_success_rate](#)

Summary

Metric Name	Type	Synopsis
oauth_nnrf_access_token_success_rate	counter	Oauth procedure success rate in last 5m

oauth_nnrf_access_token_success_rate

Description

The oauth procedure success rate. This counter indicates the average oauth responses successfully responded by the NRF in the last 5 minutes

Type

counter

Prometheus Expression

```
(( sum(increase(oauth_nnrf_access_token_response[5m]))- sum(increase(oauth_nnrf_access_token_response{status=~"5.*"}[5m])))/(sum(increase(oauth_nnrf_access_token_response[5m])))*100 >= 0 or on() vector(100))
```

Summary

1. [elastic.alert.rules Alerts](#)

elastic.alert.rules

Alert Name	Description	Alert Info
elasticsearch_cluster_health_up	Elastic-search cluster's end-point is DOWN VALUE = {{ " {{ \$value }}" }} LABELS = {{ " {{ \$labels" }}" }}	1. Summary: Elastic-search cluster's end-point is DOWN (pod {{ "{{ \$labels.pod }}" }}) Severity: critical Realert Time: 5m Expression: elasticsearch_cluster_health_up !=1
elasticsearch_cluster_health_status	Elastic-search cluster's status is RED VALUE = {{ " {{ \$value }}" }} LABELS = {{ " {{ \$labels" }}" }}. Some or all of primary shards are not ready	1. Summary: The shards are not allocated in the cluster (pod {{ "{{ \$labels.pod }}" }}) Severity: critical Realert Time: 5m Expression: elasticsearch_cluster_health_status {color = "red"} > 0
elasticsearch_process_cpu_percent	Average CPU Usage has exceeded 80 percent VALUE = {{ " {{ \$value }}" }} LABELS = {{ " {{ \$labels" }}" }}	1. Summary: Average CPU Usage (pod {{ "{{ \$labels.pod }}" }}) Severity: major Realert Time: 5m Expression: (sum (elasticsearch_process_cpu_percent) / count (elasticsearch_process_cpu_percent)) > 80

Alert Name	Description	Alert Info
Disk Usage	Disk usage has exceeded 80 percent VALUE = {{ " {{ \$value }}" }} LABELS = {{ "{{ \$labels }}" }}	1. Summary: Disk usage Percentage (pod {{ "{{ \$labels.pod }}" }}) Severity: critical Realert Time: 5m Expression: (1- (elasticsearch_filesystem_data_available_bytes/elasticsearch_filesystem_data_size_bytes)) > 0.8
Average JVM memory Used	Average JVM memory Used has exceeded 80 percent VALUE = {{ " {{ \$value }}" }} LABELS = {{ "{{ \$labels }}" }}	1. Summary: Average JVM memory Used (pod {{ "{{ \$labels.pod }}" }}) Severity: critical Realert Time: 5m Expression: (sum (elasticsearch_jvm_memory_used_bytes) / sum (elasticsearch_jvm_memory_max_bytes)) > 0.8
ElasticsearchHeapUsageTooHigh	The heap usage is over 90% VALUE = {{ " {{ \$value }}" }} LABELS = {{ "{{ \$labels }}" }} The heap size is the amount of RAM allocated to the Java Virtual Machine of an Elasticsearch node. As a general rule, you should set -Xms and -Xmx to the SAME value, which should be 50% of your total available RAM.	1. Summary: Elasticsearch Heap Usage Too High (pod {{ "{{ \$labels.pod }}" }}) Severity: critical Realert Time: 2m Expression: (elasticsearch_jvm_memory_used_bytes{area="heap"} / elasticsearch_jvm_memory_max_bytes{area="heap"}) * 100 > 90

Alert Name	Description	Alert Info
ElasticsearchHeapUsageWarning	The heap usage is over 80% VALUE = {{ " {{ \$value }}" }} LABELS = {{ "{{ \$labels }}" }}. The heap size is the amount of RAM allocated to the Java Virtual Machine of an Elasticsearch node. As a general rule, you should set -Xms and -Xmx to the SAME value, which should be 50% of your total available RAM.	1. Summary: Elasticsearch Heap Usage warning (pod {{ "{{ \$labels.pod }}" }}) Severity: warning Realert Time: 2m Expression: <code>(elasticsearch_jvm_memory_used_bytes{area="heap"} / elasticsearch_jvm_memory_max_bytes{area="heap"}) * 100 > 80</code>
ElasticsearchDiskSpaceLow	The disk usage is over 80% VALUE = {{ " {{ \$value }}" }} LABELS = {{ "{{ \$labels }}" }}	1. Summary: Elasticsearch disk space low (pod {{ "{{ \$labels.pod }}" }}) Severity: warning Realert Time: 2m Expression: <code>elasticsearch_filesystem_data_available_bytes / elasticsearch_filesystem_data_size_bytes * 100 < 20</code>

Alert Name	Description	Alert Info
ElasticsearchClusterYellow	Elastic Cluster Yellow status VALUE = {{ " {{ \$value }}" }} LABELS = {{ "{{ \$labels }}" }}. Elasticsearch has allocated all of the primary shards but some or all of the replicas have not been allocated.	1. Summary: Elasticsearch Cluster Yellow (pod {{ "{{ \$labels.pod }}" }}) Severity: warning Realert Time: 0m Expression: <code>elasticsearch_cluster_health_status{color="yellow"} == 1</code>
ElasticsearchRelocatingShards	Elasticsearch is relocating shards VALUE = {{ " {{ \$value }}" }} LABELS = {{ "{{ \$labels }}" }}	1. Summary: Elasticsearch relocating shards (pod {{ "{{ \$labels.pod }}" }}) Severity: info Realert Time: 0m Expression: <code>elasticsearch_cluster_health_relocating_shards > 0</code>
ElasticsearchRelocatingShardsTooLong	Elasticsearch has been relocating shards for 15min VALUE = {{ " {{ \$value }}" }} LABELS = {{ "{{ \$labels }}" }}	1. Summary: Elasticsearch relocating shards too long (pod {{ "{{ \$labels.pod }}" }}) Severity: warning Realert Time: 15m Expression: <code>elasticsearch_cluster_health_relocating_shards > 0</code>
ElasticsearchInitializingShards	Elasticsearch is initializing shards VALUE = {{ " {{ \$value }}" }} LABELS = {{ "{{ \$labels }}" }}	1. Summary: Elasticsearch initializing shards (pod {{ "{{ \$labels.pod }}" }}) Severity: info Realert Time: 0m Expression: <code>elasticsearch_cluster_health_initializing_shards > 0</code>

Alert Name	Description	Alert Info
ElasticsearchUnassignedShards	Elasticsearch has unassigned shards VALUE = {{ "\$value" }} LABELS = {{ "\$labels" }}. The number of shards that exist in the cluster state, but cannot be found in the cluster itself.	1. Summary: Elasticsearch unassigned shards (pod {{ "\$labels.pod" }}) Severity: critical Realert Time: 0m Expression: <code>elasticsearch_cluster_health_unassigned_shards > 0</code>
ElasticsearchPendingTasks	Elasticsearch has pending tasks. Cluster works slowly. VALUE = {{ "\$value" }} LABELS = {{ "\$labels" }}	1. Summary: Elasticsearch pending tasks (pod {{ "\$labels.pod" }}) Severity: warning Realert Time: 15m Expression: <code>elasticsearch_cluster_health_number_of_pending_tasks > 0</code>
ElasticsearchActiveShardsTooHigh	Elasticsearch has too many active shards VALUE = {{ "\$value" }} LABELS = {{ "\$labels" }}.	1. Summary: Elasticsearch too many active shards (pod {{ "\$labels.pod" }}) Severity: critical Realert Time: 2m Expression: <code>elasticsearch_cluster_health_active_shards > 2500</code>

Summary

1. [fluentd-operator.alert.rules Alerts](#)

fluentd-operator.alert.rules

Alert Name	Description	Alert Info	Diagnostics
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Alert Name	Description	Alert Info	Diagnostics
read_in_tail_container_logs	log read rate per second has exceeded 5000 lines/sec by "{{ \$labels.container }}" container	1. Summary: total number of container logs	1. Repair Action: Verify {{ "{{ \$labels.container }}" }} container if debug logs are enabled. Check the health of the container for any warning logs or error logs that might be getting continuously emitted.
		Severity: minor Realert Time: 5m Expression: sum(rate(read_in_tail_container_logs[5m])) by (container) > 5000	
fluentd_output_status_buffer_available_space_ratio	fluentd output status buffer has 20% space left	1. Summary: fluentd output status buffer available space ratio	1. Repair Action: Buffer can get filled up 80% whe it is unable to send the processed logs to the elasticsearch, Check if elasticsearch healthy and Logging is enabled in Applications with right elasticsearch host. OR Increase the buffer limit by editing the configmap fluentd-config in the namespace which is impacting the buffers.
		Severity: critical Realert Time: 5m Expression: min(fluentd_output_status_buffer_available_space_ratio) < 20	

Alert Name	Description	Alert Info	Diagnostics
write_out_container_logs_to_elasticsearch	log emmision rate per second has exceeded 5000 lines/sec by {{ "\$labels.container" }} container	1. Summary: log emmision rate per second Severity: info Realert Time: 5m Expression: <code>sum(rate(write_out_container_logs_to_elasticsearch[5m])) by (container) > 5000</code>	1. Repair Action: Fluentd will write more that 5000 logs per second when application generates 5000 logs per second. Hence verify {{ "\$labels.container" }} container, if debug logs are enabled. Check the health of the container for any warning logs or error logs that might be getting continuously emitted.

Summary

1. [istio.alert.rules Alerts](#)

istio.alert.rules

Alert Name	Description	Alert Info	Diagnostics
IstioPilotAvailabilityDrop	Pilot pods have dropped during the last 5m. Envoy sidecars might have outdated configuration	1. Summary: Istio Pilot Availability Drop Severity: Warning Realert Time: 5m Expression: <code>avg(avg_over_time(up{job="pilot"}[5m])) < 0.5</code>	1. Repair Action: Kubernetes will restart pods and No action required. Validate that the node resources are available for the pod.
IstioMixerTelemetryAvailabilityDrop	Mixer pods have dropped during the last 5m. Istio metrics will not work correctly	1. Summary: Istio Mixer Telemetry Drop Severity: Warning Realert Time: 5m Expression: <code>avg(avg_over_time(up{job="mixer", service="istio-telemetry", endpoint="http-monitoring"}[5m])) < 0.5</code>	1. Repair Action: Kubernetes will restart pods and No action required. Validate that the node resources are available for the pod.

Alert Name	Description	Alert Info	Diagnostics
IstioGalleyAvailabilityDrop	Galley pods have dropped during the last 5m. Istio config ingestion and processing will not work	1. Summary: Istio Galley Availability Drop Severity: Warning Realert Time: 5m Expression: <pre>avg(avg_over_time(up{job="galley"}[5m])) < 0.5</pre>	1. Repair Action: Kubernetes will restart pods and No action required. Validate that the node resources are available for the pod.
IstioPilotPushErrorsHigh	Pilot has too many push errors during the last 5m. Envoy sidecars might have outdated configuration	1. Summary: Number of Istio Pilot push errors is too high Severity: Major Realert Time: 5m Expression: <pre>sum(irate(pilot_xds_push_errors{job="pilot"}[5m])) / sum(irate(pilot_xds_pushes{job="pilot"}[5m])) > 0.05</pre>	1. Repair Action: Restart the pilot pod.
IstioMixerPrometheusDispatchesLow	Mixer disptaches to Prometheus has dropped below normal levels during the last 5m. Istio metrics might not be being exported properly	1. Summary: Number of Mixer dispatches to Prometheus is too low Severity: Warning Realert Time: 5m Expression: <pre>sum(irate(mixer_runtime_dispatches_total{adapter=~"prometheus"}[5m])) < 1</pre>	1. Repair Action: If Istio telemetry is enabled then restart the mixer pod.
IstioGlobalHTTP5xxRateHigh	Istio global HTTP 5xx rate is too high in last 5m. The HTTP 5xx errors within the service mesh is unusually high	1. Summary: Istio Percentage of HTTP 5xx responses is too high Severity: Major Realert Time: 5m Expression: <pre>sum(irate(istio_requests_total{reporter="destination", response_code=~"5.*"}[5m])) / sum(irate(istio_requests_total{reporter="destination"}[5m])) > 0.01</pre>	1. Repair Action: Identify the destination service and understand why it's unavailable.

Summary

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- 2. [kube-state-metrics Alerts](#)
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kube-prometheus-general.rules

Alert Name	Description	Alert Info	Diagnostics
prometheus_target_missing_alert	A Prometheus target is not working. An exporter might be in a non-responsive state. VALUE = {{ "{{ \$value }}" }} LABELS = {{ "{{ \$labels }}" }}	1. Summary: Prometheus target missing (instance {{ "{{ \$labels.instance }}" }} Severity: critical Realert Time: 2m Expression: <pre>sum_over_time(up[2m]) == 0</pre>	1. Root Cause: Target pod is down. Repair Action: Check the target pod/container is running or not, if not bring up the pod by fixing the issue.

kube-state-metrics

Alert Name	Description	Alert Info	Diagnostics
KubeStateMetricsListErrors	None	1. Message: kube-state-metrics is experiencing errors at an elevated rate in list operations. This is likely causing it to not be able to expose metrics about Kubernetes objects correctly or at all. Summary: kube-state-metrics is experiencing errors at an elevated rate in list operations. This is likely causing it to not be able to expose metrics about Kubernetes objects correctly or at all. Severity: major Realert Time: 15m Expression: <pre>(sum(rate(kube_state_metrics_list_total{job="kube-state-metrics",result="error"}[5m])) / sum(rate(kube_state_metrics_list_total{job="kube-state-metrics"}[5m]))) > 0.01</pre>	1. Root Cause: kube-state-metrics is experiencing errors at an elevated rate in list operations. This is likely causing it to not be able to expose metrics about Kubernetes objects correctly or at all. Repair Action: The Kube State Metrics collector is not able to collect metrics from the cluster without errors. This means important alerts may not fire. Check kube-state-metrics pod logs in fed-kube-prom namespace to debug the issue and fix it.

Alert Name	Description	Alert Info	Diagnostics
KubeStateMetricsWatchErrors	None	1. Message: kube-state-metrics is experiencing errors at an elevated rate in watch operations. This is likely causing it to not be able to expose metrics about Kubernetes objects correctly or at all. Summary: kube-state-metrics is experiencing errors at an elevated rate in watch operations. This is likely causing it to not be able to expose metrics about Kubernetes objects correctly or at all. Severity: major Realert Time: 15m Expression: <pre>(sum(rate(kube_state_metrics_watch_total{job="kube-state-metrics",result="error"}[5m])) / sum(rate(kube_state_metrics_watch_total{job="kube-state-metrics"}[5m]))) > 0.01</pre>	1. Root Cause: kube-state-metrics is experiencing errors at an elevated rate in list operations. This is likely causing it to not be able to expose metrics about Kubernetes objects correctly or at all. Repair Action: The Kube State Metrics collector is not able to collect metrics from the cluster without errors. This means important alerts may not fire. Check kube-state-metrics pod logs in fed-kube-prom namespace to debug the issue and fix it.

Alert Name	Description	Alert Info	Diagnostics
PodNotRunning	None	1. Message: Detects pods that are not in Running state. Summary: Detects pods that are not in Running state. Severity: critical Realert Time: {{Values.kubeStateMetrics.timeIntervalPodHealth}} Expression: <pre>sum by (pod,namespace,phase) (kube_pod_status_phase{phase!="Running",phase!="Succeeded"} and on (pod,namespace) changes(kube_pod_status_phase[5m]) != 1) >= 1</pre>	1. Root Cause: There could be various reasons for a pod to be in non running state. One of them occurs when the CPU/memory request value is invalid or is more than the limit set. Repair Action: Check the error logs of the pod for figuring out the issue. For example, for CPU/memory issue, Set reasonable pod CPU/memory requests based on the available resources.

Alert Name	Description	Alert Info	Diagnostics
PodNotHealthy	None	1. Message: Detects Running pods that are in N-x/N state. Summary: Detect Running pods that are in N-x/N state. Severity: critical Realert Time: {{Values.kubeStateMetrics.timeIntervalPodHealth}} Expression: <pre>sum by (pod,namespace,phase)(kube_pod_status_phase {phase="Running"} and on (pod,namespace)(kube_pod_container_status_ready==0) and on (pod,namespace) changes(kube_pod_container_status_ready[5m]) == 0) ==1</pre>	1. Root Cause: There could be various reasons for a container to be not created. One of them occurs when the CPU/memory request value is invalid or is more than the limit set. Repair Action: Check the error logs of the pod for figuring out the issue. For example, for CPU/memory issue, Set reasonable pod CPU/memory requests based on the available resources.

PodOOMKilledRestart	Pod {{ "{{ \$labels.pod }}" }} was restarted due to OOMKilled. Check the pod logs and resource limits to prevent this issue from happening again.	1. Message: Pod {{ "{{ \$labels.pod }}" }} restarted due to OOMKilled. Summary: Pod {{ "{{ \$labels.pod }}" }} restarted due to OOMKilled Severity: major Realert Time: 5m Expression: <pre>kube_pod_container_status_last_terminated_reason{reason="OOMKilled"} == 1</pre>	1. Root Cause: Pod was restarted due to OOMKilled. Either memory is not sufficient or load is more than expected. Repair Action: Pod was restarted due to OOMKilled. Check the pod logs and resource limits to prevent this issue from happening again.
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etcd

Alert Name	Description	Alert Info	Diagnostics
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Alert Name	Description	Alert Info	Diagnostics
K8sEtcdInsufficientMembers	None	1. Message: {{ 'etcd cluster "{{ \$labels.job }}" insufficient members ({{ \$value }}) .' }} Summary: K8s Etcd cluster has insufficient number of members. Severity: critical Realert Time: 3m Expression: <pre>count(etcd_server_id{job="k8s-etcd"}) % 2 == 0</pre>	1. N.A.
K8sEtcdNoLeader	None	1. Message: {{ 'etcd cluster "{{ \$labels.job }}" member {{ \$labels.pod }} has no leader .' }} Summary: K8s Etcd cluster has no leader. Severity: critical Realert Time: 1m Expression: <pre>etcd_server_has_leader{job="k8s-etcd"} == 0</pre>	1. N.A.
K8sEtcdHighNumberOfLeaderChanges	None	1. Message: {{ 'etcd cluster "{{ \$labels.job }}" pod {{ \$labels.pod }} has seen {{ \$value }} leader changes within the last 15 minutes.' }} Summary: K8s Etcd cluster has high number of leader changes. Severity: warning Realert Time: 15m Expression: <pre>increase(etcd_server_leader_changes_seen_total{job="k8s-etcd"}[15m]) > 2</pre>	1. N.A.

Alert Name	Description	Alert Info	Diagnostics
K8sEtcdHighNumberOfFailedGRPCRequests	None	1. Message: {{ 'etcd cluster "{{ \$labels.job }}" {{ \$value }}% of requests for {{ \$labels.grpc_method }} failed on etcd pod {{ \$labels.pod }} .' }} Summary: K8s Etcd cluster has high number of failed grpc requests. More than 1% GRPC request failure detected. Severity: warning Realert Time: 10m Expression: <pre>sum(rate(grpc_server_handled_total{job="k8s-etcd", grpc_code!="OK"}[1m])) BY (grpc_service, grpc_method, job) / sum(rate(grpc_server_handled_total[1m])) BY (grpc_service, grpc_method, job) > 0.01</pre>	1. N.A.
		2. Message: {{ 'etcd cluster "{{ \$labels.job }}" {{ \$value }}% of requests for {{ \$labels.grpc_method }} failed on etcd pod {{ \$labels.pod }} .' }} Summary: etcd cluster has high number of failed grpc requests. More than 5% GRPC request failure detected. Severity: critical Realert Time: 5m Expression: <pre>sum(rate(grpc_server_handled_total{job="k8s-etcd", grpc_code!="OK"}[1m])) BY (grpc_service, grpc_method, job) / sum(rate(grpc_server_handled_total[1m])) BY (grpc_service, grpc_method, job) > 0.05</pre>	
K8sEtcdGRPCRequestsSlow	None	1. Message: {{ 'etcd cluster "{{ \$labels.job }}" gRPC requests to {{ \$labels.grpc_method }} are taking {{ \$value }}s on etcd pod {{ \$labels.pod }} .' }} Summary: K8s Etcd grpc requests are slow. Etcd GRPC requests slow. GRPC requests slowing down, 99th percentile is over 0.15s. Severity: critical Realert Time: 10m Expression: <pre>histogram_quantile(0.99, sum(rate(grpc_server_handling_seconds_bucket{job="k8s-etcd", grpc_type="unary"}[5m])) by (job, pod, grpc_service, grpc_method, le)) > 0.15</pre>	1. N.A.
K8sEtcdMemberCommunicationSlow	None	1. Message: {{ 'etcd cluster "{{ \$labels.job }}" member communication with {{ \$labels.To }} is taking {{ \$value }}s on etcd pod {{ \$labels.pod }} .' }} Summary: K8s Etcd cluster member communication is slow. Etcd member communication slow. Etcd member communication slowing down, 99th percentile is over 0.15s. Severity: warning Realert Time: 10m Expression: <pre>histogram_quantile(0.99, rate(etcd_network_peer_round_trip_time_seconds_bucket{job="k8s-etcd"}[5m])) > 0.15</pre>	1. N.A.

Alert Name	Description	Alert Info	Diagnostics
K8sEtcdHighNumberOfFailedProposals	None	1. Message: {{ 'etcd cluster "{{ \$labels.job }}" {{ \$value }} proposal failures within the last 15m on etcd pod {{ \$labels.pod }} .' }} Summary: K8s Etcd high number of failed proposals. Etcd server got more than 5 failed proposals in last 15m. Severity: warning Realert Time: 15m Expression: <pre>increase(etcd_server_proposals_failed_total{job="k8s-etcd"}[15m]) > 5</pre>	1. N.A.
K8sEtcdHighFsyncDurations	None	1. Message: {{ 'etcd cluster "{{ \$labels.job }}" 99th percentile fync durations are {{ \$value }}s on etcd pod {{ \$labels.pod }} .' }} Summary: K8s Etcd high fsync durations. Etcd WAL fsync duration increasing, 99th percentile is over 0.01s. Severity: warning Realert Time: 10m Expression: <pre>histogram_quantile(0.99, rate(etcd_disk_wal_fsync_duration_seconds_bucket{job="k8s-etcd"}[5m])) > 0.01</pre>	1. N.A.
K8sEtcdHighCommitDurations	None	1. Message: {{ 'etcd cluster "{{ \$labels.job }}" 99th percentile commit durations {{ \$value }}s on etcd pod {{ \$labels.pod }} .' }} Summary: K8s Etcd high commit durations. Etcd commit duration increasing, 99th percentile is over 0.025s. Severity: warning Realert Time: 10m Expression: <pre>histogram_quantile(0.99, rate(etcd_disk_backend_commit_duration_seconds_bucket{job="k8s-etcd"}[5m])) > 0.025</pre>	1. N.A.

Alert Name	Description	Alert Info	Diagnostics
K8sEtcdHighNumberOfFailedHTTPRequests	None	1. Message: {{ '{{ \$value }}'% of requests for {{ \$labels.method }} failed on etcd pod {{ \$labels.pod }} .' }} Summary: K8s Etcd high number of failed HTTP requests, over 1%. Severity: warning Realert Time: 10m Expression: <pre>sum(rate(etcd_http_failed_total{job="k8s-etcd", code!="404"}[5m])) BY (method) / sum(rate(etcd_http_received_total{job="k8s-etcd"}[5m])) BY (method) > 0.01</pre>	1. N.A.
K8sEtcdHTTPRequestsSlow	None	1. Message: {{ etcd pod {{ \$labels.pod }} HTTP requests to {{ \$labels.method }} are slow.}} Summary: K8s Etcd HTTP requests slow, 99th percentile is over 0.15s. Severity: warning Realert Time: 10m Expression: <pre>histogram_quantile(0.99, rate(etcd_http_successful_duration_seconds_bucket[5m])) > 0.15</pre>	1. N.A.

node-exporter

Alert Name	Description	Alert Info	Diagnostics
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Alert Name	Description	Alert Info	Diagnostics
NodeFilesystemSpaceFillingUp	Filesystem on labels.device at labels.instance has only value available space left and is filling up.	1. Summary: Filesystem is predicted to run out of space within the next 24 hours. Severity: warning Realert Time: 1h Expression: <pre>(node_filesystem_avail_bytes{job="node-exporter",fstype!=""} / node_filesystem_size_bytes{job="node-exporter",fstype!=""} * 100 < 40 and predict_linear(node_filesystem_avail_bytes{job="node-exporter",fstype!=""}[6h], 24*60*60) < 0 and node_filesystem_readonly{job="node-exporter",fstype!=""} == 0)</pre>	1. Root Cause: Either less disk configured or data is more and fillig up fast. Repair Action: The filesystem on a particular node is filling up. Provision more space by adding a disk or mounting unused disks.
		2. Summary: Filesystem is predicted to run out of space within the next 4 hours. Severity: major Realert Time: 1h Expression: <pre>(node_filesystem_avail_bytes{fstype!="",job="node-exporter"} / node_filesystem_size_bytes{fstype!="",job="node-exporter"} * 100 < 20 and node_filesystem_readonly{fstype!="",job="node-exporter"} == 0)</pre>	
NodeFilesystemAlmostOutOfSpace	Filesystem on labels.device at labels.instance has only value available space left.	1. Summary: Filesystem has less than 5% space left. Severity: warning Realert Time: 1h Expression: <pre>(node_filesystem_avail_bytes{job="node-exporter",fstype!=""} / node_filesystem_size_bytes{job="node-exporter",fstype!=""} * 100 < 5 and node_filesystem_readonly{job="node-exporter",fstype!=""} == 0)</pre>	1. Root Cause: Either less disk configured or data is more and fillig up fast. Repair Action: The filesystem on a particular node is filling up. Provision more space by adding a disk or mounting unused disks.
		2. Summary: Filesystem has less than 3% space left. Severity: critical Realert Time: 1h Expression: <pre>(node_filesystem_avail_bytes{job="node-exporter",fstype!=""} / node_filesystem_size_bytes{job="node-exporter",fstype!=""} * 100 < 3 and node_filesystem_readonly{job="node-exporter",fstype!=""} == 0)</pre>	

Alert Name	Description	Alert Info	Diagnostics
NodeFilesystemFilesFillingUp	Filesystem on labels.device at labels.instance has only value available inodes left and is filling up.	1. Summary: Filesystem is predicted to run out of inodes within the next 24 hours. Severity: warning Realert Time: 1h Expression: <pre>(node_filesystem_files_free{job="node-exporter",fstype!=""} / node_filesystem_files{job="node-exporter",fstype!=""} * 100 < 40 and predict_linear(node_filesystem_files_free{job="node-exporter",fstype!=""}[6h], 24*60*60) < 0 and node_filesystem_readonly{job="node-exporter",fstype!=""} == 0)</pre>	1. Root Cause: Either less disk configured or data is more and fillig up fast. Repair Action: The filesystem on a particular node is filling up. Provision more space by adding a disk or mounting unused disks.
		2. Summary: Filesystem is predicted to run out of inodes within the next 4 hours. Severity: major Realert Time: 1h Expression: <pre>(node_filesystem_files_free{job="node-exporter",fstype!=""} / node_filesystem_files{job="node-exporter",fstype!=""} * 100 < 20 and node_filesystem_readonly{job="node-exporter",fstype!=""} == 0)</pre>	
NodeFilesystemAlmostOutOfFiles	Filesystem on labels.device at labels.instance has only value available inodes left.	1. Summary: Filesystem has less than 5% inodes left. Severity: warning Realert Time: 1h Expression: <pre>(node_filesystem_files_free{job="node-exporter",fstype!=""} / node_filesystem_files{job="node-exporter",fstype!=""} * 100 < 5 and node_filesystem_readonly{job="node-exporter",fstype!=""} == 0)</pre>	1. Root Cause: Either less disk configured or data is more and fillig up fast. Repair Action: The filesystem on a particular node is filling up. Provision more space by adding a disk or mounting unused disks.
		2. Summary: Filesystem has less than 3% inodes left. Severity: critical Realert Time: 1h Expression: <pre>(node_filesystem_files_free{job="node-exporter",fstype!=""} / node_filesystem_files{job="node-exporter",fstype!=""} * 100 < 3 and node_filesystem_readonly{job="node-exporter",fstype!=""} == 0)</pre>	
NodeNetworkReceiveErrs	labels.instance interface labels.device has encountered value receive errors in the last two minutes.	1. Summary: Network interface is reporting many receive errors. Severity: warning Realert Time: 1h Expression: <pre>increase(node_network_receive_errs_total[2m]) > 10</pre>	1. Root Cause: Problem with the physical network interface on the node. Repair Action: There is a problem with the physical network interface on the node. If the issues persist, it may need to be replaced.

Alert Name	Description	Alert Info	Diagnostics
NodeNetworkTransmitErrs	labels.instance interface labels.device has encountered value transmit errors in the last two minutes.	1. Summary: Network interface is reporting many transmit errors. Severity: warning Realert Time: 1h Expression: increase(node_network_transmit_errs_total[2m]) > 10	1. Root Cause: Problem with the physical network interface on the node. Repair Action: There is a problem with the physical network interface on the node. If the issues persist, it may need to be replaced.
NodeHighNumberConntrackEntriesUsed	value or humanizePercentage of conntrack entries are used	1. Summary: Number of conntrack are getting close to the limit Severity: warning Expression: (node_nf_conntrack_entries / node_nf_conntrack_entries_limit) > 0.75	1. Root Cause: Problem with the physical network interface on the node. Repair Action: There is a problem with the physical network interface on the node. If the issues persist, it may need to be replaced.
NodeClockSkewDetected	None	1. Message: Clock on labels.instanceis out of sync by more than 300s. Ensure NTP is configured correctly on this host. Summary: Clock skew detected. Severity: warning Realert Time: 10m Expression: (node_timex_offset_seconds > 0.05 and deriv(node_timex_offset_seconds[5m]) >= 0) or (node_timex_offset_seconds < -0.05 and deriv(node_timex_offset_seconds[5m]) <= 0)	1. Potential Impact: Time series data won't be in align with system time. Root Cause: Problem in system clock settings, detected one or more hosts require system clock adjustment. Repair Action: There is a problem with the clock on the node. Make sure NTP is configured correctly.
NodeClockNotSynchronising	None	1. Message: Clock on labels.instance is not synchronising. Ensure NTP is configured on this host. Summary: Clock not synchronising. Severity: warning Realert Time: 10m Expression: min_over_time(node_timex_sync_status[5m]) == 0	1. Potential Impact: Time series data won't be in align with system time. Root Cause: Problem in system clock settings. Repair Action: There is a problem with the clock on the node. Make sure NTP is configured correctly.

kubernetes-apps

Alert Name	Description	Alert Info	Diagnostics
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Alert Name	Description	Alert Info	Diagnostics
KubePodCrashLooping	None	1. Message: Pod labels.namespace/labels.pod(labels.container) is restarting value times / 5 minutes. Summary: Pod labels.namespace/labels.pod(labels.container) is restarting value times / 5 minutes. Severity: critical Realert Time: 15m Expression: <pre>rate(kube_pod_container_status_restarts_total{job="kube-state-metrics"}[15m]) * 60 * 5 > 0</pre>	1. Potential Impact: Pods will be down. Repair Action: A pod that keeps restarting unexpectedly. This can happen due to an out-of-memory (OOM) error, in which case the limits can be adjusted. Check the pod events with <code>kubectl describe</code>, and logs with <code>kubectl logs</code> to see details on possible crashes.
KubePodNotReady	None	1. Message: Pod labels.namespace /labels.pod has been in a non-ready state for longer than 15 minutes. Summary: Pod labels.namespace /labels.pod has been in a non-ready state for longer than 15 minutes. Severity: critical Realert Time: 15m Expression: <pre>(kube_pod_status_phase{job="kube-state-metrics", phase=~"Pending\ Unknown"}) > 0</pre>	1. Potential Impact: If pod is not ready then it won't allow ingress traffic. Repair Action: A pod has started, but it is not responding to the health probe with success. This may mean that it is stuck and is not able to serve traffic. You can check pod logs with <code>kubectl logs</code> to see if there is any indication of progress.
KubeDeploymentGenerationMismatch	None	1. Message: Deployment generation for labels.namespace/labels.deployment does not match, this indicates that the Deployment has failed but as not been rolled back. Summary: Deployment generation for labels.namespace/labels.deployment does not match, this indicates that the Deployment has failed but as not been rolled back. Severity: major Realert Time: 15m Expression: <pre>kube_deployment_status_observed_generation{job="kube-state-metrics"} != kube_deployment_metadata_generation{job="kube-state-metrics"}</pre>	1. Potential Impact: Upgrade has failed. Repair Action: There has been an attempted update to a deployment or statefulset, but it has failed, and a rollback has not yet occurred.
KubeDeploymentReplicasMismatch	None	1. Message: Deployment labels.namespace /labels.deployment has not matched the expected number of replicas for longer than 15 minutes. Summary: Deployment labels.namespace /labels.deployment has not matched the expected number of replicas for longer than 15 minutes. Severity: major Realert Time: 15m Expression: <pre>(kube_deployment_spec_replicas{job="kube-state-metrics"} != kube_deployment_status_replicas_available{job="kube-state-metrics"}) and (changes(kube_deployment_status_replicas_updated{job="kube-state-metrics"}[5m]) == 0)</pre>	1. Potential Impact: Deployment won't be scaled with required replicas. Repair Action: In high availability clusters with multiple replicas, this alert fires when the number of replicas is not optimal. This may occur when there are not enough resources in the cluster to schedule. Check resource utilization, and add capacity as necessary.

Alert Name	Description	Alert Info	Diagnostics
KubeStatefulSetReplicasMismatch	None	1. Message: StatefulSet labels.namespace /labels.statefulset has not matched the expected number of replicas for longer than 15 minutes. Summary: StatefulSet labels.namespace /labels.statefulset has not matched the expected number of replicas for longer than 15 minutes. Severity: major Realert Time: 15m Expression: <pre>(kube_statefulset_status_replicas_ready{job="kube-state-metrics"} != kube_statefulset_status_replicas{job="kube-state-metrics"}) and (changes(kube_statefulset_status_replicas_updated{job="kube-state-metrics"}[5m]) == 0)</pre>	1. Potential Impact: StatefulSet won't be scaled with required replicas. Repair Action: In high availability clusters with multiple replicas, this alert fires when the number of replicas is not optimal. This may occur when there are not enough resources in the cluster to schedule. Check resource utilization, and add capacity as necessary.
KubeStatefulSetGenerationMismatch	None	1. Message: StatefulSet generation for labels.namespace/labels.statefulset does not match, this indicates that the StatefulSet has failed but has not been rolled back. Summary: StatefulSet generation for labels.namespace/labels.statefulset does not match, this indicates that the StatefulSet has failed but has not been rolled back. Severity: major Realert Time: 15m Expression: <pre>kube_statefulset_status_observed_generation{job="kube-state-metrics"} != kube_statefulset_metadata_generation{job="kube-state-metrics"}</pre>	1. Potential Impact: StatefulSet won't be scaled with required replicas. Repair Action: There has been an attempted update to a deployment or statefulset, but it has failed, and a rollback has not yet occurred.
KubeDaemonSetRolloutStuck	None	1. Message: Only value humanizePercentage of the desired Pods of DaemonSet labels.namespace /labels.daemonset are scheduled and ready. Summary: Only value humanizePercentage of the desired Pods of DaemonSet labels.namespace /labels.daemonset are scheduled and ready. Severity: critical Realert Time: 15m Expression: <pre>kube_daemonset_status_number_ready{job="kube-state-metrics"} / kube_daemonset_status_desired_number_scheduled{job="kube-state-metrics"} < 1.00</pre>	1. Potential Impact: Daemonset upgrade is not successful. Repair Action: Daemonset rollout has failed. Check the events in namespace.

Alert Name	Description	Alert Info	Diagnostics
KubeContainerWaiting	None	1. Message: Pod labels.namespace /labels.pod container labels.container has been in waiting state for longer than 1 hour. Summary: Pod labels.namespace /labels.pod container labels.container has been in waiting state for longer than 1 hour. Severity: warning Realert Time: 1h Expression: <pre>sum by (namespace, pod, container) (kube_pod_container_status_waiting_reason{job="kube-state- metrics"}) > 0</pre>	1. Repair Action: A container is stuck in the waiting state. It has been scheduled to a worker node, but it cannot run on that machine. Check kubectl describe of the pod for more information. The most common cause of waiting containers is a failure to pull the image. For air-gapped clusters, this could mean that the local registry is not available.
KubeDaemonSetNotScheduled	None	1. Message: value Pods of DaemonSet labels.namespace / labels.daemonset are not scheduled. Summary: value Pods of DaemonSet labels.namespace / labels.daemonset are not scheduled. Severity: warning Realert Time: 10m Expression: <pre>kube_daemonset_status_desired_number_scheduled{job="kube- state-metrics"} - kube_daemonset_status_current_number_scheduled{job="kube- state-metrics"} > 0</pre>	1. Potential Impact: Daemonset deployment failed. Repair Action: This may indicate an issue with one of the nodes Check the health of each node, and remediate any known issues.
KubeDaemonSetMisScheduled	None	1. Message: value Pods of DaemonSet labels.namespace /labels.daemonset are running where they are not supposed to run. Summary: value Pods of DaemonSet labels.namespace /labels.daemonset are running where they are not supposed to run. Severity: warning Realert Time: 10m Expression: <pre>kube_daemonset_status_number_misscheduled{job="kube-state- metrics"} > 0</pre>	1. Potential Impact: Daemonset mis scheduled. Repair Action: This may indicate an issue with one of the nodes Check the health of each node, and remediate any known issues.
KubeCronJobRunning	None	1. Message: CronJob labels.namespace/labels.cronjob is taking more than 1h to complete. Summary: CronJob labels.namespace/labels.cronjob is taking more than 1h to complete. Severity: warning Realert Time: 1h Expression: <pre>time() - kube_cronjob_next_schedule_time{job="kube-state- metrics"} > 3600</pre>	1. Repair Action: A job takes more than 1 hours to complete. This is not expected.

Alert Name	Description	Alert Info	Diagnostics
KubeJobCompletion	None	1. Message: Job labels.namespace/labels.job_name is taking more than one hour to complete. Summary: Job labels.namespace/labels.job_name is taking more than one hour to complete. Severity: warning Realert Time: 1h Expression: <pre>kube_job_spec_completions{job="kube-state-metrics"} - kube_job_status_succeeded{job="kube-state-metrics"} > 0</pre>	1. Repair Action: A job takes more than 1 hours to complete. This is not expected.
KubeJobFailed	None	1. Message: Job labels.namespace/labels.job_name failed to complete. Summary: Job labels.namespace/labels.job_name failed to complete. Severity: warning Realert Time: 15m Expression: <pre>kube_job_failed{job="kube-state-metrics"} > 0</pre>	1. Repair Action: A job has failed; however, most jobs are retried automatically. Check the events.
KubeHpaReplicasMismatch	None	1. Message: HPA labels.namespace /labels.hpa has not matched the desired number of replicas for longer than 15 minutes. Summary: HPA labels.namespace /labels.hpa has not matched the desired number of replicas for longer than 15 minutes. Severity: warning Realert Time: 15m Expression: <pre>(kube_hpa_status_desired_replicas{job="kube-state-metrics"} != kube_hpa_status_current_replicas{job="kube-state-metrics"}) and changes(kube_hpa_status_current_replicas[15m]) == 0</pre>	1. Potential Impact: HPA failed. Repair Action: The autoscaler cannot scale the targeted resource as configured. If desired is higher than actual, then there may be a lack of resources. If desired is lower than actual, pods may be stuck while shutting down.
KubeHpaMaxedOut	None	1. Message: HPA labels.namespace /labels.hpa has been running at max replicas for longer than 15 minutes. Summary: HPA labels.namespace /labels.hpa has been running at max replicas for longer than 15 minutes. Severity: warning Realert Time: 15m Expression: <pre>kube_hpa_status_current_replicas{job="kube-state-metrics"} == kube_hpa_spec_max_replicas{job="kube-state-metrics"}</pre>	1. Potential Impact: Can't run HPA. Repair Action: The number of replicas for a given service has reached its maximum. This happens when the amount of requests being made to the cluster is very high. If high traffic is expected and temporary, you may silence this alert. However, this alert is a sign that the cluster is at capacity and cannot handle much more traffic. If more resource capacity is available on the cluster, you can increase the number of maximum replicas for the service.

kubernetes-resources

Alert Name	Description	Alert Info	Diagnostics
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Alert Name	Description	Alert Info	Diagnostics
KubeCPUOvercommit	None	1. Message: Cluster has overcommitted CPU resource requests for Pods and cannot tolerate node failure. Summary: Cluster has overcommitted CPU resource requests for Pods and cannot tolerate node failure. Severity: warning Realert Time: 5m Expression: <pre>sum(namespace:kube_pod_container_resource_requests_cpu_cores:sum{}) / sum(kube_node_status_allocatable_cpu_cores) > (count(kube_node_status_allocatable_cpu_cores)-1) / count(kube_node_status_allocatable_cpu_cores)</pre>	1. Potential Impact: Kubernetes scheduler will not be able to schedule pods as a result the workload will not start. Repair Action: Correct the resource allocation and retry deployment Or nodes should be brought back to health or replaced.
		2. Message: Cluster has overcommitted CPU resource requests for Namespaces. Summary: Cluster has overcommitted CPU resource requests for Namespaces. Severity: warning Realert Time: 5m Expression: <pre>sum(kube_resourcequota{job="kube-state-metrics", type="hard", resource="cpu"}) / sum(kube_node_status_allocatable_cpu_cores) > 1.5</pre>	
KubeMemOvercommit	None	1. Message: Cluster has overcommitted memory resource requests for Pods and cannot tolerate node failure. Summary: Cluster has overcommitted memory resource requests for Pods and cannot tolerate node failure. Severity: warning Realert Time: 5m Expression: <pre>sum(namespace:kube_pod_container_resource_requests_memory_bytes:sum{}) / sum(kube_node_status_allocatable_memory_bytes) > (count(kube_node_status_allocatable_memory_bytes)-1) / count(kube_node_status_allocatable_memory_bytes)</pre>	1. Potential Impact: Kubernetes scheduler will not be able to schedule pods as a result the workload will not start. Repair Action: Correct the resource allocation and retry deployment Or nodes should be brought back to health or replaced.
		2. Message: Cluster has overcommitted memory resource requests for Namespaces. Summary: Cluster has overcommitted memory resource requests for Namespaces. Severity: warning Realert Time: 5m Expression: <pre>sum(kube_resourcequota{job="kube-state-metrics", type="hard", resource="memory"}) / sum(kube_node_status_allocatable_memory_bytes{job="node-exporter"}) > 1.5</pre>	

Alert Name	Description	Alert Info	Diagnostics
KubeQuotaExceeded	None	1. Message: Namespace labels.namespace is using value humanizePercentage of its labels.resource quota. Summary: Namespace labels.namespace is using value humanizePercentage of its labels.resource quota. Severity: warning Realert Time: 15m Expression: <pre>kube_resourcequota{job="kube-state-metrics", type="used"} / ignoring(instance, job, type) (kube_resourcequota{job="kube-state-metrics", type="hard"} > 0) > 0.90</pre>	1. Repair Action: These alerts pertain to namespace resource quotas that only exist in the cluster if added through customization.
CPUThrottlingHigh	None	1. Message: value humanizePercentagethrottling of CPU in namespace labels.namespace for container labels.container in pod labels.pod . Summary: value humanizePercentagethrottling of CPU in namespace labels.namespace for container labels.container in pod labels.pod . Severity: warning Realert Time: 15m Expression: <pre>sum(increase(container_cpu_cfs_throttled_periods_total{container!="",}[5m])) by (container, pod, namespace) / sum(increase(container_cpu_cfs_periods_total{}[5m])) by (container, pod, namespace) > (75 / 100)</pre>	1. Potential Impact: When working with low CPU limits, spiky workloads can have low averages and still be being throttled. Repair Action: A container's CPU utilization has been throttled according to the configured limits. This is part of normal Kubernetes operation and may provide useful information when other alerts are firing. You may silence this alert.

kubernetes-storage

Alert Name	Description	Alert Info	Diagnostics
KubePersistentVolumeUsageCritical	None	1. Message: The PersistentVolume claimed by labels.persistentvolumeclaim in Namespace labels.namespace is only value humanizePercentage free. Summary: The PersistentVolume claimed by labels.persistentvolumeclaim in Namespace labels.namespace is only value humanizePercentage free. Severity: critical Realert Time: 1m Expression: <pre>kubelet_volume_stats_available_bytes{job="kubelet", metrics_path="/metrics"} / kubelet_volume_stats_capacity_bytes{job="kubelet", metrics_path="/metrics"} < 0.03</pre>	1. Potential Impact: Pods will fail to write data to storage. Repair Action: Persistence volume available space is less than 3%. Increse the PV size.

Alert Name	Description	Alert Info	Diagnostics
KubePersistentVolumeFullInFourDays	None	1. Message: Based on recent sampling, the PersistentVolume claimed by labels.persistentvolumeclaim in Namespace labels.namespace is expected to fill up within four days. Currently value humanizePercentage is available. Summary: Based on recent sampling, the PersistentVolume claimed by labels.persistentvolumeclaim in Namespace labels.namespace is expected to fill up within four days. Currently value humanizePercentage is available. Severity: major Realert Time: 1h Expression: <pre>(kubelet_volume_stats_available_bytes{job="kubelet", metrics_path="/metrics"} / kubelet_volume_stats_capacity_bytes{job="kubelet", metrics_path="/metrics"}) < 0.15 and predict_linear(kubelet_volume_stats_available_bytes{job="kubelet", metrics_path="/metrics"}[6h], 4 * 24 * 3600) < 0</pre>	1. Potential Impact: Pods will fail to write data to storage. Repair Action: The available space is less than 15% and is likely to fill up within four days. Increase the PV size or delete old data.
KubePersistentVolumeErrors	None	1. Message: The persistent volume labels.persistentvolume has status labels.phase. Summary: The persistent volume labels.persistentvolume has status labels.phase. Severity: critical Realert Time: 5m Expression: <pre>kube_persistentvolume_status_phase{phase=~"Failed Pending",job="kube-state-metrics"} > 0</pre>	1. Potential Impact: Pods will fail to write data to storage. Repair Action: PersistentVolume is not able to be provisioned. This means any service requiring the volume would not start. Check for other errors with cinder and/or Ceph storage.

kubernetes-system

Alert Name	Description	Alert Info	Diagnostics
KubeVersionMismatch	None	1. Message: There are value different semantic versions of Kubernetes components running. Summary: There are value different semantic versions of Kubernetes components running. Severity: warning Realert Time: 15m Expression: <pre>count(count by (gitVersion) (label_replace(kubernetes_build_info{job!~"kube-dns coredns"},"gitVersion","\$1","gitVersion","v[0-9]*.[0-9]*.[0-9]*.*")) > 1</pre>	1. Potential Impact: Resource may fail to get deploy if there are different versions. Repair Action: This can happen as a result of an unsuccessful Kubernetes upgrade.

Alert Name	Description	Alert Info	Diagnostics
KubeClientErrors	None	1. Message: Kubernetes API server client 'labels.job/labels.instance ' is experiencing value humanizePercentage errors.' Summary: Kubernetes API server client 'labels.job/labels.instance ' is experiencing value humanizePercentage errors.' Severity: warning Realert Time: 15m Expression: <pre>(sum(rate(rest_client_requests_total{code=~"5.."}[5m])) by (instance, job) / sum(rate(rest_client_requests_total[5m])) by (instance, job)) > 0.01</pre>	1. Potential Impact: Resource deployment will get fail. Repair Action: Kubernetes API server client is experiencing greater than 1% errors. There may be an issue with the node this client is running on, or the Kubernetes API server itself.

kube-apiserver-error-alerts

Alert Name	Description	Alert Info	Diagnostics
ErrorBudgetBurn	None	1. Message: High requests error budget burn for job=apiserver (current value:value) Summary: High requests error budget burn for job=apiserver (current value:value) Job: apiserver Severity: critical Expression: <pre>(status_class_5xx:apiserver_request_total:ratio_rate1h{job="apiserver"} > (14.4*0.010000) and status_class_5xx:apiserver_request_total:ratio_rate5m{job="apiserver"} > (14.4*0.010000)) or (status_class_5xx:apiserver_request_total:ratio_rate6h{job="apiserver"} > (6*0.010000) and status_class_5xx:apiserver_request_total:ratio_rate30m{job="apiserver"} > (6*0.010000))</pre> 2. Message: High requests error budget burn for job=apiserver (current value:value) Summary: High requests error budget burn for job=apiserver (current value:value) Job: apiserver Severity: warning Expression: <pre>(status_class_5xx:apiserver_request_total:ratio_rate1d{job="apiserver"} > (3*0.010000) and status_class_5xx:apiserver_request_total:ratio_rate2h{job="apiserver"} > (3*0.010000)) or (status_class_5xx:apiserver_request_total:ratio_rate3d{job="apiserver"} > (0.010000) and status_class_5xx:apiserver_request_total:ratio_rate6h{job="apiserver"} > (0.010000))</pre>	1. Potential Impact: Any resource deployment will get fail. Repair Action: The Kubernetes API server is burning too much error budget. Check the api server logs.

kubernetes-system-apiserver

Alert Name	Description	Alert Info
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Alert Name	Description	Alert Info
KubeAPILatencyHigh	None	1. Message: The API server has an abnormal latency of value seconds for labels.verb labels.resource. Summary: The API server has an abnormal latency of value seconds for labels.verb labels.resource. Severity: warning Realert Time: 5m Expression: <pre>(cluster:apiserver_request_duration_seconds:mean5m{job="apiserver"} > on (verb) group_left() (cluster:apiserver_request_duration_seconds:mean5m{job="apiserver"} >= 0) + 2*stddev by (verb) (cluster:apiserver_request_duration_seconds:mean5m{job="apiserver"} >= 0))) > on (verb) group_left(1.2 * avg by (verb) (cluster:apiserver_request_duration_seconds:mean5m{job="apiserver"} >= 0) and (verb,resource) cluster_quantile:apiserver_request_duration_seconds:histogram_quantile{job="apiserver",quantile="0.99"} > 1</pre>
		2. Message: The API server has a 99th percentile latency of value seconds for labels.verb labels.resource. Summary: The API server has a 99th percentile latency of value seconds for labels.verb labels.resource. Severity: critical Realert Time: 10m Expression: <pre>cluster_quantile:apiserver_request_duration_seconds:histogram_quantile{job="apiserver",quantile="0.99"} > 4</pre>
KubeAPIErrorsHigh	None	1. Message: API server is returning errors for value humanizePercentage of requests for labels.verb labels.resource labels.subresource. Summary: API server is returning errors for value humanizePercentage of requests for labels.verb labels.resource labels.subresource. Severity: critical Realert Time: 10m Expression: <pre>sum(rate(apiserver_request_total{job="apiserver",code=~"5.."}[5m])) by (resource,subresource,verb) / sum(rate(apiserver_request_total{job="apiserver"}[5m])) by (resource,subresource,verb) > 0.10</pre>
		2. Message: API server is returning errors for value humanizePercentage of requests for labels.verb labels.resource labels.subresource. Summary: API server is returning errors for value humanizePercentage of requests for labels.verb labels.resource labels.subresource. Severity: warning Realert Time: 10m Expression: <pre>sum(rate(apiserver_request_total{job="apiserver",code=~"5.."}[5m])) by (resource,subresource,verb) / sum(rate(apiserver_request_total{job="apiserver"}[5m])) by (resource,subresource,verb) > 0.05</pre>

Alert Name	Description	Alert Info
KubeClientCertificateExpiration	None	1. Message: A client certificate used to authenticate to the apiserver is expiring in less than 7.0 days. Summary: A client certificate used to authenticate to the apiserver is expiring in less than 7.0 days. Severity: warning Expression: <pre>apiserver_client_certificate_expiration_seconds_count{job="apiserver"} > 0 and on(job) histogram_quantile(0.01, sum by (job, le) (rate(apiserver_client_certificate_expiration_seconds_bucket{job="apiserver"}[5m]))) < 604800</pre>
		2. Message: A client certificate used to authenticate to the apiserver is expiring in less than 24.0 hours. Summary: A client certificate used to authenticate to the apiserver is expiring in less than 24.0 hours. Severity: critical Expression: <pre>apiserver_client_certificate_expiration_seconds_count{job="apiserver"} > 0 and on(job) histogram_quantile(0.01, sum by (job, le) (rate(apiserver_client_certificate_expiration_seconds_bucket{job="apiserver"}[5m]))) < 86400</pre>
AggregatedAPIErrors	None	1. Message: An aggregated API labels.name /labels.namespace has reported errors. The number of errors have increased for it in the past five minutes. High values indicate that the availability of the service changes too often. Summary: An aggregated API labels.name /labels.namespace has reported errors. The number of errors have increased for it in the past five minutes. High values indicate that the availability of the service changes too often. Severity: warning Expression: <pre>sum by(name, namespace)(increase(aggregator_unavailable_apiservice_count[5m])) > 2</pre>

Alert Name	Description	Alert Info
AggregatedAPIDown	None	1. Message: An aggregated API labels.name /labels.namespace is down. It has not been available at least for the past five minutes. Summary: An aggregated API labels.name /labels.namespace is down. It has not been available at least for the past five minutes. Severity: warning Realert Time: 5m Expression: <pre>sum by(name, namespace)(sum_over_time(aggregator_unavailable_apiservice[5m])) > 0</pre>

KubeAPIDown	None	1. Message: KubeAPI has disappeared from Prometheus target discovery. Summary: KubeAPI has disappeared from Prometheus target discovery. Severity: critical Realert Time: 15m Expression: <pre>absent(up{job="apiserver"} == 1)</pre>
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kubernetes-system-kubelet

Alert Name	Description	Alert Info	Diagnost
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Alert Name	Description	Alert Info	Diagnost
KubeNodeNotReady	None		1. Potent Impact: Pods won be scheduled on the no which is n ready.
			Repair Action: These ale indicate a problem with a noi In multi-machine clusters, pods wou likely be reschedul onto othe nodes. If t issue persists, y should remove a drain the node to maintain health of cluster. In clusters without e capacity, another node sho be joined the cluste first.
		1. Message: labels.node has been unready for more than 5 minutes.	
		Summary: labels.node has been unready for more than 5 minutes.	
		Severity: warning Realert Time: 1m Expression: kube_node_status_condition{job="kube-state-metrics",condition="Ready",status="true"} == 0	

Alert Name	Description	Alert Info	Diagnost
KubeNodeUnreachable	None	1. Message: labels.node is unreachable and some workloads may be rescheduled. Summary: labels.node is unreachable and some workloads may be rescheduled. Severity: warning Realert Time: 2m Expression: kube_node_spec_taint{job="kube-state-metrics",key="node.kubernetes.io/unreachable",effect="NoSchedule"} == 1	1. Potent Impact: Pods won be scheduled on the no which is n ready. Repair Action: These ale indicate a problem with a no In multi-machine clusters, pods wou likely be reschedul onto othe nodes. If t issue persists, y should remove a drain the node to maintain health of cluster. In clusters without e capacity, another node sho be joined the cluste first.
			1. Potent Impact: Node overloade with mult pods. Car deploy ar more pod Repair Action: There are too many pods running o the specif node. Joir another node to t cluster.

Alert Name	Description	Alert Info	Diagnost
KubeNodeReadinessFlapping	None		1. Potent Impact: Pods won be scheduled on the no which is n ready.
			Repair Action: These ale indicate a problem with a noi
		1. Message: The readiness status of node labels.node has changed value times in the last 15 minutes.	In multi-machine clusters, pods wou likely be reschedul onto othe nodes. If t
		Summary: The readiness status of node labels.node has changed value times in the last 15 minutes.	issue persists, y should remove a drain the node to maintain t
		Severity: warning	health of cluster. In clusters without e capacity, another node sho be joined the cluste first.
		Realert Time: 15m	
		Expression:	
		<code>sum(changes(kube_node_status_condition{status="true",condition="Ready"}[15m])) by (node) > 2</code>	

Alert Name	Description	Alert Info	Diagnost
KubeletPlegDurationHigh	None	1. Message: The Kubelet Pod Lifecycle Event Generator has a 99th percentile duration of value seconds on node labels.node. Summary: The Kubelet Pod Lifecycle Event Generator has a 99th percentile duration of value seconds on node labels.node. Severity: warning Realert Time: 5m Expression: node_quantile:kubelet_pleg_relist_duration_seconds:histogram_quantile{quantile="0.99"}>= 10	1. Potent Impact: Pods won be scheduled on the no which is n ready. Repair Action: These ale indicate a problem with a noi In multi-machine clusters, pods wou likely be reschedul onto othe nodes. If t issue persists, y should remove a drain the node to maintain health of cluster. In clusters without e capacity, another node sho be joined the cluste first.

Alert Name	Description	Alert Info	Diagnost
KubeletPodStartUpLatencyHigh	None	1. Message: Kubelet Pod startup 99th percentile latency is value seconds on node labels.node. Summary: Kubelet Pod startup 99th percentile latency is value seconds on node labels.node. Severity: warning Realert Time: 15m Expression: histogram_quantile(0.99, sum(rate(kubelet_pod_worker_duration_seconds_bucket{job="kubelet", metrics_path="/metrics"}[5m])) by (instance, le)) * on(instance) group_left(node) kubelet_node_name > 60	1. Potent Impact: Pods won be scheduled on the no which has pod start latency hi Repair Action: These ale indicate a problem with a no In multi-machine clusters, pods wou likely be reschedul onto othe nodes. If t issue persists, y should remove a drain the node to maintain health of cluster. In clusters without e capacity, another node sho be joined the cluste first.
			1. Potent Impact: Pods will to write d to storage after the memory i full and r go OOM. Repair Action: T memory consumpt has gone of threshc limits. Increase t memory limit or delete olc data.

Alert Name	Description	Alert Info	Diagnost
KubeletDown	None	1. Message: Kubelet has disappeared from Prometheus target discovery. Summary: Kubelet has disappeared from Prometheus target discovery. Severity: critical Realert Time: 15m Expression: absent(up{job="kubelet", metrics_path="/metrics"} == 1)	1. Potent Impact: Resource status wo be availat to Kuberne and Kube metrics won't be available.
			Repair Action: These ale indicate a problem with a no
			In multi-machine clusters, pods wou likely be reschedul onto othe nodes. If t
			issue persists, y should remove a drain the node to maintain
			health of cluster. In clusters without e capacity, another node sho be joined the cluste first.

prometheus

Alert Name	Description	Alert Info
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Alert Name	Description	Alert Info
PrometheusBadConfig	Prometheus labels.namespace/labels.pod has failed to reload its configuration.	1. Summary: Failure
		Severity: critical Realert Time: 10m Expression: # Without max_over_time https: //www.role details. max_over_time k8s",namespace=
PrometheusNotificationQueueRunningFull	Alert notification queue of Prometheus labels.namespace/labels.pod is running full.	1. Summary: Problem
		Severity: warning Realert Time: 15m Expression: # Without min_over_time https: //www.role details. (predicted: k8s",namespace= min_over_time(predicted: k8s",namespace=

Alert Name	Description	Alert Info
PrometheusErrorSendingAlertsToSomeAlertmanagers	value errors while sending alerts from Prometheus labels.namespace/labels.pod to Alertmanager labels.alertmanager.	1. Summary: PrometheusErrorSendingAlertsToSomeAlertmanagers.
		Severity: warning Realert Time: 15m Expression: <pre>(rate(prometheus_ .Release.Namespi rate(prometheus_ .Release.Namespi</pre>
PrometheusErrorSendingAlertsToAnyAlertmanagerKubeProm	value minimum errors while sending alerts from Prometheus labels.namespace/labels.pod to any Alertmanager.	1. Summary: PrometheusErrorSendingAlertsToAnyAlertmanagerKubeProm.
		Severity: major Realert Time: 15m Expression: <pre>min without(alei rate(prometheus_ .Release.Namespi rate(prometheus_ .Release.Namespi</pre>

Alert Name	Description	Alert Info
PrometheusNotConnectedToAlertmanagers	Prometheus labels.namespace/labels.pod is not connected to any Alertmanagers.	1. Summary: Prometheus labels.namespace/labels.pod is not connected to any Alertmanagers.
		Severity: warning Realert Time: 10m Expression: <pre># Without max_over_time, the alert is triggered by the first failed reload. https: //www.rol details. max_over_time(prometheus_reload_failures{namespace=~".+"}) > 0 k8s",namespace=</pre>
PrometheusTSDBReloadsFailing	Prometheus labels.namespace/labels.pod has detected value or humanize reload failures over the last 3h.	1. Summary: Prometheus labels.namespace/labels.pod has detected value or humanize reload failures over the last 3h.
		Severity: warning Realert Time: 4h Expression: <pre>increase(prometheus_reload_failures{namespace=~".+"}[3h]) > 0 { .Release.Name }</pre>

Alert Name	Description	Alert Info
PrometheusTSDBCompactionsFailing	Prometheus labels.namespace/labels.pod has detected value or humanize compaction failures over the last 3h.	1. Summary: PrometheusTSDBCompactionsFailing Severity: warning Realert Time: 4h Expression: <code>increase(prometheus_tsdb_compactions_failed_total{namespace="k8s"},namespace="k8s") > 10</code>
PrometheusNotIngestingSamples	Prometheus labels.namespace/labels.pod is not ingesting samples.	1. Summary: PrometheusNotIngestingSamples Severity: warning Realert Time: 10m Expression: <code>rate(prometheus_tsdb_ingested_samples_total{namespace="k8s"},namespace="k8s") < 10</code>
PrometheusDuplicateTimestamps	Prometheus labels.namespace/labels.pod is dropping value samples/s with different values but duplicated timestamp.	1. Summary: PrometheusDuplicateTimestamps Severity: warning Realert Time: 10m Expression: <code>rate(prometheus_tsdb_ingested_samples_total{namespace="k8s"},namespace="k8s") < 10</code>

Alert Name	Description	Alert Info
PrometheusOutOfOrderTimestamps	Prometheus labels.namespace/labels.pod is dropping value samples/s with timestamps arriving out of order.	1. Summary: Prometheus labels.namespace/labels.pod is dropping value samples/s with timestamps arriving out of order. Severity: warning Realert Time: 10m Expression: rate(prometheus_ k8s",namespace=
PrometheusRemoteWriteDesiredShards	Prometheus labels.namespace/labels.pod remote write desired shards calculation wants to run value shards, which is more than the max of printf prometheus_remote_storage_shards_max{instance="%s",job="prometheus-k8s",namespace="{{ .Release.Namespace }}" } labels.instance or query or first or value.	1. Summary: Prometheus labels.namespace/labels.pod remote write desired shards calculation wants to run value shards, which is more than the max of printf prometheus_remote_storage_shards_max{instance="%s",job="prometheus-k8s",namespace="{{ .Release.Namespace }}" } labels.instance or query or first or value. Severity: warning Realert Time: 15m Expression: # Without max_over_time(https: //www.rol details. (max_over_time(pi k8s",namespace= max_over_time(pi k8s",namespace=

Alert Name	Description	Alert Info
PrometheusMissingRuleEvaluations	Prometheus labels.namespace/labels.pod has missed value rule group evaluations in the last 5m.	1. Summary: Prometheus labels.namespace/labels.pod has missed value rule group evaluations in the last 5m. Severity: warning Realert Time: 15m Expression: <pre>increase(prometheus_rule_group_evaluations{namespace="k8s",namespace="k8s"},namespace="k8s") > 10</pre>

alertmanager.rules

Alert Name	Description	Alert Info	Diagnostics
AlertmanagerFailedReload	None	1. Message: Reloading Alertmanager's configuration has failed for labels.namespace/labels.pod. Summary: Reloading Alertmanager's configuration has failed for labels.namespace/labels.pod. Severity: warning Realert Time: 10m Expression: <pre>alertmanager_config_last_reload_successful{job="alertmanager-main",namespace="{ .Release.Namespace }"} == 0</pre>	1. Potential Impact: Alerts may disappear in Alertmanager. Repair Action: AlertManager has failed to load or reload configuration. Please check any custom AlertManager configurations for input errors.

general.rules

Alert Name	Description	Alert Info	Diagnostics
TargetDown	None	1. Message: value of the labels.job /labels.service targets in labels.namespace namespace are down. Summary: value of the labels.job /labels.service targets in labels.namespace namespace are down. Severity: warning Realert Time: 10m Expression: <pre>100 * (count(up == 0) BY (job, namespace, service) / count(up) BY (job, namespace, service)) > 10</pre>	1. Potential Impact: Prometheus is not able to collect metrics from the target in the alert, which means Grafana dashboards and further alerts based on metrics from that target are not be available. Repair Action: Prometheus is not able to collect metrics from the target in the alert, which means Grafana dashboards and further alerts based on metrics from that target are not be available. Check other alerts pertaining to that target.

Alert Name	Description	Alert Info	Diagnostics
Watchdog	None	1. Message: This is an alert meant to ensure that the entire alerting pipeline is functional.This alert is always firing, therefore it should always be firing in Alertmanagerand always fire against a receiver. There are integrations with various notificationmechanisms that send a notification when this alert is not firing. For example the"DeadMansSnitch" integration in PagerDuty. Summary: This is an alert meant to ensure that the entire alerting pipeline is functional. This alert is always firing, therefore it should always be firing in Alertmanager and always fire against a receiver. There are integrations with various notification mechanisms that send a notification when this alert is not firing. For example the "DeadMansSnitch" integration in PagerDuty. Severity: none Expression: vector(1)	1. Potential Impact: No impact. Repair Action: This is an alert meant to ensure that the entire alerting pipeline is functional. This alert is always firing, therefore it should always be firing in Alertmanager and always fire against a receiver. There are integrations with various notification mechanisms that send a notification when this alert is not firing. For example the "DeadMansSnitch" integration in PagerDuty.

node-network

Alert Name	Description	Alert Info	Diagnostics
NodeNetworkInterfaceFlapping	None	1. Message: Network interface "labels.device" changing it's up status often on node-exporter labels.namespace /labels.pod Summary: Network interface "labels.device" changing it's up status often on node-exporter labels.namespace /labels.pod Severity: warning Realert Time: 2m Expression: changes(node_network_up{job="node-exporter",device!~"veth.+"}[2m]) > 2	1. Potential Impact: No communication between kubernetes services. Repair Action: There is a problem with the physical network interface on the node. If the issues persist, it may need to be replaced.

prometheus-operator

Alert Name	Description	Alert Info	Diagnostics
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Alert Name	Description	Alert Info	Diagnostics
PrometheusOperatorReconcileErrors	None	1. Message: Errors while reconciling labels.controller in labels.namespace Namespace. Summary: Errors while reconciling labels.controller in labels.namespace Namespace. Severity: warning Realert Time: 10m Expression: rate(prometheus_operator_reconcile_errors_total{job="prometheus-operator",namespace="{ .Release.Namespace }"}[5m]) > 0.1	1. Potential Impact: Prometheus operator will fail to monitor Prometheus lifecycle. Repair Action: Internal errors of the Prometheus operator, which controls Prometheus resources. Prometheus itself may still be healthy while these errors are present; however, this error indicates there is degraded monitoring configurability.
PrometheusOperatorNodeLookupErrors	None	1. Message: Errors while reconciling Prometheus in labels.namespace Namespace. Summary: Errors while reconciling Prometheus in labels.namespace Namespace. Severity: warning Realert Time: 10m Expression: rate(prometheus_operator_node_address_lookup_errors_total{job="prometheus-operator",namespace="{ .Release.Namespace }"}[5m]) > 0.1	1. Potential Impact: Prometheus operator will fail to monitor Prometheus lifecycle. Repair Action: Internal errors of the Prometheus operator, which controls Prometheus resources. Prometheus itself may still be healthy while these errors are present; however, this error indicates there is degraded monitoring configurability.

time-sync-alert

Alert Name	Description	Alert Info	Diagnostics
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Alert Name	Description	Alert Info	Diagnostics
time-sync-alert.rules	The node {{ "{{ \$labels.instance }}" }} is not synchronized with an external time source.	1. Message: Time synchronization issue detected on {{ "{{ \$labels.instance }}" }} Summary: Time synchronization issue detected on {{ "{{ \$labels.instance }}" }} Severity: critical Realert Time: 5m Expression: <pre>node_timex_sync_status == 0</pre>	1. Potential Impact: Time synchronization issues can cause issues with certificate validation, logging, and other time-sensitive operations. Repair Action: Check the time synchronization configuration on the node and ensure it is synchronized with an external time source.

qdisc.rules

Alert Name	Description	Alert Info	Diagnostics
QdiscBacklogIncreaseAlert	The qdisc backlog for the network device {{ "{{ \$labels.device }}" }} has been increasing for over 2 minutes on node {{ "{{ \$labels.instance }}" }}.	1. Message: Qdisc backlog are increasing on node {{{ "{{ \$labels.instance }}" }}}} Summary: Qdisc backlog are increasing on node {{{ "{{ \$labels.instance }}" }}}} Severity: warning Realert Time: 2m Expression: <pre>increase(node_qdisc_backlog[2m]) > 0</pre>	1. Potential Impact: Network performance may degrade if the qdisc backlog is not getting cleared by itself. Repair Action: Please restart the VM if the issue persists.
QdiscPacketDropAlert	The qdisc drops for the network device {{ "{{ \$labels.device }}" }} has been increasing for over 2 minutes on node {{ "{{ \$labels.instance }}" }}.	1. Message: Qdisc packet drops are increasing on node {{{ "{{ \$labels.instance }}" }}}} Summary: Qdisc packet drops are increasing on node {{{ "{{ \$labels.instance }}" }}}} Severity: critical Realert Time: 2m Expression: <pre>increase(node_qdisc_drops_total[2m]) > 0</pre>	1. Potential Impact: Network performance may degrade if the qdisc continue to drop packets. Repair Action: Please restart the VM to mitigate the issue.

Summary

1. [prometheus.alert.rules Alerts](#)

prometheus.alert.rules

Alert Name	Description	Alert Info
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Alert Name	Description	Alert Info
prometheus_target_missing_alert	A Prometheus target is not working. An exporter might be in a non-responsive state.	1. Summary:
	VALUE = {{ "{{ \$value }}" }}	Severity: critic
	LABELS = {{ "{{ \$labels }}" }}	Realert Time: ;
		Expression: <code>sum_over_ti</code>
prometheus_target_scrape_pools_failed_total	The number of Prometheus targets which it failed to scrape	1. Summary:
	VALUE = {{ "{{ \$value }}" }}	\$labels.pod }
	LABELS = {{ "{{ \$labels }}" }}	Severity: majo
		Realert Time: ;
PrometheusJobMissing	A Prometheus job has disappeared	1. Summary:
	VALUE = {{ "{{ \$value }}" }}	Severity: warn
	LABELS = {{ "{{ \$labels }}" }}	Realert Time: ;
		Expression: <code>absent(up{j</code>

Alert Name	Description	Alert Info
PrometheusTooManyRestarts	Prometheus has restarted more than twice in the last 15 minutes. It might be crashlooping. VALUE = {{ "{{ \$value }}" }} LABELS = {{ "{{ \$labels }}" }}	1. Summary: Severity: warn Realert Time: 5m Expression: <code>changes(prometheus_core\ pushgateway) > 2</code>
PrometheusNotConnectedToAlertmanager	Prometheus cannot connect to alertmanager VALUE = {{ "{{ \$value }}" }} LABELS = {{ "{{ \$labels }}" }}	1. Summary: Severity: critic Realert Time: 5m Expression: <code>prometheus_alertmanager_status != 1</code>
PrometheusRuleEvaluationFailures	Prometheus encountered rule evaluation failures, leading to potentially ignored alerts. VALUE = {{ "{{ \$value }}" }} LABELS = {{ "{{ \$labels }}" }}	1. Summary: Severity: critic Realert Time: 5m Expression: <code>increase(prometheus_rule_evaluation_failures_total[5m]) > 0</code>

Alert Name	Description	Alert Info
PrometheusNotificationsBacklog	The Prometheus notification queue has not been empty for 10 minutes VALUE = {{ "{{ \$value }}" }} LABELS = {{ "{{ \$labels }}" }}	1. Summary:
		Severity: warn
		Realert Time: ;
		Expression: min_over_ti
PrometheusAlertmanagerNotificationFailing	Alertmanager is failing sending notifications VALUE = {{ "{{ \$value }}" }} LABELS = {{ "{{ \$labels }}" }}	1. Summary: }}")))
		Severity: critic
		Realert Time: ;
		Expression: rate(alertm
PrometheusTargetScrapingSlow	Prometheus is scraping exporters slowly VALUE = {{ "{{ \$value }}" }} LABELS = {{ "{{ \$labels }}" }}	1. Summary:
		Severity: warn
		Realert Time: ;
		Expression: prometheus_
PrometheusTsdbCheckpointCreationFailures	Prometheus encountered checkpoint creation failures VALUE = {{ "{{ \$value }}" }} LABELS = {{ "{{ \$labels }}" }}	1. Summary: }}")))
		Severity: critic
		Realert Time: ;
		Expression: increase(pr

Alert Name	Description	Alert Info
PrometheusTsdbCheckpointDeletionFailures	Prometheus encountered checkpoint deletion failures VALUE = {{ "{{ \$value }}" }} LABELS = {{ "{{ \$labels }}" }}	1. Summary: <code>}}"</code> <code>}})</code> Severity: <code>critic</code> Realert Time: <code>;</code> Expression: <code>increase(pr</code>
PrometheusTsdbCompactionsFailed	Prometheus encountered TSDB compactions failures VALUE = {{ "{{ \$value }}" }} LABELS = {{ "{{ \$labels }}" }}	1. Summary: Severity: <code>critic</code> Realert Time: <code>;</code> Expression: <code>increase(pr</code>
PrometheusTsdbHeadTruncationsFailed	Prometheus encountered TSDB head truncation failures VALUE = {{ "{{ \$value }}" }} LABELS = {{ "{{ \$labels }}" }}	1. Summary: Severity: <code>critic</code> Realert Time: <code>;</code> Expression: <code>increase(pr</code>
PrometheusTsdbReloadFailures	Prometheus encountered TSDB reload failures VALUE = {{ "{{ \$value }}" }} LABELS = {{ "{{ \$labels }}" }}	1. Summary: Severity: <code>critic</code> Realert Time: <code>;</code> Expression: <code>increase(pr</code>

Alert Name	Description	Alert Info
PrometheusTsdBWalCorruptions	Prometheus encountered TSDB WAL corruptions VALUE = {{ "{{ \$value }}" }} LABELS = {{ "{{ \$labels }}" }}	1. Summary: Severity: critic Realert Time: 5m Expression: <pre>increase(prometheus_tsdb_wal_corruptions_total[5m]) > 1</pre>
PrometheusTsdBWalTruncationsFailed	Prometheus encountered TSDB WAL truncation failures VALUE = {{ "{{ \$value }}" }} LABELS = {{ "{{ \$labels }}" }}	1. Summary: Severity: critic Realert Time: 5m Expression: <pre>increase(prometheus_tsdb_wal_truncation_failures_total[5m]) > 1</pre>
HTTPServerError5xxPercentageTooHigh	HTTP server error 5xx percentage is too high. VALUE = {{ "{{ \$value }}" }} LABELS = {{ "{{ \$labels }}" }} TROUBLESHOOTING = Use app_type, pod and namespace labels to check logs	1. Summary: \$labels.pod } Severity: major Realert Time: 5m Expression: <pre>sum without (rate(http_server_status_5xx[5m]) / rate(http_server_status_total[5m])) > 0.05</pre>

Alert Name	Description	Alert Info
HTTPServerError4xxPercentageHigh	HTTP server error 4xx percentage is high. VALUE = {{ "{{ \$value }}" }} LABELS = {{ "{{ \$labels }}" }} TROUBLESHOOTING = Use app_type, pod and namespace labels to check logs	1. Summary: \$labels.pod } Severity: major Realert Time: 5m Expression: <pre>sum without ((rate(http_4xx_total{ app_type!="api", rate(http_server_status{ .Values.http_4xx_total</pre>
HTTPClientRespRcvd5xxPercentageTooHigh	HTTP client received response error 5xx percentage is too high. VALUE = {{ "{{ \$value }}" }} LABELS = {{ "{{ \$labels }}" }} TROUBLESHOOTING = Use app_type, pod and namespace labels to check logs	1. Summary: Application pod Severity: major Realert Time: 5m Expression: <pre>sum without ((rate(http_client_resps{ [15m]) / ig rate(http_client_resps{ .Values.http_client_resps</pre>

Alert Name	Description	Alert Info
HTTPClientRespRcvd460N499PercentageHigh	HTTP client received response error 460 and 499 percentage is high. VALUE = {{ "{{ \$value }}" }} LABELS = {{ "{{ \$labels }}" }} TROUBLESHOOTING = Use app_type, pod and namespace labels to check logs	1. Summary: Application p
		Severity: majo
		Realert Time: {
		Expression:
		sum without ((rate(http_ [15m]) / ig rate(http_a .Values.htt
HTTPClientSendDataError	HTTP client observing error in sending data over connection. VALUE = {{ "{{ \$value }}" }} LABELS = {{ "{{ \$labels }}" }} TROUBLESHOOTING = Use pod and namespace labels to check logs	1. Summary: - {{ "{{ \$label
		Severity: majo
		Expression:
		increase(ht
HTTPPuntingResponse5xxPercentageTooHigh	HTTP punting response 5xx percentage is too high. VALUE = {{ "{{ \$value }}" }} LABELS = {{ "{{ \$labels }}" }}	1. Summary:
		Severity: majo
		Realert Time: {
		Expression:
		sum without ((sum_over_ [15m]) / ig sum_over_ti .Values.htt

Alert Name	Description	Alert Info
HTTTPuntingResponse408PercentageTooHigh	HTTP punting response 408 percentage is too high.	1. Summary:
	VALUE = {{ "{{ \$value }}" }} LABELS = {{ "{{ \$labels }}" }}	Severity: major Realert Time: 10m Expression: <pre>sum without (sum_over_ / ignoring(sum_over_ti .Values.htt</pre>
HTTTPuntingSuccessResponseLatencyTooHigh	HTTP punting success response latency is too high.	1. Summary:
	VALUE = {{ "{{ \$value }}" }} LABELS = {{ "{{ \$labels }}" }}	Severity: major Realert Time: 10m Expression: <pre>histogram_q sum(rate(pu canaryVersi .Values.htt</pre>

Alert Name	Description	Alert Info
IngressTLSCommunicationFailureAlert	This alert indicates that the Istio Ingress Gateway is facing TLS communication failures while establishing TLS sessions from its clients. The alert is based on failure metrics documented here ["https://www.envoyproxy.io/docs/envoy/latest/configuration/listeners/stats#tls-statistics"]. Refer to job label to identify which gateway is reporting the failures in its communication.	1. Summary: Severity: {{ .Value }} Realert Time: {} Expression: <pre>increase(in .Values.tls</pre>
EgressTLSCommunicationFailureAlert	This alert indicates that there are TLS communication failures at the Istio Egress Gateways(including LI gateway). This could indicate certificates misconfiguration in the gateways or at the TLS peers. This alert is based on failure metrics documented here ["https://www.envoyproxy.io/docs/envoy/latest/configuration/upstream/cluster_manager/cluster_stats#tls-statistics"]. Refer to job label to identify which gateway is reporting the failures in its communication.	1. Summary: Severity: {{ .Value }} Realert Time: {} Expression: <pre>increase(eg .Values.tls</pre>

Alert Name	Description	Alert Info
PrometheusErrorSendingAlertsToAnyAlertmanager	value minimum errors while sending alerts from Prometheus labels.namespace/labels.pod to any Alertmanager.	1. Summary: Alertmanage ** Expression: min without rate(promet core",names rate(promet core",names

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- Summary
- ss_nnssf_nsselection_success_rate

Summary

Metric Name	Type	Synopsis
ss_nnssf_nsselection_success_rate	counter	Slice Selection successful rate

ss_nnssf_nsselection_success_rate

Description

The slice selection procedure successful rate. This counter indicates the average slice selection successful rate on the NSSF in the last 5 minutes

Type

counter

Prometheus Expression

((sum(increase(ss_nnssf_nsselection_response_for_registration[5m])) - sum(increase(ss_nnssf_nsselection_response_for_registration{status=~"5.*"}[5m]))) / sum(increase(ss_nnssf_nsselection_response_for_registration[5m])) * 100) >= 0 or on() vector(100)

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- Summary
- pfcp_n4_interface_received
- pfcp_n4_interface_sent
- smfcc_chf_interface_timeout_count

- [smfcc_namf_n11_interface_received](#)
- [smfcc_namf_n11_interface_sent](#)
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- [smfcc_nchf_n40_interface_sent](#)
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- [smfcc_pcf_n7_interface_sent](#)
- [smfcc_pgw_bearer_setup_duration_count_total](#)
- [smfcc_pgw_bearer_setup_duration_sum_total](#)
- [smfcc_pgw_bearer_setup_rate_last_15_min](#)
- [smfcc_saegw_bearer_setup_duration_count_total](#)
- [smfcc_saegw_bearer_setup_duration_sum_total](#)
- [smfcc_saegw_bearer_setup_rate_last_15_min](#)
- [smfcc_udm_interface_timeout_count](#)
- [smfcc_udm_n10_interface_received](#)
- [smfcc_udm_n10_interface_sent](#)

Summary

Metric Name	Type	Synopsis
pfcf_n4_interface_received	counter	Sum count of all messages received over N4 interface
pfcf_n4_interface_sent	counter	Sum count of all messages sent over N4 interface
smfcc_chf_interface_timeout_count	count	Display the total number of response timeouts on chf interface for smf initiated messages
smfcc_namf_n11_interface_received	counter	Sum count of all messages received on N11 interface
smfcc_namf_n11_interface_sent	counter	Sum count of all messages sent on N11 interface
smfcc_nchf_n40_interface_received	counter	Sum count of all messages received on N40 interface
smfcc_nchf_n40_interface_sent	counter	Sum count of all messages sent on N40 interface
smfcc_nef_n29_interface_received	counter	Sum count of all messages received on N29 interface
smfcc_nef_n29_interface_sent	counter	Sum count of all messages sent on N29 interface
smfcc_pcf_interface_timeout_count	count	Display the total number of response timeouts on pcf interface for smf initiated messages
smfcc_pcf_n7_interface_received	counter	Sum count of all messages received on N7 interface
smfcc_pcf_n7_interface_sent	counter	Sum count of all messages sent on N7 interface
smfcc_pgw_bearer_setup_duration_count_total	counter	Sum of all the 4G PGW bearer setup duration
smfcc_pgw_bearer_setup_duration_sum_total	counter	Sum of all the 4G PGW bearer setup duration
smfcc_pgw_bearer_setup_rate_last_15_min	count	Rate of all the 4G PGW bearer setup duration in last 15 min
smfcc_saegw_bearer_setup_duration_count_total	counter	Sum of all the 4G SAEGW bearer setup duration
smfcc_saegw_bearer_setup_duration_sum_total	counter	Sum of all the 4G SAEGW bearer setup duration
smfcc_saegw_bearer_setup_rate_last_15_min	count	Rate of all the 4G SAEGW bearer setup duration in last 15 min
smfcc_udm_interface_timeout_count	count	Display the total number of response timeouts on udm interface for smf initiated messages
smfcc_udm_n10_interface_received	counter	Sum count of all messages received on N10 interface
smfcc_udm_n10_interface_sent	counter	Sum count of all messages sent on N10 interface

pfcf_n4_interface_received

Description

This counter indicates the count of total messages received on the N4 interface

Type

counter

Prometheus Expression

sum (pfc_p_association_setup_request_received OR on() vector(0)) + sum (pfc_p_association_setup_retransmitted_request_received OR on() vector(0)) + sum (pfc_p_association_update_request_received OR on() vector(0)) + sum (pfc_p_association_update_retransmitted_request_received OR on() vector(0)) + sum (pfc_p_association_release_request_received OR on() vector(0)) + sum (pfc_p_association_release_retransmitted_request_received OR on() vector(0)) + sum (pfc_p_association_setup_response_received OR on() vector(0)) + sum (pfc_p_association_update_response_received OR on() vector(0)) + sum (pfc_p_association_release_response_received OR on() vector(0)) + sum (pfc_p_version_not_supported_response_received OR on() vector(0)) + sum (pfc_p_heartbeat_request_received OR on() vector(0)) + sum (pfc_p_heartbeat_retransmitted_request_received OR on() vector(0)) + sum (pfc_p_heartbeat_response_received OR on() vector(0)) + sum (pfc_p_session_establishment_response_received OR on() vector(0)) + sum (pfc_p_session_modification_response_received OR on() vector(0)) + sum (pfc_p_session_deletion_response_received OR on() vector(0)) + sum (pfc_p_version_not_supported_response_received OR on() vector(0)) + sum (pfc_p_session_report_request_sent OR on() vector(0)) + sum (pfc_p_session_report_retransmitted_request_sent OR on() vector(0)) + sum (pfc_p_pfdm_request_received OR on() vector(0)) + sum (pfc_p_pfdm_retransmitted_request_received OR on() vector(0)) + sum (pfc_p_pfdm_response_received OR on() vector(0))

pfc_p_n4_interface_sent

Description

This counter indicates the count of total messages sent on the N4 interface

Type

counter

Prometheus Expression

sum (pfc_p_association_setup_request_sent OR on() vector(0)) + sum (pfc_p_association_setup_retransmitted_request_sent OR on() vector(0)) + sum (pfc_p_association_update_request_sent OR on() vector(0)) + sum (pfc_p_association_update_retransmitted_request_sent OR on() vector(0)) + sum (pfc_p_association_release_request_sent OR on() vector(0)) + sum (pfc_p_association_release_retransmitted_request_sent OR on() vector(0)) + sum (pfc_p_heartbeat_request_sent OR on() vector(0)) + sum (pfc_p_heartbeat_retransmitted_request_sent OR on() vector(0)) + sum (pfc_p_heartbeat_response_sent OR on() vector(0)) + sum (pfc_p_session_establishment_request_sent OR on() vector(0)) + sum (pfc_p_session_establishment_retransmitted_request_sent OR on() vector(0)) + sum (pfc_p_session_modification_request_sent OR on() vector(0)) + sum (pfc_p_session_modification_retransmitted_request_sent OR on() vector(0)) + sum (pfc_p_session_deletion_request_sent OR on() vector(0)) + sum (pfc_p_session_deletion_retransmitted_request_sent OR on() vector(0)) + sum (pfc_p_session_report_response_received OR on() vector(0)) + sum (pfc_p_pfdm_request_sent OR on() vector(0)) + sum (pfc_p_pfdm_retransmitted_request_sent OR on() vector(0))

smfcc_chf_interface_timeout_count

Description

This counter indicates the total number of response timeouts on chf interface for smf initiated messages

Type

count

Prometheus Expression

sum (smfcc_nchf_create_session_failures{causecode=~"failure-response-408|failure-response-460|response-timed-out"} OR on() vector(0)) + sum (smfcc_nchf_update_session_failures{causecode=~"failure-response-408|failure-response-460|response-timed-out"} OR on() vector(0)) + sum (smfcc_nchf_release_session_failures{causecode=~"failure-response-408|failure-response-460|response-timed-out"} OR on() vector(0)) + sum (smfcc_nchf_one_time_report_failures{causecode=~"failure-response-408|failure-response-460|response-timed-out"} OR on() vector(0))

smfcc_namf_n11_interface_received

Description

This counter indicates the count of total messages received on the N11 interface

Type

counter

Prometheus Expression

```
sum (smfcc_nsmf_pdu_session_create_sm_context_attempts OR on() vector(0)) + sum (smfcc_nsmf_pdu_session_update_sm_context_attempts OR on() vector(0)) + sum (smfcc_nsmf_pdu_session_release_sm_context_attempts OR on() vector(0)) + sum (smfcc_nsmf_pdu_session_create_client_attempts OR on() vector(0)) + sum (smfcc_nsmf_pdu_session_update_attempts OR on() vector(0)) + sum (smfcc_nsmf_pdu_session_update_client_attempts OR on() vector(0)) + sum (smfcc_nsmf_pdu_session_release_attempts OR on() vector(0)) + sum (smfcc_nsmf_pdu_session_release_client_attempts OR on() vector(0)) + sum (smfcc_nsmf_pdu_session_retrieve_sm_context_attempts OR on() vector(0)) + sum (smfcc_namf_evts_event_notification_attempts OR on() vector(0)) + sum (smfcc_namf_comm_n1n2_message_transfer_successes OR on() vector(0)) + sum (smfcc_namf_comm_assign_ebi_message_successes OR on() vector(0)) + sum (smfcc_nsmf_pdu_session_sm_context_status_notify_successes OR on() vector(0)) + sum (smfcc_namf_evts_subscr_create_successes OR on() vector(0)) + sum (smfcc_namf_evts_subscr_modify_successes OR on() vector(0)) + sum (smfcc_namf_evts_subscr_delete_successes OR on() vector(0)) + sum (smfcc_namf_comm_n1n2_message_transfer_failures OR on() vector(0)) - sum (smfcc_namf_comm_n1n2_message_transfer_failures{causecode="failure-response-408"} OR on() vector(0)) - sum (smfcc_namf_comm_n1n2_message_transfer_failures{causecode="failure-response-460"} OR on() vector(0)) - sum (smfcc_namf_comm_n1n2_message_transfer_failures{causecode="response-timed-out"} OR on() vector(0)) + sum (smfcc_namf_comm_assign_ebi_message_failures OR on() vector(0)) - sum (smfcc_namf_comm_assign_ebi_message_failures{causecode="failure-response-408"} OR on() vector(0)) - sum (smfcc_namf_comm_assign_ebi_message_failures{causecode="failure-response-460"} OR on() vector(0)) - sum (smfcc_namf_comm_assign_ebi_message_failures{causecode="response-timed-out"} OR on() vector(0)) + sum (smfcc_nsmf_pdu_session_sm_context_status_notify_failures OR on() vector(0)) - sum (smfcc_nsmf_pdu_session_sm_context_status_notify_failures{causecode="failure-response-408"} OR on() vector(0)) - sum (smfcc_nsmf_pdu_session_sm_context_status_notify_failures{causecode="failure-response-460"} OR on() vector(0)) - sum (smfcc_nsmf_pdu_session_sm_context_status_notify_failures{causecode="response-timed-out"} OR on() vector(0)) + sum (smfcc_namf_evts_subscr_create_failures OR on() vector(0)) - sum (smfcc_namf_evts_subscr_create_failures{causecode="failure-response-408"} OR on() vector(0)) - sum (smfcc_namf_evts_subscr_create_failures{causecode="failure-response-460"} OR on() vector(0)) - sum (smfcc_namf_evts_subscr_create_failures{causecode="response-timed-out"} OR on() vector(0)) + sum (smfcc_namf_evts_subscr_modify_failures OR on() vector(0)) - sum (smfcc_namf_evts_subscr_modify_failures{causecode="failure-response-408"} OR on() vector(0)) - sum (smfcc_namf_evts_subscr_modify_failures{causecode="failure-response-460"} OR on() vector(0)) - sum (smfcc_namf_evts_subscr_modify_failures{causecode="response-timed-out"} OR on() vector(0)) + sum (smfcc_namf_evts_subscr_delete_failures OR on() vector(0)) - sum (smfcc_namf_evts_subscr_delete_failures{causecode="failure-response-408"} OR on() vector(0)) - sum (smfcc_namf_evts_subscr_delete_failures{causecode="failure-response-460"} OR on() vector(0)) - sum (smfcc_namf_evts_subscr_delete_failures{causecode="response-timed-out"} OR on() vector(0))
```

smfcc_namf_n11_interface_sent

Description

This counter indicates the count of total messages sent on the N11 interface

Type

counter

Prometheus Expression

```
sum (smfcc_namf_comm_n1n2_message_transfer_attempts OR on() vector(0)) + sum (smfcc_namf_comm_assign_ebi_message_attempts OR on() vector(0)) + sum (smfcc_nsmf_pdu_session_sm_context_status_notify_attempts OR on() vector(0)) + sum (smfcc_namf_evts_subscr_create_attempts OR on() vector(0)) + sum (smfcc_namf_evts_subscr_modify_attempts OR on() vector(0)) + sum (smfcc_namf_evts_subscr_delete_attempts OR on() vector(0)) + sum (smfcc_nsmf_pdu_session_create_sm_context_successes OR on() vector(0)) + sum (smfcc_nsmf_pdu_session_create_sm_context_failures OR on() vector(0)) + sum (smfcc_nsmf_pdu_session_update_sm_context_successes OR on() vector(0)) + sum (smfcc_nsmf_pdu_session_update_sm_context_failures OR on() vector(0)) + sum (smfcc_nsmf_pdu_session_release_sm_context_successes OR on() vector(0)) + sum (smfcc_nsmf_pdu_session_release_sm_context_failures OR on() vector(0)) + sum (smfcc_nsmf_pdu_session_retrieve_sm_context_successes OR on() vector(0)) + sum (smfcc_nsmf_pdu_session_retrieve_sm_context_failures OR on() vector(0))
```

smfcc_nchf_n40_interface_received

Description

This counter indicates the count of total messages received on the N40 interface

Type

counter

Prometheus Expression

```
sum (smfcc_nchf_notification_received OR on() vector(0)) + sum (smfcc_nchf_create_session_success OR on() vector(0)) + sum (smfcc_nchf_update_session_success OR on() vector(0)) + sum (smfcc_nchf_release_session_success OR on() vector(0)) + sum (smfcc_nchf_one_time_report_success OR on() vector(0)) + sum (smfcc_nchf_create_session_failures OR on() vector(0)) - sum (smfcc_nchf_create_session_failures{causecode="failure-response-408"} OR on() vector(0)) - sum (smfcc_nchf_create_session_failures{causecode="failure-response-460"} OR on() vector(0)) - sum (smfcc_nchf_create_session_failures{causecode="response-timed-out"} OR on() vector(0)) + sum (smfcc_nchf_update_session_failures OR on() vector(0)) - sum (smfcc_nchf_update_session_failures{causecode="failure-response-408"} OR on() vector(0)) - sum (smfcc_nchf_update_session_failures{causecode="failure-response-460"} OR on() vector(0)) - sum (smfcc_nchf_update_session_failures{causecode="response-timed-out"} OR on() vector(0)) + sum (smfcc_nchf_release_session_failures OR on() vector(0)) - sum (smfcc_nchf_release_session_failures{causecode="failure-response-408"} OR on() vector(0)) - sum (smfcc_nchf_release_session_failures{causecode="failure-response-460"} OR on() vector(0)) - sum (smfcc_nchf_release_session_failures{causecode="response-timed-out"} OR on() vector(0)) + sum (smfcc_nchf_one_time_report_failures OR on() vector(0)) - sum (smfcc_nchf_one_time_report_failures{causecode="failure-response-408"} OR on() vector(0)) - sum
```

(smfcc_nchf_one_time_report_failures{causecode="failure-response-460"} OR on() vector(0)) - sum (smfcc_nchf_one_time_report_failures{causecode="response-timed-out"} OR on() vector(0))

smfcc_nchf_n40_interface_sent

Description

This counter indicates the count of total messages sent on the N40 interface

Type

counter

Prometheus Expression

sum (smfcc_nchf_create_session_attempts OR on() vector(0)) + sum (smfcc_nchf_update_session_attempts OR on() vector(0)) + sum (smfcc_nchf_release_session_attempts OR on() vector(0)) + sum (smfcc_nchf_one_time_report_attempts OR on() vector(0)) + sum (smfcc_nchf_notification_success OR on() vector(0)) + sum (smfcc_nchf_notification_failures OR on() vector(0))

smfcc_nef_n29_interface_received

Description

This counter indicates the count of total messages received on the N29 interface

Type

counter

Prometheus Expression

sum (evtxep_notification_successes_received_from_nef OR on() vector(0)) + sum (evtxep_notification_failures_received_from_nef OR on() vector(0)) + sum (evtxep_nef_pfd_fetch_successes OR on() vector(0)) + sum (evtxep_nef_pfd_fetch_failures OR on() vector(0)) + sum (evtxep_nef_pfd_subscribe_successes OR on() vector(0)) + sum (evtxep_nef_pfd_subscribe_failures OR on() vector(0)) + sum (evtxep_nef_pfd_delete_subscribe_successes OR on() vector(0)) + sum (evtxep_nef_pfd_delete_subscribe_failures OR on() vector(0)) + sum (evtxep_nef_pfd_subscribe_notify_received OR on() vector(0))

smfcc_nef_n29_interface_sent

Description

This counter indicates the count of total messages sent on the N29 interface

Type

counter

Prometheus Expression

sum (evtxep_notification_requests_sent_to_nef OR on() vector(0)) + sum (evtxep_nef_pfd_fetch_requests_sent OR on() vector(0)) + sum (evtxep_nef_pfd_subscribe_requests_sent OR on() vector(0)) + sum (evtxep_nef_pfd_delete_subscribe_requests_sent OR on() vector(0)) + sum (evtxep_nef_pfd_subscribe_notify_successes OR on() vector(0)) + sum (evtxep_nef_pfd_subscribe_notify_failures OR on() vector(0))

smfcc_pcf_interface_timeout_count

Description

This counter indicates the total number of response timeouts on pcf interface for smf initiated messages

Type

count

Prometheus Expression

sum (smfcc_nsmf_smpolicy_control_update_service_failures{causecode=~"failure-response-408|failure-response-460|response-timed-out"} OR on() vector(0)) + sum (smfcc_nsmf_smpolicy_control_create_service_failures{causecode=~"failure-response-408|failure-response-460|response-timed-out"} OR on() vector(0)) + sum (smfcc_nsmf_smpolicy_control_delete_service_failures{causecode=~"failure-response-408|failure-response-460|response-timed-out"} OR on() vector(0))

smfcc_pcf_n7_interface_received

Description

This counter indicates the count of total messages received on the N7 interface

Type

counter

Prometheus Expression

```
sum (smfcc_nsmf_smpolicy_control_terminate_notify_attempts OR on() vector(0)) + sum (smfcc_nsmf_smpolicy_control_update_notify_attempts OR on() vector(0)) + sum (smfcc_nsmf_smpolicy_control_create_service_success OR on() vector(0)) + sum (smfcc_nsmf_smpolicy_control_create_service_failures OR on() vector(0)) - sum (smfcc_nsmf_smpolicy_control_create_service_failures{causecode="failure-response-460"} OR on() vector(0)) - sum (smfcc_nsmf_smpolicy_control_create_service_failures{causecode="failure-response-408"} OR on() vector(0)) - sum (smfcc_nsmf_smpolicy_control_create_service_failures{causecode="response-timed-out"} OR on() vector(0)) + sum (smfcc_nsmf_smpolicy_control_update_service_success OR on() vector(0)) + sum (smfcc_nsmf_smpolicy_control_update_service_failures OR on() vector(0)) - sum (smfcc_nsmf_smpolicy_control_update_service_failures{causecode="failure-response-460"} OR on() vector(0)) - sum (smfcc_nsmf_smpolicy_control_update_service_failures{causecode="failure-response-408"} OR on() vector(0)) - sum (smfcc_nsmf_smpolicy_control_update_service_failures{causecode="response-timed-out"} OR on() vector(0)) + sum (smfcc_nsmf_smpolicy_control_delete_service_success OR on() vector(0)) + sum (smfcc_nsmf_smpolicy_control_delete_service_failures OR on() vector(0)) - sum (smfcc_nsmf_smpolicy_control_delete_service_failures{causecode="failure-response-460"} OR on() vector(0)) - sum (smfcc_nsmf_smpolicy_control_delete_service_failures{causecode="failure-response-408"} OR on() vector(0)) - sum (smfcc_nsmf_smpolicy_control_delete_service_failures{causecode="response-timed-out"} OR on() vector(0))
```

smfcc_pcf_n7_interface_sent

Description

This counter indicates the count of total messages sent on the N7 interface

Type

counter

Prometheus Expression

```
sum (smfcc_nsmf_smpolicy_control_create_service_attempts OR on() vector(0)) + sum (smfcc_nsmf_smpolicy_control_update_service_attempts OR on() vector(0)) + sum (smfcc_nsmf_smpolicy_control_delete_service_attempts OR on() vector(0)) + sum (smfcc_nsmf_smpolicy_control_terminate_notify_success OR on() vector(0)) + sum (smfcc_nsmf_smpolicy_control_terminate_notify_failures OR on() vector(0)) + sum (smfcc_nsmf_smpolicy_control_update_notify_success OR on() vector(0)) + sum (smfcc_nsmf_smpolicy_control_update_notify_failures OR on() vector(0))
```

smfcc_pgw_bearer_setup_duration_count_total

Description

This counter indicates sum of all the 4G PGW bearer setup duration

Type

counter

Prometheus Expression

```
sum without (pod, qci, arp, snssai) (smfcc_pgw_bearer_setup_duration_count)
```

smfcc_pgw_bearer_setup_duration_sum_total

Description

This counter indicates sum of all the 4G PGW bearer setup duration

Type

counter

Prometheus Expression

sum without (pod, qci, arp, snssai) (smfcc_pgw_bearer_setup_duration_sum)

smfcc_pgw_bearer_setup_rate_last_15_min

Description

This counter indicates rate of all the 4G PGW bearer setup duration in last 15 min

Type

count

Prometheus Expression

sum without (pod, qci, arp, snssai) (rate(smfcc_pgw_bearer_setup_duration_sum[15m]))/sum without (pod, qci, arp, snssai) (rate(smfcc_pgw_bearer_setup_duration_count[15m]))

smfcc_saegw_bearer_setup_duration_count_total

Description

This counter indicates sum of all the 4G SAEGW bearer setup duration

Type

counter

Prometheus Expression

sum without (pod, qci, arp, snssai) (smfcc_saegw_bearer_setup_duration_count)

smfcc_saegw_bearer_setup_duration_sum_total

Description

This counter indicates sum of all the 4G SAEGW bearer setup duration

Type

counter

Prometheus Expression

sum without (pod, qci, arp, snssai) (smfcc_saegw_bearer_setup_duration_sum)

smfcc_saegw_bearer_setup_rate_last_15_min

Description

This counter indicates rate of all the 4G SAEGW bearer setup duration in last 15 min

Type

count

Prometheus Expression

sum without (pod, qci, arp, snssai) (rate(smfcc_saegw_bearer_setup_duration_sum[15m]))/sum without (pod, qci, arp, snssai) (rate(smfcc_saegw_bearer_setup_duration_count[15m]))

smfcc_udm_interface_timeout_count

Description

This counter indicates the total number of response timeouts on udm interface for smf initiated messages

Type

count
Prometheus Expression
sum (smfcc_udm_registration_failures{causecode=~"failure-response-408 failure-response-460 response-timed-out."} OR on() vector(0)) + sum (smfcc_sdm_get_sm_data_failures{causecode=~"failure-response-408 failure-response-460 response-timed-out."} OR on() vector(0)) + sum (smfcc_sdm_subscription_failures{causecode=~"failure-response-408 failure-response-460 response-timed-out."} OR on() vector(0)) + sum (smfcc_udm_deregistration_failures{causecode=~"failure-response-408 failure-response-460 response-timed-out."} OR on() vector(0))

smfcc_udm_n10_interface_received

Description

This counter indicates the count of total messages received on the N10 interface

Type

counter

Prometheus Expression

sum (smfcc_udm_registration_successes OR on() vector(0)) + sum (smfcc_udm_registration_failures OR on() vector(0)) - sum (smfcc_udm_registration_failures{causecode="failure-response-460" } OR on() vector(0)) - sum (smfcc_udm_registration_failures{causecode="failure-response-408"} OR on() vector(0)) - sum (smfcc_udm_registration_failures{causecode="response-timed-out"} OR on() vector(0)) + sum (smfcc_sdm_get_sm_data_successes OR on() vector(0)) + sum (smfcc_sdm_get_sm_data_failures OR on() vector(0)) - sum (smfcc_sdm_get_sm_data_failures{causecode="failure-response-460"} OR on() vector(0)) - sum (smfcc_sdm_get_sm_data_failures{causecode="failure-response-408"} OR on() vector(0)) - sum (smfcc_sdm_get_sm_data_failures{causecode="response-timed-out"} OR on() vector(0)) + sum (smfcc_sdm_subscription_successes OR on() vector(0)) + sum (smfcc_sdm_subscription_failures OR on() vector(0)) - sum (smfcc_sdm_subscription_failures{causecode="failure-response-408"} OR on() vector(0)) - sum (smfcc_sdm_subscription_failures{causecode="failure-response-460"} OR on() vector(0)) - sum (smfcc_sdm_subscription_failures{causecode="response-timed-out"} OR on() vector(0)) + sum (smfcc_cancel_sdm_subscription_successes OR on() vector(0)) + sum (smfcc_cancel_sdm_subscription_failures OR on() vector(0)) - sum (smfcc_cancel_sdm_subscription_failures{causecode="failure-response-408"} OR on() vector(0)) - sum (smfcc_cancel_sdm_subscription_failures{causecode="failure-response-460"} OR on() vector(0)) - sum (smfcc_cancel_sdm_subscription_failures{causecode="response-timed-out"} OR on() vector(0)) + sum (smfcc_udm_deregistration_successes OR on() vector(0)) + sum (smfcc_udm_deregistration_failures OR on() vector(0)) - sum (smfcc_udm_deregistration_failures{causecode="failure-response-408"} OR on() vector(0)) - sum (smfcc_udm_deregistration_failures{causecode="failure-response-460"} OR on() vector(0)) - sum (smfcc_udm_deregistration_failures{causecode="response-timed-out"} OR on() vector(0)) + sum (smfcc_udm_evt_exp_create_subscription_successes OR on() vector(0)) + sum (smfcc_udm_evt_exp_create_subscription_failures OR on() vector(0)) - sum (smfcc_udm_evt_exp_create_subscription_failures{causecode="failure-response-408"} OR on() vector(0)) - sum (smfcc_udm_evt_exp_create_subscription_failures{causecode="failure-response-460"} OR on() vector(0)) - sum (smfcc_udm_evt_exp_create_subscription_failures{causecode="response-timed-out"} OR on() vector(0)) + sum (smfcc_udm_evt_exp_modify_subscription_successes OR on() vector(0)) + sum (smfcc_udm_evt_exp_modify_subscription_failures OR on() vector(0)) - sum (smfcc_udm_evt_exp_modify_subscription_failures{causecode="failure-response-408"} OR on() vector(0)) - sum (smfcc_udm_evt_exp_modify_subscription_failures{causecode="failure-response-460"} OR on() vector(0)) - sum (smfcc_udm_evt_exp_modify_subscription_failures{causecode="response-timed-out"} OR on() vector(0)) + sum (smfcc_udm_evt_exp_delete_subscription_successes OR on() vector(0)) + sum (smfcc_udm_evt_exp_delete_subscription_failures OR on() vector(0)) - sum (smfcc_udm_evt_exp_delete_subscription_failures{causecode="failure-response-408"} OR on() vector(0)) - sum (smfcc_udm_evt_exp_delete_subscription_failures{causecode="failure-response-460"} OR on() vector(0)) - sum (smfcc_udm_evt_exp_delete_subscription_failures{causecode="response-timed-out"} OR on() vector(0))

smfcc_udm_n10_interface_sent

Description

This counter indicates the count of total messages sent on the N10 interface

Type

counter

Prometheus Expression

sum (smfcc_udm_registration_attempts OR on() vector(0)) + sum (smfcc_sdm_get_sm_data_attempts OR on() vector(0)) + sum (smfcc_sdm_subscription_attempts OR on() vector(0)) + sum (smfcc_cancel_sdm_subscription_attempts OR on() vector(0)) + sum (smfcc_udm_deregistration_attempts OR on() vector(0)) + sum (smfcc_udm_evt_exp_create_subscription_attempts OR on() vector(0)) + sum (smfcc_udm_evt_exp_modify_subscription_attempts OR on() vector(0)) + sum (smfcc_udm_evt_exp_delete_subscription_attempts OR on() vector(0))